

THE OXFORD HANDBOOK OF
**COMPUTATIONAL
ECONOMICS
AND
FINANCE**

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AND
FINANCE

Edited by
SHU-HENG CHEN, MAK KABOUDAN,
and
YE-RONG DU

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CHAPTER 1

COMPUTATIONAL ECONOMICS IN THE ERA OF NATURAL COMPUTATIONALISM

*Fifty Years after The Theory of Self-Reproducing
Automata*

SHU-HENG CHEN, MAK KABOUDAN, AND YE-RONG DU

1.1 AUTOMATA AND NATURAL COMPUTATIONALISM

Fifty years after the publication of his magnum opus, *The Theory of Self-Reproducing Automata*, von Neumann's influence over the entire scientific world in computing and computation has spanned more than half a century. That includes economics. Not only has the von Neumann machine dominated the development of computing machines during that time frame, but his attempt to develop a general theory of automata has also motivated and facilitated the following interdisciplinary conversations among scientists, social scientists, and computer scientists, and nowadays, even within the humanities. The latter phenomenon is known as *natural computationalism* or *pancomputationalism*, that is, the notion that everything is a computing system.¹

We edit this book in the era of natural computationalism or pancomputationalism. In this era, the word *computation* is, indeed, everywhere. Nowadays, the name of many disciplines such as computational biology (Nussinov 2015) and biological computation

¹ Pancomputationalism is a general view of *computationalism*. Computationalism basically asserts that the mind is a computing system. This assertion plays an important role in cognitive science and the philosophy of the mind (Piccinini 2009). Natural computationalism does not go without criticisms. The interested reader is referred to Dodig-Crnkovic (2013) and Piccinini (2015).

(Lamm and Unger 2011), computational chemistry (Jensen 2013) and chemical computation (Finlayson 2012; Varghese et al. 2015), and computational physics (Landau et al. 2011) and physical computation (Piccinini 2015) can accompany this word. Economics is not an exception, and we were introduced to both computational economics and economic computation years ago. Therefore, it is high time to reflect upon the *nature and significance of economics* in the light of natural computationalism, which is the motivation behind producing this book.²

The idea of natural computationalism or pancomputationalism is not entirely alien to economists, especially those who have been exposed to von Neumann's contribution to the theory of automata (von Neumann 1966).³ The starting point of von Neumann's work is not computers or computing systems but *automata*, which include natural automata and artificial automata. Only in the latter category is a unique characterization of *computing machines* given.

Of all automata of high complexity, computing machines are the ones that we have the best chance of understanding. (von Neumann 1966, 32).

After completing his work on game theory (von Neumann and Morgenstern 1944), his main pursuit was made to overarch the studies on both types of automata, as he clearly understood that this conversation can be mutually beneficial.

Natural organisms are, as a rule, much more complicated and subtle, and therefore much less well understood in detail, than are artificial automata. Nevertheless, some regularities, which we observe in the organization of the former may be quite instructive in our thinking and planning of the latter; and conversely, a good deal of our experiences and difficulties with our artificial automata can be to some extent projected on our interpretations of natural organisms. (von Neumann 1963, 288–289)

A typical demonstration is his final work on brains and computers (von Neumann 1958).

² Here, we use the title of the milestone book by Lionel Robbins (1898–1978) (Robbins 1935) for its two connections. First, the book defines *economics* as a science of choice, a very broad subject covering almost all disciplines of the social sciences. However, choice also provides the fundamental form of computation, the Boolean function (see also Chapter 24). Second, while Robbins did not consider a universal social science, the triggered discussion of his book did lead to such a possibility and the invention of the term *economic imperialism* later by Ralph Souter (1897–1946) (Souter 1933). Hence, this book signifies a contemporaneous view of this history of economic analysis: instead of the science of choice, we have the science of computation as a new universal platform for the social sciences, and in fact, the natural sciences as well.

³ Nonetheless, we realize that this magnum opus is still largely unfamiliar to economists, compared to his book on game theory (von Neumann and Morgenstern 1944). For example, in a book published in memory of John von Neumann's contribution to economics, his work on the general theory of automata is completely ignored (Dore, Goodwin, and Chakravarty 1989). Therefore, when writing his *Machine Dreams* (Mirowski 2002), Mirowski clearly pointed out,

Unlike in the cases of the previous two phases of his career, no single book sums up von Neumann's intellectual concerns in a manner that he felt sufficiently confident himself to prepare for publication. (139)

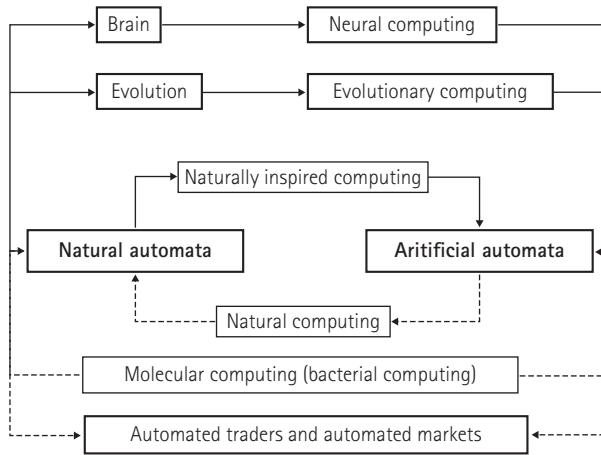


FIGURE 1.1 Natural automata and artificial automata.

Figure 1.1 represents the idea of automata in the context of modern natural computationalism. In the middle of the figure, we have the two automata and the lines between them showing their relation. As von Neumann indicated, on one hand, natural automata can inspire us regarding the design of artificial automata, or the so-called *naturally inspired computing*, as the Darwinian biological evolution has inspired evolutionary computation, neuroscience has inspired neural computing or artificial neural networks, and entomology has inspired ant-colony optimization, a part of swarm intelligence. Here, we attempt to understand the natural automata by being able to simulate them and then use the extracted algorithms as an initiative for the designs of computing systems or machines.

On the other hand, our scientific experiences of artificial automata may shed light on the underlying automata that generate the observed natural phenomena. This understanding can further endow us with the engineering knowledge required to *use* natural automata. As Amos et al. (2015) stated,

However, we can now go further than mere inspiration; instead of developing computing systems that are loosely modelled on natural phenomena, we can now directly *use* biological substrates and processes to encode, store and manipulate information. (1; *italics original*)

As shown in the bottom center panel of figure 1.1, this reverse direction is called *natural computing*, to be distinguished from *naturally inspired computing*. Profound examples are *molecular computing*, *bacterial computing* (Adleman 1994; Poet et al. 2010), and some of the forms of so-called *unconventional computing* (Dodig-Crnkovic and Giovagnoli 2012), including the well-known historical landmark the *Phillips machine* (see section 1.2).

Natural automata do not restrict themselves to the realm of natural sciences, although von Neumann (1966) might leave readers with that impression. Various behaviors, social structures, or organizations observed in entomology, zoology, psychology,

and the social sciences can enlarge the set of natural automata. This inclusiveness (pancomputationalism) enables us to see the relevance of a series of artificial automata developed in the area of economics, such as automated traders and automated markets. Both are inspired by their natural counterparts, namely, human traders and natural markets (marketplaces) and attempt to simulate them or, in many cases, outsmart them. On the other hand, we also see the development of engineering-enabling natural automata (Chapter 16).

Earlier, we mentioned that bacterial computing was a kind of natural computing. A closer look at bacterial computing shows that its essence is the massive use of natural forces, in this case, the use of *crowds*. Crowds can generate tremendous utilities, and we also make *swarm intelligence* part of naturally inspired computing toolkits. Nowadays, owing to various kinds of progress in Information and Communication Technology, such as Web 2.0, ubiquitous computing, the Internet of things, wearable devices, smart phones, and social media networks, we know better how to involve crowds in other forms of computing such as the design of a prediction market (a kind of automated futures market) to realize the wisdom of crowds and Amazon's Mechanical Turks (a kind of automated online labor market) to realize crowdsourcing. Furthermore, many new forms of automated markets such as Uber and Airbnb have been designed to enhance the discovery of trading opportunities and the success of matches. Some are not profit motivated, but they are equally important economically, since they ramp up peer production through volunteerism and other forms of pro-social behavior.

The natural computationalism reviewed above enables us to reflect on what computational economics and finance (CEF) is, and the rest of this chapter shall do so in this light. We begin with a conventional pursuit focusing more on the algorithmic or numerical aspect of CEF (section 1.3); we then move toward an automata or organism perspective of CEF (sections 1.4 and 1.5). In addition, we discuss computation or computing (section 1.3) and then turn to a view of computing systems (sections 1.4 and 1.5). However, doing so does not imply a chronological order between the two. In fact, the presence of both the Fisher machine and the Phillips machine informs us that the latter came to CEF no later than did the former (section 1.2).

1.2 COMPUTATIONAL ECONOMICS AS COMPUTING SYSTEMS

The history of computing tells us that the earliest computing machines in wide use were not digital but analog (Metropolis et al. 1980). Analog computers also played a role, albeit not an extensive one, in the early history of CEF. The prominent cases are the *Fisher machine* and the *Phillips machine*. Irving Fisher (1867–1947) was the first economist to apply analog computing to economics.⁴ In his dissertation, published

⁴ In 2005, the *American Journal of Economics and Sociology* devoted an entire issue to Irving Fisher. The Fisher machine was reviewed (Brainard and Scarf 2005; Tobin 2005). The interested reader is referred to the special issue for the details.

in 1892, Fisher presented his hydraulic-mechanical analog model in calculating the equilibrium prices and the resulting distribution of society's endowments among the agents in the economy composed of ten interrelated markets. This machine and its later version, proposed in 1925, became the forerunners of today's computable general equilibrium modeling (Chapter 5 in this book).

The second example is the Phillips machine, also known as *MONIAC*, an ingenious device invented by William Phillips (1914–1975), an electrical engineer turned economist, and his economist colleague Walter Newlyn (1915–2002). Its name stands for Monetary National Income Analogue Computer; this acronym was invented by its initial U.S. enthusiast, the economist Abba P. Lerner, in order to echo the early digital computer called ENIAC (*Fortune* 1952). Phillips demonstrated a physical model of the national economy as a series of white boxes (instead of a black box) in which each tank stands for an economic sector such as households, business, government, and exporting and importing, and colored water represents the flow of money. Although Phillips designed *MONIAC* for pedagogical purposes (Phillips 1950, 283), it is considered as a pioneering econometric computer:

The whole represented a system of nine differential equations. Ordinarily, at the time, such a system would be too complicated for anyone to solve. However, Phillips had ingeniously found an analogue solution. Not only that, but he calibrated the model for the UK economy, going as far as to estimate confidence intervals for the accuracy of results. (Bollard 2011, 4)

Oriented around monetary stocks and flows represented by colored water flowing around plastic pipes, *MONIAC* offered the opportunity for policy simulation exercises (Leeson 2000).⁵

A brief review of analog computing in economics via these two machines is not just for historical purposes, but mainly for arguing that the idea of CEF is not limited to computing per se. As a matter of fact, what these two machines demonstrated is the whole economy, regardless of the machines' being general equilibrium models or Keynesian macroeconomic models. Hence, in the earlier period the idea of CEF was about a computing system. This feature might, somehow, get lost in the later development of CEF, but it is regained in its recent extensions and becomes part of the skeleton of this handbook.

⁵ Phillips is also regarded as one of the pioneers who introduced dynamic control theory into macroeconomics when he constructed a simple model in order to illustrate the operation of stabilization policy (Phillips 1954). Later on, a more realistic version of this model was simulated on the DEUCE (Digital Electronic Universal Computing Engine) machine (Phillips 1957). An event celebrating the sixtieth anniversary of the Phillips National Income Electro-Hydraulic Analogue Machine was held by the Algorithmic Social Science Research Unit (ASSRU) at the University of Trento in December 2010. Allan McRobie (McRobie 2011) has demonstrated a few more macroeconomic simulations using the machine, and Kumararwamy Vela Velupillai (Velupillai 2011) has provided a deep reflection of analog computing by recasting the Phillips machine in the era of digital computing.

1.3 WHAT IS COMPUTED?

It has been more than sixty years since Phillips first demonstrated his MONIAC analog computing machine when the development of the computer was still in its embryonic stage. Since then, computing hardware has made substantial advances. The famous Moore's law provides an observation or conjecture regarding the advances in the semiconductor industry whereby chip density doubles every eighteen months. Although this growing speed is expected to slow down in the years to come due to the physical limitations of a chip, the recent use of massive parallelism combining high-throughput computing and graphics processor units (GPUs) will possibly prolong the exponential growth of computational speeds. The opportunity to apply such techniques to economics has also been addressed in the literature (Aldrich 2014).

The increase in computational power has broadened the range of what economists can compute.⁶ With the assistance of computation methods, some models in the standard economic framework that were previously considered as being intractable can now be solved both efficiently and reliably. The theoretical economic analysis based on standard models can now be extended from being *qualitative* to *quantitative* and from being *analytically tractable* to *computationally tractable*. Although computational analysis only offers approximate solutions, those approximations covers a far broader range of cases instead of being limited to only some special cases. It makes general theory appear to be much more comparable to reality while losing logical purity and inviting specification error as a tradeoff (Judd 1998).

From the computing or computation perspective, we address two basic questions in the first part of the handbook. First, what does computational economics intend to compute, and second, what kinds of economics make computation so hard? The first several chapters of this book provide the answers to the first question. What is computed in economics and finance are equilibrium (competitive equilibrium and general equilibrium), rational expectations, risk, and volatility. They are all fundamental concepts emanating naturally from the standard formalism of economics. In this regard, we deal with two mainstays of conventional computational economics, namely, *dynamic stochastic general equilibrium* (DSGE) models and *computational general equilibrium* (CGE) models. Chapters 2, 3, and 4 are mainly devoted to DSGE, and Chapter 5 is devoted to CGE. As for DSGE, Chapter 2 gives an overview, Chapter 3 provides a concrete application, and Chapter 4 goes further to discuss various computing approaches.

1.3.1 Dynamic Stochastic General Equilibrium

The dynamic stochastic general equilibrium is basically an equation-based approach to macroeconomics. It models the entire macroeconomy as a system of stochastic linear or

⁶ Judd (1998) provides a brief review of the potential gained from the application of computational methods.

nonlinear difference equations, with expectations as part of the system. The dynamics of the economy (the time series of the endogenous macroeconomic variables) depend on the expectations, and the expectations depend on the economic dynamics as well. This co-dependence constitutes a self-mapping between expectations and realizations, which motivates the idea of fixed points (rational expectations) as a characterization of rational agents. Two chapters of the book contribute to the computation of DSGE.

Chapter 2, “Dynamic Stochastic General Equilibrium Models: A Computational Perspective,” by Michel Juillard, provides a systematic treatment of a generic DSGE model from the estimation of the model to its solution. As for the estimation, the current practice is dominated by the Bayesian approach. Not only does it provide a balance between calibration and classical estimation, but it can effectively tackle the identification issue. This chapter provides a thorough review of various numerical approaches to Bayesian estimation, from the derivation of the posterior density to point estimation. On solving the DSGE model, the author reviews the perturbation approach from low-order approximations to high-order ones. The economic significance of the high-order Taylor expansions is addressed in terms of risk attitude and risk premium.

Chapter 2 is closely related to Chapter 4, “Solving Rational Expectations Models,” by Jean Barthélemy and Magali Marx, which provides a splendid review of the computation of rational expectations. The existence and uniqueness theorems of rational expectations are also well presented. The conditions under which multiple rational expectations can exist are also noted; interesting cases lead to sunspots. Of course, in more complex situations, these theorems are not always available. Barthélemy and Marx start from the simplest case (the benchmark), that is, linear rational expectations models, and show how this family of models can be tackled with three theoretically equivalent approaches. They then move to nonlinear rational expectations models and introduce the perturbation approach (the familiar linearization method). The limitation of the perturbation approach is addressed with respect to both the nature of economic shocks and the chaotic properties residing in small neighborhoods. A practical lesson gained from this discussion is that the constraint of the zero lower bound (ZLB) for nominal interest rates can render the perturbation approach inappropriate. Economic factors that can cause the presence of nonlinearity in the rational expectations models are nicely illustrated by the Markovian switching monetary policy and non-negativity of the nominal interest rate (the ZLB constraint). Barthélemy and Marx show how these intriguing cases of rational expectations models can be handled, including the use of global methods, for example, the projection method.

Chapters 2 and 4 are complementary to each other. They both present the perturbation method, but Chapter 4 has the first-order approximation as the main focus, whereas Chapter 2 systematically generalizes it to the second and higher orders. On the other hand, Chapter 2 focuses only on the perturbation method, whereas Chapter 4 not only gives other equivalent treatments of the perturbation method but also takes into account various global methods such as the projection method.

Chapter 3, “Tax-Rate Rules for Reducing Government Debt: An Application of Computational Methods for Macroeconomic Stabilization,” by G. C. Lim and Paul

McNelis, provides an application of Chapters 2 and 4. It demonstrates how the New Keynesian DSGE model is applied to address the design of fiscal policy related to macroeconomic adjustment from a high-debt state to a low-debt (zero-debt) state. The specific model consists of eighteen equations with only one exogenous shock attributed to government expenditure, eighteen endogenous variables, seven behavioral parameters, and six policy parameters. The DSGE model presented here involves a number of the elements reviewed in Chapters 2 and 4 that make the computation hard, such as the zero lower bound of the interest rate and wage rigidity. In order to solve this nonlinear DSGE model, the authors advocate the combined use of the perturbation method and the projection method by starting from the former as initial inputs (guesses and estimates) for the latter. As for the projection method, they suggest the use of neural networks as the kernel functions instead of the Chebyshev polynomials. This chapter, therefore, is also the first chapter to bring computational intelligence into this book. More on computational intelligence is covered in section 1.4.

1.3.2 Computational General Equilibrium

As we have mentioned, the early stage of computational economics can be represented by Irving Fisher's attempt to compute a Walrasian general equilibrium model (the Fisher machine), which is a precursor of what is currently known as the computational general equilibrium (CGE) model. The CGE model became the essential part of computational economics in its early days, which indicates that computational economics is merely just a collection of numerical and optimization techniques or just a part of operations research (OR).⁷ It has been an ambition of economists to have a structural representation of the economy. It is hoped that with this structural representation one can effectively understand the operation of the economy and have sound economic planning.

Such a serious attempt begins with the *input-output analysis* developed by Wassily Leontief (1906–1999) in the late 1930s (Leontief 1936, 1941), work for which he received the Nobel Prize in 1973 (also see section 1.5.1). The computable general equilibrium model pioneered by Leif Johansen (1930–1982) is a consolidated step in this attempt. His *Multi-Sectoral Study of Economic Growth* (Johansen 1960), normally rendered MSG, is recognized as the first work concerning CGE. Johansen's MSG model has had a profound influence on the development of the CGE literature uniquely characterized as Johansen's legacy. In 2010, the fiftieth anniversary of the publication of the original MSG model, the Lief Johansen Symposium, held in Oslo, commemorated his work, and the *Journal of Policy Modeling* has published a selection of papers presented at the symposium (Bjerkholt et al. 2016). In addition to Leif Johansen, other pioneers in CGE are Herbert Scarf (Scarf 1967, 1973), Dale Jorgenson (Hudson and

⁷ To the best of our knowledge, Thompson and Thore (1992) is the first book to use the title *Computational Economics*. The book reflects an OR perspective of computational economics, though the authors did mention the inclusion of general equilibrium theory in a future edition.

Jorgenson 1974), and two World Bank groups (Adelman and Robinson 1978; Taylor et al. 1980).⁸

Over the course of more than fifty years, CGE modeling has become the most active policy-oriented area in CEF. As pointed out by Ian Sue Wing and Edward J. Balistreri, the authors of Chapter 5, “Computable General Equilibrium Models for Policy Evaluation and Economic Consequence Analysis,” the CGE models’ key advantage is their ability to quantify complex interactions between a panoply of policy instruments and characteristics of the economy. In order to show what a CGE model is, the chapter starts with a canonical example. The application domains of CGE models have gradually expanded from public finance, economic development, industrial and trade policies, energy, the environment, and labor markets to greenhouse gas emissions, climate change, and natural disaster shocks. This reveals that the application domains have constantly been updated to deal with the pressing issues of the time, from the early policy dispute about trade liberalization to the recent global concerns with environmental calamities.

The chapter also reviews the technical modifications made to the canonical model because of either the need to answer challenging new issues such as greenhouse gas emissions mitigation or the response to progress in economic theory such as the monopolistic competition in trade. The augmented technical modifications therein include the development of intertemporal dynamic CGE, heterogeneous CGE (CGE with relevant heterogeneity of commodities, industries, and households), and finally a hybrid model incorporating bottom-up details (partial equilibrium activity analysis) in CGE models.

1.3.3 Computing Out-of-Equilibrium Dynamics

Before we proceed further, maybe this is a good place to make a brief remark and review other parts of the handbook. As Chapter 4 shows, the rational expectations models based on the device of *representative agents* are different from the expectations, learning, and adaptive behavior that *ordinary people* may face. We are not sure yet whether ordinary people will by all means solve the first-order moment equations or the Euler equations derived from an optimization formulation. Behavioral considerations from psychology, sociology, and other social sciences have motivated us and provided us with alternative formulations. In this case, the policy function as a solution for the moment equation becomes behavioral rules or heuristics that are constantly adapted over time.

In conventional computational economics, most economic systems have an equilibrium, and in this situation an essential part of economic theory refers to existence, uniqueness, and stability. However, given the “new kind of science” argued in Wolfram (2002), the equilibrium, despite its existence, may not be simple in the sense of computational irreducibility. Because of the new kind of science, many economists are

⁸ See Dixon and Rimmer (2016) for a historical account of this development.

computing the same things conventional economics is used for, except that these things are computationally irreducible. Having said that, this handbook presents two kinds of computational economics in relation to this, namely, the Walrasian kind and the Wolframian kind. Chapters 2 to 5 deal with the former and in Chapters 13 to 22 deal with the latter.

1.3.4 Volatility

In the literature, there are three approaches to financial volatility, namely, the deterministic approach, the stochastic approach, and, very recently, the multifractal approach. Chapter 6 is devoted to a general review of the multifractal approach, and Chapter 7 is devoted to the stochastic approach. Each of the three approaches has been developing for some time, and they have evolved into many different variants, which make it increasingly difficult to have a bird's-eye view. Under these circumstances, Chapter 6, “Multifractal Models in Finance: Their Origin, Properties, and Applications”, by Thomas Lux and Mawuli Segnon, provides a superb literature review enabling even those with minimum background knowledge to have a good grasp of these trees of volatility models. Their treatment of the multifractal model is both unique and applaudable. Since the idea of multifractals originated from the modeling of turbulent flow and energy dissipation in physics, the painstaking efforts made by the authors have enhanced the mobility of this knowledge from one valley to another valley. A survey like this allows the reader to have its original flavor of multifractals by tracing it to the pioneering work done by Benoit Mandelbrot (1924–2010) and hence can be better motivated with the intellectual origin of fractal geometry and its relevance to finance. Since the econometric estimation of the multifractal model of volatility can be computationally demanding, the authors review a number of computational methods and show how the simulated method of moments can be applied to make the estimation work more computationally tractable.

The Markov switching stochastic volatility (MSSV) model is continuously pursued in Chapter 7, “Particle Filters for Markov Switching Stochastic Volatility Models”, by Yun Bao, Carl Chiarella, and Boda Kang. In the MSSV model, volatility can be modeled using the state-space approach. However, since in general, when the state or the observation equation is nonlinear or has a non-Gaussian noise term, the usual analytical approach, such as the Kalman filter, is not applicable to estimate or track the state. In this case, the density estimation relies on a simulation approach. One frequently used approach is the sequential Markov Chain Monte Carlo (MCMC) approach, also known as *particle filtering*. The particle filtering method with its various extensions, such as the *auxiliary particle filters*, has been applied to the estimation of the MSSV model. After a short review of this literature, Chapter 7 proposes a new variant of auxiliary particle filters using the Dirichlet distribution to search for reliable transition probabilities rather than applying a multinormal kernel smoothing algorithm.

In sum, besides computing (estimating and forecasting) financial volatility, these two chapters can also be read as simulation-based methods in econometrics, and hence concern computational economics, specifically, computational econometrics (Gouriéroux and Monfort 1997).

1.4 NATURE-INSPIRED COMPUTING

The more familiar facet of computational economics, *computing*, involves the use of software (algorithms) and hardware (computers) to find out the answers to “standard” characterizations of economic systems such as competitive equilibria, rational expectations equilibria, risk, and volatility. However, in light of natural computationalism, the nature and significance of economics is not only about computing but about a *computing system*. The remaining chapters of the handbook concern these less familiar facets of computational economics. They focus not on an answer per se but on presenting entire computing systems. Each of them presents a computing system as either a representation of the universe or a part of the universe. These systems are naturally so complex that they are hard to comprehend and hence are frequently called *black boxes*, although they are fully transparent.

Nature-inspired computing or computational intelligence is a collection of fundamental pursuits of a good understanding of natural languages, neural systems, and life phenomena. It has three main constituents, namely, fuzzy logic, neural networks, and evolutionary computation. Each has accumulated a large number of economic and financial applications in the literature (Chen 2002a,b; Chen and Wang 2003; Chen et al. 2005; McNelis 2005; Kendrick et al. 2006; Chen et al. 2007; Brabazon and O’Neill 2008, 2009, 2010). Extensive coverage of this subject could take up another independent volume; hence, in this handbook, we only include a few representative chapters for their uniqueness and importance.

1.4.1 Genetic Programming

The idea of genetic programming (GP) can be traced back to Herbert Simon (1916–2001), the 1978 Nobel Laureate in Economics. In order to understand humans’ information-processing and decisionmaking, Simon and his colleague Allen Newell (1927–1992) co-founded a list processing language, the predecessor of LISP, one of the earliest languages in artificial intelligence. This language has been used to understand human problem-solving behavior in logic tasks and chess playing. It turns out that LISP not only generates computer programs but also provides a behavioral model of human information processing. It exemplifies a very different facet of computational economics at work in those days.

In the early 1960s Simon had already begun work on the automatic generation of LISP programs, a project called the *heuristic compiler* (Simon 1963). The human heuristic searching behavior, characterized by chunking and modularizing, can be automated and simulated by computers. In the late 1980s Nicahel Cramer and John Koza further endowed the automatic program generation process with a selection force driven by Darwinian biological evolution so that the automatically generated programs could become fitter and fitter in terms of some user-defined criteria (Cramer 1985; Koza 1989). This became what was later known as *genetic programming*.

An essential feature of genetic programming is that it is *model free*, not top-down but bottom-up. Using genetic programming, a researcher only needs to *incubate* models or rules by seeding some ingredients (see Chapter 8 for the details) and then leave these ingredients to self-organize and self-develop in a biologically inspired evolutionary process. It, therefore, brings in the idea of automata, automation, and automatic adaptation to economics, and since all these elements are reified in computer simulation, genetic programming or nature-inspired computation introduces a version of computational economics that is very different from the old-fashioned numerical stereotype (Chen et al. 2005).

“Chapter 8, “Economic and Financial Modeling with Genetic Programming: A Review”, by Clíodhna Tuite, Michael O’Neill, and Anthony Brabazon, provides a review of genetic programming in economics and finance. This is probably the only review available in the literature. It includes sixty-seven papers and hence gives a comprehensive and systematic treatment of the ways in which genetic programming has been applied to different economic and financial domains, such as the automatic evolving and discovering of trading rules, forecasting models, stock selection criteria, and credit scoring rules. On the basis of existing studies of stock markets, foreign exchange markets, and futures markets, the authors examine whether one can find profitable forecasting and trading rules. A key part of the review is based on the accumulated tests for the *efficient market hypothesis* or *adaptive market hypothesis* that take various factors such as frequencies, time periods, transaction costs, and adjusted risks into account. In addition to tree-based genetic programming, the review also covers three variants of GP, namely, linear GP, grammatical evolution, and genetic network programming, which are less well known to economists.

Genetic programming is developed on the LISP language environment, which is a construct of formal language theory (Linz 2006). Hence, when using genetic programming to generate trading rules and forecasting models, one can have a *linguistic interpretation* that is distinguished from the usual statistical interpretation. The linguistic interpretation, to some extent, is associated with the *history of ideas* (Bevir 1999) and *semiotics* (Trifonas 2015). The idea is that symbols or signs (in semiotics) within a system are both competing and cooperating with each other to generate a pattern (description) that best fits reality. A coarser description will be driven out by a finer, more precise description. From an agent-based modeling viewpoint (see section 1.5.4), actual patterns emerge from complex interactions of agents. What GP does here is recast the space of agents as the space of symbols (variables, terminals) and

replicate the patterns emerging from agents' interactions through symbols' interactions. This alternative linguistic interpretation does not prevent us from using GP to learn from a data-limited environment. Chapter 11, "Computational Spatiotemporal Modeling of Southern California Home Prices", by Mak Kaboudan, provides such an illustration.

This chapter deals with spatial interaction or the spatial dependence phenomenon, which is a typical subject in spatial econometrics. The author addresses possible spatiotemporal contagion effects in home prices. The data comprise home prices in six contiguous cities in southern California from 2010 to 2014. To deal with these data, we need dynamic spatial panel data models, which, based on Elhorst (2014), belong to the third-generation spatial econometric models, and there is no straightforward estimation method for this type of model. The daunting nature of this issue is acknowledged by the author. Therefore, the data-driven approach is taken instead of the model-driven approach. The models evolved from GP take into account both spatial dependence and spatial heterogeneity, the two main characterizations of spatial statistics or econometrics. Not surprisingly, the models discovered by GP are nonlinear, but their "linearized" version can be derived using the hints from the GP models. These linearized models, however, were generally outperformed by their nonlinear counterparts in out-of-sample forecasts.

1.4.2 Nature-Inspired Intelligence

Genetic programming is just one method (tool) in the grand family known as biologically inspired computing or nature-inspired computing. As a continuation of Chapter 8, Chapter 9, "Algorithmic Trading Based on Biologically Inspired Algorithms", by Vassilios Vassiliadis and Georgios Dounias, gives a comprehensive review of this family of tools. In addition to genetic programming, other methods reviewed in their chapter include genetic algorithms, swarm intelligence, and ant colony optimization. One can add more to this list such as particle swarm optimization, differential evolution, harmony search, bee algorithms, the firefly algorithm, the cuckoo search, bat algorithms, and the flower pollination algorithm. Not all of them are known to economists, nor are their applications to economics. The interested reader can find more of them in other books (Xing and Gao 2014; Yang 2014).

An essential general feature shared by this literature is that these methods demonstrate natural computing. Again, the point is not what the numerical solution is but how a natural process being concretized into a *computing system* can shed light on a complex problem and provide a phenomenological lift allowing us to see or experience the problem so that we can be motivated or inspired to find appropriate reactions (solutions) to the problem. As we shall see more clearly in section 1.5, solutions can be taken as emergent properties of these complex computing systems.

Nature-inspired computing facilitates the development of the idea of automation, automata, and self-adaptation in economics. A concrete demonstration made in this

chapter is the application of these ideas to build *robot traders* or *algorithmic trading systems*, which are currently involved in 30 percent to 70 percent of all transactions in the major exchanges of Europe and the United States (Beddington et al. 2012). The authors show how these biologically inspired algorithms can either individually or collectively solve portfolio optimization problems. Putting these ideas into practice actually causes *modern computational economics* to deviate from its numerically oriented predecessor and brings it back to its old form, characterized by computing systems, as manifested by Irving Fisher, William Phillips, Herbert Simon, and Allen Newell in their designed machines, grammars, or languages. These machines are, however, no longer just mechanical but have become biological; they are able to evolve and have “life”, fulfilling a dream pursued by von Neumann in his last twenty years.⁹

1.4.3 Symbiogenesis

As part of natural computationalism, computational economics is better perceived or interpreted as economic computing, and it is not just about computing, but, as mentioned earlier, the entire computing system. The recent advancement of information and communication technology (ICT) and the ICT-enabled digital society further makes it clear that the nature of modern computational economics is not only a computing system but an evolving system. Chapter 10, “Algorithmic Trading in Practice,” by Peter Gomber and Kai Zimmermann, provides the best illustration of this larger picture.

On one hand, Chapter 10 can be read as a continuation of Chapter 9. It draws our attention to the information-rich environment underpinning algorithmic trading. The ICT-enabled environment reshapes the way in which data are generated and results in a new type of data, known as *unstructured data*, to be distinguished from the conventional structured data. Unstructured data, popularly known as *big data*, are data in the linguistic, photographic, audio, and video forms. Computational intelligence such as a neural network or support vector machine is applied in order to extract information and knowledge from these data, a process known as text mining, sentiment analysis, and so on. This chapter briefly highlights the incorporation of this algorithmic innovation to algorithmic trading.

On the other hand, Chapter 10 is not so much about the technical aspect of algorithmic trading as is Chapter 9; instead, written in an unconventional way, it treats algorithm trading as an emerging (evolving) concept useful when the market is no longer metaphorized as a computing system but has actually been a computing system, thanks to the progress in ICT. Chapter 10 discusses many features demonstrated by the ICT-enabled digital society. In addition to automated trading, it includes investors’ direct and quick access to the market (low latency), and hence the disappearance of human intermediaries. This feature is not restricted to financial markets but is largely

⁹ There is a stream of the literature known as *evolvable hardware*, which is carrying on this pursuit (Trefzer and Tyrrell 2015).

applicable to the whole economy. Neologisms created for this new face of the economy include the *sharing economy* (Horton and Zeckhauser 2016), which is characterized by Uber, Airbnb, TaskRabbit, RelayRides, and Rocket Internet, among others. In all these cases, we see how information can be quickly aggregated and processed, and how that can help decision making and matching. Hence, from this viewpoint, we are still in a long evolutionary process of economic computing.

Chapter 10 also reviews the incident known as the “Flash Crash” of May 6, 2010, when, within twenty minutes, the Dow plummeted 9 percent and largely recovered the loss. This chapter brings readers’ attention to the potential threat of machines to human well-being. Related questions it evokes are, under the incessant growth of automation, what the relations between men and machines are and how men will survive and evolve (Brynjolfsson and McAfee 2014). Fortunately, thanks to the Flash Crash, men may never be replaced by machines (Beddington et al. 2012), and computational economics will become a cyborg science (Mirowski 2002) or develop with the companionship of symbiogenesis (Kozo-Polyansky et al. 2010).

1.4.4 Fuzzy Logic

Whereas most computations invoked by economic models are carried out either with numerical values or with Boolean values, real-life economic decisions are filled with computations with natural languages. Currently, it is still a challenging mission to understand the computational process underpinning our uses of natural languages. How will our cognition and decision making be affected if a modifier, say, “very,” is added to our expressions or is repeated: “very very”? What is the foundation for extracting psychological states from stakeholders through their use of natural languages?¹⁰ How our minds generate texts as outputs and how text inputs affect the operation of our minds, that is, the underpinnings of the entire feedback loop, including behavioral, psychological, and neurophysiological mechanisms, are so far not well understood. Yet various models of natural language processing have been developed as the backbones of modern automatic information extraction methods.¹¹ Computation involving natural languages is the topic of Chapter 12, “Business Applications of Fuzzy Logic,” by Petr Dostál and Chia-Yang Lin, which provides a tutorial kind of review of fuzzy logic.

Fuzzy logic, as a kind of multivalued logic (Malinowski 1993), is a computational model of natural languages proposed by Lofti Zadeh. In addition to a brief mention of some contributions by Zadeh that shaped the later development of fuzzy theory, Chapter 12 focuses on how fuzzy logic walks into economics. The authors’ review begins with a number of pioneers, specifically Claude Ponsard (1927–1990) and his

¹⁰ See, e.g., the Hedonometer project (<http://www.hedonometer.org>) and the World Well-Being project (<http://wwbp.org/>).

¹¹ A useful survey can be found in Cioffi-Revilla (2013), Chapter 3.

foundational work of bringing impreciseness to economic theory. One fundamental departure brought about by the fuzzy set is that the *preference relation* among alternatives is not precise; for example, both of the statements “A is preferred to B” and “B is preferred to A” are a matter of degree.

The tutorial part of the chapter proceeds with a number of illustrations of the economic and financial applications of fuzzy logic such as risk evaluation, economic prediction, customer relations management, and customer clustering, accompanied by the use of software packages such as *fuzzyTech* and MATLAB’s Fuzzy Logic Toolbox.

1.5 NETWORKS AND AGENT-BASED COMPUTING

Six recent developments in computational economics and finance are networks, econophysics, designs, agent-based computational economics, neurosciences, and the epistemology of simulation. Most of these subjects have not been well incorporated into the literature about CEF; agent-based computational economics is probably the only exception. Although each of these developments may deserve a lengthy separate treatment, the chapters on each of them can at least allow readers to have a comprehensive view of the forest.

1.5.1 Networks

Social and economic networks have only very recently drawn the attention of macroeconomists, but networks have a long history in economics. The idea of using a network as a representation of the whole economy started with Quesnay’s *Tableau Economique* of 1758, which depicted the circular flow of funds in an economy as a network. Quesnay’s work inspired the celebrated input-output analysis founded by Wassily Leontief in the 1950s (Leontief 1951) which was further generalized into the social accounting matrices by Richard Stone (1913–1991) in the 1960s (Stone 1961). This series of developments forms the backbone of computable general equilibrium analysis, a kind of applied micro-founded macroeconomic model pioneered by Herbert Scarf in the early 1970s (Scarf 1973).

These network representations of economic activities enable us to see the interconnections and interdependence of various economic participants. This visualization helps us address the fundamental issue in macroeconomics, that is, how disruption propagates itself from one participant (sector) to others through the network. Nowadays, network analysis is applied not only to examine the transmission of information regarding job opportunities, trade relationships, the spread of diseases, voting patterns, and which languages people speak but also to empirical works such as the World Wide Web, the Internet, ecological networks, and co-authorship networks.

From a computational economics viewpoint, network architecture or topology is indispensable not only for conceptualizing but also for carrying out economic computation. It can be applied to both hardware and software. It can be applied to both the standard computer (computation) such as the von Neumann machine or the Turing machine (those silicon-based, semiconductor-based computers) and, probably even more actively, to the nonstandard (unconventional), concurrent, agent-based computer (computation) (Dodig–Crnkovic and Giovagnoli 2012).

If we perceive a society as a computing machine and ask if the society can ever generate cooperative behavior as its outcome,¹² we certainly need to know how the social machine actually computes and what the built-in algorithm is. In fact, one of the most important findings from spatial game theory is that a cooperative outcome consists of rather robust results with respect to a large class of network topologies, either exogenously determined or endogenously evolved. In a sense, nature does compute cooperation (Nowak 2006; Nowak and Highfield 2011). Similarly, if nature also computes efficiency, then there is a network structure underpinning the market operation that delivers such an outcome. The network structure has long been ignored in formal economic analysis; however, if we want to study economic computation, then the network variable, either as architecture or as an algorithm, has to be made explicitly, and that is the purpose of the next two chapters, Chapters 13 and 14.

Both chapters treat the economic process as a computation process in which computation is carried out by a given network topology (computer architecture), and they address the way the computation process or result is affected by that topology. In Chapter 13, “Modeling of Desirable Socioeconomic Networks,” by Akira Namatame and Takanori Komatsu, the computational process is a diffusion process. Depending on the issue concerning us, for example, the spread of epidemic, technological adoptions, or bank runs, different diffusion processes have different implications for welfare. The diffusion process depends not only on the network topologies but also on the behavioral rules governing each agent (circuit gate). It is likely that the same characteristics of networks, such as density, cluster coefficients, or centrality, may have different effects on the diffusion process if the underlying behavioral rules are different. Since agents are autonomous, *designing* their behavioral rules is infeasible; instead, we can only consider various designs by taking into account the given behavioral rules. This view of the *economic computing of network design* is demonstrated in Chapter 13, which addresses the design with respect to two simple behavioral rules, the probabilistic behavioral rule and the threshold rule. The challenges regarding the centralized design and the decentralized design are also discussed.

Chapter 14, “Computational Models of Financial Networks, Risk, and Regulatory Policies,” by Kimmo Soramäki, focuses on a domain-specific diffusion process, namely, financial networking. Contagions, cascades, and emergent financial crises constitute

¹² This view has already been taken by Richard Goodwin (Goodwin 1951): “Therefore it seems entirely permissible to regard the motion of an economy as a process of computing answers to the problems posed to it.” (2).

areas of interest not only to academia but also to politicians and the public. The very foundation of the scientific and policy debates about an institution's being "too 'something' to fail" or "too 'something' to save" is rooted in our understanding of the endogenous evolution of network topologies (Battiston et al. 2012). The chapter provides a comprehensive look at the use of network theory in financial systems.

Soramäki reviews two streams of the literature concerning financial networks, namely, interbank payment networks and interbank exposure networks. The author makes it clear that the financial systems have their uniquenesses and, in order to meaningfully construct financial networks, a straightforward application of the existing network theory alone is insufficient. The uniquenesses of financial systems have brought some new elements to network research. First, network research motivates a different class of network topologies, known as the *core-peripheral network*. Second, it demands new metrics for measuring the impact of nodes from the viewpoint of system vulnerability. Third, the network is evolving because each node (financial institution) is autonomous. In order to study the complex evolution of this network, evolutionary game theory and agent-based modeling, which take the learning, adaptive, and strategic behavior into account, become pertinent. The chapter provides an excellent look at each of these three lines of development alongside the review of the literature concerning financial networks.

1.5.2 Econophysics

The relation between economics and physics has a long history and has evolved into the relatively new field of *econophysics* (Chen and Li 2012). Economists can learn not only analytical concepts and mathematical formalisms from physics but, more important, the philosophy of science as well. For the latter, one important issue is the *granularity of economic modeling*. When high-performing computation becomes possible, one can handle not only larger and larger systems, but also finer and finer details of the components of the systems. In other words, computation technology enhances our economic studies at both the macroscopic and the microscopic level. However, as long as the computing power is not unlimited, the conflict between the resources allocated to macroscopic integration and to microscopic articulation always exists. For example, consider construction of a market with one million agents, where all are homogeneous with minimal intelligence, as opposed to construction of a market with one thousand agents where all are heterogeneous in their personal traits. This may not have been an issue in the past, but when advances in ICT make both one-million-agent modeling and big data possible, such a conflict becomes evident.

Chapter 15, "From Minority Games to \$-Games," by Jorgen Vitting Andersen, provides an econophysicist's thinking about granularity. He states, "The question, then, was how much detail is really needed to describe a given system properly?" His answer is the familiar *parsimony principle*; basically, many microscopic details are irrelevant for understanding macroscopic patterns and hence can be neglected. He, however,

motivated this hallmark principle with a brief review of the essential spirit and history of *renormalization group theory* in physics. He then illustrates this principle by addressing what the minimal (agent-based) model for the financial market is and introduces his \$-games, which he adapted from the El Farol Bar game originated by Brian Arthur and taken up by physicists in the form of *minority games*.

As an agent-based financial market (see Chapter 21), the \$-game provides an explanation for predictable financial phenomena, which is also known as the “edge of the chaos” or Class IV in the taxonomy of dynamic systems (Wolfram 2002). Hence, the dynamics of financial markets are not entirely unpredictable; there is a domain of attraction in which the law governing the future depends on the recent but not immediate past. The formation of this attractor lies in two parts: first, agents’ trading strategies depend on the recent but not immediate past (decoupled strategies); second, there are paths which can synchronize agents’ strategies toward this set of strategies and make them altogether become decoupled agents.

This mechanism, once established, can also be used to account for financial bubbles. Andersen actually tests his model with real data as well as laboratory data with human subjects. The empirical work involves the estimation of the agent-based model or so-called reverse engineering. Andersen also reviews some pioneering work on reverse engineering done in the minority game and the \$-game but for a more extensive review of reverse engineering involving agent-based models, the interested reader is referred to Chen et al. (2012).¹³

1.5.3 Automated Market Mechanism Designs

In the context of automata and pancomputationalism, section 1.1 mentions the idea of automated traders and automated markets as natural expectations of the development of automata as depicted in figure 1.1. Both of these artificial automata have taken up a substantial part of modern computational economics. The idea of using automated traders (artificial agents) in economic modeling has a long history, depending on how we define *artificial agents*.¹⁴ If we mean *human-written programs* or, using the popular

¹³ There is a subtle difference between reverse engineering and the statistical estimation of agent-based models. For the former, the data are generated by the agent-based model with a given set of parameter values, and the purpose of reverse engineering is to see whether these parameter values can be discovered by any statistical or machine-learning approach. For the latter, the data are the real-world data, but they are assumed to be generated by the agent-based model with unknown parameters, and statistical or machine-learning tools are applied to estimate the models. Most of the work surveyed in Chen et al. (2012) is of the latter type, the only work belonging to the former is Chen et al. (2010), who found that statistical estimation can recover the given parameter values only poorly.

¹⁴ It also depends on how we name it. In the literature, artificial agents have also been called artificial opponents (Roth and Murnighan 1978), programmed buyers (Coursey et al. 1984), and computer-simulated buyers (Brown-Kruse 1991). In addition, the terms artificial traders, programmed traders, robot traders, and programmed trading have been extensively applied to electronic commerce. The interested reader is referred to Smith (2002), MacKie-Mason and Wellman (2006), and Beddington et al. (2012).

term, *avatars*, then we can trace their origin back to the use of the *strategy method*, initiated by Reinhard Selten (Selten 1967), in experimental economics.

In order facilitate the elicitation of strategies (trading programs) from subjects, human subjects were allowed to have on-site laboratory experiences in the first phase of the experiment and were then requested to propose their strategies on the basis of those experiences. The second phase of the experiment was run as a tournament in which those who submitted strategies (automated traders) competed. This idea was further elaborated on by Robert Axelrod in the late 1970s in his famous prisoner's dilemma tournaments (Axelrod 1984) and was continued by John Rust and his colleagues in the early 1990s in their double auction tournaments (Rust et al. 1993, 1994). In these tournaments, contestants directly submitted their strategies without passing through the on-site human-participation phase.

In modern terminology, the strategy method can be broadly understood as an application of peer production to civic sciences or open-source software projects, the typical peer production form of the Web 2.0 economy. Here, the fundamental pursuit for economists is not simply finding a winning strategy in the tournament, since if all participants can learn from their own and others' experiences, then the meaningful question to pursue must be evolution-oriented. Therefore, the point of the tournament approach is to use a large population of participants to test the viability of the incumbent strategies, to discover new strategies through the wisdom of crowds, and to search for any possible stable form of this evolutionary process. In order to do so, a platform needs to be established, and this leads to the institutionalization of running tournaments.

One of the best-known institutionalized tournaments is the *trading agent competition* (TAC), initiated by Michael Wellman and Peter Wurman and later joined by the Swedish Institute of Computer Science. The TAC has been held annually since the year 2000. The original competition was based on a travel scenario: a travel agent offers his or her clients a travel package including flights, accommodations, and entertainment programs subject to some specific travel dates and clients' preference restrictions. The travel agent, however, has to obtain these seats, rooms, and tickets by bidding in each market. This is known as the classic version of TAC (MacKie-Mason and Wellman 2006; Wellman et al. 2007). As time has passed different scenarios have been developed, including a competition between computer assemblers, which mimics the strategies used in supply-chain management (Arunachalam and Sadeh 2005; Collins et al. 2010; Groves et al. 2014), power brokers, who mediate the supply and demand of power between a wholesale market and end customers (Ketter et al. 2013; Babic and Podobnik 2014), and advertisers, who bid competitively for keywords (Hertz 2012).

The TAC focuses on the design of trading strategies, that is, on agents' behavior. There is another annual event derived from the TAC, known as CAT (just the opposite of TAC). CAT stands for market design competition, which focuses on the institutional aspect of the markets.¹⁵ A review of this tournament is given in Chapter 16, "An

¹⁵ It also stands for *Catallactics*, a term originally introduced by Richard Whatley (1787–1863) (Whatley 1832). He suggested using this term as *a science of exchanges* to replace *political economy*,

Overview and Evaluation of the CAT Market Design Competition”, by Tim Miller, Jinzhong Niu, Martin Chapman, and Peter McBurney.

Generally speaking, a competition for market or marketplace design is rather challenging if one considers different stock exchanges competing for the potential listed companies. A good design can maintain a reasonable degree of liquidity, transparency, and matching efficiency so as to enable those who want to buy and sell to find it easier to complete the exchange with a satisfying deal (McMillan 2002). This issue becomes even more challenging given the ubiquitous competition of multiple marketplaces in the Web 2.0 economy, such as the competition between Uber and Lyft, between Airbnb and HomeAway, or among many dating platforms and online job markets. The significance of market designs has been well preached by McMillan (2002), which also included some frontier issues at that time such as online auctions and combinatorial auctions of spectrums. The Nobel Prize in Economics was awarded to Lloyd Shapley and Alvin Roth in the year 2012 for their contribution to market designs, or more specifically, matching designs. With the repaid growth of the Web 2.0 economy, it turns out that the significance of these contributions will only increase.

The design of automated markets inevitably depends on the assumptions of (automated) traders. These two automata together show clearly the scientific nature of economics under natural computationalism. Chapter 16 only summarizes limited work concerning market-maker (specialist) designs, but it does not prevent us from seeing its unlimited extensions. Basically, all observed competitions of trading automata may be toyed with a counterpart tournament so that our understanding of the “natural” automata can be facilitated with the design competition of artificial automata. The further development of this kind of “scientific” tournament can be empowered by the development of online games, constituting a part of the future of economic science.

1.5.4 Agent-Based Computational Modeling and Simulation

In 2006 the long-established series *Handbooks in Economics* published its *Handbook of Computational Economics*, Vol. 2 (Tsfatsion and Judd 2006). The editors of the volume, Leigh Tsfatsion and Kenneth Judd, included the following words in their preface: This second volume focuses on *Agent-based Computational Economics (ACE)*, a computationally intensive method for developing and exploring new kinds of economic models (xi; italics original).

The reader may be interested in knowing what these new kinds of models are. In fact, according to our understanding, “new kinds of economic models” may be an understatement. That volume is actually devoted to *a new kind of economics*, as part

popular in his time, or to replace the *science of wealth* as implied by political economy, in part because wealth is an imprecise term, involving exchangeable and non-exchangeable elements. In a sense, if the ultimate objective of possessing wealth is happiness, then its existence is very subjective. This term is often mentioned together with *praxeology*, the science of human action, in the Austrian school of economics.

of the new kind of science advocated by Stephen Wolfram (Wolfram 2002) and well expounded by one of the editors of the 2006 volume in a separate article (Borrill and Tesfatsion 2011). Under natural computationalism, economic phenomena, as part of natural phenomena, are regarded as outputs of computing machines.

As we have mentioned, natural computationalism also prompts us to reflect on the physical requirements for computation: what the programs are, what the computing architectures are, and what the elementary physical units are in these computing systems. Models extended from the standard Turing model have been proposed, and the agent-based computational model is a proposal to facilitate the acceptance of natural computationalism. The elementary unit of this model can be an atom, a particle, or a physical information carrier. The carried information (bits) can be treated as part of the programs, and the interactions of these physical units are the computational processes. Since these physical units are distributed in space, interactions are concurrent. If hierarchies of elementary units are generated during the computation processes, further concurrent computation can take place at different levels. One of the earliest attempts to develop such kinds of computational models is related to *cellular automata*. In addition to John von Neumann, Konrad Zuse (1910–1995), who built the first programmable computer in 1941, also attempted to work on this problem (Zuse 1993):

I originally wanted to go back to my old ideas: to the design for a computing machine consisting of many linked parallel calculating units arranged in a lattice. Today this would be called a cellular automaton. But I did not pursue this project seriously, for who was to pay for such a device? (143)

The well-known Sakoda-Schelling model, the earliest agent-based model used in the social sciences, is a kind of cellular automaton (Sakoda 1971; Schelling 1971). The “biodiversity,” evolving complexity, and unpredictability of cellular automata have been shown by the pioneering work of John Conway (Gardner 1970) and Stephen Wolfram (Wolfram 1994). From the demonstrations of Wolfram’s *elementary cellular automata*, even economists without a technical background in computer sciences can be convinced of the unpredictability of universal computation or *computational irreducibility*, an essential characteristic of Wolfram’s new kind of science. As he wrote,

Not only in practice, but now also in theory, we have come to realize that the only option we have to understand the global properties of many *social systems* of interest is to build and run computer models of these systems and observe what happens. (Wolfram 1994, 230; italics added)

Hence, what is the nature and significance of agent-based computational economics (ACE)? It provides us with proper economic models with which to perceive economic phenomena as part of the computable universe and comprehend its unfolding unpredictability or computational irreducibility; consequently, simulation becomes an inevitable tool with which to study and understand economic phenomena.

Unlike Tesfatsion and Judd (2006), this volume is not entirely devoted to ACE, but given the increasing importance of this subject, we still include six chapters that

cover issues best demonstrating the computationally irreducible nature of the economy, namely, macroeconomics (Chapter 17), economic policy (Chapter 18), migration (Chapter 19), industry (Chapter 20), financial markets (Chapter 21), and labor markets (Chapter 22). Two of these chapters are extensions of topics that already have been broached in Tesfatsion and Judd (2006), namely, agent-based macroeconomics (Leijonhufvud 2006) and financial markets (Hommes 2006; LeBaron 2006). The other four chapters can all be seen as new elements of ACE.

1.5.4.1 *Agent-Based Macroeconomics*

Because agent-based macroeconomic models largely were developed after the year 2006, one does not expect to see much about them in Tesfatsion and Judd (2006); nonetheless, they did include one chapter on this subject (Leijonhufvud 2006). In that chapter, Axel Leijonhufvud addressed the agent-based macro from the viewpoint of the history of economic analysis. He first reviewed two kinds of neoclassical economics: Marshallian and Walrasian. The latter is the foundation of current dynamic stochastic general equilibrium models (Chapters 2 and 3), whereas the former is no longer used. Leijonhufvud argued that the Marshallian tradition was abandoned because it did not have an adequate tool to allow economists to walk into the rugged landscape of complex systems. Hence, as he stated,

Agent-based economics should be used to revive the older tradition. . . . But it is not with new problems but with the oldest that agent-based methods can help us the most. We need to work on the traditional *core* of economics—supply and demand interactions in markets—for, to put it bluntly, economists don't know much about how markets work. (1629–1630; italics original)

Chapter 17, “Agent-Based Macroeconomic Modeling and Policy Analysis: The Eurace@Unibi Model,” by Herbert Dawid, Simon Gemkow, Philipp Harting, Sander van der Hoog and Michael Neugart, documents the progress made a decade after Leijonhufvud (2006). They first give a brief review of the current state of agent-based macroeconomics by outlining the eight different branches developed since 2006, and spend the rest of the chapter detailing one of these eight, namely, Eurace@Unibi, an adaptation of the EU-funded project EURACE, made by Herbert Dawid's team at the University of Bielefeld. This chapter enables us to see many realizations of Leijonhufvud's idea in what is, basically a system of multiple markets, each driven by individual economic units using heuristics in their decisions. It is the Marshallian macroeconomics dressed in the cloth of agent-based models. The chapter summarizes some promising features delivered by agent-based macroeconomic models, including the capability to replicate stylized facts, addressing the questions that go beyond the mainstream, such as the significance of spatial specification. The model is specifically reviewed in light of its evaluation of various regional development plans and policy scenarios involving labor market openness.

1.5.4.2 Policy Analysis

Practitioners, when they come to evaluate a model or tool, are normally concerned its performance with regard to forecasting or policy recommendations. For these practitioners, ACE certainly has something to offer. The policy-oriented applications of ACE have become increasingly active (Marks 2006); the *Journal of Economic Behavior and Organization* published a special issue to address the relevance of agent-based models for economic policy design (Dawid and Fagiolo 2008). However, given the number of dramatic failures in public policy making (Schuck 2014), maybe the more pressing issue is not solely to demonstrate more applications but to inquire into what predictability and policy effectiveness may mean in such a computable universe. In fact, there is a growing awareness that the role of complexity has been frequently ignored in public policy design (Furtado et al. 2015; Janssen et al. 2015).

The failures of some policies are caused by their unintended consequences. These consequences are unintended if they can be perceived as the emergent properties of the complex adaptive system or the computable universe, or simply put, the *unknown unknowns*. Agent-based modeling may help alleviate the problems. Its flexibility and extendibility can accommodate a large amount of what-if scenarios and convert some unknown unknowns to known unknowns or even into knowns. Chapter 18, “Agent-Based Models for Economic Policy Design: Two Illustrative Examples,” by Frank Westerhoff and Reiner Franke, provides an easily accessible guide to this distinguishing feature of agent-based modeling.

Using a prototypical two-type model, the authors demonstrate the value of agent-based modeling in policy-oriented applications. This model, more generally known as the *K*-type agent-based model, has long been used to construct agent-based financial markets and is able to generate a number of stylized facts through the *endogenously evolving market fractions* of different types of agents (see also Chapter 21). These evolving market fractions are caused by mobile agents switching between a number of possible behavioral rules or heuristics, as if they are making a choice in the familiar *K-armed bandit problem* (Bush and Mosteller 1955). Based on the attractiveness (propensity, score) of each rule, they stochastically choose one of them by following a pre-specified probability distribution, say, the Boltzmann-Gibbs distribution, frequently used in statistical mechanics. The attractiveness of each rule is updated constantly on the basis of the experiences received from the environment and can be considered as a kind of *generalized reinforcement learning* (Chen 2013).

The *K*-type agent-based model has also been extended to macroeconomic modeling and has constituted part of the agent-based macroeconomic literature (Chapter 17). The authors also demonstrate a Keynesian model in this way. This model, together with the two-type agent-based financial model, is then applied to address various stabilization policies that are frequently considered in financial markets and the macroeconomy. In these two examples the *Lucas critique* (Lucas 1976), or the unknowns (whether known or unknown) are demonstrated directly by the bottom-up mechanism, that is, the endogenously evolving market fractions. Hence, when the government adopts

an intervention policy, we, as econometricians, can no longer assume that the market fraction dynamics remain unchanged (the essence of the Lucas critique). In fact, this is only the first-order Lucas critique (down to the mesoscopic level), and one can actually move further to the second or the higher order (down to the individual level). The authors suggest some extensions in this direction.

1.5.4.3 Migration

Migration has always been an essential ingredient of human history; therefore, it is not surprising that it has drawn attention from disciplines A to Z and is a highly interdisciplinary subject. Keeping track of migration dynamics can be a very highly complex task. First, we have to know why people migrated or intended to migrate; second, we have to add up their behaviors to form the foundation of migration dynamics. The first part is less difficult. Extensive surveys can help disentangle various political, economic, and environmental factors that influence migration at the individual level. The second part is more challenging because some of the determinants can be assumed to be autonomously or exogenously given, such as age and home preference, but some of them are obviously endogenously determined by such other factors as social networks and relative income. The agent-based model is very suitable for handling the complex repercussions and endogeneities among individuals.

In fact, the celebrated Schelling segregation model (Schelling 1971) can also be read as a migration model, and the only determinant for migration in the Schelling model is the tolerance for ethnic diversity. The model is simple, but it is powerful enough to demonstrate one important stylized fact in migration, namely, clustering. The migration dynamics pictured by the Schelling model basically belong to intra-urban migration, which is near-range migration. However, when it comes to intercity, rural-urban, or international migration, the behavioral rules can be further complicated by different considerations. Therefore, more realistic and sophisticated models are to be expected for different types of migration.

Spatial specificity is one of the major drivers for the agent-based models; cellular automata, checkerboard models, and landscape models are clear demonstrations of this spatial feature (Chen 2015). Nonetheless, the application of agent-based models to migration comes relatively late. It was first applied to climate-induced or environmentally induced migration (Smith et al. 1998; Dean et al. 2000), then to rural-urban migration (Silveira et al. 2006). Given the increasing importance of climate change and urbanization, both types of migration have attracted a series of studies since then, such as Hassani-Mahmooei and Parris (2012) and Cai et al. (2015). However, the research on international migration is rather limited. Chapter 19, “Computational Economic Modeling of Migration,” by Anna Klabunde, which studies the migration of Mexicans to California, can be considered to be progress in this direction.

In that chapter the author addresses not only the decision model for migration but also the reversal decision (the return decision); hence, this is the first agent-based model devoted to *circular migration*. The migration and return decisions are driven by

different considerations and determinants. These determinants were statistically tested before being selected as part of the behavioral rule of agents. Chapter 19 can be read together with Chapter 18, since both provide detailed operating procedures for building an agent-based model, from stylized fact selection to model calibration or estimation, or validation, and from robustness checks to simulations of scenarios that affect policy.

1.5.4.4 *Industrial Economics*

Industrial organization, as an extension of microeconomics, is a field with a long history and has been incorporated into many textbooks. It is mainly concerned with the structure, behavior, and performance of markets (Scherer and Ross 1990). It begins with a very fundamental issue concerning the organization of production, namely, the nature of firms, then progresses toward the interactions of firms with different scales via pricing and non-pricing competition or cooperation (collusion). These, in turn, determine observed industrial phenomena such as the size distribution of firms, lifespan of firms, pricing wars, mergers, takeovers, product differentiation, research and development expenditures, advertising, the quality of goods and barriers to entry, entry and exit.¹⁶

Over time, different research methodologies including game theory (Tirole 1988), econometrics (Schmalensee and Willig 1989b, part 3), and experimental economics (Plott 1989) have been introduced into the literature on industrial organization (IO), while more recently, bounded rationality, psychology, behavioral economics (Spiegler 2011), and social networks (Silver 2012) have all been added to the toolkit for studying IO. Despite the voluminous literature and its methodological inclusiveness, the computational aspect of industrial IO has largely been ignored in the mainstream IO literature. For example, the *Handbook of Industrial Organization* (Schmalensee and Willig 1989a,b; Armstrong and Porter 2007) has only a single chapter dealing with numerical approaches or computational models used in IO (Doraszelski and Pakes 2007). John Cross's pioneering application of reinforcement learning to oligopolistic competition (Cross 1973) has not been mentioned in any of the *Handbook's* three volumes, nor have the agent-based computational models that have subsequently arisen (Midgley et al. 1997).

It is somewhat unfortunate that agent-based models have been completely ignored in the mainstream IO literature. This absence has demonstrated the lack of biological, ecological, and evolutionary perspectives when examining industrial dynamics and has placed the study of IO strictly within the framework of rational, equilibrium, and static analysis (also see Chapter 20 for an extended discussion). In fact, the potential application of the von Neumann automata to IO was first shown in Keenan and O'Brien (1993), but this work is also ignored in the aforementioned *Handbook*. Using Wolfram's elementary cellular automata, Keenan and O'Brien were able to show that the complex competitive or collusive behavioral patterns of firms can be generated by very simple

¹⁶ For a full range of subjects, see the *Handbook of Organization* (Schmalensee and Willig 1989a,b; Armstrong and Porter 2007).

pricing rules. Later, Axtell (1999) and Axtell and Guerrero (2016) also demonstrated how the simple behavior of firms in terms of their recruitment and layoff decisions can generate the firm-size distribution empirically observed, for example, as in Axtell (2001) or Aoyama et al. (2010).

Chapter 20, “Computational Industrial Economics: A Generative Approach to Dynamic Analysis in Industrial Organization,” by Myong-Hun Chang, provides an illustrative application of agent-based computational models to the study of the entry-exit process of an industry. The constant entry and exit process, as richly observed in many markets, is best perceived as an out-of-equilibrium industrial dynamic, which can be hard to address with conventional equilibrium analysis. Dynamic models that assume highly intelligent behavior of firms, such as the concept of the Markov perfect equilibrium, are very demanding with regard to computation and suffer from the curse of dimensionality. Using the agent-based model with boundedly rational firms, the author can replicate important empirical patterns of the entry and exit process, such as the positive entry and exit correlation, and infant mortality. An important property that emerges from the proposed agent-based model is the relation between the endogenous variables such as the rate of firm turnover, industry concentration, market price, and the industry price-cost margins associated with persistent demand fluctuations.

1.5.4.5 *Financial Markets*

The financial market is one of the domains to which the earliest agent-based modeling has been applied (Palmer et al. 1994). To some extent, whether agent-based modeling can bring in a promising alternative research paradigm is often evaluated or tested on the basis of how well it has been applied to financial markets. This evaluation is two-faceted, and it raises two questions. On the modeling side, can agent-based models enhance our understanding of markets through harnessing the essential operational details? On the practical side, can agent-based models improve our forecasting and policy making? Early efforts to answer these questions have been made in the extensive reviews provided by Hommes (2006) and LeBaron (2006) in the *Handbook of CEF*. Since then, the area has remained very active or become yet more active, however; it is, therefore, desirable to keep continuous track of the progress. Chapter 21, “Agent-Based Modeling for Financial Markets,” by Giulia Iori and James Porter, provides an update.

Since 2006 there have been three major types of progress in agent-based financial markets, namely, modeling, estimation or validation, and policy designs (see also Chapter 18). Chapter 21 complements the earlier review articles by documenting the progress in each of these directions. Among the three, modeling is the pivotal one because the subsequent two are built on it. Like other markets, the financial market is complex in its unique way, but the mainstream (neoclassical) financial model tends to mask its rugged landscape with oversimplifying assumptions. One exceptional development outside this mainstream modeling is the literature known as the *market microstructure* (O’Hara 1995), which really pushes economists to ask what an exchange

is (Lee 1998) (see also Chapter 13), a question long overlooked as we often ignore what computation is.¹⁷

The early development of agent-based financial models is inclined toward the behavioral aspect of financial markets; both Hommes (2006) and LeBaron (2006) can be read as a glossary of the behavioral settings of financial agents. Since then, however, the development of empirical microfinance, specifically, in light of the use of large amounts of proprietary data, enables us to address the impacts of trading mechanism designs on market efficiency, liquidity, volatility, and transparency. They include the impact not just on the aggregate effect but also on the asymmetries distributed among different types of investors, for example, institutional and individual investors. Specific issues include finer tick sizes, extended trading hours, deeper exposure of order books, and heavier reliance on algorithmic trading. Meanwhile, financial econometrics has quickly moved into the analysis of ultra-high-frequency data, and this trend has been further reinforced by high-frequency trading, thanks to information technology (O'Hara 2015). To correspond to these developments, more and more agent-based financial markets have been built upon the order-book-driven markets, using continuous double auctions or call auctions, which are very different from the earlier Walrasian market-clearing setting. Chapter 21 documents some studies in this area.

In addition, progress in information and communication technology has led to the development of the digital society as well as social media. In order to understand the impact of such social media platforms as Twitter, Facebook, microbloggers, and online news on price dynamics, network ingredients become inevitable (see also Chapters 15 and 16). Chapter 21 also surveys the use of network ideas in agent-based financial markets, from the early lattice to social networks and from exogenous settings to endogenous formations.

1.5.4.6 Labor Markets

Among the three basic constituents in economics, the labor market has received much less attention than have the financial market (Chapter 21) and the goods market. This imbalance is somewhat puzzling, since by the degree of either heterogeneity or decentralization, the labor market tends to be no less, if not more, complex than the other two markets. Workers have different skills, ages, experiences, genders, schooling, residential locations, cultural inheritances, social networks, social status, and job-quitting or -searching strategies. Jobs have different modular structures; they differ in their weights and arrays and in the hierarchies attached to a given set of modules (capabilities). Then there are decisions about investment in maintaining and accumulating human capital, strategies to reduce information asymmetry, such as the use of the referral system, and legal restrictions on wages, working hours, job discrimination, security, and so on. Various social media and platforms introduced

¹⁷ According to Mirowski (2007), there are at least five areas of the literature that have managed to develop a fair corpus of work concerning markets as evolving computational entities, and one of them is market microstructure in finance.

to the market are constantly reshaping the matching mechanism for the two sides of the market. Intuitively, the labor market should be the arena to which agent-based modeling is actively applied, but as Michael Neugart and Matteo Richiardi point out in Chapter 22, “Agent-Based Models of the Labor Market,” the agent-based labor market may only have a marginal status in ACE. There is no independent chapter on this subject in an earlier handbook (Tsfatsion and Judd 2006). Maybe there were not enough studies in existence then to warrant a chapter. Fortunately, the first such survey is finally possible.

Chapter 22 covers these two parts of the literature and connects them tightly. The first part covers the literature on agent-based computational labor markets, starting from the late 1990s (Tsfatsion 1998, 2001, 2002). This part may be familiar to many researchers experienced with ACE. However, some earlier studies using different names when the term ACE had not yet existed, such as *microsimulation* and *micro-to-macro models*, may have been ignored. The authors review this literature by tracing its origin back to the microsimulation stage and including the work by Guy Orcutt (1917–2006), Barbara Bergmann (1927–2015), and Gunnar Eliasson.¹⁸

With this overarching framework the authors review of this development not only from the perspective of the labor market per se, but also from the perspective of so-called *macro labor*, a subject related to Chapter 17. The authors first document a series of stylized facts such as the Beveridge curve, which can be replicated by agent-based models, and then show how agent-based models can be used to address some interesting policy-oriented questions.

Like many other branches of economics, labor economics is subject to the heavy influence of both psychology and sociology. This is reflected by the emerging, sub-branches now called *behavioral labor economics* (Dohmen 2014; Berg 2015) and the *economic sociology of labor markets* (Granovetter 2005). Part of the former is concerned with various decision heuristics related to the labor market, such as wage bargaining, job-search stopping rules, human capital investment, and social network formation. Part of the latter is concerned with the significance of social structure or social networks in disseminating or acquiring job information and in the operation of labor markets (see also Chapter 19). The chapter also reviews the development of agent-based labor markets in light of these two trends.

1.5.5 Computational Neuroeconomics

At the beginning of this chapter we made a distinction between *naturally inspired computing* and *natural computing*. Natural computing has been extensively used by scientists and engineers; from Fisher Machines and Phillips Machines, automated

¹⁸ Maybe economics is not the only discipline to have experienced this two-stage development from microsimulation to agent-based modeling. Microsimulation was been used in demography long before agent-based modeling came into being. On this history and for more discussion on the relation between microsimulation and agent-based modeling, the interested reader is referred to Chen (2015), sec. 2.3.2.

markets, and prediction markets to various Web 2.0 designs of wisdom of crowds, it has also been applied to economics. Nonetheless, when talking about the computer, one cannot ignore its most salient source of inspiration the human brain. This was already manifested in von Neumann's last work, *The Computer and the Brain* (von Neumann 1958).

In this handbook, we keep asking the questions What are the computing systems, and what do they compute? These two questions become: Which part of the brain is used, and what is computed? The recent development of neuroeconomics has accumulated a large amount of literature addressing these questions. To the best of our knowledge, this topic has not been included in any part of the computational economics literature and is probably less well known to general economists as well. Chapter 23, "The Emerging Standard Neurobiological Model of Decision Making: Strengths, Weaknesses, and Future Directions," by Shih-Wei Wu and Paul Glimcher, deals with neuroeconomics article in the CEF context.

Based on the accumulation of a wealth of neurobiological evidence related to decision computation, the chapter begins with an emerging standard neurobiological model composed of two modules, namely, the valuation network and the choice (comparison) network. With regard to the former, the authors review the experiments conducted by Wolfram Schultz and his colleagues regarding the midbrain dopamine neurons in their roles of learning and encoding values. The main computational model proposed to explain the observed neural dynamics is a kind of reinforcement learning model that mathematical psychologists have investigated since the time of Robert Bush (1920–1972) and Frederick Mosteller (1916–2006) (Bush and Mosteller 1955). As for the latter or the value comparison, the authors indicate that, owing to the constraints of the choice circuitry, specifically, the limited dynamic range of lateral intraparietal neurons, what are represented and compared are the *relative* subjective values of options, rather than the original subjective values stored in the valuation network. This transformation may lead to a choice problem known as the *paradox of choice* when the number of options is large (Schwartz 2003). This standard model, the valuation network, is further extended to address some behavioral bias learned from behavioral economics, such as the reference-dependent preferences and the violation of the independence axiom (the distortion of probability information).

1.6 EPILOGUE: EPISTEMOLOGY OF COMPUTATION AND SIMULATION

One definition of computational economics is the uncovering of the computational content of the economics built on classical mathematics, just like what constructive mathematics does for classical mathematics. From the constructivists' viewpoint, there is a stronger resistance to the use of nonconstructive principles as working premises, such as the law of excluded middle and the limited principle of omniscience (Bridges

1999). Economists receive relatively little training in constructive mathematics, and they probably are not aware of the consequences of applying nonconstructive premises in order to develop their “constructive” economics. It is nice to see that some resultant confusions and in felicities are cleaned with the help of Chapter 24, “The Epistemology of Simulation, Computation, and Dynamics in Economics,” by K. Vela Velupillai, the founder of computable economics.

The chapter closes this book because it places the entire enterprise of computational economics in a rich context of history and philosophy of mathematics, and has the capacity to reveal some of its possibly promising future. The chapter begins very unusually with an accounting tradition of computational economics that originated with William Petty (1623–1687) and his *political arithmetic*. This tradition is a thread connecting many important steps made toward modern macroeconomics and applied general equilibrium. It traverses through the writings of John Hicks (Hicks 1942), Leif Johansen (Johansen 1960), Richard Stone (Stone 1961), Lance Taylor (Taylor 2004), Wynne Godley (Godley and Lavoie 2007), and many others. This tradition begins with accounting and uses the social accounting system as an implementation tool for macroeconomic dynamic analysis.

The accounting approach to the study of economic systems and computational economics is just a prelude leading to the essence of this chapter, namely, the triad of computation, simulation, and dynamics in an epistemological way. As the author states, “We economists are well on the way to an empirically implementable algorithmic research program encompassing this triad indissolubly.” Based on the development of computational physics, experimental mathematics, and Visiometrics, the author uses what he calls the “Zabusky Precept” to advocate the combined use of analysis and computer simulation as a heuristic aid for simulational serendipities and for alleviating the epistemological deficit and epistemological incompleteness. This key tone is presented with a number of illustrations not limited to economics, but including biology, engineering, physics, computer science, and mathematics. The illustrations are the famous Fermi-Pasta-Ulam problem, the Lorenz system, the Diophantine equation, the four-color problem and the computer-aided proof, program verification, and the macroeconomy characterized by the Phillips machine. In each of these cases, the author demonstrates how simulation can bring in serendipities (surprises) and how the development of theory can benefit from these experiments (simulations).

In the spirit of the Zabusky Precept, there is no reason why economists should shun simulation, and in answer to the question posed by Lehtinen and Kuorikoski (2007), “Why do economists shun simulation?” Velupillai states that “economists have never shunned simulation. However, they may have misused it, perhaps owing to a misunderstanding of the notion, nature, and limits of computation, even by an ideal machine.” Therefore, to shows faith and hope in computational economics, the author considers the five frontiers in this area, in which machine computation, in its digital mode, is claimed to play crucial roles in formal modeling exercises, and gives a critical review of them. The five are computable general equilibrium theory in the Scarf tradition, computable general equilibrium modeling in the Johansen-Stone

tradition, agent-based computational economics, classical behavioral economics, and computable economics. Some of the five research areas are critically examined using mathematical frameworks for models of computation such as constructive analysis, computable analysis, computable numbers, and interval analysis.

Finally, the author points out the need to adapt the curriculum of economics to the digital age, for example, by shifting from the original comfort zone underpinned by real analysis to a wholly different mathematics in which computing machines are built. He ends with the response of Ludwig Wittgenstein (1889–1951) to Cantor’s paradise: “You’re welcome to this; just look about you.”

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CHAPTER 2

DYNAMIC STOCHASTIC GENERAL EQUILIBRIUM MODELS

A Computational Perspective

MICHEL JUILLARD

2.1 INTRODUCTION

Dynamic stochastic general equilibrium (DSGE) models have become very popular in applied macroeconomics, both in academics and in policy institutions. This chapter reviews the methods that are currently used to solve them and estimate their parameters.

This type of modeling adopts the methodology developed by Kydland and Prescott (1982) at the origin of real business cycle analysis. It views macroeconomic models as simplified representations of reality built on behavior of representative agents that is consistent with microeconomic theory. Nowadays, DSGE models take into account a large number of frictions, real or nominal, borrowed from New Keynesian economics. Depending on the aim of the model, nominal rigidities, and money transmission mechanisms, labor market, fiscal policy, or open economy aspects are emphasized and the corresponding mechanisms developed.

Whatever the focus of a particular model, however, common features are present that define a class of models for which a particular methodology has been developed.

In this class of model, most agents (households, firms, financial intermediaries, and so on) take their decisions while considering intertemporal objective functions: lifetime welfare for households, investment value for the firms, and so on. Agents must therefore solve dynamic optimization problems.

Given the specification of utility and technological constraints usually used in microeconomics, the resulting models are nonlinear. Because of the necessity to solve

dynamic optimization problems, future values of some variables matter for current decisions. In a stochastic world, these future values are unknown, and agents must form expectations about the future. The hypothesis of *rational expectations*, according to which agents form expectations that are consistent with the conditional expectations derived from the model (see Muth 1961), provides a convenient but not very realistic solution in the absence of precise knowledge about the actual process of expectation formation.

All of that leads to mathematical models that take the form of nonlinear stochastic difference equations. Solving such models is not easy, and sophisticated numerical techniques must be used. In what follows, I present popular algorithms to find approximate solutions for such models, both in stochastic and deterministic cases. Efficient algorithms exist for the limiting case where there is no future uncertainty, and these algorithms can be used to study separately the full implication of nonlinearities in the model in the absence of stochastic components.

Most estimation of the parameters of DSGE models is currently done on the basis of a linear approximation of the model, even if several authors have attempted to estimate nonlinear approximation with various versions of the particle filter (see, e.g., Amisano and Tristani 2010; Anreassen 2011; and Chapter 7 of this handbook). Even with linear approximation, estimation of DSGE models remains very intensive in terms of computation, requiring analysts to solve the model repeatedly and to compute its log-likelihood with the Kalman filter.

It is possible to estimate DSGE models by the maximum likelihood, but I advocate instead a Bayesian approach as a way to make explicit, in the estimation process, the use of a priori information, to mitigate the problems arising from lack of identification of some parameters and to address misspecification of the model in some direction.

In the second section, I present a generic version of the model, in order to fix notations to be used later. The solution of perfect foresight models is discussed in section 2.3 and that for stochastic models in section 2.4. Estimation is presented in section 2.5. I present a list of software products implementing these methods in section 2.6 and conclude with directions for future work.

2.2 A GENERIC MODEL

A DSGE model can, in general, be represented as a set of nonlinear equations. The unknowns of these equations are the endogenous variables. The equations relate the current value of the endogenous variables to their future values, because of expectations, and to their past values, to express inertia. The system is affected by external influences that are described by exogenous variables.

The dynamics of such systems can be studied, first, by extracting away all uncertainty and making the extreme assumption that agents know the future with exactitude. One speaks then of the *perfect foresight model*. Perfect foresight, deterministic models, even

those of large size, can be studied with great accuracy and with simpler methods than can stochastic ones.

Formally, we write a perfect foresight model as

$$f(y_{t+1}, y_t, y_{t-1}, u_t) = 0$$

where $f()$ is a vector of n functions, $\mathcal{R}^{3n+p} \rightarrow \mathcal{R}^n$, and y is a vector of n endogenous variables that can appear in the model in the current period, t , as well as in the next period $t+1$ and in the previous period $t-1$. The vector u_t is a vector of p exogenous variables. Expressing the equations of the model as functions equal to zero helps the mathematical treatment.

In general, endogenous variables may appear in a model with leads or lags of more than one period and exogenous variables at periods other than the current one may also enter the equations. However, with the addition of adequate auxiliary variables and equations, it is always possible to write more complicated models in this canonical form. The algorithm for doing so is discussed in Broze et al. (1990). In a given model, not all variables are necessarily present in previous, current, and future periods. If it is important to take this fact into account for efficient implementation of computer code, it simplifies the exposition of the solution algorithms to abstract from it, without departing from generality.

When the model is stochastic, the exogenous variables are zero mean random variables. Note that this assumption is compatible with a large class of models but excludes models where exogenous processes contain an exogenous change in mean such as increase in life expectancy or policy change. When exogenous variables are stochastic, the endogenous are random as well and, in current period t , it is not possible to know the exact future values of endogenous variables in $t+1$, but only their conditional distribution, given the information available in t . With the rational expectations hypothesis, the equations of the model hold under conditional expectation

$$\mathbb{E} \{f(y_{t+1}, y_t, y_{t-1}, u_t) | \Omega_t\} = 0,$$

where Ω_t is the information set available at period t . In what follows, we assume that shocks u_t are observed at the beginning of period t and that the state of the system is described by y_{t-1} . Therefore, we define the information set available to the agents at the beginning of period t as

$$\Omega_t = \{u_t, y_{t-1}, y_{t-2}, \dots\}.$$

This convention is arbitrary, and another one could be used instead, but such a convention is necessary to fully specify a given model. Now that the information set has been made clear, we use the lighter but equivalent notation

$$\mathbb{E}_t \{f(y_{t+1}, y_t, y_{t-1}, u_t)\} = 0.$$

We make the following restrictive assumptions regarding the shocks u_t :

$$\begin{aligned}\mathbb{E}(u_t) &= 0 \\ \mathbb{E}(u_t u'_\tau) &= 0 \quad t \neq \tau.\end{aligned}$$

We take into account possible correlation between the shocks but exclude serial correlation. Note that auto-correlated processes can be accommodated by adding auxiliary endogenous variables. In that case, the random shock is the innovation of the auto-correlated process.

2.2.1 The Nature of the Solution

Although the deterministic and the stochastic versions of the model are very similar, there is an important difference that has consequences for the solution strategy.

In the deterministic case, it is a perfect foresight model where all information about the future values of the exogenous variables is known at the time of computation. On the contrary, in the stochastic case, the realizations of the exogenous variables are only learned period by period.

In the perfect foresight case, it is therefore possible to compute at once the trajectory of the endogenous variables. In the stochastic case, this approach is not available because the values of the shocks are only learned at the beginning of each period. Instead, the solution must take the form of a *solution function* that specifies how endogenous variables y_t are set as a function of the previous state y_{t-1} and the shocks observed at the beginning of period t :

$$y_t = g(y_{t-1}, u_t).$$

In most cases, there is no closed form expression for function $g()$. It is necessary to use numerical methods to approximate this unknown function. On the basis of the Implicit Function Theorem, Jin and Judd (2002) discuss the conditions for the existence of a unique solution function in the neighborhood of the steady state.

It is well known that rational expectation models entail a multiplicity of solutions, many of them taking the form of self-fulfilling dynamics (see, e.g., Farmer and Woodford 1997). Most research has focused on models that, after a shock, display a single trajectory back to steady state. Note that this convergence is only asymptotic. In such cases, agents are supposed to be able to coordinate their individual expectations on this single trajectory. Much of the literature on DSGE models has been following this approach, but attention has also been given to models with a multiplicity of stable solutions and possible sunspots as in Lubik and Schorfheide (2004).

2.3 SOLVING PERFECT FORESIGHT MODELS

In the perfect foresight case, the only approximation we make is that convergence back to the steady state takes place after a finite number of periods rather than asymptotically. The larger the number of periods one considers, the more innocuous the approximation.

One can then represent the problem as a two-boundary value problem where the initial value of variables appearing with a lag in the model is given by initial conditions and the final value of variables appearing with a lead in the model is set at their steady state value.

When one stacks the equations for all the T periods of the simulation as well as the initial and the terminal conditions, one obtains a large set of nonlinear equations:

$$f(y_{t+1}, y_t, y_{t-1}, u_t) = 0 \quad t = 1, \dots, T$$

with initial conditions $y_0 = y^*$ and terminal conditions $y_{T+1} = \bar{y}$, the steady state.

Until Laffargue (1990), the literature considered that for large macroeconomic models, the size of the nonlinear system would be too large to use the Newtonian method to solve it. Consider, for example, a multicountry model with ten countries in which the one-country model numbers forty equations and that one simulates more than one hundred periods. The resulting system of nonlinear equations would number forty thousand equations, and the Jacobian matrix, 1.6 billion elements.

Such large problems seemed better attacked by first-order iterative methods such as Gauss-Seidel as in Fair and Taylor (1983) or Gilli and Pauletto (1997). Yet Laffargue (1990) and Boucekkine (1995) show that the large Jacobian matrix of the stacked system has a particular structure that can be exploited to solve efficiently the linear problem at the heart of Newton's method.

The vectors of endogenous variables in each period, y_t , can be stacked in a single large vector Y such that

$$Y = \begin{bmatrix} y'_1 & \dots & y'_T \end{bmatrix}'$$

and the entire system for all T periods can be written as $\mathcal{F}(Y) = 0$. Using the Newtonian method to solve this system of equations entails starting with a guess $Y^{(0)}$ and obtaining iteratively a series of $Y^{(k)}$, such that

$$\left. \frac{\partial \mathcal{F}}{\partial Y} \right|_{Y=Y^{(k-1)}} \Delta Y^{(k)} = -\mathcal{F}(Y^{(k-1)}),$$

and

$$Y^{(k)} = \Delta Y^{(k)} + Y^{(k-1)}.$$

The iterations are repeated until $\mathcal{F}(Y^{(k)})$ or $\|x^{(k)} - x^{(k-1)}\|$ are small enough. As mentioned above, a practical difficulty arises when the size of the Jacobian matrix $\frac{\partial \mathcal{F}}{\partial Y}$ is very large.

As remarked by Laffargue (1990), given the dynamic nature of the system, and that, in each period, the current variables depend only on the value of the variables in the previous and in the next period, this Jacobian matrix has a block tridiagonal structure:

$$\begin{bmatrix} f_{2,1} & f_{1,1} & & & \\ & \ddots & \ddots & & \\ \vdots & f_{3,t} & f_{2,t} & f_{1,t} & \vdots \\ & & \ddots & \ddots & \\ & & & f_{3,T} & f_{2,T} \end{bmatrix} \begin{bmatrix} \Delta y_1^{(k)} \\ \Delta y_2^{(k)} \\ \vdots \\ \Delta y_{t-1}^{(k)} \\ \Delta y_t^{(k)} \\ \Delta y_{t+1}^{(k)} \\ \vdots \\ \Delta y_{T-1}^{(k)} \\ \Delta y_T^{(k)} \end{bmatrix} = \begin{bmatrix} -f_1(y_2^{(k-1)}, y_1^{(k-1)}, y^*, u_1) \\ \vdots \\ -f_t(y_{t+1}^{(k-1)}, y_t^{(k-1)}, y_{t-1}^{(k-1)}, u_t) \\ \vdots \\ -f_T(\bar{y}, y_T^{(k-1)}, y_{T-1}^{(k-1)}, u_T) \end{bmatrix}.$$

The fact that the partial derivatives with respect to the state variables appear below the main diagonal follows directly from the fact that they are, indeed, predetermined variables. This particular structure suggests that it is possible to triangularize the Jacobian by solving T linear problems of the size of the model for one period and then to find the improvement vector to the solution of the whole system through backward substitution.

For example, after triangularization in period t , the system looks like the following:

$$\begin{bmatrix} I & M_1 & & & \\ & \ddots & \ddots & & \\ & & I & M_t & \\ & & f_{3,t+1} & f_{2,t+1} & f_{1,t+1} \\ & & & \ddots & \ddots \\ & & & & f_{3,T} & f_{2,T} \end{bmatrix} \begin{bmatrix} \Delta y_1^{(k)} \\ \Delta y_2^{(k)} \\ \vdots \\ \Delta y_{t-1}^{(k)} \\ \Delta y_t^{(k)} \\ \Delta y_{t+1}^{(k)} \\ \vdots \\ \Delta y_{T-1}^{(k)} \\ \Delta y_T^{(k)} \end{bmatrix} = \begin{bmatrix} d_0 \\ d_1 \\ d_2 \\ \vdots \\ d_t \\ -f_{t+1}(y_{t+2}^{(k-1)}, y_{t+1}^{(k-1)}, y_t^{(k-1)}, u_t) \\ \vdots \\ -f_T(\bar{y}, y_T^{(k-1)}, y_{T-1}^{(k-1)}, u_T) \end{bmatrix}$$

The triangularization obeys the following recursive rules:

$$M_1 = f_{2,1}^{-1} f_{1,1}$$

$$d_1 = -f_{2,1}^{-1} f_1^{(k-1)}(y_2^{(k-1)}, y_1^{(k-1)}, y^*, u_1)$$

then

$$\begin{aligned}
M_t &= (f_{2,t} - f_{3,t}M_{t-1})^{-1}f_{1,t} \quad t = 2, \dots, T \\
d_t &= - (f_{2,t} - f_{3,t}M_{t-1})^{-1} \left(f_{3,t}d_{t-1} + f_t^{(k-1)} \left(y_{t+1}^{(k-1)}, y_t^{(k-1)}, y_{t-1}^{(k-1)}, u_t \right) \right) \\
t &= 2, \dots, T.
\end{aligned}$$

The values of $\Delta y_t^{(k)}$ are then obtained by backward substitution, starting with $\Delta y_T^{(k)}$:

$$\begin{aligned}
\Delta y_T^{(k)} &= d_T \\
\Delta y_t^{(k)} &= d_t - M_t \Delta y_{t+1}^{(k)} \quad t = T-1, \dots, 1.
\end{aligned}$$

Note that this approach at solving a large two-point boundary value problem starts showing its age. It was developed in the mid-1990s when a PC had no more than 64MB RAM. Now that 4GB RAM or more is the norm, the linear problem that is at the core of Newton's method can be more efficiently handled simply by using sparse matrix code and storing at once all the nonzero elements of the Jacobian of the entire stacked nonlinear model.

As stated above, in this approach the only simplifying assumption is to consider that, after a shock, the steady state is reached in finite time, rather than asymptotically. Usually, one is more interested by the trajectory at the beginning of the simulation; around the time when shocks are hitting the economy it is easy to verify whether the beginning of the simulation is affected by varying the horizon, T .

It is also possible to consider alternative terminal conditions such as $y_T = y_{T+1}$, or aiming at the trajectory resulting from a linear approximation of the model, among others.

2.4 SOLVING STOCHASTIC MODELS

As stated above, the stochastic version of the general model is pretty similar:

$$\mathbb{E} \left\{ f(y_{t+1}, y_t, y_{t-1}, u_t | u_t, y_{t-1}, y_{t-2}, \dots) \right\} = 0. \quad (2.1)$$

The exogenous variables u_t are now stochastic variables, and, because the future value of endogenous variables, y_{t+1} , will be affected by u_{t+1} that are still unknown in period t , the equation can only hold under conditional expectations.

Because only y_{t-1} affects the dynamics of the model, it is sufficient to retain u_t and y_{t-1} in the information set for period t .¹

Obviously, at a given date future shocks are unknown, and it is not possible to compute numerical trajectories as in the case of perfect foresight models. It is necessary

¹ The macroeconomic literature often makes different assumptions concerning the information set. For example, the assumption made here is consistent with stock of capital on an end-of-period basis. When one considers stock of capital on a beginning-of-period basis, then stock of capital at the current period is predetermined and enters the information set.

to change the focus of inquiry toward the *decision* rules used to set y_t as a function of the previous state of the system and current shocks in a way consistent with equation (2.1). In the stochastic case, we are required to search for an unknown function. It turns out that only in a very few cases does this solution function have an analytic expression. It is, therefore, necessary to use numerical methods to provide an approximation of the solution function.

Several methods exist for computing approximations of the solution function. Discretization of the state space and iterations on the policy function provide an approximation in tabular form, global methods such as projection methods control the quality of approximation over the entire support of the model, and the perturbation approach provides local approximation around a given point. Early surveys of these methods appear in Taylor and Uhlig (1990) and Judd (1996).

For medium- to large-size models, the most frequently used method is perturbation. It is possible to solve models with several hundred equations easily at first or second order. At first order, the perturbation approach is identical to linearization, which has been the dominant method used since the inception of RBC analysis. An early survey of methods used to solve dynamic linear economies can be found in Anderson et al. (1996).

Note that the approximate solution computed by the perturbation method doesn't depend on the entire distribution of the stochastic shocks but only on as many moments as the order of approximation. We write Σ_u , the covariance matrix of the shocks, and $\Sigma_u^{(k)}$, the tensor containing the k th moments of this distribution.

It is useful to introduce the *stochastic scale variable*, σ , in the model in order to take into account the effect of future uncertainty on today's decisions. We introduce also the auxiliary random variables ε_t such that

$$u_{t+1} = \sigma \varepsilon_{t+1}.$$

When $\sigma = 0$, there is no uncertainty concerning the future. The moments of ε_t are consistent with the moments of shocks u_t : $\Sigma_u^{(k)} = \sigma^k \Sigma_\varepsilon^{(k)}$.

In the perturbation approach, it is necessary to include the stochastic scale variable as an argument of the solution function:

$$y_t = g(y_{t-1}, u_t, \sigma).$$

The stochastic scale doesn't play a role at first order, but it appears when deriving the solution at higher orders.

Using the solution function, it is possible to replace y_t and y_{t+1} in the original model and define an equivalent model $F()$ that depends only on y_{t-1} , u_t , ε_{t+1} , and σ :

$$y_{t+1} = g(y_t, u_{t+1}, \sigma)$$

When using existing software for solving a DSGE model, one must be attentive to the assumption used in that software concerning the information set and rewrite the model in manner consistent with that assumption.

$$\begin{aligned}
&= g(y_{t-1}, u_t, \sigma), u_{t+1}, \sigma) \\
F(y_{t-1}, u_t, \varepsilon_{t+1}, \sigma) &= f(g(y_{t-1}, u_t, \sigma), \sigma \varepsilon_{t+1}, \sigma), g(y_{t-1}, u_t, \sigma), y_{t-1}, u_t)
\end{aligned}$$

and

$$\mathbb{E}_t \{F(y_{t-1}, u_t, \varepsilon_{t+1}, \sigma)\} = 0. \quad (2.2)$$

It is worthwhile to underscore the different roles played by the exogenous shocks depending on whether they are already realized, u_t , or still to be expected, $u_{t+1} = \sigma \varepsilon_{t+1}$. Once the shocks are observed, they are just additional elements of the state space. When there are random shocks to happen in the future, they contribute to the uncertainty faced by the agents. Their rational decision will be based on the expected value of future development. Replacing future shocks, u_{t+1} , by the stochastic scale variable and auxiliary shocks, $\sigma \varepsilon_{t+1}$, it is possible to take future uncertainty into account in the perturbation approach.

Based on the partial derivatives of $\mathbb{E}_t \{F(y_{t-1}, u_t, \varepsilon_{t+1}, \sigma)\}$, evaluated at the deterministic steady state, we will recover the partial derivatives of unknown function $g(y_{t-1}, u_t, \sigma)$. It is useful to distinguish two types of perturbation that take place simultaneously:

1. for state space points away, but in the neighborhood, of the deterministic steady state, by considering variations in y_{t-1} and u_t
2. away from a deterministic model towards a stochastic one, by increasing the stochastic scale of the model from $\sigma = 0$ to a positive value.

The deterministic steady state, $\bar{y}$, is formally defined by

$$f(\bar{y}, \bar{y}, \bar{y}, 0) = 0.$$

A model can have several steady states, but only one of them will be used for a local approximation.

Furthermore, the decision rule evaluated at the deterministic steady state must verify, in the absence of shocks and future uncertainty,

$$\bar{y} = g(\bar{y}, 0, 0).$$

The deterministic steady state is found by solving a set of nonlinear equations. Because, in practice, the steady state needs to be computed repeatedly for a great many values of the parameters in estimation, it is best to use an analytic solution when one is available, or to use analytic substitution to reduce the size of the nonlinear problem to be solved.

The perturbation approach starts with a Taylor expansion of the original model. It is necessary to proceed order by order: the first-order approximation of the solution function will enter in the second-order approximation, the first- and the second-order solution in the computation of the third order, and so on.

2.4.1 First-Order Approximation of the Model

A first-order expansion of the decision rule takes the form

$$y_t \approx \bar{y} + g_y \hat{y}_{t-1} + g_u u_t$$

where $\hat{y}_{t-1} = y_{t-1} - \bar{y}$ and the first-order derivatives of function $g(y_{t-1}, u_t)$, contained in matrices g_y and g_u , are unknown. Our task is to recover them from the first-order expansion of the original model:

$$\begin{aligned} & \mathbb{E}_t \left\{ F^{(1)}(y_{t-1}, u_t, \varepsilon_{t+1}, \sigma) \right\} \\ &= \mathbb{E}_t \left\{ f(\bar{y}, \bar{y}, \bar{y}, 0) + f_{y_+} (g_y (g_y \hat{y} + g_u u + g_\sigma \sigma) + g_u \sigma \varepsilon' + g_\sigma \sigma) \right. \\ & \quad \left. + f_{y_0} (g_y \hat{y} + g_u u + g_\sigma \sigma) + f_{y_-} \hat{y} + f_u u \right\} \\ &= 0. \end{aligned}$$

Here, we introduce the following notations: $\hat{y} = y_{t-1} - \bar{y}$, $u = u_t$, $\varepsilon' = \varepsilon_{t+1}$, $f_{y_+} = \frac{\partial f}{\partial y_{t+1}}$, $f_{y_0} = \frac{\partial f}{\partial y_t}$, $f_{y_-} = \frac{\partial f}{\partial y_{t-1}}$, $f_u = \frac{\partial f}{\partial u_t}$, $g_y = \frac{\partial g}{\partial y_{t-1}}$, $g_u = \frac{\partial g}{\partial u_t}$, $g_\sigma = \frac{\partial g}{\partial \sigma}$.

It is easy to compute the conditional expectation. Evaluated at the deterministic steady state, all partial derivatives are deterministic as well. The expectation being a linear operator, it is distributed over all the terms and reduces to $\mathbb{E}_t \{\varepsilon'\} = 0$. This disappearance of future shocks is a manifestation of the property of certainty equivalence in linear(-ized) models.

We are now faced with a deterministic equation:

$$\begin{aligned} & \mathbb{E}_t \left\{ F^{(1)}(y_{t-1}, u_t, \varepsilon_{t+1}, \sigma) \right\} \\ &= f(\bar{y}, \bar{y}, \bar{y}, 0) + f_{y_+} (g_y (g_y \hat{y} + g_u u + g_\sigma \sigma) + g_\sigma \sigma) \\ & \quad + f_{y_0} (g_y \hat{y} + g_u u + g_\sigma \sigma) + f_{y_-} \hat{y} + f_u u \\ &= (f_{y_+} g_y g_y + f_{y_0} g_y + f_{y_-}) \hat{y} + (f_{y_+} g_y g_u + f_{y_0} g_u + f_u) u \\ & \quad + (f_{y_+} g_y g_\sigma + f_{y_0} g_\sigma) \sigma \\ &= 0. \end{aligned}$$

Because the equation must hold for any value of $\hat{y}$, u , and σ , it must be that

$$(f_{y_+} g_y g_y + f_{y_0} g_y + f_{y_-}) = 0, \quad (2.3)$$

$$(f_{y_+} g_y g_u + f_{y_0} g_u + f_u) = 0, \quad (2.4)$$

$$(f_{y_+} g_y g_\sigma + f_{y_0} g_\sigma) = 0. \quad (2.5)$$

These equations will let us recover unknown g_y , g_u , and g_σ , respectively.

2.4.1.1 Recovering g_y

The first condition reveals a particular difficulty as g_y appears in a matrix polynomial equation:

$$(f_{y_+} g_y g_y + f_{y_0} g_y + f_{y_-}) \hat{y} = 0. \quad (2.6)$$

Several approaches have been proposed in the literature. One of the most robust and efficient is as follows.

First, rewrite equation (2.6) using a state space representation:

$$\begin{bmatrix} 0 & f_{y+} \\ I & 0 \end{bmatrix} \begin{bmatrix} I \\ g_y \end{bmatrix} g_y \hat{y} = \begin{bmatrix} -f_{y-} & -f_{y_0} \\ 0 & I \end{bmatrix} \begin{bmatrix} I \\ g_y \end{bmatrix} \hat{y} \quad (2.7)$$

or, using the fact that, in the absence of shocks and at first order,

$$\begin{aligned} \hat{y}_t &= g_y \hat{y}_{t-1} \\ \hat{y}_{t+1} &= g_y g_y \hat{y}_{t-1}. \end{aligned}$$

$$\begin{bmatrix} 0 & f_{y+} \\ I & 0 \end{bmatrix} \begin{bmatrix} \hat{y}_t \\ \hat{y}_{t+1} \end{bmatrix} = \begin{bmatrix} -f_{y-} & -f_{y_0} \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{y}_{t-1} \\ y_t \end{bmatrix} \quad (2.8)$$

Note that the lower submatrix block of the coefficient matrices imposes the condition that the upper half of the right-hand-side state vector be equal to the lower half of the left-hand-side one.

Given that only $y_{t-1} - \bar{y}$ is fixed by initial conditions in dynamic system (2.8), the dynamics are obviously underdetermined. This should not come as a surprise because it is well known that rational expectation models admit many solutions, most of them with self-fulfilling dynamics.

The literature about DSGE models focuses on models that have a unique stable dynamic, meaning that after a shock, there is a single trajectory back to equilibrium. The existence of a unique stable trajectory makes it easier to postulate that agents are able to coordinate their expectation on a single trajectory for the economy.

We therefore use the requirement of a single stable trajectory as a selection device to isolate one solution for g_y . Studying the stability of a linear dynamic system requires analyzing its eigenvalues. However, the presence of the D matrix on the left-hand side makes computing the eigenvalues more complicated, particularly because, in many applications, this matrix may be singular.

The theory of generalized eigenvalues and the real generalized Schur decomposition (see Golub and van Loan 1996) provides a way to handle this problem.

The real generalized Schur decomposition stipulates that for a pencil formed by two real $n \times n$ matrices, there exist orthonormal matrices Q and Z such that

$$\begin{aligned} S &= QEZ \\ T &= QDZ \end{aligned}$$

and S is upper-triangular and T is quasi-upper-triangular, with $Q'Q = Z'Z = I$. A quasi-triangular matrix is a block triangular matrix, with either 1×1 or 2×2 blocks on the main diagonal. The scalar blocks are associated with real eigenvalues, and the 2×2 blocks with complex ones. The algorithm necessary to perform the generalized Schur decomposition, often referred to as the QZ algorithm, is available in several linear

algebra libraries and matrix programming languages such as Gauss, Matlab, Octave, and Scilab.

The generalized Schur decomposition permits the computation of the generalized eigenvalue problem that solves

$$\lambda_i D x_i = E x_i.$$

When a diagonal block of matrix S is a scalar, $S_{i,i}$, the generalized eigenvalue is obtained in the following manner:

$$\lambda_i = \begin{cases} \frac{S_{i,i}}{T_{i,i}} & \text{if } T_{i,i} \neq 0 \\ +\infty & \text{if } T_{i,i} = 0 \text{ and } S_{i,i} > 0 \\ -\infty & \text{if } T_{i,i} = 0 \text{ and } S_{i,i} < 0 \\ \text{any } c \in \mathbb{C} & \text{if } T_{i,i} = 0 \text{ and } S_{i,i} = 0. \end{cases}$$

In the last case, any complex number is generalized eigenvalues of pencil $\langle D, E \rangle$. This obviously creates a problem for the stability analysis. However, this case only occurs when the model is singular, when one equation can be expressed as a linear combination of the other ones. It is nevertheless important for the software to check for this case because it is an easy mistake to make when writing a complex model.

The algorithm is such that when a diagonal block of matrix S is a 2×2 matrix of the form

$$\begin{bmatrix} S_{ii} & S_{i,i+1} \\ S_{i+1,i} & S_{i+1,i+1} \end{bmatrix},$$

the corresponding block of matrix T is a diagonal matrix,

$$(S_{i,i} T_{i+1,i+1} - S_{i+1,i+1} T_{i,i})^2 < -4 S_{i+1,i} S_{i+1,i} T_{i,i} T_{i+1,i+1},$$

and there is a pair of conjugate eigenvalues:

$$\lambda_i, \lambda_{i+1} = \frac{S_{ii} T_{i+1,i+1} + S_{i+1,i+1} T_{i,i} \pm \sqrt{(S_{i,i} T_{i+1,i+1} - S_{i+1,i+1} T_{i,i})^2 + 4 S_{i+1,i} S_{i+1,i} T_{i,i} T_{i+1,i+1}}}{2 T_{i,i} T_{i+1,i+1}} \quad (2.9)$$

In any case, the theory of generalized eigenvalues is an elegant way of solving the problem created by the possibility that the D matrix is singular: it introduces the notion of infinite eigenvalue. From the point of view of the analysis of the dynamics of a linear system, it is obvious that infinite eigenvalues, positive or negative, must be treated as explosive roots.²

² The additional complexity introduced by the emergence of quasi-triangular matrices in the real generalized Schur decomposition is the price being paid to remain in the set of real numbers. From a

The next step is to apply the real generalized Schur decomposition to the linear dynamic system while partitioning it between stable and unstable components:

$$\begin{bmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{bmatrix} \begin{bmatrix} Z'_{11} & Z'_{21} \\ Z'_{12} & Z'_{22} \end{bmatrix} \begin{bmatrix} I \\ g_y \end{bmatrix} g_y \hat{y} = \begin{bmatrix} S_{11} & S_{12} \\ 0 & S_{22} \end{bmatrix} \begin{bmatrix} Z'_{11} & Z'_{21} \\ Z'_{12} & Z'_{22} \end{bmatrix} \begin{bmatrix} I \\ g_y \end{bmatrix} \hat{y}.$$

The partitioning is such that S_{11} and T_{11} have stable eigenvalues and S_{22} and T_{22} , explosive ones. The rows of the Z matrix are in turn partitioned so as to be conformable with $\begin{bmatrix} I \\ g_y \end{bmatrix}$.

The only way to cancel the influence of explosive roots on the dynamics and to obtain a stable trajectory is to impose

$$Z'_{21} + Z'_{22}g_y = 0$$

or

$$g_y = -(Z_{22}^{-1})' Z'_{21}. \quad (2.10)$$

A unique stable trajectory exists if and only if Z_{22} is nonsingular: there must be as many roots larger than one in modulus as there are forward-looking variables in the model, and the rank condition must be satisfied. This corresponds to Blanchard and Kahn's conditions for the existence and unicity of a stable equilibrium (Blanchard and Kahn 1980).

When the condition is satisfied, equation (2.10) provides the determination of g_y . Determining g_y , while selecting the stable trajectory, is the most mathematically involved step in the solution of linear rational expectation models. Recovering g_u and g_σ is much simpler.

2.4.1.2 Recovering g_u

Given g_y , the solution for g_u is directly obtained from equation (2.4):

$$f_{y+}g_y g_u + f_{y_0}g_u + f_u = 0$$

and

$$g_u = -(f_{y+}g_y + f_{y_0})^{-1} f_u.$$

2.4.1.3 Recovering g_σ

Equation (2.5) provides the necessary condition to establish that g_σ is always null:

$$f_{y+}g_y g_\sigma + f_{y_0}g_\sigma = 0$$

computer implementation point of view, it is simpler and more efficient than having to use computations with complex numbers.

is homogeneous and

$$g_\sigma = 0.$$

This is yet another manifestation of the certainty equivalence property of first-order approximation.

2.4.1.4 First-Order Approximated Decision Function

Putting everything together, the first-order approximation of the solution function $g()$ takes the form

$$y_t = \bar{y} + g_y \hat{y}_{t-1} + g_u u_t. \quad (2.11)$$

It is a $VAR(1)$ model, but the coefficient matrices g_y and g_u are constrained by the structural parameters, the specification of the equations of the original nonlinear model, the rational expectation hypothesis, and the selection of a stable dynamics.

However, the form of the first-order approximated solution let us use all the usual tools developed for the analysis of VAR models (see, e.g., Hamilton 1994).

In particular, the first and second theoretical moments are derived as

$$\begin{aligned} \mathbb{E}\{y_t\} &= \bar{y}, \\ \Sigma_y &= g_y \Sigma_y g_y' + \sigma^2 g_u \Sigma_u g_u' \end{aligned}$$

where Σ_y is the unconditional variance of endogenous variables y_t . The variance is determined by a Lyapunov equation that is best solved by a specialized algorithm (Bini et al. 2012).

To the extent that DSGE models are used to analyze the frequencies of fluctuations, the moments are often compared to empirical moments in the data after de-trending by the Hodrick-Prescott filter. Uhlig (1999) provides formulas to compute theoretical variances after removing the Hodrick-Prescott trend.

Impulse responses functions (IRFs) can be evaluated directly, simply by running forward equation (2.11), with u_1 equal to the deterministic impulse and $u_t = 0$, for $t > 1$. This provides an *average* IRF where it is the effect of random shocks after the first period that is averaged. Because the model is linear, it is equivalent to considering the average effect of future shocks or the average shock as equal to zero. Note also that, in a linear model, the IRF is independent of the initial position at which the system sits when the deterministic impulse is hitting.

2.4.2 Second-Order Approximation of the Model

A second-order approximation brings differs from a first-order approximation in two ways: first, the decision rules have the shape of parabolic curves instead of straight lines and, more important, the certainty equivalence is broken.

In most cases, fitting the decision rules with parabolic curves instead of straight lines brings only a moderate benefit, and it is not true that a second-order approximation is

always more accurate than a first-order one. Remember also that the Taylor expansion of a function diverges outside its ratio of convergence (Judd 1998).

Breaking the certainty equivalence is the most interesting qualitative benefit of going to second order. It permits one to address issues related to attitude toward risk, precautionary motive, and risk premium, albeit in a very elementary manner: at second order, the risk premium is a constant. If one wants a risk premium that varies with the state of the system, it is necessary to consider at least a third-order approximation.

Considering a second-order approximation is a natural step in a perturbation approach. It has been discussed in Collard and Juillard (2001), Kim et al. (2008), Sims (2000), and Gomme and Klein (2011). The computation of a second-order approximation is done on the basis of the first-order approximation, adding the second-order Taylor coefficients to the solution function. As for the derivation of the first-order approximation, we start with the second-order approximation of the original model. However, the derivation is mathematically simpler, because the selection of the locally stable trajectory has been done at the first order.

A second-order approximation of model (2.2) is given by

$$\begin{aligned} \mathbb{E}_t \left\{ F^{(2)}(y_{t-1}, u_t, \varepsilon_{t+1}, \sigma) \right\} = & \mathbb{E}_t \left\{ F^{(1)}(y_{t-1}, u_t, \varepsilon_{t+1}, \sigma) \right. \\ & + 0.5 \left(F_{y-y-}(\hat{y} \otimes \hat{y}) + F_{uu}(u \otimes u) + F_{u'u'}\sigma^2(\epsilon' \otimes \epsilon') + F_{\sigma\sigma}\sigma^2 \right) + F_{y-u}(\hat{y} \otimes u) \\ & \left. + F_{y-u'}(\hat{y} \otimes \sigma\epsilon') + F_{y-\sigma}\hat{y}\sigma + F_{uu'}(u \otimes \sigma\epsilon') + F_{u\sigma}u\sigma + F_{u'\sigma}\sigma\epsilon'\sigma \right\} \end{aligned}$$

where $F^{(1)}(y_{t-1}, u_t, \varepsilon_{t+1}, \sigma)$ represents the first-order approximation in a compact manner. From the derivation of the same, we know that $\mathbb{E}_t \left\{ F^{(1)}(y_{t-1}, u_t, \varepsilon_{t+1}, \sigma) \right\} = 0$.

The second-order derivatives of the vector of functions, $F()$, are represented in the following manner:

$$\frac{\partial^2 F}{\partial x \partial x} = \begin{bmatrix} \frac{\partial^2 F_1}{\partial x_1 \partial x_1} & \frac{\partial^2 F_1}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 F_1}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2 F_1}{\partial x_n \partial x_n} \\ \frac{\partial^2 F_2}{\partial x_1 \partial x_1} & \frac{\partial^2 F_2}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 F_2}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2 F_2}{\partial x_n \partial x_n} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \vdots \\ \frac{\partial^2 F_m}{\partial x_1 \partial x_1} & \frac{\partial^2 F_m}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 F_m}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2 F_m}{\partial x_n \partial x_n} \end{bmatrix}.$$

It is easy to reduce the conditional expectation, but contrary to what happens at first order, the variance of future shocks remains after simplification:

$$\begin{aligned} \mathbb{E}_t \left\{ F^{(2)}(y_{t-1}, u_t, \varepsilon_{t+1}, \sigma) \right\} = & \mathbb{E}_t \left\{ F^{(1)}(y_{t-1}, u_t, \varepsilon_{t+1}, \sigma) + F_{y-u}(\hat{y} \otimes u) \right. \\ & + 0.5 \left[F_{y-y-}(\hat{y} \otimes \hat{y}) + F_{uu}(u \otimes u) + F_{u'u'}\sigma^2(\epsilon' \otimes \epsilon') \right. \\ & \left. + F_{\sigma\sigma}\sigma^2 \right] + F_{y-u'}(\hat{y} \otimes \sigma\epsilon') + F_{y-\sigma}\hat{y}\sigma + F_{uu'}(u \otimes \sigma\epsilon') \\ & \left. + F_{u\sigma}u\sigma + F_{u'\sigma}\sigma\epsilon'\sigma \right\} = 0.5 \left[F_{y-y-}(\hat{y} \otimes \hat{y}) + F_{uu}(u \otimes u) \right. \\ & \left. + \left(F_{u'u'}\bar{\Sigma}_\varepsilon + F_{\sigma\sigma} \right) \sigma^2 \right] + F_{y-u}(\hat{y} \otimes u) + F_{y-\sigma}\hat{y}\sigma + F_{u\sigma}u\sigma \\ & = 0 \end{aligned}$$

where $\vec{\Sigma}_\varepsilon$ represents the vectorization of the covariance matrix of the auxiliary shocks, with the columns stacked on top of each other.

The only way the above equation can be satisfied is when

$$\begin{aligned} F_{yy} &= 0 \\ F_{yu} &= 0 \\ F_{uu} &= 0 \\ F_{y-\sigma} &= 0 \\ F_{u\sigma} &= 0 \\ F_{u'u'}\vec{\Sigma}_\varepsilon + F_{\sigma\sigma} &= 0 \end{aligned}$$

Each of these partial derivatives of function $F()$ represents, in fact, the second-order derivatives of composition of the original function $f()$ and one or two instances of the solution function $g()$. The fact that each of the above partial derivatives must be equal to zero provides the restrictions needed to recover the second-order partial derivatives of the solution function.

The second-order derivative of the composition of two functions plays an important role in what follows. Let's consider the composition of two functions:

$$\begin{aligned} y &= z(s) \\ f(y) &= f(z(s)) \end{aligned}$$

then,

$$\frac{\partial^2 f}{\partial s \partial s} = \frac{\partial f}{\partial y} \frac{\partial^2 g}{\partial s \partial s} + \frac{\partial^2 f}{\partial y \partial y} \left(\frac{\partial g}{\partial s} \otimes \frac{\partial g}{\partial s} \right).$$

It is worth noting that the second-order derivatives of the vector of functions $g()$ appear in a linear manner in the final result, simply pre-multiplied by the Jacobian matrix of functions $f()$.

2.4.2.1 Recovering g_{yy}

The second-order derivatives of the solution function with respect to the endogenous state variables, g_{yy} , can be recovered from $F_{y-y-} = 0$. When one unrolls this expression, one obtains

$$\begin{aligned} F_{y-y-} &= f_{y+} (g_{yy}(g_y \otimes g_y) + g_y g_{yy}) + f_{y\sigma} g_{yy} + B_1 \\ &= 0 \end{aligned}$$

where B_1 is a term that contains not the unknown second-order derivatives of function $g()$, but only first-order derivatives of $g()$ and first- and second-order derivatives of $f()$. It is therefore possible to evaluate B_1 on the basis of the specification of the original equations and the results from first-order approximation.

This equation can be rearranged as follows:

$$(f_{y_+}g_y + f_{y_o})g_{yy} + f_{y_+}g_{yy}(g_y \otimes g_y) = -B_1.$$

It is linear in the unknown matrix g_{yy} , but, given its form, it can't be solved efficiently by usual algorithms for linear problems. Kamenik (2005) proposes an efficient algorithm for this type of equation. As noted above, matrix $f_{y_+}g_y + f_{y_o}$ is invertible under regular assumptions.

2.4.2.2 Recovering g_{yu}

Once g_{yy} is known, its value can be used to determine g_{yu} from $F_{yu} = 0$. Developing the latter gives the following:

$$\begin{aligned} F_{y_-u} &= f_{y_+} (g_{yy}(g_y \otimes g_u) + g_y g_{yu}) + f_{y_o} g_{yu} + B_2 \\ &= 0 \end{aligned}$$

where B_2 is again a term that doesn't contain second-order derivatives of $g()$.

This is a standard linear problem, and

$$g_{yu} = - (f_{y_+}g_y + f_{y_o})^{-1} (B_2 + f_{y_+}g_{yy}(g_y \otimes g_u)).$$

2.4.2.3 Recovering g_{uu}

The procedure for recovering g_{uu} is very similar, using $F_{uu} = 0$:

$$\begin{aligned} F_{uu} &= f_{y_+} (g_{yy}(g_u \otimes g_u) + g_y g_{uu}) + f_{y_o} g_{uu} + B_3 \\ &= 0 \end{aligned}$$

where B_3 is a term that doesn't contain second order derivatives of $g()$. This is a standard linear problem, and

$$g_{uu} = - (f_{y_+}g_y + f_{y_o})^{-1} (B_3 + f_{y_+}g_{yy}(g_u \otimes g_u)).$$

2.4.2.4 Recovering $g_{y\sigma}$, $g_{u\sigma}$

As for first order, the partial cross-derivatives with only one occurrence of the stochastic scale σ are null. The result is derived from $F_{y\sigma} = 0$ and $F_{u\sigma} = 0$ and uses the fact that $g_\sigma = 0$,

$$\begin{aligned} F_{y\sigma} &= f_{y_+}g_y g_{y\sigma} + f_{y_o}g_{y\sigma} \\ &= 0 \end{aligned}$$

$$\begin{aligned} F_{u\sigma} &= f_{y_+}g_y g_{u\sigma} + f_{y_o}g_{u\sigma} \\ &= 0. \end{aligned}$$

Then,

$$g_{y\sigma} = g_{u\sigma} = 0.$$

2.4.2.5 Recovering $g_{\sigma\sigma}$

Future uncertainty affects current decisions through the second derivative with respect to the stochastic scale of the model, $g_{\sigma\sigma}$. It is recovered from $(F_{u'u'}\vec{\Sigma}_\epsilon + F_{\sigma\sigma})\sigma^2 = 0$.

$$\begin{aligned} F_{\sigma\sigma} + F_{u'u'}\Sigma_\epsilon &= f_{y+}(g_{\sigma\sigma} + g_y g_{\sigma\sigma}) + f_{y_0} g_{\sigma\sigma} + (f_{y+y+}(g_u \otimes g_u) + f_{y+} g_{uu})\vec{\Sigma}_\epsilon \\ &= 0 \end{aligned}$$

taking into account that $g_\sigma = 0$. Note that g_{uu} must have been determined before one can $g_{\sigma\sigma}$.

This is a standard linear problem:

$$g_{\sigma\sigma} = - (f_{y+}(I + g_y) + f_{y_0})^{-1} (f_{y+y+}(g_u \otimes g_u) + f_{y+} g_{uu})\vec{\Sigma}_\epsilon.$$

2.4.2.6 Approximated Second-Order Decision Functions

The second-order approximation of the solution function, $g()$, is given by

$$\begin{aligned} y_t &= \bar{y} + 0.5 g_{\sigma\sigma} \sigma^2 + g_y \hat{y}_{t-1} + g_u u_t + 0.5 [g_{yy}(\hat{y}_{t-1} \otimes \hat{y}_{t-1}) + g_{uu}(u_t \otimes u_t) \\ &\quad + g_{yu}(\hat{y}_{t-1} \otimes u_t)]. \end{aligned}$$

Remember that σ and Σ_ϵ were introduced as auxiliary devices to take into account the effect of future uncertainty in the derivation of the approximated solution by a perturbation method. They are related to the variance of the original shocks by $\Sigma_u = \sigma^2 \Sigma_\epsilon$. It is, therefore, always possible to choose $\Sigma_\epsilon = \Sigma_u$ and have $\sigma = 1$.

There is no close form solution for the moments of the endogenous variables when approximated at second order because each moment depends on all the moments of higher order. As suggested by Kim et al. (2008), it is, however, possible to compute a second-order approximation of these moments, by ignoring the contribution of moments higher than 2:

$$\begin{aligned} \Sigma_y &= g_y \Sigma_y g_y' + \sigma^2 g_u \Sigma_\epsilon g_u' \\ \mathbb{E}\{y_t\} &= \bar{y} + \frac{1}{2} (I - g_y)^{-1} (g_{\sigma\sigma} + g_{yy} \vec{\Sigma}_y + g_{uu} \vec{\Sigma}_\epsilon). \end{aligned}$$

The formula for the variance, Σ_y , depends only on the first derivatives of the solution function, $g()$. It is, therefore, the same as the variance derived for the first-order approximation of the solution function. By contrast, the unconditional mean of endogenous variables is affected by the variance of y and u and $g_{\sigma\sigma}$. It is different from the mean obtained on the basis of a first-order approximation.

2.4.3 Higher-Order Approximation

Computing higher-order approximations doesn't present greater mathematical difficulties than does approximation at second order. The only computational difficulty is with the management of a very large number of derivatives.

The core of the procedure is provided by the Faà di Bruno formula for the k th order of the composition of two functions in the multivariate case (Ma 2009). As above, let's consider $f(y) = f(z(s))$. Given their high number of dimensions, we represent derivatives of arbitrary order as tensors

$$\begin{aligned}\frac{\partial^j f^i}{\partial s_{\alpha_1} \dots \partial s_{\alpha_j}} &= [F_{s^j}^i]_{\alpha_1 \dots \alpha_j}, \\ \frac{\partial^\ell f^i}{\partial y_{\beta_1} \dots \partial y_{\beta_\ell}} &= [F_{y^\ell}^i]_{\beta_1 \dots \beta_\ell}, \\ \frac{\partial^k z^\eta}{\partial s_{\gamma_1} \dots \partial s_{\gamma_k}} &= [z_{s^k}^\eta]_{\gamma_1 \dots \gamma_k},\end{aligned}$$

and, following Einsteinian notation, we use the following convention to indicate a sum of products along identical indices appearing first as subscript then as superscript of a tensor ($\beta_i, \dots, \beta_j$ in the following example):

$$[x]_{\beta_1, \dots, \beta_j}^{\alpha_1, \dots, \alpha_i} [y]_{\gamma_1, \dots, \gamma_k}^{\beta_1, \dots, \beta_j} = \sum_{\beta_1} \dots \sum_{\beta_j} [x]_{\beta_1, \dots, \beta_j}^{\alpha_1, \dots, \alpha_i} [y]_{\gamma_1, \dots, \gamma_k}^{\beta_1, \dots, \beta_j}.$$

The partial derivative of $f^i(s)$ with respect to s is written as a function of the partial derivatives of $f^i()$ with respect to y and the partial derivatives of $z()$ with respect to s :

$$[f_s^j]^i_{\alpha_1 \dots \alpha_j} = \sum_{l=1}^j [f_z^l]^i_{\beta_1 \dots \beta_l} \sum_{c \in \mathcal{M}_{l,j}} \prod_{m=1}^l [z_{s^{|c_m|}}^{\beta_m}]_{\alpha(c_m)}^{\beta_m}$$

where $\mathcal{M}_{l,j}$ is the set of all partitions of the set of j indices with l classes, $|c_m|$ is the cardinality of a set, c_m is m th class of partition c , and $\alpha(c_m)$ is a sequence of α s indexed by c_m . Note that $\mathcal{M}_{1,j} = \{\{1, \dots, j\}\}$ and that $\mathcal{M}_{j,j} = \{\{1\}, \{2\}, \dots, \{j\}\}$.

The formula can easily be unfolded by hand for second or third order. For higher order, the algorithm must be implemented in computer code, but it only requires loops of sums of products and the computation of all partitions of a set of indices.³

As already noted in the approximation at second order, the highest-order derivative z_{s^j} always enters the expression in a linear fashion and is simply pre-multiplied by the Jacobian matrix f_z .

³ Dynare++, written by Ondra Kamenik, available at <http://www.dynare.org/documentation-and-support/dynarepp>, and perturbationAIM, written by Eric Swanson, Gary Anderson, and Andrew Levin, available at <http://www.ericswanson.us/perturbation.html>, use such a formula to compute solutions of a DSGE model at arbitrary order.

Our models involve the composition of the original equation and two instances of the decision function. In order to recover the k th-order derivatives of the decision function, g_{y^k} , it is necessary to solve the equation

$$(f_{y_+}g_y + f_{y_0})g_{y^k} + f_{y_+}g_{y^k}g_y^{\otimes k} = -B$$

where $g_y^{\otimes k}$ is the k th Kronecker power of matrix g_y and B is a term that doesn't contain the unknown k -order derivatives of function $g()$, but only contains lower-order derivatives of $g()$ and first- to k -order derivatives of $f()$. It is therefore possible to evaluate B on the basis of the specification of the original equations and the results from lower-order approximation.

The other k -order derivatives are solved for in an analogous manner.

2.4.4 Assessing Accuracy

As one obtains an approximated value of the solution function, it is important to assess the accuracy of this approximation. Ideally, one would like to be able to compare the approximated solution to the true solution or an approximated solution delivered by a method known to be more accurate than local approximation. As discussed above, such solutions are only available for small models.

Judd (1992) suggests performing error analysis by plugging the approximate solution, $\hat{g}()$, into the original model (2.1) as follows,

$$\varepsilon_t = \mathbb{E}_t \left[f \left(\hat{g} \left(\hat{g}(y_{t-1}, u_t, \sigma), u_{t+1}, \sigma \right), g(y_{t-1}, u_t, \sigma), u_t \right) \right],$$

where u_{t+1} is random from the point of view of the conditional expectation at period t . The conditional expectation must be computed by numerical integration, for example, by quadrature formula when there is a small number of shocks or by monomial rule or quasi-Monte Carlo integration for a larger number of shocks.

When it is possible to specify the equations of the model in such a way that the error of an equation is expressed in an interpretable unit, it provides a scale on which one can evaluate the relative importance of errors. Judd (1992) uses the example of the Euler equation for household consumption choice that can be written so that the error appears in units of consumption.

2.5 ESTIMATION

The foregoing discussion of solution techniques for DSGE models assumed that the value of the model parameters was known. In practice, this knowledge can only be inferred from observation of the data.

In earlier real business cycle tradition, following Kydland and Prescott (1982), parameters were calibrated. A main idea of the *calibration* approach is to choose parameter values from microeconomic studies and to fix free parameters so as to reproduce moments of interest in the aggregate data. See Kydland and Prescott (1996), Hansen and Heckman (1996), and Sims (1996) for critical discussion of this approach.

The calibration method has the advantage of explicitly focusing the analysis on some aspect of the data that the model must reproduce. Its major shortcoming is probably the absence of a way to measure uncertainty surrounding the chosen calibration.

The Bayesian paradigm proposes a formal way to track a priori information that is used in estimation, and it is not surprising that it has become the dominant approach in quantitative macroeconomics. Canova (2007), DeJong and Dave (2011), and Geweke (2005) provide in-depth presentations of this approach. Schorfheide (2000) is one of the first applications of Bayesian methodology to the estimation of a DSGE model, and An and Schorfheide (2007) provides a detailed discussion of the topic.

Because the use of informative priors sidesteps the issue of identification, it facilitates computation in practice, avoiding problems encountered in numerical optimization to find maximum likelihood when a parameter is weakly identified by the data.

From a methodological point of view, one can consider that the Bayesian approach builds a bridge between calibration and classical estimation: using very tight priors is equivalent to calibrating a model, whereas uninformative priors provide results similar to those obtained by classical estimation.

Uncertainty and a priori knowledge about the model and its parameters are described by the *prior* probability distribution. Confrontation with the data leads to a revision of these probabilities in the form of the *posterior* probability distribution.

The Bayesian approach implies several steps. The first is to choose the prior density for the parameters. This requires care, because it is not always obvious how to translate informal a priori information into a probability distribution and, in general, the specification of the priors has an influence on the results.

The second step, the computation of the posterior distribution, is very demanding. Because a DSGE estimated model is nonlinear in the parameters, there is no hope for conjugate priors, and the shape of the posterior distribution is unknown. It can only be recovered by simulation, using Monte Carlo Markov Chain (MCMC) methods. Often, the simulation of the posterior distribution is preceded by the computation of the posterior mode that requires numerical optimization.

When one has obtained an MCMC-generated sample of draws from the posterior, it is possible to compute point estimates by minimizing an appropriate loss function and corresponding confidence regions. The MCMC sample is also used to compute the marginal density of the model, which is used to compare models, and the posterior distribution of various results of the model such as IRFs and forecasts.

In order to fix ideas, let's write the prior density of the estimated parameters of the model as $p(\theta_{A|A})$, where A represents the model and θ_A , the estimated parameters of that model. It helps to keep an index of the model in order to, later, compare models. The

vector of estimated parameters, θ_A , may contain not only structural parameters of the model but also the parameters describing the distribution of the shocks in that model.

The prior density describes beliefs held a priori, before considering the data. In the DSGE literature, traditional sources of prior information are microeconomic estimations, previous studies or studies, concerning similar countries. This information typically helps set the center of the prior distribution for a given parameter. The determination of the dispersion of the prior probability, which is more subjective, quantifies the uncertainty attached to the prior information.

The model itself specifies the probability distribution of a sequence of observable variables, conditional on the value of the parameters, $p(Y_T|\theta_A, A)$, where Y_T represents the sequence $y_1, \dots, y_T$. Because we are dealing with a dynamic model, this density can be written as the product of a sequence of conditional densities:

$$p(Y_T|\theta_A, A) = p(y_1|\theta_A, A) \prod_{t=1}^T p(y_t|Y_{t-1}, \theta_A, A).$$

Once we dispose of a sample of observations, Y_T , it is possible to define the *likelihood* of the model as a function of the estimated parameters, conditional on the value of the observed variables:

$$\mathcal{L}(\theta_A|Y_T, A) = p(Y_T|\theta_A, A).$$

Using Bayes's theorem, one obtains the posterior distribution of the estimated parameters:

$$p(\theta_A|Y_T, A) = \frac{p(\theta_A|A)p(Y_T|\theta_A, A)}{\int_{\Theta_A} p(Y_T, \theta_A|A)d\theta_A}.$$

The posterior distribution expresses how the prior information is combined with the information obtained from the data to provide an updated distribution of possible values for the estimated parameters.

The denominator of the posterior is a scalar, the *marginal density*, that plays the role of a normalizing factor. We write

$$\begin{aligned} p(Y_T|A) &= \int_{\Theta_A} p(Y_T, \theta_A|A)d\theta_A \\ &= \int_{\Theta_A} p(Y_T|\theta_A, A)p(\theta_A|A)d\theta_A. \end{aligned}$$

The marginal density is useful for model comparison, but its knowledge is not required for several other steps such as computing the posterior mode, running the MCMC simulation, or computing the posterior mean. In such cases it is sufficient to evaluate the posterior density *kernel*:

$$p(\theta_A|A)p(Y_T|\theta_A, A) \propto p(\theta_A|Y_T, A).$$

The essential output of the Bayesian method is to establish the posterior distribution of the estimated parameters. However, this multidimensional distribution may be too much information to handle for the user of the model, and it is necessary to convey the results of estimation in the form of point estimates.

Given the posterior density of the parameters and the loss function of the model's user, a point estimate is determined by

$$\hat{\theta}_A = \arg \min_{\tilde{\theta}_A} \int_{\Theta_A} L(\tilde{\theta}_A, \theta_A) p(\theta_A | Y_T, A) d\theta_A.$$

It minimizes the expected loss over the posterior distribution. The loss itself is defined as the loss incurred by retaining $\tilde{\theta}_A$ as point estimate when the true parameter value is θ_A .

In economics, it is often difficult to establish the exact loss function in the context of model estimation. However, there exist general results that guide common practice:

- When the loss function is quadratic, the posterior mean minimizes expected loss.
- When the loss function is proportional to the absolute value of the difference between estimate and true value of the parameter, the posterior median minimizes this loss function.
- The posterior mode minimizes a loss function of the form 0 or 1: when the estimate coincides with the true value of the parameter, the loss is null, and the loss is constant for all other values.

This justifies the common usage of reporting the posterior mean of the parameters.

It is also useful to be able to communicate the uncertainty surrounding a point estimate. This is done with *credible sets*, which take into account that the posterior distribution is not necessarily symmetrical:

$$P(\theta \in C) = \int_C p(\theta | Y_T, A) d\theta = 1 - \alpha$$

is a $100(1 - \alpha)$ percent credible set for θ with respect to $p(\theta | Y, A)$. Obviously, there is an infinity of such sets for a given distribution. It makes sense to choose the most likely.

A $100(1 - \alpha)$ percent *highest probability density* (HPD) credible set for θ with respect to $p(\theta | Y_T, A)$ is a $100(1 - \alpha)$ percent credible set with the property

$$p(\theta_1 | Y_T, A) \geq p(\theta_2 | Y_T, A) \quad \forall \theta_1 \in C \text{ and } \forall \theta_2 \in \bar{C}$$

where $\bar{C}$ represents the complement to C . When the distribution is uni-modal, the credible set is unique.

Estimating parameters is an important part of empirical research. In particular, it permits one to quantify the intensity of a given economic mechanism. But it is rarely the end of the story. Based on the estimated value of parameters, one is also interested in other quantifiable results from a model, such as moments of the endogenous variables, variance decomposition, IRFs, shocks decomposition, and forecasts. All these objects

are conditional on the value of the parameters and, for the last two, on the observed variables. Put very abstractly, these post-estimation computations can be represented as a function of parameters and observations, $\tilde{Y} = h(Y_T, \theta)$, where $\tilde{Y}$ can be either a scalar, a vector, or a matrix, depending on the actual computation. Given the uncertainty surrounding parameter estimates, it is legitimate to consider the posterior distribution of such derived statistics.

The posterior predictive density is given by

$$\begin{aligned} p(\tilde{Y}|Y_T, A) &= \int_{\Theta_A} p(\tilde{Y}, \theta_A | Y_T, A) d\theta_A \\ &= \int_{\Theta_A} p(\tilde{Y} | \theta_A, Y_T, A) p(\theta_A | Y_T, A) d\theta_A. \end{aligned}$$

2.5.1 Model Comparison

Models are compared on the basis of their marginal density. When the investigator has a prior on the relative likelihood of one model or another, the comparison should be done on the basis of the ratio of posterior probabilities or posterior odds ratio. When she considers that all the models under consideration are equally likely, the comparison can be done simply with the Bayes factor.

The ratio of posterior probabilities of two models is

$$\frac{P(A_j|Y_T)}{P(A_k|Y_T)} = \frac{P(A_j)}{P(A_k)} \frac{p(Y_T|A_j)}{p(Y_T|A_k)}.$$

In favor of the model A_j versus the model A_k :

- the *prior odds ratio* is $P(A_j)/P(A_k)$
- the *Bayes factor* is $p(Y_T|A_j)/p(Y_T|A_k)$, and
- the *posterior odds ratio* is $P(A_j|Y_T)/P(A_k|Y_T)$.

The interpretation of the last may be a delicate matter. Jeffreys (1961) proposes the following scale for a posterior odds ratio in favor of a model:

- 1–3: the evidence is barely worth mentioning
- 3–10: the evidence is substantial
- 10–30: the evidence is strong
- 30–100: the evidence is very strong
- > 100: the evidence is decisive

2.5.2 Bayesian Estimation of DSGE Models

The application of the Bayesian methodology to the estimation of DSGE models raises a few issues linked to the adaptation of the concepts presented above to the DSGE context.

2.5.2.1 Priors

Estimated parameters are the parameters of the structural models, but also the standard deviation of structural shocks or measurement errors and, sometimes, their correlation.

Independent priors are specified for each of these parameters as well as the implicit constraint that the value of the parameters must be such that Blanchard and Kahn's condition for the existence of a unique, stable trajectory is satisfied. It is important that the priors for individual parameters be chosen in such a way as to minimize the set of parameter values excluded by the constraint of a unique, stable trajectory because the existence of a large *hole* in the parameter space specified by the individual priors makes finding the posterior mode and running the MCMC algorithm much more complicated. It also creates problems for the numerical integration necessary to compute the marginal density of the model.

Some authors have tried to estimate a model while selecting solutions in the indeterminacy region. See, for example, Lubik and Schorfheide (2004).

The most common method found in the literature is the use of independent priors for parameters. Such a choice is often not without consequences for the estimation results, however. Alternatively, Del Negro and Schorfheide (2008), for example, derive joint priors for the parameters that affect the steady state of the model.

2.5.2.2 Likelihood

From a statistical point of view, estimating a DSGE model is estimating an unobserved component model: not all the variables of the DSGE model are, indeed, observed. In fact, because, in general, DSGE models have more endogenous variables than stochastic shocks, some variables are linked by deterministic relationships. It does not make sense to include several co-dependent variables in the list of observed variables.

In fact, unless the variables that are co-dependent in the model are also linked by a deterministic relationship in the real world, the relationship embodied in the model will not be reflected in the observed variables without the model's providing a stochastic shock to account for this discrepancy. This is the problem of stochastic singularity (Sargent 1989).

The unobserved components framework suggests using a state space representation for the estimated model (Soderlind 1999). The measurement equation is

$$y_t^* = \bar{y}^* + M\hat{y}_t + \eta_t$$

where y_t^* is the vector of observed variables in period t , $\bar{y}^*$, the steady state value of the observed variables; M is a selection matrix, $\hat{y}_t$ is the vector of centered endogenous variables in the model, and η_t a vector of measurement errors.

The transition equation is simply given by the first-order approximation of the model

$$\hat{y}_t = g_y(\theta)\hat{y}_{t-1} + g_u(\theta)u_t$$

where u_t is a vector of structural shocks and $g_y(\theta)$ and $g_u(\theta)$ are the matrices of reduced-form coefficients obtained via the real generalized Schur decomposition. Note that the reduced-form coefficients are nonlinear functions of the structural parameters.

We further assume that

$$\begin{aligned}\mathbb{E}(u_t u_t') &= Q, \\ \mathbb{E}(\eta_t \eta_t') &= V \quad \text{and} \\ \mathbb{E}(u_t \eta_t') &= 0.\end{aligned}$$

2.5.2.3 The Kalman Filter

Given the state space representation introduced above, the Kalman filter computes recursively, for $t = 1, \dots, T$:

$$\begin{aligned}v_t &= y_t^* - \bar{y}^* - M \hat{y}_{t|t-1}, \\ F_t &= M P_{t|t-1} M' + V, \\ K_t &= P_{t|t-1} g_y' F_t^{-1}, \\ \hat{y}_{t+1|t} &= g_y (\hat{y}_{t|t-1} + K_t v_t), \\ P_{t+1|t} &= g_y (I - K_t M) P_{t|t-1} g_y' + g_u Q g_u',\end{aligned}$$

where $g_y = g_y(\theta)$ and $g_u = g_u(\theta)$ and with $y_{1|0}$ and $P_{1|0}$ given. Here $\hat{y}_{t|t-1}$ is the one-period-ahead forecast of endogenous variables, conditional on the information contained in observed variables until period $t - 1$.

The log-likelihood is obtained on the basis of the one-step-ahead forecast errors, v_t , and the corresponding covariance matrix, F_t :

$$\ln L(\theta | Y_T^*) = -\frac{Tk}{2} \ln(2\pi) - \frac{1}{2} \sum_{t=1}^T \ln |F_t| - \frac{1}{2} v_t' F_t^{-1} v_t,$$

where k is the number of estimated parameters. The logarithm of the posterior density is then easily computed by adding the logarithm of the prior density.

The posterior mode is usually computed numerically by hill-climbing methods, but it may be difficult to compute in practice.

2.5.2.4 Metropolis Algorithm

As already mentioned, the posterior density function of DSGE models is not analytic. It must be recovered by the MCMC algorithm.

The posterior density of DSGE models doesn't have enough structure to make it possible to use Gibbs sampling, and the algorithm of choice in practice is the Metropolis algorithm.

A common implementation of the algorithm in our context is as follows. We choose first as proposal distribution a multinormal density with covariance matrix, Σ_{mode} , proportional to the one inferred from the Hessian matrix at the mode of the

posterior density. An other choice of proposal is possible. See, for example, Chib and Ramamurthy (2010) for an alternative approach.

The Metropolis algorithm consists of the following steps:

1. Draw a starting point θ^0 with $p(\theta) > 0$ from a starting distribution $p^0(\theta)$.
2. For $t = 1, 2, \dots$

(a) Draw a *proposal* θ^* from a *jumping* distribution:

$$J(\theta^*|\theta^{t-1}) = N(\theta^{t-1}, c\Sigma_{\text{mode}});$$

(b) Compute the acceptance ratio

$$r = \frac{p(\theta^*)}{p(\theta^{t-1})};$$

(c) Set

$$\theta^t = \begin{cases} \theta^* & \text{with probability } \min(r, 1) \\ \theta^{t-1} & \text{otherwise.} \end{cases}$$

The random sample generated by the Metropolis algorithm depends upon initial conditions. It is necessary to drop the initial part of the sample before proceeding with analysis. Dropping the first 30 percent or 50 percent of the sample is common in the literature.

Intuitively, one can see that an average acceptance rate that is very high or very low is not desirable. A high average acceptance rate means that the posterior density value at the proposal point is often close to the posterior density at the current point. The proposal point must not be very far from the current point. The Metropolis algorithm is making small steps, and traveling the entire distribution will take a long time.

On the other hand, when the proposal point is very far away from the current point, chances are that the proposal point is in the tail of the distribution and is rarely accepted: the average acceptance rate is very low and, again, the Metropolis algorithm will take a long time to travel the distribution. The current consensus is that aiming for an average acceptance rate close to 25 percent is nearly optimal (Roberts and Rosenthal 2001).

The scale factor of the covariance matrix of the proposal, c , helps in tuning the average acceptance rate: increasing the size of this covariance matrix leads to a smaller average acceptance rate.

It is difficult to know a priori how many iterations of the Metropolis algorithm are necessary before one can consider that the generated sample is representative of the target distribution. Various diagnostic tests are proposed in the literature to assess whether convergence is reached (e.g., Mengersen et al. 1999).

2.5.2.5 Numerical Integration

Computing point estimates such as the mean value of the estimated parameters under the posterior distribution and other statistics or computing the marginal data density

require us to compute a multidimensional integral involving the posterior density for which we don't have an analytic formula.

Once we have at our disposal a sample of N points, drawn in the posterior distribution thanks to the Metropolis algorithm, it is easy to compute the mean of the parameters or of a function of the parameters. It is simply the average mean of the function of the parameters at each point of the sample

$$\begin{aligned}\mathbb{E}(h(\theta_A)) &= \int_{\Theta_A} h(\theta_A) p(\theta_A | Y_T, A) d\theta_A \\ &\approx \frac{1}{N} \sum_{k=1}^N h(\theta_A^k)\end{aligned}$$

where θ_A^k is drawn from $p(\theta_A | Y_T, A)$.

Computing the marginal density of the model

$$\int_{\Theta_A} p(y | \theta_A, A) p(\theta_A | A) d\theta_A$$

turns out to be more involved numerically.

The first approach is to use the normal approximation provided by Laplace's method. It can be computed after having determined the posterior mode

$$\hat{p}(Y_T | A) = (2\pi)^{\frac{k}{2}} |\Sigma_{\theta_M}|^{-\frac{1}{2}} p(\theta_A^M | Y_T, A) p(\theta_A^M | A)$$

where θ_A^M is the posterior mode and k the number of estimated parameters (the size of vector θ_A). The covariance matrix Σ_{θ_M} is derived from the inverse of the Hessian of the posterior distribution evaluated at its mode.

A second approach, referred to as modified harmonic mean, proposed by Geweke (1999), makes use of the Metropolis sample:

$$\begin{aligned}p(Y_T | A) &= \int_{\theta_A} p(\theta_A | Y_T, A) p(\theta_A | A) d\theta_A \\ \hat{p}(Y_T | A) &= \left[\frac{1}{n} \sum_{i=1}^n \frac{f(\theta_A^{(i)})}{p(\theta_A^{(i)} | Y_T, A) p(\theta_A^{(i)} | A)} \right]^{-1} \\ f(\theta) &= p^{-1} (2\pi)^{\frac{k}{2}} |\Sigma_{\theta}|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (\theta - \bar{\theta})' \Sigma_{\theta}^{-1} (\theta - \bar{\theta}) \right\} \\ &\quad \times \left\{ (\theta - \bar{\theta})' \Sigma_{\theta}^{-1} (\theta - \bar{\theta}) \leq F_{\chi_k^2(p)}^{-1} \right\}\end{aligned}$$

with p an arbitrary probability and k the number of estimated parameters. In practice, the computation is done for several values of the threshold probability, p . The fact that the result remains close when p is varying is taken as a sign of the robustness of the computation.

Name	Reference and website
AIM	Anderson and Moore (1985) http://www.federalreserve.gov/pubs/oss/oss4/about.html
Dynare	Adjemian et al. (2013) http://www.dynare.org
Dynare++	Kamenik (2011) http://www.dynare.org
IRIS toolbox	IRIS Solution Team (2013) http://code.google.com/p/iris-toolbox-project
JBenge	Winschel and Krätzig (2010) http://jbenge.sourceforge.net
PerturbationAIM	Swanson et al. (2005) http://www.ericswanson.us/perturbation.html
RATS	www.estima.com
Schmitt-Grohé and Uribe	Schmitt-Grohé and Uribe (2004) http://www.columbia.edu/~mu2166/2nd_order.htm
TROLL	http://www.intex.com/troll
Uhlig's toolkit	Uhlig (1999) http://www2.wiwi.hu-berlin.de/institute/wpol/html/toolkit.htm
WinSolve	Pierse (2007) http://winsolve.surrey.ac.uk
YADA	Warne (2013) http://www.texlips.net/yada

2.6 AVAILABLE SOFTWARE

Several software products, some free, some commercial, implement the algorithms described above. Here is a partial list.

2.7 NEW DIRECTIONS

With continuous innovation, DSGE modeling is a field that advances rapidly. While linear approximation of the models and estimation of these linearized models once seemed sufficient to describe the functioning of the economy in normal times, several developments have made new demands on methods used to solve and estimate DSGE models.

Several important nonlinear mechanisms were brought in to focus by the Great Recession, such as the zero lower bound for nominal interest rate or debt deflation and sudden stops (Mendoza and Yue 2011). These developments renewed interest in nonlinear solution and estimation methods (see Chapter 4 in this handbook).

The need to integrate financial aspects into the models requires moving away from unique representative agents. Introducing a discrete number of agents with different characteristics only calls for bigger models, not different solution methods, but dealing with the distribution of an infinite number of agents is a much more complex issue. Krusell and Smith (1998), Algan et al. (2010), Den Haan and Rendahl (2010), Kim et al. (2010), Maliar et al. (2010), Reiter (2010), and Young (2010) attempt to provide solutions for the type of heterogeneous agent models in which the distribution of agents becomes a state variable.

With the multiplication of questions that are addressed with DSGE models, the size of models increased as well. Nowadays, large multicountry models developed at international institutions such as EAGLE at the European System of Central Banks (Gomes et al. 2010), GIMF at the IMF (Kumhof et al. 2010), and QUEST at the European Commission (Ratto et al. 2009) have more than a thousand equations, and it is still necessary to develop faster solution algorithms and implementations and, more important, faster estimation methods, for that is the current bottleneck.

The arrival of new, massively parallel hardware such as GPUs on the desktop of economists pushes back the frontier of computing and opens a new perspective, but many algorithms need to be reconsidered in order take advantage of parallel computing.

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CHAPTER 3

TAX-RATE RULES FOR REDUCING GOVERNMENT DEBT

*An Application of Computational Methods for
Macroeconomic Stabilization*

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3.1 INTRODUCTION

This chapter focuses on the computational approach to the analysis of macroeconomic adjustment. Specifying, calibrating, solving, and simulating a model for evaluating alternative policy rules can appear to be a cumbersome task. There are, of course, many different types of models to choose from, alternative views about likely parameter values, multiple approximation methods to try, and different options regarding simulation.

In this chapter we work through an example to demonstrate the steps of specifying, calibrating, solving, and simulating a macroeconomic model in order to evaluate alternative policies for reducing domestic public debt. The particular application is to consider macroeconomic adjustment in a closed economy following a fiscal expansion when government debt expands. Which policy combinations work best to reduce the burden of public debt? We focus on the case when the interest rate is close to the zero lower bound (so that monetary policy cannot be used to inflate away a sizable portion of the real value of government debt) and when there is asymmetry in wage rigidity (with greater flexibility in upward movements and greater rigidity in the negative direction).

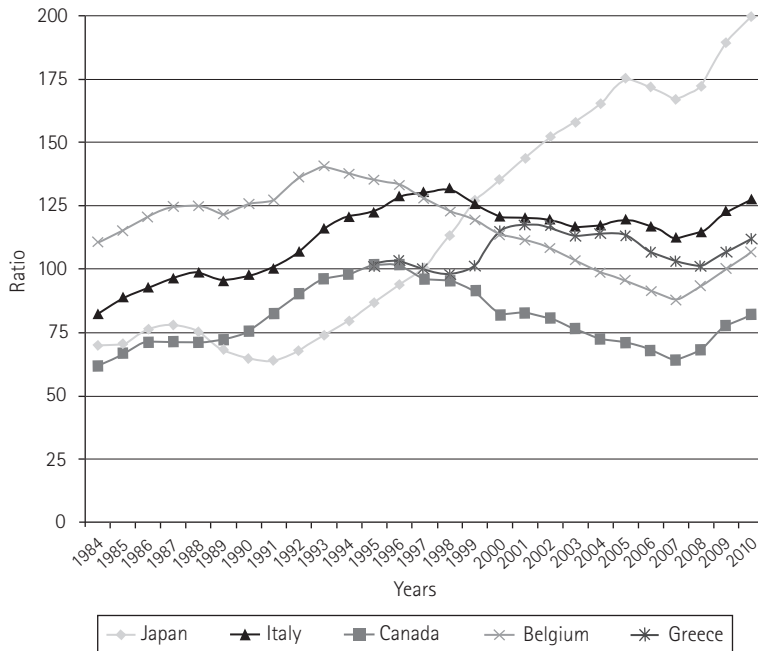


FIGURE 3.1 Government debt-to-GDP ratios, selected countries.

This question is more than just academic. Figure 3.1 shows the large increases in the debt-to-GDP ratios of selected OECD countries. Two facts emerge from this figure. First, the ratio for Japan far exceeds that of Canada and those of the smaller highly indebted OECD countries in Europe. Second, and more troubling, is that the debt-to-GDP ratio for Japan appears to be on an upward trajectory, while the ratios for Canada and the European economies appear to be much more stable. Thus, for Japan and for a number of OECD countries, stabilization of, or reduction in, the level of public debt is a pressing policy goal in its own right, apart from questions of welfare or other macroeconomic goals set in terms of inflation targets or output gaps.

We make use of a simple model, following the widely used New Keynesian framework with sticky prices and wages, but draw attention to the challenges coming from asymmetries, such as the zero lower bound for nominal interest rates and, where wage adjustment is asymmetric, in terms of greater downward nominal rigidity. We show in this chapter that the incorporation of even a little bit of additional complexity (in the form of the zero lower bound and asymmetric wage adjustment), coupled with large shocks, involves a fundamental shift in the way we go about solving and simulating models. Setting up the solution framework is a little harder (since most user-friendly programs are for linear models), and more complex computation algorithms (and more time) are needed. It is easy to understand why nonlinear models are not as popular as linear ones! But as Judd (1998) has reminded us, in the spirit of Occam's razor, good

models should be simple, but they should not be too simple, so that pressing problems are brushed aside in research.

The chapter is organized as follows. The first section presents a particular specification of a closed-economy model. This will be our base model. This is followed by a discussion about calibration and model solutions. The next section discusses solution methods, approximating functions, and optimization algorithms for solving the non-linear model. Finally, we present a few economic simulations to illustrate applications of the base model.

3.2 MODEL SPECIFICATION

The model we have chosen to work with is designed to examine alternatives to monetary policy as a debt stabilization tool when interest rates are very close to the zero lower bound. However, even when interest rates are not at the zero bound, inflating the debt away, even in part, is often not a serious option. Under central bank independence, the mandate for monetary policy is price stability. The central bank's mission is not to reduce the interest expenses of the fiscal authority by inflating away debt. This means that the reduction in debt needs to come from changes in government spending or tax revenues.

The issue is by no means straightforward; it depends on when the economy is experiencing the zero lower bound. If the economy is in a boom cycle, debt can be reduced by austerity measures (reduction in government spending) or increases in tax rates (to increase current or future tax revenues). However, if interest rates are close to the zero lower bound and the economy is in a recession, questions would be raised about the need to implement austerity measures (including increases in tax rates) solely for the purpose of managing government debt.

The application considered in this chapter is use of the base macroeconomic model to compare a number of scenarios about the effectiveness of government debt-contingent rules for tax rates (on labor income and consumption) as a means to reduce the size of the debt. Our experiment considers ways to reduce public debt (which came about because of a large initial expansion in government spending) when interest rates are close to the zero lower bound.

3.2.1 A Simple Closed-Economy Model with Bounds and Asymmetric Adjustments

The model considered is in the class of models called New Keynesian dynamic stochastic general equilibrium (DSGE) models (see also Chapter 2). The name conveys the idea that the macroeconomic model contains stickiness, adjustments, expectations,

and uncertainty, with interactions among all sectors of the economy. The simple model has three sectors: a household sector, a production sector, and a government sector with detailed policy rules about the budget deficit.

We state at the outset that this is a simple closed-economy model. There is neither an external sector nor a financial sector. The model also abstracts away from issues associated with habit persistence as well as capital accumulation and investment dynamics with adjustment costs.

3.2.1.1 Households and Calvo Wage-Setting Behavior

A household typically chooses the paths of consumption C , labor L , and bonds B to maximize the present value of its utility function $U(C, L_t)$ subject to the budget constraint. The objective function of the household is given by the following expression:

$$\begin{aligned} \text{Max}_{\{C_t, L_t, B_t\}} \mathcal{L} &= \sum_{t=0}^{\infty} \beta^t \left\{ -\Lambda_{t+l} \left[\begin{array}{c} U(C_{t+l}, L_{t+l}) \\ P_{t+l} C_{t+l} (1 + \tau_{t+l}^c) + B_{t+l} \\ -(1 + R_{t-1+l}) B_{t-1+l} \\ -(1 - \tau_{t+l}^w) W_{t+l} L_{t+l} - \Pi_{t+i} \end{array} \right] \right\} \\ U(C, L_t) &= \frac{(C_t)^{1-\eta}}{1-\eta} - \frac{L_t^{1+\varpi}}{1+\varpi}. \end{aligned}$$

Overall utility is a positive function of consumption and a negative function of labor;¹ the parameter η is the relative risk aversion coefficient, ϖ is the Frisch labor supply elasticity, and β represents the constant, exogenous discount factor. In addition to buying consumption goods C_t , households hold government bonds B_t , which pay return R_t , and receive dividends from firms Π_t . The household pays taxes on labor income $\tau_t^w W_t L_t$, and on consumption expenditures $\tau_t^c P_t C_t$. The tax rates τ_t^w, τ_t^c are treated as given policy variables.

The Euler equations implied by household optimization of its intertemporal utility with respect to C_t and B_t are

$$C_t^{-\eta} = \Lambda_t P_t (1 + \tau_t^c) \text{ and} \quad (3.1)$$

$$\Lambda_t = \beta \Lambda_{t+1} (1 + R_t). \quad (3.2)$$

The first equation (3.1) tells us that the marginal utility of consumption, divided by the tax-adjusted price level, is equal to the marginal utility of wealth Λ_t . The next equation is the Keynes-Ramsey rule for optimal saving: the marginal utility of wealth today should be equal to the discounted marginal utility tomorrow, multiplied by the gross rate of return on saving.

The Euler equation with respect to L_t is $L_t^{\varpi} = \Lambda_t (1 - \tau_t^w) W_t$ and it relates the marginal disutility of labor, adjusted by the after-tax wage, to the foregone marginal utility of wealth. However, since labor markets rarely clear, we shall replace this

¹ The coefficient of the disutility of labor is set at unity.

Euler condition (which determines labor, given the wage rate) with an alternative specification: the assumption that wages are set as staggered contracts. A fraction $(1 - \xi^w)$ of households renegotiate their contracts each period. Each household chooses the optimal wage W_t^o by maximizing the expected discounted utility subject to the demand for its labor $L_t^h : L_t^h = \left(\frac{W_t^o}{W_t}\right)^{-\zeta^w} L_t$. Taking the derivative with respect to W_t^o yields the first-order condition

$$E_t \sum_{l=0}^{\infty} \left\{ \begin{aligned} & (\xi^w \beta)^l (-L_{t+l}^w) (W_{t+l})^{\zeta^w} L_{t+l} \left[-\zeta^w (W_{t+l}^o)^{-\zeta^w-1} \right] \\ & + \Lambda_{t+l} (1 - \tau_t^w) (W_{t+l})^{\zeta^w} L_{t+l} \left[(-\zeta^w + 1) (W_{t+l}^o)^{-\zeta^w} \right] \end{aligned} \right\} = 0$$

which (assuming the usual assumption of a subsidy to eliminate the mark-up effects) can be rearranged as

$$(W_t^o)^{1+\zeta^w \varpi} = \frac{\sum_{t=0}^{\infty} (\xi^w \beta)^t (W_t)^{\zeta^w + \zeta^w \varpi} (L_t^{1+\varpi})}{\sum_{t=0}^{\infty} (\xi^w \beta) \Lambda_t (1 - \tau_t^w) (W_t)^{\zeta^w} L_t}$$

Note that, in the steady state (or when $\xi^w = 0$), this collapses to the same condition as the competitive case:

$$\begin{aligned} (W)^{1+\xi^w \varpi} &= \frac{(W)^{\zeta^w \varpi} (L^w)}{\Lambda(1 - \tau^w)} \\ W &= \frac{(L^w)}{\Lambda(1 - \tau^w)}. \end{aligned}$$

The wage equation can be rewritten using auxiliary equations N_t^w and D_t^w :

$$N_t^w = (W_t)^{\zeta^w + \zeta^w \varpi} (L_t^{1+\varpi}) + \xi^w \beta \cdot N_{t+1}^w \quad (3.3)$$

$$D_t^w = \Lambda_t (1 - \tau_t^w) (W_t)^{\zeta^w} L_t + \xi^w \beta \cdot D_{t+1}^w \quad (3.4)$$

$$(W_t^o)^{1+\xi^w \varpi} = \frac{N_t^w}{D_t^w} \quad (3.5)$$

$$W_t = \left[\xi^w (W_{t-1})^{1-\zeta^w} + (1 - \xi^w) (W_t^o)^{1-\zeta^w} \right]^{\frac{1}{1-\zeta^w}} \quad (3.6)$$

$$\xi^w = \begin{cases} \xi_{down}^w & \text{if } W_t^o \leq W_{t-1} \\ \xi_{up}^w & \text{if } W_t^o > W_{t-1} \end{cases}.$$

Since changes to wages also tend to be more sticky downwards and less sticky upwards, we have allowed the stickiness factor ξ^w to be different. More specifically, $\xi_{down}^w > \xi_{up}^w$.

3.2.1.2 Production and Calvo Price-Setting Behavior

Output is a function of labor only (that is, we abstract from issues associated with capital formation)

$$Y_t = ZL_t \quad (3.7)$$

where the productivity term Z is assumed to be fixed (for convenience at unity). Total output is for both household and government consumption

$$Y_t = C_t + G_t \quad (3.8)$$

$$G_t = \rho^g G_{t-1} + (1 - \rho^g) \bar{G} + \epsilon_t^g; \quad \epsilon_t^g \sim N(0, \sigma_g^2) \quad (3.9)$$

where government spending G_t is assumed to follow a simple exogenous autoregressive process, with autoregressive coefficient ρ^g , steady state $\bar{G}$, and a stochastic shock ϵ_t^g normally distributed with mean zero and variance σ_g^2 .

The profits of the firms are given by the following relation and distributed to the households:

$$\Pi_t = P_t Y_t - W_t L_t.$$

We assume sticky monopolistically competitive firms. In the Calvo price-setting world, there are forward-looking domestic-goods price setters and backward-looking setters. Assuming at time t that ξ^P is the probability of persistence, with demand for the product from firm j given by $Y_t \left(P_t^j / P_t \right)^{-\zeta^P}$, the optimal domestic-goods price P_t^o can be written in forward recursive formulation as

$$A = W_t / Z_t \quad (3.10)$$

$$P_t^o = \frac{N_t^P}{D_t^P} \quad (3.11)$$

$$N_t^P = Y_t (P_t)^{\zeta^P} A_t + \beta \xi^P N_{t+1}^P \quad (3.12)$$

$$D_t^P = Y_t (P_t)^{\zeta^P} + \beta \xi^P D_{t+1}^P \quad (3.13)$$

$$P_t = \left[\xi^P (P_{t-1})^{1-\zeta^P} + (1 - \xi^P) (P_t^o)^{1-\zeta^P} \right]^{\frac{1}{1-\zeta^P}} \quad (3.14)$$

where A_t is the marginal cost at time t , while the domestic price level P_t is a CES aggregator of forward- and backward-looking prices.

3.2.1.3 Monetary and Fiscal Policy

The central bank is responsible for monetary policy, and it is assumed to adopt a Taylor rule, with smoothing. We model the Taylor rule subject to a zero lower bound on the

official interest rate as

$$R_t = \max[0, \rho^r(R_{t-1}) + (1 - \rho^r)(\bar{R} + \phi^P(\pi_t - \pi^*) + \phi^Y(\theta_t - \theta^*))] \quad (3.15)$$

$$\pi_t = \frac{P_t}{P_{t-1}} - 1 \quad (3.16)$$

$$\theta_t = \frac{Y_t}{Y_{t-1}} - 1 \quad (3.17)$$

where the variable π_t is the inflation rate at time t , π^* is the target inflation rate, and θ_t is the growth rate at time t , and θ^* is the target growth rate. The smoothing coefficient is ρ^r with $0 < \rho^r < 1$. The parameter $\phi^P > 1$ is the Taylor rule inflation coefficient, ϕ^Y is the Taylor rule growth coefficient, and $\bar{R}$ is the steady-state interest rate.

The Treasury is responsible for fiscal policy, and the fiscal borrowing requirement is given as follows:

$$B_t = (1 + R_{t-1})B_{t-1} + P_t G_t - \tau_t^w W_t L_t - \tau_t^c P_t C_t. \quad (3.18)$$

3.2.1.4 Summary

In summary, the eighteen equations derived above described a simple model of a closed economy where the nominal rate is subjected to a lower bound at zero and where wage adjustments are asymmetric (more sticky downwards). The eighteen variables are $G, R, C, Y, L, \pi, \theta, \Lambda, A, P, W, N^w, D^w, W^o, N^p, D^p, P^o$, and B . In this simple model, there is only one shock (ϵ_t^g) with one unknown standard error (σ_g^2). There are seven behavioral parameters ($\beta, \eta, \varpi, \zeta^p, \zeta^w, \xi^p, \xi^w$) and six policy parameters ($\tau^w, \tau^c, \rho^g, \rho^r, \phi^P, \phi^Y$). We set $\pi^* = \theta^* = 0$.

To solve the model, we need some estimates of the parameters as well as a way to solve a model with forward-looking variables ($\Lambda_{t+1}, N_{t+1}^p, D_{t+1}^p, N_{t+1}^w$, and D_{t+1}^w). These variables, unlike the backward-looking (lagged) ones ($W_{t-1}, G_{t-1}, P_{t-1}, R_{t-1}$, and Y_{t-1}), are of course unknown at time t .

3.2.2 Calibration and Steady-State Values

The model is calibrated rather than estimated; the recent development of estimation techniques for DSGE models deserves a more detailed treatment (see Chapter 2). However, the parameters are based on estimates which are widely accepted. The calibrated base model we use is a widely shared, if not consensus, model of a closed economy that may be used for policy evaluation in the short run (fixed capital).

Table 3.1 gives the values of the parameters. The discount parameter β corresponds to an annualized risk-free rate of return equal to 2 percent. In other words, we start our analysis at the point when interest rates are low but have not yet hit the zero lower bound. Values for the Frisch elasticity of labor supply ϖ usually range from 0.5 to 1.5. We have set ϖ as equal to 1. The coefficient of relative risk aversion (equal to the

Table 3.1 Calibrated values

Symbol	Definition	Value
β	Discount factor	0.995
ϖ	Elasticity of labor supply	1
η	Relative risk aversion	2.5
ζ^w	Demand elasticity for labor	6
ζ^p	Demand elasticity for goods	6
ξ^w	Calvo wage coefficient	0.8
ξ^p	Calvo price coefficient	0.8
τ^w	Labor income tax rate	0.3
τ^c	Consumption tax rate	0.1
ρ^g	Government spending coefficient	0
ρ^r	Taylor smoothing coefficient	0.5
ϕ^p	Taylor inflation coefficient	1.5
ϕ^y	Taylor growth coefficient	0.5
σ^g	Standard deviation–spending shocks	0.01

reciprocal of the elasticity of intertemporal substitution) is usually greater than 1, since empirical estimates tend to vary between 1 and 4. We have set it to 2.5. The demand elasticity for labor ζ^w and goods ζ^p has been set at the usual value of 6 (corresponding to a markup factor of 20%) while the two Calvo stickiness parameters ξ^w, ξ^p have been set at 0.8 to capture inertia in wage and price adjustments. The tax parameters τ^w and τ^c are set at 0.3 and 0.1 respectively. The coefficients governing the Taylor rule allow for some autoregressive behavior, while satisfying the Taylor principle.

Given the parameter configuration, we can solve for the steady state. Note first that $\pi = \theta = 0$, $A = P^0 = P$, and $W^0 = W$. The values of the eight key endogenous variables (G, R, C, Y, L, P, W , and B) come from solving the following system of equations

$$\begin{aligned}
 PG &= \tau^w WL + \tau^c PC \\
 Y &= L \\
 Y &= C + G \\
 L^\varpi (1 + \tau^c)P &= C^{-\eta} (1 - \tau^w)W \\
 W &= P \\
 (1 + R) &= 1/\beta
 \end{aligned}$$

which is predicated on the assumption that there is no outstanding public debt in the steady state, $B = 0$, and the steady state price level P is normalized at unity. For the government sector, a balanced budget means that the revenue just covers government expenditure. In the steady state, equilibrium in the goods market for the closed economy requires that production of goods be equal to the demand for consumption and

Table 3.2 Steady-state values

Symbol	Definition	Value
B_0	Bonds	0
C_0	Consumption	0.7724
G_0	Government spending	0.4414
L_0	Labor	1.2137
P_0	Price level	1
W_0	Wage level	1
Y_0	Output	1.2137
R_0	Interest rate (quarterly)	0.005

government goods. For the labor market, the marginal disutility of labor should be equal to the productivity of labor, net of taxes, times the marginal utility of consumption. In this simple model, without capital, $W = P$ (because the productivity factor Z is fixed at unity). Finally, the steady-state gross interest rate, $(1 + R)$, is equal to the inverse of the social discount rate β . Solving the system of nonlinear equations gives the steady-state values (table 3.2).

For incompleteness, the values for the remaining five variables are given by the following equations:

$$\begin{aligned}
 \Lambda_o &= C_o^{-\eta} / (1 + \tau_o^c) \\
 N_o^p &= Y_o / (1 - \beta \xi^p) \\
 D_o^p &= Y_o (1 - \beta \xi^p) \\
 N_o^w &= L_o^{1+\varpi} / (1 - \xi^w \beta) \\
 D_o^w &= \Lambda_o (1 - \tau_o^w) L_o / (1 - \xi^w \beta).
 \end{aligned}$$

We also note that given the following relationship between steady-state tax rates, consumption, government spending, real labor income shares, and bond-to-GDP ratio,

$$(1 - R) \frac{B}{PY} = \frac{G}{Y} - \tau^c \frac{C}{Y} - \tau^w \frac{WL}{PY}.$$

together with our assumption that $B = 0$, and $\frac{WL}{PY} = 1$ (because labor is the only factor of production), the implied steady-state share of consumption to GDP is given by the following ratio:

$$\frac{C_o}{Y_o} = \frac{(1 - \tau_o^w)}{(1 + \tau_o^c)}.$$

In the absence of investment in this model, the government spending ratio is the remaining share.

3.3 SOLUTION METHODS

No matter how simple, DSGE models do not have closed-form solutions except under very restrictive circumstances (such as logarithmic utility functions and full depreciation of capital). We have to use computational methods if we are going to find out how the models behave for a given set of initial conditions and parameter values. However, the results may differ, depending on the solution method. Moreover, there is no benchmark exact solution for this model against which we can compare the accuracy of alternative numerical methods.

There are, of course, a variety of solution methods (see Chapters 2 and 4). Every practicing computational economist has a favorite solution method (or two). Even with a given solution method there are many different options, such as the functional form to use in any type of approximating function or the way in which we measure the errors for finding accurate decision rules for the model's control variables. The selection of one method or another is as much a matter of taste as convenience, based on speed of convergence and the amount of time it takes to set up a computer program.

Briefly, there are two broad classes of solution methods: perturbation and projection. Both are widely used and have advantages and drawbacks. We can illustrate these differences with reference to the well-known example of an agent choosing a stream of consumption (c_t) that maximizes her utility function (U) and that then defines the capital (k) accumulation, given the production function f and the productivity process z_t :

$$\begin{aligned} \max_{c_t} \sum_{t=1}^{\infty} \beta^t U(c_t) \\ k_{t+1} = f(z_t, k_t) - c_t \\ z_t = \rho z_{t-1} + \varepsilon_t, \varepsilon_t \sim N(0, \sigma^2). \end{aligned}$$

The first-order condition for the problem is

$$U'(c_t) = \beta U'(c_{t+1}) f'(k_{t+1}).$$

The system has one forward-looking variable (also known as the “jumper” or “control”) for the evolution of c_t , and one state variable, k_t which depends on the values of the forward-looking variables, c_t , and the previous-period values, k_{t-1} . The key to solving the model is to find ways to represent functional forms (“decision rules”)² for these controls, which depend on the lagged values of the state variables. Once we do this, the

² The computational literature refers to decision rules for variables that depend on their own and other expected future variables as “policy functions.” The word “policy” in this case is not to be confused with the interest rate policy function given by the Taylor rule. The terms “policy function” or “decision rule” refer to functional equations (functions of functions) that we use for the forward-looking control variables.

system becomes fully recursive and the dynamic process is generated (given an initial value for k).³

3.3.1 Perturbation Method

The first method, the perturbation method, involves a local approximation based on a Taylor expansion. For example, let $h(x_t)$ represent the decision rule (or policy function) for c_t based on the vector of state variables $x_t = [z_t, k_t]$ around the steady-state x_o :

$$h(x_t) = h(x_o) + h'(x_o)(x_t - x_o) + \frac{1}{2}h''(x_o)(x_t - x_o)^2 + \dots$$

Perturbation methods have been extensively analyzed by Schmitt-Grohé and Uribe (2004). The first-order perturbation approach (a first-order Taylor expansion around the steady state) is identical to the most widely used solution method for dynamic general equilibrium models, namely linearization or log linearization of the Euler equations around a steady state (see (Uribe 2003) for examples). The linear model is then solved using methods for forward-looking rational expectations such as those put forward by Blanchard and Kahn (1980) and later discussed by Sims (2001).

Part of the appeal of this approach lies with the fact that the solution algorithm is fast. The linearized system is quickly and efficiently solved by exploiting the fact that it can be expressed as a state-space system. Vaughan's method, popularized by Blanchard and Kahn (1980), established the conditions for the existence and uniqueness of a rational expectations solution as well as providing the solution. Canova (2007) summarizes this method as essentially an eigenvalue-eigenvector decomposition on the matrix governing the dynamics of the system by dividing the roots into explosive and stable ones. For instance, the Blanchard-Kahn condition states that the number of roots outside the unit circle must be equal to the number of forward-looking variables in order for there to be a unique stable trajectory.

This first-order approach can be extended to higher-order Taylor expansions. Moving from a first- to a second- or third-order approximation simply involves adding second-order terms linearly in the specification of the decision rules. Since the Taylor expansion has both forward-looking and backward-looking state variables, these methods also use the same Blanchard-Kahn method as the first-order approach. Collard and Julliard (2001) offer first-, second- and third-order perturbation methods in the recent versions the DYNARE system.

Log-linearization is an example of the "change of variable" method for a first-order perturbation method. Fernández-Villaverde and Rubio-Ramí (2006) take this idea

³ Taylor and Uhlig (1990) edited a special issue of the *Journal of Business and Economic Statistics* centered on the solution of the stochastic nonlinear growth model. The authors were asked to solve the model with different methods for a given set of parameters governing the model and stochastic shocks. Not surprisingly, when the shocks became progressively larger, the results of the different methods started to diverge.

one step further within the context of the perturbation method. The essence of their approach is to use a first- or second-order perturbation method but transform the variables in the decision rule from levels to power-functions. Just as a log-linear transformation is easily applied to the linear or first-order perturbation representation, these power transformations may be as well. The process simply involves iterating on a set of parameters for the power functions, in transforming the state variables, for minimizing the Euler equation errors. The final step is to back out the level of the series from the power transformations once the best set of parameters is found. They argue that this method preserves the fast linear method for efficient solution while capturing model nonlinearities that would otherwise not be captured by the first-order perturbation method.

We note that the second- and higher-order methods remain, like the first-order method, local methods. As Fernández-Villaverde and Rubio-Ramí (2006, p. 39) observe, it approximates the solution around the deterministic steady state, and it is only valid within a specific radius of convergence. Overall, the perturbation method is especially useful when the dynamics of the model consists of small deviations from the steady-state values of the variables. It assumes that there are no asymmetries, no threshold effects, no types of precautionary behavior, and no big transitional changes in the economy. The perturbation methods are local approximations, in the sense that they assume that the shocks represent small deviations from the steady state.

3.3.2 Projection Methods

The projection solution method, put forward by Den Haan and Marcet (1990, 1994), the so-called Parameterized Expectations Algorithm, or PEA, seeks decision rules for c_t that are “rational” in that they satisfy the Euler equation in a sufficiently robust way. It may be viewed intuitively as a computer analogue of the method of undetermined coefficients. The steps in the algorithm are as follows:

- specify decision rules for the forward-looking variables; for example, $\hat{c}_t = \psi(\Omega, x_t)$ where Ω are parameters, x_t contains variables known at time t (e.g., z_t, k_{t-1}), and ψ is the approximating function; and
- estimate Ω using various optimizing algorithms so that the Euler equation residual ($\epsilon_t = U'(\hat{c}_t) - \beta U'(\hat{c}_{t+1})f'(k_{t+1})$), or the difference between the left- and right-hand sides of the Euler equation, is close to zero.

3.3.2.1 Approximating Functions

The function ψ may be any approximating function, and the decision variables x_t are typically observations on the shocks and other state variables. In fact, approximating functions are just flexible functional forms that are parameterized to minimize Euler equation errors well-defined by a priori theoretical restrictions based on the optimizing behavior of the agents in the underlying model.

Neural-network (typically logistic) or Chebyshev orthogonal polynomial specifications are the two most common approximating functions used. The question facing the researcher here is one of robustness. First, given a relatively simple model, should one use a low-order Chebyshev polynomial approximation, or are there gains to using slightly higher-order expansion for obtaining the decision rules for the forward-looking variable? Will the results change very much if we use a more complex Chebyshev polynomial or a neural network alternative? Are there advantages to using a more complex approximating function, even if a less complex approximation does rather well? In other words, is the functional form of the decision rule robust with respect to the complexity of the model?

The question of using slightly more complex approximating functions, when they may not be needed for simple models, illustrates a trade-off noted by Olaf Wolkenhauer: that more complex approximations often are not specific or precise enough for a particular problem, whereas simple approximations may not be general enough for more complex models (Wolkenhauer 2001). In general, though, the “discipline” of Occam’s razor still applies: relatively simple and more transparent approximating functions should be preferred to more complex and less transparent ones. Canova (2007) recommends starting with simple approximating functions such as a first- or second-order polynomial and later checking the robustness of the solution with more complex functions.

3.3.2.2 *Logistic Neural Networks*

Sirakaya et al. (2006) cite several reasons for using neural networks as approximating functions. First, as noted by Hornik et al. (1989), a sufficiently complex feedforward network can approximate any member of a class of functions to any degree of accuracy. Second neural networks use fewer parameters to achieve the same degree of accuracy as do orthogonal polynomials, which require an exponential increase in parameters. While the curse of dimensionality is still there, its “sting,” to borrow an expression coined by St. Paul and expanded by Kenneth Judd,⁴ is reduced. Third, such networks, with log-sigmoid functions, easily deliver control bounds on endogenous variables. Finally, such networks can easily be applied to models that admit bang-bang solutions (Sirakaya et al. 2006, p. 3). For all these reasons, neural networks can serve as useful and readily available alternatives to or robustness check on the more commonly used Chebyshev approximating functions.

Like orthogonal polynomial approximation methods, a logistic neural network relates a set of input variables to a set of one or more output variables, but the difference is that the neural network makes use of one or more hidden layers, in which the input variables are squashed or transformed by a special function known as a logistic or log-sigmoid transformation. The following equations describe this form of

⁴ At the 2006 meeting of the Society of Computational Economics and Finance in Cyprus, the title of Kenneth Judd’s plenary session was “O Curse of Dimensionality, Where Is Thy Sting?”

approximation:

$$n_{j,t} = \omega_{j,0} + \sum_{i=1}^{i^*} \omega_{j,i} x_{i,t}^* \quad (3.19)$$

$$N_{j,t} = \frac{1}{1 + e^{-n_{j,t}}} \quad (3.20)$$

$$y_t^* = \gamma_0 + \sum_{j=1}^{j^*} \gamma_j N_{j,t} \quad (3.21)$$

Equation (3.19) describes a variable $n_{j,t}$ as a linear combination of a constant term, $\omega_{j,0}$, and input variables observed at time t , $\{x_{i,t}\}$, $i = 1, \dots, i^*$, with coefficient vector or set of “input weights” $\omega_{j,i}$, $i = 1, \dots, i^*$. Equation (3.21) shows how this variable is squashed by the logistic function and becomes a neuron $N_{j,t}$ at time of observation t . The set of j^* neurons are then combined in a linear way with the coefficient vector $\{\gamma_j\}$, $j = 1, \dots, j^*$, and taken with a constant term γ_0 , to form the forecast $\hat{y}_t^*$ at time t .

This system is known as a feedforward network, and when coupled with the log-sigmoid activation functions, it is also known as the “multi-layer perception” (MLP) network. An important difference between a neural network and an orthogonal polynomial approximation is that the neural network approximation is not linear in parameters.

3.3.2.3 Optimizing Algorithm

The parameters Ω are obtained by minimizing the squared residuals ϵ . A variety of optimization methods can be used to obtain the global optimum. We use an algorithm similar to the parameterized expectations approach developed by Marcet (1988) and further developed in Den Haan and Marcet (1990), Den Haan and Marcet (1994), and Marcet and Lorenzoni (1998). We solve for the parameters as a fixed-point problem. We make an initial guess of the parameter vector $[\Omega]$, draw a large sequence of shocks (ε_t) , and then generate time series for the endogenous variables of the model (c_t, k_t) . We then iterate on the parameter set $[\Omega]$ to minimize a loss function $\mathcal{L}$ based on the Euler equation errors ϵ for a sufficiently large T .⁵ We continue this process until it reaches convergence. Judd (1998) classifies this approach as a “projection” or a “weighted residual” method for solving functional equations and notes that the approach was originally developed by Wright and Williams (1984) and Wright and Williams (1991).

There are, however, drawbacks to this approach. One is that for more complex models, the iterations may take quite a bit of time for convergence to occur. Then there is the ever-present curse of dimensionality. The larger the number of state variables, the greater the number of parameters needed to solve for the decision rules. Also, the method relies on the sufficiency of the Euler equation errors. If the utility function is not strictly concave, for example, then the method may not give appropriate solutions. As

⁵ Den Haan and Marcet (1994) recommend a sample size of $T = 30,000$.

Canova (2007) suggested, minimization of Euler equations may fail when there are large number of parameters or when there is a high degree of complexity or nonlinearity.

Heer and Maußner (2005) note another type of drawback of the approach. They point out that the Monte Carlo simulation will more likely generate data points near the steady-state values than it will data points that are far away from the steady state in the repeated simulations for estimating the parameter set [Ω] p. 163]. We have used normally distributed errors here, but we note that fat tails and volatility clustering are pervasive features of observed macroeconomic data, so there is no reason not to use wider classes of distributions for solving and simulating dynamic stochastic models. As Fernández-Villaverde and Rubio-Ramí (2006) emphasize, there is no reason for a DSGE model not to have a richer structure than normal innovations. However, for the first-order perturbation approach, small normally distributed innovations are necessary. That is not the case for projection methods.

With this method, as noted by Canova (2007), the approximation is globally valid as opposed to being valid only around a particular steady-state point, as is the case for perturbation methods. While the projection method is computationally more time-consuming than the perturbation method, the advantage of using the former is that the researcher or policy analyst can undertake experiments that are far away from the steady state or involve more dramatic regime changes in the policy rule. It is also more suitable for models with thresholds or inequality constraints.⁶

Another point is that an algorithm can be specified to impose, for example, non-negativity constraints for all of the variables. The usual no-Ponzi game can only be applied to the evolution of government debt:

$$\lim_{t \rightarrow \infty} B_t \exp^{-it} = 0.$$

3.3.3 Linking Perturbation and Projection Methods

The perturbation methods, as mentioned above, involve Taylor expansions around a steady state. These methods assume that the shocks involve small deviations from a steady state and do not allow asymmetries (in the form of a zero lower bound on interest rates or greater downward rigidity in nominal wages). The advantage of these methods is that they are fast in terms of computing time. Many software programs, such as Dynare, are user-friendly. So whenever one wants to start analyzing the dynamic properties of a model, a good first step is still to implement a perturbation solution method.

Another advantage of starting with a perturbation method is that we may use it to obtain starting values for the projection method. First, run the model with a

⁶ Duffy and McNelis (2001) apply the Parameterized Expectations Algorithm with neural network specification for the solution of the stochastic growth model. Lim and McNelis (2008) apply these methods to a series of progressively more complex models of a small open economy.

perturbation method and generate time series of the variables (e.g., z_t, k_{t-1}). Next, estimate the function ψ using nonlinear methods to obtain the coefficients of the approximating equation $[\Omega]$. Good starting values go a long way toward speeding up an iterative solution method.

3.3.4 Accuracy Test: Judd-Gaspar Statistic

While the outcomes obtained using different approximating functions will not be identical, we would like the results to be sufficiently robust in terms of basic dynamic properties. Since the model does not have any exact closed-form solution against which we can compare numerical approximations, we have to use indirect measures of accuracy. Too often, these accuracy checks are ignored when researchers present simulation results based on stochastic dynamic models. This is unfortunate, since the credibility of the results, even apart from matching key characteristics of observable data, rests on acceptable measures of computational accuracy as well as theoretical foundations.

A natural way to check for accuracy is to see whether the Euler equations are satisfied, in the sense that the Euler equation errors are close to zero. Judd and Gaspar (1997) suggest transforming the Euler equation errors as follows,

$$JG_t = \frac{|\epsilon_t|}{c_t} \quad (3.22)$$

that is, they suggest checking the accuracy of the approximations by examining the absolute Euler equation errors relative to their respective forward-looking variables. If the mean absolute values of the Euler equation errors, deflated by the forward-looking variable c_t , is 10^{-2} , Judd and Gaspar note, the Euler equation is accurate to within a penny per unit of consumption or per unit spent on foreign currency.

Since consumption is an important variable in these types of models, it is conventional to substitute out the marginal utility of wealth Λ_t and work instead with the Euler equation below to generate the Judd-Gaspar statistics.

$$\frac{C_t^{-\eta}}{P_t(1 + \tau_t^c)} = \beta \frac{C_{t+1}^{-\eta}}{P_{t+1}(1 + \tau_{t+1}^c)} (1 + R_t)$$

3.3.5 Application

Consider now the model described in section 3.2. To solve the model, we started by parameterizing decision rules for C_t , N_t^p , D_t^p , N_t^w , and D_t^w :

$$\begin{aligned} C_t &= \psi^C(\mathbf{x}_t; \Omega_C) \\ N_t^p &= \psi^{Np}(\mathbf{x}_t; \Omega_{Np}) \end{aligned}$$

$$\begin{aligned}
D_t^p &= \psi^{Dp}(\mathbf{x}_t; \Omega_{Dp}) \\
N_t^w &= \psi^{Nw}(\mathbf{x}_t; \Omega_{Nw}) \\
D_t^w &= \psi^{Dw}(\mathbf{x}_t; \Omega_{Dw})
\end{aligned}$$

where the symbols Ω_C , Ω_{Np} , Ω_{Dp} , Ω_{Nw} , and Ω_{Dw} represented the parameters, and ψ^C , ψ^{Np} , ψ^{Dp} , ψ^{Nw} and ψ^{Dw} represented the expectation approximation functions. The symbol $\mathbf{x}_t$ contained a vector of observable state variables known at time t :

$$\mathbf{x}_t = [G_t, R_{t-1}, Y_{t-1}, P_{t-1}, W_{t-1}].$$

We used a neural network specification with one neuron for each of the decision variables. There were twenty-five parameters to estimate for our model, with five Euler equations and five state variables. The starting values for Ω were obtained by estimating ψ using artificial data generated by the perturbation method in DYNARE. An optimizing algorithm was then applied to minimize the Euler errors.

The accuracy of the simulations was checked using the Judd-Gaspar statistic defined for consumption C , the Calvo price P , and the Calvo wage W :

$$\begin{aligned}
JG_t^C &= \frac{1}{C_t} \left| \left(\frac{C_t^{-\eta}}{P_t(1+\tau_t^c)} - \beta \frac{C_{t+1}^{-\eta}}{P_{t+1}(1+\tau_{t+1}^c)} (1+R_t) \right) \right| \\
JG_t^P &= \frac{1}{P_t^o} \left| \left(\frac{N_t^p}{D_t^p} - \frac{Y_t(P_t)^{\zeta^p} A_t + \beta \xi^p N_{t+1}^p}{Y_t(P_t)^{\zeta^p} + \beta \xi^p D_{t+1}^p} \right) \right| \\
JG_t^W &= \frac{1}{(W_t^o)^{1+\zeta^w}} \left(\frac{N_t^w}{D_t^w} - \frac{(W_t)^{\zeta^w + \zeta^w \varpi} (L_t^{1+\varpi}) + \xi^w \beta \cdot N_{t+1}^w}{\frac{C_t^{-\eta}}{P_t(1+\tau_t^c)} (1-\tau_t^w) (W_t)^{\zeta^w} L_t + \xi^w \beta \cdot D_{t+1}^w} \right).
\end{aligned}$$

The Judd-Gaspar statistic was specified with reference to the price and the wage variables as they were easier to interpret than their respective auxiliary components N_t^p , D_t^p , N_t^w , and D_t^w . We solved and simulated the model for $T = 100$ for one thousand realizations of the stochastic process governing G_t . Figure 3.2 gives the distribution of the mean and maximum values of the Judd-Gaspar statistics. We see that the decision rules have a high degree of accuracy. Both the means and the maxima of the absolute Euler equation errors are much than 1%.

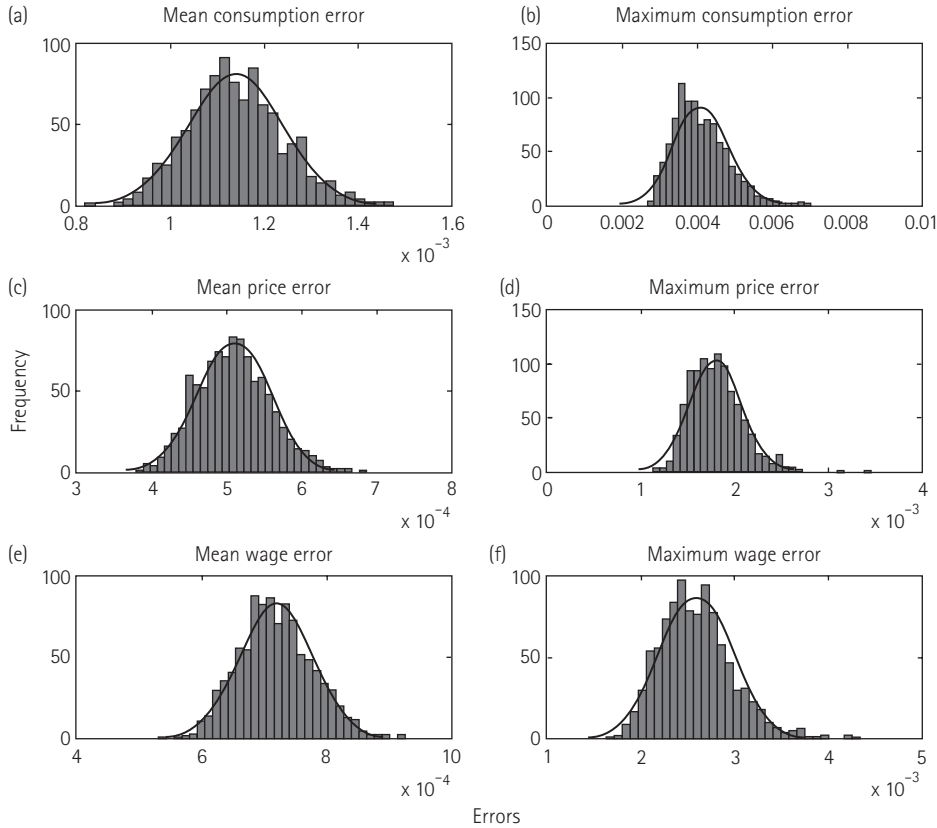


FIGURE 3.2 Distribution of mean and maximum Judd-Gaspar errors.

3.4 FISCAL SHOCK

The impulse response functions for a one-off (recall $\rho^g = 0$) shock to government spending G appear in figure 3.2(a-e). By construction, the shock takes place in period 5.

3.4.1 Simulations with Fixed Tax Rates

Two scenarios are presented: the base case, in which the nominal interest rate cannot be negative and wages are more sticky downwards (solid line), and the alternative linear case, in which the zero lower bound and asymmetric wage stickiness are not in operation (dashed line). We have dubbed the alternative case “linear” because a DSGE model without bounds and asymmetric adjustments can be rewritten equivalently in log-linearized form. Both income and consumption tax rates are predetermined and fixed in these impulses. What is reassuring about the results shown in figure 3.3 is that the adjustment paths of the linear and the PEA cases are not markedly different. This

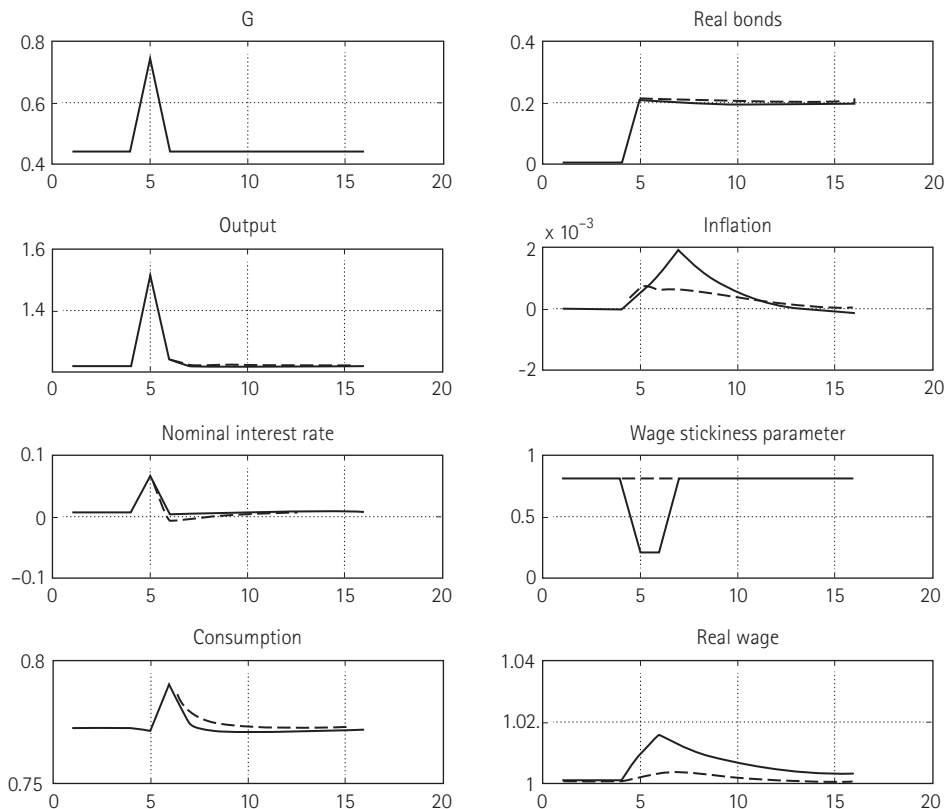


FIGURE 3.3 Impulse responses with constant tax rates.

is because the nonlinearities under study are not huge departures from the linear case. Be that as it may, the figure shows that the use of a linear model is often a good way to start before proceeding to the implementation of complex solution methods.

With respect to the results, we can see that the fiscal shock causes an increase in G , which then stimulates an increase in aggregate demand. Since the expenditure is bond-financed, B increases and there is pressure to raise the nominal interest rate. Prices rise (but the increase in inflation is trivial), and the aggregate supply increases as output expands. The increase in the demand for labor is met by an increase in wages, and consumption improves in this scenario. Note, too, that since wages are rising, the stickiness factor falls, giving less weight to the lagged wage rate. The result is a higher real wage when compared to the case with a fixed stickiness parameter.

When G drops off in the following period, the reverse process occurs. Output, consumption, prices, and wages fall. With asymmetric wage adjustment, the persistence parameter falls during the adjustment process, giving more weight to past wage rates so that the aggregate wage becomes more rigid downwards.

We also see in this figure that the interest rate in the “linear” model falls below zero. In the nonlinear base case, the zero lower bound is binding, so the fall in consumption

is greater. Allowing the nominal rate to turn negative kept consumption at a higher level than in the case when the nominal interest rate is subjected to a lower zero bound.

Finally, we want to draw attention to the trajectory of government debt. Since the evolution of bonds is determined by an accumulation equation, the stock of bonds rises initially with the increase in government spending. In these scenarios B_t remains at the higher level because the debt service $(1 + R_{t-1})B_{t-1}$ has not been offset by tax receipts $(\tau_t^w W_t L_t + \tau_t^c P_t C_t)$. Although debt service has dropped with the fall in the nominal interest rate, the economy is on a downward trajectory following the drop in G to its steady-state level. This is an example where fiscal austerity is not the policy option if the aim is to reduce the size of the government debt. Should measures be taken to keep the economy growing?

3.4.2 Simulations with Debt-Contingent Tax Rates

The question we pose is: what type of taxation should be in place to ensure that debt is gradually reduced? We suggest a number of scenarios (see table 3.3) based on tax-contingent rules as follows:

$$\begin{aligned}\tau_t^w &= \rho^w \tau_{t-1}^w + (1 - \rho^w)(\tau_0^w + \phi^w(B_{t-1} - B^*)) \\ \tau_t^c &= \rho^c \tau_{t-1}^c + (1 - \rho^c)(\tau_0^c + \phi^c(B_{t-1} - B^*)).\end{aligned}$$

In this scenario, B^* is the target debt level, and we assume that it is reduced via changes in the tax rates on labor income and consumption. The steady-state noncontingent tax rates are τ_0^w and τ_0^c . The tax rates have persistence coefficients ρ^w and ρ^c , which allow for some inertia in the adjustment of the tax rates to changes in debt.

We consider four scenarios with different values for the reaction coefficients ϕ_w and ϕ_c . The values of these reaction coefficients were chosen to keep tax changes within reasonable bounds. For example, in the absence of lagged effects ($\rho^w = \rho^c = 0$), when B jumped to 0.2 (see figure 3.2a), a positive reaction coefficient of 0.2 would bring about a change in τ^w from 0.3 to 0.34 and a change in τ^c from 0.1 to 0.14, while a negative reaction coefficient of 0.1 would bring about a change in τ^w from 0.3 to 0.28 and a change in τ^c from 0.1 to 0.08.

To obtain some idea about the likely effects of tax changes, we generated impulse responses for the scenarios and compared them to the base case (where debt increased

Table 3.3 Scenarios

	Base case (a)	case (b)	case (c)	case (d)	case (e)
ρ^w, ρ^c	0.0	0.5	0.5	0.5	0.5
ϕ_w	0.0	0.2	-0.1	0.2	-0.1
ϕ_c	0.0	0.2	-0.1	-0.1	0.2

following the fiscal stimulus). The impulses are shown in figures 3.4 to 3.7. In all cases similar dynamic patterns for output, inflation, interest rate, and the real wage prevailed. However, the key result is that there are combinations of tax rates that can be used to stimulate the economy and bring down debt.

Recall, in the base case (a) with fixed tax rates, that B increased even though government spending was back at its steady-state level. In case (b) both tax rates were increased, and we see that they were effective instruments for reducing government debt. Of course, additional austerity measures had been imposed to reduce the debt.

Since tax cuts can stimulate activity and thereby increase tax revenue, in case (c) we allowed for tax cuts in the state-contingent manner on both labor income and consumption. In this case, however, debt became destabilized because the stimulus generated by the fall in both tax rates was insufficient to generate enough tax revenue to bring down debt.

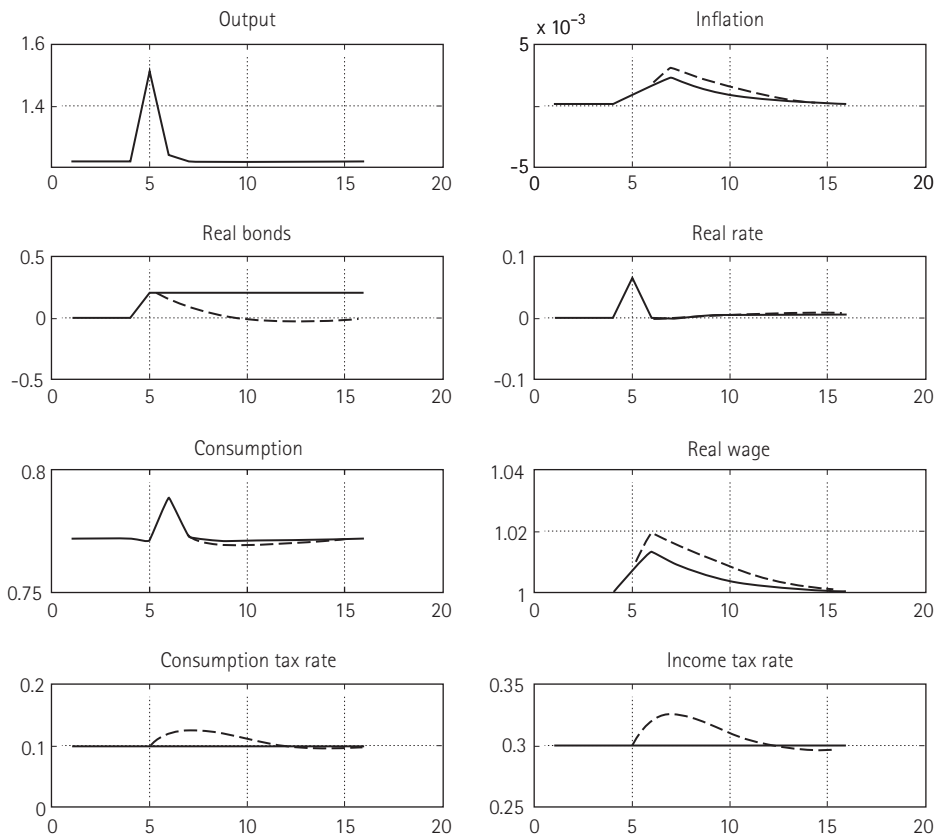


FIGURE 3.4 Impulse responses: Base case (solid line) and case with debt-contingent tax rates resulting in an increase in both income and consumption tax rates (dashed line).

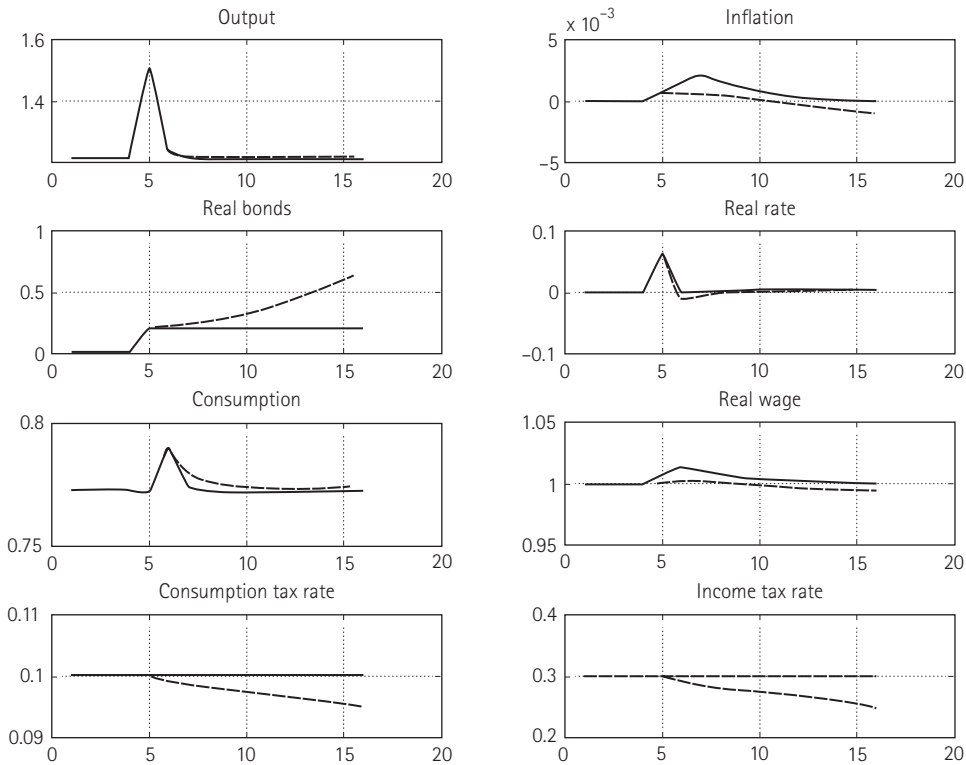


FIGURE 3.5 Impulse responses: Base case (solid line) and case with a decrease in both income and consumption tax rates (dashed line).

In cases (d) and (e) we checked out combinations of tax cuts and tax hikes. In case (d) we let the income tax increase but let the consumption tax fall. This is a program of some tax relief in a period of fiscal consolidation. We find that this combination increased consumption and tax revenue by amounts large enough to reduce government debt.

Similarly, in case (e), when we allowed the income tax rate to fall but the consumption tax rate to rise, the policy combination also reduced the debt. In this combination the increase in consumption tax revenue compensated for the loss of tax revenue from the labor income tax cut. At the same time, while the increased tax rate on consumption reduced demand in the adjustment process, the fall was more than compensated for by the stronger increase in consumption from the income tax cuts.

The reduction in case (e) is, however, not as fast as that in case (d). The main reason for this is associated with the relative size of tax revenue from income and from consumption. For most economies (and in our model), the revenue from consumption taxes is smaller than the tax revenue from income. Hence the policy combination in case (d), an income tax rise accompanied by a consumption tax cut, stabilized debt quickly

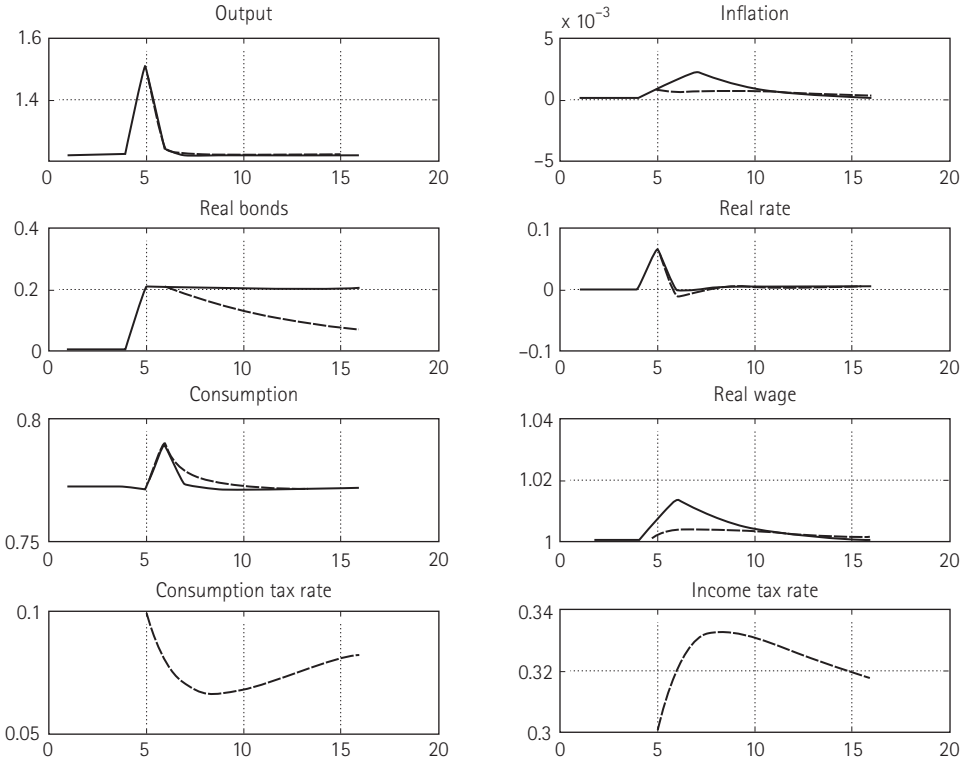


FIGURE 3.6 Impulse responses: Base case (solid line) and case with an increase in income tax rate and a decrease in consumption tax rate (dashed line).

because the reduction in tax revenue was more than compensated for by the increase in tax revenue from income tax.

Yet, the gain in tax revenue is at the expense of forgone consumption. While increases in either tax rate would lead to a negative consumption response, the negative response of consumption to a rise in the income tax is likely to be bigger. We can compute this as follows. Given our normalizing assumptions, we can compute steady-state consumption as a function of the tax rates (see below). The derived elasticities with respect to the tax rates show that the effect on consumption of a consumption tax will be smaller than the effect of an income tax:

$$C_0 = \left(\frac{(1 - \tau_o^w)}{(1 + \tau_o^c)} \right)^{(1+\varpi)/(\eta+\varpi)}$$

$$\frac{\partial C_0 / C_0}{\partial \tau_o^w / \tau_o^w} = \frac{(1 + \varpi)}{(\eta + \varpi)} \left(\frac{\tau_o^w}{(1 - \tau_o^w)} \right) (-1)$$

$$\frac{\partial C_0 / C_0}{\partial \tau_o^c / \tau_o^c} = \frac{(1 + \varpi)}{(\eta + \varpi)} \left(\frac{\tau_o^c}{(1 + \tau_o^c)} \right) (-1).$$

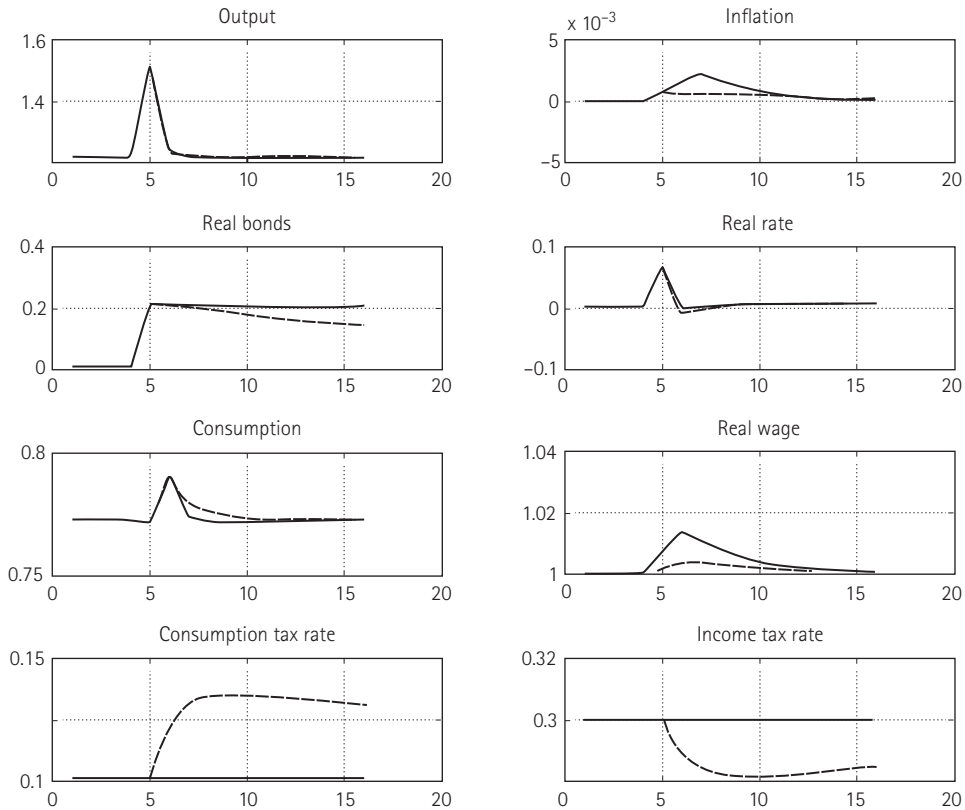


FIGURE 3.7 Impulse responses: Base case (solid line) and case with a decrease in income tax rate and an increase in consumption tax rate (dashed line).

Thus although tax-rate relief in a period of fiscal consolidation is more effective if it falls on consumption, it is also important to remember that there were negative effects on consumption from the hike in income tax. Our scenarios (d) and (e) illustrate the trade-offs between debt reduction and fall in consumption.

3.5 CONCLUDING REMARKS

This chapter has promoted the use of simulation methods to solve nonlinear models. The specific example considered a model with a zero lower bound and with asymmetric wage adjustments. The scenarios presented show that the use of debt-contingent tax cuts on labor income or on consumption can be effective ways to reduce debt by stimulating labor income, consumption demand, and tax revenue. These instruments are powerful and particularly useful when the interest rate is near or at the lower bound. Although

the example was simple, the result is profound: there is a case for considered, careful tax relief during a period of debt stabilization. But there are trade-offs, and the value of computational simulation analyses is that they are the right tools for evaluating the alternatives.

We also note the advantage of using the relatively fast perturbation method to generate the starting values for the decision rules for the forward-looking variables in the projection methods. Furthermore, while perturbation methods do not allow the imposition of the zero lower bound and asymmetric wage response, for this simple example the adjustment paths generated from the perturbation method were not very far off from those generated by the projection method. In other words, both perturbation and projection methods have their place in underpinning computational methods for economic analysis.

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CHAPTER 4

SOLVING RATIONAL EXPECTATIONS MODELS

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4.1 INTRODUCTION

This chapter presents main methods for solving rational expectations models. We especially focus on their theoretical foundations rather than on their algorithmic implementations, which are well described in Judd (1996). The lack of theoretical justifications in the literature motivates this choice. Among other methods, we particularly expound the perturbation approach in the spirit of the seminal papers by Woodford (1986) and Jin and Judd (2002). While most researchers make intensive use of this method, its mathematical foundations are rarely evoked and sometimes misused. We thus propose a detailed discussion of the advantages and limits of the perturbation approach for solving rational expectations models.

Micro founded models are based on the optimizing behavior of economic agents. Agents adjust their decisions in order to maximize their inter temporal objectives (utility, profits, and so on). Hence, the current decisions of economic agents depend on their expectations of the future path of the economy. In addition, models often include a stochastic part, implying that economic variables cannot be perfectly forecasted.

The rational expectation hypothesis consists of assuming that agents' expectations are the best expectations, depending on the structure of the economy and the information available to agents. Precisely speaking, they are modeled by the expectation operator $\mathbb{E}_t$, defined by

$$\mathbb{E}_t(z_{t+1}) = \mathbb{E}(z_{t+1}|\Omega_t)$$

where Ω_t is the information set at t and z_t is a vector of economic variables.

Here, we restrict the scope of our analysis to models with a finite number of state variables. In particular, this chapter does not study models with heterogenous agents in

which the number of state variables is infinite. We refer the reader to Den Haan et al. (2010) and Guvenen (2011) for surveys of this topic. When the number of state variables is finite, first-order conditions of maximization combined with market clearing lead to inter temporal relations between future, past, and current economic variables that often can be written as

$$\mathbb{E}_t g(z_{t+1}, z_t, z_{t-1}, \varepsilon_t) = 0 \quad (4.1)$$

where g is a function, z_t represents the set of state variables, and ε_t is a stochastic process.

The question, then, is to characterize the solutions of (4.1) and to find *determinacy conditions*, that is, conditions ensuring the existence and the uniqueness of a bounded solution.

Section 4.3 presents methods for solving model (4.1) when g is linear. The seminal paper by Blanchard and Kahn (1980), generalized by Klein (2000), gives a condition ensuring the existence and uniqueness of a solution in simple algebraic terms (see theorem 4.4). Other approaches consider methods based on undetermined coefficients (Uhlig 1999) or rational expectations errors (Sims 2002).

The linear case appears as the cornerstone of the perturbation approach when g is smooth enough. In section 4.4, we recall the theoretical foundations of the perturbation approach (theorem 4.7). The determinacy conditions for model (4.1) result locally from a linear approximation of model (4.1), and the first-order expansion of the solution is derived from the solution of the linear one. This strategy is called *linearization*. We show how to use the perturbation approach to compute higher-order Taylor expansions of the solution (lemma 4.11) and highlight the limits of such local results (sections 4.4.4.1 and 4.4.4.2).

When g presents discontinuities triggered by structural breaks or binding constraints, for instance, the problem cannot be solved by the classical perturbation approach. We describe two main classes of models for which the perturbation approach is challenged: regime switching and the zero lower balance (ZLB). Section 4.5 depicts the different methods used to solve models with Markovian transition probabilities. Although there is an increasing number of articles dealing with these models, there are only limited findings on determinacy conditions for them. We present the main attempts and results: an extension of Blanchard and Kahn (1980), the method of undetermined coefficients, and direct resolution. The topical ZLB case illustrates the problem of occasionally binding constraints. We describe the different approaches to solving models including the condition on the positivity of the interest rate, either locally (section 4.6.2) or globally (section 4.6.3).

We end this chapter with a brief presentation of the global methods used to solve model (4.1). The aim is not to give an exhaustive description of these methods (see Judd [1996]; Heer and Maussner [2005] for a very detailed exposition on this subject) but to show why and when they can be useful and fundamental. Most of these methods rely on projection methods; that is, they consist of finding an approximate solution in a specific class of functions.

4.2 THEORETICAL FRAMEWORK

We consider general models of the form

$$\mathbb{E}_t g(z_{t+1}, z_t, z_{t-1}, \varepsilon_t) = 0$$

where $\mathbb{E}_t$ denotes the rational-expectations operator conditional on the past values $(z_{t-k}, \varepsilon_{t-k})_{k \geq 1}$ of the endogenous variables and the current and past values of the exogenous shocks. The variable z denotes the endogenous variables and is assumed to evolve in a bounded set F of $\mathbb{R}^n$. We assume that the stochastic process ε takes its values in a bounded set V (containing at least two points) and that

$$\mathbb{E}_t(\varepsilon_{t+1}) = 0.$$

Notice that ε is not necessarily normally distributed even if it is often assumed in practice. Strictly speaking, Gaussian distributions are ruled out by the boundedness assumption. Nevertheless, it is often possible to replace Gaussian distributions by truncated ones.

Such an expression can be derived from a problem of maximization of an intertemporal objective function. This formulation covers a wide range of models. In particular, for models with lagged endogenous variables $x_{t-1}, \dots, x_{t-p}$, it suffices to introduce the vector $z_{t-1} = [x'_{t-1}, x'_{t-2}, \dots, x'_{t-p}]'$ to rewrite them as required.

First, we present an example to illustrate how we can put a model in the required form. Then we present the formalism behind such models. Finally, we introduce the main general concepts pertaining to the resolution.

4.2.1 An Example

We recall in this part how to cast in a practical way a model under the form (4.1). In the stochastic neoclassical growth model, households choose consumption and capital to maximize lifetime utility

$$\mathbb{E}_t \sum_{k=0} \beta^k U(c_{t+k})$$

where c_t is the consumption, U is the utility function, and β is the discount factor. Output is produced using only capital:

$$c_t + k_t = a_t k_{t-1}^\alpha + (1 - \delta)k_{t-1}$$

where k_t is the capital, a_t is the total factor productivity, α is the capital share, and $\delta \in (0, 1)$ is the depreciation rate of capital. We assume that a_t evolves recursively, depending on an exogenous process ε_t as follows:

$$a_t = a_{t-1}^{\rho_a} \exp(\varepsilon_t), \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2).$$

Using techniques of dynamic programming (Stokey et al. 1989), we form the Lagrangian $\mathcal{L}$, where λ_t is the Lagrange multiplier associated with the constraint on output:

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [U(c_t) - \lambda_t(c_t + k_t - a_t k_{t-1}^\alpha - (1-\delta)k_{t-1})].$$

The necessary conditions of optimality leads to the following:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_t} : \lambda_t &= U'(c_t) \\ \frac{\partial \mathcal{L}}{\partial k_t} : \lambda_t &= \beta \mathbb{E}_t [(\alpha a_{t+1} k_t^{\alpha-1} - (1-\delta)) \lambda_{t+1}] \\ \frac{\partial \mathcal{L}}{\partial \lambda_t} : c_t + k_t - a_t k_{t-1}^\alpha - (1-\delta)k_{t-1} &= 0. \end{aligned}$$

Then, defining $z_t = [a_t, c_t, \lambda_t, k_t]'$, the model can be rewritten as follows:

$$\mathbb{E}_t g(z_{t+1}, z_t, z_{t-1}, \varepsilon_t) = \mathbb{E}_t \begin{bmatrix} a_t - a_{t-1}^{\rho_a} \exp(\varepsilon_t) \\ \lambda_t - U'(c_t) \\ \beta \lambda_{t+1} [\alpha a_{t+1} k_t^{\alpha-1} - (1-\delta)] - \lambda_t \\ c_t + k_t - a_t k_{t-1}^\alpha - (1-\delta)k_{t-1} \end{bmatrix} = 0. \quad (4.2)$$

4.2.2 Formalism

We present two main theoretical ways of depicting solutions of model (4.1): as functions of all past shocks (Woodford 1986) or as a policy function, h , such that $z_t = h(z_{t-1}, \varepsilon_t)$ (Jin and Judd 2002; Juillard 2003). The second way is more intuitive than the first and corresponds to the practical approach of resolution. The first method appears more appropriate for handling general theoretical problems. In each case, we have to deal with infinite-dimension spaces: sequential spaces in the first case, functional spaces in the second. We will mainly adopt sequential views.

In model (4.1), we say that z is an endogenous variable and ε is an exogenous variable, since $\pi(\varepsilon_t | \varepsilon^{t-1}, z^{t-1}) = \pi(\varepsilon_t)$. Solving the model consists in finding $\pi(z_t | \varepsilon^t, z^{t-1})$.

4.2.2.1 The Sequential Approach

Following Woodford (1986), we can look for solutions of model (4.1) as functions of the history of all the past shocks. We denote by Σ the sigma field of V . Let V^∞ denote the product of an infinite sequence of copies of V , and Σ^∞ is the product sigma field (Loève 1977, p. 137). Elements $\varepsilon^t = (\varepsilon_t, \varepsilon_{t-1}, \dots)$ of V^∞ represent infinite histories of realizations of the shocks. We can represent the stochastic process ε_t by a probability measure $\pi : \Sigma^\infty \rightarrow [0, 1]$. For any sets $A \in \Sigma$ and $S \in \Sigma^\infty$, we define $AS = \{as \in V^\infty, a \in A, s \in S\}$ where as is a sequence with the first element a , the second element the first element of s , and so on.

By the Radon-Nikodym theorem (Loève 1977, p. 133), for any $A \in \Sigma$, there exists a measurable function $\pi(A|\cdot) : U^\infty \rightarrow [0, 1]$ such that

$$\forall S \in \Sigma^\infty, \quad \pi(AS) = \int_S \pi(A|\varepsilon^{t-1}) d\pi(\varepsilon^{t-1}).$$

Here $\pi(A|\varepsilon^{t-1})$ corresponds to the probability that $\varepsilon_t \in A$, given a history ε^{t-1} . For each $\varepsilon^{t-1} \in V^\infty$, $\pi(\cdot|\varepsilon^{t-1})$ is a probability measure on (V, Σ) ; thus π defines a Markov process on V^∞ with a time-invariant transition function.

We define the functional $\mathcal{N}$ by the following:

$$\mathcal{N}(\phi) = \int_V g(\phi(\varepsilon\varepsilon^t), \phi(\varepsilon^t), \phi(\varepsilon^{t-1}), \varepsilon_t) \pi(\varepsilon|\varepsilon^t) d\varepsilon. \quad (4.3)$$

Looking for a solution of model (4.1) is equivalent to finding a function Φ solution of $\mathcal{N}(\phi) = 0$.

4.2.2.2 The Recursive Approach

Following the approach presented in Jin and Judd (2002), we consider $\mathcal{S}$ the set of functions acting on $F \times V$ with values in $F \times V$. We assume that the shock ε_t follows a distribution law $\mu(\cdot, \varepsilon^{t-1})$. Then, we define the functional $\tilde{\mathcal{N}}$ on $\mathcal{S}$ by the following:

$$\tilde{\mathcal{N}}(h)(z, \varepsilon) = \int_V g(h(h(z, \varepsilon), \tilde{\varepsilon}), h(z, \varepsilon), \tilde{\varepsilon}) \mu(\tilde{\varepsilon}, \varepsilon) d\tilde{\varepsilon}. \quad (4.4)$$

In this framework, looking for a solution of model (4.1) corresponds to finding a function h in $\mathcal{S}$ such that $\tilde{\mathcal{N}}(h) = 0$. In practice, this approach is the most frequently used, since it leads to solutions' spaces with lower dimension. This approach underlies the numerical methods implemented in Dynare (Juillard 1996), a well-known software program for solving rational expectations models.

4.2.3 Definitions

Adopting the sequential approach described in Woodford (1986), we introduce the type of solutions that we are interested in.

Definition 4.1 *A stationary rational expectations equilibrium (SREE) of model (4.1) is an essentially bounded, measurable function $\phi : V^\infty \rightarrow F$ such that*

1. $\|\phi\|_\infty = \text{ess sup}_{V^\infty} \|\phi(\varepsilon^t)\| < \infty$
2. *If u is a U valued stochastic process associated with the probability measure π , then $z_t = \phi(u^t)$ is a solution of (4.1), i.e., $\mathcal{N}(\phi) = 0$.*

Furthermore, this solution is a steady state if ϕ is constant.

A crucial question is the existence and uniqueness of a bounded solution called *determinacy*.

Definition 4.2 *We say that model (4.1) is determinate if there exists a unique SREE.*

In terms of the recursive approach à la Jin and Judd (2002), it is equivalent to look for a stable measurable function h on $F \times V$ with values in $F \times V$ which is solution of the model. Notice that solutions of model (4.1) may respond to the realizations of a *sunspot variable*, that is, a random variable that conveys no information with regard to technology, preferences, and endowments and thus does not directly enter the equilibrium conditions for the state variables (Cass and Shell 1983).

Definition 4.3 *The deterministic model associated with model (4.1) is:*

$$g(z_{t+1}, z_t, z_{t-1}, 0) = 0. \quad (4.5)$$

A constant SREE of the deterministic model, $\bar{z} \in F$, is called a (deterministic) steady state. This point satisfies

$$g(\bar{z}, \bar{z}, \bar{z}, 0) = 0. \quad (4.6)$$

An equation such as (4.6) is a nonlinear multivariate equation, and solving it can be challenging. Such an equation can be solved by iterative methods, either a simple Newton method, or a Newton method by blocks, or a Newton method with an improvement of the Jacobian conditioning.

For the example presented in section 4.2.1, the computation of the steady state is simple.

$$\bar{a} = 1, \quad \bar{k} = \left[\frac{1}{\alpha} \left(\frac{1}{\beta} + (1 - \delta) \right) \right]^{\frac{1}{\alpha-1}}, \quad \bar{c} = \bar{k}^\alpha - \delta \bar{k}, \quad \bar{\Lambda} = U'(\bar{c}) \quad (4.7)$$

In the reminder of this chapter, we do not tackle issues raised by multiple steady states and focus mainly on the dynamic around an isolated steady state.

4.3 LINEAR RATIONAL EXPECTATIONS MODELS

In this part, we review some aspects of solving linear rational expectations models, since they form the cornerstone of the perturbation approach.

We consider the following model:

$$g^1 \mathbb{E}_t z_{t+1} + g^2 z_t + g^3 z_{t-1} + g^4 \varepsilon_t = 0, \quad \mathbb{E}_t \varepsilon_{t+1} = 0 \quad (4.8)$$

where g^i is a matrix for $i \in \{1, \dots, 4\}$. We present three important methods for solving these models: the benchmark method of Blanchard and Kahn (1980), the method of

undetermined coefficients of Uhlig (1999) and a method developed by Sims (2002) exploiting the rational expectations errors. The aim of this section is to describe the theory behind these three methods and to show why they are theoretically equivalent. We focus on the algebraic determinacy conditions rather than on the computational algorithms, which have been extensively depicted. Then, we illustrate these three methods in a simple example.

4.3.1 The Approach of Blanchard and Kahn (1980)

In their seminal paper, Blanchard and Kahn (1980) lay the theoretical foundations of this method.

Existence and uniqueness are then characterized by comparing the number of explosive roots to the number of forward-looking variables.

Following this method and its extension by Klein (2000), we rewrite (4.8) under the form

$$A\mathbb{E}_t \begin{bmatrix} z_{t+1} \\ z_t \end{bmatrix} = B \begin{bmatrix} z_t \\ z_{t-1} \end{bmatrix} + \begin{bmatrix} g^4 \\ 0 \end{bmatrix} \varepsilon_t \quad (4.9)$$

where $A = \begin{pmatrix} g_0^1 & g_n^2 \\ 0 & I_n \end{pmatrix}$ and $B = \begin{pmatrix} 0 & -g^3 \\ I_n & 0 \end{pmatrix}$.

Conditions on existence and uniqueness of a stable solution for this kind of linear model have been established in Blanchard and Kahn (1980) and extended to a model with non-invertible matrix g^1 by Klein (2000). They are summarized in the following result.

Theorem 4.4 *If the number of explosive generalized eigenvalues of the pencil $\langle A, B \rangle$ is exactly equal to the number of forward variables, n , and if the rank condition (4.11) is satisfied, then there exists a unique stable solution of model (4.9).*

Let us provide an insight into the main ideas of the proof. We consider the pencil $\langle A, B \rangle$ defined in equation (4.9) and introduce its real generalized Schur decomposition. When A is invertible, generalized eigenvalues coincide with the standard eigenvalues of matrix $A^{-1}B$. Following Klein (2000), there exist unitary matrices Q and Z and quasi-triangular matrices T and S such that

$$A = QTZ \quad \text{and} \quad B = QSZ.$$

For a matrix $M \in \mathcal{M}_{2n}(\mathbb{R})$, we write M by blocks of $\mathcal{M}_n(\mathbb{R})$:

$$M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}$$

We rank the generalized eigenvalues such that $|T_{ii}| > |S_{ii}|$ for $i \in [1, n]$ and $|S_{ii}| > |T_{ii}|$ for $i \in [n+1, 2n]$, which is possible if and only if the number of explosive generalized

eigenvalues is n . Equation (4.9) leads to the following:

$$TZ\mathbb{E}_t \begin{bmatrix} z_t \\ z_{t+1} \end{bmatrix} = SZ \begin{bmatrix} z_{t-1} \\ z_t \end{bmatrix} + Q' \begin{bmatrix} g^4 \\ 0 \end{bmatrix} \varepsilon_t.$$

Taking the last n lines, we see that

$$(Z_{21}z_{t-1} + Z_{22}z_t) = S_{22}^{-1}T_{22}\mathbb{E}_t(Z_{21}z_t + Z_{22}z_{t+1}) - T_{22}^{-1}Q'_{12}g^4\varepsilon_t. \quad (4.10)$$

We assume that

$$Z_{22} \text{ is full rank and thus invertible.} \quad (4.11)$$

Looking for bounded solutions, we iterate equation (4.10) to obtain

$$z_t = -Z_{22}^{-1}Z_{21}z_{t-1} - Z_{22}^{-1}S_{22}^{-1}Q'_{12}g^4\varepsilon_t. \quad (4.12)$$

By straightforward computations, we see that

$$Q'_{22} = S_{22}Z_{22} \quad Z_{22}^{-1}Z_{21} = Z'_{11}T_{11}^{-1}S_{11}(Z'_{11})^{-1}.$$

This shows that when the Blanchard-Kahn conditions are satisfied, there exists a unique bounded solution.

Reciprocally, if the number of explosive eigenvalues is strictly smaller than n , there exist several solutions of model (4.8). On the contrary, if the number of explosive eigenvalues is strictly higher than n , there is no solution. This strategy links explicitly the determinacy condition and the solution to a Schur decomposition. We notice, in particular, that the solution is linear and recursive.

The algorithm for solving that is used by Dynare relies on this Schur decomposition (Juillard 1996).

4.3.2 Undetermined Coefficients

This approach, presented in Uhlig (1999) and Christiano (2002), consists in looking for solutions of (4.8) under the form

$$z_t = Pz_{t-1} + Q\varepsilon_t \quad (4.13)$$

with $\rho(P) < 1$, where we denote by ρ the spectral radius of a matrix. The approach used in section 4.3.2 has shown that the stable solutions of a linear model can be written under the form (4.13). Introducing (4.13) into model (4.8) leads to the following:

$$(g^1P^2 + g^2P + g^3)z + (g^1PQ + g^2Q + g^4)\varepsilon = 0, \quad \forall \varepsilon \in V, \quad \forall z \in F.$$

Thus, (4.13) is satisfied if and only if

$$g^1P^2 + g^2P + g^3 = 0 \quad (4.14)$$

$$g^1PQ + g^2Q + g^4 = 0. \quad (4.15)$$

Uhlig (1999) obtains the following characterization of the solution:

Theorem 4.5 *If there is a stable recursive solution of model (4.8), then the solution (4.13) satisfies as follows:*

(i) *The matrix P is the solution of the quadratic matrix equation*

$$g^1 P^2 + g^2 P + g^3 = 0$$

and the spectral radius of P is smaller than 1.

(ii) *The matrix Q satisfies the matrix equation*

$$(g^1 P + g^2)Q = -g^4.$$

The method described in Uhlig (1999) is based on the computation of roots of the quadratic matrix equation (4.14), which is in practice done by computing generalized eigenvalues of the pencil $\langle A, B \rangle$, defined in section 4.3.2.

Higham and Kim (2000) make the explicit link between this approach and the one of Blanchard and Kahn (1980) in the following result. Introducing matrices $\langle A, B \rangle$ and the Schur decomposition as in section 4.3.2, they show that

Theorem 4.6 *With the notations of section 4.3.2, all the solutions of (4.14) are given by $P = Z_{22}^{-1} Z_{21} = Z'_{11} T_{11}^{-1} S_{11} (Z'_{11})^{-1}$.*

The proof is based on standard manipulations of linear algebra. We refer to Higham and Kim (2000) for the details. Moreover, by simple matrix manipulations, we can show that $Q'_{12}(g^1 P + g^2) = S_{22} Z_{22}$, the rank condition (4.11) implies that $(g^1 P + g^2)$ is invertible, and Q is defined uniquely by (4.15). The method of undetermined coefficients leads to manipulate matrix equations rather than iterative sequences, but the computational algorithm is similar and is depicted by Uhlig (1999).

4.3.3 Methods Based on Rational Expectations Errors

Here we depict the approach of Sims (2002) for the model (4.8) and explain how it is consistent with the previous methods.

Introducing $\eta_t = z_t - \mathbb{E}_{t-1} z_t$ and $y_t = \mathbb{E}_t z_{t+1}$, we rewrite equation (4.8) as

$$\begin{aligned} g^1 y_t + g^2 z_t + g^3 z_{t-1} + g^4 \varepsilon_t &= 0 \\ z_t &= y_{t-1} + \eta_t, \quad \mathbb{E}_t \eta_{t+1} = 0, \quad \mathbb{E}_t \varepsilon_{t+1} = 0. \end{aligned}$$

The model is rewritten in Sims's framework as

$$AY_t + BY_{t-1} + \begin{bmatrix} 0 \\ \mathbb{I}_n \end{bmatrix} \eta_t + \begin{bmatrix} \mathbb{I}_n \\ 0 \end{bmatrix} g^4 \varepsilon_t = 0 \quad (4.16)$$

where A and B are defined in section 4.3.2, and $Y_t = \begin{bmatrix} y_t \\ z_t \end{bmatrix}$.

Here the shocks η_t are not exogenous but depend on the endogenous variables Y_t . By iterating expectations of relation (4.16), we can express Y_t as a function of ε_t and η_t . Since η_t depends on Y_t , we obtain an equation on η_t . The model (4.16) is then determinate if this equation admits a unique solution.

Let us show that this approach is equivalent to the method of Blanchard and Kahn (1980). Considering the Schur decomposition of the pencil $\langle A, B \rangle$, there exists $\tilde{n} \in \{1, 2n\}$ such that $A = QTZ$ and $B = QSZ$, with $|T_{ii}| > |S_{ii}|$ for $i \in \{1, \dots, \tilde{n}\}$ and $|S_{ii}| > |T_{ii}|$ for $i \in \{\tilde{n} + 1, \dots, 2n\}$. We will show that there exists a unique solution of (4.16) if and only if $\tilde{n} = n$ and Q'_{12} is invertible.

Introducing stable and unstable subspaces of the pencil $\langle A, B \rangle$, we define $Y_t = Z' \begin{bmatrix} s_t \\ u_t \end{bmatrix}$, then the stable solutions of equation (4.16) are given by the following system:

$$s_t = 0, \quad u_t = T_{11}^{-1} S_{11} u_{t-1} + Q'_{21} \eta_t + Q'_{11} g^A \varepsilon_t, \quad Q'_{22} \eta_t = -Q'_{12} g^A \varepsilon_t. \quad (4.17)$$

The linear system (4.17) admits a unique solution if and only if Q'_{22} is a square matrix, that is, $\tilde{n} = n$, and invertible (rank condition). We find again conditions of theorem 4.4. The approach of Sims (2002) avoids the distinction between predetermined and forward-looking variables.

In this part, we have presented different approaches to solving linear models, relying on a simple determinacy condition. This algebraic condition is easy to check, even in the case of large-scale models.

4.3.4 An Example

In this part we depict these three methods in a simple example. We consider the following univariate model, a variant of an example studied in Iskrev (2008):

$$\theta^2 \kappa z_t = \theta^2 \mathbb{E}_t z_{t+1} + (\theta \kappa - 1) z_{t-1} + \varepsilon_t$$

where $0 < \theta < 1$ and $\kappa > 0$. We can rewrite this model as follows:

$$\begin{bmatrix} \mathbb{E}_t z_{t+1} \\ z_t \end{bmatrix} = \begin{bmatrix} \kappa & \frac{1-\theta\kappa}{\theta^2} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} z_t \\ z_{t-1} \end{bmatrix} + \begin{bmatrix} \frac{1}{\theta^2 \kappa} \\ 0 \end{bmatrix}$$

The matrix $\begin{bmatrix} \kappa & \frac{1-\theta\kappa}{\theta^2} \\ 1 & 0 \end{bmatrix}$ has two eigenvalues, $1/\theta$ and $(\theta\kappa - 1)\theta$. There is one predetermined variable; thus, according to Blanchard and Kahn's conditions, the model is determinate if and only if $\frac{\theta\kappa-1}{\theta} < 1$, that is, if $\kappa < (\theta + 1)/\theta$; in this case, the solution is given by

$$z_t = \frac{\theta\kappa - 1}{\theta} z_{t-1} + \frac{1}{\theta} \varepsilon_t.$$

The approach of Uhlig (1999) consists in looking for $(p, q) \in \mathbb{R}^2$ such that $y_t = py_{t-1} + q\varepsilon_t$, and $|p| < 1$. Then p and q are solutions of the equations

$$\theta^2 p^2 - \kappa \theta^2 p + (\theta \kappa - 1) = 0 \quad \theta^2 (p - \kappa) q = 1,$$

which admit a unique solution $p = \frac{\theta \kappa - 1}{\theta} \in (-1, 1)$ if $\kappa < (\theta + 1)/\theta$.

For the method of Sims (2002), we define $y_t = \mathbb{E}_t z_{t+1}$, and $\eta_t = z_t - y_{t-1}$. The model is then rewritten as

$$\begin{bmatrix} \frac{\theta^2 \kappa}{\theta \kappa - 1} & \frac{-\theta^2}{\theta \kappa - 1} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} z_t \\ y_t \end{bmatrix} = \begin{bmatrix} z_{t-1} \\ y_{t-1} \end{bmatrix} + \begin{bmatrix} \frac{\varepsilon_t}{\theta \kappa - 1} \\ \eta_t \end{bmatrix}.$$

When $(\theta \kappa - 1)\theta < 1$, the matrix on the left-hand side of the former equality has a unique eigenvalue smaller than 1 (θ). Projecting on the associated eigenspace, we get

$$\frac{\theta \kappa - 1}{\theta} z_{t-1} - y_{t-1} + \frac{\theta \kappa - 1}{\theta(\kappa \theta - 1)} \varepsilon_t - \eta_t = 0.$$

Thus, replacing η_t by $z_t - y_{t-1}$, we get

$$z_t = \frac{\theta \kappa - 1}{\theta} z_{t-1} + \frac{1}{\theta} \varepsilon_t.$$

4.3.5 Comparison of the Three Methods

From a numerical point of view, the algorithms induced by these three methods lead to globally equivalent solutions. We refer the reader to Anderson (2006) for a detailed comparison of the different algorithms.

The approaches of Blanchard and Kahn (1980) and Sims (2002) are particularly useful for building sunspot solutions when the determinacy conditions are not satisfied, as it is done, for instance, in Woodford (1986) for the first method or in Lubik and Schorfheide (2001) for the second one.

Uhlig (1999) clearly makes the link between linear rational expectations and matricial Riccati equations, which are widely used in control theory. He also allows for a more direct insight on the transition matrix. Besides, this approach lays the foundations of indeterminate coefficient methods.

4.4 PERTURBATION APPROACH

This section is devoted to the *linearization* method, which we can use to solve nonlinear, smooth-enough rational expectations models (4.1) in the neighborhood of a steady state (see definition 4.1).

We assume that the function g is smooth enough (C^1) in all its arguments, and we assume that there exists a locally unique steady state $\bar{z}$ such that

$$g(\bar{z}, \bar{z}, \bar{z}, 0) = 0.$$

We will solve model (4.1) by a perturbation approach. To this end, we introduce a scale parameter $\gamma \in \mathbb{R}$ and consider the model

$$\mathbb{E}_t g(z_{t+1}, z_t, z_{t-1}, \gamma \varepsilon_t) = 0. \quad (4.18)$$

When $\gamma = 0$, model (4.18) is the deterministic model (4.5), and when $\gamma = 1$, model (4.18) is the original model (4.1). We first explain the underlying theory of linearization, mainly developed by Woodford (1986) and by Jin and Judd (2002), and show an example. Then, we study higher-order expansions. Finally, we discuss the limits of such a local resolution.

4.4.1 From Linear to Nonlinear Models: Theory

In this section, we explain in detail how to solve nonlinear rational expectations models using a perturbation approach. Although linearization is well known and widely used to solve such models, the theory underlying this strategy and the validity domain of this approach are not necessarily well understood in practice. We rely on the works of Woodford (1986) and Jin and Judd (2002).

We define the functional $\mathcal{N}$ by the following:

$$\mathcal{N}(\phi, \gamma) = \int_V g(\phi(\varepsilon \varepsilon^t), \phi(\varepsilon^t), \phi(\varepsilon^{t-1}), \gamma \varepsilon_t) \pi(\varepsilon | \varepsilon^t) d\varepsilon. \quad (4.19)$$

By definition of the steady state, we see that the constant sequence $\phi_0(u) = \bar{z}$ for any $u \in U^\infty$ satisfies

$$\mathcal{N}(\phi_0, 0) = 0.$$

Perturbation approaches often rely on the implicit function theorem. Let us remind the reader of a version of this result in Banach spaces.

Theorem 4.7 (Abraham et al. 1988). *Let E, F , and G be 3 Banach spaces, let $U \subset E, V \subset F$ be open and $f : U \times V \rightarrow G$ be $C^r, r \geq 1$. For some $x_0 \in U, y_0 \in V$, assume $D_y f(x_0, y_0) : F \rightarrow G$ is an isomorphism. Then there are neighborhoods U_0 of x_0 and W_0 of $f(x_0, y_0)$ and a unique C^r map $g : U_0 \times W_0 \rightarrow V$ such that, for all $(x, w) \in U_0 \times W_0$,*

$$f(x, g(x, w)) = w.$$

This theorem is an extension of a familiar result in finite dimension spaces to infinite complete normed vector spaces (Banach spaces). Some statements of this theorem

require us to check that $D_y f(x_0, y_0)$ is a homeomorphism, that is, a continuous isomorphism with continuous inverse. We claim that, due to the Banach isomorphism theorem, it suffices to assume that $D_y f(x_0, y_0)$ is a linear continuous isomorphism.

We now apply this theorem to the functional $\mathcal{N} : \mathcal{B} \times \mathbb{R} \rightarrow \mathbb{R}^n$ in appropriate Banach spaces.

Because we are looking for bounded solutions, we introduce $\mathcal{B}$, the set of essentially bounded, measurable functions $\Phi : V^\infty \rightarrow \mathbb{R}^n$. $\mathcal{B}$, with the infinite norm

$$\|\Phi\|_\infty = \text{ess sup}_{u \in U^\infty} \|\Phi(u)\|.$$

The set $\mathcal{B}$ is a Banach space (see Dunford and Schwartz 1958, section III.6.14). $\mathbb{R}$ with $|\cdot|$ is also a Banach space.

The regularity of g ensures that the functional $\mathcal{N}$ is C^1 .

We introduce the operators $\mathcal{L}$ and $\mathcal{F}$, defined in $\mathcal{B}$ by the following:

$$\mathcal{F} : \Phi \mapsto ((\varepsilon^t) \mapsto \int_V H(\varepsilon \varepsilon^t) \pi(\varepsilon | \varepsilon^t) d\varepsilon) \quad (4.20)$$

$$\mathcal{L} : \Phi \mapsto ((\varepsilon^t) \mapsto \Phi(\varepsilon^{t-1})) \quad (4.21)$$

We notice that $\mathcal{F}$ and $\mathcal{L}$ have the following straightforward properties.

1. $\mathcal{F}\mathcal{L} = 1$
2. $\|\mathcal{F}\| = 1$ and $\|\mathcal{L}\| = 1$

where $\|\cdot\|$ is the operator norm associated with $\|\cdot\|_\infty$.

To apply implicit function theorem, we compute $D_{\Phi_0} \mathcal{N}(\Phi_0, 0)$.

$$D_{\Phi_0} \mathcal{N}(\Phi_0, 0)H = g^{\text{the1}} \mathcal{F}H + g^2 H + g^3 \mathcal{L}H$$

To check whether $D_{\Phi_0} \mathcal{N}(\Phi_0, 0)$ is invertible, we consider $\Psi \in \mathcal{B}$ and look to see whether there exists a unique solution of the equation

$$D_{\Phi_0} \mathcal{N}(\Phi_0, 0)H = \Psi. \quad (4.22)$$

Equation (4.22) can be rewritten as

$$g^1 \mathcal{F}H + g^2 H + g^3 \mathcal{L}H = \Psi$$

where g^1 (respectively g^2, g^3) is the first-order derivative with respect to the first variable (second, third). We refer to the method and the notations described in section 4.3.2.

Introducing the pencil $\langle A, B \rangle$ and its Schur decomposition, we rewrite (4.22) as:

$$\underbrace{\begin{pmatrix} g_2 & g_1 \\ I_n & 0 \end{pmatrix}}_A \underbrace{\begin{pmatrix} H \\ \mathcal{F}H \end{pmatrix}}_B = \underbrace{\begin{pmatrix} -g_3 & 0 \\ 0 & I_n \end{pmatrix}}_B \mathcal{L} \begin{pmatrix} H \\ \mathcal{F}H \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Psi.$$

For any $(\varepsilon^t) \in V^\infty$, defining $z_t = H(\varepsilon^t)$ and $z_{t+1} = \mathcal{F}H(\varepsilon^t)$, and $\Psi_t = \Psi(\varepsilon^t)$, we have to find bounded processes z_t such that

$$A \begin{pmatrix} z_t \\ z_{t+1} \end{pmatrix} = B \begin{pmatrix} z_{t-1} \\ z_t \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Psi_t.$$

Then, $D_{\Phi_0} \mathcal{N}(\Phi_0, 0)$ is invertible if and only if the number of explosive generalized eigenvalues of the pencil $\langle A, B \rangle$ is exactly equal to n . Moreover, the solution z_t is given by

$$z_t + Z_{22}^{-1} Z_{21} z_{t-1} = Z_{22}^{-1} \sum_{k=0}^{\infty} (S_{22}^{-1} T_{22})^k S_{22}^{-1} Q'_{12} \mathbb{E}_t \Psi_{t+k},$$

which finally gives

$$D_{\Phi_0} \mathcal{N}(\Phi_0, 0)^{-1} = (1 + Z_{22}^{-1} Z_{21} \mathcal{L})^{-1} Z_{22}^{-1} (1 - S_{22}^{-1} T_{22} \mathcal{F})^{-1} S_{22}^{-1} Q'_{12}.$$

Application of the implicit function theorem leads to the following result, which is an extension of Woodford (1986) when g^1 is non-invertible:

Theorem 4.8 *If the linearized model in $\bar{z}$ is determinate, then for γ small enough, there exists a unique SREE for model (4.18). Moreover, if not,*

- (1) *If the number of explosive generalized eigenvalues of the pencil $\langle A, B \rangle$ is smaller than n , the linearized model is indeterminate, and there is both*
 - (i) *a continuum of SREE near $\bar{z}$, in all of which the endogenous variables depend only upon the history of the exogenous shocks.*
 - (ii) *a continuum of SREE near $\bar{z}$, in which the endogenous variables respond to the realizations of a stationary sunspot process as well as to the exogenous shocks.*
- (2) *If the number of explosive generalized eigenvalues of the pencil $\langle A, B \rangle$ is greater than n , for any γ small enough, no SREE exists near $\bar{z}$.*

The foregoing result is an equivalence result; we have only added the detail that the determinacy of the linearized model implies the local determinacy of the nonlinear model. The reciprocal is a bit tricky and uses an approach similar to the Constant Rank Theorem. We refer the reader to Woodford (1986) for more details.

Theorem 4.8 shows that the determinacy condition for model (4.18) around the steady state $\bar{z}$ is locally equivalent to the determinacy condition for the linearized model in $\bar{z}$.

To expound this result in functional terms (Jin and Judd 2002), we have the following result.

Proposition 4.9 *If the linearized model is determinate, the solution of theorem (4.8) is recursive.*

For a fixed γ , let (z_t) a solution of model (4.18), there exists a unique function Φ such that, for a sequence ε^t ,

$$z_t = \Phi(\varepsilon^t, \gamma).$$

We define the following:

$$\text{Im}(\Phi(\cdot, \gamma)) = \{z \in F \mid \exists \varepsilon^0 \in V^\infty \text{ such that } z = \Phi(\varepsilon^0)\}.$$

For any $z \in \text{Im}(\Phi(\cdot, \gamma))$, we define the function $\mathcal{Z}$ by

$$\mathcal{Z}(z, \varepsilon, \gamma) = \Phi(\varepsilon \varepsilon^0, \gamma), \text{ where } z = \Phi(\varepsilon^0, \gamma).$$

We consider the sequence $\tilde{z}_t = \mathcal{Z}(z_{t-1}, \varepsilon_t)$ for $t > 0$ and $\tilde{z}_t = z_t$ for $t \leq 0$. The sequence $\tilde{z}_t$ is a solution of model (4.18). By the uniqueness of the solution, $\tilde{z}_t = z_t$ for any $t > 0$. Thus $z_t = \mathcal{Z}(z_{t-1}, \varepsilon_t)$ for any $t > 0$. In addition, computing $D_{\Phi_0} \mathcal{N}(\Phi_0, 0)^{-1} D_\gamma \mathcal{N}(\Phi_0, 0)$ leads to the first-order expansion of the solution

$$z_t = Pz_{t-1} + \gamma Q\varepsilon_t + o(\gamma)$$

where P and Q are given in equation (4.13).

4.4.2 Applications: A Fisherian Model of Inflation Determination

Consider an economy with a representative agent, living for an infinite number of periods, facing a trade-off between consuming today and saving to consume tomorrow. The agent maximizes its utility function $\mathbb{E}_t \sum_{k=0}^{\infty} \beta^k \frac{c_{t+k}^{1-\sigma}}{1-\sigma}$, where β is the discount factor, C_t the level of consumption, and σ the inter-temporal elasticity of substitution. Then maximizing utility function under the budget constraints leads to

$$\mathbb{E}_t \frac{c_t}{c_{t+1}} \frac{\sigma}{\pi_{t+1}} \frac{r_t}{\beta} = 1 \quad (4.23)$$

where r_t is the gross risk-free nominal interest rate and π_{t+1} is the inflation. In addition, we assume that $\frac{c_{t+1}}{c_t} = \exp(a_{t+1})$ where a_t is an exogenous process.

$$a_t = \rho a_{t-1} + \varepsilon_t$$

Defining $r_t = \bar{r} \exp(\hat{r}_t)$ and $\pi_t = \bar{\pi} \exp(\hat{\pi}_t)$, with $\bar{\pi} = \beta \bar{r}$, we rewrite equation (4.23) as

$$\mathbb{E}_t [\exp(\hat{r}_t - \hat{\pi}_{t+1} - \sigma a_{t+1})] = 1.$$

If we assume that $\hat{r}_t$ follows a Taylor rule,

$$\hat{r}_t = \alpha \hat{\pi}_t.$$

The vector of variables $z = [\hat{r}, \hat{\pi}, a]'$ satisfies the model:

$$\mathbb{E}_t g(z_{t+1}, z_t, z_{t-1}, \varepsilon_t) = \mathbb{E}_t \begin{bmatrix} \exp(\hat{r}_t - \hat{\pi}_{t+1} - \sigma a_{t+1}) - 1 \\ \hat{r}_t - \alpha \hat{\pi}_t \\ a_t - \rho a_{t-1} - \varepsilon_t \end{bmatrix} = 0 \quad (4.24)$$

we notice that $g(0, 0, 0, 0) = 0$ and the first-order derivatives of g in $(0, 0, 0, 0)$ are given by the following:

$$g^1 = \begin{bmatrix} 0 & -1 & -\sigma \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad g^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad g^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \rho \end{bmatrix}.$$

Then we obtain the following Taylor principle (Woodford 2003):

Lemma 4.10 *If $\alpha > 1$ and for a small variance of ε , the model (4.24) is determinate.*

The proof is immediate. It suffices to compute associated matrices A and B and the generalized eigenvalues of the pencil $\langle A, B \rangle$ and to apply theorem 4.8.

4.4.3 Computing Higher-Order Solutions

In this part we do not focus on the practical computations of high-order solutions. These aspects are developed in Jin and Judd (2002) and Schmitt-Grohé and Uribe (2004). First, we show the theoretical interest of computing expansions at higher order. Second, we show that if the linearized model is determinate, and if the model is smooth, then the solution admits an asymptotic expansion at any order (lemma 4.11), similar to the results of Jin and Judd (2002) or Kowal (2007).

4.4.3.1 Theoretical Necessity of a Quadratic Approximation: The Example of the Optimal Monetary Policy

A very important application of model (4.18) is the evaluation of alternative monetary policy rules and the concept of optimal monetary policy. There is a consensus in the literature that a desirable monetary policy rule is one that achieves a low expected value of a discounted loss function, such that the losses at each period are a weighted average of quadratic terms depending on the deviation of inflation from a target rate and in some measure of output relative to its potential. This loss function is often derived as a quadratic approximation of the level of expected utility of the representative household in the rational-expectations equilibrium associated with a given policy. This utility function is then rewritten as

$$U(z, \gamma \varepsilon) = U(\bar{z}, 0) + U_z(\bar{z}, 0)(z - \bar{z}) + \gamma U_\varepsilon(\bar{z}, 0)\varepsilon + \frac{1}{2}(z - \bar{z})' U_{zz}(\bar{z}, 0)(z - \bar{z}) \quad (4.25) \\ + \gamma(z - \bar{z})' U_{z\varepsilon}(\bar{z}, 0)\varepsilon + \gamma^2 \frac{1}{2} \varepsilon' U_{\varepsilon\varepsilon}(\bar{z}, 0)\varepsilon + O(\gamma^3).$$

Then, if we consider a first-order solution of the model $z(\gamma) = z_0 + \gamma \Phi + O(\gamma)$, owing to the properties of composition of asymptotic expansions, we obtain in general only a first-order expansion of the utility function (4.25), which is not sufficient to compute optimal monetary policy. On the contrary, a second-order expansion of z allows for computing a second-order expansion in equation (4.25). For a complete description, we refer the reader to chapter 6 of Woodford (2003) or Kim et al. (2003).

4.4.3.2 Some Insights Regarding the Expansion of the Solution

Most of the papers dealing with high-order expansions introduce tensorial notations, which are very useful but would be too much for this chapter. Thus, we illustrate the main ideas with a naive approach that stems directly from the implicit function theorem.

Lemma 4.11 *We assume that the function g in model (4.18) is C^r . If the linearized model in $\bar{z}$ is determinate, then the solution Φ admits an asymptotic expansion in γ until order r .*

$$\Phi(\gamma) = \Phi(0) + \sum_{n=1}^r \gamma^n a_n + o(\gamma^n) \quad (4.26)$$

where the functions $a_k \in C^0(V^\infty)$ are computed recursively

This result shows that local determinacy and smoothness for the model ensure an expansion at each order. Let us give some details on the proof of lemma 4.11. The real function $\eta(\alpha) = \alpha \mapsto \mathcal{N}(\Phi(\alpha), \alpha)$ is C^r ; its derivative of order $n \leq r$ is zero, and we can show, by an immediate recursion, that there exists a function η_n on $(C^0(V^\infty))^n$ such that

$$\eta^{(n)}(\alpha) = D_\Phi \mathcal{N}(\Phi(\alpha), \alpha) \Phi^{(n)}(\alpha) + \eta_n(\Phi^{(n-1)}(\alpha), \dots, \Phi(\alpha)) = 0.$$

Applying this identity for $\alpha = 0$ leads to

$$\Phi^{(n)}(0) = -D_\Phi \mathcal{N}(\Phi(0), 0)^{-1} \eta_n(\Phi^{(n-1)}(0), \dots, \Phi(0))$$

since $D_\Phi \mathcal{N}(\Phi(0), 0)$ is invertible. Thus, the functions (a_n) in formula (4.26) are given by the following:

$$a_n = \frac{\Phi^{(n)}(0)}{n!}.$$

Lemma 4.11 shows that, under a condition of determinacy, and for a smooth function, it is possible to obtain a Taylor asymptotic expansion of the solution in the scale parameter at any order. We refer the reader to Jin and Judd (2002) and Kowal (2007) for a more detailed analysis.

4.4.3.3 What are the Advantages of Higher-Order Computations?

Of course, an expansion at higher order provides a more accurate approximation of the solution in the neighborhood around the deterministic steady state, as soon as the perturbation approach is valid in this neighborhood. Notice, however, that a higher-order approximation does not change the size of the domain on which the perturbation approach is valid, and thus can be uninformative if this domain is very small.

4.4.4 Limits of the Perturbation Approach

Note that previous results are local, that is, only valid for a small neighborhood of the steady state. We illustrate these limits by two examples. The first one shows that if the size of the shocks is not very small, the conditions obtained by linearization can be evident; this is notably the case when the shocks are discrete. The second example illustrates that if the model is locally determinate, it does not exclude complex behaviors in a larger neighborhood.

4.4.4.1 Small Perturbation

The existence and uniqueness of the solution is an asymptotic result, and it remains valid for a “small” γ . Refinements of the implicit function theorem can give a quantitative condition on γ (Holtzman 1970), but the conditions of validity of the linearization in terms of size of the shocks are never checked.

As an illustration, we consider the model

$$\mathbb{E}_t(\pi_{t+1}) = \alpha_{s_t} \pi_t. \quad (4.27)$$

This model corresponds to a simplified Fisherian model of inflation determination as described in section 4.4.2: π_t is the inflation and α_{s_t} is a parameter taking two values α_1 and α_2 . We assume that the reaction to inflation evolves stochastically between these two values. A monetary policy regime is a distinct realization of the random variable s_t , and we recall that we say that a monetary policy regime is active if $\alpha_i > 1$ and passive if $\alpha_i < 1$, following the terminology of Leeper (1991). We assume that the process s_t is such that

$$p(s_t = 1) = p, \quad p(s_t = 2) = 1 - p.$$

We define

$$\bar{\alpha} = p\alpha_1 + (1 - p)\alpha_2, \quad \Delta\alpha = \alpha_2 - \alpha_1.$$

We illustrate some limits of theorem 4.8 by considering the model (4.27) as a perturbation of

$$\mathbb{E}_t(\pi_{t+1}) = \bar{\alpha} \pi_t. \quad (4.28)$$

Theorem 4. gives determinacy conditions for the following perturbed model:

$$\mathbb{E}_t(\pi_{t+1}) = \bar{\alpha}\pi_t + \gamma \Delta\alpha \frac{(\alpha_{s_t} - \bar{\alpha})}{\Delta\alpha} \pi_t \quad (4.29)$$

when the scale parameter γ is small enough.

Lemma 4.12 *Determinacy conditions for models (4.27) and (4.29) are as follows:*

1. *A sufficient condition for applying theorem 4.8 to model (4.29) for a small γ is that*

$$p\alpha_1 + (1-p)\alpha_2 > 1.$$

2. *A sufficient condition for determinacy of model (4.27) is that*

$$\frac{p}{\alpha_1} + \frac{1-p}{\alpha_2} < 1.$$

These two conditions are represented in figure 4.1.

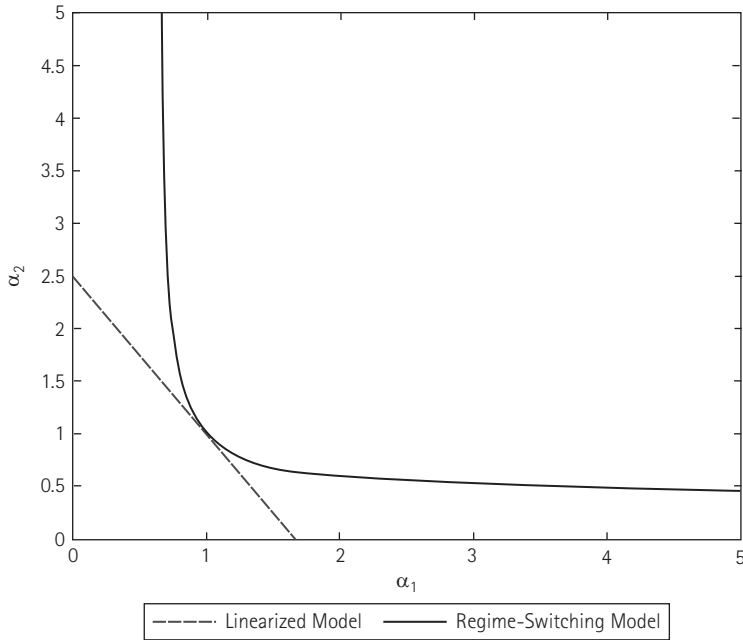


FIGURE 4.1 Determinacy conditions depending on γ .

Note: For policy parameters above (below) the solid curved line the Markov-switching Fisherian model is indeterminate (resp. determinate). For parameters above (below) the dashed black line, the linearized model is determinate (resp. indeterminate). For instance, for $\alpha_1 = 2$ and $\alpha_2 = 0.5$ the linearized model is determinate while the Markov-switching Fisherian model is not. Thus, for these parameters, the linearization is not a valid procedure for solving the Markov-switching model.

Let us give some details on the proof of lemma 4.12. To do that, following the method presented in section 4.4, we introduce the functional $\mathcal{N}$ acting on the set of continuous functions on $\{1, 2\}^\infty$ and defined by

$$\mathcal{N}(\Phi, \gamma)(s^t) = p\Phi(1s^t) + (1-p)\Phi(2s^t) - \bar{\alpha}\Phi(s^t) + \gamma \Delta\alpha(p-2+s_t)\Phi(s^t).$$

First, we find easily that, for $\bar{\alpha} > 1$, $\mathcal{N}(\Phi, 0) = 0$ admits a unique solution 0 , and that $D_\Phi \mathcal{N}(0, 0)$ is invertible. To apply theorem 4.8 for any $\gamma \in [0, 1]$, we have to find conditions ensuring that $D_\Phi \mathcal{N}(\Phi_0, \gamma)$ remains invertible. We compute $D_\Phi \mathcal{N}(\Phi, \gamma)$.

$$D_\Phi \mathcal{N}(\Phi, \gamma)H(s^t) = pH(1s^t) + (1-p)H(2s^t) - \bar{\alpha}H(s^t) + \gamma \Delta\alpha(p-2+s_t)H(s^t) \quad (4.30)$$

and look for a condition ensuring that for any $|\gamma| \in [0, 1]$, $D_\Phi \mathcal{N}(\Phi, \gamma)$ is invertible. We compute iteratively the solution of $D_\Phi \mathcal{N}(\Phi, \gamma)H(s^t) = \Psi(s^t)$:

$$H(s^t) = \frac{1}{s_t} \left[h(s^t) + \sum_{j=1}^k \sum_{s^j \in \{1, 2\}^j} \left(\frac{p}{\alpha_1} \right)^{\nu(s^j)} \left(\frac{1-p}{\alpha_2} \right)^{j-\nu(s^j)} \Psi(s^j s^t) \right]$$

where $\nu(s^j) = \#(\{s \in s^j = 1\})$. A sufficient condition for invertibility of $D_\Phi \mathcal{N}(\Phi, \gamma)$, then, is that

$$\frac{p}{\alpha_1} + \frac{1-p}{\alpha_2} < 1.$$

In figure 4.1 we display the determinacy conditions with respect to policy parameters α_1 and α_2 for model (4.27) with a in a solid line and for model (4.29) with a dashed black line. Determinacy conditions of the linearized model appear to be tangent to the determinacy conditions of the original regime-switching model at $\alpha_1 = \alpha_2 = 1$. The indeterminacy region (in the southwest) for the linearized model is included in that of the original one. However, for some policy parameters, determinacy conditions are satisfied for the linearized model whereas the regime-switching model is indeterminate. Nonetheless, there is no contradiction because determinacy conditions for the linearized model only ensure the existence and uniqueness of a stable solution for small, and hence perhaps smaller than one, γ . Thus we can not rely on such a perturbation approach for solving regime-switching models.

This example shows that, in a context of switching parameters, applying a perturbation approach around the constant parameter case is generally inadequate. Nevertheless, it does not mean that the perturbation approach cannot be employed in this context for other purposes. For example, Foerster et al. (2016) describe an algorithm for solving nonlinear Markov-switching models by a perturbation approach. In that paper the perturbation approach aims at simplifying the nonlinearity of the model and not the Markov-switching process of the parameters.¹ Barthélemy and Marx (2017) use a perturbation approach to make the link between a standard linear

¹ At this stage, there is no theoretical foundation for this method because the authors do not make explicit the exact norm underlying their perturbation approach.

Markov-switching model and a nonlinear regime-switching model in which transition probabilities may depend on the state variables.

4.4.4.2 Local Versus Global Solutions

In addition, theorem 4.8 on results in existence and uniqueness locally. Put precisely, it means that there exists a neighborhood V of the steady state in which there is a unique bounded solution; it does not exclude that there are more complex dynamics in a bigger neighborhood, as for instance the chaotic behaviors described in Benhabib et al. (2004). To give some insights into the limits they raise, we present some results for the following sequence:

$$u_{t+1} = \chi u_t^2 \times [1 - u_t^2], \quad \chi = 3.4.$$

There exists $0 < u_- < u_+ < 1$ such that

Proposition 4.13 *The sequence (u_t) has the following properties:*

- For $u_0 \in [0, u_-]$, the sequence is convergent.
- For $u_0 \in [u_-, u_+]$, the sequence (u_t) is a two-cycle, i.e., there exists $0 < c_0 < c_1 < 1$ such that

$$\lim_{t \rightarrow +\infty} u_{2t} = c_0, \quad \lim_{t \rightarrow +\infty} u_{2t+1} = c_1.$$

The proof of this proposition relies on the nonlinear theory of recurrent real sequences defined by

$$u_{t+1} = f(u_t).$$

We display the graphs of f and the iterated function $f \circ f$ in figure 4.2. In figure 4.3 we represent the adherence values of (u_t) depending on the initial value u_0 . This figure shows that if u_0 is large enough, the sequence does not converge any more.

This example reveals that, depending on the size of the neighborhood, there can be a locally unique stable solution, but more complex behaviors emerge if we enlarge the neighborhood. Benhabib et al. (2004) exhibits a similar example in which the model is locally determinate but presents chaotic features. More generally, we refer the reader to the study of the logistic map in Ausloos and Dirickx (2006) or to Benhabib et al. (2004) for more formal definitions.

4.5 DEALING WITH STRUCTURAL BREAKS: THE CASE OF REGIME-SWITCHING MODELS

A need to model behavioral changes leads us to consider rational expectations models in which parameters can switch between different values depending on the regime of economy. A way to formalize this assumption is to introduce some regimes labeled s_t ,

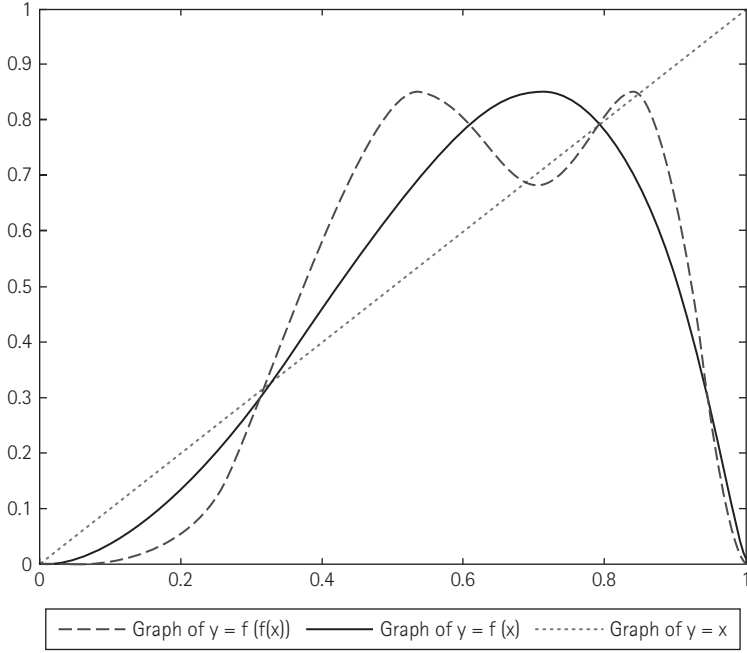


FIGURE 4.2 The functions f and $f \circ f$.

Note: We consider the function $f(x) = 3.4x^2(1 - x^2)$. The unbroken solid curve displays this function with respect to x . As this curve crosses three times the 45-degree line (the in black dotted line), f has three fixed points on $[0, 1]$. The function $f \circ f$ (the black dashed line) has five fixed points. Depending on the initial value u_0 , the sequence $u_{t+1} = f(u_t)$ admits a limit which is a fixed point of f or has subsequences converging on a fixed point of $f \circ f$.

s_t taking discrete values in $\{1, \dots, N\}$. The model can be written as

$$\mathbb{E}_t[f_{s_t}(z_{t+1}, z_t, z_{t-1}, \varepsilon_t)] = 0. \quad (4.31)$$

Let us assume that the random variables $s_t \in \{1, \dots, N\}$ follow a Markov process with transition probabilities $p_{ij} = p(s_t = j | s_{t-1} = i)$.

There is a running debate concerning good techniques for solving Markov-switching rational expectations models. The main contributions are Farmer et al. (2007, 2009a,b, 2010a,b) and Davig and Leeper (2007). The former papers focus on mean-square stable solutions, relying on some works concerning optimal control, while the latter is trying to solve the model by mimicking Blanchard and Kahn (1980). Farmer et al. (2010a) casts doubt on this second approach. We present the existing results, explain their limits, and present some complements by Svensson and Williams (2009) and Barthélemy and Marx (2017). The latter shows how to deduce determinacy conditions for a nonlinear Markov-switching model from a linear one.

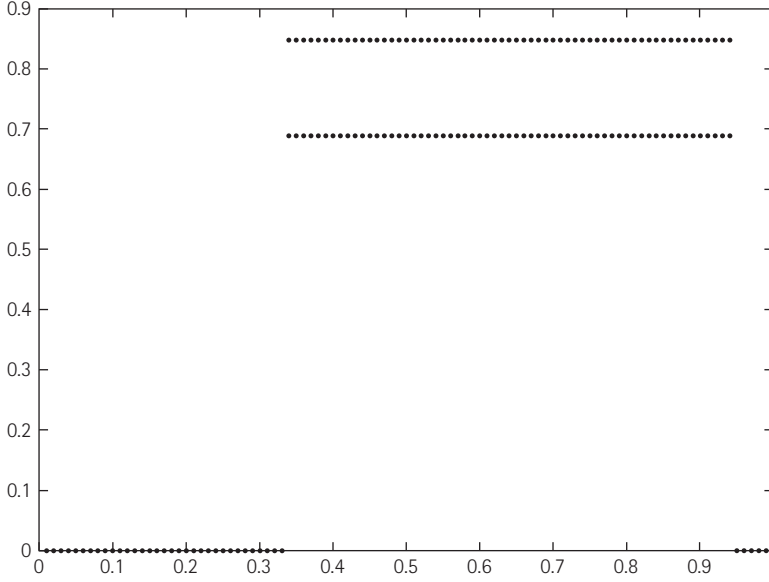


FIGURE 4.3 Adherence values of (u_t) depending on the initial value u_0 .

Note: Dark points display adherence values of $u_{t+1} = 3.4u_t^2(1 - u_t^2)$ with respect to the initial values u_0 . Depending on the initial value u_0 , the sequence either converges to 0 or admits two adherence values (i.e., two convergent subsequences).

4.5.1 A Simple Example

To begin with, we present a Fisherian model of inflation determination with regime switching in monetary policy, following Davig and Leeper (2007). This model is studied in Davig and Leeper (2007) and in Farmer et al. (2009a).

As in sections 4.4.2 and 4.4.4.1, the log-linearized asset-pricing equation can be written as

$$i_t = \mathbb{E}_t \pi_{t+1} + r_t \quad (4.32)$$

where r_t is the equilibrium ex ante real interest rate, and we assume that it follows an exogenous process:

$$r_t = \rho r_{t-1} + v_t \quad (4.33)$$

with $|\rho| < 1$, and v_t is a zero-mean IID random variable with bounded support.

Monetary policy rule follows a simplified Taylor rule, adjusting the nominal interest rate in response to inflation. The reaction to inflation evolves stochastically between regimes:

$$i_t = \alpha_{s_t} \pi_t. \quad (4.34)$$

For the sake of simplicity, we assume that $s_t \in \{1, 2\}$. Regime s_t follows a Markov chain with transition probabilities $p_{ij} = P(s_t = j | s_{t-1} = i)$. We assume that the random

variables s and v are independent. In addition, we assume that the nonlinearity induced by regime switching is more important than the nonlinearity of the rest of the model. Thus, we neglect in this section the effects of log-linearization.

In the case of a unique regime $\alpha_1 = \alpha_2$, the model is determinate if $\alpha > 1$ (see sections 4.4.2 and 4.4.4.1), and in this case, the solution is

$$\pi_t = \frac{1}{\alpha - \rho} r_t.$$

Farmer et al. (2009a) show the following result:

Theorem 4.14 *Farmer et al. (2009a). Models (4.32), (4.33), and (4.34) admit a unique bounded solution if and only if all the eigenvalues of $\begin{bmatrix} |\alpha_1|^{-1} & 0 \\ 0 & |\alpha_2|^{-1} \end{bmatrix} \times \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}$ are inside the unit circle.*

This result is explicitly derived by Davig and Leeper (2007) when $|\alpha_i| > p_{ii}$ for $i = 1, 2$. They call the determinacy conditions the Long Run Taylor Principle. Models (4.32), (4.33), and (4.34) are determinate if and only if

$$(1 - |\alpha_2|)p_{11} + (1 - |\alpha_1|)p_{22} + |\alpha_1||\alpha_2| > 1. \quad (4.35)$$

This condition, represented in figure 4.4, shows that if the active regime is very active, there is a room of maneuver for the passive regime to be passive, and this room of maneuver is all the larger as the active regime is absorbent (i.e., that p_{11} is high).

Equation (4.35) illustrates how the existence of another regime affects the expectations of the agents and thus helps stabilize the economy. Intuitively, it means that assuming that there exists a stabilizing regime, we can deviate from it either briefly with high intensity, or modestly for a long period (Davig and Leeper 2007).

4.5.2 Formalism

Linear Markov-switching rational expectations models can generally be written as follows:

$$A_{s_t} \mathbb{E}_t(z_{t+1}) + B_{s_t} z_t + C_{s_t} z_{t-1} + D_{s_t} \varepsilon_t = 0. \quad (4.36)$$

The regime s_t follows a discrete space Markov chain, with transition matrix p_{ij} . The element p_{ij} represents the probability that $s_t = j$ given $s_{t-1} = i$ for $i, j \in \{1, \dots, N\}$. We assume that ε_t is mean zero, IID and independent from s_t .

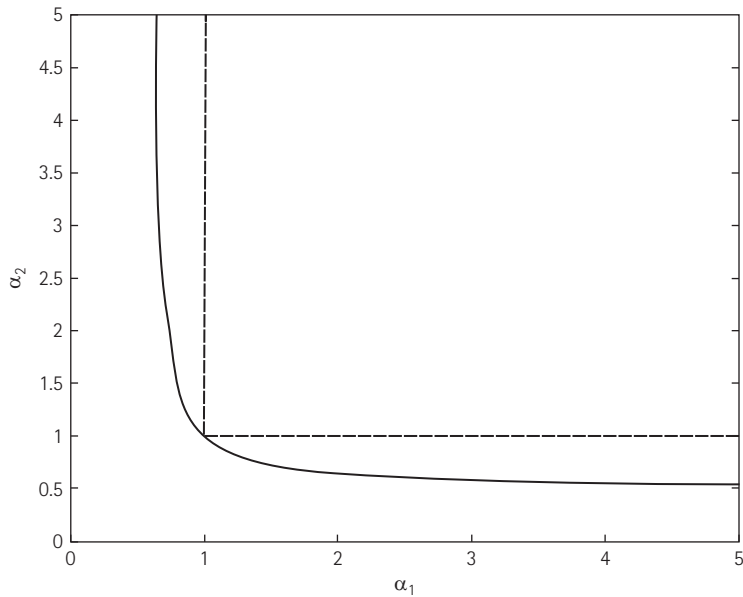


FIGURE 4.4 Long Run Taylor Principle for $p_{11} = 0.6$ and $p_{22} = 0.5$.

Note: This figure depicts the determinacy conditions for the Fisherian model with Markov switches between two regimes, depending on the reaction to inflation in each regime (α_1 and α_2). Policy parameters in the region above the solid line induce determinacy for the Markov-switching Fisherian model. The region of parameters delimited by the dashed black line corresponds to the determinacy region for models with no switches.

4.5.3 The Approach of Davig and Leeper (2007)

Davig and Leeper (2007) find determinacy conditions for forward-looking models with Markov-switching parameters ($C_i = 0$ for any i). They introduce a state-contingent variable $z_{it} = z_t \mathbb{1}_{s_t=i}$ and consider the stacked vector $Z_t = [z_{1t}, z_{2t}, \dots, z_{Nt}]'$ to rewrite model (4.36) as a linear model in variable Z_t .

In the absence of shocks ε_t , the method of the authors (equations (80) and (81), p. 631, Davig and Leeper 2007) consists in assuming

$$\mathbb{E}_t(z_{t+1} \mathbb{1}_{s_t=i}) = \sum_{j=1} p_{ij} z_{j(t+1)}. \quad (4.37)$$

They introduce this relation in model (4.36) to transform the Markov model into a linear one. Finally, they solve the transformed model by using usual linear solution methods (see section 4.4).

Equation (4.37) is not true in general, however, because its right-hand-side is not zero when $i \neq s_t$, contrary to the left-hand side. In the familiar New-Keynesian model with switching monetary policy rule, Farmer et al. (2010a) exhibit two bounded solutions for

a parameter combination satisfying Davig and Leeper's determinacy conditions. Branch et al. (2007) and Barthélemy and Marx (2012) show that these conditions are actually valid for solutions that depend on a finite number of past regimes, called Markovian. Consequently, if these conditions are satisfied but multiple bounded solutions coexist, it necessarily means that one solution is Markovian and the others are not.

The question raised by Farmer et al. (2010a) in their comment to Davig and Leeper (2007) is whether restricting oneself to Markovian solutions makes economic sense. For further details on this debate we refer the reader to Davig and Leeper (2010), Branch et al. (2007), and Barthélemy and Marx (2012).

4.5.4 The Strategy of Farmer et al. (2010b)

The contribution of Farmer et al. (2010b) consists in describing the complete set of solutions. For the sake of the presentation, we describe their results for invertible, purely forward-looking models, that is, models in which for any i , $C_i = 0$, $A_i = \mathbb{I}$, and B_i are invertible. Under these assumptions, model (4.36) turns out to be as follows:

$$\mathbb{E}_t z_{t+1} + B_{s_t} z_t + D_{s_t} \varepsilon_t = 0. \quad (4.38)$$

For this model, Farmer et al. prove the following result.

Theorem 4.15 (Farmer et al. 2009b). *Any solution of model (4.38) can be written as follows:*

$$\begin{aligned} z_t &= -B_{s_t}^{-1} D_{s_t} \varepsilon_t + \eta_t \\ \eta_t &= \Lambda_{s_{t-1}, s_t} \eta_{t-1} + V_{s_t} V_{s_t}' \gamma_t \end{aligned}$$

where for $s \in \{1, 2\}$, V_s is an $n \times k_s$ matrix with orthonormal columns, $k_s \in \{1, \dots, n\}$. The sunspot γ_t is an arbitrary process such that

$$\mathbb{E}_{t-1}(V_{s_t} V_{s_t}' \gamma_t) = 0$$

and for $(i, j) \in \{1, 2\}^2$, there exist matrices $\Phi_{i,j} \in \mathcal{M}_{k_i \times k_j}(\mathbb{R})$ such that

$$\tilde{B}_i V_i = \sum_{j=1}^2 p_{ij} V_j \Phi_{i,j}. \quad (4.39)$$

The strength of this result is that it exhaustively describes all the solutions. When the model embodies backward-looking components, solutions are recursive. The strategy of Farmer et al. (2010b) extends that of Sims (2002) in section 4.3.3 and Lubik and Schorfheide (2004). Apart from the purely forward-looking models presented above, finding all the solutions mentioned by Farmer et al. (2010b) requires employing numerical methods.

Following the influential book by Costa et al. (2005), Cho (2011), and Farmer et al. (2010b) argue that the convenient concept of stability in the Markov-switching context is mean square stability:

Definition 4.16 (Farmer et al. 2009b). *A process z_t is mean-square stable (MSS) if and only if there exists a vector m and a matrix Σ such that*

1. $\lim_{t \rightarrow +\infty} \mathbb{E}_0(z_t) = m,$
2. $\lim_{t \rightarrow +\infty} \mathbb{E}_0(z_t z_t') = \Sigma.$

This stability concept is less stringent than the boundedness concept (see definition 4.1). On one hand, checking that a solution is mean-square stable is easy (see Cho 2011; Farmer et al. 2010b). On the other, this concept does not rely on a norm and hence does not allow for applying a perturbation approach.

4.5.5 Method of Undetermined Coefficients (Svennson and Williams)

Svennson and Williams (2009) adopt an approach similar to the method of Uhlig (1999) and, consistent with the results of Farmer et al., look for solutions of model (4.36) under the form

$$Z_t = P_{s_t} Z_{t-1} + Q_{s_t} \varepsilon_t.$$

Introducing the matrices P_i and Q_i into the model (4.36) leads to a quadratic matrix system

$$\begin{aligned} A_1(p_{11}P_1 + p_{12}P_2)P_1 + B_1P_1 + C_1 &= 0 \\ A_2(p_{21}P_1 + p_{22}P_2)P_2 + B_2P_2 + C_2 &= 0. \end{aligned}$$

This system, however, is more complex to solve than the Ricatti-type matrix equation presented in Uhlig (1999) and in section 4.3. Solving such equations involves computation-based methods.

4.5.6 The Approach of Barthélemy and Marx

Barthélemy and Marx (2012) give general conditions of determinacy for purely forward-looking models with the usual definition of stability (see definition 4.1):

$$\mathbb{E}_t z_{t+1} + B_{s_t} z_t + D_{s_t} \varepsilon_t = 0. \quad (4.40)$$

Unlike Davig and Leeper (2007), the authors do not restrict the solutions space to Markovian solutions.

For a fixed operator norm on $\mathcal{M}_n(\mathbb{R})$, $|||\cdot|||$, they introduce the matrix S_p , defined for $p \geq 2$ by

$$S_p = \left(\sum_{(k_1, \dots, k_{p-1}) \in \{1, \dots, N\}^{p-1}} p_{ik_1} \cdots p_{k_{p-1}j} |||B_i^{-1} B_{k_1}^{-1} \cdots B_{k_{p-1}}^{-1}||| \right)_{ij}. \quad (4.41)$$

They give the following determinacy condition for the existence of a unique SREE of model (4.40):

Proposition 4.17 (*Barthélemy and Marx 2012*). *There exists a unique bounded solution for model (4.40) if and only if*

$$\lim_{p \rightarrow +\infty} \rho(S_p)^{1/p} < 1.$$

In this case, the solution is the solution found by Davig and Leeper (2007).

Based on eigenvalue computations, this proposition extends Blanchard and Kahn's determinacy conditions to Markov switching models following the attempt by Davig and Leeper (2007). The advantage of proposition 4.17 compared to previous methods is that it provides explicit ex ante conditions ensuring the existence and uniqueness of a bounded solution. However, this result suffers from two weaknesses. First, for some combinations of parameters, this condition is numerically difficult to check. Second, this result only applies for purely forward-looking models. (We refer the reader to Barthélemy and Marx 2012 for more details.)

To conclude this section, though major advances have been made, this literature has not yet converged toward a unified approach. This lack of consensus reflects the extreme sensitivity of the results to the definition of the solutions' space and of the stability concept. On one hand, Farmer et al. (2010b) show that the mean square stability concept leads to very applicable and flexible techniques. But mean square stability does not rely on a well-defined norm and thus, does not allow for a perturbation approach. On the other hand, the concept of boundedness is consistent with the perturbation approach (see Barthélemy and Marx 2017), and new results regarding determinacy are encouraging. However, this concept remains limited by the fact that the determinacy conditions are hard to compute and, at this stage, are not generalized to models with backward-looking components.

4.6 DEALING WITH DISCONTINUITIES: THE CASE OF THE ZERO LOWER BOUND

Solving model (4.1) with a perturbation approach such as that described above requires that the model have regularity conditions. More specifically, applying the implicit

function theorem requires that g be at least C^1 (see section 4.4). However, certain economic models including, for instance, piecewise linear functions do not fulfill this condition. One famous example of such models is one taking into account the positivity of the nominal interest rate, the so-called zero lower bound (ZLB). This part reviews existing methods to address technical issues raised by the ZLB.

4.6.1 An Illustrative Example with an Explicit ZLB

Let us first present a monetary model including an explicit ZLB. Following most of the recent literature studying the ZLB, we focus on such standard New-Keynesian models as those described in Woodford (2003). To limit the complexity of the model, most papers log-linearize the structural equations (because it is the usual procedure to solve C^1 models by using a perturbation approach), assuming that the nonlinearity of these equations is secondary compared to the nonlinearity introduced by the ZLB. This assumption is, however, a simplification that is theoretically unfounded. In such a model, the log-linear approximate equilibrium relations may be summarized by two equations, a forward-looking IS relation,

$$x_t = \mathbb{E}_t x_{t+1} - \sigma(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n), \quad (4.42)$$

and a New-Keynesian Phillips curve,

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t. \quad (4.43)$$

Here π_t is the inflation rate, x_t is a welfare-relevant output-gap, and i_t is the absolute deviation of the nominal risk-free interest rate from the steady state, r^* , the real interest rate consistent with the golden rule. Inflation and the output-gap are supposed to be zero at the steady state. The term u_t is commonly referred to as a cost-push disturbance, and r_t^n is the Wicksellian natural rate of interest. The coefficient κ measures the degree of price stickiness, and σ is the intertemporal elasticity of substitution; both are positive. The discount factor of the representative household is $0 < \beta < 1$.

To allow bonds to coexist with money, one can ensure that the return of holding bonds is positive (in nominal term). This condition translates into

$$i_t \geq -r^*. \quad (4.44)$$

This condition triggers a huge nonlinearity that violates the C^1 assumption required to use a perturbation approach and the implicit function theorem (see section 4.4). To circumvent these difficulties, one can either solve analytically by assuming an extra hypothesis concerning the nature of shocks (section 4.6.2) or use global methods (section 4.6.3).

4.6.2 Ad hoc Linear Methods

Following Jung et al. (2005), a large literature (Eggertsson and Woodford 2003, 2006; Eggertsson 2011; Christiano et al. 2011; Bodenstein et al. 2010) solves rational expectations model with the ZLB by postulating additional assumptions about the nature of the stochastic processes. In Jung et al. (2005) seminal paper, the authors find the solution of equations (4.42), (4.43) and (4.44) with the assumption that the number of periods for which the natural rate of interest will be negative is known with certainty when the disturbance occurs. Monetary policy is supposed to minimize a welfare loss function,

$$\min \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^k (\pi_{t+k}^2 + \lambda x_{t+k}^2). \quad (4.45)$$

We refer the reader to Woodford (2003) for more details about the potential microfoundation of such a loss function.

Eggertsson and Woodford (2003) show how the system can be solved when the natural interest rate is negative during a stochastic duration unknown at date t . This resolution strategy has been used by Christiano et al. (2011) to assess the size of the government spending multiplier at the ZLB.

For the sake of clarity, we present a procedure for solving equations (4.42), (4.43), and (4.44) when monetary policy authority follows a simple Taylor rule as long as it is possible instead of minimizing the welfare loss (4.45):

$$i_t = \max(-r^*, \alpha \pi_t). \quad (4.46)$$

Studying a contemporaneous Taylor rule rather than an optimized monetary policy such as (4.45) prevents us from introducing backward-looking components in the model (through Lagrange multipliers), and hence, it reduces the size of the state space (see Eggertsson and Woodford 2003 for more details).

We solve the model when the path of the future shocks is known (*perfect foresight equilibrium*) and assume a (potentially very) negative shock to the natural rate of interest during a finite number of periods τ . When the shock is small enough, the model can be solved using a standard backward-forward procedure as presented in section 4.4, because the ZLB constraint is never binding. In this case, the equilibrium is given by

$$\begin{aligned} x_t &= x_{t+1} - \sigma(i_t - r_t^n - \mathbb{E}_t \pi_{t+1}) \\ \pi_t &= \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t \\ i_t &= \alpha \pi_t, \end{aligned}$$

where $r_t^n = -r^l$ from $t = 0$ to $t = \tau$ and $r_t^n = 0$ afterwards (see figure 4.5).

Using notations from section 4.3 and denoting by Z_t the vector of variables $[x_t, \pi_t, i_t]'$, these equations can be written as follows:

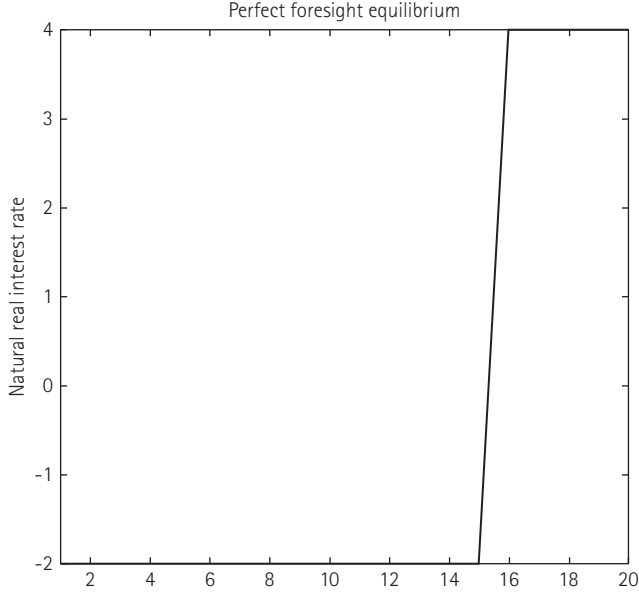


FIGURE 4.5 Scenario of a negative real natural interest rate.

Note: The black line displays the perfect foresight trajectory of the annualized real interest rate. For the first fifteen quarters the real interest rate equals -2 percent, then it turns back to its steady-state value.

$$A\mathbb{E}_t \begin{bmatrix} Z_t \\ Z_{t+1} \end{bmatrix} = B \begin{bmatrix} Z_{t-1} \\ Z_t \end{bmatrix} + Cr_t^n$$

When r^l is small enough, we find the solution by applying the methods described in section 4.3. For $t \leq \tau$,

$$\forall t \leq \tau, \quad Z_t = Z_{22}^{-1} \sum_{k=0}^{\tau-t-1} (S_{22}^{-1} T_{22})^k S_{22}^{-1} Q'_{12} [\sigma, 0, 0]' r^l \quad (4.47)$$

where Z_{22} , S_{22} , T_{22} , and Q_{12} are matrices given by the Schur decomposition (see equation (4.10) in section 4.3).

For $t > \tau$, all variables are at the steady state because the model is purely forward-looking.

$$\forall t > \tau, \quad Z_t = 0 \quad (4.48)$$

As long as i_0 given by equation (4.47) is larger than $-r^*$, the ZLB constraint is never binding, and the solution does not violate the constraint (4.44). When the shock r_l becomes large enough, the solution (4.47) is no longer valid because the ZLB constraint binds.

In this case, let us define k as the largest positive integer such that $i_{\tau-k}$, defined in equations (4.47) and (4.48), is larger than $-r^*$. Obviously, if $i_{\tau} < -r^*$ then $k = 0$, meaning that up to the reversal of the natural real interest rate shock, the ZLB constraint is binding.

Because of the forward-looking nature of the model, for $t > \tau - k$ the solution of the problem is given by equations (4.47) and (4.48), and the ZLB does not affect the equilibrium dynamic.

For $t < \tau - k$, we need to solve the model by backward induction. The solution found for $t > \tau - k$ is the terminal condition and is sufficient to anchor the expectations. Thus, for $t < \tau - k$, the policy rate is stuck to its lower bound, and the variables should satisfy the following dynamic (explosive) system:

$$x_t = x_{t+1} - \sigma(-r^* + r^l - \pi_{t+1})$$

and

$$\pi_t = \beta\pi_{t+1} + \kappa x_t.$$

This can be easily rewritten as follows:

$$\begin{bmatrix} x_t \\ \pi_t \end{bmatrix} = \begin{pmatrix} 1 & \sigma \\ \kappa & \sigma\kappa + \beta \end{pmatrix} \begin{bmatrix} x_{t+1} \\ \pi_{t+1} \end{bmatrix} + \begin{pmatrix} \sigma \\ \sigma\kappa \end{pmatrix} (r^* - r^l).$$

Thus,

$$\begin{bmatrix} x_t \\ \pi_t \end{bmatrix} = \sum_{k=0}^{\tau-k-t-1} \begin{pmatrix} 1 & \sigma \\ \kappa & \sigma\kappa + \beta \end{pmatrix}^k \begin{pmatrix} \sigma \\ \sigma\kappa \end{pmatrix} (r^* - r^l) + \begin{pmatrix} 1 & \sigma \\ \kappa & \sigma\kappa + \beta \end{pmatrix}^{\tau-k-t} \begin{pmatrix} x_{\tau-k} \\ \pi_{\tau-k} \end{pmatrix}$$

where $[x_{\tau-k}, \pi_{\tau-k}]'$ is given by equation (4.47) or is 0 if the shock is large enough ($k = 0$).

To illustrate this method, we compute the equilibrium dynamic after an unexpected shock to the negative real interest rate lasting fifteen periods (see fig. 4.5). After the initial fall in the real natural interest rate, we assume that there is no uncertainty and that the economic agents perfectly know the dynamic of this shock. We calibrate the model with common numbers: $\sigma = 1$, $\beta = 0.99$, $\kappa = 0.02$, $\alpha = 1.5$. There is no inflation at the steady state. Figure 4.6 displays the responses of output-gap, annualized inflation, and annualized nominal interest rate with and without the ZLB. The size of the natural real interest rate shock is calibrated such that the economy hits the ZLB immediately and for nine periods (thus, $\tau = 15$ and $k = 9$).

As we could expect, in the absence of the ZLB the economy would suffer from a less severe crisis with lower deflation and a higher output-gap. In our simulations the gap between the dynamic equilibria with and without the ZLB is huge, suggesting that this constraint has a potentially large effect. However, this result should be treated with care because this example is illustrative only, and the model is very stylized.

The great interest of this approach is that we can understand all the mechanisms at work and we have a proper proof of the existence and uniqueness of a stable equilibrium

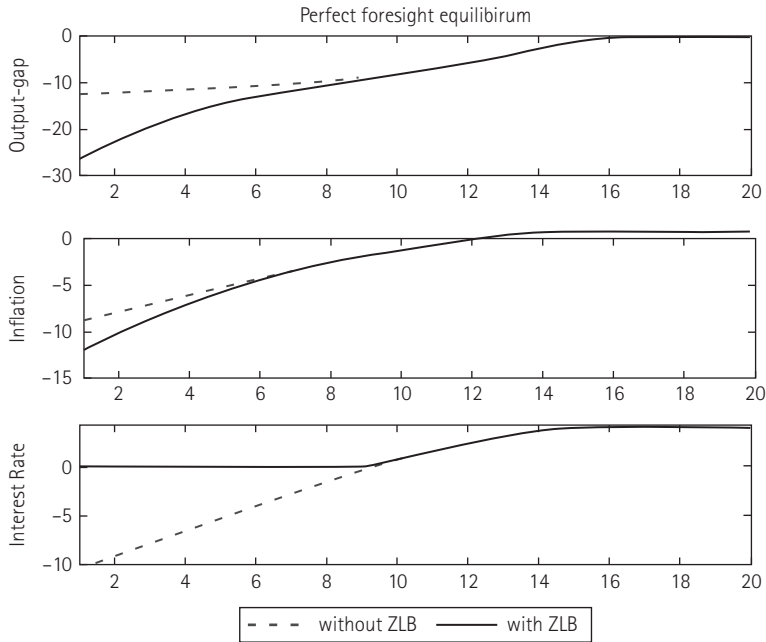


FIGURE 4.6 Responses of endogenous variables to a negative real natural interest rate.

Note: The solid lines display the evolution of the output-gap, inflation, and the nominal interest rate along the perfect foresight equilibrium incorporating the ZLB constraint. Dashed lines depict the trajectory when omitting this constraint.

(proof by construction). Yet this resolution strategy is only valid for a well-known path of shocks and in a perfect foresight equilibrium. Even if Eggertsson and Woodford (2003) extend this method for a negative natural interest rate shock of unknown but finite duration, this kind of method only addresses in part issues raised by the ZLB. Indeed, this method is mute with regard to the consequences of the ZLB in normal situations, when there are risks of a liquidity trap and a ZLB but when this risk had not materialized. In such a situation, one can expect that economic agents' decisions are altered by the risk of reaching the ZLB in the future.

4.6.3 Global Methods

To completely solve a model with a ZLB, one can turn toward global methods or nonlinear methods. Among the first to solve a model with a ZLB using a nonlinear method, we count Wolman (2005), who uses the method of finite elements from McGrattan (1996), and Adam and Billi (2004, 2006) use a functional fixed point method to tackle the ZLB for the case with policy commitment (2004) and the case with discretionary policy (2006).

The clear advantage of global methods for studying an economy that is subject to the ZLB is that the full model can be solved and analyzed without assuming a particular form of shocks. Furthermore, with this approach one can study the influence of the probability of reaching the ZLB in the future on current economic decisions and equilibria through the expectations channel.

The general strategy is to first determine the right concept of functions space in which the solution should be (here, for example, the only state variable could be the shock r_t^n). Then, one can replace the variables by some unknown functions in equations (4.42), (4.43), and (4.46). The expectations are integral to these functions (the measure is the probability distribution of shock r_t^n), and so equations (4.42), (4.43), and (4.46) translate into

$$x(r_t^n) = \int x(r_{t+1}^n) - \sigma(i(r_{t+1}^n) - r_t^n - \int \pi(r_{t+1}^n)) \quad (4.49)$$

$$\pi(r_t^n) = \beta \int \pi(r_{t+1}^n) + \kappa x(r_t^n) \quad (4.50)$$

$$i(r_t^n) = \max(\alpha \pi(r_t^n), -r^*) \quad (4.51)$$

Finally, solving for the equilibrium requires finding a fixed point in the functions space $[x(r_t^n), \pi(r_t^n), i(r_t^n)]$ of equations (4.49), (4.50), and (4.51).

To solve this kind of fixed-point problem one can either use projection methods as described in section 4.7.2 or in Judd (1996) or construct a solution of system equations (4.49), (4.50), and (4.51) as a limit of a sequence up to find a fixed point as in Adam and Billi (2004). In the latter, the authors use finite elements. They define a grid and interpolate between nodes using a linear interpolation to compute the integrals. The algorithm they propose can be summed up as follows in our context:

1. guess the initial function;
2. compute the right-hand side of equations (4.49), (4.50), and (4.51);
3. attribute a new value for the guessed solution function equal to the left hand side;
4. if the incremental gain is less than a targeted precision, stop the algorithm; otherwise go to 2.

To hasten the algorithm, it seems natural to place more nodes around the ZLB (that is, negative natural rate shocks) and fewer nodes for a large positive natural rate shock, because the model supposedly behaves in a linear fashion when the probability of reaching the ZLB is very low.

Until now, this computationally oriented strategy was the only available method for solving rational expectations models with the ZLB. This method is limited, however, since its outcome is not guaranteed by any theoretical background (no proof of existence and uniqueness results). Besides, this method is numerically costly and suffers from the curse of dimensionality. It thus prevents us from estimating a medium-scale DSGE model that uses such a strategy.

4.7 GLOBAL SOLUTIONS

When the model presents nonsmooth features or high variation in shocks, some alternatives based on purely computational approaches may be more appropriate (see section 4.6). These approaches have been presented, compared, and used in a more general framework than this chapter. In this section we present the main methods used in the context of rational expectations models. The aim is not to give an exhaustive description of all the available numerical tools—they have been extensively presented in Judd (1996), Heer and Maussner (2005) and Den Haan et al. (2010)—or to compare the accuracy and performance of different methods (Aruoba et al. 2006) but, rather, to explain what type of method can be used to deal with models that are not regular enough to apply a perturbation method. Typical examples are models with occasionally binding constraints (section 4.6) or with large shocks. Finally, numerical methods may sometimes be mixed with perturbation methods to improve the accuracy of solutions (Maliar et al. 2011).

It is worth noticing that most of these methods are very expensive in terms of computing time and do not allow for checking the existence and uniqueness of the solution. We mainly distinguish three types of methods: value function iteration, projection, and extended deterministic path methods.

4.7.1 Value Function Iteration Method

This method can be applied when the stochastic general equilibrium models are written under the form of an optimal control problem.

$$\max_{x \in \{(\mathbb{R}^n)^\infty\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(x_t, y_t) \quad \text{s.c.} \quad y_{t+1} = g(y_t, x_t, \varepsilon_{t+1}) \text{ with a fixed } y_0 \quad (4.52)$$

It is easy to see that first-order conditions for model (4.52) and changes in notations lead to an equation like (4.1), but this formulation is more general. According to the Bellman principle (see Rust 1996), such a program can be rewritten as

$$V(y_0) = \max_{x_0} [U(y_0, x_0) + \beta \mathbb{E}_0 V(y_1)]. \quad (4.53)$$

When U and g satisfy some concavity conditions, it is possible to compute by iterations the value function V and the decision function h defined by

$$h(y_t) = \arg \max_{x_t \in A} [U(y_t, x_t) + \beta \mathbb{E}_t V(g(y_t, x_t, \varepsilon_{t+1}))] \quad (4.54)$$

where A stands for the set of admissible solutions. This method consists in defining a bounded convex set of admissible values for y_t , containing the initial value and the

steady state, and considering a grid G on this set. Then, we consider the sequence of functions V^n , defined recursively for $y \in G$ by

$$V^{n+1}(y) = \max_x \{U(y, x) + \beta \mathbb{E}(V^n[g(y, x, \varepsilon)])\}.$$

We refer the reader to Judd (1996) for the theoretical formalism and to Fernández-Villaverde and Rubio-Ramírez (2006) for the algorithmic description. This method is computationally expensive since it is applied on grids and hence may be applied only to relatively small models. Theoretical properties of convergence have been studied in Santos and Vigo (1998), the illustration of the method for the growth model is presented in Christiano (1990) and in Barillas and Fernández-Villaverde (2007).

4.7.2 Projection Method

The projection method consists in looking for an approximate solution of the form $z_t = h(z_{t-1}, \varepsilon_t)$. Assuming that the shocks ε_t follow a distribution law μ , problem (4.1) can be reformulated as solving the functional equation

$$\mathcal{G}(h) = 0 \tag{4.55}$$

where $\mathcal{G}$ is defined as

$$\mathcal{G}(h)(z, \varepsilon) = \int g(h(h(z, \varepsilon'), \varepsilon), h(z, \varepsilon), z) \mu(\varepsilon') d\varepsilon'.$$

The core idea of the projection method is to find an approximate solution $\hat{h}$ belonging to a finite-dimension functional space $\mathcal{S}$ (Judd 1996).

Let us denote by $\{\Phi_i\}_{i \in \{1, \dots, P\}}$ the basis of the vector space $\mathcal{S}$. In other words, we are looking for P parameters $(c_i)_{i \in \{1, \dots, P\}}$ such that $\hat{h} = \sum_{i=1}^P c_i \Phi_i$ is close to the solution h .

There are four sets of issues that must be addressed by this approach (Judd 1996; Fackler 2005; Heer and Maussner 2005). The first is the choice of $\mathcal{S}$; for instance, $\mathcal{S}$ can be the set of polynomials of a degree smaller than d , the set of piecewise linear functions, or the set of spline functions. The second issue is the computation of the expectation operator; here, we have at our disposal all the numerical techniques dealing with integrals, mainly quadrature formulas or Monte Carlo computations. The third question is how to characterize the accuracy of the approximation, in other words, the size of the residual function $\mathcal{G}(\hat{h})$. There are three main criteria: we look for the function minimizing the L^2 norm of $\mathcal{G}(\hat{h})$ (least squares), the zero of $\mathcal{G}(\hat{h})$ in a finite number of points (collocation), or the function such that $\mathcal{G}(\hat{h})$ is orthogonal to $\mathcal{S}$. Finally, the fourth issue concerns the way we can solve such a problem. Addressing these issues leads to finding the zero $(c_i)_{i \in \{1, \dots, P\}}$ of a function; there are various

root-finding algorithms based mainly on iterative approaches (Newton, Broyden, fixed point, and so on).

This kind of method is used for solving models with occasionally binding constraints (see, e.g., Christiano and Fisher 2000) or endogenous regime-switching models (see Davig and Leeper 2008). The accuracy and the computational cost of this method depend both on the dimension of $\mathcal{S}$ and on the choice of the basis $\{\Phi_i\}_{i \in \{1, \dots, p\}}$. In practice, Christiano and Fisher (2000) and Heer and Maussner (2005) refine this method with a parameterized expectations algorithm.

4.7.3 Parameterized Expectations Algorithm

This algorithm consists in rewriting model (4.1) under the form

$$\tilde{f}(\mathbb{E}_t[\phi(y_{t+1}, y_t)], y_t, y_{t-1}, \varepsilon_t) = 0 \quad (4.56)$$

per Juillard and Ocaktan (2008). We restrict our focus to solutions depending on (y_{t-1}, ε_t) . Defining h , the expected solution, such that

$$\mathbb{E}_t[\phi(y_{t+1}, y_t)] = h(y_{t-1}, \varepsilon_t),$$

we can apply a projection method (described in section 4.7.2) to h and find an approximation $\hat{h}$ of h in an appropriate functional vector space. This method is described in Marcet and Marshall (1994) and applied in Christiano and Fisher (2000) for models with occasionally binding constraints.

4.7.4 Extended Deterministic Path Method

The extended path of Fair and Taylor (1983) is a forward iteration method for solving models with a given path of shocks. It is similar to the one presented in section 4.6. Because it does not include uncertainty, this method is unable to solve DSGE models. Let us assume that we want to get the solution on a period $[0, p]$. Fix $T > 0$ large enough, and for any $t \in \{0, \dots, p\}$, define $y_{T+s} = \bar{y}$ and $\varepsilon_{T+s} = 0$ for all $s > 0$. Then, for $t \in \{0, \dots, p\}$, by fixing the terminal condition, we can numerically solve the model

$$g(y_{t+s+1}, y_{t+s}, y_{t+s-1}, \varepsilon_{t+s}) = 0, \quad \forall s > 0, y_{t+T} = \bar{y}$$

and get $y_t^T = h^T(y_{t-1}, \varepsilon_t)$. Love (2009) shows that the approximation error of this algorithm is reasonable for the stochastic growth model. This method has also been implemented to solve models with occasionally binding constraints, such as the ZLB (see Coenen and Wieland 2003; Adjemian and Juillard 2010).

4.8 CONCLUSION

We have presented the main theories underlying the solving of rational expectations models with a finite number of state variables. We have insisted on the importance of the perturbation approach because this approach is based on a solid theoretical framework. The interest of this approach is that it allows for checking for the existence and uniqueness of a stable solution. Moreover, we have tried to give some insights regarding the limits of this approach. We have concluded with a brief review of some important numerical approaches.

This chapter raises a wide range of unsolved essential questions. What is the size of the admissible domain for applying a perturbation approach? How do we characterize the solutions of nonlinear Markov switching rational expectation models, and what are the determinacy conditions? How do we solve rational expectations models with ZLB without requiring global methods?

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CHAPTER 5

COMPUTABLE GENERAL EQUILIBRIUM MODELS FOR POLICY EVALUATION AND ECONOMIC CONSEQUENCE ANALYSIS

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5.1 INTRODUCTION

Whereas economic research has historically been dominated by theoretical and econometric analyses, computational simulations have grown to satisfy the ever-expanding demand for the assessment of policies in a variety of settings. This third approach complements traditional economic research methods by marrying a rigorous theoretical structure with an empirically informed context. This chapter offers a review of computable general equilibrium (CGE) simulations, which have emerged as the workhorse of prospective characterization and quantification of the impacts of policies that are likely to affect interactions among multiple markets.

At its core, CGE modeling is a straightforward exercise of “theory with numbers,” in which the latter are derived from input-output economic accounts and econometric estimates of key parameters. Advances in computing power and numerical methods have made it possible to specify and solve models with increasingly complex structural representations of the economy. These do far more than generate detailed information about the likely impacts of policies under consideration—their basis in theory enables researchers to pinpoint the economic processes that give rise to particular outcomes and establish their sensitivity to various input parameters.

Our goal is to rigorously document key contemporary applications of CGE models to the assessment of the economic impacts of policies ranging from tax reforms to the mitigation of, and adaptation to, global climate change. Throughout, we focus on the structural representation of the economy. In section 5.2 we begin by deriving the theoretical structure of a canonical static multiregional simulation. This model is structurally simple but of arbitrary dimension, and it is sufficiently general to admit the kinds of modifications necessary to address a wide variety of research questions and types of policies. We first demonstrate how our canonical model is a generalization of ubiquitous single-region open-economy models with an Armington structure, and show how the dynamics of capital accumulation may be introduced as a boundary condition of the economy (sections 5.2.2 and 5.2.3). In section 5.3 we illustrate the application of the canonical model in areas that are both popular and well studied—international, development, and public economics (section 5.3.1), emerging—energy economics and greenhouse gas emission abatement (section 5.3.2), and novel—climate change impacts and natural hazards (section 5.3.3). Section 5.4 moves beyond mere applications to document two prominent extensions to the canonical framework: the incorporation of discrete technological detail into representation of production in the sectors of the economy (with a focus on the electric power sector—section 5.4.1), and the representation of modern theories of trade based on heterogeneous firms and the implications for the effects of economic integration (sections 5.4.2 and 5.4.3). Section 5.5 concludes the chapter.

5.2 THE CANONICAL MODEL

The economic principles underlying a standard closed-economy CGE model are well explained in pedagogic articles such as Sue Wing (2009, 2011). To conserve space we use these studies as the point of departure to derive the theoretical structure of a static open-economy multiregional CGE model that will be the workhorse of the rest of this chapter and, indeed, has emerged as the standard platform for international economic simulations since its introduction by Harrison et al. (1997a,b) and Rutherford (2005).

5.2.1 A Static Multiregional Armington Trade Simulation

The pivotal feature of our model is interregional trade in commodities, which follows the Armington (1969) constant elasticity of substitution (CES) specification. A region's demands for each commodity are satisfied by an "Armington" composite good, which is supplied by aggregating together domestic and imported varieties of the commodity in question. The import supply composite is, in turn, a CES aggregation of quantities of the commodity produced in other regions, at prices that reflect the markup of transport

margins over domestic production costs. These bilateral commodity movements induce derived demands for international freight transport services, whose supply is modeled as a CES aggregation of regions' transportation sector outputs at producer prices.

There are six institutions in the multiregional economy: within each region, households (I₁), investment goods-producing firms (I₂), commodity-producing firms (I₃), domestic-import commodity aggregators (I₄), and import agents (I₅), and, globally, international commodity transporters (I₆). As in Sue Wing (2009, 2011), households are modeled as a representative agent who derives utility from the consumption of commodities and is endowed with internationally immobile factors of production that are rented to domestic goods-producing firms. In each region, commodity producers in a particular industry sector are modeled as a representative firm that combines inputs of primary factors and intermediate goods to generate a single domestic output.

The key departure from the familiar closed-economy model is that domestic output is sold to commodity aggregators or exported abroad at domestic prices. Regional aggregators of domestic and imported commodities are modeled as a representative firm that combines domestic and imported varieties of each commodity into an Armington composite good, which in turn is purchased by the industries and households in the region in question. The imported variety of each commodity is supplied by import agents, which are modeled as a representative firm denominated over trade partners' exports. Finally, each region exports some of the output of each of its transportation sectors to international shippers, who are modeled as a global representative firm. Interregional movements of goods generate demands for international transportation services, with each unit of exports requiring the purchase of shipping services across various modes. Thus, the economy's institutions are linked by five markets: supply and demands for domestic goods (M₁), the Armington domestic-import composite (M₂), imported commodities (M₃), international shipping services (M₄), and primary factors (M₅).

The values of transactions in these markets are recorded in the cells of interlinked regional input-output tables in the form of the simplified social accounting matrix (SAM) in figure 5.1. This input-output structure is underlain by the price and quantity variables summarized in table 5.1, in which markets correspond to the SAM's rows and institutions correspond to its columns. In line with CGE models' strength in analyzing the aggregate welfare impacts of price changes, we reserve special treatment for the households in each region, whose aggregate consumption, we assume, generates an economy-wide level of utility (u_r) at an aggregate "welfare price" given by the unit expenditure index ($\mathcal{E}_r$). The accounting identities corresponding to the SAM's column and row sums are the exhaustion of profit and supply-demand balance conditions presented in table 5.2. These make up the core of our CGE model.

To elaborate the model's algebraic structure, we assume that institutional actors are endowed with a CES technology parameterized according to table 5.2, part B, and behave in a manner consistent with consumer and producer optimization. This lets us

A. Sets		B. Arrays	
Regions	$r = \{1, \dots, \mathcal{R}\}$	Interindustry commodity flows	$\bar{X}_r$
Commodities	$i = \{1, \dots, \mathcal{N}\}$	Primary factor inputs to sectors	$\bar{V}_r$
Industries	$j = \{1, \dots, \mathcal{N}\}$	Final commodity demands	$\bar{G}_r$
Primary factors	$f = \{1, \dots, \mathcal{F}\}$	Of which:	
Domestic demands	$d = \{\text{consumption (C), investment (I)}\}$	Domestic final commodity uses	$\bar{G}_r^d$
		Aggregate commodity imports	$\bar{G}_r^M$
Transportation services	$s \subset i$	International transport service demands	$\bar{G}_r^{TM}$
		Export supplies to other regions	$\bar{G}_r^X$
		International transport service supplies	$\bar{G}_r^{TX}$

C. Benchmark interregional social accounting matrix

		$\leftarrow j \rightarrow$			$\leftarrow d \rightarrow$			$\leftarrow r' \neq r \rightarrow$			$\leftarrow r' \neq r \rightarrow$			Row
		1	...	$\mathcal{N}$	C	I	M	1	...	$\mathcal{R}$	1	...	$\mathcal{R}$	TX
$\uparrow$	1													$\bar{y}_{1,r}$
	$\vdots$													$\vdots$
	i			$\bar{X}_r$			$\bar{G}_r^d$			$\bar{G}_r^{TM}$			$\bar{G}_r^X$	$\bar{G}_r^{TX}$
$\downarrow$	$\mathcal{N}$													$\bar{y}_{\mathcal{N},r}$
$\underbrace{\hspace{10em}}_{\bar{G}_r}$														
$\uparrow$	1													$\bar{V}_{1,r}$
	$\vdots$													$\vdots$
	f			$\bar{V}_r$										
$\downarrow$	$\mathcal{F}$													$\bar{V}_{\mathcal{F},r}$
Col.														
Total		$\bar{y}_1$	...	$\bar{y}_{\mathcal{N}}$	$\bar{G}_r^C$	$\bar{G}_r^I$	$\bar{G}_r^M$	$\bar{G}_{r,1}^{TM}$	...	$\bar{G}_{r,\mathcal{R}}^{TM}$	$\bar{G}_{r,1}^X$	...	$\bar{G}_{r,\mathcal{R}}^X$	$\bar{G}_r^{TX}$

FIGURE 5.1 Multiregional accounting framework.

derive the demand functions that are the fundamental bridge between the activity levels that reflect institutional behavior and the prices that establish market equilibrium:

(I1) *Representative agents* minimize the expenditure necessary to generate each unit of utility subject to the constraint of consumption technology by allocating unit quantities of each commodity consumed ($\hat{g}_{i,r}^C = g_{i,r}^C / u_r$).

$$\min_{\hat{g}_{i,r}^C} \left\{ \mathcal{E}_r = \sum_{i=1}^{\mathcal{N}} p_{i,r}^A \hat{g}_{i,r}^C \mid 1 = \left[\sum_{i=1}^{\mathcal{N}} \alpha_{i,r}^C (\hat{g}_{i,r}^C)^{(\sigma_r^C - 1) / \sigma_r^C} \right]^{\sigma_r^C / (\sigma_r^C - 1)} \right\}.$$

The result is the unconditional demand for Armington goods inputs to consumption, $g_{i,r}^C = (\alpha_{i,r}^C)^{\sigma_r^C} (p_{i,r}^A)^{-\sigma_r^C} \mathcal{E}^{\sigma_r^C} u_r$.

Table 5.1 Summary of variables in the canonical model

A. Institutions							
Output			Inputs				
	Price	Quantity		Price	Quantity		
(1) Households	$\mathcal{E}_r$	Unit expenditure index	u_r	$p_{i,r}^A$	Armington goods price	$g_{i,r}^C$	Consumption demand for Armington good
(2) Investment goods producers	p_r^I	Investment price	G_r^I	$p_{i,r}^A$	"	$g_{i,r}^I$	Investment demand for Armington good
(3) Commodity producers	$p_{j,r}^D$	Domestic goods price	$y_{j,r}$	$p_{i,r}^A$	"	$x_{i,j,r}$	Intermediate demand for Armington good
(4) Domestic-import goods aggregators	$p_{i,r}^A$	Armington goods price	$q_{i,r}^A$	$w_{f,r}$	Factor price	$v_{f,i,r}$	Factor demand
				$p_{i,r}^D$	Domestic goods price	$q_{i,r}^D$	Domestic goods supply
(5) Import agents	$p_{i,r}^M$	Imported goods price	$q_{i,r}^M$	$p_{i,r}^M$	Imported goods price	$q_{i,r}^M$	Imported goods supply
				$p_{i,r}^D$	Domestic goods price in $r' \neq r$	$g_{i,r',r}^X$	r 's imports from other regions r'
				p_S^T	Int'l shipping services price	$g_{S,i,r',r}^{TM}$	Int'l transport services
(6) International shippers	p_S^T	Int'l shipping services price	q_S^T	$p_{S,r}^D$	r 's domestic transport price	$g_{S,r}^{TX}$	Transport service exports from r

Table 5.1 Continued

B. Markets				
	Supply		Demands	
	Price	Quantity		
(M1)	$p_{j,r}^D$	$y_{j,r}$	$q_{i,r}^D$ $x_{j,i,r,r'}^X$ $g_{j,i,r,r'}^{TX}$ $g_{s,r}^{TX}$	Domestic goods demanded by Armington aggregator Commodity exports International transport sales ($t \subset j$)
(M2)	$p_{i,r}^A$	$q_{i,r}^A$	$x_{i,j,r}$ $g_{i,r}^C$ $g_{i,r}^I$	Intermediate demand for Armington good Consumption demand for Armington good Investment demand for Armington good
(M3)	$p_{i,r}^M$	$g_{i,r}^M$	$q_{i,r}^M$	Imported goods demanded by Armington aggregator
(M4)	p_s^T	q_s^T	$g_{s,i,r,r'}^{TM}$	Margins on exports from r to other regions s
(M5)	$w_{f,r}$	$V_{f,r}$	$v_{f,j,r}$	Goods producers' demands for factors

(I2) *Investment goods producers* minimize the cost of generating a unit of output subject to the constraint of production technology by allocating unit quantities of commodity inputs ($\widehat{g}_{i,r}^I = g_{i,r}^I / G_r^I$).

$$\min_{\widehat{g}_{i,r}^I} \left\{ p_r^I = \sum_{i=1}^{\mathcal{N}} p_{i,r}^A \widehat{g}_{i,r}^I \mid 1 = \left[\sum_{i=1}^{\mathcal{N}} \alpha_{i,r}^I (\widehat{g}_{i,r}^I)^{(\sigma_r^I-1)/\sigma_r^I} \right]^{\sigma_r^I/(\sigma_r^I-1)} \right\}.$$

The result is the unconditional demand for Armington goods inputs to investment, $g_{i,r}^I = (\alpha_{i,r}^I)^{\sigma_r^I} (p_{i,r}^A)^{-\sigma_r^I} (p_r^I)^{\sigma_r^I} G_r^I$.

(I3) *Commodity-producing industry sectors* minimize the cost of creating a unit of output subject to the constraint of production technology by allocating purchases of unit quantities of intermediate commodity inputs and primary factor inputs ($\widehat{x}_{i,j} = x_{i,j}/y_j$ and $\widehat{v}_{f,j} = v_{f,j}/y_j$).

$$\min_{\widehat{x}_{i,j,r}, \widehat{v}_{f,j,r}} \left\{ p_{j,r}^D = \sum_{i=1}^{\mathcal{N}} p_{i,r}^A \widehat{x}_{i,j,r} + \sum_{f=1}^{\mathcal{F}} w_{f,r} \widehat{v}_{f,j,r} \mid 1 = \left[\sum_{i=1}^{\mathcal{N}} \beta_{i,j,r} (\widehat{x}_{i,j,r})^{(\sigma_{j,r}^Y-1)/\sigma_{j,r}^Y} + \sum_{f=1}^{\mathcal{F}} \gamma_{f,j,r} (\widehat{v}_{f,j,r})^{(\sigma_{j,r}^Y-1)/\sigma_{j,r}^Y} \right]^{\sigma_{j,r}^Y/(\sigma_{j,r}^Y-1)} \right\}.$$

The result is the unconditional demand for intermediate Armington commodity inputs and nonreproducible primary factor inputs, $x_{i,j,r} = \beta_{i,j,r}^{\sigma_{j,r}^Y} (p_{i,r}^A)^{-\sigma_{j,r}^Y} (p_{j,r}^D)^{\sigma_{j,r}^Y} y_{j,r}$ and $v_{f,j,r} = \gamma_{f,j,r}^{\sigma_{j,r}^Y} w_{f,r}^{-\sigma_{j,r}^Y} (p_{j,r}^D)^{\sigma_{j,r}^Y} y_{j,r}$.

(I4) *Domestic-import commodity aggregators* minimize the cost of producing a unit of composite output of each commodity, subject to the constraint of its CES aggregation technology, by allocating purchases of unit quantities of domestic and imported varieties of the good ($\widehat{q}_{i,r}^D = q_{i,r}^D / q_{i,r}^A$ and $\widehat{g}_{i,r}^M = q_{i,r}^M / q_{i,r}^A$).

$$\min_{\widehat{q}_{i,r}^D, \widehat{g}_{i,r}^M} \left\{ p_{i,r}^A = p_{i,r}^D \widehat{q}_{i,r}^D + p_{i,r}^M \widehat{g}_{i,r}^M \mid 1 = \left[\zeta_{i,r}^D (\widehat{q}_{i,r}^D)^{(\sigma_{i,r}^{DM}-1)/\sigma_{i,r}^{DM}} + \zeta_{i,r}^M (\widehat{g}_{i,r}^M)^{(\sigma_{i,r}^{DM}-1)/\sigma_{i,r}^{DM}} \right]^{\sigma_{i,r}^{DM}/(\sigma_{i,r}^{DM}-1)} \right\}.$$

The result is the unconditional demand for domestically produced and imported varieties of each good, $q_{i,r}^D = (\zeta_{i,r}^D)^{\sigma_{i,r}^{DM}} (p_{i,r}^D)^{-\sigma_{i,r}^{DM}} (p_{i,r}^A)^{\sigma_{i,r}^{DM}} q_{i,r}^A$ and $q_{i,r}^M = (\zeta_{i,r}^M)^{\sigma_{i,r}^{DM}} (p_{i,r}^M)^{-\sigma_{i,r}^{DM}} (p_{i,r}^A)^{\sigma_{i,r}^{DM}} q_{i,r}^A$.

(I5) **Commodity importers** minimize the cost of producing a unit of composite import good subject to the constraint of aggregation technology by allocating purchases of unit commodity inputs over trade partners' exports and associated international transport services ($\widehat{g}_{i,r',r}^X = g_{i,r',r}^X / g_{i,r}^M$ and $\widehat{g}_{s,i,r',r}^{TM} = g_{s,i,r',r}^{TM} / g_{i,r}^M$). We simplify the problem by assuming that the export of a unit of commodity i requires fixed quantities of the t types of transport services ($\kappa_{s,i,r',r}$), which enables shipping costs to be specified as mode-specific markups over the producer prices of overseas goods.

$$\min_{\widehat{q}_{i,r}^M} \left\{ p_{i,r}^M = \sum_{r' \neq r} \left(p_{i,r'}^D \widehat{g}_{i,r',r}^X + \sum_{r'} p_s^T \widehat{g}_{s,i,r',r}^{TM} \right) \middle| \widehat{g}_{s,i,r',r}^{TM} = \kappa_{s,i,r',r} \widehat{g}_{i,r',r}^X, \right. \\ \left. 1 = \left[\sum_{r' \neq r} \xi_{i,r',r} (\widehat{g}_{i,r',r}^X)^{(\sigma_{i,r}^{MM}-1)/\sigma_{i,r}^{MM}} \right]^{\sigma_{i,r}^{MM}/(\sigma_{i,r}^{MM}-1)} \right\}.$$

The result is the unconditional demand for other regions' exports and for international transshipment services, $g_{i,r',r}^X = \xi_{i,r',r}^{\sigma_{i,r}^{MM}} \left(p_{i,r'}^D + \sum_s \kappa_{s,i,r',r} p_s^T \right)^{-\sigma_{i,r}^{MM}} (p_{i,r}^M)^{\sigma_{i,r}^{MM}} g_{i,r}^M$.

(I6) **International shippers** minimize the cost of producing a unit of transport service subject to the constraint of its aggregation technology by allocating purchases of regions' transportation sector outputs ($\widehat{g}_{s,r}^{TX} = g_{s,r}^{TX} / q_s^T$).

$$\min_{\widehat{g}_{s,r}^{TX}} \left\{ p_s^T = \sum_{r=1}^{\mathcal{R}} p_{s,r}^D \widehat{g}_{s,r}^{TX} \middle| 1 = \left[\sum_{r=1}^{\mathcal{R}} \chi_{s,r} (\widehat{g}_{s,r}^{TX})^{(\sigma_r^T-1)/\sigma_r^T} \right]^{\sigma_r^T/(\sigma_r^T-1)} \right\}.$$

The result is the unconditional demand for transport services, $g_{s,r}^{TX} = \chi_{s,r}^{\sigma_s^T} (p_{s,r}^D)^{-\sigma_s^T} (p_s^T)^{\sigma_s^T} q_s^T$.

Substituting these results into the conditions for (I1) to (I6) and for (M1) to (M6) in Table 5.2, part A, yields, in table 5.3, the zero-profit conditions (5.1) to (5.6) and market clearance conditions (5.7) to (5.11). These exhibit Karush-Kuhn-Tucker complementary slackness (indicated by “ $\perp$ ”) with the activity levels and prices, respectively, presented in table 5.1. There are no markets in the conventional sense for either consumers' utility, or, in the present static framework, the investment good. The latter is treated simply as an exogenous demand (5.12). Regarding the former, u_r is the highest level of aggregate utility attainable given the values of aggregate household income ($\mathcal{I}_r$) and the unit expenditure index. This intuition is captured by the market clearance condition (5.13), with definition of the income level given by the income balance condition (5.14).

Together, (5.1) to (5.14) comprise a square system of $\mathcal{R}(5 + 7\mathcal{N} + \mathcal{F}) + 2\mathcal{T}$ nonlinear inequalities, $\Lambda(\mathcal{B})$, in as many unknowns, $\mathcal{B} = \{u_r, G_r^I, y_{i,r}, q_{i,r}^A, g_{i,r}^M, q_s^T, p_{i,r}^D, p_{i,r}^A, p_{i,r}^M, p_s^T, w_{f,r}, p_r^I, \mathcal{E}_r, \mathcal{I}_r\}$, which constitutes the pseudo-excess-demand correspondence

Table 5.2 The canonical model: Accounting identities and parameterization

A. Accounting identities based on the SAM			
Zero-profit conditions (Institutions)		Supply-demand balance conditions (Markets)	
(I1)	$\mathcal{E}_r u_r \leq \sum_{i=1}^{\mathcal{N}} p_{i,r}^A g_{i,r}^C$	(M1)	$y_{j,r} \geq q_{j,r}^D + g_{j,r}^{TX} + \sum_{r' \neq r} g_{j,r,r'}^X$
(I2)	$p_r^I G_r \leq \sum_{i=1}^{\mathcal{N}} p_{i,r}^A g_{i,r}^I$	(M2)	$q_{i,r}^A \geq \sum_{j=1}^{\mathcal{N}} x_{i,j,r} + g_{i,r}^C + g_{i,r}^I$
(I3)	$p_{j,r}^D y_{j,r} \leq \sum_{i=1}^{\mathcal{N}} p_{i,r}^A x_{i,j,r} + w_{f,r} v_{f,j,r}$	(M3)	$g_{i,r}^M \geq q_{i,r}^M p_{i,r}^M$
(I4)	$p_{i,r}^A q_{i,r}^A \leq p_{i,r}^D q_{i,r}^D + p_{i,r}^M q_{i,r}^M$	(M4)	$q_s^T \geq \sum_{i=1}^{\mathcal{N}} \sum_{r=1}^{\mathcal{R}} \sum_{r'} g_{s,i,r,r'}^{TM} p_s^T$
(I5)	$p_{i,r}^M g_{i,r}^M \leq \sum_{r' \neq r} \left(p_{i,r'}^D g_{i,r',r}^X + \sum_{r'} p_s^T g_{s,i,r',r}^{TM} \right)$	(M5)	$V_{f,r} \geq \sum_{j=1}^{\mathcal{N}} v_{f,j,r}$
(I6)	$p_s^T q_s^T \leq \sum_{r=1}^{\mathcal{R}} p_{s,r}^D g_{s,r}^{TX}$		
B. Parameters			
Institutions	Substitution elasticities	Technical coefficients	
(I1) Households	σ_r^C	$\alpha_{i,r}^C$	Armington good use: consumption
(I2) Investment goods producers	σ_r^I	$\alpha_{i,r}^I$	Armington good use: investment
(I3) Commodity producers	$\sigma_{j,r}^Y$	$\beta_{i,j,r}$	Intermediate Armington good use
		$\gamma_{f,j,r}$	Factor inputs
(I4) Domestic-import commodity aggregators	$\sigma_{i,r}^{DM}$	$\zeta_{i,r}^D$	Domestic commodity output
		$\zeta_{i,r}^M$	Imported commodities
(I5) Import agents	$\sigma_{i,r}^{MM}$	$\xi_{i,r',r}$	Exports to r from other regions s
		$\kappa_{t,i,r',r}$	International transport services
(I6) International shippers	$\sigma_{r'}^T$	$\xi_{t,r}$	Transport service exports from r

of our multiregional economy. Numerically calibrating the technical coefficients in table 5.3 on a micro-consistent benchmark multiregional input-output data set yields our CGE model in a complementarity format:

$$\Lambda(\mathcal{B}) \geq 0, \quad \mathcal{B} \geq 0, \quad \mathcal{B}' \Lambda(\mathcal{B}) = 0,$$

Table 5.3 Equations of the CGE model

$$\mathcal{E}_r \leq \left[\sum_{i=1}^{\mathcal{N}} (\alpha_{i,r}^C)^{\sigma_r^C} (p_{i,r}^A)^{1-\sigma_r^C} \right]^{1/(\sigma_r^C-1)} \perp u_r \quad (5.1)$$

$$p_r^I \leq \left[\sum_{i=1}^{\mathcal{N}} (\alpha_{i,r}^I)^{\sigma_r^I} (p_{i,r}^A)^{1-\sigma_r^I} \right]^{1/(\sigma_r^I-1)} \perp G_r^I \quad (5.2)$$

$$p_{j,r}^D \leq \left[\sum_{i=1}^{\mathcal{N}} \beta_{i,j,r}^{\sigma_{j,r}^Y} (p_{i,r}^A)^{1-\sigma_{j,r}^Y} + \sum_{f=1}^{\mathcal{F}} \gamma_{f,j,r}^{\sigma_{j,r}^Y} w_{f,r}^{1-\sigma_{j,r}^Y} \right]^{1/(1-\sigma_{j,r}^Y)} \perp y_{j,r} \quad (5.3)$$

$$p_{i,r}^A \leq \left[(\zeta_{i,r}^D)^{\sigma_{i,r}^{DM}} (p_{i,r}^D)^{1-\sigma_{i,r}^{DM}} + (\zeta_{i,j,r}^M)^{\sigma_{i,r}^{DM}} (p_{i,r}^M)^{1-\sigma_{i,r}^{DM}} \right]^{1/(1-\sigma_{i,r}^{DM})} \perp q_{i,r}^A \quad (5.4)$$

$$p_{i,r}^M \leq \left[\sum_{r' \neq r} \xi_{i,r',r}^{\sigma_{i,r'}^{MM}} \left(p_{i,r'}^D + \sum_{s=1}^{\mathcal{R}} \kappa_{s,i,r',r} p_s^T \right)^{1-\sigma_{i,r'}^{MM}} \right]^{1/(1-\sigma_{i,r}^{MM})} \perp g_{i,r}^M \quad (5.5)$$

$$p_s^T \leq \left[\sum_{r=1}^{\mathcal{R}} \chi_{r,s}^{\sigma_s^T} (p_{s,r}^D)^{1-\sigma_s^T} \right] \perp q_s^T \quad (5.6)$$

$$\begin{aligned} y_{i,r} &\geq (\zeta_{i,r}^D)^{\sigma_{i,r}^{DM}} (p_{i,r}^D)^{-\sigma_{i,r}^{DM}} (p_{i,r}^A)^{\sigma_{i,r}^{DM}} q_{i,r}^A \\ &\quad + \sum_{r' \neq r} \xi_{i,r',r}^{\sigma_{i,r'}^{MM}} \left(p_{i,r'}^D + \sum_{s=1}^{\mathcal{R}} \kappa_{s,i,r',r} p_s^T \right)^{-\sigma_{i,r'}^{MM}} (p_{i,r'}^M)^{\sigma_{i,r'}^{MM}} g_{i,r'}^M \end{aligned} \perp p_{i,r}^D \quad (5.7)$$

$$\begin{aligned} q_{i,r}^A &\geq (\alpha_{i,r}^C)^{\sigma_r^C} (p_{i,r}^A)^{-\sigma_r^C} \mathcal{E}_r^{\sigma_r^C} u_r + (\alpha_{i,r}^I)^{\sigma_r^I} (p_{i,r}^A)^{-\sigma_r^I} (p_r^I)^{\sigma_r^I} G_r^I \\ &\quad + \sum_{j=1}^{\mathcal{N}} \beta_{i,j,r}^{\sigma_{j,r}^Y} (p_{i,r}^A)^{-\sigma_{j,r}^Y} (p_{j,r}^D)^{\sigma_{j,r}^Y} y_{j,r} \end{aligned} \perp p_{i,r}^A \quad (5.8)$$

$$g_{i,r}^M \geq (\zeta_{i,r}^M)^{\sigma_{i,r}^{DM}} (p_{i,r}^M)^{-\sigma_{i,r}^{DM}} (p_{i,r}^A)^{\sigma_{i,r}^{DM}} q_{i,r}^A \perp p_{i,r}^M \quad (5.9)$$

$$q_s^T \geq \sum_{i=1}^{\mathcal{N}} \sum_{r=1}^{\mathcal{R}} \sum_{r' \neq r} \left[\xi_{i,r',r}^{\sigma_{i,r'}^{MM}} \left(p_{i,r'}^D + \sum_s \kappa_{s,i,r',r} p_s^T \right)^{-\sigma_{i,r'}^{MM}} (p_{i,r'}^M)^{\sigma_{i,r'}^{MM}} g_{i,r'}^M \right] \perp p_s^T \quad (5.10)$$

$$V_{f,r} \geq \sum_{j=1}^{\mathcal{N}} \gamma_{f,j,r}^{\sigma_{j,r}^Y} w_{f,r}^{-\sigma_{j,r}^Y} (p_{j,r}^D)^{\sigma_{j,r}^Y} y_{j,r} \perp w_{f,r} \quad (5.11)$$

$$G_r^I \text{ given} \perp p_r^I \quad (5.12)$$

$$u_r \geq \mathcal{J}_r / \mathcal{E}_r \perp \mathcal{E}_r \quad (5.13)$$

$$\mathcal{J}_r = \sum_{f=1}^{\mathcal{F}} w_{f,r} V_{f,r} \perp \mathcal{J}_r \quad (5.14)$$

which can be solved as a mixed complementarity problem (MCP)—for details, see Sue Wing (2009). Computable general equilibrium models solve for relative prices, with the marginal utility of income being a convenient numeraire. A common practice is to designate one region (say, r^*) as the numeraire economy by fixing the value of its unit expenditure index, $\mathcal{E}_{r^*} = 1$.

5.2.2 A Single-Region Open-Economy Armington Model

A noteworthy feature of this framework is that it encompasses the single-region open-economy Armington model as a special case. The latter is specified by omitting international transport, by dropping equations (5.6) and (5.10) and the corresponding variables $p_s^T = q_s^T = 0$, and collapsing bilateral exports and imports into aggregate values G^X and G^M , which are associated with the supply of and demand for an aggregate foreign exchange commodity (with price PFX). Producers in each industry allocate output between domestic and export markets according to a constant elasticity of transformation (CET) technology, while imported quantities of each commodity are a CET function of foreign exchange. The zero-profit conditions implied by these assumptions are modifications of equations (5.3) and (5.5), shown below as (5.15) and (5.16). Applying Shephard's lemma to derive the optimal unconditional supplies of domestic and imported varieties of each good yields the analogues of the market clearance conditions (5.7) and (5.9), shown below as equations (5.17) and (5.18). The model is closed through the specification of the current account, with commodity exports generating foreign exchange according to a CES technology (implying the zero-profit condition (5.19), and the price of foreign exchange exhibiting complementary slackness with the current account balance, CA_r . Equation (5.20) illustrates the simplest case in which the latter is treated as exogenous, held fixed at the level prevailing in the benchmark calibration data set.

$$\begin{aligned} & \left[\left(\delta_{j,r}^D \right)^{\sigma_{j,r}^Y} \left(p_{j,r}^D \right)^{1-\sigma_{j,r}^Y} + \left(\delta_{j,r}^X \right)^{\sigma_{j,r}^Y} \left(p_{j,r}^X \right)^{1-\sigma_{j,r}^Y} \right]^{1/(1-\sigma_{j,r}^Y)} \\ & \leq \left[\sum_{i=1}^{\mathcal{N}} \beta_{i,j,r}^{\sigma_{j,r}^Y} \left(p_{i,r}^A \right)^{1-\sigma_{j,r}^Y} + \sum_{f=1}^{\mathcal{F}} \gamma_{f,j,r}^{\sigma_{j,r}^Y} w_{f,r}^{1-\sigma_{j,r}^Y} \right]^{1/(1-\sigma_{j,r}^Y)} \quad \perp y_{i,r} \quad (5.15) \end{aligned}$$

$$\left[\sum_{i=1}^{\mathcal{N}} \left(\mu_{i,r}^M \right)^{\sigma_r^M} \left(p_{i,r}^M \right)^{1-\sigma_r^M} \right]^{1/(1-\sigma_r^M)} \leq PFX_r \quad \perp G_r^M \quad (5.16)$$

$$\left(\delta_{j,r}^D \right)^{\sigma_{j,r}^Y} \left(p_{j,r}^D \right)^{-\sigma_{j,r}^Y} \times \left[\left(\delta_{j,r}^D \right)^{\sigma_{j,r}^Y} \left(p_{j,r}^D \right)^{1-\sigma_{j,r}^Y} + \left(\delta_{j,r}^X \right)^{\sigma_{j,r}^Y} \left(p_{j,r}^X \right)^{1-\sigma_{j,r}^Y} \right]^{\sigma_{j,r}^Y/(1-\sigma_{j,r}^Y)} y_{j,r}$$

$$\begin{aligned} &\geq (\zeta_{i,r}^D)^{\sigma_{i,r}^{DM}} (p_{i,r}^D)^{-\sigma_{i,r}^{DM}} (p_{i,r}^A)^{\sigma_{i,r}^{DM}} q_{i,r}^A && \perp p_{i,r}^D \quad (5.17) \\ &(\mu_{i,r}^M)^{\sigma_r^M} (p_{i,r}^M)^{-\sigma_r^M} \left[\sum_{i=1}^{\mathcal{N}} (\mu_{i,r}^M)^{\sigma_r^M} (p_{i,r}^M)^{1-\sigma_r^M} \right]^{\sigma_r^M/(1-\sigma_r^M)} G_r^M \end{aligned}$$

$$\geq (\zeta_{i,r}^M)^{\sigma_{i,r}^{DM}} (p_{i,r}^M)^{-\sigma_{i,r}^{DM}} (p_{i,r}^A)^{\sigma_{i,r}^{DM}} q_{i,r}^A \quad \perp p_{i,r}^M \quad (5.18)$$

$$PFX_r \leq \left[\sum_{i=1}^{\mathcal{N}} (\mu_{i,r}^X)^{\sigma_r^X} (p_{i,r}^X)^{1-\sigma_r^X} \right]^{1/(1-\sigma_r^X)} \quad \perp G_r^X \quad (5.19)$$

$$G_r^X - G_r^M = CA_r \quad \perp PFX_r \quad (5.20)$$

The single-region small open-economy model is given by equations (5.1), (5.2), (5.15), (5.4), (5.16), (5.17), (5.8), (5.18), and (5.11) to (5.20), which comprise a square system of $8 + 5\mathcal{N} + \mathcal{F}$ nonlinear inequalities in as many unknowns, $\mathcal{B} = \{u, G^I, y_i, q_i^A, G_r^X, G_r^M, p_i^D, p_i^A, p_i^M, w_f, p^I, \mathcal{E}, \mathcal{J}, PFX\}$ for a given region r .

5.2.3 Introducing Dynamics

An important extension of these basic static frameworks is the introduction of a dynamic process that enables simulation of economies' time evolution. The simplest approach is to construct a "recursive dynamic" model in which factor accumulation is represented by semiautonomous increases in the primary factor endowments, and technological progress is represented by exogenous shifts in the technical coefficients of consumption and production. Letting $t = \{1, \dots, T\}$ index time, the supply of labor is typically modeled as following an exogenous trend of population increase (say, $\Psi_{r,t}^{\text{Pop}}$) combined with an increasing index of labor productivity ($\Psi_{L,r,t}^V \geq 1$):

$$V_{L,r,t} = \Psi_{L,r,t}^V \Psi_{r,t}^{\text{Pop}} \bar{V}_{L,r}. \quad (5.21)$$

Expansion of the supply of capital is semi-endogenous. Accumulation of regions' capital stocks ($KS_{r,t}$) is driven by investment and depreciation (at rate $\mathcal{D}$) according to the standard perpetual inventory formulation (5.22). Investment is determined myopically as a function of contemporaneous variables in each period's static MCP, with the simplest assumption being a constant household marginal propensity to save and invest out of aggregate income (MPS_r), in which case (5.12) is re-specified as equation (5.23). Finally, exogenous rates of return (RK_r) are used to calculate capital endowments

from the underlying stocks (5.22).

$$KS_{r,t+1} = G_{r,t}^I + (1 - \mathcal{D})KS_{r,t} \quad (5.22)$$

$$G_r^I = MPS_r \mathcal{S}_r \quad \perp \quad p_r^I \quad (5.23)$$

$$V_{K,r,t} = RK_r KS_{r,t} \quad (5.24)$$

These equations give rise to a multiregional and multisectoral Solow-Swan model, which, like its simpler theoretical counterpart, exhibits diminishing returns to accumulating factors that are offset by aggregate productivity growth. Endogenous technological progress can also be modeled by applying shift parameters that specify a decline in the values of the coefficients in table 5.3, on inputs to consumption— $\alpha_{i,r,t} = \Psi_{i,t}^C \bar{\alpha}_{i,r}$, and production— $\beta_{i,j,r,t} = \Psi_{i,j,r,t}^{YI} \bar{\beta}_{i,j,r}$ and $\gamma_{f,j,r,t} = \Psi_{f,j,r,t}^{YF} \bar{\gamma}_{f,j,r}$, with $\Psi_{i,t}^C, \Psi_{i,j,r,t}^{YI}, \Psi_{f,j,r,t}^{YF} \in (0, 1)$. Production in sector j experiences neutral technical progress when $\Psi_{i,j,r,t}^{YI} = \Psi_{f,j,r,t}^{YF} = \bar{\Psi}_{j,r,t}^Y$. A popular application of biased technical progress is energy-focused CGE models' way of capturing the historically observed non-price-induced secular decline in the energy-to-GDP ratio. This is represented via "autonomous energy-efficiency improvement" (AEEI), an exogenously specified decline in the coefficient on energy inputs ($e \subset i$) to production and consumption: $\Psi_{e,j,r,t}^{YI}, \Psi_{e,t}^C < 1$.

The ease of implementation of the recursive-dynamic approach has led to its overwhelming popularity in applied modeling work, in spite of the limitations of ad hoc savings-investment closure rules such as (5.23), which diverge sharply from the standard economic assumption of intertemporally optimizing firm and household behavior. The development and application of fully forward-looking CGE models has for this reason become an important area of research. Lau et al. (2002) derive a multisectoral Ramsey model in the complementarity format of equilibrium, using the consumption Euler equation and the intertemporal budget constraint of an intertemporal utility-maximizing representative agent. The key features of their framework are a trajectory of aggregate consumption demand determined by exogenous long-run average rates of interest and discount, the intertemporal elasticity of substitution, cumulative net income over the T periods of the simulation horizon, and an intertemporal zero-profit condition for capital stock accumulation dual to (5.22), which incorporates RK_r as a fully endogenous capital input price index. The resulting general equilibrium problem is specified and simultaneously solved for all t , which for large- T simulations can dramatically increase the dimension of the pseudo-excess demand correspondence and the associated complexity and computational cost. It is, therefore, unsurprising that single-region forward-looking CGE models¹ tend to be far more common than their multiregional counterparts.²

¹ Jorgenson and Wilcoxon (1993); Bye (2000); Bovenberg and Goulder (2001); Balistreri (2002); Diao et al. (2005); Bovenberg et al. (2005); Dellink (2005); Otto et al. (2007); Otto and Reilly (2008); Otto et al. (2008).

² Bernstein et al. (1999); Diao et al. (2003); Babiker et al. (2009); Ross et al. (2009); Tuladhar et al. (2009).

5.3 THE CANONICAL MODEL AT WORK

5.3.1 Traditional Applications: International, Development, and Public Economics

Computable general equilibrium models have long been the analytical mainstay of assessments of trade liberalization and economic integration (Harrison et al. 1997a,b; Hertel 1997). Such analysis has been facilitated by the compilation of integrated trade and input-output data sets such as the Global Trade Analysis Project (GTAP) database (Narayanan and Walmsley 2008), which include a range of data concerning protection and distortions. Incorporating these data into the canonical model allows the analyst to construct an initial tariff-ridden status quo equilibrium that can be used as a benchmark on the basis of which to simulate the impacts of a wide variety of policy reforms.

Multilateral trade negotiations are perhaps the simplest to illustrate (e.g., Hertel and Winters 2006). They typically involve reductions in and interregional harmonization of two types of distortions, which may conveniently be introduced into the canonical model as ad valorem taxes or subsidies. The former are export levies or subsidies that drive a wedge between the domestic and free on board (FOB) prices of each good and are represented using the parameter $\tau_{i,r}^X \geq 0$. The latter are import tariffs that drive a wedge between cost insurance freight (CIF) prices and landed costs, represented parametrically by $\tau_{i,r}^M > 0$. The benefits of this approach are simplicity, as well as the ability to capture the border effects of various kinds of nontariff barriers to trade where empirical estimates of these measures' "ad valorem equivalents" are available (see, e.g., Fugazza and Maur 2008). Modeling the "shocks" constituted by changes to such policy parameters follows a standard procedure that we apply throughout this chapter: first, modify the zero-profit conditions to represent the shock as a price wedge; second, specify modifications implied by Hotelling's lemma to the supply and demand functions in the market clearance conditions; and third, reconcile the income balance condition with the net revenues or captured rents. Other extensions to the model structure might be warranted, depending on the interactions of interest. Adjusting the equations for import zero profit (5.5), domestic market clearance (5.7), and income balance (5.14) in table 5.4, we obtain, respectively,

$$p_{i,r}^M \leq \left[\sum_{r' \neq r} \xi_{i,r',r}^{\sigma_{i,r}^{MM}} \left((1 + \tau_{i,r}^M)(1 + \tau_{i,r'}^X) p_{i,r'}^D + \sum_{r=1}^{\mathcal{R}} \kappa_{s,i,r',r} p_s^T \right)^{1 - \sigma_{i,r}^{MM}} \right]^{1/(1 - \sigma_{i,r}^{MM})} \perp g_{i,r}^M \quad (5.25)$$

$$\begin{aligned}
 y_{i,r} \geq & (\xi_{i,r}^D)^{\sigma_{i,r}^{DM}} (p_{i,r}^D)^{-\sigma_{i,r}^{DM}} (p_{i,r}^A)^{\sigma_{i,r}^{DM}} q_{i,r}^A + \sum_{r' \neq r} (1 + \tau_{i,r}^M)(1 + \tau_{i,r'}^X) \xi_{i,r,r'}^{\sigma_{i,r'}^{MM}} \\
 & \times \left((1 + \tau_{i,r'}^M)(1 + \tau_{i,r}^X) p_{i,r}^D + \sum_{s=1}^{\mathcal{R}} \kappa_{s,i,r,r'} p_s^T \right)^{-\sigma_{i,r'}^{MM}} (p_{i,r'}^M)^{\sigma_{i,r'}^{MM}} g_{i,r'}^M \quad \perp \quad p_{i,r}^D
 \end{aligned} \tag{5.26}$$

$$\begin{aligned}
 \mathcal{J}_r = & \sum_{f=1}^{\mathcal{F}} w_{f,r} V_{f,r} + \sum_{i=1}^{\mathcal{N}} \tau_{i,r}^X p_{i,r}^D \sum_{r' \neq r} (1 + \tau_{i,r'}^M)(1 + \tau_{i,r}^X) \xi_{i,r,r'}^{\sigma_{i,r'}^{MM}} \\
 & \times \left((1 + \tau_{i,r'}^M)(1 + \tau_{i,r}^X) p_{i,r}^D + \sum_{s=1}^{\mathcal{R}} \kappa_{s,i,r,r'} p_s^T \right)^{-\sigma_{i,r'}^{MM}} (p_{i,r'}^M)^{\sigma_{i,r'}^{MM}} g_{i,r'}^M \\
 & + \sum_{i=1}^{\mathcal{N}} \tau_{i,r}^M \sum_{r' \neq r} p_{i,r'}^D (1 + \tau_{i,r}^M)(1 + \tau_{i,r'}^X) \xi_{i,r',r}^{\sigma_{i,r}^{MM}} \\
 & \times \left((1 + \tau_{i,r}^M)(1 + \tau_{i,r'}^X) p_{i,r'}^D + \sum_{s=1}^{\mathcal{R}} \kappa_{s,i,r',r} p_s^T \right)^{-\sigma_{i,r}^{MM}} (p_{i,r}^M)^{\sigma_{i,r}^{MM}} g_{i,r}^M \quad \perp \quad \mathcal{J}_r
 \end{aligned} \tag{5.27}$$

The fact that τ^X and τ^M are preexisting distortions means that it is necessary to recalibrate the model's technical coefficients to obtain a benchmark equilibrium. Trade policies are simulated by changing elements of these vectors from their benchmark values and computing new counterfactual equilibria that embody income and substitution effects in both the domestic economy, r , and its trade partners, r' . The resulting effects on welfare manifest themselves through the new tax revenue terms in the income balance equation. Hertel et al. (2007) demonstrate that the magnitude of these impacts strongly depends on the values of the elasticities governing substitution across regional varieties of each good, $\sigma_{i,r}^{MM}$.

The breadth and richness of analyses that can be undertaken simply by manipulating distortion parameters such as tax rates—or the endowments and productivity parameters that define boundary conditions of the economy—is truly remarkable and should not be underestimated.

International economics continues to be a mainstay of the CGE literature, with numerous articles dedicated to assessing the consequences of various trade agreements and liberalization initiatives,³ as well as a variety of multilateral price support schemes (Psaltopoulos et al. 2011), distortionary trade policies (Naudé and Rossouw 2008;

³ Jean et al. (2014); Aydin and Acar (2010); Missaglia and Valensisi (2014); Engelbert et al. (2014); Braymen (2011); Kawai and Zhai (2009); Lee et al. (2009); Chao et al. (2006); Lee et al. (2004); Georges et al. (2013); Bouët et al. (2012); Gouel et al. (2011); Winchester (2009); Ghosh and Rao (2010); Brockmeier and Pelikan (2008); Ariyasajakorn et al. (2009); Ghosh and Rao (2005); Francois et al.

Narayanan and Khorana 2014), nontariff barriers to trade (Fugazza and Maur 2008; Winchester 2009), and internal and external shocks (implemented in the model of section 5.2.2 by dropping equation (5.18) and fixing the complementary variable $p_{i,r}^M$).⁴ More analytically oriented papers have investigated the manner in which the macroeconomic effects of shocks are modulated by imperfect competition (Konan and Assche 2007), agents' expectations (Boussard et al. 2006; Femenia and Gohin 2011), and international mechanisms of price transmission (Siddig and Grethe 2014). Still other studies advance the state of modeling, extending the canonical model beyond trade into the realm of international macroeconomics by introducing foreign direct investment and its potential to generate domestic productivity spillovers (Lejour et al. 2008; Latorre et al. 2009; Deng et al. 2012), and financial assets and interregional financial flows (Maldonado et al. 2007; Lemelin et al. 2013; Yang et al. 2013). Following Markusen (2002) and Markusen et al. (2005), the typical approach taken by the latter crop of papers is to disaggregate capital input as a factor of production into domestic and foreign varieties, the latter of which is internationally mobile and imperfectly substitutable for domestic capital.

A related development literature examines a broader range of outcomes in poor countries, for example, the social, environmental, and poverty impacts of trade policy and liberalization⁵ and the economic and social consequences of energy price shocks, energy market liberalization, and alternative energy promotion.⁶ Similar studies investigate the macro-level consequences for developing countries of productivity improvements generated by foreign aid (Clausen and Schürenberg-Frosch 2012), changes in the delivery of public services such as education and health (Debowicz and Golan 2014; Roos and Giesecke 2014) or domestic R&D and industrial policies to simulate economic growth (Breisinger et al. 2009; Bor et al. 2010; Ojha et al. 2013), and the growth consequences of worker protection and restrictions on international movements of labor (Ahmed and Peerlings 2009; Moses and Letnes 2004).

Yet another perspective on these issues is taken by the public economics literature, which investigates the economy-wide effects of energy and environmental tax changes (Karami et al. 2012; Markandya et al. 2013; Zhang et al. 2013), aging-related and pension policies, through either a coupled CGE-microsimulation modeling framework (van Sonsbeek 2010) or dynamic CGE models embodying overlapping generations of households,⁷ actual and proposed tax reforms in developed and developing

(2005); Lee and van der Mensbrugghe (2004); Flaig et al. (2013); Perali et al. (2012); Kitwiwattanachai et al. (2010); Bajo-Rubio and Gómez-Plana (2005).

⁴ Álvarez Martínez and Polo (2012); von Arnim (2009).

⁵ Kleinwechter and Grethe (2012); Naranpanawa et al. (2011); Pauw and Thurlow (2011); Gumilang et al. (2011); Mabugu and Chitiga (2009); Acharya and Cohen (2008); Abbott et al. (2009); O'Ryan et al. (2005); Mirza et al. (2014); Hertel and Zhai (2006); Chan et al. (2005); Naudé and Coetzee (2004).

⁶ Solaymani and Kari (2013); Naranpanawa and Bandara (2012); Al Shehabi (2012); Dartanto (2013); Arndt et al. (2011); Scaramucci et al. (2006).

⁷ Aglietta et al. (2007); Creedy and Guest (2008); Fougère et al. (2007); Rausch and Rutherford (2010); Lisenkova et al. (2013).

countries,⁸ the welfare implications of decentralized public services provision (Iregui 2005), and rising wage inequality within the OECD (Winchester and Greenaway 2007).

Common to virtually all these studies is the economy-wide impact of a change in one or more distortions. This impact is customarily measured by the marginal cost of public funds (MCF): the effect on money-metric social welfare of raising an additional dollar of government revenue by changing a particular tax instrument (Dahlby 2008). The strength of CGE models is their ability to capture the influence that preexisting market distortions may have on the MCF in real-world “second-best” policy environments. Distortions interact, potentially offsetting or amplifying one another, with the result that imposing an additional distortion in an already tariff-ridden economy may not necessarily worsen welfare, whereas removing an existing distortion is not guaranteed to improve welfare (see, e.g., Ballard and Fullerton 1992; Fullerton and Rogers 1993; Slemrod and Yitzhaki 2001). Computable general equilibrium models can easily report the MCF for any given or proposed instrument as the ratio of the money-metric welfare cost to increased tax revenue. Ranking policy instruments on the basis of their MCF gives a good indication of efficiency-enhancing reforms. An instructive example is Auriol and Warlters’ (2012) analysis of the MCF in thirty-eight African countries quantifying the welfare effects of taxes on domestic production, labor, and capital, in addition to imports and exports. Factor taxes deserve special attention because a tax on a factor that is in perfectly inelastic supply does not distort allocation, implying that the effects of distortionary factor taxes can only be represented by introducing price-responsive factor supplies, modifying the market clearance conditions (5.11). The most common way to address this matter is to endogenize the supply of labor by introducing labor-leisure choice or unemployment (for elaborations see, e.g., Sue Wing, 2011 and Balistreri, 2002).

5.3.2 Emerging Applications: Energy Policy and Climate Change Mitigation

Energy policies, as well as measures to mitigate the problem of climate change through reductions in economies’ emissions of greenhouse gases (GHGs), are two areas that are at the forefront of CGE model development and application. Sticking with the types of parametrically driven shocks discussed in section 5.3.1, the energy economics and policy literature has investigated economic consequences of changing taxes and subsidies on conventional energy,⁹ the social and environmental dimensions of energy

⁸ Radulescu and Stimmelmayer (2010); Toh and Lin (2005); Field and Wongwatanasin (2007); Giesecke and Nhi (2010); Boeters (2013); Mabugu et al. (2013).

⁹ Al Shehabi (2012); Bjertnaes (2011); He et al. (2014); Jiang and Lin (2014); Sancho (2010); Liu and Li (2011); Akkemik and Oguz (2011); Lin and Jiang (2011); Vandyck and Regemorter (2014); Solaymani and Kari (2014); He et al. (2011).

use and policy,¹⁰ macroeconomic consequences of energy price shocks (He et al. 2010; Aydin and Acar 2011; Guivarch et al. 2009), and the way energy use, efficiency, and conservation influence, and are affected by, the rate and direction of innovation and economic growth.¹¹ Further technical and methodological studies evaluate the representation of energy technology and substitution possibilities in CGE models (Schumacher and Sands 2007; Beckman et al. 2011; Lecca et al. 2011), as well as the consequences of, and mitigating effect of policy interventions on, depletion of domestic fossil fuel reserves in resource-dependent economies (Djiofack and Omgba 2011; Barkhordar and Saboohi 2013; Bretschger and Smulders 2011).

An important development in energy markets is the widespread expansion of policy initiatives promoting alternative and renewable energy supplies. This topic has been an area of particular growth in CGE assessments.¹² In most areas of the world such energy supplies are more costly than conventional energy production activities. Consequently, they typically make up little to none of the extant energy supply and are unlikely to be represented in current input-output accounts on which CGE model calibrations are based. To assess the macroeconomic consequences of new energy technologies it is therefore necessary to introduce into the canonical model new, non-benchmark production activities whose technical coefficients are derived from engineering cost studies and other ancillary data sources. They have higher operating costs relative to conventional activities in the SAM render their operation inactive in the benchmark equilibrium, but they are capable of endogenously switching on and producing output in response to relative price changes or policy stimuli.

These so-called backstop technology options—indexed by b —are implemented by specifying additional production functions whose outputs are perfect substitutes for an existing source of energy supply (e.g., electricity, e'). Their associated cost functions embody a premium over benchmark prices, modeled by the markup factor $MKUP_{e',r} > 1$, which can be offset by an output subsidy $\tau_{e',r}^b < 0$:

$$p_{e',r}^D \leq \left(1 + \tau_{e',r}^b\right) MKUP_{e',r}^b \cdot \left[\sum_{i=1}^N \left(\beta_{i,e',r}^b\right)^{\sigma_{e',r}^Y} \left(p_{i,r}^A\right)^{1-\sigma_{e',r}^Y} + \sum_{f=1}^F \left(\gamma_{f,e',r}^b\right)^{\sigma_{e',r}^Y} w_{f,r}^{1-\sigma_{e',r}^Y} + \left(\gamma_{FF,e',r}^b\right)^{\sigma_{e',r}^Y} \left(w_{FF,e',r}^b\right)^{1-\sigma_{e',r}^Y} \right]^{1/(1-\sigma_{e',r}^Y)} \perp y_{e',r}^b \quad (5.28)$$

¹⁰ Santos et al. (2013); Allan et al. (2008); Hanley et al. (2009); Bjertnaes and Faehn (2008); Shi et al. (2009); O'Neill et al. (2012).

¹¹ Allan et al. (2007); Anson and Turner (2009); Turner and Hanley (2011); Martinsen (2011); Lu et al. (2010); Otto et al. (2007); Parrado and De Cian (2014); Dimitropoulos (2007); Lecca et al. (2014); Turner (2009).

¹² Timilsina et al. (2011, 2013); Bae and Cho (2010); Proenca and Aubyn (2013); Cansino et al. (2013, 2014); Hoefnagels et al. (2013); Böhringer et al. (2013); Arndt et al. (2012); Gunatilake et al. (2014); Wianwiwat and Asafu-Adjaye (2013); Doumax et al. (2014); Ge et al. (2014); Trink et al. (2010); Kretschmer and Peterson (2010).

Note that once $\tau_{e',r}^b \leq 1/MKUP_{e',r}^b - 1$ the backstop becomes cost-competitive with conventional supply of e' and switches on, but an unpleasant side effect of perfect substitutability is “bang-bang” behavior, in which a small increase in the subsidy parameter induces a jump in the backstop’s output, which in the limit can result in the backstop capturing the entire market ($y_{e',r}^b \gg y_{e',r}$). To replace such unrealistic behavior with a smooth path of entry along which both backstop and conventional supplies coexist, a popular trick is to introduce into the backstop production function a small quantity of a technology-specific fixed factor (with price $w_{FF,e',r}^b$ and technical coefficient $\gamma_{FF,e',r}^b$) whose limited endowment constrains the output of the backstop, even at favorable relative prices. The impact is apparent from the fixed-factor market clearance condition

$$V_{FF,e',r} \geq \left(\gamma_{FF,e',r}^b\right)^{\sigma_{e',r}^Y} \left(w_{FF,e',r}^b\right)^{-\sigma_{e',r}^Y} \left(p_{e',r}^D\right)^{\sigma_{e',r}^Y} y_{e',r}^b \quad \perp w_{FF,e',r}^b \quad (5.29)$$

where, with the fixed factor endowment (V_{FF}) held constant, the quantity of backstop output increases with the fixed factor’s relative price, $(w_{FF}^b/p^D)^{\sigma^Y}$. Thus, once the elasticity of substitution between the fixed-factor and other inputs is sufficiently small, even a large increase in the backstop price results in only modest backstop activity. In dynamic models the exogenously specified trajectory of V_{FF} is an important device for tuning new technologies’ penetration in accordance with the modeler’s sense of plausibility, especially when the future character and magnitude of “market barriers,” unanticipated complementary investments or network externalities represented by the fixed-factor constraint, are unknown.

A related topic that has seen the emergence of a voluminous literature is climate change mitigation through policies to limit anthropogenic emissions of greenhouse gases. Carbon dioxide (CO_2), the chief GHG, is emitted to the atmosphere primarily from the combustion of fossil fuels. Policies to curtail fossil fuel use tend to limit the supply of energy, whose signature characteristics are being an input to virtually every economic activity and possessing few low-cost substitutes (especially in the short run), resulting in GHG mitigation policies having substantial general equilibrium impacts (Hogan and Manne 1977).

The simplest policy instrument to consider is an economy-wide GHG tax. For the sake of expositional clarity we partition the set of commodities and industries into the subset of energy goods or sectors associated with CO_2 emissions (indexed by e , as before) and the complementary subset of non-energy material goods or sectors, indexed by m . The stoichiometric linkage between CO_2 and the carbon content of the fuel being combusted implies a Leontief relation between emissions and the quantity of use of each fossil fuel. This linkage is represented using fixed emission factors ($\epsilon_{e,r}^{\text{CO}_2}$) that transform a uniform tax on emissions (τ_r^{GHG}) into a vector of differentiated markups on the unit cost of the e Armington energy commodities, as shown in equation (5.30).

This simple scheme cannot be extended to non- CO_2 GHGs, the majority of which emanate from a broad array of industrial processes and household activities but are

not linked in any fixed way to inputs of particular energy or material commodities. Non-CO₂ GHGs targeted by the Kyoto Protocol are methane, nitrous oxide, hydro- and perfluorocarbons, and sulfur hexafluoride, which we index by $o = \{1 \dots \mathcal{O}\}$, and whose global warming impact in units of CO₂-equivalents is given by ϵ_o^{GHG} . In an important advance, Hyman et al. (2003) develop a methodology for modeling non-CO₂ GHG abatement by treating these emissions as (a) *inputs* to the activities of firms and households, which (b) substitute for a composite of all other commodity and factor inputs with CES technology. The upshot is that the impact of a tax on non-CO₂ GHGs is mediated through a CES demand function whose elasticity to the costs of pollution control can be tuned to reproduce marginal abatement cost curves derived from engineering or partial-equilibrium economic studies. The key tuning parameters are the technical coefficients on emissions ($\vartheta_{o,j,r}$) and the elasticity of substitution between emissions and other inputs ($\sigma_{o,j,r}^{\text{GHG}}$). The latter indicates the relative attractiveness of industrial process changes as a margin of adjustment to GHG price or quantity restrictions. Equation (5.31) highlights the implications for production costs at the margin, which increase by the product of the unit demand for emissions of each category of pollutant, $(\vartheta_{o,j,r})^{-\sigma_{o,j,r}^{\text{GHG}}} (\tau_r^{\text{GHG}} \epsilon_o^{\text{GHG}})^{-\sigma_{o,j,r}^{\text{GHG}}} (p_{j,r}^D)^{\sigma_{o,j,r}^{\text{GHG}}}$, and its effective price ($\tau_r^{\text{GHG}} \epsilon_o^{\text{GHG}}$).

$$p_{i,r}^A \leq \left[(\zeta_{i,r}^D)^{\sigma_{i,r}^{\text{DM}}} (p_{i,r}^D)^{1-\sigma_{i,r}^{\text{DM}}} + (\zeta_{i,j,r}^M)^{\sigma_{i,r}^{\text{DM}}} (p_{i,r}^M)^{1-\sigma_{i,r}^{\text{DM}}} \right]^{1/(1-\sigma_{i,r}^{\text{DM}})} \\ + \begin{cases} \tau_r^{\text{GHG}} \epsilon_{e,r}^{\text{CO}_2} & e \subset i \\ 0 & \text{otherwise} \end{cases} \quad \perp \quad q_{i,r}^A \quad (5.30)$$

$$p_{j,r}^D \leq \left[\sum_{i=1}^{\mathcal{N}} \beta_{i,j,r}^{\sigma_{j,r}^Y} (p_{i,r}^A)^{1-\sigma_{j,r}^Y} + \sum_{f=1}^{\mathcal{F}} \gamma_{f,j,r}^{\sigma_{j,r}^Y} w_{f,r}^{1-\sigma_{j,r}^Y} \right]^{1/(1-\sigma_{j,r}^Y)} \\ + \sum_{o=1}^{\mathcal{O}} (\vartheta_{o,j,r})^{-\sigma_{o,j,r}^{\text{GHG}}} (\tau_r^{\text{GHG}} \epsilon_o^{\text{GHG}})^{1-\sigma_{o,j,r}^{\text{GHG}}} (p_{j,r}^D)^{\sigma_{o,j,r}^{\text{GHG}}} \quad \perp \quad p_{j,r}^D \quad (5.31)$$

$$\mathcal{J}_r = \sum_{f=1}^{\mathcal{F}} w_{f,r} V_{f,r} + \sum_e \tau_r^{\text{GHG}} \epsilon_{e,r}^{\text{CO}_2} q_{e,r}^A \\ + \sum_{j=1}^{\mathcal{N}} \sum_{o=1}^{\mathcal{O}} \left[(\vartheta_{o,j,r})^{-\sigma_{o,j,r}^{\text{GHG}}} (\tau_r^{\text{GHG}} \epsilon_o^{\text{GHG}})^{1-\sigma_{o,j,r}^{\text{GHG}}} (p_{j,r}^D)^{\sigma_{o,j,r}^{\text{GHG}}} y_{j,r} \right] \quad \perp \quad \mathcal{J}_r \quad (5.32)$$

A model of economy-wide GHG taxation is made up of equations (5.1), (5.2), and (5.5) to (5.13), with (5.31) and (5.30) substituting for (5.3) and (5.4), τ_r^{GHG} specified

as an exogenous parameter, and explicit accounting for recycling of the resulting tax revenue in the income balance condition (5.32), which replaces (5.14). In a domestic cap-and-trade system the tax is interpreted as the price of emission allowances and is endogenous, exhibiting complementary slackness with respect to the additional multigas emission limit (5.33). In the simplest case, rents generated under such a policy redound to households as payments to emission rights ($\mathcal{A}_r$), with which they are assumed to be endowed. The income balance condition (5.34) must then be substituted for (5.32).

$$\sum_e \epsilon_{e,r}^{\text{CO}_2} q_{e,r}^A + \sum_{j=1}^{\mathcal{N}} \sum_{o=1}^{\mathcal{O}} \left[(\vartheta_{o,j,r})^{-\sigma_{o,j,r}^{\text{GHG}}} (\tau_r^{\text{GHG}} \epsilon_o^{\text{GHG}})^{-\sigma_{o,j,r}^{\text{GHG}}} (p_{j,r}^D)^{\sigma_{o,j,r}^{\text{GHG}}} y_{j,r} \right] \leq \mathcal{A}_r \perp \tau_r^{\text{GHG}} \quad (5.33)$$

$$\mathcal{J}_r = \sum_{f=1}^{\mathcal{F}} w_{f,r} V_{f,r} + \tau_r^{\text{GHG}} \mathcal{A}_r \quad \perp \quad \mathcal{J}_r \quad (5.34)$$

A multilateral emission trading scheme over the subset of abating regions, $\mathcal{R}^\dagger$, is easily implemented by dropping the region subscript on the allowance price ($\tau_r^{\text{GHG}} = \tau^{\text{GHG}} \forall r \in \mathcal{R}^\dagger$) and taking the sum of equation (5.33) across regions to specify the aggregate emission limit (5.35). The latter, which is the sum of individual regional emission caps ($\overline{\mathcal{A}} = \sum_{r \in \mathcal{R}^\dagger} \mathcal{A}_r$), induces allocation of emissions across regions, sectors, and gases to equalize the marginal costs of abatement. The income or welfare consequences for an individual region may be positive or negative depending on whether its residual emissions are below or above its cap, inducing net purchases or sales of allowances.

$$\sum_{r \in \mathcal{R}^\dagger} \left\{ \sum_e \epsilon_{e,r}^{\text{CO}_2} q_{e,r}^A + \sum_{j=1}^{\mathcal{N}} \sum_{o=1}^{\mathcal{O}} \left[(\vartheta_{o,j,r})^{-\sigma_{o,j,r}^{\text{GHG}}} (\tau^{\text{GHG}} \epsilon_o^{\text{GHG}})^{-\sigma_{o,j,r}^{\text{GHG}}} (p_{j,r}^D)^{\sigma_{o,j,r}^{\text{GHG}}} y_{j,r} \right] - \overline{\mathcal{A}} \right\} \leq 0 \perp \tau^{\text{GHG}} \quad (5.35)$$

$$\mathcal{J}_r = \sum_{f=1}^{\mathcal{F}} w_{f,r} V_{f,r} + \tau^{\text{GHG}} \mathcal{A}_r - \sum_e \tau^{\text{GHG}} \epsilon_{e,r}^{\text{CO}_2} q_{e,r}^A - \sum_{j=1}^{\mathcal{N}} \sum_{o=1}^{\mathcal{O}} \left[(\vartheta_{o,j,r})^{-\sigma_{o,j,r}^{\text{GHG}}} (\tau^{\text{GHG}} \epsilon_o^{\text{GHG}})^{1-\sigma_{o,j,r}^{\text{GHG}}} (p_{j,r}^D)^{\sigma_{o,j,r}^{\text{GHG}}} y_{j,r} \right] \quad \perp \quad \mathcal{J}_r \quad (5.36)$$

Finally, slow progress in implementing binding regimes for climate mitigation—either an international system of emission targets or comprehensive economy-wide emission caps at the national level—has refocused attention on assessing the consequences of piecemeal policy initiatives, particularly GHG abatement and allowance trading within subnational regions and/or among narrow groups of sectors within nations. The major consequence is an inability to reallocate abatement across sources as a way of arbitraging differences in the marginal costs of emission reductions, which may be captured by differentiating emission limits and their complementary shadow prices among covered sectors (say, j') and regions (say, r'): $\mathcal{A}_{j',r'}$ and $\tau_{j',r'}^{\text{GHG}}$. The key concern prompted by such rigidities is emission “leakage,” which occurs when emission limits imposed on a subset of sources that interact in markets for output and polluting inputs actually stimulate unconstrained sources to emit more pollution. The extent of the consequent shift in emissions is captured by the leakage rate, defined as the negative of the ratio of the increase in unconstrained sources’ emissions to constrained sources’ abatement. Quantifying this rate and characterizing its precursors requires input-based accounting for emissions, because taxes or quotas apply not to the supply of energy commodities across the economy but to their use by qualifying entities.

$$\begin{aligned} & \sum_e \sum_{j'} \beta_{e,j',r'}^{\sigma_{j',r'}^Y} \left(p_{e,r'}^A + \tau_{j',r'}^{\text{GHG}} \epsilon_{e,r'}^{\text{CO}_2} \right)^{-\sigma_{j',r'}^Y} \left(p_{j',r'}^D \right)^{\sigma_{j',r'}^Y} y_{j',r'}^Y \\ & + \sum_{j'} \sum_{o=1}^{\mathcal{O}} \left[\left(\vartheta_{o,j',r'} \right)^{-\sigma_{o,j',r'}^{\text{GHG}}} \left(\tau_r^{\text{GHG}} \epsilon_o^{\text{GHG}} \right)^{-\sigma_{o,j',r'}^{\text{GHG}}} \left(p_{j',r'}^D \right)^{\sigma_{o,j',r'}^{\text{GHG}}} y_{j',r'}^{\text{GHG}} \right] \leq \mathcal{A}_{j',r'} \perp \tau_{j',r'}^{\text{GHG}} \end{aligned} \quad (5.37)$$

$$\begin{aligned} p_{j',r'}^D & \leq \left[\sum_e \beta_{e,j',r'}^{\sigma_{j',r'}^Y} \left(p_{e,r'}^A + \tau_{j',r'}^{\text{GHG}} \epsilon_{e,r'}^{\text{CO}_2} \right)^{1-\sigma_{j',r'}^Y} \right. \\ & \quad \left. + \sum_m \beta_{m,j',r'}^{\sigma_{j',r'}^Y} \left(p_{m,r'}^A \right)^{1-\sigma_{j',r'}^Y} + \sum_{f=1}^{\mathcal{F}} \gamma_{f,j,r}^{\sigma_{j',r'}^Y} w_{f,r}^{1-\sigma_{j',r'}^Y} \right]^{1/(1-\sigma_{j',r'}^Y)} \\ & \quad + \sum_{o=1}^{\mathcal{O}} \left(\vartheta_{o,j',r'} \right)^{-\sigma_{o,j',r'}^{\text{GHG}}} \left(\tau_{j',r'}^{\text{GHG}} \epsilon_o^{\text{GHG}} \right)^{1-\sigma_{o,j',r'}^{\text{GHG}}} \left(p_{j',r'}^D \right)^{\sigma_{o,j',r'}^{\text{GHG}}} \perp p_{j',r'}^D \end{aligned} \quad (5.38)$$

$$\begin{aligned} \mathcal{J}_r & = \sum_{f=1}^{\mathcal{F}} w_{f,r} V_{f,r} + \sum_e \sum_{j'} \tau_{j',r'}^{\text{GHG}} \mathcal{A}_{j',r'} \\ & \quad + \sum_{j'} \sum_{o=1}^{\mathcal{O}} \left[\left(\vartheta_{o,j',r'} \right)^{-\sigma_{o,j',r'}^{\text{GHG}}} \left(\tau_{j',r'}^{\text{GHG}} \epsilon_o^{\text{GHG}} \right)^{1-\sigma_{o,j',r'}^{\text{GHG}}} \left(p_{j',r'}^D \right)^{\sigma_{o,j',r'}^{\text{GHG}}} y_{j',r'}^{\text{GHG}} \right] \perp \mathcal{J}_r \end{aligned} \quad (5.39)$$

The foregoing models have principally been used to analyze the macroeconomic consequences of emission reduction policies at multiple scales—traditionally international,¹³ but increasingly regional and national,¹⁴ and even subnational (Zhang et al. 2013; Springmann et al. 2014) or sectoral (Rausch and Mowers 2014). Such models' key advantage is their ability to quantify complex interactions between climate policies and a panoply of other policy instruments and characteristics of the economy. Although the universe of these elements is too broad to consider in detail, key issues include the distributional effects of climate policies on consumers, firms and regions,¹⁵ mitigation in second-best settings, fiscal policy interactions, the double dividend,¹⁶ alternative compliance strategies such as emission offsets and the Clean Development Mechanism,¹⁷ interactions between mitigation and trade, emissions leakage, and the efficacy of countervailing border tariffs on GHGs embodied in traded goods,¹⁸ the effects of structural change, innovation, technological progress, and economic growth on GHG emissions and the costs of mitigation in various market settings,¹⁹ energy market interactions,²⁰ and the role of discrete technology options on both the supply side (e.g., renewables and carbon capture and storage) and the demand side (e.g., conventional and alternative-fuel transportation).²¹

5.3.3 New Horizons: Assessing the Impacts of Climate Change and Natural Hazards

Turning now to the flip side of mitigation, the breadth and variety of pathways by which the climate influences economic activity are enormous (Dell et al. 2014), and

¹³ Kallbekken and Westskog (2005); Klepper and Peterson (2005); Nijkamp et al. (2005); Böhringer and Welsch (2004); Böhringer and Helm (2008); Kallbekken and Rive (2007); Calvin et al. (2009); Magne et al. (2014).

¹⁴ Klepper and Peterson (2006); Böhringer et al. (2007); Kasahara et al. (2007); Telli et al. (2008); Thepkhun et al. (2013); Hermeling et al. (2013); Orlov and Grethe (2012); Hübner et al. (2014); Lu et al. (2010); Lim (2011); Liang et al. (2007); Loisel (2009); Wang et al. (2009); Dai et al. (2011); Hübner (2011); Meng et al. (2013).

¹⁵ Bovenberg et al. (2005, 2008); Rose and Oladosu (2002); van Heerden et al. (2006); Oladosu and Rose (2007); Ojha (2009).

¹⁶ Bor and Huang (2010); Fraser and Waschik (2013); Allan et al. (2014); Boeters (2014); Dissou and Siddiqui (2014).

¹⁷ McCarl and Sands (2007); Glomsrod et al. (2011); Michetti and Rosa (2012); Böhringer et al. (2015).

¹⁸ Babiker (2005); Babiker and Rutherford (2005); Ghosh et al. (2012); Bao et al. (2013); Branger and Quirion (2014); Hübner (2012); Alexeeva-Talebi et al. (2012); Böhringer et al. (2012); Weitzel et al. (2012); Lanzi et al. (2012); Boeters and Bollen (2012); Caron (2012); Turner et al. (2012); Kuik and Hofkes (2010); Bruvoll and Faehn (2006); Jakob et al. (2013); Egger and Nigai (2015).

¹⁹ Viguiet et al. (2006); Otto et al. (2007); Fisher-Vanden and Ho (2007); Fisher-Vanden and Sue Wing (2008); Peretto (2008); Mahmood and Marpaung (2014); Sue Wing and Eckaus (2007); Jin (2012); Qi et al. (2014); Bretschger et al. (2011); Heggedal and Jacobsen (2011).

²⁰ Hagem et al. (2006); Maisonnave et al. (2012); Daenzer et al. (2014).

²¹ Schafer and Jacoby (2005); McFarland et al. (2004); Jacoby et al. (2006); Berg (2007); Qi et al. (2014); Okagawa et al. (2012); van den Broek et al. (2011); Karplus et al. (2013b,a); Kretschmer et al. (2009); Glomsrod and Taoyuan (2005); Timilsina et al. (2011); Timilsina and Mevel (2013); Sands et al. (2014).

improved understanding of these channels has spurred the growth of a large literature on the impacts of climate change. Computable general equilibrium models have the unique ability to represent in a comprehensive fashion the regional and sectoral scope of climatic consequences—if not their detail—and can easily accommodate region- and sector-specific damage functions from the impacts of climate change. This advantage comes at the cost of inability to capture intertemporal feedbacks, however. Despite recent progress in intertemporal CGE modeling, computational constraints often limit the resolution of these machines to a handful of regions and sectors and a short time horizon. Thus, as summarized in table 5.4, a common feature of the CGE models in this area of application is that they are either static simulations of a future time period (e.g., Roson 2003; Bosello and Zhang 2006; Bosello et al. 2007) or recursive dynamic simulations driven by contemporaneously determined investment (e.g., Deke et al. 2001; Eboli et al. 2010; Ciscar et al. 2011), with 2050 being the typical simulation horizon. Consequently, they tend to simulate the welfare effects of passive market adjustments to climate shocks, or, at best, “reactive” contemporaneous averting expenditures in sectors and regions, but not proactive investments in adaptation.

Table 5.4 indicates that, apart from a few studies (Jorgenson et al. 2004; Eboli et al. 2010; Ciscar et al. 2011; Bosello et al. 2012), CGE analyses tend to investigate the

Table 5.4 CGE studies of climate change: impacts and adaptation

Studies	Regions	Sectoral focus	Models employed
Deke et al. (2001)	Global (11 regions)	Agriculture, sea-level rise	DART (Klepper et al. 2003)
Darwin (1999) Darwin et al. (2001)	Global (8 regions)	Agriculture Sea-level rise	FARM (Darwin et al. 1995)
Jorgenson et al. (2004)	U.S. (1 region)	Agriculture, forestry, water, energy, air quality, heat stress, coastal protection	IGEM (Jorgenson and Wilcoxon 1993)
Bosello et al. (2006) Bosello and Zhang (2006) Bosello et al. (2007) Bosello et al. (2007)	Global (8 regions)	Health Agriculture Energy demand Sea-level rise	GTAP-EF (Roson 2003)
Berritella et al. (2006), Bigano et al. (2008)	Global (8 regions)	Tourism, sea-level rise	Couples HTM (Hamilton et al., 2005) with GTAP-EF
Eboli et al. (2010) Bosello et al. (2010a)	Global (14 regions)	Agriculture, tourism, health, energy demand, sea-level rise	ICES Couples AD-WITCH (Bosello et al. 2010b) with ICES
Ciscar et al. (2011)	Europe (5 regions)	Agriculture, sea-level rise, flooding, tourism	GEM-E3 (Capros et al. 1997)

broad multi-market effects of one or two impact endpoints at a time. The latter are derived by forcing global climate models with various scenarios of GHG emissions to calculate changes in climate variables at the regional scale, generating response surfaces of temperature, precipitation, or sea-level rise that are then run through natural science-based or engineering-based impact models to generate a vector of impact endpoints of particular kinds. These “impact factors” are a region $\times$ sector array of exogenous shocks that are inputs to the model’s counterfactual simulations.

Shocks impact the economy in three basic ways. First, they affect the supply of climatically exposed primary factors such as land (Deke et al. 2001; Darwin et al. 2001), which we denote $IF_{f,r}^{\text{Fact.}} \in (0, 1)$, and scale the factor endowments in the model. The factor market clearance and income balance conditions (5.11) and (5.14) are then

$$IF_{f,r}^{\text{Fact.}} V_{f,r} \geq \sum_{j=1}^{\mathcal{N}} \gamma_{f,j,r}^{\sigma_{j,r}^Y} w_{f,r}^{-\sigma_{j,r}^Y} (p_{j,r}^D)^{\sigma_{j,r}^Y} y_{j,r} \quad \perp \quad w_{f,r} \quad (5.40)$$

$$\mathcal{J}_r = \sum_{f=1}^{\mathcal{F}} w_{f,r} IF_{f,r}^{\text{Fact.}} V_{f,r} \quad \perp \quad \mathcal{J}_r \quad (5.41)$$

Second, impact factors affect sectors’ transformation efficiency (see, e.g., Jorgenson et al. 2004), thereby acting as productivity shift parameters in the unit cost function, where adverse impacts both drive up the marginal cost of production and reduce affected sectors’ demands for inputs according to the scaling factor $IF_{f,r}^{\text{Prod.}} \in (0, 1)$. As a consequence, the zero profit and market clearance conditions (5.3) and (5.8) become

$$p_{j,r}^D \leq \left(IF_{j,r}^{\text{Prod.}} \right)^{-1} \left[\sum_{i=1}^{\mathcal{N}} \beta_{i,j,r}^{\sigma_{j,r}^Y} (p_{i,r}^A)^{1-\sigma_{j,r}^Y} + \sum_{f=1}^{\mathcal{F}} \gamma_{f,j,r}^{\sigma_{j,r}^Y} w_{f,r}^{1-\sigma_{j,r}^Y} \right]^{1/(1-\sigma_{j,r}^Y)} \quad \perp \quad y_{i,r} \quad (5.42)$$

$$\begin{aligned} q_{i,r}^A &\geq (\alpha_{i,r}^C)^{\sigma_r^C} (p_{i,r}^A)^{-\sigma_r^C} \mathcal{E}^{\sigma_r^C} u_r + (\alpha_{i,r}^I)^{\sigma_r^I} (p_{i,r}^A)^{-\sigma_r^I} (p_r^I)^{\sigma_r^I} G_r^I \\ &\quad + \sum_{j=1}^{\mathcal{N}} \left(IF_{j,r}^{\text{Prod.}} \right)^{\sigma_{j,r}^Y - 1} \beta_{i,j,r}^{\sigma_{j,r}^Y} (p_{i,r}^A)^{-\sigma_{j,r}^Y} (p_{j,r}^D)^{\sigma_{j,r}^Y} y_{j,r} \quad \perp \quad p_{i,r}^A \end{aligned} \quad (5.43)$$

Third, impact factors affect the efficiency of inputs to firms’ production and households’ consumption activities. Perhaps the clearest example of this is the impact of increased temperatures on demands for cooling services, and in turn electric power. Such warranted increases in the consumption of climate-related inputs can be treated as a biased technological retrogression that increases the coefficient on the relevant commodities (say, i') in the model’s cost and expenditure functions: $IF_{i',j,r}^{\text{Input}} > 1$ and $IF_{-i',j,r}^{\text{Input}} = 1$. Here, zero profit in consumption (5.1) and equations (5.3) and (5.8)

become:

$$\mathcal{E}_r \leq \left[\sum_{i=1}^{\mathcal{N}} (\alpha_{i,r}^C)^{\sigma_r^C} \left(p_{i,r}^A / IF_{i,\text{Hhold},r}^{\text{Input}} \right)^{1-\sigma_r^C} \right]^{1/(\sigma_r^C-1)} \perp u_r \quad (5.44)$$

$$p_{j,r}^D \leq \left[\sum_{i=1}^{\mathcal{N}} \beta_{i,j,r}^{\sigma_{j,r}^Y} \left(p_{i,r}^A / IF_{i,\text{Hhold},r}^{\text{Input}} \right)^{1-\sigma_{j,r}^Y} + \sum_{f=1}^{\mathcal{F}} \gamma_{f,j,r}^{\sigma_{j,r}^Y} w_{f,r}^{1-\sigma_{j,r}^Y} \right]^{1/(1-\sigma_{j,r}^Y)} \perp y_{i,r} \quad (5.45)$$

$$\begin{aligned} q_{i,r}^A &\geq (\alpha_{i,r}^C)^{\sigma_r^C} \left(p_{i,r}^A / IF_{i,\text{Hhold},r}^{\text{Input}} \right)^{-\sigma_r^C} \mathcal{E}^{\sigma_r^C} u_r + (\alpha_{i,r}^I)^{\sigma_r^I} (p_{i,r}^A)^{-\sigma_r^I} (p_r^I)^{\sigma_r^I} G_r^I \\ &\quad + \sum_{j=1}^{\mathcal{N}} \beta_{i,j,r}^{\sigma_{j,r}^Y} \left(p_{i,r}^A / IF_{i,\text{Hhold},r}^{\text{Input}} \right)^{-\sigma_{j,r}^Y} (p_{j,r}^D)^{\sigma_{j,r}^Y} y_{j,r} \perp p_{i,r}^A \end{aligned} \quad (5.46)$$

In each instance, intersectoral and interregional adjustments made in response to impacts, and the consequences for sectoral output, interregional trade, and regional welfare, can be computed.

The magnitude of damage to the economy due to climate change estimated by CGE studies varies according to the scenario of warming or other climate forcing used to drive impact endpoints, the sectoral and regional resolution of both the resulting shocks and the models used to simulate their economic effects, and the latters' substitution possibilities. Table 5.5 gives a sense of the relevant variation across six studies that focus on the economic consequences of different endpoints circa 2050. The magnitude of economic consequences is generally small, rarely exceeding one-tenth of one percent of GDP. Effects also vary in sign, with some regions benefiting from increased output while others sustain losses. Although there does not appear to be obvious systematic variation in the sign of effects, either across different endpoints or among regions, uncovering relevant patterns is complicated by a host of confounding factors. The studies use different climate change scenarios, and for each impact category economic shocks are constructed from distinct sets of empirical and modeling studies, each with its own regional and sectoral coverage, using different procedures. The influence that such critical details have on model results is difficult to discern because of the unavoidable omission of modeling details necessitated by journal articles' terse exposition. In particular, the precise steps, judgment, and assumptions involved in constructing region-by-sector arrays of economic shocks out of inevitably patchy empirical evidence tends to be reported only in a summary fashion. Strengthening the empirical basis for such input data, and documenting in more detail the analytical procedures to generate $IF_{f,j,r}^{\text{Fact}}$, $IF_{j,r}^{\text{Prod}}$, and $IF_{i',j,r}^{\text{Input}}$, will go a long way toward improving the replicability of studies in this literature. Indeed, this area of research is rich with opportunities

**Table 5.5 Costs of climate change impacts to year 2050:
Selected CGE modeling studies**

Forcing scenario and input data	Impact endpoints, economic shocks, and damage costs
Agriculture (Bosello and Zhang 2006)	
0.93°C global mean temperature rise; temperature–agricultural output relationship calculated by the FUND integrated assessment model (Tol 1996; Anthoff and Tol 2009)	Endpoints considered: temperature, CO₂ fertilization effects on agricultural productivity Shocks: land productivity in crop sectors Change in GDP from baseline: 0.006–0.07% increases in rest of Annex 1 regions, 0.01–0.025% loss in U.S. and energy-exporting countries, 0.13% loss in the rest of the world
Energy demand (Bosello et al. 2007)	
0.93°C global mean temperature rise; temperature–energy demand elasticities from De Cian et al. (2013)	Endpoints considered: temperature effects on demand for 4 energy commodities Shocks: productivity of intermediate and final energy Change in GDP from baseline: 0.04–0.29% loss in remaining Annex I (developed) regions, 0.004–0.03% increase in Japan, China/India, and the rest of the world, 0.3% loss in energy-exporting countries. Results for perfect competition only.
Health (Bosello et al. 2006)	
1°C global mean temperature rise; temperature–disease and disease–cost relation extrapolated from numerous empirical and modeling studies	Endpoints considered: malaria, schistosomiasis, dengue fever, cardiovascular disease, respiratory ailments, diarrheal disease Shocks: labor productivity, increased household expenditures on public and private health care, reduced expenditures on other commodities Direct costs/benefits (% of GDP): costs of 9% in U.S. and Europe, 11% in Japan and the remainder of Annex 1 regions, 14% in eastern Europe and Russia; benefits of 1% in energy exporters and 3% in the rest of the world Change in GDP from baseline: 0.04–0.08% increase in Annex 1 regions, 0.07–0.1% loss in energy-exporting countries and the rest of the world
Sea-level rise/Tourism (Bigano et al. 2008)	
Uniform global 25 cm sea-level rise; land loss calculated by FUND	Endpoints considered: land loss, change in tourism arrivals as a function of land loss Shocks: reduction in land endowment Direct costs (% of GDP): < 0.005% loss in most regions, 0.05% in North Africa, 0.1–0.16% in South Asia and Southeast Asia, and 0.24% in sub-Saharan Africa. Costs are due to land loss only. Change in GDP from baseline: < 0.0075% loss in most regions, 0.06% in South Asia, and 0.1% in Southeast Asia

Table 5.5 Continued

Forcing scenario and input data	Impact endpoints, economic shocks, and damage costs
Ecosystem services (Bosello et al., 2011)	
1.2°C/3.1°C temperature rise, with and without impacts on ecosystems	Endpoints considered: timber and agricultural production, forest, cropland, grassland carbon sequestration Shocks: reduced productivity of land, reduced carbon sequestration resulting in increased temperature change impacts on 5 endpoints in Eboli et al. (2010) Change in GDP (3.1°C, 2001–2050 NPV @ 3%) \$22–\$32Bn additional loss in eastern and Mediterranean Europe, \$5Bn reduction in loss in northern Europe
Water resources (Calzadilla et al., 2010)	
Scenarios of rain-fed and irrigated crop production, irrigation efficiency based on Rosegrant et al. (2008)	Endpoints considered: crop production Shocks: supply/productivity of irrigation services Change in welfare: losses in 5 regions range from \$60M in sub-Saharan Africa to \$442M in Australia/New Zealand, gains in 11 regions range from \$180M in the rest of the world to \$3Bn in Japan/Korea

for interdisciplinary collaboration among modelers, empirical economists, natural scientists, and engineers.

Recent large-scale studies define the state of the art in this regard. In the PESETA study of climate impacts on Europe in the year 2050 (Ciscar et al. 2009, 2011, 2012), estimates of physical impacts were constructed by propagating a consistent set of climate warming scenarios via different process simulations in four impact categories: agriculture (Iglesias et al. 2012), flooding (Feyen et al. 2012), sea-level rise (Bosello et al. 2012), and tourism (Amelung and Moreno 2012). These “bottom-up” results were then incorporated into the GEM-E3 model using a variety of techniques to map the endpoints to the types of effects on economic sectors (Ciscar et al. 2012). Estimated changes in crop yields were implemented as neutral productivity retrogressions in the agriculture sector. Flood damages were translated into additional unproductive expenditures by households, secular reductions in the output of the agriculture sector, and reductions in the outputs of and capital inputs to industrial and commercial sectors. Changes in visitor occupancy were combined with “per bed-night” expenditure data to estimate changes in tourist spending by country, and in turn expressed as secular changes in exports of GEM-E3’s market services sector. Costs of migration induced by land lost to sea-level rise are incurred by households as additional expenditures, and related coastal flooding is assumed to equiproportionally reduce sectors’ endowments of capital. (The direct macroeconomic effects of reduced land endowments were not considered.)

In Bosello et al. (2012), estimates of the global distribution of physical impacts in six categories in the year 2050 were derived from the results of different process simulations forced by a 1.9°C global mean temperature increase. Endpoints were expressed as shocks within the Intertemporal Computable Equilibrium System (ICES) model. Regional impacts on energy and tourism were treated as shocks to household demand. Changes in final demand for oil, gas, and electricity (based on results from a bottom-up energy system simulation; see Mima et al. 2011) were expressed as biased productivity shifts in the aggregate unit expenditure function. A two-track strategy was adopted to simulate changes in tourism flows (arrival changes from an econometrically calibrated simulation of tourist flows; see Bigano et al. 2007), with non-price climate-driven substitution effects captured through secular productivity biases that scale regional households' demands for market services (the ICES commodity that includes recreation), and the corresponding income effects imposed as direct changes in regional expenditure. Regional impacts on agriculture and forestry, health, and the effects of river floods and sea-level rise were treated as supply-side shocks. Changes in agricultural yields (generated by a crop simulation; see Iglesias et al. 2011) and forest net primary productivity (simulated by a global vegetation model; see Bondeau et al. 2007; Tietjen et al. 2009) were represented as exogenous changes in the productivity of the land endowment in the agriculture sector and the natural resource endowment in the timber sector, respectively. The impact of higher temperatures on employment performance was modeled by reducing aggregate labor productivity (based on heat and humidity effects estimated by Kjellstrom et al. 2009). Losses of land and buildings due to sea-level rise (whose costs were derived from a hydrological simulation; see Van Der Knijff et al. 2010) were expressed as secular reductions in regional endowments of land and capital, which are assumed to decline by the same fraction. Damages from river flooding span multiple sectors and are therefore imposed using different methods: reduction of the endowment of arable land in agriculture and equiproportional reduction in the productivity of capital inputs to other industry sectors, as well as reductions in labor productivity (equivalent to a one-week average annual loss of working days per year in each region) for affected populations.

An important aspect of climate impacts assessment that is ripe for investigation is the application of CGE models to evaluate the effects of specific adaptation investments. Work in this area is currently limited by a lack of information about the relevant technology options and pervasive uncertainty about the magnitude, timing, and regional and sectoral incidence of various types of impacts. The difficult but essential work of characterizing adaptation technologies that are the analogues of those discussed in section 5.3.2 will render similar analyses for reactive adaptation straightforward. Moreover, Bosello et al.'s (2010a) methodological advance of coupling a CGE model with an optimal growth simulation of intertemporal feedbacks on the accumulation of stock adaptation capacity (Bosello et al. 2010b) paves the way to modeling *proactive* investment in adaptation.

Similar issues arise in CGE analyses of the macroeconomic costs of natural and man-made hazards (Rose 2005; Rose et al. 2004; Rose and Guha 2004; Rose and Liao

2005; Rose et al. 2009; Dixon et al. 2010). The key distinction that must be made is between the three types of impact factors. The parameter $IF_{i,j,r}^{\text{Fact.}}$ captures components of damage that cause direct destruction of the capital stock (e.g., earthquakes, floods, or terrorist bombings) or reduction in the labor supply (e.g., morbidity and mortality or evacuation of populations from the disaster zone). Second, $IF_{j,r}^{\text{Prod.}}$ captures impacts that reduce sectors' productivity while leaving factor endowments intact, such as utility lifeline outages (Rose and Liao 2005) or pandemic disease outbreaks where workers in many sectors shelter at home as a precaution (Dixon et al. 2010). Third, $IF_{i',j,r}^{\text{Input}}$ can be used to model input-using biases of technical change in the post-disaster recovery phase (e.g., increased demand for construction services).

The fact that these input parameters often must be derived from engineering loss estimation simulations such as the U.S. Federal Emergency Management Agency's HAZUS software raises additional methodological issues. Principal among these are the need to specify reductions in the aggregate endowment of capital input that are consistent with capital stock losses across a range of sectors and the need to reconcile them with exogenous estimates of industry output losses for the purpose of isolating noncapital-related shocks to productivity. The broader concern, which applies equally to climate impacts, is the extent to which the methods used to derive the input shocks inadvertently incorporate the kinds of economic adjustments that CGE modeling is tasked with simulating—leading to potential double-counting of both losses and the mitigating effects of substitution. These questions are the subject of ongoing research.

5.4 EXTENSIONS TO THE STANDARD MODEL

5.4.1 Production Technology: Substitution Possibilities, Bottom-Up versus Top-down

In the vast majority of CGE models, firms' technology is specified using hierarchical or nested CES production functions, whose properties of monotonicity and global regularity facilitate the computation of equilibrium (Perroni and Rutherford 1993, 1998), while providing the flexibility to capture complex patterns of substitution among capital, labor, and intermediate inputs of energy and materials. A key consequence of this modeling choice is that numerical calibration of the resulting model to a SAM becomes a severely underdetermined problem, with the number of model parameters greatly exceeding the degrees of freedom in the underlying benchmark calibration data set.

It is common for both the nested structure of production and the corresponding elasticity of substitution parameters to be selected on the basis of judgment and assumptions, a practice that has long been criticized by mainstream empirical economists (e.g., Jorgenson 1984). Whereas econometric calibration of CGE models' technology has

traditionally been restricted to flexible functional forms such as the Translog (McKittrick 1998; McKibbin and Wilcoxon 1998; Fisher-Vanden and Sue Wing 2008; Fisher-Vanden and Ho 2010; Jin and Jorgenson 2010), there has been interest in estimating nested CES functions (van der Werf 2008; Okagawa and Ban 2008). Progress in this area continues to be hampered by lack of data, owing to the particular difficulty of compiling time-sequences of input-output data sets with consistent price and quantity series. In an attempt to circumvent this problem, various approaches have been developed for calibrating elasticity parameters so as to reproduce empirically estimated input price elasticities (e.g., Arndt et al. 2002; Adkins et al. 2003; Gohin 2005), but these have yet to be widely adopted by the CGE modeling community.

A parallel development is the trend toward modifying CGE models' specifications of production to incorporate discrete "bottom-up" technology options. This practice has been especially popular in climate change mitigation research, where it enables CGE models to capture the effects of GHG abatement measures on the competition between conventional and alternative energy technologies, and to simulate the general equilibrium incidence of policies to promote "green" energy supply or conversion options such as renewable electricity or hybrid electric vehicles. The incorporation of bottom-up detail in CGE models marries the detail of primal partial equilibrium activity analysis simulations with general equilibrium feedbacks of price and substitution adjustments across the full range of consuming sectors in the economy. This hybrid modeling approach has been used in energy technology assessments relating to transportation (Schafer and Jacoby 2005) and fuel supply (Chen et al. 2011) its most popular area of application is prospective analyses of electric power production, an example of which we provide below.

Methods for incorporating discrete technological detail and substitution in CGE models break down into two principal classes, namely the "decomposition" approach of Böhringer and Rutherford (2008, 2009) and Lanz and Rausch (2011) and the "integrated" approach of Böhringer (1998). The first method simulates a top-down CGE model in tandem with a bottom-up energy technology model iterating back and forth to convergence.²² Briefly, the representative agent in the top-down model is "endowed" with quantities of energy supplied by the various active technology options, which it uses to compute both the prices of the inputs to and outputs of the energy supply sectors, and the levels of aggregate demand for energy commodities. These results are passed as inputs to the bottom-up model to compute the aggregate cost-minimizing mix of energy supply, conversion, or demand activities, whose outputs are used to update the top-down model's endowments at the subsequent iteration. The second approach embeds activity-analysis representations of bottom-up technology options directly into a top-down model's sectoral cost functions, numerically calibrating discrete activities' inputs and outputs to be jointly consistent with ancillary statistics and the social accounts. The key requirement is a consistent macro-level representation of subsectoral

²² Examples of the latter are MARKAL (Schafer and Jacoby 2005), and economic dispatch or capacity expansion models of the electric power sector (Lanz and Rausch 2011).

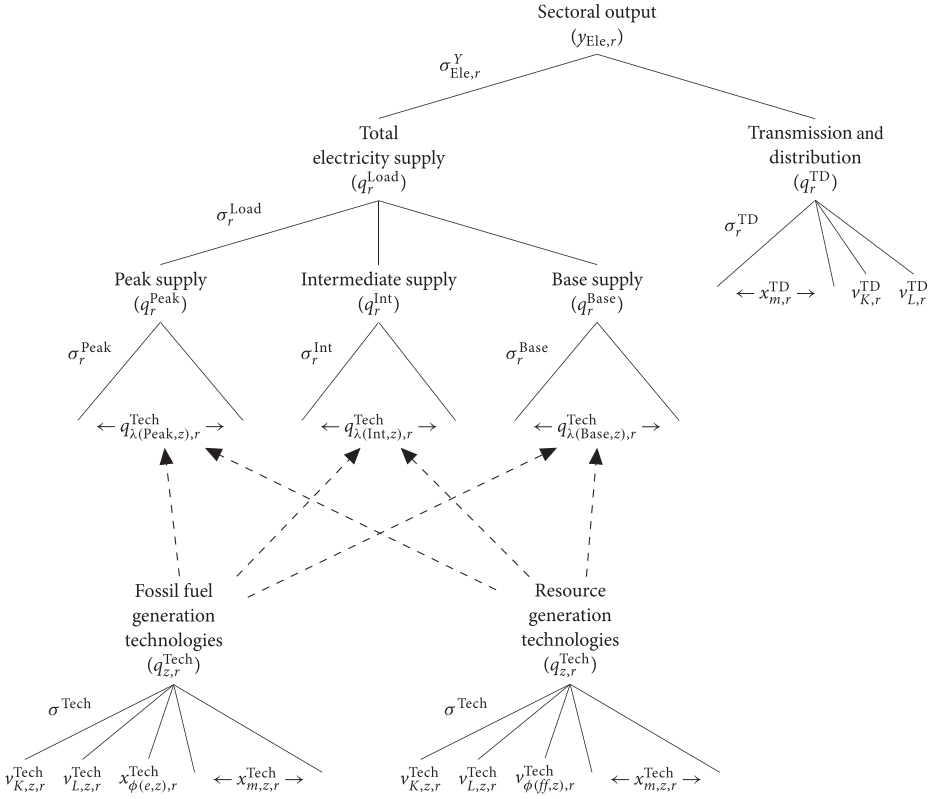


FIGURE 5.2 A Bottom-up/top-down model of electric power production.

technology options and their competition in both input and output markets, which is precisely where the nested CES model of production proves useful.

To illustrate what the integrated approach involves, we consider a simplified version of the top-down/bottom-up model of electricity production in Sue Wing (2006, 2008). Figure 5.2 shows the production structure, which divides the sector into five activities. Delivered electricity (A1) is a CES function of transmission and distribution (A2) and aggregate electricity supply (A3). Transmission and distribution is a CES function of labor, capital, and intermediate nonfuel inputs, while aggregate electricity supply is a CES aggregation of three load segments $\ell = \{\text{peak, intermediate, base}\}$. Each load segment (A4) is a CES aggregation of subsets of the z generation outputs, defined by the mapping from load to technology $\lambda(\ell, z)$. Finally, individual technologies (A5) produce electricity from labor, capital, non-energy materials, and either fossil fuels ($e \subset i$) or “fixed-factor” energy resources ($ff \subset f$) in the case of nuclear, hydro, or renewable power. The latter are defined by the fuel-to-technology mappings $\phi(e, z)$ and $\phi(ff, z)$.

Several aspects of this formulation merit discussion. First, substitutability between transmission and the total quantity of electricity generated by the sector captures the fact that reductions in congestion and line losses from investments in transmission can maintain the level of delivered power with less physical energy, at least over a small

range, suggesting the substitution elasticity value $\sigma_{\text{El},r}^Y \ll 1$. Second, although disaggregation of electricity supply—rather than demand—into load segments may seem counterintuitive, it is a conscious choice driven by the exigencies of data availability and model tractability. Demand-side specification of load segments necessitates row disaggregation of the SAM into separate accounts for individual users' demands for peak, intermediate, and base power. Often the necessary information is simply not available; this motivates the present structure, which is designed to keep delivered electric power as a homogeneous commodity while differentiating generation methods by technology. (For an exception, see Rodrigues and Linares 2014) This device is meant to capture the fact that only subsets of the universe of generation technologies compete against one another to serve particular electricity market segments. Thus, specifying relatively easy substitution among generators but not between load segments ($\sigma_r^{\text{Load}} \ll 1 < \sigma_r^\ell$) enables coal and nuclear electricity (say) to be highly fungible within base load, but largely unable to substitute for peak natural gas or wind generation.

Third, from an energy modeling standpoint, the CES aggregation technology's nonlinearity has the disadvantage of preventing generation in energetic units from summing up to the kilowatt hours of delivered electricity. But it is well known that modeling discrete activities' outputs as near-perfect substitutes can lead to "bang-bang" behavior, wherein small differences in technologies' marginal costs induce large swings in their market shares. The present formulation's strength is that it enables generation technologies with widely differing marginal costs to coexist and the resulting market shares to respond smoothly to policy shocks such as the GHG tax mentioned in section 5.3.2.

The supply-demand correspondences for the outputs of the transmission, aggregate electricity supply, load class, and generation technology activities are indicated by subsectoral allocations (S1) to (S4) in table 5.6. These are trivial—all of the action is in the allocation of factors (capital, labor, and non-fossil energy resources) as well as fuel and non-fuel intermediate Armington goods among transmission and the different technologies, (S5) to (S6). The resulting input-output structure is underlain by the price and quantity variables given in table 5.6, organized according to the exhaustion of profit and supply-demand balance identities shown in table 5.8. Table 5.7 provides the algebraic elaboration of these identities, in which (A1) to (A5) are given by the zero-profit conditions (5.57) to (5.60) and (S1) to (S6) are given by the market clearance conditions (5.61) to (5.71).

Calibration is the major challenge to computational implementation of this model. Recall that the present scheme requires a columnar disaggregation of the SAM in figure 5.1, part C, that allocates inputs to the electricity sector among the various activities. The typical method relies on two pieces of exogenous information, namely, statistics concerning the benchmark quantities of electricity generated by the different technologies ($\Upsilon_{z,r}^{\text{Gen}}$) and descriptions of the contributions of inputs of factors and Armington intermediate energy and material goods to the unit cost of generation ($\Upsilon_{f,z,r}^{\text{Tech}}$ and

Table 5.6 Summary of variables in the bottom-up/top-down model of electric power production

A. Activities									
Output					Inputs				
	Price	Quantity	Price	Quantity		Price	Quantity		
(A1) Sectoral output	$p_{Eie,r}^D$	Domestic delivered electricity price	$y_{Eie,r}$	Domestic delivered electricity supply	p_r^{TD}	Transmission & distribution Aggregate generation price	Transmission & distribution Aggregate generation	q_r^{TD}	
(A2) Transmission & distribution	p_r^{TD}	Transmission price	q_r^{TD}	Transmission supply	$p_{m,r}^A$	Armington material goods price	Armington material goods demand	$x_{m,r}^{TD}$	
(A3) Total electricity supply	p_r^{Load}	Electricity price	q_r^{Load}	Electricity supply	$w_{L,r}$	Wage	Labor demand	$v_{L,r}^{TD}$	
(A4) Load segments	$p_{\ell,r}^{Load}$	Electricity price by load segment	q_r^ℓ	Electricity supply by load segment	$w_{K,r}$	Capital rental rate	Capital demand	$v_{K,r}^{TD}$	
(A5) Generation	$p_{z,r}^{Tech}$	Generation price	$q_{z,r}^{Tech}$	Generation supply	$p_{\ell,r}^{Load}$	Electricity price by load segment	Electricity supply by load segment	q_r^ℓ	
					$p_{\lambda(\ell,z),r}^{Tech}$	Generation price	Generation supply	$q_{\lambda(\ell,z),r}^{Tech}$	
					$p_{e,r}^A$	Armington fossil fuel price	Armington fossil fuel demand	$x_{\phi(e,z),r}^{Tech}$	
					$p_{m,r}^A$	Armington material goods price	Armington material goods demand	$x_{m,z,r}^{Tech}$	
					$w_{L,r}$	Wage	Labor demand	$v_{L,z,r}^{Tech}$	
					$w_{K,r}$	Capital rental rate	Capital demand	$v_{K,z,r}^{Tech}$	
					$w_{ff,r}$	Fixed-factor price	Fixed-factor demand	$v_{\phi(ff,z),r}^{Tech}$	

Table 5.6 Continued

B. Subsectoral allocations			
	Supply		Demands
	Price	Quantity	
(S1)	p_r^{TD}	$\bar{q}_r^{\text{TD}}$	Demand for transmission services
(S2)	p_r^{load}	q_r^{load}	Demand for aggregate electric energy
(S3)	p_r^ℓ	q_r^ℓ	Demand for energy aggregated by load segment
(S4)	$p_{z,r}^{\text{Tech}}$	$q_{z,r}^{\text{Tech}}$	Load segment demands for generation technology outputs
(S5)	$w_{K,r}$	$v_{K,r}^{\text{Tech}}, v_{K,r}^{\text{TD}}$	Capital demand by technologies and transmission
	$w_{L,r}$	$v_{L,r}^{\text{Tech}}, v_{L,r}^{\text{TD}}$	Labor demand by technologies and transmission
	$w_{ff,r}$	$v_{ff,r}^{\text{Tech}}$	Fixed factor energy resource demand by technologies
		$x_{e,z,r}^{\text{Tech}}$	Generation technology demands for Armington fossil fuel inputs
(S6)	$p_{r,r}^A$	$x_{i \text{Ele},r}$	Generation and transmission demand for Armington non-energy material inputs
		$x_{m,r}^{\text{Tech}}, x_{m,r}^{\text{TD}}$	

Table 5.7 Algebraic representation of bottom-up technologies in electric power production

$p_{\text{Ele},r}^D \leq \left[(\varpi \text{Gen},r) \sigma_{\text{Ele},r}^Y (p_r^{\text{Load}})^{1-\sigma_{\text{Ele},r}^Y} + (\varpi \text{TD},r) \sigma_{\text{Ele},r}^Y (p_r^{\text{TD}})^{1-\sigma_{\text{Ele},r}^Y} \right]^{1/(1-\sigma_{\text{Ele},r}^Y)}$	$\perp$	$y_{\text{Ele},r}$	(5.56)
$p_r^{\text{Load}} \leq \left[\sum_{\ell} (\nu_{\ell,r}) \sigma_r^{\text{Load}} (p_{\ell,r}^{\text{Load}})^{1-\sigma_r^{\text{Load}}} \right]^{1/(1-\sigma_r^{\text{Load}})}$	$\perp$	q_r^{Load}	(5.57)
$p_r^{\text{TD}} \leq \left[\sum_m (\beta_{i,r}^{\text{TD}})^{\sigma_r^{\text{TD}}} (p_{m,r}^A)^{1-\sigma_r^{\text{TD}}} + (\gamma_{K,r}^{\text{TD}})^{\sigma_r^{\text{TD}}} w_{K,r}^{1-\sigma_r^{\text{TD}}} + (\gamma_{L,r}^{\text{TD}})^{\sigma_r^{\text{TD}}} w_{L,r}^{1-\sigma_r^{\text{TD}}} \right]^{1/(1-\sigma_r^{\text{TD}})}$	$\perp$	q_r^{TD}	(5.58)
$p_{\ell,r}^{\text{Load}} \leq \left[\sum_{\lambda(z,\ell)} (\eta_{z,\ell,r}) \sigma_{\ell,r}^{\text{Load}} (p_{z,r}^{\text{Tech}})^{1-\sigma_{\ell,r}^{\text{Load}}} \right]^{1/(1-\sigma_{\ell,r}^{\text{Load}})}$	$\perp$	q_r^{ℓ}	(5.59)
$p_{z,r}^{\text{Tech}} \leq \left[\begin{aligned} & (\theta_{z,r}^K)^{\sigma_r^{\text{Tech}}} w_{K,r}^{1-\sigma_r^{\text{Tech}}} + (\theta_{z,r}^L)^{\sigma_r^{\text{Tech}}} w_{L,r}^{1-\sigma_r^{\text{Tech}}} \\ & + (\theta_{m,z,r}^M)^{\sigma_r^{\text{Tech}}} (p_{m,r}^A)^{1-\sigma_r^{\text{Tech}}} \end{aligned} \right]^{1/(1-\sigma_r^{\text{Tech}})} + \left\{ \begin{aligned} & (\theta_{\phi(e,z),r}^F)^{\sigma_r^{\text{Tech}}} (p_{e,r}^A)^{1-\sigma_r^{\text{Tech}}} \\ & (\theta_{\phi(ff,z),r}^F)^{\sigma_r^{\text{Tech}}} w_{ff,r}^{1-\sigma_r^{\text{Tech}}} \end{aligned} \right\}$	$\perp$	$q_{z,r}^{\text{Tech}}$ Fossil Non-fossil	(5.60)
$q_r^{\text{Load}} \geq (\varpi \text{Gen},r) \sigma_{\text{Ele},r}^Y (p_r^{\text{Load}})^{-\sigma_{\text{Ele},r}^Y} (p_{\text{Ele},r}^D)^{\sigma_{\text{Ele},r}^Y} y_{\text{Ele},r}$	$\perp$	p_r^{Load}	(5.61)
$q_r^{\text{TD}} \geq (\varpi \text{TD},r) \sigma_{\text{Ele},r}^Y (p_r^{\text{TD}})^{-\sigma_{\text{Ele},r}^Y} (p_{\text{Ele},r}^D)^{\sigma_{\text{Ele},r}^Y} y_{\text{Ele},r}$	$\perp$	p_r^{TD}	(5.62)
$q_r^{\ell} \geq (\nu_{\ell,r}) \sigma_r^{\text{Load}} (p_{\ell,r}^{\text{Load}})^{-\sigma_r^{\text{Load}}} (p_r^{\text{Load}})^{\sigma_r^{\text{Load}}} q_r^{\text{Load}}$	$\perp$	$p_{\ell,r}^{\text{Load}}$	(5.63)
			(5.64)

Table 5.7 Continued

$q_{z,r}^{\text{Tech}} \geq (\eta_{\lambda(z\ell),r})^{\sigma_{\ell,r}^{\text{Load}}} (p_{z,r}^{\text{Tech}})^{-\sigma_r^{\text{Load}}} (p_{\ell,r}^{\text{Load}})^{\sigma_{\ell,r}^{\text{Load}}} q_r^{\ell}$	$\perp$	$p_{z,r}^{\text{Tech}}$	(5.69)
$V_{f,r} \geq \sum_{j \neq \text{Ele}} \sigma_{i,r}^Y w_{f,r}^{-\sigma_{j,r}^Y} (p_{j,r}^D)^{\sigma_{j,r}^Y} y_{j,r}$			
$+ \left\{ \begin{aligned} & \gamma_{K,r}^{\sigma_r^{\text{TD}}} w_{K,r}^{-\sigma_r^{\text{TD}}} (p_r^{\text{TD}})^{\sigma_r^{\text{TD}}} q_r^{\text{TD}} + \sum_z (\theta_{z,r}^K)^{\sigma_r^{\text{Tech}}} w_{K,r}^{-\sigma_r^{\text{Tech}}} (p_{z,r}^{\text{Tech}})^{\sigma_r^{\text{Tech}}} q_{z,r}^{\text{Tech}} \quad K \in f \\ & \gamma_{L,r}^{\sigma_r^{\text{TD}}} w_{L,r}^{-\sigma_r^{\text{TD}}} (p_r^{\text{TD}})^{\sigma_r^{\text{TD}}} q_r^{\text{TD}} + \sum_z (\theta_{z,r}^L)^{\sigma_r^{\text{Tech}}} w_{L,r}^{-\sigma_r^{\text{Tech}}} (p_{z,r}^{\text{Tech}})^{\sigma_r^{\text{Tech}}} q_{z,r}^{\text{Tech}} \quad L \in f \\ & \sum_{\phi(\text{ffz})} (\theta_{\text{ffz},r}^F)^{\sigma_r^{\text{Tech}}} w_{\text{ff},r}^{-\sigma_r^{\text{Tech}}} (p_{z,r}^{\text{Tech}})^{\sigma_r^{\text{Tech}}} q_{z,r}^{\text{Tech}} \quad \text{ff} \subset f \end{aligned} \right.$	$\perp$	$w_{f,r}$	(5.70)
$q_{i,r}^A \geq (\alpha_{i,r}^C)^{\sigma_r^C} (p_{i,r}^A)^{-\sigma_r^C} \mathcal{G}^{\sigma_r^C} u_r + (\alpha_{i,r}^I)^{\sigma_r^I} (p_{i,r}^A)^{-\sigma_r^I} (p_r^I)^{\sigma_r^I} G_r^I + \sum_{j \neq \text{Ele}} \beta_{i,j,r}^{\sigma_{j,r}^Y} (p_{j,r}^D)^{\sigma_{j,r}^Y} y_{j,r}$			
$+ \left\{ \begin{aligned} & \sum_{\phi(ez)} (\theta_{e,z,r}^F)^{\sigma_r^{\text{Tech}}} (p_{e,r}^A)^{-\sigma_r^{\text{Tech}}} (p_{z,r}^{\text{Tech}})^{\sigma_r^{\text{Tech}}} q_{z,r}^{\text{Tech}} \quad e \subset i \\ & \beta_{m,r}^{\sigma_r^{\text{TD}}} (p_{m,r}^A)^{-\sigma_r^{\text{TD}}} (p_r^{\text{TD}})^{\sigma_r^{\text{TD}}} q_r^{\text{TD}} + \sum_z (\theta_{m,z,r}^M)^{\sigma_r^{\text{Tech}}} (p_{m,r}^A)^{-\sigma_r^{\text{Tech}}} (p_{z,r}^{\text{Tech}})^{\sigma_r^{\text{Tech}}} q_{z,r}^{\text{Tech}} \quad m \subset i \end{aligned} \right.$	$\perp$	$p_{i,r}^A$	(5.71)
			(5.72)

Table 5.8 Bottom-up/top-down representation of electric power production: accounting identities and parameterization

A. Accounting identities				
Zero-profit conditions (Activities)		Supply-demand balance conditions (Subsectoral allocations)		
(A1)	$p_{\text{Ele},r}^D y_{\text{Ele},r} \leq p_r^{\text{Load}} q_r^{\text{Load}} + p_r^{\text{TD}} q_r^{\text{TD}}$	(S1)–(S4) are trivial		
(A2)	$p_r^{\text{TD}} q_r^{\text{TD}} \leq w_{K,r} v_{K,z,r}^{\text{TD}} + w_{L,r} v_{L,z,r}^{\text{TD}} + \sum_{\phi(m,z)} p_{m,r}^A x_{m,z,r}^{\text{Tech}}$			
(A3)	$p_r^{\text{Load}} q_r^{\text{Load}} \leq \sum_{\ell} p_{\ell,r}^{\text{Load}} q_r^{\ell}$	(S5)	$v_{f,\text{Ele},r} \geq v_{f,z,r}^{\text{TD}} + \sum_z v_{f,z,r}^{\text{Tech}}$	
(A4)	$p_{\ell,r}^{\text{Load}} q_r^{\ell} \leq \sum_{\lambda(\ell,z),r} p_{z,r}^{\text{Tech}} q_{z,r}^{\text{Tech}}$	(S6) $x_{i,\text{Ele},r} \geq \begin{cases} \sum x_{e,z,r}^{\text{Tech}} & \text{Fossil fuels} \\ \phi(e,z) x_{m,r}^{\text{TD}} + \sum_z x_{m,z,r}^{\text{Tech}} & \text{Materials} \end{cases}$		
(A5)	$p_{z,r}^{\text{Tech}} q_{z,r}^{\text{Tech}} \leq w_{K,r} v_{K,z,r}^{\text{Tech}} + w_{L,r} v_{L,z,r}^{\text{Tech}} + \begin{cases} \sum_{\phi(e,z)} p_{e,r}^A x_{e,z,r}^{\text{Tech}} & \text{Fossil fuel} \\ \sum_{\phi(ff,z)} w_{ff,r} v_{ff,z,r}^{\text{Tech}} & \text{Non-fossil} \end{cases}$			
B. Parameters				
Institutions		Substitution elasticities	Technical coefficients	
(A1)	Delivered electric power	$\sigma_{\text{Ele},r}^Y$	$\varpi_{\text{Load},r}$	Total electricity supply
			$\varpi_{\text{TD},r}$	Transmission
(A2)	Transmission & distribution	σ_r^{TD}	$\beta_{m,r}^{\text{TD}}$	Intermediate Armington good use
			$\gamma_{K,r}^{\text{TD}}, \gamma_{L,r}^{\text{TD}}$	Capital and labor inputs
(A3)	Total electricity supply	σ_r^{Load}	$v_{\ell,r}$	Load segments
(A4)	Load segments	σ_r^{ℓ}	$\eta_{z,\ell,r}$	Technologies' outputs
(A5)	Technologies	σ^{Tech}	$\theta_{z,r}^K, \theta_{z,r}^L$	Capital and labor
			$\theta_{m,z,r}^M$	Materials
			$\theta_{e,z,r}^F, \theta_{ff,z,r}^F$	Fossil and fixed-factor energy

$\Upsilon_{i,z,r}^{\text{Tech}}$, respectively). The calibration problem is, therefore, to find benchmark input vectors whose elements satisfy the identities given in table 5.8, part A, but yield a vector of technology outputs and input proportions that does not diverge “too far” from the exogenous data. The least squares fitting procedure presented in table 5.9 operationalizes this idea, recasting (A1) to (A5) and (S5) to (S6) as equality constraints

Table 5.9 The bottom-up/top-down calibration problem

$$\begin{aligned}
\min_{\substack{\bar{x}_{i,z,r}^{\text{Tech}}, \bar{x}_{m,z,r}^{\text{TD}}, \\ \bar{v}_{f,z,r}^{\text{Tech}}, \bar{x}_{f,z,r}^{\text{TD}}}} & \sum_z \left(\bar{q}_{z,r}^{\text{Tech}} - \gamma_{z,r}^{\text{Gen}} \right)^2 + \sum_z \sum_{f=K,L} \left(\bar{v}_{f,z,r}^{\text{Tech}} / \bar{q}_{z,r}^{\text{Tech}} - \gamma_{f,z,r}^{\text{Tech}} \right)^2 + \sum_z \sum_m \left(\bar{v}_{m,z,r}^{\text{Tech}} / \bar{q}_{z,r}^{\text{Tech}} - \gamma_{m,z,r}^{\text{Tech}} \right)^2 \\
& + \sum_z \sum_{\phi(e,z)} \left(\bar{x}_{e,z,r}^{\text{Tech}} / \bar{q}_{z,r}^{\text{Tech}} - \gamma_{e,z,r}^{\text{Tech}} \right)^2 + \sum_z \sum_{\phi(ff,z)} \left(\bar{v}_{ff,z,r}^{\text{Tech}} / \bar{q}_{z,r}^{\text{Tech}} - \gamma_{ff,z,r}^{\text{Tech}} \right)^2 \quad \text{s.t.} \\
\text{(A1')} \quad & \bar{y}_{\text{Ele},r} = \bar{q}_r^{\text{Load}} + \bar{q}_r^{\text{TD}} & \text{(A2')} \quad & \bar{q}_r^{\text{TD}} = \bar{v}_{K,z,r}^{\text{TD}} + \bar{v}_{L,z,r}^{\text{TD}} + \sum_{\phi(m,z)} \bar{x}_{m,z,r}^{\text{Tech}} \\
\text{(A3')} \quad & \bar{q}_r^{\text{Load}} = \sum_{\ell} \bar{q}_r^{\ell} & \text{(A4')} \quad & \bar{q}_r^{\ell} = \sum_{\lambda(\ell,z),r} \bar{q}_{z,r}^{\text{Tech}} \\
\text{(A5')} \quad & \bar{q}_{z,r}^{\text{Tech}} = \bar{v}_{K,z,r}^{\text{Tech}} + \bar{v}_{L,z,r}^{\text{Tech}} & \text{(S5')} \quad & \bar{v}_{f,\text{Ele},r} = \bar{v}_{f,z,r}^{\text{TD}} + \sum_z \bar{v}_{f,z,r}^{\text{Tech}} \\
& + \begin{cases} \sum \bar{x}_{e,z,r}^{\text{Tech}} & \text{Fossil fuel} \\ \sum_{\phi(e,z)} \bar{v}_{ff,z,r}^{\text{Tech}} & \text{Non-fossil} \end{cases} & \text{(S6')} \quad & \bar{x}_{i,\text{Ele},r} = \begin{cases} \sum \bar{x}_{e,z,r}^{\text{Tech}} & \text{Fossil fuels} \\ \sum_{\phi(e,z)} \bar{x}_{m,r}^{\text{TD}} & \text{Materials} \end{cases}
\end{aligned}$$

posed in terms of the SAM's benchmark quantities (indicated by a bar over a variable) with all prices set to unity. It is customary to focus on generation while allowing the inputs to—and the ultimate size of—the transmission and distribution activity to be determined as residuals to this nonlinear program.

Finally, even this systematized procedure involves a fair amount of judgment and assumptions. For example, the dearth of data about fixed-factor resource inputs in input-output accounts requires the values of $\bar{v}_{ff,z,r}^{\text{Tech}}$ to be assumed as fractions of the electric power sector's benchmark payments to capital, and engineering data about technology characteristics often lump labor and materials together into operations and maintenance expenditures, necessitating ad hoc disaggregation.

5.4.2 Heterogeneous Firms and Endogenous Productivity Dynamics

In this section we describe the radical extension of the canonical model to incorporate contemporary theories of trade, focusing on the nexus of monopolistic competition, heterogeneous firms, and endogenous productivity. The Armington trade model's assumption of perfectly competitive producers ignores the existence of monopolistic competition in a range of manufacturing and service industries. In these sectors, each firm produces a unique variety of good and faces a differentiated cost of production made up of a fixed cost and a variable cost that depends on the firm's productivity. An important limitation of the canonical model is its failure to account for the

fact that openness to trade induces competitive selection of firms and reallocation of resources from unproductive to productive producers, generating *export variety* gains from trade (Feenstra 2010) that can substantially exceed the gains-from-trade predictions of standard CGE simulations (see Balistreri et al. 2010). By contrast, the heterogeneous-firm framework is more consistent with several stylized facts (Balistreri et al. 2011): persistent intra-industry trade-related differences in firm productivities (Bartelsman and Doms 2000), the comparative scarcity and relatively high productivity of exporting firms (Bernard and Jensen 1999), and the association between higher average productivity, openness (Trefler 2004) and lower trade costs (Bernard et al. 2006).

Our heterogeneous-firm CGE simulation follows the theoretical structure developed by Melitz (2003). We consider a single industry $h \in j$ as the heterogeneous-firm sector. An h -producing firm in region r deciding whether to sell to region r' balances the expected revenue from entering that bilateral export market, (r, r') , against the expected cost. On entry, the firm's costs are sunk and its productivity is fixed according to a draw from a probability distribution. Which firms are able to sell profitably in (r, r') is jointly determined by five factors: their productivity levels $(\varphi_{h,r,r'})$, the costs of bilateral trade $(\mathcal{C}_{h,r,r'})$, the fixed operating and sunk costs associated with market entry $(\mathcal{F}_{h,r,r'}^O$ and $\mathcal{F}_{h,r,r'}^S$), and the level of demand. An individual firm takes these as given and maximizes profit by selecting which bilateral markets to supply. If fixed costs are higher in foreign markets than in the domestic market, the firm will export only if its productivity is relatively high; symmetrically, if its productivity is sufficiently low it will sell its product only in the domestic market or exit entirely. Although there are no fixed costs of *production*, the model's crucial feature is fixed costs of *trade*, which give rise to economies of scale at the sectoral level, so that the average cost of export supply declines with increasing export volume.

Table 5.10 summarizes the algebraic representation of the heterogeneous firms sector, which we now go on to derive. On the importer's side, both the aggregate demand for the relevant composite commodity and the associated price level are identical to the canonical model $(q_{h,r'}^A$ and $p_{h,r'}^A)$. The key differences are that the composite is a Dixit-Stiglitz CES aggregation of a continuum of varieties of good h , with each variety produced by a firm that may reside at home or abroad. Letting $\omega_{h,r,r'} \in \Omega_{h,r}$ index the varieties exported from r to r' , and letting $p_{h,r,r'}^H[\omega_{h,r,r'}]$ denote each variety's firm-specific price, the importer's composite price index is

$$p_{h,r'}^A = \left[\sum_r \int_{\Omega_{h,r,r'}} (p_{h,r,r'}^H[\omega_{h,r,r'}])^{1-\sigma_h^H} d\omega_{h,r,r'} \right]^{1/(1-\sigma_h^H)}$$

where σ_h^H is the elasticity of substitution between varieties. Computational implementation of this expression assumes a representative monopolistic firm in (r, r') that sets

Table 5.10 Equations of the heterogeneous firms sector (h)

$$p_{h,r,r'}^A \leq \left[\sum_r n_{h,r,r'} (\tilde{p}_{h,r,r'}^H)^{1-\sigma_h^H} \right]^{1/(1-\sigma_h^H)} \perp q_{h,r,r'}^A$$

(5.80)

$$\tilde{q}_{h,r,r'}^H \geq n_{h,r,r'} (\tilde{p}_{h,r,r'}^H)^{-\sigma_h^H} (p_{h,r,r'}^A)^{\sigma_h^H} q_{h,r,r'}^A \perp \tilde{p}_{h,r,r'}^H$$

(5.81)

$$\tilde{\varphi}_{h,r,r'} = \underline{\varphi}^{-1} \tilde{a}_h^{1/(1-\sigma_h^H)} (n_{h,r,r'} / N_{h,r})^{-1/a} \perp \tilde{\varphi}_{h,r,r'}$$

(5.82)

$$\tilde{p}_{h,r,r'}^H \leq \frac{\sigma_h^H}{\sigma_h^H - 1} \tilde{\varphi}_{h,r,r'} \mathcal{C}_{h,r,r'} p_{h,r}^D \perp \tilde{q}_{h,r,r'}^H$$

(5.83)

$$p_{h,r}^D \mathcal{F}_{h,r,r'}^0 = \tilde{p}_{h,r,r'}^H \tilde{q}_{h,r,r'}^H \tilde{a}_h / \sigma_h^H \perp n_{h,r,r'}$$

(5.84)

$$\Delta p_{h,r}^D \mathcal{F}_{h,r}^S = \frac{\sigma_h^H - 1}{a_h \sigma_h^H N_{h,r}} \sum_{r'} \tilde{p}_{h,r,r'}^H \tilde{q}_{h,r,r'}^H n_{h,r,r'} \perp N_{h,r}$$

(5.85)

$$y_{i,r} \geq \begin{cases} \Delta \mathcal{F}_{h,r}^S N_{h,r} + \sum_{r' \neq r} n_{h,r,r'} \mathcal{F}_{h,r,r'}^0 \\ + \underline{\varphi}^{-1} \tilde{a}_h^{1/(1-\sigma_h^H)} \sum_{r' \neq r} n_{h,r,r'} \mathcal{C}_{h,r,r'} \left(\frac{n_{h,r,r'}}{N_{h,r}} \right)^{-1/a} \tilde{q}_{h,r,r'}^H & h \in i \\ (\zeta_{i,r}^D)^{\sigma_{i,r}^{DM}} (p_{i,r}^D)^{-\sigma_{i,r}^{DM}} (p_{i,r}^A)^{\sigma_{i,r}^{DM}} q_{i,r}^A \\ + \sum_{r' \neq r} \xi_{i,r,r'}^{MM} (p_{i,r}^D)^{-\sigma_{i,r'}^{MM}} (p_{i,r'}^M)^{\sigma_{i,r'}^{MM}} g_{i,r'}^M & \text{otherwise} \end{cases} \perp p_{i,r}^D$$

(5.86)

an average price for its specific variety, $\tilde{p}_{h,r,r'}^H$. Given a mass of $n_{h,r,r'}$ such firms, the formula for the composite price reduces to equation (5.80), which replaces (5.4) as the zero-profit condition for composite good production. By Shephard's lemma, the demand for imports of varieties from firms located in r is given by the corresponding market clearance condition (5.81). The crucial feature is the scale effect associated with the increases in the number of available varieties, $n_{h,r,r'}$, which implies the need to keep track of the number of firms.

Faced with this demand curve for its unique variety, each h -producer maximizes profit by setting marginal cost equal to marginal revenue. To specify the profit maximization problem of the representative firm, we exploit the large-group monopolistic competition assumption that the behavior of an individual firm has no impact on the composite price. Further, we assume that sunk, fixed, and variable costs are all incurred at the marginal opportunity cost of domestic output, which for exporters in r is simply $p_{h,r}^D$. Under these conditions, a monopolistic firm with productivity $\varphi_{h,r,r'}$ has unit production cost $p_{h,r}^D/\varphi_{h,r,r'}$ and maximizes profit via the markup pricing rule:

$$\mathcal{C}_{h,r,r'} p_{h,r}^D / \varphi_{h,r,r'} \geq (1 - 1/\sigma_h^H) p_{h,r,r'}^H, \quad (5.87)$$

where $\mathcal{C}_{h,r,r'}$ is a Samuelsonian iceberg trade cost, which we treat as a market-specific calibration parameter.²³ The key to operationalizing this condition is expressing the average price ($\tilde{p}_{h,r,r'}^H$) in terms of the average productivity level ($\tilde{\varphi}_{h,r,r'}$) by identifying the marginal firm that earns zero profits and then relating the marginal firm to the average firm through the distribution of producer productivities. Melitz (2003) developed a method for doing this, which we follow below.

An individual firm's productivity is assumed to be determined by a random draw from a Pareto distribution with density $\pi[\varphi] = a\varphi^a\varphi^{-1-a}$ and cumulative probability $\Pi[\varphi] = 1 - \varphi^a\varphi^{-a}$, where a is the shape parameter and φ is a lower productivity bound. The centerpiece of the model is that every bilateral link is associated with a productivity level $\varphi_{h,r,r'}^*$ at which optimal markup pricing yields zero profit, such that a firm that draws $\varphi_{h,r,r'}^*$ is the marginal firm. Firms drawing $\varphi_{h,r,r'} > \varphi_{h,r,r'}^*$ earn positive profits and supply market (r, r') . Thus, with a mass $N_{h,r}$ of h -producing firms in region r , the share of producers entering this market is $1 - \Pi[\varphi_{h,r,r'}^*] = n_{h,r,r'}/N_{h,r}$. This property may be exploited by integrating over the density function to obtain the CES-weighted average productivity level:

$$\begin{aligned} \tilde{\varphi}_{h,r,r'} &= \left[\frac{1}{1 - \Pi[\varphi_{h,r,r'}^*]} \int_{\varphi_{h,r,r'}^*}^{\infty} \varphi^{\sigma_h^H - 1} \pi[\varphi] d\varphi \right]^{1/(\sigma_h^H - 1)} = \tilde{a}_h^{1/(1 - \sigma_h^H)} \varphi_{h,r,r'}^* \\ &= \tilde{a}_h^{1/(1 - \sigma_h^H)} \Pi^{-1}[1 - n_{h,r,r'}/N_{h,r}] = \varphi^{-1} \tilde{a}_h^{1/(1 - \sigma_h^H)} (n_{h,r,r'}/N_{h,r})^{-1/a} \end{aligned}$$

²³ Theoretical models make the simplifying assumption that variable trade and transport costs are incurred as a loss of product as the product moves through space—iceberg melt. This is generally inappropriate in CGE models, which must be consistent with transport services payments and government revenues from tariffs and other trade distortions recorded in their calibration data set. We introduce $\mathcal{C}_{h,r,r'}$, however, because it serves as an important unobserved-trade-cost parameter that facilitates a calibration of the model under symmetric demand across all varieties. Like the coefficients $\xi_{i,r}$ and $\kappa_{i,i,r',r}$ in the canonical model, the values of $\mathcal{C}_{h,r,r'}$ are fixed such that the model replicates the trade equilibrium in the benchmark. See Balistreri and Rutherford (2012) for in-depth discussion of the equivalence between strategies to calibrate the heterogeneous-firms model that employ idiosyncratic demand biases, versus those that use unobserved trade costs.

where $\tilde{a}_h = (a + 1 - \sigma_h^H)/a$ is a parameter. The above expression can be substituted into (5.87) to yield the representative monopolistic firm's zero-profit condition (5.83).

To find the level of productivity we must pin down the number of firms in (r, r') . This number is defined implicitly by the free-entry condition for the marginal firm, which breaks even with an operating profit that just covers its fixed operating cost of entering the market. By the Marshallian large group assumption, profit is the ratio of a firm's revenue to the elasticity of substitution among varieties, while the fixed cost can be expressed as the opportunity cost, $p_{h,r}^D \mathcal{F}_{h,r,r'}^O$. Equations (5.87) and (5.81) can be combined to express revenue as $p_{h,r,r'}^H q_{h,r,r'}^H = n_{h,r,r'} (p_{h,r,r'}^H)^{1-\sigma_h^H} (p_{h,r'}^A)^{\sigma_h^H} q_{h,r'}^A \propto \varphi_{h,r,r'}^{\sigma_h^H-1}$, so that the ratio of average to marginal revenue is related to the ratio of the average and marginal productivity by $(\tilde{\varphi}_{h,r,r'}/\varphi_{h,r,r'}^*)^{\sigma_h^H-1} = \tilde{a}^{-1}$. This simplification enables the free-entry condition to be recast in terms of the representative firm's average variables as the zero-profit condition (5.84).

Similar logic applies to the total mass of region- r firms, which is defined implicitly by a free-entry condition that balances a firm's sunk cost against expected profits over its lifetime. Assuming that each firm has a flow probability of disappearance, Δ , steady-state equilibrium requires an average of $\Delta N_{h,r}$ firms to be replaced each period at an aggregate nominal opportunity cost of $\Delta p_{h,r}^D \mathcal{F}_{h,r}^S N_{h,r}$. Thus, ignoring discounting or risk aversion, the representative firm's profit must be large enough to cover an average loss of $\Delta p_{h,r}^D \mathcal{F}_{h,r}^S$. On the other side of the balance sheet, expected aggregate profit is simply the operating profit in each market $(\tilde{p}_{h,r,r'}^H \tilde{q}_{h,r,r'}^H / \sigma_h^H - p_{h,r}^D \mathcal{F}_{h,r,r'}^O)$ weighted by the probability of operating in that market $(n_{h,r,r'} / N_{h,r})$. Using (5.84) to substitute out fixed operating costs, the free-entry condition equating average sunk costs with average aggregate profit is given by the zero-profit condition (5.85). With this condition the heterogeneous-firm trade equilibrium is fully specified. The final requirement for integrating the heterogeneous-firm sector into the framework of the canonical model is an elaboration of the h -sector's supply-demand balance for the domestic commodity. The market clearance condition associated with $p_{h,r}^D$ tracks the disposition of domestic output into the various sunk, fixed, and variable costs as in (5.86).

Operationalizing this model requires us to reconcile our algebraic framework for heterogeneous firms with standard trade flow accounts such as figure 5.1. To do so we need three pieces of exogenous data: the elasticity of substitution between varieties (σ_h), the Pareto distribution parameters (a and φ), and an approximation of bilateral fixed operating costs ($\mathcal{F}_{h,r,r'}^O$).²⁴ The calibration proceeds in five steps.

²⁴ Numerical results are typically not sensitive to the scale and distribution of benchmark fixed costs because only the value of the elasticity parameter determines the markup over marginal cost. Assumptions about fixed operating costs simply scale our measure of the number of firms: the larger the assumed values of $\mathcal{F}_{h,r,r'}^O$, the larger the implied per-firm revenue, and, with a given value of trade, the smaller the calibrated initial number of firms.

1. *Average firm revenue*: Plugging estimates of the fixed cost and the substitution elasticity into the zero-cutoff-profit condition (5.84), along with a typical choice of units ($p_{h,r}^D = 1 \ \forall h, r$), pins down the revenue of the average firm operating in each bilateral market ($\tilde{p}_{h,r,r'}^H \tilde{q}_{h,r,r'}^H$).
2. *The number of firms*: The fact that the total value of trade is the product of the number of firms and average revenue means that the trade account in the SAM can be divided by the result of step 1 to give the benchmark value of $n_{h,r,r'}$. Plugging this quantity into equation (5.85) enables us to derive the total mass of firms, $N_{h,r}$. The key is to treat the flow of sunk cost payments ($\Delta \mathcal{F}_{h,r}^S$) as a free composite parameter whose value is chosen to scale the measure of the total number of firms relative to those operating on each bilateral link. In performing this procedure it is necessary to ensure that $n_{h,r,r'}/N_{h,r} < 1$ for the largest market supplied by r (typically the domestic market, $r = r'$).
3. *Average firm productivity*: Substituting the shares of firms on each bilateral link from step 2 into equation (5.82) facilitates direct calculation of the average productivity level, $\tilde{\varphi}_{h,r,r'}$.
4. *Average firm price and output*: Multiplying both sides of (5.81) by the firm-level average price ($\tilde{p}_{h,r,r'}$) expresses average revenue from step 1 in terms of the average firm-level price and the composite price and quantity. By choosing composite units such that $p_{h,r'}^A = 1$ (which allows us to interpret $q_{h,r'}$ as the region r' gross consumption observed in the trade accounts), we can solve for $\tilde{p}_{h,r,r'}$, and, in turn, $\tilde{q}_{h,r,r'}$.
5. *Iceberg trade costs*: Unobserved trade costs ($\mathcal{C}_{h,r,r'}$) can be recovered from the markup pricing rule (5.83).

5.4.3 The Heterogeneous-Firm CGE Model in Action: Liberalizing Trade in Services

We illustrate how the results of the heterogeneous-firms specification can differ from those of our canonical model by considering the liberalization of trade in services in poor countries. The model discussed in section 5.4.2 is calibrated on a stylized aggregation of the GTAP version 7 database, which divides the world into three blocs that serve as the regions, (OECD, middle-income, and low-income countries) whose industries are grouped into three broad sectors (manufacturing, services, and a rest-of-economy aggregate). We use the heterogeneous-firms structure to model the manufacturing and services sectors. For our exogenous parameters we use $\sigma_h = 3.8$, taken from Bernard et al. (2003), and $a = 4.582$, $\varphi = 0.2$ and a vector of values $\mathcal{F}_{h,r,r'}^O$, taken from Balistreri et al. (2011). To capture the importance of business services, we model production using Balistreri et al.'s (2009) nested CES structure in every sector, where value added and intermediate purchases of services are combined with an elasticity of substitution of 1.25 to form a composite input commodity.

From a practical modeling standpoint, the non-convexity generated by positive feedback from expansion of exports, productivity improvement, and decline in the average cost of exporting can easily render the solution to a CGE model infeasible in the absence of countervailing economic forces that limit the general equilibrium response to increasing returns. To achieve the requisite degree of attenuation we introduce countervailing diminishing returns in the production of *h*-goods by using a specific-factor input formulation for the heterogeneous-firms sectors. This device puts a brake on output expansion by limiting the composite good's supply, which with fully reversible production would be almost perfectly elastic. Our approach is to allocate a portion of firm revenues to payments to a sector-specific primary "fixed-factor" resource. The fixed factor's benchmark cost share, as well as the elasticities of substitution between it and other components of the composite input, are numerically calibrated to be consistent with the central values given in Balistreri et al. (2009), and imply a composite-input supply elasticity value equal to four.

We investigate the impacts of the low-income-region liberalizing tariff and non-tariff barriers to trade in services. We model the latter based on Balistreri et al.'s (2009) estimates for Kenya, where barriers in *ad valorem* terms range from 0 percent to 57 percent with a median value of 13 percent, which we use as an estimate of the available savings from regulatory reforms. The service sector is calibrated so that fixed costs account for about 26 percent of revenues, which suggests that a 25 percent reduction in these costs is roughly equivalent to a 50 percent reduction in regulatory barriers. We simulate five liberalization scenarios, three that are a mix of reductions in low-income countries' import tariffs on services and manufactures and fixed costs for service-sector firms, and two that simulate bilateral trade integration with the OECD. Table 5.11 gives details of the scenarios, along with key model results. The first thing to note is the relative welfare impact of the regulatory reform scenarios in panel A. Unilateral regulatory reform that reduces the fixed costs of services firms in the low-income region generates a welfare gain of 7.9 percent. In contrast, given optimal tariff considerations, unilateral tariff reductions lead to losses of 1 percent under the heterogeneous-firms structure and 0.5 percent under the Armington structure. Considering bilateral policies of low-income countries and the OECD, combining 50 percent tariff reductions with 50 percent reductions in bilateral fixed trade costs results in an eighteen fold increase in welfare gains (from 0.4 percent to 6 percent). Moreover, even in the tariff-only bilateral scenario the heterogeneous-firm model generates far larger gains than does its Armington counterpart (0.01 percent versus 0.39 percent). The heterogeneous-firm representation of trade therefore has fundamentally important implications for measuring policy reforms. Panels B and C of table 5.11 highlight two key sources of differential economic impact: productivity shifts associated with changes in the selection of firms into export markets on the supply side, and changes in variety associated with the number and composition of services thereby produced.

Unilateral regulatory reforms generate sizable productivity and variety gains for low-income countries. We report the gains associated with new varieties of the services

Table 5.11 Liberalization of trade in services: A stylized CGE assessment

	Full unilateral reform ^a	Regulatory reform only ^b	Unilateral tariff reduction ^c	OECD free trade area ^d	OECD trade reform ^e
A. Welfare impacts under different specifications of trade (% equivalent variation)					
Armington	–	–	–0.50	0.01	–
heterogeneous firms	6.90	7.86	–1.01	0.34	6.09
B. Productivity impacts (% change in $\sum_s \tilde{\varphi}_{r,r'} N_{r,r'}$)					
Services					
OECD	–0.02	–0.03	0.01	–0.00	–0.01
Middle income	–0.03	–0.08	0.06	–0.01	–0.07
Low income	29.12	28.83	0.20	0.39	6.95
Manufacturing					
OECD	0.01	0.05	–0.04	–0.19	–0.62
Middle income	–0.07	–0.09	0.02	–0.02	–0.25
Low income	–1.74	2.19	–4.14	–0.75	17.02
C. Variety impacts (% change in Feenstra's ratio)					
Services					
OECD	0.03	0.02	0.01	0.01	0.08
Middle income	0.06	0.03	0.04	–0.02	–0.07
Low income	10.33	10.52	–0.19	0.04	2.41
Manufacturing					
OECD	0.01	0.00	0.01	0.01	0.33
Middle income	0.04	–0.01	0.05	–0.02	–0.08
Low income	0.98	0.97	–0.03	0.17	7.45

^a Low-income countries unilaterally reduce tariffs on imports of manufactures and services by 50% and reduce fixed costs of service firms operating within their borders by 25%.

^b Services firms in low-income countries see their fixed costs reduced by 25%.

^c Low-income countries unilaterally reduce tariffs on imports of manufactures and services by 50%.

^d Free trade agreement with OECD that reduces bilateral tariffs by 50%.

^e Free trade agreement with OECD that reduces bilateral tariffs by 50% and reduces bilateral fixed costs by 50%.

good, calculated according to Feenstra's (2010) expenditure-share based method (see Balistreri and Rutherford (2012) for details). Directly interpreting the change in $n_{h,r,r'}$ as an indicator of the underlying change in varieties can be misleading because liberalization may induce the replacement of high-price, low-quantity domestic varieties with foreign varieties. Indeed, though the net number of varieties has been shown to decline with trade costs (Baldwin and Forslid 2010; Feenstra 2010), this does not by itself indicate a gain or a loss because each variety has a different price.

We close this section by emphasizing that our simulations rely on a set of parameters not commonly used by CGE models. In particular, the values of the shape parameters of the distribution of firm productivities and the bilateral fixed costs of trade are drawn from the structural estimation procedures developed in Balistreri et al. (2011). It has traditionally been the case that these sorts of parameters are estimated using econometric models that are divorced from the broader theory of general equilibrium or its particular algebraic elaboration in the numerical simulation to be parameterized. By contrast, structural estimation techniques bring econometric and numerical simulation modeling together in a consistent fashion by imposing theory-based restrictions on the values of estimated parameters. An important example is Anderson and van Wincoop's (2003) widely cited study, the welfare impacts of which have been shown to be inconsistent with its estimation procedure unless the latter is properly constrained by restrictions based on the conditions of general equilibrium (Balistreri and Hillberry 2008, 2007).

Structural estimation of the parameters of our heterogeneous-firm model proceeds by isolating the complementarity conditions that characterize the trade equilibrium and imposing them as a set of constraints on econometric estimation. Following Balistreri et al. (2011), consider a vector-valued function that specifies the equilibrium in bilateral trade markets conditional on the observed benchmark demand for the regional domestic-import composites, $q_{h,r}^A$, and the domestic supply of the traded good, $y_{h,r}$. The system of equations may be stacked to generate the vector-valued function $\Xi(\mathcal{V}, \mathbf{\Gamma}) = \mathbf{o}$, which implicitly maps the vector of parameters, $\mathbf{\Gamma}$, to the vector of endogenous variables, $\mathcal{V}$. The key parameters to be estimated are the Pareto shape coefficient and the bilateral fixed costs of trade, $a, \mathcal{F}_{h,r,r'}^O \in \widehat{\mathbf{\Gamma}}$, while the endogenous variable that we are interested in reproducing is the value of bilateral trade, $\tilde{q}_{h,r,r'}^H \tilde{p}_{h,r,r'}^H n_{h,r,r'} \in \widehat{\mathcal{V}}$. Using $\overline{\mathcal{V}}$ to denote the corresponding vector of observations of the variables, we can find the best estimates of the parameters by solving the nonlinear program

$$\min_{\widehat{\mathbf{\Gamma}}, \widehat{\mathcal{V}}} \left\{ \left(\widehat{\mathcal{V}} - \overline{\mathcal{V}} \right)^2 \mid \Xi(\widehat{\mathcal{V}}, \widehat{\mathbf{\Gamma}}, \mathbf{\Gamma}^*) = \mathbf{o}, \mathbf{\Gamma}^* = \mathcal{K} \right\}$$

where $\mathbf{\Gamma}^*$ are a set of assumed parameters and $\mathcal{K}$ is a vector of constants.

This methodology has an appealing general applicability, but in practice it is severely constrained by the degrees of freedom offered by the data, which permit only a limited number of structural parameters to be estimated. For example, in their central case, Balistreri et al. (2011) only estimate the Pareto shape parameter and fixed trade costs. Furthermore, the structure of bilateral fixed costs is such that only each region's vectors of aggregate inward and outward costs can be estimated, not the full bilateral matrix. These shortcomings notwithstanding, the need to link CGE models' structural representations of the economy to underlying empirically determined parameters will likely mean that structural estimation will continue to be an area of active research in the foreseeable future.

5.5 CONCLUSIONS

This chapter documents contemporary computable general equilibrium (CGE) approaches to policy evaluation and economic consequence analysis. The algebraic formulation of a canonical multiregional model of world trade and production, followed by extensive discussion of the myriad ways in which this standard class of models is employed in the study of a wide variety of topics: economic integration and development, public economics, energy policy, and climate change mitigation, impacts, and adaptation. The standard model is then extended to consider specific advances in CGE research. First, the incorporation of detailed process-level production technologies is shown using bottom-up techniques that enhance the representation of technical progress in, and substitution among, discrete activities that play important roles at the subsectoral level. This approach is especially popular in climate change mitigation research, in which it allows CGE models to credibly represent the way policies to reduce GHG emissions impact fuel use and the penetration of alternative energy supply options.

In the context of trade policy the canonical model is extended using a trade structure that is currently favored by new theories of trade in which heterogeneous services and manufacturing firms supply differentiated products and engage in monopolistic competition. Critical to this new theory is the selection of firms with different productivities (heterogeneous firms) into different national markets, an extension which illustrates important margins for the gains from trade that are not apparent in the standard model. In particular, there are variety effects as the number and composition of available goods and services change with policy, as well as resource reallocation within an industry toward more productive firms that is capable of boosting the sector's overall productivity.

The algebraic formulations presented offer an illustrative guide to, and documentation of, key twenty-first century applications. Though technically advanced, these models offer the same advantages for policy evaluation that have been the hallmark of CGE models for the past three decades. Their theoretically grounded structure allows for an investigation of both the drivers behind specific outcomes and the sensitivity of specific outcomes to alternative assumptions. Our hope is that the stylized examples presented here, together with the rich bibliography of their recent application, prove useful as a guide to beginners seeking to understand CGE models' structure and internal interactions, as well as to experienced practitioners interested in advanced and alternative modeling techniques.

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CHAPTER 6

MULTIFRACTAL MODELS IN FINANCE

Their Origin, Properties, and Applications

THOMAS LUX AND MAWULI SEGNON

6.1 INTRODUCTION

One of the most important tasks in financial economics is the modeling and forecasting of price fluctuations of risky assets. For analysts and policy makers, volatility is a key variable for understanding market fluctuations. Analysts need accurate forecasts of volatility as an indispensable input for tasks such as risk management, portfolio allocation, value-at-risk assessment, and option and futures pricing. Asset market volatility also plays an important role in monetary policy. Repercussions from the recent financial crisis on the global economy show how important it is to take into account financial market volatility in conducting effective monetary policy.

In financial markets, volatility is a measure for fluctuations of the price p of a financial instrument over time. It cannot be directly observed but, rather, has to be estimated via appropriate measures or as a component of a stochastic asset pricing model. As an ingredient of such a model, volatility may be a latent stochastic variable itself (as it is in so-called stochastic volatility models as well as in most multifractal models) or it might be a deterministic variable at any time t (as it is in the case in so-called GARCH-type models). For empirical data, volatility may simply be calculated as the sample variance or sample standard deviation. Ding et al. (1993) propose using absolute returns for estimating volatility. Davidian and Carroll (1987) demonstrate that this measure is more robust against asymmetry and non-normality than others (see also Taylor 1986; Ederington and Guan 2005). Another way to measure daily volatility is to use squared returns or any other absolute power of returns. Indeed, different powers show slightly different time-series characteristics, and the multifractal model is designed to capture the complete range of behavior of absolute moments.

The concept of realized volatility (RV) has been developed by Andersen et al. (2001) as an alternative measure of the variability of asset prices (see also Barndorff-Nielsen and Shephard 2002). The notion of RV means that daily volatility is estimated by summing up intra-day squared returns. This approach is based on the theory of quadratic variation, which suggests that RV should provide a consistent and highly efficient nonparametric estimator of asset return volatility over a given discrete interval under relatively parsimonious assumptions about the underlying data-generating process. Other methods used for measuring volatility are the maximum likelihood method developed by Ball and Torous (1984) and the high-low method proposed by Parkinson (1980). All these measures of financial market volatility show salient features that are well documented as *stylized facts*: volatility clustering, asymmetry and mean reversion, co-movements of volatilities across assets and financial markets, stronger correlation of volatility compared to that of raw returns, (semi) heavy tails of the distribution of returns, anomalous scaling behavior, changes in shape of the return distribution over time horizons, leverage effects, asymmetric lead-lag correlation of volatilities, strong seasonality, and some dependence of scaling exponents on market structure (see section 6.2).

During the past few decades, an immense body of theoretical and empirical studies has been devoted to formulating appropriate volatility models (see Andersen et al. 2006 for a review of volatility modeling and Poon and Granger 2003 for a review of volatility forecasting). With Mandelbrot's famous work on the fluctuations of cotton prices in the early 1960s (Mandelbrot 1963), economists had already learned that the standard geometric Brownian motion proposed by Bachelier (1900) is unable to reproduce these stylized facts. In particular, the fat tails and the strong correlation observed in volatility are in sharp contrast to the "mild," uncorrelated fluctuations implied by models with Brownian random terms. The first step toward covering time variation of volatility had been taken with models using mixtures of distributions as proposed by Clark (1973) and Kon (1984). Econometric modeling of asset price dynamics with time-varying volatility got started with the generalized autoregressive conditional heteroscedasticity (GARCH) family and its numerous extensions (see Engle 1982). The closely related class of stochastic volatility (SV) models adds randomness to the dynamic law governing the time variation of second moments (see Ghysels et al. 1996 and Shephard 1996 for reviews of SV models and their applications).

In this chapter we focus on a new, alternative avenue for modeling and forecasting volatility developed in the literature at about the turn of the twenty-first century. In contrast to the existing models, the source of heterogeneity of volatility in these new models stems from the time variation of local regularity in the price path (see Fisher et al. 1997). The background of these models is the theory of multifractal measures originally developed by Mandelbrot (1974) in order to model turbulent flows. These multifractal processes initiated a broad current of literature in statistical physics refining and expanding the underlying concepts and models (Kahane and Peyrière 1976; Holley and Waymire 1992; Falconer 1994; Arbeiter and Patzschke 1996; Barral 1999). The

formal analysis of such measures and processes, the so-called multifractal formalism, has been developed by Frisch and Parisi (1985), Mandelbrot (1989, 1990), and Evertsz and Mandelbrot (1992), among others.

A number of early contributions have, indeed, pointed out certain similarities of volatility to fluid turbulence (Vassilicos et al. 1994; Ghashghaie et al. 1996; Galluccio et al. 1997; Schmitt et al. 1999), while theoretical modeling in finance using the concept of multifractality started with the adaptation to an asset-pricing framework of Mandelbrot's (1974) model by Mandelbrot et al. (1997).

Subsequent literature has moved from the more combinatorial style of the Multifractal Model of Assets Returns (MMAR) of Mandelbrot, Calvet, and Fisher (developed in the sequence of Cowles Foundation working papers authored by Calvet et al. 1997, Fisher et al. 1997, and Mandelbrot et al. 1997) to iterative, causal models of similar design principles: the Markov-switching multifractal (MSM) model proposed by Calvet and Fisher (2004) and the Multifractal Random Walk (MRW) by Bacry et al. (2001) constitute the second generation of multifractal models, and have more or less replaced the somewhat cumbersome first-generation MMAR in empirical applications.

The chapter is organized as follows. Section 6.2 presents an overview of the salient stylized facts about financial data and discusses the potential of the classes of GARCH and stochastic volatility models to capture these stylized facts. In Section 6.3 we introduce the baseline concept of multifractal measures and processes and provide an overview of different specifications of multifractal volatility models. Section 6.4 introduces the different approaches for estimating MF models and forecasting future volatility. Section 6.5 reviews empirical results of the application and performance of MF models, and section 6.6 concludes.

6.2 STYLIZED FACTS OF FINANCIAL DATA

With the availability of high-frequency time series for many financial markets from about the 1960s, the statistical properties of these series was explored in a large strand of literature to which economists, statisticians, and physicists have contributed. The two main universal features or stylized facts characterizing practically every series of interest at the high end of the frequency spectrum (daily or intra-daily) are known by the catchwords “fat tails” and “volatility clustering.” The use of multifractal models is motivated to some extent by both of these properties, but multifractality (or, as it is sometimes also called, multi-scaling or multi-affinity) proper is a more subtle feature that gradually started to emerge as an additional stylized fact since the 1990s. In the following we provide a short review of the historical development of the current stage of knowledge and the quantification of all these features, capturing in passing some less well-known statistical properties typically found in financial returns. The data format

of interest is typically returns, that is, relative price changes $\tilde{r}_t = \frac{p_t - p_{t-1}}{p_{t-1}}$, which for high-frequency data are almost identical to log-price changes $r_t = \ln(p_t) - \ln(p_{t-1})$ with p_t the price at time t (e.g., at daily or higher frequency).

6.2.1 Fat Tails

This property relates to the shape of the unconditional distribution of a time series of returns. The first “hypothesis” concerning the distribution of price changes was formulated by Bachelier (1900), who in his PhD thesis, titled “Théorie de la Spéculation,” assumed that they follow a Normal distribution. As is well known, many applied areas of financial economics such as option pricing theory (Black and Scholes 1973) and portfolio theory (Markowitz 1959) have followed this assumption, at least in their initial stages. The justification for this assumption is provided by the law of large numbers: If price changes at the smallest unit of time are independently and identically distributed random numbers (perhaps driven by the stochastic flow of new information), returns over longer intervals can be seen as the sum of a large number of such IID observations and, irrespective of the distribution of their summands, should under some weak additional assumptions converge to the Normal distribution. Although this seemed plausible, and the resulting Gaussian distribution would also come in very handy for many applied purposes, Mandelbrot (1963) demonstrated that empirical data are distinctly non-Gaussian, exhibiting excess kurtosis and higher probability mass in the center and in their tails than did the Normal distribution. As can be confirmed with any sufficiently long record of stock market, foreign exchange, or other financial data, the Gaussian distribution can always be rejected with statistical significance beyond all usual boundaries, and the observed largest historical price changes would be so unlikely under the Normal law that one would have to wait for horizons beyond at least the history of stock markets to see them occur with non-negligible probability.

Mandelbrot (1963) and Fama (1963), as a consequence, proposed the so-called Lévy stable distribution laws as an alternative for capturing these fat tails. These laws were motivated by the fact that in a generalized version of the central limit law, dispensing with the assumption of a finite second moment, sums of IID random variables converge to these more general distributions (with the Normal being a special case of the Lévy stable obtained in the borderline case of a finite second moment). The desirable stability property, therefore, indicates the choice of the Lévy stable that also has a shape which—in the standard case of infinite variance—is characterized by fat tails. In a sense, the Lévy stable model remained undisputed for about three decades (although many areas of financial economics would have preferred to continue using the Normal as their working model), and economists indeed contributed to the advancement of statistical techniques for estimating the parameters of the Lévy distributions (Fama and Roll 1971; McCulloch 1986). When physicists started to explore financial time series, the

Lévy stable law was discovered again (Mantegna 1991), although new developments in empirical finance had already allowed to reject this erstwhile time-honored hypothesis.

These new insights were basically due to a different perspective: Rather than attempting to model the entire distribution, one let the tails “speak for themselves.” The mathematical foundations for such an approach are provided by statistical extreme value theory (e.g., Reiss and Thomas 1997). Its basic tenet is that the extremes and the tail regions of a sample of IID random variables converge in distribution to one of only three types of limiting laws. For tails, these are exponential decay, power-law decay, and the behavior of distributions with a finite endpoint of their support. “Fat tails” is often used as a synonym for power-law tails, so that the highest realizations of returns would obey a law like $\text{Prob}(x_t < x) \sim 1 - x^{-\alpha}$ after appropriate normalization (i.e. after some transformation $x_t = ar_t + b$). The universe of fat-tailed distributions can, then, be indexed by their tail index α with $\alpha \in (0, \infty)$. Lévy stable distributions are characterized by tail indices α below 2 (2 characterizing the case of the Normal distribution). All other distributions with a tail index smaller than 2 would converge under summation to the Lévy stable with the same index, while all distributions with an asymptotic tail behavior with $\alpha > 2$ would converge under aggregation to the Gaussian. This demarcates the range of relevance of the standard central limit law and its generalized version.

Jansen and de Vries (1991), Koedijk et al. (1990), and Lux (1996) are examples of a literature that emerged during the 1990s using semiparametric methods of inference to estimate the tail index without assuming a particular shape of the entire distribution. The outcome of these and other studies is a tail index α in the range of 3 to 4 that now counts as a stylized fact (see Guillaume et al. 1997; Gopikrishnan et al. 1998). Intra-daily data nicely confirm results obtained for daily records in that they provide estimates for the tail index that are in line with the former (Dacorogna et al. 2001; Lux 2001a), and, therefore, confirm the expected stability of the tail behavior under time aggregation as predicted by extreme-value theory. The Lévy stable hypothesis can thus be rejected (confidence intervals of α typically exclude the possibility of $\alpha < 2$). This conclusion agrees with the evidence that the variance stabilizes with increasing sample size and does not explode. Falling into the domain of attraction of the Normal distributions, the overall shape of the return distribution would have to change, that is, get closer to the Normal under time aggregation.¹ This is indeed the case, as has been demonstrated by Teichmoeller (1971) and many later authors. Hence, the basic finding concerning the unconditional distribution is that it converges toward the Gaussian but is distinctly different from it at daily (and higher) frequencies. Figure 6.1 illustrates the very homogeneous and distinctly non-Gaussian and non-Lévy nature of stock price fluctuations. The four major South African stocks displayed in the figure could be

¹ Although, in fact, the tail behavior would remain qualitatively the same under time aggregation, the asymptotic power law would apply in a more and more remote tail region only and would, therefore, become less and less visible for finite data samples under aggregation. There is, thus, both convergence toward the Normal distribution and stability of power-law behavior in the tail under aggregation. The former governs the complete shape of the distribution, but the latter applies further and further out in the tail and would only be seen with a sufficiently large number of observations.

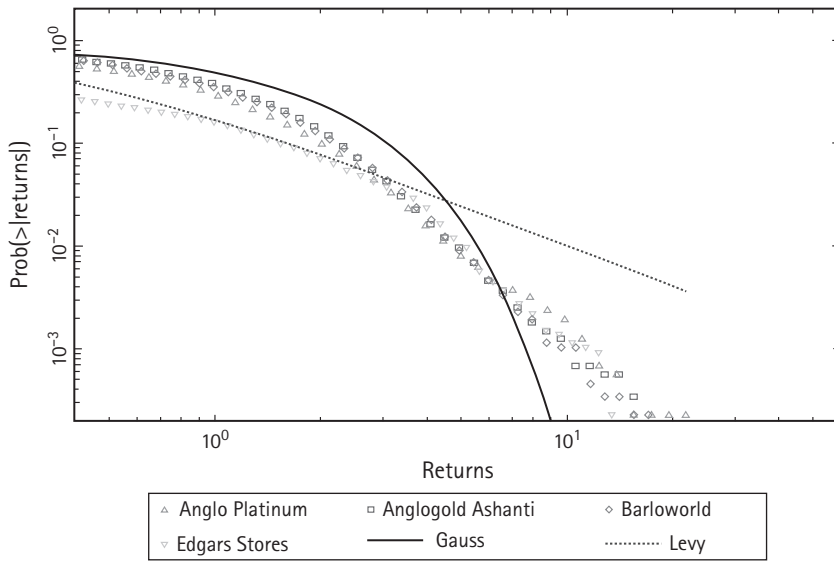


FIGURE 6.1 Cumulative distribution for daily returns of four South African stocks, 1973–2006. The solid lines correspond to the Gaussian and the Lévy distributions. The tail behavior of all stocks is different from that of both the Gaussian and the Lévy distribution (for the latter, a characteristic exponent $\alpha = 1.7$ has been chosen that is a typical outcome of estimating the parameters of this family of distributions for financial data).

replaced by almost any other time series of stock markets, foreign exchange markets, or other financial markets. Estimating the tail index α by a linear regression in this log-log plot would lead to numbers very close to the celebrated “cubic law.”

The particular non-Normal shape then also motivates the quest for the best nonstable characterization at intermediate levels of aggregation. From a huge literature that has tried mixtures of Normals (Kon 1984) as well as a broad range of generalized distributions (Eberlein and Keller 1995; Behr and Pötter 2009; Fergusson and Platen 2006) it appears that the distribution of daily returns is quite close to a Student- t with three degrees of freedom. A tail index between 3 and 4 is typically found for stock and foreign exchange markets, but some other markets are sometimes found to have fatter tails (e.g., Koedijk et al. 1992 for black market exchange rates and Matia et al. 2002 for commodities).

6.2.2 Volatility Clustering

The slow convergence to the Normal might be explained by dependence in the time series of returns. Indeed, although the limiting laws of extreme value theory would still apply for certain deviations from IID behavior, dependence could slow down

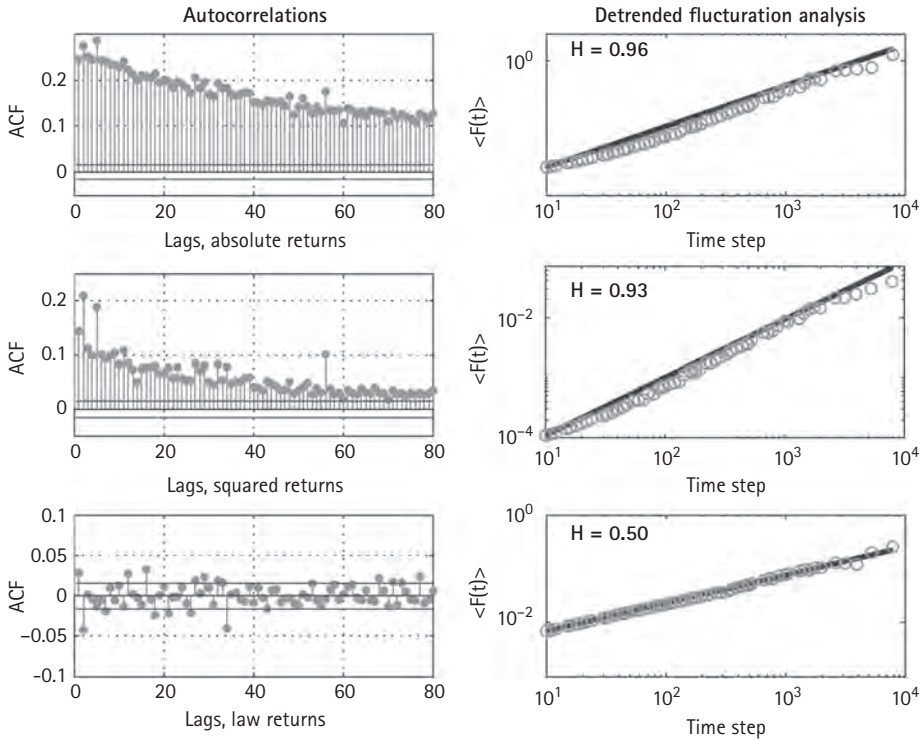


FIGURE 6.2 Long-term dependence observed in the absolute and squared returns of the S&P 500 (left upper and central panels). In contrast, raw returns (lower left panel) are almost uncorrelated. The determination of the corresponding Hurst exponent H via the so-called detrended fluctuation analysis (Chen et al. 2002) is displayed in the right-hand panels. Note that we obtain the following scaling of the fluctuations (volatility): $\mathbb{E}[F(t)] \sim t^H$. $H = 0.5$ corresponds to absence of long-term dependence, while $H > 0.5$ indicates a hyperbolic decay of the ACF, i.e., long-lasting autoregressive dependence.

convergence dramatically, leading to a long regime of pre-asymptotic behavior. That returns are characterized by a particular type of dependence has also been well known since at least Mandelbrot (1969). This dependence is most pronounced and, in fact, plainly visible in absolute returns, squared returns, or any other measure of the extent of fluctuations (volatility); see figure 6.2. In all these measure there is long-lasting, highly significant autocorrelation (see Ding et al. 1993). With sufficiently long time series, significant autocorrelation can be found for time lags (of daily data) up to a few years. This positive feedback is described as volatility clustering or “turbulent (tranquil) periods being more likely to be followed by still turbulent (tranquil) periods than vice versa.” Whether there is (additional) dependence in the raw returns is subject to debate. Most studies do not find sufficient evidence for giving up the martingale hypothesis, although a long-lasting but small effect might be hard to capture statistically. Ausloos

et al. (1999) is an example of a study claiming to have identified such effects. Lo (1991) has proposed a rigorous statistical test for long-term dependence that for the most part does not indicate deviations from the null hypothesis of short memory for raw asset returns but shows strongly significant evidence of long memory in squared or absolute returns. Similar to the classification of types of tail behavior, short memory comes with exponential decay of the autocorrelation function, while one speaks of long memory if the decay follows a power law. Evidence for the latter type of behavior has also accumulated over time. Documentation of hyperbolic decline in the autocorrelations of squared returns can be found in Dacorogna et al. (1993), Crato and de Lima (1994), Lux (1996), and Mills (1997). Lobato and Savin (1998) first claimed that such long-range memory in volatility measures is a universal stylized fact of financial markets, and Lobato and Velasco (2000) document similar long-range dependence in trading volume. Again, particular market designs might lead to exceptions from the typical power-law behavior. Gençay (2001) as well as Ausloos and Ivanova (2000) report atypical behavior in the managed floating of European currencies during the period of the European Monetary System. Presumably due to leverage effects, stock markets also exhibit correlation between volatility and raw (i.e., signed) returns (LeBaron 1992), that is absent in foreign exchange rates.

6.2.3 Benchmark Models: GARCH and Stochastic Volatility

In financial econometrics, volatility clustering has since the 1980s spawned a voluminous literature on a new class of stochastic processes capturing the dependence of second moments in a phenomenological way. Engle (1982) first introduced the autoregressive conditional heteroscedasticity model (ARCH) which has been generalized to GARCH by Bollerslev (1986). It models returns as a mixture of Normals with the current variance being driven by a deterministic difference equation:

$$r_t = \sqrt{h_t} \varepsilon_t \quad \text{with} \quad \varepsilon_t \sim N(0, 1) \quad (6.1)$$

and

$$h_t = \alpha_0 + \sum_{i=1}^p \alpha_i r_{t-i}^2 + \sum_{j=1}^q \beta_j h_{t-j}, \quad \alpha_0 > 0, \alpha_i, \beta_j > 0. \quad (6.2)$$

Empirical applications usually find a parsimonious GARCH(1,1) model (i.e., $p = q = 1$) sufficient, and when estimated, the sum of the parameters $\alpha_1 + \beta_1$ turns out to be close to the nonstationary case (or, expressed differently, for the most part only a constraint on the parameters prevents them from exceeding 1 in their sum, which would lead to nonstationary behavior). Different extensions of GARCH were developed in the literature with the objective of better capturing the stylized facts. Among them there are the exponential GARCH (EGARCH) model proposed by Nelson (1991),

which accounts for asymmetric behavior of returns, the threshold GARCH (TGARCH) model of Rabemananjara and Zakoian (1993), which takes into account the leverage effects, the regime-switching GARCH (RS-GARCH) developed by Cai (1994), and the Integrated GARCH (IGARCH) introduced by Engle and Bollerslev (1986), which allows for capturing high persistence observed in volatility of return time series. Itô diffusion or jump-diffusion processes can be obtained as a continuous time limit of discrete GARCH sequences (see Nelson 1990; Drost and Werker 1996).

To capture stochastic shocks to the variance process, Taylor (1986) introduced the class of stochastic volatility models whose instantaneous variance is driven by

$$\ln(h_t) = k + \varphi \ln(h_{t-1}) + \tau \xi_t, \quad \xi_t \sim N(0, 1). \quad (6.3)$$

This approach has been refined and extended in many ways, as well. The SV process is more flexible than the GARCH model and provides more mixing because of the co-existence of shocks to volatility and return innovations (see Gavrishchaka and Ganguli 2003). In terms of statistical properties, one important drawback of at least the baseline formalizations (6.1) to (6.3) is their implied exponential decay of the autocorrelations of measures of volatility, which is in contrast to the very long autocorrelations mentioned above. Both the elementary GARCH and the baseline SV model are characterized by only short-term rather than long-term dependence.

To capture long memory, GARCH and SV models have been expanded by allowing for an infinite number of lagged volatility terms instead of the limited number of lags appearing in (6.2) and (6.3). To obtain a compact characterization of the long-memory feature, a fractional differencing operator has been used in extensions leading to the fractionally integrated GARCH (FIGARCH) model of Baillie et al. (1996) and the long-memory stochastic volatility model of Breidt et al. (1998).² An interesting intermediate approach is the so-called heterogeneous ARCH (HARCH) model of Dacorogna et al. (1998), which considers returns at different time aggregation levels as determinants of the dynamic law governing current volatility. Under this model, equation (6.2) would have to be replaced by

$$h_t = c_0 + \sum_{j=1}^n c_j r_{t, t-\Delta t_j}^2, \quad (6.4)$$

where $r_{t, t-\Delta t_j} = \ln(p_t) - \ln(p_{t-\Delta t_j})$ are returns computed over different frequencies. The development of this model was motivated by the finding that volatility on fine time scales can be explained to a larger extent by coarse-grained volatility than vice versa (Müller et al. 1997). Hence, the right-hand side covers local volatility at various lower frequencies than the time step of the underlying data ($\Delta t_j = 2, 3, \dots$). As we will see in the following, multifractal models have a closely related structure, but they model the hierarchy of volatility components in a multiplicative rather than an additive format.

² The “self-excited multifractal model” proposed by Filimonov and Sornette (2011) appears closer to this model than to models from the class of multifractal processes discussed below.

6.2.4 A New Stylized Fact: Multifractality

Both the hyperbolic decay of the unconditional probability distribution function (pdf) as well as the similarly hyperbolic decay of the autocorrelations of many measures of volatility (squared, absolute returns) would fall into the category of scaling laws in the natural sciences. The identification of such universal scaling laws in an area like finance has spurred natural scientists to further explore the behavior of financial data and to develop models to explain these characteristics (e.g., Mantegna and Stanley 1996). From this line of research, multifractality, also called multiscaling or anomalous scaling, emerged gradually during the 1990s as a more subtle characteristic of financial data that motivated the adaptation of known generating mechanisms for multifractal processes from the natural sciences into empirical finance.

To define multifractality, or multiscaling, we start with the more basic concepts of fractality, or scaling. The defining property of fractality is the invariance of some characteristic under appropriate self-affine transformations. The power-law functions characterizing the pdf of returns and autocorrelations of volatility measures are scale-invariant properties, that is, their behavior is preserved over different scales under appropriate transformations.³ In a most general way, some property of an object or a process needs to fulfill a law such as

$$x(ct) = c^H x(t) \quad (6.5)$$

in order to be classified as scale-invariant, where t is an appropriate measurement of a scale (e.g., time or distance). Strict validity of (6.5) holds for many of the objects that have been investigated in fractal geometry (Mandelbrot 1982). In the framework of stochastic processes, such laws could only hold in distribution. In this case, Mandelbrot et al. (1997) speak of self-affine processes. An example of a well-known class of processes obeying such a scale invariance principle is fractional Brownian motion for which $x(t)$ is a series of realizations and $0 < H < 1$ is the Hurst index that determines the degree of persistence ($H > 0.5$) or anti-persistence ($H < 0.5$) of the process, $H = 0.5$ corresponding to Wiener Brownian motion with uncorrelated Gaussian increments. Figure 6.2 shows the scaling behavior of different powers of returns (raw, absolute, and squared returns) of a financial index as determined by a popular method for the estimation of the Hurst coefficient, H . The law (6.5) also determines the dependence structure of the increments of a process obeying such scaling behavior as well as their higher moments, which show hyperbolic decline of their autocorrelations with an exponent depending linearly on H . Such linear dependence is called uniscaling or unifractality. It also carries over asymptotically to processes that use a fractional process

³ For example, from the limiting power law the CDF of a process with hyperbolically decaying tails obeys $Prob(x_i > x) \approx x^{-\alpha}$, and obviously for any multiple of x the same law applies: $Prob(x_i < cx) \approx (cx)^{-\alpha} = c^{-\alpha} x^{-\alpha}$.

as a generator for the variance dynamics, for example, the long-memory stochastic volatility model of Breidt et al. (1998).⁴

Multifractality, or anomalous scaling, allows for a richer variation of the behavior of a process across different scales by only imposing the more general relationship

$$x(ct) \stackrel{d}{=} M(c)x(t) \equiv c^{H(c)}x(t), \quad (6.6)$$

where the scaling factor $M(c)$ is a random function with possibly different shape for different scales and $\stackrel{d}{=}$ denotes equality in distribution. The last equality of equation (6.6) illustrates that this variability of scaling laws could be translated into variability of the index H , which is no longer constant. We might also note the multiplicative nature of transitions between different scales: One moves from one scale to another via multiplication with a random factor $M(c)$. We will see below that multifractal measures or processes are constructed exactly in this way, which implies a combinatorial, noncausal nature of these processes.

Multiscaling in empirical data is typically identified by differences in the scaling behavior of different (absolute) moments:

$$\mathbb{E}[|x(t, \Delta t)|^q] = c(q)\Delta t^{qH(q)+1} = c(q)\Delta t^{\tau(q)+1}, \quad (6.7)$$

with $x(t, \Delta t) = x(t) - x(t - \Delta t)$, and $c(q)$ and $\tau(q)$ being deterministic functions of the order of the moment q . A similar equation could be established for uniscaling processes, for example, fractional Brownian motion, yielding

$$\mathbb{E}[|x(t, \Delta t)|^q] = c^H \Delta t^{qH+1}. \quad (6.8)$$

Hence, in terms of the behavior of moments, multifractality (anomalous scaling) is distinguished by a nonlinear (typically concave) shape from the linear scaling of unifractal, self-affine processes. The standard tool for diagnosing multifractality is, then, inspection of the empirical scaling behavior of an ensemble of moments. Such nonlinear scaling is illustrated in figure 6.3 for three stock indices and a stochastic process with multifractal properties (the Markov-switching multifractal model, introduced below). The traditional approach in the physics literature consists in extracting $\tau(q)$ from a chain of linear log-log fits of the behavior of various moments q for a certain selection of time aggregation steps Δt . One, therefore, uses regressions to the temporal scaling of moments of powers q

$$\ln \mathbb{E}[|x(t, \Delta t)|^q] = a_0 + a_1 \ln(\Delta t) \quad (6.9)$$

and constructs the empirical $\tau(q)$ curve (for a selection of discrete q) from the ensemble of estimated regression coefficients for all q . An alternative and perhaps even more widespread approach for identification of multifractality looks at the varying scaling

⁴ For the somewhat degenerate FIGARCH model, the complete asymptotics have not yet been established; see Jach and Kokoszka (2010).

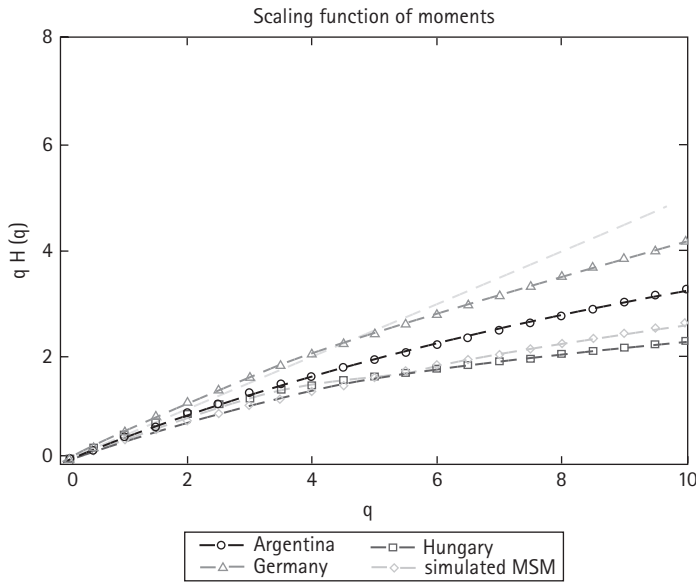


FIGURE 6.3 Scaling exponents of moments for three financial time series and an example of simulated returns from a Markov-switching multifractal process. The empirical samples run from 1998 to 2007, and the simulated series is the one depicted in the bottom panel of figure 6.5. The broken line gives the expected scaling $H(q) = q/2$ under Brownian motion. No fit has been attempted of the simulated case to one of the empirical series.

coefficients $H(q)$ in equation (6.7). Although the unique coefficient H of equation (6.8) is usually denoted the Hurst coefficient, the multiplicity of such coefficients in multifractal processes is denoted Hölder exponents. The unique H quantifies a global scaling property of the underlying process, but the Hölder exponents can be viewed as local scaling rates that govern various patches of a time series leading to a characteristically heterogeneous (or intermittent) appearance of such series. An example is displayed in figure 6.5 (principles of construction are explained below). Focusing on the concept of Hölder exponents, multifractality, then, amounts to identification of the range of such exponents rather than a degenerate single H as for unifractal processes. The so-called spectrum of Hölder exponents (or multifractal spectrum) can be obtained by the Legendre transformation⁵ of the scaling function $\tau(q)$. Defining $\alpha = \frac{d\tau}{dq}$, the Legendre transformation $f(\alpha)$ of the function $\tau(q)$ is given by

$$f(\alpha) = \arg \min_q [q\alpha - \tau(q)], \quad (6.10)$$

where α is the Hölder exponent (the established notation for the counterpart of the constant Hurst exponent, H) and $f(\alpha)$ the multifractal spectrum that describes the distribution of the Hölder exponents. The local Hölder exponent quantifies the local

⁵ The Legendre transformation is a mathematical operation that transforms a function of a coordinate, $g(x)$, into a new function $h(y)$ whose argument is the derivative of $g(x)$ with respect to x , i.e., $y = \frac{dg}{dx}$.

scaling properties (local divergence) of the process at a given point in time; in other words, it measures the local regularity of the price process. In traditional time series models, the distribution of Hölder exponents is degenerate, converging to a single such exponent (unique Hurst exponent), whereas multifractal measures are characterized by a continuum of Hölder exponents whose distribution is given by the Legendre transformation, equation (6.10), for its particular scaling function $\tau(q)$. The characterization of a multifractal process or measure by a distribution of local Hölder exponents underlines its heterogeneous nature, with alternating calm and turbulent phases.

Empirical studies allowing for such a heterogeneity of scaling relations typically identify “anomalous scaling” (curvature of the empirical scaling functions or nonsingularity of the Hölder spectrum) for financial data as illustrated in figure 6.3. The first example of such an analysis is Müller et al. (1990), followed by more and more similar findings reported mostly in the emerging econophysics literature (because the underlying concepts were well known in physics from research on turbulent flows but were completely alien to financial economists). Examples include Vassilicos et al. (1994), Mantegna and Stanley (1995), Ghashghaie et al. (1996), Fisher et al. (1997), Schmitt et al. (1999), and Fillol (2003). Ureche-Rangau and de Morthays (2009) show that both the volatility and volume of Chinese stocks appear to have multifractal properties, a finding one should probably be able to confirm for other markets as well given the established long-term dependence and high cross-correlation between both measures (see Lobato and Velasco 2000, who, among others, also report long-term dependence of volume data). Although econometricians have not been looking at scaling functions and Hölder spectrums, the indication of multifractality in the mentioned studies has nevertheless some counterpart in the economics literature: The well-known finding of Ding et al. (1993) that (a) different powers of returns have different degrees of long-term dependence and that (b) the intensity of long-term dependence varies non-monotonically with q (with a maximum obtained around $q \approx 1$) is consistent with concavity of scaling functions and provides evidence for “anomalous” behavior from a slightly different perspective.

Multifractality thus provides a generalization of the well-established finding of long-term dependence of volatility: Different measures of volatility are characterized by different degrees of long-term dependence in a way that reflects the typical anomalous behavior of multifractal processes. If we accept such behavior as a new stylized fact, the natural next step would be to design processes that could capture this universal finding together with other well-established stylized facts of financial data. New models would be required because none of the existing ones would be consistent with this type of behavior: Baseline GARCH and SV models have only exponential decay of the autocorrelations of absolute powers of returns (short-range dependence), and their long-memory counterparts (LMSV, FIGARCH) are characterized by uni-fractal scaling.⁶

⁶ For FIGARCH this is so far only indicated by simulations, but given that, as for LMSV, FIGARCH consists of a unifractal ARFIMA (autoregressive fractionally integrated moving average) process plugged into the variance equation, it seems plausible that it also has unifractal asymptotics.

One caveat is, however, in order: Whether the scaling function and Hölder spectrum analysis provide sufficient evidence for multifractal behavior is, to some extent, subject to dispute. A number of papers show that scaling in higher moments can be easily obtained in a spurious way without any underlying anomalous diffusion behavior. Lux (2004) points out that a nonlinear shape of the empirical $\tau(q)$ function is still obtained for financial data after randomization of their temporal structure, so that the $\tau(q)$ and $f(\alpha)$ estimators are rather unreliable diagnostic instruments for the presence of multifractal structure in volatility. Apparent scaling has also been illustrated by Barndorff-Nielsen and Prause (2001) as a consequence of fat tails in the absence of true scaling. It is very likely that standard volatility models would also lead to apparent multiscaling that could be hard to distinguish from “true” multifractality via the diagnostic tools mentioned above.⁷ It will always be possible to design processes without a certain type of (multi)scaling behavior that are locally so close to true (multi)scaling that these deviations will never be detected with pertinent diagnostic tools and finite sample sizes (LeBaron 2001; Lux 2001b).

On the other hand, one might follow Mandelbrot’s frequently voiced methodological premise to model apparently generic features of data by similarly generic models rather than using “fixes” (Mandelbrot 1997a). Introducing amendments to existing models (such as GARCH or SV) in order to adapt them to new stylized facts might lead to highly parameterized setups that lack robustness when applied to data from different markets, while simple generating mechanisms for multifractal behavior are available that could, in principle, capture the whole spectrum of time series properties highlighted above in a more parsimonious way. In addition, if one wants to account for multiscaling proper (rather than as a spurious property), no avenue is yet known for equipping GARCH- or SV-type models with this property in a generic way. Hence, adapting in an appropriate way some known generating mechanism for multifractal behavior appears to be the only way to come up with models that generically possess such features and jointly reproduce all stylized facts of asset returns. The next section recollects the major steps in the development of multifractal models for asset-pricing applications.

6.3 MULTIFRACTAL MEASURES AND PROCESSES

In the following, we first explain the construction of a simple multifractal measure and show how one can generalize it along various dimensions. We then move on to multifractal processes designed as models for financial returns.

⁷ There is also a sizeable literature on spurious generation of fat tails and long-term dependence; see Granger and Teräsvirta (1999) or Kearns and Pagan (1997).

6.3.1 Multifractal Measures

Multifractal measures date back to the early 1970s when Mandelbrot proposed a probabilistic approach for the distribution of energy in turbulent dissipation (e.g., Mandelbrot 1974). Building on earlier models of energy dissipation by Kolmogorov (1941, 1962) and Obukhov (1962), Mandelbrot proposed that energy should dissipate in a cascading process on a multifractal set from long to short scales. In this original setting, the multifractal set results from operations performed on probability measures. The construction of a multifractal cascade starts by assigning uniform probability to a bounded interval (e.g., the unit interval $[0, 1]$). First, this interval is split into two subintervals receiving fractions m_0 and $1 - m_0$, respectively, of the total probability mass of unity of their mother interval. In the simplest case, both subintervals have the same length (i.e., 0.5), but other choices are possible as well. Next, the two subintervals of the first stage of the cascade are split again into similar subintervals (of length 0.25 each in the simplest case), again receiving fractions m_0 and $1 - m_0$ of the probability

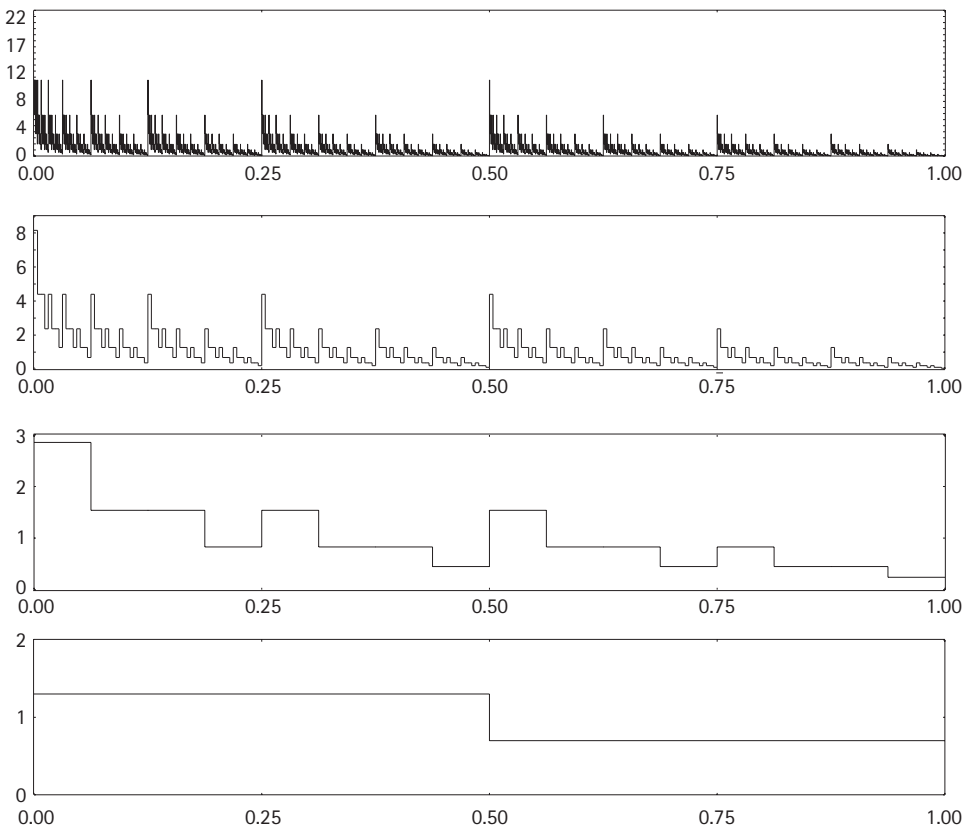


FIGURE 6.4 The baseline Binomial multifractal cascade. Displayed are the products of multipliers at steps 1, 4, 8, and 12. By moving to higher levels of cascade steps, one observes a more and more heterogeneous distribution of the mass over the interval $[0, 1]$.

mass of their mother intervals (see figure 6.4). In principle, this procedure is repeated ad infinitum. With this recipe, a heterogeneous, fractal distribution of the overall probability mass results that, even for the most elementary cases, has a perplexing visual resemblance to time series of volatility in financial markets. This construction clearly reflects the underlying idea of dissipation of energy from the long scales (the mother intervals) to the finer scales that preserve the joint influence of all the previous hierarchical levels in the buildup of the cascade.

Many variations of this generating mechanism of a simple Binomial multifractal could be thought of: Instead of always assigning probability m_0 to the left-hand descendent, one could just as well randomize this assignment. Furthermore, one could think of more than two subintervals to be generated in each step (leading to multinomial cascades) or of using random numbers for m_0 instead of the same constant value. A popular example of the latter generalization is the Log-normal multifractal model, which draws the mass assigned to new branches of the cascade from a Log-normal distribution (Mandelbrot 1974, 1990). Note that for the Binomial cascade the overall mass over the unit interval is exactly conserved at any pre-asymptotic stage as well as in the limit $k \rightarrow \infty$, while mass is preserved only in expectation under appropriately normalized Log-normal multipliers or multipliers following any other continuous function. Another straightforward generalization consists in splitting each interval on level j into an integer number b of pieces of equal length at level $j + 1$. The grid-free Poisson multifractal measure developed by Calvet and Fisher (2001) is obtained by allowing for randomness in the construction of intervals. In this setting, a bounded interval is split into separate pieces with different mass by determining a random sequence T_n of change points. Overall mass is then distributed via random multipliers across the elements of the partition defined by the T_n . A multifractal sequence of measures is generated by a geometric increase of the frequency of arrivals of change points at different levels j ($j = 1, \dots, k$) of the cascade. As in the grid-based multifractal measures, the mass within any interval after the completion of the cascade is given by the product of all k random multipliers within that segment.

Note that all the above recipes can be interpreted as implementations (or examples) of the general form (6.6), which defines multifractality on the basis of the scaling behavior across scales. The recursive construction principles are, themselves, directly responsible for the multifractal properties of the pertinent limiting measures. The resulting measures thus obey multifractal scaling analogous to equation (6.7).⁸ Denoting by μ a measure defined on $[0, 1]$, this amounts to $\mathbb{E}[\mu(t, t + \Delta t)^q] \sim c(q)(\Delta t)^{\tau(q)+1}$. Exact proofs for the convergence properties of such grid-bound cascades have been provided by Kahane and Peyrière (1976). The “multifractal formalism” that had been developed after Mandelbrot’s pioneering contribution consisted in the generalization and analytical penetration of various multifractal measures following the above principles of construction (Tél 1988; Evertsz and Mandelbrot 1992; Riedi 2002). Typical

⁸ For example, for the simplest case of the Binomial cascade one gets $\tau(q) = -\ln \mathbb{E}[M^q] - 1$ with $M \in \{m_0, 1 - m_0\}$ with probability 0.5.

questions of interest are the determination of the scaling function $\tau(\alpha)$ and the Hölder spectrum $f(\alpha)$, as well as the existence of moments in the limit of a cascade with infinite progression.

6.3.2 Multifractal Models in Continuous Time

6.3.2.1 *The Multifractal Model of Asset Returns*

Multifractal measures have been adapted to asset-price modeling by using them as a “stochastic clock” for transformation of chronological time into business (or intrinsic) time. Such a time transformation can be represented in formal terms by stochastic subordination, with the time change represented by a stochastic process, say, $\theta(t)$, denoting the subordinating process and the asset price change, $r(t)$, being given by a subordinated process (e.g., Brownian motion) measured in transformed time, $\theta(t)$. In this way, the homogenous subordinated process might be modulated so as to give rise to realistic time series characteristics such as volatility clustering. The idea of stochastic subordination was introduced in financial economics by Mandelbrot and Taylor (1967). A well-known later application of this principle is Clark (1973), who had used trading volume as a subordinator (cf. Ané and Geman 2000 for recent extensions of this approach).

Mandelbrot et al. (1997) seems to be the first paper that went beyond establishing phenomenological proximity of financial data to multifractal scaling. They proposed a model termed the Multifractal Model of Asset Returns (MMAR) in which a multifractal measure as introduced in section 6.3.1 serves as a transformation from chronological time to business time. The original paper has not been published in a journal, but a synopsis of this entry and two companion papers (Calvet et al. 1997; Fisher et al. 1997) have appeared as Calvet and Fisher (2002). Several other contributions by Mandelbrot (1997b, 1999, 2001a,b,c) contain graphical discussions of the construction of the time-transformed returns of the MMAR process and simulations of examples of the MMAR as a data-generating process. Formally, the MMAR assumes that returns $r(t)$ follow a compound process

$$r(t) = B_H[\theta(t)], \quad (6.11)$$

in which an incremental fractional Brownian motion with Hurst index H , $B_H[\cdot]$ is subordinate to the cumulative distribution function $\theta(t)$ of a multifractal measure constructed along the above lines. When investigating the properties of this process, one has to distinguish the (unifractal) scaling of the fractional Brownian motion from the scaling behavior of the multifractal measure. The behavior of the compound process is determined by both, but its multiscaling in absolute moments remains in place even for $H = 0.5$, that is, Wiener Brownian motion. Under the restriction $H = 0.5$, the Brownian motion part becomes uncorrelated Wiener Brownian motion, and the MMAR shows the martingale property of most standard asset pricing models. This model shares essential regularities observed in financial time series including long tails and long

memory in volatility, both of which originate from the multifractal measure $\theta(t)$ applied for the transition from chronological time to business time. The heterogeneous sequence of the multifractal measure, then, serves to contract or expand time and, therefore, also contracts or expands locally the homogeneous second moment of the subordinate Brownian motion.

As pointed out above, different powers of such a measure exhibit different decay rates of their auto covariances. Mandelbrot et al. (1997) demonstrate that the scaling behavior of the multifractal time transformation carries over to returns from the compound process (6.11), which would obey a scaling function $\tau_r(q) = \tau_\theta(qH)$. Similarly, the shape of the spectrum carries over from the time transformation to returns in the compound process via a simple relationship: $f_r(\alpha) = f_\theta(\alpha/H)$. If one writes $\theta(t) = \int_0^t d\theta(t)$, it becomes clear that the incremental multifractal random measure $d\theta(t)$, which is the limit of $\mu[t, t + \Delta t]$ for $\Delta t \rightarrow 0$ and k (the number of hierarchical levels) $\rightarrow \infty$, can be considered as the instantaneous stochastic volatility. As a result, MMAR essentially applies the multifractal measure to capture the time dependence and nonhomogeneity of volatility. Mandelbrot et al. (1997) and Calvet and Fisher (2002) discuss estimation of the underlying parameters of the MMAR model via matching of the $f(\alpha)$ and $\tau(\alpha)$ functions and show that the temporal behavior of various absolute moments of typical financial data squares well with the theoretical results for the multifractal model.

Any possible implementation of the underlying multifractal measure could be used for the time transformation $\theta(t)$. All examples considered in their papers built on a binary cascade in which the time interval of interest (in place of the unit interval in the abstract operations on a measure described in section 6.3.1) is split repeatedly into subintervals of equal length. The resulting subintervals are assigned fractions of the probability mass of their mother interval drawn from different types of random distributions. The Binomial, Log-normal, Poisson and Gamma distributions discussed in Calvet and Fisher (2002) each lead to a particular $\tau(\alpha)$ and $f(\alpha)$ function (known from previous literature) and similar behavior of the compound process according to the relations detailed above. Lux (2001c) applies an alternative estimation procedure minimizing a Chi-square criterion for the fit of the implied unconditional distribution of the MMAR to the empirical one, and reports that one can obtain surprisingly good approximations of the empirical shape in this way. Lux (2004) documents, however, that $\tau(\alpha)$ and $f(\alpha)$ functions are not very reliable as criteria for determination of the parameters of the MMAR because even after randomization of the underlying data, one still gets indications of temporal scaling structure via nonlinear $\tau(\alpha)$ and $f(\alpha)$ shapes. Poor performance of such estimators is also expected on the ground of the slow convergence of their variance as demonstrated by Ossiander and Waymire (2000). One might also point out, in this respect, that both functions are capturing various moments of the data, so using them for determination of parameters amounts to some sort of moment matching. It is, however, not obvious that the choice of weight of different moments implied by these functions would be statistically efficient.

Although MMAR has not been pursued further in subsequent literature, estimation of alternative multifractal models has made use of efficient moment estimators as well as more standard statistical techniques. The main drawback of the MMAR is, that despite the attractiveness of its stochastic properties, its practical applicability suffers from the combinatorial nature of the subordinator $\theta(t)$ and its non-stationarity due to the restriction of this measure to a bounded interval. These limitations have been overcome by the iterative time series models introduced by Calvet and Fisher (2001, 2004) that follow a similar principle of construction. Leövey and Lux (2012) have recently proposed a re-interpretation of the MMAR in which an infinite succession of multifractal cascades overcomes the limitation to a bounded interval, and the resulting overall process could be viewed as a stationary one.

It is interesting to relate the grid-bound construction of the MMAR to the “classical” formalization of stochastic processes for turbulence. Building on previous work by Kolmogorov (1962) and Obukhov (1962) on the phenomenology of turbulence, Castaing et al. (1990) have introduced the following approach to replicate the scaling characteristics of turbulent flows

$$x_i = \exp(\varepsilon_i)\xi_i, \quad (6.12)$$

with ξ_i and ε_i both following a Normal distribution $\xi_i \sim N(0, \sigma^2)$ and $\varepsilon_i \sim N(\ln(\sigma_0), \lambda^2)$, and ξ_i and ε_i mutually independent. This approach has been applied to various fluctuating phenomena in the natural sciences such as hadron collision (Carius and Ingelman 1990), solar wind (Sorriso-Valvo et al. 1999), and the human heartbeat (Kiyono et al. 2004, 2005). If one replaces the uniform ε_i by the sum of hierarchically organized components, the resulting structure would closely resemble that of the MMAR model. Models in this vein have been investigated in physics by Kiyono et al. (2007) and Kiyono (2009). Based on the approach exemplified in equation (6.12), Ghashghaie et al. (1996) elaborate on the similarities between turbulence in physics and in financial fluctuations but do not take into account the possibility of multifractality of the data-generating process.

6.3.2.2 The MMAR with Poisson Multifractal Time Transformation

Already in Calvet and Fisher (2001) a new type of multifractal model was introduced that overcomes some of the limitations of the MMAR as proposed by Mandelbrot et al. (1997) while—initially—preserving the formal structure of a subordinated process. Instead of the grid-based binary splitting of the underlying interval (or, more generally, the splitting of each mother interval into the same number of subintervals), they assume that $\theta(t)$ is obtained in a grid-free way by determining a Poisson sequence of change points for the multipliers at each hierarchical level of the cascade. Multipliers themselves might be drawn from a Binomial or Log-normal distribution (the standard cases) or from any other distribution with positive support. Change points are determined by renewal times with exponential densities. At each change point t_n^i a new draw $M_{t_n^i}^i$ of cascade level i occurs from the distribution of the multipliers that is standardized so as to ensure conservation of overall mass $\mathbb{E}[M_{t_n^i}^i] = 1$. In order to achieve the hierarchical

nature of the cascade, the different levels i are characterized by a geometric progression of the frequencies of arrival $b^i\lambda$. Hence, the change points t_n^i follow level-specific densities $f(t_n^{(i)}; \lambda, b) = b^i\lambda \exp(-b^i\lambda t_n^i)$, for $i = 1, \dots, k$. Similar grid-free constructions for multifractal measures are considered in Cioczek-Georges and Mandelbrot (1995) and Barral and Mandelbrot (2002). In the limit $k \rightarrow \infty$ the Poisson multifractal exhibits typical anomalous scaling, which again carries over from the time transformation $\theta(t)$ to the subordinate process for asset returns, $B_H[\theta(t)]$, in the way demonstrated by Mandelbrot et al. (1997).

The importance of this variation of the original grid-bound MMAR is that it provides an avenue toward constructing multifractal models (or models arbitrarily close to true multifractals) in a way that allows better statistical tractability. In particular, in contrast to the grid-bound MMAR, the Poisson multifractal possesses a Markov structure. Since the $t_n^{(i)}$ follow an exponential distribution, the probability of arrivals at any instant t is independent of history. As an immediate consequence, the initial restriction on its construction to a bounded interval in time $[0, T]$ is not really necessary, because the process can be continued when reaching the border $t = T$ in the same way by which realizations have been generated within the interval $[0, T]$ without any disruption of its stochastic structure. This is not the case for the grid-based approach. Although one could, in principle, append a new cascade after $t = T$ in the latter approach, the resulting new segment would not be a continuation of the cascading process before but would be completely uncorrelated with the previous one. The continuous-time Poisson multifractal has not been used itself in empirical applications, but it has motivated the development of the discrete Markov-switching multifractal (MSM) model, which has become the most frequently applied version of multifractal models in empirical finance, (see section 6.3.3).

6.3.2.3 Further Generalizations of Continuous-Time MMAR

In the foreword to the working paper version (2001) of their 2002 paper, Barral and Mandelbrot (2002) motivate the introduction of what they call “multifractal product of cylindrical pulses” because of its greater flexibility compared to standard multifractals. They argue that this generalization should be useful for capturing, in particular, the power-law behavior of financial returns. Again, in the construction of the cylindrical pulses the renewal times at different hierarchical levels are determined by Poisson processes whose intensities are not, however, connected via the geometric progression $b^i\lambda$ (reminiscent of the grid size distribution in the original MMAR). Instead, they are scattered randomly according to Poisson processes with frequencies of arrival depending inversely on the scale s , that is, assuming $r_i = s_i^{-1}$ (instead of $r_i = 2^{i-k}$ at scales $s_i = 2^{k-i}$ over an interval $[0, 2^k]$ in the basic grid-bound approach for multifractal measures). Associating independent weights to the different scales, one obtains a multifractal measure for this construction by taking a product of these weights over a conical⁹ domain in (t, s) space. The theory of such cylindrical pulses (i.e., the pertinent

⁹ The conical widening of the influence of scales can be viewed as the continuous limit of the dependencies across levels in the discrete case that proceeds with, e.g., a factor 2 in the case of binary cascades.

multipliers $M_{t_n}^i$ that rule one hierarchical level between adjacent change points t_n and t_{n+1} only requires the existence of $\mathbb{E}[M_{t_n}^i]$. Barral and Mandelbrot (2002) work out the “multifractal apparatus” for such more general families of hierarchical cascades, pointing out that many examples of pertinent processes would be characterized by nonexisting higher moments. Muzy and Bacry (2002) and Bacry and Muzy (2003) go one step further and construct a “fully continuous” class of multifractal measures in which the discreteness of the scales i is replaced by a continuum of scales.

Multiplication over the random weights is then replaced by integration over a similar conical domain in (t, s) space whose extension is given by the maximum correlation scale T (see below). Muzy and Bacry (2002) show that for this setup, nontrivial multifractal behavior is obtained if the conical subset $C_s(t)$ of the (t, s) -half plane (note that $t \geq 0$) obeys

$$C_s(t) = \{(t', s'), s' \geq s, -f(s')/2 \leq t' - t \leq f(s')/2\} \quad (6.13)$$

with

$$f(s) = \begin{cases} s & \text{for } s \leq T \\ T & \text{for } s > T, \end{cases} \quad (6.14)$$

that is, a symmetrical cone around current time t with linear expansion of the included scales s up to some maximum T . The multifractal measure obtained along these lines involves a stochastic integral over the domain $C(t)$:

$$d\theta(t) = e^{\int_{(t', s) \in C(t)} d\omega(t', s)}. \quad (6.15)$$

If $d\omega(t', s)$ is a Gaussian variable, one can use this approach as an alternative way to generate a Log-normal multifractal time transformation. As demonstrated by Bacry and Muzy (2003), subordinating a Brownian motion to this process leads to a compound process that has a distribution identical to the limiting distribution of the grid-bound MMAR with Log-normal multipliers for $k \rightarrow \infty$. Discretization of the continuous-time multifractal random walk is considered below.

6.3.3 Multifractal Models in Discrete Time

6.3.3.1 Markov-Switching Multifractal Model

Together with the continuous-time Poisson multifractal, Calvet and Fisher (2001) have also introduced a discretized version of this model that has become the most frequently applied version of the multifractal family in the empirical financial literature.¹⁰ In

¹⁰ We note in passing that for standard discrete volatility models, the determination of the continuous-time limit is not always straightforward. For instance, for the GARCH(1,1) model Nelson (1990) found a limiting “GARCH diffusion” under some assumptions, and Corradi (2000) found a limiting deterministic process under a different set of assumptions. Also, while there exists a well-known class of continuous-time stochastic volatility models, they do not necessarily constitute the limit processes of their also well-known discrete counterparts.

this discretized version, the volatility dynamics can be interpreted as a discrete-time Markov-switching process with a large number of states. In their approach, returns are modeled as in equation (6.1) with innovations ε_t drawn from a standard Normal distribution $N(0, 1)$ and instantaneous volatility being determined by the product of k volatility components or multipliers $M_t^{(1)}, M_t^{(2)}, \dots, M_t^{(k)}$ and a constant scale factor σ :

$$r_t = \sigma_t \varepsilon_t \quad (6.16)$$

with

$$\sigma_t^2 = \sigma^2 \prod_{i=1}^k M_t^{(i)}. \quad (6.17)$$

The volatility components $M_t^{(i)}$ are persistent and non-negative and they satisfy $E[M_t^{(i)}] = 1$. Furthermore, it is assumed that the volatility components $M_t^{(1)}, M_t^{(2)}, \dots, M_t^{(k)}$ at a given time t are statistically independent. Each volatility component is renewed at time t with probability γ_i , depending on its rank within the hierarchy of multipliers, and remains unchanged with probability $1 - \gamma_i$. Calvet and Fisher (2001) show that with the following specification of transition probabilities between integer time steps, a discretized Poisson multifractal converges to the continuous-time limit as defined above for $\Delta t \rightarrow 0$:

$$\gamma_i = 1 - (1 - \gamma_1)^{(b^{i-1})}, \quad (6.18)$$

with γ_1 the component at the lowest frequency that subsumes the Poisson intensity parameter λ , $\gamma_1 \in [0, 1]$, and $b \in (1, \infty)$. Calvet and Fisher (2004) assume a Binomial distribution for $M_t^{(i)}$ with parameters m_0 and $2 - m_0$ (thus guaranteeing an expectation of unity for all $M_t^{(i)}$). If convergence to the limit of the Poisson multifractal is not a concern, one could also use a less parameterized form such as

$$\gamma_i = b^{-i}. \quad (6.19)$$

Here, volatility components in a lower frequency state will be renewed b times as often as those of its predecessor. An iterative discrete multifractal with such a progression of transition probabilities and otherwise identical to the model of Calvet and Fisher (2001, 2004) had already been proposed by Breymann et al. (2000).

For the distribution of the multipliers $M_t^{(i)}$, extant literature has also used the Lognormal distribution (Liu, di Matteo, and Lux (2008); Lux 2008) with parameters λ and s , that is,

$$M_t^{(i)} \sim LN(-\lambda, s^2). \quad (6.20)$$

Setting $s^2 = 2\lambda$ guarantees $E[M_t^{(i)}] = 1$. Comparison of the performance and statistical properties of MF models with Binomial and Log-normal multipliers shows typically almost identical results (Liu, di Matteo, and Lux 2007). It thus appears that the Binomial choice (with 2^k different volatility regimes) has sufficient flexibility and cannot easily be outperformed via a continuous distribution of the multipliers.

The first three panels in figure 6.5 show the development of the switching behavior of Log-normal MSM processes at different levels. The average duration of the second-highest component is equal to 2,048 time steps. One expects this component, therefore, to switch two times, on average, during the 4,096 time steps of the simulation. Similarly, for the sixth-highest component displayed in the second panel renewal occurs about once within $2^5 = 32$ periods. The third panel shows the product of multipliers that plays the role of local stochastic volatility as described by equation (6.17). The resulting artificial time series displays volatility clustering and outliers that stem from intermittent bursts of extreme volatility.

Owing to its restriction to a finite number of cascade steps, the MSM is not characterized by asymptotic (multi-)scaling. However, its pre-asymptotic scaling regime can be arbitrarily extended by increasing the number of hierarchical components k . It is, thus, a process whose multifractal properties are spurious. Yet at the same time it can be arbitrarily close to “true” multiscaling over any finite length scale. This feature is shared by a second discretization, the multifractal random walk, whose power-law scaling over a finite correlation horizon is already manifest in its generating process.

6.3.3.2 Multifractal Random Walk

In the (econo)physics literature, a different type of causal, iterative process has been developed more or less simultaneously: the Multifractal Random Walk (MRW). Essentially, the MRW is a Gaussian process with built-in multifractal scaling via an appropriately defined correlation function. Although one could use various distributions for the multipliers as the guideline for construction of different versions of MRW replicating their particular autocorrelation structures, the literature has exclusively focused on the Log-normal distribution.

Bacry et al. (2001) define the MRW as a Gaussian process with a stochastic variance as follows:

$$r_{\Delta t}(\tau) = e^{\omega_{\Delta t}(\tau)} \varepsilon_{\Delta t}(\tau), \quad (6.21)$$

with Δt a small discretization step, $\varepsilon_{\Delta t}(\cdot)$ a Gaussian variable with mean zero and variance $\sigma^2 \Delta t$, $\omega_{\Delta t}(\cdot)$ the logarithm of the stochastic variance, and τ a multiple of Δt along the time axis. Assuming that $\omega_{\Delta t}(\cdot)$ also follows a Gaussian distribution, one obtains Log-normal volatility draws. For longer discretization steps (e.g., daily unit time intervals), one obtains their returns as

$$r_{\Delta t}(t) = \sum_{i=1}^{t/\Delta t} \varepsilon_{\Delta t}(i) * e^{\omega_{\Delta t}(i)}. \quad (6.22)$$

To mimic the dependence structure of a Log-normal cascade, these are assumed to have covariances:

$$\text{Cov}(\omega_{\Delta t}(t)\omega_{\Delta t}(t+h)) = \lambda^2 \ln \rho_{\Delta t}(h), \quad (6.23)$$

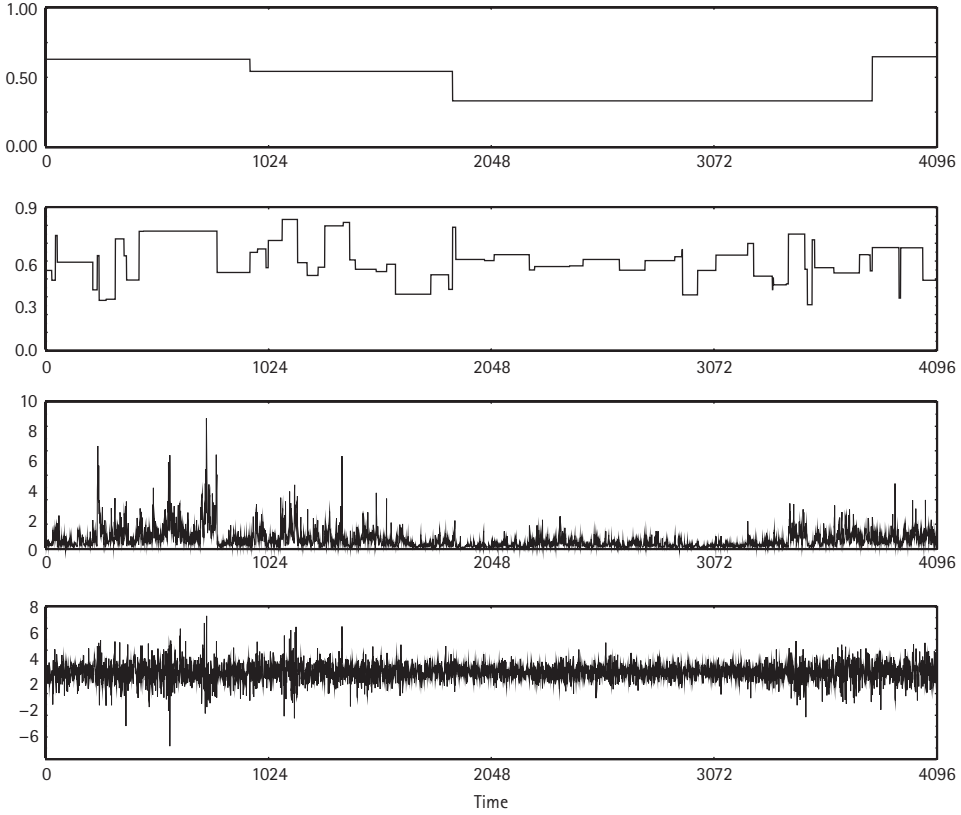


FIGURE 6.5 Simulation of a Markov-switching multifractal model with Log-normal distribution of the multipliers and $k = 13$ hierarchical levels. The location parameter of the Log-normal distribution has been chosen as $\lambda = 1.05$. The first panel illustrates the development of the second multiplier (with average replacement probability of 2^{-11}), the second panel shows the sixth level, and the third panel shows the product of all 13 multipliers. Returns in the lowest panel are simply obtained by multiplying multifractal local volatility by Normally distributed increments.

with

$$\rho_{\Delta t}(h) = \begin{cases} \frac{T}{(|h|+1)\Delta t}, & \text{for } |h| \leq \frac{T}{\Delta t} - 1 \\ 0, & \text{otherwise.} \end{cases} \quad (6.24)$$

Hence, T is the assumed finite correlation length (a parameter to be estimated) and λ^2 is called the intermittence coefficient characterizing the strength of the correlation.

In order for the variance of $r_{\Delta t}(t)$ to converge, $\omega_{\Delta t}(\cdot)$ is assumed to obey the following:

$$\mathbb{E}[\omega_{\Delta t}(i)] = -\lambda^2 \ln(T/\Delta t) = -\text{Var}[\omega_{\Delta t}(i)]. \quad (6.25)$$

The assumption of a finite decorrelation scale makes sure that the multifractal random walk process remains stationary. Like the MSM introduced by Calvet and Fisher (2001), the MRW model does not, therefore, obey an exact scaling function like equation (6.7) in the limit $t \rightarrow \infty$ or divergence of its spectral density at zero, but is characterized by only “apparent” long-term dependence over a bounded interval. The advantage of

both models is that they possess “nice” asymptotic properties that facilitate application of many standard tools of statistical inference.

As shown by Muzy and Bacry (2002) and Bacry et al. (2008), the continuous-time limit of MRW (mentioned in section 6.3.2.3) can also be interpreted as a time transformation of a Brownian motion subordinate to a log-normal multifractal random measure. For this purpose, the MRW can be reformulated in a similar way like the MMAR model.

$$r(t) = B[\theta(t)], \quad \text{for all } t \geq 0, \quad (6.26)$$

where $\theta(t)$ is a random measure for the transformation of chronological to business time and $B(t)$ is a Brownian motion independent of θ_t . Business time θ_t is obtained along the lines of the above exposition of the MRW model as

$$\theta(t) = \lim_{\Delta \rightarrow 0} \int_0^t e^{2\omega_\Delta(u)} du. \quad (6.27)$$

Here $\omega_\Delta(u)$ is the stochastic integral of Gaussian white noise $dW(s, t)$ over a continuum of scales s truncated at the smallest and largest scales Δ and T , which leads to a cone-like structure defining $\omega_\Delta(u)$ as the area delimited in time (over the correlation length) and a continuum of scales s in the (t, s) plane

$$\omega_\Delta(u) = \int_\Delta^T \int_{u-s}^{u+s} dW(v, s). \quad (6.28)$$

To replicate the weight structure of the multipliers in discrete multifractal models, a particular correlation structure of the Gaussian elements $dW(v, s)$ needs to be imposed. Namely, the multifractal properties are obtained for the following choices of the expectation and covariances of $dW(v, s)$:

$$\text{Cov}[dW(v, s), dW(v', s')] = \lambda^2 \delta(v - v') \delta(s - s') \frac{dv ds}{s^2} \quad (6.29)$$

and

$$\mathbb{E}[dW(v, s)] = -\lambda^2 \frac{dv ds}{s^2}. \quad (6.30)$$

Muzy and Bacry (2002) and Bacry and Muzy (2003) show that the limiting continuous-time process exists and possesses multifractal properties. Muzy et al. (2006) and Bacry et al. (2013) also provide results for the unconditional distribution of returns obtained from this process. They demonstrate that it is characterized by fat tails and that it becomes less heavy-tailed under time aggregation. They also show that standard estimators of tail indices are ill behaved for data from a MRW data-generating process owing to the high dependence of adjacent observations. While the implied theoretical tail indices with typical estimated parameters of the MRW would be located at unrealistically large values (> 10), taking the dependence in finite samples into account, one obtains biased (pseudo-)empirical estimates indicating much smaller values of the tail index that are within the order of magnitude of empirical ones. A similar mismatch

between implied and empirical tail indices applies to other multifractal models as well (as far as we can see, this is not explicitly reported in extant literature, but has been mentioned repeatedly by researchers) and likely can be explained in the same way.

6.3.3.3 Asymmetric Univariate MF Models

All the models discussed above are designed in a completely symmetric way for positive and negative returns. It is well known, however, that price fluctuations in asset markets exhibit a certain degree of asymmetry due to leverage effects. The discrete-time skewed multifractal random walk (DSMRW) model proposed by Pochart and Bouchaud (2002) is an extended version of the MRW that takes account of such asymmetries. The model is defined in a similar way as the MRW of equation (6.21) but incorporates the direct influence of past realizations on contemporaneous volatility,

$$\tilde{\omega}_{\Delta t}(i) \equiv \omega_{\Delta t}(i) - \sum_{k < i} K(k, i) \varepsilon_{\Delta t}(k), \quad (6.31)$$

where Pochart and Bouchaud propose to use $K(k, i) = \frac{K_0}{(i-k)^\alpha \Delta t^\beta}$ as a positive definite kernel for the influence of returns on subsequent volatility. Bacry et al. (2012) proposed a continuous-time skewed multifractal model that also incorporates the leverage effect.

Eisler and Kertész (2004) expand the MSM model in a similar way. They consider a refined version of the model in which asymmetry comes in via the renewal probabilities, and they use a term inspired by equation (6.31) to account for leverage autocorrelations.

An asymmetric MSM model has also been introduced by Calvet et al. (2015). They embed a multifractal cascade into a stochastic volatility model in which the product of multipliers enters as a time-varying long-run anchor for the volatility dynamics while at the same time governing a jump component in returns that relates positive volatility shocks to negative return shocks.

6.3.3.4 Bivariate Multifractal Models

A bivariate MF model was first introduced by Calvet et al. (2006). Consider a portfolio of two assets α and β . Let us now denote by r_t the vector of log-returns of the portfolio and r_t^α and r_t^β the individual log-returns of the two assets, respectively. Following Calvet et al. (2006) the return of the portfolio is modeled as

$$r_t = [g(M_t)]^{1/2} * \varepsilon_t, \quad (6.32)$$

where $g(M_t)$ denotes a 2×1 vector, $M_{1,t} * M_{2,t} * \dots * M_{k,t}$, $*$ denotes element-by-element-multiplication, and the column vectors $\varepsilon_t \in \mathbb{R}^2$ are IID Gaussian $N(0, \Sigma)$ with covariance matrix

$$\Sigma = \begin{bmatrix} \sigma_\alpha^2 & \rho_\varepsilon \sigma_\alpha \sigma_\beta \\ \rho_\varepsilon \sigma_\alpha \sigma_\beta & \sigma_\beta^2 \end{bmatrix}. \quad (6.33)$$

The variable ρ_ε represents the unconditional correlation between the residuals as the first source of correlation between both returns. The period t volatility state is

characterized by a $2 \times k$ matrix $M_t = (M_{1,t}; M_{2,t}; \dots; M_{k,t})$, and the vector of the components at the i^{th} frequency is $M_{i,t} = (M_{i,t}^\alpha, M_{i,t}^\beta)$. The volatility vectors $M_{i,t}$ are non-negative and satisfy $E[M_{i,t}] = \mathbf{1}$, where $\mathbf{1} = (1, 1)'$. Economic intuition behind the choice of the dynamics for each vector $M_{i,t}$ is that volatility arrivals are correlated but not necessarily simultaneous across markets. For this reason Calvet et al. (2006) allow arrivals across series to be linked by a correlation coefficient λ . Consider two random variables $I_{i,t}^\alpha$ and $I_{i,t}^\beta$, which are equal to 1 if $c \in \{\alpha, \beta\}$ is hit by an information arrival with probability γ_i , and equal to 0 otherwise. The authors specify the arrival vector to be IID and assume its unconditional distribution to satisfy three conditions. First, the arrival vector is symmetrically distributed: $(I_{i,t}^\alpha, I_{i,t}^\beta) \stackrel{d}{=} (I_{i,t}^\beta, I_{i,t}^\alpha)$. Second, the switching probabilities of both series are equal for each level i : $P(I_{i,t}^\alpha = 1) = P(I_{i,t}^\beta = 1) = \gamma_i$, with γ_i following equation (6.18) as for univariate MSM. Third, there exists $\lambda \in [0, 1]$ such that

$$P(I_{i,t}^\alpha = 1 | I_{i,t}^\beta = 1) = (1 - \lambda)\gamma_i + \lambda.$$

These three conditions define a unique distribution of $(I_{i,t}^\alpha, I_{i,t}^\beta)$ whose joint switching probabilities can be easily determined. Note that the univariate dynamics of each series coincides with a univariate MSM model. Idier (2011) proposes an extension of the bivariate MSM model by considering a time-dependent covariance for the vector of residuals $\rho_\varepsilon(t)$.

Liu (2008) considers a closely related bivariate multifractal model built on the assumption that two time series have a certain number of joint cascade levels in common, while the remaining ones are chosen independently. The returns are, then, modeled as

$$r_{q,t} = \left[\left(\prod_i^k M_{i,t} \right) \left(\prod_{l=k+1}^n M_{l,t} \right) \right]^{1/2} * \varepsilon_t, \quad (6.34)$$

where $q = 1, 2$ refers to the two time series, both having an overall number of n levels of their volatility cascades, and they share a number k of joint cascade levels which govern the strength of their volatility correlation. Obviously, the larger k is, the more correlation there is between the volatility dynamics of both series. After k joint multipliers, each series has separate additional multifractal components. The innovation, ε_t is defined as in equation (6.32) to follow a bivariate standard Normal distribution with correlation parameter ρ_ε . This model can be seen as a special case of a slightly generalized version of Calvet et al. (2006) allowing for heterogeneity of the correlation of volatility innovations, λ_i , across hierarchical levels and choosing an extreme specification in that some of the $\lambda_i (1 \leq i \leq k)$ are equal to 1 and the remaining ones are equal to 0. Liu and Lux (2015) show that allowing for different degrees of correlation between volatility innovations indeed improves the fit and performance of the bivariate MSM, but the extreme specification of Liu (2008) with alternation between full dependence and lack of correlation is dominated by a more flexible approach. They

also find that whether high- or low-frequency components are more closely correlated depends on the market.

6.3.3.5 Higher Dimensional Multifractal Models

The bivariate models presented above can be generalized for more than two assets in various ways. Liu's (2008) approach can be generalized in a straightforward way to an N -dimensional asset returns process. If one assumes that the N time series share a number of j joint cascades that govern the strength of their volatility correlation, the correlation of volatility arrivals could be generalized to the case of an arbitrary number of assets without having to add new parameters in the volatility part of the model. Additional parameters would, then, only come in via the correlation of the Gaussian innovations. If such a specification appears insufficient to capture the heterogeneity in return fluctuations across assets, one could consider a generalized framework with asset-specific multifractal parameters, m_o or λ , in the Binomial or Log-normal setting, respectively.

A generalization of the MRW in a similar vein was proposed by Bacry et al. (2000). They suggest extending the MRW model to a multivariate Multifractal Random Walk (MMRW) in order to model portfolio behavior. Let X_t be a MMRW; then, following Bacry et al. (2000), X_t is defined as

$$X(t) = \lim_{t \rightarrow 0} X_{\Delta t}(t) = \lim_{t \rightarrow 0} \sum_{k=1}^{t/\Delta t} \epsilon_{\Delta t}[k] * e^{\omega_{\Delta t}[k]}, \quad (6.35)$$

where $\epsilon_{\Delta t}$ is now a vector of Gaussians with zero mean and variance-covariance function at lag τ , $\text{Cov}[\epsilon_{i,\Delta t}(t), \epsilon_{j,\Delta t}(t+\tau)] = \delta(\tau) \Sigma_{ij} \Delta t$. The magnitude process $\omega_{\Delta t}(\cdot)$ is also Gaussian with covariance $\text{Cov}[\omega_{i,\Delta t}(t), \omega_{j,\Delta t}(t+\tau)] = \Lambda_{ij} \ln(T_{ij}/(\Delta t + |\tau|))$ for $(\Delta t + |\tau| < T_{ij})$ and 0 elsewhere. The matrix Λ , labeled “multifractal matrix,” controls the nonlinearity of the multifractal spectrum, and T_{ij} are different correlation lengths for the auto-covariances and cross-covariances characterizing the process.

6.4 ESTIMATION AND FORECASTING

Availability of efficient estimation procedures is essential for the application of theoretical asset-pricing models for practical purposes. The nonstandard format of multifractal models initially cast doubt on the applicability of many well-known statistical tools to this new family of volatility models. Fortunately, the members of the second generation multifractal models (MSM and MRW) seemed to be much more well-behaved (and have, in part, been designed to be so) in terms of asymptotic statistical behavior. Most effort has been spent so far to find stable and efficient inference methods for the discrete-time MSM model with discrete or continuous distributions for multipliers or volatility components. In the following we present the estimation methods most

often applied for MF models. We dispense with the traditional $f(\alpha)$ and $\tau(q)$ approach to inference, which was covered in detail in section 6.2.4. As it soon turned out in the pertinent literature when analysts were starting to adapt multifractal models to finance, the scaling approach provides potentially very biased and volatile estimates in applications to financial data, and owing to their fat tails, would even indicate the existence of multifractal structure after randomization of such time series. The quest for more appropriate statistical methods has been motivated to a large extent by these deficiencies. The development of the Markov-switching multifractal model and the multifractal random walk have brought forward stochastic processes with more “convenient” asymptotic properties than their predecessors had. As a consequence, they allow application of many established tools of inference. Nevertheless, their proximity to genuine long memory might still be a concern, and it motivates one to exert caution in empirical applications (e.g., while theoretical convergence of estimates might be trivially guaranteed, the pre-asymptotic regime might be much more extended than with other models).

6.4.1 Maximum Likelihood Estimation

Exact ML estimation was primarily developed for the discrete-time MSM model with a discrete distribution for the volatility components or multipliers. Calvet and Fisher (2004) introduced an ML estimation approach for the Binomial Markov-switching multifractal (BMSM) model. To show how to perform ML estimation in this context, we note that the log-likelihood function for a series of observations $\{r_t\}_{t=1}^T$ in its most general form may be expressed as

$$L(r_1, \dots, r_T; \varphi) = \sum_{t=1}^T \ln g(r_t | r_1, \dots, r_{t-1}; \varphi), \quad (6.36)$$

where $g(r_t | r_1, \dots, r_{t-1}; \varphi)$ is the likelihood function of the Markov-switching multifractal model, and φ is the vector of parameters. For Markov-switching models, the likelihood function can be decomposed in the following way: $g(r_t | r_1, \dots, r_{t-1}; \varphi) = \omega_t(r_t | M_t = m^i, \varphi)(\pi_{t-1} A)$. The three components are defined as follows: $\omega_t(r_t | M_t = m^i, \varphi)$ is a vector of dimension 2^k of conditional densities of any observation r_t for volatility regimes m^i , and A is the transition matrix, which has components $A_{ij} = P(M_{t+1} = m^j | M_t = m^i)$. The last component within the likelihood function above is π_t , which is the vector of conditional probabilities of the volatility states given observations up to time t , $\pi_t^i = P(M_t = m^i | r_1, \dots, r_t; \varphi)$. The conditional probabilities can be recursively obtained through Bayesian updating:

$$\pi_t = \frac{\omega_t(r_t | M_t = m^i, \varphi) * (\pi_{t-1} A)}{\sum \omega_t(r_t | M_t = m^i, \varphi) * (\pi_{t-1} A)}. \quad (6.37)$$

Different distributional assumptions for innovations could be embedded in this framework. The parameter vector of the BMSM with Gaussian innovations would be given by $\varphi = (m_0, \sigma)'$, while the parameter vector of a BMSM with Student- t innovations would be $\varphi = (m_0, \sigma, \nu)'$, where $\nu \in (2, \infty)$ is the distributional parameter accounting for the degrees of freedom in the density function of the Student- t distribution. The Student- t distribution for return innovations has been used by Lux and Morales-Arias (2010) in order to enhance out-of-sample forecasts of the MSM model because it may allow the MSM model to better distinguish between volatility dependence and fat-tailed innovations.

An advantage of the ML procedure is that, as a by-product, it allows one to obtain optimal forecasts via Bayesian updating of the conditional probabilities $\pi_t = P(M_t = m^i | r_1, \dots, r_t; \varphi)$ for the unobserved volatility states $m^i, i = 1, \dots, 2^k$. Maximum likelihood estimation provides good precision in finite samples (cf. Calvet and Fisher 2004).

Although the applicability of the ML algorithm greatly facilitates estimation of MSM models, it is restrictive in the sense that it is practically feasible only for discrete distributions of the multipliers and, therefore, is not applicable for, say, the case of a Log-normal distribution. Owing to the potentially large state space (we have to take into account transitions between 2^k distinct states), ML estimation also encounters practical bounds in terms of its computational demands for specifications with more than about $k = 10$ volatility components in the Binomial case. For multivariate MF models, the applicability of the ML approach is even more constrained from the computational side: In the bivariate case, the evaluation of the transition matrix with size $4^k \times 4^k$ becomes unfeasible for choices of about $k > 5$. There has also been a recent attempt to estimate the MRW model via a likelihood approach. Løvsletten and Rypdal (2012) develop an approximate maximum likelihood method for MRW using a Laplace approximation of the likelihood function.

6.4.2 Simulated Maximum Likelihood

This approach is more broadly applicable to both discrete and continuous distributions for multipliers. To overcome the computational and conceptional limitation of exact ML estimation, Calvet et al. (2006) developed a simulated ML approach. They propose a particle filter to numerically approximate the likelihood function. The particle filter is a recursive algorithm that generates independent draws $M_t^{(1)}, \dots, M_t^{(N)}$ from the conditional distribution of π_t . At time $t = 0$, the algorithm is initiated by draws $M_0^{(1)}, \dots, M_0^{(N)}$ from the ergodic distribution $\bar{\pi}$. For any $t > 0$, the particles $\{M_t^{(n)}\}_{n=1}^N$ are sampled from the new belief π_t . To this end, the formula (6.37) within the ML estimation algorithm is replaced by a Monte Carlo approximation in SML. This means that the analytical updating via the transition matrix, $\pi_{t-1}A$, is approximated via the simulated transitions of the particles. Disregarding the normalization of probabilities

(i.e., the denominator), the formula (6.37) can be rewritten as

$$\pi_t^i \propto \omega_t(r_t | M_t = m^i; \varphi) \sum_{j=1}^{2^k} P(M_t = m^i | M_{t-1} = m^j) \pi_{t-1}^j, \quad (6.38)$$

and because $M_t^{(1)}, \dots, M_t^{(N)}$ are independent draws from π_{t-1} , the Monte Carlo approximation has the following format:

$$\pi_t^i \propto \omega_t(r_t | M_t = m^i; \varphi) \frac{1}{N} \sum_{n=1}^N P(M_t = m^i | M_{t-1} = M_{t-1}^{(n)}). \quad (6.39)$$

The approximation, thus, proceeds by simulating each $M_{t-1}^{(n)}$ one step forward to obtain $\hat{M}_t^{(n)}$ given $M_{t-1}^{(n)}$. This step only uses information available at date $t-1$, and must therefore be adjusted at time step t to account for the information contained in the new return. This is achieved by drawing N random numbers q from 1 to N with probability

$$P(q = n) \equiv \frac{\omega_t(r_t | M_t = \hat{M}_t^{(n)}; \varphi)}{\sum_{n'=1}^N \omega_t(r_t | M_t = \hat{M}_t^{(n')}; \varphi)}. \quad (6.40)$$

The distribution of particles is, thus, shifted according to their importance at time t . With simulated draws $M_t^{(n)}$ the Monte Carlo (MC) estimate of the conditional density is

$$\hat{g}(r_t | r_1, \dots, r_{t-1}; \varphi) \equiv \frac{1}{N} \sum_{n=1}^N g_t(r_t | M_t = \hat{M}_t^{(n)}; \varphi), \quad (6.41)$$

and the log-likelihood is approximated by $\sum_{t=1}^T \ln \hat{g}(r_t | r_1, \dots, r_{t-1}; \varphi)$. The simulated ML approach makes it feasible to estimate MSM models with continuous distribution of multipliers as well as univariate and multivariate Binomial models with too high a number of states for exact ML. Despite this gain in terms of different specifications of MSM models that can be estimated, the computational demands of SML are still considerable, particularly for high numbers of particles N .

6.4.3 Generalized Method of Moments Estimation

Again, this is an approach that is, in principle, applicable to both discrete and continuous distributions for multipliers. To overcome the lack of practicability of ML estimation, Lux (2008) introduced a Generalized Method of Moments (GMM) estimator that is also universally applicable to all specifications of MSM processes (discrete or continuous distribution for multipliers, Gaussian, Student- t , or various other distributions for innovations). In particular, it can be used in all cases in which ML is not applicable or computationally feasible. Its computational demands are also

lower than those of SML and independent of the specification of the model. In the GMM framework for MSM models, the vector of parameters φ is obtained by minimizing the distance of empirical moments from their theoretical counterparts, that is,

$$\hat{\varphi}_T = \arg \min_{\varphi \in \Phi} f_T(\varphi)' A_T f_T(\varphi), \quad (6.42)$$

with Φ the parameter space, $f_T(\varphi)$ the vector of differences between sample moments and analytical moments, and A_T a positive definite and possibly random weighting matrix. Moreover, $\hat{\varphi}_T$ is consistent and asymptotically Normal if suitable “regularity conditions” are fulfilled (Harris and Mátyás 1999) that are satisfied routinely for Markov processes.

In order to account for the proximity to long memory that is exhibited by MSM models by construction Lux (2008) proposed the use of log differences of absolute returns together with the pertinent analytical moment conditions:

$$\xi_{t,T} = \ln|r_t| - \ln|r_{t-T}|. \quad (6.43)$$

The above variable only has nonzero auto-covariances over a limited number of lags. To exploit the temporal scaling properties of the MSM model, covariances of various moments over different time horizons are chosen as moment conditions that is,

$$\text{Mom}(T, q) = \mathbb{E} \left[\xi_{t+T, T}^q \cdot \xi_{t, T}^q \right], \quad (6.44)$$

for $q = 1, 2$ and different horizons T together with $E[r_t^2] = \sigma^2$ for identification of σ in the MSM model with Normal innovations. In the case of the MSM- t model, Lux and Morales-Arias (2010) consider moment conditions in addition to (6.44), namely, $E[|r_t|]$, $E[|r_t^2|]$, $E[|r_t^3|]$, in order to extract information on the Student- t 's shape parameter.

Bacry et al. (2008) and Bacry et al. (2013) also apply the GMM method for estimating the MRW parameters (λ, σ , and T) using moments similar to those proposed in Lux (2008). Sattarhoff (2010) refines the GMM estimator for the MRW using a more efficient algorithm for the covariance matrix estimation. Liu (2008) adopts the GMM approach to bivariate and trivariate specifications of the MSM model. Leövey (2013) develops a simulated method of moments (SMM) estimator for the continuous-time Poisson multifractal model of Calvet and Fisher (2001).

In addition, related work in statistical physics has recently considered simple moment estimators for the extraction of the multifractal intermittence parameters from data on turbulent flows (Kiyono et al. 2007). Leövey and Lux (2012) compare the performance of a GMM estimator for multifractal models of turbulence with various heuristic estimators proposed in the pertinent literature, and show that the GMM approach typically provides more accurate estimates owing to its more systematic exploitation of information contained in various moments.

6.4.4 Forecasting

With ML and SML estimates, forecasting is straightforward: With ML estimation, conditional state probabilities can be iterated forward via the transition matrix to deliver forecasts over arbitrarily long time horizons. The conditional probabilities of future multipliers, given the information set $\mathfrak{S}_t$, $\hat{\pi}_{t,n} = P(M_n | \mathfrak{S}_t)$, are given by

$$\hat{\pi}_{t,n} = \pi_t A^{n-t}, \quad \forall n \in \{t, \dots, T\}. \quad (6.45)$$

In the case of SML, iteration of the particles provides an approximation of the predictive density. Since GMM does not provide information on conditional state probabilities, Bayesian updating is not possible, and one has to supplement GMM estimation with a different forecasting algorithm. To this end, Lux (2008) proposes best linear forecasts (cf. Brockwell and Davis 1991, chap. 5) together with the generalized Levinson-Durbin algorithm developed by Brockwell and Dahlhaus (2004). Assuming that the data of interest (e.g., squared or absolute returns) follow a stationary process $\{Y_t\}$ with mean zero, the best linear h -step forecasts are obtained as

$$\hat{Y}_{n+h} = \sum_{i=1}^n \phi_{ni}^{(h)} Y_{n+1-i} = \boldsymbol{\phi}_n^{(h)} Y_n, \quad (6.46)$$

where the vectors of weights $\boldsymbol{\phi}_n^h = (\phi_{n1}^h, \phi_{n2}^h, \dots, \phi_{nn}^h)'$ can be obtained from the analytical auto-covariances of Y_t at lags h and beyond. More precisely, $\boldsymbol{\phi}_n^{(h)}$ are any solution of $\boldsymbol{\Psi}_n \boldsymbol{\phi}_n^{(h)} = \boldsymbol{\kappa}_n^{(h)}$ in which $\boldsymbol{\kappa}_n^{(h)} = (\kappa_{n1}^{(h)}, \kappa_{n2}^{(h)}, \dots, \kappa_{nn}^{(h)})'$ denote the auto-covariances of Y_t and $\boldsymbol{\Psi}_n = [\kappa(i-j)]_{i,j=1,\dots,n}$ is the variance-covariance matrix. In empirical applications, equation (6.46) has been used for forecasting squared returns as a proxy for volatility using analytical covariances to obtain the weights $\boldsymbol{\phi}_n^h$. Linear forecasts have also been used by Bacry et al. (2008) and Bacry et al. (2013) in connection with their GMM estimates of the parameters of the MRW model. Duchon et al. (2012) develop an alternative forecasting scheme for the MRW model in the presence of parameter uncertainty as a perturbation of the limiting case of an infinite correlation length $T \rightarrow \infty$.

6.5 EMPIRICAL APPLICATIONS

Calvet and Fisher (2004) compare the forecast performance of the MSM model to those of GARCH, Markov-switching GARCH, and FIGARCH models across a range of in-sample and out-of-sample measures of fit. Using four long series of daily exchange rates, they find that at short horizons MSM shows about the same performance as its competitors and sometimes a better one. At long horizons MSM more clearly outperforms all alternative models. Lux (2008) combines the GMM approach with best

linear forecasts and compares different MSM models (Binomial MSM and Log-normal MSM with various numbers of multipliers) to GARCH and FIGARCH. Although GMM is less efficient than ML, Lux (2008) confirms that MSM models tend to perform better than do GARCH and FIGARCH in forecasting volatility of foreign exchange rates. Similarly promising performance in forecasting volatility and value-at-risk is reported for the MRW model by Bacry et al. (2008) and Bacry et al. (2013). Bacry et al. (2008) find that linear volatility forecasts provided by the MRW model outperform GARCH(1, 1) models. Furthermore, they show that MRW forecasts of the VaR at any time scale and time horizon are much more reliable than GARCH(1, 1) forecasts, with Normal or Student- t innovations, for foreign exchange rates and stock indices.

Lux and Kaizoji (2007) investigate the predictability of both volatility and volume for a large sample of Japanese stocks. Using daily data of stock prices and trading volume available over the course of twenty-seven years (from January 1, 1975, to December 31, 2001), they examine the potential of time series models with long memory (FIGARCH, ARFIMA, and multifractal) to improve on the forecasts derived from short-memory models (GARCH for volatility, ARMA for volume). For volatility and volume, they find that the MSM model provides much safer forecasts than FIGARCH and ARFIMA and does not suffer from occasional dramatic failures, as is the case with the FIGARCH model. The higher degree of robustness of MSM forecasts compared to alternative models is also confirmed by Lux and Morales-Arias (2013). They estimate the parameters of GARCH, FIGARCH, SV, LMSV, and MSM models from a large sample of stock indices and compare the empirical performance of each model when applied to simulated data of any other model with typical empirical parameters. As it turns out, the MSM almost always comes in second (behind the true model) when forecasting future volatility and even dominates combined forecasts from many models. It thus appears to be relatively safe for practitioners to use the MSM even if it is misspecified and another standard model is the true data-generating process.

Lux and Morales-Arias (2010) introduce the MSM model with Student- t innovations and compare its forecast performance to those of MSM models with Gaussian innovations and (FI)GARCH. Using country data on all-share equity indices, government bonds, and real estate security indices, they find that the MSM model with Normal innovations produces forecasts that improve on historical volatility but are in some cases inferior to FIGARCH with Normal innovations. When they add fat tails to both models they find that MSM models improve their volatility forecasting, but FIGARCH worsened. They also find that one can obtain more accurate volatility forecasts by combining FIGARCH and MSM.

Lux et al. (2014) apply an adapted version of the MSM model to measurements of realized volatility. Using five different stock market indices (CAC 40, DAX, FTSE 100, NYSE Composite, and S&P 500), they find that the realized volatility-Log-normal MSM model (RV-LMSM) model performs better than non-RV models (FIGARCH, TGARCH, SV and MSM) in terms of mean-squared errors for most stock indices and at most forecasting horizons. They also point out that similar results are obtained in a certain number of instances when the RV-LMSM model is compared to the popular

RV-ARFIMA model, and combinations of alternative models (non-RV and RV) could hardly improve on forecasts of various single models.

Calvet et al. (2006) apply the bivariate model to the co-movements of volatility of pairs of exchange rates. They find that their model provides better volatility and value-at-risk (VaR) forecasts than does the constant correlation GARCH (CC-GARCH) of Bollerslev (1990). Applying the refined bivariate MSM to stock index data, Idier (2011) confirms the results of Calvet et al. (2006). In a additional, he finds that his refined model shows significantly better performance than do the baseline MSM and DCC models for horizons longer than ten days. Liu and Lux (2015) apply the bivariate model to daily data for a collection of bivariate portfolios of stock indices, foreign currencies, and U.S. one- and two-year Treasury bonds. They find that the bivariate multifractal model generates better VaR forecasts than the CC-GARCH model does, especially in the case of exchange rates, and that an extension allowing for heterogeneous dependence of volatility arrivals across levels improves on the baseline specification both in-sample and out-of-sample.

Chen et al. (2013) propose a Markov-switching multifractal duration (MSMD) model. In contrast to the traditional duration models inspired by GARCH-type dynamics, this new model uses the MSM process developed by Calvet and Fisher (2004), and thus can reproduce the long-memory property of durations. By applying the MSMD model to duration data of twenty stocks randomly selected from the S&P 100 index and comparing the result with that of the autoregressive conditional duration (ACD) model both in- and out-of-sample, they find that at short horizons both models yield about the same results, but at long horizons the MSMD model dominates the ACD model.

Žikeš et al. (2017) independently developed an Markov-switching multifractal duration model whose specification is slightly different from that proposed by Chen et al. (2013). They also use the MSM process introduced by Calvet and Fisher (2004) as a basic ingredient in the construction of the model. They apply the model to price durations of three major foreign exchange futures contracts and compare the predictive ability of the new model with those of the ACD model and the long-memory stochastic duration (LMSD) model of Deo et al. (2006). They find that the LMSD and MSMD forecasts generally outperform the ACD forecasts in terms of mean-squared errors and mean absolute errors. Though MSMD and LMSD models sometimes exhibit similar forecast performances, sometimes the MSMD model slightly dominates the LMSD model.

Segnon and Lux (2012) compare the forecast performance of Chen et al.'s (2013) MSMD model to the performances of the standard ACD and Log-ACD models with flexible distributional assumptions about innovations (Weibull, Burr, Log-normal and generalized Gamma) using the density forecast comparison suggested by Diebold et al. (1998) and the likelihood ratio test of Berkowitz (2001). Using data from eight stocks traded on the NYSE, they show empirical results that speak in favor of superiority of the MSMD model. They also find that, in contrast to the ACD model, using flexible distributions for the innovations does not exert much of an influence on the forecast capability of the MSMD model.

Option price applications of multifractal models started with Pochart and Bouchaud (2002), who show that their skewed MRW model could generate smiles in option prices. Leövey (2013) proposed a “risk-neutral” MSM process in order to extract the parameters of the MSM model from option prices. As it turns out, MSM models backed out from option data add significant information to those estimated from historical return data and enhance the ability to forecast future volatility.

Calvet, Fearnley, Fisher, and Leippold (2015) propose an extension of the continuous-time MSM process that, in addition to the key properties of the basic MSM process, also incorporates the leverage effect and dependence between volatility states and price jumps. Their model can be conceived as an extension of a standard stochastic volatility model in which long-run volatility is driven by shocks of heterogeneous frequency that also trigger jumps in the return dynamics and so are responsible for negative correlation between return and volatility. They also develop a particle filter that permits the estimation of the model. By applying the model to option data they find that it can closely reproduce the volatility smiles and smirks. Furthermore, they find that the model outperforms affine jump-diffusions and asymmetric GARCH-type models in- and out-of-sample by a sizeable margin.

Calvet, Fisher, and Wu (2017) develop a class of dynamic term-structure models in which the number of parameters to be estimated is independent of the number of factors selected. This parsimonious design is obtained by a cascading sequence of factors of heterogeneous durations that is modeled in the spirit of multifractal measures. The sequence of mean reversion rates of these factors follows a geometric progression that is responsible for the hierarchical nature of the cascade in the model. In their empirical application to a bandwidth of LIBOR and swap rates, a cascade model with fifteen factors provides a very close fit to the dynamics of the term structure and outperforms random walk and autoregressive specifications in interest rate forecasting.

Taken as a whole, the empirical studies summarized above provide mounting evidence of the superiority of the multifractal model to traditional GARCH models (MS-GARCH and FIGARCH) in terms of forecasting of long-term volatility and related tasks such as VaR assessment. In addition, the model appears quite robust and has found successful applications in modeling of financial durations, the term structure of interest rates, and option pricing.

6.6 CONCLUSION

The motivation for studying multifractal models for asset price dynamics derives from their built-in properties: Since they generically lead to time series with fat tails, volatility clustering, and different degrees of long-term dependence of power transformations of returns, they are able to capture all the universal stylized facts of financial markets. In the overview of extant applications above, MF-type models typically exhibit a tendency to perform somewhat better in volatility forecasting and VaR assessment than the

more traditional toolbox of GARCH-type models. Furthermore, multifractal processes appear to be relatively robust to misspecification, they seem applicable to a whole range of variables of interest from financial markets (returns, volume, durations, and interest rates), and they are very directly motivated by the universal findings of fat tails, clustering of volatility, and anomalous scaling. In fact, multifractal processes constitute the only known class of models in which anomalous scaling is generic; all traditional asset-pricing models have a limiting uniscaling behavior. Capturing this stylized fact may, therefore, well make a difference—even if one can never be certain that multiscaling is not spuriously caused by an asymptotically unifractal model and although the multifractal models that have become the workhorses in empirical applications (MSM and MRW) are characterized by only pre-asymptotic multiscaling.

We note that the introduction of multifractal models in finance did not unleash as much research activity as did that of the GARCH or SV families of volatility models in the preceding decades. The overall number of contributions in this area is still relatively small and comes from a relatively small group of active researchers. The reason for this abstinence might be that the first generation of multifractal models appeared clumsy and unfamiliar to financial economists. Their noncausal principles of construction along the dimension of different scales of a hierarchical structure of dependencies might have appeared too different from known iterative time-series models. In addition, the underlying multifractal formalism (including scaling functions and distribution of Hölder exponents) had been unknown in economics and finance, and application of standard statistical methods of inference to multifractal processes appeared cumbersome or impossible. However, all these obstacles have been overcome with the advent of the second generation of multifractal models (MSM and MRW), which are statistically well behaved and of an iterative, causal nature. Besides their promising performance in various empirical applications, they even provide the additional advantage of having clearly defined continuous-time asymptotics so that applications in discrete and continuous time can be embedded in a consistent framework.

Although the relatively short history of multifractal models in finance has already brought about a variety of specifications and different methodologies for statistical inference, some areas can be identified in which additional work should be particularly welcome and useful. These include multivariate MF models, applications of the MF approach beyond the realm of volatility models such as the MF duration model, and its use in the area of derivative pricing.

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to lay out in detail the historical development of the subject from its initiation in physics to its adaptation to economics. This suggestion was in stark contrast to the insistence of some referees and journal editors in economics to restrict citations to post-2000 publications in finance and economics journals and delete references to previous literature (which, for instance, amounts to not mentioning the all-important contributions by Benoît Mandelbrot, to whose 1974 model of turbulent flows all currently used multifractal models are still unmistakably related).

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CHAPTER 7

PARTICLE FILTERS FOR MARKOV-SWITCHING STOCHASTIC VOLATILITY MODELS

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7.1 INTRODUCTION

Time-varying volatilities are broadly recognized as the nature of most financial time series data. Stochastic volatility (SV) models have been considered as a practical device to capture the time-varying variance; in particular, the mean and log volatility have separate error terms.

Both autoregressive conditional heteroskedasticity (ARCH) models and stochastic volatility models are formulated under the belief of persistence of volatility to some extent. Examples of empirical studies that documented the evidence of volatility persistence include Chou (1988), French et al. (1987), Poon and Taylor (1992), and So et al. (1998). As the economic environment changes, however, the magnitude of the volatility may shift accordingly. Lamoureux and Lastrapes (1990) apply the generalized autoregressive conditional heteroskedasticity (GARCH) model to examining the persistence in volatility, while Kalimipalli and Susmel (2004) show that regime-switching SV model performs better than single-state SV models and the GARCH family of models for short-term interest rates. So et al. (1998) advocate a Markov-switching stochastic volatility (MSSV) model to measure the fluctuations in volatility according to economic forces.

Many methods have been developed to estimate Markov-switching models. Examples of expectation maximization methods include Chib (1996), James et al. (1996), Elliott et al. (1998), and Elliott and Malcolm (2008). Examples of Bayesian Markov Chain Monte Carlo (MCMC) methods include Frühwirth-Schnatter (2001), Hahn et al.

(2010), and Kalimipalli and Susmel (2004). Fearnhead and Clifford (2003) as well as Carvalho and Lopes (2007) utilize particle filters to estimate Markov-switching models.

Casarin (2004) proposes a Markov-switching SV model with heavy-tail innovations that account for extreme variations in the observed processes and apply it sequentially to the Monte Carlo approach to inference. Similarly, Raggi and Bordignon (2008) proposes a stochastic volatility with jumps in a continuous time setting and follow an auxiliary particle filter approach to inference for both the hidden state and parameters. See also Creal (2008) for a sequential Monte Carlo approach to continuous time SV models. He and Maheu (2010) apply a particle filter algorithm to a GARCH model with structural breaks. In the context of a regularized filter for SV models, Carvalho and Lopes (2007) find the regularized APF filters to be the best among several other alternatives for estimation of fixed parameters and states.

The class of filter we use in this chapter belongs to the more general class of regularized particle filters. See, for instance, Musso et al. (2001) for an introduction to regularized particle filters and Gland and Oudjane (2004) for some theoretical results concerning the convergence of this class of filters.

The transition probabilities associated with MSSV models are the crucial parameters to estimate. They not only determine the ergodic probability but also determine how long the system stays in the various regimes. Carvalho and Lopes (2007) combine a kernel-smoothing technique proposed by Liu and West (2001) and an auxiliary particle filter (Pitt and Shephard 1999) to estimate the parameters of the MSSV model. However, this method is quite sensitive to the knowledge of prior distributions.

The modification that we make here to the method of Carvalho and Lopes (2007) is to use an updated Dirichlet distribution to search for reliable transition probabilities rather than applying a multinormal kernel-smoothing algorithm. The Dirichlet distribution has been used with MCMC in Chib (1996) and Frühwirth-Schnatter (2006). The combination of auxiliary particle filters and Dirichlet distribution of transition probabilities allows for an updating path of transition probabilities over time.

As noticed in Liu and West (2001), the regularized particle filter method has an interpretation in terms of the extended model in which the model parameters are evolving under this technique. It should be noticed that in the algorithm proposed above, the use of a Dirichlet distribution with parameters depending on the past evolution of the Markov chain introduces an artificial dynamic into the model, leading to interpretation in terms of an MSSV model with time-varying transition probabilities. In this sense, this work is also related to the duration-dependent SV models in Maheu and McCurdy (2000) and to the stochastic transition Markov-switching models in Billio and Casarin (2010, 2011).

The rest of this chapter is organized as follows. Section 7.2 presents an MSSV model and the proposed method of auxiliary particle filtering. In section 7.3 the simulation results are presented and the methodology is also applied to real data, namely, the exchange rate between the Australian dollar and the South Korean won. We draw some conclusions in section 7.4.

7.2 METHODOLOGY

7.2.1 The Markov-Switching Stochastic Volatility Model

Let y_t be a financial time series with a time-varying log volatility x_t . The observations $y_1, \dots, y_t$ are conditionally independent of the latent variable x_t , and are normally distributed, so that

$$y_t = \exp\left(\frac{x_t}{2}\right) V_t,$$

and the log-volatility is assumed to follow a linear autoregressive process

$$x_t = \alpha_{s_t} + \phi x_{t-1} + \sigma W_t,$$

where V_t and W_t are independent and identically distributed random variables drawn from a standard normal distribution. The drift parameter, $\alpha = (\alpha_1, \dots, \alpha_k)$, indicates the effect of regime shifts. The elements in the set of regime switches are the labels for states, that is, $s_t = \{1, 2, \dots, k\}$, where k is the number of states. The transition probabilities are defined as

$$p_{ij} = \Pr(s_t = j | s_{t-1} = i) \quad \text{for } i, j = 1, 2, \dots, k,$$

where $\sum_{j=1}^k p_{ij} = 1$. In order to avoid the problem of identification, we assume that

$$\alpha_{s_t} = \gamma_1 + \sum_{j=1}^k \gamma_j I_{jt},$$

where $\gamma_1 \in R$, $\gamma_i > 0$ for $i > 1$, and I_{jt} is the indicator function

$$I_{jt} = \begin{cases} 1, & \text{if } s_t \geq j \\ 0, & \text{otherwise.} \end{cases}$$

In the MSSV model, the conditional probability distributions for observations y_t and state variables x_t are given by

$$p(y_t | x_t) = (2\pi e^{x_t})^{-1/2} \exp\left(-\frac{y_t^2}{2e^{x_t}}\right),$$

$$p(x_t | x_{t-1}, \theta, s_t) = (2\pi \sigma^2)^{-1/2} \exp\left(-\frac{(x_t - \alpha_{s_t} - \phi x_{t-1})^2}{2\sigma^2}\right).$$

For convenience, let the vector $\Theta = \{\alpha_1, \dots, \alpha_k, \sigma^2\}$.

In this chapter we illustrate the approach with a simple MSSV model in which there exist only two states, namely high- and low-volatility states, that is, $k = 2$. We also assume that only the mean of volatility shifts depend on the state, so that $\alpha_1 = \gamma_1$ and $\alpha_2 = \gamma_1 + \gamma_2$.

7.2.2 Auxiliary Particle Filter

Let D_t denote a set of observations, so that $D_t = \{y_1, y_2, \dots, y_t\}$. According to Bayes's rule, the conditional probability density function of x_{t+1} is given by

$$p(x_{t+1}|D_{t+1}) = \frac{p(y_{t+1}|x_{t+1})p(x_{t+1}|D_t)}{p(y_{t+1}|D_t)}. \quad (7.1)$$

As shown in equation (7.1), the posterior density $p(x_{t+1}|D_{t+1})$ consists of three components. They are the likelihood function $p(y_{t+1}|x_{t+1})$, the prior $p(x_{t+1}|D_t)$, and the denominator $p(y_{t+1}|D_t)$. The prior distribution for x_{t+1} is given by

$$p(x_{t+1}|D_t) = \int p(x_{t+1}|x_t)p(x_t|D_t)dx_t,$$

and the denominator is an integral

$$p(y_{t+1}|D_t) = \int p(y_{t+1}|x_t)p(x_t|D_t)dx_t.$$

Thus, the posterior distribution for x_{t+1} is a proportion of the numerator on the right-hand side of equation (7.1), that is,

$$p(x_{t+1}|D_{t+1}) \propto p(y_{t+1}|x_{t+1}) \int p(x_{t+1}|x_t)p(x_t|D_t)dx_t.$$

Suppose there is a set of particles $\{x_t^1, \dots, x_t^N\}$ with discrete probabilities $\{\omega_t^1, \dots, \omega_t^N\}$, and $\{x_t^j, \omega_t^j\}_{j=1}^N \sim p(x_t|D_t)$. Therefore the posterior expectation is approximated by

$$\widehat{p}(x_{t+1}|D_t) = \sum_{j=1}^N p(x_{t+1}|x_t^j)\omega_t^j. \quad (7.2)$$

Then at time $t+1$ the posterior distribution is approximated by

$$\widehat{p}(x_{t+1}|D_{t+1}) = p(y_{t+1}|x_{t+1}) \sum_{j=1}^N p(x_{t+1}|x_t^j)\omega_t^j. \quad (7.3)$$

Following Pitt and Shephard (1999), equations (7.2) and (7.3) are known as the empirical prediction density and the empirical filtering density, respectively. The auxiliary particle filter, which is also known as the auxiliary sequential importance resampling (ASIR), adds an indicator in equation (7.3) to do the resampling. The indicator can be the mean or mode, pending on the researcher's taste. Pitt and Shephard (1999) claimed that if the measure of the state variable does not vary over the particles, the ASIR is more efficient than the general SIR. Since $p(x_{t+1}|x_t)$ is more condensed

than $p(x_{t+1}|D_t)$ in terms of to their conditional likelihood, using ASIR for MSSV is a good alternative to SIR.

In addition to tracking the unobserved state variables, we adopt the kernel-smoothing approach of Liu and West (2001) to estimate the parameters, except for transition probabilities. The parameters using the kernel-smoothing estimation are the volatility levels α_1 and α_2 , the volatility variance σ^2 , and the volatility persistence ϕ .

For the case of kernel smoothing, the smooth kernel density form from West (1993) is given by

$$p(\Theta|D_t) \approx \sum_{j=1}^N \omega_t^j N(\Theta|\mathbf{m}_t^j, h^2 \mathbf{V}_t),$$

where Θ is the parameter vector, $h > 0$ is the smoothing parameter, $\mathbf{m}$ and $h^2 \mathbf{V}$ are the mean and variance of the multivariate normal density. Based on this form, Liu and West (2001) proposed the conditional evolution density for Θ according to

$$p(\Theta_{t+1}|\Theta_t) \sim N(\Theta_{t+1}|a\Theta_t + (1-a)\bar{\Theta}_t, h^2 \mathbf{V}_t),$$

where $a = \frac{3\delta-1}{2\delta}$, and $h^2 = 1 - a$. The discount factor δ is in $(0, 1]$, and $\bar{\Theta}_t$ and $\mathbf{V}_t$ are the mean and variance of the Monte Carlo approximation to $p(\Theta|D_t)$. Straight-forward calculations indicate that $\bar{\Theta}_t = \sum_{j=1}^N \omega_t^j \Theta_t^j$ and $\mathbf{V}_t = \sum_{j=1}^N \omega_t^j (\Theta_t^j - \bar{\Theta}_t) (\Theta_t^j - \bar{\Theta}_t)'$.

For the case of the transition probabilities, the parameters are updated by the Dirichlet distribution. Suppose that the matrix of transition probabilities P is $k \times k$, and the sum of each row is equal to 1. Then the i th row of P is denoted by $p_i = \{p_{i1}, \dots, p_{ik}\}$, and let p_i be the random variables of a Dirichlet distribution,¹ so that

$$p_i \sim \mathcal{D}(\lambda_{i1}, \dots, \lambda_{ik}).$$

Each prior distribution of p_i is independent of p_j , $i \neq j$. According to Chib (1996), the updated distribution of $P|S_t$ is also a Dirichlet distribution, where $S_t = \{s_1, s_2, \dots, s_t\}$ and

$$p_i|S_t \sim \mathcal{D}(\lambda_{i1} + n_{i1}, \dots, \lambda_{ik} + n_{ik}),$$

where n_{ik} is the number of one-step transitions from state i to state k in the sample S_t . In this case, we assume a two-state problem, so that $k = 2$.

Initially, the starting parameter values for each particle are drawn from their prior distributions. Afterwards, in the case of Markov-switching stochastic volatility,

¹ The Dirichlet distribution, $Y \sim \mathcal{D}(\alpha_1, \dots, \alpha_N)$, is a standard choice in the context of modeling an unknown discrete probability distribution $Y = (Y_1, \dots, Y_N)$ where $\sum_{j=1}^N Y_j = 1$ and therefore is of great importance for mixture and switching models. The Dirichlet distribution is a distribution on the unit simplex $E_N \in (R^+)^N$, defined by the constraint $E_N = \{y = (y_1, \dots, y_N) \in (R^+)^N : \sum_{j=1}^N y_j = 1\}$.

The density is given by $f_D(y_1, \dots, y_N) = y_1^{\alpha_1-1} \dots y_N^{\alpha_N-1} c$, where $c = \frac{\Gamma(\sum_{j=1}^N \alpha_j)}{\prod_{i=1}^N \Gamma(\alpha_i)}$.

the starting state variable s_0^j is determined by the ergodic probability. The ergodic probability for two states is

$$\Pr(s_0^j = 1) = \frac{1 - p_{22}^j}{2 - p_{11}^j - p_{22}^j},$$

and

$$\Pr(s_0^j = 2) = 1 - \Pr(s_0^j = 1).$$

If a random number from a uniform distribution (0,1) is less than $\Pr(s_0^j = 1)$, then $s_0^j = 1$; otherwise $s_0^j = 2$. Given the state, the starting log-volatility value can be drawn from a normal distribution, $x_0^j \sim N\left(\frac{\alpha_{s_0^j}^j}{1 - \phi_0^j}, (\sigma_0^j)^2\right)$. Below is the algorithm for the ASIR followed by an updated Dirichlet distribution.

While $t \leq T$,

Step 1: Determine the mean (by guessing initially)

For $j = 1$ to N ,

$$s_{t+1}^{\mu^j} = \arg \max_{i=1,2} \Pr(s_{t+1} = i | s_t^j)$$

$$\mu_{t+1}^j = \alpha_{s_{t+1}^j} + \phi_t^j x_t^j$$

$$\omega_{t+1}^{\mu^j} \propto p(y_{t+1} | \mu_{t+1}^j) \tilde{\omega}_t^j$$

End for

$$\text{Calculate the normalized importance weights } \tilde{\omega}_{t+1}^{\mu^j} = \frac{\omega_{t+1}^{\mu^j}}{\sum_{j=1}^N \omega_{t+1}^{\mu^j}}.$$

Step 2: Resampling

For $j = 1$ to N ,

$$\left\{ \Theta_t^{j^l}, \mu_{t+1}^{j^l}, x_t^{j^l}, s_t^{j^l} \right\} = \text{resample} \left\{ \Theta_t^j, \mu_{t+1}^j, x_t^j, s_t^j, \tilde{\omega}_{t+1}^{\mu^j} \right\},$$

$$\Theta_{t+1}^{j^l} \sim N(a\Theta_t^{j^l} + (1-a)\bar{\Theta}_t, h^2 \mathbf{V}_t^l),$$

$$\text{update } n_{t+1,ij}, \text{ and } p_{t+1,i} \sim \mathcal{D}(\lambda_{i1} + n_{t+1,i1}, \lambda_{i2} + n_{t+1,i2}), \quad i = 1, 2.$$

End for²

Step 3: Sample the hidden variables $\{s_{t+1}^j, x_{t+1}^j\}$

For $j = 1$ to N ,

Filter the conditional probability $\Pr(s_{t+1}^j = k | \mathbf{y}^{t+1}, \Theta_{t+1}^j)$, $k = 1$ or 2 .

(a) One-step-ahead prediction probabilities

$$\Pr(s_{t+1}^j = k | \mathbf{y}^t, \Theta_t^j) = \sum_{i=1}^K \Pr(s_{t+1}^j = k | s_t^j = i) \Pr(s_t^j = i | \mathbf{y}^t, \Theta_t^j).$$

² We refer the reader to the appendix for more details.

(b) Filter the s_t

$$\Pr(s_{t+1}^j = k | y^{t+1}, \Theta_{t+1}^j) = \frac{p(y_{t+1} | s_{t+1}^j = k, y^t, \Theta_{t+1}^j) \Pr(s_{t+1}^j = k | y^t, \Theta_t^j)}{\sum_{i=1}^K p(y_{t+1} | s_{t+1}^j = i, y^t, \Theta_{t+1}^j) \Pr(s_{t+1}^j = i | y^t, \Theta_t^j)}.$$

(c) Draw $\widetilde{p}_{t+1}^j \sim \text{uniform}(0, 1)$

if $\widetilde{p}_{t+1}^j \leq \Pr(s_{t+1}^j = k | y^{t+1}, \Theta_{t+1}^j)$, then $s_{t+1}^j = k$
otherwise, s_t would be another state.

Sample $x_{t+1}^j \sim p(x_{t+1}^j | x_t^j, s_{t+1}^j, \Theta_{t+1}^j)$,

$$\omega_{t+1}^j \propto \frac{p(y_{t+1} | x_{t+1}^j)}{p(y_{t+1} | \mu_{t+1}^j)}$$

End for

Normalized importance weights $\widetilde{\omega}_{t+1}^j = \frac{\omega_{t+1}^j}{\sum_{j=1}^N \omega_{t+1}^j}$

Step 4: Summarize $\Theta_{t+1} = \widetilde{\omega}_{t+1}^j \Theta_{t+1}^j$, $x_{t+1} = \widetilde{\omega}_{t+1}^j (x_{t+1}^j)'$.

Step 5: Redo from Step 1 ($t = t + 1$).

End while

7.3 RESULTS

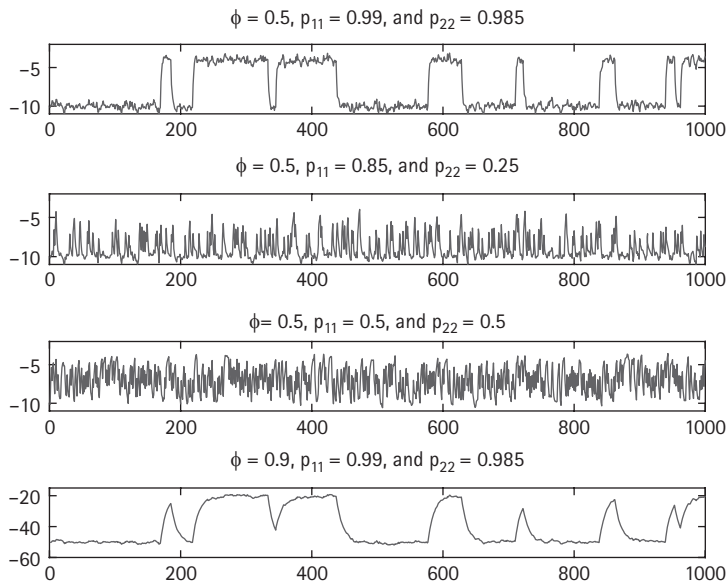
We apply the approach we have discussed in the previous section to a number of simulated prices and returns to demonstrate that our approach is able to estimate a wide range of transition probability matrices. In particular, it is able to detect the states on which the system may not stay long enough, for instance, tests 2 and 3 in the example below, which many other methods have difficulty in detecting.

7.3.1 Simulation Study

In this subsection, we use four data sets to illustrate the method. They have been generated from an MSSV model with two states. The parameters of these four data sets are shown in table 7.1, and their log-volatility is shown in figure 7.1. The parameter Θ can be updated by a multinormal distribution, so we transform some parameters such as γ_2 , σ^2 and $\frac{\phi}{1-\phi}$ to $\log(\gamma_2)$, $\log(\sigma^2)$, and $\log\left(\frac{\phi}{1-\phi}\right)$. The first and the fourth samples allow the transition probability matrix to concentrate on the diagonal, but the persistence parameter varies. Thus, the unconditional means for the volatility are different. The second and third samples have relatively lower diagonal transition probabilities, which means the volatility regime changes frequently.

Table 7.1 Parameter values for the simulated data sets

Parameter	Test 1	Test 2	Test 3	Test 4
γ_1	-5.0	-5.0	-5.0	-5.0
γ_2	3.0	3.0	3.0	3.0
σ^2	0.1	0.1	0.1	0.1
ϕ	0.5	0.5	0.5	0.9
p_{11}	0.99	0.85	0.5	0.99
p_{22}	0.985	0.25	0.5	0.985

**FIGURE 7.1** Log-volatility for four data sets.

The starting values for the estimation are determined by their prior distribution, with the central tendency close to their true values. Since $k = 2$, the transition probability matrix is

$$\begin{bmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{bmatrix},$$

where p_{11} is the probability of state 1 given that the previous state is 1, and p_{22} is the probability of state 2 given that the previous state is 2. The discount rate δ is set at 0.75, which implies that $a = 0.8333$ and $h = 0.4082$.

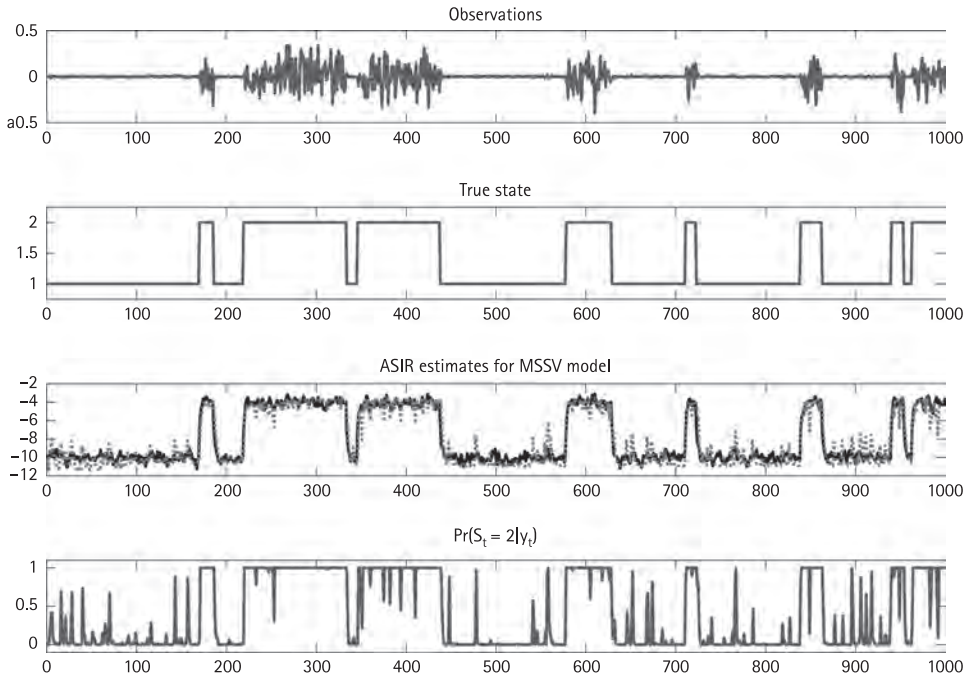


FIGURE 7.2 Simulated data set 1: the top graph shows the simulated time series y_t ; the second graph, the true regime variables s_t ; the third graph, the true (solid line) and estimated log-volatilities (dotted line); and the bottom graph, the estimated probability $\Pr(s_t = 2|y_t)$.

Figures 7.2 to 7.5 show the simulation results for the four data sets. Each figure has four graphs. The top graph shows the simulated time series data, and the second contains the simulated Markov Chain (the shifting states). The third graph compares the simulated log-volatility and the estimated log-volatility. The bottom graph shows the estimated probability that the state is in the high-volatility regime given by the previous information.

In each of the figures, the estimated states (shown in the bottom graph) match the true states (the second graph) quite well, especially for tests 2 and 3, in which the system switches quite often. The quality of the estimates can be confirmed by comparing the estimates in table 7.2 with the true parameters in each test set.

The sequential estimation of the parameters of these four simulated data sets are shown separately in figures 7.6 to 7.9. The solid lines denote the modes of the parameters, the dotted lines represent the 10 percent and 90 percent quantiles, and the dashed lines represent the true values of the parameters. In addition, the modes of the parameters are summarized in table 7.2.

These plots show the estimated posterior mode at time t for each parameter together with approximate credible 10 percent and 90 percent quantiles, along with the true value for each parameter. It can be seen that the transition probabilities are among those that are more difficult to estimate. We have fairly stable estimates for other parameters.

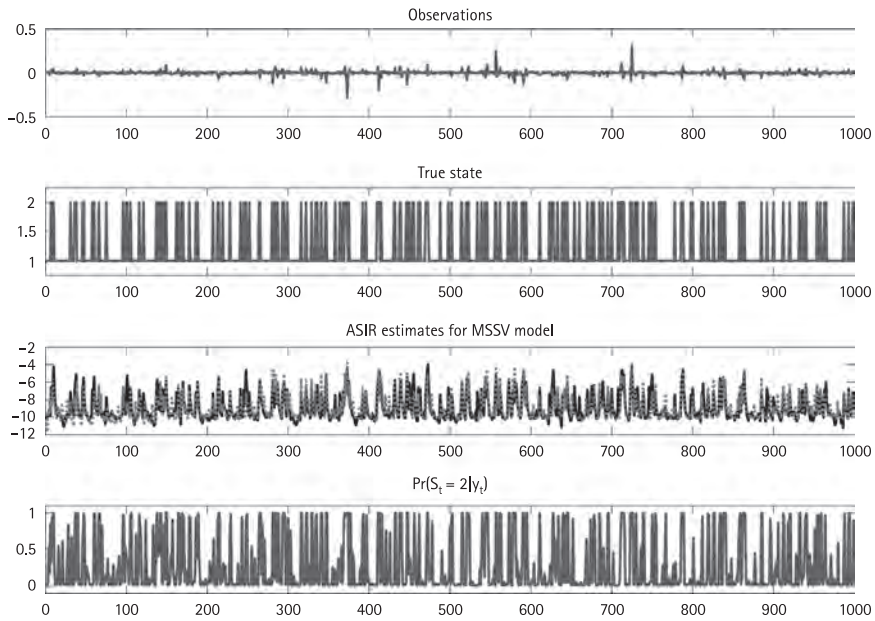


FIGURE 7.3 Simulated data set 2: the top graph shows the simulated time series y_t ; the second graph, the true regime variables s_t ; the third graph, true (solid line) and estimated log-volatilities (dotted line); and the bottom graph, the estimated probability $Pr(s_t = 2|D_t)$.

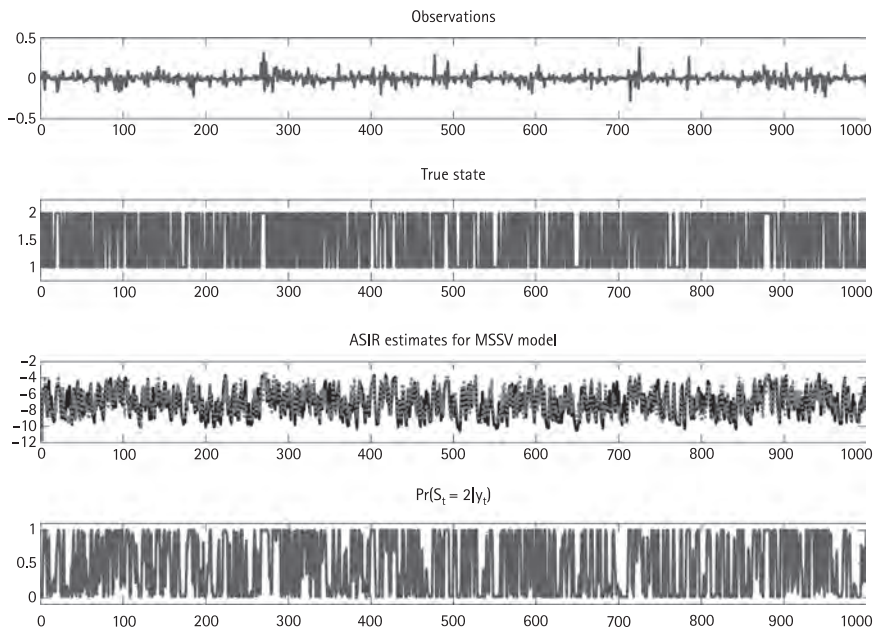


FIGURE 7.4 Simulated data set 3: the top graph shows the simulated time series y_t ; the second graph, true regime variables s_t ; the third graph, true (solid line) and estimated log-volatilities (dotted line), and the bottom graph, the estimated probability $Pr(s_t = 2|D_t)$.

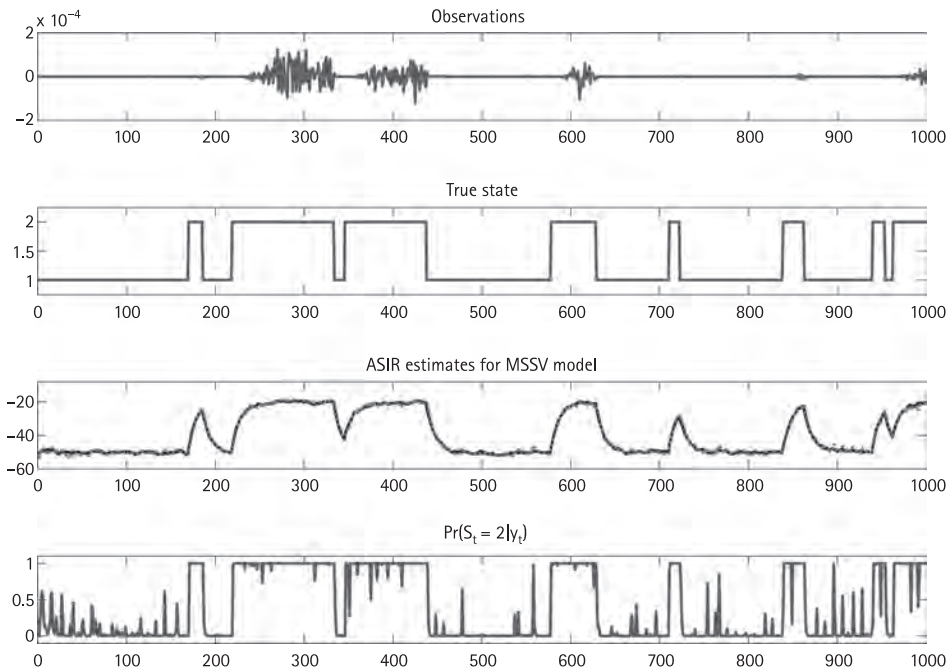


FIGURE 7.5 Simulated data set 4: the top graph shows the simulated time series y_t ; the second graph, true regime variables s_t ; the third graph, true (solid line) and estimated log-volatilities (dotted line); and the bottom graph; the estimated probability $Pr(s_t = 2|D_t)$.

Table 7.2 Posterior modes of the parameters

	Test 1 mode	Test 2 mode	Test 3 mode	Test 4 mode
γ_1	-5.0604	-4.9303	-4.9163	-5.0800
γ_2	3.2833	2.9811	3.0863	3.1540
σ^2	0.0880	0.1292	0.1332	0.1149
ϕ	0.5335	0.5170	0.4874	0.9012
ρ_{11}	0.9712	0.7841	0.5816	0.9782
ρ_{22}	0.9669	0.3690	0.5529	0.9747

7.3.2 An Application to Real Data

In this subsection, the proposed algorithm is applied to the exchange rate between the South Korean won and the Australian dollar from January 2, 1995, to December 31, 1999 (1,014 observations). This period includes the Asian financial crisis of 1997, from which South Korea suffered a great deal.

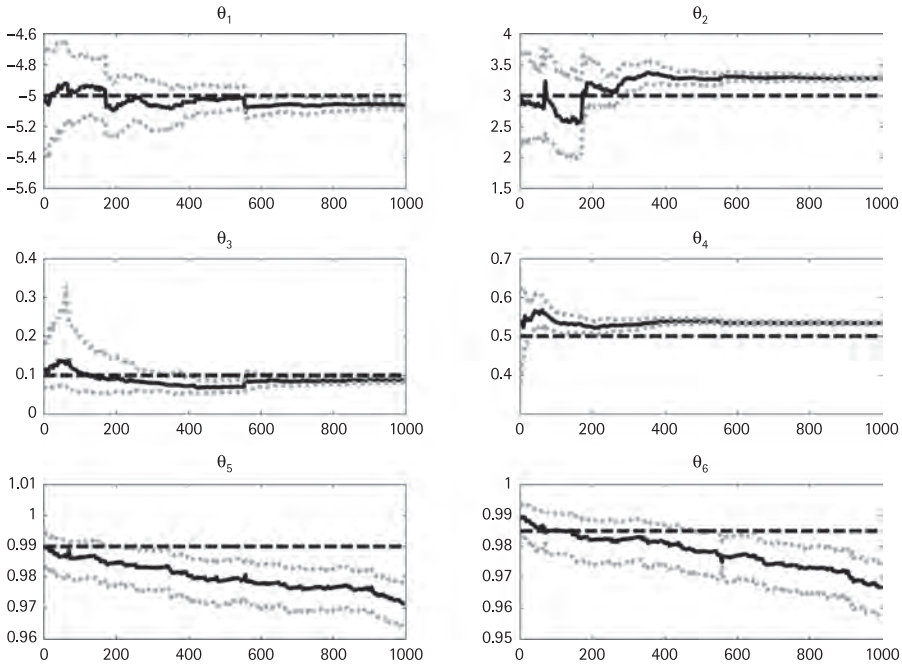


FIGURE 7.6 Posterior mode with the 10 percent and 90 percent quantiles of Θ for the first simulated data set: $\theta_1 = \gamma_1$, $\theta_2 = \gamma_2$, $\theta_3 = \sigma^2$, $\theta_4 = \phi$, $\theta_5 = p_{11}$, and $\theta_6 = p_{22}$.

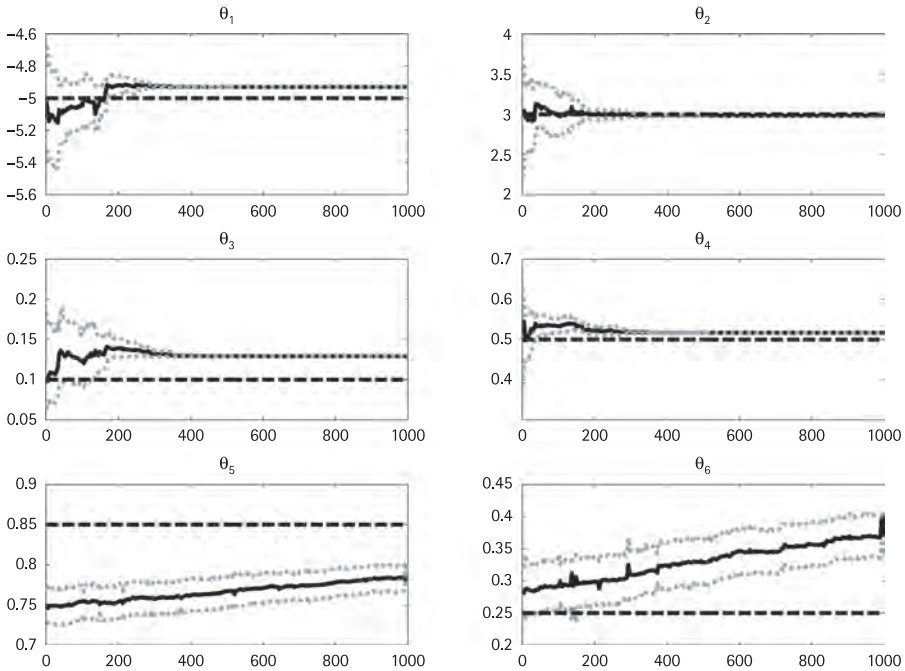


FIGURE 7.7 Posterior mode with the 10 percent and 90 percent quantiles of Θ for the second simulated data set: $\theta_1 = \gamma_1$, $\theta_2 = \gamma_2$, $\theta_3 = \sigma^2$, $\theta_4 = \phi$, $\theta_5 = p_{11}$, and $\theta_6 = p_{22}$.

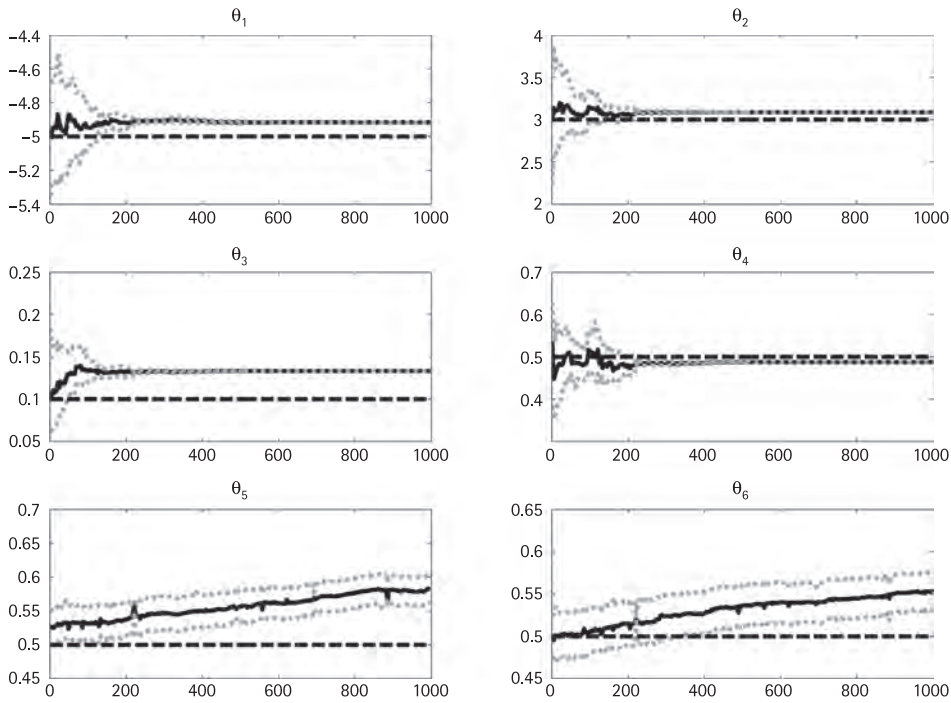


FIGURE 7.8 Posterior mode with the 10 percent and 90 percent quantiles of Θ for the third simulated data set: $\theta_1 = \gamma_1$, $\theta_2 = \gamma_2$, $\theta_3 = \sigma^2$, $\theta_4 = \phi$, $\theta_5 = p_{11}$, and $\theta_6 = p_{22}$.

Figure 7.10 shows the log difference of the exchange rate, the estimated log-volatility, and the estimated probability that state is equal to 2. According to figure 7.10, the volatility of the exchange rate becomes larger during the Asian financial crisis and switches regimes frequently afterwards. In other words, the stable movement of the exchange rate finishes when the crisis begins.

The sequential estimation of the exchange rate is shown in figure 7.11, and the updated values of mode and quantiles are shown in table 7.3. The estimate of the persistence parameter ϕ is about 0.5, which is not overestimated, according to the findings of So et al. (1998) and Carvalho and Lopes (2007). The diagonal elements of the transition probability declines over time, in particular after the Asian financial crisis. In other words, the exchange rate for the Australian dollar against the South Korean won becomes more volatile after the crisis.

7.3.3 Diagnostic for Sampling Improvement

Following Carpenter et al. (1999), we implement an effective sample size to assess the performance of the particle filter. The comparison of the effective sample size of the ASIR with multinormal kernel smoothing and the (proposed) ASIR with the updated

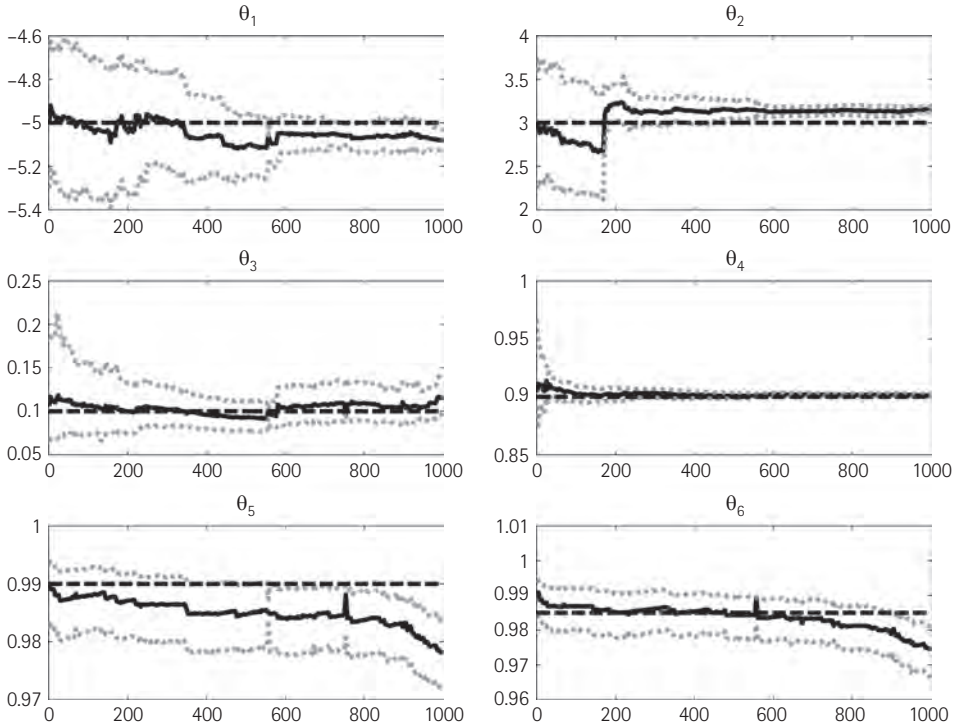


FIGURE 7.9 Posterior mode with the 10 percent and 90 percent quantiles of Θ for the fourth simulated data set: $\theta_1 = \gamma_1$, $\theta_2 = \gamma_2$, $\theta_3 = \sigma^2$, $\theta_4 = \phi$, $\theta_5 = p_{11}$, and $\theta_6 = p_{22}$.

Dirichlet distribution is presented, showing whether the proposed ASIR is more robust rather than the previous one.

The algorithm for calculating the effective sample size is shown below. Suppose $g(x_t)$ is a measure of x_t , and its expectation is

$$\theta = \int g(x_t) p(x_t | y_t) dx_t.$$

The discrete approximation to the θ is given by

$$z_t = \sum_{i=1}^N \omega_t^i g(x_t^i),$$

and the variance of the measure $g(x_t)$ is given by

$$v_t = \sum_{j=1}^N \omega_t^j g^2(x_t^j) - z_k^2.$$

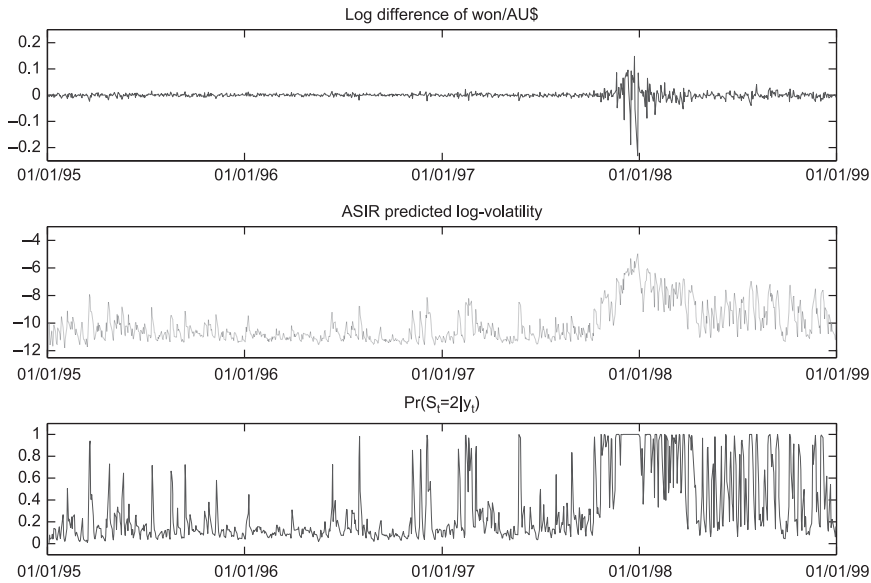


FIGURE 7.10 The exchange rate (won/AUS): the top graph shows the observed time series y_t ; the second graph, the estimated log-volatilities, and the bottom graph, the estimated probability $Pr(s_t = 2 | D_t)$.

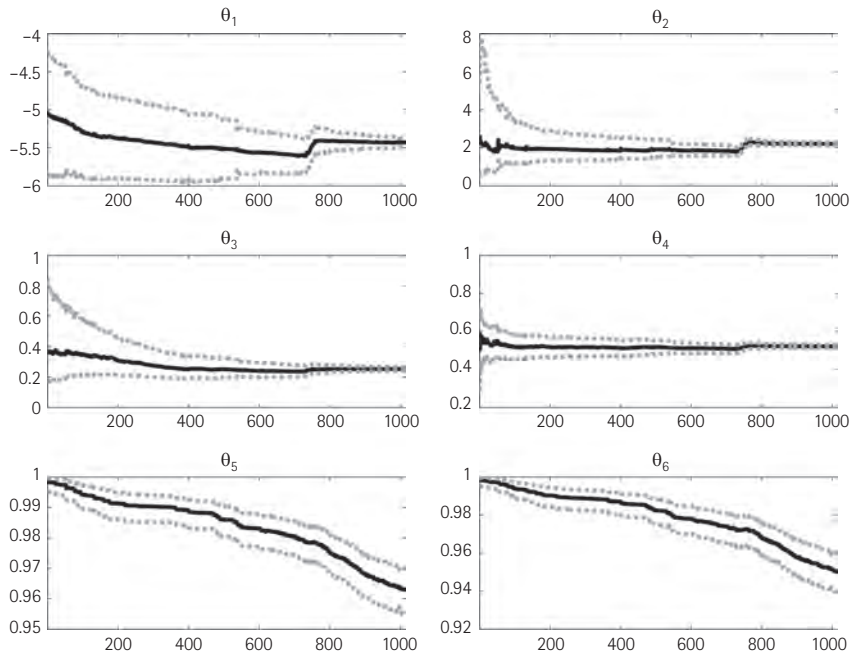


FIGURE 7.11 Posterior mode with the 10 percent and 90 percent quantiles of Θ for the exchange rate (won/AUS) from January 2, 1995, to December 31, 1999: $\theta_1 = \gamma_1$, $\theta_2 = \gamma_2$, $\theta_3 = \sigma^2$, $\theta_4 = \phi$, $\theta_5 = p_{11}$, and $\theta_6 = p_{22}$.

Table 7.3 The updated posterior modes with 10% and 90% quantiles of the parameters

	γ_1	γ_2	σ^2	ϕ	p_{11}	p_{22}
Mode	-5.4294	2.2087	0.2533	0.5216	0.9629	0.9504
10%	-5.4885	2.1534	0.2428	0.5159	0.9548	0.9396
90%	-5.3708	2.2668	0.2648	0.5272	0.9699	0.9599

Table 7.4 Comparison of effective sample sizes

Time horizon (in days)	100	200	300	400	500	600	700	800	900	1,000
Dirichlet updating	32	33	37	2	3	27	15	18	21	8
Kernel smoothing	70	27	62	1	2	20	8	2	11	1

Suppose the independent filter has been run M times, and the values of z_t and v_t can be calculated. Then the effective sample size is given by

$$N_t^* = \frac{\bar{v}_t}{\frac{1}{M} \sum_{j=1}^M (z_t^j - \bar{z}_t)^2}.$$

The greater the value of the effective sample size, the more likely the filter is to be reliable. The comparison of the proposed model and the model with kernel smoothing for transition probability is shown in table 7.4. According to the results, as time goes by, the use of Dirichlet updating is more reliable than the use of the kernel-smoothing method.

7.4 CONCLUSIONS

In this article we develop and implement an auxiliary particle filter algorithm to estimate a univariate regime-switching stochastic volatility model. The use of simulated examples was intended to show the performance of the proposed method.

In particular, in terms of estimating the transition probabilities of the Markov Chain, we modified the method given in Carvalho and Lopes (2007) to use an updated Dirichlet distribution to search for reliable transition probabilities rather than applying a multinormal kernel-smoothing method that can only give a good estimate when the probability that the system state transits from one regime to another is rather low. The combination of auxiliary particle filters and a Dirichlet distribution for transition probabilities allows for an updating path of transition probabilities over time. It also

will accommodate the case in which the probability that the system state transits from one regime to another is relatively high. This feature is often observed in the energy, commodities, or foreign exchange market.

7.5 APPENDIX: RESAMPLING

We adopt the systematic resampling method mentioned in Ristic et al. (2004). The algorithm for $\left\{ \Theta_t^{j^l}, \mu_{t+1}^{j^l}, x_t^{j^l}, s_t^{j^l} \right\} = \text{resample} \left\{ \Theta_t^j, \mu_{t+1}^j, x_t^j, s_t^j, \widetilde{\omega_{t+1}^j} \right\}$ is shown below.

Step 1: Draw $u_0 \sim \text{uniform}(0, 1)$,

construct a cdf of importance weights, i.e., $c_j = \sum_{j=1}^N \widetilde{\omega_{t+1}^j}$,
 $i = 1$.

Step 2: For $j = 1$ to N

$u_j = u_{j-1} + 1/\text{no. of particles}$,

If $u_j > c_j$,

$i = i + 1$,

else $i = i$,

End if.

$\Theta_t^{j^l} = \Theta_t^i, \quad \mu_{t+1}^{j^l} = \mu_{t+1}^i, \quad x_t^{j^l} = x_t^i, \quad s_t^{j^l} = s_t^i,$

End for.

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CHAPTER 8

ECONOMIC AND FINANCIAL MODELING WITH GENETIC PROGRAMMING

A Review

CLÍODHNA TUITE, MICHAEL O'NEILL, AND
ANTHONY BRABAZON

Genetic Programming (GP) is a stochastic optimization and model induction technique, which has found many uses in the field of financial and economic modeling. An advantage of GP is that the modeler is not required to explicitly select the exact parameters to be used in the model, before the model building process begins. Rather, GP can effectively search a complex model space defined by a set of building blocks specified by the modeler, yielding a flexible modeling approach. This flexibility has allowed GP to be used in a large number of different application areas, within the broader field. In this work, we review some of the most significant developments using GP, across several application areas: forecasting (which includes technical trading); stock selection; derivative pricing and trading; bankruptcy and credit risk assessment; and, finally, agent-based and economic modeling. Conclusions reached by studies investigating similar problems do not always agree, perhaps due to differing experimental setups employed, and the inherent complexity and continually changing nature of the application domain; however, we find that GP has proven a useful tool across a wide range of problem areas, and continues to be used and developed. Recent and future work is increasingly concerned with adapting genetic programming to more dynamic environments, and ensuring that solutions generalize robustly to out-of-sample data, to further improve model performance.

8.1 INTRODUCTION AND BACKGROUND

Financial markets have become increasingly complex in recent decades (Arewa 2010). Markets have also seen acceleration in the use of computers (Sichel 1997), and the diffusion of computers has led to the use of machine learning techniques, including those used in financial econometrics (Fabozzi et al. 2006). Machine learning is concerned with creating computer programs that improve automatically with experience, and it has been used to discover knowledge from financial and other databases (Mitchell 1997). Genetic programming (GP) is one such learning technique that does not require the solution structure to be determined a priori (Allen and Karjalainen 1999). It has been applied to a wide range of problems in economic and financial modeling, including price forecasting, stock selection, credit assessment, and it forms a component of some agent-based modeling techniques.

8.1.1 An Introduction to Genetic Programming

Genetic programming (Koza 1992) is a stochastic search-based algorithm that optimizes toward a predefined “fitness function.” It is modeled on the principles of biological evolution. The algorithm operates over a number of iterations, or generations. At each generation a “population” of “individuals,” or potential solutions, compete for survival into the next generation. Such survival is governed by the fitness function, which is a domain-specific measure of how well an individual solution has performed in terms of an objective of interest (Koza 1992; Banzhaf et al. 1998; Poli et al. 2008). In a finance application, an example fitness function to be maximized could be the return on investment.

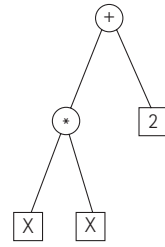
8.1.1.1 *Representations and Building Blocks*

The individuals in a GP population are typically computer programs or encodings of programs. Individuals are typically represented as trees, although string-based (and other) representations are also possible. Individuals are constructed from building blocks, which are specified by the modeler. Each building block forms part of either the terminal set or the function set. The terminal set is typically comprised of constants, variables, and zero-argument functions, and the function set could include arithmetic and Boolean functions, conditional operators, mathematical functions, and other domain-specific functions. In a finance application, such a domain-specific function could return the current price of a security of interest. Figure 8.1 shows an example of a GP individual that could be used to evolve a quadratic function, given training data consisting of points on the graph of the function.

8.1.1.2 *GP Runs*

When performing a GP run, the human modeler will specify some run parameters such as the population size, the number of generations the run should proceed for, and

FIGURE 8.1 An illustrative GP individual, which could be used to evolve a quadratic function, given training data consisting of points on the graph of the function.



probabilities that determine the application of the genetic operators of reproduction, crossover, and mutation, described below. Before the start of the run, the initial population has to be created. This is typically done in a random fashion, creating individuals from the given building blocks. However, in, for example, the tree-based approaches to GP, it is usual to ensure that (to a certain tree-depth limit) the structural space has good coverage, and this is achieved using an initialization method called ramped-half-and-half. It is important to ensure a good sampling of the structural space because we do not know a priori what the size or structure of the solution might be, and GP must search simultaneously the space of structures and the content of each structure.

After each generation, the population for the next generation is created. The first step in this process is selection, in which a subset of the population is selected to contribute to the next generation; a number of different fitness-based techniques are available for selecting individuals. A popular example is the tournament, which involves selecting a number of individuals at random from the population, the best of which, as determined by the fitness metric, “wins the tournament” and is selected (Banzhaf et al. 1998; Poli et al. 2008). Popular genetic operators include reproduction crossover, and mutation. When performing reproduction, an individual (chosen using the selection technique) is directly copied and included in the population for the next generation. Mutation, used to promote exploration of the search space, involves a minor random perturbation of a selected individual. Crossover also acts to aid in the exploration of the search space. Crossover combines pieces from two individuals. For example, if using tournament selection, two tournaments are run, one to select each parent to participate in crossover. There are various implementations of crossover for GP, a common one being the subtree. To perform this type of action, a crossover point is chosen at random in each parent, and a copy of the subtree rooted at the crossover point in one parent is inserted into a copy of the second parent, replacing the subtree rooted at the crossover point in the second parent. This process forms the offspring (Poli et al. 2008). Figure 8.2 shows crossover in operation; as in figure 8.1 the target function is a quadratic function; the crossover point is the node below the dashed line in each parent. The GP run ends after the specified number of generations have completed or when a success criterion has been satisfied (such as the best individual of a generation achieving a given desired fitness).

8.1.1.3 Further Information

Introductory textbooks on GP include those by Poli et al. (2008) and Banzhaf et al. (1998). The first book introducing what is now referred to as standard tree-based

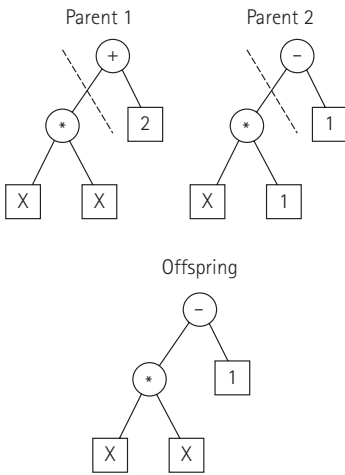


FIGURE 8.2 Crossover in genetic programming. The crossover point is the node below the dashed line in each parent.

genetic programming is that by Koza (1992). Three more books appear in Koza's series on GP (Koza 1994; Koza et al. 1999; Keane et al. 2006). Many different representations for programs have been adopted in the GP literature beyond the standard tree-based GP, including linear genetic programming (Brameier and Banzhaf 2007) and grammatical evolution (O'Neill and Ryan 2003; Dempsey et al. 2009). The latter uses a binary-string-based representation that is mapped to the solution space using a domain-specific grammar and allows the user to control the language structures under evolution, providing flexibility, whereas linear GP "evolves sequences of instructions from an imperative programming language or from a machine language" (Brameier and Banzhaf 2007, p. 6). Another technique is genetic network programming, which is useful in dynamic environments; its programs are composed of a group of connected nodes that execute simple judgment processing (Mabu et al. 2007).

8.1.2 How is Genetic Programming Useful?

Machine learning techniques such as genetic programming take a fundamentally different approach to financial modeling than do traditional approaches, which involve estimating parameters in a well-defined theoretical model (Fabozzi et al. 2006). A number of influential theoretical models have been formulated in the area of finance. For example, the capital asset pricing model (Sharpe 1964) claims that the excess expected return of a stock depends on its systematic risk (Becker et al. 2007). It was followed by arbitrage pricing theory (Ross 1976) and the Fama-French three-factor model (Fama and French 1992). In the area of option pricing there exists the seminal Black-Scholes model (Black and Scholes 1973). Given that these theoretical models may not fully reflect the real world, however, data-driven machine learning approaches such as genetic programming have been proposed to try to provide useful alternatives.

Genetic programming has a number of characteristics that make it a suitable choice for a variety of modeling tasks. It allows modelers to build on existing knowledge by allowing the incorporation of domain knowledge into the search. For example, when trying to evolve an option pricing formula, the Black-Scholes equation can be included in the initial population (Chidambaran 2003). A further advantage of GP is its ability to evolve nonlinear as well as linear models. Linear models, though easy to understand, may not capture the complexities of real-world financial markets (Becker and O'Reilly 2009). Solutions evolved using GP can be human-readable, unlike alternative machine learning techniques such as neural networks, allowing the modeler to perform further evaluation before deployment. Unlike many alternative modeling techniques, GP does not require pre-selection of the parameters that will be used to form the model. Rather, the human modeler provides GP with a terminal and function set, as described in subsection 8.1.1.1, from which it builds its model. Genetic programming does not have to include each building block, in effect setting aside those that do not increase the explanatory power of the model. Often the parameters of a model are not known before modeling begins, and so constraining the model to have a specific set of parameters can result in finding an incorrect model. Using a method like GP allows more flexibility in parameter selection and can lead to discovery of better models.

8.1.3 An Outline of the Modeling Applications Reviewed

The following sections provide an introduction and overview to the main areas where genetic programming has been applied in economic and financial modeling. We focus on five main application areas. Some of the research we review in the coming sections does not fit neatly into one category; in these cases we have chosen the most relevant categorization for the application in question. The application areas covered are as follows:

- forecasting (comprising the important forecasting activity of technical analysis)
- stock selection
- derivative price modeling and trading
- bankruptcy and credit risk assessment
- agent-based modeling

Figure 8.3 illustrates this categorization. First, a brief description is given of the Efficient Market Hypothesis; issues surrounding this hypothesis motivate much of the research undertaken in the area of financial and economic modeling using GP.

8.1.4 The Efficient Market Hypothesis

The efficient market hypothesis, or EMH (Fama 1970) has generated much debate for the past few decades. The semi-strong form of the EMH states that all public

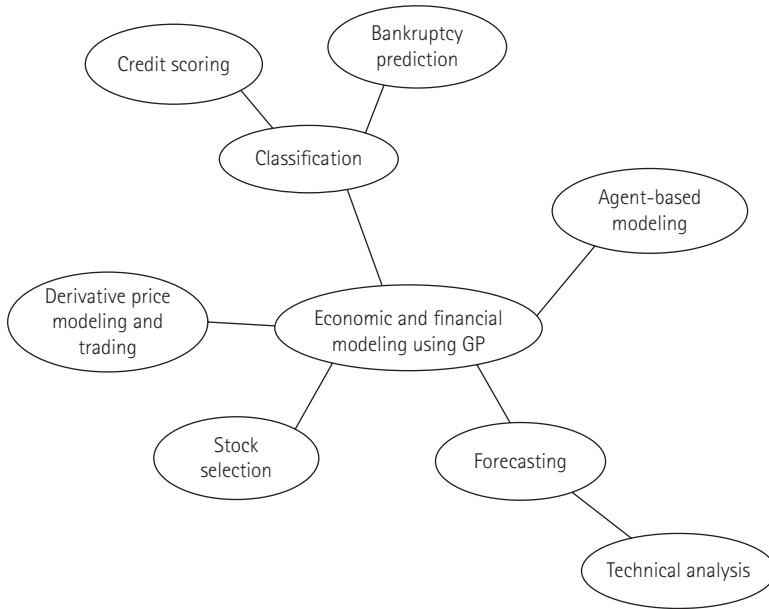


FIGURE 8.3 Application areas covered in this review.

information pertaining to a stock is incorporated in its price. Therefore, an investor cannot expect to outperform the market by analyzing information such as the firm's history or its financial statements. It is still possible to make profits by taking on risk, but on a risk-adjusted basis, the EMH implies, investors cannot expect to consistently beat the market (Mayo 2013). This hypothesis has obvious relevance for research in this broad area—indeed, investigation of this hypothesis has served as a key motivation for some of the seminal work using GP in the area of economic and financial modeling, as we discuss in the following sections.

8.2 FORECASTING AND TECHNICAL TRADING

In the mid-1990s, a number of papers appeared that used genetic programming to evaluate the profitability (or lack thereof) of technical trading rules. Underpinning much of this research was a desire to establish how well the efficient market hypothesis was supported by the data. Technical analysis runs counter to the EMH; technical analysts study past patterns of variables such as prices and volumes to identify times when stocks are mispriced. The EMH would imply that it is futile to engage in this process on an ongoing basis.

Popular categories of technical indicators include moving average indicators and momentum indicators. The simplest moving average systems compare the current price

with a moving average price over a lagged period, to show how far the current price has moved from an underlying trend (Brabazon and O'Neill 2006). A trading signal is produced when the price moves above or below its moving average. For example, if the current price moved above the moving average of prices for the past one hundred days, a trading signal to buy could be generated. The moving average convergence-divergence oscillator is slightly more involved; it calculates the difference between a short-run and a long-run moving average. If the difference is positive, it can be seen as an indication that the market is trending up. For example, if the short-run moving average crosses the long-run moving average from below, it could trigger the generation of a buy signal (Brabazon and O'Neill 2006, p. 147). Momentum traders invest in stocks whose price has recently risen significantly, or sell stocks whose price has recently fallen significantly, on the assumption that a strong trend is likely to last for a period of time. If a technical trading system composed of rules that determined when to buy or sell could consistently and significantly outperform a benchmark such as the buy-and-hold, it would provide evidence that refuted the EMH.

In the GP studies concerning technical trading rules, the basic building blocks that form the terminal and function sets vary between lower-level primitives such as arithmetic and conditional operators and constants and higher-level primitives such as moving average and momentum indicators, depending on the study in question. As stated by Park and Irwin (2007, p. 799), GP offers the following advantages in evaluating technical trading systems:

The traditional approach investigates a pre-determined parameter space of technical trading systems, while the genetic programming approach examines a search space composed of logical combinations of trading systems or rules. Thus, the fittest (or locally optimized) rules identified by genetic programming can be viewed as *ex ante* rules in the sense that their parameter values are not determined before the test. Since the procedure helps researchers avoid some of the arbitrariness involved in selecting parameters, it may reduce the risk of data snooping biases.

An illustrative example of a technical trading rule tree that could be evolved using GP is shown in figure 8.4.

8.2.1 Early Studies, Contradictory Results

One of the first widely cited papers that used genetic programming to investigate the profitability of technical trading rules in a foreign exchange setting appeared in Neely et al. (1997). The authors used GP successfully to identify profitable technical trading rules for six exchange rate series, after transaction costs. For the case of \$/DM, bootstrapping results showed that these profitable trading rules were uncovering patterns in the data that were not found by standard statistical models. The authors were curious as to whether the excess returns observed would disappear if the returns were corrected for risk. In order to investigate this possibility, they calculated betas (beta

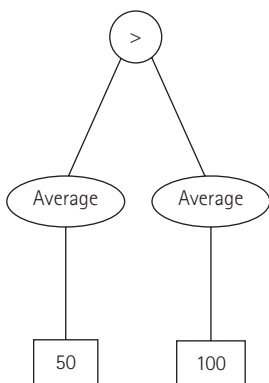


FIGURE 8.4 An illustrative example of a technical trading rule that could be evolved using GP. This rule would prompt the buying of the security if the fifty-day moving average of prices exceeded the one-hundred-day moving average of prices, or a selling of the security otherwise.

measures the relation of the return to the market index return) for the returns to their rules using a selection of benchmark portfolios and found no evidence that the excess returns acted to offset risk. Neely and Weller (1999) conducted a further study into the profitability of technical trading rules using a different set of currencies and different training, selection, and validation periods, obtaining results in line with those of their previous study.

At about the same time, GP was used to evolve technical trading rules for the S&P 500 index using daily data, taking transaction costs into consideration. Allen and Karjalainen (1999) divided their data into ten subsamples in order to prevent data snooping with respect to the chosen time period. Overall, consistent excess returns were not achieved using the evolved rules, and the authors interpreted these results in support of market efficiency. Becker and Seshadri (2003) made changes to the approach adopted by Allen and Karjalainen. They used monthly rather than daily data, and they ran two sets of experiments with modified fitness functions, in the first case including a complexity-penalising factor to reduce overfitting, and in the second employing a fitness function that took into account not only the rule's return but also the number of periods in which the rule did well. They only used one data set, with which they were able to evolve trading rules using GP, which outperformed buy-and-hold. Potvin et al. (2004) used GP to evolve technical trading rules to trade in fourteen stocks traded on the Canadian TSE 300 index, and unlike Allen and Karjalainen (1999), they allowed short as well as long positions to be taken. They found that the overall return to the rules on test data, averaged over all fourteen stocks and averaged over ten runs, was negative, indicating no improvement over a buy-and-hold approach (though they did note that GP-evolved rules were generally useful when the market was stable or falling).

Li and Tsang (1999b) used a GP-based system called financial GP (FGP), which is in turn descended from EDDIE (Evolutionary Dynamic Data Investment Evaluator, Li and Tsang (1999a)), to try to predict whether the price of the Dow Jones Industrial Average Index would rise by a certain amount during the following month. Candidate solutions were represented as decision trees, which employed technical indicators and made a positive or negative prediction at their terminal nodes. The evolved by GP performed

better than random decisions, despite the EMH's suggesting that trading rules cannot outperform random trading. Transaction costs were not accounted for, and the authors used eleven years' worth of data, split between a single training and a single test set. In Li and Tsang (1999b), EDDIE came with a pre-parameterized set of technical indicators. Kampouridis and Tsang (2010) presented an updated version of EDDIE that allowed for a larger search space of potential technical indicators. They allowed GP to search for the type of technical indicator, as well as the time-period parameterizations for the indicators (permitted periods were within a user-specified range).

Recently, Chen et al. (2009) used genetic network programming (GNP), in combination with Sarsa learning (a type of reinforcement learning), to create a stock trading model using technical indicators and candlestick charts. (Genetic network programming GNP was introduced at the end of section 8.1.1.3). The authors made no mention of transaction costs and used a single training and testing period comprising daily data; thus their work fits in with earlier, more naive approaches in terms of the structure of their financial data. Their results showed that GNP with Sarsa learning outperformed traditional approaches, including buy-and-hold, using the setup employed.

8.2.2 The Impact of Transaction Costs and Multiple Time Periods

Wang (2000) pointed out that the high liquidity in foreign exchange markets and the low value for out-of-sample transaction costs of 0.05 percent used by Neely et al. (1997) could have been responsible for the excess returns observed when using the evolved rules. In their original paper Neely et al. (1997) addressed their choice of transaction cost, quoting similar values used in other contemporary papers and noting that the increase in trading volume and liquidity in recent years had lowered transaction costs. Chen et al. (2008) replicated the process carried out by Neely et al., but used four test periods instead of one and three exchange-rate series instead of six. They found statistically significant excess returns in ten of twelve cases when they used the same level of out-of-sample transaction costs as Neely et al., had used but when they increased out-of-sample transaction costs to 0.2 percent, the trading rules that evolved only produced statistically significant excess returns in six of twelve cases. The difference in the profitability between the two sets of out-of-sample transaction costs offered support to Wang's (2000) hypothesis concerning the impact of the level of transaction costs on trading profitability. Allen and Karjalainen (1999) also found that when transaction costs were low, rules that evolved by using GP for the S&P 500 index produced better returns than when transaction costs were higher (while still not, in general, producing excess returns with respect to a buy-and-hold strategy).

Further, Wang (2000) pointed out that Neely et al. (1997) only used one data set to test the profitability of the evolved rules. In an early version of their 1999 paper, Allen and Karjalainen (1995) used a single training, validation, and test period and found excess returns to the rules evolved by GP. When they incorporated multiple time periods by

using a rolling-forward approach, they were no longer able to consistently find excess profits. This suggested that perhaps the positive results found in Neely et al. (1997) were time-period-specific. This did not appear to be the case, however, given that Chen et al. (2008) found excess returns across four different time periods when they used the same level of transaction costs when replicating the work of Neely et al. (1997). Furthermore, Neely, Weller, and Ulrich reexamined the rules from the earlier work of Neely et al. (1997) on data from the period 1995 to 2005 (Neely et al. 2009). The authors suggested that if the results obtained in Neely et al. (1997) were time-period-specific and thus due to data mining, then excess returns should not be observed after the original sample period, and a break should appear in the mean return series of the evolved rules. Mean returns, which were 2.94 percent in the original sample, had decreased to 0.15 percent in the more recent sample. They tested the hypothesis of a break in the mean return series econometrically, by fitting autoregressive integrated moving average (ARIMA) models to the rule returns data, and did not find support for a break in mean return. They surmised that average net returns declined gradually but did not disappear entirely. The authors pointed out that this gradual erosion of returns could best be explained with a model of markets as adaptive systems responding to evolutionary selection pressures, as described by the adaptive market hypothesis (Lo 2004). That is, these trading rules lost profitability gradually as markets learnt of their existence, and the excess returns found in earlier work were not an outcome of data mining.

In summary, it is important to try as far as is possible to use realistic transaction costs when investigating the profitability of technical trading rules evolved using GP, because the level of transaction costs can have an impact on whether GP can find profitable trading rules. It is equally important to test on more than one time period to ensure that the results with respect to profitability in one time period are not simply the result of chance but, rather, to establish whether evolving trading rules using GP is consistently profitable across multiple time periods.

8.2.3 High-Frequency Data

Many technical traders transact at high frequency. Thus, although many earlier GP studies aimed to evaluate the profitability of technical analysis using daily data, two studies dating from the early twenty-first century were prompted to evaluate the profitability of rules for high-frequency data (Neely and Weller 2003; Dempster and Jones 2001). When realistic transaction costs were taken into account, and trading hours were restricted to the hours of normal market activity, Neely and Weller (2003) did not find evidence of excess returns derived from technical trading rules. Dempster and Jones (2001) selected the best twenty (or fewer) in-sample trading-rule-based strategies and distributed a fixed amount of trading capital equally among them for testing out of sample. They found that in a static environment, trading using the twenty best strategies produced lower returns than the interest rate differential between the currencies being traded. In the adaptive case, the rules were loss-making. When only

the single best strategy was employed, in both the static and the adaptive cases, a profit was returned. Dempster et al. (2001) applied a maximum tree depth in order to limit complexity and prevent overfitting. They concluded that GP was capable of evolving profitable rules using out-of-sample high-frequency data, contradicting the efficient market hypothesis, though when realistic transaction costs were accounted for, profits were unremarkable.

Bhattacharyya et al. (2002) examined the impact of incorporating semantic restrictions and using a risk-adjusted measure of returns to measure fitness on the performance of high-frequency foreign exchange trading rule models evolved using GP. They found benefits with respect to improved search and improved returns-related performance on test data. Saks and Maringer (2010) were concerned with using GP to investigate the usefulness of money management in a high-frequency foreign exchange trading environment. Usually, one common trading rule is assessed for both buy and sell positions. It is used to determine whether, on one hand, a position should be entered or should continue to be held or, on the other, whether a position should be exited or should continue to be avoided. The authors took a different approach inspired by the principles of money management. They employed different rule sets depending on the current position. As an example, a negative-entry signal did not necessarily mean that a position should be exited; rather, different rules were used to find exit signals. Their findings indicated that money management had an adverse effect on performance.

8.2.4 Accounting for Risk

Neely (2003) reconsidered the profitability of *ex ante* technical trading rules generated using GP in the stock market. Given that GP rules dictated that the investor was out of the market some of the time, a trading strategy based on trading rules evolved by GP may have had less risk than a buy-and-hold strategy. Neely wished to extend the work carried out by Allen and Karjalainen (1999) by considering risk-adjusted returns, and so he evaluated risk-adjusted measures such as the Sharpe ratio on solutions evolved using a modified version of the programs and procedures similar to those used in the earlier work. He also evolved rules that maximized a number of risk-adjusted measures including the Sharpe ratio and evaluated their risk-adjusted returns out-of-sample. Neely found no evidence that, even on a risk-adjusted basis, the rules evolved using GP significantly outperformed a buy-and-hold strategy.

Fyfe et al. (2005) found no excess risk-adjusted returns over a buy-and-hold strategy when compared to technical trading rules for S&P indices evolved using GP. O'Neill et al. conducted a preliminary study using grammatical evolution (O'Neill and Ryan 2003) to evolve trading rules for the FTSE 100 index using risk-adjusted returns (O'Neill et al. 2001). They incorporated risk in the fitness function by subtracting the maximum cumulative loss from the profit over the course of the training period. This led to the evolution of risk-conservative rules. They found excess returns to the buy-and-hold benchmark, with the caveat that their building blocks included

only a single technical indicator (the moving average), fuzzy logic operators, and standard arithmetic operations. In a more recent paper, Esfahanipour and Mousavi (2011) found risk-adjusted excess returns to technical trading rules evolved using GP for ten companies listed on the Tehran Stock Exchange in Iran. The authors accounted for transaction costs and also for splits and dividends, which, they claimed, increased the accuracy of their results. They only used one training and test data set, however.

Wang (2000) tested the performance of trading rules evolved using GP in three scenarios: the S&P 500 futures market alone, simultaneous trading in both the S&P 500 index and futures markets, and rules that maximized risk-adjusted returns in both markets. The training, validation, and out-of-sample periods were moved forward each year to better model the trading environment of a real investor. Given the inability of the rules to consistently outperform buy-and-hold, Wang was unable to find evidence that rejected the efficient market hypothesis.

8.2.5 Summary

Results for trading rules evolved using daily stock data (not accounting for risk) were somewhat mixed. Allen and Karjalainen (1999) and Potvin et al. (2004) did not find that technical trading outperformed buy-and-hold overall, whereas Chen et al. (2009) found excess returns above buy-and-hold from a trading model that included technical indicators. When analysts accounted for risk, results were also mixed (O'Neill et al. 2001; Esfahanipour and Mousavi 2011; Fyfe et al. 2005). Results for trading rules evolved on daily data in foreign exchange markets were positive (Neely et al. 1997; Neely and Weller 1999). Results in foreign exchange markets were mixed when using intra-day data (Neely and Weller 2003; Dempster et al. 2001; Dempster and Jones 2001; Bhattacharyya et al. 2002). The importance of using realistic transaction costs has been highlighted, as has the need for using multiple time periods when testing the profitability of evolved technical trading rules.

8.3 A SURVEY OF THE WIDER LITERATURE ON FORECASTING ECONOMIC VARIABLES

Other studies have examined the broader area of forecasting using Genetic Programming. Attempts have been made to forecast different economic and financial variables using GP, including the value of GDP or the price of a stock or an index of stocks (in this section we focus on approaches to price forecasting that do not employ technical analysis).

8.3.1 Results from Forecasting Prices and Other Variables

Iba and Sasaki (1999) saw some success in using GP in price forecasting. They used GP to evolve both the price of the Nikkei 225 index and the difference between the price of the index currently and the price observed a minute earlier in two sets of experiments. The building blocks used in the second set of experiments, which included the absolute and directional differences between the current and the past price, allowed GP to evolve predictions that earned significant profits when used to trade on out-of-sample data. Larkin and Ryan (2008) used a genetic programming model based on news sentiment to forecast sizeable intra-day price changes on the S&P 500 up to an hour prior to their occurrence. Kaboudan (2000) attempted to forecast stock price levels one day ahead, this time for six heavily traded individual stocks, using GP. Genetic programming was compared with a naïve forecasting method which predicted that today's price would equal yesterday's price. Using the trading strategy with either the naïve method or GP for fifty consecutive trading days resulted in higher returns than a buy-and-hold strategy, with the return using the GP forecasts outperforming the return based on the naïve forecasts for five of six stocks considered. Neither this study nor that of Iba and Sasaki accounted for transaction costs. Larkin and Ryan did not simulate continuous trading based on their predictions, and thus transaction costs were not considered.

In a later work, Kaboudan (2001) used GP and neural networks to predict short-term prices for an important economic resource; oil. Prices were predicted one month and three months ahead. Neural network forecasts outperformed GP overall, but both methods showed statistically better performance than random-walk predictions for the period three months ahead. More recently, Lee and Tong (2011a) used GP as a component in their technique for modeling overall energy consumption in China. In the same year, Lee and Tong (2011b) combined GP with the commonly used ARIMA model, thus enabling nonlinear forecasting of time-series data. Among the data sets used to test their technique was data for quarterly U.S. GDP spanning more than five decades. Wagner et al. (2007) developed a dynamic forecasting genetic program model that was tested for forecasting ability on U.S. GDP and consumer price index inflation series and performed better than benchmark models. Their model was set up to adapt automatically to changing environments and preserve knowledge learned from previously encountered environments, so that this knowledge could be called on if the current environment was similar to one seen before. Dempsey et al. (2006) incorporated a similar concept of “memory” of past successful trading rules in their study on adaptive (technical) trading with grammatical evolution.

8.3.2 Summary

Genetic programming has been used—in many cases successfully—to forecast financial and economic variables. Attempts have been made to forecast the prices of individual stocks, of indexes of stocks, and of a commodity (oil), as well as the value of U.S. GDP

and the CPI Inflation series and energy consumption in China. Many papers did not explicitly treat the learning environment as being dynamic, but dynamic environments were also considered, for example, in Wagner et al. (2007).

8.4 STOCK SELECTION

Genetic programming has also been used to select stocks for investment portfolios. Restricting potential stock selection solutions to the linear domain automatically excludes a wide variety of models from consideration. Models that are linear combinations of factors may fail to reflect market complexities, and thus may fail in harnessing the full predictive power of stock selection factors. Thus, given the ability of GP to provide nonlinear solutions, models evolved using GP may offer advantages over more traditional linear models such as the CAPM (Sharpe 1964; Becker et al. 2007).

For example, GP can be used to evolve an equation that, when applied to each stock in a basket of stocks, produces a numerical output which can then be used to rank stocks in order from best to worst. This is the approach used in Yan and Clack (2007) and Yan et al. (2008). Building blocks can include fundamental indicators such as the price-to-earnings ratio alongside technical indicators such as the moving average convergence divergence (MACD) indicator. An sample tree is given in figure 8.5.

Caplan and Becker (2005) used GP to build a stock-picking model for the high-technology manufacturing industry. Becker et al. (2007) expanded on this work, this time incorporating two fitness functions to produce models for the S&P 500 index (excluding utilities and financials). Building blocks they used included the price-to-earnings ratio, earnings estimates, and historical stock returns as well as arithmetic operators, constants, and variables. One of the fitness functions was geared

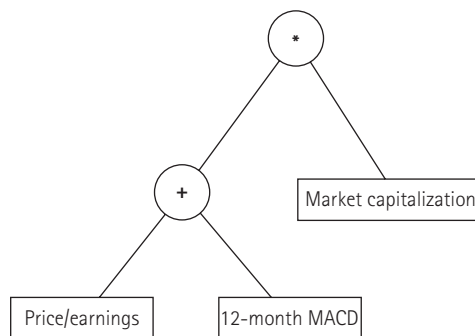


FIGURE 8.5 This illustrative individual tree could be used in a stock selection application. It adds the price/earnings ratio to the 12-month MACD indicator value and multiplies the result by the stock's market capitalization. The resulting value is used to rank the stock in question.

toward investors who favored a low-active-risk investment style. It used the information coefficient, which is the Spearman correlation between the predicted stock rankings produced by the GP model and the true ranking of stock returns. The second fitness function was geared toward investors who prioritized returns and were willing to take on risk. Both GP models outperformed a traditional model produced using linear regression and the benchmark market cap-weighted S&P 500 index (excluding utilities and financials). Furthermore, models evolved using GP were robust with respect to different market regimes.

8.4.1 Promoting Generalization and Avoiding Overfitting

Becker et al. (2008) took previous work by the authors and their colleagues, including Caplan and Becker (2005) and Becker et al. (2007), further. The authors were concerned that GP models such as those evolved in Becker et al. (2007) did not perform equally well with respect to the various criteria that investment managers consider. They were also concerned that models evolved in their earlier work did not consistently generalize to out-of-sample data. With these concerns in mind, the authors implemented multi-objective algorithms that simultaneously optimized three fitness metrics. These were the information ratio, the information coefficient, and the intra-fractile hit rate (which measured the accuracy of the GP model's ranking of stocks). The best-performing multiobjective algorithm used a constrained fitness function that incorporated the three fitness metrics in a single fitness function. Furthermore, models produced by this algorithm generalized well to out-of-sample data. Kim et al. (2009) expanded on the above-mentioned work by constraining the genetic programming search to be within a function space of limited complexity, and in so doing were able to reduce overfitting.

8.4.2 Stock Selection Solutions in Dynamic Environments

Yan and Clack (2007) used GP to evolve a nonlinear model that was used to rank thirty-three potential investment stocks from the Malaysian stock exchange. The authors were particularly concerned with examining how well GP responded to nontrivial changes in the environment. The training data consisted of a range of different environments (a bull market, a bear market, and a volatile market). Each stock was ranked using the evolved model, and an investment simulator was used to create a long-short portfolio composed of stocks that performed the best (long) and the worst (short) across four market sectors. Contracts for difference were traded instead of shares, and margin and transaction costs were accounted for. Two separate modifications were made to standard GP in order to expose the different environments to the population, and both modified GP systems outperformed a technical trading strategy and the portfolio index. In a more recent version of their paper, the authors also investigated the use of a voting mechanism, whereby a number of individuals using GP who put forward their solutions as votes, and the chosen solution was the winner

of a majority voting contest (Yan and Clack 2011). Yan et al. (2008) followed up on Yan and Clack's earlier work (Yan and Clack 2007) by comparing support vector machine and GP approaches to stock selection for hedge fund portfolios. Genetic programming performed better. The authors believed that this was because the GP system maximized the overall performance of the portfolio, as opposed to predicting individual monthly stock returns. This was due to the way fitness was evaluated in the GP system. It was applied to each stock and it produced output a number, which in turn was used to rank the stocks. The investment simulator was used to create a long-short portfolio for each month, and the Sharpe ratio of the investment simulator was calculated and used to measure the fitness of the GP model. In this way, the GP model ranked stocks for input into an optimized portfolio, which is more central to the role of stock selection than predicting stock returns.

8.4.3 Summary

In summary, stock selection solutions evolved from GP have been presented which are targeted toward changing environments, as well as solutions targeted toward displaying good generalization capabilities. Investment criteria targeting differing objectives have been optimized for. Multiobjective algorithms, which simultaneously optimize numerous objectives have been employed. Genetic programming has been compared with another machine learning technique, support vector machines, and the superior performance of GP in this instance has been analyzed.

8.5 DERIVATIVES PRICE MODELING AND TRADING

When modeling derivative prices, GP has the advantage that it can incorporate current theoretical models such as the Black-Scholes model for option pricing (Black and Scholes 1973) into its search process. The Black-Scholes model is a landmark in modern finance. Under certain assumptions, it specifies an equation for pricing options in terms of the underlying stock price, the time to maturity of the option, the exercise price of the option, the standard deviation of the continuously compounded rate of return on the stock for one year, and the continuously compounded risk-free rate of return for one year (Sharpe et al. 1999, p. 623). Genetic programming has been used to develop option pricing models, for example, in Chidambaran et al. (2002), Chen et al. (1998), and Yin et al. (2007).

Chidambaran et al. (2002) simulated underlying stock prices according to a jump-diffusion process and used GP to predict option prices. The closed-form solution to pricing options in a jump-diffusion world was based on the exposition in Merton (1976), against which the option prices produced by the GP system were compared.

Chidambaran and his co-authors found that GP outperformed Black-Scholes when estimating the option price using this setup. They also used GP to predict option prices on real-world data sets and once again found that GP performed very well. In a later paper Chidambaran considered the role of GP parameters in influencing the accuracy of the search for option pricing models and found that parameters such as the mutation rate, sample size, and population size were significant determinants of the efficiency and accuracy of the GP system (Chidambaran 2003). Yin et al. (2007) dynamically adapted the probability of mutation and crossover during the GP run and found better performance than when the probability of the application of these genetic operators remained constant throughout the run.

Other papers exploring the use of genetic programming to model or trade in derivatives include Wang (2000) and Noe and Wang (2002). Tsang et al. (2005) used EDDIE (introduced in section 8.2.1) to successfully forecast some arbitrage opportunities between the FTSE 100 index options and futures markets, though it failed to predict the majority of *possible* arbitrage opportunities.

8.6 BANKRUPTCY AND CREDIT RISK ASSESSMENT

Bankruptcy prediction can essentially be viewed as a classification task whereby the classifier, be it genetic programming or some other technique such as a neural network or support vector machine, is used to classify a set of data points into a bankrupt or nonbankrupt class. The fact that GP does not require an a priori structure to be imposed on the model is a much-trumpeted virtue of the technique, and this feature motivates some bankruptcy researchers to use it (Lensberg et al. 2006). The ability of GP to evolve nonlinear solutions to problems is a useful feature of the technique and, as noted by McKee and Lensberg (2002), allows complex interactions between predictor variables to be revealed. Credit scoring is another classification task with a use for GP that we briefly discuss in this section.

A typical GP approach to bankruptcy prediction might involve using financial ratios as terminals (comprising assets, liabilities, income, and so on), and arithmetic and other operators as part of the function set. These building blocks might be used to evolve trees such as that shown in figure 8.6, which would evaluate to a numerical value for each firm. Depending on whether the result was larger than or smaller than a predefined threshold, the firm would be classified as bankrupt or nonbankrupt. A simple fitness function could operate by counting the number of correctly classified firms. Training data could comprise a set of past bankrupt and nonbankrupt firms.

Three papers used GP as part of a multi-step bankruptcy prediction process (McKee and Lensberg 2002; Lensberg et al. 2006; Etemadi et al. 2009). In the first step, a technique such as stepwise discriminant analysis, rough sets, or GP itself reduced a set of potential predictor variables to a smaller set of variables to be used in the second-step

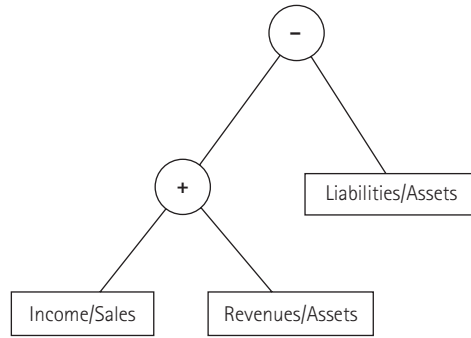


FIGURE 8.6 A bankruptcy prediction rule that could be evolved using GP. Depending on whether this tree evaluated to a number larger than or less than a threshold, the firm would be classified as bankrupt or nonbankrupt.

GP run, which produced a classification model using these variables. McKee and Lensberg (2002), the earliest among these papers, used a rough sets approach (Pawlak 1982) and GP to develop a bankruptcy prediction model. The authors built on previous work by McKee that had identified eleven bankruptcy predictors with strong support from previous academic studies, and used them as input into a rough sets bankruptcy prediction model. Genetic programming was then used in conjunction with rough sets, the latter approach having served to identify the subset of variables to use as inputs to the GP system. Genetic programming was used to evolve nonlinear algebraic expressions in these variables, which produced a numerical value. Depending on whether the result of evaluating the expression was larger than or smaller than a predefined threshold, the firm under consideration would be classified as bankrupt or nonbankrupt. The model evolved by GP had 80.3 percent accuracy for out-of-sample data.

Genetic programming was used in two steps by Lensberg et al. (2006) to classify Norwegian firms into soon-to-be-bankrupt and non-bankrupt cohorts. Twenty-eight variables were used as potential predictors. These included profitability measures, firm size, leverage, a number of potential fraud indicators, and the prior auditor's opinion. The initial set of GP runs was used to reduce the number of variables to six, and these six were used as potential predictors of bankruptcy in the second-stage GP runs. The evolved model was 81 percent accurate with regard to out-of-sample data and better at a statistically significant level than a traditional logit model. Etemadi et al. (2009) used GP to evolve a bankruptcy prediction model that employed five potential predictor variables derived from a list of forty-three variables using stepwise discriminant analysis. These five variables were the profit margin, the debt ratio, the revenue per dollar of assets, the ratio of interest payments to gross profit and a liquidity ratio. Genetic programming was used to evolve tree-based classification rules. A firm was classified as "bankrupt" when the output of the tree for that firm was above a predefined threshold. The fitness function was the hit rate, that is, the number of correctly classified firms.

Other papers that dealt with bankruptcy prediction using GP include Salcedo-Sanz et al. (2005), Alfaro-Cid et al. (2008), and Ravisankar et al. (2010). Salcedo-Sanz et al. (2005) used genetic programming to evolve decision trees to classify insurance companies as bankrupt or nonbankrupt. They compared their approach with rough sets and support vector machines and concluded that GP was a suitable decision support technique that could be used in this context by (for example) an insurance regulator. The authors controlled the depth of evolved trees and used multiple subsets of the available data for training and testing, in order to avoid overfitting. In their investigations into the classification of potential future bankrupt firms, Alfaro-Cid et al. (2008) compared an ensemble of evolutionary algorithms that optimized artificial neural networks with ensembles of GP trees, in both cases using multiobjective optimization to create the ensemble. They focused on finding solutions that balanced predictor size (in the case of GP, this meant the size of the trees) against false positives and against false negatives in classification. Finally and more recently, Ravisankar et al. (2010) used hybrids of neural networks and genetic programming, to classify dot-com companies into those predicted to fail and those predicted to survive.

Genetic programming can also be used for the related task of credit scoring, that is, assessing the creditworthiness of potential customers and classifying them as creditworthy or not creditworthy, as the case may be. Ong et al. (2005) used GP successfully to evolve discriminant functions for this classification task. Other papers that used GP for this task include Huang et al. (2006); Zhang et al. (2008), and Abdou (2009).

8.7 AGENT-BASED AND ECONOMIC MODELING

Agent-based modeling is a computational simulation-based technique that involves the modeling of individual participants in a system. The advent of large-scale computational power has allowed for the simulation of more complex systems than could be considered previously, when modeling capabilities were limited to analytical and mathematical techniques. In an agent-based model, participants can interact with each other and also with their environment. By conducting such simulations under certain assumptions, we may gain insight into how systems built up of interacting agents behave, and we can then use this information to aid in our understanding of the real world.

Computational finance modelers are interested in, for example, what we can learn about the properties of simulated groups of economic agents in terms of efficient markets and market microstructure. Computational approaches such as neural networks, genetic algorithms and genetic programming have been used to model agents' behavior (LeBaron 2000). Chen (2012) provides an overview of the type of agents used in agent-based computational economics. Genetic programming can be used to evolve functions that are used by agents as inputs into models to forecast the expected

future value of prices plus dividends, as was done by Chen and Yeh (2001, 2002). These authors included arithmetic operators, trigonometric functions, current and lagged price values, and the sum of lagged prices and dividends in their set of building blocks for the GP models.

It is important to note that any results derived in agent-based modeling simulations rely crucially on the assumptions on which the artificial market is built. For example, in describing the software used by Chen and Yeh in their 2001 work, Chen et al. (2002) state that the time series of dividends was generated using a stochastic process.

8.7.1 Investigations in Market Efficiency

Chen and Yeh (2001) examined whether the efficient market hypothesis (EMH) was borne out in an agent-based modeling context. Their approach consisted of two interacting parts. The first part involved the authors' using a technique similar to simulated annealing to model the decisions of agents in an artificial stock market. The second component used GP to co-evolve a set of forecasting models. Traders (agents) could choose to consider implementing a new forecasting model, and this choice was modeled using simulated annealing; the probability that traders would consider implementing a new model was determined by the relative performance of their trading activity. If they selected a new forecasting model, this in turn influenced their expectations with respect to future prices and future dividends. Econometric tests showed that over the long run the return series was identically and independently distributed, supporting the EMH. However, it was sometimes possible for simulated traders to profitably exploit short-term signals.

Chen and Yeh published research the following year (2002) that investigated whether the EMH and the rational expectations hypothesis could be considered emergent properties. By "emergent" they meant a property that could not be obviously predicted as resulting from an aggregation of individual behaviors. After performing econometric tests on their artificial time series, the authors found that in one subperiod, returns were independently and identically distributed, validating the EMH in the form of the martingale hypothesis in that subperiod (the martingale hypothesis states that the best forecast for tomorrow's price is today's price). In that same subperiod the number of traders who believed in the martingale hypothesis was very small, leaving the authors to conclude that the EMH is an emergent property. The authors also showed that the rational expectations hypothesis could be considered an emergent property in their artificial stock market.

Computational modeling of economic systems does not always involve the use of agents. In Chen and Yeh (1997), the authors used GP to formalize the concept of unpredictability in the EMH. They did this by measuring the probability that a GP search at a particular intensity could predict returns in the S&P 500 and the Taiwan Stock Price index better than a random walk model (search intensity increased with, among other GP parameters, the number of generations and population size). The

results showed that in approximately one-half of the runs, the GP model that was produced after generation 50 beat the random walk model in predicting out-of-sample returns. Search intensity was high in generation 50, with a population of five hundred and up to twenty-five thousand models having been evaluated. Since the random walk model was approximately as likely as the GP model to produce the better prediction of returns (even when GP was searching intensely), GP models were deemed no more efficient than a random walk model at predicting returns, given the stochastic nature of GP searches. Thus the results added support to the EMH.

In summary, none of the pieces of work detailed in this subsection found strong evidence against the efficient market hypothesis.

8.7.2 Other Applications of Agent-Based Modeling

Other papers that used GP as a part of an agent-based modeling system include Chen and Liao (2005), Ikeda and Tokinaga (2007), and Martinez-Jaramillo and Tsang (2009). Ikeda and Tokinaga (2007) analyzed price changes in an artificial double-auction electricity market. Chen and Liao (2005) used an agent-based model to investigate the causal relation between stock returns and trading volumes. Their results showed that this causal relation could be seen as a generic feature of financial markets that were composed of independent agents operating in a decentralized marketplace. Martinez-Jaramillo and Tsang (2009) used GP to model technical, fundamental, and noise traders in an artificial stock market. Their purpose was to learn about real-world financial markets. In order to do so, they uncovered what conditions applied in the artificial market when they observed statistical properties of price series that approximated those of real-world stock markets.

Moving to research in market microstructure, a number of papers have been published since the beginning of the twenty-first century that use genetic programming to model microstructure using agent-based modeling. Market microstructure is concerned with studying trading and how markets are organized with respect to such characteristics as liquidity, transaction costs, and volatility (Harris 2002). Papers using GP to model elements of microstructure include Chen et al. (2011), Kampouridis et al. (2012), and Cui et al. (2011), the last of which uses the grammar-based genetic programming variant of grammatical evolution described in O'Neill and Ryan (2003).

8.8 CONCLUDING REMARKS

This chapter has provided an overview of the areas in which the machine learning stochastic optimization technique of genetic programming has been used in the areas of financial and economic modeling. An important advantage of the technique lies in the modeler's not needing to choose in advance the exact parameters from which the model

will be built. Rather, they can specify a range of building blocks that can potentially form part of a solution, from which the technique of GP then selects the most relevant. Other advantages include its ability to incorporate known approximate solutions in the search, its ability to create human-readable output, and its ability to evolve nonlinear as well as linear solutions to problems.

8.8.1 A Historical Narrative

It is possible to trace developments in the application of GP to economic and financial modeling from its early days in the mid-1990s—when its lack of ability to consistently provide trading rules that resulted in excess returns in the stock market prompted early researchers to suggest that it be applied to more liquid markets with lower transaction costs, such as foreign exchange and futures markets (Allen and Karjalainen 1999)—to more recent times, when researchers are highlighting the potential usefulness of treating the search environment for evolution of trading rules as a dynamic environment (Chen et al. 2008). Results for trading rules evolved on daily data in foreign exchange markets were positive, and those evolved when using intra-day data were mixed. (Wang 2000) found that rules for the S&P 500 futures market were not found to be consistently profitable. Wang (2000) suggested that effort be applied to selecting stocks instead of timing the market by the use of technical trading rules, and a number of papers published since 2003, including Becker et al. (2007) and Yan and Clack (2011), applied GP to this task successfully.

8.8.2 Open Issues and Future Work

The performance of GP in dynamic environments has been documented as an open issue for researchers employing this technique (O'Neill et al. 2010). In talking about optimization using evolutionary algorithms, Branke notes that “[w]henver a change in a dynamic optimisation problem occurs ... the optimum to that problem might change as well. If this is the case, an adaptation of the old solution is necessary” (Branke 2001, p.13). Many of the applications of GP to economic and financial modeling in dynamic environments have been published since the year 2000, including Dempsey et al. (2006), Wagner et al. (2007), and Yan and Clack (2011). It is interesting to note, however, that Chidambaran et al. (1998) had taken account of this issue when they stochastically changed their training data midway through training in order to promote solutions that were robust to changing environments when they were evolving option pricing models using simulated data (Chidambaran also employed this technique in Chidambaran 2003).

Other open issues identified by O'Neill et al. (2010) include the generalization performance of GP and the matter of finding suitable representations when conducting a GP search. Finding solutions that generalize to out-of-sample data (or don't overfit

their training data) has been a research aim in a number of papers in the area of economic and financial modeling, including Dempster et al. (2001), Becker and Seshadri (2003), and Kim et al. (2009). Papers dealing with nonstandard representations include that by Bhattacharyya et al. (2002), who employ a domain-related structuring in their GP representation and incorporate semantic restrictions on the search. Future work may continue to further explore these issues in order to potentially improve on financial and economic modeling solutions using GP.

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CHAPTER 9

ALGORITHMIC TRADING BASED ON BIOLOGICALLY INSPIRED ALGORITHMS

VASSILIOS VASSILIADIS AND GEORGIOS DOUNIAS

9.1 INTRODUCTION

Today, financial markets are widely considered a complex system, with many interrelations among its components. Financial decision making is of great importance owing to the great amount of uncertainty that stems from the issue of the global economic crisis. Financial institutions, agents, managers, and individual investors are the basic members of the global financial map. The important thing is that to a great extent they have conflicting interests and goals. If one party aims at maximizing its profit, then another party suffers severe losses. What is more, the task of taking the proper decision becomes even harder if anyone realizes the amount of information available in global markets. Moreover, financial systems, as is common with all systems, are affected by worldwide developments coming about in various domains (politics, environment, and so on). One potentially difficult problem for financial decision makers to deal with is the optimal allocation of capital to a number of assets, that is, the proper construction of a fund. Two of the main challenges of burdens this task are the very large number of available assets and the appropriate formulation of the portfolio selection problem (definition of the objective function and real-world constraints). However, when the fund is constructed, a forecasting component may be applied, as well, in order to predict the future prices of this fund and provide trading signals. This is a typical example of a trading system.

Because of the large number of parameters of trading systems, as well as the incorporation of nonlinear and complex real-world objectives and constraints, traditional approaches to dealing with this issue prove to be inefficient. To begin with, humans cannot handle efficiently systems of such high complexity, with a large

number of states. Moreover, traditional methodologies from statistics and operational research offer only partial solutions to this problem, for example, when traditional methodologies are applied to the forecasting process of the fund. Finding the optimal, or near-optimal, portfolio is not an easy task, however. The solution space is quite large. Algorithmic trading (AT) provides an aggregated acceptable solution to the aforementioned problems (Jaekle and Tomasini 2009). Algorithmic trading aims at constructing decision support systems consisting of algorithms and metaheuristic techniques. The main characteristics of these algorithms are self-organizing ability, adaptation, and evolution. Thus, human intervention is limited to providing some preliminary guidance to the system. The applied algorithms should embody most of the above mentioned characteristics.

Biologically inspired intelligence (BII), or nature-inspired intelligence (NII), offers a variety of algorithmic methods whose fundamental features are evolution, self-adaptation, and self-organizing. Their main philosophy lies in the way natural systems work and evolve. Ant colony optimization and particle swarm optimization are two examples of such methods. The former depends on the way real ant colonies function and evolve in order to perform certain tasks, whereas the latter stems from the strategy implemented by the members of a swarm, which aim at reaching a certain location (e.g., bird flocking).

The main aim of this chapter is to provide evidence regarding the superiority of biologically inspired intelligent algorithms through the implementation of a financial trading problem. More specifically, the proposed algorithmic trading system comprises a portfolio selection component and a technical indicator for identifying buy-sell signals. This chapter focuses on applying biologically inspired algorithms, namely ant colony optimization (ACO) and genetic algorithm (GA) in the portfolio selection process in order to form high-quality funds for trading. It is important to note that the general strategy of biologically inspired metaheuristics fits the specificities of the optimization problem at hand. As far as the benchmark methods are concerned, two commonly used heuristics, random selection and a financial rule of thumb, are applied in order to get better insight into the performance of biologically inspired algorithms (Brabazon and O'Neill 2006).

This chapter is organized as follows. In section 9.2 we present findings from a survey of the literature. Methodological issues are covered in section 9.3. In section 9.4 we describe the experimental process and present the simulation results. In addition, some discussion of the results is provided. Section 9.5 concludes the chapter.

9.2 RELATED STUDIES

The application of biologically inspired intelligent algorithms in algorithmic trading is relatively recent in the literature; thus some steps have been made toward evolution and improvement of the performance of these techniques (hybrid schemes, alternative

encodings, and so on). What is more, based on previous studies, nature-inspired intelligent algorithms have been applied to a wide range of problems such as modeling, classification, and optimization and such domains as industry, finance, and medicine. The majority of the results indicate the great potential of these newly introduced methodologies. The aim of this section is to present selected papers regarding the problem at hand.

Lin et al. (2005) apply a genetic algorithm based on real value operators in order to solve a specific formulation of the portfolio rebalancing optimization problem. Hung and Cheung (2003) propose an extension of an adaptive supervised learning decision trading system (EASLD) combined with a portfolio optimization scheme so as to strike a balance between expected returns and risks. The proposed system has two desirable characteristics: the learning ability of the ASLD algorithm and the ability to dynamically control risk by diversifying the capital in a time-varying cardinality-constrained portfolio. Experimental results indicate that the EASLD system can considerably reduce risk in comparison to the simple ASLD (the older version), while keeping long-term returns at a reasonable level. What is more, the authors compare their intelligent hybrid scheme with individual portfolio selection strategies such as the standard portfolio optimization approach by Markowitz and the improved portfolio Sharpe ratio maximization with risk diversification component. The underlying concept of these two financial heuristics is that the portfolio is constructed in the estimation period, solving a mathematical optimization problem as defined by these methods. Then the constructed fund is bought and held for the entire trading period, that is, a typical buy and hold strategy is used.

Kuo et al. (2001) develop a hybrid algorithm, the genetic-algorithm-based fuzzy neural network, to formulate the knowledge base of fuzzy inference rules which can measure any qualitative effect on the stock market. Next, the system is further integrated with the technical indexes through the artificial neural network. The aim is to formulate in a proper manner the expert's knowledge regarding economic, political, and other news, and employ it in the decisionmaking process. Chen et al. (2009) propose an investment strategy portfolio problem using a type of "combination genetic algorithm." To be more precise, the real-number portfolio problem can be approximated by a proposed integer model. When one does so, the classical portfolio optimization problem is transformed into a combination optimization problem, and the solution space is reduced. What is more, a number of technical indicators are applied to the constructed portfolio. Experimental results have demonstrated the feasibility of the investment strategy portfolio idea and the effectiveness of the combination genetic algorithm.

A specific task of algorithmic trading is tackled by Montana et al. (2008). The authors propose a flexible least squares approach, which is a penalized version of the ordinary least squares method, in order to determine how a given stream of data depends on others. Kissell and Malamut (2006) provide a dynamic algorithmic decisionmaking framework to assist investors in determining the most appropriate algorithm for given overall trading goals and investment objectives. The approach is based on a three-step

process in which the potential investor chooses a price benchmark, selects a trading style, and specifies an adaptation tactic. Gsell and Gomber (2009) investigate the extent of algorithmic trading activity and specifically their order placement strategies in comparison to human traders in the Xetra trading system. Kim and Kaljuvee (2007) provide some analysis of various aspects of electronic and algorithmic trading.

Azzini and Tettamanzi (2008) present an approach to intraday automated trading based on a neurogenetic algorithm. More specifically, an artificial neural network is evolved so as to provide trading signals to a simple automated trading agent. The neural network receives as input high, low, open, and close quotes from the underlying asset, as well as a number of technical indicators. The positions are closed as soon as a given profit target is met or the market closes. Experimental results indicate that the proposed scheme may yield promising returns.

In another study Dempster and Jones (2001) create portfolios of trading rules using genetic programming (see Chapter 8). These combinations of technical indicators aim at emulating the behavior of individual traders. The application area refers to U.S. dollar/British pound spot foreign exchange tick data from 1994 to 1997. The performance of the genetic-based system is compared to the application of individual trading rules. The best rule found by the proposed system was found to be modestly, but significantly, profitable in the presence of realistic transaction costs. In Wilson (1994) a fully automatic trading system for common stocks is developed. The system's inputs refer to daily price and volume data, from a list of two hundred stocks in ten markets. A chaos-based modeling procedure is used to construct alternative price prediction models, and a self-organizing neural network is used to select the best model for each stock. Then, a second self-organizing network is used to make predictions. The performance of the proposed system is compared to individual investment on the market index. The results indicate that the intelligent method achieves better return and volatility.

Another approach was proposed by Chang et al. (2009). The piecewise linear representation (PLR) method was combined with a back-propagation neural network (BPN) for trading. The PLR method was applied to historical data in order to find different segments. As a result, temporary turning points can be found. The neural network is able to define buy-sell signals. Also, a genetic algorithm component is incorporated for the optimization of the neural network's parameters. The intelligent system is compared to three existing algorithms, namely a rule-based BPN, a trading solutions software package, and an older version of the PLR-BPN system. The results indicate the superiority of the newer.

In Potvin et al. (2004) genetic programming is applied in order to automatically generate trading rules (see also Chapter 8). The performance of the system is compared to the naïve buy-and-hold strategy. The application domain is the Canadian stock market. Oussaidene et al. (1997) propose a parallel implementation of genetic programming for the financial trading problem, applied to seven exchange rates. Nenortaite and Simutis (2004) combine the concepts of the artificial neural network (ANN) and swarm intelligence in order to generate one-step-ahead investment decisions. The particle

swarm optimization (PSO) algorithm aims at discovering the best ANN model for forecasting and trading. Experimental results show that the application of the proposed methodology achieves better results than the market average, as measured by the market index (the S&P 500).

Hsu et al. (2011) introduce a quite interesting approach: They propose a trading mechanism that combines particle swarm optimization and moving average techniques for investing in mutual funds. The performance of the individual moving average models is enhanced by the incorporation of the nature-inspired intelligent technique. Results from the proposed scheme are compared to the investment of a fund. Experimentation proves that the PSO-based model can achieve high profit and reduce risk to a significant degree. In a similar study, Briza and Naval Jr (2008, 2011) propose a multiobjective optimization method, namely, the particle swarm optimization (PSO), for stock trading. The system utilizes trading signals from a set of technical indicators based on historic stock prices. Then a trading rule is developed that is optimized for two objective functions, the Sharpe ratio and percent profit. The performance of the system is compared to the individual technical indicators and to the market (index) itself. Results indicate that the intelligent scheme outperformed both the set of technical indicators and the market, demonstrating the potential of the system as a tool for making stock trading decisions.

All in all, the findings from the comprehensive, but not exhaustive, survey of the literature are as follows:

- Algorithmic trading deals with various aspects of automated trading, which depends on a number of factors. This fact proves that there are vast opportunities in developing such systems. It has been shown that trading systems have been created that take advantage of such aspects of finance as the nature of technical analysis, the behavior of stocks, and the formation of portfolios.
- Biologically inspired intelligent methodologies have been applied to a specific type of trading: the portfolio optimization problem. Research has highlighted their efficiency compared to traditional approaches. What is more, these intelligent schemes, whether used alone or hybridized, have been used in parameter tuning of robust trading machines such as neural networks and genetic programming. There is clear evidence that this kind of algorithm outperforms financial heuristics, used as benchmarks, in several cases.
- Another important aspect is benchmarking. In many studies, the algorithmic trading method was compared with financial heuristics such as the buy-and-hold strategy with regard to the same underlying asset, investment in the market index, and application of individual technical indicators for trading. The literature indicates that these are the main methods traditionally used in trading by financial experts. Also, comparison with older versions of intelligent systems is an important issue that can play a guiding role in evolving trading strategies using artificial intelligence.

9.3 ALGORITHMIC TRADING SYSTEM

In this chapter we present a specific type of algorithmic trading system. It has two components. First, in the estimation interval, a biologically inspired intelligent algorithm is applied with the aim of constructing a fund. This is referred to as the portfolio optimization problem. (The mathematical formulation of the problem is presented below.) Second, after the fund has been constructed, a technical indicator is applied, in the forecasting interval, so as to produce trading signals. At this point a complete cycle of the system has been completed. Then the system moves forward in time, via the concept of the rolling window.

As mentioned above, nature-inspired intelligent algorithms stem from the way real-life systems work and evolve. Their efficiency has been demonstrated in numerous problem domains. In this study we apply two hybrid NII schemes to a specific formulation of the portfolio optimization problem. To be more precise, the portfolio problem is tackled as a dual optimization task: the first task has to do with finding the proper combination of assets (discrete optimization), whereas the second one deals with identifying the optimum weights for the selected assets (continuous optimization).

The first hybrid algorithm consists of a genetic algorithm for finding the optimal combination of assets and a nonlinear programming technique, the Levenberg-Marquardt algorithm¹ (More 1978), for finding optimal weights (Vassiliadis et al. 2011). The standard genetic algorithm was first proposed by John Holland (1929–2015). The main characteristics of the algorithm lie in the concept of evolutionary process. As in the real world, genetic algorithms apply the mechanisms of selection, crossover, and mutation in order to evolve the members of a population through a number of generations. The ultimate goal is to reach a population of good-quality solutions approaching the optimum region. In order to assess the quality of each member of the population, the concept of fitness value is introduced.

The second NII method consists of an ant colony optimization (ACO) algorithm for the selection of assets and the Levenberg-Marquardt algorithm for weight optimization (Vassiliadis et al. 2009). The ACO algorithm was first proposed by Marco Dorigo in the beginning of the 1990s (Dorigo and Stultze 2004). It belongs to a certain class of metaheuristics whose main attribute is that they yield high-quality, near-optimum solutions in a reasonable amount of time, which can be considered an advantage if the solution space is of high dimensionality. The ACO metaheuristic is mainly inspired by the foraging behavior of real ant colonies. Ants start searching for potential food sources in a random manner, owing to a lack of knowledge about the surrounding environment. When they come up against a food source, they carry some of it back to the nest. On the way back, each deposits a chemical substance called a pheromone, which is a function of the quality and the quantity of the food source. So, chemical trails are formed in

¹ This is a local search procedure based on a nonlinear programming methodology that combines the Gauss-Newton method and the steepest-descent method.

each ant's path. As time passes, however, this chemical evaporates. Only paths with strong pheromone trails, reflecting a high-quality food source, manage to survive. As a consequence, all ants from the nest tend to follow the path or paths containing the largest amount of pheromone. This indirect kind of communication is called stigmergy.

In order to solve the portfolio optimization problem, we provide specific mathematical formulation.

Objective:

$$\text{Maximize}_w : U_o$$

s.t.

$$\begin{aligned} D_o &< H \\ \sum_{i=1}^k w_i &= 1 \\ w_l \leq w_i \leq w_u, i &= 1, \dots, k \\ k &= N \end{aligned}$$

where U_o , is the upside deviation, defined as $U_o = \sqrt{\int_0^{\infty} (r)^2 * p_r(r) dr}$ for a specific distribution of the portfolio's returns. The measure of upside deviation, that is the objective function applied in our study, refers to the part of returns' distribution, which deals only with positive returns. Investors aim to maximize this objective. The

term, D_o , is the downside deviation, defined as $D_o = \sqrt{\int_{-\infty}^0 (r)^2 * p_r(r) dr}$ for a specific distribution of the portfolio's returns. In other words, it measures the deviation of negative returns from zero. This is an unwanted attribute for investors. In our case, this metric is considered as a restriction, and it must not exceed a certain threshold.

The upper and lower budget constraints $[w_l, w_u]$, are the upper and lower acceptable percentages for capital invested in each asset. k , is the cardinality constraint. It refers to the maximum allowable number of assets included in the portfolio.

So, each solution should be in agreement with the aforementioned objectives and restrictions. In time periods of extreme volatility, during a financial crisis, for example, the standard deviation of assets' returns greatly increases, and it is difficult control it. As a result, investors search for investment opportunities in which the deviation of positive returns is quite large. Furthermore, the application of NII methodologies can be justified because strict constraints are used. In our case, the cardinality constraint hardly restricts the solution problem. Also, as the cardinality of the portfolio increases, the complexity of the problem increases as well. Traditional approaches fail to provide near-optimum solutions.

When the fund has been constructed, technical indicators are applied in the forecasting interval so as to produce buy/sell signals. In this study two trading rules are applied. The first, which is based on the concept of moving averages, is a commonly used indicator, called the moving average convergence divergence, or MACD (Appel 1999). The main characteristic of this rule is that it constructs a momentum indicator based on the fund's prices. The second rule is the classical buy-and-hold strategy, according to which a fund is bought in the start of the forecasting period and is sold at the end.

At this point, it is crucial to highlight some specific aspects of the proposed trading system.

- Each forecasting (investment) time interval succeeds a specific estimation time interval.
- Estimation and forecasting intervals do not overlap, although two estimation (and two forecasting) intervals may overlap owing to the rolling window.
- The concept of the rolling window can be explained as follows. Let us consider that the first estimation time interval contains the first $1 : n$ observations. In this sample, the fund is optimally constructed. Thereafter, in time interval $n + 1 : n + 1 + m$, technical indicators are applied to the fund. The next estimation interval is defined at the time period $1 + rw : n + rw$, where rw is the length of the rolling window. The corresponding forecasting interval is $n + rw + 1 : n + rw + 1 + m$. This process is repeated until the full time period under investigation is covered.

A pseudocode of the system is shown in figure 9.1:

9.4 COMPUTATIONAL RESULTS

We present the main results from the application of the trading system in this section. The data set comprises the daily closing prices of thirty stocks on the Dow Jones Industrial Average for the period January 3, 2005, to November 11, 2011. This time frame consists of uptrends and downtrends in the U.S. and the global economies. The configuration settings of the system are presented in table 9.1. Some of the settings are the result of limited experimentation, and others have been based on previous studies.

The experiments test the performance of the proposed GA-based and ACO-based hybrid schemes when combined with buy-and-hold and MACD strategies. Furthermore, we compare the results with the random portfolio construction methodology.

```

Function Trading_System
Define system's parameters
Calculate estimation and interval time periods
    q=number of estimation (forecasting) time periods
For i = 1:q
    Apply biologically Inspired Intelligenet algorithm for
    constructing fund (i th estimation period)
    Apply technical indicators to the fund (i th forecasting period)
End
  
```

FIGURE 9.1 Pseudocode for the trading system.

Table 9.1 Parameters for the automated trading system

Genetic algorithm	
Population	200
Generation	30
Crossover probability	0.90
Mutation probability	0.10
Percentage of best parents (selected for reproduction)	10%
Ant colony optimization algorithm	
Population	200
Generation	30
Evaporation rate	70%
Percentage of best ants (applied in the solution update process)	10%
Portfolio optimization problem	
Cardinality	10
$[w_l \ w_u]$	$[-0.5 \ 0.5]$
H (downside deviation of DJIA)	0.0106
System parameters	
Estimation time interval	100
Forecasting time interval	50
Rolling window	25

More specifically, in each of the estimation periods, one-thousand randomly constructed portfolios are found, and the best one is picked. In the case of random portfolio construction, we consider the stated optimization problem (objective function and restrictions). So the randomly generated portfolios follow the same procedure as do the intelligently selected ones. The aim is to provide a framework for fair comparison between the applied methodologies. All the technical indicators are applied to the former group as to the latter. The aim is to highlight the potential differences between intelligent and non-intelligent methods.

The net result of each trading strategy is shown in table 9.2, rows 1–4. In addition, we present benchmark results using random portfolio selection methodologies compared with the buy-and-hold and MACD approaches respectively (the results are shown in table 9.2, rows 5–6).

From these prime results, it can be seen that the hybrid schemes demonstrated similar performances. The GA-LMA algorithm returned the larger profit, however. Another interesting fact is that the buy-and-hold strategy outperformed the MACD rule, which gave negative results. This can be explained in part by the fact that the MACD rule is a momentum indicator and, to some extent, requires volatile series in order to

Table 9.2 Net result of the strategies

Algorithmic trading system	Net results (in US dollars)
GA-LMA/BUY and HOLD	\$3,813.8
GA-LMA/MACD	\$ - 480.6
ACO-LMA/ BUY and HOLD	\$3475.9
ACO-LMA/MACD	\$ - 901.3
RANDOM/ BUY and HOLD	\$732.6
RANDOM/MACD	\$175.4

provide buy/sell signals. This issue can be tackled by the proper definition of the fund construction problem.

The random portfolio selection technique proved to be quite poor with regard to the buy-and-hold strategy (returning only \$732.6, in cumulative terms). However, with the application of the MACD technical rule it achieved a positive outcome, in contrast with the two intelligent methodologies. This could be explained as follows. Constructing portfolios in a random way simply means that there is no specific strategy for guiding them through the solution space.

The only topic common to the two intelligent techniques is the optimization of the same objective. So it is quite possible that a random portfolio with large values for upside and downside deviation could be found (recall that NII techniques aim at maximizing the upside deviation, while minimizing the downside deviation). This surely affects the standard deviation of the random portfolio's returns, and more specifically, it could increase its deviation. The MACD rule works better in cases (or to put it better, in assets) with large standard deviation behavior.

Some results regarding the distribution of each strategy's returns are shown in table 9.3, rows 1–4. It is important to investigate whether they have any desirable characteristics from an investor's point of view. Again, the last two rows give benchmark results obtained from the application of a random portfolio construction methodology in combination with the buy-and-hold and MACD approaches, respectively (table 9.3, rows 5–6).

Concerning table 9.3 one should note the following.

- In terms of mean return, all strategies gave similar results. The buy-and-hold indicator yielded positive returns on average. The GA-LMA approach slightly outperformed the ACO-LMA approach.
- In the case of the buy-and-hold rule, the investment's risk was high enough, justifying in part the positive average return.
- In terms of skewness, the ACO-LMA strategy offers a desirable attribute: a high value of skewness. This means that the distribution of returns leans to the right side, which is a desired feature for investors. In the other cases the distribution of results is near the mean value.

Table 9.3 Statistical analysis of distribution of returns for each strategy

	Mean	St. Dev.	Skewness	Kurtosis	Percentiles ([0.05 0.50 0.95])
GA-LMA / Buy and hold	0.0569	0.2503	0.5837	4.9979	[−0.2452 0.0302 0.5069]
GA-LMA / MACD	−0.0049	0.0840	0.0864	8.5040	[−0.1162 −0.0018 0.1268]
ACO-LMA / Buy and hold	0.0519	0.2703	1.1523	7.0839	[−0.2835 0.0067 0.4780]
ACO-LMA / MACD	−0.0090	0.0773	−0.5595	8.4119	[−0.1181 −0.0070 0.1097]
Random / Buy and hold	0.0109	0.1778	0.0271	3.1220	[−0.2832 0.0036 0.2968]
Random / MACD	−0.0130	0.1025	−3.0023	18.3252	[−0.1732 0 0.1084]

- High values of kurtosis indicate that the distribution is more outlier-prone. This is our case. So it is highly probable that our strategies could yield long-tail distributions.
- Percentiles show some direct evidence regarding the distribution of returns. For example, at the 0.05 level the corresponding return for the GA-LMA is -0.2452 . This means that 5 percent of the returns sample has a value smaller than -0.2452 . Investors should prefer distributions with large and positive values for percentiles, meaning that the whole distribution leans to the right. In our case, the GA-LMA and the ACO-LMA appear to have the wider distributions (as was shown with the measure of standard deviation). However, the value for the far right-hand side of the distribution (the 0.95 level) is large in these cases.

In terms of invested returns, the random selection technique yielded poor results when using the buy-and-hold strategy. The expected return is small while the standard deviation is quite large, which is not acceptable. What is more, based on the statistical measures of skewness and kurtosis, distribution of returns from the random portfolio construction strategy seems to approximate the normal distribution. Investors, however, seek opportunities in which distribution of returns presents large positive skewness, that is, there is a larger probability that high positive returns could be achieved. On the other hand, regarding the MACD rule, the results are in favor of the intelligent techniques in all statistical measures.

All in all, we could say that the hybrid NII schemes yield attractive results, in the case of buy-and-hold rule. What is more, the GA-LMA approach slightly outperforms the ACO-LMA. The graphs of the cumulative returns for both hybrid algorithmic approaches are shown in figures 9.2 to 9.5. They show the future worth of \$1 invested now.

As we can see from the figures, hybrid NII schemes along with the buy-and-hold rule yield the best results. In their case the initially invested capital grows greatly over time (almost six years). Again, the MACD rule fails to provide acceptable results in terms of profit.

On the basis of these results, we can make the following main points.

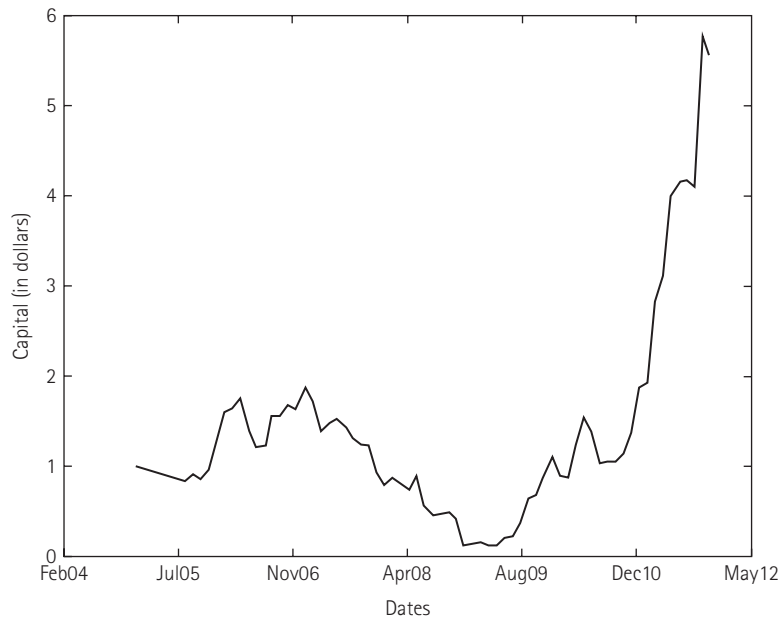


FIGURE 9.2 Cumulative returns from the GA-LMA/buy-and-hold strategy.

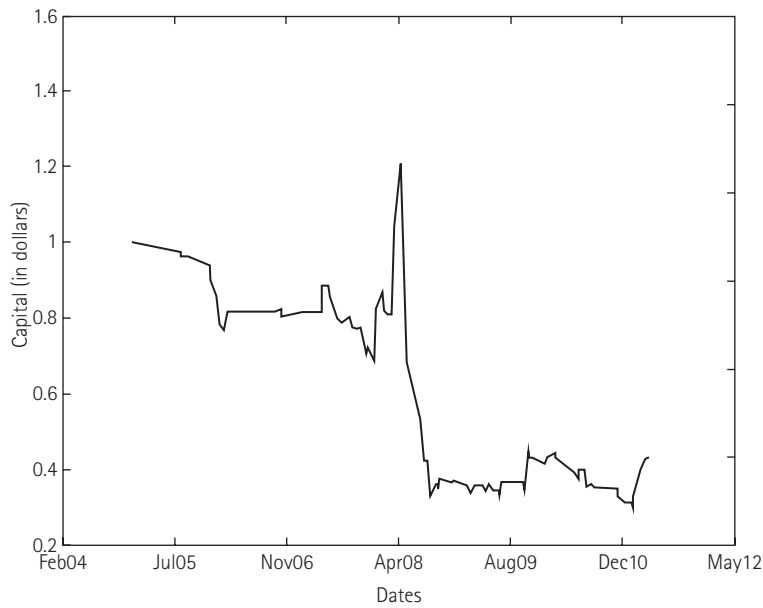


FIGURE 9.3 Cumulative returns from the GA-LMA/MACD strategy.

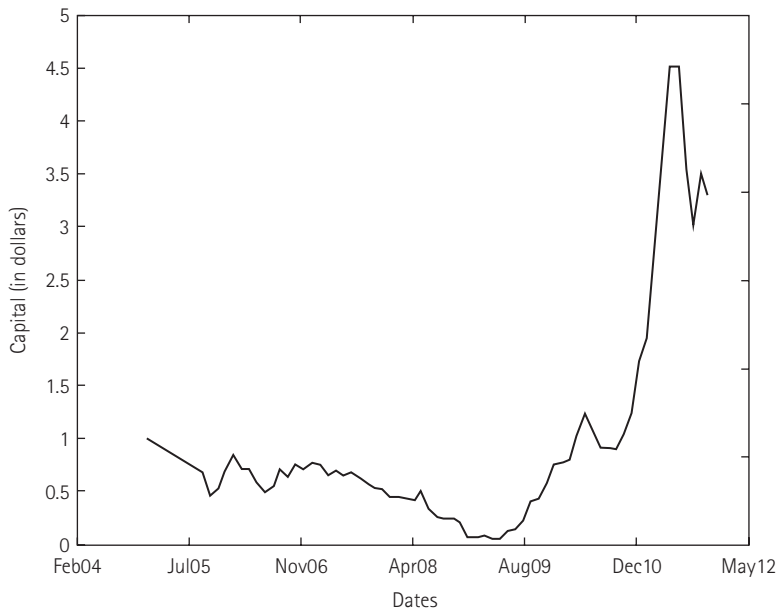


FIGURE 9.4 Cumulative returns from the ACO-LMA/buy-and-hold strategy.

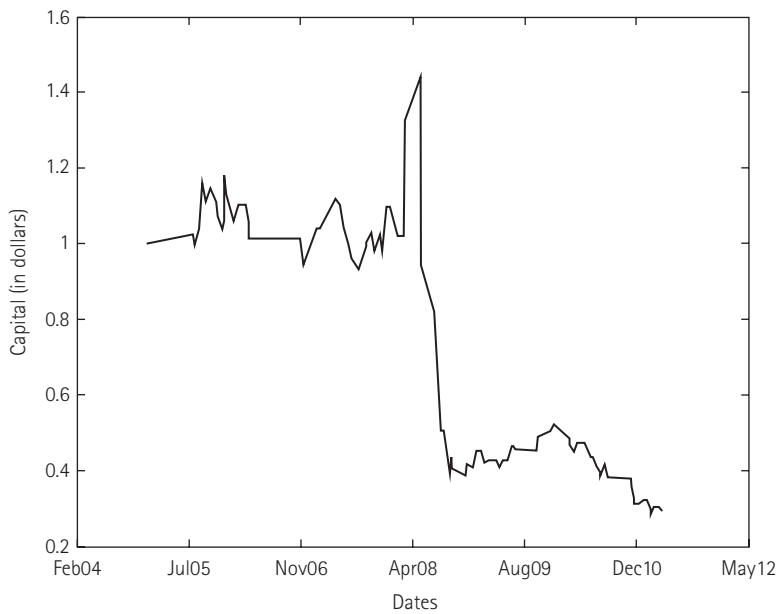


FIGURE 9.5 Cumulative returns from the ACO-LMA/MACD strategy.

- The GA-LMA system provided better results than the ACO-LMA. The basic difference between these techniques lies in the searching strategy. The genetic algorithm applies the Darwinian principles of evolution in order to make the initial population progress. In contrast, ACO implements the strategy of real ant colonies, when they search for food. These results might be an initial indication regarding the searching ability of GA. However, no safe conclusion can be drawn.
- Notice that the implemented trading rules (MACD, buy-and-hold) are independent of the fund construction process. Consequently, their performance is tested directly on the fund's price series. In this study, we tried to maximize the upside deviation of the fund. This could lead to a volatile series, thus giving an advantage to the MACD rule. Yet the buy-and-hold rule gave the best results. One possible explanation is that the time period under investigation was characterized by a number of upward trends, causing the construction of upward funds to some extent.

9.5 CONCLUSION

In this chapter we have analyzed the concept of algorithmic trading. More specifically, the main aim was to highlight the incorporation of biologically inspired intelligent algorithms into trading. These techniques are stochastic and have the unique advantage of imitating the way natural systems work and evolve, resulting in efficient search strategies. Financial trading systems deal with the automation of the trading process. The proposed trading system combined two processes: the first one had to do with the construction of a fund the (portfolio optimization problem), the second, a forecasting rule aimed at producing buy/sell signals. In the first component we applied two BII algorithms the genetic algorithm and the ant colony optimization algorithm. As far as the forecasting part is concerned, we applied two commonly used rules, the buy-and-hold rule and MACD.

Noted that results are to be considered preliminary. The goal of this study was to provide some evidence regarding the performance of NII schemes in financial-type problems. We treated the performance of the applied NII metaheuristics in terms of financial metrics (distribution of returning on investment). In finance, the aim of optimally allocating the available capital to a number of possible alternatives is of great importance. The portfolio optimization problem is quite difficult to solve in the case in which nonlinear objectives and complex constraints are imposed. A particular constraint that increases the complexity is the cardinality, which essentially restricts the number of assets included in the portfolio. Traditional methods are not able to provide attractive solutions, and definitely the optimum solution cannot be found. Nature-inspired intelligent algorithms are able to yield near-optimum solutions, approximating high-quality regions of the solution space rather efficiently. Finally, the incorporation of trading indicators for the derivation of trading signals is independent

of the BII techniques. Therefore, these indicators cannot contribute to the explanation of the behavior of these metaheuristics in terms of the financial metrics applied.

In order to obtain better insight regarding the trading performance of NII metaheuristics for this kind of problem, we indicate some directions for future research. First, a number of benchmarks should be implemented so as to compare the performance of NII algorithms with traditional techniques. Second the proposed trading system should be tested in other markets and time periods. This testing could give us clearer evidence regarding the applicability of biologically inspired intelligent algorithms to the financial (algorithmic) trading problem.

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CHAPTER 10

ALGORITHMIC TRADING IN PRACTICE

PETER GOMBER AND KAI ZIMMERMANN

10.1 INTRODUCTION

In the past few decades, decades, securities trading has experienced significant changes as more and more stages within the trading process have become automated by incorporating electronic systems. Electronic trading desks together with advanced algorithms entered the international trading landscape and introduced a technological revolution to traditional physical floor trading. Nowadays, the securities trading landscape is characterized by a high level of automation, for example, enabling complex basket portfolios to be traded and executed on a single click or finding best execution via smart order-routing algorithms on international markets. Computer algorithms encompass the whole trading process—buy side (traditional asset managers and hedge funds) as well as sell side institutions (banks, brokers, and broker-dealers) have found their business significantly migrated to an information systems-driven area where trading is done with minimum human intervention.

In addition, with the help of new market access models, the buy side has gained more control over the actual trading and order allocation processes and is able to develop and implement its own trading algorithms or use standard software solutions from independent vendors. Nevertheless, the sell side still offers the majority of algorithmic trading tools to its clients. The application of computer algorithms that generate orders automatically has reduced overall trading costs for investors because intermediaries could largely be omitted. Consequently, algorithmic trading (AT) has gained significant market share in international financial markets in recent years as time- and cost-saving automation went hand in hand with cross-market connectivity. Algorithmic trading not only has altered the traditional relation between investors and their market-access intermediaries but also has caused a change in the traders' focus as the invention of the telephone did in 1876 for communication between people.

This chapter gives an overview of the evolution of algorithmic trading, highlighting current technological issues as well as presenting scientific findings concerning the impact of this method on market quality. The paper is structured as follows: First, we characterize algorithmic trading in the light of the definitions available in the academic literature. The difference between algorithmic trading and such related constructs as high-frequency trading (HFT) is therefore illustrated. Further, we provide insights into the evolution of the trading process within the past thirty years and show how the evolution of trading technology influenced the interaction among market participants along the trading value chain. Several drivers of algorithmic trading are highlighted in order to discuss the significant impact of algorithms on securities trading. In section 10.3 we introduce the ongoing evolution of algorithmic strategies, highlighting such current innovations as newsreader algorithms. Section 10.4 outlines findings provided by academics as well as practitioners by illustrating their conclusions regarding the impact of algorithmic trading on market quality. In section 10.5 we will briefly discuss the role of this approach in the 2010 Flash Crash and explain circuit breakers as a key mechanism for handling market stress. A brief outlook will close the chapter.

10.1.1 Characterization, Definition, and Classification

A computer algorithm is defined as the execution of pre-defined instructions in order to process a given task (Johnson 2010). Transferred to the context of securities trading, algorithms provide a set of instructions on how to process or modify an order or multiple orders without human intervention.

Academic definitions vary, so we summarize the undisputed facts about which analysts agree the most. “Throughout the literature, AT is viewed as a tool for professional traders that may observe market parameters or other information in real-time and automatically generates/carries out trading decisions without human intervention” (Gomber et al. 2011, p. 11). The authors further list real-time market observation and automated order generation as key characteristics of algorithmic traders. These elements are essential in most definitions of algorithmic trading. For example, Chaboud et al. (2009) write: “[I]n algorithmic trading (AT), computers directly interface with trading platforms, placing orders without immediate human intervention. The computers observe market data and possibly other information at very high frequency, and, based on a built-in algorithm, send back trading instructions, often within milliseconds” (p. 1), and Domowitz and Yegerman (2006) state: “[W]e generally define algorithmic trading as the automated, computer-based execution of equity orders via direct market-access channels, usually with the goal of meeting a particular benchmark” (p. 1). A tighter regulatory definition was provided by the European Commission in the proposal concerning the review of the Markets in Financial Instruments Directive (MiFID) in 2011. The proposal states: “Algorithmic trading” means trading in financial instruments where a computer algorithm automatically determines individual parameters of orders such as whether to initiate the order, the

timing, price or quantity of the order or how to manage the order after its submission, with limited or no human intervention. This definition does not include any system that is only used for the purpose of routing orders to one or more trading venues or for the confirmation of orders” (European Commission 2011, p. 54). To summarize the intersection of these academic and regulatory statements, trading without human intervention is considered a key aspect of algorithmic trading and became the center of most applied definitions of this strategy. Gomber et al. (2011) further define trade characteristics not necessarily but often linked to algorithmic trading:

1. Agent trading
2. Minimization of market impact (for large orders)
3. To achieve a particular benchmark
4. Holding periods of days, weeks, or months
5. Working an order through time and across markets

This characterization delineates algorithmic trading from its closest subcategory, HFT, which is discussed in the following section.

Based on the specified design and parameterization, algorithms do not only process simple orders but conduct trading decisions in line with pre-defined investment decisions without any human involvement. Therefore, we generally refer to algorithmic as computer-supported trading decision making, order submission, and order management.

Given the continuous change in the technological environment, an all-encompassing classification seems unattainable, whereas the examples given promote a common understanding of this evolving area of electronic trading.

10.1.2 Algorithmic Trading in Contrast to High-Frequency Trading

High-frequency trading is a relatively new phenomenon in the algorithmic trading landscape, and much less literature and definitions can be found for it. Although the media often use the terms HFT and algorithmic trading synonymously, they are not the same, and it is necessary to outline the differences between the concepts. Aldridge (2009), Hendershott and Riordan (2011), Gomber et al. (2011) acknowledge HFT as a subcategory of algorithmic trading. The literature typically states that HFT-based trading strategies, in contrast to algorithmic trading, update their orders very quickly and try to keep no overnight position. The rapid submission, cancellation, and deletion of instructions is necessary in order to realize small profits per trade in a large number of trades without keeping significant overnight positions. As a prerequisite, HFT needs to rely on high-speed access to markets, that is, low latencies, the use of co-location or proximity services, and individual data feeds. It does not rely on sophisticated strategies to deploy orders as algorithmic trading does, but relies mainly on speed

that is, technology to earn small profits on a large number of trades. The concept of defining HFT as a subcategory of algorithmic trading is also applied by the European Commission in its latest MiFID proposal: “A specific subset of algorithmic trading is High Frequency Trading where a trading system analyses data or signals from the market at high speed and then sends or updates large numbers of orders within a very short time period in response to that analysis. High frequency trading is typically done by the traders using their own capital to trade and rather than being a strategy in itself is usually the use of sophisticated technology to implement more traditional trading strategies such as market making or arbitrage” (European Commission 2011, p. 25). Most academic and regulatory papers agree that HFT should be classified as technology rather than a specific trading strategy and therefore demarcate HFT from algorithmic trading.

10.2 EVOLUTION OF TRADING AND TECHNOLOGY (I)

The evolutionary shift toward electronic trading did not happen overnight. Starting in 1971, the National Association of Securities Dealers Automated Quotation (NASDAQ) becomes the first electronic stock market when it displayed quotes for twenty-five hundred over-the-counter securities. Soon competitors followed on both sides of the Atlantic. The following sections focus on the timeline of the shift and the changing relationship between the buy side and the sell side. Significant technological innovations are discussed, and the drivers of this revolution are identified.

10.2.1 Evolution of Trading Processes

Figure 10.1 presents cornerstones of the evolutionary shift in trading since the initial electronification of securities markets. From the early 1990 many of the major securities exchanges became fully electronified, that is, the matching of orders and price determination was performed by matching algorithms (Johnson 2010). The exchanges established electronic central limit order books (e-CLOB), which provided a transparent, anonymous, and cost-effective way to aggregate and store open-limit orders as well as match executable orders in real time. These advancements led to a decentralization of market access, allowing investors to place orders from remote locations, and made physical floor trading more and more obsolete. In the mid 1990s the Securities and Exchange Commission further intensified competition between exchanges by allowing electronic communication networks, computer systems that facilitate trading outside traditional exchanges, to enter the battle for order flow, leading the way to today’s highly fragmented electronic trading landscape. On the sell side, electronification proceeded to the implementation of automated price observation mechanisms, electronic eyes and

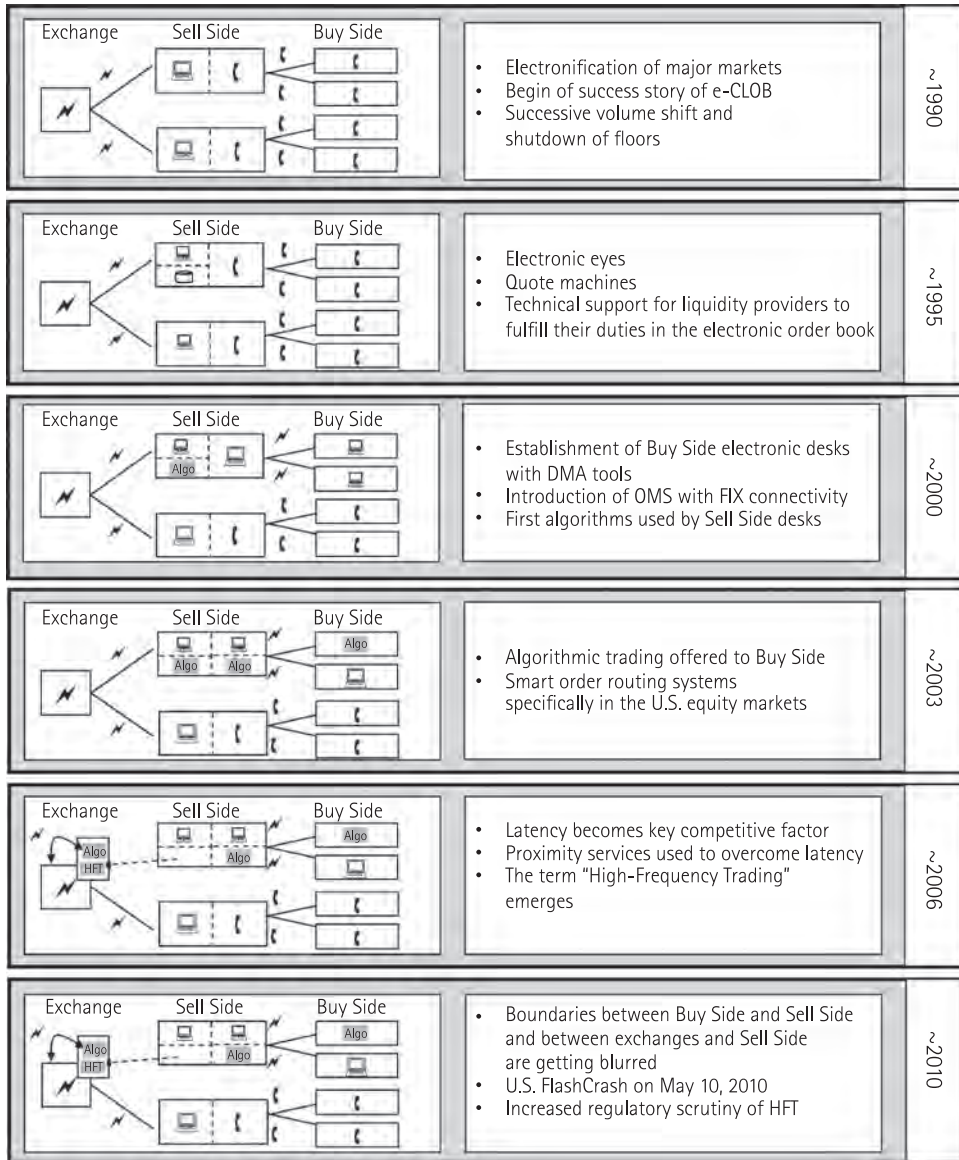


FIGURE 10.1 The evolution of trading. Technology walks up the value chain and supports an ever-increasing range of trading behaviors formerly carried out by humans.

automated quoting machines that generate quotes given pre-parameterized conditions, effectively reducing a market maker's need to provide liquidity manually. About the year 2000, buy side traders began to establish electronic trading desks by connecting with multiple brokers and liquidity sources. Trading saw significant improvements in efficiency owing to the use of order management systems (OMS), which allowed for routing automation, connectivity, and integration with confirmation, clearing, and

settlement systems. The introduction of the Financial Information eXchange (FIX) Protocol allowed for world wide uniform electronic communication of trade-related messages and became the de facto messaging standard for pre-trade and trade communication (FIX Protocol Limited 2012). About the same time, sell side pioneers implemented the first algorithms to aid and enhance their proprietary executions. Realizing that buy side clients could also benefit from these advancements, brokers started to offer algorithmic services to them shortly thereafter. Since being offered frameworks that allow for individual algorithm creation and parameterization, clients' uptake has steadily increased (Johnson 2010). The first smart order-routing services were introduced in the U.S. system to support order routing in a multiple-market system.

About 2006, the sell side started using co-location and proximity services to serve its own and the buy side's demand for lower transmission latencies between order submission and order arrival. The term "high-frequency trading" emerged. One has to keep in mind, however that, in particular, mid-sized and small buy side firms today still use the telephone, fax, or email to communicate orders to their brokers. The 2010 U.S. Flash Crash marks a significant event in the evolution of securities trading because it dramatically intensified the regulatory discussion about the benefits of this evolution (see section 10.4).

10.2.2 Evolution of Trading Processes in Intermediation Relationships

To augment and add detail to the discussion above, this section highlights major technological advancements accompanying the intermediation relationship between the buy side, the sell side, and markets in the process of securities trading. The top panel of figure 10.1 displays the traditional trading process. It reaches from the buy side investor's allocation decision to the final order arrival in the markets. In this process, the broker played the central role because he or she was responsible for management and execution of the order. Depending on order complexity and benchmark availability (both of which are driven mainly by order size and the liquidity of the traded security), the broker decided to either route the order directly to the market immediately and in full size or to split and time the order to avoid market impact. If liquidity on the market is not available, the broker executed the order against his own proprietary book, providing risk capital.

The bottom panel of figure 10.2 shows how the intermediation relationship between buy side and the sell side changed during the technological evolution. As illustrated, the responsibility for execution was shifted toward the buy side which absorbed more direct control over the order routing and execution process, and the role of the sell side changed to that of a provider of market access and trading technology. The new technologies named in the figure, direct market access and sponsored market access, as well as smart order routing are described below to show their relation to algorithmic trading. Because execution by full-service or agency broker dark pools, or electronic execution services for large institutional orders without pre-trade transparency, is

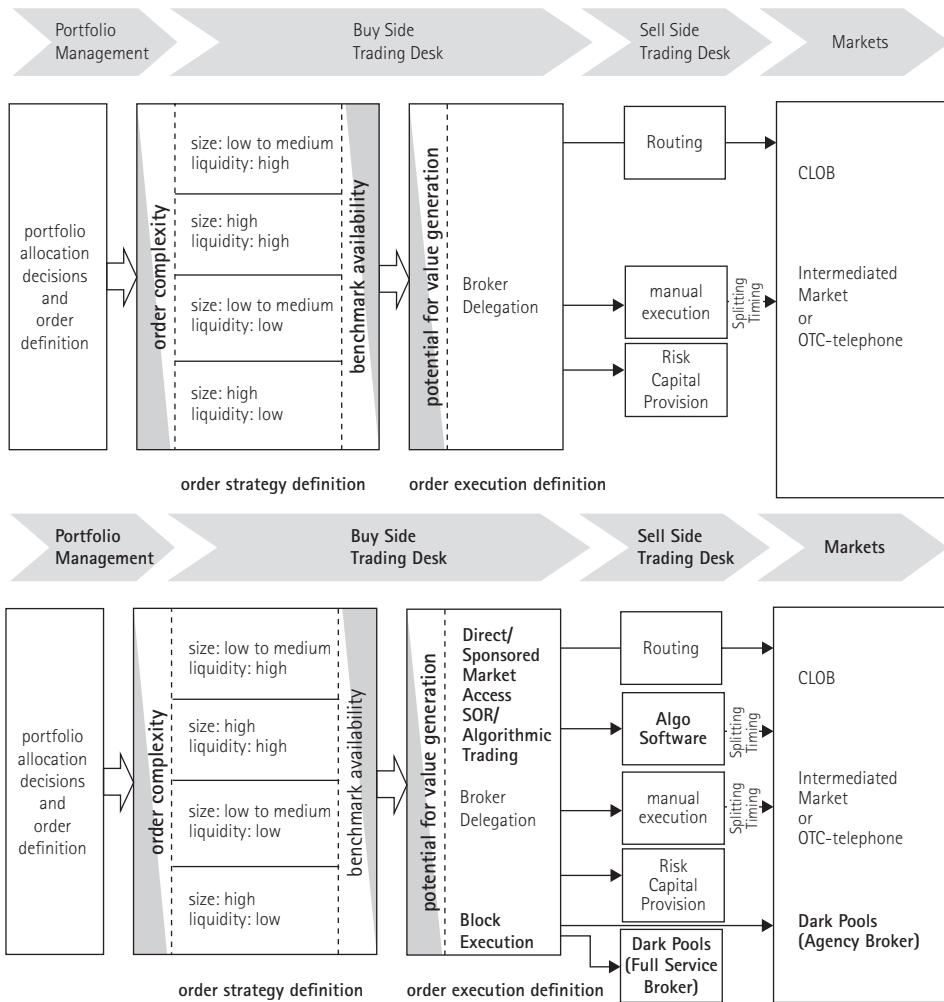


FIGURE 10.2 While traditionally the responsibility for order execution was fully outsourced to the sell side, the new technology-enabled execution services allow for full control by the buy side.

mainly focused on the direct interaction of buy side orders and only indirectly related to algorithmic trading, this technology will not be described in detail.

In markets that are organized by exchanges, only registered members are granted access to the e-CLOB. Those members are the only ones allowed to conduct trading directly; thus their primary role as market access intermediaries for investors. Market members performing that function are referred to as exchange brokers (Harris 2003). These intermediaries transform their clients' investment decisions into orders that are allocated to the desired market venues. As the buy side has become more aware of trading costs over the years brokers have begun to provide alternative market access models such as so-called direct market access (DMA). By taking advantage of DMA, an

investor no longer has to go through a broker to place an order but, rather, can have it forwarded directly to the markets through the broker's trading infrastructure. Johnson (2010) refers to "Zero-touch" DMA because the buy side takes total control over the order without direct intervention by an intermediary. Given the resulting reduction in latency, DMA models provide an important basis for algorithm-based strategies and HFT.

Sponsored market access represents a modified approach to DMA offerings. This approach targets buy side clients that focus on high-frequency strategies and therefore wish to connect to the market via their broker's identification but omit their broker's infrastructure. Sponsored access users rely on their own high-speed infrastructure and access markets using the sell side's identification; that is, they trade on the market by renting the exchange membership of their sell side broker. Afterward, intermediaries only provide automated pre-trade risk checks that are mostly implemented within the exchange software and administered by the broker, for example, by setting a maximum order value or the maximum number of orders in a predefined time period. A further extension, "naked access" or "unfiltered access," refers to the omission of pre-trade risk checks. In this process, in order to achieve further latency reduction, only post-trade monitoring is conducted, potentially allowing erroneous orders and orders submitted by flawed algorithms to enter the markets. Because of the possible devastating impacts, the SEC resolved to ban naked access in 2010. Furthermore, the SEC requires all brokers to put in place risk controls and supervisory procedures relating to how they and their customers access the market (SEC 2010b). Naked access is not allowed in the European securities trading landscape.

In a setup in which each instrument is traded only in one market, achieving the best possible price requires mainly the optimal timing of the trade and optimal order sizes to minimize price impact, or implicit transaction costs. In a fragmented market system such as those of Europe and the United States, however, this optimization problem becomes more complex. Because each instrument is traded in multiple venues, a trader has to monitor liquidity and price levels in each venue in real time. Automated, algorithm-based low-latency systems provide solutions in fragmented markets. Smart order routing (SOR) engines monitor multiple liquidity pools (that is, exchanges or alternative trading systems) to identify the highest liquidity and optimal price by applying algorithms to optimize order execution. They continuously gather real-time data from the respective venues concerning the available order book situations (Ende et al. 2009). Foucault and Menkveld (2008) analyze executions among two trading venues for Dutch equities and argue that suboptimal trade executions result from a lack of automation of routing decisions. Ende et al. (2009) empirically assess the value of SOR algorithms in a post-MiFID fragmented European securities system. They find suboptimal executed trades worth €262 billion within a four-week data set. With approximately 6.71 percent of all orders capable of being executed at better prices, they predict overall cost savings of €9.5 million within this time period, indicating an increasing need for sophisticated SOR to achieve best possible execution.

10.2.3 Latency Reduction

Among the changes in the trading process triggered by algorithmic trading, execution and information transmission latency faced the most significant adjustment. “Latency” in this context refers to the time that elapses from the insertion of an order into the trading system and the actual arrival of the order and its execution at the market. In the era of physical floor trading, traders with superior capabilities and close physical proximity to the desks of specialists could accomplish more trades and evaluate information faster than competitors and therefore could trade more successfully. Today, average latencies have been reduced to a fragment of a millisecond. This advance was driven mainly by the latest innovations in hardware, exchange co-location services, and improved market infrastructure. Such a decrease in latency translates into an increase in participants’ revenues as well as a reduction of error rates, since traders can avoid missing economically attractive order book situations due to high latency (Riordan and Stockenmaier 2012). The omission of human limitation in decision making became central in promoting algorithms for the purpose of conducting high-speed trading. Combining high-speed data access with predefined decision making, today’s algorithms are able to adapt to permanent changes in market conditions quickly. Trading venues recognize the trader’s desire for low latency, and so they intensify the chase for speed by providing more low-latency solutions to attract more clients (Ende et al. 2011).

10.2.4 Co-Location and Proximity Hosting Services

The Commodity Futures Trading Commission (CFTC) states that “[...] the term “Co-Location/Proximity Hosting Services” is defined as trading market and certain third-party facility space, power, telecommunications, and other ancillary products and services that are made available to market participants for the purpose of locating their computer systems/servers in close proximity to the trading market’s trade and execution system” (Commodity Futures Trading Commission 2010a, p. 33200). These services provide participating institutions with further latency reduction by minimizing network and other trading delays. These improvements essential for all participants conducting HFT but are also beneficial in algorithmic trading strategies. The CFTC thus acknowledges that these services should not be granted in a discriminatory way, for example, by limiting co-location space or by a lack of price transparency. In order to ensure equal, fair, and transparent access to these services, the CFTC proposed a rule that requires institutions that offer co-location or proximity hosting services to offer equal access without artificial barriers that act to exclude some market participants from accessing these services (Commodity Futures Trading Commission 2010a).

10.2.5 Fragmentation of Markets

Fragmentation of investors’ order flow has occurred in U.S. equity markets since the implementation of the Regulation of Exchanges and Alternative Trading Systems

(Reg ATS) in 2000, followed by the 2005 implementation of the Regulation National Market System (Reg NMS). Competition in European equity markets began in 2007 after the introduction of MiFID, which enabled new venues to compete with the incumbent national exchanges. Both regulatory approaches, although they differ in the explicit degree of regulation, aim to improve competition in the trading landscape by attracting new entrants to the market for markets. Because traders' interest in computer-supported trading preceded these regulations, the fragmentation of markets cannot be considered the motivating force for the use of algorithms. But considering that a multiple-market system only allows for beneficial order execution and the resulting cost savings if every relevant trading center is included in decision making, a need for algorithms to support this process is reasonable. Further, cross-market strategies (arbitrage), as well as provision of liquidity in fragmented markets can only be achieved with wide availability of cross-market data and a high level of automated decision making. Therefore, fragmentation is considered a spur for promoting algorithm use and high-frequency technologies in today's markets.

10.2.6 Market Participation and Empirical Relevance

Algorithmic trading influences not only today's trading environment and market infrastructure but also trading characteristics and intraday patterns. Although exact participation levels remain opaque owing to the anonymity of traders and their protection of their methods, a handful of academic and industry papers try to estimate overall market share. The Aite Group (2006) estimated algorithm usage from a starting point near zero around 2000, thought to be responsible for over 50 percent of trading volume in the United States in 2010 (Aite Group 2006). Hendershott and Riordan (2011) reached about the same number on the basis of a data set of Deutsche Boerse's DAX 30 instruments traded on XETRA in 2008. The CME Group (2010) conducted a study of algorithmic activity within their futures markets that indicated algorithm participation of between 35 percent (for crude oil futures) and 69 percent in (for EuroFX futures) in 2010. Because the literature is mainly based on historic data sets, these numbers may underestimate actual participation levels. Academics see a significant trend toward a further increase in use of algorithms. Furthermore, algorithmic trading as well as HFT now claim significant shares of the foreign exchange market. According to the Aite Group, the volume of trade in FX markets executed by algorithms may exceed 10% in the year 2011 (Aite Group 2011).

10.3 EVOLUTION OF TRADING AND TECHNOLOGY (II)

Not only has the trading environment adapted to technological advances, but market interaction and order management have improved with computerized support. Section 10.3 gives a comprehensive overview of the status quo in algorithmic trading strategies,

focusing on trading strategies used primarily in agent trading well as proprietary trading.

10.3.1 Algorithmic Strategies in Agent Trading

From the beginning of algorithm-based trading, the complexity and granularity of the algorithms have developed with their underlying mathematical models and supporting hard- and software. Algorithms react to changing market conditions, level their aggressiveness based on the current trading hour, and consider financial news in their trading behavior. Apart from advancements in customization, the key underlying strategies of algorithms have not changed much. Most of the algorithms today still strive to match given benchmarks, minimize transaction costs, or seek liquidity in different markets. The categorization of the various algorithms is based mainly on the different purposes or behavior of the strategies used. Domowitz and Yegerman (2005) qualify algorithms based on their complexity and mechanics, whereas Johnson (2010) suggests a classification based on their objective. We follow Johnson's proposal by illustrating the chronology of algorithm development. Impact-driven and cost-driven algorithms seek to minimize market impact costs (overall trading costs). Johnson places opportunistic algorithms in a separate category. Since both impact-driven and cost-driven algorithms are available for opportunistic modification, we give examples of opportunistic behavior in both types. We also provide a brief introduction to newsreader algorithms, among the latest developments. Section 10.3.5 focuses on algorithms used in proprietary trading.

10.3.2 Impact-Driven Algorithms

Orders entering the market may considerably change the actual market price depending on order quantity, the order limit and current order book liquidity. Imagine a large market order submitted to a low-liquidity market. This order would clear the other side of the order book to a large extent, thus significantly worsening its own execution price with every partial fill. This phenomenon is the reason why market impact costs make up one part of the implicit trading costs (Harris 2003; Domowitz and Yegerman 2005). Impact-driven algorithms seek to minimize the effect that trading has on the asset's price. By splitting orders in to sub-orders and spreading their submission over time, these algorithms characteristically process sub-orders on the basis of a predefined price, time, or volume benchmark. The volume-weighted average price (VWAP) benchmark focuses on previous traded prices relative to the order's volume. The overall turnover divided by the total volume of the order sizes indicates the average price of the given time interval and may represent the benchmark for the measurement of the performance of the algorithm. Focusing on execution time, the time-weighted average price (TWAP) benchmark algorithm generate — in its simplest implementation—equally large sub-orders and processes them in equally distributed time intervals. Trading intervals can be calculated from the total quantity, the start

time, and the end time; for example, an order to buy 120,000 shares in chunks of 5,000 shares from 10 o'clock to 12 o'clock, results in five-minute trading intervals. Both methods have substantial disadvantages. If one disregards the current market situation while scheduling the order to meet the predefined benchmark, the results of both algorithms may lead to disadvantageous execution conditions. The predictability of these algorithms may encourage traders to exploit them, so dynamization of both concepts is reasonable because actual market conditions are obviously a more efficient indicator than historical data. With real-time market data access, VWAP benchmarks are calculated trade by trade, adjusting operating algorithms with every trade. Percent-of-volume (POV) algorithms base their market participation on the actual market volume, forgo trading if liquidity is low, and intensify aggressiveness if liquidity is high to minimize market impact. Randomization is a feature of the impact-driven algorithms. As predictability decreases with randomization of time or volume, static orders become less prone to detection by other market participants.

10.3.3 Cost-Driven Algorithms

Market impact costs represent only one part of the overall costs arising in securities trading. Academic literature distinguishes between implicit cost such as market impact or timing costs and explicit costs such as commission or access fees (Harris 2003). Cost-driven algorithms concentrate on both variants in order to minimize overall trading costs. Therefore, simple order splitting may not be the most desirable mechanism, as market impact may be eventually reduced, but at the cost of higher timing risk owing to the extended time span in which the order is processed. Cost-driven algorithms must anticipate such opposing effects in order to not just shift sources of risk but instead minimize it. Implementation shortfall is one of the widespread benchmarks in agent trading. It represents the difference of the average execution price currently achievable at the market and the actual execution price provided by the algorithm. Since implementation shortfall algorithms are, at least in part affected by the same market parameters as impact-driven algorithms are, both types use similar approaches. Adaptive shortfall is a subcategory of implementation shortfall. Based on the constraints of the latter, this algorithm adapts trading to market condition changes such as price movements allowing the algorithm to trade more opportunistically in beneficial market situations.

10.3.4 Newsreader Algorithms

One of the relatively recent innovations is the newsreader algorithm. Since every investment decision is based on some input by news or other distributed information, investors feed their algorithms with real-time newsfeeds. From a theoretical perspective, these investment strategies are based on the semi-strong form of efficient markets (Fama 1970), that is, prices adjust to publicly available new information very rapidly

and in an unbiased fashion. In practical terms, information enters market prices with a certain transitory gap, during which investors can realize profits. Humans' ability to analyze a significant amount of information in short reaction times is limited, however, so newsreaders are deployed to analyze sentiment in documents. A key focus of this approach is to overcome the problem utilizing the relevant information in documents such as blogs, news, articles, or corporate disclosures. This information may be unstructured, meaning it is hard for computers to understand, since written information contains a lot of syntactic and semantic features, and information that is relevant for an investment decision may be concealed within paraphrases. The theoretical field of sentiment analysis and text-mining encompasses the investigation of documents in order to determine their positive or negative conclusion about the relevant topic. In general, there are two types of in-depth analysis of the semantic orientation of text information (called polarity mining): supervised and unsupervised techniques (Chaovalit and Zhou 2008). Supervised techniques are based on labeled data sets in order to train a classifier (for example, a support vector machine), which is set up to classify the content of future documents. In contrast, unsupervised techniques use predefined dictionaries to determine the content by searching for buzzwords within the text. Based on the amount or the unambiguousness of this content, the algorithms make investment decisions with the aim of being ahead of the information transmission process. An introduction to various approaches to extracting investment information from various unstructured documents as well as an assessment of the efficiency of these approaches is offered by Tetlock (2007) and Tetlock et al. (2008).

10.3.5 Algorithmic Trading Strategies in Proprietary Trading

Whereas the previous sections dealt with agent trading, the rest of this section will focus on strategies that are prevalent in proprietary trading, which have changed significantly owing to the implementation of computer-supported decision making.

10.3.6 Market Making

Market making strategies differ significantly from agent (buy side) strategies because they do not aim to build up permanent positions in assets. Instead, their purpose is to profit from short-term liquidity by simultaneously submitting buy and sell limit orders in various financial instruments. Market makers' revenues are based on the aggregated bid-ask spread. For the most part, they try to achieve a flat end-of-day position. Market makers frequently employ quote machines, programs that generate, update, and delete quotes according to a pre-defined strategy (Gomber et al. 2011). The implementation of quote machines in most cases has to be authorized by the market venue and has to be monitored by the user. The success of market making basically is sustained through

real-time market price observation, since dealers with more timely information about the present market price can set up quotes in a more exact manner and so generate a thinner bid-ask spread through an increased number of executed trades. On the other hand, speed in order submission, execution, and cancellation reduces a market maker's risk of misquoting instruments in times of high volatility. Therefore, market makers benefit in critical ways from automated market observation as well as algorithm-based quoting.

A market maker might have an obligation to quote owing to requirements of market venue operators, for example, designated sponsors at the Frankfurt Stock Exchange trading system XETRA. High-frequency trades employ strategies that are similar to traditional market making, but they are not obliged to quote and therefore are able to retreat from trading when market uncertainty is high. Besides the earnings generated by the bid-ask spread, HFT market makers benefit from pricing models of execution venues that rebate voluntary HFT market makers in case their orders provide liquidity (liquidity maker), that is, are sitting in the order book and get executed by a liquidity taker that has to pay a fee. This model is often called asymmetric pricing or maker/taker pricing.

10.3.7 Statistical Arbitrage

Another field that evolved significantly with the implementation of computer algorithms is financial arbitrage. Harris (2003) defines arbitrageurs as speculators who trade on information about relative values. They profit whenever prices converge so that their purchases appreciate relative to their sales. Types of arbitrage vary with the nature of underlying assumptions about an asset's "natural" price. Harris further identifies two categories: Pure arbitrage (also referred to as mean reverting arbitrage) is based on the opinion that an asset's value fundamentally tends to a long or medium average. Deviations from this average only represent momentum shifts due to short-term adjustments. The second category, speculative arbitrage, assumes a nonstationary asset value. Nonstationary variables tend to drop and rise without regularly returning to a particular value. Instead of anticipating a value's long-term time mean, arbitrageurs predict a value's future motion and base investment strategies on the expected value. The manifold of arbitrage strategies are derivatives of one of these two approaches, ranging from vanilla pair trading techniques to trading pattern prediction based on statistical or mathematical methods. For a detailed analysis of algorithm-based arbitrage strategies and insight in to current practices see, for example, Pole (2007).

Permanent market observation and quantitative models make up only one pillar essential to both kinds of arbitrage. The second pillar focuses again on trading latency. Opportunities to conduct arbitrage frequently exist only for very brief moments. Because only computers are able to scan the markets for such short-lived possibilities, arbitrage has become a major strategy of HFTs (Gomber et al. 2011).

10.4 IMPACT OF ALGORITHMIC TRADING ON MARKET QUALITY AND TRADING PROCESSES

The prevailing negative opinion about algorithmic trading, especially HFT, is driven in part by media reports that are not always well informed and impartial. Most of the scientific literature credits algorithmic trading with beneficial effects on market quality, liquidity, and transaction costs. Only a few papers highlight possible risks imposed by the greatly increased trading speed. However, all academics encourage objective assessments as well as sound regulation in order to prevent system failures without cutting technological innovation. This section concentrates on major findings regarding the U.S. and European trading landscape as it concerns. Impact, on trade modification and cancellation rates, market liquidity, and market volatility.

10.4.1 Impact on Trade Modification and Cancellation Rates

Among the first who analyzed algorithmic trading pattern in electronic order books, Prix et al. (2007) studied changes in the lifetime of cancelled orders in the XETRA order book. Owing to the characteristics of their data set, they are able to identify each order by a unique identifier and so re create the whole history of events for each order. As they focus on the lifetimes of the so-called no-fill deletion orders, that is, orders that are inserted and subsequently cancelled without being executed, they find algorithm-specific characteristics concerning the insertion limit of an order compared to ordinary trading by humans. Gsell and Gomber (2009) likewise focus on differences in trading pattern between human and computer-based traders. In their data setup they are able to distinguish between algorithmic and human order submissions. They conclude that automated systems tend to submit more, but significantly smaller, orders. Additionally, they show the ability of algorithms to monitor their orders and modify them so as to be at the top of the order book. The authors state that algorithmic trading behavior is fundamentally different from human trading concerning the use of order types, the positioning of order limits, modification or deletion behavior. Algorithmic trading systems capitalize on their ability to process high-speed data feeds and react instantaneously to market movements by submitting corresponding orders or modifying existing ones.

Algorithmic trading has resulted in faster trading and more precise trading strategy design, but what is the impact on market liquidity and market volatility? The following sections provide a broader insight to this question.

10.4.2 Impact on Market Liquidity

A market's quality is determined foremost by its liquidity. Harris (2003, p. 394) defines liquidity as "the ability to trade large size quickly, at low cost, when you want to

trade.” Liquidity affects the transaction costs for investors and is a decisive factor in the competition for order flow among exchanges and between exchanges and proprietary trading venues. Many academic articles focus on these attributes to discern possible impacts of algorithmic trading and HFT on a market’s liquidity and, therefore, on a market’s quality. Hendershott et al. (2011) provide the first event study, assessing the New York Stock Exchange’s dissemination of automated quotes in 2003. This event marked the introduction of an automated quoting update, which provided information faster and caused an exogenous increase in algorithmic trading and, on the other side, nearly no advantage for human traders. By analyzing trading before and after this event, the authors find that algorithmic trading lowers the costs of trading and increases the informativeness of quotes. These findings are influenced by the fact that the analyzed period covers a general increase in volume traded, which also contributes to market quality but is not controlled in the author’s approach. Hendershott and Riordan (2011) confirm the positive effect of algorithmic trading on market quality. They find that algorithmic traders consume liquidity when it is cheap and provide liquidity when it is expensive. Further, they conclude that algorithmic trading contributes to volatility dampening in turbulent market phases because algorithmic traders do not retreat from or attenuate trading during these times and therefore contribute more to the discovery of the efficient price than human trading does. These results are backed by findings of Chaboud et al. (2009). Based on a data set of algorithmic trades from 2003 to 2007, the authors argue that computers provide liquidity during periods of market stress. Overall these results illustrate that algorithmic trading closely monitors the market in terms of liquidity and information and react quickly to changes in market conditions, thus providing liquidity in tight market situations (Chaboud et al. 2009). Further empirical evidence for the algorithms’ positive effects on market liquidity are provided by Hasbrouck and Saar (2013) as well as Sellberg (2010).

Among the theoretical evidence on the benefits of algorithmic trading, the model presented by Foucault et al. (2011) achieved significant attention. In order to determine the benefits and costs of monitoring activities of securities markets, the authors develop a model of trading with imperfect monitoring to study this trade-off and its impact on the trading rate. In order to study the effect of algorithmic trading, the authors interpret it as a reduction of monitoring costs, concluding that algorithmic trading should lead to a sharp increase in the trading rate. Moreover, it should lead to a decrease in the bid-ask spread if, and only if, it increases the speed of reaction of market makers relative to the speed of reaction of market takers (the “velocity ratio”). Last, algorithmic trading is socially beneficial because it increases the rate at which gains from trades are realized. Yet adjustments in trading fees redistribute the social gain of algorithmic trading between participants. For this reason, automation of one side may, counter intuitively, make that side worse off after adjustments in maker/taker fees (Foucault et al. 2011).

10.4.3 Impact on Market Volatility

A high variability in asset prices indicates great uncertainty about the value of the underlying asset, thus alienating an investor's valuation and potentially resulting in incorrect investment decisions when price variability is high. Connecting automation with increased price variability seems to be a straightforward argument owing to computers' immense speed. With research, however, this prejudice proves to be unsustainable. By simulating market situations with and without the participation of algorithmic trading, Gsell (2008) finds decreasing price variability when computers act in the market. This might be explained by the fact that because there is lower latency in algorithmic trading, more orders can be submitted to the market and therefore the size of the sliced orders decreases. Fewer partial executions will occur because there will more often be sufficient volume in the order book to completely execute the small order. If fewer partial executions occur, price movements will be narrowed as the order executes at fewer limits in the order book. Assessing the foreign exchange market and basing their work on a data set that differentiates between computer and human trades, Chaboud et al. (2009) find no causal relation between algorithmic trading and increased exchange rate volatility. They state: "If anything, the presence of more algorithmic trading is associated with lower volatility" (p. 1). The authors use an ordinary least-squares approach in order to test for a causal relation between the fractions of daily algorithmic trading and to the overall daily volume. Additionally, Groth (2011) confirms this relation between volatility and algorithmic trading by analyzing data containing a specific flag provided by the respective market operator that allows one to distinguish between algorithmic and human traders. The author indicates that the participation of algorithmic traders is associated not with higher levels of volatility, but with more stable prices. Furthermore, algorithmic traders do not withdraw liquidity during periods of high volatility, and traders do not seem to adjust their order cancellation behavior to volatility levels. In other words, algorithmic traders provide liquidity even if markets become turbulent; therefore, algorithms dampen price fluctuations and contribute to the robustness of markets in times of stress.

A more critical view of algorithmic trading is provided by researchers from the London-based Foresight Project. Although they highlight its beneficial effects on market stability, the authors warn that possible self-reinforcing feedback loops within well-intentioned management and control processes can amplify internal risks and lead to undesired interactions and outcomes (Foresight 2011). The authors illustrate possible liquidity or price shock cascades, which also intensified the U.S. Flash Crash of May 6, 2010. This hypothesis is backed, in part, by Zhang (2010) and Kirilenko et al. (2017), each finding HFT to be highly correlated with volatility and the unusually large selling pressure noted during the Flash Crash.

10.5 REGULATION AND HANDLING OF MARKET STRESS

With increasing trading volume and public discussion, algorithmic trading became a key topic for regulatory bodies. The preceding major regulatory changes, Regulation NMS as well as the Dodd-Frank Act in the United States and MiFID in the European Union, had addressed the reformation of the financial market system. After crises including the collapse of the investment bank Lehman Brothers and the 2010 Flash Crash, the regulators started probing and calling the overall automation of trading into question. Since then, the SEC as well as European federal regulators have promoted dialogue with practitioners and academics in order to evaluate key issues related to algorithmic trading (IOSCO 2011; SEC 2010a; European Commission 2011). Discussion is still intense, with supporters highlighting the beneficial effects for market quality and adversaries alert to the increasing degree of computer-based decision making and decreasing options for human intervention as trading speed increases further. In the following we focus on a specific event that promoted regulators on both sides of the Atlantic to re-evaluate the contribution of algorithmic trading, the Flash Crash, when a single improperly programmed algorithm led to a serious plunge. We then present mechanisms currently in place to manage and master such events.

10.5.1 Algorithmic Trading in the Context of the Flash Crash

On May 6, 2010, U.S. securities markets suffered one of the most devastating plunges in recent history. Within several minutes equity indices, exchange-traded funds, and futures contracts significantly declined (e.g., the Dow Jones Industrial Average dropped 5.49 percent in five minutes) only to rise to their original levels again. The CFTC together with the SEC investigated the problem and provided evidence in late 2010 that a single erroneous algorithm had initiated the crash. An automated sell program was implemented to slice a larger order of E-mini S&P 500 contracts, a stock market index futures contract traded on the Chicago Mercantile Exchange's Globex electronic trading platform, in to several smaller orders to minimize market impact. The algorithm's parameterization scheduled a fixed percentage-of-volume strategy without accounting for time duration or minimum execution price. This incautious implementation resulted in a significant dislocation of liquidity, resulting in a price drop of E-mini S&P 500 futures contracts. This selling volume cascade flushed the market, resulting in massive order book imbalances with subsequent price drops. Intermarket linkages transferred these order book imbalances across major broad-based U.S. equity indices such as the Dow Jones Industrial Index and the S&P 500 Index. Finally, the extreme price movements triggered a trading safeguard on the Chicago Mercantile Exchange that stopped trading for several minutes and allowed prices to stabilize (Commodity Futures Trading Commission 2010b). In order to get a more detailed

picture of the uniqueness of the Flash Crash, a closer look at the structure of the U.S. equity market and the NMS is necessary. The U.S. trade-through rule and a circuit breaker regime that were neither targeted at individual equities nor sufficiently aligned among U.S. trading venues are also relevant causes of the Flash Crash. In Europe, a more flexible best-execution regime without re-routing obligations and a share-by-share volatility safeguard regime that have existed for more than two decades have largely prevented comparable problems (Gomber et al. 2013).

10.5.2 Circuit Breakers in Securities Trading

Automated safeguard mechanisms are implemented in major exchanges in order to ensure safe, fair, and orderly trading. In 1988 the SEC implemented a marketwide circuit breaker in the aftermath of the crash of October 19, 1987 (Black Monday). Based on a three-level threshold, markets halt trading if the Dow Jones Industrial Average drops more than 10 percent within a predefined time period (NYSE 2011). In addition, many U.S. trading venues introduced further safeguard mechanisms that are also implemented at major European exchanges. So far, the academic literature provides mixed reviews regarding the efficiency of circuit breakers. Most of the studies conclude that circuit breakers are not helping decrease volatility (Kim and Yang 2004). Chen (1993) finds no support for the hypothesis that circuit breakers help the market calm down. Kim and Rhee (1997) and likewise Bildik and Gülay (2006) observed a spillover effect of the volatility to the near future after a trading halt was put in place. Nevertheless, the importance of such automated safeguards has risen in the eyes of regulators on both side of the Atlantic. On October 20, 2011, the European Commission published proposals concerning the review of the MiFID framework and now requires trading venues to be able to temporarily halt trading if there is any significant price movement on its own market or a related market during a short period (European Commission 2011).

10.6 OUTLOOK

The demand for automation was initially driven by the desire for cost reduction and the need to adapt to a rapidly changing market environment characterized by fragmentation of order flow. Algorithmic trading as well as HFT enable sophisticated buy side and sell side participants to achieve legitimate rewards on their investments in technology, infrastructure, and know-how. To draw a picture of the future evolution of algorithmic trading, it seems reasonable that even if the chase for speed is theoretically limited to the speed of light, the continuing alteration of the international securities markets as well as the omnipresent desire to cut costs may fuel the need for algorithmic innovations. This will allow algorithmic strategies to further claim significant shares of trading volume. Considering further possible shifts to the securities trading value chain,

algorithm-based automation may continue to adopt major processes that contribute significantly to the ongoing abolishment of human traders. Richard Balarkas, CEO of Instinet Europe, an institutional brokerage firm, draws a dark future for human intermediaries: "It [algorithmic trading] signaled the death of the dealer that just outsourced all risk and responsibility for the trade to the broker and heralded the arrival of the buy-side trader that could take full control of the trade and be a more discerning buyer of sell-side services" (Trade News 2009).

So far, the academic literature draws a largely positive picture of this evolution. Algorithmic trading contributes to market efficiency and liquidity, although the effects on market volatility are still opaque. Therefore, it is central to enable algorithmic trading and HFT to unfold their benefits in times of quiet trading and to have mechanisms (like circuit breakers) in place to control potential errors at both the level of the users of algorithms and at the market level. Yet preventing use of these strategies by inadequate regulation resulting in excessive burdens may result in unforeseen negative effects on market efficiency and quality.

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CHAPTER 11

COMPUTATIONAL SPATIOTEMPORAL MODELING OF SOUTHERN CALIFORNIA HOME PRICES

MAK KABOUDAN

11.1 INTRODUCTION

Forecasting of residential home prices is both important and challenging. More accurate forecasts help decision makers (be they potential home buyers or sellers, home builders or developers, or mortgage institutions) make informed decisions that may yield gains that are perhaps more commensurate with their expectations. Accurately forecasted price decreases may encourage potential home buyers to postpone their decision to benefit from lower future prices. Home builders and developers may postpone expansions until prices are expected to move back upward. Bankers may adjust their loan policies to protect themselves from foreclosure losses. Benefits from accurately forecasted price escalations are the opposite. Potential buyers may find it advantageous to buy sooner at times of correctly forecasted rising prices, home builders may adjust their prices to reflect those expectations, and banks experience higher security in loans they extend.

Producing accurate forecasts of home-price changes is neither easy nor constant. Home prices seem to change periodically for different reasons. The challenge emanates from the mere complexity of the housing market. Residential homes are not a homogenous product. Home prices are affected by many factors, some of which do not have measurably similar impacts on property valuations at different locations in different times. This is aggravated by the fact that even measurable factors that affect home-price changes are averages with dynamics dependent on the characteristics of homes sold periodically at a given location. Exogenously changing economic conditions

add more complexity to home-price dynamics. Changes in the local unemployment rates impact home prices with varying time lags that may depend on whether those changes belong to rising or falling unemployment rates. Interdependency between changes in current prices in one city or location may be impacted by prior changes in prices in a neighboring city or cities. The existence of some of these complexities suggests that the lag between the quarter during which a decision is made to purchase a house and the actual completion of a transaction in one or more future quarters is probably not constant. Further, there is a good chance that predominantly nonlinear mathematical relationships capture the real-world interactions between the dependent variable (price changes) and one or more of the independent variables that cause quarterly home prices to change.

This chapter focuses on computing response measures of home prices in a given location to lagged home-price changes in neighboring locations. The geographic region considered encompasses six contiguous southern California cities. The six cities (shown in figure 11.1) are Anaheim and Irvine in Orange County, Riverside and Corona in Riverside County, and Redlands and San Bernardino in San Bernardino County.

As the map suggests, a minimum of two or more cities are contiguous. Home-price changes are assumed to be spatiotemporally contagious in the sense that price changes occurring in one city may affect future price changes in one or more contiguous locations. If such spatiotemporal dynamics exist, then contagious past price changes in one location may help us better forecast future price changes in neighboring locations. In addition to approximating responses of home-price changes to quarterly changes in other locations, approximating home-price change responses to changes in economic variables (such as mortgage rate and local unemployment rate) is necessary. Home-price models must also account for the impact of different home characteristics (average house footage, average number of bedrooms, and so on). All response measures used in this study are obtained using genetic programming first. The results are then compared with those obtained using ordinary least squares.

The notion that home-price movements at different locations are contagious is not new. Dolde and Tirtiroglu (1997) examined patterns of temporal and spatial dependencies in real estate price changes using GARCH-M methods. Using a simple VAR (Vector Autoregressive) model, Lennart (2002) determined that real house price changes for seven residential Swedish regions (in Stockholm, Gothenburg, and Malmö) displayed a high degree of autocorrelation, with price changes in the Stockholm area having ripple effects on the six other areas. Holmes (2007) suggested that UK regional house prices exhibit long-term convergence. Employing quarterly data from 1978 to 2008, Riddell (2011) found that contagious price and income growth from Los Angeles contributed to the bubble that formed in Las Vegas. Canarella et al. (2012) investigated the existence of ripple effects in the price dynamics of the S&P/Case-Shiller Composite 10 index and determined that shocks to house prices exhibit trend reversion. Oikarinen (2012) studied the co-movement between price changes in Finland's housing market using 1988–2008 data. Gupta and Miller (2012) examined time-series relationships

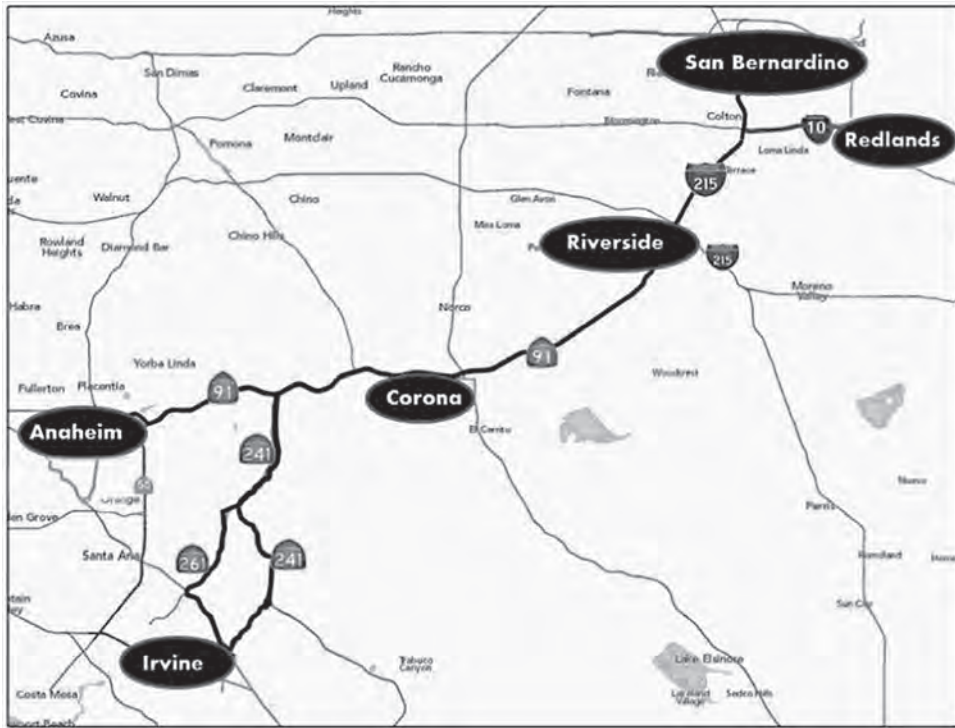


FIGURE 11.1 Spatial representation of the six southern California cities. Anaheim, Corona, and Irvine in the southwest are contiguous, Corona and Riverside are contiguous, and Riverside, Redlands, and San Bernardino to the northeast are also contiguous. (Map obtained using ArcGIS by ESRI.)

between housing prices in Los Angeles, Las Vegas, and Phoenix and forecasted 2006 prices using various VAR and vector error-correction models.

The possibility that contagious spatiotemporal price changes exist motivates specifying and estimating models that help explain quarterly residential home-price changes. If such a premise is established first, it may help deliver forecasts that may improve decision making. To establish such a premise, a basic response measure is proposed below. It is more like an “elasticity” measure that is designed to help determine whether price changes are spatiotemporally contagious. It is also helpful in estimating the temporal response of home-price changes to changes in basic economic variables that influence home-price dynamics (such as changes in local unemployment rate and the national mortgage rate) as well as the impacts of differences in home characteristics on price differentials. Because computation of the measure demands use of unique model specifications, the next section introduces a general model specification along with a brief review of the use of genetic programming (or GP) and statistical linear regressions. Section 11.3 introduces the proposed response measure. Estimation and forecasting

results of the computational spatiotemporal autoregressive model specifications follow before this chapter is concluded.

11.2 METHODOLOGY

Problems associated with use of conventional statistical or econometric analyses quickly escalate when the relation between the dependent variable and the independent variables is nonlinear or nonrecurring or when the analyses can properly represent the real-world circumstances using only a small number of observations. Modeling of why home-price changes is more likely to be better if home-price model specifications are assumed to be nonlinear as well as linear and will typically include a fairly large number of explanatory variables. The large number of variables is needed in the initial specification because the reasons home-price changes occur are governed by fluctuations in economic conditions and changes in taste due to technological changes (that affect quality of homes) or nostalgic effects, differences in home characteristics, and price changes in neighboring cities. Further, because it is important to detect whether price changes are contagious, several lagged price changes in neighboring locations must be included in each model's specification. When the number of predetermined or explanatory variables that may capture the dynamics of the dependent variable (average home-price changes) is fairly large, selecting the appropriate model becomes a daunting task. It is naturally aggravated by the underlying nonlinear or nonrecurring market dynamics. In short, residential home price changes do possess unique nonrecurring dynamics that are impacted by economic factors that do not seem to repeat themselves. This suggests that home-price changes may follow consistent dynamics for relatively short periods (two to three years) before existing conditions and the reasons for them change. Using quarterly data may help model price-change dynamics better, then. Although quarterly data provide what may be considered as a small number of observations in a period of time sufficiently short for conditions to be fairly consistent, it may yield inefficient results when traditional statistical methods are used. Therefore, the technique or method to use when modeling such dynamics must neither be affected nor restricted by the number of observations (or degrees of freedom) available when obtaining competing models.

Genetic Programming (or GP) is a computerized model evolution process capable of producing regression-like models without statistical restrictions on the number of observations or number of variables. It is utilized first to determine whether contagious price behavior exists among contiguous California cities. Then GP is used to forecast their quarterly average home-price changes, which can be used to forecast average quarterly price levels. A brief description of GP is given below after presentation of the basic model specification. Genetic programming is used first (before using linear regressions) because it may help identify (or suggest) the set of explanatory variables that would best explain the linear dynamics of residential home-price changes. When

given a large set of explanatory variables (more than ten), GP typically identifies a reasonably small number (three to five) that capture the dynamics of the dependent variable best. The resulting sets of explanatory variables may then be used as a starting point to conduct a search for competing linear regression models.

11.2.1 The Basic Model Specification

The first decision to make when analyzing home prices is the definition of the dependent variable. Given that linear regressions will be used at some point, the dependent variable's values should be stationary. This explains why the dependent variable used during modeling procedures is "home-price changes." Further, because average prices differ spatially by city (with San Bernardino home prices averaging in the low to mid \$100,000 between 2010 and 2012 and Irvine averaging between \$500,000 and \$600,000 in the same time period), computing the quarterly percentage change in home prices per city ultimately possesses two advantages: (a) the dependent variables' values are stationary and (b) a logically convenient comparison between the different locations becomes possible. Thus, the dependent variable for each location is

$$pL_t = 100 * (PL_t - PL_{t-1}) / PL_{t-1} \quad (11.1)$$

where P_t is the price level at time period t , L identifies the location, and $p = \% \Delta P =$ average percentage change in home-prices in given location L . Thus, $pAH_t = \% \Delta P_t$ in AH = the quarterly percentage changes in average prices of residential homes sold in Anaheim at time period t , for example. With six locations, there are six dependent variables for percentage quarterly home-price-changes. The data used covers the period from the first quarter of 2010 throughout the fourth quarter of 2013. After adjusting for lags, a total of sixteen observations are used to fit the models, utilizing eighteen explanatory variables in Anaheim (AH), which include six variables for house characteristics, three for mortgage rate, three for unemployment rate, four for lagged percentage price changes in Corona (CR) and Irvine (IV), and two lagged price changes for Anaheim itself, twenty in Corona (CR), eighteen in Irvine (IV), eighteen in Redlands (RL), twenty in Riverside (RS), and eighteen in San Bernardino (SB). The explanatory variables are given in table 11.1.

The explanatory variables listed in table 11.1 belong to four categories. The first has a set of house characteristics that may impact price differentials. This set includes SF , or average house size in square feet, BR , or average number of bedrooms per house sold during a given quarter, BA , or average number of bathrooms, AGE , or the average age of homes sold during a given quarter, LS , or the average lot size of homes sold at a location in a given quarter, and PL , or the probability that the average house in a city has a swimming pool. The second category has three quarterly mortgage rate lags: $MR_1 = MR_{t-1}$, $MR_2 = MR_{t-2}$, and $MR_3 = MR_{t-3}$. A minimum of three lags was selected because the price of a sold home is typically recorded at least three months

Table 11.1 Explanatory variables considered for model specifications

Location	House characteristics	Mortgage rate	Unemployment rate	Contagious location
AH	<i>SF, BR, BA, AGE, LS, PL</i>	<i>MR1, MR2, MR3</i>	<i>UR1, UR2, UR3</i>	<i>CR1, CR2, IV1, IV2</i>
CR	<i>SF, BR, BA, AGE, LS, PL</i>	<i>MR1, MR2, MR3</i>	<i>UR1, UR2, UR3</i>	<i>AH1, AH2, IV1, IV2, RS1, RS2</i>
IV	<i>SF, BR, BA, AGE, LS, PL</i>	<i>MR1, MR2, MR3</i>	<i>UR1, UR2, UR3</i>	<i>AH1, AH2, CR1, CR2</i>
RL	<i>SF, BR, BA, AGE, LS, PL</i>	<i>MR1, MR2, MR3</i>	<i>UR1, UR2, UR3</i>	<i>RS1, RS2, SB1, SB2</i>
RS	<i>SF, BR, BA, AGE, LS, PL</i>	<i>MR1, MR2, MR3</i>	<i>UR1, UR2, UR3</i>	<i>CR1, CR2, RL1, RL2, SB1, SB2</i>
SB	<i>SF, BR, BA, AGE, LS, PL</i>	<i>MR1, MR2, MR3</i>	<i>UR1, UR2, UR3</i>	<i>RL1, RL2, RS1, RS2</i>

after someone has actually decided to buy it. Mortgage rate values are identical for all locations. The same lag assumption was applied to unemployment rates (*UR*) but for each respective city. The fourth category includes lagged percentage price-change values for each *Lth* location to capture spatiotemporal interactions. The first three categories of the independent variables are representative of those typically included when modeling housing prices. Different sets of lagged percentage price changes representing the neighboring locations are included to determine whether contagious price changes do occur. Using the notation provided in table 11.1 and the map in figure 11.1, models representing the six cities are as follows:

$$pAH_t = f(SF_t, BR_t, BA_t, AGE_t, LS_t, PL_t, MR_{t-1}, MR_{t-2}, MR_{t-3}, UR_{t-1}, UR_{t-2}, UR_{t-3}, pAH_{t-1}, pAH_{t-2}, pCR_{t-1}, pCR_{t-2}, pIV_{t-1}, pIV_{t-2}) \quad (11.2)$$

$$pCR_t = f(SF_t, BR_t, BA_t, AGE_t, LS_t, PL_t, MR_{t-1}, MR_{t-2}, MR_{t-3}, UR_{t-1}, UR_{t-2}, UR_{t-3}, pCR_{t-1}, pCR_{t-2}, pAH_{t-1}, pAH_{t-2}, pIV_{t-1}, pIV_{t-2}, pRS_{t-1}, pRS_{t-2}) \quad (11.3)$$

$$pIV_t = f(SF_t, BR_t, BA_t, AGE_t, LS_t, PL_t, MR_{t-1}, MR_{t-2}, MR_{t-3}, UR_{t-1}, UR_{t-2}, UR_{t-3}, pIV_{t-1}, pIV_{t-2}, pAH_{t-1}, pAH_{t-2}, pCR_{t-1}, pCR_{t-2}) \quad (11.4)$$

$$pRL_t = f(SF_t, BR_t, BA_t, AGE_t, LS_t, PL_t, MR_{t-1}, MR_{t-2}, MR_{t-3}, UR_{t-1}, UR_{t-2}, UR_{t-3}, pRL_{t-1}, pRL_{t-2}, pRS_{t-1}, pRS_{t-2}, pSB_{t-1}, pSB_{t-2}) \quad (11.5)$$

$$pRS_t = f(SF_t, BR_t, BA_t, AGE_t, LS_t, PL_t, MR_{t-1}, MR_{t-2}, MR_{t-3}, UR_{t-1}, UR_{t-2}, UR_{t-3}, pRS_{t-1}, pRS_{t-2}, pCR_{t-1}, pCR_{t-2}, pRL_{t-1}, pRL_{t-2}, pSB_{t-1}, pSB_{t-2}) \quad (11.6)$$

$$pSB_t = f(SF_t, BR_t, BA_t, AGE_t, LS_t, PL_t, MR_{t-1}, MR_{t-2}, MR_{t-3}, UR_{t-1}, UR_{t-2}, UR_{t-3}, pSB_{t-1}, pSB_{t-2}, pRL_{t-1}, pRL_{t-2}, pRS_{t-1}, pRS_{t-2}) \quad (11.7)$$

There are four reasons that justify adopting the specification shown in equations 11.2–11.7:

1. Each equation includes the effects of typical factors that affect home prices such as the number of bedrooms and number of bathrooms.
2. Lagged mortgage rates and the local unemployment rates, which are exogenous economic variables impacting home-buying decisions, are included.
3. The specification includes lags of contiguous price changes in neighboring locations. For each location, this helps capture the effects of price changes in contiguous cities.
4. The lag structure in such a specification may help deliver a fairly accurate one-step-ahead *ex ante* forecasts of price changes in each location. These forecasts can then be used as input to forecast the quarter that follows. This helps us obtain four-quarter-ahead forecasts of price changes (and therefore price levels) for each city.

The first three categories in table 11.1 contain independent variables for which forecasts must be obtained for use as input when forecasting home-price changes one year ahead. For the first category, each variable's twelve-month moving average was computed for all available data then converted to quarterly averages to produce the four quarterly forecast values of each characteristic of each city. If those characteristics are found to impact home-price changes, their forecast values would be those of moving averages of the characteristics of prior sales. Both MR and UR were estimated using GP, assuming an autoregressive monthly specification with lags of three to twelve months. The forecast values were then used to obtain their quarterly averages.

11.2.2 Genetic Programming

Genetic programming is a computerized search technique designed to optimize a specified function (Koza 1992). It is used here to obtain nonlinear-regression-type models that minimize the fitness mean squared error (MSE). The GP computer code used is TSGP (Kaboudan 2004). It was designed such that it evolves models adopting “survival of the fittest” Darwin-like thought. The computer algorithm is capable of evolving model specifications useful in forecasting. Kaboudan (2001) established the statistical properties associated with estimation and forecasting using GP models. The code, TSGP, is written in C++ for Windows.

Users of TSGP need to provide two types of input files: data input files and a configuration file. Data values of the dependent variable and each of the explanatory variables must be provided in separate files. The configuration file contains execution information including the name given to the dependent variable, the number of observations to forecast, and other GP-specific parameters. The TSGP code produces two types of output files. One has a final model specification and the other contains actual and fitted values as well as performance statistics (R^2 and historic *MSE*).

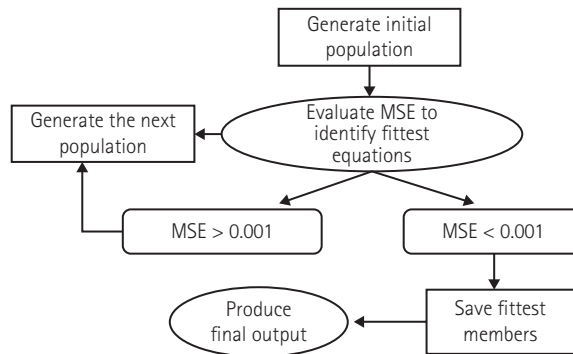


FIGURE 11.2 The basic GP architecture.

The TSGP code is programmed to assemble a user-defined fixed number of regression-like equations. Each equation is generated by randomly selecting from the given explanatory variables and a set of user-identified operators. The operators used typically include $+$, $-$, $\times$, $\div$, natural logarithm ($\ln$), exponential, sine, cosine, and square root. An equation is represented in the program by a parse tree. The tree consists of nodes and arcs. Each of the inner nodes takes one of the operators, while each of the end nodes (or terminals) takes one of the explanatory variables or a constant. Constants are obtained using a random number generator and are confined to values between -128 and $+127$. Because values of the explanatory variables or internal computations may be negative or zero, standard protections needed to compute mathematical operations during execution are programmed. The implemented protections designed to avoid computer-halting during the execution of TSGP are (a) if in X/Z , $Z = 0$, then $X/Z = 0$; (b) if in $X^{1/2}$, $X < 0$, then $X^{1/2} = -|X|^{1/2}$; (c) if in $\ln(X)$, $X < 0$, then $\ln(X) = -\ln(|X|)$.

Because evolving a single equation is random, it is highly unlikely that the program delivers a best-fit model every time it evolves an equation, and it is necessary to produce a large number of equations (typically one hundred of them) in a single run. Of the one hundred equations evolved, the best-fit equation is assumed to be that output that contains explanatory variables with logically expected signs and a strong predictive power. That best evolved model is then used to produce the final forecast. (Occasionally, however, GP software delivers unpredictable forecasts even when the computed statistics of the fitted values seem great. Thus careful evaluation of the logic of each selected equation forecast is critical before it is accepted.) Figure 11.2 shows how GP equations are evolved.

Genetic programming performs two tasks in this study: First it is used to select the variables that best explain spatiotemporal residential home-price percentage changes from each set of right-hand-side or predetermined, variables. Then GP is used to deliver the four-quarter-ahead forecasts. It takes all assumed explanatory variables as inputs and starts by randomly assembling a large population of equations according to a user-specified population size. (In this study, the population size of the equations

GP assembles was set to one thousand.) The genetic program then sorts the outcomes according to a fitness criterion (such as mean square error, mean absolute percentage error, or mean absolute deviation) the user selects. The mean square error of the fitted values was selected in this case. The equations with the ten lowest *MSEs* are stored in memory. The genetic program is designed to favor randomly selected parts of those ten equations when producing a new set of offspring (or equations). Thus, GP randomly generates another population of one thousand equations and compares their fitnesses (or the *MSE* of each) with the *MSE* values stored in memory. The equations with the lowest ten *MSEs* replace the ten equations already stored in memory. This process repeats for a specified number of times known as generations (this study contains one hundred). From each search routine, GP finally reports the best equation and its fitted values. Because selecting the variables that produce the best fitted values is random, this process must be repeated a large number of times. For this study, and for each dependent variable, GP is set to complete one hundred search routines and deliver one hundred independent equations and fitted values. The TSGP code then produces a summary of the 100 statistics to help identify the best models. The user then evaluates the best solutions, discards illogical results, and reports the forecasts delivered by the best fit equation found.

11.2.3 Linear Regressions

Ordinary least squares is used to estimate the six individual equations for finding percentage price change. Equations are specified and estimated using hints that GP provided. Through trial and error involving adding and deleting explanatory variables, more specifications are tested until the statistically and logically best explanatory variables (ones with the lowest *p*-values and correct signs) are found. For each location, then, the most logically defensible and statistically acceptable equation is selected and used for forecasting.

11.3 A SPATIOTEMPORAL CONTAGION RESPONSE MEASURE

The idea of measuring spatiotemporal contagion effects is not new. Its meaning differs from one discipline or application to another. It was used in quantifying spatial patterns of landscapes and in financial markets, for example. Regarding the former, a contagion index that quantifies spatial patterns of landscapes by measuring the degree of clumping of attributes on raster maps was first proposed by O'Neill et al. (1998). Their work was followed by Turner and Ruscher (1998), but follows earlier work by Turner (1990) and by Li and Reynolds (1993). O'Neill et al. suggested a "landscape" contagion

index computed from the frequencies by which different pairs of attributes occur as adjacent pixels on a map. The measure is useful only when dealing with adjacent cells and is not useful when investigating contiguous cells that are not exactly adjacent, as may be the case here. In attempting to quantify contagions in financial markets, Bae et al. (2003) proposed a new approach to the study of financial contagion. Their approach was influenced by Hosmer (1998), who used multinomial logistic analysis in epidemiological research. The key presumption of their approach is that a contagion phenomenon is associated with extreme returns, where small-return shocks propagate differently than do large-return shocks. Kristin et al. (2002) as well as Pesaran and Pick (2007) addressed the problems associated with measuring contagion effects in financial markets.

The spatiotemporal contagion response measure proposed here is new. Its basic definition is as follows:

The spatiotemporal contagion response coefficient (SCR) measures the average percentage changes in home prices in one location if lagged percentage price changes in a different contiguous location changed by 1%.

It may be formally defined as follows:

$$SCR = \frac{\% \Delta \text{ in current average home prices in location A}}{1\% \Delta \text{ in average lagged home prices in location B}} \quad (11.8)$$

A and B in (11.8) are two contiguous locations and $\Delta =$ change. Since the percentage change in lagged home prices in location B = 1, equation (11.8) may be written as

$$SCR = T^{-1} \sum [\{pA_t | (pB_{t-\tau} \pm 1\%) \} - \{pA_t | pB_{t-\tau} \}]/\pm 1. \quad (11.9)$$

The SCR is close to being an elasticity coefficient except that typically “elasticity” is a measure associated with computing changes in quantity traded in response to a change in price, income, or any other variable. The SCR measures the percentage change in one price relative to a 1 percent lagged change in the average price during an earlier time period in a different location. However, like any elasticity measure, the SCR may be positive or negative regardless of whether pB_t is increased or decreased. A computed $SCR \geq 1$ if the current home-price changes in location A for which the SCR is computed were contagious to lagged home-price changes in a neighboring location B. The $SCR > 0$ if current price changes in location A move in the same direction as prior price changes in location B. Estimates of the SCR using equations (11.2) to (11.7), then, leads to one of the four mutually exclusive and collectively exhaustive interpretations. They depend on whether the computed SCR is negative or positive. If the $SCR < 0$ then contagious price effects did not exist. If the $SCR > 0$, then contagious price effects exist. Their strength may then be arbitrarily (and, one hopes, logically) set. Table 11.2 presents a possible interpretation of the values that an SCR may take. If the dependent variable measures percentage price changes, estimates of the SCR involve additional computations when using GP. Additional computations are essential because it is rare to reproduce the

Table 11.2 Possible interpretations of an SCR

Possible Outcomes	Interpretation	Interpretation strength
$SCR \leq 0$	Contagious price effects are nonexistent	"strong"
$0 < SCR < 0.5$	Weak contagious lagged price effects may exist	"weak"
$0.5 \leq SCR < 1$	Likely contagious lagged price effects may exist	"likely"
$SCR \geq 1$	Strong contagious lagged price effects exist	"strong"

identical equation if runs are repeated. Further, the GP model solutions can be used to distinguish between impacts of increases in the values of an explanatory variable and impacts of decreases in the values of that same variable. To compute the SCR, it is necessary to compute solutions of the selected GP model twice to capture the upward and downward change effects of an explanatory variable. When dealing with actual data, this is done by producing and including as part of the input data two sets of augmented values for each of the explanatory variables. First, a set is increased by 1 percent; it is then decreased by 1 percent to produce the second set. The following example demonstrates how to compute the SCR using a hypothetical but typical GP output. Assume that GP's best delivered model is as follows:

$$pA_t = pC_{t-2} + (pF_{t-2})^{\frac{1}{2}} - pD_{t-4} + pF_{t-3} - \cos(pC_{t-4} + (2 * pC_{t-3} + \ln(pD_{t-4}))^{\frac{1}{2}}). \quad (11.10)$$

Equation (11.10) suggests that current percentage price changes in location A are affected by prior or lagged percentage price changes in locations C, D, and F. It is therefore possible to compute the effect of a 1 percent price change in any of the locations (C, D, or F) on the basis of the current percentage price changes in location A. This is possible since solving equation (11.10) is straightforward if the values of the right-hand-side variables are known. Accordingly, to compute the effect of a 1 percent change in prior prices in location C on current prices in location A, the measure is computed twice. First, historical percentage price changes in C are increased by 1 percent to capture the effect of such an increase on the average price changes in A. Historical percentage price changes in C are then decreased by 1 percent to capture the effect of the decrease on the average price changes in A. This distinction helps determine whether prices in two locations are contagious only when lagged prices moved upward, downward, in both directions, or in neither direction. The generation of augmented data must be repeated for the other locations as well. Thus, rather than employing sixteen observations to fit the GP model and four to forecast, and given that each location's price-change values will be augmented upward and downwards, the number of observations representing each explanatory variable would be sixty the sixteen needed to produce the model plus the four quarters to forecast, the sixteen needed to produce the model plus the four quarters to forecast when each variable is

Table 11.3 Example of computing the *SCR*

	C-U			C-D		
	<i>pA</i>	Fitted-A	Fitted-A	Difference	Fitted-A	Difference
1	0.418	0.516	1.083	0.567	−0.001	0.517
2	−0.300	−0.373	−0.037	0.336	−0.007	−0.366
3	−0.141	−0.293	0.349	0.642	−0.009	−0.283
4	−0.950	−0.256	0.176	0.432	−0.007	−0.249
5	0.040	0.777	1.343	0.566	0.002	0.774
6	0.058	0.219	0.547	0.328	−0.001	0.220
7	0.502	0.519	1.269	0.750	−0.002	0.522
8	−0.129	−0.355	0.007	0.363	−0.007	−0.348
9	1.254	1.558	2.303	0.745	0.008	1.550
10	−0.709	−0.593	0.441	1.033	−0.016	−0.577
11	3.342	3.107	4.050	0.943	0.022	3.085
12	0.940	0.929	1.700	0.771	0.002	0.927
13	1.207	1.250	2.124	0.873	0.004	1.246
14	2.690	2.568	3.419	0.851	0.017	2.551
15	1.449	1.200	1.997	0.798	0.004	1.196
16	0.535	0.818	1.577	0.759	0.001	0.817
Average =				0.672		0.724

“increased” by 1 percent, and sixteen plus four when each is “decreased” by 1 percent). To capture the complete picture, 180 augmented observations (the sum of 60 for location C, 60 for location D, and 60 for location F) are needed.

Once obtained, the GP output can then be used to compute each *SCR*. This process starts by simply taking the average difference between the original solution (that is, the one obtained before any values were augmented) and the solution capturing the impact of each augmented set of (a) sixteen fitted observations that capture the impact of an *increase* in the explanatory variable’s values, and of (b) sixteen fitted observations capturing the impact of a *decrease* in the same variable’s values. Table 11.3 presents a demonstration of how a response measure is computed.

In table 11.3, the *pA* column contains the historical values of average percentage price changes in location A. The column labeled “Fitted-A” shows GP’s solution values before any explanatory variable’s values are augmented. The column labeled “C-U Fitted-A” contains the fitted values of A after increasing price changes in location C by 1 percent. The differences between values under “Fitted-A” and values under “C-U Fitted-A” are the resulting impacts of increasing prices in C by 1 percent. The bottom of the “Differences” column contains the average response or *SCR* when prices in C increase by 1 percent. Similarly, the last column in the table shows the differences between values under “Fitted-A” and values under “C-D Fitted-A,” capturing the impacts from

decreasing prices in C by 1 percent on A's price changes. In this example, the impacts of increases in prices at location C are marginally different from the impacts of their decreases in the same period. The $SCR = 0.672$ when prices in C are rising and 0.724 when prices in C are declining, and because both are above 0.5 and less than 1, the impacts of price changes in C on A are "likely" (according to table 11.2).

Estimates of response measures are straightforward using OLS. The estimated coefficients of the lagged percentage cross-price changes in the OLS equations are spatiotemporal contagion response coefficients. Here is why. Assume that the following equation was estimated:

$$pA_t = a + bMR_{t-2} + c pA_{t-1} + d pC_{t-2} + e pC_{t-3} + f pF_{t-3}, \quad (11.11)$$

where a , b , c , d , e , and f are the OLS estimated coefficients, pA_t = percentage change in home prices at location A in time period t , pC_{t-2} = percentage change in home prices at location C in time period $t - 2$, and so on. This equation produces three spatiotemporal contagion response measure estimates:

1. $d = \Delta pA_t / \Delta pC_{t-2}$ = the average percentage change in prices in location A at time period t that occurs if prices changed by 1 percent in location C two quarters earlier.
2. $e = \Delta pA_t / \Delta pC_{t-3}$ = the average percentage change in prices in location A at time period t that occurs if prices changed by 1 percent in location C three quarters earlier.
3. $f = \Delta pA_t / \Delta pF_{t-3}$ = the average percentage change in prices in location A at time period t that occurs if prices changed by 1 percent in location F three quarters earlier.

Although reasonable they are to estimate and may be somewhat useful, there are three problems with using OLS response measure estimates: First, the estimated coefficients (d , e , and f in the example above) are constants. Second, if pC_{t-2} changes, then pC_{t-3} must also change, albeit with lag, since this is the same variable. This means that the *ceteris paribus* assumption is violated. Under such conditions, the two coefficients— d and e in equation (11.11) above—are added in order to compute the correct response measure. The third problem is that the functional form is linear, which may be far from being realistic.

11.4 ESTIMATION RESULTS

This section's aim is mainly to evaluate the estimation results to determine whether prices are in fact spatially and temporally contagious. The data for the variables listed in table 11.1 were obtained from three sources. Home prices and characteristics are available on the internet from the Chicago Title Company (2014). National (U.S.)

mortgage rates are from the Federal Housing Finance Agency (2014). Unemployment rates were obtained from the California Employment Development Department (2014).

Outputs that GP models produce are now evaluated to determine the forecasting abilities of the evolved models. Because future percentage price changes at the different locations are unknown a priori, forecasts for each of the four quarters are sequentially simulated one quarter at a time using the best models. Genetic programming is run using data from 2010 through the end of 2013 to produce one-step-ahead forecasts. The best GP models then deliver the first-quarter 2014 forecasts. The *ex ante* forecast of the first quarter from this first run is then used as input to produce an updated model that would forecast the second quarter in 2014, and so on.

11.4.1 GP Estimation

Unlike linear regression models, GP produces rather lengthy complex equations that do not lend themselves to the typical interpretations of regression estimation results. However, those equations are quite helpful in determining the explanatory variables that may approximate a possible linear relationship that may explain variations in the values of dependent variables. Given the complexity of the GP equations evolved, there is no real benefit from including all the GP-selected equations here. Using those equations, it was possible to compute each city's response measure coefficients. All the computed measures using the evolved GP equations deemed best are shown in table 11.4.

The first column in table 11.4 is a list identifying the explanatory variables. Each variable is represented by a pair, one describing the effect of a 1 percent increase in the values of the respective variable the other describing the effect of a 1 percent decrease in the values of that same variable. The stands for "upwards" to signify that the information in that particular row belongs to a scenario in which that explanatory variable's values were increased by 1 percent. The stands for "downwards" to signify that the information in that row belongs to a scenario in which that explanatory variable's values were decreased by 1 percent. The first row in the table lists the dependent variables (or quarterly average percentage price changes per location, i.e., p_{AH} , p_{CR} , etc.). Values in the second row (as well as rows that follow) of the table may be interpreted as follows: Average percentage changes in home prices are responsive to changes in square footage in five of the six cities. Only Corona's percentage changes in home prices are not responsive to changes in the average house square footage. In Anaheim, if the square footage of the average house sold increases by 1 percent, the percentage change in home price is expected to be 1.276 percent higher on average. The percentage change in price would be only 0.947 percent less if its square footage decreased by 1 percent. The information extracted from table 11.4 may be split into three sections summarized as follows.

Table 11.4 Computed response measure values using GP

	p_{AH}	p_{CR}	p_{IV}	p_{RL}	p_{RS}	p_{SB}
SF-U	1.276		1.031	1.159	0.505	1.440
SF-D	0.974		0.583	1.206	0.566	1.468
BR-U			0.085	0.540		
BR-D			0.166	0.540		
BA-U				0.140		
BA-D				0.140		
AGE-U		0.278			−0.452	
AGE-D		0.171			−0.261	
LS-U			0.021			0.124
LS-D			0.020			0.124
PL-U				0.031		0.031
PL-D				0.031		0.031
MR-U	−0.059	−0.161	−0.617		−0.169	−0.609
MR-D	−0.043	−0.283	−0.312		0.000	−0.034
UR-U	−0.080	−0.037	−0.012	−0.083	−0.011	−0.127
UR-D	−0.074	−0.042	−0.030	−0.027	−0.011	−0.060
pAH-U		0.710	0.216			
pAH-D		0.324	0.060			
pCR-U	0.553		0.254		0.996	
pCR-D	0.658		0.200		0.991	
pIV-U		0.162				
pIV-D		0.120				
pRL-U						0.000
pRL-D						0.297
pSB-U				0.538	0.195	
pSB-D				0.394	0.270	

1. Incidental response measures:

- House square footage seems to be the most influential variable affecting percentage home-price changes in five of the six cities. Generally, percentage home-price changes seem to be more responsive to larger square footage than they are to smaller square footage.
- Percentage home-price changes are likely to respond to the number of bedrooms in Redlands and weakly respond to them in Irvine.
- Percentage home-price change response to the changes in the number of bathrooms does not exist except to a minor extent in Redlands.

- d. Percentage home-price changes seem to be weakly responsive to the average home age in only two locations, Corona and Riverside. Older homes in Corona are apt to have higher percentage price-change responses, while they are apt to have lower percentage price-change responses, in Riverside.
 - e. Percentage home-price change responsiveness to homes on larger lot sizes or with pools seems to be very weak or nonexistent.
2. Economic response measures:
- a. Percentage home-price changes are responsive to changes in mortgage rate in all locations except for Redlands. Buyers are most responsive to increases in mortgage rates in Irvine and San Bernardino and generally far less responsive to decreases in MR.
 - b. Percentage home-price change responses to changes in the unemployment rate are evident in all locations but are very marginal.
3. Spatiotemporal contagion response measures:
- a. Anaheim's percentage home-price changes are contagious to Irvine's percentage home-price increases but are less contagious to Irvine's percentage home-price decreases. They marginally affect percentage home-price changes in Corona.
 - b. Corona's percentage home-price changes are most contagious. They affect three other cities (Anaheim, Irvine, and Riverside). Riverside's percentage price changes are most impacted. They are fairly responsive to increases as well as decreases in Corona's percentage home-price changes. They are also contagious to percentage home-price changes in Anaheim. They marginally impact percentage price changes in Irvine.
 - c. Irvine's percentage home-price changes marginally impact those of Corona. The percentage home-price for Redlands decreases weakly affect San Bernardino's decreases and have no effect on San Bernardino's percentage price changes when rising.
 - d. Riverside's percentage home-price changes are not contagious to changes in any location.
 - e. San Bernardino's percentage price changes marginally impact those of Redlands and Riverside. The impact is most pronounced on Redlands's percentage changes when rising.

Directional arrows in figure 11.3 show the direction of the contagious effects.

11.4.2 OLS Estimation

Using hints provided by the GP outputs, the estimated OLS models resulting from a search that delivers the best statistics yielded the following results.

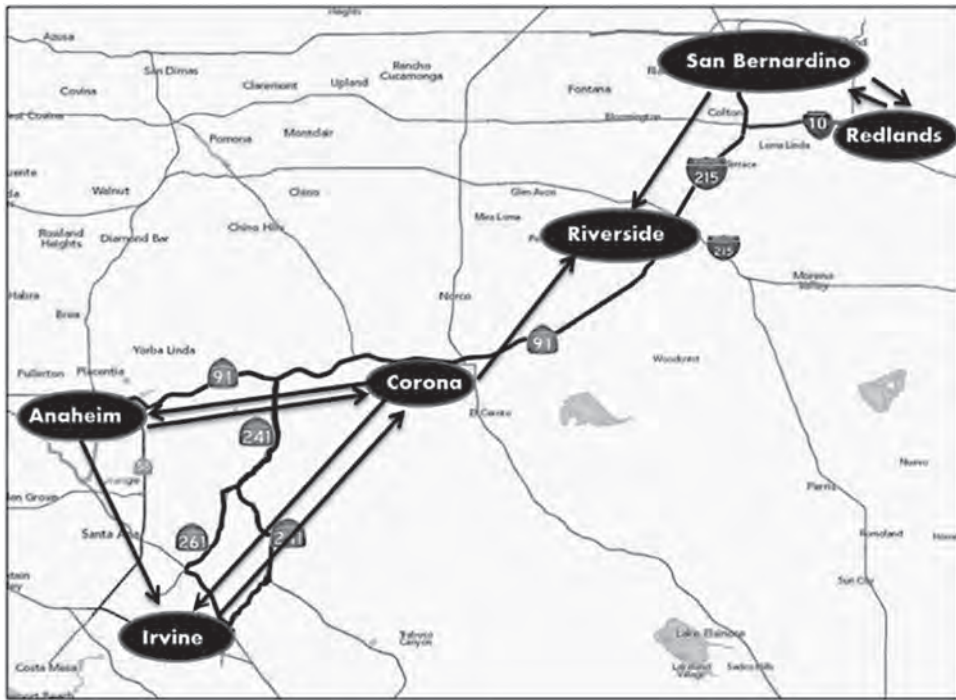


FIGURE 11.3 Map of southern California price-change contagion.

- $$pAH_t = 0.82 + 1.060 SF_t - 0.667 UR_{t-3}$$

(0.01)
(0.00)
(0.02)

$$R^2 = 0.75; MSE = 0.10; F\text{-significance} = 0.00; DW = 2.40.$$
- $$pCR_t = 0.049 + 0.493 UR_{t-1} + 0.278 pAH_{t-1}$$

(0.04)
(0.00)
(0.10)

$$R^2 = 0.64; MSE = 0.01; F\text{-significance} = 0.01; DW = 2.34.$$
- $$pIV_t = 1.012 SF_t$$

(0.04)

$$R^2 = 0.52; MSE = 0.05; F\text{-significance} = 0.01; DW = 1.75.$$
- $$pRL_t = 1.139 SF_t$$

(0.0)

$$R^2 = 0.88; MSE = 0.12; F\text{-significance} = 0.00; DW = 1.51.$$
- $$pRS_t = 0.069 + 0.869 SF_t - 0.514 UR_{t-1} + 0.274 pSB_{t-2}$$

(0.03)
(0.00)
(0.03)
(0.06)

$$R^2 = 0.91; MSE = 0.01; F\text{-significance} = 0.00; DW = 2.64.$$
- $$pSB_t = 1.183 SF_t + 0.518 pRS_{t-2}$$

(0.00)
(0.15)

$$R^2 = 0.90; MSE = 0.10; F\text{-significance} = 0.01; DW = 1.67.$$

The response measures obtained directly from the estimated equations are shown in table 11.5. The information extracted from table the may be summarized as follows.

Table 11.5 Computed response measure values using OLS

	pAH	pCR	pIV	pRL	pRS	pSB
SF	1.060		1.012	1.139	0.869	1.183
$UR_t - 1$		-0.493			-0.514	
$UR_t - 3$	-0.667					
$pAH_t - 1$		0.278				
$pRS_t - 2$						0.518
$pSB_t - 2$					0.274	

1. Incidental response measures:

House square footage seems to be the most influential variable affecting percentage home-price changes in five of the six cities. These results are similar to those obtained using GP. Generally, these changes seem to be more responsive to homes with larger square footage than they are to those with smaller square footage.

2. Economic response measures:

Percentage home-price changes responses to change in the unemployment rate are evident in three locations (Anaheim, Corona, and Riverside). Responses to changes in the unemployment rate are significantly larger than those reported using GP.

3. Spatiotemporal contagion response measures:

- Percentage home-price changes for Anaheim are marginally contagious to Corona's.
- Riverside's percentage home-price changes, lagged two quarters, are somewhat contagious to San Bernardino's current percentage home-price changes, while San Bernardino's percentage home-price changes are marginally contagious to Riverside's with the same lag.

11.4.3 Comparison of Estimation and Forecast Results

The GP and OLS estimation statistics are summarized in table 11.6. The comparison suggests that GP outperforms OLS (at least from a statistical point of view) in representing the six locations. As the table shows, the estimation root mean square errors (*RMSE*) are lower and the R^2 statistics are higher for all six locations when using GP. Although the results appear encouraging, fitting the historical values produced mixed

Table 11.6 Comparison of GP and OLS estimation results

	p_{AH}	p_{CR}	p_{IV}	p_{RL}	p_{RS}	p_{SB}
GP <i>RMSE</i>	0.52	0.32	0.76	1.53	0.62	0.79
OLS <i>RMSE</i>	0.84	0.79	2.22	3.40	0.94	3.11
GP R^2	0.89	0.93	0.77	0.98	0.96	0.93
OLS R^2	0.75	0.64	0.52	0.88	0.92	0.90

Table 11.7 Forecast consistency statistics

	p_{AH}	p_{CR}	p_{IV}	p_{RL}	p_{RS}	p_{SB}
<i>RMSD</i>	3.60	3.98	1.49	2.01	2.84	2.89
<i>Std. Dev.</i>	9.99	25.83	3.89	6.20	6.28	6.89

results. The root means square errors presented in table 11.6 reveal the weaknesses. Although GP outperformed OLS throughout, the GP as well as the OLS *RMSE* are weakest (i.e., highest) for Redlands and San Bernardino. Figure 11.4 shows plots of the fitted values as well as the four-quarter-ahead forecasts obtained using the GP and OLS results for each of the six cities.

Figure 11.4 suggests that the GP and OLS forecasts are similar in some of the locations. To determine how similar the two forecasts belonging to each location are, a consistency statistic is proposed and then computed here. The consistency statistic is

$$RMSD_L = \sqrt{4^{-1} \sum (p_{GPL_f} - p_{OLSL_f})^2} \quad (11.12)$$

where *RMSD* is the root mean square deviation between each pair of forecasts of location *L*, p_{GP} represents the forecasts GP produced over $f = 1, \dots, 4$, the four-quarter forecast period, and p_{OLS} represents the forecasts OLS produced over the same period. Computations of the *RMSD* and the standard deviations (*Std. Dev.*) are given in table 11.7. The forecasts are most similar for Irvine and Redlands. They have the lowest standard deviations. The forecasts for Corona and Anaheim are the most dissimilar. This is no surprise given that the estimation *RMSE* for Irvine and Redlands were the worst whereas they were the best for Corona and Anaheim.

Interesting new information is revealed when the price-change forecasts are converted into price-level forecasts. The forecasts are presented in Table 11.8. Given the GP estimation statistics reported above, both forecasts suggest that home prices are expected to be consistently lowest during the same quarter. Such information is helpful

Table 11.8 Comparison of GP and OLS forecast results

	AH		CR		IV		RL		RS		SB	
	GP	OLS	GP	OLS	GP	OLS	GP	OLS	GP	OLS	GP	OLS
Mar-14	461.5	472.9	397.9	402.2	740.0	746.6	366.7	370.0	298.4	299.3	166.3	165.9
Jun-14	457.9	469.2	424.6	429.2	737.0	743.6	367.4	370.7	308.4	309.3	173.0	172.7
Sep-14	484.5	496.5	425.1	429.7	733.4	739.9	379.8	383.2	305.9	306.8	174.2	173.8
Dec-14	485.8	497.8	428.2	432.8	749.0	755.8	382.3	385.8	316.4	317.3	169.2	168.9

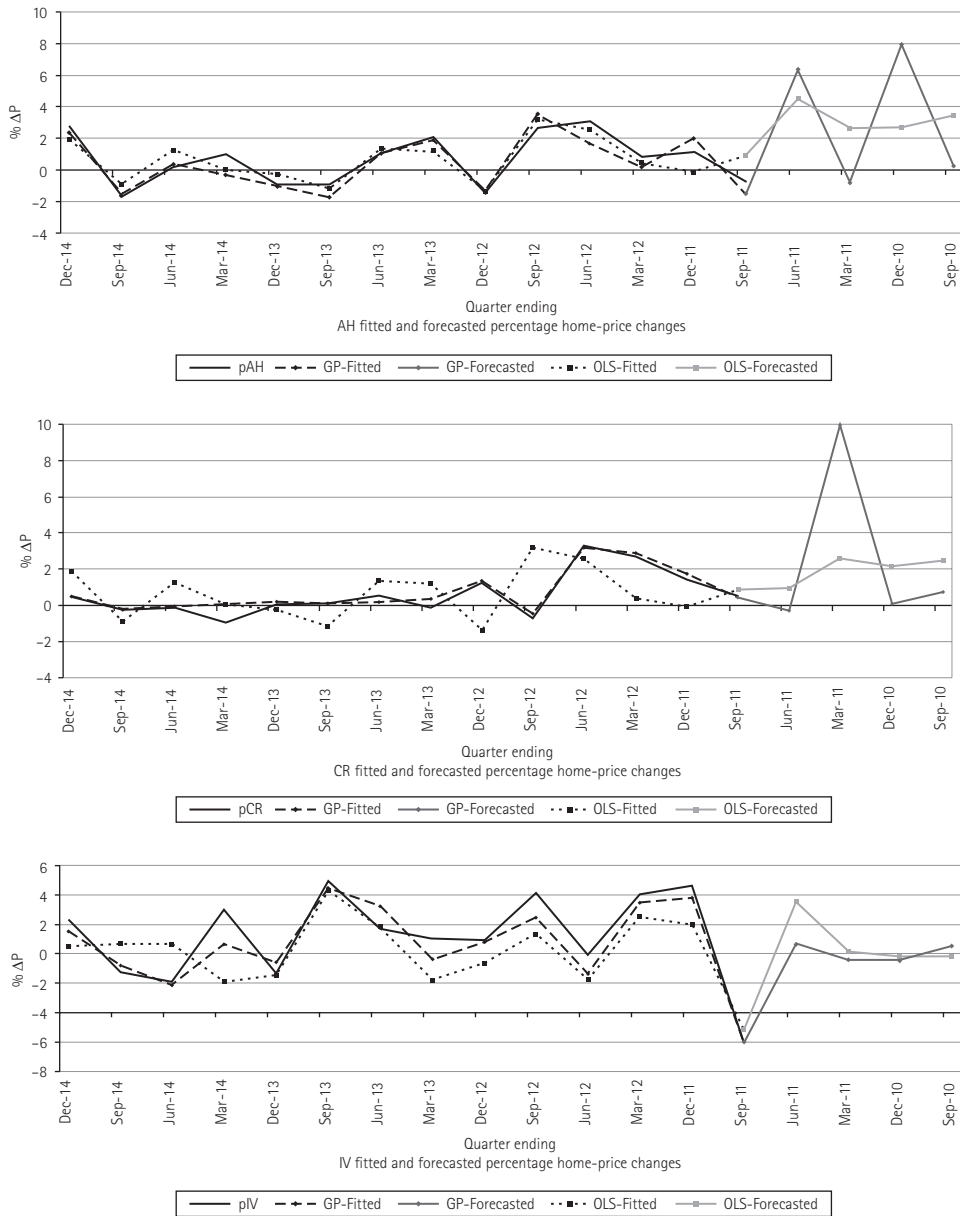


FIGURE 11.4 Percentage price-change comparisons of the fitted and forecasted values for the six cities.

to those who are contemplating purchasing a house in any of the locations. The best time to purchase in all locations except Irvine and Riverside is the second quarter in 2014. The decision-making prices are in boldface and underlined in the table. Figure 11.5 shows the fitted and forecasted price levels for all six locations.

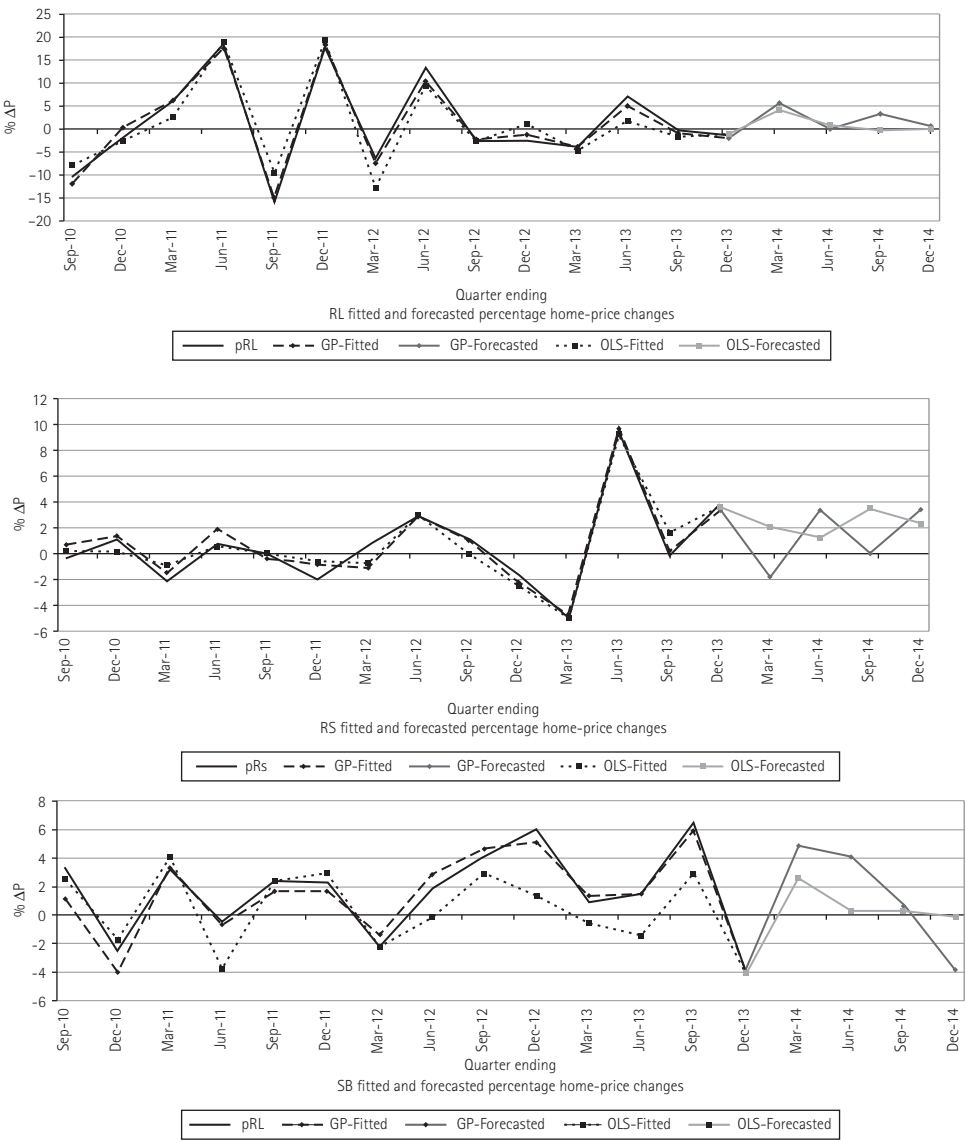


FIGURE 11.4 Continued

Table 11.9 presents a cross-sectional forecast evaluation for the first quarter of 2014. The first column in the table identifies the location. The actual and forecasted prices follow. The abbreviation *MFPE* stands for “mean forecast percentage error.” The limited computation reveals some interesting observations. First, GP and OLS forecasts are rather similar, with GP having a slight edge over OLS forecasts. The latter were slightly better for Corona and Riverside. Genetic programming, GP had the edge for the other

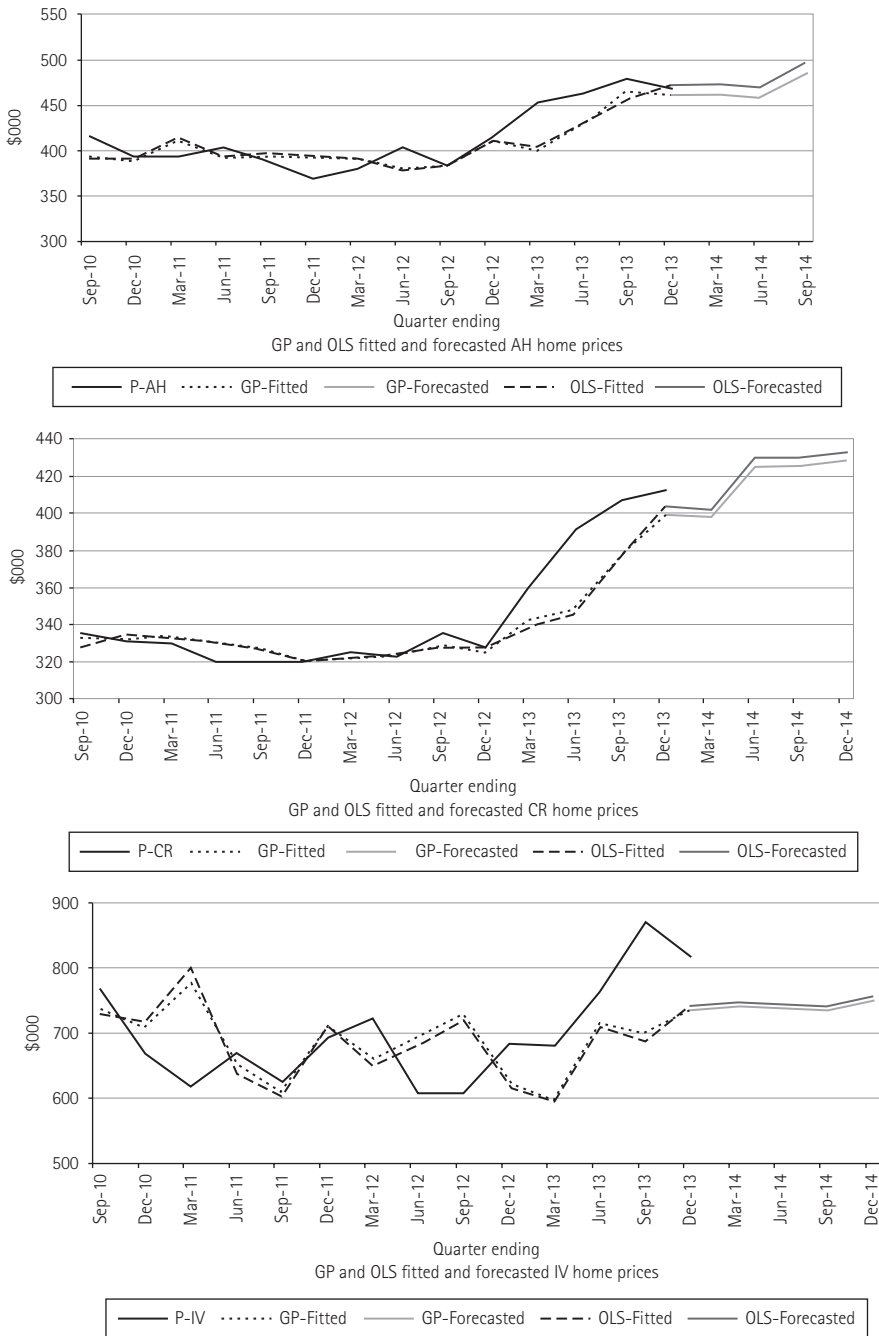


FIGURE 11.5 Fitted and forecasted home prices for the six cities.

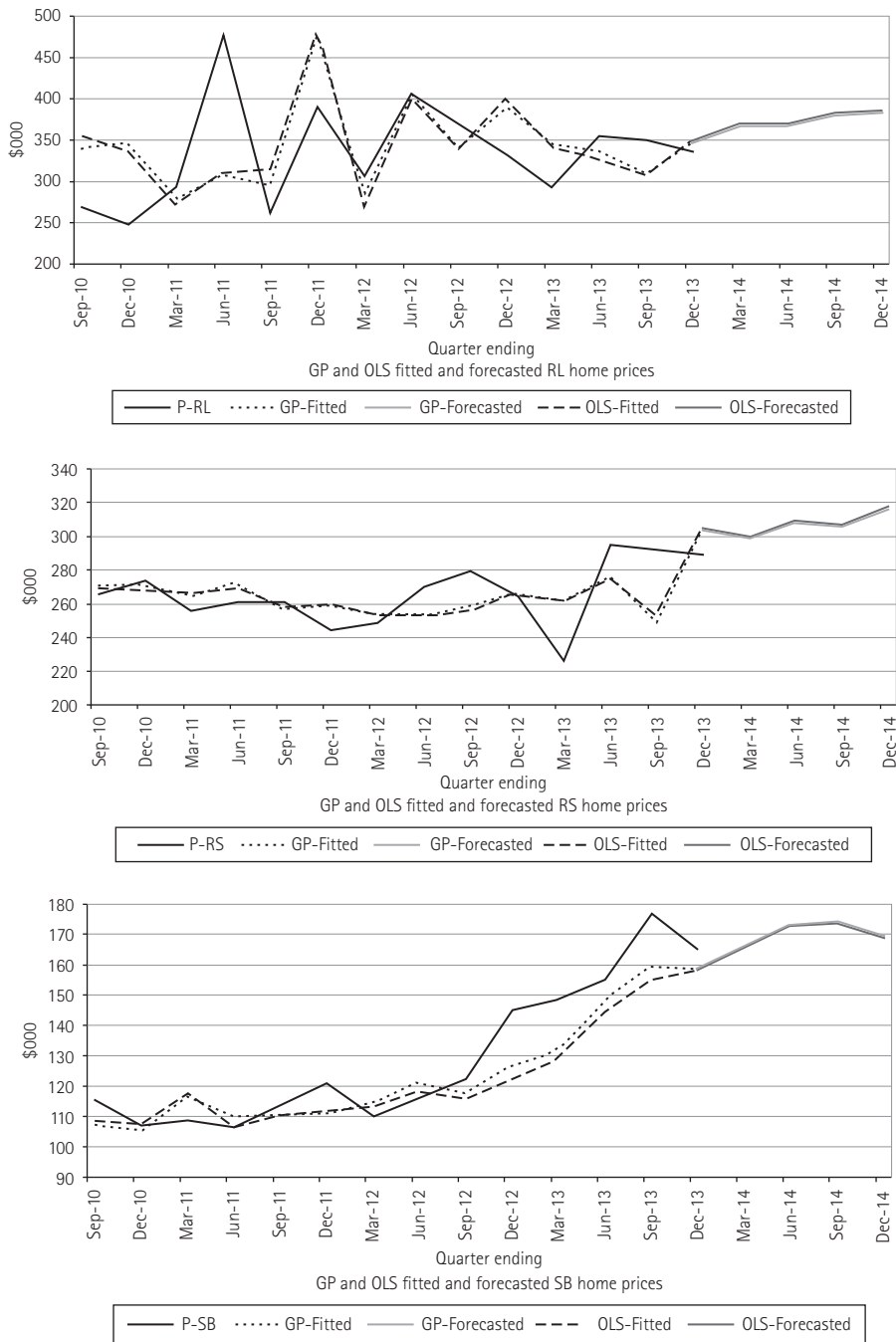


FIGURE 11.5 Continued

Table 11.9 Forecast errors for the first quarter of 2014

	P in thousands of \$			<i>MFPE</i>	
	Actual	GP	OLS	GP	OLS
AH	466.5	461.5	472.9	1.08%	−1.37%
CR	427.0	397.9	402.2	6.80%	5.80%
IV	732.2	740.0	746.6	−1.06%	−1.97%
RL	318.1	366.7	370.0	−15.27%	−16.31%
RS	312.1	298.4	299.3	4.38%	4.10%
SB	176.8	166.3	165.9	5.93%	6.15%

four locations. Generally, Anaheim and Irvine forecasts had the lowest percentage errors, and Redlands had the worst.

11.5 CONCLUDING REMARKS

Spatiotemporal models obtained using genetic programming and linear regressions were used to determine whether percentage home-price changes are contagious among neighboring contiguous cities, then used to forecast average quarterly residential home prices in those cities for one year. In order to determine whether percentage price changes are contagious between neighboring cities, a spatiotemporal contagion response measure was developed. By design, the measure is a computed coefficient that captures the average effect of a 1 percent average price change in a neighboring location on the percentage price-change in a different location. A computed spatiotemporal contagion response measure that is positive suggests contagious effects. If the computed measure is at least 1, strong effects are present. A low but positive coefficient estimate (perhaps less than 0.5) suggests the presence of weak to no contagious effects between the two locations. A negative estimate negates contagious effects.

Six cities from three neighboring counties were included in this study. The explanatory variables considered when specifying models to capture the historical dynamics of average quarterly percentage changes of home prices in a city belonged in one of three groups. One group accounts for differences in the average house characteristics within a given city. Another group accounts for the lagged average changes in economic variables (more specifically mortgage rate and local quarterly average unemployment rate). The third accounts for average quarterly home-price changes in neighboring cities. The results obtained suggested that Anaheim's lagged quarterly average percentage home-price changes are contagious to current percentage home-price changes in Irvine and Corona. Corona's lagged quarterly average percentage price changes are contagious

to current Anaheim's, Irvine's, and Riverside's average percentage price changes. Irvine's lagged percentage price changes are contagious to current Corona's percentage price changes. Further, Redland's lagged percentage price changes have marginal effects on San Bernardino's. And finally, San Bernardino's lagged percentage average quarterly price changes minimally affect current percentage price changes in Redland and Riverside.

Among the quarterly averages of measurable characteristics of the homes sold in each city, average square footage seemed to have the most impact on determining average percentage quarterly home-price changes in all cities except for Corona. The average number of bedrooms seemed to marginally impact percentage price changes in Redlands and Irvine only. Changes in average lagged quarterly mortgage rate seem to have impacted average percentage quarterly home-price changes in all cities except for Redlands. And finally, though weak, the average quarterly unemployment rate had very marginal to no effect on percentage home-price changes in all six cities. Genetic programming specifications obtained seemed more logical, and therefore the statements above focus on the results obtained using GP. The linear model specifications that were investigated using ordinary least squares yielded somewhat consistent results. Forecasts using the two techniques were similar for four of the six cities. Using a proposed forecast consistency statistic, the GP and OLS forecasts of average percentage quarterly home price-changes in the year 2014 were less consistent for Irvine and Redlands than they were for the other four cities. When compared to actual average prices in the six locations for the first quarter of 2014, the forecast errors were all reasonable except for the forecasts of Redland's prices. The GP and OLS forecast errors overestimated the prices by more than 15 percent there. The lowest forecast errors were in Anaheim and Irvine, where GP percentage forecast errors were only around 1 percent.

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CHAPTER 12

BUSINESS APPLICATIONS OF FUZZY LOGIC

PETR DOSTÁL AND CHIA-YANG LIN

12.1 INTRODUCTION

Various decisionmaking methods are used in business: classical ones and methods using soft computing (Zadeh 1994). The decisionmaking processes are very complicated because they include political, social, psychological, economic, financial, and other phenomena. Many variables are difficult to measure; they are characterized by imprecision, uncertainty, vagueness, semi-truth, approximation, and nonlinearity. Fuzzy logic differs from conventional (hard) computing in that it is tolerant of imprecision, uncertainty, partial truth, and approximation. In effect, the role model for fuzzy logic is the human mind. The guiding principle of fuzzy logic is to exploit this tolerance to achieve tractability, robustness, and low solution cost. Fuzzy logic could be used under these conditions. The basic ideas underlying soft computing in its current incarnation have links to many earlier influences, among them Zadeh's 1965 paper on fuzzy sets, the 1973 paper on the analysis of complex systems and decision processes, and the 1979 report (1981 paper) on possibility theory and soft data analysis.

12.2 BASIS OF FUZZY LOGIC

In classical logic, a theory defines a set as a collection having certain definite properties. Any element belongs to the set or not according to clear-cut rules; membership in the set has only the two values 0 or 1. Later, the theory of fuzzy logic was created by Zadeh in 1965. Fuzzy logic defines a variable degree to which an element x belongs to the set. The degree of membership in the set is denoted $\mu(x)$; it can take on any value in the range from 0 to 1, where 0 means absolute nonmembership and 1 full membership.

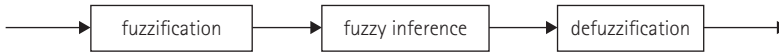
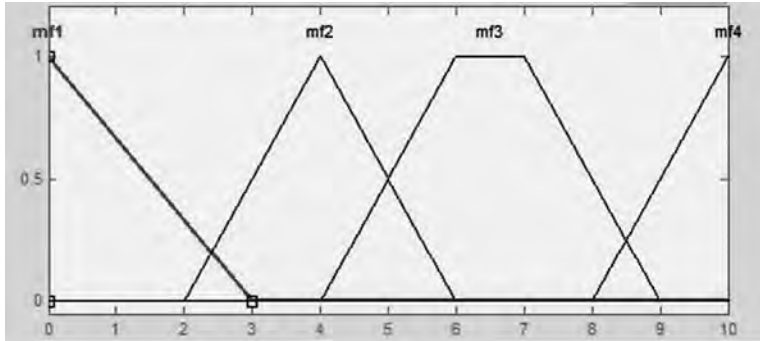


FIGURE 12.1 Fuzzy processes.

FIGURE 12.2 The types of membership functions Δ and Π .

The use of degrees of membership corresponds better to what happens in the world of our experience. Fuzzy logic measures the certainty or uncertainty of how much the element belongs to the set. People make analogous decisions in the fields of mental and physical behavior. By means of fuzzy logic, it is possible to find the solution of a given task better than by classical methods.

The proposed model is based on fuzzy logic and fuzzy sets. A fuzzy set A is defined as (U, μ_A) , where U is the relevant universal set and $\mu(x) : U \rightarrow \langle 0, 1 \rangle$ is a membership function, which assigns each element from U to fuzzy set A . The membership of the element $x \in U$ of fuzzy set A is indicated by $\mu_A(x)$. We call $F(U)$ the set of all fuzzy sets. Then the “classical” set A is the fuzzy set where $\mu(x) : U \rightarrow \langle 0, 1 \rangle$. Thus $x \in A \Leftrightarrow \mu_A(x) = 1$ and $x \notin A \Leftrightarrow \mu_A(x) = 0$. Let $U_i, i = 1, 2, \dots, n$ be universals. Then the fuzzy relation R on $U = U_1 \times U_2 \times \dots \times U_n$ is a fuzzy set R on the universal U .

The fuzzy logic system consists of three fundamental steps: fuzzification, fuzzy inference, and defuzzification. See figure 12.1.

The first step (fuzzification) means the transformation of ordinary language into numerical values. For variable risk, for example, the linguistic values can be no, very low, low, medium, high, and very high risk. The variable usually has from three to seven attributes (terms). The degree of membership of attributes is expressed by mathematical functions. There are many shapes of membership functions. Figure 12.2 provides four examples of the membership function: for $mf_1, P = [0 \ 0 \ 3]$; $mf_2, P = [2 \ 4 \ 6]$; $mf_3, P = [4 \ 6 \ 7 \ 9]$; $mf_4, P = [8 \ 10 \ 10]$; and so forth.

The types of membership functions that are used in practice are for example Δ (triangular) and Π (trapezoidal). There are many other types of standard membership functions on the list, including spline ones. The attribute and membership functions concern input and output variables.

The second step (fuzzy inference) defines the system behavior by means of rules such as <IF>, <THEN>, or <WITH>. The conditional clauses create this rule, which evaluates the input variables. These conditional clauses have the form

<IF> I_1 is mf_a <OR> I_2 is mf_b ... <OR> I_{N-1} is mf_y <OR> I_N is mf_z <THEN> O_1 is mf_{O_1} <WITH> s .

The written conditional clause could also be described verbally: If input I_1 is mf_a or I_2 is mf_b ... or I_{N-1} is mf_y or I_N is mf_z then O_1 is mf_{O_1} with the weight s , where $0 \leq s \leq 1$. These rules must be set up and then they may be used for further processing.

The fuzzy rules represent the expert systems. Each combination of attribute values that is put into the system and occurs in the condition <IF>, <THEN>, <WITH> represents one rule. Next it is necessary to determine the degree of supports for each rule; it is the weight of the rule in the system. It is possible to change the weight rules while optimizing the system. For the part of the rule behind <IF>, it is necessary to find the corresponding attribute behind the part <THEN>. The condition <AND> could be used instead of <OR>.

The third step (defuzzification) means the transformation of numerical values to linguistic ones. The linguistic values for variable risk can be, for example, very low, low, medium, high, and very high. The purpose of defuzzification is to transform fuzzy values of an output variable so as to present verbally the results of a fuzzy calculation. During the consecutive entry of data the model with fuzzy logic works as an automat. There can be a lot of variables on the input.

12.3 BUSINESS APPLICATION: A SURVEY OF THE LITERATURE

The role of fuzzy set theory in economics was recognized much later than it was in many other areas. One possible exception in economics is perhaps the writing of the British economist G. L. S. Shackle (1903–1992). Beginning in the late 1940s Shackle argued that probability theory, which is accepted in economics as the only mathematical tool for expressing uncertainty, is not enough. For Shackle, the uncertainty associated with imagined actions, whose outcomes are to some extent unknown, should be expressed in terms of degrees of possibility rather than as probabilities (Shackle 1961). However, the actual potential of fuzzy set theory to reformulate the text of economic theory and to make it more realistic was recognized by economists only in the 1980s. One of the pioneers in initiating the reformulation of economic theory by taking advantage of fuzzy set theory was the French economist Claudie Ponsard (1927–1990). A special issue of *Fuzzy Sets and Systems* that is dedicated to him (vol. 49, no. 1, 1992) provides a good overview of his contributions and the related work of other economists in this area.

Information regarding the status and the use of fuzzy set theory in economics can be found in Billot (1992). The book is mainly about microeconomics; it reformulates

the standard notion of preference by proposing a notion of fuzzy preference. The early use of fuzzy logic in business can be found in Altroc (1996). It has some successful applications, for example, evaluation of creditworthiness, fuzzy scoring for mortgage applications, fuzzy-enhanced scorecards for leasing risk assessment, fraud detection, investor classification, insider trading, cash supply optimization, supplier evaluation, customer targeting, sequencing and scheduling, optimizing research and development projects, and knowledge-based prognosis. Applications of various methods to operation research described by Chanas (1982) also deserves a mention. Approximate reasoning in decision methods that include multiperson, multicriteria, and multistage scenarios is described by Gupta and Sanchez (1982). Critical path method and Project evaluation and review technique approaches that are modified using fuzzy theory are described in Prade (1979). Ruspini (1970) describes numerical methods for fuzzy clustering. Soft computing methods in financial engineering are discussed in Ribeiro and Yager (1999). Bojadziev and Bojadziev (2007) discuss some applications concerning fuzzy averaging for forecasting, decisionmaking in fuzzy environments, fuzzy logic control for business, finance, and management, and some applications of fuzzy logic control and fuzzy queries from databases. Some applications of artificial intelligence in economics and finance are outlined in Chen and Wang (2003) and Chen et al. (2007). Some general applications of soft computing can be found in Aliev and Aliev (2002).

There are many more applications of fuzzy theory could not be detailed here owing to space limitations. A recent comprehensive survey of economic applications of fuzzy theory in macroeconomics can be found in Shin and Wang (2010). The authors examine applications of fuzzy logic to a range of variables including GDP growth rate, exchange rates, and sales potential as well as political risk, valuation, pricing, and credit risk. Given the breadth of their survey, we will limit ourselves to the additional work published after 2008. Hudec and van Grinsven (2014) apply fuzzy logic in order to improve the quality of business statistics on adaptive survey designs. With decreasing participation in surveys, this design could lead to a more robust and effective classification when we are aiming at improving unit and item response and reduce measurement error by stimulating motivation. Lazzerini and Mkrtchyan (2010) propose extended fuzzy cognitive maps (E-FCMs) to analyze the relation between risk factors and risks. The main difference between E-FCMs and conventional FCMs lies in the fact that EFCMs have non-linear membership functions, conditional weights, and time delay weights. More examples of FCMs can be found in a survey by Papageorgiou (2013).

Zhan et al. (2013) develop a fuzzy logic-based bargaining model that can dynamically update personal preferences. Other studies also discovered that fuzzy logic helps answer many questions in game theory (Roszkowska and Burns 2010; Yang et al. 2011). Fuzzy data envelopment analysis (Chen et al. 2013) can help analyze banks' business performance under market risks. Shekarian and Gholizadeh (2013) extend the adaptive network-based fuzzy inference system (ANFIS) to economic welfare analysis. Their empirical results outperform the traditional multiple regression method. Based on the evaluation by Mirbagheri and Tagiev (2011), the fuzzy-neural model fares better than

the Solow model in predicting economic growth. The fuzzy-neural model has been applied to economic and financial forecasting, such as forecasting stock prices (Jandaghi et al. 2010), stock indexes (Wei 2013) and gold prices (Yazdani-Chamzini et al. 2012).

In financial markets, to overcome the difficulty of exchange rate forecasting, Leu et al. (2009) propose a distance-based fuzzy time series model. Their results outperform the random walk model and the artificial neural network model. Gradojevic and Gençay (2013) also show that “fuzzy technical indicators” dominate the standard moving average technical indicators. To account for variation in economic performance across countries and periods, Vis et al. (2013) employ fuzzy-set qualitative comparative analysis. Ormerod et al. (2013) use fuzzy clustering to help study the relation between inflation and unemployment (the Phillips curve). In another study, Günçavdi and Kçüçk (2013) measure the instability of inflation by employing the fuzzy logic approach.

A very popular research field using fuzzy logic is multiple-criteria decision making (Erol et al. 2011; Gupta et al. 2013; Krčmar and van Kooten 2008; Liu et al. 2010; Turskis and Zavadskas 2011; Wang and Lee 2011). Multiple-period problems also benefit from the use of the fuzzy logic method (Kahraman and Behret 2010; Liu et al. 2013; Zhang et al. 2013). Advanced decisionmaking methods that are used in business and public services are described in Dostál (2011). There are various programs for processing the fuzzy logic problems, such as MATLAB and its Fuzzy Logic Toolbox of Mathworks Inc., USA (MathWorks 2013) and *fuzzyTECH* program of Inform GmbH firm, Germany (fuzzyTECH 2012).

12.4 CASE STUDIES

The chapter shall provide readers with concrete applications of fuzzy logic. Based on the survey of the literature, some of the most prominent applications in economics and finance stand out, and they are

- investment risk evaluation
- mortgage loan risk evaluation
- customer relations management
- prediction of time series
- stock market decisions
- stock trading decisions
- investment risk evaluation with ANFIS
- investment risk evaluation
- decisionmaking evaluation
- client risk evaluation
- product evaluation

The fields of application of fuzzy logic in business, economics, and finance cover a wide area. Our case studies are focussed only on applications. The programs *fuzzyTECH* and MATLAB with Global Optimization Toolbox are used for demonstrations.

12.4.1 Investment Risk Evaluation

Let us give a simple example. The inputs will be partial risks and the output will be total risk. The question is whether to invest. The example will be solved by means of the *fuzzyTECH* program.

The input variables are economic, material, political, and selling risks (R_e , R_m , R_p , and R_s). Each of these four inputs has six attributes of risk: very high, high, medium, low, very low, or none (for example, R_{eVH} , R_{eH} , R_{eM} , R_{eL} , R_{eVL} , or R_{eN}). The output represents the total risk, that is, one output with five total risk attributes: very high, high, medium, low, or very low (R_{VH} , R_H , R_M , R_L , or R_{VL}). At first it is necessary to set up the number of input and output variables with their attributes and member functions, then it is necessary to set up the number of rule blocks with their rules. See figure 12.3.

The schema of inputs and outputs with the rule box is presented in figure 12.4.

It is necessary to set up the membership functions for all inputs. We set up all inputs with six attributes together with their membership functions. Figure 12.5 shows the

Project Specification	
Input Variables:	4
Input Terms/Variable:	6
Output Variables:	1
Output Terms/Variable:	5
Rule Blocks:	1

FIGURE 12.3 Setup of parameters.

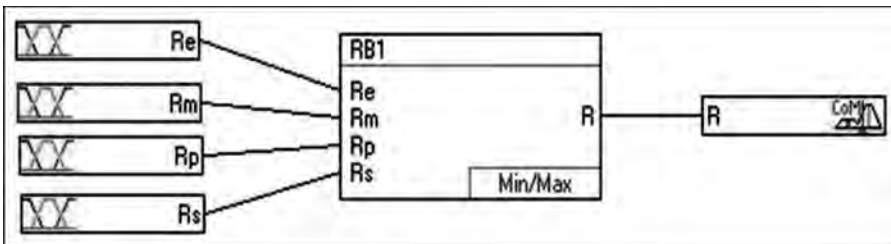


FIGURE 12.4 Scheme of a fuzzy model.

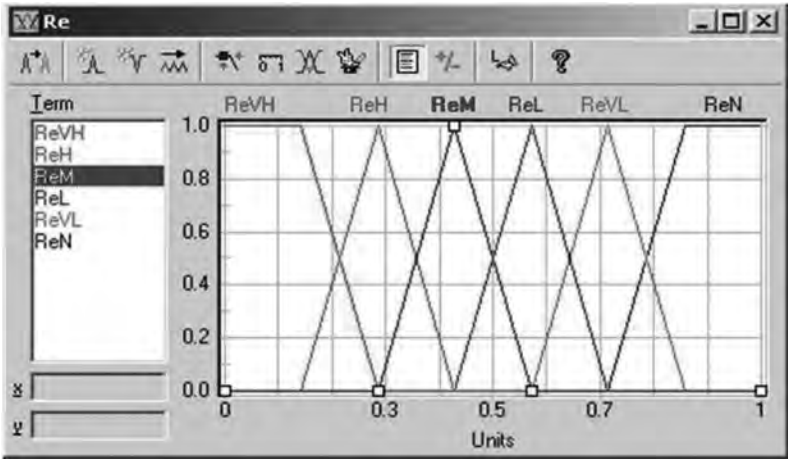


FIGURE 12.5 Definition of attributes and membership functions: input.

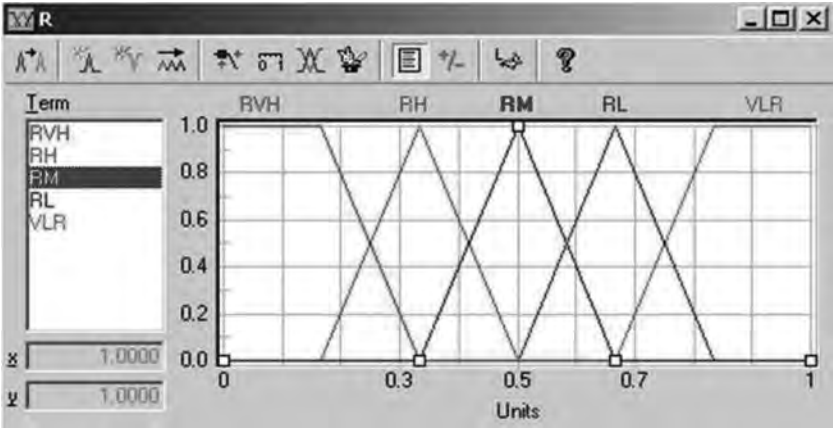
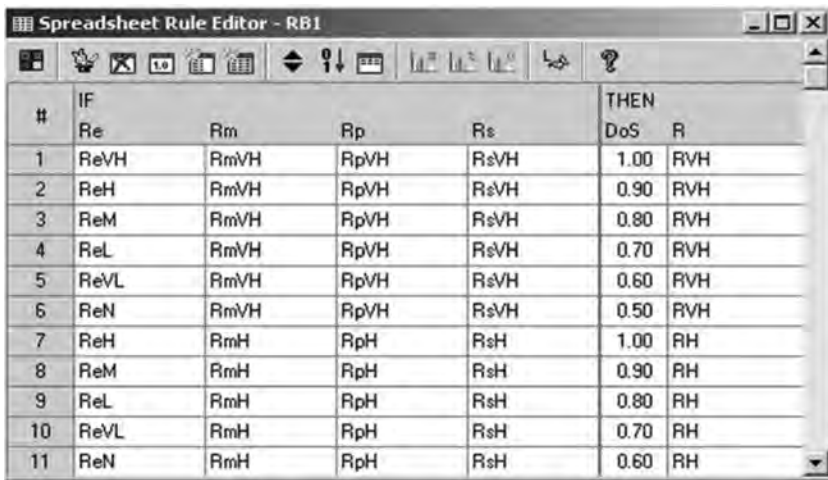


FIGURE 12.6 Definition of attributes and membership functions: output.

membership functions for economic risk. It is the same for other risks. The types S, Δ , and Z are chosen. Then we set up the membership function for output. The types S, Δ , and Z are chosen. Figure 12.6 shows the membership functions for total risk.

Finally, we must set up the rules and weights for the rule block (DoS = degree of support) among inputs and output. It is shown in figure 12.7. The first row of figure 12.7 means that if the economic risk is very high (ReVH), the material risk is very high (RmVH), the political risk (RpVH) is very high, and the selling risk is very high (RsVH), then the total risk is also very high (RVH) with the degree of support equal to 1.00.



#	IF				THEN	
	Re	Rm	Rp	Rs	DoS	R
1	ReVH	RmVH	RpVH	RsVH	1.00	RVH
2	ReH	RmVH	RpVH	RsVH	0.90	RVH
3	ReM	RmVH	RpVH	RsVH	0.80	RVH
4	ReL	RmVH	RpVH	RsVH	0.70	RVH
5	ReVL	RmVH	RpVH	RsVH	0.60	RVH
6	ReN	RmVH	RpVH	RsVH	0.50	RVH
7	ReH	RmH	RpH	RsH	1.00	RH
8	ReM	RmH	RpH	RsH	0.90	RH
9	ReL	RmH	RpH	RsH	0.80	RH
10	ReVL	RmH	RpH	RsH	0.70	RH
11	ReN	RmH	RpH	RsH	0.60	RH

FIGURE 12.7 Part of the table of fuzzy rules.

12.4.2 Mortgage Loan Risk Evaluation

Our second illustration concerns the evaluation of credit risk for a mortgage loan; the potential lender needs to decide whether the loan is granted or not. The program *fuzzyTECH* is used for the evaluation. The model has five inputs with three or five attributes, four rule blocks, and one output with five attributes. When the program is started, it is necessary to fill in the table of the model, to describe the inputs, output, and rule block, and finally to connect them.

Figure 12.8 shows the built-up model. The input variables are location, workmanship, asset, income, and interest. The outputs from rule blocks are buildings, applicant, and credit. The final output is credit. Three attributes have been set up for the input variables and five for the output variable.

While setting up the attributes, it is necessary to pay attention to selecting the number of variables. While setting up the membership functions, one must set up the shape and the course. This activity depends on the skill of those who set up the model, and also on the process of tuning the model. It is important to compare the results of models with reality. Only when the model is tuned is it possible to use it in practice. Figure 12.9 presents the attributes and membership functions for the variable building. It is similar for other input variables.

Two of four fuzzy rule boxes are presented in figure 12.10 (for the rule box Construction) and 12.11 (for the rule box Credit).

The attributes and membership functions for the output credit are functions of S, Δ , and Z types. They set up the values for granting a loan as very low, low, medium, high, very high. Figure 12.12 shows the membership functions and results of evaluation of the mortgage loan credit. The results present the situation when the granting of credit at the medium level is evaluated.

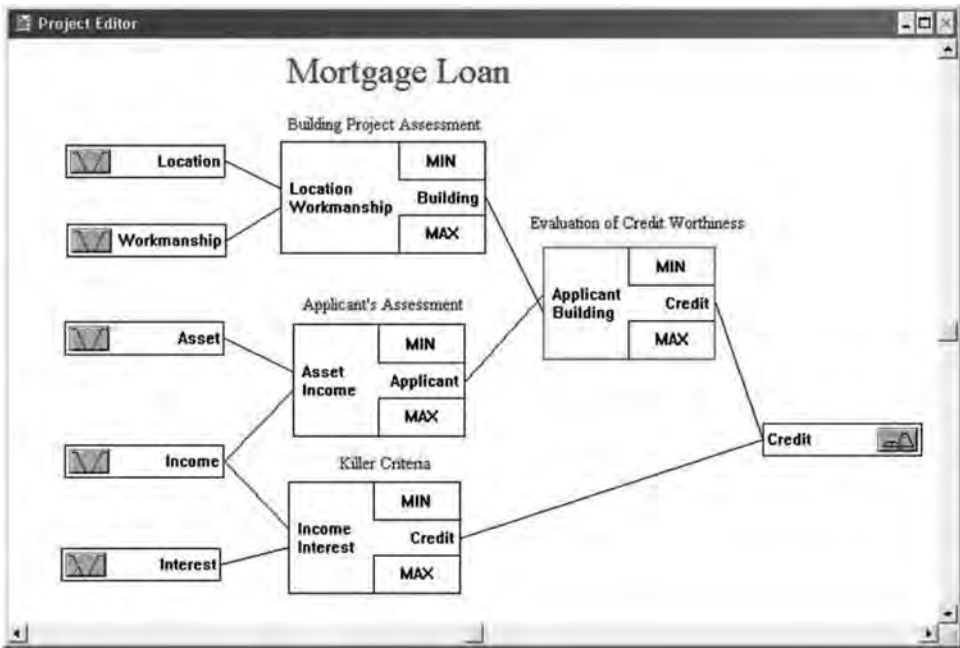


FIGURE 12.8 Built-up model.

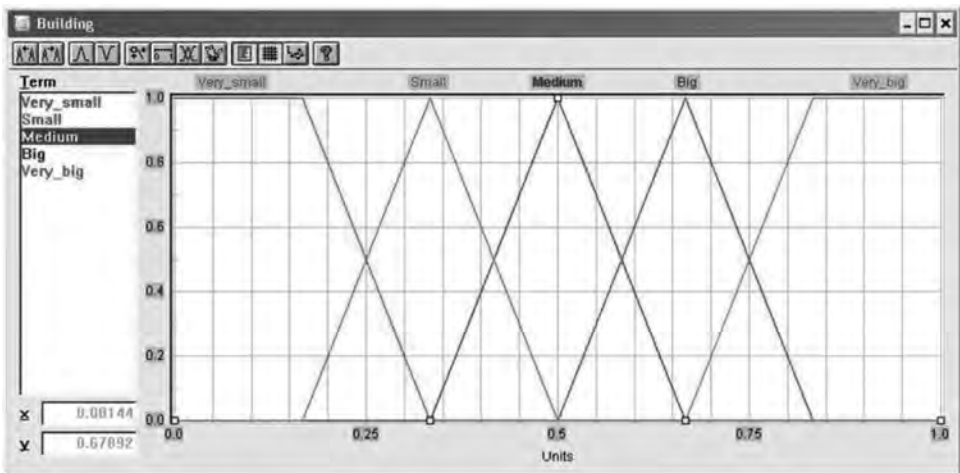


FIGURE 12.9 Attributes and membership functions for the variable Building.

When the model is built up, it has to be tuned. This means that all known input data must be substituted and a calculation must be performed, and the results compared with reality. In cases, of great deviation, changes of shape and of the course of the membership functions must be made. After the model is tuned, then the input data

	IF		THEN	
	Building	Land	DoS	Construction
1	Very_small	Small	1.00	Small
2	Very_small	Medium	1.00	Small
3	Very_small	Large	1.00	Medium
4	Small	Small	1.00	Small
5	Small	Medium	1.00	Medium
6	Small	Large	1.00	Medium

FIGURE 12.10 Rule box for construction.

	IF		THEN	
	Applicant	Construction	DoS	Credit
1	Small_quality	Small	1.00	Low
2	Small_quality	Medium	1.00	None
3	Small_quality	High	1.00	None
4	Medium_quality	Small	1.00	Medium
5	Medium_quality	Medium	1.00	Low
6	Medium_quality	High	1.00	None

FIGURE 12.11 Rule box for credit.

should be substituted, a calculation should be made, and the results should be used for the support of decision making.

12.4.3 Customer Relations Management

The fuzzy model can be used in the field of data mining. This example presents a choice in direct marketing: it evaluates whether to visit the customer personally, to send him a letter, or not to contact him. The application is solved by six input variables with three or four attributes, three rule boxes, and one output variable with three attributes.

When the program *fuzzyTECH* is started, it is necessary to fill in the table of the model, describe the inputs, output, and rule block, and connect them. See the fuzzy model in figure 12.13.

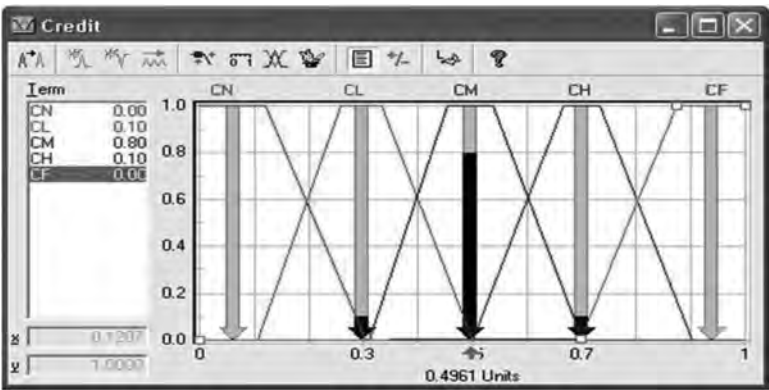


FIGURE 12.12 The results of fuzzy calculation.

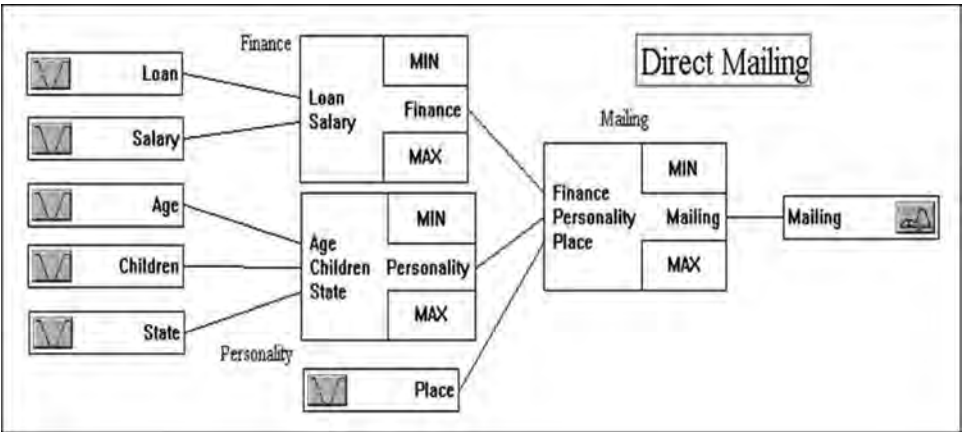


FIGURE 12.13 Fuzzy model, direct mailing.

The input variables and their attributes are salary (low, medium, high), loan (no, small, medium, high), children (none, a few, many), marital status (single, married, divorced), *age* (young, medium, old, very old), place (big city, city, village, hamlet). The output from the rule box Finance evaluates the financial situation of the as a stand customer (excellent, good, bad), and the output from the rule box Personality evaluates the personality of the man customer (excellent, good, bad). The output from the rule box Mailing has attributes (inactivity, mail, personally) that indicate whether to not contact the person, send him an e-mail, or visit him personally.

The membership functions are spline curves of shapes Λ , S, and Z. Figure 12.14 presents the attributes and membership functions of the output variable Mailing.

The model evaluates the case of a customer with a low salary, a high loan amount who lives in a village, is old, and is divorced with no children when the program indicates: No activity: Do not contact him.

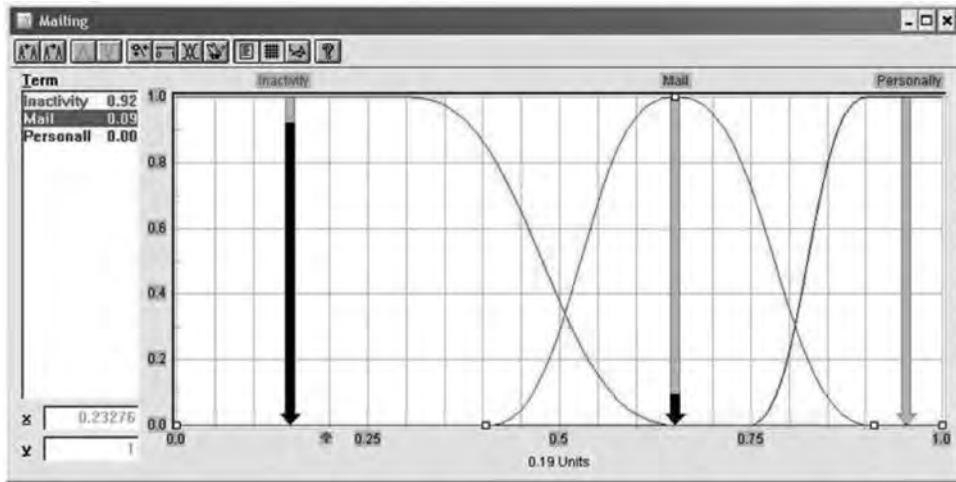


FIGURE 12.14 Membership functions of the output variable Mailing.

12.4.4 Prediction of Time Series

Let us mention an example of the use of fuzzy logic for predicting a time series. At first it is necessary to say that the time series is a sequence of values that are dependent on time t . The value at time $t = 1$ is denoted x_1 , at time $t = 2$ is denoted x_2 , and so on, and the value in time $t = N$ is denoted x_N , where N signifies the number of values in the time series. The time series can be expressed as a vector of values $x = (x_1, x_2, \dots, x_N)$. For purposes of prediction we specify that the value x_N will be the last known value of the time series and it will correspond to the present. The value $\dot{x}_{N+1}$ will be the first future value predicted, the value $\dot{x}_{N+2}$ will be the second value predicted, and so on. (The symbol $\dot{\cdot}$ is used to denote predicted values.) The interval between measurement is very often constant, then $\Delta = t_2 - t_1 = t_3 - t_2 = \dots = t_N - t_{N-1} = \text{const}$. This interval in the economy (in contrast to technical sciences) has values in the range of minutes, hours, days, weeks, months, years, and their fractions. In this respect we speak about time series with very high (minutes), high (hours), medium (days), low (weeks), and very low (year) frequencies.

The following verbal notes were chosen for the solution of predictions by means of the *fuzzyTECH* program: $\Delta_1 = x_N - x_{N-1}$, $\Delta_2 = x_{N-1} - x_{N-2}$, $\Delta_3 = x_{N-2} - x_{N-3}$, $\Delta_4 = x_{N-3} - x_{N-4}$ (the signs of these differences express the trend of the time series). The built-up model for prediction has four input variables, Δ_1 , Δ_2 , Δ_3 , and Δ_4 , one rule box, and one output variable, Prediction. See the fuzzy model shown in figure 12.15.

The input variables have five attributes defined by values of Delta, specified by their signs and the size of difference of neighbouring values (high positive, positive, zero, negative, and high negative difference). As membership functions the shapes Δ , S, and

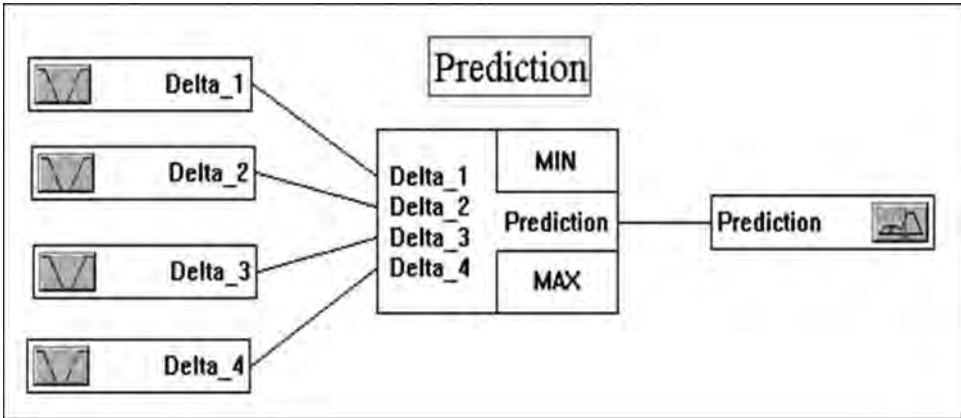


FIGURE 12.15 Fuzzy logic: prediction.

Z are used. The output variable Prediction has five attributes that evaluate the future course of the time series (high increase, increase, stagnation, decrease, high decrease), specifying the situation at time $N + 1$ (the value of prediction $\hat{x}_{N+1}$). The membership functions are spline curves of Δ , Π , S, and Z shapes.

The procedure followed by the program *fuzzyTECH* includes setup of membership functions of input variables Δ_1 , Δ_2 , Δ_3 , and Δ_4 and the latter. The latter must be set up on the basis of knowledge, preferably by the experts who understand the problem. The setup of a fuzzy rule box depends on the type of solved case. For example, a suitable rule can be similar to the following:

When inputs Δ_1 , Δ_2 , Δ_3 and Δ_4 are high negative, it means that the time series is decreasing and a large increase of the time series Prediction is expected in the future.

This situation can be verbally described in capital markets: After a great and long decrease in share values they tend to start a fast increase with 90 percent probability. The rule can be described by the form

<When> $\Delta_1 \ll 0$ <And> $\Delta_2 \ll 0$ <And> $\Delta_3 \ll 0$ <And> $\Delta_4 \ll 0$ <Then> Prediction = High increase <With> $s = 0.90$.

The rule for the opposite case can be verbally described in a capital market: After a great and long increase in share values they tend to start a fast decrease. The rule has the form

<When> $\Delta_1 \gg 0$ <And> $\Delta_2 \gg 0$ <And> $\Delta_3 \gg 0$ <And> $\Delta_4 \gg 0$ <Then> Prediction = High decrease <With> $s = 0.90$.

It is necessary to set up other rules that are combinations of these two extreme variants. Figure 12.16 presents setup attributes and membership functions for the output variable Prediction.

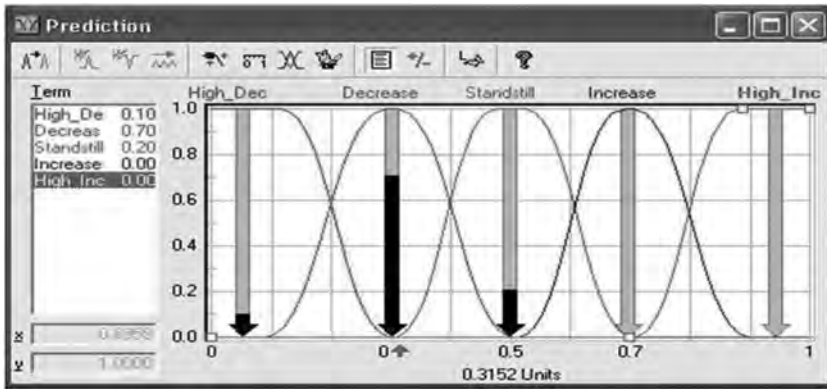


FIGURE 12.16 Membership functions of the output variable Prediction.

For another case for which an unsatisfactory model could be set up, it is necessary to define variables in different ways, including choosing other attributes and membership functions.

12.4.5 Stock Market Decision

Let us mention an example that solves the problem of decision making in capital markets: whether to trade in the stock market or not. The model for the *fuzzyTECH* program is presented in figure 12.17.

The input variables and their attributes are as follows: margin (insignificant, significant), interest rates (low, medium, high), strength of market (low, medium, high), and trend, the course of the time series (deterministic, stochastic).

The rules and attributes are as follows: the box investment (unsuitable, neutral, suitable) determines the rule of desirability of depositing money in a stock market; the first block trading (yes, no) evaluates whether trading in the market is suitable from the view point of profitability of investment and the strength of the market; the second block Trading (yes, no) gives the decision for trading when the time series is stochastic, meaning that there is no possibility of making a good prediction of future development of the time series. The output variable Trading evaluates whether to trade with share, index, commodity, or currency ratio. The membership functions used are in the shapes Δ , Π , S, and Z.

12.4.6 Stock Trading Decision

Fuzzy logic can be used in decision making in the stock market. The model is used for decisions whether to buy, sell, or hold with a share or index. The *fuzzyTECH* program

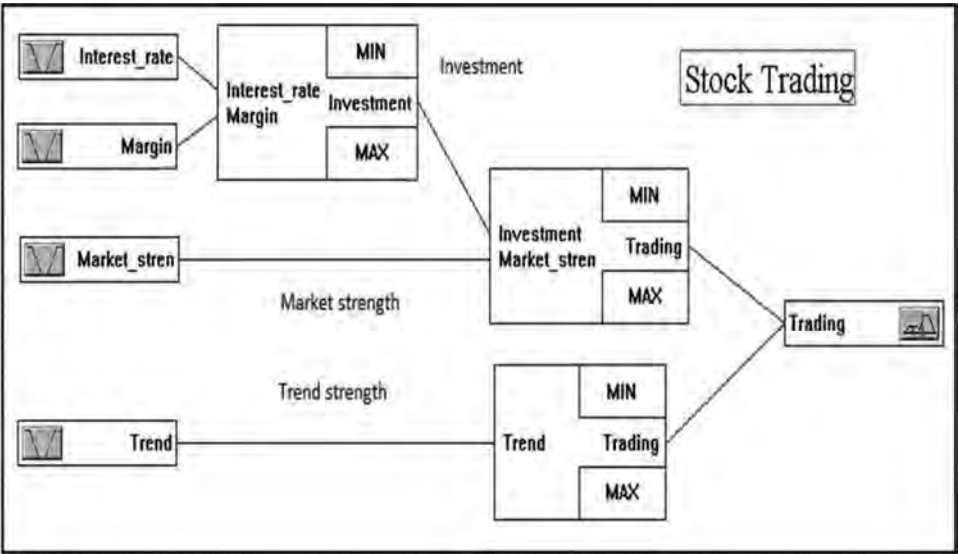


FIGURE 12.17 Diagram of a fuzzy model: stock trading.

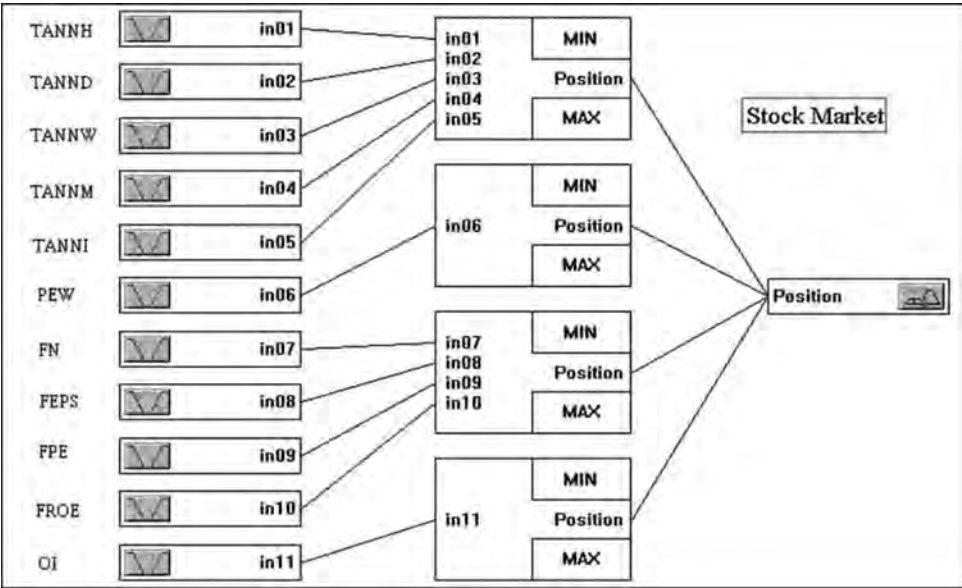


FIGURE 12.18 Diagram of a fuzzy model, stock market.

is used for this purpose when the inputs are the values from various analyses and information from the Internet. The diagram of such a model is presented in figure 12.18.

The model has eleven inputs with seven attributes, four rule boxes, and one output variable with five attributes. The input variables are the information obtained from

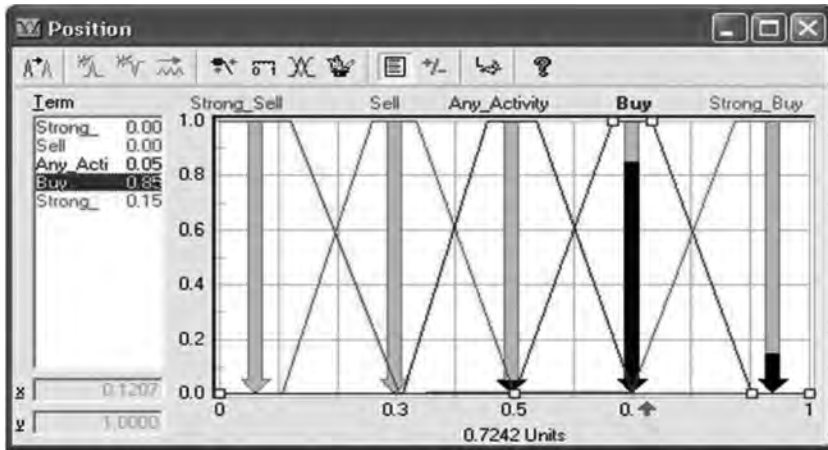


FIGURE 12.19 The attributes and membership function of output variable Position.

technical analyses represented by predictions of share prices by means of a neural network with hour TANNH, day TANND, week TANNW, and month periodicity TANNM; predictions of values of an index having influence on predictions of a searched share TANNI; psychological analyses represented by means of Elliott's waves PEW; fundamental analyses represented by information from news FN, economic indexes (earnings per share) FEPS, P/E (price-to-earnings ratio) FPE, and return on equity FROE, and other knowledge OI, such as intuition. The attributes of all input variables are the same, and they express the rate of influence on the tendency of a time series (high, medium, low positive, neutral, low, high negative). A positive influence indicates the influence of an increasing tendency of a time series, and a negative influence signifies the influence of a decreasing tendency of a time series.

The rule boxes include technical, psychological, fundamental, and other analyses. The output variable Position (Strong Sell, Sell, Hold, Buy, Strong Buy) tells the investor what he or she has to do in the stock market. As a membership function, spline curves of the shapes Λ , Π , S, and Z are used. The attributes and membership function for output variable Position are presented with the result Buy in figure 12.19.

12.4.7 Investment Risk Evaluation with ANFIS

We present the setup of fuzzy rules in the program environment MATLAB for a case study in which the program creates the fuzzy rules by means of neural networks. The Fuzzy Logic Toolbox enables setting up rules by means of neural networks using the command *ANFIS*. The inputs and outputs are defined by table 12.1, which present the logical operation $\langle \text{AND} \rangle$. We demonstrate four states. The first state represents the fact that the political risk is high (o) and economic risk is high (o), and it leads to state of

Table 12.1 The input and output values

State Order	Risk Po Input1	Risk Ec Input2	Investment Output
1	0 (H)	0 (H)	0 (H)
2	0 (H)	1 (L)	0 (H)
3	1 (L)	0 (H)	0 (H)
4	1 (L)	1 (L)	1 (L)

Table 12.2 Input and output values used to create the text file Test.dat

0	0	0
0	1	0
1	0	1
1	1	1

output of no investment (o). The second state represents the fact that the political risk is high (o) and economic risk is low (1), and it leads to state of output of no investment (o). The third state represents the fact that the political risk is low (1) and economic risk is high (o), and it leads to state of output of no investment (o). The fourth state represents the fact that the political risk is low (1) and economic risk is low (1), and it leads to state of investment (1).

The text file Test.dat is created with the mentioned data at first (see table 12.2).

The commands *fuzzy* and *File-New FIS-Sugeno* create a fuzzy model in the MATLAB environment, then the commands *Edit-Add Variable-Input* follow, which add the second input (see figure 12.20). The command *Edit-Anfis* opens the editor. The choice of menu *Type-Training* and *Load-Data* read the file *Test.dat*. (see figure 12.21).

The Fuzzy Interface System is generated from the command *FIS* with the option *Grid partition*. It is set up with the numbers of two membership functions for both inputs Number of MF's [2 2], then the Gaussian membership function is set up from the commands *MF Type gaussmf* and linear type *MF Type: linear* (see figure 12.22).

The values for training of neural networks create the rules chosen by options *Optim. method-hybrid*, error tolerance 0, and the number of epochs 20. It is desirable to watch the training error evolving over time (see figure 12.23). The command *Structure* shows the created neural network used for generation of rules (see figure 12.24). The command

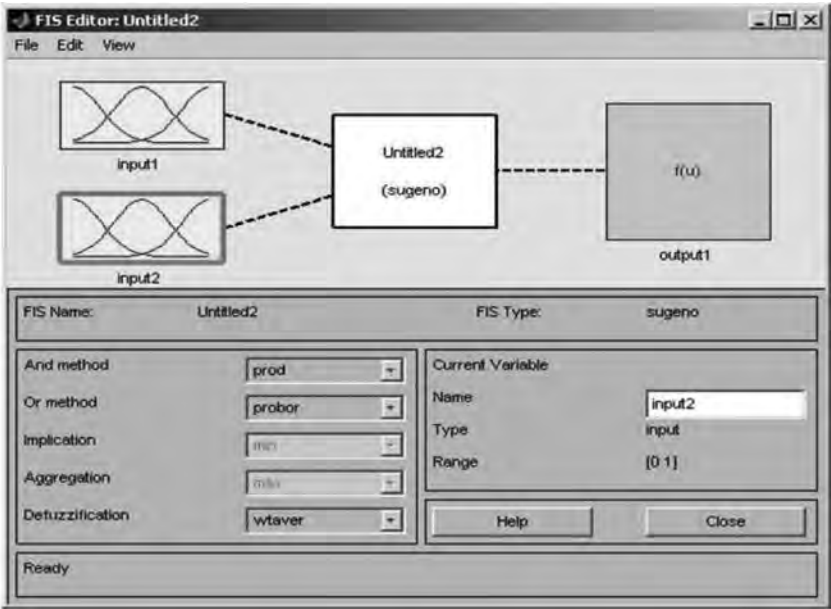


FIGURE 12.20 Fuzzy model.

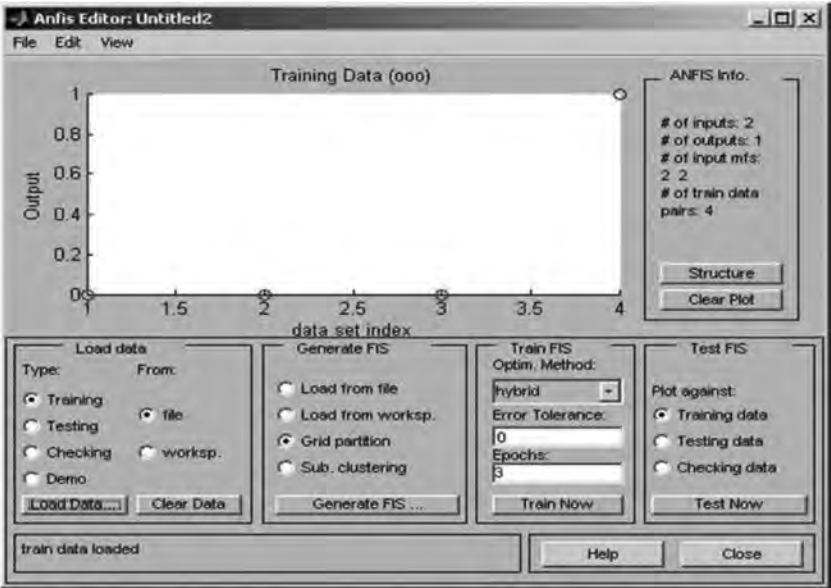


FIGURE 12.21 Reading of data and their display.

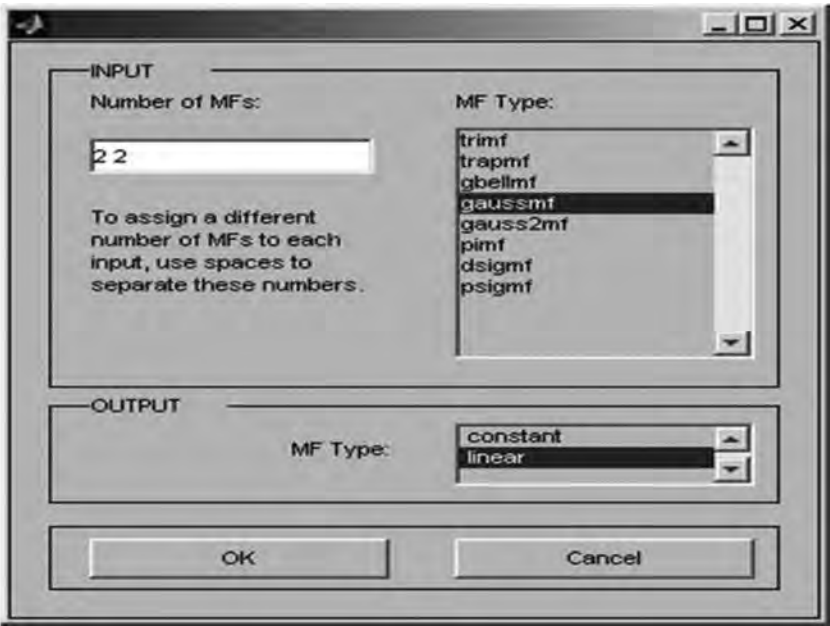


FIGURE 12.22 The setup of membership functions.

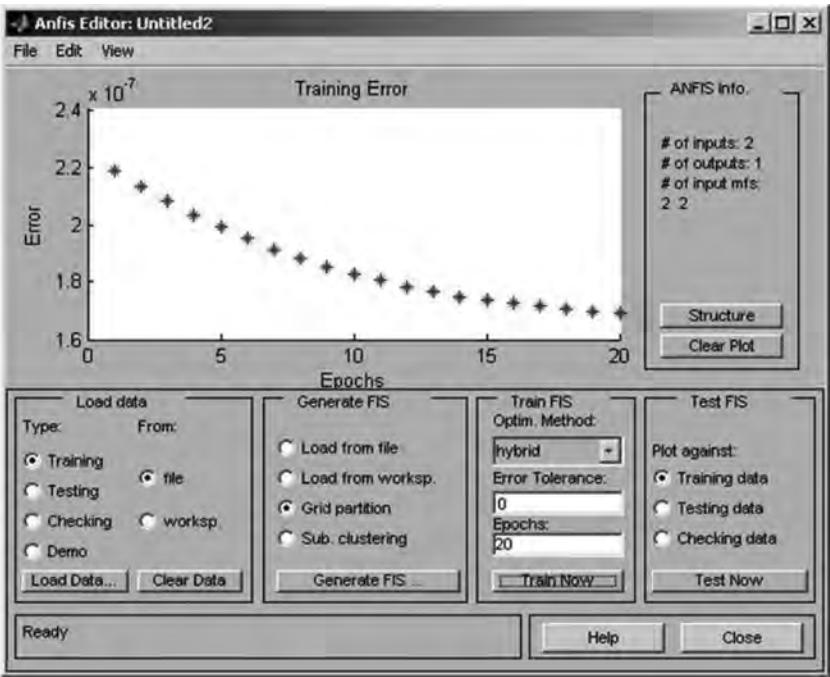


FIGURE 12.23 Error training in the ANFIS editor.

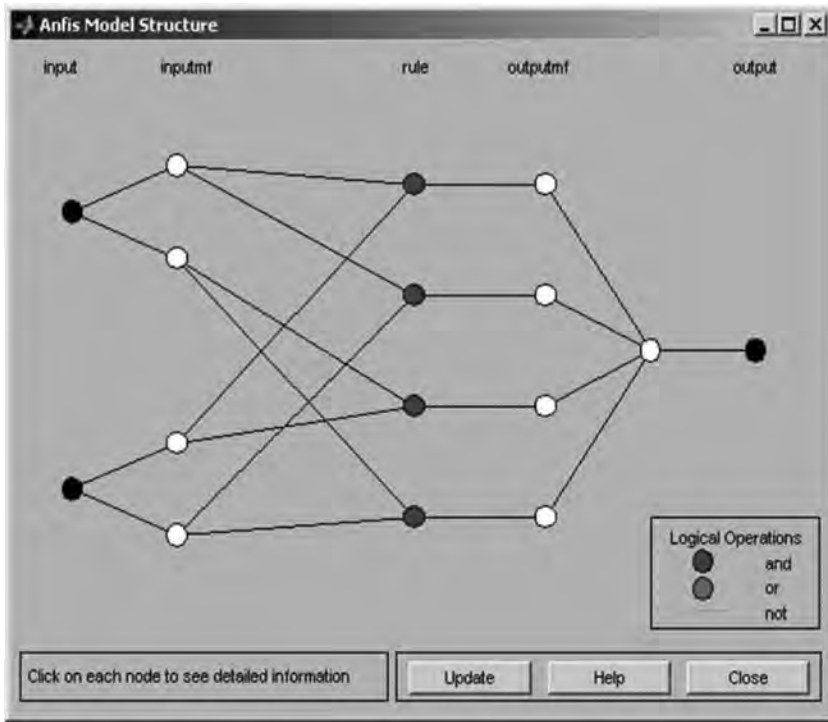


FIGURE 12.24 Created neural network.

Test FIS—Training data enables comparing training data with the real ones, possibly with testing and checking of data (see figure 12.25).

The dependence of outputs on inputs can be displayed by standard commands (see figure 12.26). The surface reflects the proper generation of rules. It is possible to display the generated rules (see figure 12.27). The command *Rules* enables the verification of created rules (see figure 12.28).

The presented case study describes the methodology of creation of rules by means of neural networks using recording data from databases. The problem could involve complicated tasks for which many rules are created. If the rules can not describe the solved problem successfully, the displayed surface shows “disturbances.” In the case of wrong generalization, the trained model does not correspond with new data.

12.4.8 Investment Risk Evaluation

There are situations when more rule boxes are necessary for using the decisionmaking process and they must be connected. The connection can be made by creating an *M*-file that enables reading the input data, but also transferring results to other blocks. The example is presented in figure 12.29. This case is an example of the investment risk

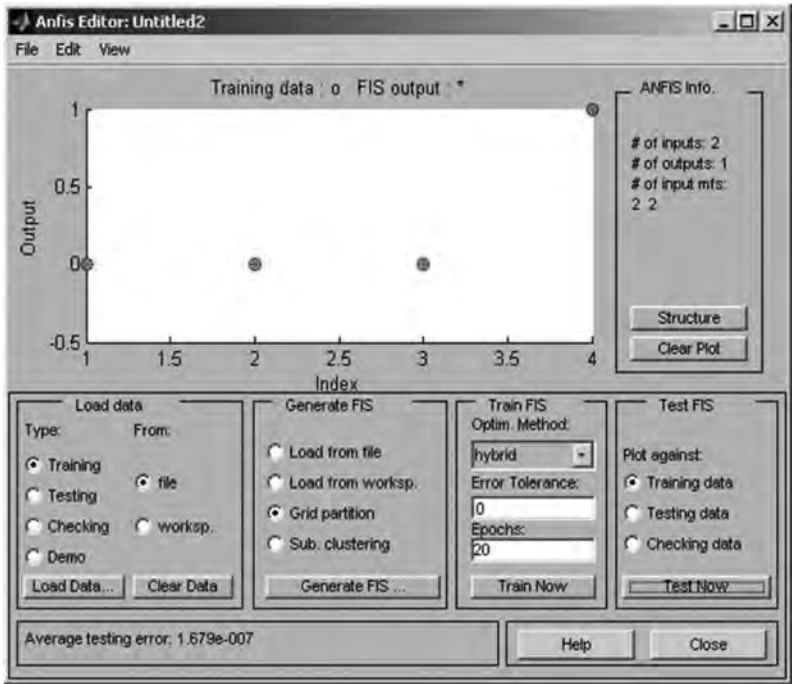


FIGURE 12.25 Evaluation of training.

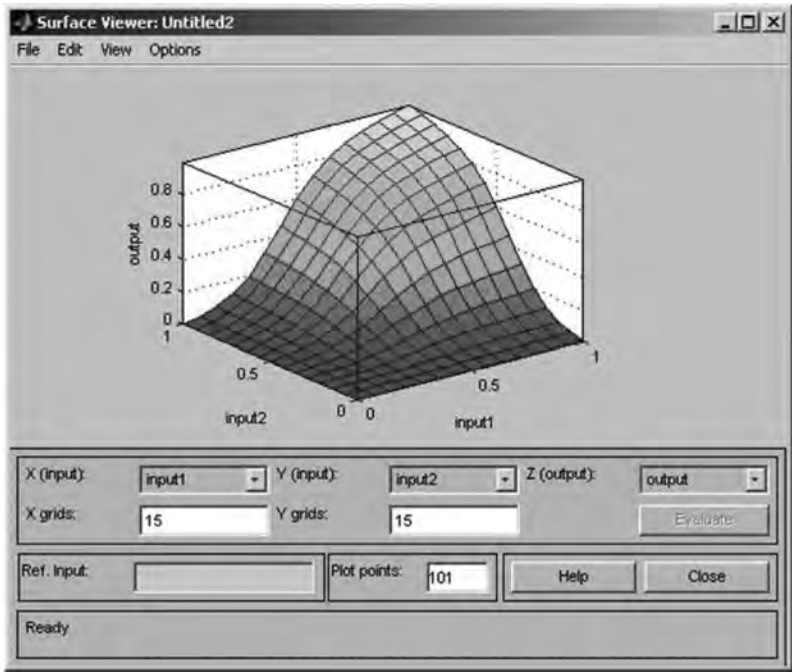


FIGURE 12.26 Dependence of output on inputs.

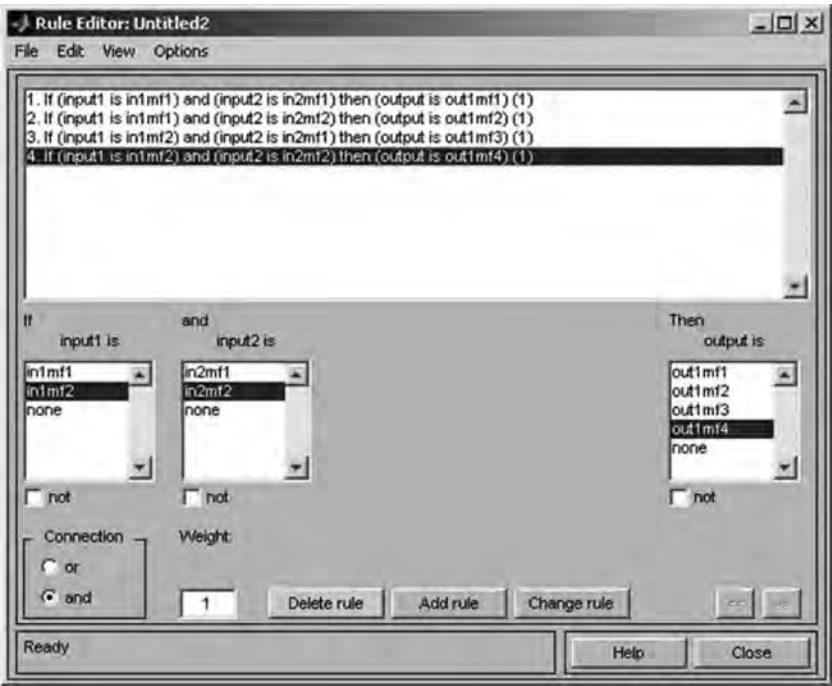


FIGURE 12.27 Display of generated rules.

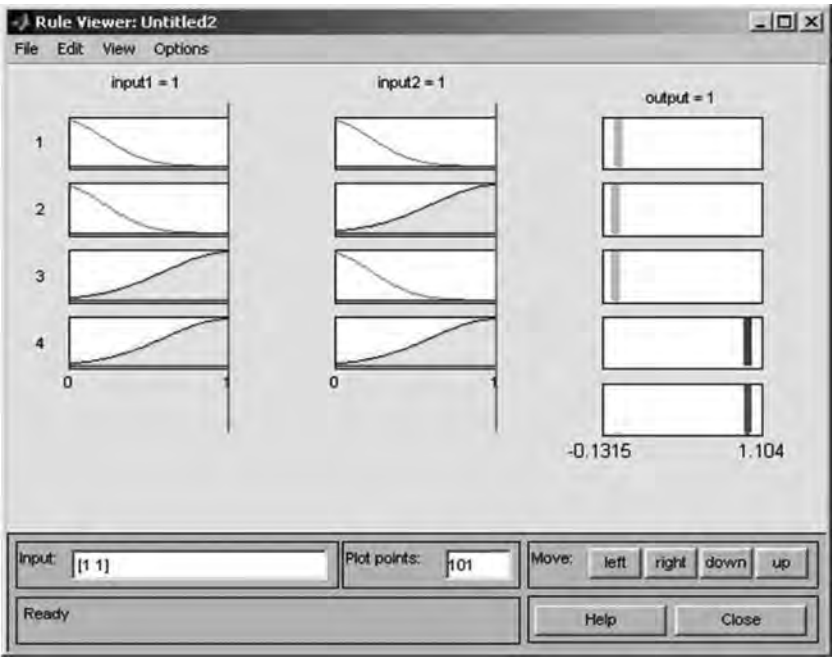


FIGURE 12.28 Verification of rules.

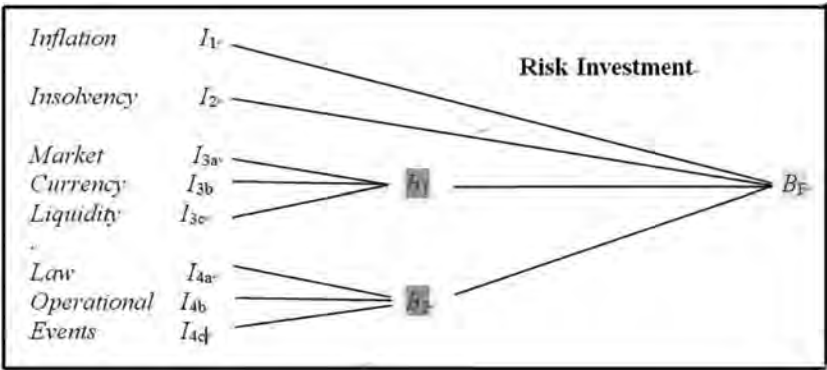


FIGURE 12.29 Connection of rule blocks.

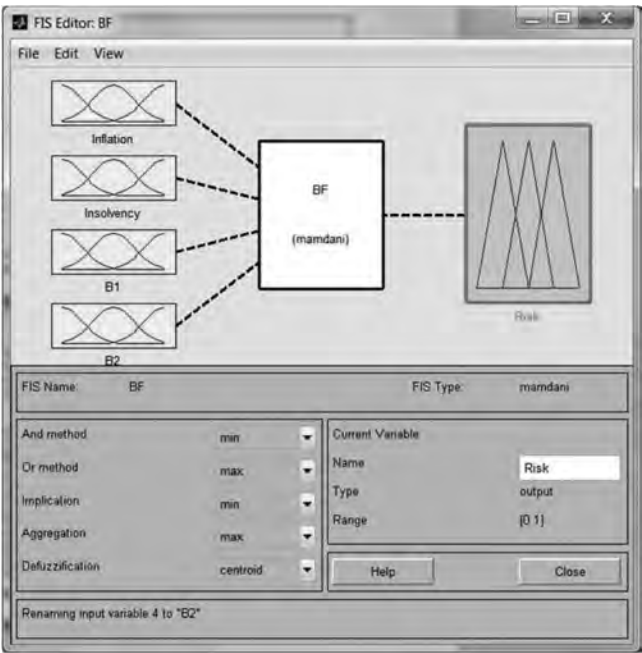


FIGURE 12.30 Box B_F .

evaluation. Inputs I_1 represent the inflation risk and input I_2 the insolvency risk. The block rule B_1 presents financial risk with inputs market I_{3a} , currency I_{3b} , and liquidity I_{3c} risk. The block rule B_2 presents other criteria such inputs law I_{4a} , event I_{4b} , and operational I_{4c} risk. The output evaluates investment risk supported by vague terms of low, medium and high risk. The variables I_1 and I_2 are inputs to the final box B_F (see figure 12.30) together with output from box B_1 (see figure 12.31), with inputs I_{3a} , I_{3b} , and I_{3c} , and output from box B_2 (see figure 12.32), with inputs I_{4a} , I_{4b} , and I_{4c} .

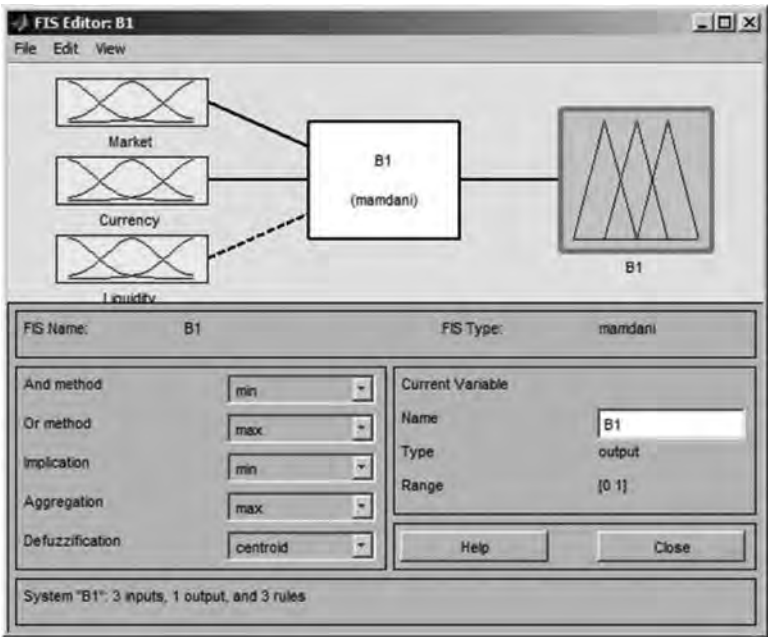


FIGURE 12.31 Box B₁.

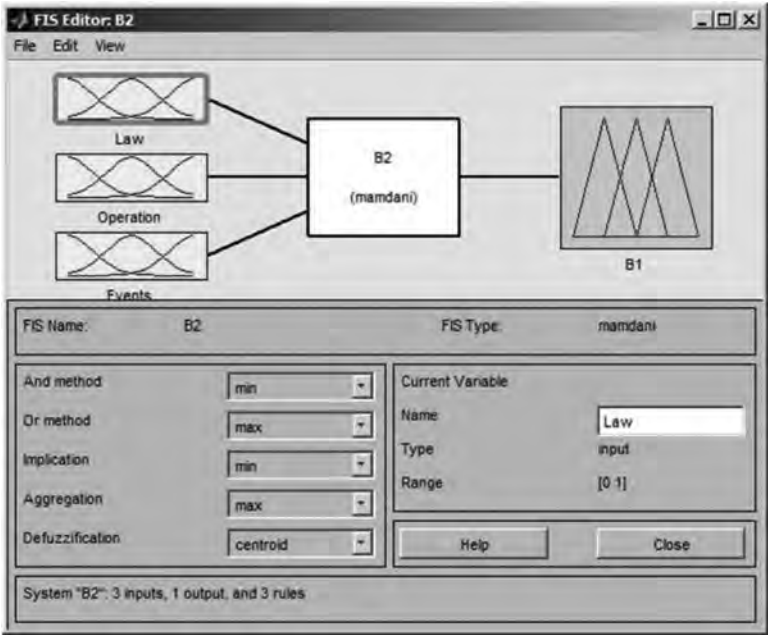


FIGURE 12.32 Box B₂.

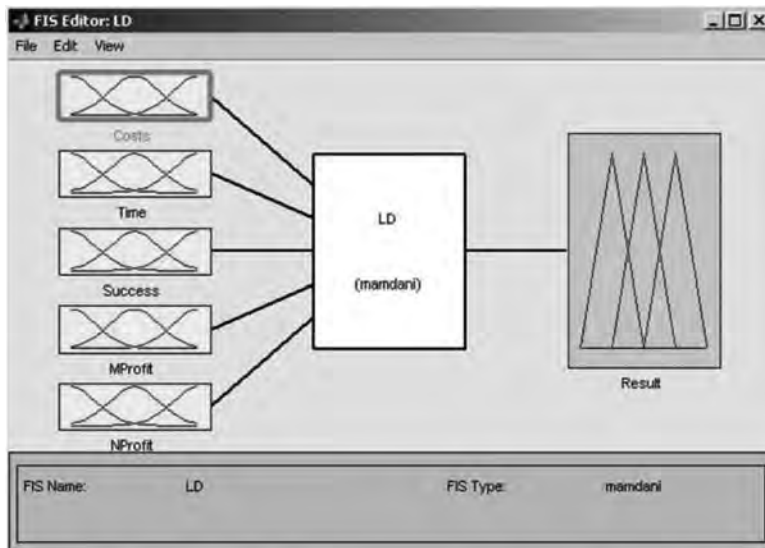


FIGURE 12.33 Fuzzy model.

The *M*-file *BF.m* provides the calculation. See the appendix, Program 12.1.

12.4.9 Decisionmaking Evaluation

There are innumerable applications of fuzzy logic in various disciplines. This section presents an application to law; the decision concerns whether to solve a legal dispute by means of judicial process, that is, whether to reject (RJP), entertains the acceptance of (EJP), or accept it (AJP) on the basis of inputs. The input variables and their attributes are costs of disputes (low, medium, high), time or duration of the dispute (low, medium, high), success of the dispute (low, medium, high), possible material profit for the client (low, medium, high), and possible nonmaterial profit for the client (low, medium, high). See figure 12.33. The MATLAB program *LD.m* was programmed. See the appendix, Program 12.2).

The rule box was set up (see figure 12.34).

When the program *LD.m* is started, the request for entering the input data is displayed in the form Costs, Time, Success, MProfit, NProfit. When the input values are written [0.7; 0.2; 0.8; 0.2; 0.5], the *Result* is to entertain the judicial process (see figure 12.35).

The results correspond to reality. Therefore, it is possible to consider the built-up model to be functional. The parameters of the built-up model are saved in a *LD.fis* file (see the appendix).

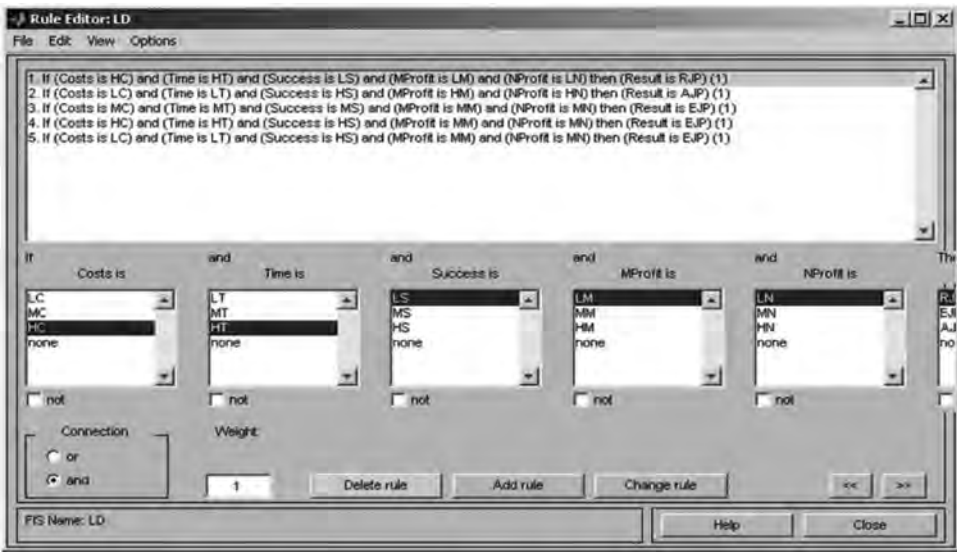


FIGURE 12.34 Rule box.

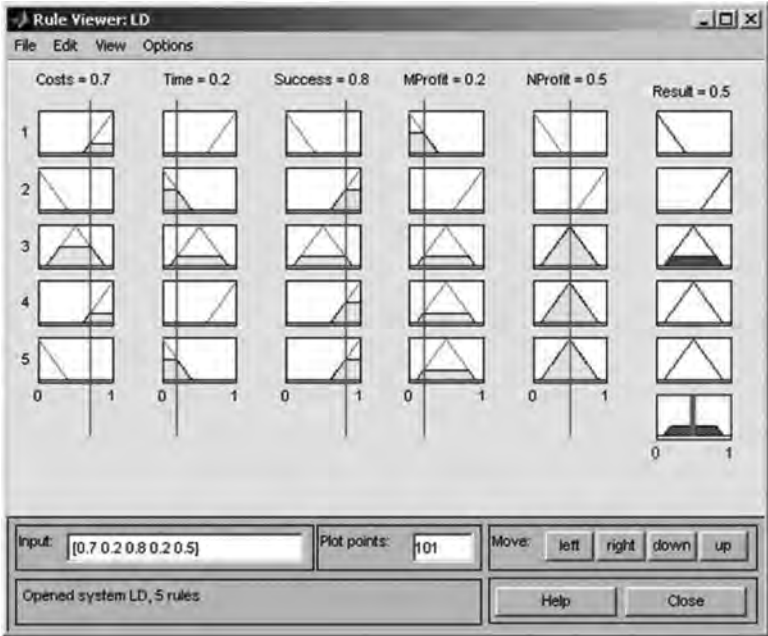


FIGURE 12.35 Results for Input [0.7;0.2;0.8;0.2;0.5].

12.4.10 Client Risk Evaluation

Application of the fuzzy logic model can be demonstrated in the case of evaluating the payment risk of active debt. The application is solved with eleven input variables,

three rule blocks, and one output variable with three attributes. The inputs and their attributes are sex (man, woman), age (young, middle, old), marital status (married, single, other), children (none, one, more), income (low, medium, high), account (none, medium, high), debt (none, medium, high), employment (short, medium, long term), contact with client (short, medium, long term), order (first, few, more), and delayed payment (none, few, more).

For these eleven inputs, where two to three attributes are selected according to the demand of the realization of project. The output from the rule box personal data evaluates the personality of the client (excellent, good, bad), the rule box financial data evaluates the financial situation of client (excellent, good, bad), the rule box quality evaluates the client from the point of view of the customer-supplier relation (excellent, good, bad). The output variable is the risk of the payment of active debt with the three attributes (low, medium, high).

It is necessary to set up the membership function for all inputs and outputs. It was used the functions in the shapes Λ , Π , Z , and S . The rule box must be set up with rules and weight they have among inputs and outputs. The weight of rules can be changed during the process of optimization. The built-up model can be used for the evaluation of the risk of the payment of active debt. On the basis of input values we find out whether the risk of payment of active debt is low, medium, or high. The course of membership function and the weight of rules DoS can be set up by means of neural networks in cases when the data are at our disposal. The fuzzy logic model was built up (figure 12.36) with boxes B_1 , B_2 , B_3 , and B_F (figure 12.37).

The M -file $BF.m$ provides the calculation (see the appendix, Program 12.3).

The results of the calculation are presented by inputs $I_1a, I_1b, I_1c, I_1d, I_2a, I_2b, I_2c, I_3a, I_3b, I_3c$, and I_3d with values 0, 1, and 0.5 (see figure 12.36). The results are low, medium, and high risk (see result 12.1 in the appendix).

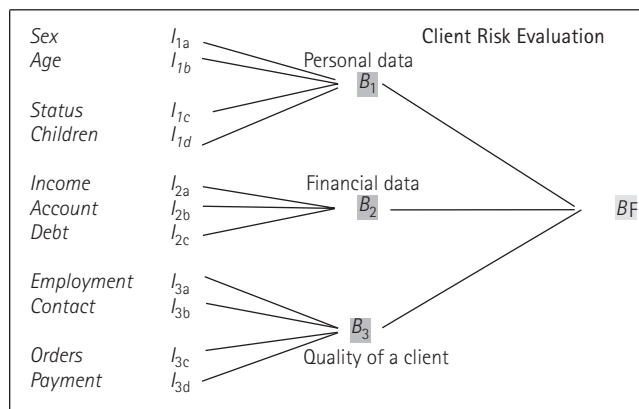
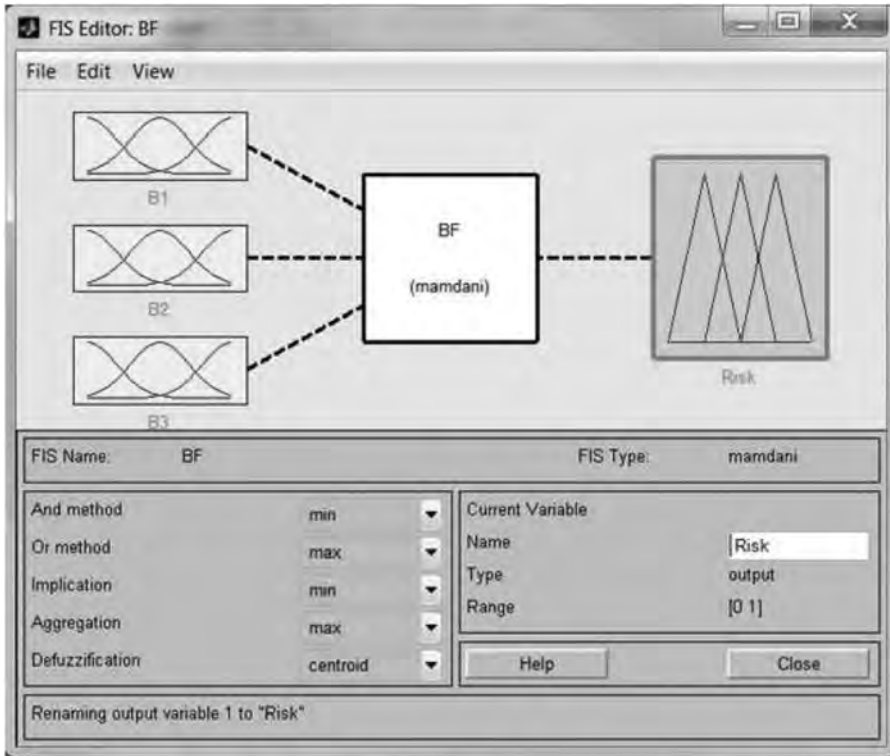


FIGURE 12.36 Model of Fuzzy logic.

FIGURE 12.37 Box B_F.

12.4.11 Product Evaluation

There are many tasks in management in which clustering helps one make a correct decision. Performing a cluster analysis or a clustering is the task of grouping a set of objects in such a way that objects in the same group (or cluster) are more similar (in some sense or another) to each other than to those in other groups (clusters). Popular notions of clusters include groups with low distances among the members. Fuzzy clustering can offer the advantages of fuzzy processing, as well as neural networks and evolutionary algorithms.

The solved marketing problem is based on sorting of products according to customer type. In other words, we have to find the right customer with the right product at the right place and time. The inputs of the case study are characterized by product parameters such as price, sell (sale) and quality. Input data are presented for seventeen objects (see table 12.3). The output will be the classification of goods according their characteristic clusters. The software program MATLAB and its Fuzzy Logic Toolbox are used for the software applications. The example presents the objects recorded in MS Excel format in the file *FC.xlsx*. This task is solved by the program *FC.m*. (see Program 12.4 in the appendix).

Table 12.3 Input and output values

Product	Price	Sell	Quality
1	8	81	8
2	310	602	84
3	78	445	5
4	93	200	64
5	60	62	2
6	40	228	40
7	180	107	25
8	196	220	57
9	46	78	46
10	80	174	21
11	120	223	18
12	107	494	46
13	384	147	39
14	42	102	31
15	131	65	32
16	81	178	17
17	10	570	15

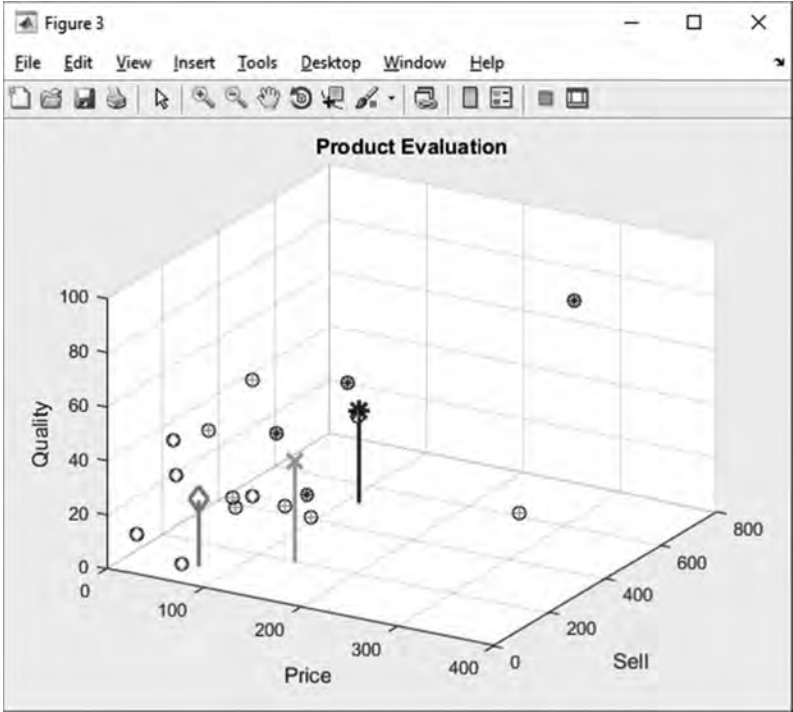


FIGURE 12.38 Three-dimensional graph of product clustering.

The program is started using the command *FC* in the MATLAB program environment. The number of clusters is set at three. During the calculation the iteration count is displayed. When the calculation is finished the output results, the coordinates of centroids, and the assignment of products to centroids are displayed (see Result 12.2 in the appendix).

The results are presented in the form of coordinates of clusters and assignment of products to the clusters. A three-dimensional stem graph is drawn (see figure 12.38). The results are represented by the centroids of three clusters marked $\blacklozenge$, $\times$, and $*$ and assignment of goods to the clusters. The cluster $\blacklozenge$ indicates the group of products with low sell, the cluster $\times$ indicates the group of products with medium sell and quality, and the cluster $*$ indicates the group of products with medium price and quality and high sell. The fuzzy clustering model enables us to sort the products for different customers. The tasks performed in this example lead to multi-dimensional ones whose graphical presentation is impossible. The image of the solution is in a hyper sphere, dependent, for example, on price, sell, quality, delivery, guarantee, discount, weight, and other factors.

12.5 CONCLUSIONS

In many ways, soft computing represents a significant paradigm shift in the aims of computing, a shift which reflects the fact that the human mind, unlike present-day computers, possesses a remarkable ability to store and process information that is pervasively imprecise, uncertain, and lacking in categoricity (Zadeh 2012).

The business applications of fuzzy logic have features specific to the field. They can help in decentralization of decisionmaking processes to make them standardized, reproducible, and documented. These methods play very important roles in companies because they help reduce costs, and that can lead to higher profits; they also can help firms compete successfully or decrease expenses. The decisionmaking processes in business are very complicated because they include political, social, psychological, economic, financial, and other factors. Many variables are difficult to measure.

Fuzzy logic is a theory that uses fuzzy sets and logic. The advantage of this approach is that the data for the processing can be imprecise, contradictory, uncertain, vague, partly true, approximated, and so forth. Fuzzy logic uses linguistic variables, the rule base or fuzzy sets are easily modified, the input and output are related in linguistic terms, easily understood, and a few rules encompass great complexity. Moreover, this model is not a black box; that the rules are clear. The disadvantages of fuzzy logic can be found in the setup of rules of complicated phenomena and its need for finer tuning and simulation before implementation. The neuro-fuzzy models could be an advantage in the setup of rules. The use of fuzzy logic is possible in numerous applications in business.

The future trends that are expected from the soft computing technologies, which may satisfy these needs, are: new fuzzy models and their combinations. Research will be focused on various applications to allow decision making in business to be

quicker and more precise because the amount of data to be processed is increasing exponentially. More and more decision making will be done by automatic systems without the influence of human beings. These automatic systems must be designed to be robust and failure-tolerant.

The development of quick, more precise, partly or fully automated decisionmaking systems is where soft computing methods will be used. They will save time, decrease wrong decisions, avoid human failures, bring cost reductions that can lead to higher profit, or decrease business expenses, and they can help firms compete successfully.

The complementarity of fuzzy logic with business needs has an important consequence: in many cases a problem can be solved most effectively by using this method. The rapid growth in the number and variety of applications of fuzzy logic methods, together with the increasing number of researchers and institutions using these methods, demonstrate their value and suggest that their impact will be felt increasingly in the coming years.

12.6 APPENDIX

12.6.1 Program 12.1

```
clear all
B1v = readfis('B1.fis');
UdajB1 = input('Input values in the form [I3a; I3b; I3c]: ');
VyhB1 = evalfis(UdajB1, B1v);
B2v = readfis('B2.fis');
UdajB2 = input('Input values in the form [I4a; I4b, I4c]: ');
VyhB2 = evalfis(UdajB2, B2v);
BFv = readfis('BF.fis');
UdajBF=input('Input values in the form [I1;I2]: ');
UdajBF(3) = VyhB1;
UdajBF(4) = VyhB2;
VyhBF = evalfis(UdajBF, BFv);
if VyhBF<0.5 'Low risk'
    elseif VyhBF<0.8 'Medium Risk'
    else 'High risk'
end
fuzzy(BFv)
mfedit(BFv)
ruleedit(BFv)
surfview(BFv)
ruleview(BFv)
```

Prog. 12.1 *M*-file *BR.m*

12.6.2 Program 12.2

```

b2 = readfis('LD.fis');
Data = input('Set up the values: Costs; Time;
            Success; MProfit;
            NProfit; [Costs; Time; Success; MProfit; NProfit]: ');
result = evalfis(Data, b2);
fuzzy(b2)
mfedit(b2)
ruleedit(b2)
surfview(b2)
ruleview(b2)
result
if result<0.20 'Reject'
    elseif result<0.50 'Entertain'
    else 'Accept'
end

```

Prog. 12.2 *M-file LD.m*

The first row reads the parameters of the fuzzy model into variable *b2* using the command *readfis* from file *LD.fis*. The second row reads the five input variables. The fourth row contains the command *evalfis*, which evaluates the input variables *Data* and *b2*. The output is in the variable *Result*. The following command *fuzzy(b2)* enables setting up a fuzzy model, the command *mfedit(b2)* enables setting up of membership functions, the command *ruleedit(b2)* enables setting up fuzzy rules, the command *surfview(b2)* enables the graphical search of dependence of input and output variables, and the command *ruleview(b2)* enables evaluating the output on the basis of inputs. The command *Result* produces the final value. The following four rows provide a vague result. If the output value *Result* is in the range $R \in (0, 20)$, the result is to reject the judicial process. If the output value *Result* is in the range $R \in (20, 50)$, the result is to entertain the judicial process, and if in the range $R \in (50, 100)$, the result is to accept the process.

12.6.3 Program 12.3

```

clear all
B1v = readfis('B1.fis');
UdajB1 = input('Input values in the form [I1a; I1b; I1c; I1d]: ');
VyhB1 = evalfis(UdajB1, B1v);
B2v = readfis('B2.fis');
UdajB2 = input('Input values in the form [I2a; I2b, I2c]: ');

```

```
VyhB2 = evalfis(UdajB2, B2v);
B3v = readfis('B3.fis');
UdajB3 = input('Input values in the form [I3a; I3b, I3c, I3d]: ');
VyhB3 = evalfis(UdajB3, B3v);
BFv = readfis('BC.fis');
UdajBF(1) = VyhB1;
UdajBF(2) = VyhB2;
UdajBF(3) = VyhB3;
VyhBF = evalfis(UdajBF, BFv);
if VyhBF<0.3 'Low risk'
    elseif VyhBF<0.6 'Medium Risk'
    else 'High risk'
end
fuzzy(BFv) mfeedit(BFv)
ruleedit(BFv)
surfview(BFv)
ruleview(BFv)
```

Prog. 12.3 M-file *BF.m*

12.6.4 Result 12.1

Input values in the form [I3a; I3b; I3c]: [0;0;0;0]

Input values in the form [I4a; I4b I4c]: [0;0;0]

Input values in the form [I1;I2]: [0;0;0;0]

ans =Low risk

Input values in the form [I3a; I3b; I3c]: [1;1;1;1]

Input values in the form [I4a; I4b I4c]: [1;1;1]

Input values in the form [I1;I2]: [1;1;1;1]

ans =High risk

Input values in the form [I3a; I3b; I3c]: [0.5; 0.5; 0.5;0.5]

Input values in the form [I4a; I4b I4c]: [0.5; 0.5; 0.5]

Input values in the form [I1;I2]: [0.5; 0.5;0.5;0.5]

ans =Medium risk

Res. 12.1 Results

12.6.5 Program 12.4

```
fd=xlsread('FC.xlsx','data');
plot3(fd(:,1),fd(:,2), fd(:,3), 'o','color','k', 'markersize',7)
```

```

title('Products Evaluation');
xlabel('Price');ylabel('Sell');zlabel('Quality')
grid
[center,U,objFcn] = fcm(fd,3);
figure
plot(objFcn)
title('Fitness Function Values')
xlabel('Iteration Count')
ylabel('Fitness Function Value')
maxU = max(U);
index1 = find(U(1, :) == maxU);
index2 = find(U(2, :) == maxU);
index3 = find(U(3, :) == maxU);
figure
centerc1='◆'
fd(index1,:)
c2='×'
fd(index2,:)
c3='*'
fd(index3,:)
plot3(fd(:,1),fd(:,2), fd(:,3), 'o','color','k', 'markersize',7)
hold on
grid
stem3(center(1,1),center(1,2),center(1,3),'marker', 'x','color','g',
'markersize',10,'LineWidth',2)
stem3(center(2,1),center(2,2),center(2,3),'marker', 'd','color','r',
'markersize',10,'LineWidth',2)
stem3(center(3,1),center(3,2),center(3,3),'marker', '*', 'color','b',
'markersize',10,'LineWidth',2)
view(30,30)
line(fd(index1, 1), fd(index1,2), fd(index1,3), 'linestyle','none',
'marker', '+', 'color','g');
line(fd(index2,1),fd(index2,2), fd(index2,3), 'linestyle','none',
'marker', 'd', 'color','r');
line(fd(index3,1),fd(index3,2), fd(index3,3), 'linestyle','none',
'marker', '*', 'color','b');
title('Products Evaluation');
xlabel('Price');ylabel('Sell');zlabel('Quality')

```

12.6.6 Result 12.2

Iteration count = 1, obj. fcn = 291532.064272
Iteration count = 2, obj. fcn = 185906.876488
Iteration count = 3, obj. fcn = 138486.939181
.....
Iteration count = 46, obj. fcn = 121026.162373
Iteration count = 47, obj. fcn = 121026.162362
Iteration count = 48, obj. fcn = 121026.162354

center =

136.1827	198.7095	37.1483
68.6142	92.1632	24.7720
111.8004	515.3442	34.4497

c1 = ♦

ans = 93	200	64
40	228	40
180	107	25
196	220	57
80	174	21
120	223	18
384	147	39
81	178	17

c2 = ×

ans = 8	81	8
60	62	2
46	78	46
42	102	31
131	65	32

*c3 = **

ans = 310	602	84
78	445	5
107	494	46
10	570	15

Res. 12.2 Results

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CHAPTER 13

MODELING OF DESIRABLE SOCIOECONOMIC NETWORKS

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13.1 INTRODUCTION

Socioeconomic networks involve large numbers of information transmissions and interactions and largely determine how individuals behave and interact. Individual behaviors are influenced by friends, acquaintances, and neighbors, and the relationships between individuals form the basis of social networks (Jackson 2008).

Understanding the relations between the different levels of macroscopic phenomena has been possible using the tools and insights generated in socioeconomic network research. In particular, much is known about measuring the topological properties of such networks. A number of works have focused on the rules of network formation and the relation between the network creation process and the structure of the resulting network (e.g., Barabási et al. 2000; Durrent 2007).

Yet little is known about the dynamic processes of networks and the dependence of these processes on network properties. Among the research concerning dynamic processes, diffusion (such as that of innovation and disease) has received the most attention (Colizza et al. 2006). Many real-world processes are diffusive in nature and as a consequence lead to optimization problems, in which the goal is to maximize or minimize the spread of some entity through a network.

Entire classes of problems ranging from the spread of epidemics to the control of cascade failures are naturally defined as optimization problems. Other aspects such as the unexpected robustness observed in some networked systems do not involve a priori optimization conditions and yet reveal enhanced properties shaped by the evolution of the system. Desirable performance or optimal behavior is most often connected to a function that the system performs. For example, positive spreads such as innovation

and useful information should be maximized. One individual's adoption of a certain marketed product may trigger his or her friends to adopt that product as well. In this case good cascading should be maximized. However, negative spreads such as rumors and infectious diseases should be minimized. The large blackouts caused by cascade failure should be also minimized.

Examples of the network optimization approach are the maximizing of good cascading and the mitigating of blackouts. In these cases, instead of looking at the details of a particular cascade of innovation or blackouts, we investigate the cascade's dynamics as a series of individual choices or blackouts using decision- or risk-propagation models. In many cases, the issue of interest depends only on certain network topologies that characterize individual interactions.

Socioeconomic networks differ from technological and physical networks in that the former are characterized as loosely coupled networks having multiple facets with multiple connections between nodes. This chapter explores an approach to designing desirable networks in multidimensional and loosely coupled networks. Intrinsic attributes of a socioeconomic network are production, trading, exchange, and communication. These attributes require a desirable property. The priority for desirable socioeconomic networks is stability, which is to be crudely modeled using the idea of network equilibrium. The condition of stability is not enough, however; we need other criteria such as efficiency and equity. In economics, efficiency means that nothing gets wasted. It follows the Pareto optimality in taking the absence of waste to be equivalent to the requirement that nobody can be made better off without someone else's being made worse off (Binmore 2001). Efficiency stands for the measurement of the desirability of socioeconomic networks at the macro level. Equity stands for the measurement of the desirability at the node level.

We examine the relation between the optimal pattern of linkages derived and the incentives of the individual nodes to form those linkages. We then explore the question whether social welfare is aligned with individual payoffs. To model these strategic considerations involved in the creation and destruction of links, and the network-formation game can be assumed (Jackson and Yariv 2014). In this chapter, however, we only focus on efficiency based on Pareto optimality or risk minimization. Since most of socioeconomic networks have particular roles, the principle for designing desirable networks should be discussed in specific network contexts.

We begin by reviewing studies that illustrate the impact of the structure of socioeconomic networks on macroscopic diffusion patterns. The discussions are divided into two classes. First, we mention probabilistic diffusion processes such as the spread of disease or rumors. Many diffusion processes are probabilistic in the sense that once a node switches from one state (non-adopted or non-infected) to another (adopted or infected), it also influences the state of each linked node independently; thus probabilistic diffusion is also referred as independent diffusion. For this probabilistic type of diffusion, we focus on the correlations between the connection patterns (network topology) and the observable diffusion rate at the macroscopic or network level.

The second class is threshold-based diffusion such as the spread of innovation or failures. Here, a few active nodes (people) efficiently trigger the activation of their neighbors. Not all propagation is the simple activation of nodes after exposure to multiple active neighbors, however. This type of diffusion is modeled using a threshold rule. Each node has a threshold that determines the proportion of neighbor nodes required to activate it.

Network interdependencies are big concerns because they can cause failure event cascades. Examples include traffic congestion, electrical power system blackouts, and financial crises. These chain reaction phenomena are known as cascade failures, avalanches, domino effects, or systemic risks. The most common feature of cascade failures is that a local failure results in a larger-scale failure. Such chain reactions of failures are ubiquitous in socioeconomic networks.

The basic mechanism behind cascade failure is a feedback loop or a vicious circle that often induces undesired side effects. Usually a failure in one node is absorbed by neighboring nodes. But if a failure of a node propagates to other nodes under certain conditions, problems increase all along the chain until the networked system ends up with a widespread failure. Another mechanism behind cascade failure is the buildup of a critical load that sharply increases the risk of reaching the threshold for cascade failure (Lorenz et al. 2009).

We use a general framework of diffusion or contagion models to describe risk propagation in networks such as epidemic spread, and we investigate the impact of network topology on epidemic propagation patterns. At the macroscopic level, systemic risk is measured as the fraction of infected nodes.

In this chapter we discuss the power of optimization for designing desirable network structures. First, we present the guided evolution of networks subject to network density constraints. To explain the formation of optimal probabilistic diffusion networks, we investigate the constrained evolution of networks toward optimal diffusion and synchrony. Link density constraints are included via a fitness function that combines the desired goal (i.e., the maximum diffusion of innovation or minimum risk propagation on the network) with the requirement of an average degree. Second, we present the comparative studies by means of simulation of the formation of optimal cascade networks based on threshold rules.

13.2 DIFFUSION OF INNOVATION AND RISKS

For decades, social scientists, economists, and physicists have been interested in the question of how infectious diseases, new technological practices, or the latest trends spread through society (Ball 2004; Buchanan 2007). Individuals pass information to other individuals. An individual who knows a piece of information will share this information with friends or colleagues. Whether the information is diffused throughout a society depends primarily on the probability that it will be passed from one individual

to his or her interaction partners. The fundamental question is under what conditions the information spreads throughout society.

Many stylized facts about the role of social network structure in information transmission are well understood in the literature about the diffusion of disease (Helbing 2010). For example, the dynamics of information transmission is also a threshold phenomenon; below a certain value the information spreads only to a limited group, but above the threshold the information penetrates throughout the entire society.

Networks increase interdependence, which creates challenges for managing risks. This is especially apparent in areas such as finance and enterprise risk management, where the actions of a single node (actor) in a network can impact all the other nodes in the network (Allen and Gale 2000). Network theory is based on a wide variety of phenomena, including networks of friends, transmitters on a power grid, and the spread of disease. In recent years, scientists have made major advances in understanding the nature of networks. We now know that the structure of a network is important in understanding how diffusion occurs.

When a new technology appears, the members of a society have the chance to become aware of it and incorporate it into their lives. A fundamental question raised in these studies is how to understand the ways in which new ideas, behaviors, and practices spread throughout socioeconomic networks (Arthur 1989).

The foremost study on diffusion modeling is the Bass model (Bass 1969). The Bass diffusion model describes the process of adopting new products through interaction between users and potential users. When an innovation is a commodity consumed by individuals, single consumers can decide whether to adopt it.

Rosenberg (1972) observed two dominant characteristics of the diffusion process: the overall slowness of the process and the wide variations in the rates of acceptance of different inventions. Empirical measurement and study have since confirmed this view, although the following questions remain to be solved. Why is diffusion sometimes slow? Why is it faster in some regions than in others? Why do rates of diffusion differ among different types of innovations? In the modern world, markets occasionally accept innovations very slowly, despite advances in technology (Chakravorti 2003).

The diffusion of innovations, new ideas, or rumors can be modeled after the spread of infectious diseases (Meyers et al. 2005). Because epidemiological models describe infection in terms of transmissibility and susceptibility, however, such models cannot adequately describe diffusion processes that typically involve individual information processing and decision making. People are usually affected by word-of-mouth information, which is referred to as a social influence. What influences people to participate in the diffusion process? How do consumers choose which items to buy? These questions are constantly in the minds of many scientists in different fields. Various models of human behavior span the extremes from simple and easily influenced behavior to interdependent and complex behavior, in which nothing is easily controlled or predicted.

Spielman (2005) defines an innovation system as a network of agents, along with the institutions, organizations, and policies that condition their behavior and performance

with respect to generating, exchanging, and utilizing knowledge. An innovation system reflects one aspect of value chain analysis, which brings actors together in the application of knowledge within a value chain. This definition highlights the need for a holistic view of the nature and the structure of interactions among agents linked to one another within networks. The adoption of an innovation by one agent in a network can have positive or negative impacts on other agents' behaviors, which are often unintentional or unpredictable.

New ideas, products, and innovations often take time to spread, a fact that is often attributed to the heterogeneity of human populations. Global markets are now larger and more complex than ever before, and so the amount of information available to consumers has increased considerably. As a result, consumers generally spend much time thinking and hesitating before making a decision. They may realize different aspects of the benefits and costs of a particular innovation, have different beliefs regarding the benefits and costs, hear about the innovation at different times, or delay in acting on the information they receive.

Young (2009) analyzed the effects of heterogeneity as three broad classes of models: contagion, social influence, and social learning. When a consumer has many neighbors, these models represent many potential sources of information, and the consumer may have problems effectively handling them all.

Some research on innovation diffusion have focused on empirical studies showing the crucial elements of the structure of social interactions and computational models for investigating the patterns of innovation diffusion via social networks (Rogers et al. 2005).

The diffusion process enhances an innovation via the feedback of information among different users. This feedback, which can then be used to improve the innovation, is similar to the micro-macro loop, which is an essential part of emergence. In this chapter we provide an overview of research that examines the impacts of the social network structure on emergent properties in diffusion processes. Most human social activities are substantially free of centralized management, and although people may care about the end result of the aggregate, individual decisions and behaviors are typically motivated by self-interest.

In banking systems, crises tend to spread from one institution to another. In one example, Allen and Gale (2000) introduced the use of network theories to enrich our understanding of financial systems and studied how financial institutions connected with different network topologies respond to contagion. This tendency is referred to as systemic risk and, at a sufficiently large scale, leads to systemic failure. Some sources of systemic failure have been noted in the literature to explain the mechanism of bank failures during financial crises. Allen and Gale also explored several critical issues related to systemic risk by studying the resilience of financial networks against contagion and the connections formed between financial institutions when they are exposed to the risk of contagion. They showed that though the existence of more links between banks might be expected to increase the risk of contagion, densely connected banking systems may be less susceptible to contagion than those with a

sparse network structure with few connections. They also showed that incomplete networks are more prone to contagion than are complete structures. For example, better-connected networks are more resilient, since the proportion of losses in one bank's portfolio is transferred to more banks through interbank agreements. In contrast, in the case of an incomplete network, the failure of one bank could trigger the failure of the entire network.

To understand the relation between microscopic behavior and macroscopic patterns of interest, we must look at the system of interactions among agents. This system can be described as interaction topology. We focus on the stylized facts of macroscopic emergent phenomena that are the results of bi-directional interactions. Social networks' topologies determine a basic and important form of social interactions among agents, and the microscopic behaviors of agents largely determine the diffusion patterns observed at the macroscopic level.

As a result, we now have a better understanding of the emergent properties of the desirable aggregate process. The actions of humans are often influenced by friends, acquaintances, or neighbors. In other words, social networks are an important form of social interaction. We conclude that the topological properties of networks should be derived from the dynamics from which the networks were created. A large number of interactions is not enough to guarantee the emergence of desirable properties. In some cases, the diffusion process may need to reach a certain critical mass or a combined threshold of diversity and connectivity before such properties appear. We discuss this issue using an optimization approach to network design.

13.3 PROBABILISTIC DIFFUSION

The susceptible-infected-recovered (SIR) model is widely used as a basic diffusion model (Colizza et al. 2006). In this model, nodes are initially susceptible to a disease and can become infected by infected neighbors. Once infected, a node continues to infect its neighbors until it is randomly removed from the system. The SIR model presumes that, once infected, a node will eventually infect all of its susceptible neighbors. Infected nodes can either recover and stop transmitting the disease or die and completely disappear from the network. Another widely used model is the susceptible-infected-susceptible (SIS) model, in which infected nodes can randomly recover, but, after recovery, they are once again susceptible to infection. The SIS model corresponds well with many real-world viral infections that cause individuals to transition back and forth between health and illness (Meyers et al. 2005).

For simple diffusion such as the spread of disease or information, in which a few active nodes (people) efficiently trigger the activation of their neighbors, hub nodes with many connections or random links connecting distant nodes allow a wide diffusion. In diffusion of innovation the probability that an individual adopts a service or product is a function of the number of neighbors who have adopted the service

or product (Immorlica et al. 2007). Naturally, if a person has more friends already using a certain service or product, he or she is more likely to adopt the service or product.

Note that while this probabilistic analysis observes that highly linked nodes are more likely to become infected, the aggregate behavior does not exhibit the S-shape common to many real-world diffusion processes. This analysis provides some insight into the kinds of social structures necessary in order for an infection to spread, but it does not provide an accurate figure of the extensiveness of the infection and the effect of the network structure on the infection rate.

Regarding the diffusion of innovation, consumers have neither awareness of nor preference for a product newly launched. After exposure to information about the product by word of mouth by peers connected on the network, each consumer becomes aware of it with some probability β . However, a consumer with awareness may forget (lose the awareness) with some probability δ .

Consumers with awareness transmit the information to each peer directly connected on the network with some probability β . For the sake of simplicity, this probability is assumed to be constant, although in a more realistic situation receptivity might vary depending on the receiver. We focus only on positive or at least neutral word of mouth. Consumers with awareness but no preference form a preference with some probability if exposed to information verbally.

This probability is assumed to be a decreasing function of the number of exposures the consumer has experienced by that time. Consumers who hold a preference in memory can make a decision without the temporal word-of-mouth stimulus. By definition, once a consumer adopts a preference, that consumer cannot do it again, even when maintaining awareness and preference. Preference is completely lost only when forgetting occurs.

Suppose that a number of consumers mutually transmit information about a newly launched product through a social network in which they are involved. Graph G can be represented by its adjacency matrix, $A = (a_{ij})$, $1 \leq i, j \leq n$, which is a real symmetric matrix where $a_{ij} = a_{ji} = 1$ if vertices i and j are connected, or $a_{ij} = a_{ji} = 0$ if these vertices are not connected. The spectrum of the graph is the set of eigenvalues of the graph's adjacency matrix $A = (a_{ij})$.

The diffusion of information, rumors, or gossip through social networks can be modeled after the spread of infectious diseases (Boyd et al. 2006). In the spread of a disease, some nodes (individuals) are infected by exogenous sources, and consequently some of these individuals' neighbors are infected through contact. A given node can be immune; however, if a node is not immune, then it is sure to catch the disease if one of its neighbors is infected.

We can model probabilistic propagation as a dynamic birth-death process (Wang and Chakrabarti 2003). For example, an informed agent i propagates the information to another agent j in a single step with probability β , while an informed agent i may forget or lose interest with probability δ (Fig. 13.1). We denote the probability that agent i is aware of the information at time t as $p_i(t)$. The column vector

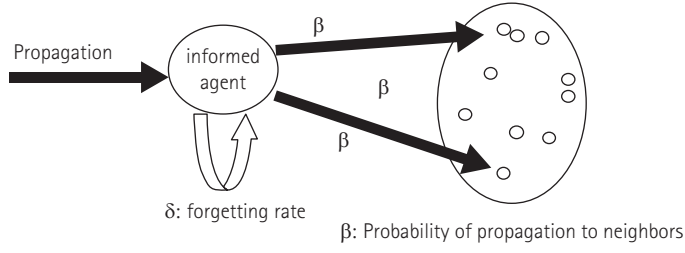


FIGURE 13.1 Probabilistic diffusion.

$p(t) = (p_1(t), p_2(t), \dots, p_n(t))$ represents the awareness probabilities of the agent population. The transition of the awareness probabilities is then described as

$$p(t+1) = (\beta A + (1-\delta)I)p(t), \quad (13.1)$$

where I is an $n \times n$ identity matrix. The long-term behavior of the above system determines the structure of the system matrix $S = \beta A + (1-\delta)I$. Wang and Chakrabarti (2003) proved that the spectrum of the system matrix S (the distribution of eigenvalues of S) is closely related to the spectrum of the adjacency matrix A , and we obtain

$$\lambda_i(S) = \beta \lambda_i(A) + (1-\delta), i = 1, 2, \dots, n, \quad (13.2)$$

where $\lambda_i(S)$ is the i th largest eigenvalue of the system matrix S . The largest eigenvalue $\lambda_1(S)$ is also called the principal eigenvalue of the system matrix. We denote the largest eigenvalue of the system matrix S as $\lambda_1(S)$, with the following properties.

1. if $\lambda_1(S) < 1$, then $p(t)$ converges to the zero vector.
 2. if $\lambda_1(S) = 1$, then $p(t)$ converges to some constant vector.
- (13.3)

From the relation shown in equation (13.2), the conditions given in (13.3) in terms of the largest eigenvalue of system matrix S can be expressed by using the largest eigenvalue of adjacency matrix A :

3. if $\beta/\delta < 1/\lambda_1(A)$, $p(t)$ converges to the zero vector.
 4. if $\beta/\delta = 1/\lambda_1(A)$, $p(t)$ converges to some constant vector.
- (13.4)

From this we know that the diffusion process is governed by the threshold phenomenon, and diffusion spreads throughout the entire network only if

$$\beta/\delta > 1/\lambda_1(A). \quad (13.5)$$

Therefore, a high relative transmission rate (β/δ) leads to the possibility of a wide diffusion, as does a degree distribution with a high largest eigenvalue. The rationale behind the high largest eigenvalue of adjacency matrix is that some densely connected

hub nodes will have a high diffusion power, which may foster the contagion and accelerate the diffusion.

13.4 THRESHOLD-BASED DIFFUSION

For probabilistic diffusion such as the spread of disease or rumors, in which a few active nodes (people) efficiently trigger the activation of their neighbors, hub nodes with many connections may allow drastic diffusion. Not all propagation works in this way, however. Another type of diffusion is modeled using a threshold rule. Each node has a threshold that determines the proportion of neighbor nodes required to activate it.

In the simplest form of threshold-based diffusion, we model a set of agents, each of which faces a binary decision: buy the product (*A*) or do not buy (*B*). These agents are endowed with their own idiosyncratic preferences concerning the product. Regarding individual preferences, we assume that each agent isolated from social influence would objectively buy the product of interest (choose *A*) with preference level h_i , which represents the willingness of agent i to pay for the product of interest. In addition, each agent receives social influence, or network externality, from other agents. A binary decision under social influence is also referred to as a binary decision with network externality. Therefore, agents' preferences evolve by receiving social influence or network externality from other agents.

Embedding idiosyncratic preferences as the willingness to pay h_i minus the product price p , and social influence or network externality, we define each agent's utility function, $i = 1, 2, \dots, n$, as

$$y_i(t) = h_i - p + \frac{w}{k_i} \sum_{k \neq i} w_{ik} x_k(t), \quad (13.6)$$

where w_{ik} takes the binary value and agent i is connected to agent k , $w_{ik}(=w_{ki}) = 1$, otherwise $w_{ik}(=w_{ki}) = 0$, and k_i is the average degree of agent i and is some positive constant. We define the binary decision variable of agent i by x_i , ($x_i = 1$ if he or she decides to buy, $x_i = 0$ if he or she decides not to buy), and we can derive the following binary decision rule of agent i :

$$x_i = \begin{cases} 1 & \text{if } h_i - p + \frac{w}{k_i} \sum_{k \neq i} w_{ik} x_k \geq 0 \\ 0 & \text{otherwise.} \end{cases} \quad (13.7)$$

If h_i , the willingness to pay of agent i , is lower than the product price p (i.e., $h_i - p$ is negative), he or she does not buy in an isolated case. When agent i also cares about neighbor agents, he or she receives social influence from them, which represents the third term in equation (13.7), in which N_i represents the neighbors and d_i is the number of the neighbors (the cardinal number of N_i or the degree) of agent i . In this case some agent decides to buy even if willingness to pay is lower than the price of the product because he or she received positive externality from the neighbors' decisions.

Table 13.1 Payoff matrix of the binary choices between A and B

Own choice \ Others' choices (ratio of choosing A or B)	A(F)	B(1 - F)
A	$1 - \theta$	0
B	0	θ

We can also describe a technology diffusion scenario based on the binary decision of agents in terms of the following network environments. Suppose two technologies A and B need to adopt the same technology to communicate. A social network of agents G exists that governs who the agents will speak with. Each edge, or a pair of agents (v, w) of G plays a coordination game with two strategies, A (new technology) or B (old technology): if v and w each choose A, then each receives a payoff of $1 - \theta$ (they can talk to each other using technology A); if they each choose B, each receives a payoff of θ . However, if they choose opposite technologies, then each receives a payoff of 0, which reflects the lack of interoperability.

Diffusion of innovation can be also analyzed by formulating the coordination games on a social network. The payoff matrix is given in table 13.1. We define the binary decision variable of agent i by x_i , ($x_i=0$ if B is chosen, $x_i=1$ if A is chosen). When the ratio of neighbor agents who adopt A is represented by $F = \sum_{k \neq i} w_{ik} x_k$, the optimal strategy of each agent is represented as

$$\begin{array}{ll} \text{If } F > \theta \text{ then} & \text{choose A} \\ \text{otherwise} & \text{choose B.} \end{array} \quad (13.8)$$

Agents choose A or B according to their own preferences (represented by a payoff θ) but at the same time they consider the influence level of the neighbor agents represented by F . This interdependence is described as the threshold rule in (13.8).

In a threshold-based diffusion model such as diffusion of innovation that involves individual decisions, the diffusion process enhances innovation via the feedback of information about the innovation's utility across users. This feedback can then be used to improve the innovation. In this case, diffusion processes are nonprogressive in the sense that as time progresses, agents switch from state A to B or from B to A, depending on the states of their neighbors.

Let us suppose that all agents (nodes) initially adopt B. Thereafter, one or a very few agents begin adopting new technology A instead of B. If we apply best-response updates

to the agents in the network, then the agents will in effect repeatedly apply the binary rule in (13.8) and switch to A if the proportion of neighbors F who have already adopted A is larger than the threshold θ .

In most settings that form the motivation for diffusion models, coexistence is the typical outcome: the coexistence of the agents may result by partitioning the agents into a set adopting new technology A and a set adopting existing technology B . These qualitatively different types of equilibrium arise in terms of the underlying network topology and the quality of $A(1 - \theta)$ relative to $B(\theta)$. An important feature that is arguably missing from the basic game-theoretic models of diffusion, however, is a more detailed picture of what is happening at the coexistence boundary, where the model assumes that nodes that adopt ($x_i = 1$) are linked to nodes that do not adopt ($x_i = 0$) (Watts and Dodds 2007).

Networks increase interdependency, creating challenges for risks management. This is especially apparent in risk management, in which the actions of a single actor in an interconnected network can impact all the other actors in the network. Diffusion of financial risk over a financial network can be also modeled as threshold-based using the diffusion rule given in (13.8) (Gai and Kapadia 2010). Therefore, the network is only as strong as its weakest link, and trade-offs are most often connected to a function that models system performance management.

A cascading sequence of agents switching to A from B may result in reaching the limit, and almost all agents in the network come to adopt A by switching from B . The condition of causing a cascade depends on both the threshold of each agent and the topology of underlying network.

A cascade threshold: θ_c^* There exists a threshold θ_c^* for occurring cascade such that a large proportion of agents choose $x = 0$ with an infinitesimally small fraction of agents choosing $x = 1$ if $\theta < \theta_c^*$.

We can investigate a cascade condition by using mean-field analysis (Watts 2002). We describe the degree distribution, the agent network connectivity, by P_k . Some typical examples of degree distributions are given in (13.19), (13.20), and (13.21). First, we define an individual who can be activated by just one contact as being active, and we define such an individual as a vulnerable node. A node having degree k and the threshold value θ is called a vulnerable node if these two values satisfy:

$$k < \frac{1}{\theta} \quad (13.9)$$

A cascade progresses through the network when vulnerable nodes are connected to each other. Therefore, we consider the condition that these nodes are connected as a cascade. Specifically, an active link, which is a link connected to an activated node, needs to lead to at least one more active link in order for a cascade to progress. The possibility of a node that is connected to an active link that has k links is given as follows:

$$P(\text{linked node has } k \text{ links}) = kP_k / \langle k \rangle, \quad (13.10)$$

where $\langle k \rangle$ is an average degree used for normalization:

$$\sum_{k=0} k P_k = \langle k \rangle \equiv z. \quad (13.11)$$

The probability β_k that the linked degree- k node is vulnerable is as follows:

$$\beta_k = \int_{\theta=0}^{1/k} f(\theta) d\theta, \quad (13.12)$$

where $f(\theta)$ is the distribution of agent threshold θ .

If a linked node is vulnerable, it produces $k - 1$ new outgoing active nodes, and if a linked node is not vulnerable, it produces no active nodes. The expected number of active nodes produced by an active node is given as follows:

$$\sum_{k=1}^{\infty} \left\{ (k-1) \beta_k \frac{k P_k}{z} + (1 - \beta_k) \frac{k P_k}{z} \right\} = \sum_{k=1}^{\infty} (k-1) \beta_k k P_k / z. \quad (13.13)$$

Then, we have the following cascade condition:

$$\sum_{k=1}^{\infty} (k-1) \beta_k k P_k / z \geq 1. \quad (13.14)$$

This cascade condition depends on the distribution of the network connectivity.

López-Pintado (2006) proved that a critical threshold can be obtained by solving the following minimizing problem:

$$\begin{aligned} \theta_c^* &= \arg \min_{\theta} \sum_{k=1}^{\lceil 1/\theta \rceil} k^2 P_k / z \\ \text{subject to } &\sum_{k=1}^{\lceil 1/\theta \rceil} k^2 P_k / z \geq 1. \end{aligned} \quad (13.15)$$

Then she obtained and compared the minimum levels of threshold that cause cascade on a homogeneous regular network with connectivity distribution $P_H(k)$, an exponential random network with connectivity distribution $P_E(k)$, and a scale-free network with connectivity distribution $P_{SF}(k)$. The minimum cascade thresholds on these three networks are represented as θ_H , θ_E , and θ_{SF} , respectively. These minimum threshold have the following relations:

$$\theta_H \leq \theta_{SF} \leq \theta_E. \quad (13.16)$$

(The equality holds only for $k = 2$). Therefore, a scale-free network, which is extremely important in epidemiology, does not properly support threshold-based contagion. This type of contagion is better accomplished by a more intermediate variance network such as an exponential random network. On the other hand, a homogeneous network with the lowest threshold is the most unsuccessful network for a cascade.

13.5 DESIRABLE NETWORKS UNDER PROBABILISTIC DIFFUSION

In section 13.3 we formulated the probabilistic diffusion based on the epidemiological model, which describes information spread in terms of transmissibility and forgetting at the individual level. We showed that the largest eigenvalue of the adjacency matrix A of the underlying network should be maximized if we want to maximize the diffusion. Similarly, we also must seek the desirable network topologies for minimizing diffusion of such things as diseases and bad rumors. The network diffusion property is determined by the largest eigenvalue of the adjacency matrix $A = (a_{ij})$ of the underlying network, and it should be minimized if we want to minimize the diffusion.

The task of designing desirable networks is complex and may pose multiconstraint and multicriterion optimization problems. In what follows we present a genetic optimization approach to designing an optimal network for minimizing risk contagion and for maximizing propagation. Traditionally, optimization has a particular mathematical definition that focuses on obtaining solutions that strictly optimize a well-defined function. Here we adopt a looser definition of the word by extending it to include the tendency of networks to improve their diffusion performance. Socioeconomic networks are large and complex, and accordingly, most attempts to find the optimum solution are not realistic. Finding a “good enough” solution or a “better” solution in an iterative manner is a workable alternative.

The principal eigenvalue of the adjacency matrix $A = (a_{ij})$, where $a_{ij}(=a_{ji}) = 1$, if node i and node j are connected and $a_{ij}(=a_{ji}) = 0$ if they are disconnected, provides useful information for determining how quickly an epidemic-type diffusion progresses. Maximum diffusion is realized if the principal eigenvalue of the adjacency matrix $A = (a_{ij})$ is large. The maximum eigenvalue increases with the increase of the average degree of the network, and the obvious network to maximize the maximum eigenvalue is the completely connected network. Therefore, we formalize the optimization problem with the constraint of the average degree. The average degree of a network with the adjacency matrix $A = (a_{ij})$ is defined as

$$\langle k \rangle = \frac{1}{n} \sum_{i,j=1}^n a_{ij}. \quad (13.17)$$

The function to be minimized is defined as

$$E_1(w) = w(N-1)/(\lambda_1(A)) + (1-w)\langle k \rangle \quad (13.18)$$

eigenvalue, and the average degree. If we increase the weight parameter w by putting emphasis on maximizing the spread, which is equivalent to obtaining the network with the largest eigenvalue $\lambda_1(A)$, we have a dense network with many high-degree nodes.

We obtain and compare the largest eigenvalues of several widely used networks and an evolutionary optimized network. First, a random regular network (RR):

$$p_{RR}(k) = 1, \text{ if } k = \langle k \rangle; \quad (13.19)$$

Second, an exponential regular network (ER):

$$p_E(k) \propto \frac{e^{-\langle k \rangle} \langle k \rangle^k}{k!}; \quad (13.20)$$

and third, the Barabási-Albert scale-free network (BA)

$$p_{BA}(k) \propto k^{-3}. \quad (13.21)$$

In order to obtain the network with the maximum largest eigenvalue, which is defined as KN network under the constraint of the average degree of the adjacency matrix, we can use a heuristic approach. The formation of a KN network consists of two steps at each time step. The first step adds a node to the network with a preferential attachment. The second step adds m links to the network with the preferential attachment. Therefore, at each time step, the network has one new node with a link and m new links (Komatsu and Namatame 2011). The degree distribution of the KN network also obeys power law with the power index $\alpha = 2 + 1/(\langle k \rangle - 1)$. For the KN network with the maximum largest eigenvalue (KN),

$$p_{KN}(k) \propto k^{-(2 + \frac{1}{\langle k \rangle - 1})}. \quad (13.22)$$

We show the largest eigenvalues of the above four networks—a random regular network ($\lambda_1(A_{RR})$), a random network ($\lambda_1(A_E)$), a scale-free network ($\lambda_1(A_{BA})$) and the KN network, $\lambda_1(A_{KN})$ —as the functions of the average degree $\langle k \rangle$, and we have the following relations:

$$\lambda_1(A_{RR}) < \lambda_1(A_E) < \lambda_1(A_{BA}) < \lambda_1(A_{KN}). \quad (13.23)$$

We also consider the opposite optimization, in which we want to design a network that minimizes diffusion such as risk or failure propagation. In this case the network with the minimum largest eigenvalue $\lambda_1(A)$ is preferred.

Designing desirable networks for minimizing diffusion also poses multicriterion optimization problems. To illustrate, we also consider optimization under link constraints. The objective function in this case is to define the inverse of the function for maximizing diffusion. Specifically, we define the following function to be minimized.

$$E_2(\omega) = \omega(\lambda_1(A)/(N-1)) + (1-\omega)1/\langle k \rangle \quad (13.24)$$

Management strategies can be risk-reducing as well as risk-sharing measures, and $w(0 \leq w \leq 1)$ is a parameter controlling the two minimizing objects: largest eigenvalue

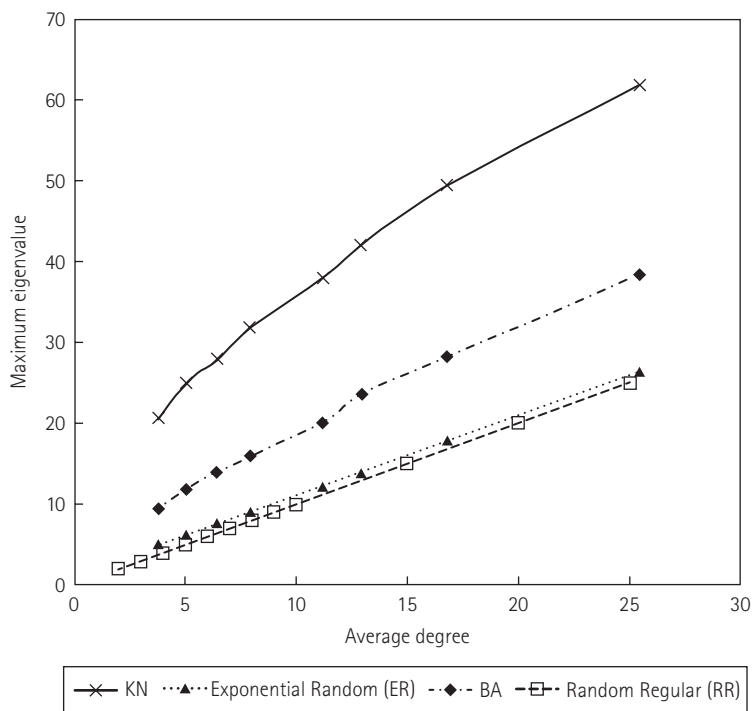


FIGURE 13.2 Largest eigenvalues of different network topologies as the function of the average degree: the KN network (KN), the B-A scale-free network (BA), an exponential random network (ER), and a random regular network (RR).

and the average degree. For a symmetrical matrix A , all eigenvalues are real numbers and they can be arranged as

$$\lambda_{\min} = \lambda_n \leq \lambda_{n-1} \leq \lambda_{n-2} \leq \dots \leq \lambda_1 = \lambda_{\max}. \quad (13.25)$$

Furthermore, we have the following theorem known as Rayleigh-Ritz inequality:

$$\lambda_n \mathbf{x}^* \mathbf{x} \leq \mathbf{x}^* \mathbf{A} \mathbf{x} \leq \lambda_1 \mathbf{x}^* \mathbf{x}, \quad (13.26)$$

where $A = (a_{ij})$ denotes the adjacency matrix and $\mathbf{x}$ is the arbitrary d -dimensional vector. Therefore, the smallest and largest eigenvalues may be thought of as the solution to a constrained minimum and maximum problem, respectively.

Then, we obtain the following inequality about the average degree $\langle k \rangle$, the maximum largest $\lambda_1(A)$, and the maximum degree $k_{\max}$:

$$\langle k \rangle \leq \lambda_1(A) \leq k_{\max} \quad (13.27)$$

From this result it is easily seen that a random regular network (RR) has the minimum largest eigenvalue and that this type of network is the most desirable for minimizing probabilistic diffusion. Comparative results of the simulation are shown in figure 13.2

for the different network topologies: the KN network with the maximum largest eigenvalue, the scale-free network (BA), the exponential random network, and the random regular network with the minimum largest eigenvalue.

The diffusion on a graph is a threshold phenomenon: if the relative diffusion probability exceeds some tipping point, it starts to diffuse. To compare the diffusion process near a tipping point, on various network topologies, we introduce the score defined as follows (Wang and Chakrabarti 2003):

$$s = \lambda_1(A_{KN})(\beta/\delta). \quad (13.28)$$

This score specifies the condition under which diffusion fades out ($s < 1$) or starts to expand ($s > 1$) on KN network with maximum largest eigenvalue.

Figure 13.3 shows the overall diffusion rates ($\sum_{i=1}^n x_i/n$) of the various network topologies over time for various values of the score s defined in (13.28) in log-log scales. We can observe a clear trend at and below the threshold ($s = 1$), and the diffusion dies out if the diffusion starts above the threshold ($s = 1$) on the KN network with the maximum largest eigenvalue. The diffusion starts with higher scores than the KN network on the B-A scale-free network and on the random network. However, the diffusion starts with the highest score on the exponential random regular network with the minimum largest eigenvalue.

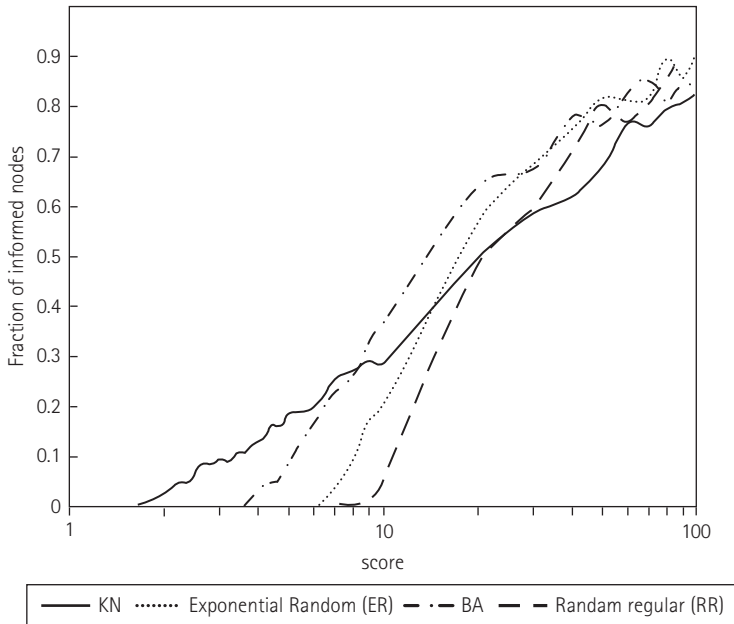


FIGURE 13.3 The ratios of informed nodes ($\rho = \sum_{i=1}^N p_i/N$) as a function of the score defined in (13.23) in different network topologies: a KN network (KN), a B-A scale-free network (BA), an exponential random network (ER), and a random regular network (RR).

Therefore, we observe that diffusion is easier to start with the lowest relative propagation rate β/δ in the network with the maximum largest eigenvalue. However, diffusion requires a much higher score in the random regular network with the minimum largest eigenvalue. Thus diffusion is the most difficult to start and requires a higher relative propagation rate β/δ in the network with the lower largest eigenvalue. These simulation results are consistent with the relations among largest eigenvalues in (13.27).

13.6 DESIRABLE NETWORKS UNDER THRESHOLD-BASED DIFFUSION

To maximize or minimize diffusion in a network under probabilistic diffusion, we need a network (adjacency matrix) with the maximum or minimum largest eigenvalue. In the previous section, we compared the four network types: the KN network with the maximum eigenvalue, the scale-free network (BA), the exponential random network, and the random regular network with the same average degree. We use these four types for the comparative study.

In this section we obtain the desirable diffusion network by comparing cascade based on the threshold rule observed on various network topologies. In threshold-based diffusion, the activation of each node occurs after exposure to multiple active neighbor nodes. Threshold-based diffusion is modeled using a threshold rule, and each node has a threshold that determines the proportion of neighbor nodes required in order for it to spread. Based on the threshold concept, we define a critical threshold as follows.

There are two types of cascade: good cascade and bad cascade. An example former is diffusion of innovation, and an example of the latter is when everybody makes the wrong choice, or cascade failure. The network topology is crucial for determining the dynamics of the cascade. Then we can obtain the most desirable networks, the successful networks for cascade with the maximum cascade threshold θ_c^* in (13.15), or the least successful networks for cascade with the minimum cascade threshold θ_c^* in (13.15). However, the optimization problems for obtaining the maximum and minimum thresholds in (13.15) are analytically hard to solve, and we obtain nearly optimal networks by implementing a genetic algorithm.

Using a genetic algorithm (GA), we obtain the approximately maximum threshold by solving the minimizing problem in (13.15) under the constraint of the average degree. The results for the GA are shown as circles in figure 13.4. By solving the minimizing problem we can obtain the cascade thresholds in different network topologies: a homogeneous random regular network (θ_{RR}) with the degree distribution in (13.19), an exponential network (θ_E) with the degree distribution in (13.20), a scale-free network (θ_{SF}) with the degree distribution in (13.21), and a KN network (θ_{KN}) with the degree distribution in (13.22). These are shown as the functions of the average degree in figure 13.4. Based on these results, there are two main regimes for desirable networks depending on the average degree.

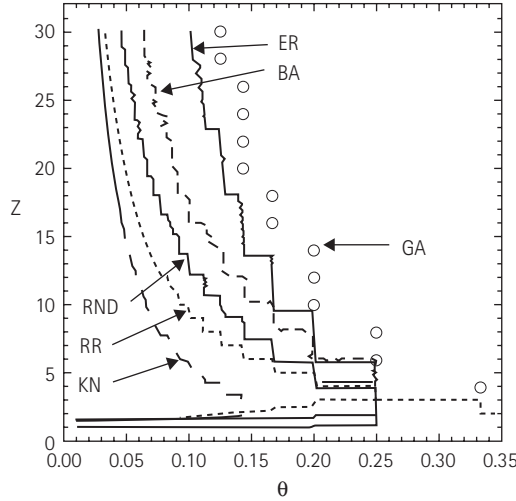


FIGURE 13.4 The cascade window as a function of the threshold value and the networks in different network topologies: an optimized network derived by genetic algorithm (GA), a KN network (KN), a B-A scale-free network (BA), an exponential random network (ER), a random network (RND), and a random regular network (RR).

Regime 1 is a sparse network with a low average degree less than $\langle k \rangle = 4$.

The cascade thresholds of different network topologies have the following inequalities:

$$\theta_{KN} < \theta_{BA} \cong \theta_{ER} < \theta_{RR} \quad (13.29)$$

Therefore, homogeneous networks such as random regular networks are the most successful networks for cascade, and KN network with the maximum eigenvalue is the least successful network for cascade.

Regime 2 is a dense network with a high average degree greater than $\langle k \rangle = 4$.

The cascade thresholds of different network topologies have the following inequalities:

$$\theta_{KN} < \theta_{RR} < \theta_{BA} < \theta_{ER} < \theta_{GA} \quad (13.30)$$

where θ_{GA} is the maximum cascade threshold obtained by GA. Therefore, an exponential network is not the most successful type for cascade, as suggested by López-Pintado (2006) or Watts (2002). There is another class of networks generated by GA with the maximum cascade threshold for cascade with the maximum cascade area. The type of network which maximizes cascade has a log-normal degree distribution. We further extend the previous results in some important respects:

1. A network with a log-normal degree distribution has a larger cascade area (pairs of the cascade threshold and the average degree in the region of dense networks). However, unlike probabilistic diffusion models, in which contagion

events occurring between pairs of individuals are stochastic and independent, a KN network has a smaller cascade area and is the least successful network for cascade in the region of sparse networks.

2. A homogeneous network such as a random regular network has a larger cascade area in the region of sparse networks.

Cascade phenomena depend on the vulnerability of interconnected systems to global cascades on the network of interpersonal influences governing the information that individuals have about the world and, therefore, their decisions. The rationale behind these results is the existence of a giant cluster of vulnerable nodes: a network with the log-normal degree distribution has such a cluster in a region of dense networks. In contrast, a homogeneous network in which all agents are connected to the same number of neighbors has the largest component of vulnerable nodes in the region of sparse networks. On the other hand, cascade-type diffusion is the most difficult to effect on a KN network, and this result differs from the probabilistic diffusion based on epidemic models shown in figure 13.2.

The cascade condition also depends on the initial seed: an initial proportion of those who choose *A*. Then we perform a simulation study by changing the threshold value (θ) of each agent and the initial seed who choose *A* under different network topologies with the same average degree $\langle k \rangle = 4$. In threshold-based cascade, we define the penetration ratio $F(t)$, the ratio of agents who choose *A* at time period t , and determine the agents that choose *A* or *B*. As the initial condition, let us suppose that a very few agents choose *A* (initial value of $F(0)$) and almost all agents choose *B*. If we apply the updated rule in (13.8) to agents in the network, then agents will in effect repeatedly apply the threshold rule: switch to *A* if the proportion of neighbors who have chosen *A* is larger than the threshold θ .

Figure 13.5 shows the simulation results for the different network topologies: the z-axis represents the final ratio of agents who choose *A* (the penetration rate (F^*)), the x-axis represents the threshold θ (all agents have the same value), and the y-axis represents $F(0)$: the ratio of agents who choose *A* initially.

In figure 13.5 we see that the trajectories of $F(t)$, the proportions of agents to choose *A*, stay close to zero in a wide area of thresholds (x-axis) without regarding the ratio of initial adopters (y-axis). We can observe the phase transitions, however, in the diffusion in the network topologies, that is, the situations in which small changes in the threshold of the agent behavioral rule result in big changes in the macroscopic diffusion patterns.

Based on these results, the macroscopic diffusion is a threshold phenomenon in terms of agent threshold and the network density. If the threshold of each agent is smaller than a critical value, cascade spreads and a large proportion of agents choose to adopt; otherwise, almost no agent adopts and it fails to diffuse. The network density is also crucial for determining the cascade dynamics. In the region of dense networks, a heterogeneous network with the log-normal degree distribution promotes good cascade such as diffusion of innovation, whereas in the region of sparse networks a homogeneous regular network is the most successful in promoting cascade.

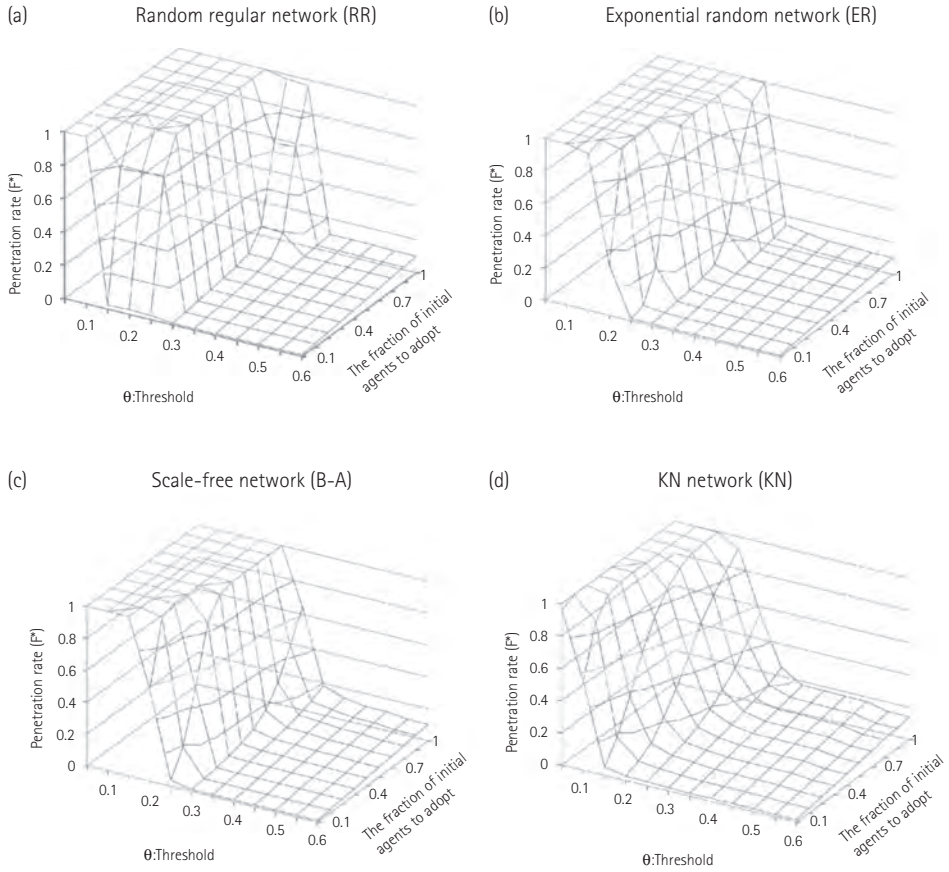


FIGURE 13.5 Figures showing the final penetration rates F^* on each network.

13.7 CONCLUSIONS

Diffusion is the process by which new products and practices are invented and successfully introduced into a society. Which network structures favor the rapid spread of new ideas, behaviors, or technologies? This question has been studied extensively by using epidemic models or cascade models with threshold rules that involve the binary choices of individuals. Diffusion processes have both good and harmful aspects in many real-world phenomena. Therefore diffusion control is a challenging issue.

In this chapter we studied diffusion processes in different network topologies and focused on the design of optimal networks for maximizing good diffusion as well as minimizing bad diffusion. It is important to design good intervention strategies to control both useful and harmful diffusions. Many studies shed light on how network topology interacts with the structure of social networked systems such as those of financial institutions to understand systemwide crises.

We considered two types of diffusion, probabilistic and threshold-based, by focusing on two facts of diffusion on networks. Usually we want to devise efficient strategies that are easy to implement to help the spread of positive or useful information. We refer to these processes as positive diffusions. Earlier works studied this for a variety of models, but the models were primarily based on static networks. The major point that separates our research from previous works is that we consider dynamically changing networks, which extend previous models to a larger range of real-life situations. We especially studied ways to design optimal networks controlled by an evolutionary approach.

Performance and robustness are common properties of socioeconomic systems that we seek to optimize. In this context, entire classes of problems, ranging from fostering diffusion of innovation to the control of cascade failures, fall naturally under the definition of optimization problems. Usually, optimization has a strict mathematical definition that concerns obtaining solutions to well-defined problems. In this chapter we extending that definition to include the tendency of a system to improve its performance as a result of properly imposed objective functions. We proposed a heuristic optimization approach to designing desirable network topologies and verified their desirability with modeling and simulations.

One standard approach to designing desirable networks for maximizing (or minimizing) diffusion is to study the problem of adding edges to a graph so as to maximize its largest eigenvalue. This is a difficult combinatorial optimization, so we sought a heuristic for approximately solving the problem. The largest eigenvalue of the adjacency matrix of a graph is the good measure for designing diffusion networks. We proposed a heuristic method for network based on maximizing or minimizing the largest eigenvalue under the constraint of the average degree. We also proposed a heuristic method that can be applied to very large graphs and investigated cascade dynamics in growing networks.

Network topology also has an impact on other aspects of network performance including robustness and resilience. The increasing scale, complexity, and interdependence between policies and configurations of emerging networks call for a shift from the current human-centric network management toward more autonomic and adaptive mechanisms. However, we still lack the proper methodologies for controlling network topology. Another challenging issue is developmet of an autonomic topology control approach that builds on the concept of self-organization to evolve and adapt the network topology to satisfy dynamically changing requirements. Adaptation may be performed in a decentralized manner in which local nodes maintain neighborhood information and apply local neighbor selection policies in order to shift to the desirable network topology. It is hoped that the combined framework of analyzing diffusion and cascades will stimulate theoretical and empirical efforts to explore network management methodologies.

We can extend our results to further study of questions such as how to devise good strategies for controlling diffusion processes. This problem is crucial when we deal with harmful information such as human diseases or computer viruses. We distinguish between centralized and decentralized intervention strategies. In centralized

intervention strategies, a controller has a limited amount of intervention resources (vaccinations, in the case of diseases). We can also extend this work to the problem of allocating limited resources among the network agents so that the spread of the diffusion process is minimized. In decentralized intervention strategies, individuals in the network make their own decision about protecting themselves, based on their individual utility and local knowledge. In such settings, we are interested in questions such as, Is there a stable set of intervention strategies? What is the cost of decentralized solutions compared with an optimal centralized one? We can also augment the study of intervention strategies with consideration of individual behavior changes that would lead to a new kind of network dynamics. Earlier work showed that the combination of behavioral change and intervention failure can lead to perverse outcomes in which less intervention is more effective. Yet, the extent of the perversity, its dependence on network structure, and the precise nature of the behavior change remain largely unknown.

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CHAPTER 14

COMPUTATIONAL MODELS OF FINANCIAL NETWORKS, RISK, AND REGULATORY POLICIES

KIMMO SORAMÄKI

14.1 INTRODUCTION

The financial crisis of 2007–2008 revealed the intertwined nature of modern financial systems. While the events unfolded, it became clear that the consequences of such interconnected and complex systems are particularly hard to predict. As a consequence, researchers and policy makers have recently developed a stronger awareness of the need for new analytical methods for monitoring the financial system; see, for example, Colander et al. (2009) and Haldane (2009). Recognition of the fact that the stability of the financial system depends on the collective behavior of market participants and on their interconnectedness also underpins the recent emphasis on the adoption of a macroprudential view of financial supervision, that is, supervision focused not only on individual institutions but also on their position and relevance in the financial system as a whole.

The intricate structure of linkages between financial institutions, among sectors of the economy, and across entire financial systems can conveniently be captured by using a network representation. The general concept of a network is very intuitive: a network describes a collection of nodes or vertices (e.g., financial institutions) and the links between them, which can be directed (arcs) or undirected (edges). The links can denote different relationships between the nodes, depending on the domain of analysis. In finance, these linkages may be related to lending relationships, derivatives trading, insurance agreements, asset correlations, or joint ownership of (e.g., toxic) assets. The

linkages can exist in different time scales from intraday liquidity requirements to term loans.

The main premise of network analysis is that the structure of the links between the nodes has an effect on the performance of each node and the system as a whole. A particular node cannot be analyzed on the basis of its own properties and behavior alone, because these may be affected by nodes that have links to it, and also by other nodes that have no direct links but are linked to its neighbors. Thus, in order to understand the behavior of one node one must analyze the behavior of many nodes, including those that are, perhaps, several nodes apart in the network.

In the past the focus of financial risk analysis has been based on the balance sheets of individual institutions, and the relationships between institutions have generally not been formally considered. This changed during the financial crisis. Regulators and researchers are now beginning to understand that financial institutions cannot be looked at in isolation. In order to understand the riskiness of a particular institution, one must consider the riskiness of the institutions that it is exposed to. For example, when extending emergency funding to Bear Stearns the Federal Reserve System justified its decision in a network context as follows:

[G]iven the fragile condition of the financial markets at the time, the prominent position of Bear Stearns in those markets, and the expected contagion that would result from the immediate failure of Bear Stearns, the best alternative available was to provide temporary emergency financing to Bear Stearns. Minutes of the Board of Governors of the Federal Reserve System, 14 March 2008

In the language of networks, the Federal Reserve Board believed that Bear Stearns was a central node (i.e., it held a prominent position) in the network and its failure could have led to a cascade of failures (contagion) had it been allowed to fail. The risk of such a chain of events is generally called systemic risk. Although systemic risk has many definitions, they usually refer to a chain of events that is initiated or magnified in the finance sector and adversely impacts the broader economy. De Bandt and Hartmann (2000) provide a survey of different forms that contagion can take.

Network models have been used to explain a wide range of natural and societal systems, ranging from the World Wide Web to cellular, ecological, and citation networks, to name a few. Jackson (2008) provides a comprehensive synthesis of several strands of network science in sociology, physics, mathematics, computer science, and economics. Social network analysis, which was developed in the 1950s within the field of sociology (Freeman 2006), has brought forth a number of important findings related, for instance, to the diffusion of ideas, the spread of habits and behaviors, the efficiency of groups, based on their social network properties, the origins of power among groups, and the concepts of centrality (or importance) of nodes in a network.

The approach that has been taken in physics has focused more on the statistical properties of networks, the resilience of different structures, and the processes that take place in networks. Moreover, physicists have tried to explain how networks grow over time and exhibit the complex nonrandom structure that has been uncovered for

many empirical networks. Newman (2003) and Albert and Barabási (2002) review advances in the modeling of complex networks, focusing on the statistical mechanics of networks. More recently, Newman (2010) provides an approachable introduction to several strands of network research.

The remainder of this chapter reviews research in three of the most active research areas of financial systems: payment networks, interbank exposure networks, and correlation networks. I describe and present empirical research on these networks and approaches for modeling them.

14.2 PAYMENT NETWORKS

Interbank payment systems have been an area of financial infrastructures where economy-wide simulations have been carried out for research and policy purposes probably longer than in other areas of financial research. Interbank payment systems provide the backbone for all financial transactions. Virtually all economic activity is facilitated by transfers of claims by financial institutions. In turn, these claim transfers generate payments between banks whenever they are not settled across the books of a single bank.

In 2010 the annual value of interbank payments made in the pan-European system TARGET2 was \$839 trillion; in the corresponding U.S. system Fedwire, the amount was \$608 trillion—more than forty times the annual U.S. GDP (BIS 2010). Owing to the sheer size of the transfers, and their pivotal role in the functioning of financial markets and the implementation of monetary policy, payment systems are central for policy makers and regulators.

Soramäki et al. (2007) present an early analysis applying concepts of network theory in empirical research of large-scale financial systems, in particular, the payment flows between banks in the United States. The empirical findings of the paper were similar to those of Boss et al. (2004), who analyzed interbank exposures in Austria, and of De Masi et al. (2006), who analyzed the Italian interbank market. They were, however, in marked contrast to the interbank networks that had usually been considered in economic and financial models. The empirical networks were found to be complex, with a small number of highly connected nodes that had connections with a large number of nodes with few links. The cores of the networks, composed of the most connected banks, processed a very high proportion of the total value (see figure 14.1). In the figure, the network subset focuses on the links of the network that comprise 75 percent of the value transferred. This network consists of 66 nodes (representing banks) and 181 links (representing payment flows) of a total of over 6,600 nodes and more than 70,000 links. The prominent feature of this network is that 25 nodes form a complete sub-graph to which the remaining nodes connect. In other words, we find that only a small number of banks and the links between them constitute the majority of all payment values sent over the network. These networks in fact shared many characteristics with other empirically

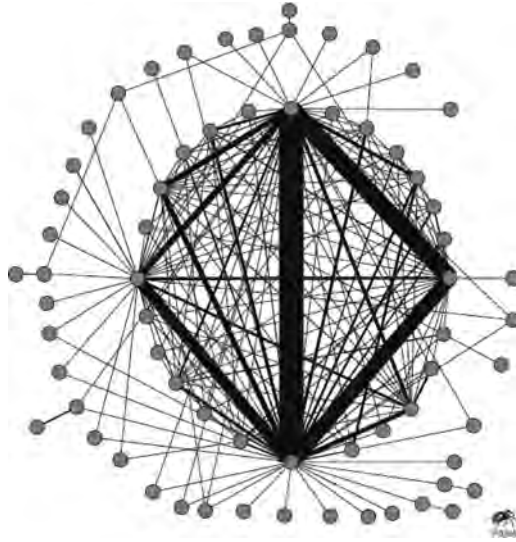


FIGURE 14.1 Subset of a network showing payment flows on a representative day in the Fedwire interbank payment system. This subset focuses on those links of the network that comprise 75 percent of the value transferred. This network consists of 66 nodes (representing banks) and 181 links (representing payment flows) of a total of over 6,600 nodes and more than 70,000 links. Note the prominent feature of this network is that 25 nodes form a complete sub-graph to which the remaining nodes connect. In other words, we find that only a small number of banks and the links between them constitute the majority of all payment values sent over the network.

Source: Soramäki et al. (2007).

complex networks, such as a scale-free degree distribution, high clustering coefficient, the small world phenomenon, and disassortativeness (Newman 2003).

More recently, a number of other studies have looked at national interbank networks, reconstructed using interbank payment flows and enabled by access to data from interbank payment systems operated by central banks. These include Lubloy (2006) on flows in Hungary, Becher et al. (2008) on those in the United Kingdom, Boss et al. (2008) on those in Austria, Pröpper et al. (2009) on those in the Netherlands, Embree and Roberts (2009) on those in Canada, Akram and Christophersen (2010) on those in Norway, and Huang and Jia (2010) on those in China.

The analysis and visualization of the network structures is used by regulators to better understand the interdependencies in the system and as a first step in identifying systemically important banks (ECB 2010b,c). Traditional metrics for the systemic importance of banks in a payment system tend to focus on local measures such as the number or value of payments made or received and ignore the system's network structure. Network visualizations and network-based centrality measures can give a more complete description of the importance of banks in the network. However, most centrality metrics have been developed with other applications in mind and do not capture the features of payment systems. Soramäki and Cook (2013) develop the

SinkRank metric, which captures a bank's importance as the impact the entire system would suffer in the event of an operational failure by the bank. The measure effectively calculates the speed at which liquidity in the system would flow into the troubled bank, making this liquidity unavailable to the other banks. The more important a participant is, the faster this speed and the higher the participant's SinkRank.

The payment system is a network in which each bank is dependent on behavior of each other bank, even if there are no direct links between the two banks. Payment systems are often complex networks, and the liquidity dynamics in the network follow convoluted patterns that are analytically intractable. Beyeler et al. (2011) show how simple rules of settlement can create complex system-level behavior. The model also shows that the mechanics of the system become unpredictable at low liquidity levels when each participant's ability to settle payments is dependent on receiving funds from other participants, whose ability to settle depends in turn on receiving funds from other banks upstream.

A natural way to model the complex dynamics in these networks is via counterfactual simulations. In this approach the details of the system are replicated in a computational model, and real transaction data are used to replicate the behavior of the real system. In the second step, desired aspects of the system are altered to see how, *ceteris paribus*, the system performs under the new circumstances. These models have generally been used to investigate systemic risk present in payment networks and to evaluate the effectiveness of alternative payment system setups.

Systemic risk in payment systems has been studied by counterfactual simulations since Humphrey (1986), who found significant risk in the U.S. Fedwire system in the mid-1980s. Subsequent studies by Angelini et al. (1996), Bech and Soramäki (2001), and Galos and Soramäki (2005) found the risks to be limited. Since then, however, most payment systems have switched from net settlement to real-time gross settlement (RTGS), transforming credit risk into liquidity risk, because gross settlement eliminates the former but at the cost of the latter (Bech et al. 2008). Various works have since used simulations to study risks and liquidity needs in RTGS systems, either by creating entirely simulated systems or by introducing changes in data from real payment systems. A growing body of work (Schulz 2011; Grat-Osinka and Pawliszyn 2007; Arjani 2006) uses simulation to study the relation between liquidity requirements and delays in payment systems. Simulations of failures in payment systems generally focus on systemwide risks and liquidity effects (Glaser and Haene 2009; McAndrews and Wasilyew 2005; Ledrut 2007; Ball and Engert 2007; Docherty and Wang 2010). Schmitz and Pühr (2009) studied network structure in payment systems with induced shocks but found that network properties were of limited use for stability analysis.

Much of the literature on counterfactual simulations has been subject to a form of "Lucas Critique" namely, that the behavioral response of banks in the system has not been endogenized in the research and thus can lead to wrong conclusions. Hence, understanding how the participants interact and react when faced with operational adversity will assist operators and regulators in designing countermeasures, devising policy, and providing emergency assistance, if necessary. The first approach to studying

bank behavior in interbank payment systems was to use standard game theory. Angelini (1998) uses a setup derived from earlier literature on precautionary demand for reserves and shows that in an RTGS system, where banks are charged for intraday liquidity, payments will tend to be delayed and that the equilibrium outcome is not socially optimal.

Bech and Garratt (2003) analyze strategic incentives under different intraday credit policy regimes employed by central banks and characterize the way the Nash equilibria depend on the underlying cost parameters for liquidity and delays. They use a stylized two-period, two-player “liquidity management game” to analyze intraday liquidity management behavior of banks in an RTGS environment. It turns out that two classical paradigms in game theory—the Prisoner’s Dilemma and the Stag Hunt—emerge from the analysis, thus permitting policy questions to be understood in terms of well-known conflicts and dilemmas in economics.

More recently, Martin and McAndrews (2008) develop a two-period model in which each bank in a continuum has to make and receive exactly two payments of unit size. Banks have to choose when and how to make payments. They can choose to pay either via an RTGS stream or via a “liquidity saving mechanism” (LSM). Delayed payments and the use of liquidity generate costs, and banks may be hit by liquidity shocks; that is, the urgency of certain payments is unknown *ex ante*. The model is solved analytically under assumptions about the pattern of payments that may emerge. Analytically solvable models need by design to be simple in their mechanics and thus cannot capture the rich interaction among the decisions of the system’s participants, its internal mechanics, and its feedback loops. Other approaches that have been applied to similar problems of repeated interaction among a large number of players stem from evolutionary game theory, reinforcement learning, or agent-based modeling (ABM) in general.

In ABM, the system under study is modeled as a collection of autonomous decisionmaking entities called agents. Each agent assesses its situation individually and makes decisions on the basis of a set of rules. The behavior of the system is an emergent property of the behavior of its constituents. The rules by which the agents operate can be static, or the agents may learn over time to improve their actions.

The literature developing agent-based models for understanding financial systems dates to the Santa Fe Artificial Stock Market in the early 1990s (Arthur et al. 1997). Modeling financial markets and prices has remained an active field. As for models concerning financial stability, Martinez-Jaramillo and Tsang (2007) and Martinez-Jaramillo and Tsang (2009) study extensively evolution in agent-based financial markets. Markose et al. (2010) develop an agent-based model for contagion from credit default swaps. Adams et al. (2010) and Galbiati and Giansante (2010) look at the formation of interbank payment systems.

Agents who learn about each others’ actions through repeated strategic interaction are a topical theme in evolutionary game theory (see discussion in Kirman 2010). Often the existing literature looks at the players’ asymptotic behavior in situations where the payoffs are some known function of players’ strategies. In the theory of repeated games,

this knowledge is a prerogative of the players, who can therefore use adaptive rules of the type “choose a best reply to the current strategy profile.” In a reinforcement learning tradition (e.g., Sutton and Barto 1998; and Fudenberg and Levine 1998), the rules do not require knowledge of the payoff function on the part of the learners. Such rules are instead of the kind “adopt more frequently a strategy that has given a high payoff.”

Galbiati and Soramäki (2011) use fictitious play (Brown 1951) to numerically solve a model with interactions among a number of banks that settle payments on a continuous basis under conditions of imperfect information, stochastic payoffs, and a finite but long sequence of settlement days. It is a dynamic multiagent model of an interbank payment system in which payments are settled on the basis of pre-committed funds. In the model, banks choose their level of committed funds on the basis of private payoff maximization.

Simulations are often also used in policy and financial infrastructure oversight, starting from work at the Bank of Finland in the late 1990s on the BOF-PSS2 Simulator (Koponen and Soramäki 1998). Simulation studies played an integral part in the regulatory approval of the Continuous Linked Settlement (CLS) system, which launched in 2003. It is the world’s largest settlement system, settling on peak days in 2013 almost \$9 trillion worth of foreign exchange transactions on the books of seventeen central banks (and currencies). Owing to its high and time-critical liquidity requirements for settlement, as well as the interlinkages to many currencies and economies, regulators placed special emphasis on assessing its impact on liquidity markets (ECB 2003). For this purpose CLS carried out and overseers evaluated a wide range of simulation scenarios (*ibid.*).

Other examples include the Bank of Japan, which used simulations to evaluate alternative liquidity saving mechanisms before implementing them (Imakubo and McAndrews 2006). The Eurosystem has embraced payment system simulations as an ongoing oversight tool by specifying how transaction-level data may be used (ECB 2010a) and has developed a TARGET2 simulation platform. More recently, Denbee and McLafferty (2012) used real payment data to quantify the liquidity efficiency that could be obtained in CHAPS, the United Kingdom’s large-value payment system, by the implementation of a liquidity-saving mechanism.

To better understand the functioning and risks in payment systems, network approaches have been applied to their visualization, to the identification of important banks by existing and new centrality algorithms, and by counterfactual network simulations. In particular, simulations are used on an ongoing basis in policy and regulation by the major central banks.

14.3 INTERBANK EXPOSURE NETWORKS

Starting with the seminal papers by Allen and Gale (2000) and Freixas et al. (2000), the economics literature has looked at the implications that a higher or lower degree

of completeness of interbank structures (that is, of interconnectedness generated by cross-holdings of deposits) could have for financial stability. These papers evaluate the potential for contagion that follows an aggregate or an idiosyncratic liquidity shock or a bank's failure and analyze the role of the central bank in preventing systemic repercussions. Although the results depend strongly on the assumptions about the process taking place in the network, the common lesson learnt from these models is the importance of understanding the structure of financial flows in order to understand the functioning of the system, and thus to be able to assess systemic stability. In fact, Allen and Babus (2009) argue that a network approach to financial systems is particularly important for assessing financial stability and can be instrumental in capturing the externalities that the risk associated with a single institution may create for the entire system.

The unsecured overnight money market (broadly called the interbank market) is one of the segments of financial markets to which network analysis has been applied intensively as well. This is due to the key role that money markets play in modern financial systems and to the availability of granular data that can be extracted from payment systems for their analysis. Money markets constitute the locus where banks exchange deposits, which allows the efficient redistribution of liquidity in the system and the effective implementation of the monetary policy stance, and it represents a possible channel of contagion.

In order to gain insights into unsecured interbank loan networks, variations of a methodology proposed by Furfine (1999) for matching two temporally separated payments (advance and repayment) as a loan have been applied to payment data. This allows the construction of network time series of the unsecured money market. Loan data of this level of granularity are generally not available from other sources. A representative paper following this approach is Atalay and Bech (2008), who use data from Fedwire to recover federal funds loans and analyze the network properties of this market. Examples of other applications are shown in Bech and Bonde (2009) for the Danish interbank market, Wetherlit et al. (2009) for the United Kingdom, and Heijmans et al. (2010) for the Netherlands. Iazzetta and Manna (2009) identify banks that are important in terms of a liquidity crisis, based on the distribution of liquidity among Italian banks since 1990, and Iori et al. (2008) study the Italian overnight lending markets based on data from eMID.

Empirical research on the other parts of the financial system was initially less common owing to the restricted nature of sufficiently detailed data. Degryse and Nguyen (2007) investigate the extent of systemic risk and network structure in the Belgian banking system over a ten-year period. Castrén and Kavonius (2009) build a model for understanding links and contagion between sectors of the economy based on flow-of-funds data. Recently, however, the literature has expanded quickly into many areas such as credit networks among banks (Bastos e Santos and Cont 2010), global banking networks (Garratt et al. 2011; Minoiu and Reyes 2011), stock trading networks, (Adamic et al. 2017; Jiang and Zho 2011) and Granger causality networks for contagion in banking (Billio et al. 2011), to name a few.

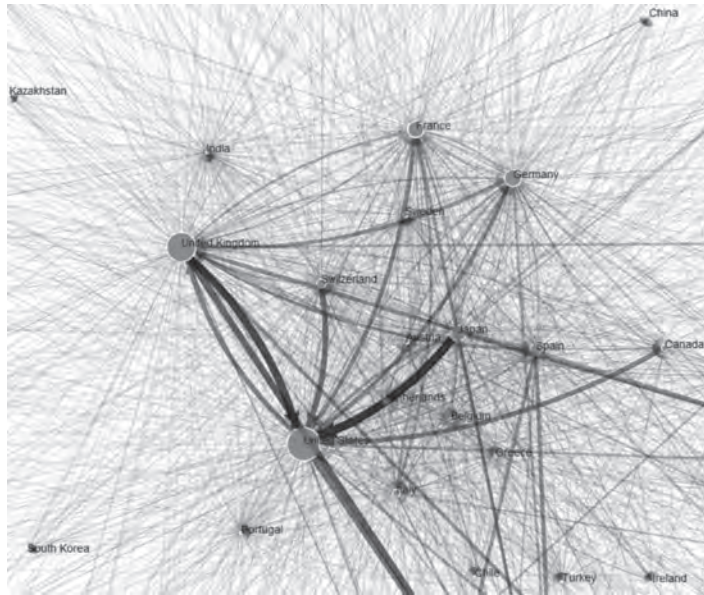


FIGURE 14.2 Global network of bilateral banking system exposures (core of the complete network). Countries are denoted as nodes, and arrows between nodes denote exposures of banking sectors of link source countries to creditors in link target countries. Thicker and darker arrows denote higher bilateral exposure, and larger nodes a higher total exposure. Data consist of exposures among 21 reporting countries and exposures to 209 nonreporting countries.

Source of data: Bank for International Settlements consolidated banking statistics: <http://www.bis.org/statistics/consstats.htm>.

Empirical network analysis, as reviewed so far, contributes to the existing theoretical results on systemic risk in the interbank market by considering the overall structure of the network (thus going beyond the earlier focus on its degree of completeness, as illustrated in Figure 14.2). It also provides a stronger basis for the assessment of contagion risk by means of counterfactual simulations. Counterfactual simulation models often have roots in various percolation and cascade models developed within statistical mechanics and network theory. One branch of this literature has investigated the resilience of processes in different network topologies in terms of a connectivity (or percolation) threshold (examples are shown by Bollobas 1985, Moore and Newman 2000, and Callaway et al. 2000) at which a network dissolves into several disconnected components. A well-known finding is that scale-free networks are more robust to random failures than other types of networks, although they are very susceptible to the removal of the very few highly connected nodes. However, such static failure models are more applicable to networks in which the interest is the availability of paths between nodes in the network (e.g., in transportation) but are less applicable to networks of monetary flows that contain both flows via the shortest paths as well as any walks within the

network.¹ Another branch of the literature has studied the impact of perturbations that cascade through the network on the basis of established theoretical or domain-specific rules (see Sachtjen et al. 2000 and Kinney et al. 2005 for power networks). In these dynamic models nodes generally have a capacity to operate at a certain load and, once the threshold is exceeded, some or all of the node's load is distributed to neighboring nodes in the network. While the detailed dynamics depend on the rules applied for the cascades, generally the most connected nodes (or nodes with highest load in relation to overall capacity) are more likely than an average node to trigger cascades.

Cascade models have been applied to various systems in fields such as geology, biology, and sociology, as shown by Jensen (1998). This research has demonstrated that models made of very simple agents, interacting with neighboring agents, can yield surprising insights about system-level behavior. Models applied to financial systems have been developed by, for example, Nier et al. (2007), who also use simulated exposure data, and Bastos e Santos and Cont (2010), who use data from the Brazilian credit registry. Others (Lorenz et al. 2009; Georg 2011) have applied cascade models to systemic risk analysis. Recently Battiston et al. (2012) proposed the DebtRank metric to quantify the systemic importance of banks. DebtRank is based on the structure of an exposure network and represents the proportion of the network's total value that could be affected by distress or default in a given bank. Aldasoro and Angeloni (2013) develop six indicators that capture different aspects of systemic importance. Acemoglu et al. (2013) provide a new framework for studying the relation between the financial network architecture and the likelihood of systemic failures due to contagion of counterparty risk.

Craig and von Peter (2014) introduce the concept of a core-periphery, or tiered, structure for banking systems. The concept entails a small group of highly connected core banks and a larger group of less-connected periphery banks. Many central banks and researchers have since applied their methodology, for example, in pursuit of identifying a set of systemically important banks. A core-periphery system has the following properties:

- Core banks lend to each other.
- Periphery banks do not lend to each other.
- Core banks lend to (some) periphery banks.
- Core banks borrow from (some) periphery banks.

Real banking systems deviate somewhat from this perfect core-periphery structure, some by having core banks not linked to all other core banks some by having links between periphery banks. Nevertheless, the core-periphery concept has proven useful for understanding markets and banking systems and for identifying important banks. For example, Craig and von Peter (2014) find a core-periphery structure in the German banking system and also find that bank-level features such as connectedness and

¹ A walk is a traversal of nodes along links without any constraints. In contrast, a trail is a walk in which a given link is not visited twice and a path is a walk in which a given node is not visited twice.

balance sheet size are helpful for predicting a bank's classification as core or periphery. The finding that large, well-connected banks are more likely to be in the core lends support to the core-periphery structure as a realistic model for banking systems. Craig and von Peter also find that the core-periphery classification of banks in the German system is stable over time; Fricke and Lux (2012) report similar findings for the e-MID trading platform, and Langfield et al. (2014) report a core-periphery structure in the UK interbank system.

A rich literature examining interbank exposure networks has emerged in recent years. While each network of interbank exposures is different, common features have been uncovered. In order to simplify the complex network structure and dynamics, the literature has recently developed metrics that summarize information about aspects that are important to policy. In other fields where network models have been used, the centrality of nodes has been an active area of research for decades. The new metrics and algorithms particularly suited for interbank networks to identify systemically important banks or sets of banks are based on cascade models and the identification of core-periphery structures.

14.4 CORRELATION NETWORKS

Today there is a focus on how network theory can be used to explain and better understand financial markets. A main area is the study of correlation-based networks. These networks can be used to reduce the complexity of financial dependencies, to enhance the understanding of dynamics in financial markets, and to identify market anomalies and outliers.

A seminal paper is Mantegna (1999), which develops a methodology for representing asset correlations as trees. The paper introduces a method for extracting a subgraph (asset tree) of the most relevant links from a complex network, where links represent observed correlations between asset prices and returns. By constructing the minimum spanning tree (MST) Mantegna finds that the U.S. stocks are grouped on the basis of their industry sectors. Onnela et al. (2003) introduce a new network type known as the dynamic asset graph. This is similar to the asset tree introduced by Mantegna (1999), but the function for mapping correlations into link lengths (the distance function) is nonlinear. Like MSTs, dynamic asset graphs have $N-1$ links in total (where N = number of nodes); however, in a dynamic asset graph the links may form loops, and unconnected nodes are allowed.

The network also includes gold and silver (XAU, XAG – squares) and Brent and West Texas Oils (XCB, XCT – diamonds). The MST was constructed as follows. First calculate pairwise Pearson correlations between the returns of all currencies. This correlation matrix is transformed to a distance matrix using Gower's (1971) distance function. The distances map to correlations so that a short distance denotes a high correlation and a long distance a low correlation. The MST of the distance matrix can be extracted by,

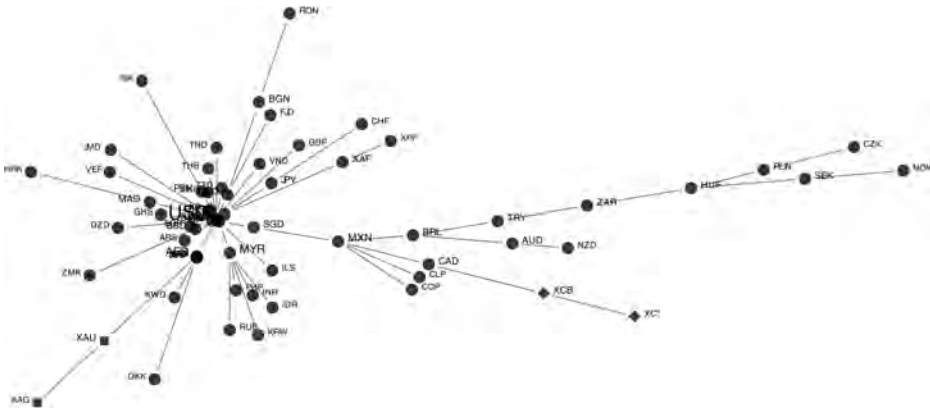


FIGURE 14.3 Minimum spanning tree built from correlations of foreign exchange returns against the euro, 2004–2011.

Source of data: Exchange Rate Service hosted by Sauder Business School at the University of British Columbia (fx.sauder.ubc.ca/).

for example, using Kruskal's algorithm (Kruskal 1956). The resulting tree is shown in figure 14.3, where the link lengths scale with the distance. The layout algorithm is based on visualization of phylogenetic trees developed in Bachmaier et al. (2005). We can see that the USD is the most central currency, with many other currencies linked to it, and we can identify clusters, such as a cluster on the right with many Scandinavian currencies. For a more in-depth study on correlations of currency returns, see Naylor et al. (2007).

Bonanno et al. (2004) consider how the returns of stocks traded in a market are affected by varying the time horizons used to compute the correlation coefficients. They find that as the time horizon decreases the graph structure progressively changes from a complex organization to a simple form (where clusters are sparser). The paper also discusses volatility time series of stock prices and global financial markets.

A problem with MSTs is that they are very fragile to small changes in underlying correlations. Tumminello et al. (2005) extracts a subgraph of the most relevant links from a complex network using the correlation coefficient matrix and Gower's (1971) metric distance. The paper outlines the method for constructing an MST and then directly builds from this a planar maximally filtered graph (PMFG, as illustrated in Figure 14.4). The main constraint in the construction of an MST is that each new link must not form a cycle, whereas a PMFG is a slightly more complex graph that allows loops and cliques of up to four elements. It can be drawn on a planar surface without link crossings, and it is more robust to small changes in underlying correlations. The paper develops MSTs and PMFGs of U.S. stock correlations and also considers the economic properties of the four-clique structures in the PMFGs.

Another approach is taken by Kennet et al. (2010), who analyze stationary correlations between stocks and introduce a new tool to investigate the structure and dynamics of the market through the study of correlations. Financial markets are often driven by common factors resulting in correlations between two assets that are driven

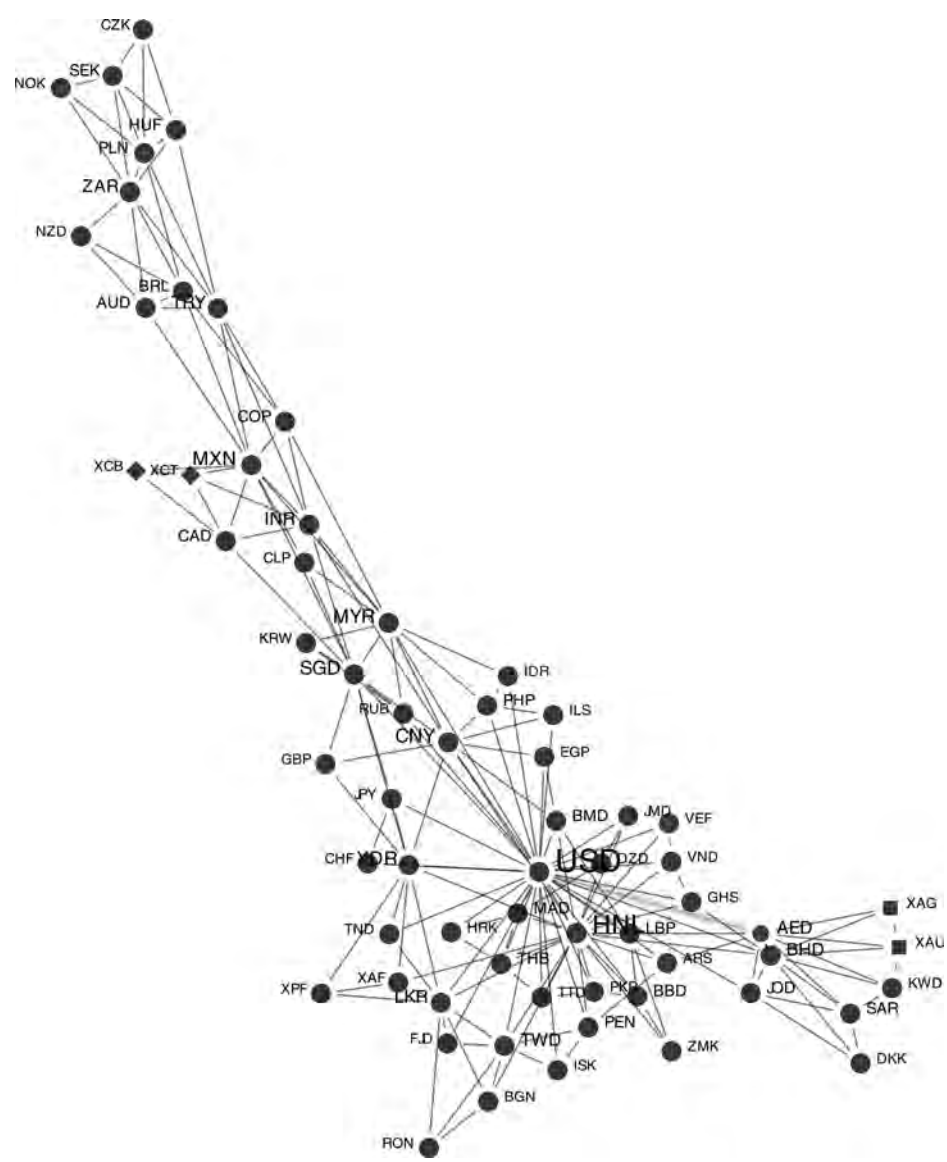


FIGURE 14.4 Planar maximally filtered graph of correlations of FX returns against the euro, 2004–2011. The network also includes gold and silver (XAU, XAG – squares) and Brent and West Texas Oils (XCB, XCT – diamonds).

Source of data: Exchange Rate Service hosted by Sauder Business School at the University of British Columbia (fx.sauder.ubc.ca/).

by a dependency on a third asset, or on the market in general. Kennet et al. (2010) discuss the use of partial correlations (the correlation between two variables adjusted for their relation with other variables) when studying correlation-based networks. A partial correlation network can be used as a tool for filtering the most relevant

dependencies between assets. The PMFG is adapted so that it can be constructed from partial correlations to form a partial correlation planar maximally filtered graph. The paper also gives details on constructing the partial correlation threshold network, a type of dependency network in which the inclusion of links depends on means and standard deviations of nodes. All methods are demonstrated on stock data from the New York Stock Exchange.

Correlations measure an unobserved link that may be caused by a multitude of underlying factors. As such they are very different from the links normally considered by network analysis, which are observed and measured rather than inferred. On the other hand, correlation networks are quite flexible and can be constructed from any time-series data to uncover linkages.

14.5 CONCLUDING REMARKS

This section provides some views concerning how financial network research can be further operationalized to become policy and what challenges lie ahead.

Payment system research has been an area of financial research in which simulation and network models were applied first, probably because data of the necessary granularity has long been available from interbank payment systems. Results gained in this field have also, in some cases, been used to argue for regulatory change in others (see, e.g., Lo 2009). Better data about other infrastructures and markets will allow regulators to develop more complete maps of the financial system. These maps will be invaluable for decision support of crisis management, when time is too short to develop more robust models and when intuition can be assisted by data mining. Financial network analysis and network visualizations are likely to be good tools for gaining this intuition and for developing more objective network-based metrics. The challenge for regulators is to develop systems that allow the monitoring of the financial system over time—ideally, in real time. Financial relationships are in a continuous flux and can rapidly change form in the event of a financial crisis.

Since the 2007–2009 financial crisis, the architecture of financial regulation has been subject to an overhaul. Within financial regulation, systemic risk is currently being addressed by enhancing macroprudential supervision, and many supervisors have been mandated to carry out systemic risk monitoring. New systemic risk regulators such as the Office of Financial Research in the United States and the European System of Financial Supervisors have been set up.

Two policy innovations agreed on by the G20 are particularly noteworthy from the aspect of networks. First, the G20 agreed that systemically important banks should be subject to additional capital requirements. Higher requirements for loss absorbency for important nodes should reduce the risk that contagion propagates through the network of interbank exposures. Second, the G20 agreed that all standardized over-the-counter

(OTC) derivative contracts should be traded on exchanges or electronic trading platforms, where appropriate, and cleared through central counterparties. Central clearing of derivative contracts should reduce the quantity of counterparty risk, information asymmetry, and complexity in the financial system.

These policy innovations also give rise to research challenges in the field of financial networks. First, the Basel Committee's methodology (2013) for identifying systemically important banks does not incorporate requirements or models based on information on bilateral exposures. Such data is, however, already collected by many regulators who are developing methodologies that make use of this more granular data. Second, mandatory clearing of OTC derivatives potentially increases the systemic importance of central counterparties. Investigating this possible unintended consequence should be a priority for research on financial networks.

Before the financial crisis of 2007–2008, regulators had little access to network data beyond the payment system. As a consequence of the crisis they have much broader mandates to collect data and as a consequence are facing the opposite challenge: the management and summarization of large data sets. In this context, researchers should continue to develop methods to summarize voluminous data in accurate and economically meaningful ways. This includes improvements to visualizations of complex data, which would allow policy makers to view systemic vulnerabilities in real time.

A rapidly developing area of current financial network research is based on correlation networks. Using data reduction algorithms, large correlation matrices can be filtered to produce simpler networks that still retain much relevant information and allow quicker visual identification of the main dependence structures. Various degrees of data reduction are possible. For example, the minimum spanning tree, presenting the bare minimum, can be modified to form the planar maximally filtered graph, which includes more links and also cliques, without displaying the full complex network. Information-theoretically better methods based on the use of partial correlations and statistical models instead of graph theoretic concepts are also being developed. Correlation networks summarize and can visually display vast amounts of market information and together with new algorithms for outlier and regime detection give regulators easier access to the information-processing capabilities of the market.

Tools for network analysis have improved substantially over the past few years, and many open-source projects and tools are available for free for research purposes. The main general-purpose tools for network analysis include Pajek (Batagelj and Mrvar 2003) and Gephi (Bastian et al. 2009). The iGraph (Csardi and Nepusz 2006) library for the statistical software R (R Development Core Team 2008) contains many algorithms for analyzing complex networks. These programs contain some visualization modules. For more advanced programming needs, JavaScript libraries like d3.js (Bostock et al. 2011) and Sigma.js (<http://simgajs.org>) provide the needed flexibility. The application of network analysis to monitor and assess systemic risk and contagion in financial systems should benefit from important progress made in other sciences. Keep in mind, however, that the results depend on the process and behavior of the particular network and may not be directly applicable to the financial context. An online analytics platform

implementing many of the centrality algorithms and visual layouts suitable for financial networks is FNA (Financial Network Analytics, www.fna.fi; Soramäki and Cook 2014).

In addition to mapping the financial system at a given point in time, researchers and policy makers need to take into account the mechanics of the financial system in order to be able to make informed forecasts of likely courses of events. These mechanics have become more complex and automated with continuous advances in technology. Human decision making, and therefore economic behavior, is in many cases pre-programmed as algorithms and, as a consequence bounded not only by rationality but also by the operational IT framework. Modern financial systems involve a high degree of automation both in how decisions are made and how they are executed, including algorithmic trading, straight-through processing of settlement, and automated rules for liquidity management of the settlement process. These automated infrastructures have become ubiquitous but are not well understood. For example, under stressed market conditions the automated execution of trades can trigger extreme price movements, and the interactions between automated orders and algorithmic trading strategies can be unpredictable—and once the rules have been set up they are part of the system mechanics. The system is highly complex and, even in the absence of any human decision making, exhibits many feedback loops that affect performance. Counterfactual simulations, as developed for payment systems and contagion of interbank exposures, can bring improved understanding of the interaction of the mechanics of a system.

In spite of larger amounts of data and better ways to summarize and use it as a basis for computational economic and financial models, financial networks will likely be limited to providing views of the functioning of particular subsystems of the whole economy. The interdependencies between networked systems (multilayer networks) are only very recently being considered by researchers. Also, network models will likely serve relatively well for understanding the consequences of systemic events, and will be somewhat limited for foreseeing their occurrence, because they suffer from the same limitation (access only to historical data about rare events) as do other empirical models. Overall, network-based methods that take into account the complex connections in the financial system and allow for their visual presentation in intuitive ways will help the industry, regulators, and researchers better understand financial markets and infrastructures.

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CHAPTER 15

FROM MINORITY GAMES TO \$-GAMES

JORGEN VITTING ANDERSEN

15.1 INTRODUCTION

It is probably not an exaggeration to say that the theory of rational expectations really is at the very core, hiding deep down in the framework, of most theories used in finance since the mid-1970s. The idea is very simple and very elegant, and has very deep implications.

Unfortunately, it is also very unrealistic. This chapter aims to go beyond such a traditional view of finance but also beyond the more recent behavioral finance view, which as its starting point takes the case of individual investor psychology. The main theme will be price formation in financial markets, seen as a sociological phenomenon. For a background on the ideas presented here, see Andersen and Nowak (2013).

If we reject the the main building block of financial models, the rational expectations hypothesis, as unrealistic, the undeniable question is then, How can we do better? The lack of established and convincing alternatives is probably one of the main reasons why macroeconomists keep on using rational expectations in their models, for the simple fear of the “wilderness” of alternatives. If we are not to use the sharply defined hypothesis of rational expectations, then what exactly do we use?

15.1.1 Chaos Theory and Financial Markets

One of the main issues with rational expectations theory is the assumption that all investors act alike, analyzing the market in a cool and pervasive manner by using all available information relevant to future cash flows of a given asset. This description sounds far from reality, and is far from reality, as probably most economists would

agree. Let us instead for one moment take the extreme opposite view. Assume for a moment, in a thought experiment, that we know all investors, their wealth, and their investment strategies independent of the condition of the markets. Let us also assume that we were somehow able to put this huge amount of information into a fast supercomputer. Would we then be able to understand and predict future market movements?

This line of thinking goes back to the tradition from Pierre Simon de Laplace, a French scientist born in the eighteenth century, who in his book *A Philosophical Essay on Probabilities* (1814) wrote:

We may regard the present state of the universe as the effect of the past and cause of the future. An intellect which at any given moment knew all the forces that animate nature and the mutual positions of the beings that compose it, if this intellect was vast enough to submit the data to analysis, could condense into a single formula the movement of the greatest bodies of the universe and that of the lightest atom; for such an intellect nothing could be uncertain and the future just like the past would be present before its eyes.

One century later a compatriot of Laplace's, Henri Poincaré, would discover that having just the equations is not enough since we also need to know the initial conditions of a system in exact detail: given the slightest change in initial conditions, the same equations could give completely different results. With the invention of computers and the computing power they obtained in the 1970s, it became possible to formalize this point and now name it "chaos theory." Most readers probably will recognize chaos theory because of the meteorologist Edward Lorenz, who coined the saying that the flap of a butterfly's wings in Brazil sets off a tornado in Texas.

Let us, then, step back and reconsider the question posed two paragraphs above concerning the predictability of future market movements. We can summarize the situation as follows. According to rational expectations theory used in traditional finance, financial market movements are random and consequently impossible to predict, full stop! Laplace, on the other hand, opens up the exact opposite scenario with everything in this universe described by determinism, from the movement of an atom to that of a galaxy, and of course in between these two extremes, all other activity including the movement of prices in a stock market. Even though Laplace's considerations are more philosophical in nature than of practical relevance, the idea is that everything ultimately could be understood deterministically if we could just grasp the right underlying variables governing all forces in nature and analyze the positions of beings in it. Poincaré agrees with the determinism in the equations but points out that in practice understanding the world deterministically is forever beyond our reach since changing the initial conditions of the equations changes their outcome. This, however, has not held people back from trying to understand market movements using chaos theory, assuming some underlying variables governing cycles of price formation in financial markets much as they do business cycles governing the economy. The goal in that case is to determine a so-called strange attractor, or finding out which dimensions

give the number of relevant variables in question. The principle of a parsimonious description of a system in terms of some few underlying relevant variables is the hallmark of most theories coming from the field of physics, and it has proved successful when imported to other domains such as bio-physics and geo-Physics. It is probably fair to say that the jury is still out when it comes to the possibility of gaining insight by describing financial markets or economic developments in terms of chaos theory. The complexity approach to be presented in the following, rather than looking for some equations with relevant variables, stresses the impact that each constituent can have on a system. By treating each constituent of a system, complexity theory in a sense appears more “chaotic” than to chaos theory, which, instead of considering all the constituents, only considers a few. If we return to the question of price formation in a financial market, each constituent is an entity that takes part in making the price by trading, be it a computer or a human. The question, then, is how to find order in this seeming cacophony of voices of the constituents.

15.1.2 Complexity Theory and the “Symphony” of the Market

In order to visualize the complexity models to be introduced in the following, let us for a moment try to make the analogy between the way that prices are formed in a financial market, and the creation of a symphony by an orchestra. One can think of each market participant as a musician in such an orchestra, playing a different tunes (trading frequency and direction of trade) with different volume (amount of cash behind a trade). The role of the “maestro” that the market participants watch would be played by the central bankers worldwide. The question is whether we are somehow able to decode the symphony of price movements played by such an orchestra.

Even though the analogy might sound a bit far-fetched, it does illustrate the collective behavior of a system in which humans have to perform an act that depends on the outcome of what other people do. Just as different traders use different trading strategies, so do different musicians in an orchestra have different sheet music that they use in the act of playing. Sheet music tells the musician how to play, but he or she has to wait for his or her colleagues to play the instrument in the right manner. It is similar for decisions in a trading strategy played by a trader. A trader never acts in complete isolation but takes his or her decisions depending on the “tune” played by the market. Trend-following strategies depend on the momentum of the market, whereas fundamental value strategies depend of the price level of the market; traders buy when the stock has become under valued and selling when the stock has become over valued. In addition, anticipation of how future tunes will be played matters because of new policy making by central bankers or new economic statistics.

The complex but also fascinating, way the market symphony is created, becomes clearer when we consider its hierarchical structure. At the first level of the hierarchy, we can imagine each given stock and its price movements being played by a local

orchestra of individual traders. This local tune (price movements of the stock) is not made in complete isolation but depends (in part) on the how other local orchestras (other stocks) play their melodies. Going up one level, we can then consider the tune-of-a-stock created by the aggregate action of traders, now itself as created by a single instrument, but playing in a national orchestra composed of all stocks in an index. Adding up the contributions of the local orchestras creates what one could call a national hymn, illustrated by a given country's stock index. As discussed in Andersen et al. (2011), the hierarchical structure does not stop at the level of a given country, but goes into a super-hierarchical structure where a global symphony is played by a worldwide orchestra of all nationalities. Each national orchestra does not play its hymn in isolation but has to wait for the outcome of the other orchestras in order to know how the play its hymn properly.

15.1.3 Agent-Based Modeling: Search for Universality Classes in Finance?

Having discussed the way the diversity in human decision making collectively somehow creates a symphony of price movements, the question is, How one can formalize such an image? One step in this direction is to use agent-based models which are built, so to speak, to allow for diversity when it comes to decision making.

The concept of agent-based models dates from the 1940s and the ideas of John von Neumann and Stanislaw Ulam, who were interested in self-replicating machines and came up with the idea of the cellular automata. Like cellular automata, agent-based models cover a very broad range mostly computational models, that simulate the actions and interactions of some set of autonomous entities called agents. With this description the link to complex systems should be clear. In the context of finance and economics the great advantage of agent-based models is that they give us a means to introduce behavioral traits at the micro level of each agent, and then study how such traits and rules of interaction among the agents generate complex behavior at the system level.

Originating in the domain of physics, several types of agent-based models have been proposed to describe the dynamics of pricing in financial markets. Typically these kinds of models consider a financial market as composed of agents that use strategies to buy or sell shares. The decision could, for example, be based on an estimation of fundamental values or, in order to encompass technical analysis, based on the last price movements of the market. The agents are flexible since they adapt to the price movements of the market, choosing the most successful strategy at each time step. In this perspective such models offer a ground-breaking view of financial markets since the agents are bounded rational with, in some cases, exact solutions obtained via techniques borrowed from physics.

Before going into the specifics of the different agent-based models, I begin with a word of caution. It is easy to fall into the trap of considering narrowly each model as a

complete description of a given financial market. Clearly no model will ever accomplish the full task of explaining in every detail the price dynamics of a financial market, just as no map ever can tell all details of a location. Still, maps are useful for getting an overview, and my claim in the following is similarly that we need models to guide us through the wilderness of the price dynamics of financial markets. I would rather emphasize a view in which the models serve as probes or handles used to get a successively better descriptions of what one could call “generic” behavior of financial markets. Generic behavior could, for example, be the point of identifying certain moments when the markets are framed, as discussed below. Let us stress realism as the main advantage of the complexity models compared to the traditional rational expectations models. We know from psychology that people do not behave rationally (Kahneman 2011). Incorporating nonrationality via complexity models therefore appears as an appealing alternative to models based on the rational expectations hypothesis.

At the lowest level of description, a complexity model of a financial market could be defined as follows: “A financial market is a place where a number of actors trade, using different strategies.” Even though this statement does not say a lot, it is hard to dispute, and it can serve as a foundation for what follows. The fear of traditional economists is, however, that this will never bring us very far, since the next refinement in such a description can be taken in an infinite number of different directions.

Physics faced what at first appeared to be somewhat similar concerns in the 1950s, 1960s, and 1970s. The question then was, How much detail is really needed to describe a given system properly? This question is relevant also in the present context, since if a detailed description of the price dynamics of a financial market is needed with all its interwoven facets, then there will not be much hope of gaining insight on financial markets using the rather simple models introduced below. It is, however, noteworthy that the problems encountered in physics eventually led to the development of tools now used in different areas of physics that go under the general name of “renormalization group theory.” Back then there were concerns in the domain of particle physics (atomic physics) that in order to understand the interactions of the forces in nature one needed a *detailed* description of the forces described at a certain scale (distance). It turns out that, for example, an electron, which looks like a singular solid object when seen from afar, looks differently as one zooms in. As one decreases the viewing distance (goes to higher energies) an electron appears to be composed of self-similar copies of yet other electrons, positrons, and photons. Similarly a stock index isn’t just a stock index but has its own life in terms of the stocks that constitutes it. Also the forces that acts between different stock indices appear different than the forces acting between the individual stocks.

The development of the renormalization group theory was the result of successive contributions. First Stueckelberg and Petermann (1953), then Gell-Mann and Low (1954), followed by Feynman, Schwinger and Tomonaga (who won the Nobel Prize in physics in 1965 for their contribution) showed how details, depending on scale, were irrelevant. A deeper understanding of this problem came from condensed matter physics (Kadanoff 1966; Wilson 1975), the technique was used to understand the

general nature of phase transitions. This led to yet another Nobel Prize, illustrating how important the technique has become in modern physics.

The lesson from physics tells us that insight can be gained as one studies how forces change when varying the typical scale of a phenomenon. The question is whether similar insight could be used in understanding how market forces change, when instead of looking at correlations between different stocks in an index, one goes to a different scale and considers the forces acting now between different stock indices, with currency and commodity indices adding complexity to such an understanding. In a similar vein, it would be interesting to have a theoretical understanding of how market forces change as one change the time scale in question, relating events at the time scale of seconds and minutes to what happens at the hourly, daily and monthly time scale. I mention this in part because use of an analogy to gain insight into market forces by looking at what happens when one changes the scale is appealing, but, more important, because the renormalization group theory explains the concept of “universality” seen in physical systems. It turns out that phase transitions seen in, for example, magnetic systems, alloy materials, and superfluid and superconducting transitions, despite their apparent different nature, have similar critical exponents, meaning that near the transition phase their behavior is identical. The phase transitions of these very different systems can therefore all be described by the same set of relevant variables, leaving out (as mentioned in the paragraph above) a detailed description for each system as irrelevant to understanding the physics governing the phase transitions. It should be mentioned that the notion of universality extends to systems that are out of equilibrium. In that case it has been shown how domain growth of, for example, growing crystals, or growth of domains of oxygen in high-temperature superconductors, again is determined by a few relevant variables (such as the dimensions of the system and whether the so-called order parameter is conserved or not). Given the success of renormalization group theory, physicists are therefore brought up to look for the relevant variables of a system, knowing that details ultimately should not matter. It is through such a lens that the models introduced below should be seen: as primitive models of financial markets that, despite their simplicity, are meant as probes to search for what could be the equivalent of relevant variables in financial markets.

15.2 METHODOLOGY: DEFINING AGENT-BASED MODELS AND EXPLORING THEIR PROPERTIES

15.2.1 The El Farol Bar Game and the Minority Game

The El Farol Bar Game was invented by the economist Brian Arthur (Arthur 1994). In the game N people have to decide independently each week whether go to at their

favorite bar, which once a week has some very popular live folk music. The problem, however, is that the bar only has $L < N/2$ chairs. Now since each person will only be happy if seated, the people try to use the past week's attendance to predict future attendance. If a person predicts that the number of people who will attend is larger than L , that person will stay home. Notice that the El Farol Bar Game describes adaptive behavior, since the audience in the bar uses past data to predict future attendance.

One reason to discuss this model is that it gives a clear example of a situation in which rational expectations theory cannot be used to solve the problem. Suppose a rational expectations prediction machine existed and that all the agents possessed a copy of it. If the machine predicted, for instance, that a number larger than L would attend the bar, nobody would show up, thereby negating the prediction of the machine.

Inspired by the El Farol Bar Game, Ye-Cheng Zhang and Damien Challet introduced the Minority Game (Challet and Zhang 1997) as an agent-based model proposed to study market price dynamics. The model follows a leading principle in physics, namely, that in order to solve a complex problem one should first identify essential factors at the expense of trying to describe all aspects in detail. The Minority Game (MG) therefore should be considered as a minimal model of a financial market. It is described by just three parameters:

- N , the number of agents (market participants)
- m_i , the memory of agent i
- s_i , the number of strategies available to agent i

in the case where all agents use the same memory $m_i \equiv m$ and the same number of strategies $s_i \equiv s$ (to be assumed in the following). The name Minority Game was inspired by the El Farol Bar Game; the agents are rewarded whenever their decision is in the minority. I explain this point in detail below where I define the payoff function.

In order to perform technical analysis, the MG agents use lookup tables presenting the last m directional moves of the market. Representing an upward movement of the market as a 1 and a downward movement of the market as a 0, a strategy can be determined via tables, of which an example for $m = 3$ is shown in table 15.1.

A strategy therefore tells you what to do whatever the market did in the past. If the market went down over the past three days, the strategy represented in table 15.1 tells you that now is a good moment to buy ($000 \rightarrow 1$). If instead the market went down over the past two days and then up today, the same strategy tells you that now is a good moment to sell ($001 \rightarrow -1$ in table 15.1).

The model may appear simple, but its complexity is revealed when we consider the total number of possible strategies. Representing just the up and down movements, there are 2^m possible price histories when we look at the time period of the past m days. Since for each possible price history (for each entry on the left-side of table 15.1) a strategy gives a recommendation of what to do (buy = 1, sell = -1) the total number of strategies S is given by $S = 2^{2^m}$. Even for a relatively short time period of, say, two weeks, with ten trading days, $S = 2^{1024}$. This number exceeds 10^{300} . To grasp the magnitude of such a number, notice that the total number of elementary particles in the universe is

Table 15.1 Example of a strategy

Signal	Prediction
000	1
001	-1
010	-1
011	1
100	-1
101	-1
110	1
111	-1

“only” supposed to be about 10^{90} . So in a simulated financial market in which traders only needed to make decisions based on the directional moves of the market over the past two weeks, without taking into account their magnitude, they would have by far more choices than there are particles in the universe! If nothing else, this little exercise shows that it is no wonder that people gets overwhelmed in making decisions about when to invest. Trading is complex.

The possible set of different price histories for a given fixed value of memory m of course always stays the same, that is, the left-hand side of the table 15.1 is the same for all possible strategies. What characterizes a strategy is therefore the action it recommends given this constant set of price histories, the action given by the right-hand side of table 15.1. Formally speaking, therefore, a strategy can be seen as represented by a 2^m binary vector of length m . As mentioned above, there are $S = 2^{2^m}$ of such vectors. Many of these vectors are however very similar in the trading strategy they recommend: simple change the last entry in table 15.1 so that instead of telling us to sell if the past three days were up, it would tell us to buy. The two strategies are represented by the vectors $\vec{V}_1 = (1, -1, -1, 1, -1, -1, 1, -1)$ respectively, and $\vec{V}_2 = (1, -1, -1, 1, -1, -1, 1, 1)$. Clearly these two vectors, or trading strategies, are highly correlated. It can be shown that a qualitative understanding of the model can be obtained from, a much smaller set of just 2^m independent vectors (strategies) instead of the total set of $S = 2^{2^m}$ strategies (Challet and Zhang 1998).¹ So to get a qualitative description of the MG one only needs a small subset of the total number of strategies.

It turns out that a qualitative understanding of the MG for a given fixed value of S can be obtained from just one parameter (Savit et al. 1999), α , given by the ratio of the size of the set of uncorrelated strategies to the total number of N agents, $\alpha \equiv \frac{2^m}{N}$. For a general value of S it is, however, more natural to define the control parameter instead

¹ An example of independent strategies could be two strategies that the opposite, conditioned on a given realization of the price history. Such strategies are supposed to play a major role in the “crowded” phase, when two groups of almost equal size are formed with opposite views of the market behavior.

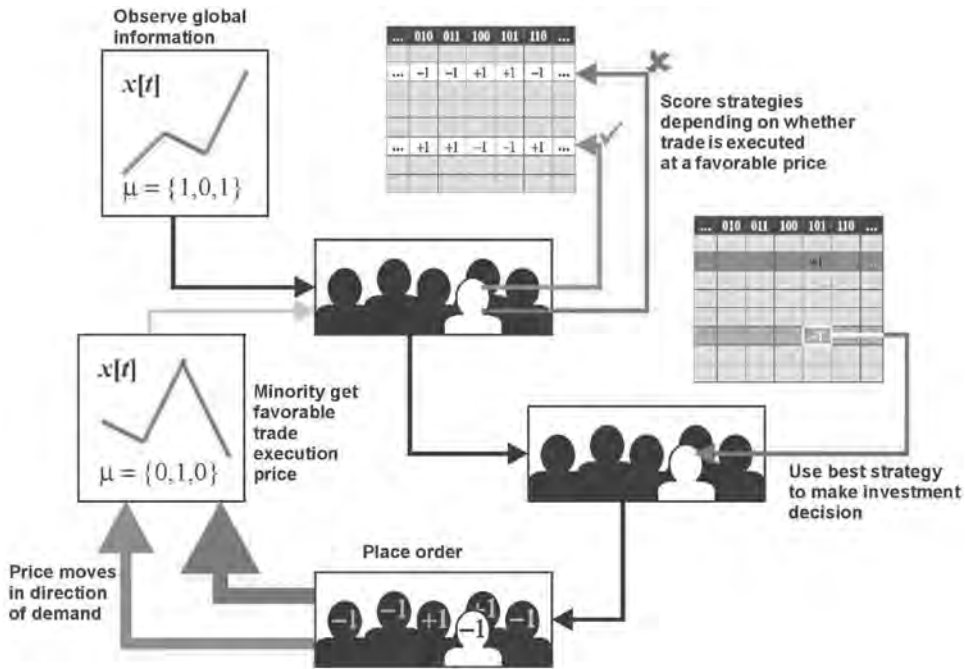


FIGURE 15.1 Graphic illustration of the Minority Game.

Source: Johnson et al. (2003).

as $\alpha' \equiv \frac{2^m}{S \cdot N}$ (Challet and Zhang 1998; Zhang 1998) since this parameter gives the ratio of the size of the set of uncorrelated strategies to the total number of strategies at play for any value of S . The distinction between the two definitions is not important for the results discussed in the following since S is taken as a constant, so we will stick to the notation used in the literature: $\alpha \equiv \frac{2^m}{N}$.

Readers interested in the MG model can find more information in Challet et al. (2004), which is specifically dedicated to the research on the MG.

The price dynamics of the model enters in quite a special way, and a word of caution is necessary since the details given in the following turn out to be absolutely crucial understanding of the model. Figure 15.1 gives a schematic representation of the model.

At each time step the agents update the score of their strategies according to the MG payoff function (see equation 15.1 below). Based on the actual history of the market each agent chooses the best-performing strategy from a set of S available strategies then uses it to decide whether to buy or sell an asset. The performance of the i th agents j th strategy is determined via its payoff function $f_{ij}^{MG}(t)$, which in the MG is updated according to:

$$\delta f_{i,j}^{MG}(t) = -a_i^j(t) \sum_{k=1}^N a_k^*(t), \quad (15.1)$$

where $a_i^j(t)$ denotes the action of the i th agent's j th strategy. It can be found in the strategy's lookup table as the right-hand side corresponding to the price history that occurred at time t . Let us take the example in which, over the past three days, the market went up, then up again, and today finally down; that is, the price history was (110). If the i th agent's, j th strategy was given by the information shown in table 15.1, then $a_i^j = 1$, as can be seen from that table. The strategy $a_k^*(t)$ represents the best-performing (optimal) strategy of agent k at time t . In other words, a_k^* is the strategy (of S strategies) actually used to trade with for agent k .

The reason for the name of the model should now be clear, since (1) says that every time a given strategy takes the opposite of the position taken by the majority, given by the sum in (1), it gains, whereas if it takes the same position as the majority it loses. The gain or loss in the representation (1) is proportional to the cumulative action of the agents.

Note the nonlinearity that enters (1) because of the (*) notation on $a_k^*(t)$. The nonlinearity is created since the agents try to cope with market changes by using their optimal strategy (of the available pool of s strategies) at each time step. Therefore, as the market changes, the optimal strategies of the agents change, which in turn (see figure 15.1) leads to changes of the market. This illustrates a highly nonlinear and nontrivial temporal feedback loop in the model, something that, it is argued, takes place in the way people trade in real markets when they try out optimal strategies depending on the present market behavior.

15.2.2 Some Results for the Minority Game

In the basic version of the MG the agents can either buy or sell a unit of stock at each time step, and they are assumed to have an unlimited amount of money and stock. The dynamics of the return of the market at time t , $r(t)$, is a function of the difference between the number of market participants who buy, and those who sell. Specifically:

$$r(t) = \log [P(t)/P(t-1)] \quad (15.2)$$

$$= \sum_{k=1}^N a_k^*(t - \frac{1}{2}) / \lambda, \quad (15.3)$$

with λ a constant describing the liquidity or market depth. Equation 15.3 expresses the fact that the return is proportional to the order imbalance given by $\sum_{k=1}^N a_k^*(t - \frac{1}{2})$.

The " $t - 1/2$ " notation in $a_k^*(t - 1/2)$ is meant to stress that the decisions made by the agents take place between the announcement of the price at the two moments $t - 1$ and t (see figure 15.1). In the MG literature, however, this fact is usually little stressed and an "abuse" of notation writes r and a_k^* taken at the same time t , but what is meant is that the action is taken first and then the price is announced. The sequence of events

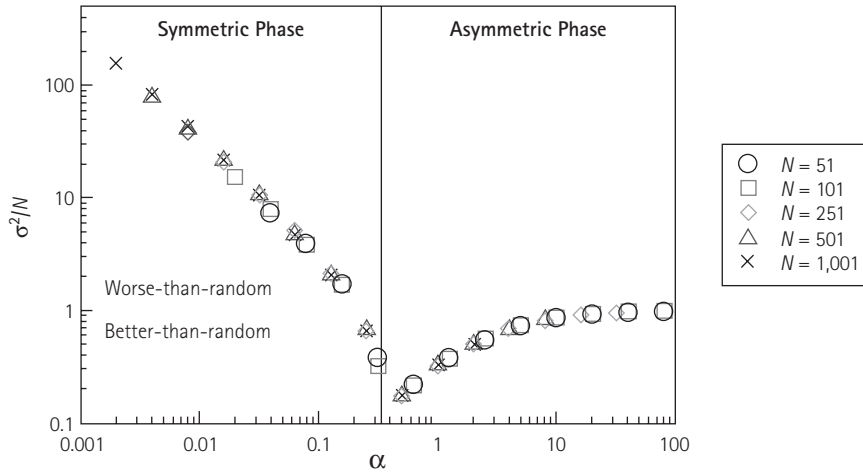


FIGURE 15.2 Volatility versus control parameter $\alpha = 2^m/N$ for fixed $S = 2$.

Source: Yeung and Zhang (2009).

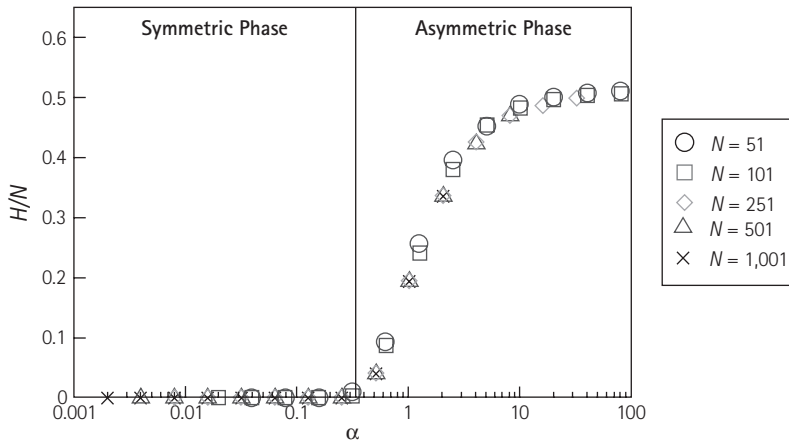


FIGURE 15.3 Predictability parameter H versus control parameter $\alpha = 2^m/N$ for fixed $S = 2$.

Source: Yeung and Zhang (2009).

is clearly illustrated in figure 15.1. Now we will, for simplicity's sake, stick to the same notation and take r and a_k^* at same instant of time.

Figures 15.2–15.3 illustrate the main results for the MG. They show the behavior of the model for different values of the control parameter $\alpha = 2^m/N$ but for a fixed value of S ($S = 2$). The figures are constructed as an average over many different games. Figure 15.2 shows the volatility of the model for a given fixed value of α plotted on the x-axis. The first thing to notice is the data “collapse” seen by the overlap of the five curves for the five different values of N . It tells us that α is indeed the relevant parameter to describe

the model. The small α case can be understood in terms of the crowd–anti-crowd theory of Johnson et al. (Johnson et al. 1999; Hart et al. 2000). In that case two groups of agents form, holding strategies with opposite predictions. This gives rise to huge fluctuations, seen by the large value of σ in figure 15.2 for small values of α . On the other hand, for large values of α , it becomes more and more unlikely that any two agents will hold the same optimal strategy. In this case the MG becomes a random game and the variance per agent approaches asymptotically the value 1, as would be the case in a game where the agents sold and bought randomly at each time step. Most interesting is the fact that for $\alpha \approx 0.34$ the agents, somehow, organize into a state where the market has lower volatility than in the case where the agents trade completely randomly. The minimum separates the state of the system into two different phases: First the crowded symmetric phase for small values of α where the market is unpredictable, as seen by the predictability parameter $H \equiv \frac{1}{P} \sum_{\mu} \langle a^* | \mu \rangle^2$, where $\langle a^* | \mu \rangle$ denotes the conditional average of a^* given the price history μ and the mean $\langle \rangle$ is calculated over all P price histories. $H = 0$ in figure 15.3. Second is the uncrowded phase, where, on the contrary, the actions of the agents leaves information that can be traded on. This is seen by the predictability parameter $H > 0$. Finally, the study by Cavagna (1999) should be mentioned, since it shows the perhaps a priori surprising result, that it is not the feedback that is important in the MG but rather that the agents react to the same information. This is seen from the fact that the results in figures 15.2 and 15.3 remain almost unchanged (though there are some minor modifications in the asymmetric phase) if the agents use a randomly generated sequence of price moves instead of their own generated price history (see figure 15.1), in the MG literature this is called the exogenous games case). This means that the agents, sharing the same information is what matters, and not the feedback they generate in the formation of the price history (in the MG literature this is called the endogenous game case). This is an important result for analytical approaches to understanding the MG, since analytical results can be obtained for exogenous games but not in general for endogenous games. Note, however, that this is not a general result for agent-based models; the best counterexample will be presented in the next section, where feedback will prove all-important in the so-called \$-Game.

15.2.3 The \$-Game

The main problem with the Minority Game is, of course, that it is hard to imagine people trading in real markets just because they would like to be in the minority!

People trade either because they try to make money or because they need to hedge their positions, but even when trying merely to hedge a position traders will try to do so at the best possible price. This elementary fact is not taken into account in the Minority Game, and this was the main criticism that led to another agent-based model, called the \$-Game (\$G), introduced to stress the fact that people trade to make a profit (Andersen and Sornette 2003).

Actually, sometimes it is advantageous to be on the minority side of trades, especially when you enter a new position, since when opening a position you want to be opposite the rest of the market in order to get a favorable price: if most people buy but few people (including you) sell, the imbalance will push up the price, ensuring that you sold at a favorable price and vice versa, entering a position by buying an asset. Yet in the time that follows after entering a position, whether long or short, you no longer want your bet to be a minority bet. Let's say you bought some gold today. You want this bet to be on the majority side following your purchase, since then the imbalance created by the majority of trades pushes up the price in your favor. The best strategy in terms of profiting is therefore not always targeting the minority but shifting opportunistically between the minority and the majority. Specifically, the payoff function taking these considerations into account was proposed in Andersen and Sornette (2003) to be given by the following:

$$\delta f_{i,j}^{\$G}(t) = a_i^j(t-1) \sum_k^N a_{k(t)}^*(t) \quad (15.4)$$

$$= a_i^j(t-1)r(t)\lambda. \quad (15.5)$$

Notice that the payoff function now depends on two different times. To understand this point imagine you only have access to the daily close of a given market. You decide to invest in this market at time $t-1$ and enter a position determined by how the market closes at that instant. Then it is not until the day after you know how the market closed (time t) that you will know if the decision you made the previous day was a wise one. This is especially clear from equation (15.5), where one can see the link between the decision made at day $t-1$, given by $a_i^j(t-1)$, and the return of the market between day $t-1$ and day t , given by $r(t)$. So a strategy gains (losses) the return of the market over the following time step, depending on if it was right (wrong) in predicting the market movement. Therefore, in the \$G agents correspond to speculators trying to profit from predicting the direction of price change.

One should note that the payoffs for both the MG (15.1) and the \$G (15.4) are assigned to strategies that are not active. This means all strategies except the optimal one at a given time t counts a virtual return since its contribution is not counted in the term $\sum_k^N a_k^*(t)$. The lack of this "self-impact" has been shown to have important theoretical consequences, but note that from a practical point of view it makes very much sense that agents assign a virtual return to their strategies. This is the reality in practice when investors try out new arbitrage ideas in a financial market, they first use price data as information in the search for arbitrage possibilities without actually entering the market.

The emergent properties of the price dynamics and the wealth of agents are strikingly different from those found in the MG. Most remarkable is that, as it is seen in figure 15.4, the wealth (the dotted line in the lower graph) of the best-performing MG agent—that is, (i.e. optimal as described by equation (15.1)—consistently performs badly whereas the relatively good performance in terms of wealth for the worst MG agent (the solid

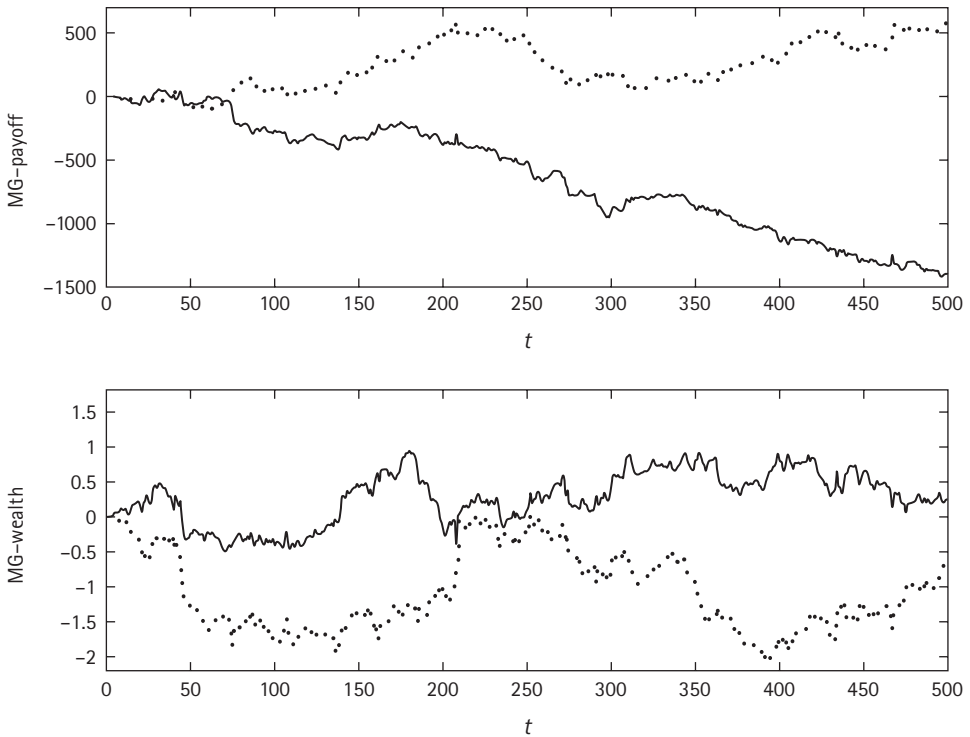


FIGURE 15.4 Payoff function (15.5) (upper figure) and wealth (lower figure) for the MG showing the best- (dotted line) and worst- (solid) performing agent for a game using $N = 501$ agents, memory of $m = 10$ and $s = 10$, strategies per agent.

Source: Andersen and Sornette (2006).

line in the lower graph) shows clearly that a minority strategy will perform poorly in a market where agents compete to make money. In contrast, $\$G$ agents, as defined in terms of (15.5), match by definition the performance of the wealth of the agents. This does not, however, preclude the potential usefulness of MG strategies in certain situations, in particular for identifying extrema, as is shown below. As with the MG, the dynamics of the $\$G$ contains nonlinear feedback, which is thought to be an essential factor in real markets, because each agent uses his or her best strategy at every time step. This attribute makes agent-based models highly nonlinear and, in general, unsolvable. As the market changes, the best strategies of the agents change, and in turn change the market. As shown in Roszczyńska-Kurasinska et al. (2012), the Nash equilibrium for the $\$G$ without constraints is given by Keynes's "Beauty Contest," in which it becomes profitable for the subjects to guess the actions of the other participants, and the optimal state is one for which all subjects cooperate and take the same decision (either to buy or to sell). A plain way of saying the same thing is that in the $\$G$ without any constraints on the amount of money or stocks that the agents can hold, bubbles are generated spontaneously. This intrinsic bubble-generating property is specific to the $\$G$ and

completely absent in the MG since agents seeking the minority generate mean-reverting behavior. As we will see we can use this intrinsic tendency of the \$G to generate bubbles to actually detect bubble generation in real markets. For the moment, notice that the price in the bubble state deviates exponentially in time from the fundamental value of an asset, which is assumed to be constant. All subjects profit from further price increases or decreases in the bubble state, but it requires coordination among the subjects to enter and stay in such a state.

15.2.4 A Scientific Approach to Finance

I would like to advocate for what could be best be called a scientific approach to finance. Scientific, because researchers propose experiments to go hand in hand with theory, a tradition that comes from the hard sciences but is a rare approach in traditional finance. It has been argued that the reason for the success of the hard sciences is precisely the link between proposing theories or models against which one can test via experiments. This lies at the heart of the philosophy of science suggested by the Austro-British philosopher Karl Popper (1902–1994). He introduced the idea that a scientific theory can never be completely verified no matter how many times experimental testing confirms the assumptions of the theory, but one single case in which the theory is shown not to hold experimentally, can falsify the theory. The term “falsifiable,” introduced by Popper, simply means that if a theory is wrong one should be able to show this by experiments. By the same logic this also means that if somebody introduces a theory that cannot be falsified it is not a scientific theory. The efficient market hypothesis is an example of a theory that can never be falsified by experiments, and therefore in Popper’s view, is not a scientific theory. Popper placed falsifiable theories at the root of the apparent progress of scientific knowledge over time, and therefore they also seem like the best candidates for a way forward in finance.

Having stressed the need for experiments according to Popper, it is noteworthy that experimental finance took a big step forward with the awarding of the Nobel Prize in economics to Vernon Smith in 2002. His most significant work was concerned with market mechanisms and tests of different forms of auction. By far the major part of the experimental work in finance (including that by Vernon Smith), however, has considered human rationality and the ability of markets to find the proper price close to an equilibrium setting. Contrary to this approach, behavioral finance uses a much more realistic description of how actual decision making takes place in financial markets. It would therefore seem like a very natural approach to take the insight gained from behavioral finance and apply it to experiments performed on financial markets.

Very little effort has been made in this direction. The main reason is may be that the major part of research done in behavioral finance is concerned with how individual decision making takes place (prospect theory included) in a *static* setting, whereas price setting in financial markets is clearly a *collective* and *dynamic* phenomenon.

Ultimately, one should see the actual pricing taking place in a given financial market as one big experiment performed in real time. So if you have a model for a financial market such as the Minority Game or the $\$$ -Game, you should be able to falsify at least parts of your model against real market data. Most important, however, is the following minimal requirement, or test, for any models claiming to describe price dynamics in financial markets: randomly assign values to the parameters of the model, and generate a history of price time series according to the model with these given values of the parameters. Imagine you didn't know the parameters used to generate the price time series, and launch a search to try to determine their value. If this is not possible, then for sure you will never be able to estimate the parameters describing any real-time series of financial data.

15.2.5 Agent-Based Model Used as a Thermometer: Taking the Temperature of the Market to Predict Big Swings

Just as one can get information about the internal state of water by inserting a thermometer into the liquid, the idea of applying models of agent-based simulations to real financial market data and looking at how agents react via their decision making, would be a way to probe the internal state of the market. The hope would be to get a new way of characterizing markets by obtaining general information about, say, whether the market would be in a "hot" or "cold" state. I turn to discussion of how to implement this idea in practice and how sudden large swings in markets seem to have precursors that can be used in attempts to predict such swings.

Before making any claims that agent-based models should be applicable for extracting information about real market data, it is clear that a first minimal requirement needs to be met. Imagine, therefore, a "black-box" time series generated by a multi agent game for a fixed *unknown* parameter set of the three parameters (N , m , and s) as well as an *unknown* specific realization of initial strategy choices assigned to the agents in the game. If you perform reverse engineering using the black-box time series as a blind test, you should be able to extract the parameters used to create this time series. If you cannot estimate the parameters of a theoretical defined model it would be useless spending time on real market data, which are not created according some theoretical framework.

A reverse engineering test for the MG was proposed by Lamper et al. (2002). A priori it seems an almost impossible task trying to estimate parameters in a model with an astronomically large number of different initial conditions given by the 2^{2^m} different strategies. Nonetheless, Lamper et al. (2002) showed how it is possible to perform this task for a moderate number of strategies, meaning for a sufficiently small memory (say, $m = 2 - 5$) used by the agents in the MG. This is indeed very good news since we see that it is possible, at least in the case of the MG, to reverse engineer and find the parameters of a black-box-generated time series *without knowing in detail the theoretical setup of the model!* Similarly, the possibility of reverse engineering was later found to hold for

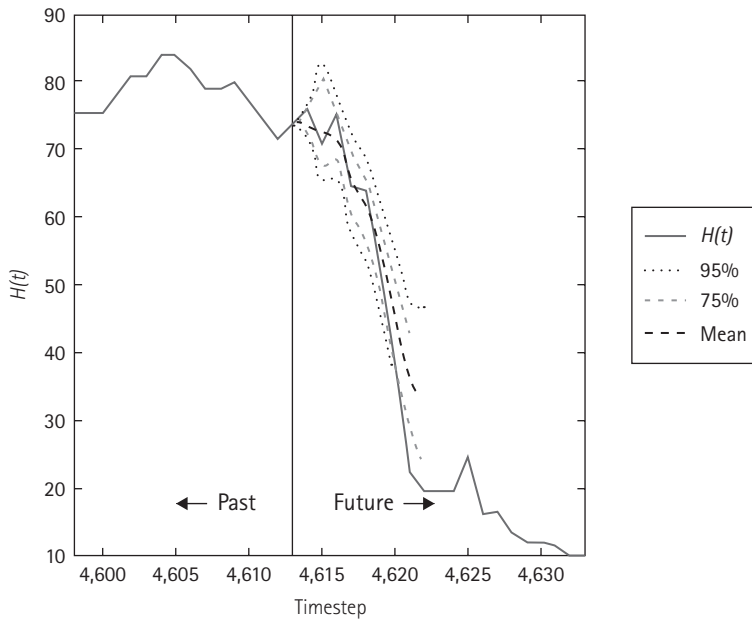


FIGURE 15.5 Comparison between the forecast density function and the realized time series $H(t)$ for a typical large movement. The large, well-defined movement is correctly predicted.

Source: Lamper et al. (2002).

moderate values of m ($m = 2 - 5$) in the case of the $\$G$ as well (Andersen and Sornette 2006).

I end this section with an appetizer to the discussion in the next section. Lamper et al. (2002) give an example of how large changes can be predicted in a multi agent population described by the MG. The remarkable fact is that the predictability actually increases prior to a large change. It was shown how predictable corridors appear at certain time periods, a typical example of which is shown in figure 15.5. In the next section I identify the mechanism that gives rise to such large market swings.

15.3 RESULTS

15.3.1 The Idea of Decoupling

I start by presenting a method by which soft human decision heuristics can be precisely formalized as rules in the agent's decision making in agent-based models of financial markets. The agent-based models that I have in mind are games such as the Minority

Table 15.2 Example of an $m = 2$ strategy table

Signal	Action
00	+1
01	-1
10	+1
11	+1

Game or the $\$$ -Game. In this formalism, cognitive closure is defined as the decoupling of an agent's strategies from the market feedback.

In what follows I describe decoupling in the context of the $\$$ G but should note that this is a general property related to agent-based games that use look-up tables like table 15.2. It is important for the understanding of the following to note that the optimal state of the $\$$ G is the solution in which the price deviates exponentially in time from the fundamental value of the asset, enabling all agents to profit from constant price increases or decreases in the bubble or anti-bubble states. Finding the optimal solution in the $\$$ G requires coordination among the agents to enter and stay in such states. Coordination, however, is not driven by intentionally cooperative behavior of all agents; rather, it emerges from independent decisions of the majority of agents who choose the optimal strategies from the entire set of strategies. These optimal strategies, presented in the reference tables, happen to lead to the same action, which on an aggregate level is seen as synchronization. The question is whether this mathematical formalism can adequately describe the process of human decision making.

Agents' strategies, at first glance, are very different from what we know about decision heuristics of humans (Tversky and Kahneman 1974). Decision heuristics provide rules for human decision making. Heuristics are formed in terms of verbally (or rather propositionally) formulated conditional rules. Clearly, humans are cognitively incapable of precisely representing the many vectors and exact sequences of market dynamics needed to represent and value the strategies. The reverse formalization of human decision heuristics by lookup tables is simple, however. Any conditional rule of human reasoning can be represented by a lookup table. In order to accept the notion that agents' strategies represent human decision heuristics, we only need to assume that each agent's strategy depicts in an algorithmic way implementation of a decision heuristic that for humans would be specified in higher-level language.

In this vein, cognitive closure in market players may be interpreted as setting up the mind for what will happen in relatively distant future, regardless of what happens in the near future. In terms of decision heuristics, after observing certain patterns of market dynamics investors may come to the conclusion that the market trend is set and, further, the temporary market reversals are not indicative of the real market trend. For example,

if the market player judges that the market trend is upward, then the increase in price serves as a confirmation of the expected trend, so the decision is to buy. A drop in price may be perceived as a momentary deviation from the governing trend, which indicates immediate correction, so the decision is also to buy. This may be translated as decoupling of agents' strategies, something that will be crucial in understanding how speculative bubbles are formed.

As long as the majority of investors are reacting to incoming information the market dynamic is unpredictable. If, however, a large enough proportion of investors makes the same decision about the direction, the market will evolve regardless of what happens next. The market may become temporarily predictable since investors are in fact locked into their decisions, which are temporarily decoupled from information concerning the market. A prolonged locked-in decision of investors to buy results in bubbles, and locking in the decision to sell will result in market crashes.

Some strategies represented by reference tables have a unique property: the actions that they recommend are decoupled from the incoming information. A decoupling of the strategy means that the different patterns of market history lead to the same decision (e.g., to buy), regardless of whether the market went up or down at the previous time step. The main interest in the mechanism of decoupling is, as we will see, that it gives us a way to predict formation of speculative bubbles before they are seen in the price data.

15.3.2 The Formalism of Decoupling

The simplest example of decoupling in agent-based models is the case in which an agent uses a strategy like the one presented in table 15.2, but with each item in the action column being. In this case, the strategy is trivially decoupled because, no matter what the price history is, this strategy will always recommend buying. In the notation used in Andersen and Sornette (2006), such a strategy would be referred to as an infinite number of time steps decoupled, conditioned on any price history. In plain words: If somebody ends up holding a strategy having only 1s in the decision column, such a strategy would always make the decision to buy at every time step, independently of what goes on in the price history of the market. Note that the probability of an agent's possessing such a strategy is very small and is given by 2^{-2^m} because 2^{2^m} is the total number of strategies. The strategy presented in table 15.2 is one time step decoupled, conditioned on the price history being $\mu = (01)$ at time t because in cases where the market at time $t + 1$ went up— $(01) \rightarrow (11)$ —or down $(01) \rightarrow (10)$ —the strategy in both cases recommends buying at time $t + 2$; that is, for both (10) and (11) , buying is recommended. In plain words: Every time we see an occurrence of the price history where the market first went down (0), then up (1), we know for sure what action this strategy will recommend in two time steps. In order to visualize this, imagine that the market following the down-up movement goes down (0). Then the updated price history to be used by the strategy at the next time step is up-down (10) , and

here the strategy recommends buying. If instead the market following the down-up movement goes up (1), the updated price history at the next time step is would be up-up (11), in which case the strategy also recommends buying. So this means that whatever happens in the time step after the up-down movement of the market, we can tell that following this time step the strategy will always recommend buying. Likewise, the same strategy is seen to be decoupled by one time step, conditioned on the price history (11), since, independent of the next market movement at time $t + 1$, the strategy will always recommend buying at time $t + 2$. In a game with only one agent and only one strategy, such as the one given in table 15.2, we could therefore know with certainty what the agent would do at time $t + 2$ if the price history at time t was either (01) or (11), independent of the price movement at time $t + 1$.

As discussed above we have seen how sometimes the strategies of the agents can lead to momentary pockets of predictability of that action a given agent will take in near future. But even knowing what one or several agents will do, does not mean that we necessarily know what will happen at the level of the market. To know for sure how the market will behave, we need to encounter the situation in which not only are a majority of agents but they are decoupled *in the same direction*. To see how such a condition can arise, I introduce the following formalism. A strategy is deemed coupled to the price time series if, conditioned on a given price history at time t , we need to know the price movement at time $t + 1$ in order to determine what the strategy will recommend at time $t + 2$. Conditioned on having a price history of either (00) or (10) at time t , the strategy shown in table 15.2 is coupled to the price time series because we cannot know what it will recommend at time $t + 2$ without first knowing the price history at time $t + 1$. At any time t , one can therefore ascribe the actions of agents to two contributions, one from coupled strategies and one from decoupled strategies, as:

$$A^{\mu(t)} = A_{coupled}^{\mu(t)} + A_{decoupled}^{\mu(t)} \quad (15.6)$$

The condition for certain predictability one time step ahead is therefore

$$\left| A_{decoupled}^{\mu(t)}(t + 2) \right| > N/2 \quad (15.7)$$

because in that case we know that, given the price history at time t , the sign of the price movement at time $t + 2$ will be determined by the sign of $\left| A_{decoupled}^{\mu(t)}(t + 2) \right|$.

15.3.3 Decoupling Seen in Computer Simulations and in Real Market Data

A priori, it is highly nontrivial whether one should ever find the condition for decoupling to be fulfilled at any instant of time, even in the computer simulations of the agent-based models. As shown in Andersen and Sornette (2006), if the agents in the

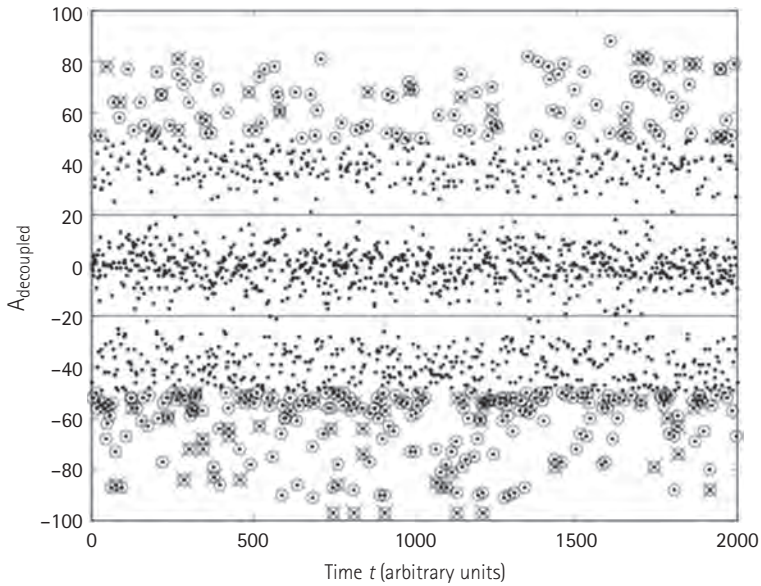


FIGURE 15.6 $|A_{decoupled}|$ defined in (15.6) as a function of time for the MG with $N = 101$, $s = 12$, $m = 3$. Circles indicate one-step “prediction days” defined via the condition (15.7). Note that “prediction day” means prediction with certainty of the price direction of the market two time steps ahead. Crosses correspond to the subset of days with two or more consecutive one-step prediction days.

Source: Andersen and Sornette (2006).

MG and \$G play their strategies randomly, the condition (given the parameters used in the simulations) is never fulfilled. When the agents choose their strategies randomly there is no feedback between the price movements of the market and the decision making of the agents, so it’s important to note that in this case we cannot use decoupling to predict what the agents will do next. The way decoupling arises therefore has to be related to feedback and the dynamics of the pricing, which somehow imposes the requirement that the optimal strategies of agents will be attracted to regions in the phase space of strategies that contains decoupled strategies. In the \$G the natural candidates for attractors are the two most trivial strategies with actions either all $+1$ or all -1 . However, because it is very unlikely that an agent will possess these two strategies, an attractor would have to consist of regions in the phase space of strategies that are highly correlated to such strategies. In the MG it seems even less likely that decoupling should ever take place, since there seem to be no natural attractors for decoupling in that game.

The MG nonetheless shows predictable behavior, as can be seen from figure 15.6. Each dot in this figure shows the value of $A_{decoupled}$ versus time. Given the parameter values of the MG used in the simulations, the condition of (15.7) means that whenever the $A_{decoupled}$ value becomes larger than 50, or smaller than -50 , we can predict the direction of the market two time steps ahead independent of the market move one time step ahead. As can be seen from the figure, most of the dots lie within the $[-50 : 50]$

interval, and for such events we have no prediction power. However, the dots enclosed by a circle illustrate “prediction days” given by the condition (15.7), which means that standing at time t one can predict for sure the outcome of the market at time $t + 2$ independent of which direction the market takes at time $t + 1$. Crosses in the figure correspond to the events concerning which two or more consecutive price movements can be predicted ahead of time.

At first it might seem like a somewhat theoretical exercise to predict ahead of time how agents in a computer simulation will behave. One could argue that since the setup of the computer program is entirely determined by the parameters of the simulations, all information is already encoded in the program, and the only thing needed to predict movement two time steps ahead would be to let the program run two additional steps and then see what happened. This suggestion, however, misses the point concerning the more interesting situations of practical applications where simulations of the agent-based models are “locked” to real market data, as explained in section 15.2. This corresponds to considering real predictions in real time. In the case of applying real market data as input to the computer simulations of the agent-based models, we encounter the situation of knowing, for example, the closure of the markets today. Observing a moment of decoupling in the computer simulations given the market closure of today means that we know what the agent-based model will predict, not for tomorrow, but for the day after tomorrow. In that case we don’t need to wait and see how the markets close tomorrow to be able to make the prediction tomorrow—it can be made today.

The mechanism of decoupling can now be seen as a natural to understand and define the process that leads to observed big swings in the market. To see how the method of decoupling works when applied to real market data, consider figure 15.7, which shows the Nasdaq Composite price history (the thick dashed line) as a function of time in units of days over a half-year period. The sample period was chosen so as to show no apparent direction of the market in the first half of the period. As described in section 15.2, when “slaving” a game to real market data, one uses the last m price directions of the real market data as input to the $\$G$ agents in the computer simulations. The agents therefore adjust their optimal strategies according to the real price history. The $A_{decoupled}$ defined in (15.6) can in this way be calculated from the optimal strategies of the agents, which are determined dynamically via the price history of the real data. The data from the in-sample period were first used to fix the parameters of the $\$$ -Games that could best describe the Nasdaq Composite during the in-sample period. The $\$$ -games that best fitted the Nasdaq data in-sample were found using genetic algorithms that explored the three different parameters of the $\$$ -Games as well as different initial strategies attributed to the agents in the game. The parameters of the $\$G$ that supposedly give the best representation of the Nasdaq data in-sample were fixed, and, the remaining half of the data set was then used out-of-sample. The ten thin solid lines in shown figure 15.7 show third-party $\$$ -Games obtained in this manner. They were all constructed with the same optimal parameters found in the genetic algorithm search, but each game had agents using different initial realizations of the strategies attributed to them in the beginning

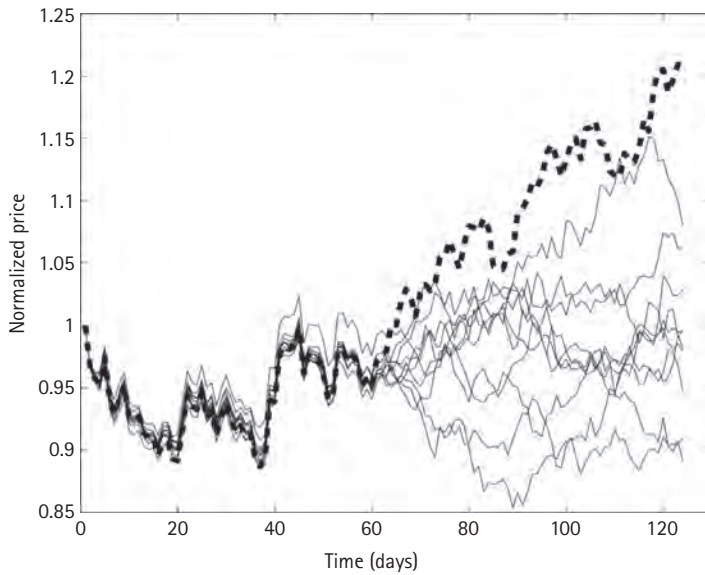


FIGURE 15.7 Nasdaq Composite price history (thick dashed line) as a function of time (days). Thin solid lines predict trajectories obtained from third-party games. The first in-sample 61 days are used to calibrate ten third-party games. Days 62–123 are out of sample. The third-party games do a poor job of predicting the out-of-sample prices of the Nasdaq Composite index, while table 15.1 shows that they have specific pockets of predictability associated with the forecasted “prediction days.”

Source: Andersen and Sornette (2006).

of the time period. Note that the ten thin solid lines all follow closely the real market data (the thick dashed line) for the in-sample period. It shows that different games with the same parameters, all slaved to the same input data (here the Nasdaq Composite), perform similarly despite having agents with different pools of strategies assigned to them at the beginning of each game. Given the size of the pool of strategies (2^{2^m} this is may be a priori a surprising result. It lies at the core of the reason why one can perform reverse engineering and find the parameters of a given time series generated by an agent-based model as shown by Lamper et al. (2002). Of course, the best way to try out a method is to test it on real data and see how it works. In order to see if predictions could be made using the idea of decoupling, the ten third-party games were then fed the Nasdaq price history for the second half of the period, and predictions were issued at each close of day. The ten games were expected issue a prediction when detecting a prediction day.

It turns out that using only the majority decision of the third-party games results in poor prediction of the out-of-sample prices of the Nasdaq Composite index, while, as table 15.3 shows, they predict specific pockets of predictability associated with the forecasted prediction days. As can be seen from the table, the larger the threshold (measured by the parameter $A_{decoupled}$) for prediction, the larger the success rate in

Table 15.3 Percentage of out-of-sample success rate (second row) using different thresholds for the predicted global decoupled action (first row) of the third-party \$-Games calibrated to the Nasdaq Composite index. Nb (third row) is the number of days from $t = 62$ to $t = 123$ that have a predicted global decoupled action $|A_{decoupled}|$ larger than the value indicated in the first row

$ A $	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5
%	53	61	67	65	82	70	67	67	100	100
Nb	62	49	39	23	17	10	6	3	2	1

Source: Andersen and Sornette (2006).

predicting the direction of the price movement of the Nasdaq Composite will be. The most important point to note is that the success rate increases with the amplitude of $A_{decoupled}$. The larger the value of $A_{decoupled}$, the more confident one should be in the prediction. A large value also comes at a price, however, since there are fewer such events, meaning that we obtain worse statistics.

Having glimpsed how to apply the idea behind decoupling to real data, we now turn our attention to the detection of speculative financial bubbles. The following section will give additional insight into when decoupling could be a mechanism driving real price dynamics, and in particular, under which circumstances this mechanism could be used in the prediction of the onset of speculative bubbles.

15.3.4 Using Decoupling to Detect Speculative Financial Bubbles

This example of decoupling involves experiments with human subject trading in a market. The results shown are taken from Roszczynska-Kurasinska et al. (2012), which gives a detailed description of the experiments.

Figure 15.8 illustrates the price histories generated by humans in three experiments. The circles illustrate the evolution of the price. At the beginning of all three experiments the price was normalized to 1. The two first plots give examples of the creation of a bubble whereas the last plot illustrates an anti-bubble. The time $t - t_b$ was chosen so that the onset of the bubble/anti-bubble (defined by t_b) happened at time 0. The dashed vertical lines show in all three cases the time $t_b + m$ when the presence of a bubble/anti-bubble becomes evident from the price history itself. This is the time when the last m directions of the price movements had the same sign (positive: up; negative: down). The solid lines illustrate the percentage of optimal decoupled strategies along the direction of the bubble (plots a and b) or anti-bubble (plot c); whereas the dotted lines represent the optimal decoupled strategies along the opposite direction. For experiments in which synchronization is due to decoupling, one can see a clear

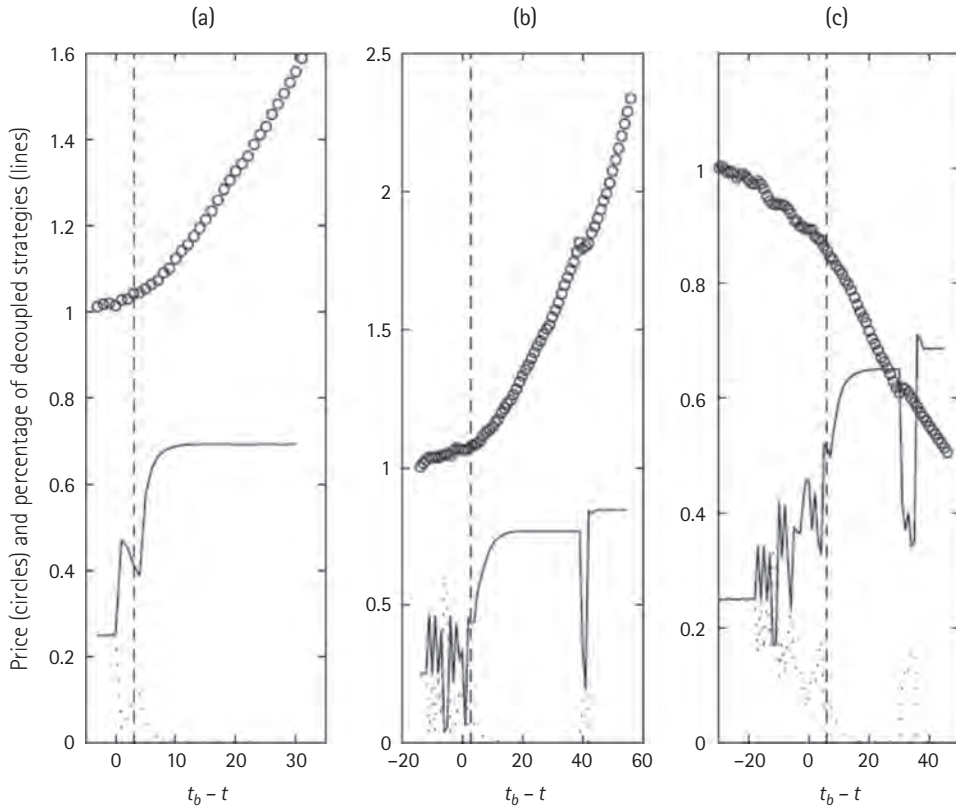


FIGURE 15.8 Splitting of the optimal strategies used by agents when acting on the basis of prices generated by the subjects in experiments with lengths of memory of $m = 3$ (shown in a and b) and $m = 6$ (shown in c) (vertical dashed line).

Source: Roszczynska-Kurasinska et al. (2012).

split of positive decoupled versus negative decoupled strategies happening before the onset of the bubble. As seen in figure 15.8a, even when a bubble is created very rapidly (with small $m = 3$), we see a split. However, as expected, this split becomes clearer over a longer time period for the larger memory length $m = 6$, as seen in figure 15.8c. In this case the subjects had a longer period over which they traded in a descending market before the final synchronization occurred. Such a condition resembles features seen in real markets, with a typical run-up or run-down before the first stages of a bubble/anti-bubble sets in, which gives us confidence in applying our method to real market data. To further test the idea of decoupling, a manipulation of the price was done one time step after the subject had entered a decoupled state. This is seen by the spikes in the prices in figure 15.8b–c. If truly decoupled, the subjects should continue in a state of decoupling independent of the manipulation of the price. This was indeed found to be the case. These results reveal that decoupling can explain the synchronizations of people. Moreover, it appears that decoupling enables the prediction of the future market

state, not only during the emergence of bubbles/anti-bubbles but also in a short-lived synchronization (for several time steps) of investors.

15.4 CONCLUSIONS

I have tried to argue for uses of game theory and agent-based modeling that go beyond the standard methods used in traditional approaches to finance. The idea is that we need models that more realistically describe the decision making of investors in financial markets. Two such models have been introduced, the Minority Game and the $\$$ -Game, and their main results have been discussed. It has been argued that a more scientific approach to finance is needed, with theories and with experiments carried out on trading floors to falsify and thereby select relevant properties of theoretical models. Some preliminary results in this direction have been shown by the introduction of the idea of decoupling of investor decision making. The hope is that the present article can help spark interest in such a scientific approach to finance with the introduction of more models that can be tested by experiments performed on trading floors.

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CHAPTER 16

AN OVERVIEW AND EVALUATION OF THE CAT MARKET DESIGN COMPETITION

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16.1 INTRODUCTION

Since 2007 a team of academics from the University of Liverpool, City University of New York, and the University of Melbourne have run the annual CAT Market Design Tournament, or CAT Competition. This competition, which is part of the larger Trading Agents Competition (TAC), aims to encourage research into the design and application of automated market mechanisms that are able to adapt to changing competitive environments.

In computer science, there are many research areas where experimental platforms have been developed for comparing and benchmarking research, and open competitions have been built around some of these platforms. Examples include the MiniZinc competition for finite domain constraint solvers¹ and the various TAC competitions for automated trading, supply-chain networks, online ad exchanges, and brokerage of electric power.² From the TAC competitions, the research arising on automated trading, price prediction, and adaptive bidding has been summarized in the book by Wellman et al. (2007). Perhaps the most famous series of research competitions in computer science is Robocup, which since 1997 has organized annual competitions for soccer-playing robots, in separate leagues comprising physical robots of various

¹ See <http://www.minizinc.org/challenge.html>.

² See <https://www.sics.se/projects/trading-agent-competition>.

shapes and sizes, as well as virtual players and managers.³ This crowd-sourced series has generated no fewer than nineteen annual volumes of refereed research papers, each published in Springer's Lecture Notes in Computer Science series, as well as numerous other publications, and advanced computer science considerably.

The TREC conference on information retrieval⁴ is another example of a competition and online repository, in this case aimed at encouraging and comparing research into information retrieval. This annual competition, started in the early 1990s, aims to promote research into a range of information retrieval tasks and provides sets of data for researchers to use. However, work from Armstrong et al. (2009) casts doubt on the success of the TREC tournaments. Their research shows that the baselines used in information retrieval experiments are typically uncompetitive in many cases they perform worse than the first TREC experimental systems—and there is no upward trend in scores, with results reported in 2008 showing no improvement over results from 1999, and many 2008 scores falling below the median of the original TREC experimental systems. Entrants are not requested to upload their entries, as they are in the CAT competition.

The aim of this chapter is to present our experience with the CAT competition (section 16.2), to present details of research that has arisen from the tournaments (section 16.3), and to evaluate whether the CAT competition has helped improve the state of the art in automated mechanism design (section 16.4), so that research in this area does not go the way of information retrieval. To evaluate the CAT competition, we experimentally compared the five annual winning entrants of the competition (2007–2011) against each other in a series of five-player experiments, as well as a series of two-player round-robin experiments. The results indicate that later entrants perform better on aggregate, especially in the five-player experiments. In the two-player round-robin experiments, the most notable improvement came from the winner in the second year of the competition, which was the first year that entrants had access to a previous year's entrants to run their own experiments.

16.2 THE CAT TOURNAMENT

16.2.1 Motivation

The economic world we inhabit has an enormous diversity of marketplaces. From garage sales of unwanted household goods to local bodegas and corner stores, to supermarkets and international chains, through automated online exchanges for the trading of complex derivative and futures products, this diversity is astonishing. The response of mainstream economic theory to this diversity until recently, however, was to abstract all the particular features of markets away to create a generic model,

³ See <http://www.robocup.org/>.

⁴ See <http://trec.nist.gov/>.

a model of a marketplace having independent buyers and sellers who interact in a uniform manner (which is often left implicit). This generic model ignores many of the specific features of particular markets, for example, that in markets for medical services, the provider and the purchase decision maker may often be the same person (e.g., a doctor acting on behalf of a patient), or that in higher-education markets, the consumer—the student—may also constitute the product being purchased, or that in management consultancy markets, there may be no way for the purchaser to assess the quality of the provided service, neither ahead of time nor even afterwards. Most important, a generic model also abstracts away from the “ecology” of all the different markets operating in broad parallel with one another, some competing directly with others, some competing indirectly, some parasitic on others, some complementary to others, and some generating information used by others (Mirowski 2002). Ignoring this ecology of markets means that, for example, with a single generic model of a marketplace we cannot explain why the world’s most sophisticated futures trading exchanges in Chicago or New York or London operate within only short walking distances from local household garage sales, and yet use very different rules and procedures. Of course, such complex futures markets usually target different trading participants than do garage sales, but this is precisely why the rules of the two types of markets are different. A generic model of all markets does not allow for different markets to have different rules, nor does it seek to understand the impacts and dynamics of these rules and how they may interact with one another.

Although mainstream economics has long studied some exceptions to the generic model of perfect competition—for example, monopoly, oligopoly, and monopsony markets—the discipline has mostly ignored competition *between* marketplaces. But competition between markets has grown in importance with the rise of the Internet, the Web, and electronic trading. While there are some exceptions in economic theory that analyze competition between platforms, such as Ellison et al. (2004) and Armstrong (2006), the abstract nature of these models makes them unable to capture some important, finer-level details of particular marketplace designs that have a major influence on the emergent behavior of the market as a whole. With buyers, sellers, and trading platforms now able to be located anywhere, so too may their transactions take place anywhere. As a result, the precise rules and procedures of a particular market may have important consequences for the attraction and retention of buyers or sellers, with each trader able to choose between multiple competing marketplaces, and with each marketplace having its own institutional rules and architecture.

Mirowski (2007) argues that markets are fundamentally collections of diverse, evolving algorithms and that the field of computer science is required if we are to better represent the complexity of economic processes in order to understand markets. We believe that this is true for studying competition between heterogeneous marketplaces.

One key motivation, therefore, for establishing the CAT Market Design Tournament was to use computational simulations involving such algorithms to better understand the nature of competition between heterogeneous marketplaces, in the absence of any mature economic theories on this topic. In particular, such an understanding should provide scientific knowledge and guidance on how specific market rules and

procedures would impact market and trader performance, and how they would interact with or influence one another via competing markets. For example, Cai et al. (2012) ran experiments using JCAT that restricted traders from moving between markets that are somehow “connected” to neighboring marketplaces. Their results indicate that the specific network topology of connections between marketplaces influences the allocation of traders to marketplaces and the amount of time it takes to find an equilibrium point. It is likely that the network topology between markets enforcing other constraints, such as allowing exchange of information (e.g., unfilled orders or fee structures), would also affect marketplace and trader actions and performance. The operators of, and traders in, the world’s stock exchanges have anecdotal understanding of these relationships and impacts, but no coherent, grounded, and granular theory exists for the specific rules that exchange operators typically consider and deploy. Such a theory would provide guidance to the designers, implementers, and managers of electronic marketplaces, people who are increasingly information technologists rather than economists.

Second, given the rise of electronic trading and the globalization of trading, it is easy to imagine that operators of exchanges would need to respond in real time, or close to it, to changes in their business environments: If a competing exchange introduces a new rule or new fee structure and quickly attracts more traders or traders who are more desirable, then rival exchanges that are losing customers will need to respond as quickly as possible. Thus, our second key motivation was to develop the science and software engineering of adaptive market mechanism design, aiming ultimately for fully automated mechanism design.

Traditionally, the design of market mechanisms has been undertaken by human economists using a complex mix of mathematical analysis (chiefly using game theory), computer simulation, experiments with human subjects, and trial operations. Nik-Khah (2005), for example, observes that all of these methods were used in the design of the specific rules of the U.S. Federal Communications Commission’s auctions of the PCS radio frequency spectrum in the mid-1990s. Given the subtlety and complexity of actual market design engineering, it was not obvious how, or even whether, automated market mechanism design would be possible. We developed the CAT Market Design Tournament to encourage multidisciplinary research in this area and to provide a standard experimental platform by which such research could be compared and benchmarked. In terms of the refereed research publications generated by the game participants, our goal of encouraging research here has certainly been achieved, and several promising approaches to adaptive and automated mechanism design proposed, for example, Niu et al. (2012), Pardoe et al. (2010), and Phelps et al. (2010).

In designing the CAT Tournament and its supporting software platform JCAT, we sought to model practice in actual (spot) financial exchanges, while keeping the game rules straightforward. Thus, the marketplaces in CAT are called *specialists*, representing that they are specific agents that take orders for particular assets. Similarly, the performance criteria used in the tournament reflect those used by managers to evaluate performance of real-world providers of exchange services: profits, market shares, and transaction success rates (i.e., matching intending traders with appropriate

counterparties). The strategies used by software traders in the JCAT platform are four automated trading strategies commonly studied in the trading agents literature, and used, to a greater or lesser extent, by automated traders in financial markets. Further details are presented in the next subsection.

Full details of the organization of the CAT Market Design Tournament can be found in the technical documents describing the game and participation (Cai et al. 2009; Niu et al. 2009). In addition, the various research published by those involved in the CAT Tournament provide useful information about the game operations as well as reports of the resulting research; the most important of these research publications are discussed in section 16.3.

16.2.2 Operation of the Tournament

We now briefly outline the structure of the CAT Tournament. Entrants to the competition design and implement software agents, called specialists, which operate as matchmakers between automated traders. In the JCAT platform, specialists represent competing marketplaces in the same way that, say, the New York Stock Exchange, NASDAQ, the London Stock Exchange, the Paris Bourse, and others all compete. The traders, adaptive software agents provided by the tournament organizers, buy and sell a single generic commodity. Traders are either buyers or sellers but not both. Each trading day, each trader first selects a specialist to trade through on that day and then decides what level of shout to bid or ask in that specialist. For the shout decision, each trader is endowed with one of four standard automated trading strategies:

- ZI-C: Zero Intelligence—Constrained (Gode and Sunder 1993), a strategy that essentially involves random shouting (bidding by potential buyers or asking by potential sellers) within the budget constraint of each trader. This strategy ignores any trading history or experience of the trader.
- ZIP: Zero Intelligence Plus (Cliff and Bruten 1997), a strategy that takes account of the trader's experience to extend ZI-C and adapts the shouting price according to changing market conditions.
- RE: Roth and Erev (Erev and Roth 1998), a strategy that mimics reinforcement learning by humans in game-playing.
- GD: Gjerstad-Dickhaut (Gjerstad and Dickhaut 1998), a strategy in which traders form beliefs about the expected success and profits from particular shouts based on their trading experience.

These strategies were selected because they have been extensively studied in the automated trading literature, and some have also been shown to work well in practice. Within the CAT Tournament, the strategy used by each trader is private knowledge, and is not revealed to other traders nor to the specialists. Also, the proportion of traders executing each of the four strategies is not revealed to specialists.

Each trading day in the CAT Tournament begins with specialists deciding on and posting their fee levels for all traders to see. The fee structure and fee levels inform the traders' decisions as to which specialist to select that day. Specialists have the following goals:

- To attract traders to their marketplace, in competition with the other specialists;
- To match potential buyers with potential sellers through a market matching and clearing algorithm; and
- To earn profits by deciding a daily fee structure and fee levels, such as fees for placing a shout or a commission fee based on a trader's profit.

Entrants to the tournament are scored on three equally weighted criteria each trading day: share of all profits earned by specialists that day, share of traders attracted to their marketplace that day, and the specialists' transaction success rate, which is the percentage of traders that were successfully matched with counterparties by the specialist. These daily scores are summed over the course of the tournament, with the winner being that specialist having the highest cumulative score across all scored days. To eliminate the possibility of specialists' taking tactical advantage of edge effects, scoring in the tournament does not commence until some days after the first trading day, and ends some days before the last trading day.

16.2.3 History of the Tournament: 2007–2011

The CAT Market Design Tournament was run five times, once each year from 2007 to 2011. The tournament was operated as part of the annual Trading Agent Competitions (TAC), hosted each northern summer by the International Trading Agent Research Association.⁵ The first TAC game, now called TAC Classic, invited entrants to program trading agents able to buy and sell various electronic goods in multiple complementary marketplaces. The name *CAT* was adopted for the Market Design Competition for two reasons: first, because it is short for *catallactics*, the science of economic exchange, and second, because the game operates in reverse to the way in which TAC Classic operates: in CAT, the entrants provide marketplaces and the competition organizers provide automated traders, rather than the reverse.

The TAC Competitions are usually held alongside a leading computer science research conference in artificial intelligence, multiagent systems, or e-commerce, and so they change location each year. The CAT Tournament was hosted in Vancouver, Canada (2007), Chicago (2008), Pasadena, California, (2009), Boston (2010) and Barcelona (2011). The games in 2007, 2008, and 2009 were operated from a server in the Department of Computer Science at the University of Liverpool, UK, while those

⁵ See <http://tradingagents.org/>.

Table 16.1 Summary of CAT Market Design Tournaments, 2007–2011

Year	Entries	Winner (Country)	Countries Entering
2007	10	IAMWildCat (UK)	Australia, Croatia, France, Greece, Iran, Romania, UK, US
2008	11	PersianCat (Iran)	Australia, Croatia, Egypt, France, Greece, Iran, UK, US
2009	14	jackaroo (Australia-France)	Australia, China, Croatia, France, Greece, Iran, Mexico, Taiwan, UK, US
2010	9	Mertacor (Greece)	Australia, Croatia, Egypt, France, Greece, Iran, Poland, Taiwan, UK, US
2011	8	jackaroo (Australia-France)	Australia, France, Greece, Mexico, Poland, Taiwan, UK

in 2010 and 2011 were operated from a machine in what is now the Department of Computing and Information Systems at the University of Melbourne, Australia.⁶

Entrants to the CAT tournament came from around the world and included both large academic research teams and individuals. It is a measure of the intellectual challenges provided by the game that individual entrants from developing countries having clever ideas may compete effectively against first-world teams having access to the large-scale computational processing power needed for full testing of entrants. Table 16.1 shows the number and nationalities of entrants in each year of the CAT Tournament.

As with the other TAC games, entrants to CAT each year were encouraged to upload their entries (either as raw code or compiled binaries) to an online repository following the tournament.⁷ Doing this enables future entrants to test their intended entries against those of past entrants and helps ensure that the research community as a whole learns from the crowd-sourced research experiment that the CAT Tournament effectively constitutes. We have drawn upon the entries available in this repository to run the Hall-of-Fame simulation experiments we report in section 16.4.

16.3 RESEARCH ARISING FROM THE TOURNAMENT

Much research work has arisen from the design and the organization of the CAT Tournaments, exploring issues such as the modeling and engineering of electronic

⁶ The authors of this chapter include the CAT Gamemaster for 2007–2009 (McBurney), the CAT Gamemaster for 2010–2012 (Miller), and the lead software developer of the CAT platform (Niu). All three of these authors were involved in the design of the tournament, along with others, principally Enrico Gerding, Simon Parsons, and Steve Phelps.

⁷ See <http://www.sics.se/tac/showagents.php>.

market mechanisms, the design of novel market rules, the systematic evaluation of their performance, the dynamics of the interactions between multiple marketplaces and between markets and traders, and what can be learned from these dynamics to adapt market mechanisms to achieve better performance.

In this section we summarize some of the most important research that has resulted from the CAT Tournament. We broadly divide it into two categories: research on market mechanisms and research on automated traders using market mechanisms. In addition to the publications discussed below, no fewer than four doctorates have been awarded for research arising from the CAT Tournament or which uses the JCAT platform: Furuhata (2009), Niu (2011), Robinson (2011), and Shi (2011).

16.3.1 Modeling Market Mechanisms

Niu et al. (2008) presented a modular model of market mechanisms in preparing a reference implementation of market mechanism for entrants of the CAT Tournaments. In this modular view, which extends prior understandings of market mechanisms (e.g., Wurman et al. 1998), a market mechanism is composed of multiple auction rules, or what they called *policies*, each regulating one aspect of the marketplace. There are seven different types of policy:

1. *matching policies* define how a market matches shouts made by traders;
2. *quote policies* determine the quotes issued by markets;
3. *shout accepting policies* judge whether a shout made by a trader should be permitted in the market;
4. *clearing policies* define when to clear the market and execute transactions between matched asks and bids;
5. *pricing policies* set transaction prices for matched ask-bid pairs;
6. *charging policies* determine the charges imposed by a market; and
7. *subscribing policies* control the collection of market information so that the market mechanism may adapt itself.

A more detailed description of this model and numerous policies that can be classified in such a way and are implemented in JCAT are given by Niu (2011). According to the entries that were uploaded into the TAC repository, this modular model was generally adopted by the entrants of the CAT tournaments, and the policies in JCAT were the primitive building blocks for the entrants to design more advanced components of their market mechanisms.

For example, a common shout accepting policy is what Niu (2011) calls a *quote-beating accepting policy* (also known as the NYSE shout improvement rule owing to its use by the New York Stock Exchange). This policy is typically used in the *continuous double auction* (CDA), where traders may submit shouts at any time during trading, transactions are executed whenever a pair of shouts can be matched, and a shout is allowed to be placed in the market only when it is more competitive than the corresponding market

quotes; that is, proposed asks need to be lower than the ask quote, and proposed bids need to be higher than the bid quote. The market quotes are typically determined by the least-competitive matched shout from the opposite side and the most-competitive unmatched shout from the same side for example, the ask quote being based on the lowest matched bid and the lowest unmatched ask (McCabe et al. 1993). The quote-beating policy is believed effective in speeding up the convergence to the market competitive equilibrium in a CDA market (Cliff and Bruten 1997). Although this policy is simple and can be easily understood by human traders in traditional markets, it relies upon limited information and may lead to volatile transaction prices when traders are sub-rational and the shouts they place do not cluster around the market competitive equilibrium price. In fact, the market quotes can be viewed as estimates of the market competitive equilibrium price, the ask quote being the upper bound and the bid quote being the lower bound.

Niu et al. (2006) proposed what they called the *equilibrium-beating accepting policy* using the average of transaction prices in a sliding window of the transaction history as the estimated equilibrium price. They showed that this simple method was effective in reducing price fluctuations as well as improving the allocative efficiency in a stand-alone CDA market populated by ZI-C traders.

Various CAT entries devised their variants to obtain desirable results in the CAT setting. PersianCat in CAT 2008 (Honari et al. 2008), for instance, estimated the market equilibrium price with a running average of the minimum price of matched bids and the maximum price of matched asks on each day over a certain number of recent game days. This variant of equilibrium estimation samples from a relatively long history of transactions and helps maintain an even smoother estimate over time. An improved version of AstonCAT in CAT 2010 (Chang et al. 2011) combined the original equilibrium-beating accepting policy and the PersianCat variant in a linear manner, where the weighted parameters were carefully chosen to balance the information learned from the intraday trading as well as trading spanning multiple days. PersianCat won the 2008 CAT tournament and the improved AstonCAT, or AstonCAT-plus, performed well in controlled experiments, taking a close second place in 2010 after Mertacor (the CAT 2010 champion) and beating all other entries by a large margin. The designers of both PersianCat 2008 and AstonCAT-plus attributed their specialist's good performances to their effective market equilibrium estimation algorithms, which were the core component of many policies in their respective market mechanisms rather than merely the accepting policies. For further details on the design of these entries and some others, see the references listed in table 16.2.

The complex interactions between multiple marketplaces and various types of traders and the limited coverage during actual tournaments made it necessary to run post-tournament analyses to understand the multi-marketplace dynamics, identify causal effects, and provide guidance for further improvements. Niu et al. (2010) systematically carried out two types of analyses, *white-box analysis* and *black-box analysis*, of the CAT 2007 entries. The white-box analysis ran experiments in the same way the actual tournament ran and tried to relate observed market dynamics

Table 16.2 Publications on the design of entries of the CAT tournaments

Year	Entry	Publications
2007	CrocodileAgent	Petric et al. (2008)
	IAMwildCAT	Vytelingum et al. (2008)
	PersianCat	Honari et al. (2008)
2008	Mertacor	Stavrogiannis and Mitkas (2010)
	PersianCat	Honari et al. (2008)
	PSUCAT	Sohn et al. (2009b)
2009	jackaroo	Furuhata et al. (2009)
	Mertacor	Stavrogiannis and Mitkas (2010)
2010	AstonCAT	Chang et al. (2011)

to the policies adopted by those entries, particularly how intramarginal traders (those traders that would transact if all traders place truth-telling shouts and the market is cleared at the equilibrium) and extra-marginal traders (all non-intramarginal traders) migrated between marketplaces as the entries interacted and adapted their strategies over time. The black-box analysis considered the behaviors of entries as atomic actions in the game-theoretic sense and played these actions repeatedly across many games in an ecological process in which highly rewarded actions became more and more popular and less-rewarded actions died out gradually, in a way similar to the simulated ecological evolution of entries in the Iterated Prisoner's Dilemma Tournament of Axelrod (2006). This ecological process helped identify weaknesses of certain entries in varying competitive environments. For example, IAMwildCAT, the CAT 2007 champion, could lose the competition against PersianCat through one-on-one games in certain experimental setups. Indeed, the actual tournaments reflected only limited trajectories of interaction dynamics and were not able to tell the full story.

Robinson et al. (2010), in particular, examined whether the performance of entries from CAT 2008 and 2009 could be *generalized* when the competitive environment varied, as when facing different market competitors or when the distribution of strategies varied across the trader population. They showed that the ranking of entries was sensitive to changes in these competitive settings. This result was further confirmed by results obtained in other post-tournament analyses (Vytelingum et al. 2008; Stavrogiannis and Mitkas 2010,?; Chang et al. 2011). Chen and Yu (2012) have also studied the effect of market policies on market performance indicators, using the JCAT platform.

Such results are not surprising because CAT games involve complex interactions between adaptive marketplaces and traders, and may not allow solution concepts in game theory like dominant strategies or Nash equilibrium strategies. Nevertheless, empirical analysis still has merit in examining the relative robustness of entries facing varying situations and helps inspire and foster further improvements.

As stated above, one of the ultimate goals of running the CAT Tournaments was to promote research on automated auction mechanism design. Here we describe two pieces of work that exemplify early steps toward achieving this goal. Gruman and Narayana (2010) devised a method, based on machine learning, for estimating the distribution of different trading strategies among traders in CAT Tournaments. The specific classes of trading strategies used by CAT traders are known to CAT entrants, but not the strategy used by any particular trader, nor the distribution of the different strategy types across the trader population. Entrants may be able to optimize their policies if the distribution can be accurately estimated. In this work the authors showed that a classifier based on hidden Markov models was able to estimate the distribution of trading strategies with an accuracy of around 60 percent, better than other classifiers they evaluated (including support vector machines and conditional random fields) and in contrast to the 25 percent accuracy expected when guessing randomly. Using this result they designed a framework based on Markov decision processes for entries in CAT Tournaments. Although this framework turned out to be not as successful, this effort helped shed light on the difficulty of modeling the multi-marketplace scenario and encouraged further exploration in this direction.

Another piece of work along this avenue is the gray-box approach of Niu et al. (2012), based upon their prior work using a white-box analysis and a black-box analysis. Combining the two analysis methods, this gray-box approach tackled the problem of automated mechanism design through automated search in a tree-based structure of possible auction mechanisms. The gray-box approach used reinforcement learning to explore the composition of the structure, related the performance of auction mechanisms to that of auction rules that formed the mechanisms, and utilized evolutionary computation to maintain viable auction mechanisms. Experiments showed that the gray-box approach was able to produce significantly better performance than a classic genetic algorithm. This approach is particularly useful because CAT entries typically adopt a modular design framework, as this not only makes it easier to run post-tournament empirical analysis and evaluate the performance of entries on a common basis, but also makes it possible to combine policies from different entries and to generate and search for better variants of these entries.

Finally, the CAT Tournament environment provided a suitable setting to study economies involving competing or networked marketplaces (Cai et al. 2010, 2012; Moyaux et al. 2010). These many publications show the value of the CAT Tournament and the supporting software platform, JCAT, in defining a common set of research problems and a shared basis for exploring them.

16.3.2 Modeling Traders in Multimarketplace Environments

Faced with multiple competing marketplaces, a trader faces two decision problems: which market to select and what amount to bid or ask in the selected market. For self-interested traders, both decisions can be undertaken with the intention of maximizing expected profits. These two questions have led to two lines of research.

16.3.2.1 *Marketplace Selection*

The CAT game designers modeled each trader as an N -armed bandit problem solver whose daily profits were used as rewards for the trader to learn to optimize its selection between alternative action options, or marketplaces (Niu et al. 2007). They examined the model through experiments in running marketplaces that charge differently and showed that profitable traders consistently favored the marketplace that charged the lowest fees. This configuration of market selection was used in all CAT Tournaments; similar results were observed in the post-tournament analysis (Niu et al. 2008), in which the movement of intramarginal and extramarginal traders between CAT 2007 entries were examined.

Sohn et al. (2009a) created a simple game-theoretic model and explored the Nash equilibrium strategies for a few pairs of intramarginal and extramarginal truth-telling traders choosing between competing marketplaces. Their results demonstrated that extra-marginal traders are more sensitive than intra-marginal traders to registration fees, fees that traders need to pay when they choose a marketplace to trade in no matter whether they are able to transact or not. In later work, Sohn et al. (2012) obtained similar results when software trading agents were replaced by human traders. Shi et al. (2013) further investigated the problem of market selection, presenting a formal description of the problem and Bayes-Nash Equilibrium (BNE) solutions. In particular, after certain assumptions that made the problem analytically tractable, they showed that there existed BNEs in which all traders choose the more expensive marketplace of two or in which traders choose the more expensive marketplace stochastically with a nonzero probability. Further analysis based on evolutionary game theory revealed that the latter BNE was neither stable nor easy to reach. On the other hand, reaching the former BNE depended on particular pairings of sellers with buyers, and these pairings depended on chance factors such as the numbers of traders, their initial distribution, and the fees charged by the marketplaces.

Miller and Niu (2012) explored the parameter spaces of several N -armed bandit algorithms for the purpose of market selection. They ran two sets of experiments, one containing only adaptive specialists and one containing only nonadaptive specialists, and found that the softmax strategy (Sutton and Barto 1998) is the most successful at maximizing trader profit and overall market efficiency in both adaptive and non-adaptive markets.

16.3.2.2 *Trading Strategies*

Although most research on trading agents in the CAT Tournaments has focused on market selection, there is some work that paid attention to strategies that traders are equipped with for making offers in marketplaces. Sohn et al. (2009b) argued that trading strategies for traders in CAT tournaments, such as ZIP, originally designed to perform in a single stand-alone marketplace using some presumed market mechanism, may not work well when they need to accomplish additional decision making about market selection and when specialists' entries typically adopt varying customized

market mechanisms. They discussed the misalignment of ZIP with the pricing policy of PSUCAT, their entry in CAT 2007, and showed that a revised version of ZIP could achieve higher profits and do better at choosing marketplaces.

Scope exists for considerably more research in this area, particularly looking at trading strategies other than the four used in the tournaments, and in developing strategies that adapt to or exploit the particular mechanisms adopted by the selected specialist.

16.4 HALL-OF-FAME SIMULATIONS

In this section we present a lightweight evaluation of the CAT Tournament. The hypothesis of CAT is that running competitive tournaments helps improve the state of the art in automated mechanism design. It is not possible to directly test this hypothesis, because other factors will also influence research into this topic. Instead, we indirectly tested this hypothesis by measuring the relative success of participants in the CAT Tournament. These experiments were not intended to be conclusive or complete, but instead are meant to give us an indication of the effectiveness of the competition.

We present two sets of experiments in this section:

- Hall-of-fame grand tournament: the first set of experiments consisted of twenty CAT games, in which the entrants were the five winners of the CAT tournament during the period 2007–2011 (see table 16.1).
- Hall-of-fame round-robin: the second set of experiments took the same five entrants but played each of them off against an other twenty times in a round-robin tournament of two-player CAT games; that is, we had ten different pairs, each playing twenty games against each other.

By measuring various outcomes over the course of these two sets of experiments, we would expect to see a positive correlation between specialist performance and the year in which they won the tournament; that is, winners from later years should perform better on aggregate than those from earlier years.

It is not possible to undertake valid, formal statistical tests of the presence or absence of trends in these data, for two reasons. First, we have only five time-points' worth of observations (corresponding to the five CAT Tournaments). Second, our observations are highly correlated. Some participants entered the tournament in multiple years, and most participants uploaded their code following each tournament for others to study for future tournaments. At least one participant used the uploaded code to create large-scale Monte Carlo simulations of participant interactions as part of the design process for their subsequent CAT entries. Thus, this correlation between observations is not straightforward, and there is no obviously appropriate model for it. In addition, the five measures (corresponding to 2007–2011) are all correlated in that they run as part of the same JCAT simulations, so, for example, an increase in market share of one specialist

will take market share away from others. Given these features of the data available to us, only exploratory data analysis procedures are appropriate (not asymptotic statistical tests), and that is what we report here.

16.4.1 Experiment Design

We used the JCAT platform version 0.17⁸ to run CAT games as part of the experiments. Our experiments were set up to replicate a CAT Tournament game, meaning that each game consisted of five hundred trading days, with ten rounds per day, and four hundred traders, evenly split between buyers and sellers. The distribution of the four trader types described in section 16.2 was also uniform (25 percent each). Private values of the generic commodity being traded were randomly drawn from a uniform distribution between 50 and 150. These parameters were varied during CAT tournaments, with the entrants not knowing their values or even their distributions, but for the experiments, we maintained the same values.

To measure the performance of each specialist, we used the same three measures used to score the CAT Tournament (Cai et al. 2009). On each trading day, the following three measures were taken for each specialist:

- **Profit share:** the profit obtained by a specialist as a proportion of the total profit made by all specialists in the market, between 0 and 1 inclusive.
- **Market share:** the market share obtained by a specialist as a proportion of all traders that registered with a specialist on that day, between 0 and 1 inclusive.
- **Transaction success rate:** the number of shouts that were matched on that day, as a proportion of the total number of shouts placed with that specialist on that day. Note that this is independent of the number of shouts placed with other specialists.

These three measures were added together to produce an overall score for the trading day. The total game score was simply the summation across all trading days.

In addition, we measured the efficiency of individual specialists and the overall market. *Global allocative efficiency* measures how close the entire market is to trading at the equilibrium price, where the *equilibrium price* is defined as the price at which demand equals supply when all traders offer to buy or sell at their private values, assuming that all traders in the market can trade with each other. The global allocative efficiency is calculated using,

$$E = \frac{\sum_j \sum_i |v_i^j - p_i^j|}{\sum_j \sum_i |v_i^j - p_o|} \quad (16.1)$$

in which p_o is the equilibrium price of the market, v_i^j is the private value of trader i in specialist market j , and p_i^j is the price paid by trader i in specialist market j .

⁸ See <http://jcat.sourceforge.net/>.

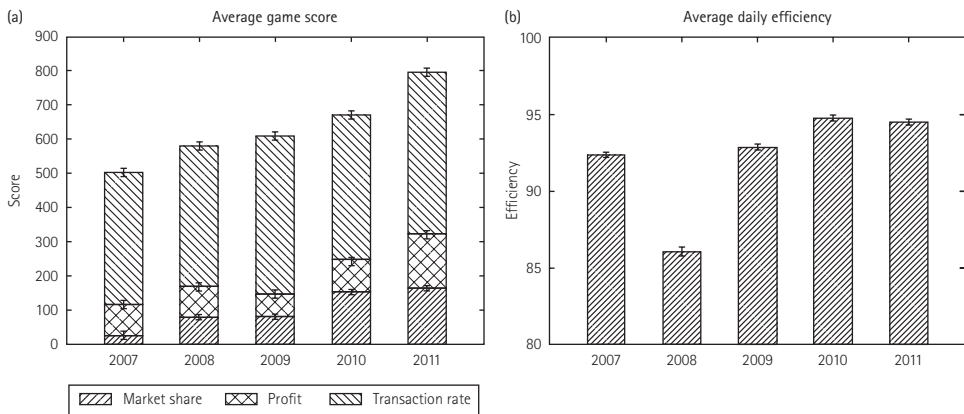


FIGURE 16.1 Average score and allocative efficiency for each entrant for all grand tournament experiments.

Allocative efficiency of individual specialists is similar to the global allocative efficiency defined in equation 16.1 but is measured only for transactions in the individual specialist's market.

Note that there is a loose (and nonlinear) correlation between game score and global allocative efficiency. A market clearing close to the equilibrium price (high global allocative efficiency) is likely to have a higher transaction rate, thereby increasing the overall score.

16.4.2 Results

Figure 16.1 shows the average game score and average allocative efficiency for each winning entrant (2007–2011) for all for grand-tournament experiment runs. Error bars represent the 99 percent confidence interval. From this figure one can see a clear and consistent upward trend in the scores.

Table 16.3 shows the proportion of placings that the specialists received in the twenty experiments that were run. This also demonstrates a clear and consistent trend of more recent specialists outperforming previous specialists. The most recent specialist won all twenty games in the experiments, and the earliest came in last in all games.

Figure 16.2 shows the average game score and average allocative efficiency for each winning entrant (2007–2011) for all round-robin experiment runs. Error bars represent the 99 percent confidence interval. From this figure one can see a clear upward trend in the data, although the 2008 and 2009 winners both outsourced the 2010 and 2011 winners. All scored higher than the 2007 winner. The average efficiency also shows an upward trend, with the notable exception that the 2008 winner traded well below the allocative efficiency of all other specialists. The 2011 entrant achieved an allocative efficiency marginally below that of the 2010 entrant.

Table 16.3 Number of placings for each specialist over all twenty grand tournament experiments

Specialist	1 st	2 nd	3 rd	4 th	5 th
2007	0	0	0	0	20
2008	0	0	3	17	0
2009	0	1	16	3	0
2010	0	19	1	0	0
2011	20	0	0	0	0

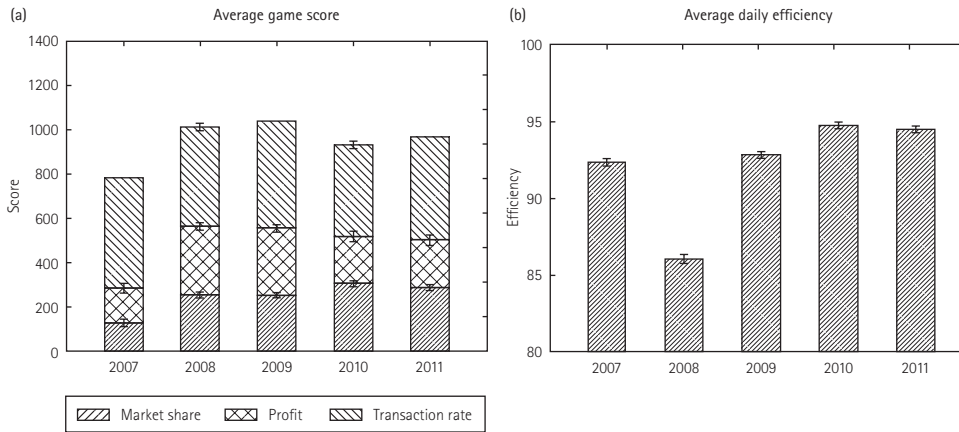


FIGURE 16.2 Average score and allocative efficiency for each entrant in all round-robin experiments.

Figure 16.3 provides a further breakdown of the data from figure 16.2, showing the average scores and allocative efficiency for each pair of specialists playing against each other. Two adjacent bars represent a pair, with the left bar representing the “earlier” specialist. The final column in each, the average, is the average of all pairs.

From this figure we can see that the 2007 winner was substantially outscored by all other entrants and that, consistent with figure 16.2, the 2009 winner beat all other specialists in the round-robin experiments. Allocative efficiency is consistent with Figure 16.2, with later specialists outperforming earlier specialists, with the exception of the 2008 winner, which traded below the other specialists in all round-robin experiments. While there is a clear upward trend in the data, the 2008 and 2009 entrants both outscored the 2010 and 2011 entrants.

16.4.3 Discussion

These results support our hypothesis that the research in automated mechanism design improved during the tournament and given the consistent rise in scores shown in figure 16.1 and table 16.3, we believe that the CAT Tournament helped drive this progress.

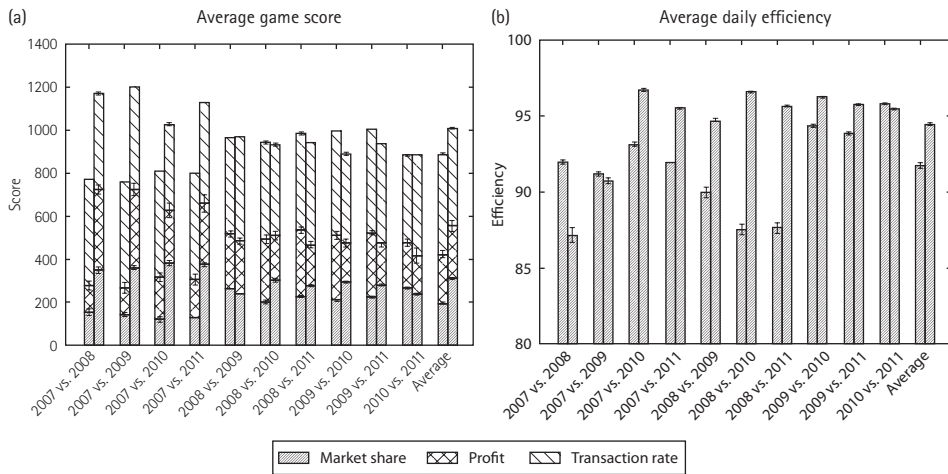


FIGURE 16.3 Average total score for specialists in two-player round-robin experiments.

The grand tournament experiments demonstrate a clear and consistent upward trend in the scores of specialists, supporting our hypothesis. The round-robin experiments support the hypothesis, but to a lesser extent, with the trend being upward though some earlier tournament winners outscored later winners. This is consistent with the findings of Robinson et al. (2010), who demonstrated that entrants in the 2008 and 2009 competitions were sensitive to the other specialists with whom they competed. It is likely that the specialists we used were tuned to games with several competitors, rather than two-player games.

The graphs in figures 16.1–16.3 show that obtaining a large market share is correlated with a high overall score. In addition to being one-third of the total score, increasing market share provides more opportunities for successful transactions. Assuming the transaction rate remains the same, this results in higher profits. This is particularly pronounced in the round-robin games containing the 2007 entrant, which, despite having a comparable transaction rate, is well behind with regard to market share and profit (figure 16.3).

A larger market share may also help to increase the transaction rate. Cai et al. (2010) showed that dividing traders between many small markets results in lower efficiency than one larger market owing to the reduced likelihood of successful matches. Therefore, we can conclude that a larger market share will increase the transaction rate in each round, increasing profit and overall score.

Transaction rates are fairly evenly matched over all surveyed specialists. However, the 2008 and 2009 entrants scored well in the round-robin tournaments with increased profits: both had lower market share and similar transaction rates than the 2010 and 2011 specialists yet outscored them overall. One phenomenon we observed anecdotally during runs of the actual CAT competitions is that specialists set their fees low (often charging no fees) early in the tournament to obtain a strong market share, and then

start charging fees once they have the market share they desire. Thus, the policies for pricing and charging are closely linked to policies such as shout accepting.

16.4.4 Threats to Validity

There are several threats to the validity of the conclusions stated above, which we present in this section.

First, to evaluate whether the CAT competition has fulfilled its aims, we have used only the winning entrants over the life of the competition, ignoring other entrants.

Second, we explore only a slice of the possible parameter space. Such a restriction is necessary because the experiments take weeks to run. Robinson et al. (2010) note that CAT entrants are sensitive to many factors such as the types of traders and the other specialists with whom they compete. It is therefore possible that, were we to vary the competitive parameters, the results would show less support for our hypothesis.

Third, while we use experiments to assess whether the CAT competition has achieved our aim of encouraging research into automated mechanism design, we cannot measure this directly. Instead, we have measured the entrants in the competition only. Clearly, there are other external factors that influence this research area and influence the quality of the entrants. Despite this, research arising from this area has made use of specialists in the TAC repository, and we are confident that this has helped improve this research field.

Finally, the CAT competition aims to encourage research into automated mechanism design in general, but the specialists entered into the tournaments are geared towards the CAT Tournaments in particular, and it may be that the CAT competition is not helping achieve progress outside this particular niche of the tournament itself. Robinson et al.'s (2010) finding that entrants are sensitive to specific parameters of the tournament demonstrate that this is possible. While some strategies employed by participants are more specific to the tournament (e.g., learning the distribution of specific types of traders), it is evident from the resulting research that much work, such as improving transaction efficiency (Chen and Yu 2012), is more generally applicable. For example, Sohn et al. (2009a) reported results using JCAT on attracting intra-marginal traders across multiple markets, and then later found similar results when automated traders were replaced with humans (Sohn et al. 2012).

Our opinion is that the research arising from this tournament is more general than the tournament itself, but solutions would need to be tuned to particular application domains, in the same way the CAT specialists have been tuned to the CAT Tournament.

16.5 CONCLUSIONS

In this chapter we presented an overview of the CAT Market Design Tournament, including the history of the tournament and key research that has resulted from it. The primary aim of the CAT Tournament is to encourage research into automated mechanism design in online marketplaces. We evaluated whether the competition

is achieving this aim using a series of experiments in which the winners of five tournaments played off against each other. In the first set of experiments, the five winners played a series of grand hall-of-fame tournaments against each other, and in the second set, the winners were pitched against each other in a set of round-robin two-player games.

Results indicate that the design of automated mechanisms for the CAT Tournament improved consistently over the five years of the competition, both from the perspective of individual markets and from a global perspective. We believe that these results support the conclusion that the CAT Tournament achieved its aims.

We believe that one reason the tournament was successful is that, after each competition, entrants uploaded binary versions of their specialist, allowing other researchers to use these as benchmarks in their own experiments. Such a repository is missing from the TREC repository, which, analysis indicates, is less successful than the CAT tournament at improving the state of the art (Armstrong et al. 2009).

This chapter has presented a review of the CAT Tournament from 2007 to 2011. We are confident that the CAT tournament has supported important state-of-the-art research in automated mechanism design, automated trading, and competition between multiple marketplaces. We hope that the CAT Tournament and the other TAC tournaments serve as models for competitions in other research areas.

ACKNOWLEDGMENTS

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CHAPTER 17

AGENT-BASED MACROECONOMIC MODELING AND POLICY ANALYSIS

The Eurace@Unibi Model

HERBERT DAWID, SIMON GEMKOW, PHILIPP HARTING,
SANDER VAN DER HOOG, AND MICHAEL NEUGART

17.1 INTRODUCTION

Predicting short- and long-run effects of economic policy measures and of institutional changes in an economy is a challenging task. The difficulties of this task are largely amplified by the close linkages between the dynamics of financial markets, credit markets, and the real side of the economy as well as by the growing interrelation between developments in different regions of a globalized world economy. Ideally, predictions about policy effects and the design of policy measures should be informed by rigorous analyses based on calibrated economic models. But capturing the main aspects of the different linkages mentioned above is often beyond the scope of standard economic models. During the financial and economic crises which started in 2008, several policy makers voiced concerns that it is hard for them to base their decisions on existing often very simple policy models, which might not be able to capture the mechanisms that seem to be mainly responsible for the problems. To quote former ECB president Jean-Claude Trichet: “When the crisis came, the serious limitations of existing economic and financial models immediately became apparent... Macro models failed to predict the crisis and seemed incapable of explaining what was happening to the economy in a convincing manner. As a policy-maker during the crisis, I found the available models of limited help. In fact, I would go further: in the face of the crisis, we

felt abandoned by conventional tools.¹ Trichet also sketched properties of economic models that, in his opinion, would make them more suitable to capture crucial properties of economic systems and make them more appealing to policy makers: “We need to deal better with heterogeneity across agents and the interaction among those heterogeneous agents. We need to entertain alternative motivations for economic choices. [...] Agent-based modelling dispenses with the optimization assumption and allows for more complex interactions between agents. Such approaches are worthy of our attention.”

The aim of this chapter is to describe the Eurace@Unibi agent-based macroeconomic model and review research that has been carried out within this framework in recent years in order to develop a useful tool for the rigorous analysis of economic policy measures. Agent-based simulation models are relatively new to the toolbox of economists, and in spite of the growing effort that has been invested in developing this research area, the field is still in its adolescence. However, the research reported here highlights that new kinds of questions can be asked and new kinds of insights can be obtained using such an approach, thereby showing the potential of agent-based modeling as an instrument that complements established modeling approaches. This potential has also been demonstrated by a number of other agent-based macroeconomic models that have been developed in recent years. Although space constraints prevent us from discussing this research in great detail, we briefly review it in section 17.2.

Agent-based analyses have been carried out for numerous policy areas, where issues related to financial market dynamics, banking regulation, credit linkages, and monetary policy have attracted considerable attention. In this chapter we concentrate on research that is concerned with the facilitation of economic growth on a local and global scale in an economy consisting of different regions with (potentially) heterogeneous characteristics. Understanding how different policy measures affect the economic dynamics in different regions of an integrated economic area is not only crucial for proper design of cohesion policies, but is also an important contribution to the prevention of deep economic crises. For example, the government debt crisis in the Euro area was not only due to excessive spending. One of the major underlying causes was the persistent gap in terms of economic competitiveness between the central European economies and the countries in the southern periphery. As a result, a sustainable solution to such a crisis should not only entail cuts in government spending; rather, there has to be a coordinated and specifically targeted policy that increases the competitiveness of the lagging countries and fosters economic convergence. One keystone of the Economic and Monetary Union (EMU) is, however, the integration of the European economies where integration is associated with a reduction of barriers for a free flow of goods, labor, and capital. But this might imply that it is almost impossible to isolate the effects

¹ Jean-Claude Trichet, “Reflections on the nature of monetary policy non-standard measures and finance theory,” opening address at the ECB Central Banking Conference Frankfurt, November 18, 2010. The text of the full speech is available at <http://www.ecb.int/press/key/date/2010/html/sp101118.en.html>.

of an economic policy in the target region only. In fact, it is often *ex ante* not clear how a policy affects the economic performance of neighboring regions and whether there are also feedback effects, either positive or negative, on the target region.

This chapter presents research addressing these important issues based on simulations with an agent-based macroeconomic model with spatial structure called the Eurace@Unibi model. The model has been developed with the particular aim to gain a better understanding of the processes that foster the development and diffusion of new technologies in an economy, of the interplay between industrial dynamics and the development of the workforce, and the contribution of these dynamics to economic growth and business cycles. It provides a general framework for the examination of economic mechanisms and policy effects on labor, goods, and credit markets. The underlying modeling philosophy is to generate economic dynamics on the industry and macroeconomic level as the aggregate outcome of interactions on the micro-level that are modeled with recourse to strong empirical foundations.

To put this work into perspective we first present in section 17.2 a brief overview of recent research on agent-based macroeconomics. In section 17.3 we provide a brief description of the Eurace@Unibi model and discuss some of its properties. Section 17.4 demonstrates the kind of economic mechanisms that can be identified by careful consideration of simulation results by considering the dynamics of technological diffusion in the Eurace@Unibi economy. Several spatial policy analyses that have been carried out using (previous versions of) the model are discussed in section 17.5. The value-added of the proposed framework and the results that can be derived from it are discussed in more general terms in section 17.6. We conclude in section 17.7 with an outlook on high-potential research areas for the application of agent-based policy models.

17.2 RECENT DEVELOPMENTS IN AGENT-BASED MACROECONOMIC MODELING

In recent years a number of projects have been initiated that aim at the development of closed macroeconomic models using an agent-based approach. Although these different models share the general approach of describing macroeconomic dynamics as the outcome of aggregation of (local) interaction on the micro-level, their focus differs quite significantly. This difference in focus implies that the care and detail with which certain markets and institutions are modeled varies between these models. At this point no generic standard agent-based macroeconomic modeling framework has evolved that can be used as a starting point for more specific models aiming at certain areas of policy analysis. Nevertheless, a number of common denominators can be identified in the modeling choices, and also a number of qualitative features of simulation results, which also match empirical stylized facts, are generated in several of the models described below. Many of these models are able to replicate the main

stylized facts concerning business cycle fluctuations and to highlight the economic processes that generate these fluctuations. At the same time they attempt to match the micro-dynamics to reproduce the heterogeneity in firm investment behavior, firm distributions, and income distributions.

The model that is perhaps closest to Eurace@Unibi has been developed by Giovanni Dosi, Giorgio Fagiolo, Mauro Napoletano, and Andrea Roventini, and has been documented in a series of papers (Dosi et al. 2008, 2010, 2013, 2015). Their model consists of two sectors, in one of which firms perform R&D producing a heterogeneous capital good that is used together with labor input for the production of a homogenous consumption good. A government levies wage and profit taxes to finance an unemployment benefit system. Overall the model can replicate an impressively large set of empirical stylized facts such as typical growth paths, business cycle properties, macroeconomic correlates, and cross-sectional distributions. Using this model, the authors have analyzed the (long-run) effects of policies aimed at the strengthening of demand and of policies that facilitate the speed of technical change as well as the interaction of these policies. In Dosi et al. (2013) a more explicit representation of the credit market and of bank behavior is added to the model with the aim of exploring the interactions between the functional income distribution and monetary and fiscal policies. They find that fiscal policies are becoming more effective as the income distribution is more skewed toward profits. Moreover, there appears to be an interaction between monetary policies and income distribution in the sense that a change in the interest rate does not affect macroeconomic variables very strongly when profit rates are higher.

Domenico Delli Gatti, Mauro Gallegatti, and colleagues have been working on agent-based macro models with a particular emphasis on the role of credit markets for aggregate fluctuations (see, e.g., Delli Gatti et al. 2007; Gaffeo et al. 2008; Delli Gatti et al. 2008, 2010; Assenza and Delli Gatti 2013; Assenza et al. 2015). This line of research has been heavily influenced by the work of Greenwald and Stiglitz (1993). At their core these models typically feature a credit market and a goods market in which a homogenous good is produced. Firms are heterogeneous with respect to their financial robustness, captured by firm-specific equity ratios. They adjust their capital stock by maximizing expected profits. Credit supply is mostly rule-based, mimicking capital requirement regulations. Typical simulations reveal clustered volatility of output growth rates, and other variables show distributional properties similar to those of empirical findings. Among these are skewed firm size distributions with power law properties. An important aspect of several of these researchers' models is that they capture the linkages based on credit relationships between firms and banks, among firms, and among banks. These properties allow us to gain important insights into relevant mechanisms responsible for default avalanches and to explore implications of different types of bank and credit market regulations.

The exploration of the effects of bank regulation is also an important aim of Ashraf et al. (2016, 2017). Their agent-based macroeconomic model puts strong emphasis on the institutional role of banks in a market economy as the providers of liquidity. Behavior of agents is determined by simple rules, and market interactions are governed by

search and matching processes. The authors provide a careful calibration of the model using U.S. data and based on this calibration examine in detail the role of banks in the economy in normal times as well as in times of crisis. They show that banking regulation in normal times hardly affects macroeconomic stability, while in bad times more stringent regulation has an effect on the economy as it suppresses lending to firms in need.

The focus of the agent-based macroeconomic model by Gintis (2007) is less policy-oriented, and no particular empirical regularities or policy issues are addressed. Rather, the main concern is to investigate (disequilibrium) dynamics in a standard Walrasian economy under the assumption that a certain fraction of the agents uses private prices. The model has several sectors, where different consumption goods are produced using labor and capital. Consumers have heterogeneous utility functions and make their consumption decisions in accordance with utility maximization based on price information about a (random) subset of producers in each sector. Firms alter their behavior owing to imitation and mutation as well as to myopic adjustment dynamics. It is shown that with private prices the dynamics of the economy is stable overall with persistent small volatility. Neither shocks nor structural changes induce a loss of these stability properties. However, if a certain fraction of agents use public prices for their trades the price dynamics becomes less stable, and if this fraction is sufficiently large the dynamics exhibits price explosions in which prices deviate from their equilibrium values.

The Lagom model (Mandel et al. 2010; Wolf et al. 2012) extended the Gintis model to include capital accumulation, inter-industry trade, and life-cycle saving behavior. A monetary authority lends money to new firms and poorly performing firms until they go bankrupt. The same agent (called monetary authority) takes over government activities. It levies profit and wage taxes in order to finance unemployment benefits. In the Lagom model, technology choice, household consumption, firms' mark-ups, and wages are determined using genetic algorithms. The different sectors of the model are linked by input-output tables, which are initialized on the basis of real world data. In addition, some of the agents' production and investment decisions are rule-based. The authors of the Lagom model see their first main contribution in providing a "proof of concept" that large-scale agent-based macro-models are feasible. But the authors also have more policy-oriented applications of the model in mind. In particular, there is a spatial version of the model, which, among other aspects, includes a representation of local emissions induced by economic activity. This opens up the possibility to consider different types of economic policy measures from an environmental perspective as well. As pointed out in Wolf et al. (2012), a systematic calibration of the model using data from European and Mediterranean countries is on the research agenda of the authors.

Apart from these contributions several other agent-based macroeconomic models have been developed more recently. For example, Lengnick (2013) presents a more minimalistic agent-based macroeconomic model than the previously introduced ones. It consists of firms and households that are connected by temporarily fixed networks. Wages and prices adjust according to a stochastic process. Firms produce with labor input only, and the production technology is fixed. No government agencies are modeled because the main purpose of the contribution is to demonstrate that major stylized

facts can be replicated with a bottom-up approach of interacting agents. Further, more policy-oriented contributions include Oeffner (2009), Greiff et al. (2011), Seppecher and Salle (2015), Caiani et al. (2016), Chiarella and Di Guilmi (2017), and the research by Cincotti et al. (2010) and Teglioni et al. (2012) that also originates from the Eurace project and mainly focuses on the macroeconomic effects of financial regulations. Space constraints do not allow us to discuss all these contributions in more detail here.

17.3 A SKETCH OF THE EURACE@UNIBI MODEL

The Eurace@Unibi model is based on the agent-based macroeconomic simulation platform developed within the Eurace project.² After the completion of the Eurace project in 2009 the authors of this chapter extended and altered the model substantially in numerous directions, leading to the current version of the model. In particular, substantial changes and extensions have been made to the production technology, to the model of the decision processes of consumption good producers, to the investment goods production sector, and to the bankruptcy procedures of firms. Furthermore, interest payments of banks have been introduced on the credit market and the structure of the balance sheets has been completely overhauled. Finally, the financial market of the original Eurace model has been replaced by a much simpler model following a slightly different approach than the original one.

The policy orientation and the specific interest in the spatial dimension of policy effects imply a partly different focus of our model when compared to the agent-based macroeconomic models reviewed in the previous section. Because the Eurace@Unibi model features regions, it allows for the analysis of regionally differentiated policies and interactions arising through goods and factor flows between the entities. These flows include the movement of labor, which made it necessary to impose more structure on modeling the labor market. The interaction protocol between workers and firms features a frictional labor market with adapting wage offers of firms and adapting reservation wages of workers, rendering labor supply endogenous. Besides the spatial dimension of policies, the interest has been in the policy effects arising from subsidies to firms purchasing capital goods and human capital policies. To this end, a more distinct production technology is implemented as compared to other agent-based macroeconomic models. This allows for an interplay between a firm's vintage structure and the skill endowment of its workers. Moreover, the detailed modeling of agents' behavior entails a need to find meaningful decision rules. Therefore, the Eurace@Unibi

² The EU-funded project was carried out by a consortium led by Silvano Cincotti (University of Genova, Italy), Herbert Dawid (Bielefeld University, Germany), Christophe Deissenberg (Aix-Marseille University, France), Kaan Erkan (TUBITAK National Research Institute of Electronics and Cryptology, Turkey), Mauro Gallegati (Marche Polytechnic University, Italy), Mike Holcombe (University of Sheffield, UK), Michele Marchesi (University of Cagliari, Italy), and Christopher Greenough (STFC–Rutherford Appleton Laboratory, UK).

model relies on rules with strong empirical micro-foundations, as explained below. Finally, the model is a closed model and a balance-sheet approach is systematically used, making all flows in the modeled economy transparent and ensuring stock-flow consistency. These features highlight the main differences between this approach and other models in this literature. We now explain the setup of the Eurace@Unibi model in more detail.

17.3.1 Overall Structure

The model describes an economy with markets for labor, consumption goods, investment goods, financial assets, and credit. It is possible to specify a geography with multiple regions. The economy is populated by firms (consumption goods producers and investment goods producers), households, and banks. For each of these agent types there are multiple instances. Each of these agents is located in one of the regions. In addition, there is a single central bank and there is a government that collects taxes and finances social benefits as well as, potentially, economic policy measures that might differ between regions. Finally, there is a statistical office (Eurostat) that collects data from all individual agents in the economy and generates aggregate indicators according to standard procedures. These indicators are communicated to the agents in the economy (who might use them as input for their decision rules) and also stored in order to facilitate the analysis of the simulation results. An illustration of the crucial parts of the model is given in figure 17.1.

Investment goods of different quality (i.e., potential productivity) are provided by investment goods producers with infinite supply. The technological frontier (the quality of the best currently available investment good) improves over time, and technological change is driven by a stochastic (innovation) process. Firms in the consumption goods sector use investment goods combined with labor input to produce consumption goods. The labor market is populated with workers who have a distinct general skill level. They acquire on the job specific skills, which they need to fully exploit the technological advantages of the capital employed in the production process. Each time a consumption goods producer invests in new investment goods, he decides which quality level to select. The quality (or vintage) determines the diffusion speed, the pace at which new technologies spread throughout the economy. Consumption goods are sold at local market platforms (called malls), where firms store and offer their products and consumers come to buy goods at posted prices. Labor market interaction is described by a simple multi-round search-and-matching procedure in which firms post vacancies, searching workers apply, firms make offers, and workers accept or reject the offers. Wages of workers are determined, on one hand, by the expectation the employer has at the time of hiring about the level of specific skills of the worker, and, on the other hand, by a base wage variable, which is influenced by the (past) tightness of the labor market and which determines the overall level of wages paid by a particular employer. Banks collect deposits from households and firms and supply credit to firms. The interest rate

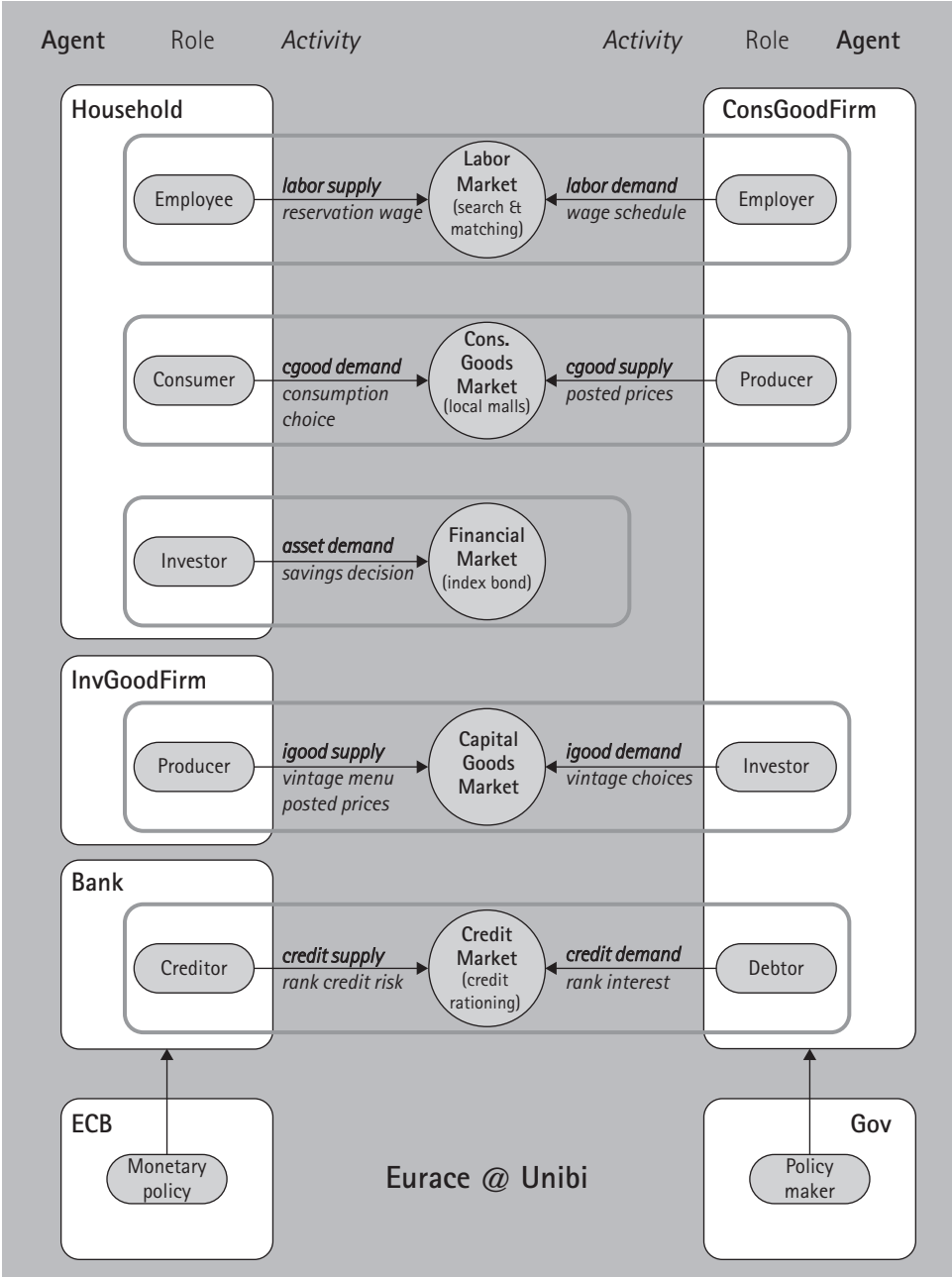


FIGURE 17.1 Overview of the Eurace@Unibi model.

depends on the financial health of the firm, and the size of the loan might depend on the bank's liquidity and risk exposure. Firms can be credit-rationed, in which case they need to remake their original investments and production plans and rescale the level of output. There is a financial market where shares of a single asset are traded, namely

an index bond representing all firms in the economy. The dividend is determined by the total dividends currently paid by all firms, divided by the total number of index bonds outstanding. This simple representation of a financial market is not meant to be suitable to describe speculative bubbles in the financial market, but it captures potentially important feedback effects between firms' profits and households' wealth dynamics.

The central bank provides overdraft and emergency liquidity facilities for the banks at a given base rate. It pays interest on banks' overnight deposits and might provide fiat money to the government.

Firms that are not able to meet their financial commitments are declared to be illiquid and enter into an illiquidity bankruptcy procedure. Furthermore, if at the end of the production cycle the firm has negative net worth, the firm is declared insolvent and enters into a bankruptcy procedure. In both cases it goes out of business and stops all productive activities, and all employees lose their jobs. The firm writes off a fraction of its debt with all banks with which it has a loan and stays idle for a certain period before it becomes active again.

The spatial extensions of the markets differ depending on the type of goods traded. The investment goods market is global, that is, firms in all regions buy from the same global investment goods producer. Therefore every consumption goods firm has access to the same production technology. On the consumption goods market, demand is determined locally in the sense that all consumers buy at a mall that is located in their region. Supply of consumption goods is global' each firm can sell its product in every regional mall. Labor markets are characterized by spatial frictions determined by commuting costs that arise if workers accept jobs outside their own region. It is assumed that firms have access to all banks in the economy and, therefore, that credit markets operate globally. Firms select banks ranked on the lowest interest rate offered, whereas banks select firms ranked on the lowest risk exposure. Rationing is possible on all markets except on the investment goods market, where the supply is fully elastic.

17.3.2 Decision Making and Sequence of Activities

In contrast to dynamic equilibrium models—where it is assumed that the behavior of all actors is determined by the maximization of their own (intertemporal) objective function using correct expectations about the behavior of all others—in an ABM the model description needs to provide explicit rules that describe how different agents build expectations and take decisions based on the available (local, private) information.

To explicitly specify these rules for all agents and decisions is an important modeling decision. As put forward by Sims (1980), the “Wilderness of Bounded Rationality” is a serious concern since a large number of different approaches to modeling boundedly rational behavior and its adaptation have been put forward in the literature. At this point there is little indication of the emergence of a widely accepted consensus that provides empirically or theoretically well-founded concepts for tackling this issue.

The choice of decision rules in the Eurace@Unibi model is based on a systematic attempt to incorporate rules that resemble empirically observable behavior documented in the relevant literature. Concerning households, this means that empirically identified rules for consumption and savings behavior are used. Purchasing choices are described by models from the marketing literature that have a strong empirical support. Regarding firm behavior we follow the management science approach, which implements relatively simple decision rules matching standard procedures of real-world firms. There is a rich literature on (heuristic) managerial decision rules in many areas of management science. This includes pricing (see, e.g., Nagle and Hogan 2006), production planning (see, e.g., Silver et al. 1998), and market selection (see, e.g., Wind and Mahajan 1981; Kotler and Keller 2009). Although it certainly cannot be assumed that all firms in the economy rely on such standard managerial heuristics, capturing the main features of such heuristics when modeling the firm adds a strong empirical micro-foundation to the agent-based modeling approach.

Cyert and March's *Behavioral Theory of the Firm* (1963) could be considered a precursor to the management science approach. In this seminal work many operating procedures of firms' decision making are detailed.³ For an assessment of the impact of *A Behavioral Theory of the Firm* and its methodological stance, see Argote and Greve (2007, p. 339):

The general methodological point was that theory should model organizational processes, and should be generated through systematic observation of processes in actual organizations. One component of this point is that organizational theory should not oversimplify. Although parsimony is needed in theory building, parsimony that throws out basic insights—like replacing a process model with a maximization assumption—can be harmful.

Generally speaking, the management science approach implies that a parametrized decision rule is developed for each decision an agent makes. Whereas the parameter values that determine the exact form of the rule might differ between individuals and might change over time, it is assumed that all individuals have the same (empirically motivated) structure for every rule and that this structure is time-invariant. A more extensive discussion of the management science approach can be found in Dawid and Harting (2012).

Concerning the activation of agents, the actions can be calendar-based (time-driven) or event-based; the former can follow either subjective or objective time schedules (agent time vs. clock time). Furthermore, the economic activities take place on a hierarchy of time scales: yearly, monthly, weekly, and daily activities all take place following the calendar time or subjective agent time. Agents are activated asynchronously according to their subjective time schedules, which that are anchored on an individual activation day. These activation days are uniformly and randomly distributed among the agents at the start of the simulation but may change endogenously.

³ Besides the emphasis on organizational processes and decisionmaking routines, a further aim of the theory was to link empirical data to the models by considering results from case studies of real firms.

Table 17.1 Consumption goods producers: sequence of economic activities

Time scale	Activity
Yearly	determine retail price
Monthly	production planning: decide planned output
	determine input demand for capital and labor
	perform financial management
	interact in credit market
	interact in investment goods market
	interact in labor market
	produce output and distribute to malls
	complete accounting: revenues, income statement and balance sheet
	pay taxes, dividends
Event-based	apply bankruptcy protocol: entry of exit (if equity < 0)

Table 17.2 Households: sequence of economic activities

Time scale	Activity
Monthly	receive labor income (unemployment benefits if unemployed)
	receive capital income (dividends)
	pay taxes
	make consumption/savings decisions for upcoming month
	carry out financial market trading
Weekly	interact in consumption goods market
Event-based	interact in labor market (if unemployed)

For example, an employee of a firm is paid on the day the firm starts its production, but each firm starts production on a different day of the month. This induces a heterogeneous schedule of activities on all input markets. When the employee is fired he or she receives unemployment benefits on the same day of the month as the salary was received. When re-employed by a different firm, however, his or her subjective monthly time schedule is synchronized with the activation day of the new employer because of the wage payments. Now, the employee receives his or her salary on a different day of the calendar month and shifts consumption accordingly. This modeling approach is supposed to capture the decentralized and typically asynchronous nature of decisionmaking processes and activities of economic agents. It should avoid artificial lock-in and overshooting effects that quite naturally arise in many purely clock-driven simulation models.

In tables 17.1 and 17.2 we provide an overview of firms' and households' decisions and activities and their frequencies. A more extensive discussion can be found in Dawid

et al. (2017), and a full description of the C code is available online as a user manual (Dawid et al. 2011).

17.3.3 Parameterization, Baseline Scenario, and Sensitivity Analysis

Given the complexity of agent-based macroeconomic simulation models, the induced large number of modeling choices, and the typically large set of model parameters,⁴ it is important to provide empirical and conceptual arguments for the usefulness of the developed model for economic analysis. In accordance with the recent literature concerning the empirical grounding of agent-based models in the social sciences (see, e.g., Windrum et al. 2007; Poteete et al. 2010) we follow an approach that can be characterized as follows:

1. Incorporate empirical and experimental evidence about the principles underlying real-world behavior of the agents.
2. Identify a set of stylized facts to be reproduced and explained at the industry or macro level.
3. Restrict the parameter space and initial conditions to sets for which the simulation output matches the stylized facts.
4. Deepen the understanding of the causal mechanisms that underlie the selected stylized facts.

Using this method, parameter constellations have been identified where a large set of stylized facts can be reproduced. In figure 17.2, panel (a) shows the annual GDP growth rate with an approximate mean of 1.5 percent. Panel (b) features the unemployment rate, and panels (c) and (d) contain band-pass filtered time series.⁵ As can be seen in panel (c), investment is more volatile than consumption. Moreover, consumption and investment are pro-cyclical. Panel (d) presents the countercyclical movement of unemployment and firms' price mark-ups. All these features are typically brought forward as the stylized facts describing business cycles. Moreover, the relative size of the amplitudes qualitatively matches those reported in, for example, Stock and Watson (1999).

We should point out that stochastic productivity shocks are not the driving force of the business cycles generated by the model. However, the GDP growth rate can be steered by setting the rate of technological change. Detailed analysis of the dynamics

⁴ In the Eurace@Unibi model overall about forty parameters have to be determined, and for some the values can be heterogeneous across agents. The exact number of parameters depends on the setting considered in the particular policy analysis under consideration.

⁵ Since we are using quarterly data, we use a band-pass filter that admits frequency components between eight and thirty-six quarters.

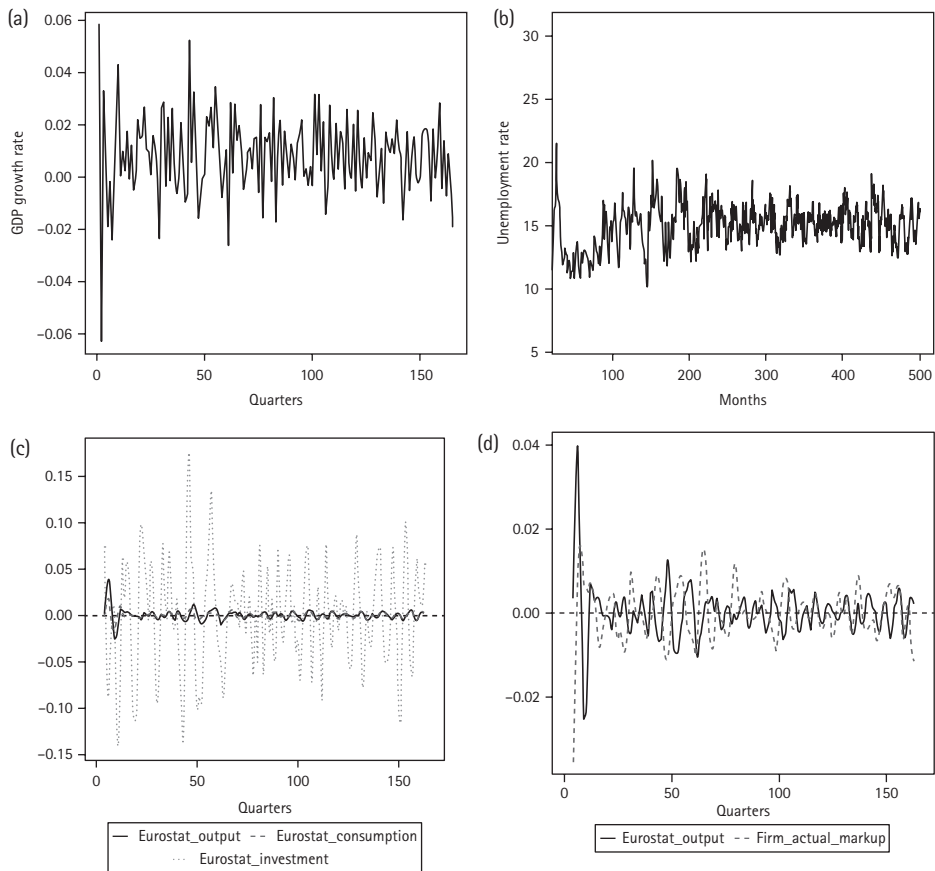


FIGURE 17.2 Time series of macroeconomic variables. (a) annual GDP growth rate (quarter-on-quarter); (b) unemployment rate; (c) band-passed filtered time series for output, consumption, and investment; (d) band-passed filtered time series for output and average price markups.

of such micro-level variables as demand expectations, investments, costs, prices, and household income and consumption budgets show that the cycles are generated by the interplay of optimistic demand estimations of firms in the upswing, the resulting investment patterns, the induced cost and price effects, the implications of these effects on sales, mark-ups and firm earnings, and the reinforcement of such implications through the demand channel due to changes in the real consumption budget of households.

As can be seen in figure 17.3, panels (a) and (b), the model is also able to generate persistent heterogeneity in firm sizes as documented, for example, in Dosi et al. (1997) and substantial persistence in firm market shares. Finally, we show a Beveridge curve in panel (c) derived from the time data series about on vacancies and unemployment rates that resembles real-world data. Note that several of the stylized facts, in particular,

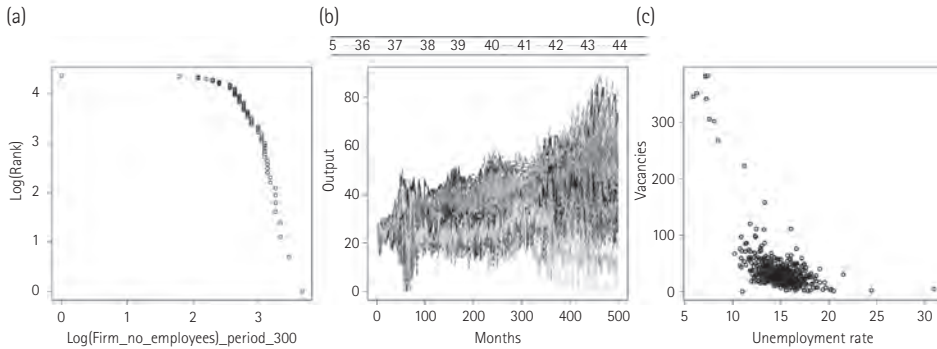


FIGURE 17.3 Stylized facts with regard to firm heterogeneity. (a) Firm size distribution (measured as the number of employees) after 300 months; (b) Time series of output for all 80 firms; (c) Beveridge curve.

all the distributional ones, are reproduced by the model, although the reproduction of these empirical patterns was targeted when determining the parameter settings.⁶

In addition to the variables shown here, the model also reproduces other empirical findings; see Dawid et al. (2017). Rather than going through a long list of simulation variables, in this chapter we would like to illustrate how economic mechanisms can be identified and how to interpret the results of the model in an economically plausible fashion. Adding to the credibility of the model and the trust that it is based on plausible and empirically relevant economic mechanisms is, in our understanding, the main reason to check how the model's output compares to stylized facts. Given this goal of gaining a qualitative understanding of economic mechanisms driving dynamics and policy effects, no attempt has been made to calibrate the model more precisely to specific regions or sectors and to quantitatively target certain time series data. Nevertheless, in different papers based on this model, parameter settings have been identified to distinguish between, for example, technologically more and less advanced regions (see Dawid et al. 2012, 2013).

In order for observations at the aggregate level to be useful, it is crucial to gain an understanding of the (dynamic) mechanisms at the micro and the industry level that are responsible for the observed changes in the aggregate dynamics. Gaining such an understanding is also of particular importance for the analysis of the implications of economic policy measures in different economic environments and in different time windows. In the next section we illustrate one such economic mechanism by considering in more detail the driving forces of technology diffusion and productivity change.

⁶ In a recent paper based on a two-region version of the model it has also been demonstrated that the model in its standard calibration reproduces typical patterns of income inequality dynamics across regions with different technological levels, although such patterns have never been targeted at the parameter determination. See Dawid et al. (2013).

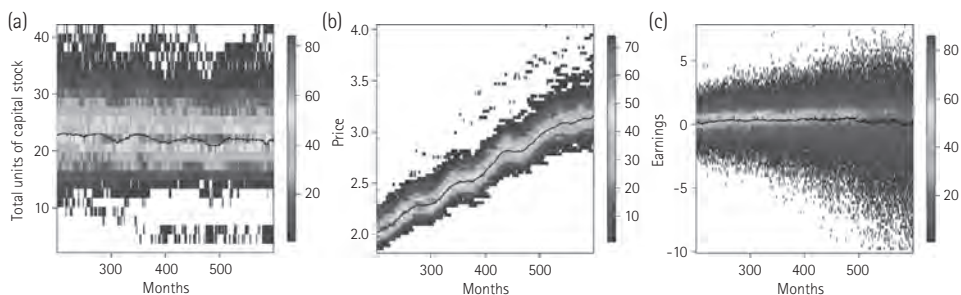


FIGURE 17.4 Evolution of population distributions after an initial transient phase (500 months) of (a) units of capital stock, (b) prices, and (c) earnings per unit of capital stock in a single simulation run. The grey code indicates the density of the population distribution in the considered range of the depicted variable. The black line shows the median of the population distribution.

17.4 IDENTIFYING ECONOMIC MECHANISMS

As alluded to above, persistent heterogeneity of firms with respect to a number of key variables is an empirical stylized fact in many industries. In figure 17.4 we show heat maps depicting the dynamics of the evolution of population distributions of prices, total units of capital stock, and earnings per unit of capital for a single run of the model with the standard parameter constellation.

We can clearly see that the model indeed generates persistent heterogeneity. The shape of the distributions seems to be quite robust over time, although for prices and earnings the concentration of the distribution is slightly decreasing. More detailed analyses of the dynamics of these variables for single firms, not shown here, further highlight that not only do the population distributions exhibit persistent heterogeneity, but also the position of individual firms within the distribution changes only slowly over time (compare figure 17.3b). These observations are appealing from an empirical perspective, but from a more theoretical point of view they raise the questions of how such heterogeneity can survive in the long run and how far such persistent heterogeneity influences the aggregate economic dynamics.

Our discussion here focuses on the role of heterogeneity in the mechanisms driving the diffusion of new technologies and thereby the dynamics of technological change in the economy. Figure 17.5 shows the evolution of the relation between firm size (measured by units of capital stock) and the average quality of the capital stock as well as the actual productivity of a firm. Firm size is measured on the vertical axis, and at each point in time the grey code of the block at a certain firm size indicates the average quality of the capital stock. Notice that, on average, larger firms have capital stocks of higher quality and that new technologies spread into the economy in waves starting at the large firms and then moving to the smaller ones. On the other hand, actual productivity seems to be largely independent of firm size.

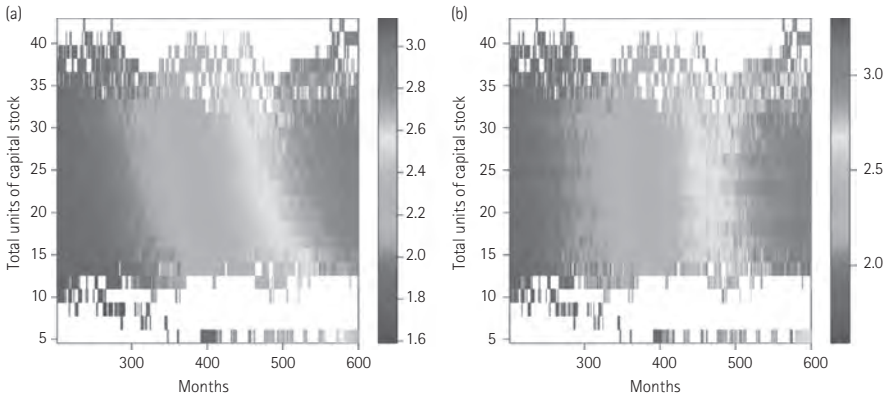


FIGURE 17.5 Time evolution of (a) average quality of capital stock and (b) output per worker sorted by firm size (measured as total units of capital stock).

In order to understand this pattern one first has to take into account that actual productivity of a firm depends not only on the average quality of its capital stock but also on the degree of capital utilization (because firms first use their best vintages and then have to move down to older and worse ones) and the skills of its employees. Indeed, as can be seen in panel (a) of figure 17.6 the degree of capacity utilization of large firms is persistently higher than for small firms. Intuitively, both size and degree of capacity utilization are driven by the relative competitiveness of firms. Those firms that charge the lowest prices tend to expand over time in the presence of demand increases. Expansion is triggered if firms hit their capacity constraints, which implies that the most competitive firms are, on average, characterized by large size (as explained below) and high capacity utilization. The correlation between size and capacity utilization explains, on one hand, the observation that there is no systematic relation between actual productivity and firm size in spite of the systematic difference in average capital stock quality. On the other hand, it makes clear why the diffusion of new capital vintages tends to start at large firms. Owing to the lower average amount of unused capacity, it is more likely that single large firms hit their capacity constraints and decide to expand their capital stock by investing in new investment goods. As can be seen in figure 17.6b, the average investment (measured in terms of quality units) by firms whose size is above the population median is in fact almost twice as large as the average investment by small firms whose size is below the population median.

Despite these systematic differences in firm behavior and characteristics between large and small firms, actual labor productivity and unit costs (not depicted here) do not differ between these groups. The reason for the persistence of differences in firm sizes and market shares is a systematic difference in prices—large firms charge lower prices (see figure 17.6c)—which is due to differences in markups (see figure 17.6d), rather than in costs. Note that, like empirical studies, we define markups relative to observable *unit costs*, rather than to *marginal costs* as is typically done in the theoretical industrial organization literature. To understand these price differences it is important to realize

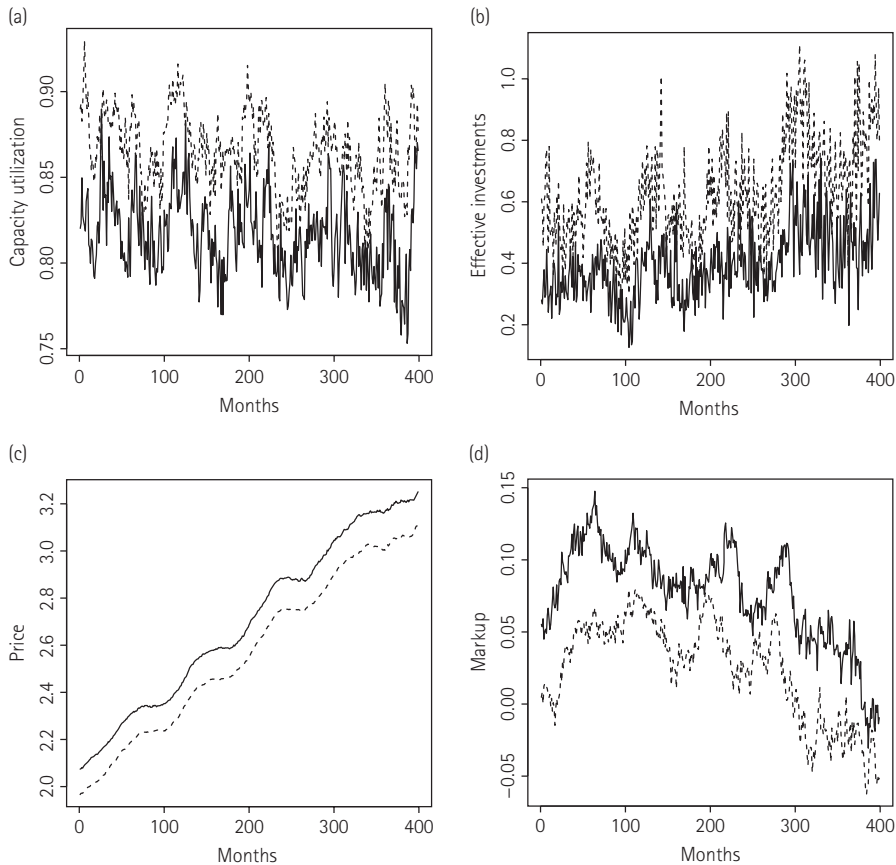


FIGURE 17.6 Comparison of time evolution of the means of key economic variables of large (dashed line) and small (solid line) firms, where the group of large firms consists of all firms whose size (measured in units of capital stock) is above the population median: (a) capacity utilization; (b) investment measured in quality units; (c) price of consumption good; (d) markup.

that the pricing rules used by firms in the Eurace@Unibi model, which according to the management science approach resemble real-world pricing procedures in simplified form, take into account estimations of both demand elasticities and cost functions. The estimated (marginal) cost curves of firms with large capital stocks are much flatter than those of their smaller competitors,⁷ which by standard economic arguments implies that these firms choose lower prices. The fact that profit-oriented pricing rules are considered, rather than, for example, fixed markups, is crucial for the emergence of this phenomenon and highlights the more general point that considering forward-looking and goal-oriented behavior of agents enhances the richness of economic insights that can be obtained from agent-based models.

⁷ This holds in particular in the high-output regions, where small firms would have to invest heavily in order to generate the necessary capacities.

Overall, this brief discussion shows that owing to path dependencies, firms persistently differ in their investment and pricing behavior. Moreover, because of these differences a certain class of firms (namely, the large firms that charge lower prices) is of particular importance for the dispersion of new capital vintages, and therefore accelerates the speed of diffusion of new technologies in the economy. Insights of this kind are highly useful when considering the effect of policy measures on the aggregate dynamics and on the behavior and performance of different groups of agents. One might, for example, infer from our discussion that fiscal policy measures that try to foster technological change by providing subsidies for investment in new vintages should have a redistribution bias toward large firms.

Although a good understanding of the different forces at work at the different levels of aggregation is very useful for developing such intuition for the implications of different policies, a rigorous statistical analysis of the effect of a policy measure on key economic variables and a careful examination of the induced economic mechanisms are needed to confirm such conjectures. For a given parameter constellation and a given policy environment the time series observed in a simulation is a realization of a stochastic process whose exact properties typically are not fully understood. Hence, statistical methods are appropriate to evaluate the effects of changes in the policy environment on the properties of the time series under consideration. Particularly challenging in this respect is evaluating in a statistically rigorous way the evolution of policy effects over time.

In Dawid et al. (2013) and Dawid et al. (2013) we have introduced a penalized spline approach as a tool to unravel the dynamic effects of policies on the economic variables of interest. According to this approach the time series resulting from batch runs with different policy settings are pooled and based, on these data a nonparametric model using penalized spline functions, is estimated. For each of the considered policy options a spline function is associated with a dummy variable indicating whether the policy has been used in a particular run. These spline functions then capture the dynamic effects of the corresponding policy measure. Standard techniques allow the derivation of confidence intervals for those spline functions, and therefore insights about the statistical significance of policy effects at a given point in time can be obtained. We refer to Kauermann et al. (2009) for a more extensive treatment of penalized spline methods.

In the following section we briefly discuss several examples of policy analyses that have been carried out using (previous versions of) the Eurace@Unibi model. Several of these analyses rely on the penalized spline method described above.

17.5 SPATIAL POLICY ANALYSIS

Explicit consideration of heterogeneity is not only important on an individual level but also on more aggregated levels such as regions or nations. Macroeconomic processes are characterized by the aggregation of economic interactions that to a large degree

take place on a local level, where local economic environments and properties of individual agents might differ significantly between regions. Such regional differences are of particular importance if it is taken into account that the flow of goods and production factors between regions is affected by spatial frictions that hinder equilibrating forces between different local markets. Such considerations have important normative implications since they suggest that policy makers should not only care about the type and dose of policy interventions but also about the spatial distribution of the measures taken. Agent-based models are particularly well suited to explore the implications of (local) policy measures in economies consisting of regions with different characteristics because they allow us to understand how individuals with different characteristics (such as skills for workers, capital endowment for firms) in different regions are affected over time by the policy and how the resulting varying reactions of these individuals influence regional and aggregate economic dynamics. Detailed analyses of the evolution of the population distributions of key economic variables, similar to the one sketched in the previous section, are an important instrument for policy studies of this kind. They should complement statistical tests of the implications of policy measures on simulation results.

The Eurace@Unibi model has been used to examine in a multiregion setting the design of economic policy measures that facilitate technical change and economic growth. In the political debate the balance between policies fostering the catching-up of weaker regions and the reinforcement of strengths of advanced regions is a recurrent theme, not least in the context of policy design for the European Union. However, theoretical studies of the effects of different spatial distributions of policy measures that also take into account the interplay of local dynamics and cross-regional flows of factors and goods are limited. In particular, careful consideration of the implications of spatial frictions on different markets for the effects of different policy measures are rare. This seems to be mainly due to the fact that most theoretical policy models do not explicitly take into account heterogeneity on the individual *and* regional levels.

The analysis in Dawid et al. (2008) contributes to the debate about whether activities to strengthen technological change should be targeted at stronger regions of weaker regions, or whether it should be uniformly distributed. The concrete policy measure under consideration is an increase in the level of general skills of workers in a region. Our interest in the effects of skill-improving policies was driven by two main motivations. First, there is strong empirical evidence that the skill distribution of the workforce has substantial influence on the speed of technological change, the employment and wage dynamics, and growth in the economy (e.g., Bassanini and Scarpetta 2002). In order to efficiently use new technologies the workforce of the industrial firms has to be able to build up the required level of specific skills, and the ability to do so depends on the general skill levels of the employees. Therefore, policies that aim at a change in the local skill distribution may play an important role in fostering technological change and growth. Furthermore, we observe that high-skill employees are in many instances strongly concentrated in a few areas. The Eurace@Unibi model can capture these aspects and is therefore well suited to explore the different implications of the described policy variants.

Under the assumption that the flow of workers between regions is hindered by substantial spatial frictions (which might be due to commuting costs or legal restrictions), the simulation experiments show that the concentration of policy measures in one region in the short run triggers stronger overall growth in the economy than a uniform allocation of policy measures across both regions. But a policy maker with a long-term growth perspective in mind should refrain from such spatial concentration of the policy effort since the uniform policy shows better performance in the long run.

The economic mechanisms behind the results are intricate and serve to underline the valuable insights that can be gained from an agent-based approach. The finding is driven by the relatively low mobility of labor compared to that of consumption goods, which in the long run leads to an incomplete substitution of production in the low-skill (less supported) region with the production in the high-skill (more supported) region. In the case of a concentration of skill-upgrading in one region (the high-skill region) in the short run, this region gains a competitive advantage due to faster productivity increases, which induces a shift of demand toward this region. This shift has a moderate overall growth-enhancing effect in the economy, since production is shifted toward the firms where productivity increases fastest. However, eventually the additional demand cannot be covered with the available workforce in the high-skill region. This leads to rationing of firms on the labor market with associated wage increases that open a gap between wages and productivity of labor. The combination of these frictions and wage distortions, which do not occur in case of a uniform application of the policy in both regions, are the main reason for the lower long-run growth effects of the spatially concentrated policy. Furthermore, the falling demand for goods of producers in the low-skill region and labor mobility costs prevent these firms from hiring high-skilled workers from the other region. Consequently, no technological spillovers from the high- to the low-skill region emerge, and this also contributes to the observation that the geographically focused policy is less effective.

This policy analysis is developed further in Dawid et al. (2009), in which where the role of labor market frictions is analyzed more deeply. Again we compare policies that moderately increase general skill levels of workers uniformly across regions with an alternative approach in which all efforts are concentrated in one region. Contrary to Dawid et al. (2008), however, we look into the opposite case of low spatial frictions. It turns out that the size of the commuting costs indeed has a crucial impact on the relative performance of the different policy types. In the empirically hardly relevant benchmark case of zero commuting costs (i.e., workers are completely indifferent regarding working in their own or some other region as long as the same wage is offered), no significant differences between the effects of the policy types emerge. For small frictions in labor mobility, however, the performance of the spatially concentrated policy is better than that of the uniform one. With positive but low commuting costs, contrary to the previous findings, a self-reinforcing cycle of capital and labor investments emerges that leads to faster overall growth. The origin of this cycle is an initial asymmetry in labor costs and prices induced by the combination of a geographically concentrated skill-upgrading policy and (small) spatial labor market frictions. The price asymmetry

leads to demand asymmetry, which in the case of low commuting costs induces spatial worker flows rather than local wage increases. Therefore, firms facing the increased demand in this scenario are not rationed on the labor market and are able to expand their production and have incentives to invest in additional physical capital. The additional investment induced by this demand shift leads to a faster diffusion of technology. This faster diffusion, together with the policy-induced improvement in workers' ability to acquire the specific skills needed to work with the new machines, leads to a speedup of productivity and output growth.

In Dawid et al. (2012) the relation between spatial labor market frictions and (regional) economic growth is considered from a slightly different policy angle. The main question addressed in that paper is to which extent different policies of opening up labor markets that accompany an integration of goods markets affect output and consumption dynamics in regions that start from different levels of economic development. This policy question has clear real-world relevance. For example, for countries that became part of the EU in May 2004, during a transition phase of up to seven years restrictions on labor mobility of workers could be imposed. Along the "2+3+2" formula restrictions had to be reviewed after two years and another three years. In fact, these policies raise important and so far not well-understood research questions, in particular, to which extent spatial frictions with respect to labor mobility may have positive or detrimental effects on overall and region-specific variables related to the well-being of their citizens in the medium and long run. It is a complex task to understand the most likely consequences of various labor market integration policies in a world where regions differ with respect to distributions of skills and the endowments of firms' physical capital and where productivity dynamics are influenced by complementarities between skill adjustments, investment dynamics, and potentially skill-selected labor flows. Moreover, the mobility of workers interacts with (1) important feedback effects through the wage dynamics (accompanying productivity growth), (2) demand shifts driven by changes in households' consumption, and (3) the investment behavior of firms.

The policy experiments on labor market integration conducted with the Eurace@Unibi model indeed yield strikingly different outcomes depending on the variable of interest (regional output or consumption), the regional level of analysis, and the time horizon. Total output is lowest for closed regional labor markets and equally higher for all policies that mimic an opening up of the labor markets. Regional output, however, differs along all four policies.

Thus, when the objective is to maximize overall output the policy advice would be to choose either of the policies that at least gradually opens up labor markets. The advice to a policy maker who cares about convergence of regions would go differently. One who is willing to trade off some output on the aggregate for more convergence should rather not integrate labor markets. If, however, a policy maker is not willing to give up overall output, then the advice is to fully open up labor markets because among all the policies that promote labor market integration, this was the one that resulted in the least inequality between regions with respect to output levels.

Furthermore, in a world in which there are considerable flows of people who work abroad but still consume at home, results in terms of convergence effects of the various labor market integration policies differ, and so would the policy advice if a policy maker's objective is to reduce inequality between regions regarding per capita consumption. In this case, no trade-off between convergence of the regions and overall performance arises. Consequently, it is advisable to open up labor markets because this policy yields better results in terms of overall consumption and convergence of regions than not allowing workers to commute between regions. A detailed discussion of the dynamic mechanisms yielding these policy implications can be found in Dawid et al. (2012).

In Dawid et al. (2013) we again analyze a concrete European regional policy issue using the Eurace@Unibi model. The focus of the paper is on the effectiveness of different types of cohesion policies with respect to convergence of regions. In particular, two types of policies are considered, which in a stylized form represent two main instruments of the EU cohesion policy.

The first is a technology policy that resembles activities carried out under the European Fund for Regional Development (ERDF), which provides subsidies to firms in an economically lagging region in order to invest in technologies at the technological frontier. Thereby, the incentive for firms in that region to acquire the latest available vintages of physical capital increases. This should lead not only to an increase in the quality of the physical capital in that region, but, owing due to the learning-by-doing effect captured in the Eurace@Unibi model, also to an improvement of the specific skills of the workers.

The second policy is a human capital policy, which represents measures within the European Social Fund (ESF), that induces an improvement of the distribution of general skills in the workforce in the target region. This improvement implies that on average workers in the target region acquire specific skills more quickly and, since the technology choice of firms is influenced by the skills of their workforce, it should further be expected that this policy also leads to an increase in general skills and to an improvement of the quality of physical capital.

The dynamic implications of these two types of policies are studied in a two-region version of the Eurace@Unibi model, where firms in region 1 (the high-tech region) are initially endowed with a capital stock whose technological level is close to the frontier, while the capital in region 2 (the low-tech region) shows a considerable gap. The human capital differs in that the labor force in the first region is better educated and, by working with the most recent technology, has acquired higher specific skills than workers in the other region. Two different setups are considered. In the first setup the labor markets are fully integrated such that there are small frictions and all workers have almost unhindered access to both local labor markets. In the other setup the labor markets are completely separated and workers can only work in their home region. These are two extremes where the former may be seen as the political aim of an integrated European labor market, whereas the second is close to the current status quo in the EU, where actual labor mobility across larger distances and language barriers is still relatively small.

The main results of the analysis are that the human capital policy is only effective in terms of fostering cohesion if labor markets are separated. If labor markets are integrated, output actually falls in the lagging region at which the policy is targeted. Technology policies speed up convergence for integrated and separated labor markets, and their effect is substantially stronger if they are specified such that they incentivize firms to purchase the newest vintages.

In summary, these findings strongly make the point that the spatial distribution of policy measures matters and the degree of frictions in labor mobility is crucial for the convergence patterns to appear. This inherently relates to the regional differences of the interaction patterns between the agents that lead to macroscopic outcomes and that again feed back on the individuals' and firms' choices. The Eurace@Unibi model has proved to be a useful tool to capture various aspects that are relevant for spatial policy design. However, the structure of the model also allows us to use it to gain insights into issues that are not directly related to spatial policy questions. For example, the explicit representations of the processes generating heterogeneity of firms as well as of workers makes the model suitable for the examination of dynamic mechanisms influencing (income) inequality in an economy. An example in this respect is Dawid and Gemkow (2013), which studies the impact of labor market characteristics, in particular the importance of social networks and referral hiring, on wage inequality.

17.6 HOW MUCH COMPLEXITY FOR POLICY ANALYSIS?

Macroeconomic policy issues, including questions related to the policy studies described above, have of course been analyzed extensively using macroeconomic models of different types. In particular, in recent work dynamic stochastic general equilibrium (DSGE) models have been frequently used to address such issues. Hence, it is important to understand how the agent-based approach and the policy analyses carried out using this approach relate to DSGE-based studies. The overall structure of the Eurace@Unibi model is in fact quite similar to that of typical dynamic equilibrium models that capture (endogenous) processes of technical change (see, e.g., Acemoglu 2008).⁸ In particular, there is a consumption goods sector producing a final consumption good using labor input provided by households, and there is physical capital (machinery) provided by upstream firms in an investment goods sector. The driver of economic growth is the growing quality of the physical capital employed in production. Also similar to general equilibrium models, the Eurace@Unibi model is a closed macroeconomic model in the sense that there are no monetary or real flows into and out of the economy. Both types of models are parameterized to replicate the empirical stylized facts of the macroeconomy.

⁸ Similar statements hold true for other agent-based macroeconomic models.

In contrast to their DSGE counterparts, however, in agent-based models the agents' decisions do not follow from the assumption of a representative agent who is intertemporally optimizing an objective function under rational expectations. Rather, the behavior of the agents is modeled using rules that have well-established (empirical) foundations. The microbehavior may differ between agents, and they do not fully foresee all the consequences of their own actions and that of the other agents. While this exposes agent-based models to the Lucas Critique, it seems a more appropriate assumption about firm and household behavior within the context of a complex dynamic economy (cf. Cyert and March 1963). Heterogeneity is allowed to a large degree, and there is an explicit aggregation of individual behavior. One could even go further and claim that the agent-based models are truly micro-founded because aggregation is not achieved by adopting the simplifying and possibly erroneous assumption of a representative agent. The aggregation of individual decisions could end up in situations where markets do not clear. Therefore, contrary to the equilibrium-based DSGE framework, agent-based models capture economic frictions that result from the fact that actual economic agents are not fully coordinated in their (market) actions. Agent-based models are able to address how such frictions affect the aggregate economic dynamics or the effects of policy measures. With that said, we refrain from discussing in more detail the motivations and merits for those differences in modeling philosophy (see Fagiolo and Roventini 2012 for an extensive discussion of DSGE- versus agent-based modeling).

In terms of economic policy analysis the agent-based approach provides possibilities for analysis that do not seem feasible in a DSGE framework. We illustrate this point with the policy experiments discussed in the previous section. As we pointed out, the mechanisms responsible for the effects of the policies rely on the interaction of distinct actions by agents exposed to heterogeneous market situations. Path dependencies might emerge as a result of these heterogeneous actions, from the (selected) factor flows across regions, or from the frictions arising in markets owing to the explicit interaction protocols. In a representative agent equilibrium framework it is not possible to capture policy effects that are associated with such phenomena. We are convinced that the effects of actual policy measures are strongly influenced by such path dependencies, heterogeneities, and market frictions. Hence, in many policy domains it seems essential to work with models that are able to capture these effects.

Moreover, our analyses highlight the ability to rigorously study the effects of policy measures at different time scales. This is something that is very difficult (if not impossible) to do in a dynamic general equilibrium framework, where the transitional dynamics between equilibria—that arise after an (unanticipated) shock to the economic environment—typically cannot be characterized. Hence, such transient effects after a policy shock are subsequently ignored in the DSGE framework, while the ABM approach takes full account of them.

A challenge of working with agent-based macroeconomic models is the complexity of interrelated effects and dynamics that result from the explicit modeling of interactions on different markets and the representation of (heterogeneous) rules to determine

the behavior of agents. For an economic analysis carried out within an agent-based model we can ask whether the mechanisms in focus could be represented in a more parsimonious, analytical model. The approach taken in the development and analysis of the Eurace@Unibi model is to build a framework that captures main (empirically motivated) factors that are responsible for technological change and associated skill dynamics on the micro-level of an economy. Hence, the model exhibits complementarities between the skill levels of workers and the quality of the capital stock inside the firms, and both variables are determined endogenously and influenced by the emerging patterns of the matching between workers and firms. Furthermore, the modeling approach integrates the dynamics of labor, consumption, and investment goods markets in order to capture the feedbacks between technology choices by firms, the skill acquisition by workers, and the evolution of demand for consumption goods. These aspects seem crucial to understanding the (regional) technological change and related policy issues and therefore have been modeled with considerable micro structure. The parts of the model that are not crucial for the issues we are focusing on, in particular the financial and credit markets, as well as the interactions between consumers and producers on the consumption goods market, are kept as simple as possible and are mainly used for model closure.

For several features of the model, such as the endogenous determination of firm mark-ups or the vintage choice of firms, simplified alternatives including fixed constant mark-ups have been tested during the development of Eurace@Unibi and then dismissed because they generated economically implausible dynamics either on the micro or the macro level. So, in this respect as well the development of the model is based on the intention to avoid complexity that is not essential for a sound understanding of the economic mechanisms under consideration.

As has been shown above, the developed framework is sufficiently general to be used to address different policy issues in our focus area. A common modeling strategy is to develop a separate (partial) model for each of the issues, capturing a single economic mechanism that seems especially important to the investigator. In contrast, our approach has been to analyze in a full-fledged model the linkages between different agents and markets that turn out to be of particular relevance to understanding the effects of a certain policy.

17.7 CONCLUDING REMARKS

There have been convincing attempts in terms of proof-of-concept to use agent-based models for macroeconomic research. These models are often simultaneously reproducing a large set of empirical stylized facts, including correlations, and lag structures between macroeconomic variables or distributions.

The Eurace@Unibi model and other contributions in this field have gone well beyond showing that stylized macro features can be replicated by agent-based models. Standard

macroeconomic models such as Real Business Cycle (RBC) models and New Keynesian DSGE models require a continuous stream of exogenous shocks to produce business cycle fluctuations. Agent-based models, on the other hand, produce self-sustaining business cycle fluctuations by relying on the endogenous generation of shocks at the micro level. In addition, there have been insightful contributions using agent-based macroeconomic models in the area of ex ante policy simulations. It has been shown how policy measures related to the improvement of the quality of human capital or to the flow of production factors change the behavior of firms and workers in a spatial context, and how the interaction of agents unfolds at the more aggregated level. Nontrivial improvements have been achieved in terms of the technical implementation of such large-scale simulation models. In terms of reliability of the results, stock-flow consistent models have been important.⁹

Besides these very promising steps forward, many things remain to be done or are currently at an unsatisfactory stage. In particular, the patchwork of decision rules that we typically find across variants of agent-based macroeconomic models and sometimes even within a single model is not a convincing approach to modeling agent behavior. Also, the development of feasible techniques for systematic calibration or estimation of ABMs is to a large extent still an open issue. Furthermore, the way simulation output is analyzed and policy experiments are carried out should be made more rigorous and unified across the various agent-based analyses to allow for a better comparison of model properties and results.

We see various possibilities for building on existing work in the area of agent-based macro. One route for future research is the inclusion and analysis of the role and consequences of networks in ABMs. There have been various attempts to build into agent-based macroeconomic models networks with regard to the credit and financial sector (e.g., Battiston et al. 2007, 2012). What has been neglected so far are labor market networks in which the links between workers and firms evolve endogenously, and in which the links between workers vary along different characteristics to varying degrees. Similarly, the role of firm networks or sectoral networks such as production chains, which are typically at the center of empirical studies such as input-output analysis, is not, to our knowledge, included in (agent-based) macroeconomic models. We know little about how characteristics of networks between suppliers of investment goods or other input factors and downstream firms would alter the diffusion of new technologies, business cycle characteristics, or the efficacy of policies in general.

Another route is to study in much more detail the effects of particular policies on the interaction between the real and the financial economy. Interesting questions here focus on the transmission channels of risk through, for instance, the financial accelerator mechanism. We see in particular the strength of agent-based models in linking short-run policy analyses of the business cycle with the longer run because policy measures may involve trade-offs over time. For example, policies that lead to

⁹ For early statements about the importance of stock-flow consistency, see Lavoie and Godley (2006) and Lavoie (2001).

the expansion of the financial system may in the short run allow more innovative projects to be financed, and hence may increase the speed of technological change and technological diffusion. In itself, from a partial model point of view, this would enhance long-run growth. However, it also leads to more risky financial positions of firms, opening up the possibility of endogenous crises through increased financial fragility, firm bankruptcy, and debt write-offs. Banking regulations aimed at preventing such endogenous increases of risk may dampen the short-run volatility, but could also reduce the speed of technological change and thereby depress long-run growth. It is only in models having real and financial markets that are sufficiently detailed and integrated that such second-order effects of policies can be studied. As such, agent-based macroeconomic models could form a bridge between the literature on financial fragility and endogenous growth models by using integrated models instead of partial market models.

As a final hint about future directions of research we would like to emphasize that with its very strong micro-foundations, which stress heterogeneity at the agent level, agent-based modeling opens up more interesting ways to study the distributional outcomes of policy. With ABMs we have the chance to study income distributions across agents or groups of agents as well as along the spatial dimension or the time dimension (see, e.g., Dosi et al. 2013; Dawid and Gemkow 2013). All these topics are exciting areas for future research with agent-based macroeconomic models, and we are looking forward to more interesting insights produced by a steadily growing community of researchers in the future.

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CHAPTER 18

AGENT-BASED MODELS FOR ECONOMIC POLICY DESIGN

Two Illustrative Examples

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18.1 INTRODUCTION

Agent-based models have made substantial progress in various areas of economic research in recent years (for surveys see Tesfatsion and Judd 2006; Hens and Schenk-Hoppé 2009; Rosser 2009). In this contribution, we discuss the usefulness of these models for economic policy design by means of two examples. Our first example is concerned with financial markets. We present an agent-based financial market model in which a central authority, say, a central bank, applies simple intervention rules to stabilize the evolution of prices. In our second example, we develop an agent-based goods market model to study how a central authority, say, a government, may stabilize national income by adjusting its expenditure. Overall, we conclude that agent-based models may serve as artificial laboratories to study how markets function and the effects of various stabilization policies. Moreover, we provide advice about how to design and implement such policy experiments and point out their shortcomings, drawbacks, and limitations. To be able to pin down the effects of regulatory policies, our survey focuses primarily on small-scale agent-based models.

Let us turn to our first example. From an economic point of view, financial markets are quite volatile and frequently display severe bubbles and crashes. Since the dynamics of financial markets may be harmful to the real economy, policy makers regularly debate about the stronger regulation of financial markets. However, it is often unclear how the implementation of policy measures such as transaction taxes, trading halts, or short-selling constraints will affect the markets, especially if they have never been applied before. Fortunately, a number of agent-based financial market models (e.g., Day and Huang 1990; Kirman 1991; Chiarella 1992; De Grauwe et al. 1993; Lux 1995; Brock

and Hommes 1998; LeBaron et al. 1999; Farmer and Joshi 2002; Westerhoff and Dieci 2006; He and Li 2007) exist that have improved our understanding of financial markets and that may be used to address these questions (see Chapter 21 for more details).

The dynamics of agent-based financial market models is driven by the trading behavior of heterogeneous speculators who follow simple trading strategies. As we know from many empirical studies, as summarized by Menkhoff and Taylor (2007), financial market participants rely on technical and fundamental analysis to predict the future direction of the market. Whereas technical analysis (Murphy 1999) derives trading signals from past price movements, fundamental analysis (Graham and Dodd 1951) presumes that prices will return toward their fundamental values. Note that trend-extrapolating behavior and mean-reversion behavior have also been observed in laboratory experiments, as summarized by Hommes (2011), so there is widespread empirical evidence to back the main cornerstones of these models.

Overall, technical analysis destabilizes financial markets. In periods when many speculators opt for technical analysis, strong price fluctuations and significant bubbles may be observed. In contrast, fundamental analysis lends stability to financial markets. In a market in which a sufficiently large number of speculators follow fundamental analysis, prices are relatively stable and tend to move toward fundamental values. Of course, alternating periods of turbulent and calm market dynamics may emerge if both forces act together. Since agent-based financial market models are quite powerful, they have already been used as artificial laboratories to scrutinize the effects of various policy measures (see Westerhoff 2008 for an early survey).

In the current study we use an agent-based financial market model by Franke and Westerhoff (2012). In this model a market maker adjusts prices with respect to excess demand and speculators select between technical and fundamental trading rules to determine their orders. The speculators' choice of trading rules is endogenous and depends on predisposition effects, herding behavior, and market circumstances. As it turns out, this model is quite successful at mimicking the dynamics of financial markets. Since prices adjust in this model with respect to excess demand, it is easy to add a central authority that also participates in the trading process and, by manipulating excess demand, seeks to direct the development of prices. In particular, we explore two strategies. According to the first, the central authority buys in undervalued markets and sells in overvalued markets, thus trading in the same direction in which fundamental traders usually trade. According to the second strategy, the central authority trades against past price trends and thus counters the orders of chartists. As we will see, agent-based models may help us understand how such interventions may influence the market dynamics.

Small-scale agent-based business cycle models such as those by Westerhoff (2006a), Tuinstra and Wagener (2007), Franke (2008), Branch and McGough (2009), Lines and Westerhoff (2010), De Grauwe (2010), Anufriev et al. (2013), and Franke (2012) have also been proposed in recent years. Although these models are not yet as well developed as their financial market counterparts, they are already able to explain business cycle dynamics, at least to some degree. In most of these models, market participants have

to form expectations, say, inflation expectations or output expectations. For this task, they endogenously select between different forecasting rules (the analogy between agent-based financial market models and agent-based business cycle models will be clarified below), and it is this switching behavior that creates endogenous dynamics in these models. The use of extrapolative expectations tends to amplify business cycle fluctuations, whereas the use of regressive expectations exerts a dampening effect on the dynamics. If both forces gain and lose popularity over time, intricate business cycles may emerge. We are currently observing several new exciting results with respect to the effectiveness of monetary and fiscal policy rules in this research area.

The agent-based business cycle model employed in our study is quite simple. Essentially, we have a Keynesian-type goods market model in which firms' investment expenditure depends on national income expectations. Investment expenditure increases if firms expect a boom, and vice versa. A firm forms its expectations on the basis of an extrapolative rule or a regressive prediction rule, between which the firm dynamically selects with respect to predisposition effects, herding behavior, and market assessments. Simulations reveal that the model is able to produce complex business cycles. We therefore study whether simple fiscal policy rules may stabilize the evolution of national income. According to one strategy, the central authority increases (decreases) its expenditure if national income is below (above) a desired target value. According to the other strategy, it increases (decreases) its expenditure if national income decreases (increases). As it turns out, agent-based business cycle models give us an idea of how these strategies may affect the dynamics.

The remainder of this chapter is organized as follows. In section 18.2 a simple agent-based financial market model is introduced in which a central authority applies different strategies to stabilize the price dynamics. In section 18.3 a simple agent-based goods market model is proposed in which a central authority relies on different strategies to stabilize the evolution of national income. We close with a summary and an outlook.

18.2 FINANCIAL MARKETS

In this section we present a simple agent-based financial market model (for surveys see Chiarella et al. 2009; Hommes and Wagener 2009; Lux 2009; Westerhoff 2009; Chen et al. 2012). Since the setup is quite stylized, it may be interpreted as a model for stock markets, commodity markets, or foreign exchange markets or, more generally, as a model for speculative markets. For simplicity, however, we call it a model for financial markets. We will use this model to analyze whether a central authority has the ability to stabilize the dynamics of financial markets by applying basic intervention strategies. To this end, we introduce some statistics that allow us to measure the success or failure of these interventions and to check their viability.

18.2.1 A Simple Agent-Based Financial Market Model

The agent-based financial market model of Franke and Westerhoff (2012) serves as our workhorse. In this model a market maker quotes prices with respect to speculative excess demand and the speculators determine their orders following a representative technical or a representative fundamental trading rule. According to the technical trading rule, prices move in trends and buying (selling) is suggested when prices increase (decrease). The fundamental trading rule predicts that prices will return toward their fundamental value. In undervalued markets, fundamental trading rules recommend buying and in overvalued markets they propose selling. In each time step, speculators have to select one of the two strategies. The choice made by speculators depends on three socioeconomic principles: a predisposition toward one of the two strategies, herding behavior, and an evaluation of the current market circumstances.

Since the model of Franke and Westerhoff (2012) has a remarkable ability to match some important stylized facts of financial markets, it seems reasonable to use it as an artificial laboratory to conduct a number of policy experiments. Therefore, we add a central authority to the model. The central authority seeks to stabilize the dynamics by applying simple feedback strategies. One intervention strategy aims to push the price toward a desired target value. If the target value corresponds with the fundamental value, then the central authority trades in the same direction as the fundamentalists. The other strategy seeks to counter the behavior of chartists by trading against past price trends. As reported by Neely (2005), such strategies are frequently applied by central banks in foreign exchange markets to stabilize exchange rate fluctuations. We assume that all interventions by the central authority are secret. As a result, speculators cannot strategically exploit the interventions, nor it is clear to them that the course of the price is influenced by the central authority. The latter observation is important since technical traders may otherwise lose their trust in these price signals.

Let us now turn to the formal part of the model and start with the behavior of the market maker. As usual, the market maker collects all individual orders from speculators and changes the price with respect to the resulting excess demand. Therefore, we write

$$P_{t+1} = P_t + a(NW_t^C D_t^C + NW_t^F D_t^F + D_t^A), \quad (18.1)$$

where P_t is the log of the price (at time step t), a is a positive price adjustment parameter, N is the total number of speculators, W_t^C is the market share of speculators following a technical trading rule, D_t^C is the order of a (single) chartist, W_t^F is the market share of speculators following a fundamental trading rule, D_t^F is the order of a (single) fundamentalist, and D_t^A is the order (intervention) of a central authority. According to (18.1), excess buying drives prices up and excess selling drives them down. Without loss of generality, we set the price adjustment parameter to $a = 0.01$ and normalize the population size of speculators to $N = 1$.

The orders placed by chartists and fundamentalists depend on their price expectations, given by $E_t^C[P_{t+1}]$ and $E_t^F[P_{t+1}]$, respectively. If speculators expect a price increase (decrease), they buy (sell), as indicated by the first term on the right-hand side of trading rules (18.2) and (18.3). However, the trading motives of speculators may be more complicated—which is why we add random terms to their trading rules. Jointly, this gives us

$$D_t^C = b_0^C(E_t^C[P_{t+1}] - P_t) + \varepsilon_t^C, \quad (18.2)$$

and

$$D_t^F = b_0^F(E_t^F[P_{t+1}] - P_t) + \varepsilon_t^F, \quad (18.3)$$

where b_0^C and b_0^F are positive reaction parameters and $\varepsilon_t^C \sim N(0, \sigma_0^C)$ and $\varepsilon_t^F \sim N(0, \sigma_0^F)$ are normally distributed random variables with mean zero and constant standard deviation.

Chartists and fundamentalists rely on different expectation formation schemes. Since technical analysis predicts that prices move in trends, we express chartists' expectations as

$$E_t^C[P_{t+1}] = P_t + b_1^C(P_t - P_{t-1}) + \eta_t^C. \quad (18.4)$$

Accordingly, chartists pay attention to the most recent price trend, where b_1^C is a positive extrapolation parameter. To capture part of the variety of the many existing technical prediction rules, we add a normally distributed random variable $\eta_t^C \sim N(0, \sigma_1^C)$ with mean zero and constant standard deviation to (18.4).

In contrast, fundamentalists believe that prices will revert toward their fundamental values. Let P^* be the log of the fundamental value, known to all market participants. Fundamentalists' expectations are formalized as

$$E_t^F[P_{t+1}] = P_t + b_1^F(P^* - P_t) + \eta_t^F, \quad (18.5)$$

with $0 < b_1^F < 1$ as a mean reversion parameter. Since fundamentalists may display digressions from pure mean reversion expectations, we also add a normally distributed random variable $\eta_t^F \sim N(0, \sigma_1^F)$ with mean zero and constant standard deviation to (18.5).

Combining (18.2) to (18.5), and assuming that the random variables in these equations are independent, we obtain the simplified trading rules

$$D_t^C = b^C(P_t - P_{t-1}) + \delta_t^C, \quad (18.6)$$

and

$$D_t^F = b^F(P^* - P_t) + \delta_t^F, \quad (18.7)$$

where $b^C = b_0^C b_1^C$, $b^F = b_0^F b_1^F$, $\delta_t^C = b_0^C \eta_t^C + \varepsilon_t^C$ and $\delta_t^F = b_0^F \eta_t^F + \varepsilon_t^F$. Obviously, $\delta_t^C \sim N(0, \sigma^C)$ with $\sigma^C = \sqrt{(b_0^C \sigma_1^C)^2 + (\sigma_0^C)^2}$ and $\delta_t^F \sim N(0, \sigma^F)$ with

$\sigma^F = \sqrt{(b_0^F \sigma_1^F)^2 + (\sigma_0^F)^2}$. Summing up, chartists follow the current price trend, fundamentalists bet on a fundamental price correction, and both strategies are subject to exogenous noise.

We now come to the central part of the model. In each time step, speculators have to decide which trading rule to follow. This decision depends on the attractiveness of the technical and fundamental trading rule, formalized as

$$A_t^C = c_p^C + c_h W_t^C - c_m^C (P^* - P_t)^2, \quad (18.8)$$

and

$$A_t^F = c_p^F + c_h W_t^F + c_m^F (P^* - P_t)^2. \quad (18.9)$$

First of all, speculators may have a behavioral preference for one of the two trading rules. Such a predisposition effect is expressed by parameters c_p^C and c_p^F . Second, speculators may follow the crowd. The attractiveness of a trading rule thus increases with the size of its market share. Parameter $c_h > 0$ captures the strength of speculators' herding behavior. Finally, speculators may assess market circumstances. Although speculators believe in the persistence of bubbles, they know that all bubbles will eventually burst. In particular, speculators perceive a higher probability that a fundamental price correction is about to set in if the price deviates from its fundamental value. How sensitively the attractiveness of the two trading rules reacts to distortions is controlled by parameters $c_m^C > 0$ and $c_m^F > 0$.

To reduce the number of parameters, we compute from (18.8) and (18.9) a relative attractiveness function

$$A_t = A_t^F - A_t^C = c_p + c_h (W_t^F - W_t^C) + c_m (P^* - P_t)^2, \quad (18.10)$$

where $c_p = c_p^F - c_p^C$ and $c_m = c_m^F + c_m^C$. In short, the relative attractiveness of fundamentalism over chartism depends on speculators' predisposition, herding behavior, and market assessment.

The market shares of speculators following the representative technical trading rule and the representative fundamental trading rule are determined by

$$W_t^C = \frac{\text{Exp}[dA_{t-1}^C]}{\text{Exp}[dA_{t-1}^C] + \text{Exp}[dA_{t-1}^F]} = \frac{1}{1 + \text{Exp}[d(A_{t-1}^F - A_{t-1}^C)]} = \frac{1}{1 + \text{Exp}[dA_{t-1}]}, \quad (18.11)$$

and

$$W_t^F = \frac{\text{Exp}[dA_{t-1}^F]}{\text{Exp}[dA_{t-1}^C] + \text{Exp}[dA_{t-1}^F]} = \frac{1}{1 + \text{Exp}[-d(A_{t-1}^F - A_{t-1}^C)]} = \frac{1}{1 + \text{Exp}[-dA_{t-1}]}. \quad (18.12)$$

Parameter $d > 0$ is the so-called sensitivity of choice parameter. Since d is a scaling parameter in this model, it can, without loss of generality, be set to $d = 1$. Equations

(18.11) and (18.12) imply that if the relative attractiveness of fundamentalism over chartism increases, the market share of chartists decreases and the market share of fundamentalists increases. In this sense, speculators exhibit a kind of learning behavior.

To be able to understand the basic functioning of the model, we abstain for the moment from involving interventions by a central authority; that is, we impose

$$D_t^A = 0. \quad (18.13)$$

In sections 18.2.3 and 18.2.4 we introduce simple feedback strategies that may or may not stabilize the model dynamics.

Since we assume that the log fundamental value is equal to zero in all simulations, that is, $P^* = 0$, seven parameters remain to be specified. Franke and Westerhoff (2012) apply the Method of Simulated Moments to estimate these parameters and obtain the following results:

$$\begin{aligned} b^C &= 1.500, \sigma^C = 2.147, b^F = 0.120, \sigma^F = 0.708, \\ c_p &= -0.336, c_h = 1.839, c_m = 19.671. \end{aligned}$$

In general terms, the idea behind the Method of Simulated Moments is to match a predefined set of summary statistics, or moments, that capture the main stylized facts of financial markets (for a review of the properties of financial markets see Lux and Ausloos 2002; Sornette 2003; Shiller 2005). For this reason, the model parameters are determined via a multidimensional grid search such that the moments of a simulated time series come close to the moments of a real financial market time series. What is meant by “close” has to be specified by an objective function. Here it suffices to note that the current model produces—in a quantitatively acceptable manner—excess volatility, fat tails for the distribution of returns, uncorrelated price changes, volatility clustering, and long memory effects which, according to Chen et al. (2012), belong to the most prominent stylized facts of financial markets.

Figure 18.1 depicts a representative simulation run. Since the model is calibrated to daily data, the five thousand observations displayed correspond to a time span of about twenty years. The top panel shows the evolution of log prices. In the long run, prices fluctuate around the fundamental value. In the short run, however, prices may deviate substantially from the fundamental value and exhibit severe bubbles and crashes. Despite these long-run price swings, the day-to-day evolution of prices resembles a random walk. The third panel contains the corresponding return time series. Although the fundamental value is constant, price changes are substantial on average and include a number of extreme movements. In addition, periods in which the market is rather calm alternate with periods in which the market is rather volatile. In the fourth panel, the Hill tail index estimator is plotted as a function of the largest returns (as a percentage). At the 5 percent level, for instance, the Hill tail index is given at 3.65, which corresponds well to observations in real markets. Finally, the bottom two panels depict autocorrelation functions for raw returns and absolute returns, respectively. Although returns are essentially uncorrelated and price changes are thus virtually unpredictable,

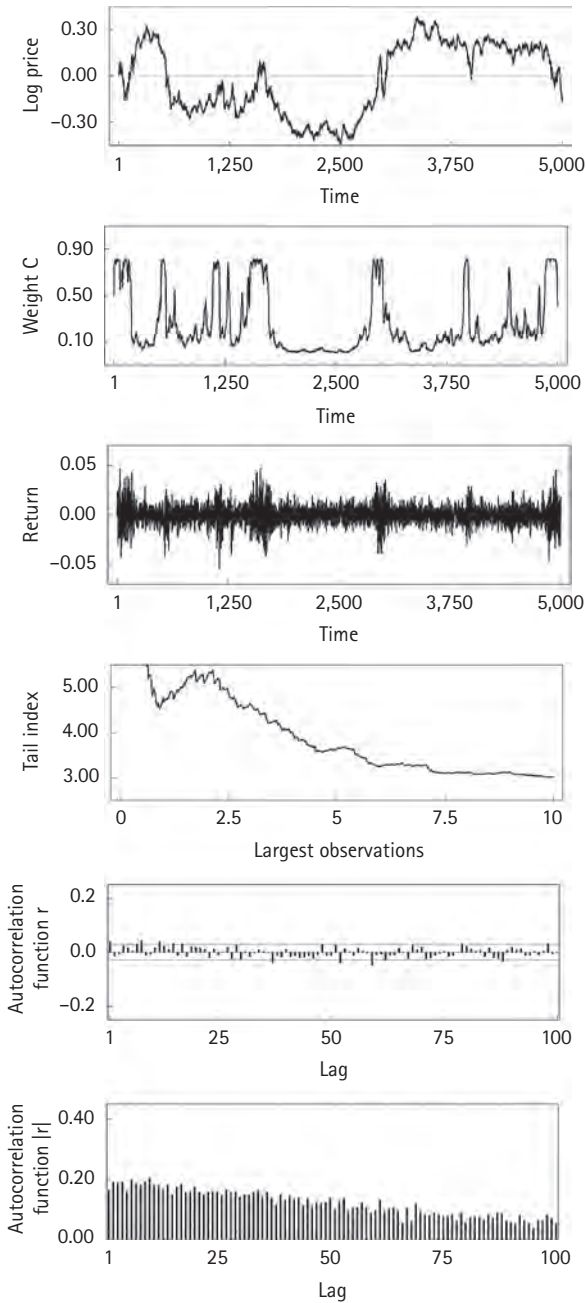


FIGURE 18.1 The dynamics of the agent-based financial market model without regulations. Parameter setting as in section 18.2.

the autocorrelation coefficients of absolute returns are clearly significant and decay slowly over time, witnessing volatility clustering and long memory effects.

In a nutshell, the model functions as follows. As revealed by the second panel, speculators permanently switch between technical and fundamental analysis. Accordingly, there are periods in which technical traders dominate the market. During these periods the market is highly volatile and significant bubbles may emerge. However, fundamental analysis becomes increasingly attractive as bubbles grow. If speculators switch from technical analysis to fundamental analysis, then volatility decreases and prices gradually retreat toward their fundamental values. Of course, this development decreases the attractiveness of fundamental trading and may, together with speculators' behavioral preference for technical analysis, lead directly to a new wave of chartism and instability. While there is permanent ongoing competition between the two trading rules, speculators' herding behavior lends both regimes a degree of persistence, which is responsible for the marked volatility clustering and long memory effect.

18.2.2 Policy Objectives

Since the dynamics of the model is close to the dynamics of actual financial markets, we feel confident that we can use it as an artificial laboratory to run a number of policy experiments. First of all, however, we have to think about how to define the success or failure of an intervention strategy. One advantage of agent-based modeling is that this task is usually quite easy, at least in a technical sense. It seems natural to assume that policy makers prefer prices to be relatively stable and to be close to a desired target value, for example, the fundamental value. In addition, we have to check the viability of an intervention strategy. For instance, the central authority should not build up a larger position, nor should the size of an intervention be unreasonably high. In total, we define four statistics to quantify these aspects. These statistics also allow us to compare the consequences of different intervention strategies.

Let T be the underlying sample length for the statistics. We measure the variability of prices by the average absolute log price change. Expressed as a percentage, this gives us

$$volatility = 100 \frac{1}{T} \sum_{t=1}^T |P_t - P_{t-1}|. \quad (18.14)$$

To quantify the market's mispricing, one may estimate the average absolute distance between log prices and the log fundamental value. However, the central authority's desired target value for the market price may deviate from the fundamental value. Taking this into account, we define

$$distortion = 100 \frac{1}{T} \sum_{t=1}^T |P^A - P_t|, \quad (18.15)$$

where P^A is the desired log target value of the central authority. Hence, this measure estimates the average absolute distance between the log price and the central authority's log target value, and is again expressed as a percentage.

The application of an intervention strategy should not result in the accumulation of too large a position. Therefore, we monitor the growth rate of the central authority's position by calculating

$$growth = \frac{1}{T} \sum_{t=1}^T D_t^A. \quad (18.16)$$

Ideally, the growth rate is zero. Moreover, we identify the average (absolute) intervention size of a strategy by

$$size = \frac{1}{T} \sum_{t=1}^T |D_t^A|. \quad (18.17)$$

Naturally, other measures, such as the profitability of an intervention strategy, may be computed. To limit the analysis, we abstain from such extensions.

As we will see, interventions by a central authority will affect the evolution of the trading rules' market shares. Since these market shares are the key to understanding how the model operates, we furthermore report

$$weight\ C = \frac{1}{T} \sum_{t=1}^T W_t^C, \quad (18.18)$$

and

$$weight\ F = \frac{1}{T} \sum_{t=1}^T W_t^F, \quad (18.19)$$

that is, the average market shares of chartists and fundamentalists. Equipped with these statistics, we can now start our policy experiments.

18.2.3 Targeting Long-Run Fundamentals Interventions

Let us start with a simple and, at least at first sight, plausible intervention strategy. Recall that P^A is the central authority's desired log target value. The central authority may seek to drive the price toward the desired target value by submitting buying orders if the price is below the target value and by submitting selling orders if it is above the target value. Based on these considerations, we obtain the intervention strategy

$$D_t^A = e(P^A - P_t), \quad (18.20)$$

where parameter $e > 0$ denotes the central authority's intervention force. If the central authority's target value is equal to the fundamental value, we call this strategy the (unbiased) targeting long-run fundamentals strategy. Otherwise, we call it the (biased) targeting long-run fundamentals strategy.

Figure 18.2 illustrates how the (unbiased) targeting long-run fundamentals strategy may affect the model dynamics. Figure 18.2 is constructed in the same way as figure 18.1, except that we set the central authority's intervention force to $e = 0.4$. With respect to market stability, two aspects become immediately apparent. First, prices are now less disconnected from the fundamental value. And indeed, computing the distortion, we find that the average absolute distance between the log price and the log target value decreases from 20.2 percent (figure 18.1, no interventions) to 13.3 percent (figure 18.2, current setup). Second, the price variability has increased. The volatility of the time series depicted in figure 18.1 is 0.72 percent, whereas the volatility of the time series depicted in figure 18.2 is 0.86 percent.

How do these changes come about? Of course, by buying when the price is low and selling when the price is high, the central authority manages to reduce the distortion. But this is not the end of the story. There are indirect effects in addition to this direct effect. These indirect effects that become apparent from our computer experiments may amplify or diminish the (first) direct effect of the interventions. Comparing the second panel of figure 18.1 with the second panel of figure 18.2 reveals that the market share of chartists increases (in numbers from 27 percent to 44 percent). With distortion reduced, the relative attractiveness of fundamentalism over chartism decreases. Since the market impact of chartists increases, volatility naturally increases. Of course, this offsets part of the central authority's stabilizing interventions. Note that the fat tail property, the unpredictability of prices, and the volatility clustering phenomenon remain essentially unaffected by interventions. We clarify why it is also important to scrutinize these statistics in the sequel.

Since the results represented in figure 18.2 are just an example, we have to generalize our analysis. Figure 18.3 shows how statistics (18.14) to (18.19) depend on the central authority's intervention force. To be precise, parameter e is increased in fifty discrete steps from 0 to 0.5. Moreover, all six statistics are computed for each of the fifty values of parameter e , as averages over fifty simulation runs with five thousand observations each. The results of this exercise, which reveal another advantage of agent-based policy analysis—namely the generation of vast amounts of data for different policies—may be summarized as follows. While volatility continuously increases as the central authority becomes more aggressive, the distortion decreases at the same time. Note that the effects are quite significant. Volatility increases from 0.72 percent for $e = 0$ to 1.00 percent for $e = 0.5$ and the distortion, in turn, decreases from 20 percent for $e = 0$ to 11 percent for $e = 0.5$.

Our model allows these changes to be explained. The market shares of chartists and fundamentalists depend on the central authority's intervention force. As parameter e increases from 0 to 0.5, the market share of chartists increases from about 27 percent to around 55 percent, implying that the market share of fundamentalists decreases

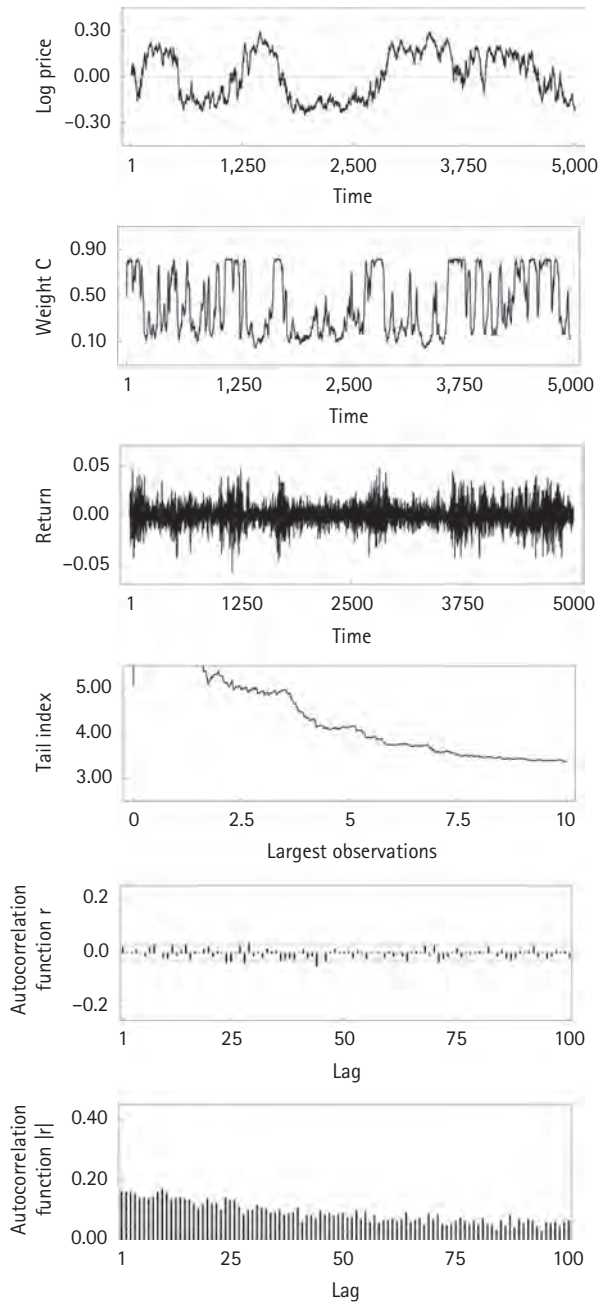


FIGURE 18.2 The dynamics of the agent-based financial market model with (unbiased) targeting long-run fundamentals interventions. Parameter setting as in figure 18.1. In addition, parameter $e = 0.4$ and log target price $P^A = 0$.

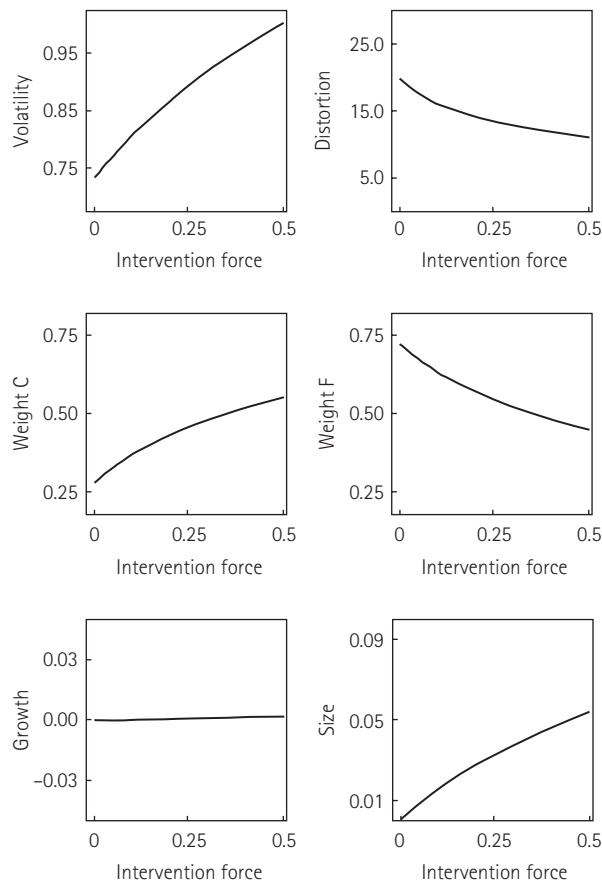


FIGURE 18.3 Some effects of (unbiased) targeting long-run fundamentals interventions. Parameter setting as in figure 18.1. In addition, parameter e is increased from 0 to 0.5 and the log target price is $P^A = 0$.

from approximately 73 percent to about 45 percent. Owing to the presence of more destabilizing chartists, the price variability increases. Of course, the reduction in the market share of fundamentalists is disadvantageous for the central authority since the orders placed by fundamentalists help drive the price toward the fundamental value. At least for the underlying parameter setting, however, the central authority is able to compensate for the loss of these orders. In fact, the central authority brings the price closer to the fundamental value. Moreover, it seems that interventions are feasible. At least, the position of the central authority remains more or less balanced (as the next experiment reveals, this need not always be the case). To sum up, the (unbiased) targeting long-run fundamentals strategy turns out to be a mixed blessing. Prices may be driven closer to the fundamental value, yet only at the expense of stronger price fluctuations.

Let us now suppose that the central authority tries to push prices above the fundamental value by raising the log target value to $P^A = P^* + 0.25 = 0.25$. Figure 18.4 gives us an example of what may happen. The reaction parameter of the central authority is $e = 0.4$. Apparently, the central authority is able to shift the price dynamics upward, with a few occasional interruptions. It also becomes obvious that the price variability decreases. What is going on here? Since the price is now further away from its fundamental value, technical analysis appears to be less attractive, and more speculators opt for fundamental analysis. As a result, the price variability decreases. However, the impact of fundamentalists is not strong enough to push the price towards the fundamental value. According to the bottom three panels, the fat tail property, the unpredictability of prices, and the volatility clustering phenomenon remain more or less unaffected by interventions.

Figure 18.5 explores the impact of this strategy more systematically by varying the central authority's intervention force between $e = 0$ and $e = 0.5$. The design of figure 18.5 is as in figure 18.3, except that $P^A = P^* + 0.25 = 0.25$. At first sight, there may be enthusiasm for this strategy. As the intervention force increases, volatility and distortion decrease, a result that is in line with the central authority's goals. Unfortunately, this strategy will most probably be unviable in the long run. Although the magnitude of the average size of interventions is comparable to that in the previous experiment, the central authority ends up with a massive positive position. The reason for this is as follows. On average, fundamentalists now perceive an overvalued market and continuously submit selling orders. In order to prevent a price decrease, the central authority has to offset these orders and, over time, builds up a significant positive position. To sum up, the second strategy also appears to be a mixed blessing. The central authority may decrease the variability of prices and may push prices toward a desired target value for some time, but eventually it has to abort this policy because its position will otherwise become unbounded.

18.2.4 Leaning-Against-the-Wind Interventions

Let us now study a different, equally simple intervention strategy. In the previous experiments, the central authority attempted to stabilize the dynamics by mimicking the behavior of fundamentalists. In the next experiments, the central authority seeks to stabilize the dynamics by countering the behavior of chartists. In the following, the central authority thus trades against the current price trend, that is, it employs the intervention strategy

$$D_t^A = f(P_{t-1} - P_t), \quad (18.21)$$

where parameter $f > 0$ denotes the central authority's intervention force. For obvious reasons, this strategy is called the leaning-against-the-wind strategy.

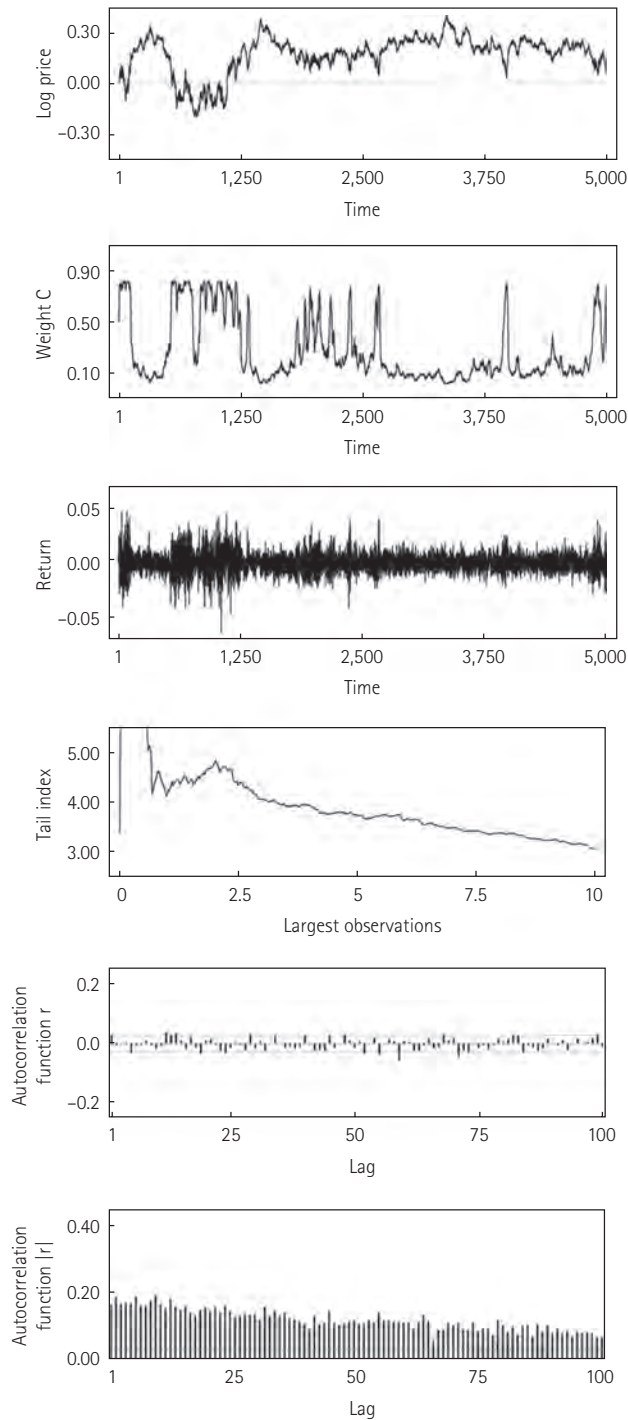


FIGURE 18.4 The dynamics of the agent-based financial market model with (biased) targeting long-run fundamentals interventions. Parameter setting as in figure 18.1. In addition, parameter $e = 0.4$ and log target price $P^A = 0.25$.

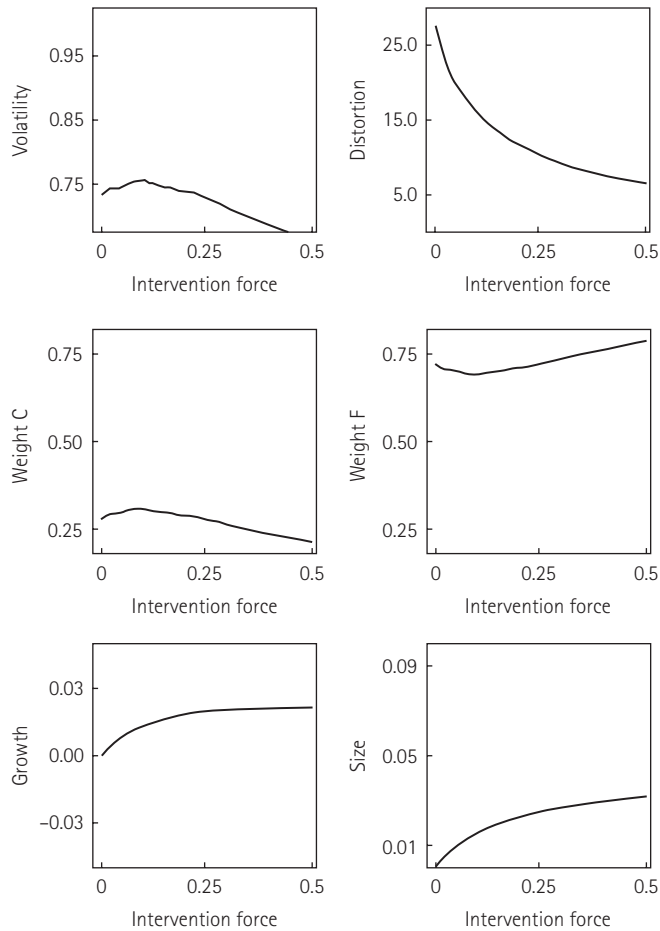


FIGURE 18.5 Some effects of biased targeting long-run fundamentals interventions. Parameter setting as in figure 18.1. In addition, parameter ϵ is increased from 0 to 0.5 and the log target price is $P^A = 0.5$.

Figure 18.6 gives us a first impression of how this strategy may affect the model dynamics. The design of figure 18.6 is again as in figure 18.1 (and as in figures 18.2 and 18.4), except that the central authority's intervention force is now given by $f = 12$. This strategy's performance appears to be rather disappointing. The price does not seem to be closer to the fundamental value, nor has the variability of prices decreased. Also, speculators' switching behavior seems to be unchanged. Inspecting the other three panels, we realize that there is a change in the autocorrelation function of the returns. In particular, at the first lag it is significantly negative, which is unrealistic (we will return to this issue in section 18.2.5).

Unfortunately, figure 18.7 gives no cause for greater hope. Overall, the leaning-against-the-wind strategy may reduce the distortion somewhat, but at the cost of greater

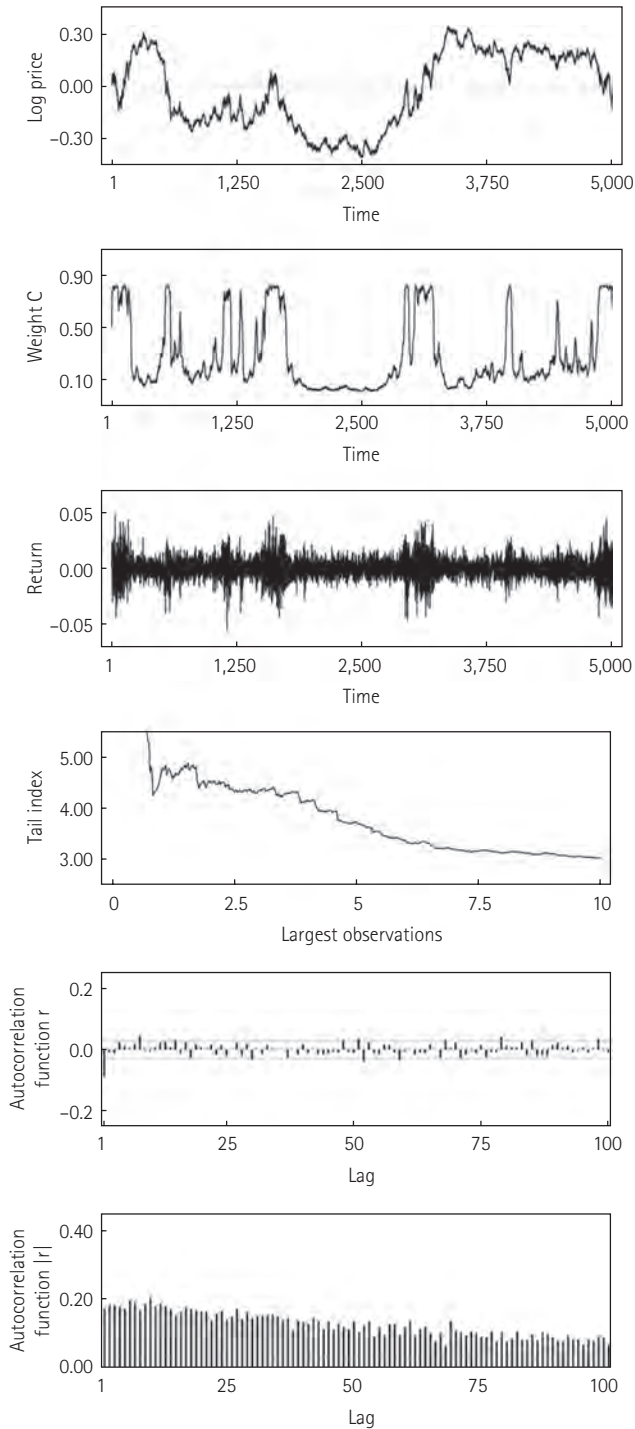


FIGURE 18.6 The dynamics of the financial market model with leaning-against-the-wind interventions. Parameter setting as in figure 18.1. In addition, parameter $f = 12$ and log target price $p^A = 0$.

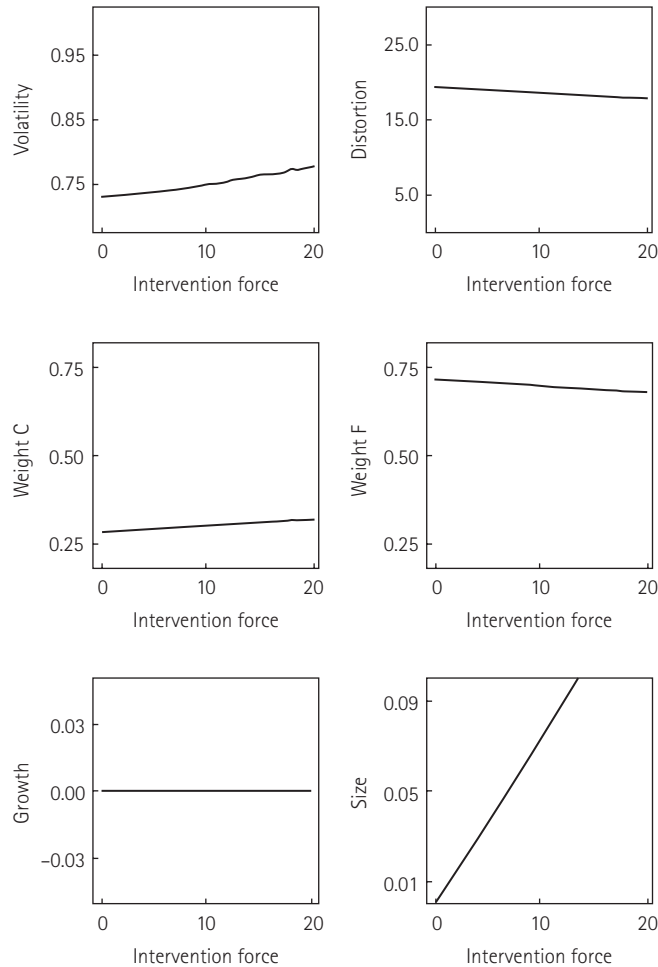


FIGURE 18.7 Some effects of leaning-against-the-wind interventions. Parameter setting as in figure 18.1. In addition, parameter f is increased from 0 to 20 and the log target rate is $P^A = 0$.

volatility. Why is this the case? For low values of parameter f , the central authority indeed offsets part of the destabilizing (trend-based) orders of chartists, which may reduce the distortion slightly. As a result, the market share of chartists increases to some extent, and with it volatility. Moreover, the central authority may itself increase volatility, as its interventions increase excess demand. A major problem of this strategy is that it frequently generates trading in the wrong direction. Since financial market returns are essentially uncorrelated, it is obviously hard to counter price trends.

If the central authority increases its intervention force, its orders eventually overcompensate the (trend-based) orders of chartists and start to induce a mean reversion effect, which further dampens the distortion. The chartists' market share and, thus, volatility increase. But all in all, the effects are very moderate and almost negligible. From the last

two panels we see that at least the central authority does not build up a larger position. However, the size of its interventions may, for higher values of parameter f , be larger than in the case of the two previous experiments.

18.2.5 Discussion

We have seen that agent-based models help us understand how financial markets function and assess the effects of regulatory policies. Our approach may appear too mechanical, however, which may cause an uneasy feeling. It is therefore time to critically review our results. Let us return to the first intervention strategy, the (unbiased) targeting long-run fundamentals strategy. Recall that parameter c_m captures how strongly the relative attractiveness of fundamentalism over chartism increases as the price deviates from its fundamental value, and that this parameter has been estimated for a market environment in which there are no such interventions. Now, speculators who observe that mispricing in a market decreases substantially over time may change their behavior. In particular, they may switch more rapidly to fundamental analysis during the build-up of a bubble, implying, technically, that parameter c_m increases. Should this be the case, then there are on average more fundamentalists in the market and both volatility and distortion are lower. In this sense, the stabilizing effect of this intervention strategy would be underestimated. Of course, in our model speculators adapt to market developments in their strategy selection. But this effect, with constant parameters, may be too weak.

There is also a problem with the second intervention strategy, the (biased) targeting long-run fundamentals strategy. Recall that the central authority manages to drive the price toward a desired target value. In the short run, our results may be reasonable. But this is not the case in the long run. Owing to the accumulation of a large open position, the strategy is simply not viable. Moreover, speculators may change their behavior. At some point in time, they ought to realize that prices will not return toward the fundamental value and, as a result, may base their strategy selection on the central authority's target value and not on the fundamental value. In addition, speculators who opt for fundamental analysis should then condition their orders on the distance between the target value and the current price. Should such a learning behavior occur, the impact of the biased intervention strategy becomes more or less identical to the impact of the unbiased strategy. A model in which speculators learn long-run equilibrium prices may do a better job here.

The third strategy, the leaning-against-the-wind strategy, is also associated with a problem. It is clear from the penultimate panel of figure 18.6 that the central authority introduces a correlation into the price dynamics. To be precise, the autocorrelation coefficient of the returns at the first lag is significantly negative. In reality, we would expect speculators to try to exploit such a pattern. In particular, technical traders should switch from a trend-extrapolation strategy to a contrarian strategy. However, the current model is too simple to be able to take account of this.

Also, we learn that it may be dangerous to perform Monte Carlo studies as in figure 18.7. One should at least inspect some of the time series behind the depicted statistics and check whether the model dynamics still makes sense. In case of figure 18.7, for instance, the analysis should be stopped for values of parameter f larger than 10 (for values of f smaller than 10 the problem with the autocorrelation function does not occur).

All in all, this gives us clear warnings: agent-based models should not be used too mechanically. But since theoretical reasoning, human subject experiments, and empirical studies have their challenges, too, agent-based models may nevertheless serve as a valuable tool to help policy makers determine economic policies. Finally, we mention a few other applications to illustrate some of the areas that have already been addressed using agent-based models: Westerhoff (2001) studies the effects of central bank interventions in foreign exchange markets, He and Westerhoff (2005) investigate the impact of price floors and price ceilings on agricultural commodity markets, Pellizzari and Westerhoff (2009) explore the consequences of transaction taxes, Westerhoff (2003) and Yeh and Yang (2010) are concerned with trading halts and price limits, respectively, Hermesen et al. (2010) investigate the role of disclosure requirements, Anufriev and Tuinstra (2013) model short-selling constraints, Scalas et al. (2005) inspect insider trading and fraudulent behavior, and Brock et al. (2009) examine the problems associated with the appearance of an increasing number of hedging instruments in financial markets.

18.3 GOODS MARKETS

In this section we present a simple agent-based goods market model. Note that many agent-based macro models are inspired by agent-based financial market models. For instance, expectations of the market participants play a crucial role in both research fields, and their modeling is often quite similar. Since our agent-based goods market model is able to produce business cycles, we use it to explore whether simple intervention strategies may stabilize fluctuations in economic activity. For this reason, we have to consider again how to quantify the success or failure of the interventions and how to check their viability.

18.3.1 A Simple Agent-Based Goods Market Model

The model we present in this section stands in the tradition of Samuelson's (1939) famous multiplier-accelerator model. National income adjusts to aggregate demand, which, in turn, is composed of consumption, investment, and governmental expenditures. For simplicity, consumption expenditure is proportional to national income.

Following Westerhoff (2006a), investment expenditure depends not on past output changes, as in the Samuelson's (1939) model, but on expected future output changes. In our model firm managers rely on two prediction rules to forecast the course of the economy. If they rely on extrapolative expectations, they believe that the current trend of national income is persistent. Consequently, they increase their investment expenditure during an upswing and decrease it during a downswing. If they rely on regressive expectations, they believe that national income will return to its long-run equilibrium value. As a result, they increase their investment expenditure if national income falls below this value and decrease it otherwise. Firm managers select their prediction rule at the beginning of every period, with respect to predisposition effects, herding behavior, and market circumstances. The use of extrapolative and regressive expectations as well as switching behavior is well documented in empirical studies (Branch 2004; Hommes 2011).

Since the model is able to produce business cycles, we use it as an artificial laboratory to study two common intervention strategies. According to the first—the trend-offsetting strategy—the government aims at offsetting income trends and thus increases (decreases) its spending when national income has just fallen (risen). According to the other—the level-adjusting strategy—the government seeks to reduce the gap between the actual level of national income and a desired target value. Government expenditure is thus high (low) if national income is below (above) the target value. Note that Baumol (1961) already studied the effects of these two strategies using Samuelson's original model.

Let us now turn to the details of the model (we apply a new notation). National income adjusts to aggregate demand with a one-period production lag. If aggregate demand exceeds (falls short of) production, then production increases (decreases). Therefore, we write

$$Y_{t+1} = Y_t + a(Z_t - Y_t), \quad (18.22)$$

where Y_t represents national income (at time step t), Z_t denotes aggregate demand, and a indicates the goods market's adjustment speed. For simplicity, we set $a = 1$. As a result, national income at time step $t + 1$ equals aggregate demand at time step t , that is, $Y_{t+1} = Z_t$. Since our focus is on a closed economy, aggregate demand is defined as

$$Z_t = C_t + I_t + G_t, \quad (18.23)$$

where C_t , I_t , and G_t stand for consumption, (gross) investment, and governmental expenditure, respectively. Consumers' expenditure is proportional to national income

$$C_t = bY_t. \quad (18.24)$$

The marginal propensity to consume is limited to $0 < b < 1$. Of course, much more interesting specifications including, for instance, income expectations or consumer sentiment effects may be assumed in (18.24).

The interesting part of the model concerns firms' investment behavior. Firms are boundedly rational and select between two representative investment rules. Aggregate investment expenditures are given as

$$I_t = NW_t^C I_t^C + NW_t^F I_t^F, \quad (18.25)$$

where N is the number of firms, W_t^C is the market share of firms with extrapolative expectations, I_t^C is the investment expenditure of a single firm with extrapolative expectations, W_t^F is the market share of firms with regressive expectations, and I_t^F is the investment expenditure of a single firm with regressive expectations (we stick to the labels C and F to represent the two different kinds of behavior). To simplify matters, we normalize the population size of firms to $N = 1$.

Both (representative) investment rules depend on three components: an autonomous component, an expectations-based component, and a random component. These are formalized as

$$I_t^C = \bar{I}^C + i_o^C (E_t^C[Y_{t+1}] - Y_t) + \varepsilon_t^C, \quad (18.26)$$

and

$$I_t^F = \bar{I}^F + i_o^F (E_t^F[Y_{t+1}] - Y_t) + \varepsilon_t^F. \quad (18.27)$$

Autonomous investments are denoted by $\bar{I}^C$ and $\bar{I}^F$. Parameters $i_o^C > 0$ and $i_o^F > 0$ indicate how strongly investment expenditures react to expected changes in national income, and $\varepsilon_t^C \sim N(0, \sigma_o^C)$ and $\varepsilon_t^F \sim N(0, \sigma_o^F)$ reflect additional random influences.

Firm managers form either extrapolative or regressive expectations. Extrapolative expectations are given by

$$E_t^C[Y_{t+1}] = Y_t + i_1^C (Y_t - Y_{t-1}) + \eta_t^C, \quad (18.28)$$

where i_1^C is a positive extrapolation parameter. The random variable $\eta_t^C \sim N(0, \sigma_1^C)$ allows for digressions from purely trend-based expectations. Regressive expectations result in

$$E_t^F[Y_{t+1}] = Y_t + i_1^F (Y^* - Y_t) + \eta_t^F. \quad (18.29)$$

Followers of the regressive rule expect national income to return to its long-run equilibrium level, perceived as Y^* , at adjustment speed $0 < i_1^F < 1$. Random deviations from this principle are summarized by the random variable $\eta_t^F \sim N(0, \sigma_1^F)$.

Combining (18.26) to (18.29) and assuming that the random variables in these equations are independent, we obtain the simplified investment functions

$$I_t^C = \bar{I} + i^C (Y_t - Y_{t-1}) + \delta_t^C, \quad (18.30)$$

and

$$I_t^F = \bar{I} + i^F (Y^* - Y_{t-1}) + \delta_t^F, \quad (18.31)$$

where $\bar{I} = \bar{I}^C = \bar{I}^F$, $i^C = i_0^C i_1^C$, $i^F = i_0^F i_1^F$, $\delta_t^C = i_0^C \eta_t^C + \varepsilon_t^C$ and $\delta_t^F = i_0^F \eta_t^F + \varepsilon_t^F$. Note that $\delta_t^C \sim N(0, \sigma^C)$ with $\sigma^C = \sqrt{(i_0^C \sigma_1^C)^2 + (\sigma_0^C)^2}$ and that $\delta_t^F \sim N(0, \sigma^F)$ with $\sigma^F = \sqrt{(i_0^F \sigma_1^F)^2 + (\sigma_0^F)^2}$.

How do firms select their investment rules? Analogous to the financial market model, we assume that firms compare the attractiveness of the rules and that the mass of them select the most attractive investment rule. Let us first discuss the attractiveness of the rules, which are defined as

$$A_t^C = c_p^C + c_h W_t^C - c_m^C (Y^* - Y_t)^2, \quad (18.32)$$

and

$$A_t^F = c_p^F + c_h W_t^F + c_m^F (Y^* - Y_t)^2. \quad (18.33)$$

Accordingly, firms may have different behavioral preferences for the rules, indicated by parameters c_p^C and c_p^F . In addition, firms may be sensitive to herding dynamics; parameter $c_h > 0$ controls the strength of this effect. Finally, firms assess current market circumstances. The more extreme the business cycle becomes, the more attractive the regressive forecasting rule appears to them. Parameters $c_m^C > 0$ and $c_m^F > 0$ calibrate how quickly firms switch from extrapolative to regressive expectations as the business cycle develops.

We employ a relative attractiveness function to reduce the number of parameters. Taking the difference between (18.33) and (18.32) yields

$$A_t = A_t^F - A_t^C = c_p + c_h (W_t^F - W_t^C) + c_m (Y^* - Y_t)^2, \quad (18.34)$$

where $c_p = c_p^F - c_p^C$ and $c_m = c_m^F + c_m^C$. To sum up, the relative attractiveness of regressive expectations over extrapolative expectations depends on firms' predisposition, herding behavior, and market assessment.

The market shares of firms forming extrapolative and regressive expectations is formalized by

$$W_t^C = \frac{\text{Exp}[dA_{t-1}^C]}{\text{Exp}[dA_{t-1}^C] + \text{Exp}[dA_{t-1}^F]} = \frac{1}{1 + \text{Exp}[d(A_{t-1}^F - A_{t-1}^C)]} = \frac{1}{1 + \text{Exp}[dA_{t-1}]}, \quad (18.35)$$

and

$$W_t^F = \frac{\text{Exp}[dA_{t-1}^F]}{\text{Exp}[dA_{t-1}^C] + \text{Exp}[dA_{t-1}^F]} = \frac{1}{1 + \text{Exp}[-d(A_{t-1}^F - A_{t-1}^C)]} = \frac{1}{1 + \text{Exp}[-dA_{t-1}]}. \quad (18.36)$$

Parameter $d > 0$ describes how sensitively firms react to changes in the investment rules' relative attractiveness. Again we set, without loss of generality, $d = 1$.

For the moment, governmental expenditure is constant, that is,

$$G_t = \bar{G}, \quad (18.37)$$

where $\bar{G} > 0$. In sections 18.3.3 and 18.3.4 we discuss the impact of countercyclical intervention strategies on the model dynamics and their long-run viability.

Firm managers believe that the long-run equilibrium value of national income is given by the Keynesian multiplier solution, $Y^* = (\bar{I} + \bar{G})/(1 - b)$. In total, therefore, ten parameters remain to be specified. We choose:

$$b = 0.9, \bar{I} = 0.3, i^C = 1.4, \sigma^C = 0.015, i^F = 0.8, \sigma^F = 0.015, \bar{G} = 0.2, \\ c_p = -0.45, c_h = 1.2, c_m = 500.$$

This time, we selected the model parameters by hand such that the dynamics of the model resembles actual business cycles, at least to some degree. Obviously, a calibration approach is more informal than an estimation approach, such as the use of the Method of Simulated Moments in the previous example. One should therefore be careful when interpreting the results. Although quantitative statements are virtually impossible to justify, qualitative reasoning may nevertheless be possible.

Figure 18.8 contains an example of the model dynamics. The depicted time series last one hundred periods, which should be interpreted as a time span of one hundred years. The top panel shows that national income oscillates around $Y^* = 5$ (horizontal gray line) and that business cycles last, on average, around eight years. The second, third, and fifth panels depict the paths of consumption, investment, and governmental expenditure, respectively. Note that the first three time series evolve procyclically, as is the case in reality. In addition, the consumption changes and investment changes are roughly comparable in magnitude. Since total investment expenditure is lower than total consumption expenditure, the relative variability of the former is larger than the relative variability of the latter (empirical properties of actual business cycles are summarized in Stock and Watson 1999). Although the power of the model should not be overstated, given its simplicity it does the job adequately.

The penultimate panel of figure 18.8 presents the market shares of extrapolating firms and helps us comprehend how the model functions. If the majority of firms opt for extrapolative expectations, the economy is unstable and national income drifts away from its long-run equilibrium value. As a result, the attractiveness of regressive expectations increases. Since more and more firms turn to regressive expectations, the economy becomes stabilized and national income returns to its long-run equilibrium value. Owing to the prevailing conditions and firms' behavioral preference for extrapolative expectations, the economy, however, receives its next destabilizing impulse. This pattern repeats itself in a more or less complex manner. For instance, herding effects may easily prolong stable and unstable periods, thereby affecting the amplitude and frequency of business cycles.

Despite the random shocks, there is still some regularity in the business cycles, in particular in the first fifty periods. We briefly add here that, in the absence of

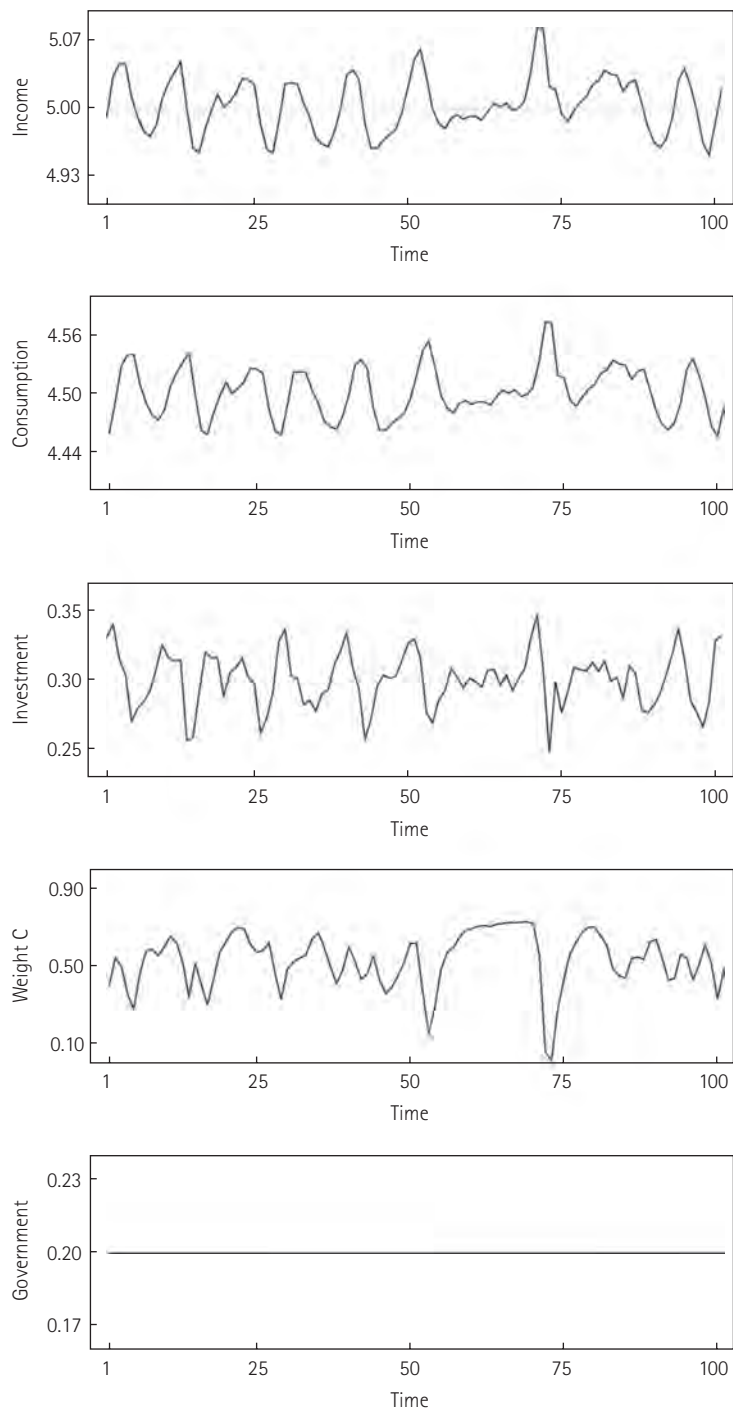


FIGURE 18.8 The dynamics of the agent-based goods market model without regulations. Parameter setting as in section 18.3.

exogenous shocks, our deterministic goods market model produces quasi-periodic dynamics whereas our deterministic financial market model is characterized by fixed point dynamics. Both models are highly nonlinear, however, and exogenous shocks trigger irregular transients.

18.3.2 Policy Objectives

We introduce four statistics to evaluate the success of the intervention strategies. It seems reasonable to assume that policy makers prefer national income to be relatively stable and close to a desired target value. In addition, the interventions should on average be balanced and the size of the interventions should be not too large. The sample length for which we compute these statistics is again given by T .

The variability of national income is measured by its average absolute relative change. Expressed as a percentage, we have

$$volatility = 100 \frac{1}{T} \sum_{t=1}^T |Y_t - Y_{t-1}| / Y_{t-1}. \quad (18.38)$$

Let Y^A be the policy makers' desired target value for national income. In this sense, a natural measure for an "undesired" output gap, also expressed as a percentage, is

$$distortion = 100 \frac{1}{T} \sum_{t=1}^T |Y^A - Y_t| / Y^A. \quad (18.39)$$

Since the policy makers aim at a balanced net position, we report the growth rate of their position by calculating

$$growth = \frac{1}{T} \sum_{t=1}^T (G_t - \bar{G}). \quad (18.40)$$

In addition, the average (absolute) size of an intervention strategy is given by

$$size = \frac{1}{T} \sum_{t=1}^T |G_t - \bar{G}|. \quad (18.41)$$

Note that by subtracting $\bar{G}$ in (18.40) and (18.41), these two measures take only the active part of the intervention strategies into account. Although, of course, other measures may be computed, we abstain from such extensions.

We also keep track of the average market shares of the two investment rules by defining

$$weight\ C = \frac{1}{T} \sum_{t=1}^T W_t^C, \quad (18.42)$$

and

$$\text{weight } F = \frac{1}{T} \sum_{t=1}^T W_t^F. \quad (18.43)$$

These statistics are, again, the key to understanding how the model functions.

18.3.3 Level-Adjusting Interventions

The first rule we explore is the so-called level-adjusting rule. According to this rule, the government increases (decreases) its expenditure if national income is below (above) its desired target value in the hope that national income will thereby be pushed upward (downward). Formally, we have

$$G_t = \bar{G} + e(Y^A - Y_t), \quad (18.44)$$

where parameter $e > 0$ indicates the government's aggressiveness. In analogy to the financial market examples, let us refer to an (unbiased) level-adjusting rule if the government's target value is equal to Y^* , that is to the long-run equilibrium value of national income as perceived by firms, and to a (biased) level-adjusting rule otherwise.

Figure 18.9 illustrates how an (unbiased) level-adjusting rule may affect the model dynamics. Its design is as in figure 18.8, except that $e = 0.5$ and $Y^A = 5$. A comparison of figures 18.8 and 18.9 reveals that the interventions reduce the amplitude of business cycles but simultaneously increase their frequency. Although this placates consumption expenditures, investment expenditure appears to be more volatile. What is going on here? By increasing expenditure when national income is low and decreasing it when national income is high, the government indeed manages to drive national income closer to $Y^A = Y^* = 5$. On average, however, extrapolative expectations now appear more attractive to firms. Since the central authority also reverses the course of national income more frequently via its interventions, the length of business cycles is shortened.

Figure 18.10 captures the effects of (unbiased) level-adjusting interventions in more detail by showing how our policy measures respond to stronger intervention forces. Parameter e is increased in fifty discrete steps from 0 to 1 and all statistics are computed as averages over fifty simulation runs with five thousand observations each. The main results are as follows. First of all, volatility increases: national income changes more strongly if the government reacts more aggressively. The good news is that the distortion decreases: the government manages to drive national income closer toward its target value. For $e = 0$, for instance, the distortion is about 0.43 percent while for $e = 1$ it is only about 0.33 percent, a reduction of about one-fourth.

Note that firms switch from regressive expectations to extrapolative expectations as parameter e increases. This change is most likely to dampen the stabilizing effect of (unbiased) level-adjusting interventions, contributing to greater volatility. As indicated

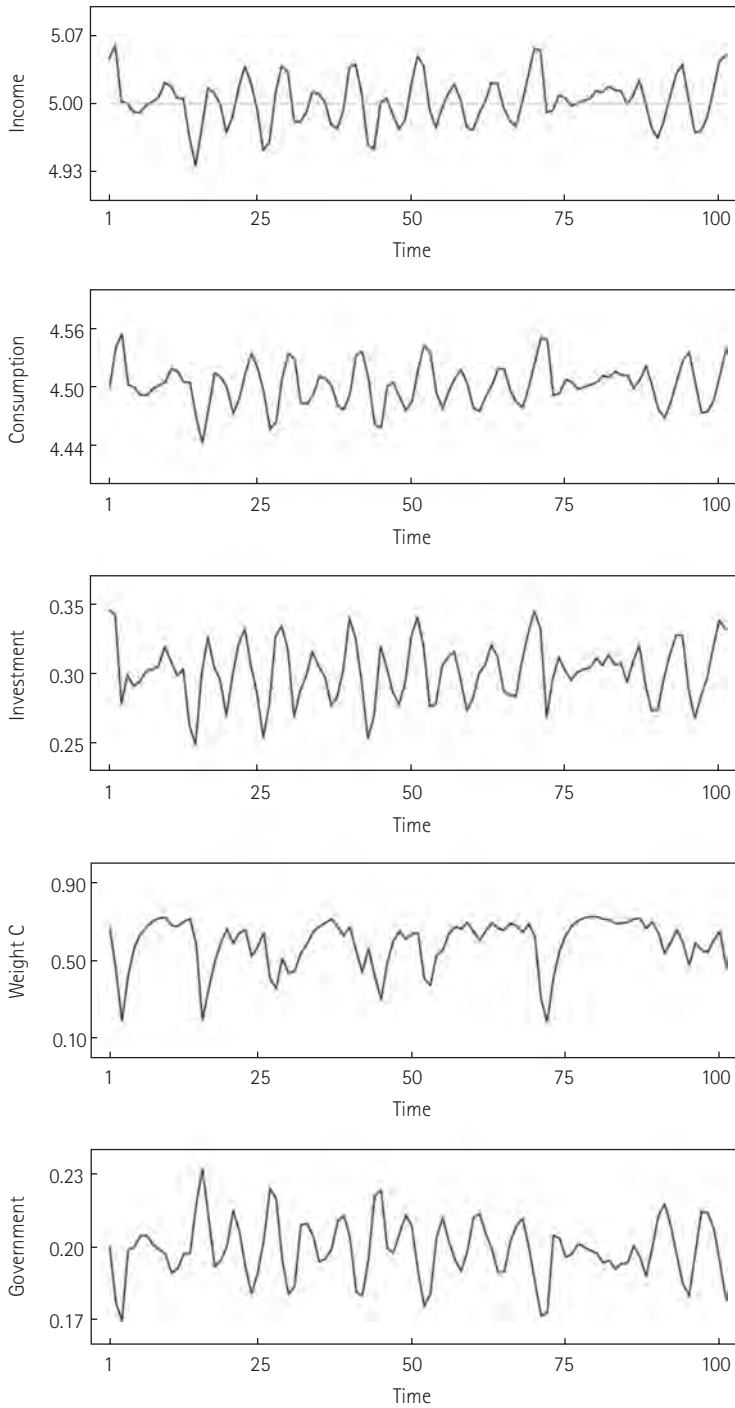


FIGURE 18.9 The dynamics of the agent-based goods market model with (unbiased) level-adjusting interventions. Parameter setting as in figure 18.8. In addition, parameter $e = 0.4$ and target value of national income $Y^A = 5$.

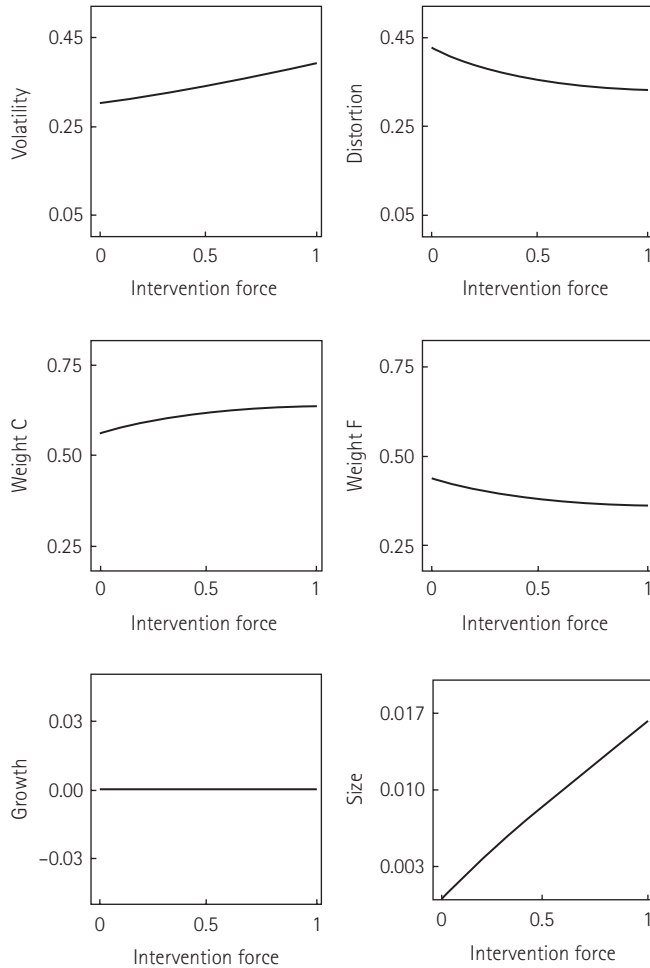


FIGURE 18.10 Some effects of (unbiased) level-adjusting interventions. Parameter setting as in figure 18.8. In addition, parameter e is increased from 0 to 1 and the target value of national income is $Y^A = 5$.

by the bottom two panels of figure 18.10, the government's average net position is close to zero, yet the size of its interventions increases with intervention force. To put the numbers into perspective, for $e = 0$, for instance, the average size of the interventions is about 0.01, which corresponds to a stimulus of 0.2 percent of national income per year, a number which should be accomplishable.

Figure 18.11 presents a simulation run for the (biased) level-adjusting rule. The design of figure 18.11 is as in figure 18.8, except that $e = 0.5$ and $Y^A = 5.05$ (that is, the target value of the government is one percent above $Y^* = 5$). Obviously, the government manages to increase the average level of national income. In addition, although the amplitude of business cycles has decreased, the frequency of business cycles is higher

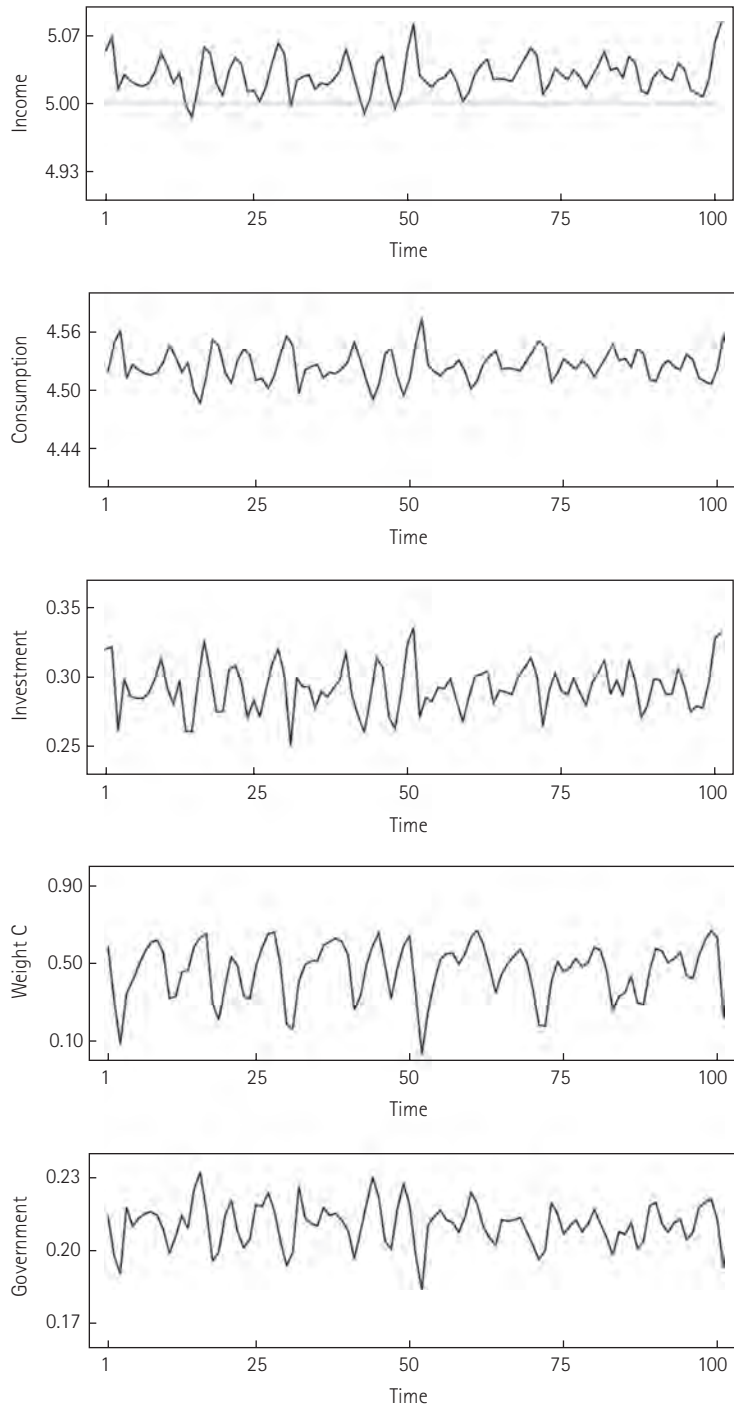


FIGURE 18.11 The dynamics of the agent-based goods market model with (unbiased) level-adjusting interventions. Parameter setting as in figure 18.8. In addition, parameter $e = 0.4$ and target rate of national income $Y^A = 5.05$.

than before (see figure 18.8). Also, the relevance of extrapolative expectations has diminished.

Let us examine figure 18.12 to understand what is happening here. By supporting the target value $Y^A = 5.05$, the average level of national income increases and thus the distortion decreases. Moreover, more firms now rely on regressive expectations. Since this creates a downward pressure for national income (investment expenditures based on regressive expectations are lower), the government has to stimulate the economy in most time steps. Indeed, as revealed by the bottom left panel of figure 18.12, the government's average net position is no longer neutral and, over time, this strategy is virtually unviable (although the average absolute size of the interventions is roughly

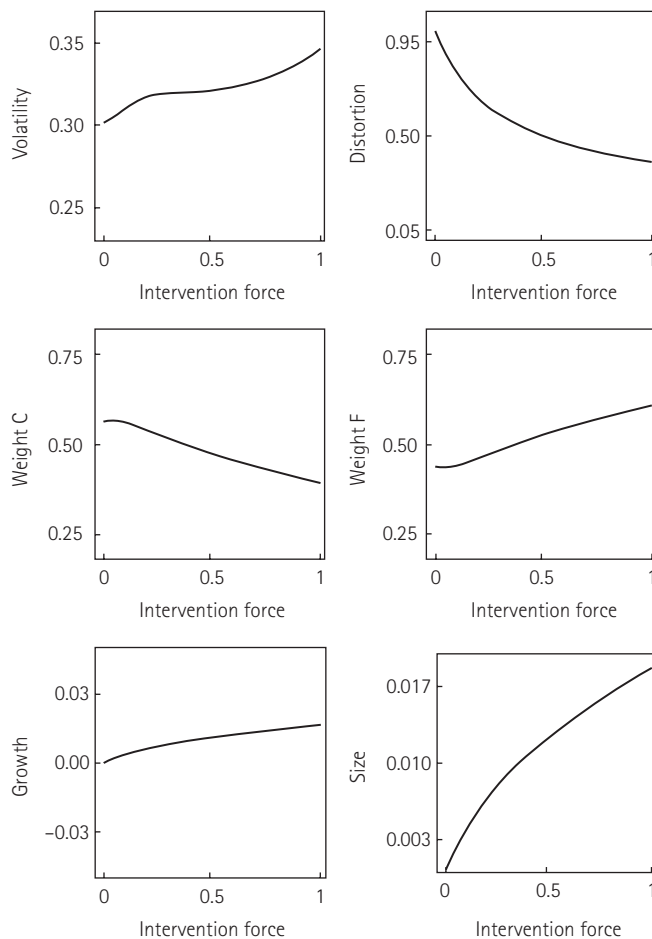


FIGURE 18.12 Some effects of (biased) level-adjusting interventions. Parameter setting as in figure 18.8. In addition, parameter e is increased from 0 to 1 and the target rate of national income is $Y^A = 5.05$.

comparable to those of the previous strategy). For completeness, note that there is also a slight increase in volatility.

18.3.4 Trend-Offsetting Interventions

The last strategy we investigate is a so-called trend-offsetting strategy, given by

$$G_t = \bar{G} + f(Y_{t-1} - Y_t). \quad (18.45)$$

Since f is a positive parameter, the government increases its expenditure if national income decreases, and vice versa.

Figure 18.13 depicts how this strategy may affect the dynamics. The difference between figure 18.8 and figure 18.13 is that $f = 0$ in the former, and $f = 0.5$ in the latter. We may finally have a strategy that is capable of stabilizing the dynamics. As suggested by the first panel of figure 18.13, both the amplitude and the frequency of business cycles decrease because of the interventions. Although more firms now rely on extrapolative expectations, changes in national income have not increased. Of course, a government that manages to weaken the trend of national income automatically tames the destabilizing impact of extrapolative expectations.

Figure 18.14 confirms these promising results. If the government increases its intervention force, then volatility and distortion decrease. The market share of firms forming regressive expectations decreases, but the dynamics nevertheless becomes stabilized. As already mentioned, the destabilizing impact of extrapolative expectations decreases if trends in the business cycle decrease. But why does this strategy work in the business cycle model but not in the financial market model? Recall that price changes are unpredictable in the financial market model, and a central authority that trades against the direction of the most recent price trend trades with a probability of 50 percent in the direction of the current price trend. In case of business cycles, the government makes errors less frequently. The intervention by the government only goes with the trend at the turning point of a business cycle. Otherwise, it effectively counters the trend.

18.3.5 Discussion

Let us critically review the results of the preceding two sections and start with the (unbiased) level-adjusting strategy. By changing the features of business cycles, firms may be expected to adjust their behavior. In our model, firms adjust their behavior via the selection process of the prediction rules. If the attractiveness of regressive expectations decreases, for instance, firms will opt for extrapolative expectations more frequently. However, the way in which firms perceive the attractiveness of the prediction rules remains constant (the parameters are fixed). One may expect in reality that if the amplitude of business cycles decreases, firms will take this into account and will

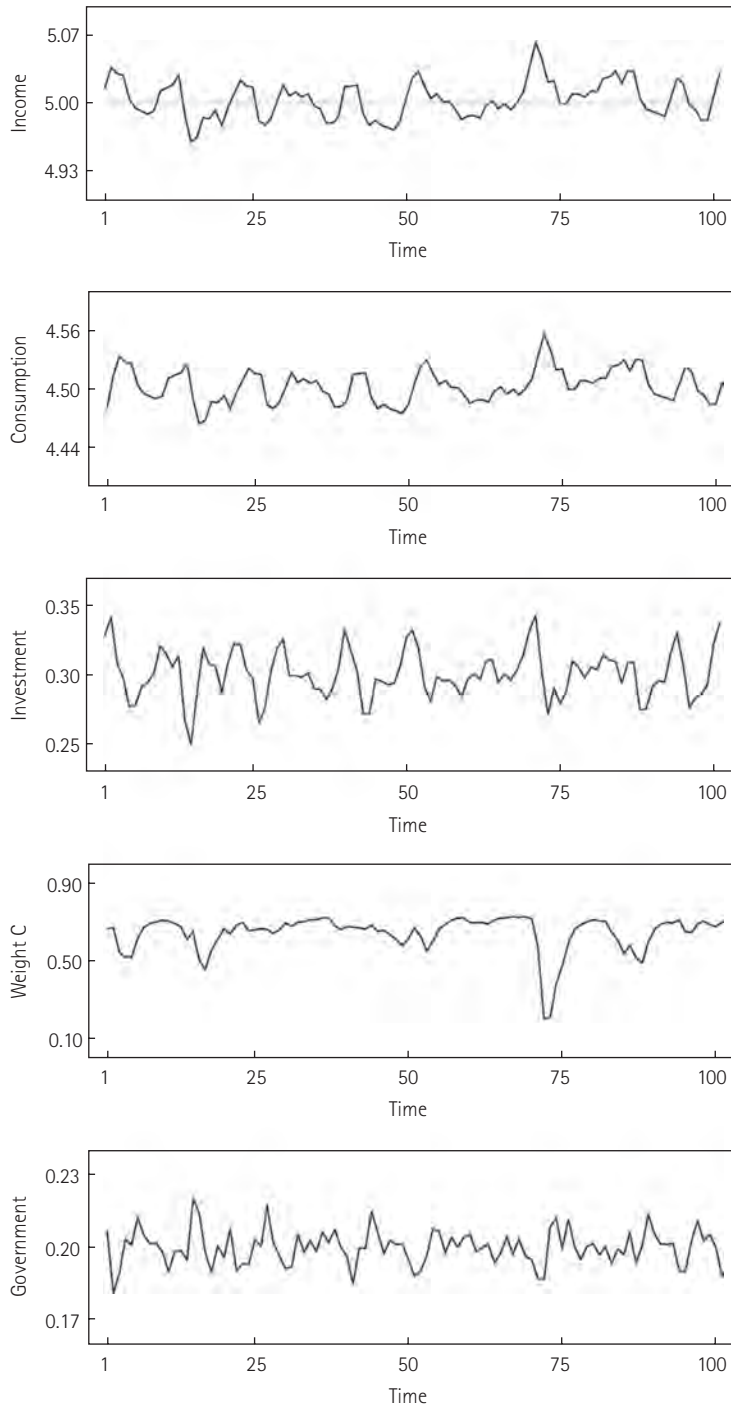


FIGURE 18.13 The dynamics of the agent-based goods market model with trend-offsetting interventions. Parameter setting as in figure 18.8. In addition, parameter $f = 0.5$ and target rate of national income $Y^A = 5$.

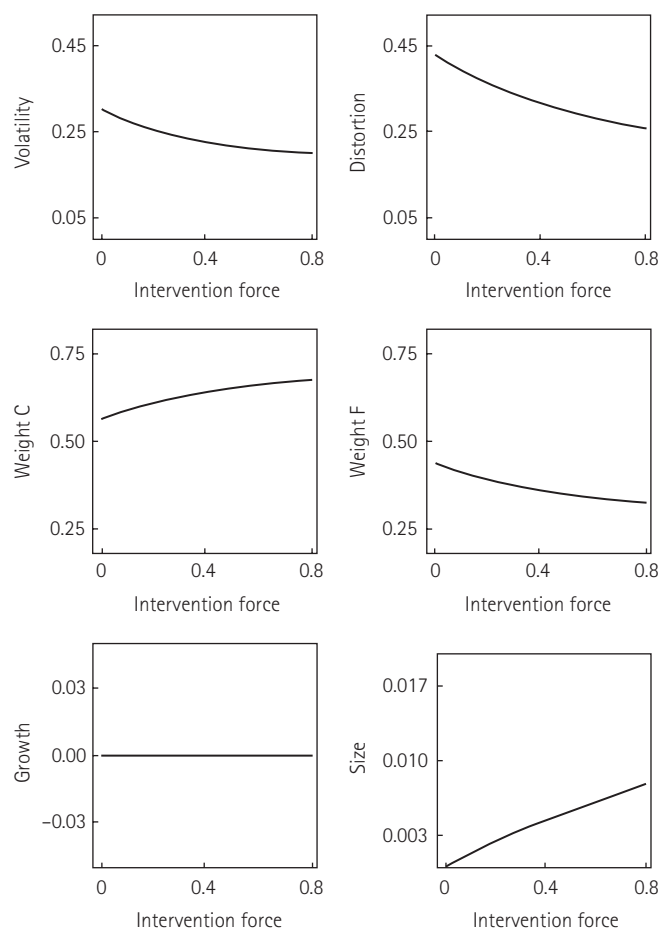


FIGURE 18.14 Some effects of trend-offsetting interventions. Parameter setting as in figure 18.8. In addition, parameter f is increased from 0 to 0.8 and the target rate of national income is $Y^A = 5$.

consequently switch earlier to regressive expectations during the build-up of a boom or a recession. This would presumably further stabilize the dynamics; that is, we may currently underestimate the effectiveness of this strategy.

What about the (biased) level-adjusting strategy? If firms realize that national income fluctuates above their perceived long-run equilibrium value, they should take this observation into account when selecting their prediction rules and forming their expectations. In our simple model, this is not the case. If firms did this, the results of the (biased) strategy would ultimately converge toward the results of the (unbiased) strategy. On the other hand, learning in a macroeconomic context may take a considerable length of time. For instance, the duration of three business cycles, after which a permanent change in the level of national income may become apparent, may be twenty-five years. From this perspective, this aspect should presumably not be

regarded as too critical. The illustrated results may be reasonable for a few years, at least, but it comes at a cost, namely, that the government's position becomes non-neutral.

The trend-offsetting strategy faces a problem similar to that of the (unbiased) level-adjusting strategy. In the end, firms will realize that the amplitude of business cycles has declined and will thus adapt their behavior. If they switch more rapidly to regressive expectations, the dynamics may even be more stabilized. On the other hand, note that the effectiveness of this strategy depends on the regularity of business cycles. The more irregular they are, the less effective the trend-offsetting strategy will be. Since actual business cycles may be more irregular than the business cycles represented in figure 18.8, we may be overstating the functioning of the trend-offsetting strategy.

Small-scale agent-based macro models are emerging; there are not yet many policy applications. Further examples include Westerhoff (2006b), Brazier et al. (2008), Westerhoff and Hohnisch (2010), De Grauwe (2010), Lines and Westerhoff (2010), and Anufriev et al. (2013). Finally, some authors have started to connect agent-based models from the real sector with agent-based financial market models. This research area seems to be promising, also with a view to policy applications. Examples include Lengnick and Wohltmann (2013) and Westerhoff (2012).

18.4 SUMMARY AND OUTLOOK

Agent-based modeling may serve as a valuable tool for economic policy analysis in addition to theoretical reasoning, experiments with human subjects, and empirical studies. In this chapter, we review the extent to which agent-based models are currently suitable for evaluating the effectiveness of certain regulatory policies. As our analysis shows, agent-based modeling is associated with a number of natural advantages, some of which are as follows.

- Agent-based models give us fresh insights into how economic systems function and, thereby, how regulatory policies may shape their dynamics. For instance, regulatory policies frequently have an obvious direct effect, but their indirect effects are often much less clear. By disentangling direct effects and indirect effects, agent-based models help us grasp the impact of regulatory policies in more detail. Agent-based models also reveal the limits of regulatory policies. We can learn what to expect and what not to expect from regulatory policies.
- Agent-based models can also be used to pre-test the effectiveness of newly proposed policies. Moreover, we can even use agent-based models to improve these policies or to design alternative ones. For instance, an agent-based model may reveal that a nonlinear intervention rule does a better job of stabilizing markets than its linear counterpart.
- Agent-based models enable us to control for all exogenous shocks and to simulate extreme events. If we are interested in how a certain policy works during a

particular type of crisis, say, a dramatic downward shift of the equilibrium value of national income, we may simply add such a crisis to our analysis.

- Agent-based models allow us to generate as many data as necessary. Compared to empirical studies and human subject experiments, in which data are typically limited (or simply not available), modern computers enable vast amounts of observations to be computed. This is quite important: because of the appearance of infrequent market frenzies, many statistics such as volatility measures converge relatively slowly, and wrong conclusions may be drawn if the data are collected in a rather calm or turbulent period.
- Agent-based models enable all variables to be measured precisely. In reality, it is often difficult to identify the exact equilibrium value of a financial market or of the real economy. In agent-based models, this task is usually quite simple. Other variables, such as expectations and transactions of market participants, may also be recorded.
- In agent-based models, the intensity of a policy can be varied smoothly. In contrast, many empirical studies are based on periods in which a policy with a constant strength was applied. By gradually increasing the intensity of a policy, agent-based models may reveal nontrivial effects that are precluded at coarser scales. Other types of sensitivity analysis, such as the use of different model parameters or functional specifications of the model's building blocks, are also possible.

Of course, the starting point of the analysis should always be an appropriate agent-based model. The appropriateness of the underlying model decides whether results are obtained that are interesting from a quantitative perspective, as in our financial market example, or which have a more qualitative outlook, as in our goods market example. In this respect, three aspects deserve attention:

- First, the model ideally possesses an empirical microfoundation, meaning that its main building blocks should be supported by empirical evidence. For instance, speculators' use of technical and fundamental analysis, as assumed in our agent-based financial market model, is well documented in the empirical literature. At first sight, the setup of agent-based models may seem ad hoc to outsiders. In our view, however, this is not the case, at least not as long as the models' main building blocks are in line with reality.
- Second, the internal functioning of the underlying model should be convincing. For instance, complex dynamics emerge in our agent-based financial market model since speculators switch between destabilizing technical trading rules and stabilizing fundamental trading rules. Bubbles are thus initiated by a wave of chartism and crashes are triggered by a surge of fundamentalism. Clearly, it would not sound very plausible if this were the other way around.
- Third, the data generated by the model should be as realistic as possible. In particular, an agent-based model should be able to mimic the main stylized facts of a market for which it is developed. Here it is encouraging to see that more and

more agent-based models are being estimated rather than merely calibrated (by hand), which lends these models a considerable amount of additional support.

If these requirements are fulfilled, agent-based models may be well suitable for conducting policy experiments. The following steps have to be considered:

- One important task is to define a set of measures that allow a policy's success or failure to be evaluated. However, it is equally important to determine whether a policy is viable in the long run. A policy that is able to stabilize the markets but causes unsustainable costs is obviously of not much use.
- Moreover, a policy has to be properly implemented in the agent-based model. In our two examples, the intervention policies alter the markets' excess demands. Other policies may affect the attractiveness of the market participants' strategies or other model components. Fortunately, agent-based models are relatively flexible in this respect, yet this modeling aspect is crucial and should not be underestimated.
- Finally, we have seen that it may be dangerous to apply agent-based models too mechanically. It is thus vital to check whether the agents' behavior still makes sense after the imposition of a new policy and whether the resulting dynamics is still reasonable.

To conclude, our economy is a complex adaptive system, and nonlinearities often make it very difficult to anticipate the consequences of regulatory policies. The use of agent-based models offers us insights into the functioning of economic systems that are otherwise precluded. We are convinced that agent-based models are an excellent tool for enhancing our understanding of regulatory policies, in particular, if they are empirically supported, and look forward to more exciting research in this area.

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CHAPTER 19

COMPUTATIONAL ECONOMIC MODELING OF MIGRATION

ANNA KLABUNDE

19.1 INTRODUCTION

Fifty-nine percent of Mexican migrants to the United States surveyed in the Mexican Migration Project (MMP128, described below) make more than one move; that is, after returning to Mexico they go back to the United States at least once. The phenomenon makes it difficult to forecast stocks of migrants in the United States at any point in time and to make estimates of where they are likely to go and when, if at all, they are going to return. So far, research on so-called circular migration has mostly been empirical, using multinomial logit, count data models, duration models, or Markov transition matrices to estimate migration and return probabilities controlling for characteristics of individuals and of the home or host country. Examples are Constant and Zimmermann (2003, 2011), Bijwaard (2010), Vadean and Piracha (2009), Reyes (2001) and, using the MMP, Massey and Espinosa (1997). Hill (1987) is an attempt at formalizing duration of stay and frequency of trips in a life-cycle model. A more recent theoretical model of circular migration is Vergalli (2011), who studies the phenomenon in a real option framework.

When developing a model that is sufficiently realistic to be used for policy analysis or, eventually, forecasts and that is empirically founded, one has to take into account some important aspects of the issue at hand.

First, a migrant's decision is not independent of that of other migrants and potential migrants. Other migrants support the newly arrived in their job search, and home-community members help return migrants to reintegrate into the home country's labor market. The role of social networks in migration decisions has been the subject of substantial research; Radu (2008) provides a survey of the literature. Networks are

often thought to be the reason migration is concentrated in a certain number of places and that people from one neighborhood tend to go to the same few places.

Since a migrant expands his or her network with every migration move and network ties possibly become weaker over time, different parts of the migration cycle should not be seen separately.

An individual's decision to move creates two externalities: one on the network at the location of origin, and the other one on the network in the destination country. If one migrant leaves, others might leave as well. Whether or not the migrant returns, the size of the destination country network will have changed because of his or her move. Thus when he or she considers migrating again, the conditions have changed from those of the previous move, in part because of his or her own behavior. Hence, there is a recursive process, with the network influencing the migrant, the migrant influencing the network, and the new network influencing the migrant. This has been dubbed the "reflection problem" by Manski (1993). This chapter investigates how large the effect of networks is on both migration and return decisions, and what other possible determinants of circular migration exist.

In order to approach this question and to create a space for policy experiments related to (circular) migration, an agent-based model is proposed that allows for the necessary modeling flexibility and for the spatial dimension of the problem. Its central component is the role of networks that evolve endogenously from migration decisions. Links decay over time and physical distance. The migration behavior of one generation of heads of household is modeled over a period of thirty-three years.

There are some rather simple, uncalibrated agent-based models concerning different aspects of migration (Silveira et al. 2006; Espíndola et al. 2006; Biondo et al. 2012; Barbosa Filho et al. 2011). The present model, in turn, is one of the few examples of completely calibrated and empirically founded agent-based models that deal with migration. Related empirical models included Fonseca Feitosa (2010) on urban segregation, Sun and Manson (2010) on the search for housing in Minnesota, Haase et al. (2010) and Fontaine and Rounsevell (2009) on residential mobility, and Mena et al. (2011) and Entwisle et al. (2008), who model changes inland use. A recent paper by Kniveton et al. (2011) replicates climate-induced regional migration flows in Burkina Faso using an agent-based model with networks as information transmission mechanisms. Rehm (2012) provides a very sophisticated agent-based model to study remittances of Ecuadorian rural-urban and international migrants. A different computational approach to empirically founded models of Mexican circular migration has been introduced recently in which discrete choice dynamic programming models are estimated using Maximum Likelihood (Lessem 2011) or the Simulated Method of Moments (Thom 2010).

In the present model the Mexican Migration Project (MMP) and other data sources were used for parameterization. Parameters that cannot be found easily in econometric models owing to endogeneity problems and the spatial dimension are calibrated such that parameter values are found that create a close match between simulated and observed data. By proceeding in this way a common criticism of agent-based models, namely many degrees of freedom and the resulting possibility of creating almost any

desired output, is avoided. All of the parameters but except four are fixed. The remaining four—the distribution of number of trips of migrants, the distribution of migrants across U.S. cities, and the time series of percent age of agents migrating and returning per year—are calibrated indirectly by matching the simulated data to real data. It is then possible to perform experiments with the model.

The chapter is structured as follows: Section 19.2 describes the methodology used and the main data set. Section 19.3 introduces three stylized facts about circular migration that the model should match. Section 19.4 derives and tests hypotheses about behavioral motives to include in the model. Section 19.5 describes the model, which is parameterized in section 19.6. The indirect calibration procedure is described in section 19.7. Section 19.8 describes the model, which is an example of how to use the model for policy experiments. Section 19.9 concludes.

19.2 METHODOLOGY AND DATA

19.2.1 Methodology

The methodology used is the following. First, for a model to be adequate for policy analysis, it has to be “true” in the sense that it represents a plausible candidate for the true data-generating process of the phenomenon of interest. To find out whether this is the case, it is indispensable to have some empirical measure against which to check the model’s output, that is, some means of (external) validation. Therefore three stylized facts are introduced in section 19.3 that the model has to match in order to be considered useful, two of which are distributions of empirical data (number of migrants in each city and distribution of number of trips). Furthermore, the model will be matched against two time series of migration and return flows. For validation, this study follows largely Cirillo and Gallegati (2012) and Bianchi et al. (2008). It is assumed that migrants maximize a utility function that is implicit in the behavioral rules introduced in section 19.5, rather than explicitly stated. They use heuristics to cope with the high level of uncertainty they face in terms of future earnings, others’ migration behavior, and future levels of border control. In every period t , agent i ’s payoff depends on the vector of players’ actions in that particular period and on the current (payoff-relevant) state of the system only (Maskin and Tirole 2001). Behavioral motives for migration and return are chosen from the literature as candidates to be included as behavioral heuristics in the model, similar to those in Rehm (2012). However, instead of systematically varying the behavioral parameters to calibrate the model so that it generates reasonable outputs as in Rehm’s model, the behavioral parameters here were directly estimated from microdata wherever possible. Comparable models in this respect are da Fonseca Feitosa (2010), Kniveton et al. (2011), and Entwisle et al. (2008). Sometimes, as in the case of the effect of the network, there were clear endogeneity problems, so these four parameters were calibrated later to match the stylized facts. Next, the model was built in NetLogo (Wilensky 1999), all parameters

were set to fixed, empirically determined values, and the four remaining free parameters were set to reasonable values. After verifying that the model roughly matches most of the stylized facts for most of the settings of the free parameters, those were calibrated performing simple grid searches in the parameter space. The resulting match of model output and empirical data was considered satisfactory, given that this is a much simpler model than, for instance, Rehm (2012), and given that it has only four degrees of freedom. Finally, robustness checks were performed and it was demonstrated how the model can be used for policy analysis. The model code, all data files needed for running the model, the MATLAB code for estimating the properties of the power law distribution, and a full description are available on the “Open ABM platform” at <http://www.openabm.org/model/3893/version/3/view>.

19.2.2 Data

For estimating the behavioral rules and for setting most of the other model parameters, the MMP128 version of the Mexican Migration Project was used. It is a large event-history microsurvey data set of Mexican migrants and non-migrants from 128 different Mexican communities. Respondents were interviewed once in waves, starting in 1982 and ending (in the version used) in 2008. Both heads of households and spouses were asked to indicate their migration history (time spent in the United States) and labor market experience (employed or not and type of job) as well as family events for every year since they were born. Additional information is available for the time of the interview and for the first and last migration, such as whether it was a legal migration, the type of visa used, income, wealth, and health status. The full sample comprises 1,004,825 person-year observations. The simulation was run with 2,860 agents, the number of heads of household in the MMP128 data set born between the years 1955 and 1965 who—if they migrated—went to California and who were interviewed (or had lived, in the case of migrants) in the central and western Mexican states of Sinaloa, Durango, Zacatecas, Nayarit, Jalisco, Aguascalientes, Guanajuato, Colima, and Michoacán. Those states together form an area approximately the size of California and at the same time they comprise the most important states for out-migration. All population distribution measures refer to this subset of the data. The model therefore simulates migration behavior of one generation from one region over the course of thirty-three years.

19.3 STYLIZED FACTS ABOUT CIRCULAR MIGRATION

From the literature and the MMP128, three stylized facts about circular migration can be derived that the model should match. If it succeeds in re-creating these prominent

characteristics of circular migration behavior, it is a plausible candidate for the true data-generating process.

19.3.1 The Distribution of Migrants Across Cities is Heavy-Tailed

In order to calibrate the model to the empirical distribution and to have a means to validate the model, the distribution of migrants across cities is determined.

One sees in the complete MMP₁₂₈ sample that the distribution across cities is very similar to that of the western Mexico–California subsample. Therefore, both the subsample and the full sample are used in order to avoid bias in the estimates due to small sample size. The bulk of migration originates in a few places, and it concentrates on a fairly small number of places in the country of destination. In the case of Mexican migration to the United States, the communities with the highest percentage of adults with migrant experience are in the states of Guanajuato, Durango, Jalisco, and Michoacán (MMP₁₂₈). The percentage varies from just above 1 percent to almost fifty percent across communities. Of the migrating heads of household surveyed in the MMP₁₂₈, 20 percent went to the Los Angeles area on their last trip; this was by far the highest number, followed by the Chicago region (8 percent) and the San Diego region (5 percent).

Distributions that result from social interaction often follow a power law; that is,

$$Pr[X \geq x] \sim cx^{-\gamma}. \quad (19.1)$$

Examples are Axtell (2001) for the distribution of size of cities, Redner (1998) for the distribution of scientific citations, and Liljeros et al. (2001) for the distribution of number of sexual partners. One of the generative mechanisms of power law distributions is preferential attachment: cases in which it is more likely for a new node in a network to attach to a node that already has many links to other nodes, rather than to a random node. Mitzenmacher (2004) provides an intuitive example that can explain the often-found power-law distribution of links to a Web site: If a new Web site appears and it attaches to other sites not completely at random, but, rather, links to a random site with probability $\alpha < 1$, and with probability $1 - \alpha$ it links to a page chosen in proportion to the number of links that already point to that site, then it can be shown that the resulting distribution of links to and from a Web site approaches a power law in the steady state. See Mitzenmacher (2004) for a simple derivation and Cooper and Frieze (2003) for a more general proof. There is a small number of cities that attract a very large proportion of migrants, and many cities attract only one migrant. Furthermore, the typical formation of migrant networks—joining friends and family in the host country—suggests a case of preferential attachment. Therefore, a power law is the first distributional candidate to check. The methodology is taken from Goldstein et al. (2004) and Clauset et al. (2007). The MATLAB routines that were provided by that latter,

include estimating a minimum value for x above which the power law applies. First, it is assumed that the distribution of migrants across cities does indeed follow a power law, and its parameters γ , the power law exponent, and x_{min} , the value above which the power law applies, are estimated. Then, it is checked for whether synthetic power-law distributions with the same exponent and the empirical distribution are likely to belong to the same distribution. The most commonly used power-law distribution for discrete data is the discrete Pareto distribution. It takes the form

$$p(x) = \frac{x^{-\gamma}}{\zeta(\gamma, x_{min})} \quad (19.2)$$

where x is a positive integer measuring, in this case, number of migrants in a city, $p(x)$ is the probability of observing the value x , γ is the power law exponent, $\zeta(\gamma, x_{min})$ is the Hurwitz or generalized zeta function defined as $\sum_{n=0}^{\infty} (n + x_{min})^{-\gamma}$, and x_{min} is a minimum value for k above which the power law applies. The maximum likelihood estimator is derived by finding the zero of the derivative of the log-likelihood function, which comes down to solving

$$\frac{\zeta'(\hat{\gamma}, x_{min})}{\zeta(\hat{\gamma}, x_{min})} = -\frac{1}{n} \sum_{i=1}^n \ln(x_i) \quad (19.3)$$

numerically for $\hat{\gamma}$, with x_i as the number of migrants in city i and n as the total number of cities in the sample; see Goldstein et al. (2004) for the derivation. Usually, if empirical data follow a power-law distribution at all, they do so only for values larger than some minimum value (Clauset et al. 2007). This value should be estimated in order to not bias the estimated $\hat{\gamma}$ by fitting a power law to data that are not actually power-law distributed. In accordance with Clauset et al. (2007), this x_{min} is chosen so that the Kolmogorov-Smirnov (KS) statistic, which measures the maximum distance between two cumulative distribution functions (CDFs), is minimized. The KS statistic is

$$D = \max_{x \geq x_{min}} |S(x) - P(x)| \quad (19.4)$$

where $S(x)$ is the CDF of the empirical observations with a value of at least x_{min} and $P(x)$ is the CDF of the estimated power-law distribution that best fits the data for the region $x \geq x_{min}$. This yields a minimum value for x of 34 and a scale parameter $\hat{\gamma}$ of 1.9. Although visually the power law seems to be a good fit (not shown), it is checked to see whether the distribution might actually follow a power law above $x_{min} = 34$. In order to do this, the KS statistic is computed for the distance between the empirical CDF and the best-fit power law. Then, a large number of artificial data sets distributed according to the power law with $\gamma = 1.9$ and $x_{min} = 34$ is created, a power-law model is fitted to each artificial data set again, and the KS statistic (the distance from that data set to its own power-law model) is computed. Then the proportion p of artificial data sets is determined in which the KS statistic is larger than the one from the empirical distribution. If the proportion p is such that $p < .1$, a power law can be ruled out because extremely rarely the artificial data sets are a worse fit to a power-law distribution than

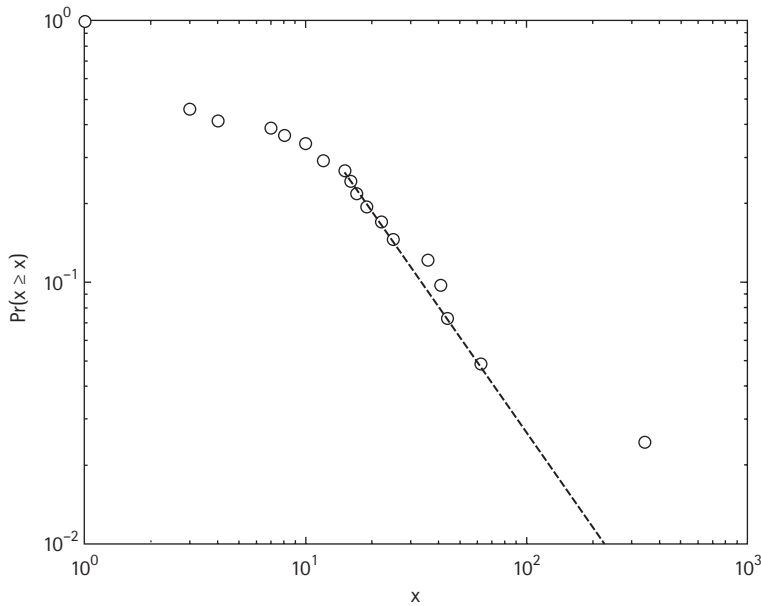


FIGURE 19.1 Log-log plot of the cumulative distribution function of numbers of migrants per city in the small subsample and fitted values using MLE, with $\gamma = 2.2$.

are the observed data. In the present case, however, $p = .4250$, so a power law seems a reasonable description of the data.

The same procedure is followed for the smaller subsample that is used as a basis for the simulation. The results indicate that even for the small subsample the distribution might follow a power law for values larger than 15 with $\gamma = 2.2$ (see figure 19.1).

The p -value of .845 is even higher for the smaller sample, indicating that the artificial distributions are, on average, a worse fit to a power law than the empirical one. This result has to be taken with caution, however, owing to the small sample size.

For comparing the empirical to the simulated distribution at the end of the calibration procedure, the mean, standard deviation, and median of the two distributions are compared. Whether the simulated distribution resembles a power law, both for single runs and for the overall distribution after ten thousand model runs, will be checked (see section 19.7.2).

19.3.2 Tendency of People from one Neighborhood to Migrate to the Same Place

People tend to settle where people from the same region of origin have settled previously; for example, 65 percent of the migrant heads of household surveyed in the community with the highest percentage of migrants, a village in Michoacán, went to the

Chicago region. Patterns in most other communities are very similar. For additional evidence see Munshi (2003) and Bauer et al. (2007). The reasons for this are positive network externalities.

19.3.3 Migration-Specific Capital

Several studies reveal the importance of migration-specific capital, that is, experience and knowledge that facilitate every subsequent move. This capital is closely related to migrant networks as well: with every move, migrants build up new links that facilitate job searches, (re)integration, and information flow (DaVanzo 1981). Therefore, once a move has taken place, migrants are more prone to move (again) than they were before their first move (Constant and Zimmermann 2011).

Because some of the individuals in the subsample were interviewed before the last year considered (2007) and therefore their migration histories are not complete, the full sample is used for measuring the distribution of number of trips. The total number of trips is measured at age forty-seven, which corresponds to the last year in the lives of the simulated agents. The distribution of number of trips displays overdispersion (mean = .964, standard deviation = 2.785) and “excess zeros” as compared to a Poisson distribution. The observed distribution fairly closely resembles a negative binomial (see figure 19.2). In fact, the null hypothesis that it is equal to a negative binomial one could not be rejected in a Kolmogorov-Smirnov test ($p = .12$).

The overdispersion and “excess zeros” could be due to either the heterogeneity of individuals or to the existence of two different data generating-mechanisms creating zero and nonzero counts of trips (Greene 2003, 744–752). Both explanations would be in line with the argument by DaVanzo (1981) and Constant and Zimmermann (2011): Migrants could have characteristics that distinguish them from nonmigrants, so the degree of heterogeneity between people who do not migrate at all (number of trips = 0) and people who make one trip would be much larger than that between migrants who make one trip and those who make two trips. Alternatively, the conditions for making the first trip are much different from those for subsequent trips owing to the above-mentioned build-up of migration-specific capital. Therefore, the mechanism “generating” 0 moves is different from the one generating a positive number of trips.

The model developed here is useful if it succeeds in re-creating these three stylized facts.

19.4 SELECTION OF BEHAVIORAL MOTIVES

Several behavioral motives can be found in the literature that might influence the migration or return decision. We determine which ones to include in the model by

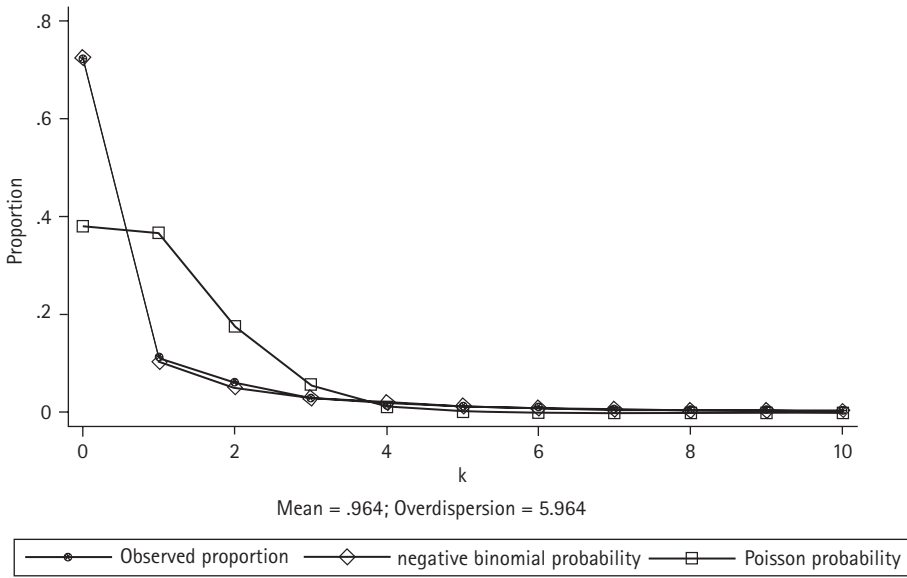


FIGURE 19.2 Observed distribution of number of trips compared to a Poisson and a negative binomial distribution.

running logit and probit regressions on the MMP128 data set for the probability of migrating and returning in a person-year. The full sample of individuals for the years 1970–2008 was used, thereby implicitly assuming that they are not systematically different from the western Mexican subsample that is used for the simulation. All of the hypotheses mentioned below are included in the regressions, as well as controls for family status, community of origin, profession, and current job. All results are displayed in table 19.1.

Hypothesis 1: Expected Earnings

The first hypothesis is as follows: Migrants are attracted by a higher expected income in the host country than in the home country, taking into account the unemployment rate (Harris and Todaro 1970). The higher the expected income as compared to the current income, the more likely someone is to migrate.

It is not a straightforward matter to find the effect of the difference between expected earnings and current earnings on the migration and return decision with the data available from the MMP128, for two reasons: First, the data do not contain information about earnings for every person-year, but only about the year of the survey and of the first and last migration. Second, it is unclear how to compute expected earnings without knowing how those expectations are formed. In section 19.5 it is suggested that they

Table 19.1 Probability of moving, 1970–2007. County and occupation dummies were used. Robust standard errors are in parentheses; * significant at least at the 1% level, **significant at least at the 5% level, * significant at least at the 10% level**

Variables	Probit Probability for trip	Probit Probability for return if in U.S.	Logit Probability for return if in U.S.
Coefficients			
Sex (female = 1)	-.265*** (.033)	-.313*** (.064)	-.550*** (.113)
Age 18 to 30 (reference: < 18)	.135*** (.020)	.248*** (.051)	.464*** (.087)
Age 31 to 45	-.173*** (.023)	.258*** (.055)	.491*** (.094)
Age 46 to 60	-.509*** (.027)	.121* (.064)	.225** (.110)
> 60	-.998*** (.047)	.168* (.095)	.231 (.163)
No. family of origin ever migrated	.076*** (.003)	-.025*** (.004)	-.040*** (.008)
Married	-.033*** (.012)	.087*** (.022)	.158*** (.038)
Number of children	-.00001 (.00005)	.017*** (.004)	.031*** (.007)
Property index categories (reference = 0)			
Property index = 1	-.049*** (.012)		
Property index = 2	-.109*** (.019)		
Property index = 3	-.171*** (.024)		
Property index = 4	-.106*** (.028)		
Green card		-.00004** (.00002)	-.00006** (0.00003)
Documentation used for trip (ref. "unknown")			
Legal resident		.342** (.172)	.537* (.303)
Contract: Bracero		.944*** (.352)	1.680*** (.653)
Contract: H2A (agricultural)		.595*** (.204)	.982*** (.358)
Temporary: Worker		.471** (.210)	.765** (.362)
Temporary: Tourist		.739*** (.177)	1.235*** (.311)

Table 19.1 Continued

Variables	Probit Probability for trip	Probit Probability for return if in U.S.	Logit Probability for return if in U.S.
	Coefficients		
Citizen		.366* (.207)	.473 (.368)
Undocumented or false documents		.680*** (.171)	1.101*** (.302)
Years since last trip	-.058*** (.002)	-.077*** (.003)	-.151*** (.006)
Number of previous trips	.084*** (.002)		
Property index larger in $t + 1$ (1 = yes)		-.599*** (.039)	-1.093*** (.069)
Last U.S. wage Purchasing Power Parity (PPP) (thou.)		-.0009*** (.00005)	-.002*** (.00008)
Last U.S. wage Purchasing Power Parity (PPP) (thou.) *		.0009*** (.00009)	.0002*** (.00002)
Property index larger in $t + 1$			
Exp. ann. wage-diff.	.033*** (.001)	-.005 (.003)	-.013** (.006)
US-Mexico (thou. USD)			
Before first migration? (yes = 1)	-.875*** (.016)		
Constant	-2.907*** (.093)	-.712** (.317)	-1.098* (.582)
Number of observations	510578	32709	32709
Pseudo R^2	0.426	0.292	0.296

are formed by averaging over network neighbors' earnings in the host country. As a proxy for expected wage, we use the difference between real GDP per capita in Mexico, and United States GDP per capita multiplied by the United States employment rate. The coefficient of the expected annual wage difference between Mexico and the United States is positive and highly significant for the probability of making a trip. I checked whether the marginal effect of the wage difference on the probability of going on a trip differs by whether someone is a potential first-time migrant or has gone on at least one migration before, which is the case. The increase in probability of migrating in a person-year per thousand USD expected wage difference is .0011 for those who have never migrated before and .0035 for those who have.

The effect of the expected wage difference on the return decision should be opposite: The higher the expected wage difference, the lower the probability of return. Indeed,

the coefficient for expected wage difference is negative, but only significant in the logit model and not in the probit model (see table 19.1). Therefore, it is not included as a behavioral parameter for the return decision.

Hypothesis 2: Number of Previous Migrants

Workers with a network are both less likely to be unemployed and to have higher wages (Munshi 2003). Therefore, migrants tend to go where they know somebody, as shown by Lindstrom and Lauster (2001), Flores-Yeffal and Aysa-Lastra (2011), and Massey and Aysa-Lastra (2011). Previous migrants have an incentive to help the newly arrived find jobs because this increases the flow of information and trade among migrants, as argued by Stark and Bloom (1985). The help of others decreases assimilation costs for new migrants, as shown for Mexican migrants by Massey and Riosmena (2010). Previous migrants influence potential migrants' decisions through the policy channel as well: Immigration policy often includes a family reunification element that permits family members of migrants to immigrate as well. However, Beine et al. (2011) estimate the relative importance of the different channels for immigrants to the United States in a recent paper and find that the immigration policy channel is much less important than the assimilation cost channel and has decreased in importance since the 1980s.

In sum, the more previous migrants somebody knows, the more likely he or she is to migrate.

This seems to be true for the sample here as well; the coefficient for the number of family members in the United States is highly significant (table 19.1). The influence of the number of previous migrants on the migration decision is calibrated in section 19.7.

Hypothesis 3: Home Preference

Migrants are often assumed to have a preference for consuming home amenities (a home bias, as in Faini and Venturini 2008 and Hill 1987). Everything else held constant, utility is always higher if he or she is at home. The hypothesis is therefore: The stronger someone's home preference, the less likely he or she is to migrate.

Assuming that people are heterogeneous in their home preference, each individual is assigned an idiosyncratic home preference parameter. Property ownership in Mexico before first migration is used as a proxy because people who consider it likely that they will spend their life in the home country are more likely to invest in property there than in the host country. Logit and probit regressions of the probability of ever migrating on property ownership, individual controls, and community fixed effects before first migration (table 19.2) show that property ownership before first migration is significantly negatively correlated with becoming a migrant. This confirms the findings by Massey and Espinosa (1997).

Table 19.2 Probability of ever becoming a migrant. County and occupation dummies were used. Robust standard errors are in parentheses

	Logit (1) Property categories	Logit (2) Continuous property index	Probit (3) Continuous property index	Logit (4) Discrete property index
Variables	Coefficients			
Sex (female = 1)	−1.007*** (.028)	−1.015*** (.027)	−.519*** (.014)	−1.001*** (.027)
Family members ever migrated	.316*** (.005)	.315*** (.005)	.178*** (.003)	.319*** (.005)
Hectares owned	−.003*** (.001)		—	—
Pieces of land owned	−.365*** (.019)	—	—	—
Pieces of property owned	−.870*** (.012)	—	—	—
Number of businesses owned	−.528*** (.024)	—	—	—
Property index	—	−.599*** (.008)	−.304*** (.004)	—
Property index (ref. = 0)	—	—	—	—
Property index = 1	—	—	—	−.876*** (.012)
Property index = 2	—	—	—	−1.390*** (.023)
Property index = 3	—	—	—	−1.67*** (.033)
Property index > 3	—	—	—	−1.620*** (.042)
Constant	−.276*** (.065)	−.326*** (.064)	−.250*** (.036)	−.286*** (.065)
Number of observations	452675	452675	452675	452675
Pseudo- R^2	0.221	0.219	0.216	0.220

An index was created from hectares, properties, and businesses owned. The number of hectares owned is transformed to a logscale, then the values from the categories are added. The coefficient of the property index is also negative and significant.

The relative frequencies of the property index in the central and western Mexico subsample are used as relative frequencies for the home preference parameter h_i . The analysis is confined to values for the property index from 0 to 4, because the proportion of individuals with a property index larger than 4 is less than 1 percent. Of the subsample, 57.85 percent of individuals have a property index of 0, 29.96 percent

Table 19.3 Average probability of migrating in a person-year at different levels of the property index. Predictive margins were obtained via the delta method. Standard errors are in parentheses. For the simulation it is assumed that the probability of migration decreases by .003 for people with home preference = 1, by .005 home preference = 2, by .007 if it = 3, and by .003 if it = 4

Property index	Average probability	<i>z</i>	<i>P</i> > <i>z</i>
0	.034 (.0004)	79.70	0.00
1	.031 (.0005)	68.06	0.00
2	.029 (.0008)	34.50	0.00
3	.027 (.0011)	23.63	0.00
4	.031 (.0015)	20.12	0.00

have a value of 1, 7.83 percent have a value of 2, 2.57 percent have a value of 3, and 1.09 percent have a value of 4.

The probability of migrating in a person-year negatively depends on the property index, as can be seen in table 19.1. The average probability of migrating in a person-year was subsequently computed at every level of the property index (see table 19.3).

It is interesting that a property index of 4 increases the probability as compared to a property index of 3.

Hypothesis 4: Ties to Home

Constant and Zimmermann (2003) find that family is a driving force of repeat migration. Ties to the home country can be understood as relationship capital. It is helpful for the migrant's reintegration into the home community on return. However, the longer a migrant is away from the home country, the stronger might be the depreciation of that form of capital because of physical distance. This phenomenon is studied analytically by McCann et al. (2010) and found to be empirically relevant for the return decision by de Haas and Fokkema (2011). This yields the following hypothesis: The more family and friends someone has at home, and the stronger the links are with them, the more likely someone is to return.

The decrease in likelihood of returning to the home country (for people in the host country) or of migrating again (for people in the home country) is measured for

migrants with at least one trip to the host country, taking time since last migration move as an explanatory variable. This illustrates the diminishing importance of ties across physical distance over time. A probit regression of the likelihood of making a move in a year on the number of years since the last move yields a negative coefficient that is significant at the 1 percent level for both the migration decision and the return decision (see table 19.1). The links connecting physically distant network neighbors are, therefore, assumed to become weaker in each period by an amount a . The probability of making a move in a person-year (migration or return) starts out at 3.3 percent when the last trip took place in the previous year. It decreases on average by 1.9 percent with each additional year that has passed since the last move. After thirty-two years without a trip the probability is 1.8 percent. The relationship capital associated with links between physically distant neighbors is, therefore, assumed to decrease by 2 percent every year. The coefficient of the size of the effect of relationship capital in the home country on the probability to return home is calibrated in section 19.7.1.

Hypothesis 5: Purchasing Power

Migrants's savings have a higher purchasing power in their home country than in the host country, as modeled by Dustmann (2001). This might be a motive to return. Lindstrom (1996) follows a similar argument: He tests whether Mexican migrants from areas that provide dynamic investment opportunities stay longer in the United States in order to accumulate more savings that they can put to productive use in their home country, and he finds some evidence in favor of his hypothesis. Reyes (2004) shows that devaluation of the peso relative to the dollar leads to more return migration, providing another piece of evidence in favor of the purchasing power motive. A related argument is brought forward by Berg (1961) and Hill (1987), who discuss the case in which migrants aim to achieve a level of lifetime income, and once that is achieved they return home because they have a preference for living there. Either argument yields the same conclusion: Holding everything else constant, the higher someone's savings are, the more likely he or she is to return.

Unfortunately, the MMP₁₂₈ does not provide information about savings. Therefore the supposed purchasing power effect is captured by including the last wage in the United States, multiplied by the exchange rate for that year and by the consumer price index from the Bank of Mexico, and a dummy that is 1 if property ownership was larger in $t + 1$ than in t in the return regression (see table 19.1). An interaction term of the dummy and the last wage is also included. The ownership dummy is significant and negative, which implies that people who lived in the United States in year t and bought property the same or the following year are *less* likely to have returned that year than are people who did not buy property. That somewhat contradicts the hypothesis and indicates that people seem to buy property in the United States rather than in Mexico. The coefficient for the last wage in the United States is negative and significant for return,

albeit the coefficient is extremely small in size. The interaction term has a positive and significant effect, in line with the hypothesis. This implies that if property ownership in $t + 1$ is larger than in t , the probability of return increases with the wage. The size of the coefficient, however, is very small as well. For that reason and because the proxy for the purchasing power motive is imperfect, it is not included in the model.

Hypothesis 6: Education

Education and heterogeneity in skill levels have been found to be important determinants of self-selection of migrants in a wide range of theoretical papers originating from Borjas (1987) and in empirical studies (Brücker and Trübswetter 2007).

The evidence in the literature about skill selection of Mexican migrants, however, is mixed: Borjas and Katz (2007), Fernández-Huertas Moraga (2011), and Ibarra and Lubotsky (2007) find that Mexican migrants to the United States are mostly from the lower tail of the Mexican earnings distribution. Other studies find that migrants tend to have a medium position in the country's skill distribution because returns to skill are higher in Mexico, making migration less attractive for highly skilled individuals, while low-skilled individuals are likely to be more credit-restrained and not able to afford the moving costs (Chiquiar 2005; Lacuesta 2006; Orrenius and Zavodny 2005). There is, furthermore, evidence that there is a self-selection process for migrants who move to a different region within Mexico (Michaelsen and Haisken-DeNew 2011) but not for international migrants between Mexico and the United States (Boucher et al. 2005).

In the simulation model it is difficult to take different levels of education and skills into account without significantly increasing the complexity of the problem. The fact that the individuals in the subsample are pre-dominantly low-skilled (81 percent of migrants born between 1955 and 1965 had completed nine years of schooling or less), in combination with the very mixed evidence in the literature, suggests that it does not seem to bias the results dramatically to leave out education and assume a uniform level of education across individuals. This path is chosen here.

Hypothesis 7: Age

All cohorts display a similar migration behavior during their life cycle (see figure 19.3).

Migration behavior starts about age eighteen, reaches a peak between the ages of twenty-five and thirty, and then decreases, with small peaks in both migration and return behavior, at about age seventy. Age might, therefore, have an effect on migration and return moves, independent of the other motives.

Age is significant in all regressions, except for the fifth age group in the return regression. All in all, the results confirm the inverted U-pattern shown in Figure 19.3.

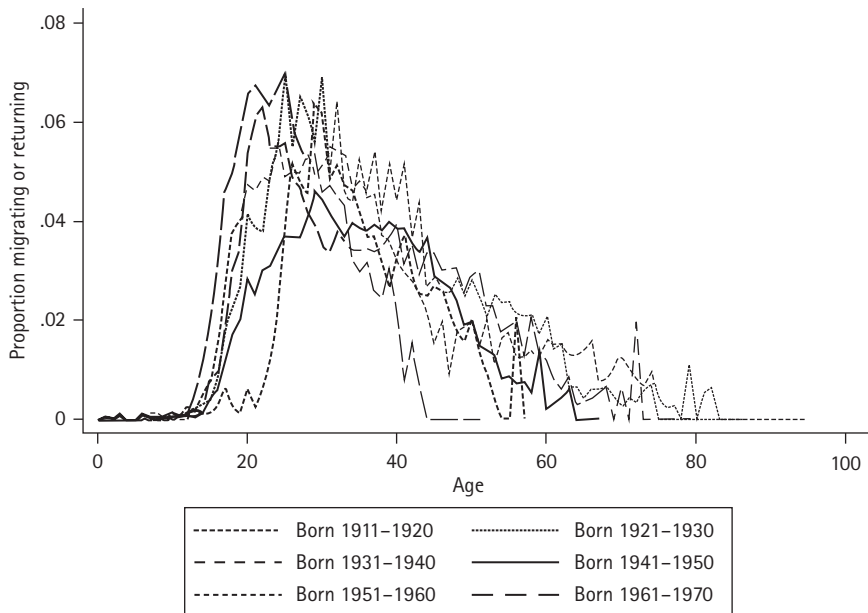


FIGURE 19.3 Proportion of MMP128 full sample who make a trip at a certain age, for different cohorts.

Considering marginal probabilities, the probability of migrating increases by .8 percentage points when entering the age-group of 18 to 30, then decreases by 1.1 percentage points between the ages of thirty-one and forty-five, and so on (see table 19.4).

The behavioral parameters that were included in the model are summarized in table 19.5.

19.5 THE MODEL

The model assumes two types of agents—workers and firms—which are spread out randomly on a grid. Workers are heterogeneous only in a home preference parameter (fixed throughout the simulation) and in a savings parameter (time-specific). There are two countries: one with high productivity of labor (the host country) and one with low productivity of labor (the home country). Workers can move, but firms cannot.

The model is initiated via a setup procedure. During setup, the following happens:

- The world with two countries separated by a wall is initialized.
- Workers are created. A number that is equal to the initial percentage of workers in the home country is assigned a random spot in the home country. The remainder is assigned a random spot in the host country.

Table 19.4 Marginal probabilities of migrating a trip at different ages. Predictive margins are obtained via the delta method. Standard errors are in parentheses

Age	Marginal probability of migrating if in Mexico	<i>z</i>	<i>P</i> > <i>z</i>	Marginal probability of returning if in the US	<i>z</i>	<i>P</i> > <i>z</i>
< 18	.031 (.001)	26.42	0.00	.304 (.017)	17.53	0.00
18–30	.039 (.001)	73.92	0.00	.367 (.005)	76.26	0.00
31–45	.028 (.0004)	69.70	0.00	.352 (.005)	68.27	0.00
46–60	.022 (.001)	28.57	0.00	.316 (.013)	23.75	0.00
> 60	.009 (.002)	3.96	0.00	.279 (.010)	2.82	0.01

Table 19.5 Overview of behavioral parameters used

Parameter	Description	Relevant for	Hypothesis no.	Direction of effect
$p_{1,i}$	Difference between expected and current earnings	Migration	1	+
p_2	Number of previous migrants in network	Migration	2	+
$p_{3,i}$	Home preference	Migration	3	—
q_1	Ties to home	Return	4	+
p_4, q_2	Age	Migration and return	7	mixed

- Workers receive their individual values for the home preference and the savings parameters.
- Workers in the home country create links with other workers in their Moore neighborhood, whereas workers in the host country create links with all other workers within a radius s .
- Firms are created in both home and host country and assigned a random spot and a random initial wage.

In every step of a model run the following happens:

- Workers form links to all other workers in their Moore neighborhood (home country) or within a small radius of size s (host country).

- Links between workers that are not immediate neighbors get weaker by amount a (relationship capital diminishing over time owing to physical distance). Through migration and renewed physical closeness, the relationship capital associated with those links can be replenished, as in McCann et al. (2010).
- All other variables are updated.
- Workers consume their earnings of the previous period minus savings determined by their individual savings rate. Savings are added to wealth.
- Workers without earnings consume a minimum consumption.
- Workers use the information about earnings of network neighbors in the host country to compute their expected earnings in the host country:

$$w_{exp,i,t} = \frac{1}{N} \sum_{n=1}^N w_{n,t} \quad (19.5)$$

where $n = 1, \dots, N$ are all the worker's network neighbors in the host country, measured at time t .

- Migration is a three-step procedure. First, workers in the home country compute whether their wealth is larger than the moving costs and whether their expected earnings in the host country are larger than their current earnings. If so, they next compute their individual probability of moving. The probability of worker i to migrate at time t is assumed to have the following functional form:

$$p_{i,t}(\text{migrate} | K_{i,t} > m_1, w_{exp,i,t} > w_{i,t}) = p_0 + p_{1,i}(w_{exp,i,t} - w_{i,t}) + p_2 N_i - p_{3,i} + p_{4,t} \quad (19.6)$$

where $K_{i,t}$ is the worker's wealth in time t , m_1 are the migration costs, p_0 is the baseline migration probability, $p_{1,i,t}$ is the behavioral parameter for the difference between expected and current earnings that depends on whether it is a first migration or not, p_2 is the behavioral parameter for the number of network neighbors in the host country (N_i), $p_{3,i}$ is the individual home preference parameter, and $p_{4,t}$ is the age parameter. They draw a random number $\in (0, 1)$. If this number is smaller than their individual probability, they migrate. Their wealth K decreases by the amount of moving costs m_1 . In the last step, the probability that somebody who is willing to migrate can do so is determined by the level of border control.

- Migrants become unemployed and decide where to go: If they have any network neighbors in the host country, they move to the network neighbor with the highest wage. If not, they move to a random spot in the host country.
- Unemployed workers in both the host and the home country move to the network neighbor in the same country who is employed and has the highest wage. If they do not have any network neighbors, they move one step in a random direction in search of employment (but never across the border).

- Firms hire unemployed workers who are on their patch. All workers receive the firm's current wage rate. In order to keep the model as simple as possible, firms are assumed to pay a fixed, uniform, idiosyncratic wage to all of their employees. At every time step the wage is adjusted exogenously to account for inflation.
- Analogous to the potential migrants, potential return migrants in the host country first determine whether their wealth is larger than the return costs and then decide to return according to their individual probability. The probability of worker i to return at time t given that his or her wealth K is larger than the return costs m_2 is thus assumed to have the following functional form:

$$q_{i,t}(\text{return}|K_{i,t} > m_2) = q_0 + \sum_{r=1}^R \frac{q_1}{a_{r,t}} + q_{2,t} \quad (19.7)$$

where q_0 is the baseline return probability, q_1 is the behavioral parameter for ties to the home country, $r = 1, \dots, R$ are the worker's network neighbors in the home country, $a_{r,t}$ is the age of a link, and $q_{2,t}$ is the age parameter.

- Return migrants' wealth decreases by the amount of return costs m_2 . They become unemployed and return to the spot in the home country they were assigned in the setup procedure.
- All measurements of model output take place.

The model is run for thirty-three time steps, with each step representing one year.

19.6 PARAMETERIZATION OF NONBEHAVIORAL PARAMETERS

All nonbehavioral parameters in the model were fixed to empirically determined values (summarized in table 19.6).

Parameters of the model were set to sample population parameters that were estimated using the MMP128, the Encuesta Nacional de Ingresos y Gastos de los Hogares (ENIGH) and the Encuesta Nacional de la Dinámica Demográfica (ENADID).

The number of firms in the home country is determined by dividing the number of workers initially in the home country (2,700) by the average firm size in Mexico, which, according to Laeven and Woodruff (2007), is 13.6 employees per firm. That yields 199 firms. For the host country, the number of firms is assumed to be 58, which is the number of counties in California. In this way, the distribution of migrants across cities can be measured conveniently (see section 19.7.2). Values reported in pesos are converted to USD using the annual average of the official exchange rate for the year the data were measured (reported by the World Bank). In order to obtain moving costs, an average of legal and illegal crossings weighted according to the proportions of legal and illegal crossings in the MMP128 data set was computed. Return costs are assumed

Table 19.6 Fixed parameters that were derived from data and used for all simulation runs

Parameter	Value used for simulation	Source
City size	6.2	Average county size in California (from National Association of Counties)
Number of people	2,860	Number of individuals interviewed in the MMP128 survey born between 1955 and 1965 and living in central and western Mexico
Initial percentage at home	94.4	Proportion of people from the subset of the MMP that was in Mexico in the year 1975
Border control	Annual border enforcement budget normalized to [0, 1]	U.S. Immigration and Naturalization Service (until 1998), Homeland Security Digital Library (after 1998), through MMP128 supplementary files
Saving rate	Skew-normal distribution with $\xi = 0.616$, $\omega = 0.721$, and $\alpha = -7.5$	Encuesta Nacional de Ingresos y Gastos de los Hogares
Moving costs	1,110.26	MMP128, Instituto Nacional de Estadística y Geografía
Return costs	1,715.65	U.S. Bureau of Labor Statistics, MMP128
Wage in home country	GNI p.c. PPP	World Bank
Wage in host country	Annual average wage of production and nonsupervisory employees on private nonfarm payrolls	Bureau of Labor Statistics
α : Decrease in relationship capital	2% every period	MMP128

to be travel costs plus loss of one month's American wages, which are determined by a weighted average of illegal immigrants' and legal workers' wages. For details, please refer to <http://www.openabm.org/model/3893/version/3/view>.

Firms' wages are determined in the following way: In the setup procedure firms are assigned an idiosyncratic productivity parameter $\alpha \sim \mathcal{N}(0, \sigma^2)$ for the host country. The standard deviation $\sigma = .28$ is the standard deviation of the average wage in a county as a percentage of the overall average per capita personal income in California in 2007 from the U.S. Census Bureau. For Mexico, the standard deviation of wages across states for the usual western and central states in 2001 from Chiquiar (2005) was used, which is 22 percent of the overall average wage. Accordingly, for each period, a firm's wage is

set in the following way:

$$w_{j,t} = \bar{w}_t + \bar{w}_t \alpha_j, \quad (19.8)$$

where $\alpha \sim \mathcal{N}(0, \sigma^2)$ and $\bar{w}_t$ is the time-specific average wage for the country. For this value, the time series are updated in each step of the model run. For the United States, data from 1975 to 2007 are taken from the average hours and earnings of production and nonsupervisory employees on private nonfarm payrolls by major industry sector data set from the Bureau of Labor Statistics. For Mexico, Gross National Income per capita in purchasing power parity, 1975–2007, from the World Bank was used because wage data for the subsample are not available for all years.

For minimum consumption in the United States, the average annual expenditure on food and housing of a household in the lowest income quintile of the population in 2010 from the Consumer Expenditure Survey was used, which is US \$17,290.81 (in 2002 prices). For Mexico, the average annual overall expenditure of a household at the bottom income decile in 2006 from the ENIGH, which is US \$4,819 (in 2002 prices) was chosen. For both cases the percentage of average income that this value constitutes is calculated for the respective year in which it was measured. Assuming that the relation between minimum consumption and average income remains constant over time, the minimum consumption is updated by multiplying the average wage each year by .3 for the home country and by .7 for the host country.

To determine the savings parameter, data from the 2008 ENIGH were used. The data set was restricted to 2,860 random observations from western Mexico, thereby assuming that the sample surveyed for the ENIGH is not different in relevant ways from the one surveyed for the MMP128.

Since only 17 percent of respondents make any deposits in saving and other accounts, the difference between current income and current expenditure is used as the measure for savings. The distribution of the saving rate in the population is approximately skew normal with parameters $\xi = 0.616$, $\omega = 0.721$, and $\alpha = -7.5$. This distribution is used for the simulation, drawing a savings rate for each worker in each period from this distribution.

A set of correlated border enforcement indicators were checked using a principal components analysis (line watch hours, probability of apprehension, visa accessibility, real border enforcement budget, and number of border patrol officers) for principal components in order to find a good proxy for the threshold of border control b , which is the probability of actually being successful when wanting to migrate. Three factors account for 87 percent of the variance. The border enforcement budget contributes the most to the first factor, which in turn accounts for 54 percent of the overall variance. The unique variance of the border enforcement budget is one of the lowest as well. Therefore, that variable is chosen as a proxy for border control. The annual values from 1975 to 2007 are normalized to $[0, 1]$ so that the probability that an agent who wants to migrate can do so is inversely proportional to the level of border enforcement in the respective year. Of course, there is a clear endogeneity problem here: if the level of border

enforcement is low, a lot of people will decide to try their luck and migrate. That might increase border protection, which in turn influences whether migrants choose to try to cross the border. For this reason, the way this is modeled here—migration decision and independent random draw whether migration is permitted—is not realistic. Therefore, a baseline probability of migrating is estimated within the final calibration procedure (section 19.7.1) with the border enforcement in place as it is.

19.7 CALIBRATION AND MATCH

19.7.1 Determination of Remaining Parameters

The first remaining parameter to be calibrated via simulation is the baseline probability of moving in any given year. This cannot be obtained from the data because the data set does not contain information about failed migration attempts of people who end up not migrating at all (only of those who, after failed attempts, finally succeed). The baseline return probability is also calibrated via simulation, as well as the two network-related parameters p_2 and q_1 . In order to find the best values for the remaining free parameters, 27,951 combinations of parameter were run; that is, every parameter was set to values between 0 and 1 (for p_0 , q_0 , and p_2) or between 0 and 2 (for q_1), in steps of .1. Using a simple grid search, the parameter combination is determined that is closest to fulfilling three criteria: causing an emergence of the mean, standard deviation, and median of the distribution of migrants across cities, causing the emergence of the negative binomial distribution of number of trips of migrants, and yielding a similar time series of flows of migrants and return migrants. For each of the three criteria, a distance function was minimized. For the flows of migrants, the function was

$$f_1 = \left(\sum_{t=1}^{t=4} (m_{t,emp} - m_{t,sim})^2 + (r_{t,emp} - r_{t,sim})^2 \right) : 4 \quad (19.9)$$

where $m_{t,emp}$ is the empirical number of migrants at time t , $m_{t,sim}$ is the number of migrants in the simulation, $r_{t,emp}$ is the number of empirical return migrants at time t , and $r_{t,sim}$ is the simulated number of return migrants. The four points t_1 up to t_4 represent the first, the twelfth, the fourteenth, and the thirty-second years of the simulation. For the distribution of migrants across cities, the distance function to be minimized was

$$f_2 = (\bar{x}_{emp} - \bar{x}_{sim})^2 + \left(\sqrt{\frac{1}{n_{emp}} \sum_{i=1}^{n_{emp}} (x_{i,emp} - \bar{x}_{emp})^2} - \sqrt{\frac{1}{n_{sim}} \sum_{i=1}^{n_{sim}} (x_{i,sim} - \bar{x}_{sim})^2} \right)^2 + (\tilde{x}_{emp} - \tilde{x}_{sim})^2 \quad (19.10)$$

where $\bar{x}_{emp}$ is the empirical average number of migrants in a city, $\bar{x}_{sim}$ is the simulated equivalent, n_{emp} is the total number of cities in the data, n_{sim} is the simulated equivalent, $x_{i,emp}$ is the number of empirical migrants in city i , $x_{i,sim}$ is the simulated equivalent, and $\tilde{x}_{emp}$ and $\tilde{x}_{sim}$ are the empirical and simulated median values. For the distribution of numbers of trips, a distance function very similar to the one above was minimized, however, this time without using the median. In a next step, the parameter combinations which were among the top decile of matches for all objective functions were selected. This was the case for two parameter combinations. The search was refined around those values in steps of .01, then the above procedure was repeated. The overall best match turned out to be $p_0 = .1$, $p_2 = .2$, $q_0 = .38$, and $q_1 = .12$ (details of derivation and sensitivity analysis are available from the author).

Subsequently ten thousand simulations were run with the best parameter combination found, using different random seeds each time, to see how much the resulting distributions and time series differed from one another and from the empirical ones. All of the following is based on these ten thousand runs with the combination listed above.

19.7.2 Stylized Facts Revisited: The Distribution of Migrants Across Cities

The mean, standard deviation, and median of the distributions of survey respondents' last U.S. trip and of the last trip of the same number of computer agents were directly compared and checked to see whether the power law hypothesis can be rejected for the simulated data.

To determine the simulated distribution, all patches on the left-hand side of the grid containing at least one worker were brought in a random order. Then, in a radius of city size s , the number of workers who chose this radius as the destination for their final migration move was counted. I moved on to the next random patch until all workers who migrated at some point were counted. Finally, the distribution of number of migrants per radius of city size s was determined.

Some of the individual runs were extremely close to the empirically observed mean and standard deviation (e.g., mean = 17.6, standard deviation = 54.9, and median = 2 compared to mean = 17.5, standard deviation = 54, and median = 1 for the empirical observation). As with the empirical distribution of migrants across cities from the small sample, most of the simulated ones also seemed to follow a power law (see figure 19.4 and figure 19.1 for comparison).

As for the empirical distributions, however, one has to be cautious because of the small sample size. Furthermore, the overall distribution after ten thousand model runs had a mean of 27.2, standard deviation of 60.1, and median of 4, which are slightly too high.

The facts that not all simulated distributions follow a power law and that often the median is too high is because there are, on average, more medium-sized cities in the

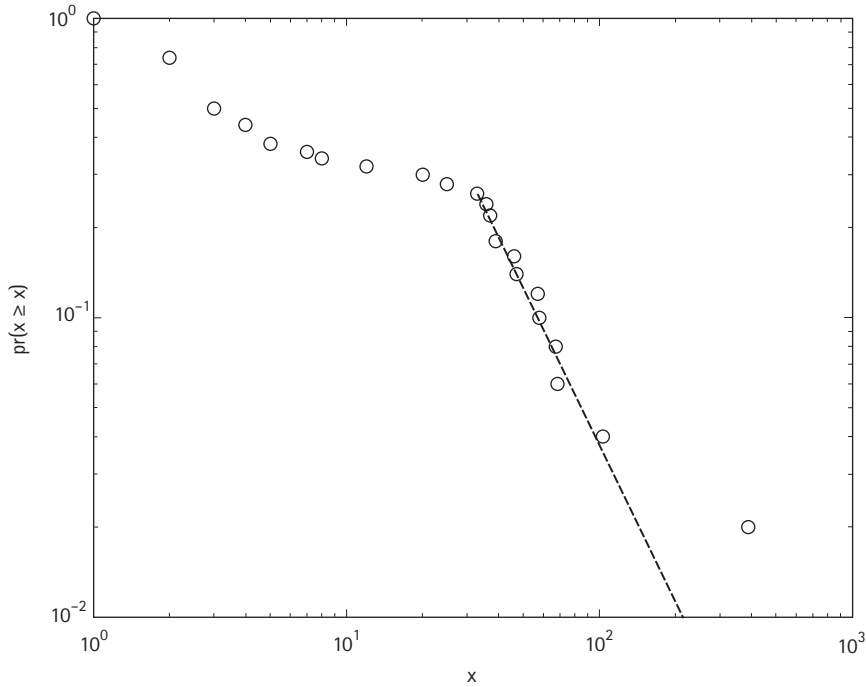


FIGURE 19.4 Example of a log-log plot of the cumulative distribution function of numbers of migrants per radius of city size in the simulation with best parameter settings, and fitted values using MLE, with $\gamma = 2.7$ and $x_{min} = 33$. In a Kolmogorov-Smirnov test, $p = 0.68$, so the power-law hypothesis is not rejected.

simulation than in reality. The simulated distribution is not as skewed as the empirical one. This is probably because, for reasons of simplicity, the model does not take into account that some cities attract many more migrants than others, not just because of network effects but simply because they are larger and provide better job and other opportunities. Bauer et al. (2007) find that the probability that migrants choose a particular U.S. location increases with the total population in that location for almost all groups of migrants. Future versions of should take this model this factor into account.

19.7.3 Stylized Facts Revisited: Migration-Specific Capital

The distribution of the number of trips in the sample was negative binomial. The simulated distribution is not exactly negative binomial because even-numbered counts of trips are much more frequent than odd-numbered ones in the simulation, but not in reality. That is to say, moving to the host country and moving back at some point is more frequent in the simulation than in reality. That might be because survey respondents have more degrees of heterogeneity than do computer agents: The people

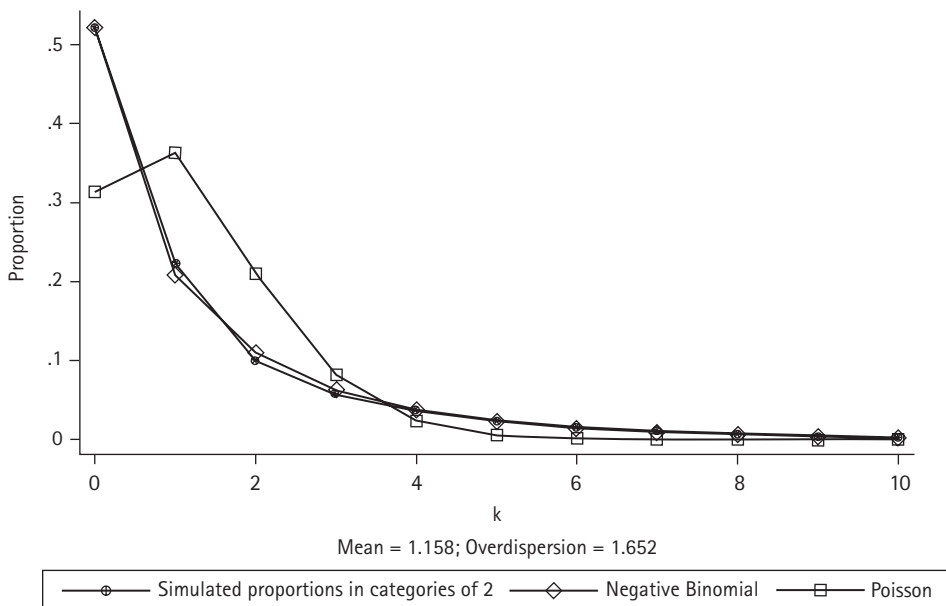


FIGURE 19.5 Smoothed distribution of number of trips after ten thousand model runs compared to a Poisson and a negative binomial distribution.

who stay in the United States are different from the ones who return, with respect to a set of characteristics that were not considered here. Furthermore, in reality, some of the migrants have family in the United States and others do not; this factor might fundamentally alter the psychic costs of separation (Lindstrom 1996). Therefore, their behavioral rules might also differ. In the simulation, everyone makes the same type of decision, albeit with different idiosyncratic parameter values such as the home bias $p_{3,i}$. To correct this inaccuracy of the model in a satisfactory way will be a subject of further research. When smoothing the distribution of number of trips by forming categories of two values each to correct for this inaccuracy (0, 1 – 2, 3 – 4, etc.), the distribution is very close to being negative binomial (see figure 19.5).

19.7.4 Match of Empirical and Simulated Time Series

The observed empirical time series of migration and return are depicted in figure 19.6.

In order not to overcalibrate the model, the mean squared error between simulated and empirical data was minimized in four points only. The focus was on matching the overall pattern: an inverted U-shape. The results of the ten thousand Monte Carlo runs with the best parameter setting $p_0 = .1$, $p_2 = .2$, $q_0 = .38$, and $q_1 = .12$ are depicted in figures 19.7 and 19.8. The curves that indicate mean, standard deviation, and quantiles

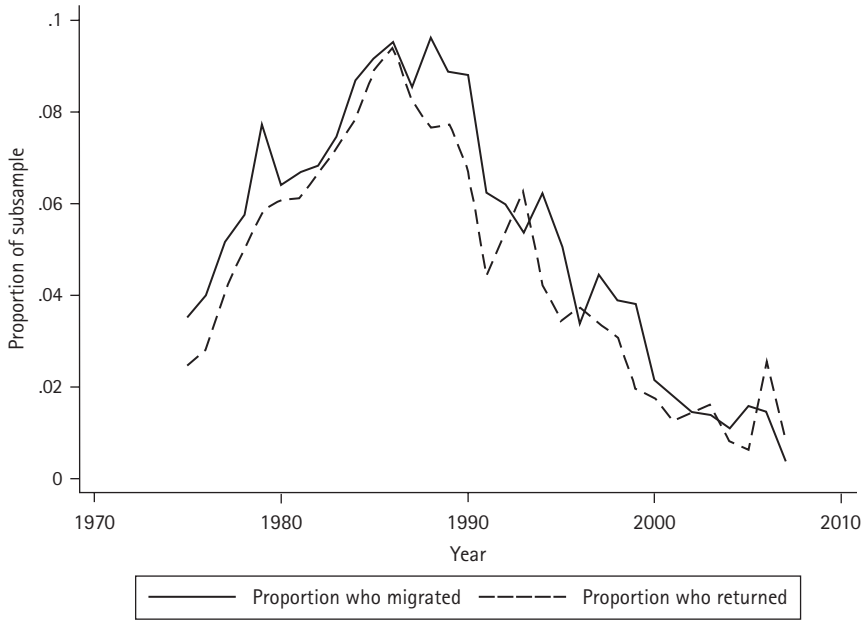


FIGURE 19.6 Proportion of MMP128 subsample survey respondents who migrated and returned in a given year between 1975 and 2007.

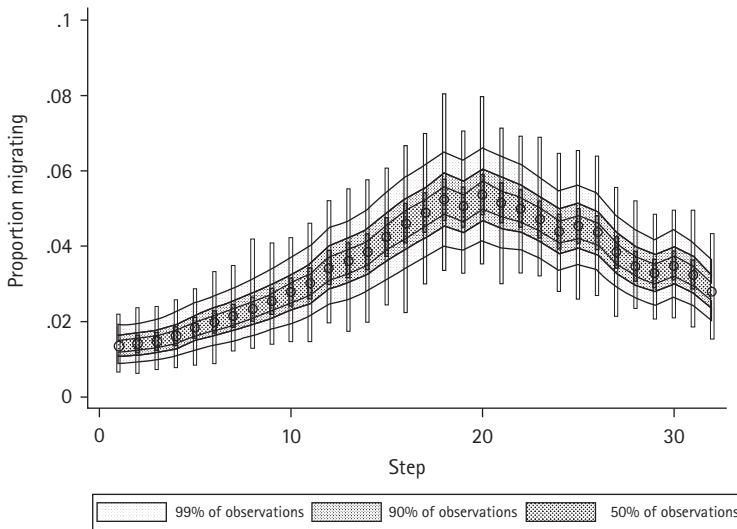


FIGURE 19.7 Result of Monte Carlo simulations for proportion of agents migrating. Dark bars show the mean $\pm$ the standard deviation; empty bars show range.

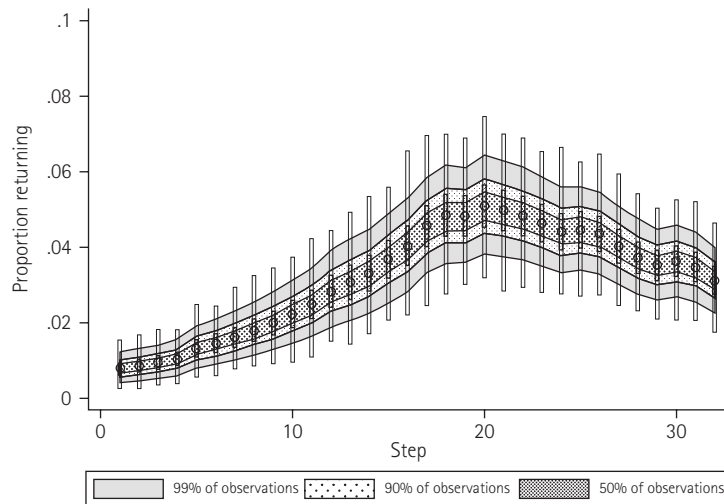


FIGURE 19.8 Result of Monte Carlo simulations for proportion of agents returning. Dark bars show the mean $\pm$ the standard deviation; empty bars show range.

can be used to classify particular simulation results in the context of the conceptual population of simulated scenarios, similar to Voudouris et al. (2011).

Most of the simulation runs are within a fairly narrow range. The overall pattern—both migration and return movement behavior increase and then decrease over time—follows the pattern of the empirical data.

19.8 ROBUSTNESS CHECKS AND POLICY ANALYSIS

A robustness analysis of an agent-based model serves to check whether the model reacts as expected when parameter changes are introduced that should alter the results in an unambiguous way. By increasing the home-country wage relative to the host country wage it shall now be demonstrated that the model at hand passes a test for robustness. Afterwards it will be illustrated how the model can be used for policy analysis. It is shown how the effect of a tightening of border control depends on the level of foresight of potential return migrants.

Some potential migrants can probably not afford to migrate and would therefore be enabled to overcome a “poverty trap” if wages increased slightly (McKenzie and Rapoport 2007). A larger increase in home-country wages should decrease stocks of migrants in the host country. To check whether this result is produced by the model, the home-country wage is increased by multiplying each value in the time series by 1.1, 1.2,

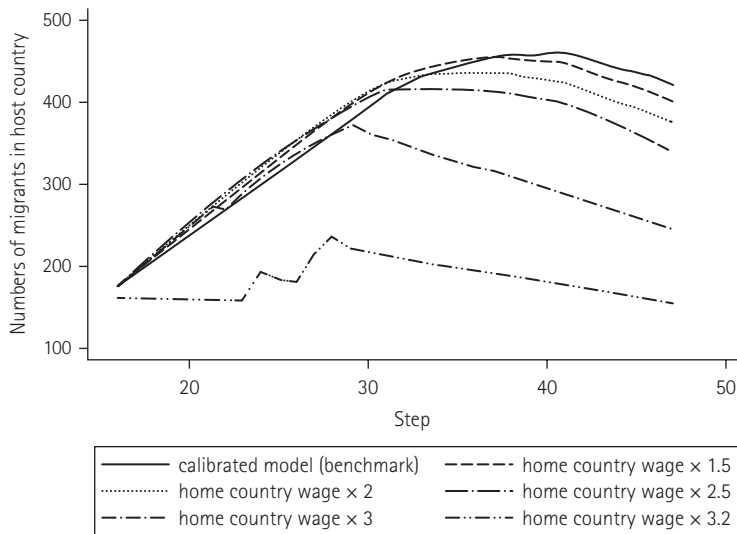


FIGURE 19.9 Average stocks of migrants at each model step (1,000 model runs), at different values of the average home-country wage.

and so on up to 3.2 and running the model one thousand times for each treatment. An increase in average stocks is observed in early periods for increases in the average home wage. At some point every potential migrant has gathered sufficient wealth to overcome the poverty trap. The higher the home-country wage, the earlier that point is reached. Beyond that point, the higher the home-country wage, the lower are the average migrant stocks in the host country (see figure 19.9). At values larger than 3.2, migration ceases almost completely as the home-country wages are, on average, as high as the host-country wages.

This is the effect that was expected, and it is reproduced by the model. Whether this estimate can be trusted quantitatively depends on whether one believes that the behavioral rules, and in particular, the impact of the wage difference on the migration decision, are stable if the wages increase substantially. Further research is needed to verify this assumption.

The next experiment that is performed concerns the level of border control. It is unclear whether increasing border protection increases or decreases the stock of migrants in a country. Kossoudji (1992) observes that tighter regulation increases stocks of migrants because it decreases out-migration. Angelucci (2005) finds an ambiguous answer: Tighter border control clearly decreases inflow but also decreases outflow. Thom (2010) does not find that stricter border control increases stocks of migrants. Clearly, the net effect depends on how far migrants are deterred from returning since they take into account the lower probability of being able to migrate again.

In order to test the impact on stocks, it is assumed that the level of border control increases by 10 percent. Figure 19.10 depicts the average stocks across a period of

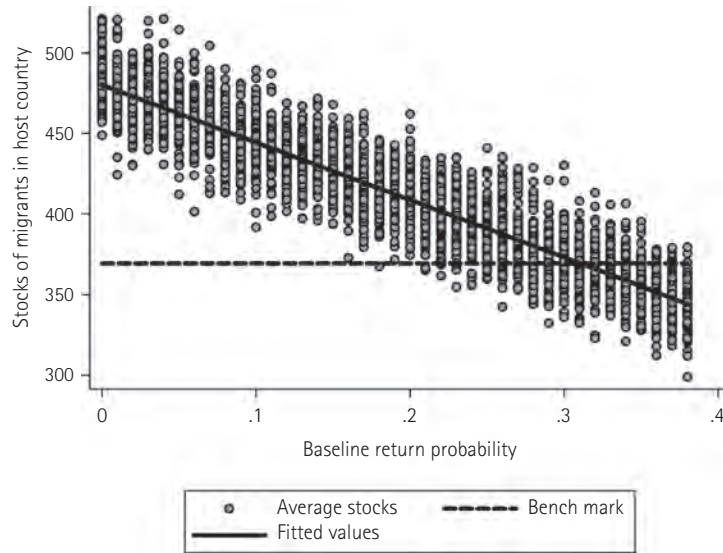


FIGURE 19.10 Increase of border control by 10 percent: Average stocks of migrants in the host country across 33 years at different levels of baseline return probability. The simulation was run 100 times for each level. The horizontal line indicates the average stocks after 10,000 runs of the benchmark simulation (369.17). The intersection of the fitted values and the benchmark scenario indicates at which level of decrease in baseline return probability the average stocks at the higher level of border control actually start to be higher than in the benchmark scenario.

thirty-three years at levels of baseline return probability of .38 and at lower levels, showing how stocks increase with a decrease in return probability.

The relation between average stocks and baseline return probability is almost linear. Average stocks increase by 3.58 individuals with every percentage point decrease in baseline return probability. It can be concluded that, on average, stocks increase after an increase in border control by 10 percent if, of one hundred migrants in the United States of whom thirty-eight would have returned in a given year, seven (or 18 percent) or more take into account the reduced migration probability and refrain from returning.

19.9 CONCLUSIONS

In this study the phenomenon of circular migration is analyzed in an agent-based model. To the author's knowledge it is the first completely empirically founded and spatially explicit model of the phenomenon that is able to take account of the whole cycle of migration and the role of networks. Three stylized facts about circular migration are introduced that the model can match, despite its being fairly simple: Migration concentrates on a certain number of places, people from one neighborhood tend to

go to the same few places, and migration-specific capital makes subsequent migration more likely. A set of hypotheses is derived from the literature concerning influential factors in the decision to migrate or to return in a given year. These hypotheses are tested using the Mexican Migration Project (MMP128). The behavioral motives that survived the empirical check are included in the model. It is found that expected earnings, an idiosyncratic home bias, network ties to other migrants, strength of links to the home country, and age have a highly significant impact on circular migration patterns over time. A model is presented that includes two countries with differing average wages, workers who search for employment, and firms. Workers can migrate and return according to probabilistic behavioral rules estimated from the MMP128. Four remaining parameters are calibrated by running Monte Carlo simulations. Thus, avoiding a common criticism of agent-based models, this model has only four degrees of freedom and yet is able to replicate two distributions and two time series from the data fairly well.

Computational experiments are performed in order to demonstrate the robustness of the model. Finally, it is shown how the model can be used to perform policy analysis. It has the potential to help answer the much-debated question whether increasing border protection increases or decreases the stock of migrants in a country. It is found that if 18 percent or more of migrants who would have returned at the lower level of border control take into account that they might not be able to migrate again and therefore refrain from returning, stocks increase. Otherwise, they decrease.

Promising avenues for future research are making the model spatially accurate using a geographic information system or introducing more sophisticated behavioral rules and additional degrees of heterogeneity to account for existing mismatches between data and simulation.

Moreover, with further calibration and sensitivity analysis, the model can be used for forecasting flows of migration and return in certain regions or cities, possibly by combining it with local border enforcement data, and to estimate the effect of labor market shocks or changes in immigration law.

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CHAPTER 20

COMPUTATIONAL INDUSTRIAL ECONOMICS

A Generative Approach to Dynamic Analysis in Industrial Organization

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20.1 INTRODUCTION

I propose a computational modeling framework that can form the basis for carrying out dynamic analysis in industrial organization (IO). The main idea is to create an artificial industry within a computational setting, which can then be populated with firms that enter, compete, and exit over the course of the industry's growth and development. These actions of the firms are driven by pre-specified decision rules, the interactions of which then generate a rich historical record of the industry's evolution. With this framework, one can study the complex interactive dynamics of heterogeneous firms as well as the evolving structure and performance of the industry.

The base model of industry dynamics presented here can be extended and refined to address a variety of standard issues in IO, but it is particularly well suited for analyzing the dynamic process of adjustment in industries that are subject to persistent external shocks. The empirical significance of such processes is well reflected in the literature that explores patterns in the turnover dynamics of firms in various industries. The seminal work in this literature is Dunne et al. (1988). They found that the turnovers are significant and persistent over time in a wide variety of industries, though they also noticed differences in their rates: “[W]e find substantial and persistent differences in entry and exit rates across industries. Entry and exit rates at a point in time are also highly correlated across industries so that industries with higher than average

entry rates tend to also have higher than average exit rates” (p. 496). To the extent that firm turnovers are the manifestation of the aforementioned adjustment process, this literature brings to light the empirical significance of the out-of-equilibrium industrial dynamics. Furthermore, it identifies patterns to this adjustment process that the standard equilibrium models (static or dynamic) are not well equipped to address. For instance, Caves (2007) states: “Turnover in particular affects entrants, who face high hazard rates in their infancy that drop over time. It is largely because of high infant mortality that rates of entry and exit from industries are positively correlated (compare the obvious theoretical model that implies either entry or exit should occur but not both). The positive entry-exit correlation appears in cross-sections of industries, and even in time series for individual industries, if their life-cycle stages are controlled” (pp. 9-10). The approach proposed here addresses this uneasy coexistence of equilibrium-theorizing and the non-equilibrium empirical realities in industrial organization.

The model entails a population of firms; each of which is endowed with a unique technology that remains fixed over the course of its life. The firms go through a repeated sequence of entry-output-exit decisions. The market competition takes place amid a technological environment that is subject to external shocks, inducing changes in the relative production efficiencies of the firms. Because the firms’ technologies are held fixed, there is no firm-level adaptation to the environmental shifts. However, an industry-level adaptation takes place through the process of firm entry and exit, which is driven by the selective force of the market competition.

The implementation of the adjustment process in the proposed model rests on two assumptions. First, the firms in the model are myopic. They do not have the foresight that is frequently assumed in the orthodox industrial organization literature. Instead, their decisions are made on the basis of fixed rules motivated by myopic but maximizing tendencies. Second, the technological environment within which the firms operate is stochastic. How efficient a firm was in one environment does not indicate how efficient it will be in another. Thus, there is always a possibility that the firms may be reshuffled when the environment changes. These two assumptions—myopia (or, more broadly, bounded rationality) of firms and stochastic technological environment—lead to persistent firm heterogeneity. It is this heterogeneity that provides the raw materials over which the selective force of market competition can act. Firm myopia drives the entry process; the selective force of market processes, acting on the heterogeneous firms, drives the exit process; all the while the stochastic technological environment guarantees that the resulting process never settles down, giving us an opportunity to study patterns that may emerge along the way.

I offer two sets of results in this chapter. The first is obtained under the assumption of stable market demand; hence the only source of turbulence is the shocks to the technological environment. The results show that the entry and exit dynamics inherent in the process generate patterns that are consistent with empirical observations, including the phenomenon of infant mortality and the positive correlation between entry and exit. Furthermore, the base model enables the analysis of the

relations between the industry-specific demand and technological conditions and the endogenous structure and performance of the industry along the steady state, thereby generating cross-sectional implications in a fully dynamic setting. The second set of results is generated under the assumption of fluctuating demand; hence, the demand shocks are now superimposed on the technological shocks. The basic relation between the rate of entry and the rate of exit continues to hold with this extension. In addition, the extension enables a characterization of the way fluctuating demand affects the turnover as well as the structure and performance of the industry. Finally, I discuss a potential extension of the base model in which the R&D activities of firms can be incorporated.

20.2 INDUSTRY DYNAMICS: AN OVERVIEW OF THE THEORETICAL LITERATURE

Industrial organization theorists have used analytical approaches to explore the entry and exit dynamics of firms and their impact on the growth of the industry. Most of these works involve dynamic models of firm behavior, in which heterogeneous firms with perfect foresight (but with incomplete information) make entry, production, and exit decisions to maximize the expected discounted value of future net cash flow. Jovanovic (1982) made the first significant attempt in this line of research. He presented an equilibrium model in which firms learn about their efficiency through productivity shocks as they operate in the industry. The shocks were specified to follow a nonstationary process and represented noisy signals that provided evidence to firms about their true costs. The “perfect foresight” equilibrium leads to selection via entry and exit in this model. However, there is no firm turnover in the long run, once learning is completed; hence, the patterns in the persistent series of entry and exit, identified in the empirical literature, cannot be explored with this model.

Hopenhayn (1992) offers a tractable framework in which perpetual and simultaneous entry and exit are endogenously generated in the long run. His approach entails a dynamic stochastic model of a competitive industry with a continuum of firms. The movements of firms in the long run are part of the stationary equilibrium he develops in order to analyze the behavior of firms along the steady state as they are hit with individual productivity shocks. This model assumes *perfect competition* and, hence, the stationary equilibria in the model maximize net discounted surplus. In this sense, it is similar to Lucas and Prescott (1971), who looked at a competitive industry with aggregate shocks but with no entry and exit. Hopenhayn (1992) finds that changes in aggregate demand do not affect the rate of turnover, though they raise the total number of firms. As Asplund and Nocke (2006) point out, this result hinges on the assumption of perfect competition in which there is no price competition effect induced by the increase in the mass of active firms. For this reason, the price-cost margins are independent of market size in this model.

Asplund and Nocke (2006) adopt the basic framework of Hopenhayn (1992) and extend it to the case of imperfect competition. (Melitz (2003) is another extension of Hopenhayn's approach to monopolistic competition, though his focus is on international trade.) It is an equilibrium model of entry and exit that utilizes the steady-state analysis developed in Hopenhayn (1992). The model uses the reduced-form profit function and, hence, avoids specifying the details of the demand system as well as the nature of the oligopoly competition (e.g., output competition, price competition, or price competition with differentiated products). The key difference between Hopenhayn (1992) and their model is that they assume the existence of the *price competition effect*, which implies a negative relation between the number of entrants and the profits of active incumbents. The assumptions made about the reduced-form profit function lead to two countervailing forces in the model: First, the rise in market size increases the profits of all firms proportionally by means of an increase in output levels, holding prices fixed; second, the distribution of firms also increases with the market size, reducing the price and the price-cost margin via the *price competition effect*. The main result is that an increase in market size leads to a rise in the turnover rate and a decline in the age distribution of firms in the industry. They provide empirical support for these results using data about hair salons in Sweden.

Asplund and Nocke (2006) make a significant contribution in that they are able to generate persistent entry and exit along the steady state of the industry and perform comparative statical analysis with respect to market-specific factors such as market size and the fixed cost. However, their equilibrium approach, though appropriate for the study of the steady-state behavior, is inadequate when the focus is on the behavior of firms along the non-equilibrium transitory path. In contrast to Asplund and Nocke (2006), the model presented in this chapter specifies a linear demand function and Cournot output competition. The price competition effect, instead of being assumed, is generated as an endogenous outcome of the entry-competition process. The functional forms specified in this model are more restrictive than the reduced-form profits approach used by Asplund and Nocke (2006), but they are necessary for the task at hand: Given that the research objective is to examine the effects of the market-specific factors on the evolving structure and performance of the industry, it is imperative that the model fully specify the demand and cost structure so that the relevant variables such as price, outputs, and market shares can be endogenously derived. This allows for a detailed analysis of the firms' behavior along the transitory phase, identifying and interpreting the patterns that exist in the non-equilibrium state of the industry, further enhancing our understanding of the comparative dynamics' properties.

Another important development is the concept of Markov Perfect Equilibrium (MPE), which incorporates strategic decision making by firms with perfect foresight in fully dynamic models of oligopolistic interactions with firm entry and exit; see Pakes and McGuire (1994) and Ericson and Pakes (1995). The MPE framework is directly imported from the game-theoretic approach to dynamic oligopoly, which models strategic interactions among players (firms) who take into account each other's strategies and actions. Firms in these models are endowed with perfect foresight:

They maximize the expected net present value of future cash flows, taking into account all likely future states of its rivals conditional on all possible realizations of industry-wide shocks. The assumption of perfect foresight implies that the firms' decisionmaking process uses recursive methodology; hence, solving for the equilibrium entails Bellman equations. Given the degree of complexity in the model specification and the solution concept, this approach requires extensive use of computational methodologies. Although this approach is conceptually well positioned to address the central issues of industry dynamics, its success thus far has been limited owing to the well-known "curse of dimensionality" as described by Doraszelski and Pakes (2007):

The computational burden of computing equilibria is large enough to often limit the type of applied problems that can be analyzed. There are two aspects of the computation that can limit the complexity of the models we analyze; the computer memory required to store the value and policies, and the CPU time required to compute the equilibrium.... [I]f we compute transition probabilities as we usually do using unordered states then the number of states that we need to sum over to compute continuation values grows exponentially in both the number of firms and the number of firm-specific state variables. (pp. 1915–1916)

As they point out, the computational burden is mainly due to the size of the state space that needs to be explored in evaluating the value functions and the policy functions. The exponential growth of the state space that results from increasing the number of firms or the set of decision variables imposes a significant computational constraint on the scale and scope of the research questions one can ask.¹ Furthermore, if the objective is to generate analytical results that can match data, the MPE approach often falls short of meeting it.

The agent-based computational economics (ACE) approach taken in this chapter offers a viable alternative to the MPE approach. Tesfatsion and Judd (2006) provide a succinct description:

ACE is the computational study of economic processes modeled as dynamic systems of interacting agents who do not necessarily possess perfect rationality and information. Whereas standard economic models tend to stress equilibria, ACE models stress economic processes, local interactions among traders and other economic agents, and out-of-equilibrium dynamics that may or may not lead to equilibria in the long run. Whereas standard economic models require a careful consideration of equilibrium properties, ACE models require detailed specifications of structural conditions, institutional arrangements, and behavioral dispositions. (p. xi)

In the same vein, my model does away with the standard assumption of perfect foresight on the part of the firms. By assuming myopia and limited rationality in their decision making, I eliminate the need to evaluate the values and policies for all possible future

¹ See Weintraub, Benkard, and Van Roy (2008, 2010) for attempts to circumvent this problem while remaining within the general conceptual framework of the MPE approach.

states for all firms, thus effectively avoiding the curse of dimensionality that is inherent in the MPE approach. The computational effort thus saved is utilized in studying the complex interactions among firms and tracking the movements of relevant endogenous variables over time for a large number of firms. The trade-off, then, is the reduction in the degree of firms' rationality and foresight in return for the ability to analyze in detail the non-equilibrium dynamics of the industry. In view of the increased scale and scope of the research questions that can be addressed, the use of the agent-based computational model is deemed well justified.

Given the current state of the literature as outlined above, the present study makes two contributions. First, it offers a computational model that can be used as a test bed for large-scale experiments that involve generating and growing artificial industries. By systematically varying the characteristics of the environment within which the industry evolves, I am able to identify the patterns inherent in the growth process of a given industry as well as any difference in those patterns that may exist across industries. Such computational experiments allow reevaluation of the results concerning comparative dynamics obtained in the previous analytical works, but they also offer additional results and insights, including the cyclical industry dynamics that may result from fluctuations in market demand. The in-depth investigation of the non-equilibrium adjustment dynamics offered here is not feasible with the standard equilibrium models.

The second contribution is in generating predictions that can match data better than the existing models can. The agent-based computational approach taken here greatly reduces the demand for computational resources at the level of the individual firm's decision making. Instead, the saved resources are allocated to tracking and analyzing the complex interactions among firms over time. Compared to the numerical models formulated in the MPE framework, the model described in this chapter offers a flexible platform for generating predictions that can better match empirical data by incorporating a larger number of firms and their decision variables.

20.3 METHODOLOGY

The base model entails an evolving population of firms which interact with one another in repeated market competition.² Figure 20.1 captures the overall structure of the model.

Each period t starts out with a group of surviving incumbents from the previous period, and consists of three decision stages. In stage 1, each potential entrant makes a decision to enter. A new entrant is assumed to enter with a random (but unique) technology and a start-up fund that is common for all entrants. In stage 2, the new entrants and the incumbents, given their endowed technologies, compete in the market.

² Note that while the surviving incumbents may be engaged in "repeated interaction" with one another, the assumption of myopia precludes the possibility of collusion in this model.

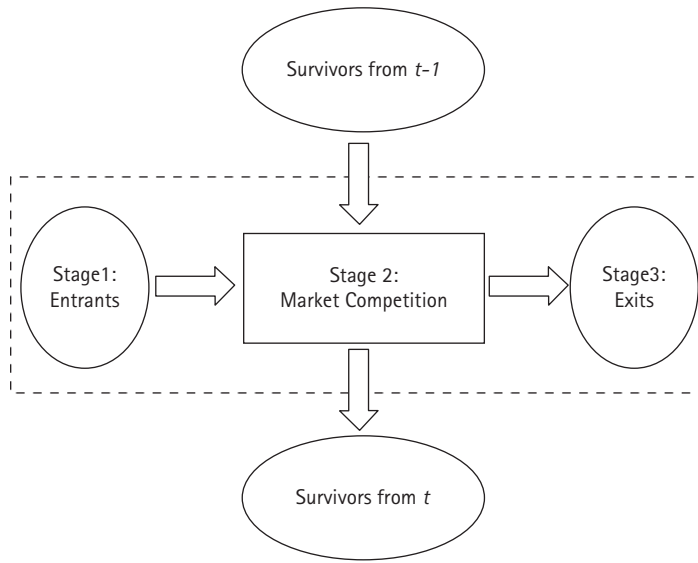


FIGURE 20.1 Base model of industry dynamics.

The outcome of the competition in stage 2 generates profits (and losses) for the firms. In stage 3, all firms update their net wealth based on the profits earned (and losses incurred) from the competition and decide whether to exit the industry. Once the exit decisions are made, the surviving incumbents with their technologies and their updated net wealth move on to period $t + 1$, in which the process is repeated.

Central to this process are the heterogeneous production technologies held by the firms (which imply cost asymmetry) and the selective force of market competition that leads to entry and exit of firms. The next subsections describe the nature of technology and the market conditions, followed by a detailed description of the multi-stage decision process.

20.3.1 Modeling Technology

In each period, active firms in the market produce and sell a homogeneous good. The good is produced in a process that consists of N distinct tasks. Each task can be completed using one of two different methods. Though all firms produce a homogeneous good, they may do so using different combinations of methods for the N component tasks. The method used by a firm for a given task is represented by a bit (0 or 1) such that there are two possible methods available for each task and thus 2^N variants of the production technology. To the extent that the production technology is viewed as a combination (vector) of methods (techniques), my modeling of technology is similar to that of Jovanovic and Rob (1990). However, the nature of the technology space is

substantially different. In their model, technology is a countable infinity of technology types, with a continuum of techniques within each type. In contrast, a production technology in my model entails a fixed number of “tasks,” each of which is carried out using a specific method (from a finite number of alternatives).

The base model assumes that a firm enters the industry with a technology with which it is endowed. The firm stays with it over the course of its life; that is, it is committed to using a particular method for a given task at all times.³ A firm’s technology, then, is fully characterized by a binary vector of N dimensions that captures the complete set of methods it uses to produce the good. Denote it by $\underline{z}_i \in \{0, 1\}^N$, where $\underline{z}_i \equiv (z_i(1), z_i(2), \dots, z_i(N))$ and $\underline{z}_i(h) \in \{0, 1\}$ is firm i ’s method in task h .

In measuring the degree of heterogeneity between any two technologies (i.e., method vectors), $\underline{z}_i$ and $\underline{z}_j$, we use “Hamming distance,” which is the number of positions for which the corresponding bits differ:

$$D(\underline{z}_i, \underline{z}_j) \equiv \sum_{h=1}^N |z_i(h) - z_j(h)|. \quad (20.1)$$

How efficient a given technology is depends on the environment in which it operates. In order to represent the *technological environment* that prevails in period t , I specify a unique methods vector, $\hat{\underline{z}}^t \in \{0, 1\}^N$, which is defined as the *optimal technology* for the industry in t . How well a firm’s technology performs in the current environment depends on how close it is to the prevailing optimal technology in the technology space. More specifically, the marginal cost of firm i in period t is specified to be a direct function of $D(\underline{z}_i, \hat{\underline{z}}^t)$, the Hamming distance between the firm’s endowed technology, $\underline{z}_i$, and the current optimal technology, $\hat{\underline{z}}^t$. The optimal technology in t is common for all firms; that is, in a given industry all firms face the same technological environment at a given point in time. Thus, once it is defined for a given industry, its technological environment is completely specified for all firms because the efficiency of any technology is well defined as a function of its distance from this optimal technology.⁴

20.3.2 Modeling Market Competition

In each period there exist a finite number of firms that operate in the market. In this subsection I define the static market equilibrium among such firms. The technological environment and the endowed technologies for all firms jointly determine the marginal costs of the firms and, hence, the resulting equilibrium. The static market equilibrium

³ There is, hence, no adaptation at the firm level in this model, although adaptation at the industry level is possible by the selection of firms via market competition.

⁴ Given that an entering firm is endowed with a fixed technology that cannot be modified, it does not matter whether the optimal technology is known to the firms or not. In a more general setting where the technology can be modified, any knowledge of the optimal technology, however imperfect, will affect the direction of the firm’s adaptive modification via its R&D decisions.

defined here is then used to *approximate* the outcome of market competition in each period. In this subsection I temporarily abstract away from the time superscript for ease of exposition.

Let m be the number of firms operating in the market. The firms are Cournot oligopolists that choose production quantities of a homogeneous good. In defining the Cournot equilibrium in this setting, I assume that all m firms produce positive quantities in equilibrium. The inverse market demand function is specified as

$$P(Q) = a - \frac{Q}{s} \quad (20.2)$$

where $Q = \sum_{j=1}^m q_j$ and s denotes the size of the market.

Each operating firm has its production technology, $\underline{z}_i$, and faces the following total cost:

$$C(q_i; \underline{z}_i, \hat{\underline{z}}) = f_i + c_i(\underline{z}_i, \hat{\underline{z}}) \cdot q_i. \quad (20.3)$$

For simplicity, the firms are assumed to have identical fixed cost: $f_1 = f_2 = \dots = f_m = f$. The firm's marginal cost, $c_i(\underline{z}_i, \hat{\underline{z}})$, depends on how different its technology, $\underline{z}_i$, is from the optimal technology, $\hat{\underline{z}}$:

$$c_i \equiv c_i(\underline{z}_i, \hat{\underline{z}}) = 100 \cdot \frac{D(\underline{z}_i, \hat{\underline{z}})}{N}. \quad (20.4)$$

Hence, c_i increases the Hamming distance between the firm's technology and the optimal technology for the industry. It is at its minimum of zero when $\underline{z}_i = \hat{\underline{z}}$ and at its maximum of 100 when all N bits in the two technologies are different from one another.⁵ The total cost can then be rewritten as

$$C(q_i; \underline{z}_i, \hat{\underline{z}}) = f + 100 \cdot \frac{D(\underline{z}_i, \hat{\underline{z}})}{N} \cdot q_i. \quad (20.5)$$

Given the demand and cost functions, firm i 's profit is

$$\pi_i(q_i, Q - q_i) = \left(a - \frac{1}{s} \sum_{j=1}^m q_j \right) \cdot q_i - f - c_i \cdot q_i. \quad (20.6)$$

Taking the first-order condition for each i and summing over m firms, we derive the equilibrium industry output rate, which gives us the equilibrium market price, $\bar{P}$, through equation (20.2):

$$\bar{P} = \left(\frac{1}{m+1} \right) \left(a + \sum_{j=1}^m c_j \right). \quad (20.7)$$

⁵ For concreteness, suppose $N = 10$ and the current technological environment is captured by the optimal vector of $\hat{\underline{z}} = (1100101100)$. If a firm i entered the industry with a technology, $\underline{z}_i = (0010011100)$, then $D(\underline{z}_i, \hat{\underline{z}}) = 5$ and the firm's marginal cost is 50.

Given the vector of marginal costs defined by the firms' technologies and the optimal technology, $\bar{P}$ is uniquely determined and is independent of the market size, s . Furthermore, the equilibrium market price depends only on the sum of the marginal costs and not on the *distribution* of $c_i s$.

In deriving the equilibrium output rate, $\bar{q}_i$, I assume that all m firms are active and, hence, $\bar{q}_i > 0$ for all $i \in \{1, 2, \dots, m\}$. This assumption is relaxed in the simulations reported later in the chapter, given that the cost asymmetry inherent in this model may force some of the firms to shut down and become inactive (the algorithm used to identify inactive firms is discussed in section 20.3.3.2). The equilibrium firm output rate is then

$$\bar{q}_i = s \cdot \left[\left(\frac{1}{m+1} \right) \left(a + \sum_{j=1}^m c_j \right) - c_i \right]. \quad (20.8)$$

A firm's equilibrium output rate depends on its own marginal cost and the equilibrium market price such that $\bar{q}_i = s \cdot (\bar{P} - c_i)$. Finally, the Cournot equilibrium firm profit is

$$\pi(\bar{q}_i) = \bar{P} \cdot \bar{q}_i - f - c_i \cdot \bar{q}_i = \frac{1}{s} (\bar{q}_i)^2 - f. \quad (20.9)$$

Note that $\bar{q}_i$ is a function of c_i and $\sum_{j=1}^m c_j$, where c_j is a function of $\underline{z}_j$ and $\hat{z}$ for all j . It is then straightforward that the equilibrium firm profit is fully determined, once the vectors of methods are known for all firms. Further note that $c_i \leq c_k$ implies $\bar{q}_i \geq \bar{q}_k$ and, hence, $\pi(\bar{q}_i) \geq \pi(\bar{q}_k) \forall i, k \in \{1, \dots, m\}$.

20.3.3 The Base Model of Industry Dynamics

In the beginning of any typical period t , the industry opens with two groups of decision makers: a group of incumbent firms surviving from $t-1$, each of which enters t with its endowed technology, $\underline{z}_i$, and its net wealth, w_i^{t-1} , carried over from $t-1$, and a group of firms ready to consider entering the industry in t , each with an endowed technology of $\underline{z}_j$ and its start-up wealth. All firms face a common technological environment within which its technology will be used; this environment is fully represented by the prevailing optimal technology, $\hat{z}^t$.

Central to the model is the assumption that the production environment is inherently stochastic; that is, the technology which was optimal in one period is not necessarily optimal in the next. This is captured by allowing the optimal technology, $\hat{z}^t$, to vary from one period to the next in a systematic manner. The mechanism that guides this shift dynamic is described next.

20.3.3.1 Random Shifts in the Technological Environment

Consider a binary vector, $\underline{x} \in \{0, 1\}^N$. Define $\delta(\underline{x}, l) \subset \{0, 1\}^N$ as the set of points that are exactly Hamming distance l from $\underline{x}$. The set of points that are within Hamming distance l of $\underline{x}$ is then defined as

$$\Delta(\underline{x}, l) \equiv \bigcup_{i=0}^l \delta(\underline{x}, i). \quad (20.10)$$

The following rule drives the shift dynamic of the optimal technology:

$$\hat{\underline{z}}^t = \begin{cases} \hat{\underline{z}}'^t & \text{with probability } \gamma \\ \hat{\underline{z}}^{t-1} & \text{with probability } 1 - \gamma \end{cases} \quad (20.11)$$

where $\hat{\underline{z}}' \in \Delta(\hat{\underline{z}}^{t-1}, g)$ and γ and g are constant over all t . Hence, with probability γ the optimal technology shifts to a new one within g Hamming distance from the current technology, while with probability $1 - \gamma$ it remains unchanged at $\hat{\underline{z}}^{t-1}$. The volatility of the technological environment is captured by γ and g , where γ is the frequency and g is the maximum magnitude of changes in technological environment.

An example of the type of changes in the technological environment envisioned in this model is the series of innovations in computers and digital technologies that have occurred in recent decades. Although many of these innovations originated in the military or in academia (e.g., Arpanet and the Internet from the Department of Defense and email from MIT's Compatible Time-Sharing System), the gradual adoption of these technologies by the suppliers and customers in the complex network of market relationships fundamentally and asymmetrically affected the way firms operate and compete. As a timely example, consider the challenges the online business models pose for the retail establishments entrenched in the traditional brick-and-mortar modes of operation: Firms using a set of practices deemed acceptable in the old environment can no longer compete effectively in the new one. It is these innovations created outside of the given industry that are treated as exogenous shocks in my model. In a framework more similar to the neoclassical production theory, one could also view an externally generated innovation as a shock that affects the relative input prices for the firms. If firms, at any given point in time, are using heterogeneous production processes with a varying mix of inputs, such a change in input prices will have diverse impacts on the relative efficiencies of firms' production processes—some may benefit from the shock; some may not.

The change in technological environment is assumed to take place *in the beginning of each period before firms make any decisions*. Although the firms do not know what the optimal technology is for the new environment, they are assumed to receive accurate signals of their own marginal costs based on the new environment when making their decisions to enter. They, however, do not have this information about the incumbents. Nevertheless, because they observe $\bar{P}^{t-1}$ and $\bar{q}_j^{t-1}$ for each j in $t - 1$, they can infer c_j^{t-1} for all firms. In calculating the attractiveness of entry, therefore, a potential entrant i relies on c_i^t and c_j^{t-1} for all j in the set of surviving incumbents from $t - 1$.

20.3.3.2 Three-stage Decision Making

Denote by S^{t-1} the set of surviving firms from $t-1$, where $S^0 = \emptyset$. The set of surviving firms includes those which were active in $t-1$ in that their outputs were strictly positive as well as those which were inactive, with their plants shut down during the preceding period. The inactive firms in $t-1$ survive to t if and only if they have sufficient net wealth to cover their fixed costs in $t-1$. Each firm $i \in S^{t-1}$ possesses a production technology, z_i , with which it entered the industry and which gives rise to its marginal cost in t of c_i^t as defined in equation (20.4). It also has a current net wealth of w_i^{t-1} that it carries over from $t-1$.

Let R^t denote the finite set of firms that contemplate entering the industry in the beginning of t . I assume that the size of the pool of potential entrants is fixed and constant at r throughout the entire horizon. I also assume that this pool of r potential entrants is renewed each period. Each potential entrant j in R^t is endowed with a technology, z_j , randomly chosen from $\{0, 1\}^N$ according to uniform distribution. In addition, each potential entrant enters the market with a fixed start-up level of wealth.

Stage 1: Entry Decisions. In stage 1 of each period, the potential entrants in R^t first make their decisions to enter. Just as each firm in S^{t-1} has its current net wealth of w_i^{t-1} , we will let $w_j^{t-1} = b$ for all $j \in R^t$ where b is the fixed start-up fund common to all potential entrants. The start-up wealth, b , may be viewed as a firm's available fund that remains after paying for the one-time set-up cost of entry. For example, if one wishes to consider a case in which a firm has zero funds available but must incur a positive entry cost, it would be natural to consider b as having a negative value.

The entry rule takes the simple form that an outsider will be willing to enter the industry if and only if it perceives its post-entry net wealth in period t to be strictly above a threshold level representing the opportunity cost of operating in this industry. The entry decision then depends on the profits that it expects to earn in the periods following entry. Realistically, this would be the present discounted value of the profits to be earned over some foreseeable future starting from t . In the base model presented here, I assume the firms to be completely myopic such that the expected profit is simply the static one-period Cournot equilibrium profit based on three things: first, the marginal cost of the potential entrant, accurately reflecting the new technological environment in t ; second, the marginal costs of the active firms from $t-1$; and, third, the potential entrant's belief that it is the only new entrant in the market.⁶ In terms of rationality, the extent of myopia assumed here is obviously extreme. The other end of the spectrum is the strategic firm with perfect foresight, as typically assumed in the MPE

⁶ This requires: (1) that a potential entrant (correctly) perceives its own marginal cost from its endowed technology and the prevailing optimal technology; and 2) that the market price and the active firms' production quantities in $t-1$ are common knowledge. Each active incumbent's marginal cost can be directly inferred from the market price and the production quantities as $\bar{q}_i = s(\bar{P} - c_i)$.

models.⁷ The realistic representation of firm decision making would lie somewhere between these two extremes. The assumption of myopia is made here to focus on computing the finer details of the interactive dynamics that evolve over the growth and development of an industry. Relaxing this assumption in ways that are consistent with the observations and theories built up in the behavioral literature will be an important agenda for the future.

The decision rule of a potential entrant $k \in R^t$ is then

$$\begin{cases} \text{Enter, if and only if } \pi_k^e(\underline{z}_k) + b > \underline{W} \\ \text{Do not enter, otherwise} \end{cases} \quad (20.12)$$

where π_k^e is the static Cournot equilibrium profit the entrant *expects* to make in the period of its entry and $\underline{W}$ is the threshold level of wealth for a firm's survival (common to all firms). Once every potential entrant in R^t makes its entry decision on the basis of the above criterion, the resulting set of actual entrants, $E^t \subseteq R^t$, contains only firms with technologies efficient enough to guarantee some threshold level of profits given its belief about the market structure and the technological environment.

Denote by M^t the set of firms ready to compete in the industry: $M^t \equiv S^{t-1} \cup E^t$. I will denote by m^t the number of competing firms in period t such that $m^t = |M^t|$. At the end of stage 1 of period t , we then have a well-defined set of competing firms, M^t , with their current net wealth, $\{w_i^{t-1}\}_{i \in M^t}$, and their technologies, $\underline{z}_i$ for all $i \in S^{t-1}$ and $\underline{z}_j$ for all $j \in E^t$.

Stage 2: Output Decisions and Market Competition. The firms in M^t , with their technologies and current net wealth, engage in Cournot competition in the market, where we “approximate” the outcome with the standard Cournot-Nash equilibrium defined in section 20.3.2.⁸ The use of Cournot-Nash equilibrium in this context is admittedly inconsistent with the “limited rationality” assumption employed in this model. A more consistent approach would be to explicitly model the process of market experimentation. Instead of doing so, which would further complicate the model, I implicitly assume that it is done instantly and without cost. Cournot-Nash equilibrium is then assumed to be a reasonable approximation of the outcome from that process. Support for this assumption is provided in the small body of literature in

⁷ Weintraub et al. (2008) and Weintraub et al. (2010) represent attempts to relax the assumption of perfect foresight while remaining within the MPE framework.

⁸ As an alternative mode of oligopoly competition, Bertrand price competition may be considered. The computational implementation of this alternative mode is straightforward if firms are assumed to produce homogeneous product, though its ultimate impact on industry dynamics will require a whole new set of simulations and analyses. On the other hand, price competition with “differentiated products” will require an innovative approach to modeling the demand system since new entrants are now expected to enter with products that are differentiated from those offered by the incumbents. The linear city model of Hotelling (1929) or the circular city model of Salop (1979) may be considered for this purpose, but how the modeling of demand affects the industry dynamics will have to be left for future research.

which experimental studies are conducted to determine whether firm behavior indeed converges to the Cournot-Nash equilibrium: Fouraker and Siegel (1963), Cox and Walker (1998), Theocharis (1960), and Huck et al. (1999) all show that the experimental subjects that play according to the best-reply dynamic do indeed converge on the Nash equilibrium. (See Armstrong and Huck (2010) for a survey of this literature.) In contrast, Vega-Redondo (1997), using an evolutionary model of oligopoly, showed how introducing imitation of successful behavior into the Cournot games leads to the Walrasian outcome in the long run; the price converges to the marginal cost. An imitative dynamic, hence, intensifies the oligopolistic competition, driving the market price below the Nash equilibrium level. I should also note, however, that Apesteguia et al. (2010) found that the theoretical result of Vega-Redondo (1997) is quite fragile in that a minor degree of cost asymmetry—an inherent part of my model—can lead to outcomes other than the Walrasian outcome.⁹

Note that the equilibrium in section 20.3.2 was defined for m firms under the assumption that all m firms produce positive quantities. In actuality, given asymmetric costs, there is no reason to think that all m^t firms will produce positive quantities in equilibrium. Some relatively inefficient firms may shut down their plants and stay inactive. What we need is a mechanism for identifying the set of active firms among M^t such that the Cournot equilibrium among these firms will indeed entail positive quantities only. This is done in the following sequence of steps. Starting from the initial set of active firms, compute the equilibrium outputs for each firm. If the outputs for one or more firms are negative, then deactivate the least efficient firm, that is, set $q_i^t = 0$ where i is the least efficient firm. Re-define the set of active firms (as the previous set of active firms minus the de-activated firm) and recompute the equilibrium outputs. Repeat the procedure until all active firms are producing non-negative outputs. Each *inactive* firm produces zero output and incurs economic loss equivalent to its fixed cost. Each active firm produces its equilibrium output and earns the corresponding profit. We then have π_i^t for $i \in M^t$.

Stage 3: Exit Decisions. Given the single-period profits or losses made in stage 2 of the game, the firms in M^t consider exiting the industry in the final stage. Each firm's net wealth is first updated on the basis of the profit (or loss) made in stage 2:

$$w_i^t = w_i^{t-1} + \pi_i^t. \quad (20.13)$$

The exit decision rule for each firm is then

$$\begin{cases} \text{Stay in, if and only if } w_i^t \geq \underline{W} \\ \text{Exit, otherwise} \end{cases} \quad (20.14)$$

⁹ Their experiments allowing subjects access to information about their rivals generated outcomes that are much more competitive than the prediction of the Cournot-Nash equilibrium, demonstrating the need for further refined theory on firm behavior.

where $\underline{W}$ is the threshold level of net wealth (as previously defined). Denote by L^t the set of firms that leave the market in t . Once the exit decisions are made by all firms in M^t , the set of surviving firms from period t is then defined as

$$S^t \equiv \{all\ i \in M^t \mid w_i^t \geq \underline{W}\}. \quad (20.15)$$

The set of surviving firms, S^t , their technologies, $\{z_i\}_{i \in S^t}$, and their current net wealth, $\{w_i^t\}_{i \in S^t}$, are then passed on to $t + 1$ as state variables.

20.4 DESIGN OF COMPUTATIONAL EXPERIMENTS

The values of the parameters used here, including those for the baseline simulation, are provided in table 20.1.

I assume that there are ninety-six separate tasks in the production process, where the method chosen for each task is represented by a single bit. This implies that there are $2^{96} (\cong 8 \times 10^{28})$ different combinations of methods for the complete production process. In each period, there are exactly forty potential entrants to the industry, where a new firm enters with a start-up wealth b of 0. An incumbent firm will exit the industry if its net wealth falls below the threshold rate $\underline{W}$ of 0. The demand intercept is fixed at 300. The time horizon is five thousand periods; in period 1 the market starts out empty.

Table 20.1 Parameters and their values

Notation	Definition	Baseline value	Parameter values considered
T	Time horizon	5,000	5,000
N	Number of tasks	96	96
r	Number of potential entrants per period	40	40
b	Start-up capital for a new entrant	0	0
$\underline{W}$	Threshold level of net wealth for survival	0	0
a	Demand intercept	300	300
s	Market size	4	{4, 6, 8, 10}
f	Fixed production cost	200	{100, 200, 300, 400}
γ	Rate of change in technological environment	0.1	{0.1, 0.2, 0.3, 0.4}
g	Maximum magnitude of change in technological environment	8	8

Table 20.2 Definitions of Endogenous Variables

Variable	Definition
$ E^t $	Number of firms actually entering the industry in the beginning of t
$ M^t $ or m^t	Number of firms in operation in t (including active and inactive firms)
$ L^t $	Number of firms leaving the industry at the end of t
$ S^t $	Number of firms surviving at the end of t ($= M^t - L^t $)
p^t	Market price at which goods are traded in t
$\{c_i^t\}_{\forall i \in M^t}$	Realized marginal costs of all firms that were in operation in t
$\{q_i^t\}_{\forall i \in M^t}$	Actual outputs of all firms that were in operation in t
$\{\pi_i^t\}_{\forall i \in M^t}$	Realized profits (losses) of all firms that were in operation in t

I focus my analysis on the impacts of the market size s and the fixed cost f , as well as of the turbulence parameter γ . I consider four different values for each of these parameters: $s \in \{4, 6, 8, 10\}$, $f \in \{100, 200, 300, 400\}$, and $\gamma \in \{0.1, 0.2, 0.3, 0.4\}$. The maximum magnitude of change, g , is held fixed at 8. Note that a higher value of γ reflects more frequent changes in the technological environment.

Starting from an empty industry with a given parameter configuration, I evolve the industry and trace its growth and development by keeping track of the endogenous variables listed in table 20.2.

Using these endogenous variables, I next construct the following variables, which are useful for characterizing the behavior of firms in the industry:

- Q^t : Industry output, such that $Q^t = \sum_{\forall i \in M^t} q_i^t$
- H^t : Herfindahl-Hirschmann Index in t , where $H^t = \sum_{\forall i \in M^t} \left(\frac{q_i^t}{Q^t} \cdot 100 \right)^2$
- WMC^t : Industry marginal cost, where $WMC^t = \sum_{\forall i \in M^t} \left(\frac{q_i^t}{Q^t} \cdot c_i^t \right)$
- PCM^t : Industry price-cost margin, where $PCM^t = \sum_{\forall i \in M^t} \left(\frac{q_i^t}{Q^t} \cdot \frac{p^t - c_i^t}{p^t} \right)$

The Herfindahl-Hirschmann index, H^t , captures the concentration of the industry at any given point in time. This is important in this model because firms, in general, have asymmetric market shares that evolve over time owing to persistent entries and exits. The industry marginal cost, WMC^t , reflects the overall level of production (in)efficiency because it is the weighted sum of the marginal costs of all operating firms, where the weights are the market shares of the individual firms. To the extent that a firm that is inactive (produces zero output) has zero impact on this measure, the industry marginal cost captures the average degree of production inefficiency of the active firms. Likewise, the industry price-cost margin, PCM^t , is the market-share-weighted sum of the firms' price-cost margins. It is a measure of the industry's performance in terms of its allocative inefficiency, that is, the extent to which the market price deviates from the marginal costs of firms in operation.

20.5 RESULTS I: THE BASE MODEL WITH STABLE MARKET DEMAND

20.5.1 Firm Behavior Over Time: Technological Change and Recurrent Shakeouts

I start by examining the outcomes from a single run of the industry, given the baseline set of parameter values as indicated in table 20.1. To see the underlying source of the industry dynamics, I first assume an industry that is perfectly protected from external technological shocks—an industry with $\gamma = 0$ whose technological environment never changes.

The industry starts out empty in $t = 0$. A pool of firms considers entry into this industry given their endowed technologies. Figure 20.2(a) shows the time series of the number of entries that occur for the first sixty-eight periods of the horizon. The time series data of the number of exits is captured in figure 20.2(b). Clearly, there is a rush to get into the industry in the beginning, followed by a large number of exits. Moves into and out of the industry quickly slow down and stabilize toward zero. The interaction of entries and exits generates a time series of the number of operating firms as depicted in figure 20.2(c). It shows the existence of a shakeout in which the initial increase in the number of firms is followed by a severe reduction, ultimately converging toward about thirty firms. These results are in line with the empirical observations made by Gort and Klepper (1982), Klepper and Simons (1997), Klepper and Simons (2000a), Klepper and Simons (2000b), Carroll and Hannan (2000), Klepper (2002), and Jovanovic and MacDonald (1994). Also, consistent with Gort and Klepper (1982) and Jovanovic and MacDonald (1994), the market price in figure 20.2(d) declines gradually over time, while the industry (aggregate) output in figure 20.2(e) tends to increase. That the selective force of the market competition is the source of these patterns is shown in figure 20.2(f), where the time series of the market-share-weighted industry marginal costs falls.

With no technological change ($\gamma = 0$), the industry will eventually come to rest with no entry or exit taking place. Although it is a good benchmark, it does not provide us with an environment in which we can examine the persistent turnover patterns in the long run. For this reason, we move on to a setting in which the technological environment is subject to occasional shocks. This is accomplished in my model by setting $\gamma > 0$. In figure 20.3(a–c), I plot the number of entries, the number of exits, and the number of operating firms in each t over the entire horizon of five thousand periods when $\gamma = 0.1$ and $g = 8$, that is, in each period the technological optimum changes with a probability of 0.1 and, when it does, up to eight tasks (of ninety-six) can have their optimal methods switched from the previous ones. Contrary to the case of no technological change, there is now a series of entries and exits that persist throughout

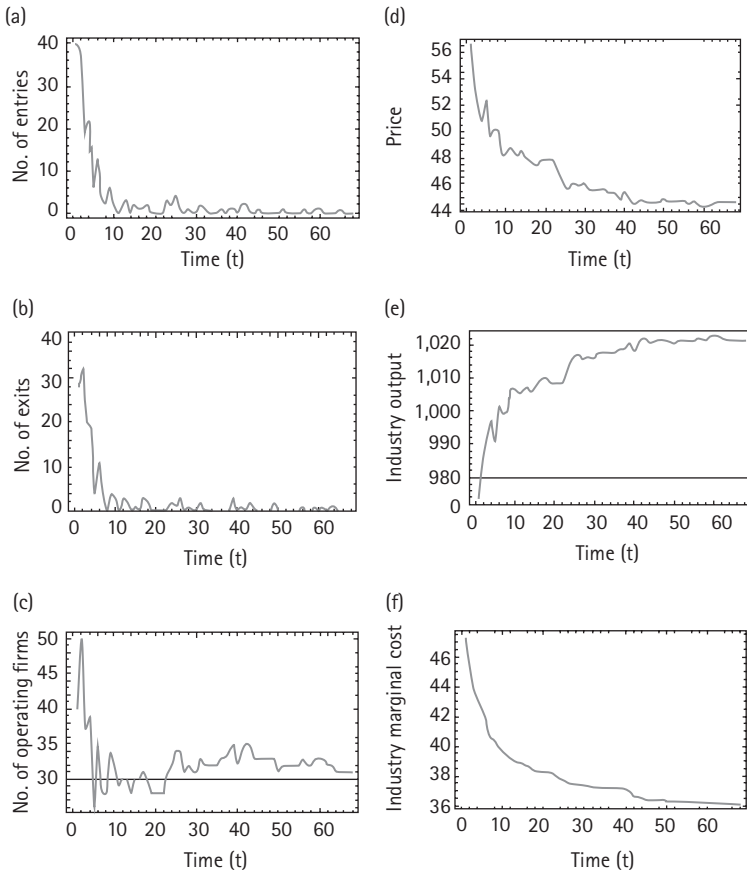


FIGURE 20.2 Shakeout in an infant industry.

the horizon. The number of operating firms, as shown in figure 20.3(c), fluctuates considerably, but it appears to move around a steady mean of about forty-three firms. Panels (d)–(f) of the figure capture the same information, the time series now reflecting the averages over one hundred independent replications of the model. For instance, denote by m_k^t the number of operating firms in period t in replication k . We would have the time series $\{m_k^t\}_{t=1}^{5,000}$ from the k th replication $\forall k \in \{1, \dots, 100\}$. Such a time series from a single replication is what is captured in figure 20.3(c). For figure 20.3(f), we perform one hundred independent replications and then compute the time series of the replication-means as $\{\frac{1}{100} \sum_{k=1}^{100} m_k^t\}_{t=1}^{5,000}$. Each of the one hundred replications used a fresh set of random numbers, although the parameter values were kept at the baseline level for all replications. Figure 20.3(f) clearly shows that after about one thousand periods the industry achieves a steady state in which the distributions of the values for the endogenous variables remain invariant over time. Thus, I consider the industry to be in the steady state for $t > 1,000$.

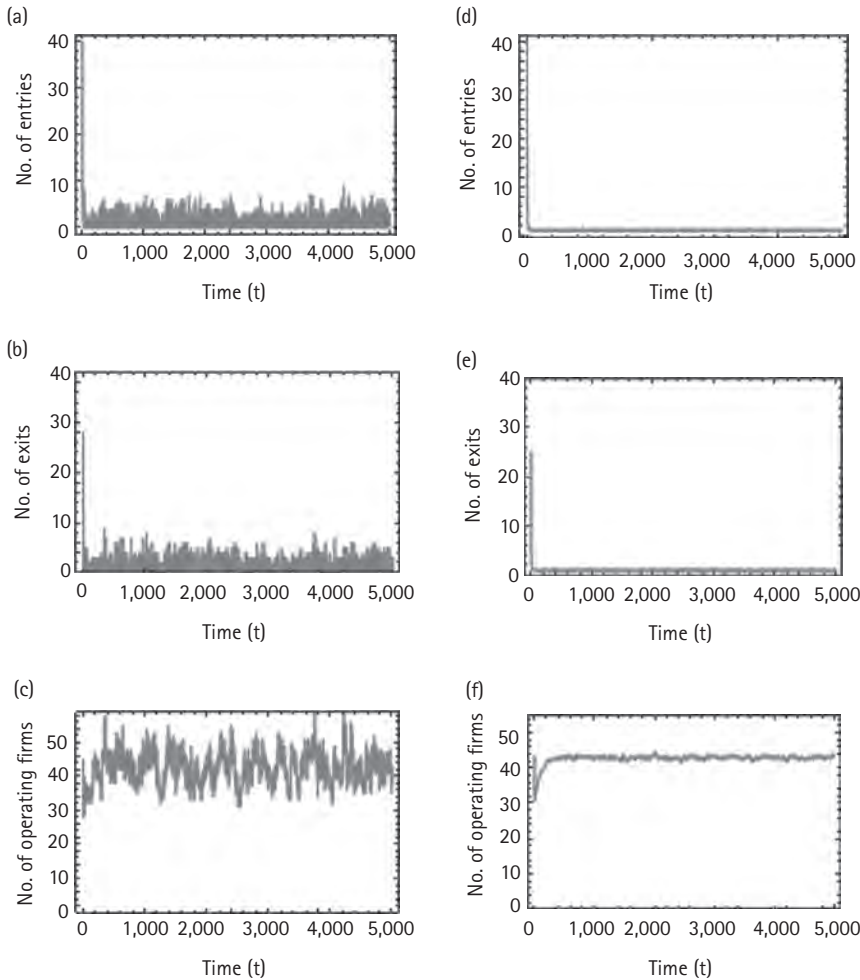


FIGURE 20.3 Firm turnovers in a turbulent technological environment. Panels (a)–(c): single replication. Panels (d)–(f): average over 100 replications.

The shakeout phenomenon identified in the case of a static technological environment (figure 20.2) was due to the opening of a new industry with myopic firms overestimating their profitability on entry. In the stochastic case, a shift in the technological environment is likely to have a similar effect, in that a new environment will invite entry by new firms while forcing some unfortunate incumbents to exit.¹⁰ In other words, with technological shocks we should see a persistent series of mini-shakeouts

¹⁰ This exit can have two causes, one direct and one indirect. The direct cause is the change in the technological environment that negatively affects the firm's production efficiency. The indirect cause is the new entrants, favored by the new environment, putting competitive pressure on the existing firms.

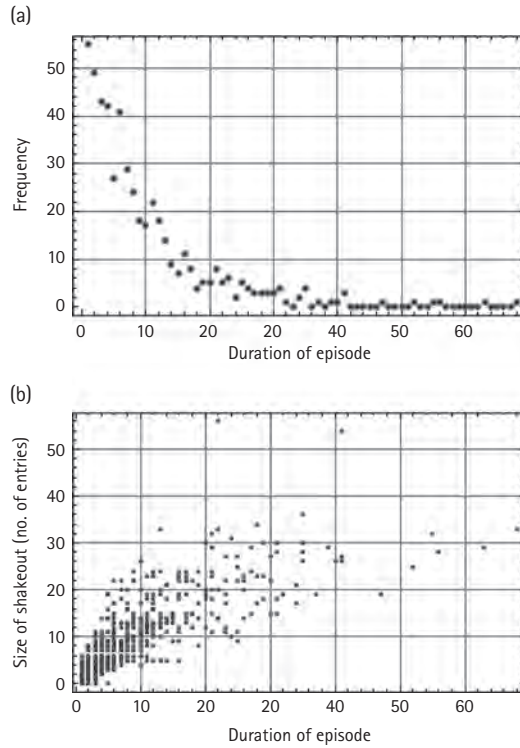


FIGURE 20.4 Episodes of technological change.

that give rise to the turnover dynamics as observed empirically. To show this, I take the benchmark replication captured in figure 20.3(a–c) and examine in detail the time series data on technological changes, that is, the timing of their occurrences. There were 514 instances of technological shift over the horizon of five thousand periods. For each occurrence, I examine how many periods of stability followed it, that is, the duration of technological stability for each episode. Figure 20.4(a) plots the frequencies of various durations of technological stability. For instance, there were fifty-five cases of technological change that only lasted one period, forty-nine cases that lasted two periods, and so on. The frequency is much higher for technological changes of shorter duration than for ones of longer duration; in fact, a log-log plot (not reported here) of frequency versus duration shows the presence of power law in this process. Figure 20.4(b) presents the relation between the duration of an episode and the size of the corresponding shakeout as measured by the total number of entries that took place over the duration of the given episode; note that the same relation exists between the duration and the total number of exits. Clearly, there is a positive correlation between them such that an episode of longer duration entails a bigger shakeout.

For each period t on the horizon, I also ask how long it has been since the last technological shift. This will tell us where the firms are in a given (mini-)shakeout. In figure 20.5(a–b) I plot for each period the numbers of entries and exits against the

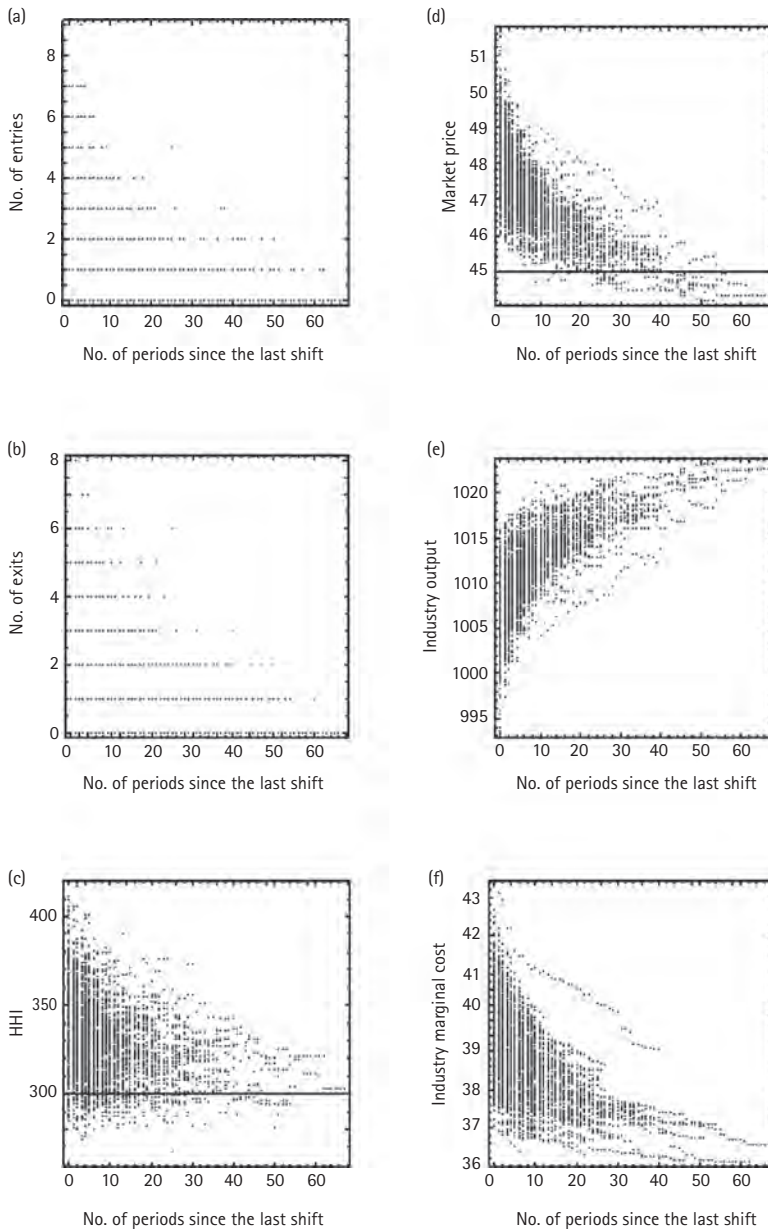


FIGURE 20.5 Time since the last technological shift and the values of endogenous variables (a) for top left plot; (b) for middle left plot; (c) for bottom left plot; (d) for top right plot; (e) for middle right plot; (f) for bottom right plot.

number of periods the industry has been in the current technological environment. Note that in the time periods for which the technological environment has been stable for a longer duration, the numbers of entries and of exits tend to be lower on average. These results suggest that the periods immediately following a technological change

should have high occurrences of entries and exits; such turnovers should diminish over time as the industry stabilizes around the new technological environment until the next technological shift. The co-movement of entry and exit, then, implies a positive correlation between entry and exit over time for any given industry.

Property 1: In any given industry, a period with a larger (smaller) number of entries is also likely to have a larger (smaller) number of exits.

Figure 20.5(c) shows that the industry tends to be more concentrated immediately following the technological change; as the industry gradually adjusts to the new environment, concentration tends to fall. Figure 20.5(d) shows that the market price generally is higher in the early periods following the technological shift and it tends to decline as the industry stabilizes. The industry output, of course, moves in the opposite direction from the price. Both of these observations are consistent with what was observed in a single shakeout case of the benchmark industry's having no technological change, as seen in figure 20.2(d–e). It is also worthwhile to point out that the decline in the market price shown in figure 20.5(d) is largely due to the selection effect of the market competition, which tends to exert downward pressure on the industry's marginal cost. This is shown in figure 20.5(f). As a given technological environment persists, inefficient firms are gradually driven out by new entrants having technologies that are better suited for the environment.

The observations presented in figure 20.5 imply that we are likely to observe certain relations between these endogenous variables. Table 20.3 reports these correlations as the average over one hundred replications. The correlations are for the steady-state time series between $t = 3,001$ and $t = 5,000$. As expected on the basis of figure 20.5, there exists positive correlation between the market price (P^t) and the number of entries (exits) as well as between the industry marginal cost (WMC^t) and the number of entries (exits). Hence, a high degree of firm turnover is likely to occur simultaneously with a high price and low production efficiency. The tendency of the high turnover to be induced by a technological shift implies that the high market price

Table 20.3 Correlations (the case of stable demand)

	$ E^t $	$ L^t $	P^t	H^t	PCM^t	WMC^t
$ E^t $	1	.377893	.229091	−.146535	−.253699	.313877
$ L^t $		1	.178438	−.123290	−.207645	.249140
P^t			1	.324510	−.094246	.872944
H^t				1	.909583	−.174888
PCM^t					1	−.565350
WMC^t						1

is mainly due to the productive inefficiency of firms faced with a new technological environment.

The number of entries and the industry price-cost margin (PCM^t) are negatively correlated. Hence, there is a (strong) negative correlation between the industry marginal cost and the industry price-cost margin such that the period in which the firms are relatively efficient tends to generate a relatively high price-cost margin for the industry and vice versa. An industry that has been in technological tranquility for a longer period of time is, then, more likely to have greater production efficiency and a higher price-cost margin.

Although the industry concentration (H^t) is negatively correlated with the number of entries, it is positively correlated with the market price and the industry price-cost margin: The period of high concentration is likely to show high market price and high price-cost margins. These correlations were for the baseline parameter values, but when they are computed for all $s \in \{4, 6, 8, 10\}$ and $f \in \{100, 200, 300, 400\}$, the results (not reported here) confirm that these relations are robust to varying the values of the two parameters.

20.5.2 Comparative Dynamics Analysis: Implications for Cross-sectional Studies

There are two parameters, s and f , that affect the demand and cost conditions of the firms in the industry and two parameters, γ and g , that affect the volatility of the technological environment of the firms. The main objective in this section is to examine how these parameters, mainly s and f , affect the long-run development of an industry. The approach is to run five hundred independent replications for each parameter configuration that represents a particular industry. For each replication, I compute the steady-state values (average values over the periods from 3,001 to 5,000) of the endogenous variables and then average them over the five hundred replications.¹¹ The resulting mean steady-state values for the relevant endogenous variables are denoted as follows: $\bar{H}$, $\bar{P}$, $\bar{WMC}$, and $\bar{PCM}$. For instance, $\bar{H}$ is computed as

$$\bar{H} = \left(\frac{1}{500} \right) \sum_{k=1}^{500} \left(\frac{1}{2,000} \right) \sum_{t=3,001}^{5,000} H_k^t \quad (20.16)$$

where H_k^t is the HHI in period t from replication k .

Note that the steady-state number of operating firms is likely to vary as a function of the parameters s and f . Whereas the absolute numbers of entries and exits are sufficient for capturing the degree of firm turnovers for a *given* industry, they are not adequate when we carry out comparative dynamics analysis with implications for cross-industry

¹¹ The number of replications is raised from one hundred to five hundred for the analysis of steady states in this section in order to reduce as much as possible the variance in the distribution of the steady state-values (of endogenous variables).

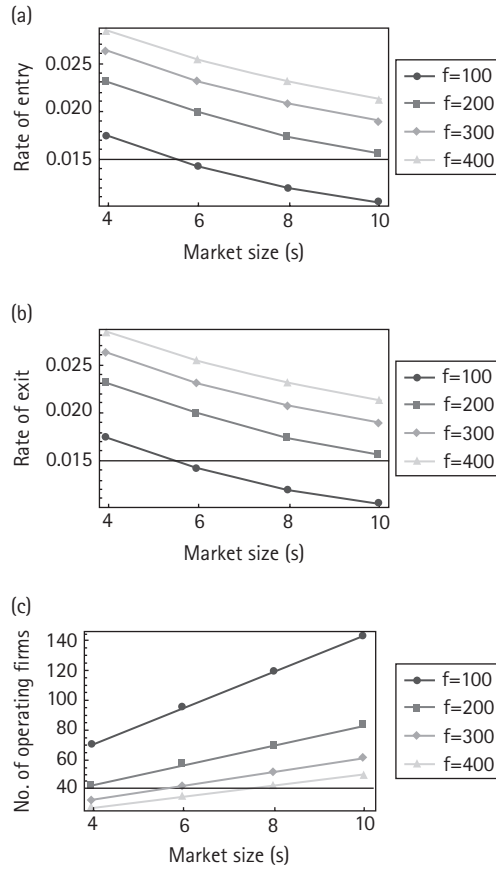


FIGURE 20.6 Firm turnovers in steady state.

differences in firm behavior. In this section, I use the rates of entry and exit to represent the degree of turnovers as follows:

- ER^t : Rate of entry in t , where $ER^t = \frac{|E^t|}{|M^t|}$
- XR^t : Rate of exit in t , where $XR^t = \frac{|L^t|}{|M^t|}$

These time series are again averaged over the five hundred independent replications, and the resulting mean steady-state values are denoted as $\overline{ER}$ and $\overline{XR}$.

We start the analysis by first looking at the above endogenous variables for various combinations of the demand and cost parameters s and f , given $\gamma = 0.1$ and $g = 8$. I plot in figure 20.6 the steady-state values of (a) the rate of entry, (b) the rate of exit, and (c) the number of operating firms for all $s \in \{4, 6, 8, 10\}$ and $f \in \{100, 200, 300, 400\}$. They show that both $\overline{ER}$ and $\overline{XR}$ decline with the size of the market, s , but increase with the fixed cost, f . Since the two rates move in the same direction in response to the changes

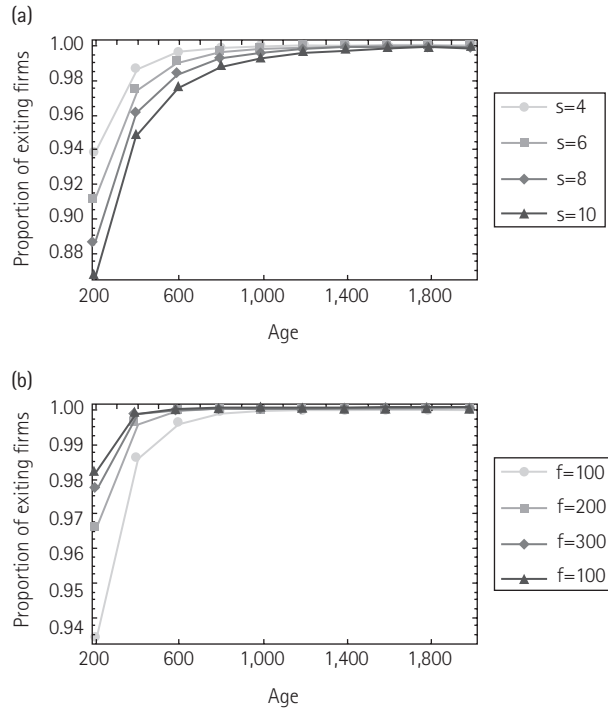


FIGURE 20.7 Firm age at the time of exit.

in the two parameters, I will say that the rate of firm turnover increases or decreases as either of these rates goes up or down, respectively.

Closely related to the rates of entry and exit is the age distribution of the dying firms. In order to investigate the relation between the rate of turnovers and the severity of infant mortality, I collected the age (at the time of exit) of every firm that exited the industry between $t = 3,001$ and $t = 5,000$. The proportion of those firms that exited at a given age or younger (of all exiting firms) was then computed and averaged over the five hundred independent replications for each parameter configuration of s and f . The results are plotted in figure 20.7. Panel (a) shows the cumulative proportions for varying market size, $s \in \{4, 6, 8, 10\}$, and panel (b) shows the same for varying values of the fixed cost, $f \in \{100, 200, 300, 400\}$. As shown, a larger proportion of exiting firms exit at a younger age in industries having a smaller market size or a larger fixed cost.

Property 2: (a) The steady-state rate of firm turnovers is *negatively* related to the size of the market and *positively* related to the size of the fixed cost. (b) The steady-state proportion of exiting firms that are of a given age or younger at the time of exit is *negatively* related to the size of the market and *positively* related to the size of the fixed cost.

Hence, both the rate of turnover and the degree of infant mortality are higher in markets of smaller size or higher fixed cost. Not surprisingly, it is also found that the rate of turnovers and the degree of infant mortality are both higher when γ , the rate of change in the technological environment, is greater: A more volatile technological environment induces more frequent reversal of fortune for the firms leading to a higher rate of turnover and a lower rate of survival for the firms. Because these properties are intuitively straightforward, I omit the plots showing these results.

Property 2 also implies that the steady-state rate of firm survival, $1 - \overline{XR}$, increases with market size and decreases with fixed cost. As expected, the number of operating firms in the steady state increases with the market size and decreases with the fixed cost, as shown in figure 20.6(c). This implies that the industry concentration, $\overline{H}$, should decrease with the market size and increase with the fixed cost. This result is shown in figure 20.8(a). Differences in the concentration reflect differences in the steady-state

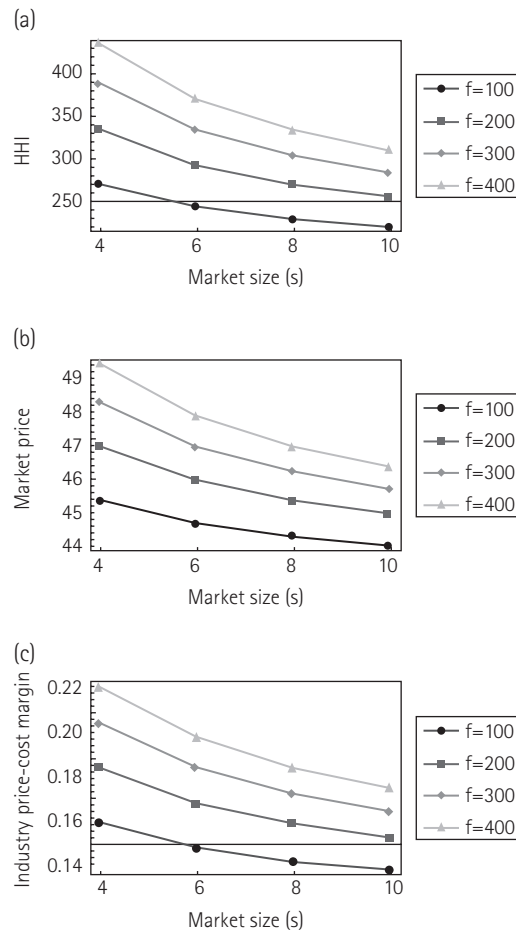


FIGURE 20.8 Industry structure and performance in the steady state.

degree of market competition. This leads to the steady-state market price as well as the industry price-cost margin decreasing with the market size and increasing with the fixed cost of production, as shown in figure 20.8(b–c).

Note that the absolute numbers of entries and exits (not reported here) actually *increase* with the size of the market and *decrease* with the fixed cost. Because the number of operating firms increases with the size of the market and decreases with the fixed cost by a larger proportion, however, the rates (which divide the numbers of of firms entering and exiting by the number of operating firms) tend to *decrease* with the market size and *increase* with the fixed cost. The increase in the number of operating firms that comes with larger market size and lower fixed cost, then, intensifies market competition, exerting downward pressure on the market price as well as the price-cost margin. Hence, the *price competition effect* assumed in Asplund and Nocke (2006) is endogenously generated in my model.

Note that the effect of market size on the market price or the price-cost margin is due to the relative strength of the two countervailing forces: the direct positive impact on the profits from larger demand and the indirect negative impact via the price competition effect. The negative relation between the market size and the price (or the price-cost margin) thus implies that the price competition effect dominates the demand effect; this feature is absent in the competitive market model of Hopenhayn (1992) but assumed as part of the imperfect competition model in Asplund and Nocke (2006).

The impact of fixed costs on the turnover rate is the same in both the Asplund and Nocke model and this model, but the impact of the market size on the turnover rate is not: Asplund and Nocke (2006) predict the impact to be positive, whereas my model predicts it to be negative. It is difficult to pinpoint the exact source of the discrepancy, since the two models are substantially different in terms of both how demand and technology are specified and how the degree of firms' foresight is modeled (perfect foresight myopia). What is clear, however, is the chain of endogenous relationships that are affected by the way the demand function is specified. To be specific, note that the entry decisions (and consequently the rate of turnovers) are based on the post-entry expected profits, computed as a present value of discounted profits under perfect foresight and as a single-period profit under myopia. In both cases, main determinants of the post-entry profit are the price and price-cost margin, the levels of which depend on the magnitude of the price competition effect identified above. Because the magnitude of the price competition effect is likely to be influenced by the shape of the demand curve (price elasticity of demand, to be specific), the relation between the market size and the rate of turnovers may depend on the precise specification of the demand function. Although this is an important issue that deserves a thorough investigation, it is beyond the scope of the present study and is left for future research.

An important property that emerges from this comparative dynamics analysis is the relation between the endogenous variables such as the rate of firm turnover, industry concentration, market price, and the industry price-cost margins. As can be inferred from figure 20.6 and figure 20.8, all these variables tend to decrease with the market

size and increase with the fixed cost. The following property emerges from these relations.

Property 3: An industry with a high turnover rate is likely to be concentrated and have a high market price and high price-cost margins.

Hence, if one carries out a cross-sectional study of these endogenous variables across industries having different market sizes and fixed costs, such a study is likely to identify positive relations between these variables; that is, the market price, as well as the industry price-cost margin, is likely to be higher in more concentrated industries (those with a smaller market size or a larger fixed cost). Finally, I find that the properties identified in figures 20.6 through 20.8 are robust to all $\gamma \in \{0.1, 0.2, 0.3, 0.4\}$.

20.6 RESULTS II: THE BASE MODEL WITH FLUCTUATING MARKET DEMAND

How do cyclical variations in market demand affect the evolving structure and performance of an industry? Is the market selection of firms more effective (making firms more efficient on average) when there are fluctuations in demand? What are the relations between the movements of demand and those of endogenous variables such as industry concentration, price, aggregate efficiency, and price-cost margins? Are they pro-cyclical or countercyclical? The proposed model of industry dynamics can address these issues by computationally generating the time series of these variables in the presence of demand fluctuation.

These issues have been explored in the past by researchers from two distinct fields of economics: macroeconomics and industrial organization. A number of stylized facts have been established by the two strands of empirical research. For instance, many papers find pro-cyclical variations in the number of competitors. Chatterjee et al. (1993) find that both net business formation and new business incorporations are strongly pro-cyclical. Devereux et al. (1996) confirm this finding and further report that the aggregate number of business failure is countercyclical. Many researchers find that markups are countercyclical and negatively correlated with the number of competitors; see Bils (1987), Cooley and Ohanian (1991), Rotemberg and Woodford (1990), Rotemberg and Woodford (1999), Chevalier and Scharfstein (1995), Warner and Barsky (1995), MacDonald (2000), Chevalier et al. (2003), and Wilson and Reynolds (2005).¹² Martins et al. (1996) cover different industries in fourteen OECD countries and find markups to be countercyclical in fifty-three of the fifty-six cases

¹² A deviation from this set of papers is Domowitz et al. (1986), who suggested that markups are pro-cyclical. Rotemberg and Woodford (1999) highlight some biases in these results, as Domowitz et al. use measures of average variable costs and not marginal costs.

they consider, with statistically significant results in most. In addition, these authors conclude that entry rates have a negative and statistically significant correlation with markups. Bresnahan and Reiss (1991) find that an increase in the number of producers increases the competitiveness in the markets they analyze. Campbell and Hopenhayn (2005) provide empirical evidence to support the argument that firms' pricing decisions are affected by the number of competitors they face; they show that markups react negatively to increases in the number of firms. Rotemberg and Saloner (1986) provide empirical evidence of countercyclical price movements and offer a model of collusive pricing when demand is subject to IID shocks. Their model generates countercyclical collusion and predicts countercyclical pricing.¹³

The model presented here has the capacity to replicate many of the empirical regularities mentioned above and explain them in terms of the selective forces of market competition in the presence of firm entry and exit. In particular, it can incorporate demand fluctuation by allowing the market size s to shift from one period to the next in a systematic fashion. As a starting point, the market size parameter can be shifted according to a deterministic cycle such as a sine wave. Section 20.6.1 investigates the movement of the endogenous variables with this demand dynamic. In section 20.6.2, I allow the market size to be randomly drawn from a fixed range according to uniform distribution. Using the correlations between the time series of the market size and that of several endogenous variables, I study the impact of demand fluctuation on the evolving structure and performance of the industry.

20.6.1 Cyclical Demand

I run one hundred independent replications of the base model with the baseline parameter configuration. The only change is in the specification of the market size s . I assume that it stays fixed at the baseline value of 4 for the first two thousand periods and then follows a deterministic cycle as specified by the rule

$$s^t = \hat{s} + s_a \cdot \sin \left[\frac{\pi}{\tau} \cdot t \right] \quad (20.17)$$

for all $t \geq 2,001$, where $\hat{s}$ is the mean market size (set at 4), s_a is the amplitude of the wave (set at 2), and τ (set at 500) is the period for half-turn (hence, one full period equals 2τ).

Note that the demand fluctuation is not introduced until $t = 2,001$. This is to ensure that the industry reaches its steady state before demand fluctuation occurs. Given the assigned values of $\hat{s}$, s_a , and τ , the market size then fluctuates between the maximum of 6 and the minimum of 2 with a full cycle of 1,000 periods. In examining the evolution

¹³ See Haltiwanger and Harrington Jr. (1991) and Kandori (1991) for further support. In contrast, Green and Porter (1984) develop a model of trigger pricing that predicts positive co-movements of prices and demand.

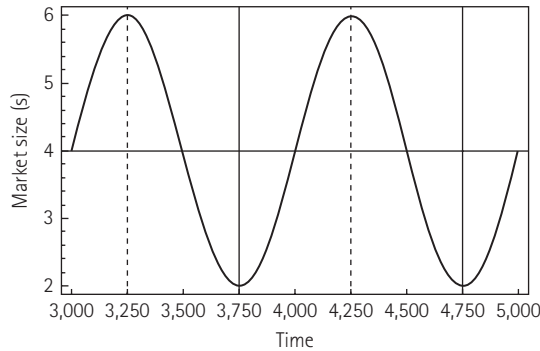


FIGURE 20.9 Deterministic demand cycle.

of the industry in the midst of fluctuating demand, I focus on the last 2,000 periods from $t = 3,001$ to $t = 5,000$. The demand cycle during this time period is shown in figure 20.9. The points at which the market size reaches its maximum are indicated by the dotted lines at $t = 3,251$ and $t = 4,251$, whereas the points at which the market size reaches its minimum are indicated by the solid lines at $t = 3,751$ and $t = 4,751$. I use the same indicator lines when analyzing the movements of the endogenous variables.

Recall that one hundred independent replications are performed. Each replication generates the time series values for the relevant endogenous variables. For each endogenous variable, I then average its time series values for the one hundred replications.

The number of entries and the number of exits during the relevant horizon are captured in figure 20.10(a–b).¹⁴ Both numbers tend to move together in a pro-cyclical pattern, though the exit cycle tends to lag slightly behind the entry cycle. This indicates that the degree of turnover is stronger during a boom than during a bust. The number of net entries—that is, the number of entries minus the number of exits—is presented in figure 20.10(c). It is generally positive during the upswing, or the periods between step 3,751 and step 4,250, and negative during the downswing of the market, or the periods between step 3,251 and step 3,750. This generates the *pro-cyclical* movement in the number of operating firms, as shown in figure 20.11(a).

Property 4: The number of entries, the number of exits, and the number of operating firms are pro-cyclical.

The aggregate output at the industry level, as shown in figure 20.11(b), follows the movement of market demand s perfectly, while the market price follows the countercyclical whereas path as shown in figure 20.11(c).

¹⁴ Because we are focusing on the intertemporal movement for a *given* industry, we use the numbers (rather than the rates) of entry and exit.

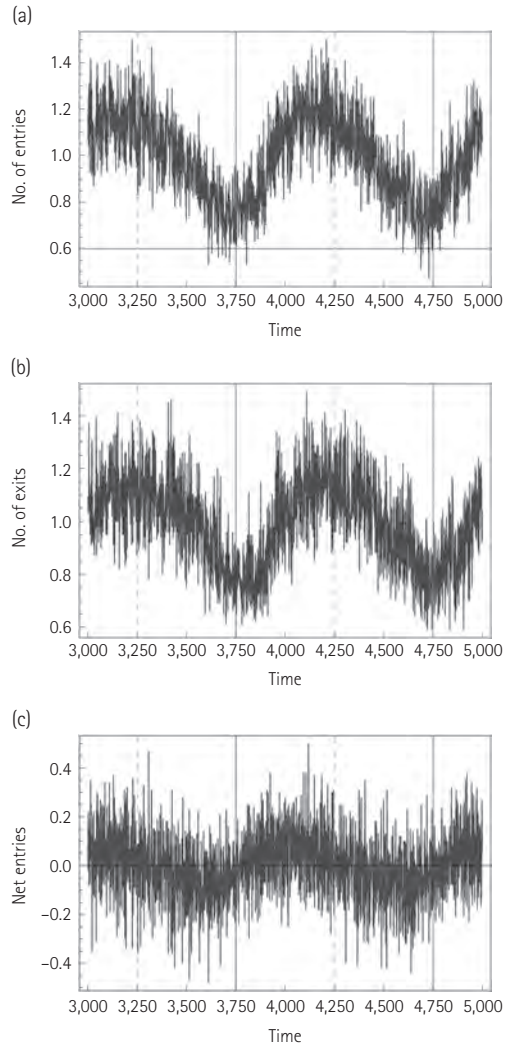


FIGURE 20.10 Turnover of firms with cyclical demand.

Figure 20.12 tracks the movement of other structural and performance variables for the industry. Fully in line with the pro-cyclical number of firms, the industry concentration (HHI) is countercyclical. The industry marginal costs do not show any recognizable pattern, but the industry price-cost margin exhibits countercyclicity. Given the relative steadiness of the industry marginal costs, it is clear that the countercyclical price-cost margin implies countercyclical market price. The time series plot for the market price, therefore, is omitted here.

Property 5: The industry concentration is countercyclical.

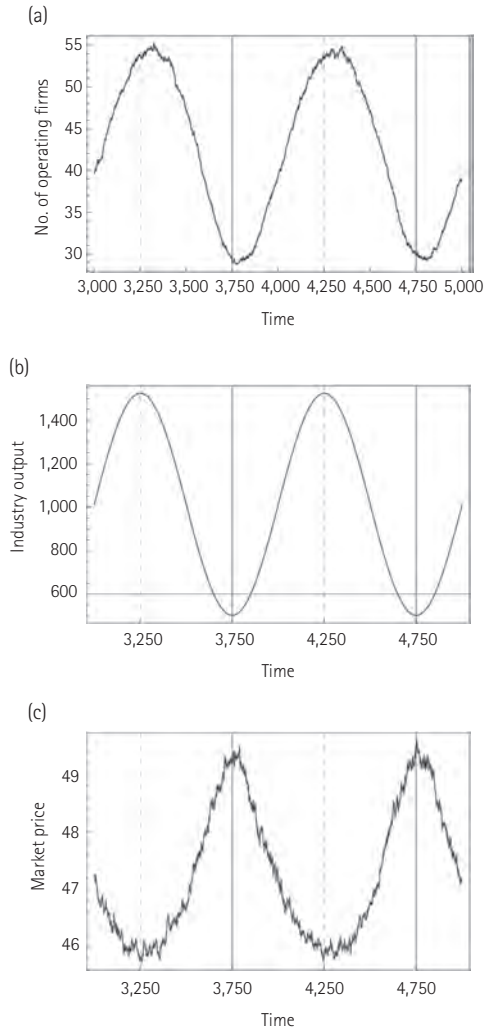


FIGURE 20.11 Evolving industry structure with cyclical demand.

Property 6: The market price and the industry price-cost margin are countercyclical.

The aggregate profit for the industry, though not quite pro-countercyclical, shows that it tends to be positive during the upswing of market demand and negative during the downswing. The negative industry profit during the downswing is due to there being many inactive firms that suffer economic losses in the form of fixed costs.

To summarize, we observe that the level of entries and exits, the number of firms, and the industry output are pro-cyclical, whereas market price, industry concentration, and industry price-cost margin are countercyclical.

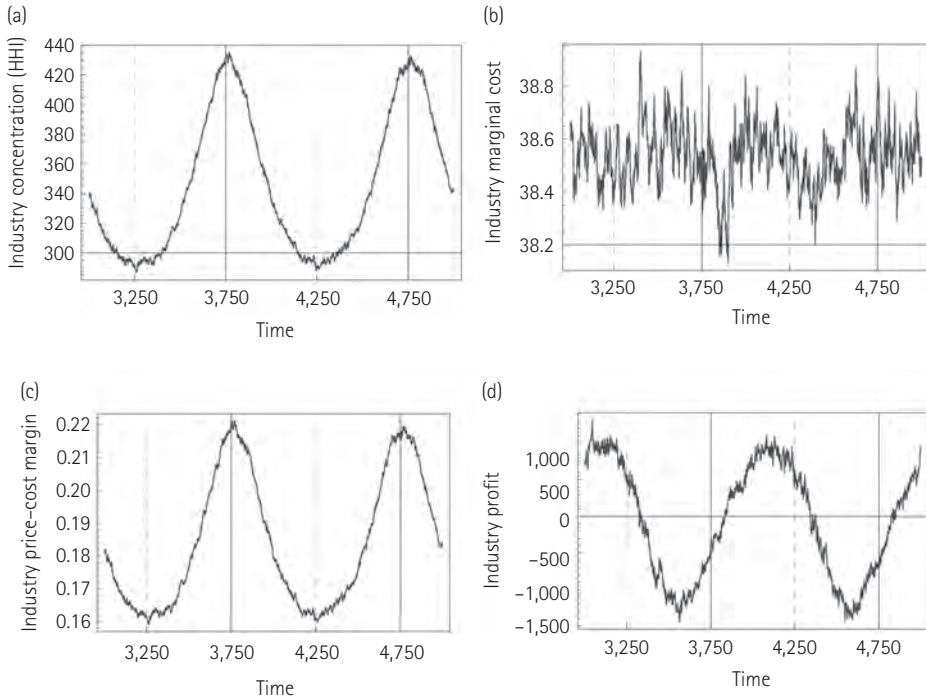


FIGURE 20.12 Industry-level endogenous variables with cyclical demand.

20.6.2 Stochastic Demand

I now replace the deterministic demand cycle for the periods $t = 2,001$ to $t = 5,000$ with a stochastic demand. Starting from $t = 2,001$, let the market size s^t take a random value from the range of $[2, 6]$ according to uniform distribution. Hence, the mean value is still at 4, but the market size can randomly fluctuate anywhere between 2 and 6. Again, we focus on the time series values for the last 2,000 periods from $t = 3,001$ to $t = 5,000$. The relation between the size of the market and the endogenous variables is not going to exhibit clear patterns, as it did in the case of the deterministic demand cycle. However, we expect the same type of relations to exist here if we examine the correlations between the market size and those variables, that is a positive correlation for pro-cyclicality and a negative correlation for countercyclicality.

Table 20.4 presents the correlations between the market size s and each of the endogenous variables. First consider the turnover variables: the number of entries, the number of exits, and the number of operating firms. Note that the size of the market s^t is positively correlated with the number of entries $|E^t|$ and negatively correlated with the number of exits $|L^t|$. It is also positively correlated with the number of operating firms $|M^t|$. The positive correlation between the market size and the number of entries is consistent with the pro-cyclicality found for the case of a deterministic demand cycle.

Table 20.4 Correlations (the case of stochastic demand)

	s^t	$ E^t $	$ L^t $	$ M^t $	P^t	Q^t	Π^t	PCM^t	WMC^t	HHI^t
s^t	1	.3815	-.1706	.1153	-.0838	.9999	.8695	-.1250	-.0043	-.1522
$ E^t $		1	.2513	.3372	.1757	.3779	.1866	-.2825	.2892	-.2003
$ L^t $			1	.3398	.1410	-.1727	-.3173	-.2390	.2380	-.1718
$ M^t $				1	.1426	.1129	-.3469	-.5743	.4113	-.4886
P^t					1	-.0988	-.0848	-.0788	.8587	.3190
Q^t						1	.8696	-.1235	-.0174	-.1567
Π^t							1	.2927	-.2193	.2441
PCM^t								1	-.5766	.9185
WMC^t									1	-.2093
HHI^t										1

However, the negative correlation between the market size and the number of exits is not in line with the result from the case of deterministic cyclical demand. This may be due to the lagged nature of the exits, as we observed in figure 20.10.

With respect to other variables, the market size is almost perfectly correlated with the industry output Q^t , but only weakly correlated with the market price P^t . It also has a strongly positive correlation with the industry profit Π^t and negative correlation with both the industry price-cost margin PCM^t and the industry concentration H^t . These results are consistent with what we observed in the case of deterministic demand cycle. Note that the correlations between any two of the endogenous variables are fully consistent with what we observed in the case of stable market demand, hence providing us with a robustness check on those results.

Finally, I ask how the presence of fluctuating demand itself affects the industry. Does it increase the turnover of firms? How does it affect the structure and performance of the industry along the steady state? In order to address these questions, I take the steady-state mean of the time series values for each endogenous variable, where the steady state is specified as the last 2,000 periods from $t = 3,001$ to $t = 5,000$. I do this for each of the one hundred independent replications under the two separate assumptions about the market size: when the demand is stable (as in section 20.5), and when the demand is stochastic (as in this section).

In figure 20.13(a–b) I plot histograms of the one hundred observations of the steady-state mean numbers of entries and exits with and without demand fluctuation. The dark bars show results with fluctuations and the white bars show results without fluctuations. The gray sections represent the overlaps between the two histograms. The plots show that the numbers of entries and exits both tend to be higher with fluctuation than without. *Hence, the fluctuation in market demand tends to raise the degree of firm turnovers.* Figure 20.13(c) shows that the number of operating firms is generally higher

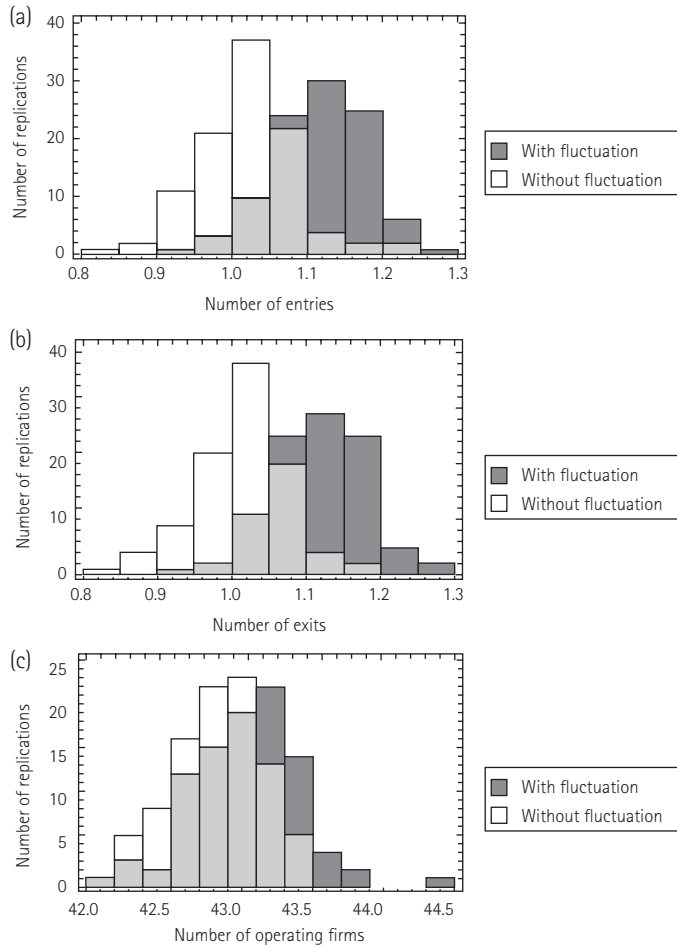


FIGURE 20.13 Impact of demand fluctuation on turnover.

with fluctuation; this is likely to be due to the higher number of inactive firms that results from increased turbulence in the market.

In figure 20.14 I plot similar information concerning other structural and performance variables. Both market price and industry price-cost margin tend to be higher with demand fluctuation. The industry marginal cost is also likely to be higher with fluctuation, thereby implying a loss of efficiency that results from the fluctuation in market demand. Finally, the industry is more concentrated with demand fluctuation. It is interesting that the presence of demand fluctuation raises both the number of operating firms and the industry concentration. This seeming contradiction is due to the fact that the set of operating firms includes active and inactive firms. The presence of fluctuating demand, while it increases the number of inactive firms (and the number of operating firms), actually reduces the number of active firms. Since the industry

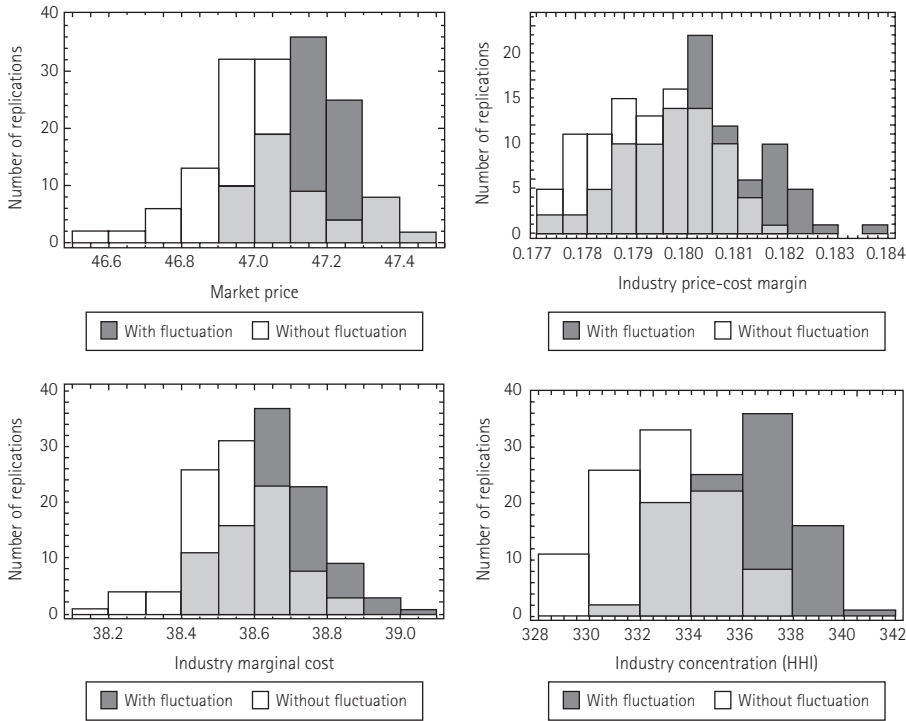


FIGURE 20.14 Impact of demand fluctuation on industry structure and performance.

concentration takes into consideration the market shares of the active firms only, it is likely to increase as demand fluctuation is introduced into the market.

20.7 POTENTIAL EXTENSION: ENDOGENIZING R&D

Variants of the base model, as described here, have been used in two earlier publications, Chang (2009) and Chang (2011). In both papers, an extra stage of decision making, that for R&D, was added to the base model. Specifically, the R&D stage was inserted in between stage 1 (entry) and stage 2 (competition). Figure 20.15 shows the overall structure.

In Chang (2009) and Chang (2011), the R&D activity was assumed to be exogenous and costless.¹⁵ More specifically, R&D was viewed as serendipitous discovery in

¹⁵ Note that many of the results obtained in the base model with the stable demand also hold with these extensions. That they hold in the base model without the R&D activities implies that it is the

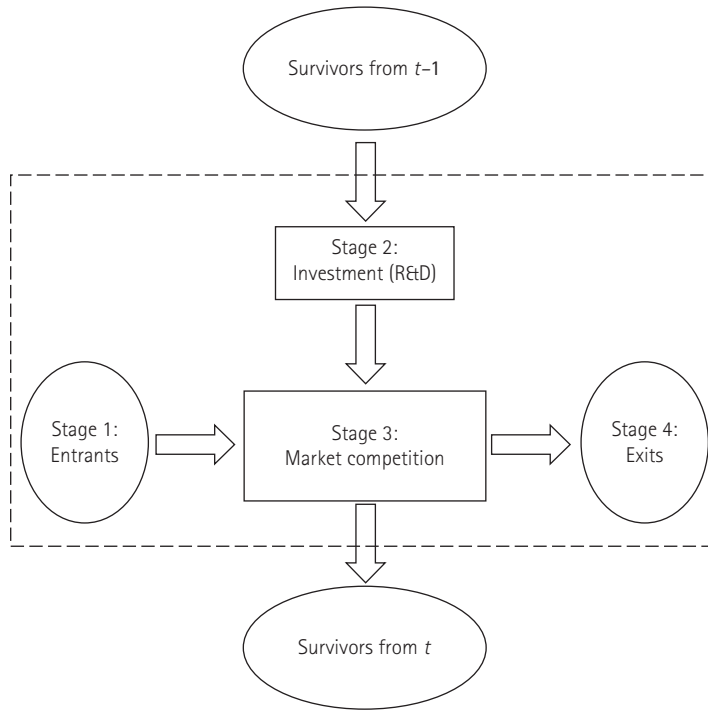


FIGURE 20.15 Base model of industry dynamics with R&D.

which the methods used in one or more of the tasks were randomly altered for experimentation. Chang (2009) assumes a stable technological environment in which the optimal technology did not change from one period to the next, that is, $\gamma = 0$. Instead, the technology itself was assumed to be complex in nature such that there were multiple optima on which a firm could converge. The main focus was on the determinants of the shakeout phase of an industry's life cycle. Chang (2011) allows turbulence in the technological environment, as I do, that is, $\gamma > 0$. The focus was on the steady state in which continual series of entries and exits were observed.

A more satisfying approach is to *endogenize* the process of R&D in the base model of industry dynamics. Let me sketch one possible way to pursue this extension. [An attempt in this direction, as described here, has been made and reported in Chang (2015).] Suppose we assume that the R&D-related decisions of a firm are driven by a pair of choice probabilities, α_i^t and β_i^t , that evolve over time on the basis of a reinforcement learning mechanism. Specifically, a firm i chooses to pursue R&D in period t with probability α_i^t . If it chooses not to do so (which happens with probability $1 - \alpha_i^t$), its technology stays the same as the previous period's. If a firm decides to pursue R&D, it

market selection mechanism, and not the firm-level adaptation, that is central to the emergent patterns in the non-equilibrium industry dynamics.

can do so through either *innovation* or *imitation*. The firm chooses innovation mode with probability β_i^t and imitation mode with probability $1 - \beta_i^t$. The firm incurs an R&D expenditure, the size of which depends on which of the two modes it chooses. The R&D intensity and the innovation-to-imitation intensity at the firm level can be inferred from the time paths of these two probabilities. Similar intensities at the industry level can be inferred from the aggregate expenditure for the overall R&D as well as that for innovation or imitation.

Modeling R&D as an endogenous search within the finite technology space allows for precise representation of the firms' knowledge acquisition process; it also captures the possibility of two different firms' pursuing two distinct paths of technological development that may prove equally adaptive over time. Jovanovic and Rob (1990) also viewed innovation as a search within a well-defined technology space. In their model, a firm could engage in two types of search: an intensive search and an extensive search. The former focuses on searching within a given technology type, whereas the latter encompasses searching across different types. Given that the number of tasks in my model is known and held fixed, R&D in this setting only involves the intensive form—searching among multiple methods for each well-defined task.

Incorporating R&D into the base model opens up a number of issues. First, it allows us to examine the two Schumpeterian hypotheses: (1) innovation activity is more intense in large than in small firms and, hence, *firm size matters*; (2) innovation activity is more intense in concentrated than in unconcentrated industries and, hence, *industry structure matters*. Given the process of *creative destruction* as envisioned by Schumpeter, both firm size and industry structure co-evolve with R&D activities of the firms over the course of the industrial development. The model presented here can explore such evolving relationships in sufficient detail that the two hypotheses can be tested in a fully dynamic framework.

The second issue of interest is to relate the empirical regularities concerning firm turnovers to the endogenous R&D intensities. The point is that the persistence of entry and exit by firms, as noted in this chapter regarding the base model, indicates the underlying instability in the cost positions of the firms. Such instability may be due to the R&D activities of the firms as well as their changing fortunes in the competitive and turbulent market environment. What are the theoretical connections between the degree of turnovers, the intensities of R&D activities, and the degree of technological turbulence (as captured by γ or g) in industries that are differentiated in terms their structural characteristics (for example, s and f)? Geroski (1995, p. 431) states: "High rates of entry are often associated with high rates of innovation and increases in efficiency. ... Numerous case studies have suggested that entry is often used as a vehicle for introducing new innovations (frequently because incumbents are more interested in protecting existing rents than in seeking out new profit opportunities), and many show that entry often encourages incumbents to drastically cut slack from their operations." Lying underneath such a relation are the unpredictable external shocks to the technological environment surrounding the firm. The base model with endogenous R&D should enable us to perform a detailed analysis of the way γ

and g affect the endogenous relation between the turnover rate and the level of aggregate R&D.

20.8 CONCLUDING REMARKS

I proposed a computational model of industry dynamics in which a population of firms could interact with one another via market competition. The entries and exits of firms are generated endogenously in this model, giving us an opportunity to study the historical patterns in the turnover dynamics for industries having different demand and technological conditions. The *generative* approach taken here allows a detailed study of the historical path an industry may take from birth to maturity. The results obtained from the model could be compared to those from the voluminous empirical literature about time series and cross-sectional studies of industrial organization, providing substantive understanding of the evolutionary dynamics of industries. Although there have been other attempts to explain some of the stylized facts using standard analytical approaches, the agent-based computational approach taken here makes a unique contribution by generating the out-of-equilibrium industry dynamics with a realistically large number of firms, potentially providing a better fit to empirical data.

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CHAPTER 21

AGENT-BASED MODELING FOR FINANCIAL MARKETS

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21.1 INTRODUCTION

An agent-based model (ABM)¹ is a computational model that can simulate the actions and interactions of individuals and organizations in complex and realistic ways. Even simple ABMs can exhibit complex behavioral patterns and provide valuable information about the dynamics of the real-world system that they emulate. Such models are not limited by the numerous restrictive and empirically problematic assumptions underlying most mainstream economic models. They can create emergent properties arising from complex spatial interaction and subtle interdependencies between prices and actions, driven by learning and feedback mechanisms. They replace the theoretical assumption of mathematical optimization by agents in equilibrium by explicit agents with bounded rationality adapting to market forces.

Many approaches have been adopted in modeling agent behavior for financial ABMs. Agents can range from passive automatons with no cognitive function to active data-gathering decision makers with sophisticated learning capabilities. Indeed, agents might not only be heterogeneous and interacting but adaptive; they can have different states and different histories and adapt continuously to the overall economy. Agents can engage in comprehensive forms of learning that include inductive reasoning (experimentation with new ideas) as well as reinforcement learning, imitation, and forecasting. When agent interaction is contingent on experience, analytical approaches may be very limited in their ability to predict outcomes.

Traditional economic models can get reasonably good insights into economic scenarios in which the assumptions that human behavior leads to stable, self-regulating

¹ From now on we use the abbreviation ABM for both “agent-based model” and “agent-based modeling”; it should be clear from the context which is intended.

markets and prices never depart far from equilibrium are realistic. But although a dynamic system may have an equilibrium state, its basin of attraction can be very narrow, and the system may rarely settle there. The equilibrium might also be very sensitive to small perturbations, and therefore it becomes less relevant for the understanding of the system. The highly stylized, analytically tractable traditional models in economics and finance are not well suited to study crises (Bouchaud 2008; Farmer and Foley 2009; Kirman 2010); in particular, because of its focus on equilibrium solutions there is no framework in classical economics for the understanding of crises. By contrast, ABMs can represent unstable systems with crashes and booms that develop from nonlinear responses to proportionally small changes.

Economists have developed powerful tools to understand the role of strategic interaction among a limited number of agents, but the social settings in which economic activity is embedded were largely ignored by the economic profession until the early 1990s. Since then the study of socioeconomic networks has exploded, with the main focus on the development of models of strategic network formation. The underlying assumption in these models is that the payoffs to each individual provide the incentives to form or sever links and can form the basis for welfare evaluation (comparison of outcomes). By focusing on the optimal behavior of agents in forming links, these theoretical models provide useful insights into why certain network structures emerge. Nonetheless, the networks that emerge as stable or efficient are too simple (for example, star networks) and rarely observed in reality. Thus, these models are not well suited in terms of matching the properties of observed large social networks, with considerable structural heterogeneity. As suggested by Jackson (2007), agent-based simulation could be a valuable tool to study more realistic network formation models that could capture more node heterogeneity and randomness in behavior, issues of particular importance for understanding financial markets.

There has been significant popular coverage of the potential of agent-based models for preventing financial crises and better understanding the economy. The *Economist*, in *Economist* (2010), argues that ABMs might do better than conventional economic models such as dynamic stochastic general equilibrium models in foreseeing financial crises. In an interview in *Institutional Investor* Farmer (2012) advocates the use of ABM simulations for understanding the economy and financial markets as complex, evolving systems. Buchanan (2009) asks whether, in analogy to traffic forecasting models, it may be possible to build a control center (or “war room”) for financial markets where policy makers could be alerted to potential crises and run appropriate simulations in order to understand how best to respond to events. Regulators and policy makers (Trichet 2010; Haldane 2009) have also been calling for novel approaches and tools to monitor the state of the economy, tools that recognize its complex and interconnected nature.

Despite the widespread interest in ABM approaches, agent-based models remain at the fringe of mainstream economics. Some critics argue that ABMs are too narrow in focus. Agent-based modeling in financial markets has devoted a lot of attention to providing a behavioral explanation for a number of universally observed facts (or

stylized facts) of financial time series that were inconsistent with standard asset-pricing models and, though ABM has achieved considerable success in this area, there has been less engagement with the more general topics of interest to the traditional financial market research communities.

Another criticism of agent-based modeling is the lack of clarity about how one can do policy with it. Farmer and Foley (2009) describe the substantial progress that has been made using ABMs to model large parts of an economy; however, they acknowledge the need to go further and to apply the ABM methodology to the creation of larger models that can incorporate multiple markets. The Eurace project (Deissenberg et al. 2008) was an attempt to create a large-scale model of the European economy; owing to the complex nature of the model, however, this goal has so far been technically unrealizable. More recently, the CRISIS project² has undertaken the challenging task of building an integrated finance and macroeconomic ABM to produce a quantitative understanding of financial crises. A visionary project, the FuturICT Knowledge Accelerator³ is another major effort toward large-scale ABM and foresees, among its goals, the development of a sophisticated simulation platform with models driven and calibrated by data aggregated in real time. This might be used to address issues such as risk, trust, resilience, and sustainability and to support policy making, along with business and individual decisions.

The most *fundamental* critique by economists is that ABMs lack microfoundations for agents' economic activities, unlike traditional intertemporal optimization models.⁴ An aim of this review is to show how ABMs in financial markets have evolved from simple zero-intelligence agents⁵ that follow arbitrary rules of thumb into more sophisticated agents described by better microfounded rules of behavior. We then look at the key issue of model calibration. Finally, we look at some cases in which ABMs have been successful at providing insight for policy making.

21.2 EARLIER ABM REVIEWS

A number of reviews of ABM have been published since about 2005, testifying to the growing academic interest in this methodology, both in the economics and physics communities. In this review we focus on the development of models of financial markets, the increasing structural and behavioral sophistication of models, thinking

² <http://www.crisis-economics.eu/home>.

³ <http://www.futurict.eu/>.

⁴ Of course, many traditional microfounded models have their own significant limitations.

⁵ We look at zero-intelligence models in more detail in section 21.4.1, but it is useful to note here that what various authors mean by "zero intelligence" varies substantially: Those with backgrounds in physics will typically mean something like random behavior, whereas more traditional economists often mean something like random parameters for agents, though these agents may still be trying to optimize a utility function, albeit perhaps in a boundedly rational way.

about empirical issues for such models, thinking about policy issues, and some future possibilities.

A relatively early and comprehensive survey of agent-based modeling for finance is LeBaron (2006). LeBaron concentrates on questions of design before surveying the types of existing models and some empirical issues. The design section is of particular interest to those pursuing ABM of financial markets from an economics perspective. It covers issues such as preferences, time, price formation, evolution, learning, how to represent information, and social learning. The importance of having benchmarks, or parameters for which the model is well understood, is highlighted. LeBaron's survey covers a range of models running from "few type" models to very dynamic, heterogeneous models. The few type models analyze a small number of strategies, typically technical or fundamentalist, that are used by agents to trade a risky asset. The proportion of agents adopting different strategies is determined by the strategies' past performance. These models tend to be more analytic than computational. In "many type" models the small sets of tractable trading rules are replaced with larger sets of strategies. Models remain close to a well-defined theoretical framework but extend the framework by including learning agents. The next set of artificial market models moves away from testing specific theoretical models. These models are characterized by a dynamic ecology of trading strategies, and simulations are used to determine which strategies will emerge and survive, and to which will fail. The Santa Fe Artificial Stock Market, to which we return below, is one of the earliest examples of such models.

Hommes (2006) surveys heterogeneous agent models (HAMs) with an emphasis on models that are at least somewhat tractable by analytical methods. These models are simple, stylized versions of the more complicated and computationally oriented ABM but share with them the paradigm shift from the representative agent approach toward a behavioral approach in which heterogeneous, boundedly rational agents follow rule-of-thumb strategies. Such strategies, though simple, perform well and lead to sophisticated macro-level structure. Attention is initially focused on early models that include "fundamentalist" and "chartist" agents, the former forming their expectations on the basis of market fundamentals and the latter on trends in historical price patterns. Other topics covered include examples of disequilibrium HAMs, which present complex market dynamics such as cycles or chaotic fluctuations, systems of agents with stochastic or social interactions, and financial market models with herding behavior. Hommes and Wagener (2009a) survey simple HAM models in which financial markets are viewed as complex evolutionary systems. They introduce the main features of adaptive belief systems and discuss a number of examples with their empirical implications, and confront the models with data from laboratory experiments with human subjects. A number of chapters in the same edited collection (Hens and Schenk-Hoppé 2009) provide of overviews cutting-edge research about financial markets that model the dynamics of asset prices as driven by the heterogeneity of investors.

In Kirman (2002), and in greater length in Kirman (2011), consideration is given to the way agent-based modelers build models of economic systems (and more generally how economic modelling should be done). An argument is made for models that take

into account direct interactions of agents and, in particular, for approaches that utilize a network to model these interactions.

Samanidou et al. (2007) outline the main ingredients of some influential early models in financial markets and a number of more recent contributions appearing in the physics and the economics literature. In particular, they focus on models that formalize the description of financial markets as multiagent systems and can reproduce empirically observed universal scaling laws.

A more recent survey is Cristelli et al. (2011), which discusses a number of influential ABMs for finance with the objective of identifying possible lines of convergence. Models are compared in term of their realism and their tractability. The authors argue that it doesn't make sense to try to identify the best model because the stylized facts are relatively limited and not too difficult to reproduce in an ABM framework. In a similar way to Samanidou et al. (2007), Cristelli et al. (2011) point out that in most ABM models that are considered, the stylized facts do not correspond to a genuine asymptotic behavior but can only be obtained for a specific number of agents and in a limited region of parameters. A similar point could be made, however, about the parameterization of *all* financial models.

An extensive review of econophysicists' work in ABMs is provided in Chakraborti et al. (2011). The authors examine three key areas; models of order-driven markets, kinetic theory models for wealth distribution, and game-theoretic models (particularly for the minority game). The authors conclude that existing models either are simple toy models that cannot be calibrated with real data or more realistic models, suitable for calibration but poorly tractable and whose sensitivity to the various parameters is particularly difficult to understand. Finally, they observe that the cancellation of orders is the least realistic mechanism implemented in existing models and that no agent-based model of order books deals with the multidimensional case. Thus, fully reproducing empirical observations about correlation and dependence is still an open challenge for ABM.

A broader perspective can be found in Chen (2012), which gives a historical overview of how agent-based computational economics has developed by looking at four points of origin: the market, cellular automata, tournaments (or game theoretic) and experiments. In thinking about financial markets the first is of most obvious relevance, but work stemming from all four approaches has played a role in the agent-based modeling of financial markets. The market, understood as a decentralized process, has been a key motivation for agent-based work; Chen argues that the rise of agent-based computational economics can be understood as an attempt to bring the ideas of *many* and *complex heterogeneous* agents back into economic consideration. Zero-intelligence (ZI) agents, which we look at in detail below, have been a key part of finance research. The intuition behind these models is that, given the law of large numbers, no matter the individual motivations behind agents' behavior, their aggregate behavior appears equivalent to that of randomly behaving agents. Other simple programmed agents have included features such as swarming, social intelligence, and regime switching. Computer tournaments have been used to solicit human-programmed behaviors for

complicated dynamic games and to test computer-generated solutions. Experiments, or less formal observations of human behavior, have been important for agent-based modeling and calibration. We return to the issue of calibration in section 21.5.

A recent survey focused on ZI approaches for finance is Ladley (2012). Zero-intelligence models have allowed researchers to gain insight into market dynamics without having to make diverse behavioral assumptions regarding the strategies of traders. By removing strategy from market participants, the researcher may gain insight into the effect of the market mechanism on the overall market dynamics. The simplicity of these models has the additional benefit, in some cases, of making them analytically tractable. Ladley shows how ZI models may do poorly where there are opportunities for learning (ZI agents don't typically explicitly learn) and where feedback loops between agents' actions and the state of the environment they operate generate complex dynamics.

21.3 TRADITIONAL APPROACHES AND EMPIRICAL EVIDENCE

Much research in financial markets has focused on thinking about fully rational agents (perhaps with some learning) processing information (which may be imperfect) to infer the correct "fundamental" value of an asset. There is no scope in these models for chartist agents or herding behavior. Typically it is argued that the predictability of prices should be reduced to zero by rational investors who should earn higher profits and drive less rational traders out of the market. However, artificial stock market models show that the market does not generally select the rational, fundamentalist strategy and that simple technical trading strategies may survive.

The idea of market efficiency, central to much financial research, has a strong theoretical and empirical literature. In empirical terms, models such as the random walk are arguably pretty good approximations for evidently unpredictable financial markets. A benchmark (theoretical) model for thinking about efficiency in which agents can purchase a signal about an asset is outlined in Grossman and Stiglitz (1980). In an efficient world with a small cost on the signal, no one would buy the signal; but how then could it be (informationally) efficient? This paradoxical character of information efficiency is a theme of much of the criticism of such concepts.

Although relating market efficiency to empirical studies is controversial, it is generally accepted that there are many empirical financial phenomena that are difficult to explain using traditional models. As many authors have noted, the empirical distributions of returns of many market indices and currencies, over different but relatively short time intervals, show an asymptotic power-law decay (Mandelbrot 1963; Pagan 1996; Guillaume et al. 1997; Gopikrishnan et al. 1999). A Gaussian form, as predicted by the random-walk hypothesis, is recovered only on time scales longer

than a month. Moreover, while stock market returns are uncorrelated on lags larger than a single day, the correlation function of the volatility is positive and slowly decaying, indicating long-memory effects. This phenomenon is known in the literature as volatility clustering (Ding et al. 1993; DeLima and Crato 1994; Ramsey 1997; Ramsey and Zhang 1997). The empirical evidence also points to persistency in trading volume and positive cross-correlation between volume and volatility (Tauchen and Pitts 1983; Ronalds et al. 1992; Pagan 1996). There is also evidence that both the moments of the distribution of returns (Ghashghaie et al. 1996; Baviera et al. 1998) and the volatility autocorrelations (Baviera et al. 1998; Pasquini and Serva 2000) display multiscaling.

The empirical analysis of limit order data has revealed a number of intriguing features in the dynamics of placement and execution of limit orders. In particular, Zovko and Farmer (2002) found a fat-tailed distribution of limit order placement from the current bid/ask. Bouchaud et al. (2002) and Potters and Bouchaud (2003) found a fat-tailed distribution of limit order arrivals and a fat-tailed distribution of the number of orders stored in the order book. The analysis of order book data has also added to the debate about what causes fat-tailed fluctuations in asset prices. Gabaix et al. (2003) put forward the proposition that large price movements are caused by large order volumes. A variety of studies have suggested that the mean market impact is an increasing function of the order size. Nonetheless, Farmer et al. (2004) have shown that large price changes in response to large orders are very rare. Order submission typically results in a large price change when a large gap is present between the best price and the price at the next best quote (see also Weber (2005) and Gillemot et al. (2006)). Bouchaud et al. (2009) gives a comprehensive survey of stylized facts of financial markets, including, for example, the observation of long memory in the signed order flow.

21.4 MODELING APPROACHES

In this section we survey the range of models used for financial ABM in which the market mechanism is a major area of interest. In building these kinds of models two key areas are pertinent: understanding the structure of the market and understanding the modeling of behavior. In order to focus on questions of market structure behavior can be modeled in very simple ways, ranging from leaving it out entirely (for example, having market orders placed randomly) to zero-intelligence trading (in which we have individuals but they essentially trade randomly, though typically subject to some constraints such as a budget) to more sophisticated models that include ideas such as bounded rationality, game-theoretic principles, or approaches from behavioral sciences. Another key element is the way in which heterogeneity is featured in the model; for the simpler models it often arises purely from the random behavior of (statistically) identical individuals, but for the more complex models the heterogeneity often plays a more central role. Approaches from across the range are surveyed below. We focus initially on three categories of models for agents in markets: ZI agents,

heterogeneous agents interacting through a market mechanism, and heterogeneous agents interacting directly, including network models. This approach allows us to think about the level of sophistication in agent behavior and in the structural detail of the interactions in the models.

Direct interactions, or social interactions, are meant to capture how the choice of each agent is influenced by the choices of others. Various alternatives have been considered in the social utility literature including global interaction, in which individuals tend to conform to the average behavior of the entire population, and local interactions, in which individuals have an incentive to conform to a specific population subgroup of have information about it. Interactions could be heterogeneous (with different strengths and signs between pairs) and asymmetric (Iori and Koulovassilopoulos 2004), but the literature has focused mainly on pairwise symmetric spillover, in which case the payoff of a particular choice increases when others behave similarly. Positive social interaction models generate polarized group behavior even when agents' characteristics are uncorrelated. Models allow for the neighborhood composition to evolve over time, possibly in a self-organized way. Typically agents can form new alliances according to some fitness maximization scheme.

Agent-based models challenge the neoclassical hypothesis of agents' relying on perfect knowledge of the economy and infinite computing capabilities to form rational expectations. Rather, they embrace the bounded rationality paradigm, according to which the expectation formation process is driven by adaptive learning or evolutionary selection via genetic algorithms (Chen et al. 2008). Agents may use technical trading rules or artificial neural networks (ANNs) (Terna 2002) to forecast market prices. In most of the behavioral models surveyed below, agents face discrete choices (submit buy or sell orders, switch between different strategies, sever or form new links, and so on). Bounded rationality in this context may enter via the assumption that, whereas utility is deterministic, the agents' choice process is stochastic. This formulation captures the difficulty agents face in evaluating different features of the various alternatives and do not necessarily select what is best for them.

21.4.1 Zero-Intelligence Agents

We start with zero-intelligence agent models; these models actually vary substantially in what is meant by the term "zero intelligence" and range from very random behaviors, perhaps constrained by a budget, to those in which some kind of strategy may be specified. However, one can identify two key features of most such models: lack of (explicit) learning and a minimalist approach to agent behavior. In addition to more obvious agent-based economic models, we consider some approaches from physics that are closely related.

Research in ZI trading for financial markets was started by Gode and Sunder (1993), although there is some related earlier work by Becker (1962) on "irrational" agents and aggregate outcomes showing how budget constraints can play an important role

and Föllmer (1974), which uses techniques from statistical physics to look at random economies. In Gode and Sunder (1993) ZI traders are compared to a set of human traders and aggregate outcomes are contrasted. These zero intelligence traders are not intended as descriptive models of individual behaviour, instead, the point of the model is to think about efficiency arising from the structure. The key result is that for their model, in terms of the aggregate property of allocative efficiency,⁶ ZI (“irrational”) traders perform nearly as well as human traders. The mechanism studied is that of a double auction in which buyers and sellers submit limit order asks or bids and can accept these asks or bids. Each bid is independently and uniformly drawn from the range $(1, 2, \dots, 200)$. When they match or cross, they are accepted. Each buyer has a valuation v_i and each seller a reserve cost c_i . For a unit sold at price p the profit is thus $p - c_i$. Two variants are investigated and compared to results from experiments: one in which agents trade randomly over the full range of possible values and one in which agents have a budget constraint, that is, they must make a profit or pay less than their valuation. The latter achieves an average efficiency extremely close to that of human traders.

In Gode and Sunder (1997) this research is continued with an exploration of explanations for allocative efficiency for markets. Again ZI traders are constrained to avoid losses, but otherwise bid randomly. A larger number of markets are investigated, including modifications such as limited collection of bids, a limit of trading to only one round, and current bid and ask prices not being made public. The results suggest that for many market structures the simple rules, rather than complex behaviors, may give rise to most of the efficiency.

In the spirit of Gode and Sunder (1993), Duffy and Ünver (2006) ask whether a simple agent-based model can generate the kind of bubbles and crashes that have been observed in experimental settings; the question here is not a matter of efficiency but of replication and understanding of observed behavior. The experimental setup is a round-based trading market with cash and a single asset, one unit of which at most could be sold or bought in each round via an order book. The agent-based model used is somewhat more complex than that of Gode and Sunder (1993); indeed, Duffy and Ünver term it “near-zero-intelligence.” The key difference is that, rather than having purely random prices (albeit constrained to profitable prices), the average transaction price of the previous round is known. This provides a mechanism for the generation of price bubbles that does not depend on the more sophisticated strategies of many later models.

Another relatively recent example of the kind of work which shows that zero-intelligence trading can give rise to observed market phenomena (in contrast to the experimental results, which are the point of comparison for the above research) is Ladley and Schenk-Hoppé (2009), which looks at a limit-order-driven market using a ZI approach similar to the original style (there is a no-loss constraint on orders), and each buyer or seller has a reservation price as in Gode and Sunder (1993). Orders

⁶ Allocative efficiency is total profit divided by maximum total profit, or the sum of consumer and producer surplus.

are randomly drawn pairs (p, q) taken from the set of feasible trades (price p must not incur a loss and q units must be available for sale), and when an order is placed, the previous order from that trader is removed. In contrast to the original version, traders randomly enter and exit the market. The goal is to determine whether characteristics of the order book are the results of the market mechanism or of trader strategy. An “average” order book is constructed by looking at the best five bid-and-ask prices and then compared to empirical findings. The bid-ask spread is found to be about twice the width of either adjacent spread, and the volume available is almost constant across prices; both of these correspond to empirical findings. As tick size is reduced the volume offered at the best price is reduced, according with another empirical observation. The model’s lack of sophisticated behavior on the part of traders may suggest that these and other properties of such markets may in large part arise from the mechanism.

The models discussed above take a minimal approach to ABM, but it is possible to go to the extent of having implicit agents with actions such as the placing of market orders occurring randomly at a market rather than at an explicit individual level. A major contribution in this area is Daniels et al. (2003), which is explored in more detail in Farmer et al. (2005). In this work order arrival and cancellations are modeled as Poisson random processes (rather than being the explicit actions of agents). Orders arrive in chunks of size σ at rate μ shares per unit time with equal probability of being a buy or sell order. Offers are placed with uniform probability at multiples of a tick size over an infinite interval. At time t the best asks and bids are $a(t)$ and $b(t)$, with spread $s(t) = a(t) - b(t)$. The shape of the order book, in particular, the distribution of stored market orders, is a key consideration. Market orders are matched against limit orders, in order of price, and removed. Based on this model predictions can be made for key properties of the market, especially for the diffusion rate of prices and the spread (difference between best buying and selling prices) and price impact functions. This is possible because the model is simple enough to characterize these properties through dimensional analysis. This kind of model is tested against data from the London Stock Exchange in Farmer et al. (2005), where it can explain more than 95 percent of the spread and more than 75 percent of the variance of the price diffusion rate with a single free parameter. These and similar results suggest that there may be simple laws connecting price and properties of the market which do not depend on sophisticated strategies on the part of agents.

21.4.2 Heterogeneous Agents with Market-Mediated Interactions

Although ZI models can replicate many stylized facts about financial markets, they cannot address many questions about modeling behavior (which, for good reason, they omit) and, given that we are using an ABM methodology, they are less comprehensive than is necessary. The models we explore below have richer behaviors

and agent interactions, though for now we restrict our attention to those with purely market-mediated interactions.

An example of early research in building models of financial markets where traders have strategies and speculate on endogenous price fluctuations is Caldarelli et al. (1997). As in many later models, each trader switches between cash and a single stock. All traders start with the same quantity of each, and all have access to the complete price history of the stock. At each time period agent's strategy is an amount of stock $S_{i,t}$ to buy or sell. Each agent has a mapping from the history of prices to a fraction of stock $S_{i,t}$ to buy or sell, somewhat in the style of Arthur (1994). In this case strategies are moving averages of combinations of derivatives of logged prices. This simple model generates a complex price history with the scaling of price variations close to that observed in real financial markets.

The distribution of returns is something poorly captured by traditional financial market models. Research that explores this issue from a behaviorally minimal but structurally detailed way is LiCalzi and Pellizzari (2003). The model consists of an economy with two assets, a bond and a stock.⁷ The price of the stock depends on demand and supply of agents. The total supply of cash and stock is constant, though not all traders may be active at once. Traders enter market with cash c_i and stock s_i and can buy additional stock or sell stock through an order book process that allows both market orders (which can be fully or partly filled) and limit orders. All agents are fundamentalists (we look at richer models elsewhere in this section) who try to buy low and sell high relative to their individual estimate of the fundamental value, v_i , of the stock. Upon entering the market at time t each agent wishes to maximize his or her gains over the time period $h_i - t$, where h_i is the end of his or her investment activity. Bonds have a risk-free return of r , so i requires a sufficient risk premium π_i to invest in the stock, or

$$\frac{v_i}{p} \geq 1 + (r + \pi_i)(h_i - t), \quad (21.1)$$

and the agent will invest in the bond (sell stock) if

$$\frac{v_i}{p} \leq 1 + r(h_i - t). \quad (21.2)$$

Based on the above, agents buy or sell stock at the best prices available in the order book when it makes sense (given their valuation) to buy or sell. When such orders are not available they place their own limit orders. This trading takes place in a number of sessions ("days"). Simulations include both the approach above with its risk attitude and knowledge of r and ZI trading in which only v_i is known. Even in the latter case we see fat-tailed logged returns, suggesting that this is a result of the structure rather than behavior. This phenomenon may be largely due to structural causes, but additional

⁷ We go into a little detail about the basic specification because this model is representative at a basic level of many models we look at later.

properties such as volatility clustering and short-term correlations cannot be explained by the market structure alone.

Lux and Marchesi (1999) describe a model of a financial market with chartists and fundamentalists that gives rise to scaling laws is described.⁸ Here a market maker balancing demand and supply of n_c agents with chartist strategies and n_f agents with fundamentalist strategies is explored. The total number of agents is kept as a constant $N = n_c + n_f$. There is further heterogeneity in that within the chartist group agents may be optimistic or pessimistic about the near future; we have n_+ and n_- of each and an opinion index

$$x = \frac{n_+ - n_-}{n_c}, x \in [-1, 1]. \quad (21.3)$$

The chartists buy or sell (a fixed number of units) if they are optimistic or pessimistic, respectively. Fundamentalists buy or sell if the market value is below or above the fundamental value. Agents endogenously switch between these groups with the transition probabilities arising from economywide average profits and parameters for the inertia between groups. The switching probability from positive to negative is

$$\pi_{+-} = \nu \left(\frac{n_c}{N} \exp U \right) \quad (21.4)$$

and from negative to positive is

$$\pi_{-+} = \nu \left(\frac{n_c}{N} \exp -U \right) \quad (21.5)$$

where

$$U = \alpha_1 x + \alpha_2 \frac{\dot{p}}{\nu} \quad (21.6)$$

and ν is a frequency of opinion revaluation parameter, and α_1 and α_2 are parameters for the relative importance of majority and price trend. The simulation results give consistent statistical characteristics for the market, including both fat tails and volatility clustering that correspond to empirical observations. Usually the macro behavior of this model is stable, but outbreaks of volatility can occur, and the stylized facts of simulated time series data correspond to those observed in real markets. The market behavior is related in Lux and Marchesi (2000) to the concept of “on-off intermittency”: there is an attracting state that becomes temporarily unstable owing to the crossing of some local stability threshold; in this model it is the fraction of traders adopting chartist rather than fundamentalist strategies.

LeBaron (2005) looks at the development of the Santa Fe artificial stock market, one of the first major attempts to build a (somewhat) detailed ABM of a financial market.

⁸ A closely related model is investigated in more detail in Lux and Marchesi (2000). Brock and Hommes (1998) and Chiarella and He (2001) are similarly simple models that capture similar dynamics.

As in many of the models considered elsewhere in this survey, the initial artificial stock market model is one with a risk-free asset and a risky stock paying a dividend

$$d_t = d + \rho(d_{t-1} - d) + \mu_t \quad (21.7)$$

where d and ρ are fixed parameters and $\mu \sim \mathcal{N}(0, \sigma_\mu^2)$. Agents have individual expectations about the price change of the stock and about its variance. The more modern versions of the Santa Fe artificial stock market use a constant relative risk-aversion preference for the formation of individual demands for the stock. Agents use a classifier system to estimate the returns. The classifier is based on the presence of a number of properties, for example, “price greater than five-period moving average.” These properties are mapped to estimation parameters. Each agent has an *individual* evolving set of one hundred rules such that periodically the twenty worst-performing rules are removed and replaced with new rules via both crossover and mutation from their existing rules. This kind of individual selection between rules is similar in style to that in Arthur (1994). The model generates many features of real financial data, specifically excess kurtosis in returns, low linear autocorrelation, and persistent volatility.

The models discussed above use some kind of switching, either between classes or strategies. An alternative is to think about agents using a mixed strategy, giving different weightings to different components. In Chiarella and Iori (2002) a model of a simplified limit-order-book market is built in order to investigate the effects of differing combinations of strategies on aggregate outcomes. Attention is also given to how structural details of the market (tick size and order lifetime) affect these aggregate outcomes. In the model, weightings are given to fundamentalist and chartist components of an agent’s strategy, and a return is individually estimated via the following:

$$\hat{r}_t = g_1 \frac{(p^f - p_t)}{p_t} + g_2 \bar{r}_L + n_t. \quad (21.8)$$

The sign of g_2 indicates a trend-chasing (> 0) or contrarian (< 0) chartist component.

Building on Chiarella and Iori (2002), LeBaron and Yamamoto (2007) introduce learning to the order-driven market. The parameters for weightings in the equation from Chiarella and Iori (2002), that is, g_1, g_2 and n_i , are initially assigned randomly and are updated via a genetic algorithm in which the fitness function f_i is based on the mean squared deviation of realized prices from predicted prices given by

$$f_i = \frac{1}{\sum_{5 \text{ rounds}} (p_t - E^i(p_t))^2} \quad (21.9)$$

where the $E^i(p_t)$ was based on the individual i ’s weighted estimations of return, and the probability of a strategy’s being copied into the next round is

$$P_i = \frac{f_i}{\sum_j^N f_j}. \quad (21.10)$$

In addition to this copying, there is also a small probability of mutation, in which one of the parameters of the price prediction function is replaced by a new value drawn from the original distribution. The key stylized fact they are able to capture are long memory in trading volume, volatility of return and in sign of market orders;⁹ this entails that future values of these quantities are (significantly) predictable on the basis of past values. A modified version of the rescaled range statistic is used to test the simulated data, and the authors are able to reject their null hypothesis of short-range dependence (or lack of long memory) in a majority of simulations for all volume, volatility, and sign of market order quantities. When the simulations were carried out without evolution, there was insufficient evidence to reject the null hypothesis in most cases.

Most of the work concerning heterogeneous agents has focused on strategy. However, it is important to also understand how the market structure may determine outcomes. Bottazzi et al. (2005) and Anufriev and Panchenko (2009) look at this kind of question. The former fixes a proportion of chartists and noise traders and shows how in that model the market architecture plays a bigger role than behavioral heterogeneity in determining the aggregate outcome. The latter is closer to many of the other models described in this section, in that the heterogeneous behavior evolves on the basis of past performance. The authors show, in contrast to Bottazzi et al. (2005), how behavioral features are also important for aggregate properties.

In the style of Chiarella and Iori (2002), Chiarella et al. (2009) look at more sophisticated agents with heterogeneous strategies. Again agents have components of each return forecasting strategy (with heterogeneous weights) and different parameters for each component; however, now utility functions are introduced for each agent and further heterogeneity is facilitated by varying risk aversion. The agents maximize their expected utilities via their individual estimates of stock return, based on their weighted average of the three components. Simplifying a little, this estimate is

$$r = \frac{1}{g_1 + g_2 + n} \left[g_1 \frac{1}{\tau_f} \ln(p_f/p) + g_2 \bar{r} + n\epsilon \right] \quad (21.11)$$

where these estimates are individual, g_1, g_2 , and n are the heterogeneous weightings given to each of the fundamentalist, chartist, and noise components, τ_f is the time scale for mean reversion to the fundamental price, p_f is the fundamental price, and $\bar{r}$ is the chartist component based on estimated return over previous time steps. The agents can place limit or market orders in the order book depending on the best prices available and their estimate of return. The results suggest that chartist strategies generate longer tails in the distribution of orders (in keeping with empirical findings). The increase in volatility following a large price movement can be explained by the large and opposite contributions to price expectations from the chartist and fundamentalist components.

⁹ This is the case whether most orders are buyer- or seller-initiated.

21.4.3 Heterogeneous Agents with Direct Interactions

Although for many purposes it is ideal to have parsimonious models, because we are adopting a computational approach we do not have to be limited to models that can be approached analytically or even those that attempt to create the simplest possible model of an agent for a particular scenario. Drawing on insights from behavioral sciences and knowledge of market structures, we can build considerably more comprehensive models; they may include sophisticated learning behaviors and explicit modeling of the direct interactions of agents. Traditional approaches tend to suggest that an irrational departure from market fundamentals should not be sustainable. However, empirical evidence of repeated financial market bubbles (and subsequent crashes) suggests that ruling out such behavior means omitting a major feature of financial markets from models. There is some more traditional work that includes the idea of herding, such as Banerjee (1992), in which, because agents are making a choice sequentially and basing their choice on choices already made by others, agents may overrule their own (better) information (see Hirshleifer and Teoh (2009) for a review of herding behavior in financial markets).

It is possible to identify three stages in the modeling of direct interactions. The first is global interactions, in which an agent uniformly randomly interacts with another agent. The second is local interactions on a lattice, where interactions are constrained to a set of neighbors but in a regular way. The final stage is local interactions on a network; we examine this stage in the next subsection.

An early work dealing with the interactions of individual agents and the macro consequences is Kirman (1993); many works in ABM for financial markets build on this analytic foundation of recruitment. The basic idea is that we have a system such as an anthill with N agents (ants) that retrieve food from two sources, “black” and “white.” The state is just the number of ants k using the black source. Ants switch between the two resources via an individual process of recruitment: two ants meet, and one switches to the other’s source with probability $(1 - \delta)$. There is an additional ϵ probability that an ant switches its food source without an interaction. So at each time step the system evolves from state k to $k + 1$ with probability

$$P(k, k + 1) = \left(1 - \frac{k}{N}\right) \left(\epsilon + (1 - \delta) \frac{k}{N - 1}\right) \quad (21.12)$$

and from k to $k - 1$ with probability

$$P(k, k - 1) = \left(\frac{k}{N}\right) \left(\epsilon + (1 - \delta) \frac{N - k}{N - 1}\right). \quad (21.13)$$

We can characterize the long-term behavior of this Markov chain with particularly interesting results when most of the time the system is at the extreme values. If we think of the choice of source as choice of opinion or strategy for trading, then these states are ones where we may see herding.

Another early approach to crowding behavior, or herding, is Bak et al. (1997), which adopts methods from physics. There are N traders, each of which can own one share; if they do then they are potential sellers; if they don't then they are potential buyers. Each has a price $p_s(j)$ or $p_b(j)$, the price at which they are willing to sell or buy respectively, which is determined by their individual strategy. Scenarios considered include markets with only fundamental traders and with noise traders (random valuations at first uniformly distributed within a range, then fluctuating randomly). In the most interesting version of the model imitation is introduced for the noise traders such that when they are selecting a new price they randomly copy a price from another agent (of either type). What happens as the proportion of rational traders is varied? When there are few rational traders (about 2 percent) they can be priced out of the market in a "bubble." When there are many (about 20 percent), the prices are kept within their range.

Vriend (2000) distinguishes between learning at an individual and at a population level: in the former case the agent learns exclusively on the basis of his experience, and in the latter case the agent also bases his learning on the experience of other players. The example that forms the focus of Vriend (2000) is a Cournot oligopoly game in which two ways of implementing a genetic algorithm for this kind of scenario are identified. In the first, each individual firm has an output rule, and after many periods some kind of crossover or mutation is applied based on the relative success of the rules. The second (or individual) kind of learning, as in Arthur (1994) and LeBaron (2005), has a set of rules for each agent and those that were most successful recently are more likely to be used. These approaches result in completely different aggregate outcomes, with social learning producing a much higher average output than individual learning. It is suggested that this kind of effect may occur for much more complex models.

Building on the approaches in papers such as Kirman (1993) and Lux and Marchesi (2000), Westerhoff (2009) builds a model with strategy switching between fundamental and technical trading. In contrast with Kirman (1993), above, the switching of opinions is now more sophisticated: the probability of adopting the rule used by another trader now depends on past profitability of the rule and no longer has the same symmetric, random specification. This is accomplished via fitness variables A^C and A^F for the chartist and fundamentalist strategies respectively, each a discounted sum of the past returns. The Kirman (1993) style dynamics are modified by including a weighting on transition probabilities given by $0.5 + s\lambda$ where $s \in \{-1, 1\}$ and the sign of s reflects the relative fitness of the strategy to be changed to. This switching leads to periods dominated by a fundamentalist rule, but with major shifts toward technical rules that increase volatility and may result in bubbles and crashes.

A major area of research for agent-based models of financial markets is the minority game, in which an odd number of agents choose between two options independently and want to be in the minority. There have been hundreds of research papers on this topic because it is seen as a good model for thinking about issues relating to financial market. It developed from Arthur (1994), which considered a model of individual inductive reasoning about aggregate outcomes for attendance at the El Farol bar. This

kind of model, in which every agent wishes to be in the minority, is believed to encapsulate key qualities of financial markets. Challet et al. (2005) includes both an introduction to and a comprehensive collection of many major papers on the minority game.

In Challet et al. (2000) the basic formulation of the minority game is built on in multiple ways such as the introduction of heterogeneity (in the form of different classes of agent), an increase in memory length, the possibility of having more strategies, and agents getting the information of other agents. The idea is to think about various issues concerning financial markets in the context of this well-understood model. The initial formulation is standard, with agents $i = 1, \dots, N$ and actions $a_i(t) = \pm 1$. The gain of agent i at time t is $g_i(t) = -a_i(t)A(t)$ where

$$A(t) = \sum_{j=1}^n a_j(t). \quad (21.14)$$

A particularly interesting section is one that looks at what happens if an agent knows ahead of time the actions of a subset of other agents. This agent can then adopt a different strategy depending on the aggregate decision of the subset. This agent always gains at least as much as the average. If there are more agents with this extra information, the gain of the informed agents is reduced.

A model with interacting agents that can give rise to volatility clustering is presented in Iori (2002). A modified random field Ising model is used to model the behavior of agents in a financial market. There is an $L \times L$ lattice, with each node i being an agent connected to his four nearest neighbors. Initially each agent owns the same amount of capital with $M_i(0)$ units of cash and $N_i(0)$ units of stock. At each time step three actions, $S_i(t)$, are possible: -1 if they sell a unit of stock, 0 if they do nothing, and 1 if they buy. A market maker clears orders and adjusts prices. Agents make decisions based on an idiosyncratic signal $v_i(t)$ —a shock to personal preferences—and through exchanges of information between neighbors. The aggregate signal is

$$Y_i(\tilde{t}) = \sum_{i,j} J_{ij} S_j(\tilde{t}) + A v_i(t) \quad (21.15)$$

where J_{ij} captures influence. For simple cases of J_{ij} this model is well understood in statistical physics; for example, if $J_{ij} = 1$ then this is the Ising model and traders would all agree (with large resultant fluctuations in price). In addition to the above formulation, friction is introduced; otherwise, agents would sell given any positive or negative signal, however small. Synchronization effects (which generate large fluctuations in returns) are shown to arise purely from imitation among these simple traders. These fluctuations exhibit the mutliscale phenomena observed empirically (Pasquini and Serva 2000).

Traditional models of financial markets have particular difficulties in explaining the presence of bubbles. Föllmer et al. (2005) look at a model of a financial market in which the demand of agents for assets is determined by their forecasts of prices. The agents switch between rules in a way that is driven by the success of the rules and influenced

by other traders. Expectations of prices can be heterogeneous, though agents are not, in contrast to related approaches, systematically wrong. The prices can move far from the fundamentals, but the fundamentals do determine the long-run behavior. These rules are supplied by the recommendations R_t^i of a guru or financial expert i . Agents choose from the available experts randomly, with choices weighted by the discounted average of past profits for those recommendations. This model allows clear investigation of the effects of different kinds of rules (or gurus). In particular, it is seen that the switching of forecasting methods can actually be self-fulfilling and may result in bubbles. Chartist experts increase both variance and kurtosis of the limiting empirical distribution of logarithmic prices; they cause (temporary) bubbles and crashes in the model.

In Stauffer and Sornette (1999) clusters of agents aggregate and shatter via variation of a parameter p for connectivity. These clusters act together, and the idea is that there may be times when traders act very individually and times when herding is strong. This is developed from a model in Cont and Bouchaud (2000) that is simple enough (in contrast to, say, Bak et al. 1997) to allow for some analytical results. There is a market with N agents and a single asset with a price at time t of $x(t)$. Demand for agent i is a random variable $\phi_i \in [-1, 0, 1]$ where a positive represents a bullish agent (wanting to buy) and a negative, a bearish agent (wanting to sell). If $\phi_i = 0$ then the agent doesn't trade in that period. So the excess demand for the market is

$$D(t) = \sum_{i=1}^N \phi_i(t) \quad (21.16)$$

and because demand is assumed to be symmetric, or

$$P(\phi_i = +1) = P(\phi_i = -1) = a \text{ and } P(\phi_i = 0) = 1 - 2a, \quad (21.17)$$

then average excess demand is 0. Price change is assumed to be proportionate to the excess demand with a parameter λ for market depth (controlling how sensitive the price is to excess demand). The key element of the model is communication between agents, which is modeled here by a set of clusters that coordinate individual demand, so the excess demand is now the weighted sum over cluster demands, or

$$\frac{1}{\lambda} \sum_{\alpha=1}^k W_{\alpha} \phi_{\alpha}(t). \quad (21.18)$$

Modeling clustering of agents via a random graph model for links between agents allows us to characterize the distribution of cluster sizes, depending on a single parameter for overall willingness of agents to coordinate their demand. Once we have distribution of cluster sizes it is possible to characterize the distribution of aggregate demand and hence price changes. In particular, two key results are derived: there is heavy-tailed density of price changes, and the heaviness of tails (kurtosis of price change) is inversely proportional to order flow; these results hold for a range of parameter values. In Stauffer and Sornette (1999) the stylized facts of interest arise not

from a parameter value being within a certain range but from the variation in herding strength.

21.4.4 Heterogeneous Agents and Networks

With respect to interactions on a network, more recent work takes seriously issues such as how interagent structures arise. We can see the evolution from early models, in which the network is assumed, to later models, in which the network structure may arise endogenously.

The kind of herding models we saw above (in the style of Kirman 1993) have been connected to network structure. A review of the mean field approximation approach to determining transition rates and resulting equilibrium distribution is provided in Alfarano (2008). Transition probabilities are reformulated for individuals based on their neighbors; now the probability of switching for an individual is

$$p_i = \frac{a + \lambda n(i,j)}{a + \lambda N} \quad (21.19)$$

where a is the idiosyncratic parameter, λ is the global herding intensity, and $n(i,j)$ is the number of i 's neighbors in the opposite state. The probability of not switching is simply given by $1 - p_i$. Results for regular, scale-free, random,¹⁰ and small-world networks are compared to the mean field results, which are a reasonable approximation of the simulated results on networks. Only the random network captures the stylized fact of constant variance in particular states for any system size. When heterogeneity in behavior is introduced it has little effect on outcomes, in contrast to introducing a heterogeneous network structure, which can have a major effect on the aggregate outcomes.

Endogenous network formation for financial markets is considered in Tedeschi et al. (2012). Unlike work such as Föllmer et al. (2005), where the idea of gurus is something built into the market via the availability of a set of rules, the idea of a guru here arises as an endogenous result of an information network. Each agent has an outward connection to another agent, and the guru is the agent with the greatest number of incoming links.

Agents have cash and stocks and a wealth relative to the wealthiest agent, which is used as a measure of fitness:

$$f_t^i = \frac{W_t^i}{W_t^{\max}}. \quad (21.20)$$

Agents may randomly rewire (choose another agent to be connected to), and they do this based on the fitness of agents. The probability of i rewiring to agent j from k is

$$p_r^i = \frac{1}{1 + e^{-\beta^i(f_t^j - f_t^k)}}. \quad (21.21)$$

¹⁰ This means the uniform random probability of any particular link existing.

Agents' expectations are a combination of their own individual expectations and those of the agent to whom they are connected. In the model gurus emerge endogenously, rise and fall in popularity over time, and are possibly replaced by new gurus. Traders have an incentive to imitate and a desire to be imitated since herding turns out to be profitable. The assumption that noise traders quickly go bankrupt and are eliminated from the market is unrealistic in the presence of herding and positive feedback. It is shown that more sophisticated strategies underperform the guru and his followers, and positive intelligence agents cannot invade a market populated by noise traders when herding is high.

We have focused on agent-based network models in which the network is modeling informational connections between agents. There has been a lot of work on modeling credit market interrelationships via a network approach, and there is great potential in using ABMs for this kind of modeling. Upper (2011) provides a review of interbank contagion simulation work, much of which could be extended to a fuller ABM approach, in place of simple distress-spreading mechanisms.

There has been quite a lot of activity within more traditional economics regarding networks. One of the earliest examples of work concerning contagion can be found in Allen and Gale (2000), which looks at a stylized model of a small number of financial institutions. Allen and Babus (2008) surveys economic work on applying networks to finance. The books Goyal (2009) and Jackson (2010) introduce networks from a more general economic and social perspective. However, most of the network research within economics has focused on formation driven by incentives rather than network properties. It would be desirable to be able to combine these two kinds of research, allowing policy makers to set incentives to achieve desirable aggregate network properties; ABMs offer a natural way to approach this.

21.5 CALIBRATION OF AGENT-BASED MODELS OF FINANCIAL MARKETS

Relating ABM and empirical knowledge is a key challenge of ABM in general. In the case of financial markets we typically have large volumes of high-resolution data that should be helpful both for calibrating and evaluating models. The typical approach taken for ABM, as with most of the work surveyed above, is to replicate stylized facts from financial markets. However, in addition to this kind of qualitative replication, attempts have been made to more fully calibrate models using empirical data about particular financial markets. We consider some general issues, then look at specific examples of calibration of financial market models.

A general guide to empirical validation of economic agent-based models can be found in Fagiolo et al. (2007). It highlights key issues facing modelers attempting empirical validation, tries to classify models, and identifies unresolved issues. Problems and solutions are split into three categories: relating theory and empirical research,

relating models and real world systems, and discovering how the empirical validation deals with the first two issues. A key aspect of the approach adopted here is to think of there being a real-world data generating process (rwDGP) and a model data-generating process (mDGP). The latter must be simpler than the former, and its “goodness” is to be evaluated by comparing simulated outputs and real-world observations. The lack of consensus about validation is remarked on, and four categories of heterogeneity in approaches are identified: those relating to the nature of the object studied, the goal of the analysis, the assumptions used in modeling, and the method of sensitivity analysis. Three methods of validation with particular relevance to modeling of markets are examined: the indirect approach, the Werker-Brenner approach, and the history-friendly approach. Indirect calibration is the process by which stylized facts are identified and a model is built with reference to known microeconomic description; then the stylized facts are used to restrict parameters. The Werker-Brenner approach combines Bayesian inference (retaining only the parameters that provide the highest likelihood estimation), and an attempt to identify structure on the basis of remaining models. The history-friendly approach uses case studies (for example, those of particular financial markets) and, for it, a good model is one that generates stylized facts for those studies.

Richiardi (2012) is a recent introduction to agent-based computational economics with an emphasis on interpretation of results and estimation. In addressing estimation, the necessary approach is contrasted with that for an analytical model. One must compare artificial data with real data and should change the structural parameters of the model so that these two sets of data become as close as possible. There are various ways in which one might measure this closeness and form an objective function for the optimization algorithm. The method of simulated moments, in which different orders of moments can be weighted by their uncertainty, is suggested. For real data this can be estimated, and for simulated data the uncertainty can be reduced by repeated simulation.

Judd (2006) offers a general overview of methodological computational issues related to agent-based economic modeling. It outlines the main appeal of computational approaches, namely, that the elements of economic investigations previously sacrificed for simplicity can be investigated. Two common objections to numerical approaches are examined. The first is the lack of generality, regarding which it is argued that theories look at a “continuum of examples” but perhaps a measure zero set of plausible or interesting examples. Viewed this way, Judd argues, the relevance and robustness of examples are more important than their number. The second common objection, the presence of errors, is dismissed, because when handled carefully these can be negligible. The main question for Judd is how we can systematically do computational (economic) research. We can’t prove theorems using computers (in the conventional sense), but we can search for counterexamples to a proposition, use Monte Carlo sampling methods (which can be clearly expressed in terms of classical or Bayesian statistics), use regression methods to obtain a “shape” of some distribution, and perhaps straightforwardly adapt a computer model to a new case (something that is often not at all straightforward for a theorem).

In Chen et al. (2012) the development of agent-based (computational) modeling is described from the viewpoint of econometrics. Of particular relevance to this section are the accounts of ABM and stylized facts, and the use of econometric methods for estimation for ABM. The first of these gives a comprehensive description of thirty stylized facts observed in the literature (we looked at research focusing on many of these in section 21.4) and how these relate to the number of agent types in the model. The second offers a clear account of the major options in estimating an ABM. Essentially, one can carry out direct or indirect estimation. The former case may be possible for simpler agent-based models. One uses statistical techniques to estimate the probability of parameters. The latter approach will typically be necessary for more complex models. We will see examples of both approaches below.

One of the first examples of validation and estimation of an ABM of financial markets is Gilli and Winker (2003). This work uses a stochastic approximation of an objective function for estimating the parameters of a foreign exchange model. Bianchi et al. (2007) looks at a case study of validating the Complex Adaptive Trivial System model of Mauro et al. (2005). This model has reproduced many stylized facts about financial markets with ad hoc parameter values. The calibration process takes a sample of Italian firms and estimates parameters for the model using this real-world data. The model has been modified, mostly to make it more realistic (introducing realistic heterogeneity for firms), though the new model uses a homogeneous market interest rate (because micro-level data are not available to do otherwise). The process used is one of indirect inference, minimizing the distance between the actual and simulated distributions of the model, thus allowing for a close match of the simulated results to the empirical data. A “simple” ABM of order flow is validated in Mike and Farmer (2008), where we find a random order placement process. The original model is from Daniels et al. (2003), introduced above. Now additional empirical regularities are modeled; specifically, the order signs, the order price, and order cancellations all now have empirically motivated models, allowing for a more realistic model of order placement than the previous approach. The model is constructed on a single stock and tested on twenty-four others. For those with small tick sizes and low volatility, the model works particularly well.

A dynamic asset price model with heterogeneous agents, who base their choice of forecasting strategies on past profitability, is estimated from U.S. stock price data in Boswijk et al. (2007). A fundamentalist regime and a trend-following regime are identified. Chiarella et al. (2009) continues this kind of approach, identifying the existence of fundamentalist and chartist agents from empirical financial market data. Kouwenberg and Zwinkels (2010) looks at the U.S. housing market and shows that while in general there are a roughly equal numbers of fundamentalists and trend followers, from 1992 to 2005 there were more trend followers. They also compare their approach to time series models and show how they generate boom-bust cycles endogenously.

In Ghonghadze and Lux (2010) a framework for collective opinion formation is created and compared to two more standard time series models when applied to EU business and consumer survey data. Specifically, the model's performance in out-of-sample forecasting is compared to $\text{ARMA}(p, q)$ and $\text{ARFIMA}(p, d, q)$ univariate

time series models. In the model we have two opinion states, positive and negative, with n^+ and n^- of agents holding each view; let $n_t = (n_t^+ - n_t^-)/2$ be the configuration, and assuming we have $2N$ agents, the aggregate expectation is the ratio $x_t = n_t/N$. A Master equation can be formulated for this system and a continuous approximation can be numerically solved, allowing us to calibrate the system with the EU data. It typically does better than the ARMA models and performs similarly to the ARFIMA models for individual series (looking at the performance across all the data, it does better than the ARFIMA models in a majority of cases).

Housing bubbles had an important role to play in the 2008 financial crisis, though this is an area that has historically been of little interest for macroeconomics. Geanakoplos et al. (2012) provides retrospective model of housing that includes large amounts of actual data from Washington, DC. Because it is a detailed agent-based model it can include a great deal of heterogeneous individual-level data that many models would have to omit or aggregate. This includes information about race, income (from detailed IRS income data for that area), wealth, age, and household position. In addition, demographic trends such as population size, death rates, and migration patterns can be included along with economically relevant parameters such as loan-to-value ratios.

21.6 POLICY AND ABMs

Below we look at several applications of agent-based modeling of financial markets to policy. Such modeling seems to be particularly useful when experimenting with policy rules, for ABMs may capture details an analytical model cannot and may be more acceptable to policy makers because they are less abstract (Dawid and Neugart (2011)). As we mentioned above, building large-scale forecasting models, though a difficult task, is a goal of many agent-based modelers.

A publication by Agentlink, Luck (2005), attractively presents numerous uses of ABM for commercial purposes, including practical application in financial markets. It notes, for example, the ability of agent-based traders to outperform human traders by 7 percent and the use of agent-based auctioning systems for the decentralized allocation of resources in many substantial real-world settings. Even at the time of its publication a large proportion of trades on many financial markets were carried out by some kind of automated trader (potentially exactly corresponding to a trading agent in a financial market ABM).

One early and successful commercial application of ABM was developed by Bios Group for the National Association of Security Dealers Automated Quotation (NASDAQ) Stock Market.¹¹ In 1997 the NASDAQ was about to implement a sequence of apparently small changes, namely, reductions in tick size from one-eighth to one-sixteenth and so on down to pennies. In the agent-based NASDAQ model, market

¹¹ www.cbi.cgey.com/journal/issue4/features/future/future.pdf.

makers and investor agents (institutional investors, pension funds, day traders, and casual investors) buy and sell shares using various strategies. The agents' access to price and volume information approximates that in the real-world market, and their behaviors range from very simple ones to complicated learning strategies. Neural networks, reinforcement learning, and other artificial intelligence techniques were used to generate strategies for agents. The model produced some unexpected results. Specifically, the simulation suggests that a reduction in the market's tick size can reduce the market's ability to perform price discovery, leading to an increase in the bid-ask spread. A spread increase in response to tick-size reduction is counterintuitive because tick size is a lower bound on the spread.

The impact of Tobin-style transaction taxes on an artificial financial market is explored in Mannaro et al. (2008). The motivation for this tax comes from the proposal by James Tobin to charge a tax of 0.1 percent on all foreign exchange transactions in order to discourage short-term speculation while leaving longer-term investors relatively unaffected; this, it is widely believed, would reduce market volatility.¹² A model similar to many of those above, that is, with one stock, cash, and various classes of traders, is considered under a transaction tax regime. A number of computational experiments are carried out. It seems that in this model transaction taxes increase volatility and that when both a taxed and an untaxed market are available the volume traded in the taxed market decreases, and that may increase volatility. Another examination of transaction taxes in an ABM is Pellizzari and Westerhoff (2009), in which two micro-structures are considered: a continuous double auction and a central dealership. In the former case, while volume decreases with the transaction tax, so, too, does liquidity, eliminating gains in stability from the reduced volume of trading. In the latter case, as liquidity is provided by the dealership the volatility of the market can be significantly reduced via the imposition of a transaction tax.

Hommes and Wagener (2009b) study the effects of financial innovation on price volatility and welfare. They introduce hedging instruments in an asset-pricing model with heterogeneous beliefs and show that increased use of hedging instruments may destabilize markets and decrease welfare when agents are boundedly rational and choose investment strategies based on reinforcement learning.

Gsell (2008) incorporates algorithmic trading into the ABM of Chiarella et al. (2009). Two strategies of order splitting are implemented: (1) a simple static execution strategy in which the overall volume is executed linearly over time, and (2) a dynamic execution strategy whose aggressiveness varies over time depending on the current market situation and the algorithm's previous performance. The results of the simulation show that algorithmic trading affects market outcome in terms of both price impact and market volatility.

In more recent work related to the understanding of current economic issues, Anand et al. (2013) adopt a rule-based approach with a focus on modeling credit derivative markets. When considering the purchase of an asset-backed security (ABS), agents

¹² This is a controversial view, and many have argued the opposing viewpoint. This kind of transaction tax has, at the time of writing, not been fully implemented in practice.

can choose whether to rely on a signal from a rating agency or they can carry out independent risk analysis. If many other agents also believe the rating agency, then it is rational to believe that the ABS is liquid, irrespective of its underlying quality. In this model this simple but rational approach can result in a highly fragile state of the market as rules spread through the economy.

Cincotti et al. (2010) gives an account of how the EURACE model was used to examine the provision of credit. The model includes detailed financial markets, credit markets, and a central bank that can pursue quantitative easing. Two policy options, quantitative easing and fiscal tightening, are explored across multiple runs of the model, and the results suggest that while quantitative easing increases inflation in both the short and the long run it leads to a better macroeconomic result (higher output).

In Thurner (2011) leverage is connected to systemic financial risk. Here the kind of heavy-tailed fluctuations that have arisen in some of the models discussed above owing to strategies such as trend following, are shown to arise from the effects of leverage. The focus is on collateralized loans with margin calls, a type of loan in which a loan-to-value (of collateral) ratio must be maintained alongside interest payments by repaying rather than rolling over debt. This has a feedback effect (selling collateral reduces the value of collateral, which demands further sale of collateral, and so on) that increases as the level of leverage increases. In this context the policy of leverage restriction may have unintended consequences, causing a local failure to become systemic.

The policy of constraining short selling is investigated in Anufriev and Tuinstra (2013), which deals with a model with chartists and fundamentalists. They examine look at restricting short selling via the addition of trading costs and find that doing so results in higher levels of mispricing with respect to the fundamental values. They note the potential for making the model more realistic, in particular, via aggregate constraints on stock quantities, which could be done in a large-scale ABM.

21.7 CONCLUSIONS

As this survey has shown, much ABM research has concentrated on proof of concepts rather than the development of robust tools to control and forecast complex real-world financial markets. Nevertheless, stylized models are extremely useful for understanding how complex macro-scale phenomena emerge from micro-rules. This kind of exercise has allowed for the testing of existing economic theories and their refinement in attempting to achieve greater realism.

The ABMs discussed in this chapter show a clear evolution toward better micro-founded behavioral approaches to modeling agents' decision processes, although this may not always be micro-foundation in the traditional sense. In many models, agents, while behaving purposefully, use rules of thumb and inductive reasoning to make decisions. Although the fully rational utility maximizer of classical economics is not a faithful empirical model of real people, more sophisticated models of behavior may be important for a fuller understanding of financial market dynamics. Brenner (2006)

surveys various learning processes that could guide modeling the behavior of economic agents to be as close as possible to that of humans. Choosing between the various approaches can be difficult, and there is not consensus yet about which approaches are best in which situations. It is argued that evolutionary approaches are good for population-level results, but they may not capture individual dynamics well. Fictitious play is both simple and supported by evidence. Where more information about beliefs is available, stochastic belief learning may be a good approach. In short, the set of behavioral models that could be applied to financial market ABMs is rich and growing.

When it come to modeling interactions, much work has been done both in physics (Newman 2010) and in economics (Goyal 2009; Jackson 2010) on the subject of networks. Financial systems are networks that have become increasingly complex and interlinked. Nonetheless, the literature concerning financial networks is still at an early stage (Allen and Babus 2009), with most of the research concentrating on financial stability and contagion. Network ABMs typically focus on understanding the system dynamics of a given structure. From a regulatory perspective, questions such as optimal network design and optimal design of incentives that lead to the formation of networks with desirable characteristics offer interesting opportunities for ABM research.

The calibration and validation of ABM are challenging. One key advantage for ABM, with its explicit modeling of heterogeneous individuals, is the possibility of calibration using fine-grained microeconomic data (see, e.g., Geanakoplos et al. 2012) and using evidence from laboratory experiments with human subjects (Duffy 2006; Hommes and Lux 2009; Heckbert 2009). By using experimental techniques, well-defined decision scenarios can be reproduced, and strategies that humans actually use in dealing with complex situations may be revealed. This approach offers a way to capture the heuristics of decision making in a model that is grounded in empirical data.

In section 21.6 we gave multiple examples of the use of ABM for policy related to financial markets. Yet the application of ABM to this area and, indeed, to macroeconomic policy in general is still in its infancy. Increasingly sophisticated modeling techniques, detailed structural modeling, and better calibration methods offer great promise for future research.

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CHAPTER 22

AGENT-BASED MODELS OF THE LABOR MARKET

MICHAEL NEUGART AND MATTEO RICHIARDI

22.1 INTRODUCTION

The labor market is in some respects a very special market, insofar as it interacts with many other economically meaningful domains. It is not just firms and workers meeting together to trade time for money. It is crucial to understand production, on one side, and income, hence consumption and savings, on the other side. As such, it is a key ingredient to any macro model of the economy.

In this chapter we provide an original perspective on the agent-based (AB) approach to the modeling of labor markets.¹ We start from a broad definition of the AB computational approach to economic modeling, according to which AB models are characterized by three features: there are a multitude of objects that interact with each other and with the environment, these objects are autonomous, that is, there is no central or “top-down” control over their behavior and more generally of the dynamics of the system, and the outcome of their interaction is numerically computed (Gallegati and Richiardi 2009; Richiardi 2012). To be able to compute the evolution of the system without resorting to external coordination devices, a basic requirement is that the system be specified in a recursive way (Leombruni and Richiardi 2005; Epstein 2006). This feature not only is of technical relevance for modeling purposes—as Bergmann

¹ In previous attempts to take stock of AB modeling in economics, as in the *Handbook of Computational Economics*, edited by Tesfatsion and Judd (2006) or in the two special issues edited by Leigh Tesfatsion (Tesfatsion 2001a,b), labor market issues were touched on only marginally. In particular, the *Handbook* contained no special chapter devoted to the labor market.

(1990, p. 101) puts it, “The elimination of simultaneous equations allows us to get results from a simulation model without having to go through a process of solution”—but bears a substantive resemblance to how the real systems behave: “The world is essentially recursive: response follows stimulus, however short the lag” (Watts 1991, p. 174).

Now, if we keep to this definition, the roots of AB models of the labor market must be traced back to two early studies that generally are not even recognized as belonging to the AB tradition: Barbara Bergmann’s microsimulation of the U.S. economy (Bergmann 1974) and Gunnar Eliasson’s microsimulation of the Swedish economy (Eliasson et al. 1976). Both authors developed a macro model with production, investment, and consumption (Eliasson had also a demographic module). As in the dynamic microsimulation literature that was emerging at the time, the labor market was only one of the markets they reproduced in their models. Yet they introduced two basic innovations with respect to the standard approach put forward by the father of microsimulation, Guy Orcutt (Orcutt 1957, 1961), that make the labor market module a fundamental block in the microsimulation: they explicitly considered the interaction between the supply and demand for labor, and modeled the behavior of firms and workers in a structural sense. On the other hand, Orcutt’s approach to microsimulation, or, as he called it, the “microanalytic approach for modeling national economies” (Orcutt 1990, p. 29), was based on the use of what he considered atheoretical conditional probability functions whose changes over time, in a recursive framework, describe the evolution of the different processes that were included in the model. This is akin to reduced-form modeling, in which each process is analyzed on the basis of the past determination of all other processes, including the lagged outcome of the process itself.

Bergmann and Eliasson had a complete and structural, though relatively simple, model that they calibrated to replicate many features of the U.S. and Swedish economy, respectively. However, their approach, summarized in Bergmann et al. (1977), passed relatively unnoticed in the dynamic microsimulation literature,² which evolved along the lines identified by Orcutt mainly as reduced-form, probabilistic partial equilibrium models, with limited interaction between the micro units of analysis and with abundant use of external coordination devices in terms of alignment with exogenously identified control totals.

The AB approach thus emerged with a focus on the analysis of evolving economic systems populated by heterogeneous interacting agents. This occurred at the expense of the empirical grounding of AB models that developed mainly as theoretical tools used to identify and study specific mechanisms that are supposed to work in real systems. Hence, the work of Bergmann and Eliasson could be interpreted as a bridge between the (older) dynamic microsimulation literature and the (newer) AB modeling literature, a bridge that has so far remained unnoticed (Richiardi 2013).

² In his influential review of dynamic microsimulation models, O’Donoghue (2001) classifies Eliasson’s work as a microsimulation of labor demand, with firms as the (only) micro unit of analysis, and makes no mention of Bergmann’s model.

The evolution of the AB approach to the modeling of the labor market can be further understood by referring to Ricardo Caballero's distinction between a *core* and a *periphery* in mainstream macroeconomics (Caballero 2010). The core, as he suggests, is the dynamic stochastic general equilibrium (DSGE) approach, and the periphery lies at the intersection of macroeconomics and other strands of the literature such as corporate finance with the investigation of issues ranging from bubbles to crises, panics, and contagion. According to Caballero, "The periphery has focused on the details of the subproblems and mechanisms but has downplayed distant and complex general equilibrium interactions. The core has focused on (extremely stylized) versions of the general equilibrium interactions and has downplayed the subproblems" (p. 87).

In their struggle with the mainstream approach, AB models have evolved along similar lines. The works by Bergmann and Eliasson were first attempts at replacing the core of macroeconomics with an AB alternative. Their goal of providing an AB macroeconomic model to be calibrated empirically was, indeed, very ambitious. After having languished for a few decades, the core approach to AB modeling has recently revived, with a key role played by the European Commission, which has funded ambitious projects such as EURACE (Deissenberg et al. 2008), aimed at developing an AB software platform for European economic policy design, and CRISIS (Farmer et al. 2012), aimed at understanding systemic instabilities.³ These projects developed closed macroeconomic models (no real or monetary flows are lost), in the same vein as Bergmann's and Eliasson's early work. The focus is on the interaction between different (possibly differentiated) markets—typically labor, goods, and credit with some attempts to include financial markets—with the goal of replicating the behavior of a real economy and qualitatively tracking the evolution of major economic time series. These approaches offer artificial labs for what-if studies of distant and complex general disequilibrium interactions, rather than forecasting tools as in the dynamic microsimulation tradition.

Parallel to the analytical tradition, a more peripheral approach has also emerged, with the aim of developing single-purpose rather than multi-purpose models. These models, which we label *partial models*, focus on heterogeneous and interacting agents in a particular market and are kept as simple as possible to isolate and investigate specific mechanisms of interest, possibly at the expense of abstracting altogether from or offering an over simplified representation of other dimensions and their feedback mechanisms. We review a selection of this AB periphery, in Caballero's parlance, below. The partial modeling approach, which often is identified with the AB modeling paradigm itself, gained popularity as a way to illustrate the Santa Fe complexity paradigm (Gallegati and Richiardi 2009).

An additional distinction can be drawn according to the research objectives of the models. Two main goals can be identified. The first is to replicate a set of well-known stylized facts, possibly wider than those replicated by traditional analytical models

³ EURACE was produced under the FP6-STREP grant, 2006–2009; CRISIS, the FP7-ICT grant, 2012–2015.

(e.g., the wage curve, the wage distribution, and the Beveridge curve). The second is to analyze the effects of specific policies (for instance, training policies, employment protection legislation, and unemployment benefits). In recent decades, accelerated by the advent of econometric software packages with ready-to-use techniques for causal analyses of policy effects, most of the labor market policy evaluation focused on micro analysis. It is well acknowledged that though the analysis of micro data gives valuable insights into the effect of policies, these evaluations only yield a partial picture (OECD 2005). Aggregated effects might be smaller than analyses at the level of micro data suggest because of deadweight losses, substitution, or displacement effects. Aggregate analyses that have the potential to capture the overall effect very often lack sufficient institutional details to be valuable for policy makers or are incapable of addressing the magnitude of countervailing effects because many channels of interaction are shut down. Agent-based models have been offering valuable insights into the mechanisms that reduce aggregate effects with respect to a simple aggregation of individual changes in behavior and the kind of institutional details that can be incorporated.

The two goals of factual replication and counterfactual analysis are by no means exclusive. Indeed, a model that reproduces behavior of the labor market realistically is a priori a good candidate for investigating the effects of a given policy. Yet understanding of the causal mechanisms triggered by a policy can sometimes benefit from a simpler model, cast at a higher level of abstraction. For this reason, whereas models in the core typically pursue both objectives, some models in the periphery are restricted to the latter.

In what follows we first elaborate on the value-added of modeling the labor market via AB simulations (section 22.2). We sketch in section 22.3 the contributions by Bergmann (1974) and Eliasson et al. (1976) and review the literature that has developed since then by classifying the models according to their scope, from partial models used for analyzing particular policies and addressing stylized facts of labor markets (section 22.4) to models in which an AB labor market is embedded in a macroeconomic framework aimed at reproducing the behavior of multiple interacting markets (section 22.5). We then discuss the main methodological features of all these models, which relate to the way individual behavior (section 22.6) and the interaction structure (section 22.7) are modeled. Section 22.8 offers our conclusions.

22.2 WHY AB LABOR MODELS?

It is legitimate to ask why we need AB labor models at all, or what value is added of AB modeling with respect to the mainstream (analytical) approach?

In short, the added value lies in the ability to weaken many of the standard assumptions at the same time. *Flexibility in model design*, which allows for richer and more complex specifications to address unexplored economic mechanisms and empirical phenomena, is the selling point of the methodology. And, indeed, this is how

the discipline seems to have evolved ever since. As researchers have become more and more demanding, standard approaches were abandoned, and the models, our main tools for thinking about the workings of economies, were re-engineered to gain new insights.

Let us elaborate on this argument a bit more. A textbook about labor economics typically introduces students to the decisions of households about how to optimally allocate time. Aggregation yields a market labor supply as a function of the going wage rate. Next, a firm's optimal demand for labor is derived, which, aggregated up, yields a market demand for labor. Particular assumptions concerning the households' preferences and the firms' production technologies make sure that labor supply is upward-sloping and that labor demand is downward-sloping with regard to wages so that there is a market-clearing wage.⁴ We can go a long way toward explaining wage and employment patterns with a simple demand-and-supply model, and we can apply it to analyze the effects of various policy interventions, very often in a meaningful way. But the model has its shortcomings; the most important is that, contrary to our everyday observation, it does now allow for unemployment.

Consequently, the search for an explanation of unemployment has been the driving force in theoretical development. Various proposals have been made, but the unifying approach was always to give up one or more of the simplifying assumptions. In efficiency wage models, unemployment arises because firms do not lower wages to clear the market. This might happen because (1) firms want to avoid shirking (Shapiro and Stiglitz 1984) and unemployment works as a disciplinary device because the cost of job loss, and hence the threat of firing, increases, (2) firms want to minimize turnover (Stiglitz 1974; Schlicht 1978; Salop 1979), (3) with above-market wages, the worker's motivation to quit is reduced, (4) firms want to attract better workers (Stiglitz 1976; Weiss 1980; Malcomson 1981), or (5) firms do not want to lower the morale of the workforce and hence its effort (Akerlof 1982; Akerlof and Yellen 1990). In another strand (or textbook chapter) we learn that wages are not set as they are in a spot market but rather are set by powerful unions or are bargained over by unions and employer associations (see, e.g., Oswald 1985), again allowing for an explanation of unemployment. In the third route the assumption that vacancies are filled instantaneously has been lifted. The search and matching models (Diamond 1981; Mortensen 1982; Pissarides 1990) allow us to cope with the simultaneous occurrence of vacancies and job searchers, as illustrated in Beveridge curves and the movements along the Beveridge curve over the business cycle. All these strands of the literature have paved the way for what we nowadays would accept as the mainstream explanations by working out the effects on labor market outcomes after one or more of the standard assumptions are given up.

⁴ Only under very restrictive and implausible assumptions can the aggregate supply and demand curves be derived as the result of the optimal choice of a representative household and a representative firm (Kirman 1992). In order to obtain comparative static results, however, most models implicitly pretend that these assumptions hold.

Analytical models have come a long way along this route. A good example of the sophistication they can attain is the work by Elsby and Michaels (2013), who introduce a notion of firm size into a search and matching model with endogenous job destruction. In our view, a model of this kind is, indeed, quite an achievement. Still, in this example, there is no firm creation or on-the-job search and empirically the model is able to generate only one-quarter of the positive wage–firm size effect observed in the data. The authors dismiss this shortcoming by acknowledging that many other channels might be present, in addition to the interaction of surplus sharing with heterogeneity in employer productivity on which the model is focusing on. They cite efficiency wages, market power, and specific human capital, and we could add union power, worker heterogeneity, and the endogenous sorting of workers and firms into temporary jobs, as in Berton and Garibaldi (2012), or the assumption of a constant return to scale matching function that is already criticized in Neugart (2004) and Richiardi (2006).

Extensions into any of these dimensions are likely to prove analytically hard, and computational techniques would have to be employed at some stage. As model complexity increases, not only do analytical solutions of the aggregate steady state-behavior have to be abandoned, but the aggregation problem itself becomes intractable.

Moreover, all these models still rest on the hypothesis of rational expectations (the assumption that individuals make no systematic errors), which can be considered the watershed between mainstream and more heterodox approaches. The hypothesis is not without rationale: for one thing, the ability of individuals to act optimally based on rational expectations provides a well-defined benchmark for economic analysis. However, the plausibility of rational expectations has been criticized from within the mainstream camp itself. As Caballero (2010, pp. 91–92) notes,

Rational expectations is a central ingredient of the current core; however, this assumption becomes increasingly untenable as we continue to add the realism of the periphery into the core. While it often makes sense to assume rational expectations for a limited application to isolate a particular mechanism that is distinct from the role of expectations formation, this assumption no longer makes sense once we assemble the whole model. Agents could be fully rational with respect to their local environments and everyday activities, but they are most probably nearly clueless with respect to the statistics about which current macroeconomic models expect them to have full information and rational information.

...In trying to add a degree of complexity to the current core models, by bringing in aspects of the periphery, we are simultaneously making the rationality assumptions behind that core approach less plausible.

On the contrary, decision making in AB models generally consists of learning processes based on adaptive behavior with respect to expectations formation and strategy exploration and sequential, rather than simultaneous, problem solving. Agents do not maximize inter temporal utility under conditions of perfect information and

unlimited computing abilities.⁵ Optimal behavior can be obtained by various conscious or unconscious learning processes for which a large array of formalizations drawing on the psychology literature and experimental evidence exists (Brenner 2006), or by evolutionary selection (Arifovic 1994).

Also, by not building on the rational expectations paradigm, AB models can be scaled up much more easily. Adaptive expectations generate a relation between the dimensions of the decisionmaking problem and its complexity that is roughly linear, rather than exponential, as in the rational expectations paradigm. In order to see this, suppose there are n binary choices to be made (or one binary choice to be repeated over n periods). If the problem is solved simultaneously (inter temporally), the choice set is composed of 2^n elements. If, on the other hand, the problem is solved sequentially, conditional on past choices, the choice set only includes $2n$ elements. Of course, the result in the latter case could be quite suboptimal, but with a decentralized selection mechanism such as market competition or some sort of social or individual learning, the extent of suboptimality can be greatly reduced without increasing too much the complexity of the overall optimization problem.

The bottom line is that AB modeling gives us a tool with which to analyze patterns of behavior in the labor market that would not have been analyzable in the past. Will this tool by itself give us a better understanding of the labor market or of which policies to apply? We side with Richard Freeman, who claims: “Of course not. Computer tools do not solve anything. You need ideas and data” (Freeman 1998, p. 19). But he continues, “Still the new tools can sharpen our thinking about competing models of capitalism and allow us to assess alternative theories or explanations about which we could previously only hand wave.” We will try to assess how far AB labor researchers have gone in this respect and where they may want to (have to) go in the future.

22.3 EARLY MICRO-TO-MACRO MODELS

22.3.1 Bergmann's Model of the U.S. Economy

Barbara Bergmann was deeply influenced by Orcutt's lessons while a graduate student at Harvard University (Olson 2007). Yet her microsimulation model (Bergmann 1974) departs from Orcutt's approach in significant ways. The behavior of all actors is modeled in a structural sense: workers, firms, banks, financial intermediaries, work-long government, and the central bank act on the basis of pre-defined decision rules, rather than being described in terms of probabilities of transition between different states. In each period, (1) firms make production plans based on past sales and inventory position; (2) firms attempt to adjust the size of their workforce, wages

⁵ Giving agents rational expectations in an AB model does not seem to be feasible. Critics may argue that this is the true reason why AB modelers resort to learning behavior—a view that we do not support.

are set, and the government adjusts public employment; (3) production occurs; (4) firms adjust prices; (5) firms compute profits, pay taxes, and buy inputs for the next period; (6) workers receive wages, government transfers, and property income, pay taxes, make payments on outstanding loans; (7) workers decide how much to consume and save, choose among different consumption goods, and adjust their portfolios of assets; (8) firms invest; (9) the government makes public procurements from firms; (10) firms make decisions about seeking outside financing; (11) the government issues public debt; and (12) banks and financial intermediaries buy or sell private and public bonds, the monetary authority buys or sells government bonds, and interest rates are set. In the early 1974 version only one bank, one financial intermediary, and six firms, “representative” of six different types of industrial sectors and consumer goods (motor vehicles, other durables, nondurables, services, and construction) are simulated. In the labor market, firms willing to hire make offers to particular workers, some of which are accepted; some vacancies remain unfilled, with the vacancy rate affecting the wage-setting mechanism. Unfortunately, the details of the search process are described only in a technical paper that is not easily available anymore (Bergmann 1973). Admittedly, the model was defined by Bergmann herself as a work in progress, completed years later (Bennett and Bergmann 1986). The assumption of representative firms is particularly questionable from an AB perspective, although it is not engraved in the model’s architecture. The model still is noteworthy for its complexity and for the ample relevance given to rule-based decision making.

22.3.2 Eliasson’s Model of the Swedish Economy

Eliasson et al. (1976) micro-to-macro model, which eventually came to be known as MOSES (“model of the Swedish economy”), is a dynamic microsimulation with firms and workers as the units of analysis. A concise description of the model can be found in Eliasson (1977). The labor market module, which is of central importance to the model, is firm-based insofar as the search activity is led by the firms that look for the labor force they require to meet their production targets. Labor is homogeneous, and a firm can search the entire market and raid all other firms, subject only to the constraint that searching takes time (a limited number of search rounds are allowed in each period). Firms scan the market for additional labor randomly, the probability of hitting a source being proportional to the size of the firm (number of employed) and the size of the pool of unemployed persons. If a firm meets another firm with a wage level that is sufficiently below its own, it gets the people it wants, up to a maximum proportion of the other firm’s labor force. The other firm then adjusts its wage level upward with a fraction of the difference observed, and it is forced to reconsider its production plan. If a firm raids another firm with a higher wage level it does not get any people, but it upgrades its wage offer for the next attempt. Firms then produce, sell their products, make investment decisions, and revise their expectations. Individuals allocate their income to savings and consumption of durables, nondurables, and services. Each year the population evolves with flows into and out of the labor force.

The model was designed to offer a micro explanation for inflation and study the relation between inflation, profits, investment, and growth. It was populated partly with real balance-sheet firms and partly with synthetic firms whose balance sheets were calibrated in order to obtain sector totals. After its original formulation, the model was updated and documented in a series of papers (Eliasson 1991).

22.4 AGENT-BASED LABOR MARKET STAND-ALONE MODELS

22.4.1 Policy Evaluations

The first AB “toy” model of the labor market, which considers only limited actors and actions, is presented in Bergmann (1990). At that time the AB modeling approach was not shaped yet, and the author clearly stated that her goal was to provide an introduction to microsimulation. Her model is so simple that no more than fifty rows of BASIC code are needed to program it. Workers are homogenous, labor demand is exogenous, matching is random, and the unemployed always accept an offer (with the exception of those who have just been laid off and who have to wait one period to reenter the labor market). Wages are not modeled; this is equivalent to assuming exogenous and homogeneous salaries. The fact that such a paper, with a whole paragraph devoted to explaining what random numbers are and how they can be obtained in BASIC, appeared in such a prestigious journal as the *Journal of Economic Perspectives* is a reminder of how recent the diffusion of personal computers is. At the same time, having what now looks like a basic tutorial in AB modeling published so early and so well marks a point (albeit not a decisive one) in mainstream economics, which is often criticized for obstructing the development of new ideas and approaches in the profession (Krugman 2009).

Within this simple framework, Bergmann envisaged a stylized policy experiment: she added an unemployment insurance program (with time-limited benefits) and analyzed its effects on individual spells of unemployment and aggregate unemployment during recessions and recovery in the labor market. Her main result is that an unemployment insurance system might not increase unemployment during recessions. The reason is that, although a particular worker may, on the basis of being eligible for unemployment benefits, refuse a job offer, doing so paves the way for another worker who is offered that vacancy.

Although one might call this a crowding-in effect, the major finding in Neugart (2008) stems from a crowding-out of workers not being part of the policy treatment, which in this particular AB model is a training policy. The model consists of heterogeneous workers and firms that are allocated across different sectors. Workers differ with respect to their skills, and firms located across sectors have distinct skill requirements. Workers may acquire skills that equip them with the necessary knowledge to work

in sectors other than their current one. Thus, should they become unemployed they may also apply for jobs outside their current sector. In order to spur outflows from unemployment, the government introduces a training policy that subsidizes workers' acquisition of skills. On aggregate the policy has a positive effect on the outflow rate from unemployment, but it also has distributional consequences. Those who receive government transfers and thus increase the marketability of their skills find jobs more easily. But this occurs at the cost of workers who would have found a job in their current sector if they had not faced competition from the trained workers entering that sector. Much as in Bergmann's model, nontreated workers are crowded out by treated workers, reducing the aggregate effect of the policy with respect to a simple aggregation of the shorter unemployment spells of the treated workers.

Matching between heterogeneous workers and firms is also analyzed in Boudreau (2010) in which firms pay different wages, and workers have initial skills and an endowment that they may invest to improve their productivity. The most productive workers are matched with the firms paying the highest wages. Those firms grow faster because they employ the more skillful workers. As for the workers, the higher wages are passed on to the descendant of each worker as the new wealth endowment. Then the model analyzes how a redistributive tax changes inequality. Besides the results' being dependent on the specification of the technological growth, some interesting and counteracting mechanism related to the incentive to invest in skills can be detected. With the transfer of funds to workers with high initial skills but low endowment, competition for well-paying jobs becomes fiercer, increasing the incentives of other workers to invest. At the same time their funds are lower because of the redistribution scheme, making the overall effect on investments in skills ambiguous.

Ballot and Taymaz (2001) looked into three different training policies, all of which can be considered suitable proxies for actually conducted efforts spurring the acquisition of human capital: a subsidy to education and training activities, a policy that forces firms to spend a certain share of their wage bill on training activities, and a policy by which firms receive subsidies for training if they hire unemployed workers. Results are that the first policy and to a smaller extent the third policy may improve long-run economic performance, whereas the second policy does not. The effect of the first policy runs via an increase in the likelihood of a successful innovation. This effect is less powerful if the training policy is only for hired unemployed workers. The second policy is ineffective on the aggregate because it drives less profitable firms out of the market.

In most policy evaluations that use an AB approach, the policy is exogenously varied. The policies may change as market outcomes change the payoffs for voters. In Martin and Neugart (2009) an attempt was made to endogenize policy choices. An AB labor market model is set up wherein voters cast their vote for the type of employment protection system they prefer. It is shown that employment protection is neutral with respect to employment on average. However, employment rates decrease if at the onset of a more volatile economic environment the deregulation party was in power, because backward-looking voters blame the current party for the mal-performing economy and vote for the alternative, which further depresses labor demand.

22.4.2 Addressing Stylized Facts of Labor Markets

There is a strand of AB models that have been developed in order to replicate some stylized facts of real labor markets and to understand the emergence of aggregate regularities from the micro behavior of individual units. In these models labor markets are still central, and although efforts were made to incorporate possible feedback processes from other markets, we still tend to classify them as being partial models because only certain other markets are typically incorporated.

The stylized facts that are most often targeted are the wage and Beveridge curves, Okun's law, the form of the aggregate matching function, and the shape of the wage and firm size distributions. The wage curve (WC) postulates a negative relation between the wage level and the unemployment rate (Blanchflower and Oswald 1994; Card 1995). The Beveridge curve (BC) describes a negative relation between the unemployment rate and the vacancy rate, and Okun's law (OL) posits a negative relations between the changes in the unemployment rate and the GDP growth rate (Prachowny 1993; Attfield and Silverstone 1997). The matching function (MF) relates the number of matches to unemployed job searchers and the number of vacancies (Blanchard and Diamond 1990; Petrongolo and Pissarides 2001), and is often assumed to show constant returns to scale. Finally, the income and firm size distribution, as with many other economic variables, have been shown to be highly skewed, as predicted by a log-normal or power-law functional form (Growiec et al. 2008; Gabaix 2009).

Fagiolo et al. (2004) were able to reproduce the WC, BC, and OL with an AB model focusing on the interactions of the firms with the output market. Building on their work, Tavares Silva et al. (2012) present an AB model with technologically neutral progress that features rising wage inequality. In a series of papers, Gallegati and coauthors (Bianchi et al. 2007; Delli Gatti et al. 2005; Delli Gatti et al. 2004; Russo et al. 2007) worked in the direction of filling the gap between firm demography and unemployment theory by focusing on the interactions of the firms with the financial system.

Richiardi (2004, 2006) modeled the matching process between workers and firms with on-the-job search, entrepreneurial decisions, and endogenous wage determination. He showed that a negatively sloped WC and an MF with constant returns to scale emerge only out-of-equilibrium during the processes of adjustment toward the steady state. In the steady state, the WC is upward-sloped, while the coefficients of unemployment and vacancies in the MF do not even have the right sign. These results call into question equilibrium models that take these aggregate empirical regularities as starting assumptions.

Ballot (2002) models a dual labor market in the spirit of Doeringer and Piore (1971). He distinguishes between open-ended and temporary positions. Some firms have an internal labor market (ILM) for permanent positions where employees compete for promotions (seven levels are considered), while other firms do not. Promotions have two roles in the model. First, they are one way to fill a vacant permanent job because they enlarge the pool of candidates for a job. Second, they operate as a screening device. Nominal wages are fixed, but given that workers differ in their productivity, the

quality-adjusted wages are endogenous. Jobs require a minimal level of human capital, and firms have to invest in training if the hired workers are below that level. Moreover, firms can set a hiring standard for their vacancies. The higher the standard, the higher the expected quality of the selected worker will be, but the longer the expected duration of the vacancy. In setting their standards, firms look at labor market tightness and take the expected duration of the position offered into account. Hiring under a temporary contract involves paying the intermediation cost of a temporary help agency. Apart from that, temporary jobs have a linear cost in duration, while permanent jobs have nonlinear costs in duration because of a seniority premium and redundancy payments. On-the-job search on the part of the workers is considered, at the cost of deferred leisure. Individuals and firms learn in the market and adapt their behavior according to their experience. Although the model only comprises forty firms and seventeen hundred individuals belonging to eight hundred households, it is roughly calibrated to the French labor market over the period 1972–1977, that is, around the time of the first oil shock. It is able to reproduce the changes in mobility patterns of some demographic groups when the oil crisis in the 1970s occurred, and in particular, the sudden decline of good jobs. Moreover, ILMs for permanent positions increase unemployment, an effect which is mitigated by the existence of a secondary labor market (made up of temporary jobs or of open-ended jobs in firms without an ILM). In line with the microsimulation literature the model is given a name (ARTEMIS). With a household composition and expenditure module that is, however, simpler than the labor market matching module, it more than a partial model.⁶

Similarly, Dosi et al. (2006) may be considered as lying somewhere in between the partial AB models focusing on labor markets and those trying to incorporate feedback processes from other markets. They developed a model with an intermediate sector that produces machine tools, engages in R&D activity, and has a final consumption good sector. The model is able to replicate a number of aggregate empirical regularities: investment is more volatile than GDP, consumption is less volatile than GDP, investment, consumption, and change in stocks are pro-cyclical and coincident variables, employment is pro-cyclical, the unemployment rate is countercyclical, firm size distributions are skewed (but depart from log-normality), and finally, firm growth distributions are tent-shaped.

22.5 LABOR MARKET MODULES EMBEDDED IN AB MACROECONOMIC MODELS

Embedding an AB labor market in a macroeconomic model allows us to analyze feedback processes arising from goods, financial, or credit markets regarding the labor

⁶ ARTEMIS has now evolved into WorkSim, which at the moment of writing this chapter appears to be at an advanced stage of development.

demands of firms and the labor supply decisions of workers. These models pave the way for investigating policies that cannot be addressed in partial models in a meaningful way. In his prototypical model, Eliasson studied the effects of a regulation aimed at preventing layoffs without ample advance notice (Eliasson 1977). He showed that such a device actually fostered growth during the first years after implementation, as firms choose to make use of the workers they cannot lay off. In the longer run, however, wages are lowered and prices increase permanently, with possible adverse effects on welfare. But if the business sector is highly profit-oriented, as he showed, the latter effect is only marginal, as competitive pressure forces firms to step up efficiency.

A major effort in attaining an AB model of the whole economy was put forward by the EU-funded EURACE project (Deissenberg et al. 2008), which aimed at a proof of concept that an AB macroeconomic model including capital, goods, credit, financial, and labor markets within a spatial context can be developed and simulated. The resulting model has been used, with a focus on different submarkets, in a number of papers addressing policy-related questions (see also Chapter 17).

Among these, Dawid et al. (2008, 2009) analyzed the regional allocation of funding of human capital investments in the presence of labor market frictions in a closed AB macroeconomic model. When commuting costs for workers between regions are high, a uniform distribution of funds to promote general skills for workers creates larger effects on output than a spatially unequal distribution. In the absence of commuting costs for workers, regional output levels evolve similarly no matter what spatial distribution of funds to promote general skills is chosen. For positive and low commuting costs, however, a spatially concentrated policy performs better than a uniform approach, and furthermore, the region that receives fewer funds outperforms the regions receiving the larger fraction of funds. These effects are due to the technological spillovers through the labor market and demand-induced investment incentives for producers in that region. Using an augmented framework, Dawid et al. (2012) also looked into labor market integration policies establishing a trade-off between aggregate output and convergence of regions. There, it is shown that closed labor markets result in relatively high convergence but generate low output whereas more integrated labor markets yield higher output but lower convergence.

In another AB macroeconomic model also originating from the EURACE project, Teglio et al. (2012) studied the impact of banks' capital adequacy regulation on GDP growth, the unemployment rate, and aggregate capital stock. They found that allowing for a higher leverage gives a boost to the economy in the short run but can be depressing in the longer run because firms become more fragile, possibly triggering credit crunches.

These examples, as well as other attempts to build AB macroeconomic models reviewed in other chapters of this handbook, are promising with respect to a meaningful inclusion of labor market modules into a larger framework. They have also shown that particular policies targeted at submarkets might gain from being studied in closed macroeconomic models because they can trigger important and nontrivial feedback processes that drive aggregate outcomes.

22.6 BEHAVIORAL RULES

In describing the set-up of AB models and the insights that have been gained, we hardly went into the specific modeling choices concerning how agents make decisions and interact. Both are crucial assumptions that necessitate a closer look. Let us first elaborate on the choice of behavioral rules.

Giving up rational expectations as the prime input for modeling agents' behavior, as it is done in the AB approach, opens up a whole range of possible ways to model agents' behavior. This is reflected in how firms' and workers' choices are modeled in existing AB labor market models. We find examples of firms choosing (among applicants) randomly (Tassier and Menczer 2008) as well as more sophisticated behavior. Ballot and Taymaz (1997, 2001) modeled firms' searches for more efficient technologies using genetic algorithms. The same approach is taken in Tesfatsion (2001c), in which where firms and workers adjust their worksite behavior, leading to recombination, mutation, and elitism operations favoring more suitable strategies. Similarly, Tassier and Menczer (2001) apply a local selection algorithm with which they allowed more successful agents to reproduce themselves. A rule-based approach has been followed by Boudreau (2010), who let workers choose their level of investment in human capital such that the labor market prospects of higher-ranked workers are matched. The rule-based approach also features prominently in Dawid et al. (2008, 2009, 2012), who model agents' behavior using rules about firms' choices coming from the management science literature. Rules with adaptive behavior of agents have been used by Richiardi (2006) to model the decision whether to search for a new job. Here workers compare present and future expected income, with expected income being formed adaptively to arrive at a decision. Fagiolo et al. (2004) used adaptive rules for the adjustment of firms' vacancies based on past profit growth, wage setting, and updating of workers' satisficing wages. In some contributions, for example, Axtell and Epstein (1999), there is a mix of behavioral rules, with some workers choosing randomly, others imitating, and yet others just doing the right thing. Discrete choice models have been used by Neugart (2008) and Martin and Neugart (2009), and Gemkow and Neugart (2011) employ reinforcement learning to model agents' choice concerning how much to invest in the size of a network of friends.

As we can see from these examples, AB (labor) market modelers have imposed quite distinct assumptions about agents' choice behavior. Although they are sometimes backed by empirical evidence, there remains a flavor of arbitrariness. Also, the extent to which results are sensitive to these modeling choices is not always apparent.

There are several ways to proceed in the future. Contributions could be extended by further robustness tests that exchange parts of the model and rerun simulations in order to validate that, qualitatively speaking, at least, the results do not change. Another approach that has been pursued in part in the EURACE project is to implement rules as they are typically applied in firms for standard decisions such as stocking up. Actually, these rules are very often already implemented in standard software to which firms refer in organizing their production processes. In that sense, it would constitute a modeling

choice that mimics firms' behavior very closely. Finally, we would like to see more attention to having the findings of experimental economists or modeling choices made in AB contributions backed up by laboratory experiments (Duffy 2006; Contini et al. 2007).

22.7 INTERACTION STRUCTURE

In labor markets social interaction plays a prominent role. Access to information about job opportunities is embedded in an individual's social network. Moreover, in a world of asymmetric information, matching of vacancies to job searchers is alleviated by the social capital of a network in which one worker refers another. Thus, it does not come as a surprise that the role of networks in labor market outcomes is increasingly acknowledged (Ioannides and Loury 2004; Ioannides 2013). What is surprising, however, is that AB contributions to this literature are sparse. The AB approach seems to be a natural candidate to address these research questions, with a focus on labor market outcomes from heterogeneous agents interacting within and across social groups and with group formation possibly being endogenized.

An early contribution to this strand of AB models comes from Tassier and Menczer (2001), who set up a labor market model assuming an economy with a fixed number of jobs and randomly assigned wages. Agents search for these jobs by two means: they may devote part of their resources to directly finding a job or they may expend effort in making friends who eventually may tell them about job openings. Simulations reveal the emergence of small-world networks, but they do not inhibit the transfer of information. In addition, it is found that individuals exert too much effort on finding jobs. Though the job search is optimal on an individual level, this strategy is suboptimal from a social point of view.

The network of agents is also endogenous in the labor market model developed by Gemkow and Neugart (2011). Whereas Tassier and Menczer (2001) zoomed in on the role of networks for the transmission of information about job openings, the emerging network in Gemkow and Neugart helps workers who apply for a job overcome the asymmetric information problem faced by prospective employers and employees. It is shown that workers who build up a network of employed friends experience shorter spells of unemployment at the cost of the agents who have less elaborate networks and therefore rank low on the applicant's lists of prospective employers because they lack referrals. It is interesting that the unequal allocation of unemployment durations diminishes with a more volatile labor market because workers allocate fewer resources to network on building as it becomes more likely that their friends are themselves unemployed and cannot act as referees for prospective employers.

As in their earlier contribution, Tassier and Menczer (2008) focus on the role of a networked labor market in transferring information about job openings. Contrary to the two contributions just described, however, in this work they fix the network structure. The aim is to investigate the extent to which the randomness of two overlying

networks influences the labor market success of individuals as measured by their employment rates. In particular, there is a social network that comprises agents of the same ethnicity or the same gender, and a network of jobs, say, for engineers, within which information about vacancies is spread. It turns out that employment rates of social groups with a more random network are larger if connections in the job network are random. Yet if the job information flows are nonrandom, a less random social network fares better. Behind these results is the fact that higher randomness means better access to information that is outside one's social group. Higher randomness, however, also implies that information within a social group is more likely to leak outside, to the advantage of the members of the other social groups.

In an AB model concerning worker protest Kim and Hanneman (2011) place agents with limited sight in a neighborhood. Workers relate their wage to those of their neighbors and are more likely to protest as the difference becomes larger. As they protest they run the risk of being arrested, which is a function of similarly acting agents in the local area. It is shown that if wages are more unequally distributed protests become more frequent, intensive, and persistent, and that the group identity on the local level contributes to the global synchronization of the uprisings.

These attempts to implement a network structure in AB labor market models are also promising points of departure for future work. What we have in mind are AB models of labor markets with network structures that are used to evaluate the effectiveness and efficiency of policies in the light of agents' having social preferences or being embedded in distinct neighborhoods so that positive as well as negative feedback on outcome variables might arise. Such a research agenda would depart from the usual microeconomic evaluation exercise, which focuses on individual effects only by completely abstracting from the social environment of agents. We believe that applied in this way, AB labor market models may give important insights for policy making.

22.8 CONCLUSIONS

We reviewed the development of AB models of the labor market largely along the lines of the number of links to other markets. We now ask to which degree the integration of the same into broader frameworks is possible and desirable. Should we aim for models of the whole economy, with additional features steadily implemented (and possibly sequentially tested against the real data)? Or should we, rather, become acquainted with papers that provide an answer to a specific research question with exogenously given links to other parts of the economy?

From within the mainstream approach, Caballero (2010, p. 90) warns us against "an El Dorado of macroeconomics where the key insights of the periphery are incorporated into a massive dynamic stochastic general equilibrium model." To him, the core should remain "a benchmark, not a shell or a steppingstone for everything we study in macroeconomics," to be used "as just one more tool to understand a piece of the complex

problem, and to explore some potentially perverse general equilibrium effect which could affect the insights isolated in the periphery”.

We argued that one of the main advantages of the AB approach is that it allows the inclusion of features that account for potentially important economic mechanisms to a larger extent than do the analytical approaches. The flexibility, however, has its own drawbacks. Most important, the small cost of growing large models can produce black boxes that are difficult to calibrate, analyze, and interpret. Interpretation of an AB model can be done either with the help of an analytical model that gives some benchmark behavior in a simplified setting, or by means of extensive statistical testing and sensitivity analysis (Grazzini et al. 2013). When a model becomes too large, the probability of having an analytical benchmark quickly drops to zero. At the same time, the complexity of a sensitivity analysis rapidly increases. Also, the structural estimation of AB models is problematic, and only models with few parameters have so far been properly estimated (Grazzini and Richiardi 2015). Describing the behavior of a system via simulations provides a more tarnished picture than showing first derivatives. Moreover, the difficulty of the task inevitably increases as the model grows bigger.

The importance of having relaxed many of the assumptions of simpler models into a more general framework can be assessed only by reinstating them one by one, which suggests a gradualist approach to model development, and that small- or medium-scale AB versions of the periphery type are here to stay and prosper. However, AB models are more amenable to extensions and scaling up than are their analytical counterparts. The technology is ready for the big effort of combining many insights from research into larger models, much in the same way as climatologists increasingly do. This will not eradicate the uncertainty we face with respect to the behavior of our economies, but it will reasonably offer a much better alternative to the models currently used by governments and central banks all over the world. In the words of Orcutt (1987, pp. 167–168), “[M]uch remains to be achieved before the dream of combining research results, gleaned at the micro-level, into a powerful system that is useful for prediction, control, experimentation and analyses, on the aggregate level, is realized.” Yet the premises are there for big improvements to be obtained in the near future.

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CHAPTER 23

THE EMERGING STANDARD NEUROBIOLOGICAL MODEL OF DECISION MAKING

Strengths, Weaknesses, and Future Directions

SHIH-WEI WU AND PAUL W. GLIMCHER

23.1 OVERVIEW

In the 1930s Samuelson famously demonstrated that consistent human choosers behave as if they had an internal representation of an idiosyncratic subjective value, or the *utility*, of choice objects under current consideration and selected from these internal representations the single choice object that had the highest utility. Taking that proof as a starting point, many neuroeconomists have argued for a very literal and mechanistic reinterpretation of Samuelson's insight: That consistent choosers behave as they do *because* they have an internal representation of subjective value encoded in units of physical action potentials per second in their brains (e.g., Dorris and Glimcher 2004; Kim et al. 2008; Hayden et al. 2009; Glimcher 2010, 2011). They hypothesize that the brain performs an *argmax* operation on this internal representation based on the ordering of these action potential rates to select the most desirable option from a choice set. Of course, it is critical to note that these action potential rates are physical objects that are fully cardinal and unique, a property that makes them quite different from an economist's notion of utility. For that reason, and for others related to the causal relation between this activity and choice, this physical correlate of utility is typically referred to as *subjective value*.

This basic construct has led naturally to the notion that the human choice mechanism can be usefully divided into two subcomponents. The first is presumed to learn, represent, and store the values of goods and actions. The network in the brain involved in these computations is generally referred to as the *valuation network*. It is this

mechanism that explains, for any given choice set, how humans assign values to choice objects that are unique to the individual decision maker. The second of these subcomponents is presumed to allow the direct comparison of two or more valued objects and results in the selection of the option associated with higher levels of neural activation through a “winner-take-all” computational process—an algorithmic instantiation of the mathematician’s *argmax*. The brain network that performs this algorithmic operation is typically referred to as the *choice network*. Although our current evidence suggests that these processes, and the networks that embody them, cannot be seen as entirely separate, there is good evidence that these processes are instantiated as, at least in part, separable and sequentially executed algorithms (for an alternative view see Padoa-Schioppa 2011). It is this mechanism that explains how humans select the best option from any given choice set based on the values computed, stored, and represented in the antecedent valuation mechanism. Our goals in this chapter are to provide a more detailed overview of these two components and to discuss the strengths and weaknesses of this general two-stage model.

23.2 STAGE 1: THE VALUATION MECHANISM

23.2.1 Ordinal Utility to Cardinal Subjective Value

Perhaps the first critical challenge faced by any theory which assumes that humans choose the way they do because of an underlying utility-like representation in the nervous system is that of ordinality. Since Pareto, nearly all economists have acknowledged that measurements of utility are largely ordinal. Although we can say that a chooser prefers apples to oranges based on, for example, the Strong Axiom of Revealed Preference (Houthakker 1950), we cannot meaningfully say either that an apple produces twice as much utility as an orange for some chooser or that an apple specifically produces 2 utils and an orange 1 util. Von Neumann and Morgenstern (1944) elaborated on this issue when they introduced the independence axiom, but even their approach only specifies utilities to within a linear transform. Ordinality (or perhaps weak cardinality in the case of vNM utilities) is a fundamental feature of economic utility derived by choice.

This is a challenge because the measurements neurobiologists make are fundamentally cardinal and necessarily unique. When neurobiologists measure activity in the nervous system, they typically employ one of two techniques: direct measurements of the times of occurrence of electrochemical action potentials in single nerve cells or indirect measurements of this activity using functional magnetic resonance imaging (fMRI). In either case, neurobiologists measure (with error) a unique and fully cardinal object. If, as all neurobiologists believe, all of human behavior is generated through transformations of this cardinally specified activity, then measurements in the nervous

system cannot be measurements of utility itself. In practice, neuroeconomists address this issue by searching for neural signals that are linearly correlated with economically specified expected utilities or which correlate ordinally with less cardinal systems of utility. These signals are typically referred to as subjective values and are defined as real numbers ranging from zero to one thousand (the range of physically possible action potential rates). Mean subjective values are the mean firing rates of specific populations of neurons and are linearly proportional to fMRI measurements of these activities (Heeger and Ress 2002).¹ Note that mean subjective values predict choice stochastically, which reflects the fact that these action potential rates are stochastic.

Notice as well that the features of this stochasticity are reasonably well understood (e.g., Tolhurst et al. 1983; Glimcher 2005). This indicates that subjective value theory will be most closely allied with random utility-type models from economics, a parallel now being carefully explored by a number of economists. Finally, subjective values, because of their causal relation to action, should be *always* consistent with choice, though stochastically, even when choice is not consistent with utility theory. This is, of course, a critical point that may turn out to have profound implications for welfare theory.

23.2.2 Primary Sensory Transformation and Subjective Value

Whence do these subjective values arise? To some degree they must arise from the algorithmic mechanisms that transform physical events in the outside world into neural activities that guide choice. The fact that all human choosers prefer sugar to quinine, to take one obvious example, must reflect innate properties of the mechanisms by which we sense the external world.

¹ A brief introduction to two principal techniques often used to measure activity in the brain and the signals they measure is in order here. (A) "Single-neuron recording" measures activity from a single neuron by placing a tiny electrical probe very near to a targeted neuron. This technique measures the electrochemical state of a single neuron. Because the probe must be inserted into the brain, however, the technique cannot be applied to humans except in rare surgical environments. (B) Functional magnetic resonance imaging typically measures a physicochemical signal called the "blood-oxygen-level-dependent (BOLD) response" using an MRI scanner. The BOLD signal reflects changes in blood flow, blood volume, and blood oxygenation caused by changes in the metabolic demands neurons. Because changes in metabolic demand closely parallel the electrochemical states of nearby neurons, the BOLD signal is an indirect measure of neuronal activity. This measurement technique is entirely noninvasive and thus has revolutionized the study of brain and behavior in humans. However, the precise mapping between neural activity and the BOLD signal has not yet been specified with complete accuracy. To a first approximation, fMRI yields a measurement that is a linear transform of mean action potential rates across a population of nerve cells over a spatial extent of several millimeters and over a period of several seconds (Heeger and Ress 2002). We caution, however, that the precise mapping of the fMRI signal to underlying activity is a subject of intense current scrutiny. There is no significant doubt that this signal is monotonic with mean action potential rates, but it may well be that it maps more linearly to aggregate membrane depolarization than to the action potential rates derived physically from this quantity. See Logothetis et al. (2001); Heeger and Ress (2002); Logothetis (2008) for more about this issue.

In fact, these processes have now been widely studied, and the transformations that relate such external properties as sugar or quinine concentration to the internal (or endogenous) representations of these quantities tend to be strictly concave functions that bear a striking resemblance to utility functions in some regards (Fechner 1912; Stevens 1957; Glimcher 2011). This set of observations has led quite naturally to the suggestion that at least one source of concavity in subjective values involves processes that simply learn, through repeated sampling, the action potential rates associated with repeatedly consumed goods.

23.2.3 Learning and Storing the Values of Actions

To understand how neurobiologists think about this process, consider what is typically called a Pavlovian conditioning task. In such a situation a cue, for example a visual stimulus, is presented to a subject and is followed by the delivery of a reward (a positively valued good). After experiencing several times this cue-reward pairing, the subjects begin to exhibit a direct response to the cue, suggesting (sometimes indirectly) that he or she views it as a positive utility shock. This is, of course, the famous salivating dog of Pavlov's experiments (Pavlov 1927).

In 1997 Wolfram Schultz and his colleagues measured the mean action potential rates of a class of nerve cells in the base of the brain, midbrain dopamine (DA) neurons, while this process unfolded and found the following: When there was no cue associated with reward, there was a burst of DA firing at the time of reward delivery (figure 23.1). When a cue consistently preceded the reward, the activity of DA neurons at reward delivery would remain at their unique baseline (or zero) firing rate. But under these conditions, the DA neurons would fire at the time of cue presentation. Perhaps even more interesting, they observed a decrease in action potential rates (a uniquely negative number) when an apparently expected reward was omitted. These observations (figure 23.1) suggested that the dopamine signal could be seen as encoding a kind of "utility shock" that related expectations about future positive outcomes to the properties of directly sensed rewards.

This work has triggered enormous interest in the neuroscientific (and neuroeconomic) community. The primary question of interest is how to model the dynamics of neuronal activity that change with experience and how to relate these changes to choice behavior. A number of models have emerged, but the dominant class appears to be the temporal-difference (TD) learning model developed by computer scientists Richard Sutton and Andrew Barto (Sutton and Barto 1981, 1988). This is an algorithmic model that provides clear ties to normative theories of learning.

Consider modeling the dynamics of DA activity in the Pavlovian conditioning task using the TD model. In this model, an agent computes an estimate of value separately at each moment in time within a trial (a trial being a stereotyped multi-period learning problem of finite duration that is repeatedly encountered). Suppose that there are n time points within each trial. The consumption value available at time t within a trial

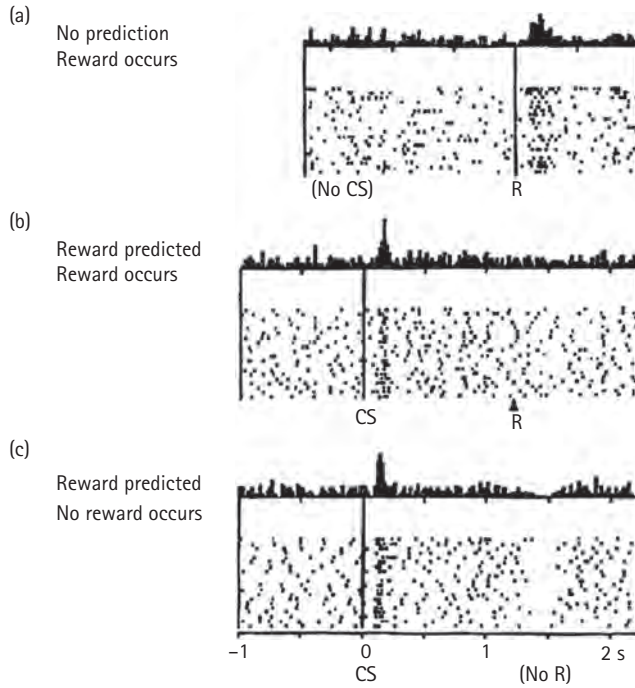


FIGURE 23.1 Activity of midbrain dopamine neurons in the Pavlovian conditioning task: (A) The DA neurons exhibit an increase in activity immediately after delivery of a reward (unconditioned stimulus, US) when no conditioned stimulus (CS) is presented prior to the reward. (B) After repeated CS–US pairings, when a CS is paired with a US, DA neurons start to show an increase in activity when CS is presented. At US delivery, DA activity remains at baseline level, when US was delivered. (C) At US delivery, DA activity goes below baseline when US is not delivered. Adapted from Schultz et al. (1997).

is defined by the sum of expected, temporally discounted future rewards during one entire trial:

$$V(t) = E[\gamma^0 r(t) + \gamma^1 r(t+1) + \gamma^2 r(t+2) + \dots] \quad (23.1)$$

where $E[\cdot]$ denotes expectation, $r(t)$ is the physical reward experienced at time t , and $\gamma = [0, 1]$ is the discount parameter. By definition, $V(t)$ can be further written as the sum of expected reward at time t and the value at $t+1$ weighted by γ :

$$V(t) = E[r(t)] + \gamma V(t+1) \quad (23.2)$$

Hence, the expected reward at time t is the difference between the estimate of value at time t , or $V(t)$, and the estimate of value at $t+1$, $V(t+1)$. The learning agent updates the value estimate by computing the prediction error $\delta(\cdot)$ —the difference between actual reward received at t and expected reward at t , $r(t) - E[r(t)]$ —by the following equation

$$V_{\text{updated}}(t) = V_{\text{current}}(t) + \alpha \delta(t), \quad (23.3)$$

where the updated value estimate at time t , $V_{updated}(t)$, is the current value estimate at time t , $V_{current}(t)$, plus the weighted prediction error $\alpha\delta(t)$. The weight $\alpha = [0, 1]$ assigned to the prediction is a parameter that determines how fast the agent learns and is often referred to as the learning rate. Some normative conditions can be placed on this term, but a discussion of those details lies outside the scope of this presentation (see Sutton and Barto 1981 for details). Given equation (23.2), we can rewrite equation (23.3) as

$$V_{updated}(t) = V_{current}(t) + \alpha [r(t) + \gamma V_{current}(t+1) - V_{current}(t)] \quad (23.4)$$

The attractive property of this model lies in the fact that the expected reward at any given point in time within a trial is the difference between the value estimate at that time point and the discounted value estimate *at the next time point* per equation (23.2). Because of this property, an update on the value estimate at the time of reward delivery would subsequently affect the value estimate of the preceding time point. Eventually (after a sufficient number of repeated trials), the value of any moment in time propagates back to the point where that future reward can first be anticipated. This backward propagation of expectation thus causes the agent (in at least some environments) to form correct (rational) expectations about all future reward deliveries with a nonzero probability of occurrence on cue presentation. The model thus learns what an economist might call “consumption paths” and responds to any event that signals a change in current or future consumption path with a learning signal. Of course, the goal of this learning is to develop a policy for choosing among possible consumption paths the one that maximizes the discounted sum of future rewards, but the details of the policy element would take us too far from the dopamine neurons that form our principle subject.

What is striking about the TD model is that it quite accurately describes the dynamics of DA neuron activity and how this activity changes over time in the Pavlovian learning tasks. These were the subjects of early DA studies. To summarize those empirical findings, DA neurons at the beginning of an experimental session do not fire when a visual cue is presented that, unknown to the subject, signals a future reward. Instead, these neurons fire when a reward is delivered. After repeated trials in which the cue-reward association consistently happens, DA neurons come to fire at the time of cue presentation, that is, at the time of the utility shock. And this is, of course, exactly what is predicted by TD-class algorithms.

Perhaps it is not surprising that different learning models have been proposed that vary in some ways from this basic template but produce quantitatively similar results. For a discussion of TD-class algorithms and their limitations in explaining DA activity, see Niv and Montague (2008) and Daw and Tobler (2013). For recent advances in modeling reinforcement learning (RL) and, in particular, on dissociating the contributions of model-free RL (e.g., the TD model) and model-based RL to choice and neural activity, see Gläscher et al. (2010) and Daw et al. (2011). Using methods from neoclassical economics, Caplin and Dean (2008) proposed an axiomatic description of

this class of learning algorithm which has been more widely influential in economic circles and that has been tested empirically by Caplin et al. (2010). They identified a set of axioms that are necessary and sufficient conditions for representing utility shocks and found that brain activity (mean action potential rates measured with fMRI) in at least one brain region, the ventral striatum, met the axiomatic conditions in a way that could drive this kind of near-normative learning of the subjective values of consumable rewards. This brain area is rich in the neurotransmitter dopamine and receives direct projections from the midbrain DA system.

23.2.4 An Overview of the Network for Valuation

There is now accumulating evidence that neural circuitry including the midbrain DA neurons, mentioned above, and a series of other brain areas including the striatum, the ventromedial prefrontal cortex (vmPFC), and the orbitofrontal cortex (OFC) are involved in the representation of the subjective values of consumable goods and monetary rewards (Padoa-Schioppa and Assad 2006; Lau and Glimcher 2008; Plassmann et al. 2007; Chib et al. 2009; Levy and Glimcher 2011). Neuronal action potential rates in these areas have been widely shown to be both linearly proportional to the utilities (or expected utilities in probabilistic lotteries) and predictive of choice even when subjects behave inconsistently (e.g., Kable and Glimcher 2009). Moreover, a critical feature of these brain areas is that activity elicited by a given option, although stochastic in nature, is independent of what the other available options are (Padoa-Schioppa and Assad 2008). For reviews and meta-analysis on the valuation system see Bartra et al. (2013) and Clithero and Rangel (2014).

23.3 STAGE 2: THE CHOICE MECHANISM

The choice stage refers to the algorithmic processes that compare the subjective values associated with different objects in a choice set so as to guide the chooser. In principle, the neural circuits involved in the choice process should be able to represent the subjective value associated with each available option in any given choice set. Hence, the choice circuit should receive information about subjective value from the valuation circuit, but in a way restricted to the current choice set (figure 23.2). However, one should remain cautious when thinking algorithmically about the interaction between valuation and choice circuits as purely “feed forward” in the sense that subjective-value signals are passed unidirectionally from the valuation circuit to the choice circuit. In fact, these two systems are heavily and reciprocally interconnected, suggesting that as we come to understand the algorithmic process more completely, the logical separability of these two systems will come to be reduced. Indeed, there is already evidence that

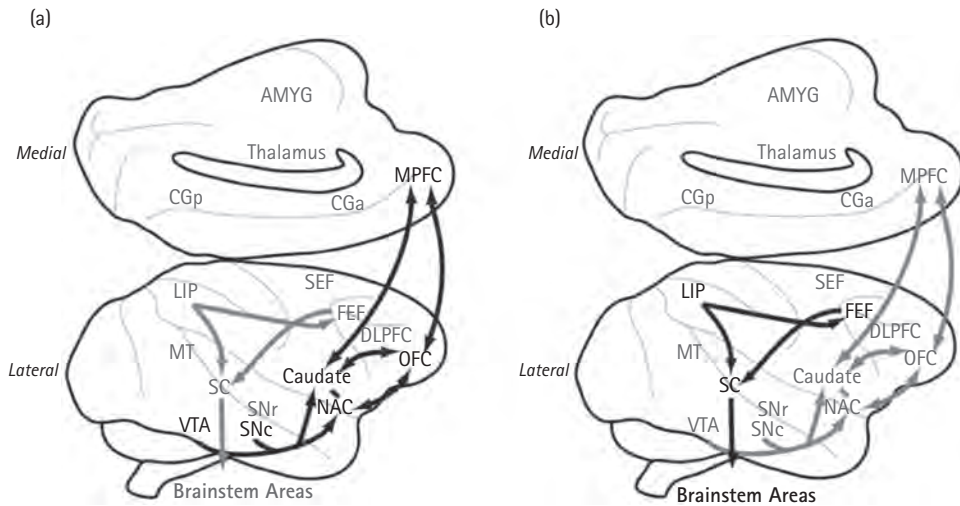


FIGURE 23.2 Neural circuitry of valuation and choice. (A) Valuation circuitry. (B) Choice circuitry. Structures in these networks and the directions of their connections are highlighted in black. Adapted from Kable and Glimcher (2009).

choice and valuation circuits may interact algorithmically. Although the implications of this for reduced-form models are currently unclear, several models have been proposed. In one model, Padoa-Schioppa (2011) proposed that subjective value is being computed and compared in the space of “goods” in the OFC and that vmPFC and that these computations are done in a fashion that is independent of the sensorimotor contingencies of choice. After a choice is made, a transformation that maps the chosen good onto the appropriate course of action originates in the OFC and vmPFC and culminates in the planning and execution of motor action. Other models, such as the one proposed by Glimcher (e.g., Louie et al. 2011) and by Shadlen and colleagues (Gold and Shadlen 2007), emphasize value coding in the space of motor actions that are required to obtain the desirable goods. The latter view is reviewed in detail in the next section.

23.3.1 An Overview of the Network for Choice

Our current understanding of the value comparison process at the theoretical, algorithmic, and circuit levels is largely based (for technical reasons) on studies of a well-understood model system of decisionmaking in monkeys. In these *awake-behaving monkey electrophysiology studies*, monkeys choose between two lotteries by making an eye movement (saccade) to one of two possible visual targets that vary in the magnitude or probability of reward, sometimes under conditions of partial information. This model system consists of a heavily interconnected network of brain areas that participate in both the encoding of the subjective value of the lotteries under consideration

and the winner-take-all, or argmax, process. The brain areas that participate in this process include the lateral intraparietal area (LIP), the frontal eye field (FEF), and the superior colliculus (SC) (figure 23.2). There is now accumulating evidence that this circuitry is involved in representing the *relative subjective value* (RSV) associated with different options (Platt and Glimcher 1999; Gold and Shadlen 2001; Louie et al. 2011). The current data suggest that at any moment in time neurons in the LIP represent the instantaneous RSV of each lottery (e.g., Dorris and Glimcher 2004; Rorie et al. 2010), a representation that is believed to be derived (algorithmically) from the representation of SV localized in the valuation network, particularly in the vmPFC, OFC, and ventral striatum.

Note that RSV would serve to map SV onto the limited dynamic range of the LIP neurons. Such neurons are limited in number, fire over a roughly 100Hz dynamic range, and have (errors that are drawn from) Poisson-like distributions. This means that the representation of RSV, rather than SV, in this structure may solve an important problem. The shift to RSV guarantees a distribution of the SVs of the current choice set over the limited dynamic range of these neurons. Unfortunately, the finite dynamic range and noise associated with these neurons may also impose a constraint. As the choice set becomes larger, noise may swamp the signal, leading to profound failures to deterministically identify the preferred option when selecting among large numbers of possible movements (Louie et al. 2013). In summary, the available data suggest that at all three of these areas, LIP, FEF and SC, carry signals encoding RSV and that movements occur when activity associated with one of the positively valued options drives its associated collicular neurons past a fixed numerical threshold, triggering the physical action that instantiates choice.

23.4 FUTURE DIRECTIONS

Although our current model incorporates many existing data and provides a useful framework for thinking about the neurobiological mechanisms of decision making, the model still lacks descriptions of certain concepts that have been identified by economists and psychologists as critical in the decisionmaking processes. In this section, we seek to expand the current model in two directions. The first direction is motivated by the notion of reference dependence, which, as many psychologists have argued (e.g., Kahneman and Tversky 1979), is a core feature of the valuation process. It has been observed not only in economic decision making but also in a wide variety of judgment tasks. Economists have also begun to incorporate this concept into newly developed models of decision making (Sugden 2003; Koszegi and Rabin 2006). This concept has not received much attention in the neuroeconomics community, but as we mention later, there is a close tie between what dopamine neurons encode and reference dependence. Our goal must therefore be to incorporate reference dependence to the computational algorithm implemented during valuation.

The second direction is motivated by the fact that in many decision scenarios we face, information about probability associated with potential outcomes is not explicitly given and often needs to be estimated by the decision maker. This feature makes these decisions unlike the classical lottery tasks studied in a typical economic laboratory, where probability information is explicitly revealed to the subjects in numerical or graphical form. We introduce recent studies concerning the way information about probability appears to be distorted (more formally: how the independence axiom is violated) in classical economic lottery tasks and in mathematically equivalent “motor” and “perceptual” lottery tasks. Our goal is to expand the standard model to include the violations of the independence axiom in different tasks and to search for the algorithmic sources of this distortion at a neural level.

23.4.1 Incorporating Reference Dependence into Value Computation

Kahneman and Tversky (1979) defined the choice-related “value” (a utility-like construct) of potential outcomes as gains or losses relative to a reference point. The reference point, as the authors put it, can be viewed as an adaptation level, status quo level, or expectation level defined by the past and present experiences of the decision maker. They argued that the evaluation of monetary changes from this reference point shares many of the mathematical properties of perceptual judgments about such things as sugar concentration, temperature, or brightness, and they noted that many of these perceptual experiences show shifting unique zero-levels that impact perception. For example, it is well known that in a room that is 15°C, it is easier to discriminate 17°C from 19°C than it is to discriminate 27°C from 29°C but the reverse is true when the room is 25°C. In more economic terms, the discriminability of temperature change decreases as the distance from the reference point increases (Weber 1850). This is relevant to economic choice because the neural mechanisms that underlie these phenomena are now fairly well understood and turn out to be ubiquitous. The second feature of Kahneman and Tversky’s reference-dependent value function is that it captures simultaneous risk aversion in the gain domain and risk-seeking in the loss domain (although there are, of course, other ways to capture this, e.g., Friedman and Savage 1948). The third feature of the value function is that it captures aversion to losses. As these authors often put it, losses loom larger than gains, for “the aggravation that one experiences in losing a sum of money appears to be greater than the pleasure associated with gaining the same amount (Kahneman and Tversky 1979, p. 279).” To understand loss aversion, consider a lottery $(0.5, \$x; 0.5, -\$x)$ with a 50-50 chance of gaining $\$x$ or losing $\$x$. Empirically, it has been observed that most people find this lottery very unattractive. Furthermore, for $x > y \geq 0$, $(0.5, \$y; 0.5, -\$y)$ is often preferred to $(0.5, \$x; 0.5, -\$x)$, according to Kahneman and Tversky. They used these two observations to motivate a value function for losses that is steeper than the value function for gains.

Thus the value function they proposed was

$$v(x) = \begin{cases} x^\alpha, & x > 0 \\ -\lambda(-x)^\beta, & x < 0 \end{cases} \quad (23.5)$$

where x denotes outcomes relative to the reference point, α and β characterize the curvature of the function in the gain domain and loss domain respectively, and λ is used to represent the degree of loss aversion.

In a seminal paper, Tom et al. (2007) attempted to study the neural basis of loss aversion using fMRI in humans. In their experiment, on each trial the subjects had to decide whether to accept or reject a mixed lottery ($0.5, \$x; 0.5, -\y), a 50-50 chance of winning $\$x$ or losing $\$y$. The amounts of gain and loss were independently manipulated throughout the experiment. This is critical because in the fMRI analysis, gains and losses could be implemented as separate and uncorrelated parametric regressors of interest. The authors found that regions including the ventromedial prefrontal cortex and the ventral striatum encode both the gains and the losses associated with any given lottery (figure 23.3a). Activity in these regions was positively correlated with gains and negatively correlated with losses. This result was consistent with Kahneman and Tversky's value function if one assumes that the reference point was fixed at zero throughout the experiment for each subject and remained so across all subjects. When treating the value function as linear and only modeling the loss aversion parameter, Tom and colleagues found that their behavioral measure of λ was highly correlated with the neural measure of λ (the asymmetry of the gain and loss regression slopes) in regions including the ventral striatum (figure 23.3b). This pointed out the possibility of a neural representation of a simplified version of the value function in the valuation circuitry

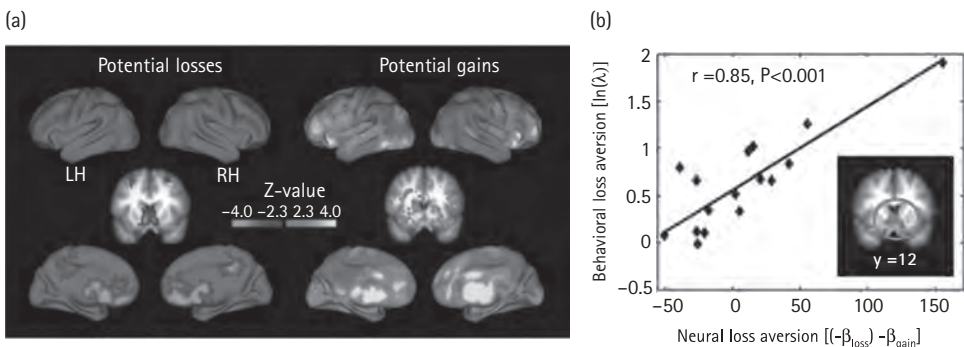


FIGURE 23.3 Neural correlates of gains, losses, and loss aversion - Neural correlates of gains, losses, and loss aversion. **A.** Neural correlates of gains and losses. Two separate, thresholded statistical parametric maps showing regions whose activity was negatively correlated with losses (blue), and regions whose activity was positively correlated with gains (orange-yellow). **B.** Neural measure of loss aversion obtained from ventral striatum (red) was correlated with behavioral measure of loss aversion. Adapted from Tom et al. (2007).

that respects both the unique zero imposed by Kahneman and Tversky's model and the gain-loss asymmetry that they hypothesized.

These advances at the neural-algorithmic level aside, Koszegi and Rabin (2006, 2007) have, in related work, begun to develop a behavioral model of reference-dependent preferences in which they have defined the reference point as the decision maker's rational beliefs about future outcomes. Instead of a fixed point or status quo, such rational beliefs are stochastic. Let $r = [r_1, \dots, r_k] \in R^k$ denote the set of possible reference-level consumptions and $G(r)$ denote the probability distribution over r . For an option with a set of potential consumption outcomes $c = [c_1, \dots, c_k] \in R^k$ with support $F(c)$, the utility of this option is given by

$$u(F|G) = \int \int u(c|r) dG(r) dF(c) \quad (23.6)$$

where $u(c|r)$ denotes the utility of a consumption given a reference level. What is unique about these authors' definition is that of $u(c|r)$. It is defined as the sum of two components: the consumption utility $m(c)$ —the pure and absolute pleasure derived from consuming an outcome—and a reference-dependent gain-loss utility $n(c|r)$ similar to Koszegi and Rabin's value function

$$u(c|r) = m(c) + n(c|r). \quad (23.7)$$

This implies that the utility of consuming an outcome has both absolute and relative components. The absolute component is the consumption utility $m(c)$. The relative component is the reference-dependent gain-loss utility $n(c|r)$. Specifically, $n(c|r)$ is derived from comparing $m(c)$ with the consumption utility of a reference level

$$n(c|r) = u(m(c) - m(r)). \quad (23.8)$$

Here, $u(\cdot)$ is the gain-loss utility function. In this definition, gain or loss associated with an outcome is based on comparing its consumption utility with that of a reference level. The utility of gain or loss is equivalent to saying "how a person feels about gaining or losing depends in a universal way on the changes of consumption utility associated with such gains or losses" (Koszegi and Rabin 2006, p. 1139).

Several intriguing features in Koszegi and Rabin's formalization of the reference-dependent utility deviate from Kahneman and Tversky's reference-dependent value function. First, unlike Kahneman and Tversky's model where reference point is a fixed value, reference point in Koszegi and Rabin is defined as probability distribution over potential outcomes. Specifically, such probability distribution reflects rational beliefs established on the basis of recent experience. Second, Koszegi and Rabin defined the utility associated with an outcome as the sum of its consumption utility, which is absolute and context-independent, and its gain-loss utility, which is relative and reference-dependent. Third, gain-loss utility is strictly defined on the basis of consumption utility as in equation (23.8). Finally, gain-loss utility associated with a

potential outcome depends on the deviation of the consumption utility of that outcome from the consumption utility of the reference level.

These different models thus provide exciting and interesting hypotheses for testing reference-dependent value computations performed at the neural level. We outline several questions of interest and testable hypotheses below.

1. What is the neurobiological nature of consumption utility? Consumption utility is not required in the Kahneman and Tversky definition, but it is the core of the Koszegi and Rabin utility function. Hence, to be able test the two models at the neural level, one needs to first establish neurobiological evidence for the presence of consumption utility. The notion that it is absolute, strictly increasing, and represents the hedonic value of consuming a good or reward provides important search criteria for relevant neural signals. However, neurobiologists and economists need to work together to axiomatize the necessary and sufficient conditions for the neural consumption utility. For example, one key question is whether consumption utility necessarily and sufficiently requires one to measure neural signals at the time of consumption. Can neural signals associated with a cue that predicts a reward or neural signals right before consumption of the reward be labeled as consumption utility of that reward?
2. What are the neurobiological natures of gains and losses? Here, the two models provide different hypotheses because of their differences in defining the reference point. Kahneman and Tversky defined gain and loss with respect to something like a fixed status quo that is based on animals' current state. This state can be the current wealth level (e.g., monetary rewards) or the satiation or consumption level (e.g., a primary reward such as food or juice). In contrast, Koszegi and Rabin defined a reward as a gain or loss by comparing its consumption utility with the expectations about rewards established by recent experience with the environment. Such expectations are summarized by a probability distribution on possible rewards. To test the two models, one needs to independently manipulate status quo and expectation. For example, status quo manipulation can be achieved via endowment (for monetary rewards) or selective satiation (for food or juice rewards) prior to an experimental session, while one can manipulate expectation by systematically varying rewards in terms of, (e.g., quantity, size, or type) that the subjects would experience in an experimental session.
3. If the answer to the previous question supports an expectation-based reference point, then one needs to further investigate whether the neural representation of the reference point is stochastic and how reference-dependent value computations take into account such a feature. Koszegi and Rabin provided a specific algorithm for such computations in equation (23.6). In summary, the utility of a potential reward is computed by the sum of the gain-loss utility associated with each possible reward weighted by its probability of occurrence.
4. Finally, a key feature of Koszegi and Rabin's notion of "utility" is that it is composed of two components: the consumption utility of a potential reward and the

gain-loss utility of that reward. To test this conjecture, one needs to independently manipulate these two components. For example, we can manipulate consumption utility on a trial-by-trial basis by varying the amount of juice reward associated with an option and, independent of this manipulation, we can manipulate the gain-loss utility by changing the reference-point distribution across different experimental sessions.

The neurobiological evidence for valuation and choice accumulated since the beginning of the twenty-first century has only provided some very preliminary results for resolving the questions outlined above. For example, we now know that the valuation system represents the expected value of a cue that predicts the delivery of a reward (e.g., Schultz et al. 1997; Tremblay and Schultz 1999). But do such signals reflect consumption utility? At the time of reward consumption, dopamine neurons and the ventral striatum encode prediction error. This seems to support the notion that the valuation network does not represent consumption utility-like signals. Rather, it encodes the deviation of actual outcome from expectation at the time of consumption. In contrast, there is evidence suggesting that the OFC might be a candidate region for representing consumption utility. Neurons in the OFC are known to selectively respond to different rewards. For example, a population of OFC neurons may respond only to grape juice, while another population may respond only to apple juice. Moreover, these neurons respond (1) at the time when the cue that predicts the specific reward is presented, (2) prior to reward delivery, and (3) immediately after reward delivery (Schultz et al. 2000). These results indicate distinct computation roles in the subcomponents of the valuation system and provide key insights into the formation of the reference point and how it relates to subjective value and choice at the neural algorithmic level. Another example of the challenge we face is the stochastic, expectation-based reference-point hypothesis. Although we know that the valuation network represents the expected value of a stimulus that predicts a reward at the time of stimulus presentation, we know very little about how such expected-value signals would be modulated by the expected value of other stimuli present in the recent past. What we know is that valuation signals in the OFC are context-dependent. It adapts to the range of subjective values associated with rewards experienced in the recent past (Tremblay and Schultz 1999; Padoa-Schioppa 2009).

23.4.2 Decision Under Risk: Neural Representations and Distortions of Different Sources of Probability Information

In this section we focus on the representation of probability information, another key variable when people are engaged in decision making under risk. We begin with a review of past research on probability distortion in standard lottery tasks. Then we review recent behavioral and neural studies investigating how differences in the way information about probability is revealed to choosers affects how it is distorted. Finally, we discuss recent fMRI studies that employ these behavioral approaches to shed light

on the neural representation of probability and the neural mechanisms underlying probability distortion (FitzGerald et al. 2010; Wu et al. 2011).

23.4.2.1 *The Allais Paradox and the Common Consequence Effect*

Traditionally, in decision under risk, people choose between lotteries, expressed as $(p_1, x_1; \dots; p_n, x_n)$, $\sum_{i=1}^n p_i = 1$ where x denotes the outcome and p denotes the probability of occurrence associated with the outcome (e.g., Von Neumann and Morgenstern 1944). In expected utility theory (EUT), the desirability of a lottery is, of course, specified by its expected utility, the sum of the utility associated with each outcome weighted by its objective probability of occurrence. Formally, von Neumann and Morgenstern (1944) expressed this with their independence axiom: for any lottery L_1 and L_2 , if L_1 is preferred to L_2 , then $\alpha L_1 + (1 - \alpha)L_3$ should be preferred to $\alpha L_2 + (1 - \alpha)L_3$, and vice versa, $0 < \alpha \leq 1$. Yet a wealth of empirical evidence now suggests that human choosers often violate this axiom (e.g., Allais 1953). Consider the following violation of independence from Kahneman and Tversky (1979):

1. Problem 1: Choose between A: (0.33, \$2,500; 0.66, \$2,400) and B: (1, \$2,400)
2. Problem 2: Choose between C: (0.33, \$2,500) and D: (0.34, \$2,400)

In problem 1, 82 percent of subjects chose B, while 83 percent of subjects chose C in problem 2. This violates the independence axiom because adding a common consequence of a 66 percent chance of winning \$2400 to C and D in problem 2 to construct A and B in problem 1 should not alter subjects' choice. If subjects prefer A in problem 1, then they should prefer C in problem 2, and vice versa. If subjects prefer B in problem 1, then they should prefer D in problem 2 and vice versa.

In order to interpret the violation of the independence axiom in decision under risk, Kahneman and Tversky hypothesized that people assign nonlinear weights to probabilities when making risky decisions. The function that characterizes these nonlinear weights is often referred to as the weighting function $\pi(\cdot)$ (e.g., Kahneman and Tversky 1979; Tversky and Kahneman 1992) or the probability weighting function (Wu and Gonzalez 1996). Note that the weighting function is at least somewhat conceptually distinct from the notion of subjective probability advanced by Savage (Savage 1954). In the original prospect theory (Kahneman and Tversky 1979), the decision value (or prospect) over a lottery was hypothesized to be derived by choosers at an algorithmic level by the sum of the value associated with each outcome in the lottery (defined in equation [23.5] in the preceding section) weighted by its decision weight. Note that the decision weight associated with an outcome might not be the weight associated with its probability of occurrence. In the original prospect theory, this is the case. In the subsequent version, cumulative prospect theory, however, it is not. For example, in the original theory (Kahneman and Tversky 1979), the prospect of a lottery with two nonzero outcomes $(p, \$x; q, \$y)$, $x > y \geq 0$, $p + q = 1$ is $v(y) + \pi(p)[v(x) - v(y)]$ where $\pi(\cdot)$ is the weighting function and $v(\cdot)$ is the value function. In cumulative prospect theory, Tversky and Kahneman (1992) incorporated

rank-dependence into this framework such that the prospect of a lottery became $\pi(p)v(x) + [\pi(p+q) - \pi(p)]v(y)$.

Broadly speaking, when models of this type are parameterized in any number of ways (e.g., Tversky and Kahneman 1992; Gonzalez and Wu 1999; Wu and Gonzalez 1996), $\pi(\cdot)$ is found to be well described by an inverse-S-shaped function, that is, a function that is concave at small probabilities and convex at moderate-to-large probabilities. Prelec (1998) developed an axiomatic foundation for these functions, deriving axioms that were necessary conditions for the probability weighting function proposed by Kahneman and Tversky (Kahneman and Tversky 1979; Tversky and Kahneman 1992). Since that time a host of studies have examined human choice behavior with prospect theory and the Prelec function and almost universally found that parameterizations indicate this inverse S-shaped structure for the Prelec function.

Notice, however, that nearly all of these parameterizations have been based on data gathered from human subjects in what might be called classical lottery tasks. In these kinds of experiments information about probability distributions on possible outcomes are explicitly described in numerical or graphical form to subjects who then express their preferences. What is worth noting is that this kind of decision making scenario describes only a subset of the risky decision making scenarios we face in everyday life. What is surprising is that a growing body of evidence now suggests that the parameterized probability weighting function extracted outside classical lottery tasks looks quite different from that extracted in these more classical situations.

23.4.2.2 *Motor Decision Making and Probability Distortion*

A baseball player deciding whether to swing a bat at an incoming ball is not given explicit numerical estimates of the probability of producing a base hit, a home run, or a miss. In situations like these, decision makers typically estimate probability based on experience. And of course these estimates must take into account a number of sources of variance including errors in neurobiologically derived estimates of the speed and position of the ball and estimates of the movement error associated with a plan to swing the bat toward a fixed location in space and time. How humans estimate probability in these situations, and how well they do it, are currently under intense investigation. There is accumulating evidence that in these domains humans achieve near-normative performance, taking into account noise coming from the perceptual and motor systems in a way that seems to obey the independence axiom (Geisler 1989; Trommershäuser et al. 2003b,a; Körding and Wolpert 2004; Najemnik and Geisler 2005; Tassinari et al. 2006; Battaglia and Schrater 2007; Dean et al. 2007). These findings present a sharp contrast to results from economic decision under risk, in which information about probability is explicitly stated. Few studies, however, directly compared decision making in classical lottery tasks with perceptual or motor tasks.

To formally compare decision making under different modalities, Wu et al. (2009) developed a method for translating a classical lottery to a mathematically equivalent “motor” (or movement-based) lottery (figure 23.4). They then asked the subjects to perform identical sets of incentive-compatible classical and motor lotteries. Information

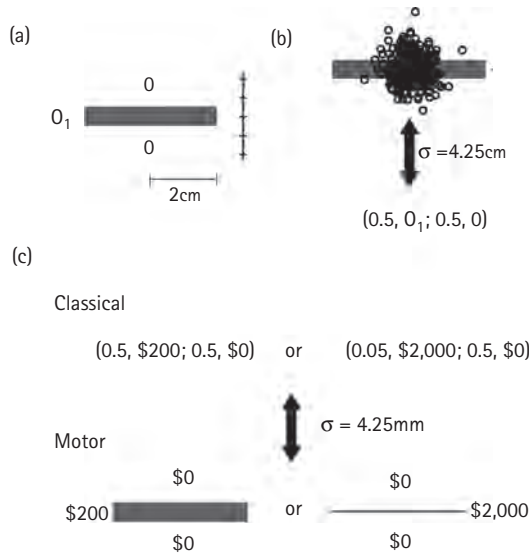


FIGURE 23.4 Construction of a motor lottery task. (A) In a rapid point task, the subjects were trained to hit a single target within a very short time window (usually $< 0.7s$). (B) Here we superimposed the distribution of movement end points from an actual subject. We verified that the distribution of movement end points is bivariate Gaussian and characterized motor noise by the estimated standard deviation (σ) of the distribution. For this subject ($\sigma = 4.25mm$), this target was equivalent to a lottery $(0.5, O_1; 0.5, 0)$. (C) Given the motor noise separately estimated for each subject, we could translate a binary classical lottery task, for example, choosing between $(0.5, \$200; 0.5, \$0)$ and $(0.05, \$2000; 0.5, \$0)$, to a mathematically equivalent motor lottery task. In a later, decision-making phase of the experiment, we asked the subjects to choose between lotteries in classical tasks and in motor tasks. We emphasized that subjects during the motor lottery task only indicated which lottery they preferred. They did not execute any pointing movement during the decision tasks. Adapted from Wu et al. (2011).

about the probability of winning in the motor lotteries depended on both the size of the target that the subjects had to successfully hit and the intrinsic variability of the subject's movement (which subjects had to learn from experience).

In an initial training session aimed at teaching subjects about their movement variability, the subjects were asked to repeatedly and quickly (within < 0.7 seconds) hit with their finger a rectangular target that appeared on a computer touchscreen. The size of that target was varied independently to control the probability of “winning” a given lottery (or trial). Hitting the target (p) resulted in a small monetary gain (v), and hitting anywhere else on the screen ($1 - p$) won nothing (o).² The critical idea of these lotteries,

² We note that, during training, hitting the screen after the 0.7-second time limit resulted in a monetary loss five times greater than the gain. This manipulation served to train the subjects to response within 0.7 seconds. In practice, the probability of this occurring in a trained subject is negligible.

what makes them lotteries, is that subjects do not have perfect control over their movements owing to intrinsic noise in the motor system introduced by the very short time window. After extensive training under the same time constraint, the motor noise often becomes stable at a within-subject level (Trommershäuser et al. 2003a,b). For the experimenter, this means that any binary lottery can be constructed once the movement variability, or “motor noise” has been measured, although the approach assumes that subjects can estimate the probability of their hitting a target given knowledge of their own motor noise.

The questions raised by this line of research are whether decisions made in this way show different patterns of rationality, particularly with regard to the independence axiom, and whether they show different risk preferences. In fact, Wu and colleagues found that subjects violated the independence axiom in motor lotteries just as much as they violated the axiom in the classical lottery task. The pattern of violation was markedly different, however. Their parametric analysis suggested that this difference could be attributed to a change in the probability weighting function. Rather than the typical overweighting of small probabilities and underweighting of moderate-to-large probabilities, subjects in the motor task tend to *underweight small probabilities and overweight moderate-to-large ones*.

This pattern of inferred probability distortion is of particular interest because the ability of subjects to estimate the probability of reward in the motor lottery task depends on the subjects’ previous experience hitting targets on the touchscreen. Hence, it is an experience-based lottery task in which knowledge about probability associated with hitting motor targets was established by experience. Behavioral studies have begun to reveal that, as opposed to overweighting rare events when probability information is revealed explicitly, people tend to underweight rare events when information about probability associated with rare monetary gains is acquired by sampling experience (Hertwig et al. 2004; Jessup et al. 2008; Ungemach et al. 2009); for a review, see Hertwig and Erev (2009). This difference is often called the description-experience gap. Despite accumulating evidence suggesting the existence of such difference at the behavioral level, very few studies (FitzGerald et al. 2010; Wu et al. 2011) have directly compared the neural representation of probability in decision under risk when information about probability comes from different sources, for example, when it is described explicitly versus when it is learned from experience. That seems important because the neural measurements might give insight into the algorithmic constraints that shape these two classes of decision.

Neurobiological studies of decisions involving risk and uncertainty have identified the neural systems that correlate with these economic variables (Platt and Huettel 2008). In reinforcement learning tasks, dopamine neurons have been shown to represent the probability of both reward and risk (defined as the variance) associated with reward-predicting stimuli (Fiorillo et al. 2003). In humans, fMRI studies have reported that the striatum, the anterior insula, the medial prefrontal cortex, lateral prefrontal cortex, and posterior parietal cortex represent these variables as well (FitzGerald et al.

2010; Hsu et al. 2009; Huettel et al. 2005, 2006; Knutson et al. 2005; Paulus et al. 2003; Preusschoff et al. 2006; Tobler et al. 2008; Wu et al. 2011).

Unfortunately, the neural results available today are not entirely consistent. Wu and colleagues (2011) found that the medial prefrontal cortex (mPFC) encodes “probability weight” in a classical lottery task and in a motor lottery task. In that study, mPFC activity only showed correlation with probability of reward in the motor lottery but was not correlated with physical size of the target in a size judgment task in which the physical properties of the stimuli were identical to those in the motor lottery task (figure 23.5a). Together, the results suggest a convergence of two mechanisms for probability encoding and pushes neuroeconomists to search upstream (in the algorithmic sense) for these two probability-encoding mechanisms. Others, however, have found that activity in the dorsolateral prefrontal cortex (Tobler et al. 2008) and in the striatum (Hsu et al. 2009) is correlated with probability distortion in similar tasks. FitzGerald and colleagues (2010) had subjects choose between lotteries for which the probability of reward associated with one lottery was revealed explicitly and the probability of reward associated with the other was acquired by sampling experience. They found that at the time when the subjects were asked to choose between the lotteries, activity in the medial

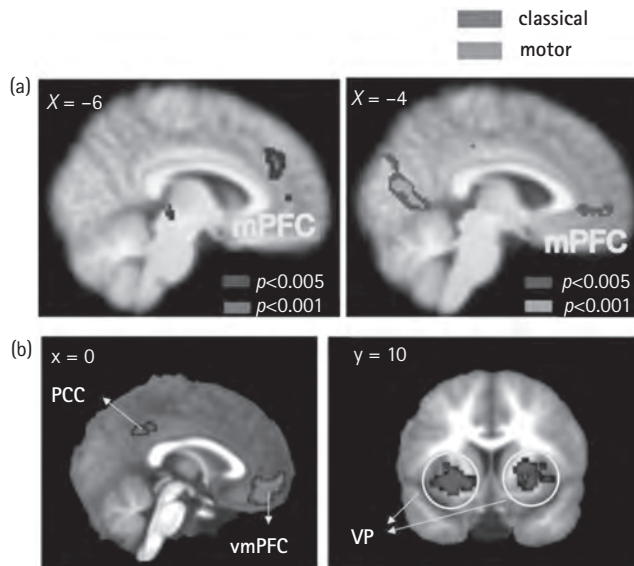


FIGURE 23.5 Comparisons of neural correlates of different sources of probability in described and in experienced lottery tasks. (A) Neural correlates of probability weight in classical (described) lottery tasks (red) and in motor lottery task (green); mPFC: medial prefrontal cortex. (B) Neural correlates of subjective value (estimated probability of reward) in classical (described) lottery tasks (red) and in experienced or learned lottery tasks (blue); vmPFC: ventromedial prefrontal cortex; PCC: posterior cingulate cortex; VP: ventral putamen. Adapted from Wu et al. (2011) and FitzGerald et al. (2010).

prefrontal cortex and another area (the posterior cingulate cortex) was correlated with the decision-value of the lottery learned by experience, while activity in yet another brain area, the ventral putamen, was correlated with the decision-value of a classically described lottery (figure 23.5b).

In summary, since Allais (1953), converging evidence at the behavioral level suggests that humans distort information about probability in decision under risk in a highly characteristic manner. That is, people tend to overweight small probabilities and underweight moderate-to-large ones. However, such a pattern of distortion has been found primarily in situations where information about outcomes and their associated probabilities of occurrence are described explicitly in numerical or graphical form to the subjects. This line of results was recently challenged, however, by two other lines of research. The first pointed out the description-experience gap in decision making and found that the pattern of distortion was markedly different in situations where people acquired information about probability by sampling experience (Barron and Erev 2003; Hertwig et al. 2004; Jessup et al. 2008; Ungemach et al. 2009). The second challenge came from research about perception and action suggesting that people are near-optimal EUT maximizers in perceptual and motor tasks that are formally equivalent to decision making under risk (Geisler 1989; Trommershäuser et al. 2003b,a; Körding and Wolpert 2004; Najemnik and Geisler 2005; Tassinari et al. 2006; Battaglia and Schrater 2007; Dean et al. 2007). A direct comparison of the classical descriptive lottery tasks against mathematically equivalent motor lottery tasks (Wu et al. 2009) suggested that, similar to the findings in the description-experience gap, people tend to underweight small probabilities and overweight large probabilities in motor tasks. Together, these results pointed out that in more naturalistic settings where probability information has to be inferred, either from experience or from knowledge about the environment or the nervous system, the way such information is distorted is markedly different from descriptive scenarios. Neuroimaging studies have just begun to address these phenomena observed in behavior by investigating the underlying neural correlates of probability distortion. We believe that the ultimate success or failure of the contribution of neuroeconomics to understanding probability representations would depend on identifying the algorithms involved in probability distortion at the neural level and how their neural implementations would tie to current understanding about reward learning systems, particularly the midbrain dopamine system.

In addition, fundamental questions such as how the reference-dependent utility of a reward is integrated with probability weight at the time of choice and the computational algorithm for such integration also need to be addressed. In doing so it is important to first separately examine the neural systems involved in computing the reference-dependent utility and probability, in either choice or nonchoice situations. Once the neural representations in these domains have been established, one can then create a decision task that independently manipulates rewards and probability. This will then allow for the possibility of investigating the integration computations and will allow for an assessment of how neural systems involved in either kind of computation

(reference-dependent utility or probability weight) contribute to the systems that represent the integrated products.

23.5 CONCLUDING REMARKS

Since Samuelson (1938), neoclassical economics has largely relied on revealed-preference theory in conducting economic analysis. What is central in this approach is that the relation between internal representations of subjective value and choice behavior is established on the basis of axioms about revealed preference. We began the chapter by arguing that subjective value is represented by neuronal activity in units of action potential and that choosers behave as they do *because*, not *as if*, there is an internally consistent cardinal representation of subjective value. Armed with this assumption, neuroeconomists have accumulated a wealth of neurobiological evidence related to decision computations. As a result of this rapid accumulation of findings, a standard neurobiological model emerged. This emerging model proposed that the neurobiological mechanisms of decision making could be subdivided into two components, valuation and choice. Valuation refers to computing subjective value for each option available in a choice set. The network involved in this component includes the dopamine neurons in the midbrain, the medial prefrontal cortex, the orbitofrontal cortex, and the striatum. We reviewed evidence suggesting SV computation and algorithms for learning and representing SV in the valuation circuitry. The second component, referred to as choice, is involved in integrating SV computed in the valuation circuitry so as to compute relative subjective value associated with different options. The RSV serves to guide action planning for the animal to obtain the desired option. In the framework of saccadic decision making, the choice network involves the lateral intraparietal area, the frontal eye field, and the superior colliculus. To summarize, this emerging model placed a particular emphasis on identifying algorithms involved in decision-related computations at different stages and their implementations at the system level.

As part of an attempt to expand the model at both the algorithmic and the neural-implementation levels, we identified two novel future directions for value computations raised by economic theorists and psychologists. The first direction was related to the notion of the reference point. Different models in economics and psychology for how to formally model reference points have been proposed. The models in essence provided precise algorithms for reference-dependent value computations that could be tested and compared at the neural level. The theoretical notion of reference point proposed by Koszegi and Rabin (2006) drew particular attention to neurobiologists. In their view, the reference point is the rational belief, represented by probability distribution over possible outcomes, established by experience from the recent past. This is closely related to what neurobiologists found in midbrain dopamine neurons in reward learning tasks, and several laboratories have begun to investigate the algorithmic form of reference-dependent computations in the valuation network.

The second direction was related to different sources of probability information and how they might differentially affect choice in decision under risk. Classical results indicated that humans tend to distort probability information, but these results tended to be replicated in situations where information about probability was explicitly revealed to the chooser. More recently, accumulating evidence suggests that the way probability information is distorted is different from descriptive scenarios in motor decisions and in scenarios where knowledge of probability was acquired via sampling experience. However, the neural correlates associated with different sources of probability information and the precise neural mechanisms responsible for differences in probability distortion remain largely unknown and are under investigation. We believe that the differences lie in the systems representing probability information and the algorithms they are capable of in computing probability information. Hence, the key contribution that neuroeconomics could possibly make to this field would be to identify the systems involved in representing different sources of probability information and the potential differences in the algorithms implemented in those systems.

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CHAPTER 24

THE EPISTEMOLOGY OF SIMULATION, COMPUTATION, AND DYNAMICS IN ECONOMICS

K. VELA VELUPILLAI

Tobin exclaimed at Nozick: “There’s nothing more dangerous than a philosopher who’s learned a little bit of economics.” To which Nozick immediately responded: “Unless it’s an economist who hasn’t learned any philosophy”

T. Hutchinson

In what other way, if not simulation by a Turing machine, can we understand the process of making free choices? By making them, perhaps.

R. Nozick (1981)

What must be achieved is in fact this: That every paralogism be recognised as an error of calculation, and that every sophism when expressed in this new kind of notation . . . be corrected easily by the laws of this philosophical grammar . . . Once this is done, then when a controversy arises, disputation will no more be needed between two philosophers than between two computers. It will suffice that, pen in hand, they sit down . . . and say to each other: Let us calculate.

G. Leibniz (1965)¹

¹ My friend Brian Hayes began his respected computing science column in the *American Scientist* with a reference to this “Panglossian” optimism by Leibniz (Hayes 2008). My own interpretation of Leibnizian confidence in computation was expressed in (Velupillai 2000, p. 16, n. 8).

24.1 SIMULATION, DYNAMICS, AND COMPUTATION: EPISTEMOLOGICAL DEFICITS

Ulam first used “synergesis” in the context of the “Computing machine as a heuristic aid” in mathematical research. . . . This was a new mode of working. . . .

From “synergesis” I formulated the *visiometrics* approach. . . . Visiometrics is the process of producing cogent 2D and 3D images and parameter-scaled (normalized) graphs *for developing intuition and aiding in mathematical model formation*.

N. J. Zabusky (2005)

The art of gleaning epistemological lessons from simulating a mathematical model generating dynamic behavior on a (digital) computer was elegantly outlined by Weissert (Weissert 1997), in the context of the famous Fermi-Pasta-Ulam problem (Fermi et al. 1955), which was the setting for Zabusky’s path from synergesis to visiometrics:

The goal of *simulation in dynamics* is to learn about the true solution by simulating the trajectories of the associated differential equations. . . . Whether or not the true solution of a model actually relates to physical reality depends upon the fundamental laws used and the *approximations* made to obtain the model. As the simulation teaches us about the true solution, we make decisions about the adequacy of the model. Simulation’s function is to reveal the properties of the true solution and to aid our decisions about how well the model suits the physical system. As doubts arise about the model’s adequacy, but we are reasonably confident of the simulation, we must turn our attention to the approximations made in the model. Several times . . . a researcher indicates an implicit belief that *the simulation does tell us something about the physical reality*. Such an [indication] implies a *trust in a long chain of inferences and approximations* beginning from the adequacy of the fundamental theory itself, to the equivalency of an infinite series expansion of terms, then to the approximations that are made to obtain a model, and finally to the simulation and what it might be telling us about the model. (Weissert 1997, p. 106, italics added)

Weissert’s concise, yet elegant, encapsulation of a research program on the epistemology of simulation in dynamics leaves a few crucial questions unbroached. First, if there is such a thing as one true solution of a model, the means—that is, the methods of proof—used to demonstrate the existence of such a true solution, in theory, may or may not provide any information about the adequacy of the model and how well suits the physical system it aims to model. Second, the approximations to the physical system used to obtain the model may not be definable uniquely. Third, the methods of approximation may not be independent of the methods of proof used to demonstrate the existence of a true solution. Fourth, unless the chosen methods of approximation are very carefully selected, most models that are even reasonably faithful (in any one of many possible ways of defining this tortuous term) cannot be forced to yield unique solutions. Finally, and most disturbing, from the point of view of simulation of the

dynamics of the trajectories of the differential equations associated with the model by machine computation—that is, by, say, a digital computer—we must ask the following question: Can a dynamical system, whose *equilibrium existence*, or true solution, is proved by nonfinite means, be simulated by finite means to obtain epistemological answers to questions about the true solution of a model?

Let me illustrate this last point more explicitly. Consider one possible elementary statement of the *Peano existence theorem* for the initial value problem (IVP) of ordinary differential equations, or ODEs (Iserless 1996, pp. 364–365, italics added):

Let the function $f : [t_0, t_0 + a] \times U \rightarrow \mathbb{R}^d$ (where: $U \subseteq \mathbb{R}^d$) be continuous in the cylinder:

$$S = \{(t, \mathbf{x}) : t \in (t_0, t_0 + a), \mathbf{x} \in \mathbb{R}^d, \|\mathbf{x} - \mathbf{y}_0\| \leq b\} \quad (24.1)$$

where $a, b > 0$ and the vector norm $\|\cdot\|$ is given. Then, the ODE:

$$\mathbf{y}' = f(t, \mathbf{y}), t \in (t_0, t_0 + \alpha), \quad \mathbf{y}(t_0) = \mathbf{y}_0 \in \mathbb{R}^d \quad (24.2)$$

where $\alpha = \min \left\{ a, \frac{b}{\mu} \right\}$ and $\mu = \sup_{(t, \mathbf{x}) \in S} \|f(t, \mathbf{x})\|$ *posseses at least one solution*.

It can be shown, however, that $\exists f(t, \mathbf{y})$, satisfying the hypotheses of the Peano existence theorem such that there is no solution to the IVP (Aberth 2001). Why is this so? This is because the existence of a solution violates a cardinal theorem of computable calculus: the unsolvability of the halting problem for Turing machines. More specifically, there is a series of problems that are unsolvable by finite means in the computable calculus, some of which have to be made solvable *by nonfinite* means for the Peano existence theorem to be satisfied. In the case of this theorem, the relevant unsolvable problems are the following:

Proposition 24.1 *It is undecidable (by finite means) whether $\forall a \in \mathbb{R}, a$ or $\sim a$ is rational.*

Proposition 24.2 *It is undecidable (by finite means) whether $\forall a \in \mathbb{R}, a \geq 0$ or $a \leq 0$.*

Thus, implicit in any standard proof of the Peano existence theorem are appeals to nonfinite means to decide (intrinsically undecidable) disjunctions. No amount of simulation, even by an ideal computer—say, a Turing machine (or any of its equivalents, by the Church-Turing thesis)—can contribute epistemologically to any question about the true solution of a model that in analytical terms is fully characterized but uses standard real analysis underpinned by set theory plus the axiom of choice.

However, it must be noted that Peano went out of his way to avoid any appeal to the axiom of choice in his proof. Furthermore, Peano dropped the assumption of the Lipschitz condition in his original paper of 1890 (thereby losing uniqueness, Ince 1944, p. 66). Indeed, it was during the course of this proof that the ubiquity of this controversial axiom was first recognized and avoided.²

² This was long before Zermelo christened it the *axiom of choice*, fourteen years later.

This is, by the way, an instance of the ignorance of alternative mathematical traditions and careless understanding of the underpinning axioms, which leads, in turn, to ignorant assertions about the interactions between simulation, mathematical formalisms, formal solutions, approximations, and so on.³

The above conundrums of the formal, mathematical ambiguities involved in understanding the deep structure of such a simple and ubiquitous theorem as Peano's existence theorem for the IVP problem surely is a case in point for invoking Jeremy Avigad's (2009) perceptive observation:

[The] adoption of the infinitary, nonconstructive, set theoretic, algebraic, and structural methods that are characteristic to modern mathematics ... were controversial, however. At issue was not just whether they are consistent, but, more pointedly, whether they are meaningful and appropriate to mathematics. After all, if one views mathematics as an essentially computational science, then arguments without computational content, whatever their heuristic value, are not properly mathematical ... [At] the bare minimum, we wish to know that the universal assertions we derive in the system will not be contradicted by our experiences, and the existential predictions will be borne out by calculation. This is exactly what Hilbert's program was designed to do. (pp. 64–65, italics added)

Even if one does not view mathematics as “an essentially computational science,” appealing to nonfinite means to demonstrate the existence of a solution to a dynamic system and, then, to study its dynamic behavior by simulation on a computer to be able to characterize the nature of the solution—say, its basin of attraction—surely illegitimate from any epistemological point of view. This is easily illustrated by taking the example of Goodstein's theorem and the associated Goodstein sequence (Goodstein 1948; Paris and Tavakol 1993). They provide an elementary example of a *finite* combinatorial result that cannot be proved without an appeal to *infinite* sets, in the form of invoking the use of infinite ordinals. An illuminative perspective on the theorem, especially in view of the above example concerning Peano's existence theorem, is provided by the observation that it is expressible in first-order arithmetic but cannot be given a proof in Peano Arithmetic (PA), which is generally considered to coincide with finite arithmetic.⁴

The Goodstein sequence itself can be interpreted as a dynamic system of the type $X_{t+1} = f(X_t, t)$, that is, discrete, deterministic, and nonautonomous, from $\mathbb{N} \rightarrow \mathbb{N}$, with a very simple global attractor but with extremely long transients and possessing the property of super sensitive dependence on initial conditions (SSDIC). Most pertinently, it has not been possible, so far, to associate it with an ODE capable of the kind of rapid transient growth it exhibits.

³ An egregious example of this kind of infelicity can be found in Boumans (2001), especially pp. 74, 81.

⁴ This does not mean at all that I subscribe to anything like the *ultrafinitist* philosophy of mathematics, especially in view of the fact that even the first-order validity for finite models is not recursively enumerable owing to Trakhtenbrot's theorem (cf. Boolos et al. 2002, p. 198, problem 15.7).

But what is a computation? The lucid, elementary answer to this question, entirely in terms of computability theory, was given by Martin Davis in a masterly exposition (Davis 1978). Here I am interested in an answer that links the triad of computation, simulation, and dynamics in an epistemological way. As a path toward a tentative answer, it may be useful to recall the way Kant tackled his legendary question: What is man? He subcategorized it into the three following questions: What can I know? What must I do? What may I hope?

However, I shall assume the following claims so that I can answer the question What is a computation? in an epistemologically meaningful sense.

Claim 24.3 *Every computation is a dynamic system.*

Claim 24.4 *Every simulation of a dynamic system is a computation.*

Analogously, then: What can we know, what must we do, and what can we hope *from a computation*, which is, by the above claim, a dynamic system? This, in turn, means, What can we know, what must we do, and what can we hope from studying the behavior of a dynamic process during a computation? Since, however, not everything can be computed it follows that not every question about a dynamic system can be answered unambiguously. But by the second of the above claims, I have expressed an identity between a *simulation* and a *computation* via the intermediation of a dynamic system, which implies that not everything can be learned about the behavior of a dynamic system by simulating it on (even) an ideal device that can compute anything that is theoretically computable (e.g., a Turing machine, assuming the Church-Turing thesis). Above all, we cannot delineate in any meaningful sense (i.e., in an algorithmically decidable sense) between what can be known or learned and that which lies beyond this undecidable, indefinable border, on one side of which we live our scientific lives.

Now, observe the following:

An argument deriving the truth of a universal arithmetical sentence from that of its numerical instances suggests that the truth of the numerical instances has some kind of *epistemological priority* over the truth of the sentence itself: our knowledge of the truth of the sentence stems from the fact that we know all its numerical instances to be true ... I shall show that it is just *the other way around*... [T]he source of *our knowledge* of the truth of the totality of its numerical instances is the truth of the sentence itself. (Serény 2011, p. 48, italics added).

Juxtapose this inversion of epistemological priority between the truth of a universal arithmetical sentence and the totality of its numerical instances—remembering that the latter is obtained by computations—with Cohen's causal conjecture:

Hilbert's vision of a universal algorithm to solve mathematical theorems⁵ required a unification of Logic, Set Theory and Number Theory. This project was initiated

⁵ I suspect Cohen means "solve mathematical problems," since solving mathematical theorems seems a meaningless phrase.

by Frege, rerouted by Russell, repaired by Whitehead, derailed by Gödel, restored by Zermelo, Frankel, Bernays and von Neumann, shaken by Church and finally demolished by Turing. Hence, to say that the interest in algorithmic methods in mathematics or the progress in logic was engendered by the computer is wrong way around. For these subjects it is more correct to observe *the revolution in computing that was inspired by mathematics*. (Cohen 1991, p. 323, italics added)

There are *at least* five frontier research fields in economics, encompassing both micro and macro aspects of economic theory, in which machine computation,⁶ in its digital mode, is claimed to play crucial roles in formal modeling exercises.⁷

1. Computable general equilibrium theory (CGE) and its ‘extensions,’ recursive competitive equilibrium (RCE) and dynamic stochastic general equilibrium (DSGE) theories—the Scarf tradition
2. Computable general equilibrium modeling—the Johansen-Stone tradition (Johansen 1974; Dixon and Parmenter 2009)
3. Agent-based computational economics (see Tesfatsion 2006; Epstein 2006)⁸
4. Classical behavioral economics (CBE, as distinct from MBE, modern behavioural economics; see Kao and Velupillai 2015)
5. Computable economics⁹

⁶ The claims about computation in standard advanced microeconomics (e.g., Mas-Colell et al. 1995, especially sec. 20.D) and general equilibrium theory (e.g., Arrow and Hahn 1971, especially chap. 5 and sec. 5 in appendix C) are easily shown to be untenable from the point of view of any of the six mathematical frameworks for a model of computation, listed in the concluding section.

⁷ The discerning reader would easily recognize that I disagree quite comprehensively with the discussion, characterization, and visions of “computational economics” in Mirowski (2002, especially pp. 523 ff.). There are also five varieties of game theory: (von Neumann–Morgenstern/Nash orthodox) game theory, algorithmic game theory, constructive game theory, arithmetic games, and combinatorial games. Of these, the second is really only an attempt at “algorithmizing” orthodox game theory in the same sense in which computable general equilibrium theory tried to make computable sense of orthodox general equilibrium theory. The last three are intrinsically computational, with the third squarely in the constructive mathematics fold and the fourth (mostly) underpinned by classical recursion theory. Combinatorial game theory could be underpinned by either constructive or computability theory.

⁸ The word “constructive” in the title of the article by Testfatsion (2006) has nothing whatsoever to do with any variety of constructive mathematics. The claims, definitions, and characterizations of constructive mathematical aspects of agent-based economics in Epstein (2006, chap. 1) are technically incorrect. In particular, the characterization of nonconstructive existence proofs, in terms of acceptance of the *tertium non datur*, implying that the contrary in the case in constructive existence proofs (ibid. pp. 11–12), is not accurate (Brouwer 1975). This is because the author has not stated the kind of *tertium non datur* that is unacceptable in constructive mathematics. Above all, conflating computability theoretic statements, themselves imprecisely stated, with constructive ones makes the whole argument meaningless from a mathematical point of view. In Velupillai (2012) these unwarranted claims are discussed in greater and more precise technical detail.

⁹ Once again I am in profound disagreement with the views expressed in Mirowski (2002) on the origins and current status of this area of research. In particular, his technical discussions of the work of Alain Lewis (see Velupillai et al. 2011 for many of the published classics by Lewis and Introduction, loc cit., for a summary of his contribution to the “proto-history” of the subject) is technically incorrect and

They broach, without resolving, many interesting methodological and epistemological issues in economic theorizing in (alternative) mathematical modes.

That CGE in the Scarf tradition is not underpinned by either computable or constructive mathematics is, by now, fairly well known (Velupillai 2010). But it is less well documented that the claims of computability of the Johansen-Stone tradition are equally untenable. Therefore some comments may be in order.

Somewhat surprisingly, the adherents and aficionados of Leif Johansen's classic work *A Multi-Sectoral Study of Economic Growth* (Johansen 1974) claim that this was "the first CGE model" (Dixon and Parmenter 2009, p. 6). Their rationale for this claim is the following (p. 6, all but the first two italics added):

[The Johansen model] was *general* in that it contained . . . cost minimizing industries and utility-maximizing household sectors . . . His model employed market *equilibrium* assumptions in the determination of prices. Finally, it was computable (and *applied*). It produced a numerical, multi-sectoral description of growth in Norway using Norwegian input-output data and estimates of household price and income elasticities derived using Frisch's . . . additive utility method.

This is an untenable claim, especially since it is very specific: that the Johansen model is of the applied general equilibrium variety, which is also *computable* and, hence, *numerically* meaningful.¹⁰ The Johansen model has no underpinning whatsoever in any formal model of computation, least of all a computable (or constructive) one.

The main claim of this chapter, albeit stated implicitly and expressed indirectly, is that there is a serious epistemological deficit in all of the approaches, but it can be discovered only in the last two, precisely because they are underpinned by computability and constructivity theories, in their strict mathematical senses, and the first three are not. This claim does not imply, however, that classical behavioral economics and computable economics are complete from an epistemological perspective, especially from the point of view of natural or intrinsic dynamics of formal models. The epistemological deficit and the epistemological incompleteness, it is suggested, can be resolved by a theory of simulation, itself based on recognizing the double "duality" between dynamic systems and numerical analysis, on one hand, and that between computation and simulation, mediated by dynamic systems, on the other. Both dualities are to be interpreted

misleading in many instances. Moreover, the technical contents of the letters by Lewis to Debreu, quoted on pp. 431–432 and pp. 526–527 of Mirowski (2002), contain serious errors, which make the implications drawn in the text meaningless. In particular, the notions of combinatorial, finite, and totally finite models, as well as continuity and Peano Arithmetic are used and referred to by Lewis in unfortunately imprecise, indeed incorrect, ways. The juxtaposition of combinatorial and recursive analysis, on one hand, and combinatorial and totally finite models, on the other, is simply incorrect, from a purely mathematical point of view.

¹⁰ My stance on this issue is reflected exactly by the view held by my friend Lance Taylor. After attending the fiftieth anniversary celebrations of the Johansen Model, held in Oslo, Lance wrote as follows (email, August 27, 2010): [Most participants at the] conference in honor of the 50th anniversary of Johansen's MSG model, were thinking that Leif was taking off from Arrow-Debreu when in fact he was doing disaggregated macro planning, moving around the numbers in a set of accounts that they had been constructed to satisfy. There is certainly no mention of A-D in his book.

computably or constructively, leading to the core triad of computation, simulation, and dynamics (because numerical analysis can be interpreted, equivalently, in terms of dynamic systems or computably) and, it is hoped, paying heed to Turing's precept: "the inadequacy of 'reason' unsupported by common sense."

Let me end this preliminary reflection on a computable approach to a discussion of the epistemological deficit intrinsic to the interlinked triad of computation, simulation, and dynamics by invoking Feynman: "Computer science touches on a variety of deep issues . . . It naturally encourages us to ask questions about the limits of computability, about what we can and cannot know about the world around us (Feynman 1996, p. xiii).

The rest of this chapter is organized as follows. In the next section I tell a contemporary story of simulational serendipities, just to set a light-hearted tone for what may appear to be a too technically demanding paper making a case for underpinning the triad rigorously and, then, economic theorizing concerning the triad. Section 24.3 contains the full setting for computation, simulation, and dynamics to interact fruitfully in an epistemological sense, along research frontiers encapsulated by what I call the Zabusky precept at the end of the section. Section 4 is titled Computation, Discretization, Proof and Other Mathematical Infelicities. In the final section I attempt to summarize, as concisely as possible, the core areas of economics that initiated and maintained what we call "the noble tradition of simulation in economics" with reflections and hopes for what must be done to keep the bright shining light of computation, simulation, and dynamics focused on economic theory and its formalization.

24.1.1 Visiometrics and the Graph Theorems

The problems of scientific computing often arise from the study of continuous processes, and questions of computability and complexity over the reals are of central importance in laying the foundations for the subject. The first step is defining a suitable computational model for functions over the reals. M. Braverman and S. Cook.

Of the five frontier research fields in economics claiming computational underpinnings, mentioned above, only the first and the last two are explicitly based on "a suitable computational model for functions over the reals." Moreover, both classical behavioral economics and computable economics are either explicitly or implicitly defined over the computable, constructive, or nonstandard reals. Yet computable general equilibrium theory (in the Scarf mode) does not develop its theory on a model of computation that is consistent with its basis on all of the standard reals (a fortiori for RCE and DSGE). Neither computable general equilibrium economics (in the Johansen-Stone mode) nor agent-based computational economics (and finance) is based on any kind of formal (or informal) computational model over the reals. None of the five frontier research fields, except every kind of agent-based computational economic (and financial) practice,¹¹

¹¹ The field is a paradigmatic example of the classic approach that Koopmans called "measurement without theory."

relies overwhelmingly on an elementary and undefined kind of visiometrics, appealing to computer graphics without the slightest basis in the mathematics of the computer (of any variety). In this brief section I outline the kind of care needed to make sense of the excessive claims of the practitioners of agent-based computational economics (and finance),¹² so that they can be brought into the fold of serious visiometrics.¹³

Zabusky's more discursive description—I shall not call it a formal definition—of what he means by “visiometrics” is the following (Zabusky 2005, p. 12, italics in the original):

VISIOMETRICS is the process of: *Visualization, projection, identification and juxtaposition of evolving amorphous coherent structures and statistical backgrounds in massive multidimensional data sets.* The goal is to produce cogent images and specific, parameter-scaled (normalized) graphs for intuition building and mathematization.

The crucial terms are “cogent images” and “graphs for intuition building and mathematization,” especially in the case of *economic visiometrics*, in which the massive multidimensional data sets reside, at best, in the set of rational numbers.

Essentially, all of the frontier computational economic research areas theorize in the domain of the real numbers and real number functions (of arbitrary high dimensions), but agent-based computational economics, in particular, seeks to extract patterns from the projected dynamics on the screens of digital computers (there are, of course, enlightened exceptions to this general rule). If the theorizing is in terms of real analysis, then the graphs of theory are subject to the classical graph theorem.

Theorem 24.5 (the classical graph theorem.) *A function (or mapping or transformation),*

$\psi : A \rightarrow B$ is any subset $\psi \subseteq (A \times B)$ such that $(\forall x \in A) (\exists! y \in B)$ and $[(x, y) \in \psi]$ and $(\forall x \in A) (y \in B), [(x, y) \in \psi]$ and $[(x, y') \in \psi \implies y = y']$,

where A denotes the domain set, B denotes the range set, and $\exists!$ denotes “exists exactly one.”

Theorem 24.6 (the recursion theoretic graph theorem.) *Let φ and ξ be, respectively, a partial and a total function. Then:*

¹² An egregious example of such an excessive claim, completely unfounded in any kind of practice or theory, claim is the one made by Leijonhufvud in his chapter titled “Agent-Based Macro” (Tefatsion and Judd 2006, p. 1626): “Agent-based computational methods provide the only way in which the self-regulatory capabilities of complex dynamic models can be explored so as to advance our understanding of the adaptive dynamics of actual economies.” Quite apart from the many undefined and formally (unambiguously) undefinable concepts in this statement, the extraordinary claim that agent-based computational methods “provide the only way” to understand anything, let alone the adaptive dynamics of actual economies, must rank with the most foolish claims, even in a field with many assertions of this type (see, e.g., Delli Gatti et al. 2008)

¹³ Zabusky's imaginative Visiometric approach was developed independent of Ralph Abraham's equally illuminating development of the Vismath vision for studying the geometry of dynamic behavior (see, e.g., Abraham and Shaw 1984).

1. φ is partial recursive iff its graph is a recursively enumerable set;
2. ξ is recursive iff its graph is a recursive set.

To these theorems I would like to add, and invoke, what I call Brattka's Theses (Brattka 2008).

Criterion 24.7 *Brattka's thesis I.* A function $\zeta : \mathbb{R}^n \rightarrow \mathbb{R}$ is computable iff it can be evaluated on a physical computer with arbitrary given precision.

Criterion 24.8 *Brattka's thesis II.* A closed subset $\Phi \subseteq \mathbb{R}^2$ is recursive iff it can be displayed by a physical computer for an arbitrary given resolution.

Conjecture 24.9 *No function employed in any agent-based computational model satisfies either of Brattka's theses.*

Remark 24.10 *This is too strong a conjecture; I should be more specific about the particular agent-based model. I want only to give a flavor of the kind of conditions that have to be satisfied in order for some of the enthusiastic, vague, and imprecise claims to be valid.*

From these theorems and criteria any discerning reader can understand why I claim that classical behavioral economics and computable economics are eminently suitable for visiometric explorations. Moreover, these theorems and criteria are the reasons why I made the main claim of this chapter above.

24.2 SIMULATIONAL SERENDIPITIES

The May–June, 2009, issue of *American Scientist* (vol. 97, no. 3) contained, serendipitously, four (possibly five)¹⁴ articles about the fundamental role played by *simulation*—in its synergetic interactions with *theoretical analysis*, *experiment*, *computation*, *prediction*, and *dynamics*—in macroeconomics (Hayes 2009), physics (Porter et al. 2009), engineering (Petroski 2009)¹⁵ and (Hall and Day 2009). In particular,

¹⁴ This is because the article “The Origin of Life” in that issue (Trefil et al. 2009) traces the emergence of “experimental research in origin-of-life studies” in the analogue device with which Harold Urey and Stanley Miller studied, via simulations of hypothetical conditions satisfying the Oparin-Haldane hypothesis of chemical evolution, the “chemical processes that might have occurred on the planet soon after its birth”. The ultimate methodological message of Trefil, Morowitz, and Smith, the article's, appears to be an intensive experimental research program tied to the development of the appropriate theory, itself guided by experimental results. The experimental program of studying “complex cooperative networks,” is, surely, carried out via simulational studies of the fruitful interaction of analogue experimental setups and digital computing methods.

¹⁵ It is most interesting, that Dr. Petroski's visit to Japan, which resulted in his fascinating article on the Akashi Kaikyo Bridge, was sponsored by the Association for the Study of Failure (Shippai Gakkai),

the repository of simulation in the example of macroeconomics by Brian Hayes is an exemplary exposition of the Phillips machine devised and constructed as an electro-mechanical-hydraulic *analogue computing machine*, encapsulating early Keynesian monetary macrodynamics and capable of interacting with macroeconomic theory and even settling controversial theoretical debates decisively. The workings of the machine, entirely transparent, were such that almost any nonlinear dynamic system then current in macroeconomic theory could have been exactly simulated,¹⁶ without any recourse to approximations or discretizations, normally required in a *digital* computer—unless, of course, continuous data were available (which was not the case) to exploit the Phillips machine's full analogue potential. Moreover, the machine was capable of displaying, in all its transparent detail, the propagation mechanisms of policy and shocks, whether monetary or real, confirming and disconfirming, as the case may be, orthodox and non-orthodox propositions concerning policy and even inculcating a sense of humility in the emerging consensus on the feasibility of what came to be known as *fine-tuning* (see Barr 2000, in particular, sec. 5, pp. 108 ff). The machine was also capable of generating *surprises*, a sine qua non of an experimental device or design, in its interaction with theories that underpin and interact with it. At the request of Nicholas Stern at the LSE, when attempts were being made to resurrect one of their two Phillips machines, Richard Goodwin wrote a memoir¹⁷ about his experiences in working and teaching with it. In a postscript to the cover letter (dated August, 16, 1991) he sent to Stern together with the memoir, Goodwin noted, "I was very pleased that Phillips had 2 machines in London and I was able to show him (which he had doubted) that we could produce aperiodic, 'chaotic', motion with the two interconnected."¹⁸ In other words, every desideratum specified, implicitly, explicitly, or even vaguely, from an epistemic, epistemological, or methodological point of view as economists and economists is handsomely satisfied by a Phillips machine simulation, if structured and implemented in any traditional experimental sense (in more senses, in fact, than in the famous Fermi-Pasta-Ulam exercise, to which I now turn, albeit briefly).

In their fascinating recapitulation of the circumstances under which Enrico Fermi, John Pasta, and Stanislaw Ulam tried to resolve a theoretical conundrum—still completely unresolved—with a discrete approximation of a continuum model implemented

as its invited speaker at their international conference in November 2008. As economists we are painfully aware—simulations or not—of the need for such a society in economics.

¹⁶ This is subject, of course, to precision constraints in the manufacture of the electrical, mechanical, and hydraulic components. The constraints of natural laws, in the processing of data, for example, are common to any physical mechanism, analogue or digital.

¹⁷ A truncated version of this appears in Leeson (2000) pp. 118–119. Copies of the full memoir and the cover letter to Nicholas Stern were sent by Goodwin to me; I will be happy to make them available to interested readers.

¹⁸ In the unabridged memoir, this P.S. appears:

"Furthermore, I was very excited to find Phillips had two of his magical machines in London, so I could reproduce what I had analyzed back in 1947 in my dynamical coupling paper. If I remember correctly, Phillips did not believe we could produce erratic behaviour by coupling his machines—but we did."

on one of the first available digital computers, the MANIAC, Porter et al. (2009) point out the many ways in which the *simulations* interacted with the analytical theory to enrich both in surprising ways. The epistemic, epistemological, and methodological implications of the series of simulations that have been implemented with increasing precision, detail, and generalizations since the original experiment are exhaustively discussed and dissected in the admirable monograph by Thomas Weissert (Weissert 1997).¹⁹ As they perceptively and clearly note,²⁰ (Porter et al. 2009):

Fermi had long been fascinated by a fundamental mystery of statistical mechanics that physicists call the “arrow of time” [irreversibility]. . . . Fermi believed that the key [to the unlocking of the mystery of irreversibility] was nonlinearity. . . . He knew that it would be far too complicated to find solution to nonlinear equations using pencil and paper. Fortunately, because he was at Los Alamos in the early 1950s, he had access to one of the earliest digital computers [the MANIAC]. . . . The FPU problem was one of the first open scientific investigations carried out with the MANIAC, and it ushered in the age of what is sometimes called experimental mathematics [by] which we mean computer-based investigations designed to give insight into complex mathematical and physical problems that are inaccessible, *at least initially*, using more traditional forms of analysis. . . . With Pasta and Ulam, Fermi proposed to investigate what he assumed would be a very simple nonlinear dynamical system. The key question FPU wanted to study was how long it would take the oscillations of the masses and nonlinear springs to come to equilibrium. . . . They were absolutely astonished by the results.” (pp. 214–216, italics added)

The FPU problem exemplifies every aspect of an epistemic, epistemological, and methodological issue that can be conceived, not all of which, though confronted, have been adequately resolved after deeply serious theoretical and empirical attempts. I return to the FPU problem briefly below, from an epistemic and epistemological point of view, especially in conjunction with a computational dynamic macroeconomic problem that (see Zambelli 2011) has attempted to resolve by structured simulation studies, in close combination with established macrodynamic and interindustrial economics.

Petroski’s brief but illuminating description and general discussion of the analogue model—the 40-meter long replica—that was used in the wind-tunnel tests emphasizes the elements that were neglected in the construction of London’s Millennium Bridge.

¹⁹ This admirable monograph is now out of date, given the massive research and results concerning variations of the Fermi-Pasta-Ulam (FPU) problem that have been, and are being, conducted at the frontiers of what has come to be called “experimental nonlinear dynamics.”

²⁰ One of the authors of this crystal-clear exposition of the FPU problem, Norman Zabusky, was himself a pioneer in extracting new theoretical directions of research (and, indeed, together with his co-author, Martin Kruskal, to whose memory this particular article is dedicated) in (re-)discovering and giving a mathematical formalism to solitary waves, now called *solitons*. By retaining the original continuum domain of the FPU theoretical framework, and eschewing the discretizations necessary for digital computer implementation, they were able to predict the existence of solitons (see, in particular, Zabusky 2005).

As emphasized by Allan McRobie during personal conversations with me,²¹ the construction of a bridge is less about physics and engineering than about people because when a bridge is in use, especially a pedestrian-dominated suspension bridge, it becomes a “nonlinear *biological system*”. This implies an analogue computation model for simulation that is *a coupled system of the interaction between engineering structures and human beings*. Failing to take this into account in the analogue computing simulation of the Millennium Bridge construction in its design and testing stages led to the bridge’s having to be closed within twenty minutes of the long-awaited opening, due to the fearful wobbling that occurred when pedestrians began to make their presence felt. In other words, the analogue simulation (buttressed, of course, by various uses of the digital computer) failed to see the design problem as one that should have been studied as a nonlinear, coupled oscillator just as the FPU problem was, and just as it still remains a mystery, so will bridge building be, in the sense that there is, at present, no complete characterization of the dynamics of nonlinear, coupled oscillators. Every epistemic, epistemological, and methodological conundrum faced by the FPU problem, many of which were solved, has to be faced in the construction of every bridge, especially if it is a suspension bridge. Every model of an economy to be studied by computer simulations, whether digital or analogue, and underpinned by economic theory is naturally and intrinsically coupled, but nonlinearly. It is this latter fact that is often neglected in much of the recent simulation-dominated literature.

The fourth of the serendipitous articles, Hall and Day (2009), revives and reminds us, in this age of increasing environmental concerns, of the simple but powerful message of Malthus. The much-maligned dichotomy between an entity growing exponentially while relying for its growth on something else growing arithmetically was made (in)famous, particularly in economics and public policy, by the well-meaning Malthus, to be revived, particularly in economics, whenever the shadow of an exhaustible resource was seen in the horizon. Perhaps the most spectacularly dramatic example of a neo-Malthusian apocalyptic scenario for economic societies, smug in their reliance on the manna of exogenous technological factors to propel them through the golden era of Keynesian prosperity and beyond, was the ill-timed release of the Club of Rome document *Limits to Growth* (Meadows et al. 1972). It was ill-timed in both the positive and the negative sense: the first oil price hikes, the great stagflation of the 1970s, the

²¹ The phrase used by Allan McRobie to refer to the Millennium Bridge is the Rainbow Bridge, also referred to as the Blade of Light (cf. Sudjic 2001) during a personal conversation with me in Cambridge on May 17, 2010. McRobie was instrumental in detecting the source of the dramatic failure of the Millennium Bridge via an inspiring simulation study using a homemade analogue device in his university laboratories at Cambridge. I had gone to Cambridge to view a demonstration of the Phillips machine, an analogue computing machine constructed by the economist A. W. H. Phillips to model, simulate, and study Keynesian business cycle theories in policy contexts. The Phillips machine, also known as the MONIAC (clearly a play on the name given to the first large-scale digital computing device, MANIAC) in Cambridge was resurrected single-handedly by McRobie. I had been educated with it as a computing tool for simulating Keynesian Nonlinear Multiplier-Accelerator models by its first and only economist custodian, Richard Goodwin, whose theoretical work informed decisively the construction of the analogue computing machine by Phillips (Phillips 1950).

collapse of the Bretton Woods compromises, the demise of the Neoclassical Synthesis, the rise of varieties of Monetarism and, eventually, the emergence of endogenous growth theory and the miracle economies of East Asia, together with the all-embracing dominance attained by Newclassical economics, discrediting any attempt at active policy in any domain, were all on the horizon.

With hindsight it may be remarked that—although Hall and Day (2009) show, convincingly, that the reasoning and analysis underpinning *Limits to Growth* have stood the test of time most admirably—almost nothing discredited the intellectual credibility of simulation-based analysis and projections than this single work, directed by the founding father of *system dynamics*, Jay Forrester.²² It is little remembered or recounted that Forrester's initial fame owed as much to his use of “hand-held simulations” to resolve an internal conundrum of employment stability, independent of the economy-wide business cycle, at General Electric, as to his insight into the need for understanding corporate dynamism in terms of the interaction between *engineering and management synergies*. In my opinion, however, the main reason for the discrediting could be found in the philosophy underpinning the *Limits to Growth* methodology, which was unanchored in theory. It was, to the economist at least, a case of not even “measurement without theory” for a generation of economists who were to extol the virtues of “theory ahead of measurement.” Moreover, the epistemological justification of the *Limits to Growth* policy prescription would have to be made on inferring general propositions from *induction*, forgetting Hume's dictum and Popper's strictures against this noble practice by distinguished empiricists and scientists all the way from Newton to Darwin.

In the spirit of the disciplining criterion for simulation modeling of complex, intractable dynamic systems mentioned above, I locate the weakness of the case made in the *Limits to Growth* literature—the original and the revision (Meadows et al. 2004)—in its eschewing a *nonlinear, coupled, dynamics* framework in modeling the interaction between a natural system and its dependent human economic link. The dynamics of such coupled nonlinear dynamic systems cannot be breached, provably so, by known analytical approaches and require, as one learns from the saga of the FPU problem, the helping hand of computer simulations to get a handle on plausible dynamic evolutions and possible policy responses, if only in limited senses. Of course, there was also the problem of the lack of theoretical underpinnings, above all in economic theory.

If anything is to be learned from the four examples, serendipitously brought together in just celebrations of their various anniversaries, it is that the *synergies* between

²² This is the general view, even of those who were, and remain, sympathetic to the message, if not the full paraphernalia of methods, of the Club of Rome report (Gilbert and Troitzsch 1999, p. 6, italics added): “The Club of Rome simulations which predicted global environmental catastrophe made a major impact, but also gave *simulation an undeservedly poor reputation* as it became clear that the results depended very heavily on the specific quantitative assumptions made about the model's parameters. Many of these assumptions were backed by rather little evidence.” With hindsight, too, it was regrettable that the Club of Rome team was not familiar with the art, science, methodology, and epistemology of simulation that were being learned as the attempt to solve the FPU problem was being played out.

intractable coupled nonlinear dynamics, underlying theoretical conundrums, and exploitation of the ubiquity of the emerging power of new and innovative paradigms of computations, could at best be exploited for advancing the respective disciplines only with an attitude of modesty in epistemological aims, humility in the face of methodological (in the limited sense of methods of mathematics) limitations and a generosity of spirit in the light of philosophical confusions. Heavy reliance and untrammelled confidence in the power of one kind of mathematical analysis, if coupled to unreflective confidence in the power of the emerging paradigms of computation to solve the unsolvable, has caused much mischief in the natural as well as the social sciences. Shunning simulation, an attitude not confined to economists, is akin to the precept that one should not throw away the baby with the bathwater.

24.3 DISCOVERING DYNAMICS BY SIMULATION IN MACROECONOMICS

Progress in our understanding of the natural sciences has always depended upon the give-and-take between modelling (or theorizing), analysis, and experiment. With large-scale computers, we can process experimental data from a variety of sensors and juxtapose them readily with large-scale simulations—numerical solutions of ordinary and partial differential equations, etc. The insights gained from attempting to bring these results into agreement can synergize²³ the rate of improvement of *models, algorithms, analytical methods, and experimental procedures*”

N. J. Zabusky (1981).

At least five kinds of standard dynamic systems have been routinely used in economic theory: ordinary differential equations (linear and nonlinear), standard & nonstandard,²⁴ partial differential equations (linear and nonlinear), stochastic differential equations (linear, nonlinear and partial), mixed-difference differential equations, difference equations (linear & nonlinear).

²³ Zabusky has almost from the outset of his sustained work on the Fermi-Pasta-Ulam problem used the word “synergize” in the sense in which it was first used, in a similar context, by Ulam (Ulam 1960, chap. 8, sec. 10): I use the word ‘synergetic’ here to mean the enhancement in the rate and depth of mathematical understanding through the combined use of analysis and computer simulation.” (Zabusky 1981, p. 196) May one be forgiven for wondering, in Nozick mode. *If not simulation by a computer, then simulation by what?*

²⁴ I am referring to formal nonstandard differential equations (cf., e.g., Diener and Diener 1995, especially chap. 10, or Fenstad 1987). To the best of my knowledge nonstandard stochastic differential equations have not been applied in any area of economics or finance theory. In finance theory, in particular, they should be the natural framework for modeling dynamics. A particularly lucid and pedagogical exposition of this virgin field can be found in Albeverio et al. (1986).

Remark 24.11 *The above is a descriptive delineation. I am able to reduce them to two classes or perhaps even one, and all the other forms can be dealt with as special cases. But this would make the discussion somewhat un-illuminating.*

In economics, formal nonlinear dynamics, in terms of explicit and standard Rayleigh–van der Pol equations, were first introduced within the context of an explicit Keynesian model of aggregate fluctuations in the late 1940s (Goodwin 1949).²⁵ From the outset simulation studies of variants of these equations were used to explore their dynamic behavior owing to the difficulties of taming them by purely analytical means (Strotz et al. 1951, 1953), but these were conducted on electro-analogue computers (or with pencil and paper in classic geometric mode, using the auxiliary device of the *characteristic* first described with almost exquisite pedagogical elegance in Le Corbeiller 1931).

It seems to me that it is possible to describe all nonlinear, endogenous (i.e., non-stochastic) dynamic equations that have so far been used in modeling aggregate fluctuations (and, often, also nonaggregate dynamics) as special cases of the following canonical mixed-difference differential equation:

$$F[t, u(t), u(t - \omega_1), \dots, u(t - \omega_m), u'(t), u'(t - \omega_1), \dots, u'(t - \omega_m), \dots, u^{(n)}(t), u^{(n)}(t - \omega_1), \dots, u^{(n)}(t - \omega_m)] = 0 \quad (24.3)$$

This is an n^{th} -order difference-differential equation, and it is a function of $1 + (m + 1)(n + 1)$ variables where F and u are real functions and $\omega_i \in \mathbb{R}$ and $n \in \mathbb{Z}$.

In particular, the more well-known theories of aggregate fluctuations, modeled as nonlinear, endogenous dynamic systems, have encapsulated the underpinning economic theories in terms of the following

The van der Pol equation,

$$\ddot{x} - k(1 - x^2)\dot{x} + x = 0, \quad (24.4)$$

equations of the Liénard type,

$$\ddot{x} + f(x)\dot{x} + g(x) = 0 \quad (24.5)$$

studied in the Liénard plane,

$$\dot{x} = y - F(x), \dot{y} = -g(x), \quad (24.6)$$

the generalized, forced van der Pol equation,

$$\ddot{x} + f(x, \dot{x})\dot{x} + g(x) = p(t), \quad (24.7)$$

²⁵ Like all such attributes of origins, a great deal of caution should be exercised in interpreting this claim, as clearly outlined in Ragupathy and Velupillai (2012).

the Rayleigh equation,

$$\ddot{x} + \eta \left(-\dot{x} + \frac{\dot{x}^3}{3} \right) + x = 0, \{0 < \eta < \infty\}, \quad (24.8)$$

the logistic map,

$$x_{n+1} = \lambda x_n (1 - x_n), \quad (24.9)$$

the difference-differential equation,

$$\sum_{\mu=0}^m \sum_{v=0}^n a_{\mu v} y^{(v)}(x + \mu) = 0, \quad (24.10)$$

and the second-order difference equation,

$$y_{n+1} = F(y_n, y_{n-1}) \quad \forall n = 0, 1, 2, \dots \quad (24.11)$$

where $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ and given initial conditions $y_0, y_{-1} \in \mathbb{R}$.

The first five encapsulated the business cycle theories of Fisher, Keynes, Harrod, Schumpeter, and Hawtrey in the pioneering works of Kaldor, Hicks, Goodwin, Yasui, Ichimura, Morishima, and others (Ragupathy and Velupillai 2012); the sixth, models of the “cobweb” type, as in Leontief (1934); the seventh, in various specialized forms, the business cycle theories of Tinbergen, Kalecki, and Frisch; the last one, with variously specified functional forms for F , variations on the dynamics of Swedish sequence analysis (although I do not fully subscribe to this interpretation of their dynamic method), on one hand, and the Hicks version of a Keynes-Harrod model of the trade cycle, on the other.

The example I shall discuss in this section is an appropriate version of any one of (24.5), (24.6), (24.7), (24.8), or (24.9), all of which are formally reducible to one another, from particular economic, geometric, or numerical points of view. The example encapsulates a noble tradition of computation in economics in every sense of this concept, to study a precisely specified mathematical system on both analogue and digital computers. It is, in a precise sense, also a substitute for an analytical study (because such a study is provably unlikely to succeed in any meaningful way). Moreover, it can be viewed as an explicit example of an epistemological tool for interpreting the results (most of which were unexpected) in order to gain insight into the link between a computing machine and *its theory* and the theory of nonlinear dynamic systems. The latter point is turning out to be the most significant from the point of view of the epistemology of computation, since the interaction can only be explored by representing one system by the other and, therefore, exploring a new domain: studying the repertoire of digital machine behavior with analogue computing machines, and vice versa.

Consider, therefore, the following equation, representing a classical Keynesian nonlinear multiplier-accelerator model of the dynamics of national income, y :

$$\varepsilon \dot{y}(t) + (1 - \alpha)y(t) = \phi[\dot{y}(t - \theta)] + [\beta(t) + l(t)] \quad (24.12)$$

Now, there are at least six different ways to investigate dynamic behavior and solutions (or the basins of attraction) to this nonlinear difference-differential equation:

- In old-fashioned analytical modes
- Using nonstandard analysis
- Graphically, that is, in terms of the geometry of dynamic behavior, as is usually done in the qualitative theory of differential equations (cf. Goodwin 1951)
- By the method of equivalent linearization (cf. Bothwell 1952)
- Using an electro-analogue computer (cf. Strotz et al. 1953)
- Using digital computers (cf. Thalberg 1966; Zambelli 2011)

Assuming, for example, that $\beta(t) + l(t)$ is a constant and reinterpreting $y(t)$ as a deviation from the *unstable equilibrium*, $\frac{\beta(t)+l(t)}{(1-\alpha)}$ of (24.12), one obtains a mixed nonlinear difference-differential equation:

$$\varepsilon \dot{y}(t+\theta) + (1-\alpha)y(t+\theta) = \phi[\dot{y}(t)]. \quad (24.13)$$

Now, expanding this equation by a Taylor series approximation and retaining only the first two terms, one obtains the famous (unforced) Rayleigh (and van der Pol)-type equation:

$$\ddot{y} + \left[\frac{\chi(\dot{y})}{\dot{y}} \right] \dot{y} + y = 0 \quad (24.14)$$

With this approximated reformulation began an “industry” in the endogenous theory of the business cycle, in which the cardinal desideratum was the existence of a unique, stable limit cycle independent of initial conditions. For $\chi(\dot{y})$ displaying a “cubic characteristic” such as figure 24.1.

The unique, stable limit cycle, independent of initial conditions, was shown to display planar dynamics such as those shown in figure 24.2.

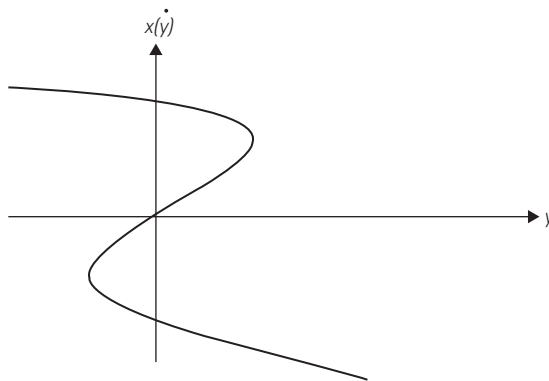


FIGURE 24.1 Behavior of $\chi(\dot{y})$ of equation (24.14).

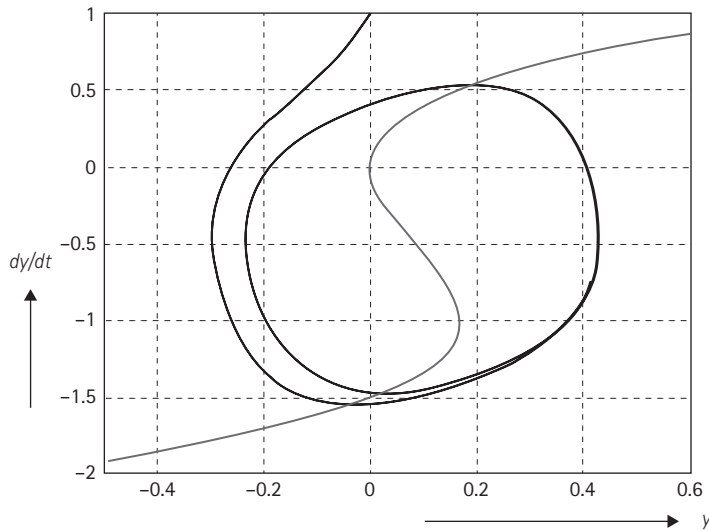


FIGURE 24.2 Planar dynamics of equation (24.14).

All four desiderata were violated when the approximations were more precise—in a purely technical sense—and the analysis proceeded via studies by means of analogue and digital computing machines. Using an electro-analog computer, it was found, in Strotz et al. (1953), that the approximation of (24.12) retaining the first four terms of a Taylor series expansion generated twenty-five limit cycles and had the potential for a *countable infinity* of limit cycles with further higher-order terms included in the approximations. Thus, for

$$\begin{aligned} & \epsilon \frac{\theta^4}{24} y^{(5)}(t) + C_4 y^{(4)}(t) + C_3 y^{(3)}(t) + C_2 y^{(2)}(t) + C_1 y^{(1)}(t) \\ & - \phi \left[y^{(1)}(t) \right] + (1 - \alpha) y(t) = 0. \end{aligned} \quad (24.15)$$

The displayed phase-plane dynamics are shown in figure 24.3.

Even more interesting, the insights obtained from an analogue computing machine study provided hints for setting up a computing study of (24.13) by means of digital computing machines.

Next, coupling two equations of type (24.15) via the Monetary National Income Analogue Computer (MONIAC, Velupillai 2011), Goodwin and Phillips were unexpectedly able to generate the quasi-periodic paradox (Abraham 1985). Neither Goodwin, nor Phillips, who did the coupled-dynamics computation on the Phillips machine, had any clue, theoretical or otherwise, about interpreting and encapsulating this outcome in any economic theoretical formalization. The key point is that they were *surprised* by the outcome and did not know how to interpret it when it emerged. This is where the richness of the epistemology of computation manifests itself most dramatically. There

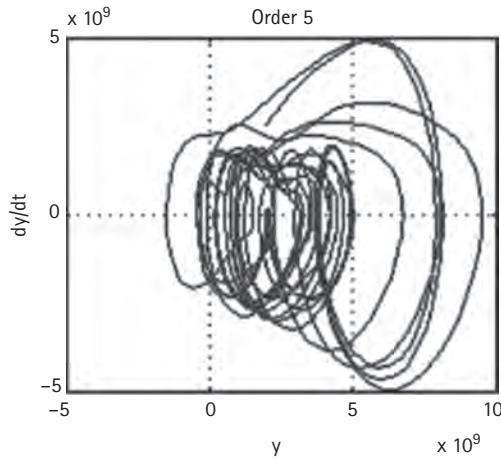


FIGURE 24.3 Phase-plane dynamics of equation (24.15)

was no macrodynamic theory to which they could relate the observed behavior, which was contrary to expected behavior.

Relaxation oscillations encompass two-phase dynamics in the sense that there is an interaction between slow and fast variables in the system, rather like one set of markets (financial?) clearing “infinitely fast” and another set (real?) clearing in relatively slow mode. The problem, of course, is that “infinitely fast” is a meaningless concept in standard analysis but an eminently sensible notion in nonstandard analysis; analogously, the infinitesimal is a fully viable concept in nonstandard analysis but not in standard analysis. What kind of dynamics can one expect for (24.13) when the mathematical underpinning of infinitely fast and infinitesimally slow variables are made to work in the nonstandard analytic phase-plane?

Consider, now, the following nonlinear equation, a variant of (24.13):

$$\epsilon \frac{d^2 x}{dt^2} + (x + x^2) \frac{dx}{dt} + x + \alpha = 0. \quad (24.16)$$

This can be represented as

$$\frac{dx}{dt} = \epsilon^{-1} (y - f(x)) \quad (24.17)$$

$$\frac{dy}{dt} = -(x + \alpha) \quad (24.18)$$

and the “characteristic,” $f(x)$, is given by

$$f(x) = \frac{1}{3}x^3 + \frac{1}{2}x^2. \quad (24.19)$$

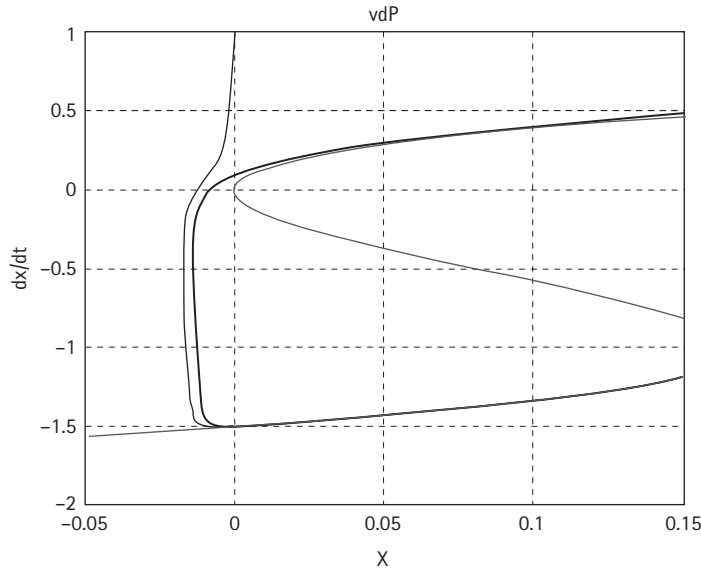


FIGURE 24.4 Phase-plane dynamics of equation (24.16): $\alpha = .5, \epsilon = 1,000$.

The phase-plane dynamics depicted in figures 24.4 to 24.6 are for the following numerical values of α and ϵ (the lighter curves, in all cases, are the graphs of the cubic characteristic):

1. The values used to obtain figure 24.2 were chosen so that the phase-plane dynamics would resemble, as closely as possible, that given in the pioneering nonlinear trade cycle model of Goodwin; here, $\alpha = .45, \epsilon = 10$.
2. Figure 24.4, similar in its geometry to figure 24.2, has $\alpha = .5, \epsilon = 1,000$.
3. Figure 24.5 (the “Duck Headed vdP” dynamics) were obtained for $\alpha = .001012345, \epsilon = 1,000$.
4. Finally, figure 24.6, the “Unheaded Duck vdP” dynamics, were obtained for: $\alpha = .00001025; \epsilon = 1,000$.

For $\alpha = 0$ and if $\frac{1}{2}x^2$ is replaced by x , the system reduces to the van der Pol equation on the Liénard plane²⁶

It is clear that y is the “slow” variable; in other words, it is finite for all finite points of the domain of the plane; x , then, is the “fast” variable and takes infinitely large values for some finite values of its domain. If the trajectory of y is defined to be on that curve at which $\dot{x} = 0$, then its graph is given by $f(x)$.

²⁶ I chose the characteristic with the $\frac{1}{2}x^2$ term instead of the x term only because I had severe difficulties of precision to get the kind of phase-plane dynamics I could have obtained with a computer capable of more precise computations.

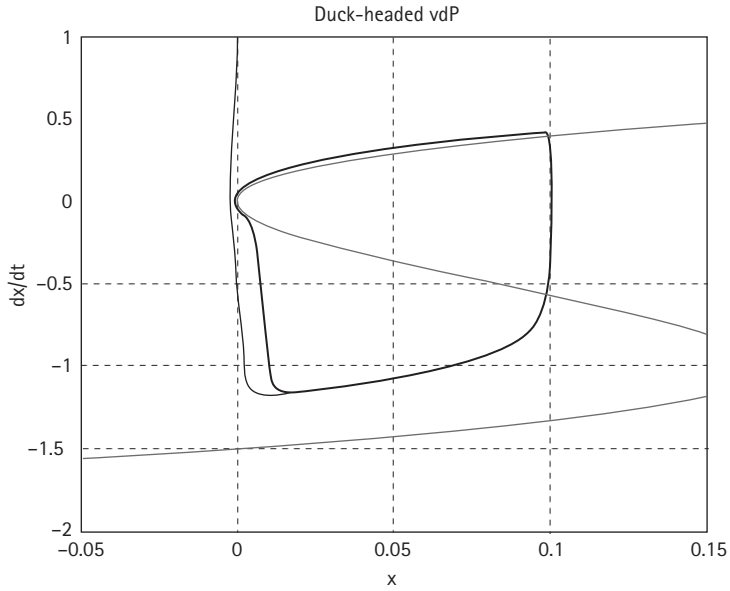


FIGURE 24.5 The phase-plane dynamics of equation (24.16): $\alpha = .001012345$, $\epsilon = 1,000$.

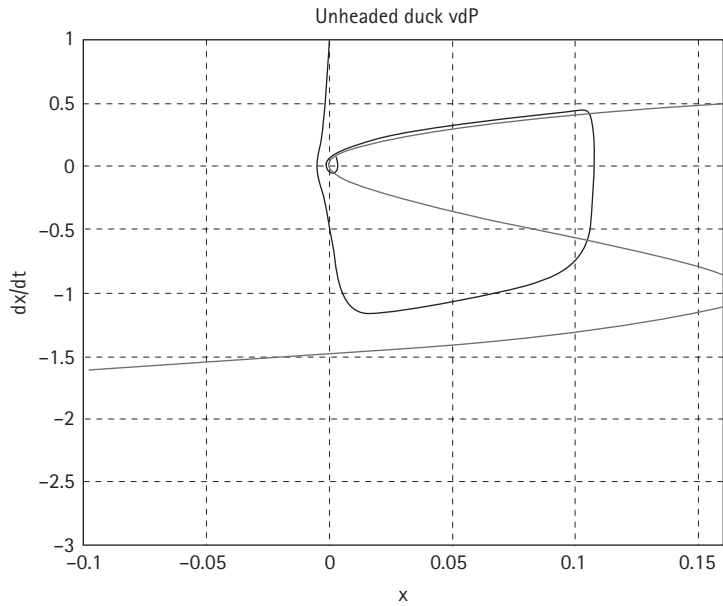


FIGURE 24.6 The phase-plane dynamics of equation (24.16): $\alpha = .00001025$, $\epsilon = 1,000$.

As for the “duck” terminology, the idea should be obvious from a perusal of the diagrams (and a bit of imagination!).

The proof of the existence of unheaded ducks, or *counter intuitive cycles* being *attracted to unstable manifolds*, for the van der Pol system is extremely simple provided that one learns a bit of nonstandard analysis or, at least, nonstandard terminology. Let me simply state it, in as heuristic and intuitive a way as possible, to illustrate what I mean; the interested reader can get a good idea from the exceptionally clear and detailed article by Zvonkin and Shubin (1984). The only thing to keep in mind is that α is an *infinitesimal in the sense of nonstandard analysis*. Then,²⁷ referring to figures 24.5 and 24.6, and Zvonkin and Shubin (1984), sec. 4.2), we have the following:

Definition 24.12 *An admissible form for the characteristic $f(x)$.*

$f(x)$ has an admissible form on a closed interval, say, $[\beta_1, \beta_2]$, if

- (1) *$f(x) \in [\beta_1, \beta_2]$ is standard and $\mathbb{C}^2$;*
- (2) *$f(x) \in [\beta_1, \beta_2]$ has exactly two isolated extremum points, say, a minimum at x_0 and a maximum at x_1 , and $\beta_1 < x_1 < x_0 < \beta_2$, so that $f'(x) > 0$ on $[\beta_1, x_1]$ and $(x_0, \beta_2]$ and $f'(x) < 0$ on (x_1, x_0) ;*
- (3) *$f(\beta_1) < f(x_0)$ and $f(\beta_2) > f(x_1)$.*

Theorem 24.13 *Existence of duck cycles in the van der Pol system [figures 24.4–24.6].*

Suppose $f(x)$ has an admissible form on $[\beta_1, \beta_2]$; if $x_\beta \in \Re$ and $x_\beta \in (x_1, x_0)$, then $\exists$ value of the infinitesimal α , for which the van der Pol system has a duck cycle such that x_β is the value on the x -coordinate corresponding to the “beak” of the “duck”.

The point of the exercise is that a knowledge of the possibilities for exploring a dynamic system with parameters and variables taking *infinitesimal* and *infinite* values is indispensable, not simply for reasons of pure mathematical aesthetics, but also for eminent economic reasons, where financial market variables move infinitely fast, at least relative to real variables; reactions in market sentiments to infinitesimal variations in parameters is a nonnegligible factor in turbulent markets. An economist who is narrowly trained in standard mathematics will always have to resort to *ad hockeries* to handle the infinitesimal and the infinity—for example, in models capable of relaxation oscillations. Quite apart from aesthetics and pragmatics, it is also the case that the mathematics of nonstandard analysis is intuitively natural and much simpler, without all the artificial paraphernalia of the $\epsilon - \delta$ calisthenics.

But the purist may wonder whether the mathematics of the digital computer, on which the above simulations helped me “discover” counter intuitive dynamics, is faithful to the underpinning logic of algorithmic mathematics. Without the space to go into details, smooth or infinitesimal analysis (see, e.g. Bell 1998) is entirely based on

²⁷ This is only a sufficient condition, and the admissible curve is simply a formalization of the traditional cubic characteristic for the van der Pol equation. I conjecture that Duck Cycles can be shown to exist even without a cubic characteristic with, say, a characteristic of the form $\tau(e^x - 2)$. Such a form would have only one isolated maximum or minimum.

intuitionistic logic and its source is category theory. These two facts alone are sufficient to answer any skeptical purist.

Suppose, now, we are made aware of Ralph Abraham's conjectures about stability: "The ubiquity of structurally unstable motions . . . suggests that structural stability is not an appropriate concept for experimental systems. . . . Here we may hazard a conjecture: all natural systems are dynamically stable. In fact, we will probably evolve the definition of stability until this conjecture becomes true" (Abraham 1985, pp. 120–121; underlined emphasis in the original). Simultaneously we, as economists, recall Leontief's characteristically perceptive observation (Goodwin 1953, p. 68): "Professor Leontief does not accept [that instability is an unrealistic hypothesis] and maintains that we may utilize dynamical systems that are unstable throughout and cites capitalism as an example."

The question, then, is whether dynamic economics (aggregative or not), modeled as a (nonlinear) dynamic system, is a *natural system* (Abraham), an *experimental system* (Abraham), or an *empirical system* (Leontief). There is no a priori reason for any of these kinds of dynamic systems to be stable for observational, simulational, or experimental purposes (Plott and Smith 1999) – especially since it is easy to show that only dynamical systems incapable of being underpinned by any notion of maximization (Samuelson 1972, p. 12) are capable of computation universality and, hence, consistent with the standard assumption of rationality in economics. A concise summary of the relevant results follows:

Definition 24.14 *A dynamic system capable of computation universality is one whose defining initial conditions can be used to program and simulate the actions of any arbitrary Turing machine, and in particular that of a universal Turing machine.*

Proposition 24.15 *Dynamic systems characterizable in terms of limit points, limit cycles, or chaotic attractors, called "elementary attractors", are not capable of universal computation.*

Theorem 24.16 *There is no effective procedure to decide whether a given observable trajectory is in the basin of attraction of a dynamic system capable of computation universality.*

Claim 24.17 *Only dynamic systems capable of computation universality can generate behavior that cannot be encapsulated in, or rationalized by, any notion of maximization.*

Finally, the simulated dynamics, done on only a laptop computer, illustrate unambiguously the precepts for the synergy enunciated by Norman Zabusky with which I started this section: "The insights gained from attempting to bring these results into agreement can synergize the rate of improvement of models, algorithms, analytical methods, and experimental procedures: The two most important, unexpected, or surprising improvements were the generation, entirely by the simulation of an orthodox

dynamic system, of the counterintuitive notion of a trajectory attracted to an unstable manifold, justified by an alternative mathematical formalism. Economists, in particular, schooled for centuries in a particularly narrow vision of what Paul Samuelson quite perceptively referred to as the “dogma of stability” (Samuelson 1974, p. 10) in business cycle theorizing and almost totally ignorant of alternative mathematical formalisms, have much to gain from what I should henceforth refer to as the Zabusky Precept.²⁸

24.4 COMPUTATION, DISCRETIZATION, PROOF AND OTHER MATHEMATICAL INFELICITIES

[T]here is, strictly, no such thing as mathematical proof; that we can, in the last analysis, do nothing but *point*; that proofs are what Littlewood and I call *gas*, rhetorical flourishes designed to affect psychology, pictures on the board in the lecture, devices to stimulate the imagination of pupils. This is plainly not the whole truth, but there is a good deal in it.

G. H. Hardy

In their highly slippery discussion of the interaction between individual rational behavior and equilibrium, Lehtinen and Kuorikoski (2007) point out that “the equilibrium that is used to solve an analytical problem is based on mutual expectations,” which, in turn, require a resolution of an infinite regress in mutual expectations (pp. 310–311).²⁹ They continue, “The role of equilibrium is that of breaking this and thus enabling the derivation of a definite solution. In equilibrium, none of the agents has a unilateral incentive to change behaviour, and hence the equilibrium ‘determines’ how the agents will act. Computers cannot model such an infinite regress of expectations because, being based on constructive mathematics, they cannot handle it” (p. 311).

The non sequiturs in this paragraph are embarrassingly many, and covering one of these by enclosing “determines” within quotation marks does not hide the obfuscation. Essentially, L&K are groping toward a definition of a Nash equilibrium, but the economic environment and the basis of agents’ behavior are not fully and well enough

²⁸ I must hasten to add that what I have referred to here as the “counterintuitive notion of a trajectory attracted to an unstable manifold” is not an entirely new or fanciful notion for economics. It is a partial property of the famous Keynes-based Hicksian, piecewise linear, endogenous model of the trade cycle (cf. Sedaghat 1997).

²⁹ This section may appear to be motivated by the loose, dangerous, and thoroughly uninformed opinions—not reasoned arguments—concerning simulation, proof, and computation made in Lehtinen and Kuorikoski (2007), henceforth referred to as L&K. Although it is ostensibly motivated, by the themes in a thoughtless and unreflective paper replete with terrible inaccuracies of a conceptual, factual, and technical nature, I have chosen a place for it in this chapter partly because the substance of this section was written before I became aware of the absurdities in L&K. The content of the section may appear to be defined by the themes in L&K, but it was conceived and constructed independent of it.

specified for one to be very certain of this; the whole setup could easily be for some kind of more dynamic game-theoretic project, but let that pass. The key objection I have to the claim in this statement can be divided into two sub-objections:

- “Computers cannot model such an infinite regress” “based on constructive mathematics”³⁰ and
- Constructive mathematics “cannot handle” infinite regress.

I take these claims in the above order. First, it is plainly incorrect to write that computers cannot handle an infinite regress, particularly the kind of infinite regress they define (admittedly in a very loose way). They fail to point out that the infinite regress in expectation to which they refer, in economics and game theory, are “broken” by the utilization of one or another fixed-point theorem, usually a *nonconstructive* and *uncomputable*³¹ one. There are eminently respectable constructive and computable fixed points that can be utilized to break the infinite regress emerging from the potential indeterminacies of mutual expectations. In Velupillai (2007, , sec. 4) I have, in fact, devised and derived a perfectly well-defined rational expectations equilibrium using the standard mathematics of the computer, namely, recursion theory. Moreover, there is an eminently rigorous fixed-point theorem in constructive mathematics, derived and proved constructively on the basis of intuitionistic logic, by no less an authority than Brouwer (Brouwer 1952), which can be used to define the kind of equilibrium with the infinite regress of mutual expectations.

The caveat to these objections and my counter claims is, of course, the implication that one must go back to the proverbial drawing board and formalize the basic closures of economic theory—especially preferences and endowments, but technology, too, eventually—either in recursion theoretic terms or constructively. Both of these enterprises are feasible, have been achieved successfully, and are, then, entirely consistent with using either a computer running on recursion theoretic principles or on constructive mathematical foundations.³²

Second, it is simply not true that standard computers and their associated workings are “based on constructive mathematics”. In every standard working, computer, the mathematical basis is recursion theory (computability theory), if anyone cares to think deeply enough about it. It is not as if one cannot make a working computer on the basis of constructive mathematics, or even make the standard Turing machine realization implement programs written in a language adapting some version of constructive mathematics.³³

³⁰ Presumably, they mean “computer behavior”, i.e., the underlying *program* on the basis of which the computer processes data, whether numerical or not.

³¹ It is not clear that L&K understand the difference between recursion theory, or computability theory, and constructive mathematics.

³² For a representative, but not exhaustive, sample of such work, see Velupillai et al. (2011).

³³ See the eminently readable text by Nordström, Petersson, and Smith (Nordström et al. 1990) for an elegant and accessible introduction to Martin-Löf’s *type theory*, developed “with the aim of being a clarification of constructive mathematics” (ibid. p. 1). Several of the essays in Crosilla and Schuster

Third, it is absolutely false that constructive mathematics cannot handle infinite regress, especially the kind needed in the L&K framework of individual behavior. Even apart from this, it is entirely feasible to handle varieties of infinite regress within constructive mathematics, except that the kinds of infinities are more carefully defined and invoked and, therefore, the nature of economics in a constructive mode would be very different from the orthodox theory of individual rational behavior, equilibria (whether game-theoretic or not) and, above all, the associated solutions (particularly via existence proofs, typically of equilibria).³⁴

Incidentally, L&K, like many in economics, seem to contrast a *digital proof* with a so-called analytical proof. If what is meant by a “digital proof”, is a theorem that is provable by programming a digital computer, then every analytical proof, say, in standard textbooks on constructive analysis (Bishop 1967, Bishop and Bridges 1985, or Bishop and Cheng 1972)³⁵ is a digital proof. If by “proof” is meant, say, those sanctioned by intuitionistic logic only, then practically every so-called proof in almost any kind of formalized economic theory fails to be acceptable. Even if not underpinned by intuitionistic logic, in many varieties of constructive analysis (e.g., in Bishop (1967)) no appeal will be made to the *tertium non datur* in cases in which infinitary instances have to be considered. Hence, any proof of a theorem in mathematical economics or game theory derived with appeal to the Bolzano-Weierstrass theorem cannot and *will not be considered a valid proof existence*. By now many are aware that both the Nash equilibrium, as derived by John Nash, and the Arrow-Debreu equilibrium are theorems whose proofs are based on the Brouwer (or Kakutani) fixed-point theorem,³⁶ but few note that this is mainly because the so-called proofs invoked the Bolzano-Weierstrass theorem, which makes them unimplementable in a digital computer underpinned by computable, constructive, or nonstandard mathematics.

On the other hand, every proof is a logico-mathematical argument. Now, every valid computer program is a kind of logico-mathematical argument, but what kind of *logic* and which branch of *mathematics* underpin a computer program? Above, I have belabored this point with mention of constructive mathematics, recursion theory,

(2005), particularly chapters 1 and 6, are equally illuminative with respect to the kind of approach to programming practicable computers with program languages that was developed for the specific purpose of encapsulating constructive mathematics.

³⁴ Unfortunately, L&K compound the above infelicities with a further absurdity when they continue (2007, p. 311): “However, they [i.e., the computers] can be programmed to check for each possible strategy combination whether it constitutes an equilibrium.” How does a computer, whether based on recursion theory or on constructive mathematics, “check” for an equilibrium that is provably uncomputable and nonconstructive?

³⁵ See also the elegant and illuminative discussion in *Algorithm in Modern Mathematics and Computer Science* by Donald Knuth (Knuth 1981), where he states unambiguously (p. 94): “The interesting thing about [Bishop’s constructive mathematics] is that it reads essentially like ordinary mathematics, yet it is entirely algorithmic in nature if you look between the lines.”

³⁶ Of course Brouwer, forty years after he first derived the nonconstructive version of the fixed point theorem that bears his name, did derive a fixed-point theorem based on intuitionistic logic that had the express aim of avoiding any reliance on the Bolzano-Weierstrass theorem (see the first footnote in Brouwer 1952).

and intuitionistic logic. In this precise sense, therefore, every computer program is a proof in a strict mathematical sense. However, Tymoczko's discussion of surveyability of proofs suggests (Tymoczko (1979), p. 59) three characteristics of proofs: they should be *convincing*, *surveyable*, and *formalizable*. He goes on to claim that surveyability and formalizability "are the deep features" of a proof and that

It is because proofs are surveyable and formalizable that they are convincing to *rational agents*.

Surveyability and formalizability can be seen as two sides of the same coin. ... Can there be surveyable proofs that are not formalizable or formal proofs that cannot be surveyed? Are all surveyable proofs formalizable? ... Given any sufficiently rich theory, we can find a surveyable proof of a statement of that theory which has no formal proof. ...

Are all formalizable proofs surveyable? ... Here the answer is an easy no.

However, if we stop to think about this situation, it appears unlikely that this logical possibility can ever be realized. ...

In summary, although formal proofs outrun surveyable proofs, it is *not at all obvious* that mathematicians could come across formal proofs and *recognize them* as such without being able to survey them. (pp. 61–62, italics added)

One cannot let these interesting remarks pass unchallenged! First, who or what is a "rational agent"? It is entirely conceivable and formally demonstrable (see, e.g., Putnam 1975; Velupillai 2000, especially chap. 3; and Velupillai 2010, especially parts 2 & 4) that an effective characterization of the behavior of a rational agent in the sense of economic theory is formally equivalent to the computing activity of a Turing machine. Next, Tymoczko is admirably clear in defining the concept of formalizability and formal proof—by appealing to results in model and proof theory and to Gödel numberings of formal proofs considered as mathematical objects—but does not define or characterize the meaning, formal or not, of surveyability. In the case of surveyability he falls back on intuitive concepts such as "rational agents", "humanly surveyable", "recognition". Suppose, however, Tymoczko did formally define or characterize formally the notion of surveyability, in the same sense in which the intuitive notion of effective calculability was encapsulated in the formal notion of a Turing machine, the λ -calculus, or partial recursive functions, all formally equivalent to each other by the Church-Turing thesis. Then, it will be possible to show that a rational agent will not be able to recognize as surveyable the proof of some theorems by appealing to the halting problem for Turing machines. Tymoczko's admirable and informal discussion is valid—formally, of course—only on the basis of an invalid asymmetry between his way of defining, implicitly, the notion of formalizability and formal proofs but leaving to the intuitive domain the characterization of surveyability. This makes the rest of his philosophical arguments against accepting the Appel-Haken proof of the four-color theorem much less than formally convincing. There are many other ways in which I can cast rigorous doubts on Tymoczko's loose, allegedly philosophical arguments *against*

considering computer-assisted proofs as mathematical proofs, but this must suffice for the moment.³⁷ Within this context of the discussion of computer-aided proofs, L&K have thoughts about “program verification” but fail to state exactly what they mean by the term³⁸ It is not as if there are no rigorous formal definitions, even in graduate textbooks with impeccable credentials (e.g., Davis et al. 1994, p. 536). However, a serious discussion of program verification requires much more depth and understanding of the issue as in Platek’s (1990) almost pedagogic article: “It goes without saying that program verification is more difficult in practice than verifying an analytical proof: there are simply more factors that can go humanly wrong” (p. 320).

Concerning the repeated appeals to something undefined but called “humanly” doable, I—in my human, intellectual capacity—would be very happy to provide, for every “difficult in practice” computer-aided proof (or digital proof, if L&K give a formal definition of this term, as regarding nonconstructive analytical proofs), an equally difficult analytical proof, constructed by ordinary human beings, that has gone quite seriously wrong. Here, too, I must rely on one or more persons of saintly disposition to “sacrifice themselves to the tedium of eternal verification” of the validity of analytical proofs!³⁹ Just off the cuff, I have in mind something that is of concern for me in my own research on economic dynamics: the (in)famous example of Dulac’s theorem, which claims to have proved a theorem contributing to the resolution of the second part of

³⁷ Incidentally, everything that L&K write about program verification, in this context, never rises above the trivial, even the banal. They are, very clearly, without any grounding in the rich and burgeoning literature on program verification and its inherent formal undecidabilities. Here, too, they get away with loose claims, using even looser references, simply because they do not bother to state clearly (i.e., formally) exactly what they mean by “program verification.” On the other hand, every time they do try to define any concept rigorously, they make a mess of their claims, anyway. For example, it is all very well to refer to “De Millo, Lipton and Perlis, 1979” (De Millo et al. 1979), on p. 319, when suggesting that ‘program verification’ is an academically and financially unrewarding exercise. Quite apart from this being untrue—the financial and academic rewards for usable results regarding program verification are considerable, given its importance in cryptology, patent codification, and similar security-related fields—one would have expected L&K to refer also to the counter argument to De Millo et al. (1979), given with pungency and clarity by Fetzer (Fetzer 1988, p. 1062, italics added): “The fact that one or more persons of saintly disposition might sacrifice themselves to the tedium of eternal verification of tens of millions of lines of code for the benefit of the human race is beside the point. The limitations involved here are *not merely practical*. . . . In maintaining that program verification cannot succeed as a generally applicable and completely reliable method of guaranteeing the performance of a program, De Millo, Lipton and Perlis thus *arrived at the right conclusion for the wrong reasons*.” Actually, however, the nuanced discussion in De Millo et al. (1979) is far richer and more persuasive than the caricature of the message in it summarized by L&K.

³⁸ Except they state that (p. 319) “[i]t is also, in principle, possible to check computer codes for errors because from the *syntactic* perspective the code is comparable to mathematical symbolism.” They forget that program verification is a part of denotational *semantics* (see Davis et al. 1994). But even if we grant them this definition, what is the scope of “comparable”?

³⁹ Witness the brouhaha surrounding the recent claim by Vinay Deolalikar, in no less a medium than the World Wide Web, that he had solved one of the Clay Millennium Problems, that of resolving the $P \stackrel{?}{=} NP$ conundrum. One supposes that his claim was motivated entirely by a sense of intellectual achievement. But the hundreds who seem to have engaged themselves in verifying the validity of the proofs are obviously of a ‘saintly disposition’.

the sixteenth of Hilbert's famous twenty-three problems.⁴⁰ It was published by Dulac in 1923;⁴¹ it was more than half a century later that the errors in Dulac's original proofs were corrected by Yulij Ilyashenko and, independently, by J. P. Écalle (see, e.g., Dulac 1923; Perko 1991; chap. 3 and Ilyashenko and Yakovenko 1995).⁴²

More pertinently, I would like to provide two examples of "computer-aided" proofs, both executed with full cognizance of the difficulty of program verification but, at the same time, with rigorous and transparent criteria explicitly given to make sure that any factors that can go humanly wrong can be detected and corrected, if anyone wishes to do so. But more important, the first example shows the intimate way mathematical theory, experimental simulation, and deep numerical analysis were brought to bear to resolve a long-standing paradox. It is the very recent proof of the existence of the Lorenz attractor (Tucker 1999). In 1985 no less an authority on dynamic systems theory than Morris Hirsch (1985) observed regarding for the Lorenz system,⁴³

chaotic behaviour has not been *proved*. As far as I am aware, practically nothing has been proved about this particular system. . . . It is of no particular importance to answer this question; but the lack of an answer is a sharp challenge to dynamicists, and considering all the attention paid to this system, it is something of a scandal.
(p. 191, italics in original)

In the same volume in which Hirsch's article appeared, another distinguished dynamic system theorist, Ralph Abraham, added his nuanced opinion, in softer phrases, to this so-called scandal (Abraham 1985, p. 117; italics added):

The chaotic attractor of mathematical theory began with Birkhoff in 1916. The chaotic attractor of simulation experiment arrived with Lorenz in 1962. . . . The identification of these two objects has not yet succeeded, despite many attempts during the past twenty years. Of course, everyone (including myself) expects this to happen soon.

Abraham's take on the scandal was expressed in another way a little further down (p. 118; underlined phrase in original):

However, most of the time experimentalists observe not braids (rationally related frequencies) but quasi-periodic motions (apparently irrationally related frequencies). That is the quasi-periodic paradox. More than one scientists has lost faith in mathematics because of the ubiquity of this illegal motion in the natural world.

The most interesting point here is that the scientists lost faith in mathematics because *it* was not able to make sense of the simulation experimentalists' observation.

⁴⁰ The second part of Hilbert's Sixteenth Problem remains unsolved to this day.

⁴¹ I must confess that I have never actually read the original by Dulac, mainly because I am incapable of reading any intricate mathematical text in French. My own favourite text on this problem, which also gives a complete report on the rich Chinese tradition of research in this area, is the monograph by Ye Yan-Qian and his many collaborators (Ye and Others 1986).

⁴² Incidentally, Écalle's proof of Dulac's conjecture was constructive; see Écalle (1993).

⁴³ The Lorenz system is as follows: $\frac{dx}{dt} = -10x + 10y$; $\frac{dy}{dt} = 28x - y - xz$; $\frac{dz}{dt} = -\frac{8z}{3} + xy$

Now, the Lorenz system is the paradigmatic repository of the property that almost characterizes so-called chaotic dynamical systems: *sensitive dependence on initial conditions* (SDIC). In such a system, then, what can L&K mean by saying, “[I]n discretizations it is necessary to check that the computer model is presented in exactly the same way as the analytical model upon which it is based”? (p. 320) Since L&K require the presentation to be to (presumably) a digital computer, discretization presupposes that the “analytical model upon which it is based” is *continuous* in some rigorous, well-defined sense. However, what does “exactly the same way” mean? Do they mean that the presentation of the analytical model is to be in its original continuous form? For example, in the case of the Lorenz system, are they expecting the analyst, experimenter, simulator, or whoever to be able to use the digital computer to faithfully replicate the dynamics of the continuous time-space Lorenz system’s nonlinear dynamics—SDIC and all—exactly the same way? No serious experimenter, simulator, numerical analyst, or mathematically competent dynamic system theorist would forget that *the digital computer has its own way of truncating floating point representation of real numbers, depending on its built-in precision*. A system of nonlinear equations such as Lorenz’s, susceptible to the problems of SDIC, cannot, therefore, almost by definition, be represented exactly the same way, if we interpret the phrase in its obvious, intuitive way (for lack of a formal definition).

On the other hand, suppose we interpret “exactly the same way” to mean that the numerical method that is implemented on the digital computer to simulate, experiment with, or analyze the Lorenz system should be mathematically equivalent to it; then we must ask what mathematical equivalence entails. This is one kind of frontier research in the interface between nonlinear dynamics and theoretical numerical analysis, elegantly summarized in Stuart and Humphries (1996).

Another way to make sense of this thorny issue would be to construct a Turing machine equivalent of, in this case, the Lorenz system. Then, of course, the question becomes: What is the meaning of “Turing machine equivalent”? Again, a precise answer can be given (e.g., Velupillai (2000), Velupillai (1999)), so that one avoids the pitfalls of discretizations and the rounding errors due to the computer’s internal floating point representations and truncations.

Finally, there is the fairly straightforward alternative of using Interval analysis (see Moore 1966) for the numerical method that is implemented in the digital computer to analyze, experiment, or simulate the continuous time-space system (in this case, of course, the Lorenz system). It is this alternative that is chosen in Tucker’s computer-aided proof of the existence of the Lorenz attractor (Tucker 1999, especially pp. 1200–11).

But, surely, the existence of the Lorenz attractor should be a classic analytical proof, perhaps utilizing one or another (nonconstructive) fixed-point theorem. Why, then, this preoccupation with discretizations and presentations of exactly the same model to a digital computer? For the same reasons that the proof of the four-color theorem was achieved by Appel and Haken with the aid of a digital computer (for accessible but quite complete details, see Saaty and Kainen 1986, chap. 3). The parallels are more than simply the recourse to a digital computer to evaluate complex numerical

calculations. (Tucker 1999, p. 1199) begins with the classic mathematical method of an *Ansatz*, an intuitive hunch, that, it is hoped, be confirmed by the results of the complete analysis and necessary evaluations. The intuitive hunch is not a frivolous guess; it is an educated guess of the right starting point, based on a thorough knowledge of all possible aspects of an unsolved problem, in this case that of finding a correspondence between a mathematical object and an experimentally discovered one. In Tucker's *Ansatz*, normal form theory combined with rigorously implemented digital computations is brought to bear so as to obtain on getting the desired result. In deriving the normal form, an analytic change of coordinates leads to a classic small-divisor problem, which, in order to complete a necessary element of the analytic proof, requires the numerical evaluation of 19,386 low-order divisors.⁴⁴ It is here that the computer-aided part of the proof acts as a "scratch pad."

It is possible to identify the *Ansatz* in the Appel-Haken proof: it is a particularly well-informed probabilistic argument establishing, with *almost* absolute certainty, that there must exist some discharging procedure producing an unavoidable set all of whose configurations are reducible. That is, they showed that the computer-assisted reducibility proof was overwhelmingly likely to succeed" (Saaty and Kainen 1986, p. 83; italics added)⁴⁵

However, in these kinds of hybrid proofs, in which the analytic (usually nonconstructive) and the numerical or combinatorial elements are brought to bear on a procedure or a thought experiment, the dividing line between the two domains has to be carefully distinguished. In Tucker's case, "a change of variables . . . in a small cube centered at the origin, transforms the Lorenz equations . . . into a carefully selected normal form.... Inside the cube, we can then estimate the evolution of trajectories *analytically*, and *thereby we avoid the problem of having to use the computers in regions where the flow times are unbounded*:" (p. 1199, italics added). The construction of the small cube, via the change of variables, entails "an analytic change of coordinates," which introduces a small-divisor problem, all 19,386 of them, which then necessitates recourse to a digital computer and to interval analysis to numerically compute their estimates. The versatile Tucker wrote a "small C-program, SMALLDIV.C" (ibid. p. 1200) to estimate these small divisors.

An illustration of this point is made in Ruelle's report of one of Oscar Landford's computer-aided proofs (Ruelle 2007, p. 100 emphasis added):

My colleague Oscar Landford reported once on a theorem [whose] proofs [were] computer aided, which means that it consists of some mathematical preliminaries and then a computer program. The program (or code) uses *interval arithmetic* to

⁴⁴ The knowledgeable reader would immediately recognize the similarity with the origins of what eventually became the celebrated Kolmogorov-Arnold-Moser (KAM) theorem in dynamical systems theory. Small divisors, quasi-periodic orbits, perturbations (of Hamiltonians)—all issues I have had to mention in my various discussions, above—play significant parts in the motivation and the eventual formalization of Kolmogorov's original conjecture.

⁴⁵ This was after having first produced 1,936 reducible configurations, at least one of which had to occur in any planar triangulation.

check various inequalities; if these are found to be correct, the theorem is proved. The complications of the problem forced Landford to write a relatively long program, about 200 pages. . . . Oscar Landford is a very careful person, and he took pains to check that, when the code is fed into the computer, the computer does exactly what it is supposed to do. In this manner—after the computer has agreed with the inequalities in the code—the proof of the theorem is complete.⁴⁶

Scarf's elegant, clear, and complete exposition of the genesis of the CGE research program (Scarf 1973), admirable though it is— it resides as the fountainhead of the core of current orthodoxy in the neo-Walrasian cloisters, the Real Business Cycle (RBC) model's Recursive Competitive Equilibrium (RCE)—simply does not confront the conflict between the analytical and the constructive or computable domains. The interplay between the analytical and the numerical was bridged by the Uzawa equivalence theorem, and the parallels with discharging procedures, unavoidable sets, and reducibility can be identified with the *construction* of a specific sequence of primitive sets, replacement operations, labeling, and so on. In fact, a study of the precise identity of the computer-aided nature of the establishment of Scarf's theorem (ibid. p. 45, theorem 2.5.1) and its utilization in demonstrating the original Brouwer fixed-point theorem would be the starting point for finding a way to reduce the remaining indeterminacy in this research program: the constructive or computable determination of "a convergent subsequence of subsimplices. . . which tend in the limit to a single vector x^* " (p. 51). The missing link is an *imaginative* Ansatz.⁴⁷ In its absence, the CGE program, followed by its uncritical application by the AGE practitioners and then taken up even more uncritically by the RBC theorists, remains unfinished because, according to Scarf (1973), "The passage to the limit is the nonconstructive aspect of Brouwer's theorem, and we have no assurance that the subsimplices determined by a fine grid of vectors on [the price simplex] *contains or is even close to a true fixed point of the mapping*" (p. 52, italics added).

Yet the whole program has been accepted as having been successful in determining constructive and computable methods to locate Walrasian equilibria, proved to exist by Arrow and Debreu, of course, nonconstructively. This magic transformation of a non-constructively derived uncomputable equilibrium, via an algorithm that appeals to an

⁴⁶ In the spirit of complete honesty and candor with which I have written this chapter, in fairness to the skeptics of the mathematical purity of computer-aided proofs, I must inform the reader of the following fact: the continuation of the above-quoted paragraph by Ruelle may be rather disheartening to people like me, who believe that such proofs are on an equal footing, mathematically speaking, with so-called analytical proofs.

⁴⁷ The Ansatz will have to find a way either to avoid any appeal to the Bolzano-Weierstrass theorem or to work directly with constructive mathematics without undecidable disjunctions. In fact, Brouwer's intuitionistically corrected proof of his original theorem (Brouwer 1952) is the solution—but only if the foundations of the theory developed in the neo-Walrasian cloisters is redone in terms of constructive mathematics. My own intuition is that ordinary economic theory, formalized on $\mathbb{N}$, $\mathbb{Q}$, or $\mathbb{Z}$, would avoid reliance on fix point theorems for the proof of equilibrium and, hence, would be amenable to a fruitful interaction of the analytic and the combinatorial to prove, with the aid of the digital computer, the existence of an equilibrium.

undecidable disjunction during its execution, is uncritically accepted by neo-Walrasian and is taken to define—implicitly, of course—the “perfect model” of the economist. This kind of economist knows, perhaps instinctively, that there is no point in simulating anything, using a nonconstructive algorithm, to find an uncomputable equilibrium.

Finally, it may be apposite to remind L&K of another aspect of computer-aided proofs: the candor and care with which those who appeal to the computer, at any particular stage of a proof, make available the codes and the kind of Ansatz that may have forced them to seek the aid of the computer, so that any interested person could examine the procedures adopted in the interface and by the computer. How many analytical proofs are made transparent in this way? How many years has it taken to “prove” Ramanujam’s results?

24.5 REFLECTIONS AND BRIGHT HOPES

The vital discovery which made possible the analysis of a process of change, in properly economic terms, was the introduction of accounting procedure. While economists were fumbling around to find a set of categories by which they could make a formal analysis of economic change, other people were doing the job in a professional manner. In all its main forms, modern economic dynamics is an accounting theory. It borrows its leading concepts from the work which had previously been done by accountants (with singularly little help from economists); and it is in accordance with this that social accounting should be its main practical instrument of application.

J. R. Hicks

It was fitting that Hicks, who considered himself an accountant among economists, made these observations in the *Festschrift* for Lindahl, himself the founding father of macroeconomics underpinned by social accounting. There is no better exponent of this particular vision of modern economic dynamics as an accounting theory and social accounting as a main instrument of application than Lance Taylor.

It is in this particular sense that Taylor, I believe, carries on the noble tradition of Petty, the Swedes of the period of Shackle’s High Theory (in particular, Lindahl, Myrdal, and Svernilsson) and the parallel work of Frisch in the same period, inherited by Leif Johansen, and the Keynes of *How to Pay for the War* (Keynes 1940), from which the political arithmetic of Stone emerged.⁴⁸ It was profoundly misleading for the computational economists who underpinned their economics with equilibrium and

⁴⁸ The third of the many books Sir John Hicks wrote (Hicks 1942) was titled *The Social Framework: An Introduction to Economics*, of which the two chapters of Part 5 are entirely devoted to social accounting systems. Hicks began writing it around the time *How to Pay for the War* was completed and Stone began his work, under Keynes, in putting together the accounts to implement the Radical Plan for the Exchequer advocated in it. The Swedes had a head start on this, and Hicks was intimately aware of the Swedish work on social accounting as an underpinning for the emerging field of

rationality to seek to claim they were working in the traditions of Leif Johansen, who was balancing accounts in a social accounting framework.

The numbers that make up a social accounting system—whether in formal matrix format or in any other scheme that allows the accounting balance, over time, of credits and debts, retrospectively or prospectively—are, at best, *rational numbers*. The founding father of political economy, William Petty, laid the foundations for political arithmetic, in the opening statement of his *Political Arithmetick*.⁴⁹

The Method I take to [carefully examin[e]] whatever tends to lessen my hopes of the publick Welfare, is not yet very usual. . . for instead of using only comparative and superlative Words, and intellectual Arguments, I have taken the course (as a Specimen of the Political Arithmetick I have long aimed at) to express my self in Terms of Number, Weight, or Measure; to use only Arguments of Sense, and to consider only such Causes, as have visible Foundations in Nature; leaving those that depend upon the mutable Minds, Opinions, Appetites, and Passions of particular Men, to the Consideration of others.

The natural data types of the numbers, weights, and measures in economics and finance are the integers or rational numbers. But almost without exception all theorising in these fields assumes that observable data or even digitally generated data from these two spheres are real numbers (or, in moments of enlightened weakness, one or another kind of the nonstandard reals). How the real numbers of theory are related to the numbers that a digital computer can process is never specified, at least not in any of the standard advanced (or elementary) textbooks in economics, finance, IO, game theory, or other areas. Maury Osborne's (1995) warning to traders in the stock market not to approximate by the continuous that which is intrinsically discrete⁵⁰ has never been heeded by the traders nor, worse, those who sometimes provide the *Hilfskonstruktion* of computer programs for their mechanized responses to actual events, represented by the ostensibly patterned data on computer screens:

As for the question of replacing rows of closely spaced dots by solid lines, you can do that too if you want to, and the governors of the exchange and the community of brokers and dealers who make markets will bless you. If you think in terms of solid lines while the practice is in terms of dots and little steps up and down, this misbelief

macroeconomics. I want to add here that social accounting is one thing; so-called stock-flow-consistent modeling is quite another animal. Whereas the latter is nontrivial, which is a rare event, it is part of some meaningful social accounting system. Lindahl and Myrdal, in their work in the 1920s and 1930s developing what later came to be called macroeconomics, knew and worked within the discipline such awareness entailed.

⁴⁹ I do not know why, when, and how the *k* in *Political Arithmetick* disappeared!

⁵⁰ I want to make it crystal clear that nowhere have I argued that all economic theorizing should be in terms of so-called discrete models, nor have I ever supported the ridiculous suggestion that continuity is irrelevant in economic theorizing. Nothing in the kind of economics called computable economics, which I practice, has anything to do with these kinds of absurd claims. I add this caveat to circumvent assertions, as in Durlauf (2007), regarding the former, and observations by several of my senior colleagues in the department of economics at the University of Trento.

on your part is worth, I would say conservatively, to the governors of the exchange, at least eighty million dollars per year. (p. 34)

Brian Hayes (Hayes 2003) reminded us—at least those of us concerned with respecting the discrete and finite nature of digital computers—of the dangers of arbitrary approximations and routinized truncations of standard computations:

On February 25, 1991, a Patriot missile battery assigned to protect a military installation at Dahrahn, Saudi Arabia, failed to intercept a Scud missile, and the malfunction was blamed on an error in computer arithmetic. ... In combination with other peculiarities of the control software, the inaccuracy caused a miscalculation of almost 700 meters in the predicted position of the incoming missile. Twenty-eight soldiers died. (p. 484)

What was this tragic “error in computer arithmetic”? It is simply that the binary fraction for the decimal fraction $10^{-1} = 0.1$ is *not terminating*:

$$10^{-1} = (0.1)_{10} = (.0001100110011...)_{2} = (0\ 0011\ 0011\ 0011\ ...)_{2} \quad (24.20)$$

In other words, the decimal fraction, in its binary notation, cycles and is nonterminating and will have to be truncated with unpredictable consequences, unless a serious approximation analysis is included in the software that truncates automatically for some predetermined instruction. But there is another alternative, in the case of pitfalls due to the discrete and finite nature of the digital computer and its arithmetic. This alternative would be to use interval analysis, in which an ‘interval of real numbers is treated as a new kind of number, represented by a pair of real numbers, namely its right and left end points’ (Moore 1966, p. vii). Had such numbers been used in the software that was built into the operation of the control software referred to above, the error would have been eliminated.

There are, in fact, (at least) six types of mathematical theorizing in which a computational basis is naturally embedded:

- varieties of constructive mathematics and constructive analysis (Bridges and Richman 1987)
- computability theory and varieties of computable analysis (Cooper 2004; Weihrauch 2000)
- interval analysis (Russian constructivism; Aberth 2001; Moore 1966; Moore et al. 2009)
- real computation (the “Smale School”; Blum et al. 1998)
- smooth (infinitesimal) analysis (Bell 1998)
- numerical analysis (Blum et al. 1998; Stuart and Humphries 1996)

In a series of works I have (see, in particular, the introduction to Velupillai et al. 2011, many of the essays in Velupillai 2010 and Velupillai 2009) reflected on the nuanced differences between these mathematical theories and their applications in economics. The interested reader is referred to them for more detailed information

on these mathematical theories, their alternative logical foundations, the nature of their algorithmic underpinnings, and much else—for example, the role (or non-role, as the case may be) of the Church-Turing thesis in the different theories. If economic theorizing eschews an underpinning in one or another of these mathematical theories, the computational claims and, hence, the feasibility and meaning of *simulation* by machine computation to explore their non-analytical aspects borders on the absurd (in the noble sense of *The Theatre of the Absurd*, *pace* Esslin).

I have argued that economic relationships, whether micro or macro, whether in game-theoretic or IO modes, whether in finance or interindustrial analysis, should be formulated as Diophantine equations. Such equations, or systems of equations, are rarely algorithmically solvable, in general, without extensive simulational experiments which lead to new theretical insights, encapsulated in innovative conceptual analysis. Essentially, the case I am putting forward is that the framework developed for studying and resolving Hilbert's tenth problem should be the paradigmatic one for any research field in economics whose theretical basis is underpinned by digital machine computation. Consider, for example, the following three apparently simple Diophantine equations (Poonen 2008, p. 344), and ask the seemingly straightforward question whether they have solutions in integers:

$$X^3 + Y^3 + Z^3 = 29 \quad (24.21)$$

$$X^3 + Y^3 + Z^3 = 30 \quad (24.22)$$

$$X^3 + Y^3 + Z^3 = 33 \quad (24.23)$$

To date, we know that the answer is in the affirmative for (24.21) and that for (24.22), that the answer is yes has been known only since 1999 (the solution is: -283059965 , -2218888517 , 22204229320). However, (24.23) is still to be cracked! Of course, in a strictly economic context, interpreting the right-hand side as the available 'budget' for spending and the left-hand side as the allocation of spending on three items—without bringing into play the motivational base for the alternative choices—it is clear that one seeks solutions that are not only integer valued but also nonnegative. One must ask, in this context, why the mathematization of economics, which claimed to go beyond the "Walrasian" method of simply counting equations and variables to ensure consistency and meaningful solutions, did not go further than considering nonnegative real values for economic variables. I have my own answers to this quasi-hypothetical question, but this is not the place to express and attempt to justify it.

In any case, continuing the theme of advocating a Diophantine approach to economic theorizing and modeling, it may well be apposite at this point to state the Diophantine conundrums of dynamics from the point of view of computability, and hence the importance of simulation to help resolve some of the conundrums. Indeed, to take up the subject discussed above, the Peano existence theorem for the IVP of ODEs, one of the frontier research results in applied recursion theory, is the following (see Matiyasevich 1993, chap. 9):

Theorem 24.18 *There is no effective method for determining, for an arbitrary system of differential equations of the form*

$$\begin{aligned} P_1(x, \Xi_1(x), \dots, \Xi_k(x), \Xi'_1(x)) &= 0 \\ &\dots \\ P_k(x, \Xi_1(x), \dots, \Xi_k(x), \Xi'_k(x)) &= 0 \end{aligned} \quad (24.24)$$

where $P_1, \dots, P_k$ are polynomials with integer coefficients, whether the system has a solution on the interval $[0, 1]$.

This is just one representative result, in an important applied domain, derived using a uniform method of proof.

Essentially, this means that one has to resort to what I have called the Zabusky precept to breach the algorithmic walls and make conceptual advances in economics in the Diophantine mode. Nothing less than simulation by machine computation of Diophantine dynamics should be the research program of economics.

Richard Stone was prescient, as he always had the penchant to be, when dealing with issues that had to do with computations, simulations, numerical collation of statistical data, and quantitative modelling: “Our approach is quantitative because economic life is largely concerned with quantities. We use computers because they are the best means that exist for answering the questions we ask. It is our responsibility to formulate the questions and get together the data which the computer needs to answer them” (Stone and Brown 1962, p. viii).

If we supplement this admonishment to the empirically oriented economic theorist, with David Deutsch’s equally important and eminently rigorous insight, keeping in mind the Zabusky precept, we as economists are well on the way to an empirically implementable algorithmic research program encompassing the triad of computation, simulation, and dynamics indissolubly. According to the Turing principle as annunciated by Deutsch (1985, p. 9): “Every finitely realizable physical system can be perfectly simulated by a universal model computing machine operating by finite means.”

Deutsch enunciated the Turing principle on the basis of a searching analysis of the meaning of the Church-Turing thesis. He came to the conclusion that underpinning the thesis was a *physical principle*, which he stated as the Turing principle. Naturally, this principle, as given above, requires a precise statement of what is to be meant by “perfectly simulated.” Deutsch, being the serious scientist he is, did not forget to add a definition (ibid. p. 99):

Definition 24.19 *“A computing machine M is capable of perfectly simulating a physical system S , under a given labelling of their inputs and outputs, if there exists a program $\Pi(S)$ for M that renders M computationally equivalent to S under the labelling. In other words, $\Pi(S)$ converts M into a ‘black box’ functionally indistinguishable from S .”*

Much of my discussion has had as a backdrop the precise theory of computation, which is underpinned by the Church-Turing thesis. But I have also consciously adopted—although, I hasten to add, only *pro tempore*⁵¹—Deutsch's important extensions, all of which seem to be consistent with the stand taken also by (Gandy 1980), concerning the physical principles that will have to be the basis on which the Church-Turing thesis is interpreted in computability theory. Even though I worked with an informal (but, it is hoped, precise) definition of “simulation”, my arguments have been made with the precise definitions of Deutsch in mind. It is, therefore, just as well that I state them precisely at this concluding stage.

It is my belief that economists with aims and ambitions to construct models and theorize with mathematical tools should be exposed to a variety of mathematics and the corresponding logical bases for them. To be taught mathematical economics as if real analysis and set theory are the be-all and end-all is absurd, especially when the next step is to use the mathematical models built on such foundations for computation by a digital computer, which is based on wholly different mathematical and logical principles: constructive mathematics and proof theory, on one hand, or the theory of computation and recursion theory, on the other, with category theory in the wings to help us dispense with set theory altogether.

From my own experiences in teaching and interaction with colleagues, I am painfully aware that economists are, in general, blissfully ignorant of any notion of limits to computation, even with ideal machines. But worse is the equally blissful ignorance which respect to the intrinsic limits to the results obtained with real analysis, underpinned by set theory plus the axiom of choice, let alone the impossibility of adapting such results from such domains, for computation on machines built on a wholly different mathematics—even with the most rigorous and careful notion of “approximation”.

Economists have never shunned simulation. However, they may have misused it, perhaps owing to a misunderstanding of the notion, nature, and limits of computation, even by an ideal machine. Engineers do not attempt to design perpetual motion machines that violate the laws of thermodynamics or mechanics, although cranks, over the centuries, have claimed to have done so; most of the models emanating from work in economic theory belong to *the Museum of Unworkable Devices*,⁵² at least when viewed from the vantage point of constructive mathematics or recursion theory, that is, from the point of view of computation. How those in the neowalrasian cloisters make their unworkable devices perform the tasks that need to be done just for survival is beyond our commonsense comprehension.

⁵¹ This is in part because I am *not* completely convinced that the particular physical principle that Deutsch derives and states as the Turing principle encapsulates entirely, for example, Gandy's Principles for Mechanisms. It is the latter that I have usually worked with and have referred to it by that name. Of course, this also requires a precise definition of simulation, which turns out to be slightly more complicated and lengthy to formulate than the admirably succinct definition derived by Deutsch. A deep and persuasive critique of Deutsch's Turing principle can be found in Timpson (2004).

⁵² See the illuminating website dedicated to the Museum of Unworkable Devices:
<http://www.lhup.edu/~dsimanek/museum/unwork.htm>.

Surely a strong case can be made for making economists, at least at the level of graduate pedagogy, aware of the Museum of Unworkable Devices and the Association for the Study of Failure. An imposing catalogue of unworkable devices and their failures can easily be composed entirely of the products of the neowalrasian cloisters, without any mediation from constructive mathematics or recursion theory.

Only common sense of a universal variety and the ability to think logically (preferably, but not necessarily, along the natural lines outlined by the Brouwerian intuitionists) are required to understand and work with the above-mentioned concepts, all of which are elementary in a deep mathematical sense. I have never found any advanced undergraduate or graduate student of reasonable maturity to have had any difficulty with understanding the case I make for an economic theory framed in a mathematics that can handle these concepts. Since an economic theory that encapsulates the possibility of, say, a computationally universal dynamical system can only be explored by *actual simulation*—“to collect specimens, to describe them with loving care, and to cultivate them for study under laboratory conditions” (Temple 1958, p. 233)—of the relevant system, it is natural for such students to realize that there is a wholly different world of economics than the one peddled by the purveyors of the ideas and tools emanating from the neowalrasian cloisters.

No one equipped with the concepts discussed above and their mathematical and epistemological underpinnings would dream of thinking that bounded rationality is some special subset of the economist's notion of rationality, the quintessential unworkable device. No one who understands the ubiquity of non-maximum dynamic systems and understands the notion of computation universality would try to anchor a norm in equilibrium dynamics. No student of economics equipped with these concepts would feel comfortable in the neo-Walrasian cloisters, themselves located in Cantor's paradise. It may be apposite to end this long essay, a paean (I hope) to the triad of computation, simulation, and dynamics in their epistemological settings, with the thoughts of two giants of twentieth-century mathematics and philosophy, David Hilbert and Ludwig Wittgenstein: Hilbert (1926), p. 191: “No one shall drive us out of the paradise which Cantor has created for us.” Wittgenstein (1974), p. 103: “I would say, ‘I wouldn't dream of trying to drive anyone out of this paradise. I would try to do something quite different: I would try to show you that it is not a paradise—so that you'll leave of your own accord. I would say, You're welcome to this; just look about you.’”

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and friend Richard Goodwin, who underlined the importance of computation and simulation in nonlinear, interdependent, multisectoral economic analysis in almost all his professional writings and teaching. Joe McCauley's works in nonlinear dynamics and financial market dynamics—the latter of which followed the tradition of Maury Osborne (Osborne 1995)—have been important inspiration for my work and belief in synergy, in particular, the significance he has increasingly come to attach to Poincaré's "recurrence theorem" (McCauley 1993, chap. 3; McCauley 2009). This theorem, implicitly and explicitly, lies at the heart of Fermi's inspiration in formulating what came to be known, justly and famously, as the Fermi-Pasta-Ulam Paradox (Fermi et al. 1955; Ford 1992) and the genesis of simulation in dynamics (Weissert 1997). This paradox *should* be my paradigmatic case study for substantiating a belief in the validity of an extended synergetics (between *simulation*, *computation*, and *analysis*), but space limitations constrain me. Finally, nothing reported in this paper would have been possible without the sterling intellectual support of my ASSRU colleagues and friends, Stefano Zambelli, Selda Kao, and Ragu Ragupathy. They are not absolved from responsibility for remaining infelicities; all the others are.

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